Exposé I. Presheaves

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In numbers 0 through 5 of this exposé we present the elementary, and most often well-known, properties of categories of presheaves.[1] These properties are used constantly throughout the rest of the seminar, and knowing them is essential for understanding the subsequent exposés. The proofs are immediate; for the most part they are omitted. In numbers 6 through 9 we take up a few themes that are used several times later on. The reader in a hurry may omit them on a first reading. Number 10 fixes the terminology used. Appendix 11 is due to N. Bourbaki.

1 Universes

A *universe* is a nonempty set[2] $\mathcal{U}$ satisfying the following properties:

($\mathcal{U}$1) If $x \in \mathcal{U}$ and $y \in x$, then $y \in \mathcal{U}$.

($\mathcal{U}$2) If $x, y \in \mathcal{U}$, then $\{x, y\} \in \mathcal{U}$.

($\mathcal{U}$3) If $x \in \mathcal{U}$, then $\mathcal{P}(x) \in \mathcal{U}$.

($\mathcal{U}$4) If $(x_i)_{i \in I}$ is a family of elements of $\mathcal{U}$ and $I \in \mathcal{U}$, then $\bigcup_{i \in I} x_i \in \mathcal{U}$.

From the preceding axioms one easily deduces the following properties:

- if $x \in \mathcal{U}$, then the singleton $\{x\}$ belongs to $\mathcal{U}$;
- if x is a subset of some $y \in \mathcal{U}$, then $x \in \mathcal{U}$;
- if $x, y \in \mathcal{U}$, the ordered pair $(x, y) = \{\{x, y\}, x\}$, in Kuratowski's definition, is an element of $\mathcal{U}$;
- if $x, y \in \mathcal{U}$, the union $x \cup y$ and the product $x \times y$ are elements of $\mathcal{U}$;
- if $(x_i)_{i \in I}$ is a family of elements of $\mathcal{U}$ indexed by an element $I \in \mathcal{U}$, then the product $\prod_{i \in I} x_i$ is an element of $\mathcal{U}$;
- if $x \in \mathcal{U}$, then $\operatorname{card}(x) < \operatorname{card}(\mathcal{U})$ strictly. In particular, the relation $\mathcal{U} \in \mathcal{U}$ is not satisfied.

Thus one may perform all the usual operations of set theory starting from elements of a universe without the final result ceasing to be an element of that universe.

The first use of the notion of universe is that it supplies a definition of the usual categories: the category of sets belonging to the universe $\mathcal{U}$ ($\mathcal{U}$-**Set**), the category of topological spaces belonging to $\mathcal{U}$, the category of commutative groups belonging to $\mathcal{U}$ ($\mathcal{U}$-**Ab**), the category of categories belonging to $\mathcal{U}$, and so on.

However, the only known universe is the set of symbols of the type $\{\{\varnothing\}, \{\{\varnothing\}, \varnothing\}\}$, etc. All the elements of this universe are finite sets, and this universe is countable. In particular, no

[1] The reader may also consult SGA 3, I, §§1–3.

[2] This definition contradicts Bourbaki's appendix below. However, the interest of empty universes seems debatable.

universe is known that contains an element of infinite cardinality. We are therefore led to add to the axioms of set theory the axiom

$(\mathcal{U}A)$ *For every set x there exists a universe $\mathcal{U}$ such that $x \in \mathcal{U}$.*

Since the intersection of a family of universes is a universe, it follows immediately that every set is an element of a smallest universe. One can show that axiom $(\mathcal{U}A)$ is independent of the axioms of set theory.

We shall also add the axiom:

$(\mathcal{U}B)$ *Let $R\{x\}$ be a relation and let $\mathcal{U}$ be a universe. If there exists an element $y \in \mathcal{U}$ such that $R\{y\}$, then*

The consistency of axioms $(\mathcal{U}A)$ and $(\mathcal{U}B)$ relative to the other axioms of set theory is not proved and, it seems, not provable.

Let $\mathcal{U}$ be a universe and let $c(\mathcal{U})$ be the least upper bound of the cardinals of elements of $\mathcal{U}$; thus $c(\mathcal{U}) \leq \operatorname{card}(\mathcal{U})$. The cardinal $c(\mathcal{U})$ has the following properties:

(FI) If $a < c(\mathcal{U})$, then $2^a < c(\mathcal{U})$.

(FII) If $(a_i)_{i \in I}$ is a family of cardinals strictly smaller than $c(\mathcal{U})$, and if $\operatorname{card}(I)$ is strictly smaller than $c(\mathcal{U})$, then $\sum_{i \in I} a_i < c(\mathcal{U})$.

Cardinals satisfying properties (FI) and (FII) are called *strongly inaccessible cardinals.*

The cardinal 0 and the countably infinite cardinal are strongly inaccessible.

Axiom $(\mathcal{U}A)$ implies:

$(\mathcal{U}A')$ *Every cardinal is strictly bounded above by a strongly inaccessible cardinal.*

One can show (11) that, conversely, the consistency of $(\mathcal{U}A')$ implies the consistency of $(\mathcal{U}A)$, and that the consistency of these axioms implies the consistency of axiom $(\mathcal{U}B)$.

Call a set E *Artinian* if there is no infinite family $(x_n)_{n \in \mathbb{N}}$ such that $x_0 \in E$ and $x_{n+1} \in x_n$. One can then show [*loc. cit.*] that there is a one-to-one correspondence between strongly inaccessible cardinals and Artinian universes, defined as follows: to every strongly inaccessible cardinal c one associates the unique Artinian universe $\mathcal{U}_c$ such that

$$\operatorname{card}(\mathcal{U}_c) = c.$$

If $c < c'$ are two strongly inaccessible cardinals, then

$$\mathcal{U}_c \in \mathcal{U}_{c'}.$$

In particular, the Artinian universes of cardinal smaller than a given cardinal form a well-ordered set for the membership relation. Axiom $(\mathcal{U}A')$ is equivalent to the axiom:

$(\mathcal{U}A'')$ *Every Artinian set is an element of an Artinian universe.*

We note that all the usual sets $(\mathbb{Z}, \mathbb{Q}, \ldots)$ are Artinian sets.

2 $\mathcal{U}$-categories. Presheaves of sets

1.0

In the rest of the seminar, unless the contrary is expressly stated, the universes considered will possess an element of infinite cardinality. Let $\mathcal{U}$ be a universe. A set is said to be *$\mathcal{U}$-small*, or simply *small* when no confusion can result, if it is *isomorphic* to an element of $\mathcal{U}$. We also use the terminology small group, small ring, small category,[3] etc. We shall often suppose, without explicit mention, that the schemes, topological spaces, indexing sets, etc. with which we work are in $\mathcal{U}$, or at least have cardinality in $\mathcal{U}$; however, many of the categories with which we shall work will not themselves be in $\mathcal{U}$.

Definition (1.1)**.** Let $\mathcal{U}$ be a universe and let C be a category. We say that C is a $\mathcal{U}$-category if, for every pair (x, y) of objects of C, the set $\mathrm{Hom}_C(x, y)$ is $\mathcal{U}$-small.

1.1.1

Let C and D be two categories, and let $\mathrm{Fun}(C, D)$ denote the category of functors from C to D. The following assertions are immediate:

(a) If C and D are elements of a universe $\mathcal{U}$ (respectively are $\mathcal{U}$-small), then the category $\mathrm{Fun}(C, D)$ is an element of $\mathcal{U}$ (respectively is $\mathcal{U}$-small).

(b) If C is $\mathcal{U}$-small and D is a $\mathcal{U}$-category, then $\mathrm{Fun}(C, D)$ is a $\mathcal{U}$-category.

Remark (1.1.2)**.** Let D be a category satisfying the following properties:

(C1) the set $\mathrm{Ob}(D)$ is contained in the universe $\mathcal{U}$;

(C2) for every pair (x, y) of objects of D, the set $\mathrm{Hom}_D(x, y)$ is an element of $\mathcal{U}$.

The usual categories constructed from a universe $\mathcal{U}$ have these two properties: $\mathcal{U}$-**Set**, $\mathcal{U}$-**Ab**, and so on. Let C be a category belonging to $\mathcal{U}$. Then the category $\mathrm{Fun}(C, D)$ does not in general have properties (C1) and (C2). For example, the category $\mathrm{Fun}(C, \mathcal{U}\text{-}\mathbf{Set})$ has neither property (C1) nor property (C2). This justifies the definition adopted for a $\mathcal{U}$-category, rather than the more restrictive notion given by conditions (C1) and (C2) above.

Definition (1.2)**.** Let C be a category. The category of presheaves of sets on C relative to the universe $\mathcal{U}$—or, when no confusion can result, the category of presheaves on C—is the category of contravariant functors on C with values in the category of $\mathcal{U}$-sets.

We denote by $\widehat{C}_{\mathcal{U}}$, or more simply by $\widehat{C}$ when no confusion can result, the category of presheaves of sets on C relative to the universe $\mathcal{U}$. The objects of $\widehat{C}_{\mathcal{U}}$ are called $\mathcal{U}$-presheaves, or simply presheaves, on C. When C is $\mathcal{U}$-small, the category $\widehat{C}_{\mathcal{U}}$ is a $\mathcal{U}$-category. When C is a $\mathcal{U}$-category, $\widehat{C}_{\mathcal{U}}$ is not necessarily a $\mathcal{U}$-category.

Construction–Definition (1.3)**.** Let x be an object of a $\mathcal{U}$-category C. The *$\mathcal{U}$-functor represented by x* is the functor

$$h_{\mathcal{U}}(x) : C^{\mathrm{op}} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

constructed as follows.[4] Let y be an object of C.

[3] A category C is viewed as a set of arrows, equipped with the subset of objects, viewed as the set of identity arrows, and with the source and target maps. Thus the expressions 'C is an element of $\mathcal{U}$' and 'C is $\mathcal{U}$-small' make sense.

[4] C^{op} will always denote the category opposite to C.

(a) If $\mathrm{Hom}_C(y,x)$ is an element of $\mathcal{U}$, then

$$h_{\mathcal{U}}(x)(y) = \mathrm{Hom}_C(y,x).$$

(b) Suppose that $\mathrm{Hom}_C(y,x)$ is not an element of $\mathcal{U}$, and let $R(Z,x,y)$ be the relation: 'The set Z is the target of an isomorphism $\mathrm{Hom}_C(y,x) \xrightarrow{\sim} Z$.' We then put

$$h_{\mathcal{U}}(x)(y) = \tau_Z R(Z).$$

By axiom $(\mathcal{U}B)$, $h_{\mathcal{U}}(x)(y)$ is an element of $\mathcal{U}$. Let $R'(u,x,y)$ be the relation: 'u is a bijection

$$u : \mathrm{Hom}_C(y,x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y)'.$$

We put

$$\varphi(y,x) = \tau_u R'(u).$$

Notice that in both cases (a) and (b) we have a canonical isomorphism

$$\varphi(y,x) : \mathrm{Hom}_C(y,x) \xrightarrow{\sim} h_{\mathcal{U}}(x)(y).$$

In case (a), $\varphi(y,x)$ is the identity. Now let $u : y \to y'$ be an arrow of C. Composition of morphisms defines a map

$$\mathrm{Hom}_C(u,x) : \mathrm{Hom}_C(y',x) \longrightarrow \mathrm{Hom}_C(y,x).$$

We set

$$h_{\mathcal{U}}(x)(u) = \varphi(y,x)\ \mathrm{Hom}_C(u,x)\, \varphi(y',x)^{-1}.$$

One verifies immediately that $h_{\mathcal{U}}(x)$, so defined, is a functor $C^{\mathrm{op}} \to \mathcal{U}$-**Set**.

1.3.1

Similarly, using the isomorphisms φ, for every morphism $v : x \to x'$ one defines a morphism of functors

$$h_{\mathcal{U}}(v) : h_{\mathcal{U}}(x) \longrightarrow h_{\mathcal{U}}(x'),$$

and one verifies immediately that this defines a functor

$$h_{\mathcal{U}} : C \longrightarrow \widehat{C}_{\mathcal{U}}.$$

1.3.2

Let $\mathcal{V}$ now be a universe such that $\mathcal{U} \subset \mathcal{V}$. We then have a canonical fully faithful functor

$$\mathcal{U}\text{-}\mathbf{Set} \hookrightarrow \mathcal{V}\text{-}\mathbf{Set},$$

and hence a fully faithful functor

$$\widehat{C}_{\mathcal{U}} \hookrightarrow \widehat{C}_{\mathcal{V}}.$$

The diagram

$$\begin{tikzcd} C \arrow[r, "h_{\mathcal{U}}"] \arrow[dr, "h_{\mathcal{V}}"'] & \widehat{C}_{\mathcal{U}} \arrow[d] \\ & \widehat{C}_{\mathcal{V}} \end{tikzcd}$$

commutes up to canonical isomorphism. When C is an element of $\mathcal{U}$, this canonical isomorphism is the identity.

1.3.3

In practice, the universe $\mathcal{U}$ is fixed once and for all and is not mentioned. We then use the notation $\widehat{C}$ for the category of $\mathcal{U}$-presheaves of sets and

$$h : C \longrightarrow \widehat{C}.$$

For every object x of C, the presheaf $h(x)$ is called the *presheaf represented by* x, and we shall always identify its value at an object $y \in \mathrm{Ob}(C)$ with $\mathrm{Hom}_C(y, x)$.

Proposition (1.4)**.** Let C be a $\mathcal{U}$-category, let F be a presheaf on C, and let X be an object of C. There is an isomorphism, functorial in X and in F,

$$i : \mathrm{Hom}_{\widehat{C}}(h(X), F) \xleftarrow{\sim} F(X),$$

where, when F is of the form $h(Y)$, the map i is just the map

$$h : \mathrm{Hom}(h(X), h(Y)) \longleftarrow \mathrm{Hom}(X, Y).$$

In particular, the functor h is fully faithful.

1.4.1

This proposition justifies the usual abuse of language that identifies an object of C with the corresponding contravariant functor. A presheaf isomorphic to an object in the image of h, or, using the above abuse of language, isomorphic to an object of C, is called a *representable presheaf*.

1.4.2

Let $\mathcal{V}$ be a universe containing a universe $\mathcal{U}$. The category of $\mathcal{U}$-presheaves is a *full* subcategory of the category of $\mathcal{V}$-presheaves; consequently, a $\mathcal{U}$-presheaf is representable if and only if its image in the category of $\mathcal{V}$-presheaves is a representable $\mathcal{V}$-presheaf.

Continuation anchor. The next untranslated source section is Exposé I, Section 2, ‘Projective and inductive limits’.

Exposé I. Presheaves

Sections 2–5

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 2–5, continuing the preceding translated fragment through Section 1.4. Obvious typographical slips in the working French TeX source have been corrected where the mathematics fixes the reading.

2 Projective and inductive limits

Let C be a $\mathcal{U}$-category and let I be a small category. Write $\mathrm{Fun}(I,C)$ for the category of functors from I to C. The category $\mathrm{Fun}(I,C)$ is a $\mathcal{U}$-category. To every object X of C associate the subcategory $\mathcal{X}$ of C having X as its only object and the identity of X as its only arrow. Let $i_X : \mathcal{X} \to C$ be the inclusion functor. There is a unique functor $e_X : I \to \mathcal{X}$, and we shall denote by $k_X : I \to C$ the functor $i_X \circ e_X$. We shall say that k_X is the constant functor with value X. The correspondence $X \mapsto k_X$ is visibly functorial in X, which enables us to define, for every functor $G : I \to C$, the presheaf

$$X \longmapsto \mathrm{Hom}_{\mathrm{Fun}(I,C)}(k_X, G).$$

Definition 2.1. The *projective limit* of G, denoted $\varprojlim_I G$, is the presheaf

$$X \longmapsto \mathrm{Hom}_{\mathrm{Fun}(I,C)}(k_X, G).$$

When the presheaf $\varprojlim_I G$ is representable, we again denote by $\varprojlim_I G$ an object of C representing it. The object $\varprojlim_I G$ is therefore defined only up to isomorphism. When no confusion can result, we use the abbreviated notation $\varprojlim G$.

As G varies, $\varprojlim G$ is a functor from $\mathrm{Fun}(I,C)$ to $\widehat{C}$.

2.1.1

By symmetry, reversing the direction of the arrows in C, one defines similarly the inductive limit of a functor: it is a covariant functor on C with values in the category of $\mathcal{U}$-sets. We shall use the notations $\varinjlim_I G$ and $\varinjlim G$.

Products, fiber products, and kernels are projective limits. Similarly, sums, amalgamated sums, and cokernels are inductive limits.

Definition 2.2. Let I and C be two categories, and let $G : I \to C$ be a functor. We say that the projective limit of G is *representable* if there exists a universe $\mathcal{U}$ such that:

1) the category I is $\mathcal{U}$-small;

2) the category C is a $\mathcal{U}$-category;

3) the presheaf $\varprojlim G$, with values in the category of $\mathcal{U}$-sets, is representable.

It follows from no. 1 that the object $\varprojlim G$ representing the presheaf $\varprojlim G$ does not depend, up to isomorphism, on the universe $\mathcal{U}$. Note also that there is a smallest universe $\mathcal{U}'$ having properties (1) and (2), and that the presheaf $\varprojlim G$ necessarily takes values in the category of $\mathcal{U}'$-sets.

2.2.1

Let I and C be two categories. We say that *I-projective limits in C are representable* if, for every functor $G : I \to C$, the projective limit of G is representable. Finally, let C be a category and let $\mathcal{U}$ be a universe. We say that *$\mathcal{U}$-projective limits in C are representable* if, for every $\mathcal{U}$-small category I and every functor $G : I \to C$, the projective limit of G is representable.

Proposition 2.3. Let C be a category and let $\mathcal{U}$ be a universe. The following assertions are equivalent:

i) $\mathcal{U}$-projective limits in C are representable.

ii) Products indexed by a small set are representable, and kernels of pairs of arrows are representable.

iii) Products indexed by a small set are representable, and fiber products are representable.

Proof. It suffices to observe that there is an isomorphism, functorial in G,

$$\varprojlim G = \operatorname{Ker}\left(\prod_{i\in \mathrm{Ob}(I)} G(i) \rightrightarrows \prod_{u\in \mathrm{Fl}(I)} G(\mathrm{target}(u))\right),$$

where the pair of arrows is defined by the morphisms

$$\prod_{i\in \mathrm{Ob}(I)} G(i) \xrightarrow{\mathrm{pr}_{\mathrm{target}(u)}} G(\mathrm{target}(u)),$$
$$\prod_{i\in \mathrm{Ob}(I)} G(i) \xrightarrow{\mathrm{pr}_{\mathrm{source}(u)}} G(\mathrm{source}(u)) \xrightarrow{G(u)} G(\mathrm{target}(u)).$$

Moreover, it is clear that

$$\operatorname{Ker}\big(X \overset{u}{\underset{v}{\rightrightarrows}} Y\big) = X \times_{X\times Y} X,$$

where the two morphisms from X to $X \times Y$ are $\mathrm{id}_X \times u$ and $\mathrm{id}_X \times v$.

2.3.1

There are, of course, analogous definitions and assertions for inductive limits, which we shall not spell out. The same applies to finite projective and inductive limits, that is, those relative to a finite category I.

Corollary 2.3.2. Let $\mathcal{U}$-**Set** denote the category of $\mathcal{U}$-sets, and let $\mathcal{U}$-**Ab** denote the category of abelian-group objects of $\mathcal{U}$-**Set**. The $\mathcal{U}$-projective and $\mathcal{U}$-inductive limits in $\mathcal{U}$-**Set** and in $\mathcal{U}$-**Ab** are representable.

Proposition 2.3.3. Let I be a small category, let $F : I \to \mathcal{U}$-**Set** be a functor, and let $G \subset \mathrm{Ob}(I)$ be a small set of objects of I such that, for every object X of I, there exists an object $Y \in G$ and a morphism $Y \to X$ (respectively $X \to Y$). Then $\varprojlim_I F$ (respectively $\varinjlim_I F$) is representable, and

$$\mathrm{card}(\varprojlim_I F) \leq \prod_{Y\in G} \mathrm{card}(F(Y))$$

respectively

$$\operatorname{card}(\varinjlim_I F) \leq \sum_{Y \in G} \operatorname{card}(F(Y)).$$

Proof. One checks immediately that $\varprojlim_I F$ (respectively $\varinjlim_I F$) is isomorphic to a subobject of $\prod_{Y \in G} F(Y)$ (respectively to a quotient of $\coprod_{Y \in G} F(Y)$), whence the proposition.

Definition 2.4.1. Let C be a category in which finite projective (respectively inductive) limits are representable, and let $u : C \to C'$ be a functor. We say that u is *left exact* (respectively *right exact*) if it "commutes" with finite projective (respectively finite inductive) limits. A functor that is both left and right exact is called *exact*.

2.4.2

It follows from 2.3 that, for a functor to be left exact, it is necessary and sufficient that it carry the final object, i.e. the empty product, to the final object, the product of two objects to the product of their images, and the kernel of a pair of arrows to the kernel of the image pair; equivalently, that it carry the final object to the final object and fiber products to fiber products. Here it is assumed that finite projective limits in C are representable.

2.5.0

Let I, J, and C be three categories, and let $G : I \times J \to C$ be a functor, i.e. a functor from I with values in the category of functors from J to C. Suppose that the projective limits of the functors

$$\begin{aligned} G_i : J &\longrightarrow C \qquad (i \in \operatorname{Ob}(I)) \\ j &\longmapsto G(i,j) \end{aligned}$$

are representable, and that the functor

$$i \longmapsto \varprojlim_J G_i$$

admits a representable projective limit. Then it is clear that the functor G admits a representable projective limit and that there is a canonical isomorphism

$$\varprojlim_{I \times J} G \xrightarrow{\sim} \varprojlim_I \varprojlim_J G_i,$$

and hence a canonical isomorphism

$$\varprojlim_I \varprojlim_J G_i \xrightarrow{\sim} \varprojlim_J \varprojlim_I G_j.$$

We shall say more briefly that projective limits commute with projective limits. One sees in the same way that inductive limits commute with inductive limits. But it is not true in general that inductive limits commute with projective limits.

Definition 2.5. Let C be a category with representable fiber products, let I be a category, let $G : I \to C$ be a functor, let $g : G \to X$ be a morphism from G to an object of C, i.e. a morphism from G to the constant functor associated with X, and let $m : Y \to X$ be a morphism of C. Let

$$G \times_X Y : I \longrightarrow C$$

be the functor

$$i \longmapsto G(i) \times_X Y.$$

We say that the inductive limit of G is *universal* if, for every object X, every morphism $g : G \to X$, and every morphism $m : Y \to X$,

a) the inductive limit of the functor $G \times_X Y$ is representable;

b) the canonical morphism
$$\varinjlim (G \times_X Y) \longrightarrow (\varinjlim G) \times_X Y$$
is an isomorphism.

Proposition 2.6. Let $\mathcal{U}$ be a universe. Inductive limits in $\mathcal{U}$-**Set** that are representable, in particular $\mathcal{U}$-inductive limits (2.2.1) in $\mathcal{U}$-**Set**, are universal.

We shall also use another commutation result between projective and inductive limits, which we now present.

Definition 2.7. A category I is *pseudo-filtered* if it has the following properties:

PS 1) Every diagram of the form

$$\begin{tikzcd} & j \\ i \arrow[ur] \arrow[dr] & \\ & j' \end{tikzcd}$$

can be inserted into a commutative diagram

$$\begin{tikzcd} & j \arrow[dr] & \\ i \arrow[ur] \arrow[dr] & & k \\ & j' \arrow[ur] & \end{tikzcd} .$$

PS 2) Every diagram of the form

$$i \underset{v}{\overset{u}{\rightrightarrows}} j$$

can be inserted into a diagram

$$i \underset{v}{\overset{u}{\rightrightarrows}} j \xrightarrow{w} k$$

such that $w \circ u = w \circ v$.

A category I is called *filtered* if it is pseudo-filtered, nonempty, and connected, i.e. if any two objects of I can be joined by a finite chain of arrows, with no condition imposed on the directions of the arrows. Equivalently, in the presence of PS 2, this means that $I \neq \varnothing$ and that, for any two objects a, b of I, there is always an object c of I and arrows $a \to c$ and $b \to c$. A category I is called *cofiltered* if I^{op} is filtered.

Example 2.7.1. If, in I, amalgamated sums (respectively sums of two objects) and cokernels of double arrows are representable, then I is pseudo-filtered (respectively filtered).

Proposition 2.8. Let $\mathcal{U}$ be a universe. Filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-**Set** commute with finite projective limits.

One is immediately reduced to proving that filtered $\mathcal{U}$-inductive limits commute with fiber products. The proof is left to the reader. One may use the following description of the limit.

Lemma 2.8.1. Let I be a small filtered category, and let $i \mapsto X_i$ be a functor from I to $\mathcal{U}$-**Set**. On the sum set $\coprod_{i \in \mathrm{Ob}\, I} X_i$, let R be the following relation:

(R) Two elements $x_i \in X_i$ and $x_j \in X_j$ are related if there exist an object $k \in \mathrm{Ob}\, I$ and two morphisms $u : i \to k$ and $v : j \to k$ such that the images in X_k of x_i and x_j under the transition maps u and v, respectively, are equal.

Then:

1) R is an equivalence relation.

2) The quotient $\coprod_{i \in \mathrm{Ob}\, I} X_i / R$ is canonically isomorphic to $\varinjlim_I X_i$.

3) Two elements $x_i \in X_i$ and $x_j \in X_j$ are respectively equivalent, modulo R, to two elements of one and the same X_k.

4) For every $i \in \mathrm{Ob}\, I$, two elements α and β of X_i are equivalent modulo R if and only if there exists a morphism $u : i \to j$ such that $u(\alpha) = u(\beta)$.

Assertion 1) follows from PS 1 (2.7). To prove 2), one checks that $\coprod_{i \in \mathrm{Ob}\, I} X_i / R$ has the universal property of the inductive limit; here only PS 1 is used. Assertion 3) follows from the fact that I is connected. Assertion 4) follows from PS 2.

Corollary 2.9. Let γ be a species of algebraic structure "defined by finite projective limits." The reader is asked to give a mathematical meaning to the preceding phrase; let us merely note that the structures of groups, abelian groups, rings, modules, etc. are structures of this kind. Let $\mathcal{U}$-γ denote the category of γ-objects of $\mathcal{U}$-**Set**. The functor that associates with every object of $\mathcal{U}$-γ its underlying set commutes with filtered $\mathcal{U}$-inductive limits. Consequently filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-γ commute with finite projective limits.

Corollary 2.10. Pseudo-filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-**Ab** commute with finite projective limits.

One is reduced to filtered limits by decomposing the indexing category into its connected components.

We now record a result that we shall use constantly. Let C and C' be two $\mathcal{U}$-categories, and let

$$C \underset{v}{\overset{u}{\rightleftarrows}} C'$$

be two functors, with u *left adjoint* to v. Recall that this means that there is an isomorphism of bifunctors with values in $\mathcal{U}$-**Set**,

$$\mathrm{Hom}_{C'}(u(X), X') \xrightarrow{\sim} \mathrm{Hom}_C(X, v(X')).$$

Proposition 2.11. The functor u commutes with representable inductive limits; the functor v commutes with representable projective limits.

This assertion means that, for every category I and every functor $G : I \to C'$ such that the projective limit of G is representable, the functor $v \circ G$ admits a representable projective limit and there is a canonical isomorphism

$$v(\varprojlim G) \longrightarrow \varprojlim(v \circ G).$$

The reader will write out for himself the assertion concerning the functor u.

Finally, let us point out a computation of projective limits in the case where the indexing category admits products.

Proposition 2.12. Let I and C be two categories, and let $\mathcal{V}$ be a universe. Suppose that $\mathcal{V}$-projective limits in C are representable and that there exists in I a family of objects $(i_\alpha)_{\alpha\in A}$, with $A \in \mathcal{V}$, such that:

1) the products $i_\alpha \times i_\beta$ are representable for every pair $(\alpha, \beta) \in A \times A$;

2) every object of I maps to at least one of the i_α.

Then, for every contravariant functor

$$F : I^{\mathrm{op}} \longrightarrow C,$$

the projective limit of F exists and there is an isomorphism, functorial in F,

$$\varprojlim F \xrightarrow{\sim} \mathrm{Ker}\left(\prod_{\alpha\in A} F(i_\alpha) \rightrightarrows \prod_{(\alpha,\beta)\in A\times A} F(i_\alpha \times i_\beta)\right),$$

where the two arrows are defined by the projections of the products $i_\alpha \times i_\beta$ onto the factors.

3 Exactness properties of the category of presheaves

Let C be a $\mathcal{U}$-category, and let $\widehat{C}$ be the category of presheaves on C. The exactness properties of $\widehat{C}$ are all deduced from the following proposition.

Proposition 3.1. The $\mathcal{U}$-projective and $\mathcal{U}$-inductive limits in $\widehat{C}$ are representable. For every object X of C, the functor on $\widehat{C}$

$$F \longmapsto F(X), \qquad F \in \widehat{C},$$

commutes with inductive and projective limits.

In other words, inductive and projective limits in $\widehat{C}$ are computed argument by argument. We list a few corollaries.

Corollary 3.2. Let γ be an algebraic structure defined by finite projective limits. The category of contravariant functors with values in $\mathcal{U}$-γ is equivalent to the category of γ-objects of $\widehat{C}$.

Corollary 3.3. A morphism of $\widehat{C}$ that is both a monomorphism and an epimorphism is an isomorphism. A morphism of $\widehat{C}$ factors uniquely as an epimorphism followed by a monomorphism. Representable inductive limits in $\widehat{C}$ are universal. Filtered $\mathcal{U}$-inductive limits in $\widehat{C}$ commute with finite projective limits. The canonical functor $C \to \widehat{C}$ commutes with representable projective limits.

More generally, one may say that the category $\widehat{C}$ inherits all the properties of the category of $\mathcal{U}$-sets that involve inductive and projective limits.

3.4.0

Let F be an object of $\widehat{C}$. We denote by C/F the following category. Its objects are pairs consisting of an object X of C and a morphism u from X to F. Given two objects (X, u) and (Y, v), a morphism from (X, u) to (Y, v) is a morphism $g : X \to Y$ such that the following diagram is commutative:

$$\begin{tikzcd} X \arrow[rr, "g"] \arrow[dr, "u"'] & & Y \arrow[dl, "v"] \\ & F & \end{tikzcd}$$

Proposition 3.4. With the notation of (3.4.0), the source functor $C/F \to \widehat{C}$ admits a representable inductive limit in $\widehat{C}$. The canonical morphism

$$\varinjlim_{C/F} \operatorname{source}(.) \longrightarrow F$$

is an isomorphism.

Corollary 3.5. Let F and H be two objects of $\widehat{C}$. There is a canonical isomorphism

$$\operatorname{Hom}(F, H) \xrightarrow{\sim} \varprojlim_{(X,u) \in \operatorname{Ob}(C/F)} H(X).$$

Proof. The corollary follows immediately from 3.4 and 1.2.

Proposition 3.6. Let C be a $\mathcal{U}$-category, let $\mathcal{V}$ be a universe containing $\mathcal{U}$, and let

$$i : \widehat{C}_{\mathcal{U}} \longrightarrow \widehat{C}_{\mathcal{V}}$$

be the natural inclusion functor between the corresponding categories of presheaves (1.3). The functor i commutes with inductive and projective limits. Moreover, for every object F of $\widehat{C}_{\mathcal{U}}$, every subobject of $i(F)$ is isomorphic to the image under i of a unique subobject of F; thus one obtains a bijection from the set of subobjects of F to the set of subobjects of $i(F)$.

4 Sieves

Definition 4.1. Let C be a category. A *sieve* of the category C is a full subcategory D of C having the following property: every object of C admitting a morphism to an object of D belongs to D. If X is an object of C, the sieves of the category C/X will be called, by abuse of language, *sieves of* X.

Let $\mathcal{U}$ be a universe such that C is a $\mathcal{U}$-category, and let $\widehat{C}$ be the corresponding category of presheaves. To every sieve of X one associates a subobject of X in $\widehat{C}$ as follows: to every object Y of C one assigns the set of morphisms $f : Y \to X$ such that the object (Y, f) belongs to the sieve.

Proposition 4.2. The map defined above establishes a bijection between the set of sieves of X and the set of subobjects of X in $\widehat{C}$.

Proof. We only indicate the inverse map. To every subfunctor R of X one associates the category C/R of objects of C over R (3.4). One checks immediately that C/R is a sieve of X.

Remark 4.2.1. One sees in the same way that the sieves of C are in canonical one-to-one correspondence with the set of subfunctors of the final functor on C, the final object of $\widehat{C}$.

4.3

Let C be a $\mathcal{U}$-category. By abuse of language, we shall also call the subobjects of X in the category $\widehat{C}$ the sieves of X. This abuse of language allows us, for every presheaf F and every sieve R of X, to define $\mathrm{Hom}_{\widehat{C}}(R, F)$ as the set of morphisms from the functor R to F. There is, moreover, an isomorphism canonical and functorial in F (3.5):

$$\mathrm{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_{C/R} F(.),$$

which also gives a direct definition.[1] Similarly, Proposition 4.2 allows us to transpose the usual operations on functors to sieves. We mention the following.

4.3.1 Base change

Let R be a sieve of X, and let $f : Y \to X$ be a morphism of objects of C. The fiber product $R \times_X Y$ is a sieve of Y, called the *sieve deduced from R by base change*. The corresponding subcategory of C/Y is the inverse image of the subcategory of C/X defined by R, under the canonical functor $C/Y \to C/X$ defined by f.

4.3.2 Order relation, intersection, union

The inclusion relation on subfunctors of X is an order relation. One may define the union and intersection of a family of sieves indexed by an arbitrary set as the least upper bound and greatest lower bound of the corresponding family of subpresheaves.

[1] More simply, C/R identifies with R, so that one has the formula $\mathrm{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_R F$, where F abusively denotes the composite of F with the tautological source functor $R \to C/X \to C$.

4.3.3 Image, generated sieve

Let $(F_\alpha)_{\alpha\in A}$ be a family of presheaves, and for each $\alpha \in A$ let $f_\alpha : F_\alpha \to X$ be a morphism, where X is an object of C. The *image* of this family of morphisms is the union of the images of the f_α. The image of this family is therefore a sieve of X. In particular, if all the F_α are objects of C, the image sieve will be called the sieve generated by the morphisms f_α. The category C/R is the full subcategory of C/X formed by those objects $X' \to X$ over X for which there exists an X-morphism from X' to one of the F_α.

As an exercise, the reader may translate the relations and operations just defined on subfunctors into terms of the categories C/R. He will then see that these relations and operations do not depend on the universe $\mathcal{U}$ such that C is a $\mathcal{U}$-category, and that they are therefore defined for every category, without requiring one to specify the universe to which the sets of morphisms belong. This was in any case already foreseeable *a priori* from 3.6.

5 Functoriality of categories of presheaves

5.0

Let C, C', and D be three categories, and let $u : C \to C'$ be a functor. We denote by u^* the functor obtained by composition with u:

$$u^* : \mathrm{Hom}((C')^{\mathrm{op}}, D) \longrightarrow \mathrm{Hom}(C^{\mathrm{op}}, D), \qquad G \longmapsto G \circ u.$$

The functor u^* commutes with inductive and projective limits.

Proposition 5.1. Suppose that C is small and that, in D, $\mathcal{U}$-inductive limits (respectively projective limits) are representable. The functor u^* admits a left adjoint $u_!$ (respectively a right adjoint u_*). Thus one has an isomorphism

$$\mathrm{Hom}_{\mathrm{Hom}(C^{\mathrm{op}},D)}(F, u^*G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{Hom}((C')^{\mathrm{op}},D)}(u_!F, G),$$
$$\big(\text{respectively } \mathrm{Hom}_{\mathrm{Hom}(C^{\mathrm{op}},D)}(u^*G, F) \simeq \mathrm{Hom}_{\mathrm{Hom}((C')^{\mathrm{op}},D)}(G, u_*F)\big).$$

Proof. We indicate only the proof of the existence of the left adjoint. The corresponding part of the proposition follows formally from the isomorphisms

$$\mathrm{Hom}(C^{\mathrm{op}}, D) \xrightarrow{\sim} \mathrm{Hom}(C, D^{\mathrm{op}})^{\mathrm{op}} \xrightarrow{\sim} \mathrm{Hom}((C^{\mathrm{op}})^{\mathrm{op}}, D^{\mathrm{op}})^{\mathrm{op}}.$$

Let Y be an object of C'. Denote by I_u^Y the following category. Its objects are pairs (X, m), where X is an object of C and m is a morphism $Y \to u(X)$. If (X, m) and (X', m') are two objects of I_u^Y, a morphism from (X, m) to (X', m') is a morphism $\xi : X \to X'$ such that $m' = u(\xi)m$. Composition is defined in the evident way.

Let $f : Y \to Y'$ be a morphism of C'. By composition, f defines a functor $I_u^f : I_u^Y \to I_u^{Y'}$.

We also have a functor $\mathrm{pr}_Y : I_u^Y \to C$ which sends the object (X, m) to the object X. Then the evident diagram

$$\begin{tikzcd} I_u^Y \arrow[r, "I_u^f"] \arrow[d, "\mathrm{pr}_Y"'] & I_u^{Y'} \arrow[dl, "\mathrm{pr}_{Y'}"] \\ C & \end{tikzcd} \tag{*}$$

is commutative.

Now let F be a presheaf on C and set

$$u_!F(Y) = \varinjlim_{I_u^Y} F \circ \mathrm{pr}_Y \,. \tag{5.1.1}$$

The commutativity of diagram $(*)$ and the functoriality of inductive limits make $u_!F$ into a presheaf on C'. Let us show that the functor $u_!$ is left adjoint to u^*. For this it is enough to show that, for every presheaf G on C', there is a functorial isomorphism

$$\mathrm{Hom}(u_!F, G) \xrightarrow{\sim} \mathrm{Hom}(F, u^*G).$$

Let $\xi \in \mathrm{Hom}(u_!F, G)$. For every object X of C, we have a morphism

$$\xi_{u(X)} : u_!F(u(X)) \longrightarrow G(u(X)).$$

But $(X, \mathrm{id}_{u(X)})$ is an object of $I_u^{u(X)}$, and by the definition of the inductive limit there is a canonical morphism

$$F(X) \longrightarrow u_!F(u(X)).$$

We obtain, for every object X of C, a morphism

$$\eta_X : F(X) \longrightarrow G(u(X)),$$

which is visibly functorial in X. Hence a morphism

$$\eta : F \longrightarrow u^*G.$$

Conversely, let $\eta \in \mathrm{Hom}(F, u^*G)$. One obtains, for every object Y of C', a morphism of functors

$$\eta_Y : F \circ \mathrm{pr}_Y \longrightarrow (u^*G) \circ \mathrm{pr}_Y,$$

and hence, by composing with the evident morphism from the functor $(u^*G) \circ \mathrm{pr}_Y$ to the constant functor $G(Y)$, a morphism

$$\xi_Y : u_!F(Y) \longrightarrow G(Y),$$

which is functorial in Y. Hence a morphism

$$\xi : u_!F \longrightarrow G.$$

The reader will check that the two maps thus defined are inverse to one another, completing the proof.

Proposition 5.2. Suppose that in D the $\mathcal{U}$-inductive limits are representable, the finite projective limits are representable, and filtered $\mathcal{U}$-inductive limits commute with finite projective limits. Suppose moreover that, in C, finite projective limits are representable and that the functor u is left exact (2.3.2). Then finite projective limits are representable in $\mathrm{Hom}(C^{\mathrm{op}}, D)$ and in $\mathrm{Hom}((C')^{\mathrm{op}}, D)$, and the functor $u_!$ is left exact.

Proof. The first assertion is trivial. Let us prove the second. By the proof of 5.1, for every presheaf F on C and every object Y of C' one has

$$u_!F(Y) \xrightarrow{\sim} \varinjlim_{I_u^Y} F \circ \mathrm{pr}_Y \,.$$

It is therefore enough to show that the category $(I_u^Y)^{\mathrm{op}}$ satisfies axioms PS 1 and PS 2 (2.7), and that this category is connected. The verification is left to the reader.

5.3

Specializing these results to the case where D is the category of $\mathcal{U}$-sets, one obtains a sequence of three functors

$$u_!, \quad u^*, \quad u_*,$$

which is a "sequence of adjoint functors" in the sense that, for two consecutive functors in the sequence, the one on the right is right adjoint to the other. Their essential properties are summarized in the following proposition.

Proposition 5.4. Let C be a small category, let C' be a $\mathcal{U}$-category, and let $u : C \to C'$ be a functor.

1) The functor $u^* : \widehat{C'} \to \widehat{C}$ commutes with inductive and projective limits.
2) The functor $u_* : \widehat{C} \to \widehat{C'}$ commutes with projective limits. For every presheaf F on C and every object Y of C', one has

$$u_*F(Y) \xrightarrow{\sim} \operatorname{Hom}_{\widehat{C}}(u^*(Y), F).$$

3) The functor $u_! : \widehat{C} \to \widehat{C'}$ commutes with inductive limits. The functor $u_!$ is defined only up to isomorphism, but it can always be chosen so that the diagram

$$\begin{array}{ccc} C & \xrightarrow{u} & C' \\ {\scriptstyle h}\big\downarrow & & \big\downarrow{\scriptstyle h'} \\ \widehat{C} & \xrightarrow[u_!]{} & \widehat{C'} \end{array}$$

is commutative, where h and h' are the canonical inclusion functors.

For every presheaf F on C, one has

$$u_!F \xrightarrow{\sim} \varinjlim_{C/F} h' \circ u.$$

4) If finite projective limits are representable in C and if u is left exact (2.3.2), then the functor $u_!$ is left exact.

Proof. Assertion (1) is trivial. Assertion (2) follows from the fact that u_* is a right adjoint, by 2.11 and 1.4. The same applies to assertion (3), but one also uses 3.4. Finally, assertion (4) is precisely 5.2, which applies by 2.7.

Proposition 5.5. Let C and C' be two small categories, and let

$$C \underset{v}{\overset{u}{\rightleftarrows}} C'$$

be a pair of functors, where v is left adjoint to u. Then there exist isomorphisms, compatible with the adjunction isomorphisms,

$$v^* \xrightarrow{\sim} u_!, \qquad v_* \xrightarrow{\sim} u^*.$$

Proof. It is enough to exhibit an isomorphism $v^* \xrightarrow{\sim} u_!$; the other isomorphism will follow by adjunction. Let F be a presheaf on C and let Y be an object of C'. Then

$$v^*F(Y) \xrightarrow{\sim} \mathrm{Hom}(v(Y), F).$$

Using 3.4, we get

$$\mathrm{Hom}(v(Y), F) \xrightarrow{\sim} \varinjlim_{C/F} \mathrm{Hom}(v(Y), .).$$

Since v is left adjoint to u, it follows that

$$\varinjlim_{C/F} \mathrm{Hom}(v(Y), .) \xrightarrow{\sim} \varinjlim_{C/F} \mathrm{Hom}(Y, u(.)).$$

Using 5.4.3), we then obtain

$$\varinjlim_{C/F} \mathrm{Hom}(Y, u(.)) \xrightarrow{\sim} \mathrm{Hom}(Y, u_!F) \xrightarrow{\sim} u_!F(Y).$$

Thus, for every object Y of C', we have determined an isomorphism $v^*F(Y) \xrightarrow{\sim} u_!F(Y)$, visibly functorial in Y and in F.

Corollary 5.5.1. Let $u : C \to C'$ be a functor that admits a left adjoint. The functor $u_! : \widehat{C} \to \widehat{C'}$ commutes with projective limits; recall that it commutes with inductive limits by 5.4.3.

Remark 5.5.2. One thus obtains a "sequence of *four* adjoint functors" (cf. 5.3):

$$v_!, \quad v^* = u_!, \quad v_* = u^*, \quad u_*,$$

of which the first three (respectively the last three) commute with inductive limits (respectively projective limits).

Proposition 5.6. The hypotheses are those of 5.4. The following conditions are equivalent:

i) the functor u is fully faithful;

ii) the functor $u_!$ is fully faithful;

iii) the adjunction morphism $\mathrm{id}_{\widehat{C}} \to u^*u_!$ is an isomorphism;

iv) the functor u_* is fully faithful;

v) the adjunction morphism $u^*u_* \to \mathrm{id}_{\widehat{C}}$ is an isomorphism.

Proof. It is clear that ii) $\Leftrightarrow$ iii) and iv) $\Leftrightarrow$ v), by general properties of adjoint functors, and that ii) $\Rightarrow$ i), by 5.4.3). Let us prove that i) $\Rightarrow$ iii). The functors $\mathrm{id}_{\widehat{C}}$, u^*, and $u_!$ commute with inductive limits. By 3.4, it is therefore enough to prove that $H \to u^*u_!H$ is an isomorphism when H is representable, which is evident.

It remains to show that iii) is equivalent to v). For every object H (respectively K) of $\widehat{C}$, denote by $\Phi(H) : H \to u^*u_!H$ (respectively by $\Psi(K) : u^*u_*K \to K$) the adjunction morphism. Then one has a commutative diagram

$$\begin{tikzcd}
\mathrm{Hom}_{\widehat{C}}(H, u^*u_*K) \arrow[r, "\mathrm{Hom}(H,\Psi(K))"] \arrow[d, "\sim"'] & \mathrm{Hom}_{\widehat{C}}(H, K) \\
\mathrm{Hom}_{\widehat{C'}}(u_!H, u_*K) \arrow[d, "\sim"'] & \\
\mathrm{Hom}_{\widehat{C}}(u^*u_!H, K) \arrow[uur, "\mathrm{Hom}(\Phi(H),K)"'] &
\end{tikzcd}$$

Hence $\Phi(H)$ is an isomorphism for every H if and only if $\Psi(K)$ is an isomorphism for every K.

Remark 5.7.

a) The equivalences ii) $\Leftrightarrow$ iii) $\Leftrightarrow$ iv) $\Leftrightarrow$ v) are general facts about adjoint functors.

b) The explicit form of $u_!$, respectively u_*, given in the proof of 5.1 shows immediately that, under the general hypotheses of 5.1, if u is fully faithful, then the adjunction morphism $\mathrm{id} \to u^*u_!$ (respectively $u^*u_* \to \mathrm{id}$) is an isomorphism; equivalently, $u_!$ (respectively u_*) is fully faithful.

5.8.0

Let γ be a species of algebraic structure defined by finite projective limits. Let $\mathcal{U}$-γ-$\mathbf{Set}$ be the category of γ-objects of $\mathcal{U}$-$\mathbf{Set}$, and let

$$\mathrm{und} : \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

be the underlying-set functor. For simplicity, we assume that the species of structure under consideration has a single underlying set. Let C be a category. Composition with und gives a functor denoted

$$\widehat{\mathrm{und}} : \mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}) \longrightarrow \widehat{C}.$$

Since in $\widehat{C}$ projective limits are computed argument by argument, the functor $\widehat{\mathrm{und}}$ factors as an equivalence

$$\mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}) \xrightarrow{\approx} \widehat{C}_\gamma, \tag{5.8.1}$$

where $\widehat{C}_\gamma$ is the category of γ-objects of $\widehat{C}$, followed by a functor again denoted

$$\widehat{\mathrm{und}} : \widehat{C}_\gamma \longrightarrow \widehat{C},$$

and called the *underlying presheaf of sets* functor.

5.8.2

Suppose that the functor

$$\mathcal{U}\text{-}\gamma\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

admits a left adjoint

$$\mathrm{Free} : \mathcal{U}\text{-}\mathbf{Set} \longrightarrow \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set},$$

the functor of "freely generated γ-objects." Examples are the free group, free commutative group, free A-module, and so on. In fact one can show that this condition is always satisfied. Composition with Free gives a functor

$$\widehat{\mathrm{Free}} : \widehat{C} \longrightarrow \mathrm{Hom}(C^{\mathrm{op}}, \mathcal{U}\text{-}\gamma\text{-}\mathbf{Set}),$$

and, by composing with the equivalence (5.8.1), a functor again denoted

$$\widehat{\mathrm{Free}} : \widehat{C} \longrightarrow \widehat{C}_\gamma,$$

and called the functor "presheaf of freely generated γ-objects." The functor $\widehat{\mathrm{Free}}$ is left adjoint to $\widehat{\mathrm{und}}$.

Proposition 5.8.3. Let γ be a species of algebraic structure defined by finite projective limits such that, in the category of γ-objects of $\mathcal{U}$-**Set**, $\mathcal{U}$-inductive limits are representable.[2] Let C be a category belonging to $\mathcal{U}$, let C' be a $\mathcal{U}$-category, and let $u : C \to C'$ be a functor. Denote by $\widehat{C}_\gamma$ (respectively $\widehat{C'}_\gamma$) the category of γ-objects of $\widehat{C}$ (respectively $\widehat{C'}$), and by $(u^*)^\gamma$ the functor on γ-objects deduced from u^*. It follows from 5.1 and the equivalence 5.8.1 that there exists a functor left adjoint (respectively right adjoint) to $(u^*)^\gamma$. This functor is denoted $u_{!\gamma}$ (respectively $u_{*\gamma}$).

1) The functor $(u^*)^\gamma$ commutes with inductive and projective limits. The diagram

$$\begin{tikzcd} \widehat{C'}_\gamma \arrow[r, "(u^*)^\gamma"] \arrow[d, "\widehat{\mathrm{und}'}"'] & \widehat{C}_\gamma \arrow[d, "\widehat{\mathrm{und}}"] \\ \widehat{C'} \arrow[r, "u^*"'] & \widehat{C} \end{tikzcd} \tag{*}$$

is commutative; here $\widehat{\mathrm{und}'}$ and $\widehat{\mathrm{und}}$ denote the underlying-set functors.

2) The functor $u_{*\gamma}$ commutes with projective limits. The diagram

$$\begin{tikzcd} \widehat{C}_\gamma \arrow[r, "u_{*\gamma}"] \arrow[d, "\widehat{\mathrm{und}}"'] & \widehat{C'}_\gamma \arrow[d, "\widehat{\mathrm{und}'}"] \\ \widehat{C} \arrow[r, "u_*"'] & \widehat{C'} \end{tikzcd} \tag{**}$$

is commutative up to isomorphism.

[2] One can show that this condition is always satisfied.

3) The functor $u_{!\gamma}$ commutes with inductive limits. Suppose that $\widehat{\text{und}}$ (respectively $\widehat{\text{und}'}$) admits a left adjoint $\widehat{\text{Free}}$ (respectively $\widehat{\text{Free}'}$), as in (5.8.2). The diagram

$$\begin{tikzcd} \widehat{C} \arrow[r, "u_!"] \arrow[d, "\widehat{\text{Free}}"'] & \widehat{C'} \arrow[d, "\widehat{\text{Free}'}"] \\ \widehat{C}_\gamma \arrow[r, "u_{!\gamma}"'] & \widehat{C'}_\gamma \end{tikzcd} \tag{***}$$

is commutative up to isomorphism.

Suppose that $u_!$ commutes with finite projective limits (5.4 and 5.6). Then the diagram

$$\begin{tikzcd} \widehat{C}_\gamma \arrow[r, "u_{!\gamma}"] \arrow[d, "\widehat{\text{und}}"'] & \widehat{C'}_\gamma \arrow[d, "\widehat{\text{und}'}"] \\ \widehat{C} \arrow[r, "u_!"'] & \widehat{C'} \end{tikzcd} \tag{****}$$

is commutative up to isomorphism, and $u_{!\gamma}$ commutes with finite projective limits.

Proof. Assertion (1) is evident. Assertion (2) is also evident, since u_* is a right adjoint and hence, by 2.11, commutes with projective limits and in particular with finite projective limits; this gives the commutativity of diagram (**). The commutativity of diagram (***) follows from the uniqueness, up to isomorphism, of the left adjoint functor. Finally, the commutativity of diagram (****) follows immediately from the fact that $u_!$ commutes with finite projective limits.

Notation 5.9. By abuse of notation, the functors $(u^*)^\gamma$ and $u_{*\gamma}$ will often be denoted by u^* and u_*; this risks no confusion in view of the commutativity of diagrams $(*)$ and $(**)$. In contrast, when $u_!$ does not commute with finite projective limits, diagram $(****)$ is not commutative up to isomorphism, and the notations $u_!$ and $u_{!\gamma}$ must be used to avoid confusion.

5.10

Let C be a small category. For every object X of $\widehat{C}$, let C/X denote the category of arrows whose target is X and whose source is an object of C. The source functor defines a functor $j_X : C/X \to C$. Let $m : Y \to X$ be a morphism of $\widehat{C}$. Composition of morphisms defines a functor $j_m : C/Y \to C/X$. The diagram

$$\begin{tikzcd} C/Y \arrow[rr, "j_m"] \arrow[dr, "j_Y"'] & & C/X \arrow[dl, "j_X"] \\ & C & \end{tikzcd}$$

is commutative. It follows from 5.1 that, for every object F of $\widehat{C/X}$ and every object Y of C, one has

$$j_{X!}F(Y) = \coprod_{u \in \mathrm{Hom}_{\widehat{C}}(Y,X)} F(u). \tag{5.10.1}$$

Formula (5.10.1) allows one to define $j_{X!}$ when C is a $\mathcal{U}$-category, and one verifies that the functor $j_{X!}$ thus defined is always left adjoint to the functor

$$j_X^* : \widehat{C} \longrightarrow \widehat{C/X}.$$

Proposition 5.11. Let C be a $\mathcal{U}$-category, and let X be a presheaf on C.

1) The functor

$$j_{X!} : \widehat{C/X} \longrightarrow \widehat{C}$$

factors through the category $\widehat{C}/X$:

$$\widehat{C/X} \xrightarrow{e_X} \widehat{C}/X \longrightarrow \widehat{C}.$$

The functor e_X is an equivalence of categories.

2) The functor

$$e_X \circ j_X^* : \widehat{C} \longrightarrow \widehat{C}/X$$

is canonically isomorphic to the functor

$$H \longmapsto (H \times X \xrightarrow{\mathrm{pr}_2} X).$$

Proof.

1) Let f be the final object of $\widehat{C/X}$. One has a canonical isomorphism

$$f \xrightarrow{\sim} \varinjlim_{Y \in \mathrm{Ob}(C/X)} Y,$$

and hence $j_{X!}(f) = \varinjlim_{Y \in \mathrm{Ob}(C/X)} j_X(Y)$ by 5.4. But $j_{X!}(f) \simeq X$, which gives the factorization. To show that e_X is an equivalence, we merely exhibit a quasi-inverse: to every object $H \to X$ of $\widehat{C}/X$ associate the presheaf on C/X

$$(Y \to X) \longmapsto \mathrm{Hom}_{\widehat{C}/X}((Y \to X), (H \to X)).$$

2) The functor $e_X \circ j_X^*$ is right adjoint to the forgetful functor, and therefore $e_X \circ j_X^*$ is canonically isomorphic to the functor $H \mapsto (H \times X \xrightarrow{\mathrm{pr}_2} X)$.

5.12

Let $m : Y \to X$ be a morphism of $\widehat{C}$. By (5.11), the morphism m is canonically isomorphic to the image under e_X of an object of $\widehat{C/X}$, which we shall denote by $[m]$. By restriction to subcategories, the functor e_X defines an equivalence

$$(C/X)/[m] \xrightarrow{e_m} C/Y.$$

The diagram

$$\begin{tikzcd}
(C/X)/[m] \arrow[rr, "e_m"] \arrow[dr, "j_{[m]}"'] & & C/Y \arrow[dl, "j_m"] \\
& C/X &
\end{tikzcd}$$

is commutative up to canonical isomorphism.

5.13

We record a result that will be useful in Exposé VI. Let $u : C \to C'$ be a functor between small categories. For every object H of $\widehat{C}$, denote by

$$u/H : C/H \longrightarrow C'/u_!H$$

the functor which associates to every morphism $m : X \to H$ the morphism

$$u_!X \xrightarrow{u_!m} u_!H$$

(one knows from 5.4 that one may always set $u_!X = uX$). The following diagram is commutative:

$$\begin{tikzcd} C/H \arrow[r, "u/H"] \arrow[d, "j_H"'] & C'/u_!H \arrow[d, "j_{u_!H}"] \\ C \arrow[r, "u"'] & C'. \end{tikzcd} \tag{5.13.1}$$

We therefore have a diagram commutative up to isomorphism; and, since $(u/H)_!$ carries the final object of $\widehat{C/H}$ to the final object of $\widehat{C'/u_!H}$, the diagram

$$\begin{tikzcd} \widehat{C/H} \arrow[r, "(u/H)_!"] \arrow[d, "e_H"'] & \widehat{C'/u_!H} \arrow[d, "e_{u_!H}"] \\ \widehat{C}/H \arrow[r, "u_!/H"'] & \widehat{C'}/u_!H \end{tikzcd} \tag{5.13.3}$$

is commutative up to isomorphism.

Proposition 5.14. Let $u : C \to C'$ be a functor between small categories. Suppose that u has the following property:

(PPF) For every object X of C, the functor

$$u/X : C/X \longrightarrow C'/uX$$

is fully faithful.

Then:

1) Let f be the final object of $\widehat{C}$. The functor u factors as

$$C = C/f \xrightarrow{u/f} C'/u_!f \xrightarrow{j_{u_!f}} C'.$$

The functor u/f is fully faithful.

2) The functor $u_! : \widehat{C} \to \widehat{C'}$ factors as

$$\widehat{C} = \widehat{C/f} \xrightarrow{(u/f)_!} \widehat{C'/u_!f} \xrightarrow{e_{u_!f}} \widehat{C'}/u_!f \longrightarrow \widehat{C'},$$

where the functor $(u/f)_!$ is fully faithful, the functor $e_{u_!f}$ is an equivalence, and $\widehat{C'}/u_!f \to \widehat{C'}$ is the forgetful functor.

3) In particular, the functor $u_!$ is faithful and consequently the adjunction morphism $\mathrm{id} \xrightarrow{\Phi} u^* u_!$ is a monomorphism. Moreover, for every morphism $\alpha : H \to K$ of $\widehat{C}$, the diagram

$$\begin{tikzcd} H \arrow[r, hook, "\Phi(H)"] \arrow[d, "\alpha"'] & u^* u_! H \arrow[d, "u^* u_!(\alpha)"] \\ K \arrow[r, hook, "\Phi(K)"'] & u^* u_! K \end{tikzcd}$$

is cartesian.

Proof.

1) The factorization comes from diagram (5.13.1). The functor u is faithful. Hence u/f is faithful. Let us show that it is fully faithful. Let X and Y be two objects of C, let $\mathrm{can}_X : uX \to u_! f$ and $\mathrm{can}_Y : uY \to u_! f$ be the canonical morphisms, and let

$$\begin{tikzcd} uX \arrow[rr, "m"] \arrow[dr, "\mathrm{can}_X"'] & & uY \arrow[dl, "\mathrm{can}_Y"] \\ & u_! f & \end{tikzcd}$$

be a morphism of $C'/u_! f$. We have $u_! f = \varinjlim_{Z \in \mathrm{Ob}\, C} uZ$, hence by 3.1,

$$\mathrm{Hom}_{\widehat{C'}}(uX, u_! f) = \varinjlim_{Z \in \mathrm{Ob}\, C} \mathrm{Hom}_{C'}(uX, uZ).$$

By the definition of the inductive limit, to say that $\mathrm{can}_Y\, m = \mathrm{can}_X$ is equivalent to saying that there exist

a) a finite sequence of objects X_i of C, $i \in [0, n]$, with $X_0 = X$ and $X_n = Y$;

b) for every i, a morphism $m_i : uX \to uX_i$, with $m_0 = \mathrm{id}_X$ and $m_n = m$;

c) for every pair $(i, i+1)$, a morphism $f_i : X_i \to X_{i+1}$ or a morphism $f_i : X_{i+1} \to X_i$;

such that the corresponding triangular diagrams

$$\begin{tikzcd} & uX \arrow[dl, "m_i"'] \arrow[dr, "m_{i+1}"] & \\ uX_i \arrow[rr, "u(f_i)"'] & & uX_{i+1} \end{tikzcd} \quad \text{or} \quad \begin{tikzcd} & uX \arrow[dl, "m_i"'] \arrow[dr, "m_{i+1}"] & \\ uX_i & & uX_{i+1} \arrow[ll, "u(f_i)"] \end{tikzcd}$$

are commutative. One then proves immediately, by induction on i and using property (PPF), that m_i is of the form $u(p_i)$. In particular $m = u(p)$, and hence u/f is fully faithful.

2) The factorization is immediate. The functor $(u/f)_!$ is fully faithful by 5.6. The other assertions follow from 5.11.

3) The functor $u_!$ is the composite of the forgetful functor, which is faithful, and fully faithful functors. It is therefore faithful. It follows, from the general properties of adjoint functors, that the adjunction morphism $\mathrm{id} \to u^* u_!$ is a monomorphism. By 2), the functor $u_!$ appears

as the composite of a fully faithful functor $v : \widehat{C} \to \widehat{C'}/u_! f$ and the forgetful functor. The functor u^*, right adjoint to $u_!$, is therefore the composite of the "product by $u_! f$" functor, right adjoint to the forgetful functor, and a functor w right adjoint to v. Moreover, since v is fully faithful, the adjunction morphism $\mathrm{id} \to wv$ is an isomorphism. The last assertion follows easily.

Exposé I. Presheaves

Sections 6–7

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 6–7. Obvious typographical slips in the working French TEX source have been corrected where the mathematics fixes the reading; they are recorded in the accompanying worklog.

6 Faithful functors and conservative functors

Definition 6.1. Let E be a category, and let

$$(\varphi_i : E \to F_i)_{i \in I}$$

be a family of functors. We say that the family (φ_i) is *faithful* if, for every pair of objects X, Y of E and every pair of arrows $u, v : X \rightrightarrows Y$, the equalities $\varphi_i(u) = \varphi_i(v)$ for all $i \in I$ imply $u = v$. In other words, for every X, Y the map

$$\mathrm{Hom}_E(X, Y) \longrightarrow \prod_i \mathrm{Hom}_{F_i}(\varphi_i(X), \varphi_i(Y))$$

defined by the φ_i is injective. We say that (φ_i) is *conservative* if every arrow u of E such that $\varphi_i(u)$ is an isomorphism for all $i \in I$ is itself an isomorphism. We say that (φ_i) is conservative for monomorphisms, respectively for epimorphisms, etc., if the preceding condition is verified whenever u is a monomorphism, respectively an epimorphism, etc.

6.1.1

If one introduces the single functor

$$\varphi : E \longrightarrow F = \prod_{i \in I} F_i$$

defined by the family (φ_i), then it is clear that the family is faithful, respectively conservative, respectively conservative for monomorphisms, etc., if and only if φ is faithful, respectively conservative, etc. Here, of course, this means that the one-object family consisting of φ has the corresponding property. Thus there would be no serious loss in subsequently restricting to the case of a family consisting of a single functor. For convenience of later reference, however, we give the following statements for families.

The notions of 6.1 are especially useful when the φ_i satisfy suitable exactness properties; in that case they tend to coincide.

Proposition 6.2. The notation is that of 5.1.

(i) Suppose that kernels of pairs of arrows, or cokernels of pairs of arrows, are representable in E, and that the φ_i commute with them. Then

$$(\varphi_i) \text{ conservative} \quad \Longrightarrow \quad (\varphi_i) \text{ faithful}.$$

(ii) Suppose that fiber products, respectively amalgamated sums, are representable in E, and that the φ_i commute with them. Suppose that (φ_i) is faithful or conservative. Then, for every arrow u of E, the arrow u is a monomorphism, respectively an epimorphism, if and only if, for every $i \in I$, the arrow $\varphi_i(u)$ has the corresponding property.

(iii) Suppose that in E fiber products and amalgamated sums are representable, that the φ_i commute with them, and that every arrow in E which is a bimorphism, i.e. both a monomorphism and an epimorphism, is an isomorphism. Then

$$(\varphi_i) \text{ faithful} \quad \Longrightarrow \quad (\varphi_i) \text{ conservative.}$$

(iv) Suppose that in E fiber products, respectively amalgamated sums, are representable, and that the φ_i commute with them. If (φ_i) is conservative for monomorphisms, respectively for epimorphisms, then (φ_i) is conservative.

(v) Let D be a diagram type, let $F : d \mapsto F(d)$ be a diagram of type D in E, let X be an object of E, and let $u = (u_d)_{d \in D}$ be a family of arrows

$$X \longrightarrow F(d) \qquad (\text{respectively } F(d) \longrightarrow X).$$

Suppose that (φ_i) is conservative, that projective limits, respectively inductive limits, of type D are representable in E, and that the φ_i commute with them. Then u makes X a projective limit, respectively an inductive limit, of F in E if and only if, for every $i \in I$, $\varphi_i(u)$ makes $\varphi_i(X)$ a projective limit, respectively an inductive limit, of $\varphi_i(F)$ in F_i.

Proof. For (i), in the non-dual statement, given a pair of arrows $u, v : X \rightrightarrows Y$, it suffices to express the equality $u = v$ by the condition that the inclusion

$$\operatorname{Ker}(u, v) \longrightarrow X$$

is an isomorphism. Here, and below, we do not repeat the dual argument for the dual statement.

For (ii), if (φ_i) is faithful, one expresses the condition that $u : X \to Y$ be a monomorphism by the equality $\mathrm{pr}_1 = \mathrm{pr}_2$ for the fiber product $X \times_Y X$. If (φ_i) is conservative, one expresses it by the condition that the diagonal morphism

$$\delta : X \longrightarrow X \times_Y X$$

be an isomorphism.

For (iv), in the latter argument the morphism δ is a monomorphism, so it would actually have sufficed to assume that (φ_i) is conservative for monomorphisms. This then implies that (φ_i) is conservative tout court. Indeed, if $u \in \mathrm{Fl}(E)$ is such that the $\varphi_i(u)$ are isomorphisms, then by what precedes they are monomorphisms, and hence are isomorphisms by the hypothesis on (φ_i).

Assertion (iii) is a trivial consequence of (ii), and (v) is a trivial consequence of the definitions.

We note the following consequence of (i), (ii), and (iv).

Corollary 6.3. Suppose that in E fiber products and amalgamated sums are representable and that the φ_i commute with them, and that kernels of pairs of arrows or cokernels of pairs of arrows are representable and that the φ_i commute with them. It suffices, for example, that finite projective limits and finite inductive limits be representable in E, and that the φ_i be exact functors. Then the following conditions are equivalent:

a) (φ_i) is faithful;

b) (φ_i) is conservative;

c) (φ_i) is conservative for monomorphisms;

(c') (φ_i) is conservative for epimorphisms.

We also record the following fact.

Proposition 6.4. Let $\varphi : E \to F$ be a functor admitting a right adjoint ψ, so that

$$\mathrm{Hom}_F(\varphi(X), Y) \simeq \mathrm{Hom}_E(X, \psi(Y)).$$

For φ, respectively ψ, to be faithful it is necessary and sufficient that, for every object X of E, respectively every object Y of F, the adjunction morphism

$$X \longrightarrow \psi\varphi(X)$$

be a monomorphism. For φ, respectively ψ, to be fully faithful it is necessary and sufficient that the preceding adjunction morphism be an isomorphism.

Indeed, if X', X are two objects of E, the map

$$\mathrm{Hom}_E(X', X) \longrightarrow \mathrm{Hom}_F(\varphi(X'), \varphi(X)) \tag{$*$}$$

is identified with the map obtained from

$$X \longrightarrow \psi\varphi(X) \tag{$**$}$$

by applying the functor $\mathrm{Hom}_E(X', -)$. Thus, for $(*)$ to be a monomorphism, respectively an isomorphism, for all X', with X fixed, it is necessary and sufficient that the adjunction morphism $(**)$ be a monomorphism, respectively an isomorphism.

Proposition 6.5. Let $p : E' \to E$ be a functor. The following conditions are equivalent:

(i) p is faithful, conservative, and fibred *(SGA 1, VI, 6.1)*.

(ii) p is a fibred functor whose fibers are discrete categories.

(iii) For every $X' \in \mathrm{Ob}(E')$, the functor

$$E'_{/X'} \longrightarrow E_{/p(X')}$$

induced by p is an equivalence of categories, surjective on objects.

(iv) If E is a $\mathcal{U}$-category, there exists an object $F \in \mathrm{Ob}(\widehat{E})$ and an equivalence of categories over E *(SGA 1, VI, 4.3)*

$$E' \xrightarrow{\sim} E_{/F},$$

where $E_{/F}$ is the full subcategory of $\widehat{E}_{/F}$ consisting of the arrows $X \to F$ whose source is in E.

6.5.1

Recall that a category C is said to be *discrete* if it is a *groupoid*, i.e. every arrow is invertible, and if it is *rigid*, i.e. the automorphism group of every object is reduced to the unit group. Equivalently, the category is equivalent to the category C' defined by a set I, with $\mathrm{Ob}(C') = I$ and with identity arrows as its only arrows. When C is already assumed to be a groupoid, saying that C is discrete amounts to saying that, for any two objects X, Y of C, there is at most one arrow from X to Y, i.e. that C is isomorphic to the category defined by a preordered set.

The equivalence of conditions (i) and (ii) in 6.5 is an immediate consequence of these reminders and of the following lemma.

Lemma 6.5.2. Let $p : E' \to E$ be a fibred functor. Then:

(i) for p to be conservative it is necessary and sufficient that its fiber categories be groupoids;

(ii) for p to be faithful it is necessary and sufficient that its fiber categories be ordered categories.

Proof. [Proof of 6.5.2] Suppose first that p is conservative. For every arrow u in a fiber E'_X, one has $p(u) = \mathrm{id}_X$, an isomorphism; hence u is an isomorphism in E', and also in E'_X, since an inverse of u in E' is evidently an inverse in E'_X. Thus E'_X is a groupoid. Conversely, suppose that the E'_X are groupoids, and let u' be an arrow of E' such that $p(u')$ is an isomorphism. We prove that u' is an isomorphism. Since p is fibred, one can factor

$$u' : X' \longrightarrow Y'$$

as a composite

$$X' \longrightarrow u^*(Y') \longrightarrow Y',$$

where the first arrow is an X-morphism, with $X = p(X')$ and $u = p(u')$, and the second is a cartesian morphism over u. The first arrow is an isomorphism because E'_X is a groupoid, and the second is an isomorphism because a cartesian morphism in a fibred category is evidently an isomorphism as soon as its projection is an isomorphism.

Now suppose that p is faithful, and let X', Y' be two objects of a fiber category E'_X. Two arrows from X' to Y' lie over the same arrow id_X of E, hence are identical; therefore E'_X is ordered. Conversely, suppose that the fiber categories are ordered, and prove that p is faithful. Let $u', v' : X' \rightrightarrows Y'$ be arrows of E' over the same arrow $u : X \to Y$ of E. They factor as

$$X' \rightrightarrows u^*(Y') \longrightarrow Y',$$

where the two arrows $X' \rightrightarrows u^*(Y')$ are arrows of E'_X with the same source and the same target. These are therefore equal, and hence $u' = v'$. □

We return to the proof of 6.5. We have proved (i) $\Leftrightarrow$ (ii). On the other hand, (ii) $\Leftrightarrow$ (iv) is fairly clear: indeed, on one hand the category $E_{/F}$ is fibred over E with fibers the discrete categories defined by the sets $F(X)$, as follows at once from the definitions; on the other hand, if p is as in (ii), then by the sorites of SGA 1, VI, 8, the category E' fibred over E is E-equivalent to the split category over E defined by the functor $F : E^{\mathrm{op}} \to \mathbf{Set}$ which sends an object X to the set of isomorphism classes of objects of E'_X. This split category is E-isomorphic to $E_{/F}$.

Since (iv) $\Rightarrow$ (iii) is clear, it remains only to prove (iii) $\Rightarrow$ (i). But it is clear that, for p to be faithful, respectively conservative, it is necessary and sufficient that the induced functors

$$E'_{/X'} \longrightarrow E_{/p(X')}$$

be faithful, respectively conservative; *a fortiori*, it suffices that these functors be fully faithful. It remains only to prove that (iii) implies that p is fibred. But again one sees that a functor p is fibred if and only if the induced functors $E'_{/X'} \to E_{/p(X')}$ are fibred. This is so, in particular, if they are equivalences of categories surjective on objects.

7 Generating and cogenerating subcategories

Definition 7.1. Let E be a category and let C be a full subcategory of E. We say that C is a subcategory of E *generating by strict epimorphisms*, respectively *generating by epimorphisms*, if, for every object X of E, the family of arrows of E with target X and with source $X' \in \mathrm{Ob}(C)$ is strictly epimorphic, respectively epimorphic (1.3). We say that C is a *generating subcategory*, respectively *generating for monomorphisms*, respectively *generating for strict monomorphisms*, if, for every arrow $u : Y \to X$, respectively every monomorphism of E, respectively every strict monomorphism of E (10.5), such that for every $X' \in \mathrm{Ob}(C)$ the corresponding map

$$\mathrm{Hom}_E(X', Y) \longrightarrow \mathrm{Hom}_E(X', X)$$

is bijective, u is an isomorphism. Finally, we say that a family (X_i) of objects of E generates by strict epimorphisms, respectively etc., if the full subcategory C of E generated by this family generates by strict epimorphisms, respectively etc.

7.1.1

In terms of the family $(h_{X'})_{X' \in \mathrm{Ob}(C)}$ of functors represented by the objects $X' \in \mathrm{Ob}(C)$,

$$h_{X'} : E \longrightarrow \mathbf{Set} \qquad (X' \in \mathrm{Ob}(C)),$$

one can express the condition that C be generating, respectively generating for monomorphisms, respectively generating for strict monomorphisms, by saying that the family $(h_{X'})$ is conservative, respectively conservative for monomorphisms, respectively conservative for strict monomorphisms (6.1). It follows immediately from the definitions as well that C generates by epimorphisms if and only if the family $(h_{X'})_{X' \in \mathrm{Ob}(C)}$ is faithful (6.1). We shall also give below, in 7.2(i), an analogous interpretation of the condition that C generate by strict epimorphisms.

7.1.2

As with the notions introduced in 6.1, the notions in 7.1 are especially useful when E has suitable exactness properties; then the various notions introduced have a marked tendency all to be equivalent (7.3). For this reason the question of which of the notions in 7.1 should be considered the most important hardly arises: in the most important cases they coincide, and the term "generating subcategory" may therefore be interpreted indifferently as referring to any of the properties considered in 7.1, for example the first one, which is the strongest of all, as we shall see in 7.2(ii).

7.1.3

Suppose that E is a $\mathcal{U}$-category, and consider the canonical functor

$$\varphi : E \longrightarrow \widehat{C} = \mathrm{Hom}(C, \mathcal{U}\text{-}\mathbf{Set}), \tag{7.1.3.1}$$

obtained by composing $E \to \widehat{E} \to \widehat{C}$, where the first functor is the canonical functor (1.3.3), and the second is the restriction functor to C. It is evident that saying that φ is conservative, respectively faithful, is the same as saying that the family of functors

$$h_{X'} : X \longmapsto \mathrm{Hom}_E(X', X) = \varphi(X)(X')$$

for variable $X' \in \mathrm{Ob}(C)$ is a conservative, respectively faithful, family, i.e. also, by 7.1.2, that C is generating, respectively generating by epimorphisms. Similarly, φ is conservative for monomorphisms, respectively for strict monomorphisms, if and only if the family of the $h_{X'}$ is conservative for monomorphisms, respectively for strict monomorphisms; equivalently, if and only if the subcategory C of E is generating for monomorphisms, respectively for strict monomorphisms.

Proposition 7.2. Let E be a $\mathcal{U}$-category, and let C be a full subcategory.

(i) The following conditions are equivalent:

a) C is a subcategory generating by strict epimorphisms.

b) For every $X \in \mathrm{Ob}(E)$, if $C_{/X}$ denotes the full subcategory of $E_{/X}$ formed by the arrows $X' \to X$ with $X' \in \mathrm{Ob}(C)$, then the natural arrow from the inclusion functor $j : C_{/X} \to E$ to the constant functor on $C_{/X}$ with value X makes X an inductive limit of j:

$$X \xleftarrow{\sim} \varinjlim_{C_{/X}} X'.$$

c) The canonical functor φ of (7.1.3.1) is fully faithful.

(ii) Among the notions of 7.1 one has the following implications:

1) C generates by strict epimorphisms (φ fully faithful)

$\Downarrow \quad \searrow$

2) C generates by epimorphisms (φ faithful) 3) C is generating (φ conservative)

$\Downarrow$

4) C generates for monomorphisms (φ conservative for monomorphisms)

$\Downarrow$

5) C generates for strict monomorphisms (φ conservative for strict monomorphisms).

(iii) One has the following conditional implications:

a) If in E epimorphic families of arrows are strictly epimorphic, then 2) $\Rightarrow$ 1). If in E monomorphisms are strict, then 5) $\Rightarrow$ 4).

b) If in E kernels of pairs of arrows, respectively fiber products, are representable, then 3) $\Rightarrow$ 2), respectively 4) $\Rightarrow$ 3).

c) If in E every family of morphisms $X_i \to X$ with common target X factors as a strictly epimorphic, respectively epimorphic, family $X_i \to Y$ followed by a monomorphism, respectively by a strict monomorphism, $Y \to X$, then 4) $\Rightarrow$ 1), respectively 5) $\Rightarrow$ 2).

We immediately record the following corollary.

Corollary 7.3. All the notions considered in 7.1, and recalled in the implication diagram of 7.2(ii), are equivalent in each of the following two cases:

(i) In E, kernels of pairs of arrows and fiber products are representable, monomorphisms are strict, and epimorphic families of arrows are strictly epimorphic.

(ii) In E, every family $(X_i \to X)_{i \in I}$ of arrows with common target X factors as an epimorphic family $(X_i \to Y)$ followed by a monomorphism $Y \to X$, every monomorphism of E is strict, and every epimorphic family of arrows of E is strict.

Indeed, in case (i) one has 4) $\Rightarrow$ 3) and 3) $\Rightarrow$ 2) by (b), and 5) $\Rightarrow$ 4) and 2) $\Rightarrow$ 1) by (a). In case (ii) one has, by (c), the implications 4) $\Rightarrow$ 1) and 5) $\Rightarrow$ 2). The conclusion follows from the implication diagram 7.2(ii).

Proof. [Proof of 7.2] For (i), the implication b) $\Rightarrow$ a) follows at once from the definitions. We prove a) $\Rightarrow$ b). Under hypothesis a), one must prove that, for every $X, Y \in \mathrm{Ob}(E)$, every system of arrows

$$u_{X'} : X' \longrightarrow Y$$

indexed by the $X' \in \mathrm{Ob}(C_{/X})$ such that $u_{X'} f = u_{X''}$ for every arrow $f : X'' \to X'$ in $C_{/X}$ factors through an arrow, necessarily unique by hypothesis a), $X \to Y$. By a), it suffices to check that, for every object Z of $E_{/X}$ and every pair of morphisms $v' : Z \to X'$ and $v'' : Z \to X''$ in $E_{/X}$, with X' and X'' in $C_{/X}$, one has $u_{X'} v' = u_{X''} v''$. But by hypothesis a), the family of arrows $w : X''' \to Z$, with $X''' \in \mathrm{Ob}(C)$, is epimorphic; hence it suffices to check that for every such w one has

$$(u_{X'} v') w = (u_{X''} v'') w,$$

which is also written $u_{X'}(v' w) = u_{X''}(v'' w)$ and follows immediately from the hypothesis on the family u.

We now prove the equivalence of b) and c). For every $X \in \mathrm{Ob}(E)$, the object $\varphi(X)$ of $\widehat{C}$ is the inductive limit in $\widehat{C}$ of the canonical functor $\widehat{C}_{/\varphi(X)} \to \widehat{C}$ (3.4). But $\widehat{C}_{/\varphi(X)}$ is canonically isomorphic to $C_{/X}$, so that in $\widehat{C}$ one has

$$\varphi(X) = \varinjlim_{C_{/X}} X',$$

where the object X' of C is identified with the functor in $\widehat{C}$ that it represents. Consequently, for a second object Y of E, there is a canonical isomorphism

$$\begin{aligned} \mathrm{Hom}_{\widehat{C}}(\varphi(X), \varphi(Y)) &\simeq \varprojlim_{C_{/X}} \mathrm{Hom}_{\widehat{C}}(X', \varphi(Y)) \\ &\simeq \varprojlim_{C_{/X}} \varphi(Y)(X') \\ &\simeq \varprojlim_{C_{/X}} \mathrm{Hom}_E(X', Y). \end{aligned} \tag{$*$}$$

Thus the map

$$\operatorname{Hom}_E(X,Y) \longrightarrow \operatorname{Hom}_{\widehat{C}}(\varphi(X),\varphi(Y))$$

defined by φ is, through the isomorphism between the extreme terms of $(*)$, exactly the map

$$\operatorname{Hom}_E(X,Y) \longrightarrow \varprojlim_{C_{/X}} \operatorname{Hom}_E(X',Y)$$

induced by the inductive system of arrows $X' \to X$ indexed by $C_{/X}$ considered in 7.2(i)b). Therefore the first map is bijective for every Y, with X fixed, if and only if X is an inductive limit of the inclusion functor $j : C_{/X} \to E$. This proves the equivalence of b) and c).

For (ii), the implications 1) $\Rightarrow$ 2) and 3) $\Rightarrow$ 4) $\Rightarrow$ 5) are trivial by the definitions. The implication 1) $\Rightarrow$ 3) is obtained by interpreting property 1) as the full faithfulness of φ, by (i), and observing that a fully faithful functor is conservative. We have already observed in 7.1.1 that 3) means that φ is conservative.

For (iii), assertion (a) is a tautology. Assertion (b) follows from 6.2(i), respectively 6.2(iv), applied to the functor φ, noting that φ is left exact. Finally we prove (c). For $X \in \operatorname{Ob}(E)$, consider the family of all morphisms $X_i \to X$ with $X_i \in \operatorname{Ob}(C)$. By the hypothesis on E, it factors as a strictly epimorphic, respectively epimorphic, family $X_i \to Y$ followed by a monomorphism, respectively by a strict monomorphism, $Y \to X$. Then hypothesis 4), respectively 5), implies that $Y \to X$ is an isomorphism. Thus the family in question is strictly epimorphic, respectively epimorphic. □

Proposition 7.4. Let E be a category, let C be a full subcategory of E, and let X be an object of E. Suppose that C is generating in E, respectively that C generates for monomorphisms and that the fiber product over X of two subobjects of X is representable in E. Then a strict subobject (10.11), respectively a subobject, X' of X is known once, for every $T \in \operatorname{Ob}(C)$, one knows the subset of $\operatorname{Hom}_E(T,X)$ which is the image of $\operatorname{Hom}_E(T,X')$. Consequently, the cardinal of the set of strict subobjects, respectively of the set of subobjects, of X is bounded above by

$$\prod_{T \in \operatorname{Ob}(C)} 2^{\operatorname{card} \operatorname{Hom}_E(T,X)},$$

and if $X \in \operatorname{Ob}(C)$, it is bounded above by $2^{\operatorname{card} \operatorname{Fl}(C)}$.

We first prove the statement with "respectively". Let X' and X'' be two subobjects of X such that, for every $T \in \operatorname{Ob}(C)$, the images of $\operatorname{Hom}_E(T,X')$ and $\operatorname{Hom}_E(T,X'')$ in $\operatorname{Hom}_E(T,X)$ are equal. They are therefore also equal to the image of $\operatorname{Hom}_E(T,X''')$, where X''' is the fiber product of X' and X'' over X. Since C generates for monomorphisms, it follows that the monomorphisms $X''' \to X'$ and $X''' \to X''$ are isomorphisms. Hence X' and X'' are equal, each being equal to the subobject X''' of X.

We now prove the non-respective assertion. By the definition of strict subobject, it suffices to check that knowing the subset $\operatorname{Hom}_E(T,X')$ of $\operatorname{Hom}_E(T,X)$ for every $T \in \operatorname{Ob}(C)$ determines the set of pairs of arrows $u, v : X \rightrightarrows T$ such that $ui = vi$, where $i : X' \to X$ is the canonical injection. Since C is generating, the relation $ui = vi$ is equivalent to the relation $(ui)f = (vi)f$ for every $f \in \operatorname{Hom}_E(T,X')$, i.e. to $ug = vg$ for every $g \in \operatorname{Hom}_E(T,X)$ coming from $\operatorname{Hom}_E(T,X')$ (that is, of the form if with $f \in \operatorname{Hom}_E(T,X')$). This proves the assertion.

Corollary 7.5. Let E be a category, let C be a full generating subcategory, and let X be an object of C. Then a strict quotient (10.8) X' of X is known once, for every $T \in \operatorname{Ob}(C)$, one knows the subset of $\mathrm{Hom}_E(T,X)^2$ formed by the pairs (u,v) such that $qu = qv$, where $q : X \to X'$ is the canonical morphism. Thus the cardinal of the set of strict quotients of X is bounded above by

$$\prod_{T \in \mathrm{Ob}(C)} 2^{\mathrm{card}\,\mathrm{Hom}_E(T,X)^2}.$$

Indeed, by definition, a strict quotient X' of X is known once, for every object Y of E, one knows the subset of $\mathrm{Hom}_E(Y,X) \times \mathrm{Hom}_E(Y,X)$ formed by the pairs (u,v) such that $qu = qv$, where $q : X \to X'$ is the canonical morphism. But the relation $qu = qv$ is equivalent to $(qu)f = (qv)f$ for every morphism $f : T \to Y$ whose source T lies in C, since C is generating. This relation may also be written $q(uf) = q(vf)$, which proves the first assertion of 7.5. The second follows immediately.

7.5.1

One can generalize 7.5 by introducing, for a family of objects $(X_i)_{i\in I}$ of E, the notion of a *strict quotient in E of the family*. By this we mean a strictly epimorphic family $(p_i : X_i \to X')_{i\in I}$ of morphisms of E, with the convention, as for ordinary quotients, that two such families $(p_i : X_i \to X')$ and $(q_i : X_i \to X'')$ are identified if there exists an isomorphism $v : X' \to X''$ (necessarily unique) such that $vp_i = q_i$ for every $i \in I$. With this terminology, the proof of 7.5 applies *mutatis mutandis* to give the following variant.

Variant 7.5.2. Let E and C be as in 7.5, and let $(X_i)_{i\in I}$ be a family of objects of E. Then a strict quotient X' of $(X_i)_{i\in I}$ in E (7.5.1) is known once, for every $T \in \mathrm{Ob}(C)$ and every pair $(i,j) \in I \times I$, one knows the subset of $\mathrm{Hom}_E(T,X_i) \times \mathrm{Hom}_E(T,X_j)$ formed by the pairs (u,v) such that $p_iu = p_jv$, where $p_i : X_i \to X'$ denotes the canonical morphism for each $i \in I$. Consequently, the cardinal of the set of strict quotients of $(X_i)_{i\in I}$ in E is bounded above by

$$\prod_{T\in\mathrm{Ob}(C)} \prod_{i,j\in I} 2^{\mathrm{card}\,\mathrm{Hom}_E(T,X_i)\times\mathrm{card}\,\mathrm{Hom}_E(T,X_j)}.$$

7.5.3

One sees at once that the analogous conclusion is true if one assumes only that C generates for strict monomorphisms, provided one assumes that the products $X_i \times X_j$ are representable and restricts to *effective* quotients of the family $(X_i)_{i\in I}$, i.e. to strict quotients X' such that the fiber products $X_i \times_{X'} X_j$ are representable in E; this is no restriction if E is stable under fiber products.

Proposition 7.6. Let E be a category, let C be a full subcategory of E generating by epimorphisms (7.1), let Π_0 be an infinite cardinal $\geq \mathrm{card}\,\mathrm{Fl}(C)$, let $\Pi \geq \Pi_0$ be a cardinal, and let $(u_i : X_i \to X)_{i\in I}$ be a universal epimorphic family (10.3) in E formed by squareable morphisms (10.7), with sources $X_i \in \mathrm{Ob}(C)$, and such that $\mathrm{card}(I) \leq \Pi$. Then

$$\mathrm{card}\,\mathrm{Fl}(C_{/X}) \leq \Pi^{\Pi_0}.$$

Let $T \in \mathrm{Ob}(C)$. It will be enough to prove that

$$\mathrm{card}\,\mathrm{Hom}_E(T,X) \leq \Pi^{\Pi_0}. \tag{7.6.1}$$

for then one successively obtains

$$\operatorname{card}\operatorname{Ob}(C_{/X}) \leq (\Pi^{\Pi_0}) \times \operatorname{card}\operatorname{Ob}(C) \leq (\Pi^{\Pi_0}) \times \Pi_0 = \Pi^{\Pi_0},$$

and finally

$$\operatorname{card}\operatorname{Fl}(C_{/X}) \leq ((\Pi^{\Pi_0})^2) \times \Pi_0 = \Pi^{\Pi_0},$$

since for two objects S, T of $C_{/X}$ one has

$$\operatorname{card}\operatorname{Hom}_{C_{/X}}(S,T) \leq \operatorname{card}\operatorname{Hom}_C(S,T) \leq \Pi_0.$$

To prove (7.6.1), first note the following lemma.

Lemma 7.6.2. Let J be a set such that $\operatorname{card}(J) = \Pi_0$. For every object T of C and every morphism $f : T \to X$, there exists an epimorphic family $(v_j : S_j \to T)_{j \in J}$, with sources $S_j \in \operatorname{Ob}(C)$, and for every $j \in J$ an element $i(j) \in I$ and a morphism $g_j : S_j \to X_{i(j)}$, such that

$$u_{i(j)} g_j = f v_j.$$

Indeed, the family of projections

$$X_i \times_X T \longrightarrow T$$

is epimorphic by hypothesis. On the other hand, since C generates by epimorphisms, for every i the family of arrows $S \to X_i \times_X T$ with source $S \in \operatorname{Ob}(C)$ is epimorphic. Thus, by transitivity, the family of arrows $v : S \to T$ with source in C which factor through one of the $X_i \times_X T$ is epimorphic. These are precisely the arrows $v : S \to T$ with source in C for which there exist $i \in I$ and $g : S \to X_i$ such that $u_i g = f v$. Since the set of these arrows $v : S \to T$ is contained in $\operatorname{Fl}(C)$ and therefore has cardinal $\leq \Pi_0$, it can be indexed by the index set J. This proves the lemma.

Now a morphism $f : T \to X$ is known once the $f v_j$ are known, and these are known once the g_j are known. Thus $\operatorname{card}\operatorname{Hom}_E(T, X)$ is bounded above by the cardinal of the set of families $(i(j), S_j, v_j, g_j)_{j \in J}$. The cardinal of the set of maps $j \mapsto i(j)$ from J to I is $\leq \Pi^{\Pi_0}$; and for such a map fixed, the cardinal of the set of the corresponding families S_j, g_j, v_j is bounded above by the cardinal of the set of maps from J to $\operatorname{Fl}(C) \times \operatorname{Fl}(C)$, which is $\leq \Pi_0^{\Pi_0}$, since $\operatorname{card}(\operatorname{Fl}(C) \times \operatorname{Fl}(C)) = \Pi_0^2 = \Pi_0$. Thus the left hand side of (7.6.1) is bounded above by

$$\Pi^{\Pi_0} \times \Pi_0^{\Pi_0} = \Pi^{\Pi_0},$$

which completes the proof of 7.6.

Proposition 7.7. Let E be a category in which fiber products are representable, let $(X_\alpha)_{\alpha \in A}$ be a generating family of objects of E, and let $(\varphi_i)_{i \in I}$ be a family of functors $\varphi_i : E \to E_i$ commuting with fiber products. For (φ_i) to be conservative (6.1), it is necessary and sufficient that for every $\alpha \in A$ and every subobject X' of X_α distinct from X_α, there exist $i \in I$ such that the morphism

$$\varphi_i(X') \longrightarrow \varphi_i(X_\alpha)$$

is not an isomorphism. In this case, if E is a $\mathcal{U}$-category and A is $\mathcal{U}$-small, there exists a $\mathcal{U}$-small subset J of I such that $(\varphi_j)_{j \in J}$ is already a conservative family of functors.

The necessity of the condition of conservativity under consideration is evident; we prove its sufficiency. By 6.2(iv), it suffices to prove that (φ_i) is conservative for monomorphisms. Let $u : Y' \to Y$ be a monomorphism in E which is not an isomorphism. We must prove that there is an $i \in I$ such that $\varphi_i(u)$ is not an isomorphism. By the hypothesis on (X_α), there exists an α and a morphism $X_\alpha \to Y$ which does not factor through Y'. In other words, the inverse image X' of Y' is a subobject of X_α distinct from X_α. By hypothesis, there exists an $i \in I$ such that

$$\varphi_i(X') \longrightarrow \varphi_i(X_\alpha)$$

is not an isomorphism. Since

$$\varphi_i(X') \simeq \varphi_i(X_\alpha) \times_{\varphi_i(Y)} \varphi_i(Y'),$$

it follows that $\varphi_i(Y') \to \varphi_i(Y)$ is not an isomorphism.

The second assertion of 7.7 follows at once from the first, taking 7.4 into account.

Corollary 7.7.1. Let E be a $\mathcal{U}$-category in which fiber products are representable, and suppose that E admits a generating family of objects which is $\mathcal{U}$-small. Then, for every generating family $(Y_i)_{i\in I}$ of E, there exists a $\mathcal{U}$-small generating subfamily $(Y_j)_{j\in J}$, with $\mathrm{card}(J) \in \mathcal{U}$.

It suffices to apply 7.7 to a $\mathcal{U}$-small generating family $(X_\alpha)_{\alpha\in A}$ of E and to the family of functors φ_i represented by the Y_i.

Proposition 7.8. Let E be a category, let C be a full subcategory generating by strict epimorphisms, let D be a category, and let $\mathrm{Hom}'(E, D)$ be the full subcategory of $\mathrm{Hom}(E, D)$ consisting of the functors which commute with inductive limits of type $C_{/X}$, where X ranges over the objects of E. Then the functor $F \mapsto F|_C$ induces a fully faithful functor

$$\mathrm{Hom}'(E, D) \longrightarrow \mathrm{Hom}(C, D).$$

Consequently, if $\mathrm{Hom}(C, D)$ is a $\mathcal{U}$-category (1.1), for example by (1.1.1 b)) if C is $\mathcal{U}$-small and D is a $\mathcal{U}$-category, then $\mathrm{Hom}'(E, D)$ is also a $\mathcal{U}$-category.

Let $F, G : E \to D$ be two functors in the source category, and let $u : F \to G$ be a morphism. If $X = \varinjlim X_i$ in E and if F and G commute with the inductive limit in question, then $u(X) : F(X) \to G(X)$ is identified with the limit of the morphisms $u(X_i) : F(X_i) \to G(X_i)$, and hence is known once the $u(X_i)$ are known. This shows that the functor considered in 7.8 is faithful, by the implication a) $\Rightarrow$ b) in 7.2(i). Conversely, let $v : F|_C \to G|_C$ be a morphism; we prove that it comes from a morphism $u : F \to G$. For every $X \in \mathrm{Ob}(E)$ define

$$u(X) : F(X) = \varinjlim_{C_{/X}} F(X_i) \longrightarrow G(X) = \varinjlim_{C_{/X}} G(X_i)$$

as the inductive limit of the $v(X_i)$. It is immediate that this gives a morphism functorial in X, hence a morphism $u : F \to G$, and finally that the morphism induced by u from $F|_C$ to $G|_C$ is v. This completes the proof.

7.1 Families and subcategories cogenerating

Let E be a category, and let C be a full subcategory of E. We say that C is cogenerating by strict monomorphisms, respectively cogenerating by monomorphisms, respectively cogenerating, respectively cogenerating for epimorphisms, respectively cogenerating for strict epimorphisms, if the full subcategory C^{op} of E^{op} generates by strict epimorphisms, respectively etc. The terminology for families is analogous. Thus all the results in this section concerning the notion of generating subcategory and its variants (7.1) give corresponding results for the dual notions, which we leave to the reader to formulate for personal satisfaction.

Proposition 7.10. Let E be a category, let C be a full generating subcategory (7.1), and let D be a full subcategory of E. For D to be cogenerating (7.9), it is necessary and sufficient that, for every pair of arrows

$$T \underset{v}{\overset{u}{\rightrightarrows}} X$$

in E with source $T \in \mathrm{Ob}(C)$ and with $u \neq v$, there exist an arrow $w : X \to I$ with target $I \in \mathrm{Ob}(D)$ such that $wu \neq wv$.

By definition, to say that D is cogenerating means that for every pair of arrows $Y \rightrightarrows X$ in E such that $u \neq v$, there exists an arrow $w : X \to I$, with $I \in \mathrm{Ob}(D)$, such that $wu \neq wv$. Thus 7.10 simply says that it suffices to test this property when $Y \in \mathrm{Ob}(C)$. Since C is generating, the hypothesis $u \neq v$ implies that there exists $f : T \to Y$ such that $uf \neq vf$. By the hypothesis there exists $w : X \to I$, with target $I \in \mathrm{Ob}(D)$, such that $w(uf) \neq w(vf)$, and hence $wu \neq wv$. □

Corollary 7.11. With the notation of 7.10, suppose that the objects I of D are injective objects of E, i.e. that for every monomorphism $X \to Y$ in E, the corresponding map

$$\mathrm{Hom}_E(Y, I) \longrightarrow \mathrm{Hom}_E(X, I)$$

is surjective. Suppose moreover that every pair of arrows

$$T \underset{v}{\overset{u}{\rightrightarrows}} X$$

in E factors as an epimorphic, respectively effective epimorphic, pair $T \rightrightarrows X'$ followed by a monomorphism $i : X' \to X$. Then in the criterion 7.10 for D to be cogenerating, one may restrict to pairs (u, v) which are epimorphic, respectively effective epimorphic.

We conclude:

Corollary 7.12. Let E be a $\mathcal{U}$-category admitting a small generating subcategory C, and such that every object of E is the source of a monomorphism into an injective object of E. Suppose moreover that, for every $T \in \mathrm{Ob}(C)$, the sum $T \amalg T$ in E is representable, and that every morphism with source $T \amalg T$ factors as an effective epimorphism followed by a monomorphism. Then E admits a small full cogenerating subcategory D. More precisely, one may take D such that

$$\operatorname{card} \mathrm{Ob}(D) \leq \prod_{T, T' \in \mathrm{Ob}(C)} 2^{\operatorname{card}(\mathrm{Hom}_E(T', T \amalg T))^2}.$$

Indeed, by 7.11 it is enough, for every $T \in \mathrm{Ob}(C)$ and every effective quotient X of $T \amalg T$, to choose an embedding of X into an injective object I of E, and to take for D the full subcategory of E generated by these I. The conclusion then follows from 7.5.

Examples 7.13. To construct small cogenerating subcategories in terms of small generating subcategories, one is therefore led to look for conditions ensuring that a $\mathcal{U}$-category E has "enough injectives", i.e. that every object embeds into an injective object by a monomorphism. It is well known [Tohoku] that this condition is satisfied in a $\mathcal{U}$-abelian category with exact small filtered inductive limits (axiom AB5 of *loc. cit.*) admitting a small generating family. The construction of *loc. cit.* is not, moreover, essentially tied to abelian categories, and also works in the category of sheaves of $\mathcal{U}$-sets on a topological space $X \in \mathcal{U}$. We shall not state here the exactness properties which make the construction in question work, and shall limit ourselves to observing that in the special case of the category of sheaves of sets on X, one immediately reduces to the case of *loc. cit.* as follows.

Note that if $\mathcal{O}_X$ is a ring on X, then every injective object I of the category of $\mathcal{O}_X$-modules is also injective as an object of the category of sheaves of sets. Indeed, if F is a sheaf of sets and $\mathcal{O}_X[F]$ is the "free $\mathcal{O}_X$-module generated by F", then by definition there is a homomorphism of sheaves of sets

$$F \longrightarrow \mathcal{O}_X[F] \tag{$*$}$$

giving an isomorphism, functorial in F,

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{O}_X[F], I) \xrightarrow{\sim} \mathrm{Hom}(F, I).$$

Since the functor $F \mapsto \mathcal{O}_X[F]$ plainly carries monomorphisms to monomorphisms, it follows at once that, if I is an injective $\mathcal{O}_X$-module, then it is also an injective sheaf of sets. It follows that if the morphism $(*)$ is a monomorphism, as happens if one takes for $\mathcal{O}_X$ a constant ring with value a ring $A \neq 0$, then an embedding of $\mathcal{O}_X[F]$ into an injective $\mathcal{O}_X$-module I gives an embedding of F into the underlying injective sheaf of sets of I.

The same argument applies, without change, to the category of sheaves of $\mathcal{U}$-sets on a $\mathcal{U}$-site, which will be introduced in the next exposé.

The interest of the existence of a small cogenerating subcategory lies chiefly in the representability criterion 8.12.7 below.

Exposé I. Presheaves

Section 8.1–8.6

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 8.1 through Section 8.6. The source has one visibly truncated sentence in the proof of Proposition 8.6.4; the translation marks the mathematically forced completion as referring to the immediately following Corollary 8.6.5.

8 Ind-objects and pro-objects

8.1 Cofinal functors and cofinal subcategories

Definition 8.1.1. Let

$$\varphi : I \longrightarrow I' \tag{8.1.1.1}$$

be a functor. We say that φ is a *cofinal functor* if, for every category C and every functor $u : I' \to C$, considering $\varinjlim u$ and $\varinjlim u \circ \varphi$ as objects of $\mathrm{Hom}(G(\mathcal{V}\text{-}\mathbf{Set}))^{\mathrm{op}}$ (2.1, 2.3.1), where $\mathcal{V}$ is a universe such that $I, I' \in \mathcal{V}$ and C is a $\mathcal{V}$-category, the canonical morphism

$$\varinjlim u \circ \varphi \longrightarrow \varinjlim u \tag{8.1.1.2}$$

is an isomorphism. When φ is the inclusion functor of a subcategory of I', we say that I is a *cofinal subcategory* if φ is cofinal.

8.1.2

Notice that, for fixed φ and u, the bijectivity of (8.1.1.2) does not depend on the choice of the universe $\mathcal{V}$.

Returning to the definition of the terms appearing in (8.1.1.2), one sees that to say that φ is cofinal is also to say that, for every universe $\mathcal{V}$ such that I and I' are $\mathcal{V}$-small, and for every functor

$$v : I'^{\mathrm{op}} \longrightarrow (\mathcal{V}\text{-}\mathbf{Set}),$$

the canonical homomorphism

$$\varprojlim v \longrightarrow \varprojlim v \circ \varphi \tag{8.1.2.1}$$

is an isomorphism, that is, a bijection. To see that this condition is indeed necessary, observe, putting $v = v^{\mathrm{op}}$, that it also says that the condition of Definition 8.1.1 is fulfilled when one takes $C = (\mathcal{V}\text{-}\mathbf{Set})^{\mathrm{op}}$; this implies that the inductive limits considered in (8.1.1.2) are representable in C.

8.1.2.2

It is immediate from the definition that *the composite of two cofinal functors is cofinal.*

Proposition 8.1.3. Let $\varphi : I \to I'$ be a functor.

a) For φ to be cofinal, it is necessary that φ satisfy the condition:

F 1) For every object i' of I', there exists an object i of I such that $\varphi(i)$ dominates i', i.e. such that $\mathrm{Hom}(i', \varphi(i)) \neq \emptyset$.

b) Suppose that I is filtered. For φ to be cofinal, it is necessary and sufficient that φ satisfy condition (F 1) and the following condition:

F 2) For every object i of I and every double arrow

$$i' \overset{f'}{\underset{g'}{\rightrightarrows}} \varphi(i)$$

in I' with target $\varphi(i)$, there exists an arrow $h : i \to j$ in I such that $\varphi(h)f' = \varphi(h)g'$.

Moreover, if φ is cofinal, then I' is filtered.

c) Suppose that I' is filtered and that φ is fully faithful. For φ to be cofinal, it is necessary and sufficient that it satisfy condition F 1) of a). This implies that I is filtered.

Proof. For necessity in a) and b), one uses Definition 8.1.1 only in the case where u is the canonical inclusion functor

$$u : I' \hookrightarrow \widehat{I'}$$

(choosing a universe $\mathcal{U}$ such that I and I' are $\mathcal{U}$-small). Then (8.1.1.2) may be interpreted as an arrow of $\widehat{I'}$, whose target is the *final* functor on I'. One sees immediately that condition F 1) expresses that this arrow is an epimorphism of $\widehat{I'}$, i.e. that it is surjective on each argument, since inductive limits in $\widehat{I'}$ are calculated argument by argument. This proves a).

Now suppose that I is filtered and that φ is cofinal. Then I' is filtered: indeed, the condition that two objects of I' be dominated by a third follows at once from the corresponding condition on I and from F 1). It remains only to prove condition PS 2) of 2.7, i.e. that for every double arrow

$$f', g' : i' \rightrightarrows j'$$

in I', there exists an arrow $h' : j' \to k'$ in I' such that $h'f' = h'g'$. By F 1) we may suppose that k' is of the form $\varphi(i)$. But by the hypothesis that φ is cofinal, for every object i' of I' the set $\varinjlim_i \mathrm{Hom}(i', \varphi(i))$ has one element. It follows from the standard description of filtered inductive limits in **Set** (2.8.1) that there is an arrow $h : i \to j$ in I such that $\varphi(h)f' = \varphi(h)g'$. This finishes the proof that I' is filtered, and proves F 2) at the same time.

To prove that the conditions stated in b) are sufficient for φ to be cofinal, use the form 8.1.2 of the definition. One immediately checks that F 1) implies that (8.1.2.1) is a monomorphism, i.e. injective on each argument, while F 2), together with F 1), ensures that it is bijective, since projective limits in $\widehat{I'^{\mathrm{op}}}$ are calculated argument by argument. This proves b).

Finally, if I' is filtered and φ is fully faithful, it is immediate that F 1) implies that I is filtered and implies F 2). Thus c) follows from a) and b).

8.1.4

When I' is a filtered category and I is a full subcategory of I', Proposition 8.1.3 c) shows that the condition that I be cofinal in I' depends only on the subset $\mathrm{Ob}\, I$ of the preordered set $\mathrm{Ob}\, I'$ (for the preorder relation "$x \leq y$" iff "$\mathrm{Hom}(x, y) \neq \emptyset$"). If J' is a preordered set and J is a subset of J', we shall sometimes say that J is a *cofinal subset of* J' when every element of J' is dominated by an element of J. When J' is filtered, this therefore means that the inclusion functor $\mathcal{J} \hookrightarrow \mathcal{J}'$ of the associated categories is cofinal.

8.1.5

In what follows, we shall use the notion of cofinal functor only in cases where the categories I and I' are filtered. Classically, one even restricted oneself to categories associated with preordered sets, i.e. categories in which there is at most one arrow with given source and target. It turns out, however, that this restriction is inconvenient in applications, because the "natural" filtered categories which arise in many applications are *not* ordered categories. The following result, due to P. Deligne, nevertheless shows that there is no essential difference between the two points of view.

Proposition 8.1.6. Let I be a small filtered category. Then there exists a small ordered set E, and a cofinal functor

$$\varphi : \mathcal{E} \longrightarrow I,$$

where $\mathcal{E}$ denotes the category associated with E.

First suppose that the preordered set $\mathrm{Ob}\, I$ has no greatest element. By a subdiagram of I we mean a pair $D = (\mathcal{O}, F)$ consisting of a subset F of $\mathrm{Fl}(I)$ and a subset $\mathcal{O}$ of $\mathrm{Ob}\, I$, such that for every $f \in F$, the source and target of f belong to $\mathcal{O}$. An element e of $\mathcal{O}$ is called a *final object* of the subdiagram D if, for every $x \in D$, the set $\mathrm{Hom}(x, e) \cap F$ of arrows of D with source x and target e has exactly one element f_x, if for every arrow $g : x \to y$ of D one has $f_x = f_y \circ g$, and if $f_e = \mathrm{id}_e$.

Let E be the set of *finite* subdiagrams D of I having a *unique* final element $\varphi(D)$, and order E by inclusion. If D, D' are elements of E and $D \subset D'$, then there is a unique arrow $\varphi(D', D)$ of D' with source $\varphi(D)$ and target $\varphi(D')$; if $D \subset D' \subset D''$ are inclusions in E, then of course

$$\varphi(D'', D) = \varphi(D'', D')\varphi(D', D),$$

and also $\varphi(D, D) = \mathrm{id}_{\varphi(D)}$. Thus one obtains a functor $\varphi : \mathcal{E} \to I$, and it remains to prove that E is filtered and that φ is cofinal. By Proposition 8.1.3 b), three verifications are required.

1) Condition F 1) of 8.1.3. For every $i \in I$, take D to be the subdiagram of I reduced to the object i and its identity arrow. Then $i \leq \varphi(D) = i$. This condition F 1) also shows that $E \neq \emptyset$.

2) Condition F 2). For every $i \in \mathrm{Ob}\, I$, every $D \in E$, and every double arrow $i \rightrightarrows \varphi(D)$, one must find a diagram $D' \in E$, containing D, such that

$$\varphi(D', D) : \varphi(D) \longrightarrow \varphi(D')$$

equalizes the double arrow. Since I is filtered, there exists an object j of I and an arrow $f : \varphi(D) \to j$ equalizing the double arrow. Since I has no greatest element, we may suppose that j is a *strict* upper bound of $\varphi(D)$, which implies that it is distinct from all the other objects of D. Let D' be the subdiagram of I whose set of objects is $\mathcal{O} \cup \{j\}$, and whose set of arrows consists of the arrows of D, together with the composites

$$x \xrightarrow{f_x} \varphi(D) \xrightarrow{f} j,$$

where x is an object of D and f_x is the unique arrow of D with source x and target $\varphi(D)$, and together with id_j. It is then clear that D' is a finite subdiagram of I, that it has j as its unique final object, hence that $D' \in E$, and that D' satisfies the required condition.

3) Let D and D' be two subdiagrams in E. They are contained in some common $D'' \in E$. Indeed, one can find a strict upper bound j of $\varphi(D)$ and $\varphi(D')$,

$$\begin{tikzcd} \varphi(D) \arrow[dr, "f"] & \\ & j \\ \varphi(D') \arrow[ur, "f'"'] & \end{tikzcd}$$

and take D'' to be the subdiagram whose set of objects is the union of the sets of objects of D and D' and of $\{j\}$, and whose set of arrows is the union of the sets of arrows of D and D', the composites $f \circ f_x$ with x an object of D, the composites $f' \circ f'_{x'}$ with x' an object of D', and $\{\mathrm{id}_j\}$. This is indeed a finite subdiagram of I. We show that, after replacing j, f, f' by j', gf, gf' with a suitable $g : j \to j'$, we may arrange for j to be a final object of D''. This is the same as saying that, for every x which is an object of both D and D', one has

$$ff_x = f'f'_{x'}.$$

But this is only a *finite* set of double arrows to be equalized by some $g : j \to j'$, and this is possible because I is filtered.

This completes the proof in the case considered. The general case is reduced to it by introducing the filtered category $\mathbb{N}$ associated with the ordered set $\mathbb{N}$ of natural numbers, and observing that the category $\mathbb{N} \times I$ is filtered, has no greatest element, and that the projection $\mathbb{N} \times I \to I$ is a cofinal functor.

Corollary 8.1.7. Let I be a $\mathcal{U}$-category. For there to exist a small filtered ordered set E and a cofinal functor $\mathcal{E} \to I$, it is necessary and sufficient that I be filtered and that $\mathrm{Ob}\, I$ admit a small cofinal subset I' (8.1.4).

This is necessary by 8.1.3 a), and sufficient by 8.1.6 applied to the full subcategory of I defined by I'.

Definition 8.1.8. Let I be a filtered category. We say that I is *essentially small* if I is a $\mathcal{U}$-category and if it satisfies the equivalent conditions of 8.1.7.

8.2 Ind-objects and ind-representable functors

8.2.1

In what follows we fix a $\mathcal{U}$-category C, which we always regard as embedded in the category $\widehat{C}$ of $\mathcal{U}$-presheaves by means of the canonical functor (1.3.1)

$$h : C \hookrightarrow \widehat{C}. \tag{8.2.1.1}$$

We shall call an *ind-object of* C (or an *inductive system* of C) any functor

$$\varphi : I \longrightarrow C, \tag{8.2.1.2}$$

where I is a *filtered* category (2.7), called the *index category* of the ind-object. Apart from this, I is arbitrary; in particular we do not require I to be a $\mathcal{U}$-category. It will often be convenient to denote an ind-object φ by the indexed notation $(X_i)_{i \in \operatorname{Ob} I}$, or even, by a further abuse of notation, by $(X_i)_{i \in I}$, $(X_i)_i$, or (X_i), where $X_i = \varphi(i)$ for $i \in \operatorname{Ob} I$. This notation is no more and no less abusive than the classical notation for inductive systems indexed by filtered ordered sets, where the transition morphisms are likewise not mentioned. The X_i are called the *component objects* of the ind-object $\mathcal{X}$. Notice that, in the general case considered here, the "*transition morphisms*" $\varphi(f)$ of the inductive system are indexed by the set $\operatorname{Fl}(I)$ of arrows of I, which in general is not identified with a subset of $\operatorname{Ob} I \times \operatorname{Ob} I$. In other words, for given i and j, with j dominating i, there may exist more than one transition morphism from X_i to X_j.

8.2.2

The most useful ind-objects $\mathcal{X} = (X_i)_{i \in I}$ of C are those for which the index category I is essentially small (8.1.8); such an ind-object is called *essentially small.* If $(X_i)_{i \in I}$ is such an object, using the fact that small inductive limits in $\widehat{C}$ are representable (3.1), one may consider

$$L(\mathcal{X}) := \varinjlim h \cdot \mathcal{X} = \varinjlim_i X_i \in \operatorname{Ob} \widehat{C}, \tag{8.2.2.1}$$

where the last limit is taken *in* $\widehat{C}$. Thus $L(\mathcal{X})$ is the presheaf

$$L(\mathcal{X}) : Y \longmapsto \varinjlim_i \operatorname{Hom}(Y, X_i) \tag{8.2.2.2}$$

on C. We say that this presheaf is the *presheaf ind-represented by the ind-object* $\mathcal{X} = (X_i)_{i \in I}$ *of* C. A presheaf F on C is called an *ind-representable presheaf* if it is isomorphic to a presheaf $L(\mathcal{X})$ ind-represented by an essentially small ind-object. It follows from 8.1.6 that, in this definition, one may suppose that I is a small category, and even that I is associated with an ordered set belonging to $\mathcal{U}$.

8.2.3

Consider an ind-object $\mathcal{X} = (X_i)_{i \in I}$, given by a functor $\varphi : I \to C$, and let

$$u : I' \longrightarrow I$$

be a functor, where I' is a second filtered category. Then the functor $\varphi \circ u$ is an ind-object $\mathcal{X}'$ of C with index category I', written in indexed notation as $(X_{u(i')})_{i' \in I'}$. It is called the ind-object of C *deduced from* $(X_i)_{i \in I}$ *by change of index categories* using the functor u. If I and I', equivalently $\mathcal{X}$ and $\mathcal{X}'$, are essentially small, one obtains a canonical morphism

$$L(\mathcal{X}') = \varinjlim_{i'} X_{u(i')} \longrightarrow L(\mathcal{X}) = \varinjlim_i X_i \tag{8.2.3.1}$$

between the presheaves ind-represented by $\mathcal{X}'$ and $\mathcal{X}$. When u is a cofinal functor (8.1.1), $\mathcal{X}$ is essentially small if and only if $\mathcal{X}'$ is so (8.1.7), and the preceding homomorphism (8.2.3.1) is an isomorphism

$$L(\mathcal{X}') \simeq L(\mathcal{X}). \tag{8.2.3.2}$$

8.2.4

Let

$$\mathcal{X} = (X_i)_{i\in I}, \qquad \mathcal{Y} = (Y_j)_{j\in J} \tag{8.2.4.1}$$

be two essentially small ind-objects of C, indexed by filtered essentially small categories I and J. A *morphism* from the ind-object $\mathcal{X}$ to the ind-object $\mathcal{Y}$ is by definition a morphism

$$L(\mathcal{X}) \longrightarrow L(\mathcal{Y}) \tag{8.2.4.2}$$

between the presheaves they ind-represent. We put

$$\mathrm{Hom}_{\text{ind-ob}}(\mathcal{X}, \mathcal{Y}) = \mathrm{Hom}_{\widehat{C}}(L(\mathcal{X}), L(\mathcal{Y})). \tag{8.2.4.3}$$

We suppress the subscript "ind-ob" on Hom if no confusion is likely. Composition of morphisms of ind-objects of C is defined as composition of morphisms of the presheaves they ind-represent. If $\mathcal{V} \supset \mathcal{U}$ is a universe containing $\mathcal{U}$, and if one restricts to essentially small ind-objects which belong to $\mathcal{V}$, these form the set of objects of a category whose arrows are triples $(\mathcal{X}, \mathcal{Y}, u)$, where $\mathcal{X}$ and $\mathcal{Y}$ are essentially small ind-objects belonging to $\mathcal{V}$ and $u : \mathcal{X} \to \mathcal{Y}$ is a morphism of ind-objects, i.e. a morphism $L(\mathcal{X}) \to L(\mathcal{Y})$. The category so obtained is denoted

$$\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U}). \tag{8.2.4.4}$$

When $\mathcal{V} = \mathcal{U}$, it is denoted simply

$$\mathrm{Ind}_{\mathcal{U}}(C) \quad \text{or} \quad \mathrm{Ind}(C), \tag{8.2.4.5}$$

and is called the *category of ind-objects of C* relative to $\mathcal{U}$, the mention of $\mathcal{U}$ being suppressed when no confusion is likely.

Of course, if $\mathcal{V} \subset \mathcal{V}'$ are two universes containing $\mathcal{U}$, then $\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U})$ is a full subcategory of $\mathrm{Ind}_{\mathcal{V}'}(C,\mathcal{U})$. It follows from 8.2.3 that the inclusion functor is an *equivalence of categories*

$$\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U}) \xrightarrow{\approx} \mathrm{Ind}_{\mathcal{V}'}(C,\mathcal{U}). \tag{8.2.4.6}$$

This shows in particular that *every essentially small ind-object of C defines an element of* $\mathrm{Ind}(C)$*, uniquely up to isomorphism.* This observation justifies the fairly common practical abuse of language which consists in identifying any essentially small ind-object of C with an object of $\mathrm{Ind}(C)$.

It follows at once from the definitions that for every universe $\mathcal{V}$ we have a canonical functor

$$L : \mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U}) \longrightarrow \widehat{C} \tag{8.2.4.7}$$

which is fully faithful. These functors are of course known up to unique isomorphism, by (8.2.4.6), once one of them is known, in particular once one knows the canonical functor

$$L : \mathrm{Ind}(C) \longrightarrow \widehat{C}. \tag{8.2.4.8}$$

Since this functor is fully faithful, it is often used to identify an object of the first category with the functor it pro-represents, or even to identify $\mathrm{Ind}(C)$ with its essential image in $\widehat{C}$, consisting of the ind-representable presheaves. Notice, however, that this identification has distinctly more drawbacks than the analogous identification of C with a full subcategory of $\widehat{C}$ via the functor h (8.2.1.1), because unlike the latter functor, the functor (8.2.4.8) is not in general injective on objects.

8.2.5

Let us spell out definition (8.2.4.3) in terms of the indexed expressions (8.2.4.1) of the ind-objects under consideration. Taking account of the definition (8.2.2.1), one obtains

$$\mathrm{Hom}(\mathcal{X},\mathcal{Y}) \simeq \varprojlim_i \mathrm{Hom}(X_i,\mathcal{Y}) := \varprojlim_i \mathrm{Hom}(X_i, L(\mathcal{Y})) = \varprojlim_i \varinjlim_j \mathrm{Hom}_C(X_i, Y_j),$$

the last equality coming from (8.2.2.2) applied to $\mathcal{Y}$. Thus there is a canonical bijection

$$\mathrm{Hom}((X_i)_{i\in I},(Y_j)_{j\in J}) \simeq \varprojlim_{i\in I} \varinjlim_{j\in J} \mathrm{Hom}_C(X_i, Y_j). \tag{8.2.5.1}$$

We leave to the reader the task of making the composition of morphisms of ind-objects explicit using this formula. Notice that it follows at once from this formula that the set $\mathrm{Hom}(\mathcal{X},\mathcal{Y})$ of homomorphisms of ind-objects is *$\mathcal{U}$-small*; equivalently, the categories (8.2.4.4) are $\mathcal{U}$-categories. Equivalently again, by (8.2.4.6), the *category* $\mathrm{Ind}(C)$ *is a $\mathcal{U}$-category*; that is, the right hand side of (8.2.5.1) is small when $I, J \in \mathcal{U}$, which follows immediately from the fact that C is a $\mathcal{U}$-category, hence the $\mathrm{Hom}_C(X_i, Y_j)$ are small.

8.2.6

Let I be an essentially small filtered category. Then one sees either from (8.2.5.1), or from the functoriality of the inductive limit of a functor $I \to \widehat{C}$ with respect to that functor, that there is a canonical functor

$$\mathrm{Hom}(I, C) \longrightarrow \mathrm{Ind}(C). \tag{8.2.6.1}$$

Notice that this functor is *not* in general fully faithful, or even faithful.

Remark 8.2.7. The definitions of 8.2.4 make clear the notion of *isomorphism* of two essentially small ind-objects, which also means the isomorphism of the presheaves they ind-represent. Notice that this is a very weak notion of isomorphism when compared with the notion of isomorphism in categories of the form $\mathrm{Hom}(I, C)$. Thus many fairly natural relations one can impose on an ind-object, such as having its components in a given strictly full subcategory, or being strict (8.12 below), are *not* stable under isomorphism. Therefore, if one wants to work with notions stable under isomorphism of ind-objects, one must "saturate" the notions under consideration by passing to the strictly full subcategory of $\mathrm{Ind}(C)$ generated by the subcategory of $\mathrm{Ind}(C)$ consisting of objects satisfying the condition in question. See, for example, the notion of essentially constant ind-object introduced below (8.4).

Exercise 8.2.8. Let $\mathcal{U} \subset \mathcal{V}$ be two universes and let C be a $\mathcal{U}$-category belonging to $\mathcal{V}$. Let $\mathrm{SysInd}(C)$ denote the following category.

a) The objects of $\mathrm{SysInd}(C)$ are functors $\varphi : I \to C$, where I is an essentially $\mathcal{U}$-small filtered category belonging to $\mathcal{V}$.

b) If (I,φ) and (J,ψ) are two objects of $\mathrm{SysInd}(C)$, a morphism in $\mathrm{SysInd}(C)$ from (I,φ) to (J,ψ) is a pair (m,u), where $m : I \to J$ is a functor and $u : \varphi \to \psi \circ m$ is a morphism of functors. The composition of morphisms in $\mathrm{SysInd}(C)$ is defined in the evident way.

Let S be the set of morphisms $(m, u) : (I, \varphi) \to (J, \Psi)$ of $\mathrm{SysInd}(C)$ such that m is cofinal and u is an isomorphism. Let (I, φ) be an object of $\mathrm{SysInd}(C)$. The category $\mathrm{Fl}(I)$ of arrows of I maps to I by two natural functors: source and target. Moreover these two functors are related by the canonical morphism of functors $v : \mathrm{source} \to \mathrm{target}$. Hence there are two morphisms in $\mathrm{SysInd}(C)$ from $(\mathrm{Fl}(I), \varphi \circ \mathrm{source})$ to (I, φ):

$$p_1(I, \varphi) = (\mathrm{source}, \mathrm{id}),$$

and

$$p_2(I, \varphi) = (\mathrm{target}, \varphi^* v : \varphi \circ \mathrm{source} \to \varphi \circ \mathrm{target}).$$

Let $p : \mathrm{SysInd}(C) \to \mathrm{Ind}(C)$ be the evident functor. Let B be a category. Show that the functor $F \mapsto F \circ p$,

$$\mathrm{Hom}(\mathrm{Ind}(C), B) \longrightarrow \mathrm{Hom}(\mathrm{SysInd}(C), B),$$

is fully faithful, and that a functor $G : \mathrm{SysInd}(C) \to B$ belongs to its essential image if and only if it has the following two properties:

1) For every $s \in S$, $F(s)$ is an isomorphism of B.

2) For every object (I, φ) of $\mathrm{SysInd}(C)$, $F(p_1(I, \varphi)) = F(p_2(I, \varphi))$.

8.3 Characterization of ind-representable functors

Proposition 8.3.1. Let F be an ind-representable presheaf on C. Then F is left exact, i.e. (2.3.2), for every finite category J and every functor $\varphi : J \to C$ such that $\varinjlim \varphi$ is representable (i.e. $\varprojlim \varphi^{\mathrm{op}}$ is representable), the canonical map

$$F(\varinjlim \varphi) \longrightarrow \varprojlim F \circ \varphi$$

is bijective.

Indeed, representable presheaves are plainly left exact; hence by 2.8 so is every filtered inductive limit of such functors. $\square$

8.3.2

Given a presheaf F on C, we shall often have to work with the category

$$C_{/F} \hookrightarrow \widehat{C}_{/F}, \tag{8.3.2.1}$$

which denotes the full subcategory of the second category consisting of arrows $X \to F$ *whose source lies in* C. It follows from 1.4 that the objects of this category are identified with pairs (X, u), where X is an object of C and $u \in F(X)$. A morphism from (X, u) to (X', u') is then interpreted as an arrow $f : X \to X'$ in C such that $F(f)(u') = u$. The "forget-the-target" functor from $\widehat{C}_{/F}$ to $\widehat{C}$ induces a functor, called the *canonical* functor,

$$C_{/F} \longrightarrow C, \tag{8.3.2.2}$$

already considered in 3.4, where it is proved that the inductive limit of this functor exists in $\widehat{C}$ (with no smallness condition on $C_{/F}$) and is canonically isomorphic to F.

8.3.2.3

Notice that if finite inductive limits are representable in C, and if F is left exact, i.e. carries them to finite projective limits in **Set**, then the same is true in $C_{/F}$, and *a fortiori* (2.7.1) $C_{/F}$ is filtered. More generally, if sums of two objects and cokernels of double arrows are representable in C, and if F carries them respectively to products and to kernels, then $C_{/F}$ is stable under the same types of finite inductive limits. Hence it is filtered if and only if it is nonempty, i.e. if and only if the functor F is not the constant functor with value $\emptyset$.

Theorem 8.3.3. Let C be a $\mathcal{U}$-category and let $F \in \operatorname{Ob} \widehat{C}$, say $F : C^{\mathrm{op}} \to (\mathcal{U}\text{-}\mathbf{Set})$ is a $\mathcal{U}$-presheaf on C. The following conditions are equivalent:

(i) F is ind-representable (8.2.2).

(ii) The category $C_{/F}$ (8.3.2) is filtered and essentially small.

(iii) If finite inductive limits are representable in C: the functor F is left exact, and $\operatorname{Ob} C_{/F}$ admits a small cofinal subset (8.1.4).

(iii bis) If sums of two objects and cokernels of double arrows are representable in C: the functor F carries sums of two objects of C to products and cokernels of double arrows of C to kernels, the category $C_{/F}$ is nonempty, i.e. the functor F is not the constant functor with value $\emptyset$, and finally there exists a small family of objects of $C_{/F}$ such that every object X of $C_{/F}$ is dominated by some X_i, i.e. admits an F-morphism $X_i \to X$.

(iv) If the category C is equivalent to a small category: the category $C_{/F}$ is filtered.

(v) If the category C is equivalent to a small category and if filtered inductive limits in C are representable: the functor F is left exact.

Proof. (i) $\Rightarrow$ (ii). Suppose that F is ind-represented by $(X_i)_{i \in I}$, with I small. We prove that $C_{/F}$ is filtered. Let X, X' be two objects of $C_{/F}$, i.e. objects of C equipped with morphisms $X \to F$ and $X' \to F$. We prove that they are dominated by a third object of $C_{/F}$. The morphisms $X \to F$ and $X' \to F$ come from morphisms $X \to X_i$ and $X' \to X'_{i'}$, and after replacing i, i' by a common upper bound in $\operatorname{Ob} I$, we may suppose $i = i'$. We then take as common upper bound of X and X' the object X_i equipped with the canonical morphism $X_i \to F$.

Now let

$$X \underset{g}{\overset{f}{\rightrightarrows}} X' \xrightarrow{h} F$$

be a double arrow in $C_{/F}$. We prove that it is equalized by an arrow $X' \to X''$ of $C_{/F}$. The morphism $h : X' \to F$ is given by a morphism $h_i : X' \to X_i$, and the condition $hf = hg$ says that there exists $\alpha : i \to j$ in I such that

$$\varphi(\alpha)(h_i f) = \varphi(\alpha)(h_i g).$$

Thus, after replacing h_i by $\varphi(\alpha)h_i$, we equalize f and g. This proves that $C_{/F}$ is filtered. It is then immediate that it is essentially small, since the objects $(X_i \to F)_{i \in \operatorname{Ob} I}$ form a small cofinal family in $\operatorname{Ob} C_{/F}$.

(ii) $\Rightarrow$ (i). Put $I = C_{/F}$ and consider the canonical functor (8.3.2.1)

$$\varphi : I = C_{/F} \longrightarrow C.$$

The presheaf represented by this ind-object of C is known to be F (3.4), so F is ind-representable by definition (8.2.2).

(ii) $\Leftrightarrow$ (iii). Indeed, (ii) $\Rightarrow$ (iii) because an ind-representable functor is left exact (8.3.1), and (iii) $\Rightarrow$ (ii) because we have observed (8.3.2.3) that left exactness of F implies that $C_{/F}$ is filtered, and one applies Definition 8.1.8 to conclude that this category is essentially small.

The equivalence (ii) $\Leftrightarrow$ (iii bis) is proved in the same way as (ii) $\Leftrightarrow$ (iii).

The equivalences (ii) $\Leftrightarrow$ (iv) and (iii) $\Leftrightarrow$ (v) are trivial, since if C is equivalent to a small category, then $C_{/F}$ is evidently also equivalent to a small category and *a fortiori* is automatically essentially small as soon as it is filtered. This completes the proof of 8.3.3.

Remark 8.3.4. Let $F : C' \to C''$ be a functor between $\mathcal{U}$-categories. It appears that the notion of left exactness (2.3.2) is of practical interest only when finite projective limits are representable in C'. In the case where $C'' = (\mathcal{U}\text{-}\mathbf{Set})$ and C' is $\mathcal{U}$-small, writing C' in the form C^{op} (so $C = C'^{\mathrm{op}}$), the "right notion" which in the general case replaces left exactness is that of ind-representability of F when it is considered as a presheaf on C^{op}; equivalently, it is the pro-representability of F when it is considered as a functor on C' (cf. 8.10). This notion does coincide with left exactness when finite projective limits are representable in C', i.e. when finite inductive limits are representable in C (8.3.3). When one no longer assumes that C' is $\mathcal{U}$-small, or at least equivalent to a $\mathcal{U}$-small category, the two notions for a functor F, left exactness and pro-representability, no longer necessarily coincide, even if finite projective limits are representable in C' (cf. 8.12.9). In fact, it seems that in this case left exact functors which are not ind-representable should be regarded as pathological; the "good" objects remain the ind-representable functors. Compare 8.13.3. For the case of functors $F : C' \to C''$, with C' and C'' again arbitrary $\mathcal{U}$-categories, we shall develop below (8.11.5), more generally, a notion which "improves" that of left exactness, namely the notion of a functor admitting a pro-adjoint functor.

8.4 Constant and essentially constant ind-objects

Choose a final category e, i.e. one such that $\mathrm{Ob}\, e$ and $\mathrm{Fl}\, e$ are each reduced to a single element; for example, one may take for e the full subcategory of **Set** consisting of the empty set, if one wants a canonical choice. This is plainly a filtered and small category. If X is an object of C, we associate with it the constant ind-object indexed by e, with value X, denoted $c(X)$. It is clear that, as X varies, one obtains a fully faithful functor

$$c : C \longrightarrow \mathrm{Ind}(C), \tag{8.4.1}$$

which is moreover injective on objects, and through which we shall identify C with a full subcategory of $\mathrm{Ind}(C)$. Notice that the composite functor

$$C \xrightarrow{c} \mathrm{Ind}(C) \xrightarrow{L} \widehat{C} \tag{8.4.2}$$

is the canonical functor h (8.2.1.1).

8.4.3

More generally, let I be a filtered category. The constant functor on I with value an object X of C is also called *the constant ind-object of value X indexed by I*. If I is essentially small, it is clear that this ind-object ind-represents the functor $h(X)$ represented by X, hence this ind-object is isomorphic to $c(X)$.

8.4.4

An ind-object $\mathcal{X}$ of C is called *essentially constant* if it is isomorphic (in a category $\mathrm{Ind}_{\mathcal{V}}(C, \mathcal{V}')$ for suitable $\mathcal{V}, \mathcal{V}'$) to an ind-object of the form $c(X)$, with $X \in \mathrm{Ob}\, C$. The object X, which is then determined up to canonical isomorphism, is called the *value* of the essentially constant ind-object in question. Thus, by definition, the functor c of (8.4.1) induces an equivalence of C with the full subcategory of $\mathrm{Ind}(C)$ consisting of essentially constant ind-objects; this subcategory is the essential image of the functor c.

Obviously a constant ind-object is essentially constant; the converse is true only in the trivial case where C is empty or a one-point category.

8.5 Filtered inductive limits in $\mathrm{Ind}(C)$

Proposition 8.5.1. Let C be a $\mathcal{U}$-category. In $\mathrm{Ind}(C)$, small filtered inductive limits are representable, and the canonical functor (8.2.4.8)

$$L : \mathrm{Ind}(C) \longrightarrow \widehat{C}$$

commutes with them.

Taking account of the fact that L is fully faithful, the assertions of the proposition are equivalent to the following one.

Corollary 8.5.2. Every small filtered inductive limit, in $\widehat{C}$, of ind-representable presheaves is ind-representable.

This follows easily from criterion 8.3.3(ii). The details of the verification are left to the reader.

8.5.3

The "tautological" case of filtered inductive limits in $\mathrm{Ind}(C)$ is the case where one starts from an essentially small ind-object $\mathcal{X} = (X_i)_{i \in I}$ of C. One then has, in $\widehat{C}$ and hence also in $\mathrm{Ind}(C)$,

$$\mathcal{X} = \varinjlim_{I} X_i. \tag{8.5.3.1}$$

In writing this formula, one should take care that this is not an inductive limit in C, and that even when the latter exists, it is not isomorphic in $\mathrm{Ind}(C)$ to $\mathcal{X}$: indeed, the canonical functor (8.4.1) $c : C \to \mathrm{Ind}(C)$ *does not in general commute with filtered inductive limits.* See Exercise 8.5.5 below.

To avoid this possible confusion in writing (8.5.3.1), "certain authors" (= P. Deligne) prefer to write it

$$\mathcal{X} = \text{“}\varinjlim_{I}\text{”}\, X_i, \tag{8.5.3.2}$$

the role of the quotation marks being to indicate that the inductive limit is taken in a category of ind-objects $\mathrm{Ind}(C)$. By extension, one should then denote by “ lim ” every inductive-limit operation in $\mathrm{Ind}(C)$, without requiring the components of the inductive system considered in $\mathrm{Ind}(C)$ to lie in the image, or in the essential image, of $c : C \to \mathrm{Ind}(C)$.

8.5.4

To compute the inductive or projective limit of a functor

$$\varphi : J \longrightarrow \operatorname{Ind}(C), \tag{8.5.4.1}$$

where J is now an arbitrary category, it is often convenient to make φ explicit by means of a functor

$$\Phi : J \times I \longrightarrow C, \tag{8.5.4.2}$$

where I is an essentially small, or preferably small, filtered category. Such a Φ indeed defines a functor

$$j \longmapsto (i \longmapsto \Phi(j,i)) : J \longrightarrow \operatorname{Hom}(I,C),$$

hence, after composing with the canonical arrow (8.2.6.1) $\operatorname{Hom}(I,C) \to \operatorname{Ind}(C)$, a functor

$$j \longmapsto (i \longmapsto \Phi(j,i)) : J \longrightarrow \operatorname{Ind}(C). \tag{8.5.4.3}$$

We may say that Φ is an *indexed expression of φ, indexed by the filtered category I*, if the corresponding functor (8.5.4.3) is isomorphic to φ. We shall study below (8.8) general existence conditions for an indexed expression of a given functor φ. Here we shall merely start from a functor φ given in indexed form (8.5.4.3), in the case where J is a small filtered category, in order to indicate the calculation of

$$\text{“}\varinjlim\text{”}\,\varphi = \text{“}\varinjlim_{j}\text{”}\,\varphi(j).$$

Formula (8.5.3.2), together with the associativity formula for inductive limits (2.5.0), then immediately gives the *tautological calculation* of “$\varinjlim$” φ:

$$\text{“}\varinjlim_{J}\text{”}\,\varphi = \text{“}\varinjlim_{J\times I}\text{”}\,\Phi, \tag{8.5.4.4}$$

i.e.

$$\text{“}\varinjlim_{j}\text{”}\,(i \longmapsto \Phi(j,i)) = \text{“}\varinjlim_{j,i}\text{”}\,\Phi(j,i). \tag{8.5.4.5}$$

Thus the inductive system sought is none other than Φ itself, the index category being $J \times I$, which is indeed filtered since J and I are so.

Exercise 8.5.5. Let C be a $\mathcal{U}$-category.

a) Prove that the following conditions are equivalent:

   (i) Small inductive limits are representable in C, and the functor $c : C \to \operatorname{Ind}(C)$ commutes with them.

   (ii) Small inductive limits are representable in C, and for every $X \in \operatorname{Ob} C$, the functor $Y \mapsto \operatorname{Hom}(X,Y)$ commutes with these limits.

   (iii) The functor c is an equivalence of categories.

   (iv) Small filtered inductive limits are representable in C, and the functor $\lim c : \operatorname{Ind} C \to C$ (cf. 8.7.1.5) is fully faithful, or equivalently an equivalence.

b) If C is a finite category, prove that the preceding conditions are equivalent to the following one: for every projector in C (10.6), its image is representable in C.

8.6 Extension of a functor to ind-objects

8.6.1

Let
$$f : C \longrightarrow C' \tag{8.6.1.1}$$
be a functor between $\mathcal{U}$-categories. For every ind-object
$$\varphi : I \longrightarrow C$$
of C, where I is a filtered category, the composite functor $f \circ \varphi$ is an ind-object of C', also denoted $\mathrm{Ind}(f)(\varphi)$. In indexed notation, if φ is written as $\mathcal{X} = (X_i)_{i\in I}$, one obtains
$$\mathrm{Ind}(f)(X_i)_{i\in I} = (f(X_i))_{i\in I}. \tag{8.6.1.2}$$
Let $\mathcal{Y} = (Y_j)_{j\in J}$ be a second inductive system of C. Then one obtains an evident map, also denoted $u \mapsto \mathrm{Ind}(f)(u)$:
$$\begin{aligned} \mathrm{Hom}(\mathcal{X},\mathcal{Y}) &= \varprojlim_i \varinjlim_j \mathrm{Hom}(X_i, Y_j) \\ &\longrightarrow \mathrm{Hom}(\mathrm{Ind}(f)(\mathcal{X}), \mathrm{Ind}(f)(\mathcal{Y})) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}(f(X_i), f(Y_j)). \end{aligned} \tag{8.6.1.3}$$
One immediately checks that these maps are compatible with the composition of morphisms of ind-objects, referring to the unwritten formula for composition of morphisms of ind-objects (8.2.5). We have thus obtained, for every universe $\mathcal{V}$, a functor
$$\mathrm{Ind}(f) : \mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U}) \longrightarrow \mathrm{Ind}_{\mathcal{V}}(C',\mathcal{U}) \tag{8.6.1.4}$$
(notation of (8.2.4.5)), and in particular a functor
$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C'). \tag{8.6.1.5}$$
If one has a second functor
$$g : C' \longrightarrow C'',$$
then of course there is an identity of functors between categories of ind-objects, or of $\mathrm{Ind}_{\mathcal{V}}$-objects if desired,
$$\mathrm{Ind}(gf) = \mathrm{Ind}(g) \circ \mathrm{Ind}(f), \tag{8.6.1.6}$$
and for $C' = C$ one has the perennial formula
$$\mathrm{Ind}(\mathrm{id}_C) = \mathrm{id}_{\mathrm{Ind}(C)}\,. \tag{8.6.1.7}$$
Thus, figuratively speaking, one may say that the category $\mathrm{Ind}(C)$ *depends functorially on* C, covariantly.[1] We leave to the reader the task of making explicit even a 2-*functorial* dependence, by defining, for every pair of $\mathcal{U}$-categories C, C', a canonical functor
$$\mathsf{Ind} : \mathrm{Hom}(C, C') \longrightarrow \mathrm{Hom}(\mathrm{Ind}(C), \mathrm{Ind}(C')), \tag{8.6.1.8}$$
and by specifying the functorial nature of the identity isomorphisms (8.6.1.6) and (8.6.1.7).

[1] For a contravariant dependence, under certain conditions, cf. 8.11.

8.6.2

Return to the functor (8.6.1.1), and consider the corresponding functor (5.1)

$$f_! : \widehat{C} \longrightarrow \widehat{C'}, \tag{8.6.2.1}$$

and the diagram of functors

$$\begin{tikzcd} \mathrm{Ind}(C) \arrow[r, "\mathrm{Ind}(f)"] \arrow[d, "L_C"'] & \mathrm{Ind}(C') \arrow[d, "L_{C'}"] \\ \widehat{C} \arrow[r, "f_!"'] & \widehat{C'}. \end{tikzcd} \tag{8.6.2.2}$$

Here the vertical arrows denote the canonical functors L of (8.2.4.8). Since the functor $f_!$ "extends f" and commutes with inductive limits (5.4.3), and since the functors L_C and $L_{C'}$ commute with small filtered inductive limits (8.5.1), it follows from (8.5.3.2) that for every ind-object $(X_i)_{i\in I}$ of C there is, in $\widehat{C'}$, a canonical isomorphism

$$f_!(L_C((X_i)_{i\in I})) \simeq \varinjlim_i f_! L_C(X_i) \simeq \varinjlim_i L_{C'} f(X_i) = L_{C'}(f(X_i)_{i\in I}),$$

i.e. a canonical isomorphism

$$f_! L_C(\mathcal{X}) \simeq L_{C'}(\mathrm{Ind}(f)(\mathcal{X})).$$

We leave to the reader the verification that the latter is functorial, i.e. that it corresponds to a canonical isomorphism

$$f_! L_C \simeq L_{C'} \circ \mathrm{Ind}(f). \tag{8.6.2.3}$$

In other words, *the square (8.6.2.2) is commutative up to canonical isomorphism.* Taking account of the fact that $f_!$ commutes with small inductive limits, and of 8.5.1, we conclude as follows.

Proposition 8.6.3. Let $f : C \to C'$ be a functor between $\mathcal{U}$-categories. Then the functor

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C')$$

commutes with filtered inductive limits. Moreover, it makes the following diagram commutative:

$$\begin{tikzcd} C \arrow[r, "f"] \arrow[d, "c_C"'] & C' \arrow[d, "c_{C'}"] \\ \mathrm{Ind}(C) \arrow[r, "\mathrm{Ind}(f)"'] & \mathrm{Ind}(C'), \end{tikzcd}$$

where the vertical arrows $c_C, c_{C'}$ are the canonical functors (8.4.1).

The last assertion is trivial from the definitions and has been included for ease of reference. Note moreover that the properties stated in 8.6.3 *characterize the functor* $\mathrm{Ind}(f)$ up to unique isomorphism as being induced by the functor $f_!$ (8.6.2.1), as follows from the proof just given of (8.6.2.3).

Proposition 8.6.4. The notation is that of 8.6.3.

a) For $\mathrm{Ind}(f)$ to be faithful, respectively fully faithful, it is necessary and sufficient that f be so.

b) For $\mathrm{Ind}(f)$ to be an equivalence of categories, it is necessary and sufficient that f be fully faithful and that every object of C' be isomorphic to a direct factor (10.6) of an object in the image of f.

Proof.

a) Necessity follows immediately from the fact that f is induced by $\mathrm{Ind}(f)$. For sufficiency, it is enough to use the form (8.6.1.3) of $\mathrm{Ind}(f)$ on the sets $\mathrm{Hom}(\mathcal{X}, \mathcal{Y})$, remembering that filtered inductive limits of sets, and arbitrary projective limits, carry monomorphisms to monomorphisms and isomorphisms to isomorphisms.

b) We may already suppose f, hence $\mathrm{Ind}(f)$, fully faithful. Since every object of $\mathrm{Ind}(C')$ is a small filtered inductive limit of objects of C', full faithfulness of $\mathrm{Ind}(f)$ implies that for this functor to be essentially surjective it is equivalent that every object X' of C' lie in its essential image. If one has an isomorphism

$$X' \xrightarrow{\sim} \text{“}\varinjlim\text{”}\, f(X_i),$$

then this isomorphism factors through one of the $f(X_i)$; this shows that X' is a direct factor of $f(X_i)$, proving necessity in b). For sufficiency, using the full faithfulness of $\mathrm{Ind}(f)$, the assertion follows at once from the following corollary.

Corollary 8.6.5. In $\mathrm{Ind}(C)$ the images of projectors (10.6) are representable, and the functor $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$ commutes with the formation of these images.

Indeed, this follows from the fact that in $\mathrm{Ind}(C)$ small filtered inductive limits are representable (8.5.1) and that $\mathrm{Ind}(f)$ commutes with them (8.6.3), taking account of the fact that the image of a projector $p : X \to X$ may be interpreted as the inductive limit of the filtered inductive system indexed by $\mathbb{N}$

$$X \xrightarrow{p} X \xrightarrow{p} X \longrightarrow \cdots,$$

or, equivalently, as the inductive limit of the evident functor on the filtered category P having a single object and a nonidentity arrow Π such that $\Pi^2 = \Pi$.

Exposé I. Presheaves

Section 8.7–8.8

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 8.7 through Section 8.8. A few evident typographical slips in the source notation are corrected where the mathematical context forces the correction.

8 Ind-objects and pro-objects

8.7 The functor $\varinjlim_C : \mathrm{Ind}(C) \to C$. Universal characterizations of $\mathrm{Ind}(C)$

8.7.1

Let C again be a $\mathcal{U}$-category, and let

$$c : C \longrightarrow \mathrm{Ind}(C) \tag{8.7.1.1}$$

be the canonical functor. Let $\mathcal{X} = (X_i)_{i \in I}$ be an object of $\mathrm{Ind}(C)$. It follows immediately from the definition of representable inductive limits (2.1, 2.1.1) that $\varinjlim_i X_i$ is *representable in C if and only if the left adjoint of c is defined at $\mathcal{X}$*; in that case $\varinjlim_i X_i$, computed in C, is precisely the value of that left adjoint at $\mathcal{X}$:

$$\mathrm{Hom}_C(\varinjlim_i X_i, Y) \simeq \mathrm{Hom}_{\mathrm{Ind}(C)}((X_i)_{i \in I}, c(Y)). \tag{8.7.1.2}$$

Let

$$\mathrm{Ind}(C)' \subset \mathrm{Ind}(C) \tag{8.7.1.3}$$

denote the full subcategory of $\mathrm{Ind}(C)$ formed by the ind-objects of C which admit an inductive limit *in* C. The preceding observation implies that this subcategory is *strictly full*, and that one has a canonical functor

$$\varinjlim_C : \mathrm{Ind}(C)' \longrightarrow C, \tag{8.7.1.4}$$

whose value on an object $(X_i)_{i \in I}$ of $\mathrm{Ind}(C)'$ is its inductive limit in C. In the special case in which small filtered inductive limits are representable in C, one therefore obtains a natural functor

$$\varinjlim_C : \mathrm{Ind}(C) \longrightarrow C. \tag{8.7.1.5}$$

8.7.1.6

Of course, the preceding functors may also be defined for categories of the form $\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U})'$ and $\mathrm{Ind}_{\mathcal{V}}(C,\mathcal{U})$; but, in view of (8.2.4.6), they are already determined, up to unique isomorphism, by the corresponding functors in the case $\mathcal{V} = \mathcal{U}$.

8.7.1.7

From the preceding construction of $\varinjlim_C$ as a left adjoint, it follows immediately that this functor *commutes with arbitrary inductive limits*, and in particular with small filtered inductive limits, the latter being representable in $\mathrm{Ind}(C)$ by 8.5.1. Notice also that $\mathrm{Ind}(C)'$ always contains the essential image of c, formed by the essentially constant ind-objects, and that one has a canonical isomorphism, functorial in $X \in \mathrm{Ob}\,\mathrm{Ind}(C)'$,

$$\varinjlim_{C} c(X) \simeq X. \tag{8.7.1.8}$$

Equivalently, in the favorable case where C is stable under small inductive limits, one has a canonical isomorphism

$$\varinjlim_{C} \circ c \simeq \mathrm{id}_C\,. \tag{8.7.1.9}$$

8.7.2

Now consider a functor

$$f : C \longrightarrow E, \tag{8.7.2.1}$$

where E is a $\mathcal{U}$-category in which small inductive limits are representable, so that there is a functor

$$\varinjlim_{E} : \mathrm{Ind}(E) \longrightarrow E.$$

Since (8.6) also defined

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(E),$$

one may consider the composite

$$\bar{f} = \varinjlim_{E} \circ \mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow E, \tag{8.7.2.2}$$

which is sometimes called the *canonical extension of f to ind-objects*, although it should not be confused with $\mathrm{Ind}(f)$. Since it is a composite of two functors commuting with small filtered inductive limits (8.6.3, 8.7.1.7), it also commutes with small filtered inductive limits. Moreover, from the isomorphism (8.7.1.9), applied to E, and from (8.6.2.3), it follows that $\bar{f}$ *extends* f up to isomorphism; that is, there is a canonical isomorphism

$$\bar{f} \circ c_C \simeq f. \tag{8.7.2.3}$$

It is also rather clear, taking into account the fact that every object of $\mathrm{Ind}(C)$ is a small filtered inductive limit of objects of C, that the two preceding properties still *characterize* $\bar{f}$ up to unique isomorphism. This can be made precise so as to obtain, at the same time, a universal characterization, up to equivalence, of $\mathrm{Ind}(C)$ among the $\mathcal{U}$-categories E in which small filtered inductive limits are representable. If E and F are two such $\mathcal{U}$-categories, let

$$\mathrm{Hom}(E, F)' \subset \mathrm{Hom}(E, F) \tag{8.7.2.4}$$

denote the full subcategory of $\mathrm{Hom}(E, F)$ formed by the functors which commute with small filtered inductive limits. Then one has the following result.

Proposition 8.7.3. Let C be a $\mathcal{U}$-category, and use the notation above (8.7.2.4). Then the canonical functor

$$c_C : C \longrightarrow \mathrm{Ind}(C)$$

is 2-universal among functors with source C and target a $\mathcal{U}$-category in which small filtered inductive limits are representable. In other words, $\mathrm{Ind}(C)$ is such a category (8.2.5, 8.5.1), and if E is such a category, the functor

$$g \longmapsto g \circ c_C : \mathrm{Hom}(\mathrm{Ind}(C), E)' \longrightarrow \mathrm{Hom}(C, E) \tag{8.7.3.1}$$

is an equivalence of categories.

To prove the last assertion, it is enough to verify that the assignment $f \mapsto \bar{f}$ defined by (8.7.2.2) can be made into a *functor*

$$f \longmapsto \bar{f} = \varinjlim_E \cdot \mathrm{Ind}(f) : \mathrm{Hom}(C, E) \longrightarrow \mathrm{Hom}(\mathrm{Ind}(C), E)', \tag{8.7.3.2}$$

and that this functor is a quasi-inverse of (8.7.3.1). The details of the verification, which are essentially trivial, are left to the reader. One may also invoke 7.8 to conclude first that (8.7.3.1) is fully faithful, and then use (8.7.2.3) to conclude that it is essentially surjective.

8.7.4

Let us return to a functor between $\mathcal{U}$-categories

$$f : C \longrightarrow E, \tag{8.7.4.1}$$

where E is a $\mathcal{U}$-category in which small filtered inductive limits are representable. This gives a functor

$$\bar{f} : (X_i)_{i \in I} \longmapsto \varinjlim_i f(X_i) : \mathrm{Ind}(C) \longrightarrow E. \tag{8.7.4.2}$$

We shall study conditions on f which ensure that $\bar{f}$ is fully faithful, respectively an equivalence of categories. Since f is isomorphic to the composite $\bar{f} \circ c_C$, and since $c_C : C \to \mathrm{Ind}(C)$ is fully faithful, it is *necessary*, for $\bar{f}$ to be fully faithful, that f be so. Replacing C by its essential image in E, we therefore lose no essential generality by assuming, as we shall do below in order to simplify notation, that f is the inclusion functor of a subcategory C of E.

Proposition 8.7.5. The notation is that of 8.7.4.

a) For the functor $\bar{f}$ to be fully faithful, it is necessary and sufficient that the following condition hold:

   (PF) Every object X of C satisfies the following property: for every small filtered inductive system $(Y_i)_{i \in I}$ in C, with inductive limit $\varinjlim_i Y_i$ in E, the canonical map

$$\varinjlim_i \mathrm{Hom}(X, Y_i) \longrightarrow \mathrm{Hom}(X, \varinjlim_i Y_i) \tag{8.7.5.1}$$

   is bijective.

b) For the functor $\bar{f}$ to be an equivalence of categories, it is necessary and sufficient that C satisfy condition (PF) and the following two conditions:

(ii) C is a subcategory of E generating by strict epimorphisms (7.1).

(iii) For every object X of E, the full subcategory $C_{/X}$ of $E_{/X}$ formed by objects X' over X whose source is in $\operatorname{Ob} C$ is filtered and essentially small.

Condition (iii) holds in particular if C is equivalent to a small category, if finite inductive limits are representable in C, and if the inclusion functor $f : C \to E$ commutes with them; that is, for every finite category J and every functor $\varphi : J \to C$, the inductive limit of $f\varphi$ is representable and is isomorphic in E to an object of C.

Proof.

a) With the notation of condition (PF), if $\mathcal{Y}$ denotes the ind-object (Y_i) and $\mathcal{X}$ the constant ind-object defined by X, then (8.7.5.1) is precisely the canonical map

$$\operatorname{Hom}(\mathcal{X}, \mathcal{Y}) \longrightarrow \operatorname{Hom}(\bar{f}(\mathcal{X}), \bar{f}(\mathcal{Y})).$$

Thus, if $\bar{f}$ is fully faithful, this map is bijective. Conversely, if $\mathcal{X} = (X_j)_{j \in J}$ and $\mathcal{Y} = (Y_i)_{i \in I}$ are arbitrary ind-objects of C, then the map

$$u : \operatorname{Hom}(\mathcal{X}, \mathcal{Y}) \longrightarrow \operatorname{Hom}(\bar{f}(\mathcal{X}), \bar{f}(\mathcal{Y}))$$

is the projective limit over j of the maps

$$u_j : \operatorname{Hom}(\mathcal{X}_j, \mathcal{Y}) \longrightarrow \operatorname{Hom}(\bar{f}(\mathcal{X}_j), \bar{f}(\mathcal{Y})),$$

where $\mathcal{X}_j$ is the constant ind-object with value X_j. Hence u is bijective if the u_j are, and condition (PF) implies that $\bar{f}$ is fully faithful.

b) Suppose that (PF), (ii), and (iii) hold, and prove that $\bar{f}$ is an equivalence. It remains only to prove that it is essentially surjective. Let $X \in \operatorname{Ob} E$. Hypothesis (ii) means precisely that

$$\varinjlim_{C_{/X}} Z \longrightarrow X$$

is an isomorphism, while (iii) says that the category $C_{/X}$ is filtered and essentially small. Thus X is the image of the ind-object of C defined by the natural functor $C_{/X} \to E$.

Conversely, suppose that $\bar{f}$ is an equivalence. In view of a), it remains to check (ii) and (iii). We may suppose that $E = \operatorname{Ind}(C)$, with C identified with the subcategory $c(C)$ of E. But it is known (8.3.3(ii)) that, for every $X \in \operatorname{Ind}(C)$, identified if desired with the functor F on C^{op} which it ind-represents, $C_{/X} = C_{/F}$ is a filtered essentially small category. This proves (iii). Moreover, the inductive limit in $\widehat{C}$ of the functor $C_{/F} \to \widehat{C}$ is F. A fortiori the same is true in the full subcategory E of $\widehat{C}$, since $F = X$ lies in E. This proves (ii).

It remains to prove the last assertion of b). But the hypothesis made on C plainly implies that finite inductive limits are representable in the category $C_{/X}$; a fortiori $C_{/X}$ is filtered.

Remark 8.7.5.2. The preceding argument shows more generally that, when condition (PF) holds, the essential image of the fully faithful functor $\bar{f}$ (8.7.4.2) consists of the objects $X \in \operatorname{Ob} E$ such that the category $C_{/X}$ is filtered and essentially small–a condition automatically satisfied if C satisfies the conditions stated at the end of b)–and such that the inductive limit of the canonical functor $C_{/X} \to E$ is X.

Corollary 8.7.6. Let E_{PF} be the full subcategory of E formed by the objects satisfying condition (PF) of Proposition 8.7.5 a). Then, for every functor $J \to E_{PF}$, with J a finite category, which admits an inductive limit X in E, one has $X \in \operatorname{Ob} E_{PF}$. The canonical functor

$$(X_i)_{i \in I} \longmapsto \varinjlim_{i,E} X_i,$$

obtained from the inclusion $g : E_{PF} \hookrightarrow E$,

$$\bar{g} : \operatorname{Ind}(E_{PF}) \longrightarrow E \tag{8.7.6.1}$$

is fully faithful. If finite inductive limits are representable in E and E_{PF} is equivalent to a small category, then the essential image of $\bar{g}$ is formed by the objects X of E for which the canonical morphism

$$\varinjlim_{(E_{PF})_{/X}} Y \longrightarrow X$$

is an isomorphism. If, moreover, the subcategory E_{PF} of E generates by strict epimorphisms (7.1), then the functor (8.7.6.1) is an equivalence of categories.

All the assertions are evident, in the order in which they are stated, from 8.7.5, using 2.8 for the first assertion.

Corollary 8.7.7. Suppose that finite inductive limits are representable in E. Let C be a full subcategory of E, equivalent to a small category, and consider the functor

$$\bar{f} : \operatorname{Ind}(C) \longrightarrow E, \qquad (X_i)_{i \in I} \longmapsto \varinjlim_{i,E} X_i. \tag{8.7.7.1}$$

Let E_{PF} be as in 8.7.6. The following conditions are equivalent:

(i) The functor $\bar{f} : \operatorname{Ind}(C) \to E$ is an equivalence.

(ii) The category C is a subcategory of E generating by strict epimorphisms, is contained in E_{PF}, and every object of E_{PF} is isomorphic in E (or in E_{PF}, which is the same thing by 8.7.6) to a direct factor of an object of C.

When the subcategory C of E is stable under direct factors (10.6), the preceding conditions are also equivalent to the following:

(ii bis) $C = E_{PF}$, and C generates in E by strict epimorphisms.

(iii) The subcategory C of E generates by strict epimorphisms, is contained in E_{PF}, and finite inductive limits are representable in C.

When, moreover, finite inductive limits are representable in E, these conditions are also equivalent to

(iii bis) The subcategory C of E generates by strict epimorphisms, is contained in E_{PF}, and is stable in E under finite inductive limits; equivalently, under finite sums and cokernels of double arrows.

We already know (8.7.5) that condition (i) implies that $C \subset E_{PF}$ and that C generates by strict epimorphisms. Let us also prove that then every object X of E_{PF} is isomorphic to a direct factor of an object of C. Indeed, X is a filtered inductive limit $\varinjlim_i X_i$ in E of objects of C; by the definition of E_{PF}, for a suitable i the canonical homomorphism $X_i \to X$ admits a left inverse, so that X is indeed a direct factor of the object X_i of C. This proves (i) $\Rightarrow$ (ii), without any further hypothesis on C or E.

Let us prove (ii) $\Rightarrow$ (i). Since C is equivalent to a small category and every object of E_{PF} is a direct factor of an object of C, it follows that E_{PF} is also equivalent to a small category. Hence, by 8.7.6, the functor (8.7.6.1) is an equivalence of categories, and one concludes by applying 8.6.4 b) to the inclusion $C \hookrightarrow E_{PF}$.

When every direct factor in E of an object of C lies in C, it is clear that (ii) is equivalent to (ii bis). On the other hand, (ii bis) implies (iii), respectively (iii bis), by 8.7.6. Finally, (iii), and *a fortiori* (iii bis), implies (i) by 8.7.5. This completes the proof.

Exercise 8.7.8 (Karoubi envelopes). Let C be a $\mathcal{U}$-category, let $E = \mathrm{Ind}(C)$, and let $E_{PF} \supset C$ be the full subcategory of E defined in 8.7.6.

a) Prove that images of projectors are representable in E_{PF}, and that every object of E_{PF} is a direct factor of an object of C.

b) Call a category F *Karoubian* if images of projectors are representable in it. Let

$$\varphi : C \longrightarrow K$$

be a fully faithful functor, with K Karoubian, such that every object of K is a direct factor of an object in the image of C. Prove that φ is 2-*universal* among functors from C to Karoubian categories, more precisely: for every Karoubian category F, the functor

$$g \longmapsto g \circ \varphi : \mathrm{Hom}(K, F) \longrightarrow \mathrm{Hom}(C, F)$$

is an equivalence of categories. Use the fact that every functor commutes with images of projectors. The category K equipped with φ, determined up to equivalence, itself determined up to unique isomorphism, by the preceding properties, is called the *Karoubi envelope* $\widetilde{C}$ of C. Compare also IV, 7.5.

c) Deduce from a) and b) that $\mathrm{Ind}(C)_{PF}$, equipped with the functor $C \to \mathrm{Ind}(C)_{PF}$ induced by $c : C \to \mathrm{Ind}(C)$, makes $\mathrm{Ind}(C)_{PF}$ a Karoubi envelope of C.

d) Show that every functor $f : C \to C'$ extends, essentially uniquely, to a functor

$$\widetilde{f} : \widetilde{C} \longrightarrow \widetilde{C'}$$

between the Karoubi envelopes, and that if one takes these Karoubi envelopes as in c), then $\widetilde{f}$ is the functor induced by $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$. Show that the conditions of 8.5.4 b) are also equivalent to the following condition: $\widetilde{f}$ is an equivalence of categories.

Exercise 8.7.9. Let E be a $\mathcal{U}$-category.

a) Show that the following conditions are equivalent:

(i) E is equivalent to a category of the form $\mathrm{Ind}(C)$, with C a $\mathcal{U}$-category.

(i bis) As in (i), with C moreover Karoubian (8.7.8 b)).

(ii) The subcategory E_{PF} of E (8.7.6) generates by strict epimorphisms, and for every object X of E, $(E_{PF})_{/X}$ is a filtered essentially small category.

Show that, if C is a *Karoubian* $\mathcal{U}$-category such that E is equivalent to $\mathrm{Ind}(C)$, then C is equivalent to E_{PF}, by an equivalence determined up to unique isomorphism. Establish a 2-equivalence between the 2-category formed by the categories E satisfying the preceding conditions, with functors between them *commuting with small filtered inductive limits*, and the 2-category formed by the *Karoubian* $\mathcal{U}$-categories and arbitrary functors between them.

b) Show that the following conditions are equivalent:

(i) E is equivalent to a category of the form $\mathrm{Ind}(C)$, where C is a $\mathcal{U}$-category in which finite inductive limits are representable.

(i bis) As in (i), but with C moreover Karoubian.

(ii) Finite inductive limits are representable in E, the subcategory E_{PF} of E generates by strict epimorphisms, and for every object X of E there exists a small subset I of $\mathrm{Ob}\, E_{PF/X}$ such that every object of $E_{PF/X}$ is dominated by an object of I. Use 8.9.5 b).

8.8 Indexed representation of a functor $J \to \mathrm{Ind}(C)$

8.8.1

Let

$$\varphi : J \longrightarrow \mathrm{Ind}(C), \qquad \varphi \in \mathrm{Ob}\,\mathrm{Hom}(J, \mathrm{Ind}(C)),$$

be a functor, where C is a $\mathcal{U}$-category. Recall (8.5.4) that an *indexed representation of* φ is, by definition, an ind-object of $\mathrm{Hom}(J, C)$,

$$\psi : I \longrightarrow \mathrm{Hom}(J, C),$$

where I is a filtered essentially small category, whose inductive limit in $\mathrm{Hom}(J, \mathrm{Ind}(C))$ is isomorphic to φ, by a specified isomorphism. Thus φ admits an indexed representation if and only if it is isomorphic to an essentially small filtered inductive limit in $\mathrm{Hom}(J, \mathrm{Ind}(C))$ of objects of the full subcategory $\mathrm{Hom}(J, C)$. We shall say that the category J is *admissible for* C if every functor $\varphi : J \to \mathrm{Ind}(C)$ admits an indexed representation.

Since small filtered inductive limits are representable in $\mathrm{Ind}(C)$ (8.5.1), the same is true in $\mathrm{Hom}(J, \mathrm{Ind}(C))$, and they are computed argument by argument. Consequently, the inclusion functor

$$\mathrm{Hom}(J, C) \hookrightarrow \mathrm{Hom}(J, \mathrm{Ind}(C)) \tag{8.8.1.1}$$

extends canonically to a functor

$$\mathrm{Ind}(\mathrm{Hom}(J, C)) \longrightarrow \mathrm{Hom}(J, \mathrm{Ind}(C)) \tag{8.8.1.2}$$

by 8.7.2. The elements of the essential image of this functor are then precisely the φ which admit an indexed representation. Thus saying that J is admissible for C is the same as saying that the functor (8.8.1.2) is essentially surjective. In this connection we record the following result.

Proposition 8.8.2. If the category J is equivalent to a finite category, then the canonical functor (8.8.1.2) is fully faithful. Hence J is admissible for C if and only if this functor is an equivalence of categories.

By 8.7.5 a), everything reduces to proving that, for every small filtered inductive system $(\varphi_i)_{i\in I}$ in $\mathrm{Hom}(J, C)$ and every object φ of $\mathrm{Hom}(J, C)$, the canonical map

$$\varinjlim_i \mathrm{Hom}(\varphi, \varphi_i) \longrightarrow \mathrm{Hom}(\varphi, \varinjlim_i \varphi_i) \tag{8.8.2.1}$$

is bijective, where $\varinjlim_i \varphi_i$ denotes the inductive limit taken in $\mathrm{Hom}(J, \mathrm{Ind}(C))$. But, if φ and ψ are two functors $J \to C$, one has an exact diagram of sets, functorial in φ and ψ,

$$\mathrm{Hom}(\varphi, \psi) \longrightarrow \prod_{X\in \mathrm{Ob}\, J} \mathrm{Hom}(\varphi(X), \psi(X)) \rightrightarrows \prod_{\substack{f\in \mathrm{Fl}\, J \\ f: X\to Y}} \mathrm{Hom}(\varphi(X), \psi(Y)).$$

Since filtered inductive limits commute with kernels of double arrows and with finite products, the bijectivity of (8.8.2.1) follows when J is finite. The case in which J is equivalent to a finite category immediately reduces to this one.

Proposition 8.8.3. Let $\varphi : J \to \mathrm{Ind}(C)$ be a functor, with J equivalent to a small category. For φ to admit an indexed representation, it is sufficient that the following two conditions hold; these conditions are also necessary when J is equivalent to a finite category.

a) The category $\mathrm{Hom}(J, C)_{/\varphi}$, formed by the arrows of $\mathrm{Hom}(J, \mathrm{Ind}(C))$ with target φ and source in $\mathrm{Hom}(J, C)$, is filtered.

b) For every object j of J, the functor

$$\psi \longmapsto \psi(j) : \mathrm{Hom}(J, C)_{/\varphi} \longrightarrow C_{/\varphi(j)} \tag{8.8.3.1}$$

is cofinal, i.e. satisfies conditions F 1) and F 2) of 8.1.3.

Proof. Suppose a) and b) hold, and prove that φ admits an indexed representation. We may of course suppose that J is small. Put

$$I = \mathrm{Hom}(J, C)_{/\varphi}, \tag{8.8.3.2}$$

which is a filtered category by hypothesis, and first suppose that I is essentially small. Then the "inclusion" of I into $\mathrm{Hom}(J, C)$ defines an ind-object of $\mathrm{Hom}(J, C)$, say $i \mapsto \psi_i$. Moreover, one has a canonical homomorphism

$$\varinjlim_i \psi_i \longrightarrow \varphi, \tag{8.8.3.3}$$

given argument by argument by the homomorphism

$$\varinjlim_I \psi_i(j) \longrightarrow \varphi(j) = \varinjlim_{C_{/\varphi(j)}} X$$

deduced from the functor (8.8.3.1). Since the latter functor is cofinal, it follows that (8.8.3.3) is an isomorphism, and hence the conclusion follows.

In the case where I is not assumed essentially small, it is enough to construct an essentially small full subcategory I' such that (8.8.3.3) remains an isomorphism when $\varinjlim_{I'}$ is used instead of $\varinjlim_{I}$. For this, it is enough that the composite functors

$$I' \longrightarrow I \longrightarrow C_{/\varphi(j)}$$

induced by the functors (8.8.3.1) all be cofinal. One then concludes by the following lemma, whose proof is immediate from criterion (8.1.3 b)) and is left to the reader.

Lemma 8.8.3.4. Let I be a filtered $\mathcal{U}$-category, and let

$$f_j : I \longrightarrow I_j \qquad (j \in J)$$

be a small family of cofinal functors from I to essentially small filtered categories. Then there exists a full subcategory I' of I which is filtered and essentially small, and such that the functors induced by the f_j are cofinal.

Let us finally prove the necessity in 8.8.3 when J is assumed equivalent to a finite category. An indexed representation of φ by means of an essentially small filtered index category I defines a functor ψ and, for each j, a functor ψ_j:

$$\begin{tikzcd} I \arrow[r, "\psi"] \arrow[dr, "\psi_j"'] & \mathrm{Hom}(J,C)_{/\varphi} \arrow[d] \\ & C_{/\varphi(j)}. \end{tikzcd} \tag{8.8.3.5}$$

It is enough to prove that ψ and each of the resulting functors ψ_j are cofinal. By 8.1.3 b), it will then follow that $\mathrm{Hom}(J,C)_{/\varphi}$ is filtered, and by Definition 8.1.1 that the vertical arrow in (8.8.3.5) is also cofinal. By 8.8.2 one may identify φ with an object of $\mathrm{Ind}(E)$, where $E = \mathrm{Hom}(J,C)$, and the fact that the functor

$$\psi : I \longrightarrow E_{/\varphi}$$

deduced from the ind-object φ of E indexed by I is cofinal is a general fact, verified immediately by means of the criteria F 1) and F 2) of 8.1.3. The same argument, with E and φ replaced by C and $\varphi(j)$, shows that ψ_j is cofinal. This completes the proof.

Remark 8.8.4.

a) In the case where J is equivalent to a finite category, if φ admits an indexed representation, we have seen that (8.8.3.5) is a cofinal functor. This implies that the category $\mathrm{Hom}(J,C)_{/\varphi}$ itself is essentially small.

b) Suppose that finite inductive limits are representable in C. Then the same is true in $E = \mathrm{Hom}(J,C)$, where these limits are computed argument by argument; they are also inductive limits in $\mathrm{Hom}(J,\widehat{C})$ and *a fortiori* in $\mathrm{Hom}(J,\mathrm{Ind}(C))$. It follows immediately that condition a) of 8.8.3 is then automatically satisfied, and everything reduces to condition b). I do not know whether this condition is automatically satisfied when, in addition, J is assumed small.

We now come to the principal result of the present subsection.

Proposition 8.8.5. Let J be a category equivalent to a finite category, and suppose moreover that J is rigid, i.e. that for every $j \in \operatorname{Ob} J$, every endomorphism of j is the identity. Then J is admissible for C, whatever the $\mathcal{U}$-category C may be; equivalently, every functor $J \to \operatorname{Ind}(C)$ admits an indexed representation.

Replacing J by an equivalent category, we may suppose that J is *reduced*, i.e. that two isomorphic objects of J are identical. Then J is finite. We proceed by induction on $\operatorname{card} \operatorname{Ob} J$, the case where this cardinal is ≤ 0 being trivial; hence we suppose it is at least 1. Let j_0 be a *maximal* object of J, i.e. such that for every arrow $j_0 \to j$ there exists an arrow $j \to j_0$. Since J is rigid, this implies that $j_0 \to j$ is an isomorphism, and since J is reduced, this implies $j_0 = j$. Let J' be the full subcategory obtained from J by removing the object j_0, and let J'' be the full subcategory reduced to the object j_0. The proposition then follows from the following lemma.

Lemma 8.8.5.1. Let J be a finite category, and let J' and J'' be two full subcategories such that

$$\operatorname{Ob} J = \operatorname{Ob} J' \cup \operatorname{Ob} J'',$$

and such that, for every $j' \in \operatorname{Ob} J'$ and $j'' \in \operatorname{Ob} J''$, one has

$$\operatorname{Hom}(j'', j') = \varnothing.$$

If J' and J'' are admissible for C, then so is J.

Everything reduces to proving criteria a) and b) of 8.8.3. This amounts to six elementary verifications: the last two conditions for filtered categories for $\operatorname{Hom}(J, C)_{/\varphi}$, and the conditions F 1) and F 2) for the functors (8.8.3.1), successively in the case $j \in \operatorname{Ob} J'$ and in the case $j \in \operatorname{Ob} J''$; these latter conditions also imply that $\operatorname{Hom}(J, C)_{/\varphi}$ is nonempty. The rather tedious verification presents no difficulty and is left to the reader. The editor searched in vain for an elegant proof bypassing these unpleasant verifications.

Exercise 8.8.6.

a) Let J and J' be two categories admissible for C, one of them finite. Prove that $J \times J'$ is admissible for C. Use 8.8.2.

b) Prove that a small discrete category is admissible for every category C.

c) Let J be a category satisfying the following conditions: (i) J is rigid; (ii) $\operatorname{Ob} J$ is countable; (iii) for any two objects j, j' of J, $\operatorname{Hom}(j, j')$ is finite; (iv) for every object j of J, the set of objects of J which are dominated by j, i.e. which are sources of an arrow with target j, is finite. Show that J is admissible for every $\mathcal{U}$-category C. First show that J can be written as a filtered union of finite full subcategories J_n such that every object of J dominated by an object of J_n lies in J_n; then verify criteria a) and b) of 8.8.3.

Exercise 8.8.7. In the notation, we identify an ordered set with the category it defines.

a) Let C be an ordered set. Let $\widetilde{C}$ be the set of subsets A of C which are filtered and such that $x \in A$ and $y \leq x$ imply $y \in A$. Let $L_0 : C \to \widetilde{C}$ be the map which sends $x \in C$ to the set $L_0(x)$ of all $y \in C$ such that $y \leq x$. Show that L_0 is injective and that the order on C

is induced by that on $\widetilde{C}$. For every $A \in \widetilde{C}$, consider the inclusion $f(A) : A \to C$, which is an ind-object of C. For every ind-object $\varphi : I \to C$ of C, let $g(\varphi) \in \widetilde{C}$ be the subset of C formed by the elements of C dominated by an element of the form $\varphi(i)$. Show that in this way one obtains two quasi-inverse equivalences

$$\mathrm{Ind}(C) \underset{g}{\overset{f}{\rightleftarrows}} \widetilde{C}$$

which transform the functor L_0 into the canonical functor $L : C \to \mathrm{Ind}(C)$.

b) Suppose that, for every $x \in C$, the set of elements majorizing, respectively minorizing, x is finite. Show that every ind-object, respectively pro-object, of C is essentially constant; more precisely, for every such $\varphi : I \to C$, respectively $\varphi : I^{\mathrm{op}} \to C$, there exists $i_0 \in \mathrm{Ob}\, I$ such that the induced functor on $I_{/i_0}$ is constant, i.e. sends every arrow to an isomorphism.

c) Let $C \subset \mathbb{N}^{\mathrm{op}} \times \mathbb{N}$ be the ordered set formed by the pairs of natural numbers (j, i) such that $i \geq j$. Show that $\widetilde{\mathbb{N}}$ is isomorphic to the ordered set obtained from $\mathbb{N}$, identified with a subset of $\widetilde{\mathbb{N}}$ as in a), by adjoining a greatest element ∞. Show that $\widetilde{C}$ is isomorphic to $\mathbb{N}^{\mathrm{op}} \times \widetilde{\mathbb{N}}$. Consider the functor

$$\varphi : \mathbb{N}^{\mathrm{op}} \longrightarrow \widetilde{C}, \qquad \varphi(j) = (j, \infty).$$

Show that:

1) the projective limit of φ is not representable in $\widetilde{C}$, although filtered projective limits in C are representable, and are essentially constant by b);

2) for every functor $\psi : \mathbb{N}^{\mathrm{op}} \to C$, one has $\mathrm{Hom}(\psi, \varphi) = \varnothing$, and *a fortiori* the functor φ admits no representation in indexed form.

Exercise 8.8.8. Let $X = \mathbb{N} \amalg \mathbb{N}$ be the disjoint union of two copies of $\mathbb{N}$, let s be the symmetry of X, and for every $i \in \mathbb{N}$, let X_i be the subset $\Delta_{i+1} \amalg \Delta_i$ of X, where Δ_i denotes the subset of $\mathbb{N}$ formed by the j such that $j \leq i$. Let C be the subcategory of the category of subsets of X whose objects are the X_i and whose arrows are the maps between the X_i induced either by the identity of X or by its symmetry s. Let $\widetilde{C}$ be the category defined analogously, but in which the object X is also allowed.

a) Show that $\mathrm{Ind}(C)$ is equivalent to $\widetilde{C}$, with the canonical functor $C \to \mathrm{Ind}(C)$ corresponding to the inclusion $C \to \widetilde{C}$. Show that there is no pair (Y, f), consisting of an object Y of C and an endomorphism f of Y, together with a morphism of objects-with-endomorphism in $\mathrm{Ind}(C)$ from (Y, f) to (X, s).

b) Conclude that, if J is the category with one object and, besides the identity arrow, a single arrow of order 2, then the functor $J \to \mathrm{Ind}(C)$ defined by (X, s) admits no indexed representation.

Exposé I. Presheaves

Sections 8.9–8.13

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Translator's status note. This is a working English translation of SGA 4, Exposé I, Sections 8.9 through 8.13. A few evident typographical slips in the source notation are corrected where the mathematical context forces the correction.

8 Ind-objects and pro-objects

8.9 Exactness properties of $\mathrm{Ind}(C)$

Proposition 8.9.1. Let C be a $\mathcal{U}$-category.

a) The canonical functors L (8.2.4.8) and c (8.4.1)

$$C \xrightarrow{c} \mathrm{Ind}(C) \xrightarrow{L} \widehat{C}$$

commute with projective limits.

b) Suppose that finite inductive limits are representable in C, and that C is equivalent to a small category. Then small projective limits are representable in $\mathrm{Ind}(C)$.

c) If small projective limits (respectively finite projective limits) are representable in C, then the same is true in $\mathrm{Ind}(C)$. Let J be a category which is finite and rigid, or discrete. If projective limits of type J are representable in C, the same type of projective limits is representable in $\mathrm{Ind}(C)$.

d) Small filtered inductive limits in $\mathrm{Ind}(C)$, which are representable by 8.5.1, are left exact, that is, they commute with finite projective limits.

Proof.

a) The fact that L commutes with projective limits follows from the fact that it is fully faithful and from the computation of projective limits in $\widehat{C}$ argument by argument. Thus, in order to verify that a projective system of morphisms $F \to F_i$ $(i \in I)$ in $\widehat{C}$ makes F the projective limit of the F_i, it is enough to verify the analogous assertion for the projective systems of sets

$$\mathrm{Hom}(X, F) \longrightarrow \mathrm{Hom}(X, F_i)$$

for all $X \in \mathrm{Ob}\, C$. But $C \subset \mathrm{Ind}(C)$.

The fact that c commutes with projective limits follows formally from the fact that L does so, together with the fact that the composite $L \circ c : C \to \widehat{C}$ does so.

b) In view of a), the assertion says that every projective limit of ind-representable presheaves is ind-representable. This follows immediately from criterion 8.3.3(v), since a projective limit of left exact functors is left exact; this is just the fact that projective limits commute with projective limits (2.5.0).

c) This follows formally from the analogous property of $\widehat{C}$ (3.3), together with the facts that L is conservative, being fully faithful, and that it commutes with the limits of the type in question, by a) and 8.5.1.

d) It is well known (cf. 2.3) that representability of finite projective limits is equivalent to representability of projective limits of the special finite ordered types consisting of the empty type and the elementary fiber-product type. Representability of small projective limits is equivalent to representability of products and finite projective limits. Hence the second assertion made in c) implies the first.

First suppose that I is finite. Using 8.8.5 on the representability of functors $\varphi : J \to \mathrm{Ind}(C)$ in indexed form (8.5.4)

$$\Phi : J \times I \longrightarrow C, \tag{8.9.1.1}$$

the existence of $\varprojlim \varphi$ is a special case of the following more precise and more general result.

Corollary 8.9.2. Let $\varphi : J \to \mathrm{Ind}(C)$ be a functor given in indexed form Φ as in (8.9.1.1), where J is a finite category. If, for every $i \in \mathrm{Ob}\, I$, the partial functor $j \mapsto \Phi(j,i)$ has a projective limit (respectively an inductive limit) representable in C, then φ has a projective limit (respectively an inductive limit) representable in $\mathrm{Ind}(C)$, and there is a canonical isomorphism

$$\varprojlim \varphi \simeq \text{“}\varinjlim_i\text{”} \varprojlim_j \Phi(j,i), \tag{8.9.2.1}$$

respectively

$$\varinjlim \varphi \simeq \text{“}\varinjlim_i\text{”} \varinjlim_j \Phi(j,i), \tag{8.9.2.2}$$

where $\varprojlim_j$ (respectively $\varinjlim_j$) is computed in C.

For the first formula, one uses only that in $\mathrm{Ind}(C)$ filtered inductive limits commute with $\varprojlim_j$, and that c commutes with $\varprojlim_J$ by 8.9.1(d) and a):

$$\begin{aligned} \varprojlim \varphi &= \varprojlim_j \text{“}\varinjlim_i\text{”}\, \Phi(j,i) \\ &\xleftarrow{\sim} \text{“}\varinjlim_i\text{”} \varprojlim_j{}^{\mathrm{Ind}(C)} \Phi(j,i) \simeq \text{“}\varinjlim_i\text{”} \overset{C}{\varprojlim_j} \Phi(j,i). \end{aligned}$$

The second is proved in the same way, using the commutation of inductive limits of type I with inductive limits of type J (2.5.0) and the fact that c commutes with inductive limits of type J, proved in 8.9.4(a) below.

The case where J is discrete. This case is contained in the following more precise assertion.

Corollary 8.9.3. Let I_α be essentially small filtered categories indexed by a small set A, and, for each $\alpha \in A$, let

$$X(\alpha) = (X(\alpha)_{i_\alpha})_{i_\alpha \in I_\alpha}$$

be an ind-object of C indexed by I_α. Let I be the product category of the I_α, and suppose that, for every $\mathbf{i} = (i_\alpha)_{\alpha \in A} \in I$, the product

$$\prod_{\alpha \in A} X(\alpha)_{i_\alpha}$$

is representable in C. Then the product $\prod_{\alpha\in A} X(\alpha)$ is representable in $\mathrm{Ind}(C)$, and is canonically isomorphic to the ind-object indexed by I given by the formula

$$\prod_{\alpha\in A} \text{“}\varinjlim_{i_\alpha\in I_\alpha}\text{”}\, X(\alpha)_{i_\alpha} \simeq \text{“}\varinjlim_{\mathbf{i}=(i_\alpha)_{\alpha\in A}\in I}\text{”}\left(\prod_{\alpha\in A} X(\alpha)_{i_\alpha}\right), \tag{8.9.3.1}$$

where the product on the right is computed in C. Notice that I is filtered and essentially small because the I_α are.

Indeed, it is well known, and immediate after reducing to the category of sets, that the displayed formula is valid when the limits are computed in $\widehat{C}$. The conclusion then follows from the fact that L commutes with the limits under consideration (8.9.1(a) and 8.5.1).

Remarks 8.9.4. The proof just given for 8.9.2 and 8.9.3 shows, more generally, that if, for every $i \in \mathrm{Ob}\, I$, the limit $\varprojlim_j \Phi(j,i)$ (respectively $\varinjlim_j \Phi(j,i)$), computed in $\mathrm{Ind}(C)$, is representable, then the corresponding limit of φ is representable. This and the argument in c) show that, for a given category J arising from a finite ordered set or a discrete category, or more generally one which is C-admissible in the sense of 8.3.1, projective limits (respectively inductive limits) of type J are representable in $\mathrm{Ind}(C)$ if and only if, for every functor $\varphi : J \to C$, the projective limit (respectively inductive limit) of φ, computed in $\mathrm{Ind}(C)$, is representable. In the non-parenthesized case this also means, by a), that every projective limit of type J of representable presheaves on C is ind-representable.

Proposition 8.9.5. Let C be a $\mathcal{U}$-category.

a) The canonical functor
$$c : C \longrightarrow \mathrm{Ind}(C)$$
is right exact, hence exact in view of 8.9.1(a).

b) If finite inductive limits (respectively finite sums) are representable in C, then small inductive limits (respectively small sums) are representable in $\mathrm{Ind}(C)$. Let J be a finite or discrete preordered set. If inductive limits of type J are representable in C, then the same is true in $\mathrm{Ind}(C)$.

Proof.

a) Suppose that in C one has $X \simeq \varinjlim_J X_j$, with J finite and with X and the X_j in C. Then, for every ind-object $\mathcal{Y} = (Y_i)_{i\in I}$ of C,

$$\begin{aligned}\mathrm{Hom}(X,\mathcal{Y}) &\simeq \varinjlim_i \mathrm{Hom}(X, Y_i) \simeq \varinjlim_i \varprojlim_j \mathrm{Hom}(X_j, Y_i)\\ &\simeq \varprojlim_j \varinjlim_i \mathrm{Hom}(X_j, Y_i) \simeq \varprojlim_j \mathrm{Hom}(X_j, \mathcal{Y}),\end{aligned}$$

where the middle isomorphism is 2.8. The desired conclusion follows by comparing the extreme terms.

b) It was already observed in 8.8.4 that the assertions follow from a) and 8.9.2, at least for finite inductive limits. To prove the conclusions in the infinite cases, one is reduced to the case of sums, which may be interpreted as a filtered inductive limit of finite sums; the conclusion follows from 8.5.1.

Remark 8.9.6. We have already observed that the functor c does not in general commute with filtered inductive limits, and therefore not with infinite sums either, in contrast with what happens for $L : \mathrm{Ind}(C) \to \widehat{C}$. Conversely, apart from commutation with filtered inductive limits (8.5.1), L practically never has commutation properties with any other type of inductive limit: initial object, sum of two objects, or cokernels of double arrows. Thus, if C has an initial object $\varnothing_C$, which is then an initial object of $\mathrm{Ind}(C)$ by a), $\varnothing_C$ is never an initial object of $\widehat{C}$, that is, never identical with the constant presheaf of value $\varnothing$, since $\mathrm{Hom}(\varnothing_C, \varnothing_C) \neq \varnothing$.

Proposition 8.9.7. Let

$$f : C \longrightarrow C'$$

be a functor between $\mathcal{U}$-categories, and let J be a finite and rigid category (8.8.5). If inductive limits (respectively projective limits) of type J are representable in C and if f commutes with them, then

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C')$$

also commutes with this type of limit.

This follows immediately from 8.8.5 and from the calculation, in 8.9.2, of finite limits in a category of ind-objects, for a functor represented in indexed form.

Corollary 8.9.8. If finite inductive limits (respectively finite projective limits) are representable in C, and if f is right exact (respectively left exact), then the same is true of $\mathrm{Ind}(f)$. In the non-parenthesized case, $\mathrm{Ind}(f)$ even commutes with arbitrary small inductive limits.

The first assertion follows from 8.9.7. The second follows from the first, taking into account that $\mathrm{Ind}(f)$ commutes with filtered inductive limits by 8.6.3.

Exercise 8.9.9. Let C be a $\mathcal{U}$-category.

a) Suppose finite sums are representable in C. Show that if finite sums in C are disjoint (respectively universal; cf. II 4.5), then small sums in $\mathrm{Ind}(C)$ are disjoint (respectively universal).

b) Suppose every morphism in C factors as an epimorphism followed by a monomorphism (respectively as an effective epimorphism followed by a monomorphism, respectively as an epimorphism followed by an effective monomorphism, respectively as an effective epimorphism followed by an effective monomorphism). Show that $\mathrm{Ind}(C)$ satisfies the same property. In the last parenthesized case one concludes that every epimorphism of $\mathrm{Ind}(C)$ is effective, every monomorphism of $\mathrm{Ind}(C)$ is effective, and every bimorphism of $\mathrm{Ind}(C)$ is an isomorphism. Show that, in this case, every epimorphism (respectively monomorphism) of $\mathrm{Ind}(C)$ can be represented by an inductive system of epimorphisms (respectively monomorphisms) of C. If, moreover, every epimorphism (respectively every monomorphism) in C is universal, then the same property holds in $\mathrm{Ind}(C)$.

c) Suppose C is additive (respectively abelian). Then $\mathrm{Ind}(C)$ is additive (respectively abelian).

d) Let $X = \text{“}\varinjlim\text{”}_I X_i$ be an ind-object of C, and let $R \rightrightarrows X$ be an equivalence relation in X. For every $i \in \mathrm{Ob}\, I$, let $R_i \rightrightarrows X_i$ be the equivalence relation in X_i, considered as an object of $\mathrm{Ind}(C)$, induced by R via $X_i \to X$. Here one assumes that the fiber product

$$R_i = R \times_{X \times X} (X_i \times X_i)$$

in $\mathrm{Ind}(C)$ is representable by an object of $\mathrm{Ind}(C)$, which is the case if finite projective limits are representable in C. Show that, for R to be effective, it is enough that the R_i be effective.

e) Let X be an object of C, and let R be an equivalence relation in $c(X)$, that is, in X considered as an object of $\mathrm{Ind}(C)$. Suppose fiber products are representable in C. For R to be effective, it is necessary that R be of the form "$\varinjlim$"$_i R_i$, where the R_i are equivalence relations in X, viewed as an object of C; this condition is sufficient if equivalence relations are effective in C.

f) Suppose every morphism in C factors as an effective epimorphism followed by an effective monomorphism, and suppose finite projective limits are representable. Let X be an object of C, and let R be a subobject of $X \times X$, viewed as an object of $\mathrm{Ind}(C)$. Write R in the form "$\varinjlim$"$_i Z_i$, where $(Z_i)_{i \in I}$ is a filtered increasing family of subobjects of $X \times X$ in C; this is possible by b). Show that, for R to be an equivalence relation in X, it is necessary and sufficient that for every $i \in \mathrm{Ob}\, I$ there exist $j \in \mathrm{Ob}\, I$ containing $s(Z_i)$ and $Z_i \circ Z_i$, where s is the symmetry of $X \times X$.

g) Suppose finite projective limits and finite sums are representable in C, every equivalence relation in C is effective, and every morphism in C factors as an effective epimorphism followed by an effective monomorphism. Show that equivalence relations in $\mathrm{Ind}(C)$ are universal if and only if C satisfies the following condition:

ST) For every object X of C and every subobject Z of $X \times X$ in C, if one defines recursively a sequence of subobjects Z_n $(n \geq 0)$ of $X \times X$ in C by $Z_0 = Z$ and

$$Z_{n+1} = \mathrm{Sup}(Z_n, s(Z_n), Z_n \circ Z_n),$$

where the supremum is taken in the set of subobjects of $X \times X$, which exists by the hypotheses on C, then the sequence $(Z_n)_{n \geq 0}$ is stationary.

k) Show that in $\mathrm{Ind}(\mathbf{Set})$ equivalence relations are not necessarily effective.

*NB. For criteria ensuring that $\mathrm{Ind}(C)$ is a topos, see VI 8.9.9.$_*$

Exercise 8.9.10. Let C be a $\mathcal{U}$-category. Put $\mathrm{Pro}(C) = (\mathrm{Ind}(C^{\mathrm{op}}))^{\mathrm{op}}$ (cf. 8.11).

a) Suppose finite sums (respectively small sums) are representable in C. Show that if they are disjoint (cf. II 4.5), then $\mathrm{Pro}(C)$ satisfies the same condition.

b) Let J be a small set such that sums indexed by J are representable in C, hence also in $\mathrm{Pro}(C)$. Let $(X(\alpha))_{\alpha \in J}$ be a family of elements of $\mathrm{Pro}(C)$, with

$$X(\alpha) = \text{“}\varprojlim_{I_\alpha}\text{”} X_{i_\alpha}.$$

Show that, for the sum of the $X(\alpha)$ in $\mathrm{Pro}(C)$ to be universal, it is enough that the same be true for each of the families $\varprojlim_J X_{i_\alpha}$, where $(i_\alpha)_{\alpha \in J} \in \prod_{\alpha \in J} I_\alpha$. Deduce that, if sums of type J are universal in C, then for the same to be true in $\mathrm{Pro}(C)$ it is necessary and sufficient that, for every family $(X(\alpha))_{\alpha\in J}$ as above with $I_\alpha = I$ for all $\alpha \in J$, the canonical homomorphism in $\mathrm{Pro}(C)$

$$\text{“}\varprojlim_{I}\text{”}\ \amalg_\alpha X(\alpha)_i \longleftarrow \text{“}\varprojlim_{I^J}\text{”}\ \amalg_{\alpha\in J} X(\alpha)_{i_\alpha}$$

be an isomorphism. Deduce that in $\mathrm{Pro}(\mathbf{Set})$ sums of type J are universal if and only if J is finite.

Exercise 8.9.11. Let C be a $\mathcal{U}$-category.

a) Let $(X_i \to X)_{i\in I}$ be a family of morphisms in C. Show that for it to be epimorphic in $\mathrm{Ind}(C)$, it is enough that it admit a finite subfamily which is epimorphic in C. Prove that this condition is also necessary if one assumes that every finite family of morphisms with target X in C factors as an epimorphic family followed by an effective monomorphism (10.5). For the latter assertion, let P be the set of finite subsets of I, ordered by inclusion, let $X' = (X'_J)_{J\in P}$ be the ind-object formed by the images of the finite subfamilies of the given family, and finally let

$$X'' = X \amalg_{X'} X = \text{“}\varinjlim\text{”}_J\, X \amalg_{X'_J} X.$$

Show that the two canonical morphisms $X \rightrightarrows X''$ are distinct but coincide on the X'_J.

b) Suppose that in C every finite family of morphisms with the same target factors as an epimorphic family followed by an effective monomorphism. Prove that $\mathrm{Ind}(C)$ has a small subcategory generating by epimorphisms (7.1) if and only if C has a small subcategory C' such that every object of C is the target of a finite epimorphic family with source in C'. If one assumes that C has a small generating subcategory (7.1), then $\mathrm{Ind}(C)$ has a small subcategory generating by epimorphisms if and only if C is equivalent to a small category. For this last statement, use 7.5.2.

c) $\mathrm{Ind}(\mathbf{Set})$ has no small generating subcategory.

Exercise 8.9.12. Let C be a $\mathcal{U}$-category, and consider

$$\mathrm{Pro}(C) = \mathrm{Ind}(C^{\mathrm{op}})^{\mathrm{op}}.$$

a) Show that if $\mathrm{Pro}(C)$ has a small subcategory P' generating by epimorphisms (7.1), then there is a small subcategory C' of C which generates by epimorphisms in $\mathrm{Pro}(C)$. Take the category of components of the objects of P'.

b) Show that the category $\mathrm{Pro}(\mathbf{Set})$ has no small subcategory generating by epimorphisms. If C' is as in a), choose a set X whose cardinal strictly dominates the cardinals of the elements of C', and consider the pro-set $\mathcal{X}$ formed by the complements in X of the subsets of cardinal $< \mathrm{card}(X)$. Show that, for every nonempty set Y of cardinal $< \mathrm{card}(X)$, one has $\mathrm{Hom}(Y, \mathcal{X}) = \varnothing$, but that $\mathcal{X}$ is not isomorphic to the constant pro-set $\varnothing$.

8.10 Dual notions: pro-objects and pro-representable functors

Let C be a $\mathcal{U}$-category. A *pro-object* of C is a functor

$$\varphi : I^{\mathrm{op}} \longrightarrow C, \tag{8.10.1}$$

where I is an essentially small filtered category, called the index category, and where, as usual, I^{op} denotes the opposite category. Let $\mathcal{V}$ be a universe such that $\mathcal{V} \supset \mathcal{U}$. Notice that pro-objects of C indexed by $I \in \mathcal{V}$ are in bijective correspondence with ind-objects of C^{op} indexed by $I \in \mathcal{V}$: to such an ind-object $\psi : I \to C^{\mathrm{op}}$ one associates the pro-object $\varphi = \psi^{\mathrm{op}} : I^{\mathrm{op}} \to C$. Following this correspondence, one defines, by transport of structure and reversal of arrows, the notion of morphism between pro-objects of C from the analogous notion (8.2.4.2, 8.2.5.1) for ind-objects. One accordingly obtains the category of pro-objects of C indexed by $I \in \mathcal{V}$, together with a canonical isomorphism

$$\mathrm{Pro}_{\mathcal{V}}(C,\mathcal{U}) \simeq \mathrm{Ind}_{\mathcal{V}}(C^{\mathrm{op}},\mathcal{U})^{\mathrm{op}}. \tag{8.10.2}$$

For $\mathcal{V} = \mathcal{U}$ one simply speaks of the *category of pro-objects of* C, denoted $\mathrm{Pro}_{\mathcal{U}}(C)$ or simply $\mathrm{Pro}(C)$; it is therefore defined by

$$\mathrm{Pro}(C) = \mathrm{Pro}_{\mathcal{U}}(C,\mathcal{U}) \simeq \mathrm{Ind}(C^{\mathrm{op}})^{\mathrm{op}}. \tag{8.10.3}$$

If two pro-objects are given in indexed form

$$\mathcal{X} = (X_i)_{i\in I}, \qquad \mathcal{Y} = (Y_j)_{j\in J}, \tag{8.10.4}$$

then formula (8.2.5.1) takes here the form

$$\mathrm{Hom}(\mathcal{X},\mathcal{Y}) \simeq \varprojlim_{j} \varinjlim_{i} \mathrm{Hom}(X_i, Y_j). \tag{8.10.5}$$

The functor (8.4.1), applied to C^{op}, gives, after passage to opposite categories, a canonical functor, denoted by the same letter c when no confusion is possible, which allows one to identify C with a full subcategory of $\mathrm{Pro}(C)$:

$$c : C \longrightarrow \mathrm{Pro}(C). \tag{8.10.6}$$

The arrows were reversed in the definition of morphisms of pro-objects precisely in order to obtain such a functor, rather than a functor $C \to \mathrm{Pro}(C)^{\mathrm{op}}$. Pro-objects in the essential image of (8.10.6) are again called *essentially constant pro-objects*.

Put

$$\check{C} = (C^{\mathrm{op}})\widehat{\ } = \mathrm{Hom}(C, \mathbf{Set}). \tag{8.10.7}$$

Then the functor (8.2.4.7) may be regarded as a canonical fully faithful functor

$$L : \mathrm{Pro}_{\mathcal{V}}(C,\mathcal{U}) \longrightarrow \check{C}^{\mathrm{op}}, \tag{8.10.8}$$

and in particular one obtains

$$L : \mathrm{Pro}(C) \hookrightarrow \check{C}^{\mathrm{op}}. \tag{8.10.9}$$

We shall leave to the reader the task of translating, as needed, the results of the preceding sections and of the following ones from the language of ind-objects into that of pro-objects. Depending on the mathematical context, one or the other language is the more useful, not to mention the cases in which both notions occur simultaneously, for instance when one has to consider complex categories such as $\mathrm{Pro}(\mathrm{Ind}(C))$ or $\mathrm{Ind}(\mathrm{Pro}(C))$. We content ourselves with giving a few additional indications, in order to fix terminology and notation and to clarify the yoga.

8.10.10

A *covariant functor* $F : C \to \mathbf{Set}$ is called a *pro-representable functor* if it lies in the essential image of (8.10.9), or equivalently of 8.10.8. The full subcategory of $\check{C}$ consisting of such functors is therefore equivalent to $\mathrm{Pro}(C)^{\mathrm{op}}$. One generally prefers to work in the opposite category, as the essential image as a subcategory of (8.10.9), hence equivalent to $\mathrm{Pro}(C)$ itself. In accord with this usage, it is often preferable to regard covariant functors

$$F : C \longrightarrow \mathbf{Set}$$

as objects of $\mathrm{Hom}(C, \mathbf{Set})^{\mathrm{op}} = \check{C}^{\mathrm{op}}$, and consequently to write the category of morphisms

$$X \longrightarrow F \quad \text{in } \check{C}^{\mathrm{op}},$$

that is, the pairs (X, u) consisting of an object X of C and an element $u \in F(X)$, as

$$_F\backslash C. \tag{8.10.11}$$

With this convention one obtains a covariant functor, the source functor,

$$_F\backslash C \longrightarrow C, \tag{8.10.12}$$

and 3.4 is rewritten in the form

$$F \xrightarrow{\sim} \text{“}\varprojlim_{(_F\backslash C)^{\mathrm{op}}}\text{”} X. \tag{8.10.13}$$

8.10.14

Criterion 8.3.3, applied to C^{op}, becomes a criterion of pro-representability for F: it is necessary and sufficient that $(_F\backslash C)^{\mathrm{op}}$ be filtered and essentially small. The latter condition is superfluous if C is equivalent to a small category. If, moreover, finite projective limits are representable in C, this is equivalent to saying that F commutes with them, that is, that F is *left exact*.

8.10.15

In $\mathrm{Pro}(C)$, small filtered projective limits are representable, and the functor (8.10.9) commutes with them. In other words, this functor transforms projective limits in $\mathrm{Pro}(C)$ into inductive limits in $\check{C} = \mathrm{Hom}(C, \mathbf{Set})$, just as does its composite $C \to \check{C}^{\mathrm{op}}$ with (8.10.6), which shares with it the unfortunate property of being contravariant when it is regarded as taking values in $\check{C}$. Notice carefully that pro-representable functors from C to $\mathbf{Set}$ are functors which are small filtered inductive limits of representable functors, and not projective limits, as the terminology might perhaps suggest.

8.11 Ind-adjoints and pro-adjoints

8.11.1

Let

$$f : C \longrightarrow C' \tag{8.11.1.1}$$

be a functor between $\mathcal{U}$-categories. It induces the functor $F \mapsto F \circ f$

$$f^* : \widehat{C'} \longrightarrow \widehat{C}. \tag{8.11.1.2}$$

We say that f *admits an ind-adjoint* if this functor sends ind-representable functors to ind-representable functors. Since f^* commutes with inductive limits, and since the full subcategory of $\widehat{C'}$ formed by the ind-representable functors is stable under small filtered inductive limits, it is equivalent to say that f admits an ind-adjoint, or that f^* sends the representable objects of $\widehat{C'}$ to ind-representable functors on C, that is,

$$X \longmapsto \operatorname{Hom}(f(X), Y') \quad \text{is ind-representable for every } Y' \in \operatorname{Ob} C'. \tag{8.11.1.3}$$

Another way to express the condition that f admit an ind-adjoint is to say that there is a functor

$$g : \operatorname{Ind}(C') \longrightarrow \operatorname{Ind}(C) \tag{8.11.1.4}$$

which is essentially induced by f^*, that is, such that there is an isomorphism of bifunctors

$$\operatorname{Hom}_{\operatorname{Ind}(C)}(X, g(\mathcal{Y}')) \simeq \operatorname{Hom}_{\operatorname{Ind}(C')}(f(X), \mathcal{Y}'), \tag{8.11.1.5}$$

where $X \in \operatorname{Ob} C$ and $\mathcal{Y}' \in \operatorname{Ob} \operatorname{Ind}(C')$. In fact it is enough to have a functor

$$g_0 : C' \longrightarrow \operatorname{Ind}(C) \tag{8.11.1.6}$$

and an isomorphism of bifunctors

$$\operatorname{Hom}_{\operatorname{Ind}(C)}(X, g_0(Y')) \simeq \operatorname{Hom}_{C'}(f(X), Y') \tag{8.11.1.7}$$

for $X \in \operatorname{Ob} C$ and $Y' \in \operatorname{Ob} C'$. The functor g, respectively g_0, is plainly determined by f up to unique isomorphism, and conversely f can be reconstructed up to canonical isomorphism from g, respectively from g_0. Formula (8.11.1.5) makes it clear that g commutes with inductive limits and extends g_0; this determines it up to unique isomorphism in terms of g_0 by 8.7.2. The functor g, and sometimes also the functor g_0 that it extends, is called the *ind-adjoint functor of* f. There is no need here to specify "right", since the other one, if it exists, will be called the pro-adjoint (cf. 8.11.5 below).

Of course, if f admits a right adjoint

$$f^{\text{ad}} : C' \longrightarrow C,$$

then it admits an ind-adjoint g, and this is canonically isomorphic to the canonical extension of f^{ad} to ind-objects:

$$g \simeq \operatorname{Ind}(f^{\text{ad}}). \tag{8.11.1.8}$$

The notion of ind-adjoint is therefore a natural generalization of the notion of right adjoint.

Now consider the canonical extension

$$\operatorname{Ind}(f) : \operatorname{Ind}(C) \longrightarrow \operatorname{Ind}(C'). \tag{8.11.1.9}$$

We then have the following proposition.

Proposition 8.11.2. For the functor $f : C \to C'$ to admit an ind-adjoint, it is necessary and sufficient that the functor $\operatorname{Ind}(f)$ in (8.11.1.9) admit a right adjoint. This right adjoint is canonically isomorphic to the ind-adjoint (8.11.1.4).

The sufficiency and the last assertion are trivial from the ind-adjunction formula (8.11.1.5). For necessity, note that, by passage to the projective limit in this formula over the X_i, one deduces an adjunction isomorphism for the ind-objects $\mathcal{X} = (X_i)_{i\in I}$ and $\mathcal{Y}'$:

$$\mathrm{Hom}_{\mathrm{Ind}(C)}(\mathcal{X}, g(\mathcal{Y}')) \simeq \mathrm{Hom}_{\mathrm{Ind}(C')}(\mathrm{Ind}(f)(\mathcal{X}), \mathcal{Y}'). \tag{8.11.2.1}$$

□

Corollary 8.11.3. If f admits an ind-adjoint, then $\mathrm{Ind}(f)$ commutes with inductive limits, and the ind-adjoint g commutes with projective limits.

Proposition 8.11.4. For the functor $f : C \to C'$ to admit an ind-adjoint, it is necessary that f be right exact, and this condition is sufficient when C is equivalent to a small category.

This follows from criterion (8.11.1.3), and from 8.3.1 and 8.3.3(iv). For another criterion in terms of the notion of accessible functor, see 8.13.3.

8.11.5

Now consider the functor $F \mapsto F \circ f$

$$f^{\mathrm{op}*} : \check{C}' \longrightarrow \check{C} \tag{8.11.5.1}$$

induced by f. We shall say that f *admits a pro-adjoint* if the preceding functor sends pro-representable functors to pro-representable functors, that is, if there exists a functor, called the *pro-adjoint functor of* f,

$$g : \mathrm{Pro}(C') \longrightarrow \mathrm{Pro}(C), \tag{8.11.5.2}$$

and an isomorphism of bifunctors

$$\mathrm{Hom}_{\mathrm{Pro}(C)}(g(\mathcal{Y}'), \mathcal{X}) \simeq \mathrm{Hom}_{\mathrm{Pro}(C')}(\mathcal{Y}', \mathrm{Pro}(f)(\mathcal{X})). \tag{8.11.5.3}$$

Of course, saying that f admits a pro-adjoint means that f^{op} admits an ind-adjoint, so the notions and results for ind-adjoints translate trivially into terms of pro-adjoints. Let us merely point out that f admits a pro-adjoint if and only if

$$\mathrm{Pro}(f) : \mathrm{Pro}(C) \longrightarrow \mathrm{Pro}(C')$$

admits a left adjoint, and that in this case the preceding functor g is such a left adjoint of $\mathrm{Pro}(f)$; and that for this it is necessary that f be left exact, this condition also being sufficient when C is equivalent to a small category. In this case, f is exact if and only if it admits both an ind-adjoint and a pro-adjoint.

Example 8.11.6. Consider the case of a functor

$$f : C \longrightarrow \mathbf{Set}.$$

If this functor admits a pro-adjoint, then it is pro-representable. The converse is true if and only if the full subcategory of $\check{C}$ formed by pro-representable functors is stable under small products; this is the case in particular if C is equivalent to a small category (8.10.14), or if small products are representable in C (8.9.5(b) applied to C^{op}). Thus, up to minor qualifications, one may say that, for a functor $f : C \to C'$, the existence of a pro-adjoint is the *natural generalization of pro-representability of* f, which is defined when $C' = \mathbf{Set}$.

8.12 Strict ind-objects and pro-objects. Application to a representability criterion

8.12.1

Let

$$\mathcal{X} = (X_i)_{i \in I}$$

be an ind-object of the $\mathcal{U}$-category C, and let F be the presheaf which it ind-represents. It is immediate that the following two conditions are equivalent: the canonical morphisms

$$X_i \longrightarrow F = \text{“}\varinjlim_i\text{”} X_i$$

are monomorphisms of $\operatorname{Ind}(C)$, or equivalently of $\widehat{C}$, that is, monomorphisms of functors argument by argument; and, for every arrow $i \to j$ of I, the corresponding transition morphism

$$X_i \longrightarrow X_j$$

is a monomorphism. When these conditions hold, and when moreover I is an ordered category, $\mathcal{X}$ is called a *strict ind-object*. Notice that this condition is not invariant under isomorphism of ind-objects. An ind-object will be called *essentially strict* if it is isomorphic to a strict ind-object. A presheaf will be called *strictly ind-representable* if it is ind-representable by a strict ind-object; thus $F = \text{“}\varinjlim_i\text{”} X_i$ is strictly ind-representable if and only if $\mathcal{X} = (X_i)_{i \in I}$ is essentially strict.

8.12.1.1. Let F be a presheaf on C, and consider the full subcategory of $C_{/F}$ formed by the arrows $X \to F$ with source in C which are monomorphisms. We shall call it the *category of representable subfunctors of* F; it is the category associated with the ordered set of representable subfunctors of F, ordered by the order induced from the set of subobjects of F. The following then follows immediately from the definitions.

Proposition 8.12.2. For a presheaf F on C to be strictly ind-representable, it is necessary and sufficient that the ordered category I of representable subfunctors of F be filtered and essentially small and that one have

$$\varinjlim_I X_i \xrightarrow{\sim} F. \tag{8.12.2.1}$$

When, for every object X of C, the set of subobjects of X in C is small, for example when C admits a small generating subcategory (7.4), this implies that the category of representable subfunctors is even small.

8.12.2.2. If F is strictly ind-representable, there is therefore a preferred way to ind-represent it by an ind-object, and that ind-object is even strict: take the representation (8.12.2.1). A strict ind-object $\mathcal{Y}$ is called *saturated* if it is isomorphic to an ind-object of the form (X_i) appearing in (8.12.2.1), for a suitable F. Thus there exists, up to unique isomorphism, only one saturated strict ind-object isomorphic to the given strict ind-object, namely the one considered in 8.12.2, taking $F = \varinjlim \mathcal{Y}$ as a limit in $\widehat{C}$.

8.12.2.3. Suppose one already knows that a small cofinal subset can be found in $\operatorname{Ob} C_{/F}$, which is the case in particular if F is ind-representable. Then the same condition follows in the full subcategory I considered in 8.12.2, so in the criterion 8.12.2 one may omit the condition that I be essentially small.

8.12.3

Let F be a presheaf on C, and consider an object

$$u : X \longrightarrow F$$

of $C_{/F}$. We say that u, or the pair (X, u), is *minimal* if, for every factorization of u as

$$X \xrightarrow{p} X' \xrightarrow{u'} F, \tag{8.12.3.1}$$

with p a strict epimorphism (10.2), the morphism p is an isomorphism. Considering u as an element of $F(X)$ (1.4), to say that u is minimal therefore means that every strict epimorphism $p : X \to X'$ such that

$$u \in \operatorname{Im}\bigl(F(p) : F(X') \to F(X)\bigr)$$

is an isomorphism. This notion is clarified by part a) of the following lemma.

Lemma 8.12.4. Let F be a presheaf on C.

a) Suppose that F transforms cokernels into kernels, and consider a morphism

$$u : X \longrightarrow F$$

with $X \in \operatorname{Ob} C$. For u to be a monomorphism, it is sufficient, when cokernels of double arrows are representable in C, that u be minimal; this condition is also necessary if fiber products are representable in C.

b) Suppose finite inductive limits are representable in C. For the subcategory of representable subfunctors of F (8.12.1.1) to be filtered, not necessarily small, and to have inductive limit F, it is sufficient that F be left exact and that every morphism $u : X \to F$, with $X \in \operatorname{Ob} C$, factor as

$$X \longrightarrow X' \xrightarrow{u'} F$$

with u' minimal. This condition is also necessary if fiber products are representable in C.

c) Suppose that C admits a small generating subcategory (7.1), and that I, the category of representable subfunctors of F, is filtered. Then I is small.

Proof.

a) Suppose u is minimal; we prove that u is a monomorphism, that is, that for every double arrow $v, v' : Y \rightrightarrows X$ such that $uv = uv'$, one has $v = v'$. Indeed, if $X' = \operatorname{Coker}(v, v')$, then u factors as $X \xrightarrow{p} X' \xrightarrow{u'} F$, since F transforms cokernels into kernels. As p is a strict epimorphism by construction, it follows that p is an isomorphism, that is, $v = v'$. Conversely, suppose u is a monomorphism, and prove that it is minimal. Consider a factorization (8.12.3.1). Since p is a strict epimorphism, it is the cokernel of the canonical double arrow

$$v, v' : X \times_{X'} X \rightrightarrows X.$$

Since $(u'p)v = (u'p)v'$ and $u'p = u$ is a monomorphism, we have $v = v'$, hence p is an isomorphism.

b) Sufficiency: since F is left exact, $C_{/F}$ is filtered. Since we know that $F = \varinjlim_{C_{/F}} X$, we are reduced by 8.1.3(c) to proving that the full subcategory I of $C_{/F}$ formed by the subobjects of F is cofinal in $C_{/F}$. By the sufficiency assertion in a), this is exactly what is ensured by the hypothesis that every object of $C_{/F}$ is dominated by a minimal object. Necessity: since every filtered inductive limit of left exact functors is again left exact, the first condition is trivially necessary. The second then follows from the necessity assertion in a).

c) A representable subfunctor $X \hookrightarrow F$ of F is known once one knows the full subcategory $C'_{/X}$ of $C'_{/F}$, where C' is a fixed small generating subcategory of C. Use the hypothesis that I is filtered. Since $C'_{/F}$ is small, the set of its full subcategories is small, whence the conclusion.

Proposition 8.12.5. Let C be a $\mathcal{U}$-category in which finite inductive limits are representable and which admits a small generating subcategory (7.1). Let F be a presheaf on C. For F to be strictly ind-representable, it is sufficient that it satisfy the following two conditions; and these conditions are also necessary if fiber products are representable in C:

a) F is left exact.

b) Every pair (X, u), with $X \in \operatorname{Ob} C$ and $u \in F(X)$, is dominated in $C_{/F}$ by a minimal pair (8.12.3). In other words, there is a minimal pair (X', u'), with $X' \in \operatorname{Ob} C$ and $u' \in F(X')$, and a morphism $f : X \to X'$ such that $u = F(f)(u')$.

Sufficiency follows from 8.12.2 and 8.12.4(b),(c); necessity follows from 8.3.1 and 8.12.4(a).

Remark 8.12.6. When F transforms amalgamated sums in C into fiber products, one sees immediately that, for a given pair (X, u) as in b), the set of strict quotients X' of X such that

$$u \in \operatorname{Im}\bigl(F(X') \to F(X)\bigr)$$

is decreasing filtered. It follows that if the set of strict quotients of X is Artinian, then condition b) of 8.12.5 is automatically satisfied. Thus, if the preceding condition on X is satisfied for every object X of C – in which case one sometimes also says that the objects of C^{op} are *Artinian* – then 8.12.5 implies that F is strictly ind-representable if and only if F is left exact. In this case, F is strictly ind-representable as soon as it is ind-representable (8.3.1).

Corollary 8.12.7. Let C be a $\mathcal{U}$-category in which small projective limits are representable, and which admits a small cogenerating subcategory (7.9, 7.13). Then a functor $F : C \to \mathbf{Set}$ is representable if and only if F commutes with small projective limits.

Necessity is clear. For sufficiency, apply 8.12.5 to the opposite category C^{op}. To prove first that F is strictly pro-representable, it remains to prove that every pair (X, u), with $X \in \operatorname{Ob} C$ and $u \in F(X)$, is dominated by a minimal pair. But the family $(X_i)_{i \in I}$ of strict subobjects of X such that

$$u \in \operatorname{Im}\bigl(F(X_i) \to F(X)\bigr)$$

is small (7.5 in its dual form). Since F is left exact, this family is cofiltered (8.12.6), and since F commutes with small projective limits one sees that $X' = \varprojlim_i X_i$ is a smallest object of this family. Use the fact that F, being left exact, transforms monomorphisms into monomorphisms.

If $u' \in F(X')$ is the unique element whose image in $F(X)$ is u, it follows that (X', u') is a minimal pair dominating (X, u). This proves that F is pro-representable, and the conclusion follows from the following lemma.

Lemma 8.12.8.1. Let $F : C \to \mathbf{Set}$ be a functor, where C is a $\mathcal{U}$-category in which small projective limits are representable. For F to be representable, it is necessary and sufficient that it be pro-representable and commute with small projective limits.

Necessity is clear. We prove sufficiency. To say that F is representable plainly means that ${}_F\backslash C$ has an initial object; indeed, the initial objects of ${}_F\backslash C$ are precisely the isomorphisms $F \to X$, that is, the data of a representation for F. Since F is pro-representable, $({}_F\backslash C)^{\mathrm{op}}$ is filtered and equivalent to a small category. Thus

$$X = \varprojlim_{({}_F\backslash C)^{\mathrm{op}}} X$$

is representable in C, and since F commutes with the limit in question, it follows that X is an initial object of ${}_F\backslash C$. □

Corollary 8.12.8. With C satisfying the hypotheses of 8.12.7, let $f : C \to C'$ be a functor from C to a $\mathcal{U}$-category C'. For f to admit a left adjoint, it is necessary and sufficient that f commute with small projective limits.

This reduces trivially to 8.12.7 by applying that statement to the composite functors of the form

$$X \longmapsto \mathrm{Hom}(Y', f(X)).$$

Examples 8.12.9. As noted in 7.13, the hypotheses on C in 8.12.7 and 8.12.8 are satisfied when C is the category of $\mathcal{U}$-sheaves of sets on a topological space $X \in \mathcal{U}$, or more generally on a $\mathcal{U}$-site (II 3.0.2). We give an instructive example, due to H. Bass, showing that the hypothesis of existence of a small cogenerating subcategory D of C is not redundant in 8.12.7. Take C to be the category of groups belonging to $\mathcal{U}$. Let J be the set of isomorphism classes of simple groups in C. For each $j \in J$, choose a simple group G_j in the class j, and let I be the filtered ordered set of $\mathcal{U}$-small subsets of J. For $i \in I$, let

$$X_i = \amalg_{j \in i} G_j.$$

The X_i then form a projective system $(X_i)_{i \in I}$ in C, with epimorphic transition morphisms, and the corresponding functor

$$F(X) = \varinjlim_i \mathrm{Hom}(X_i, X) \tag{$*$}$$

takes values in $\mathcal{U}$-sets, although the index set I plainly does not have cardinal in $\mathcal{U}$. More precisely, let us show that for every small full subcategory C_0 of C, there exists $i_0 \in I$ such that the restriction of F to C_0 is represented by X_{i_0}. This will prove both that F is $\mathcal{U}$-set-valued and that it commutes with small projective limits. To prove the assertion, it is enough to note that, for every $X \in \mathrm{Ob}\, C_0$, the cardinal of the set $J(X)$ of $j \in J$ for which there exists a nontrivial homomorphism from G_j to X is necessarily small, since such a morphism is necessarily a monomorphism, G_j being simple. Hence, if i_0 is the subset of J which is the union of the $J(X)$ for $X \in \mathrm{Ob}\, C_0$, then i_0 is small, that is, $i_0 \in I$, and it does the job. On the other hand, it is clear that F is not representable, since $\mathrm{card}\, I \notin \mathcal{U}$. From this and 8.12.7 one therefore concludes that *the category C of groups in $\mathcal{U}$ admits no small full cogenerating subcategory.* Since the object $\mathbb{Z}$ of C is nevertheless a generator, it follows from the proof of 7.12 that there exists a two-generator group G which does not embed into an injective object of the category C of groups in $\mathcal{U}$. It also seems plausible that C has no injective objects other than the unit groups.

8.13 Pro-representable functors and accessible functors

8.13.1

In this section we use some notions and results from the following paragraph, notably 9.11 and 9.13, in order to obtain a criterion of pro-representability which will be used incidentally in IV 9.16. In what follows, C denotes a $\mathcal{U}$-category satisfying condition L of 9.1(b). This condition is satisfied, for example, if small filtered inductive limits are representable in C.

Proposition 8.13.2. Let C be as above, and let

$$f : C \longrightarrow \mathbf{Set}$$

be a functor.

a) Suppose every object of C is accessible (9.3). If f is pro-representable, then f is accessible (9.2) and left exact.

b) Suppose finite projective limits are representable in C, and that C admits a cardinal filtration (9.12). If f is accessible and left exact, then f is pro-representable.

Proof.

a) The hypothesis on C means that the covariant representable functors from C to **Set** are accessible. Therefore the same is true of every small inductive limit of such functors (9.6(i)), and hence of every pro-representable functor.

b) By 8.3.3(iii), it remains to prove that $\mathrm{Ob}({}_F\backslash C)^{\mathrm{op}}$ contains a small cofinal subcategory. By hypothesis there exists a cardinal Π such that f is Π-accessible. Let (X, u), $u \in F(X)$, be an object of ${}_F\backslash C$. With the notation of 9.12(c), one then has $X = \varinjlim_i X_i$, with I filtered and large relative to Π, and with the X_i in $C' = \mathrm{Filt}^{\Pi}(C)$. Hence

$$F(X) \xleftarrow{\sim} \varinjlim_i F(X_i).$$

This shows that the small subcategory $({}_F\backslash C')^{\mathrm{op}}$ is cofinal in $({}_F\backslash C)^{\mathrm{op}}$, and completes the proof.

Corollary 8.13.3. Let C be a $\mathcal{U}$-category satisfying the following conditions:

a) finite projective limits are representable in C;

b) small filtered inductive limits are representable in C;

c) the functor Ker on the category of double arrows of C is accessible, for example it commutes with small filtered inductive limits;

d) every strict epimorphism of C is strictly universal (10.2);

e) there exists a small subcategory of C generating by strict epimorphisms (7.1).

Under these conditions, a functor $f : C \to \mathbf{Set}$ is pro-representable if and only if it is left exact and accessible (9.2).

Indeed, the hypotheses on C in 8.13.2(a) and (b) are satisfied by 9.11 and 9.13, respectively.

Corollary 8.13.4. Let $f : C \to C'$ be a functor between $\mathcal{U}$-categories satisfying conditions a) through e) of 8.13.3. For f to admit a pro-adjoint, it is necessary and sufficient that f be left exact and accessible.

Since the representable functors

$$h_{Y'} : X' \longmapsto \mathrm{Hom}(Y', X')\,, \qquad C' \to \mathbf{Set},$$

are left exact and accessible (9.11), if f has these same properties then so do their composites with f, and these composites are therefore pro-representable by 8.13.3; that is, f admits a pro-adjoint. Conversely, suppose that f admits a pro-adjoint, and let C'_1 be a small generating subcategory of C'. Choose a cardinal Π such that the objects $Y' \in \mathrm{Ob}\, C'_1$ are Π-accessible. To prove that f is Π-accessible, it is enough to prove this for its composites with the $h_{Y'}$, $Y' \in \mathrm{Ob}\, C'_1$. But by hypothesis these composites are pro-representable, hence accessible by 8.13.3, and therefore Π-accessible after increasing Π if necessary. $\square$

Continuation anchor. The next section is Section 9, "Accessible functors, cardinal filtrations, and construction of small generating subcategories."

Exposé I. Presheaves

Section 9, beginning through Lemma 9.13.1

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 9 from the start through Lemma 9.13.1. The notation follows the previous batches: $\mathcal{U}$ denotes the fixed universe; $\mathcal{U}$-**Set** is written as $\mathcal{U}$-**Set**; and $\operatorname{card} \operatorname{Fl} C$ denotes the cardinal of the set of arrows of a category C.

9 Accessible functors, cardinal filtrations, and construction of small generating subcategories

The present section is more technical than the other sections of this exposé. In this seminar it will be used only in IV 9 and VI 4, which themselves are not used elsewhere in the Seminar. It is therefore advisable to omit the present section on a first reading.

9.0

All categories considered in this number are assumed to be $\mathcal{U}$-categories. Except for the small indexing categories $I, J, \ldots$ that we shall use, the developments below will mainly concern "large" categories $E, F, \ldots$ that are stable under small filtered inductive limits. Most often, however, it will be enough that a slightly weaker condition hold, namely condition L of Definition 9.1 below. All cardinals considered in this number are assumed to belong to $\mathcal{U}$.

Following a suggestion of P. Deligne, we shall study, for a functor $f : E \to F$ between large categories, a condition expressing commutation of f with certain types of filtered inductive limits. This condition is remarkably stable and is satisfied by the most important functors that occur naturally. The applications we have in view for this seminar are 9.13.3 and 9.13.4, used in VI 4, and above all 9.25, which gives, in a non-trivial case, the existence of a small generating family in a category of sections of a fibred category; that result will be used in IV 9.16.

Definition 9.1.

a) Let I be a preordered set and let π be a cardinal. We say that I is *large with respect to* π if I is filtered and if every subset of I of cardinal $\leq \pi$ has an upper bound in I.

b) Let E be a category. If π is a cardinal, we say that E satisfies condition L_π if, for every small ordered set I large with respect to π, E is stable under inductive limits of type I. We say that E satisfies condition L if there exists a cardinal $\pi \in \mathcal{U}$ such that E satisfies L_π.

9.1.1

When, in Definition 9.1(a), one has $\pi \geq 2$, the second stated condition already implies that I is filtered. If π is finite, saying that I is large with respect to π simply means that I is filtered. In what follows we shall be interested almost exclusively in the case where π is infinite. Notice that if π and π' are cardinals with $\pi' \geq \pi$, then I large with respect to π' plainly implies I large with respect to π.

9.1.2

As announced in 9.0, the conditions L_π and L should be regarded as technical variants of the stronger condition of stability under small filtered inductive limits. It is clear that if π and π' are cardinals with $\pi' \geq \pi$, then condition L_π implies condition $L_{\pi'}$.

Definition 9.2. Let $f : E \to F$ be a functor. If Π is a cardinal, we say that f is Π-*preaccessible* (respectively Π-*accessible*) if E satisfies L_Π (9.1) and if, for every ordered set $I \in \mathcal{U}$ large with respect to Π, and every inductive system $(X_i)_{i\in I}$ in E of type I, the canonical morphism

$$\varinjlim f(X_i) \longrightarrow f(\varinjlim X_i)$$

is a monomorphism (respectively an isomorphism). We say that f is *preaccessible* (respectively *accessible*), relative to the universe $\mathcal{U}$, if there exists a cardinal $\Pi \in \mathcal{U}$ such that f is Π-preaccessible (respectively Π-accessible).

The category of Π-accessible (respectively accessible) functors from E to F will be denoted by $\mathrm{Hom}(E, F)_\Pi$ (respectively $\mathrm{Hom}(E, F)_{\mathrm{acc}}$).

9.2.1

Plainly, a functor which commutes with small filtered inductive limits, for example a functor admitting a right adjoint, is Π-accessible for every cardinal $\Pi \geq 2$.

Definition 9.3. Let E be a category, let X be an object of E, and let

$$h^\circ_X : E \longrightarrow (\mathcal{U}\text{-}\mathbf{Set})$$

be the covariant functor it represents. Let $\Pi \in \mathcal{U}$ be a cardinal. We say that X is a Π-preaccessible (respectively Π-accessible) object of E if the functor h°_X is Π-preaccessible (respectively Π-accessible). We say that X is preaccessible (respectively accessible) if there exists a cardinal $\Pi \in \mathcal{U}$ such that X is a Π-preaccessible (respectively Π-accessible) object of E.

9.3.1

We denote by E_Π the strictly full subcategory of E consisting of the Π-accessible objects of E.

When Definition 9.3 is applied to a category of the form $\mathrm{Hom}(C, F)$, the terminology just introduced is, *a priori*, ambiguous with the analogous terminology introduced in 9.2, when one interprets the objects of $\mathrm{Hom}(C, F)$ as functors. There does not, however, seem to be any serious risk of confusion.

Definition 9.4. Let E be a category and let $\pi \in \mathcal{U}$ be a cardinal. We say that E is a π-preaccessible (respectively π-accessible) category if there exists in E a small full subcategory C which is generating (7.1) and whose objects are π-preaccessible (respectively π-accessible) (9.3). We say that E is preaccessible (respectively accessible) if there exists a cardinal $\pi \in \mathcal{U}$ such that E is π-preaccessible (respectively π-accessible).

For important examples, see 9.11.3 below.

Proposition 9.5. Let $f : E \to F$ be a functor between categories such that F is accessible and every object of E is accessible. If f admits a left adjoint, then f is accessible.

Indeed, by hypothesis, F admits a small conservative family of representable functors $F \to (\mathcal{U}\text{-}\mathbf{Set})$ which are accessible. Thus one is immediately reduced to showing that the composite of f with each of these functors is accessible. Since f admits a left adjoint, these composites are representable functors $E \to (\mathcal{U}\text{-}\mathbf{Set})$, hence are accessible by the hypothesis on E.

Proposition 9.6. Let E and F be two categories, let $\Pi \in \mathcal{U}$ be a cardinal, and consider the full subcategory $\mathrm{Hom}(E,F)_\Pi$ of $\mathrm{Hom}(E,F)$ formed by the Π-accessible functors (9.2) from E to F.

(i) This subcategory is stable under every type of inductive limit which is representable in F.

(ii) Suppose that F is Π-accessible (9.4), and let J be a category such that $\operatorname{card}\operatorname{Fl} J \leq \Pi$ and such that projective limits of type J are representable in F; then projective limits of type J are representable in $\mathrm{Hom}(E,F)$. The subcategory $\mathrm{Hom}(E,F)_\Pi$ is stable under projective limits of type J.

Corollary 9.7. Let E and F be two categories, with F accessible (9.4). Then the full subcategory $\mathrm{Hom}(E,F)_{\mathrm{acc}}$ of $\mathrm{Hom}(E,F)$ formed by accessible functors is stable under every inductive or projective limit, relative to a small indexing category J, which is representable in F and hence in $\mathrm{Hom}(E,F)$.

Proof. [Proof of Proposition 9.6] Assertion (i) follows trivially from the commutation of the functor $\varinjlim$ with arbitrary inductive limits. For (ii), let $(f_j)_{j\in J}$ be a projective system of Π-accessible functors $E \to F$, and let

$$f = \varprojlim_j f_j$$

be its projective limit, computed argument by argument. We prove that f is Π-accessible; in other words, for every ordered set $I \in \mathcal{U}$ large with respect to Π, and every inductive system $(X_i)_{i\in I}$ in E, the canonical morphism

$$\varinjlim_i((\varprojlim_j f_j)(X_i)) \longrightarrow (\varprojlim_j f_j)(\varinjlim_i X_i)$$

is an isomorphism. The argument-by-argument computation of the functor $\varprojlim_j f_j$ identifies this canonical morphism with

$$\varinjlim_i \varprojlim_j f_j(X_i) \longrightarrow \varprojlim_j \varinjlim_i f_j(X_i),$$

the canonical morphism attached to the bifunctor

$$(i,j) \longmapsto f_j(X_i) : I \times J \longrightarrow F.$$

Thus Proposition 9.6 follows from the following more general assertion.

Corollary 9.8. Let F be a Π-accessible category, let I be an ordered set large with respect to Π, let J be a small category such that $\operatorname{card}\operatorname{Fl} J \leq \Pi$ and such that projective limits of type J are representable in F, and let $h : I \times J \to F$ be a functor. Then the canonical morphism

$$\varinjlim_i \varprojlim_j h(i,j) \longrightarrow \varprojlim_j \varinjlim_i h(i,j) \tag{9.8.1}$$

is an isomorphism. In other words, the functor

$$\varinjlim_{i} : \mathrm{Hom}(I, F) \longrightarrow F \tag{9.8.2}$$

commutes with projective limits of type J, for every small category J such that $\operatorname{card} \mathrm{Fl}\, J \leq \Pi$ and such that projective limits of type J are representable in F. Equivalently, for every J as above, the functor

$$\varinjlim_{j} : \mathrm{Hom}(J, F) \longrightarrow F \tag{9.8.3}$$

is Π-accessible.

Since, by hypothesis, F admits a conservative family of covariant representable Π-accessible functors $g : F \to (\mathcal{U}\text{-}\mathbf{Set})$, one is immediately reduced to proving 9.8 in the case $F = \mathcal{U}\text{-}\mathbf{Set}$. We shall then prove the assertion in the form that (9.8.2) commutes with projective limits of type J, with $\operatorname{card} \mathrm{Fl}\, J \leq \Pi$. Since I is filtered, (9.8.2) is known to commute with finite projective limits (2.8). A standard argument (cf. 2.3) therefore reduces us to proving that it commutes with products indexed by a set J of cardinal $\leq \Pi$. This reduces to proving the bijectivity of (9.8.1) when J is discrete.

For injectivity, consider two elements a, b of the left-hand side. They come from $\prod_j h(i_0, j)$ for a suitable $i_0 \in I$, say from elements $\prod_j a(i_0, j)$ and $\prod_j b(i_0, j)$. Suppose that the elements $\prod_j h(\infty, j)$ of the right-hand side which they define are equal; that is, for every j there exists $i(j) \geq i_0$ such that $a(i_0, j)$ and $b(i_0, j)$ have the same image in $h(i, j)$. Since I is large with respect to $\operatorname{card} J$, there exists a common upper bound $i_1 \in I$ for all the $i(j)$. It follows that the two elements under consideration in $\prod_j h(i_0, j)$ have the same image in $\prod_j h(i_1, j)$, and therefore define the same element of the left-hand side of (9.8.1). This proves injectivity.

For surjectivity, let $\prod_j a(\infty, j)$ be an element of the right-hand side. For each $j \in J$, $a(\infty, j)$ comes from an element of $h(i(j), j)$ for a suitable $i(j) \in I$. As before, one can find a common upper bound $i \in I$ for all the $i(j)$. Then $\prod_j a(\infty, j)$ comes from an element $\prod_j a(i, j)$ of $\prod_j h(i, j)$, hence lies in the image of (9.8.1). This completes the proof.

Corollary 9.9. Let F be a Π-accessible (respectively accessible) category. Then, for every category J such that $\operatorname{card} \mathrm{Fl}\, J \leq \Pi$ (respectively every small category J), and for every functor $(X_j)_{j \in J} : J \to F$ whose inductive limit in F is representable, if the X_j are Π-accessible (respectively accessible), then so is $\varprojlim X_j$.

Indeed, the functor $F \to (\mathcal{U}\text{-}\mathbf{Set})$ represented by $\varinjlim X_j$ is the projective limit of the functors represented by the X_j, and one applies Proposition 9.6(ii) to the resulting projective system of functors.

Remark 9.10. In the statements 9.6, 9.7, 9.8, and 9.9 one may everywhere replace the words "Π-accessible", "accessible" by "Π-preaccessible", "preaccessible". The proof just given proves this variant of the preceding statements as well.

Proposition 9.11. Let E be a category satisfying condition L (9.1), and let C be a small full generating subcategory (7.1). Suppose that the following condition is satisfied:

a) E is stable under kernels of double arrows, and the functor Ker on the category of double arrows of E is accessible; for example, it commutes with small filtered inductive limits.

Then every object of E is preaccessible (9.3), and *a fortiori* E is preaccessible. Suppose moreover that C is generating by strict epimorphisms (7.1), and suppose that E satisfies the following conditions:

b) E is stable under fibre products.

c) Every small strict epimorphic family in E is universally strict epimorphic (10.3); or else small coproducts are representable in E and every strict epimorphism in E is a universal strict epimorphism.

Then every object of E is accessible, and *a fortiori* E is accessible.

The fact that, under (a), every object of E is preaccessible follows from the fact that, for every object X of E, the set of strict subobjects of X is small (7.4), and from the following lemma.

Lemma 9.11.1. Under the hypotheses of 9.11(a), let $X \in \operatorname{ob} E$ and let Π be a cardinal such that E satisfies L_Π (9.1), the functor Ker in 9.11(a) is Π-accessible, and the set of strict subobjects of X has cardinal bounded by Π. Then X is Π-preaccessible.

Indeed, let I be a set large with respect to Π, let $(Y_i)_{i\in I}$ be an inductive system in E with inductive limit Y, and let

$$(u_i, v_i : X \rightrightarrows Y_i)$$

be a double arrow such that the composite double arrow $u, v : X \rightrightarrows Y$ satisfies $u = v$. We prove that there exists $j \geq i$ in I such that $u_j = v_j$. For this, consider the inductive system of double arrows $(u_j, v_j : X \rightrightarrows Y_j)_{j\geq i}$, whose inductive limit is (u, v). By hypothesis,

$$X = \operatorname{Ker}(u, v) = \varinjlim_j \operatorname{Ker}(u_j, v_j) = \varinjlim_j X_j, \qquad X_j = \operatorname{Ker}(u_j, v_j).$$

The X_j are strict subobjects of X, so there exists a subset I' of I, consisting of indices $j \geq i$, with $\operatorname{card} I' \leq \Pi$, such that every X_j is equal to one of the $X_{i'}$ for $i' \in I'$. Since I is large with respect to Π, there is an upper bound j of I' in I. Then X_j contains all the $X_{j'}$ for $j' \geq i$, hence $X_j \to X$ is an epimorphism, since the family of maps $X_{j'} \to X$ is epimorphic. It is therefore an isomorphism, since it is a strict monomorphism. Thus $u_j = v_j$, proving 9.11.1.

The second assertion of 9.11 follows from the first and from the following lemma.

Lemma 9.11.2. Under the hypotheses of 9.11(a), (b), and (c), let X be an object of E, and let Π be an infinite cardinal such that E satisfies L_Π (9.1), such that $\operatorname{card} \operatorname{ob} C_{/X} \leq \Pi$, such that for two objects $X' \to X$ and $X'' \to X$ of $C_{/X}$ the fibre product $X' \times_X X''$ is Π-preaccessible, and finally such that X is Π-preaccessible. Then X is Π-accessible.

With the notation of the proof of 9.11.1, it remains only to prove that every morphism $u : X \to Y$ comes from a morphism $u_i : X \to Y_i$. Consider the family of morphisms $Y_i \to Y$, which is a strict epimorphic family. By (b) and (c), the family

$$Y_i \times_Y X \longrightarrow X$$

is also strictly epimorphic. On the other hand, for every i, since C is generating by strict epimorphisms, the family of maps

$$T \longrightarrow Y_i \times_Y X$$

with source in C is strictly epimorphic. In the second alternative in (c), one can find such a strict epimorphism with source a small coproduct of objects of C. By transitivity of universally strict epimorphic families (II 2.5), it follows that the family of maps $T_\alpha \xrightarrow{f_\alpha} X$ with source in C which factor through one of the $Y_i \times_Y X$ is strictly epimorphic. Let J be the set of indices of this family; its cardinal is bounded by $\operatorname{card} \operatorname{ob} C_{/X} \leq \Pi$.

For each $\alpha \in J$, choose an $i = i(\alpha) \in I$ and an X-morphism

$$T_\alpha \longrightarrow Y_i \times_Y X,$$

or equivalently a morphism $v_\alpha : T_\alpha \to Y_i$ such that $p_i v_\alpha = u f_\alpha$, where $p_i : Y_i \to Y$ is the canonical morphism. Since I is large with respect to Π, one may choose $i(\alpha)$ independently of α; call it i. For every pair of indices $\alpha, \beta \in J$, consider the two composites

$$\operatorname{pr}_1 v_\alpha, \operatorname{pr}_2 v_\beta : T_\alpha \times_X T_\beta \rightrightarrows Y_i.$$

Their composites with $Y_i \to Y$ are equal. Since $T_\alpha \times_X T_\beta$ is Π-preaccessible by hypothesis, there exists an index $i' = i(\alpha, \beta) \geq i$ such that the composites of the arrows under consideration with $Y_i \to Y_{i'}$ are equal. The set of pairs (α, β) has cardinal $\leq \Pi^2 = \Pi$, because Π is infinite; hence one may again choose i' independently of α and β. Plainly we may suppose $i' = i$. But then, since the family $(f_\alpha : T_\alpha \to X)$ is strictly epimorphic, one can find a morphism $u_i : X \to Y_i$ such that $u_i f_\alpha = v_\alpha$. We then have $p_i u_i = u$, because for every α,

$$(p_i u_i) f_\alpha = p_i (u_i f_\alpha) = p_i v_\alpha = u f_\alpha,$$

and the family of the f_α is epimorphic. This completes the proof of 9.11.2.

Remark 9.11.3. As a special case of 9.11, one sees that every object of E is accessible in each of the following two cases. First, E is an abelian $\mathcal{U}$-category with exact filtered inductive limits and admitting a small generating subcategory; here it is the second alternative in (c) which applies. Second, E is the category of sheaves of sets on a topological space $X \in \mathcal{U}$. More generally, it is enough that E be the category of sheaves of sets on a $\mathcal{U}$-site (II 2.1), or that E be a $\mathcal{U}$-topos (IV 1.1).

Definition 9.12. Let E be a category. A *cardinal filtration* of E is an increasing filtration $(\operatorname{Filt}^\Pi(E))_{\Pi \geq \Pi_0}$ of E by strictly full subcategories $\operatorname{Filt}^\Pi(E)$, indexed by the cardinals $\Pi \in \mathcal{U}$ with $\Pi \geq \Pi_0$, where Π_0 is a fixed infinite cardinal depending on the cardinal filtration under consideration, and satisfying the following conditions:

a) For every $\Pi \geq \Pi_0$, $\operatorname{Filt}^\Pi(E)$ is equivalent to a small category.

b) E satisfies L_{Π_0} (9.3), and for every $\Pi \geq \Pi_0$, $\operatorname{Filt}^\Pi(E)$ is stable in E under filtered inductive limits indexed by ordered sets I which are large with respect to Π_0 and such that $\operatorname{card} I \leq \Pi$.

c) For every $\Pi \geq \Pi_0$ and every $X \in \operatorname{ob} E$, one can find an isomorphism

$$X \simeq \varinjlim_I X_i,$$

where $(X_i)_{i \in I}$ is an inductive system in E indexed by an ordered set I large with respect to Π (9.1), and where the X_i belong to $\operatorname{Filt}^\Pi(E)$. Moreover, if $X \in \operatorname{ob} \operatorname{Filt}^\Pi(E)$ and $\Pi' \geq \Pi$, one can take I with $\operatorname{card} I \leq (\Pi')^\Pi$.

9.12.1

It follows from (b) and (c) that every $X \in \operatorname{ob} E$ belongs to some $\mathrm{Filt}^{\Pi}(E)$ for Π sufficiently large.

Example 9.12.2. Take $E = (\mathcal{U}\textbf{-Set})$, $\Pi_0 = \aleph_0$, and let $\mathrm{Filt}^{\Pi}(E)$ be the full subcategory of E consisting of the sets X such that $\operatorname{card} X \leq \Pi$. More generally:

Proposition 9.13. Let E be a $\mathcal{U}$-category. Suppose that E is stable under small filtered inductive limits, under sums of two objects, and under cokernels of double arrows; that E is stable under fibre products; and that epimorphic morphisms in E are universal strict epimorphisms. Let C be a small full subcategory of E which is generating by strict epimorphisms. Let Π_0 be an infinite cardinal $\geq \operatorname{card} \mathrm{Fl}\, C$. For every cardinal $\Pi \geq \Pi_0$, let $\mathrm{Filt}^{\Pi}(E)$ be the strictly full subcategory of E consisting of the objects X of E for which there exists a strict epimorphic family $(X_i \to X)_{i \in I}$ with target X, such that $\operatorname{card} I \leq \Pi$ and $X_i \in \operatorname{ob} C$ for every $i \in I$. Then $(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0}$ is a cardinal filtration of E. Moreover, for every $X \in \operatorname{ob} \mathrm{Filt}^{\Pi}(E)$, the cardinal of the set of arrows of $C_{/X}$, and *a fortiori* of the set of objects of $C_{/X}$, is bounded by Π^{Π_0}.

The last assertion is precisely 7.6.

To prove that this is a cardinal filtration, it remains to prove conditions (a), (b), and (c) of 9.12. Condition (a) follows at once from 7.5.2. For (b), suppose that

$$X = \varinjlim_{I} X_i,$$

with $\operatorname{card} I \leq \Pi$ and $X_i \in \operatorname{ob} \mathrm{Filt}^{\Pi}(E)$. The family of canonical morphisms $X_i \to X$ is therefore strictly epimorphic. By the hypothesis on the X_i, for each $i \in I$ there is a strict epimorphic family $(X_{ij} \to X_i)_{j \in J_i}$ with $\operatorname{card} J_i \leq \Pi$. Passing to the sums $\coprod_i X_i$ and $\coprod_{i,j} X_{ij}$, one sees, by transitivity of universal strict epimorphisms (II 2.5), that the family of composites

$$X_{ij} \longrightarrow X_i \longrightarrow X$$

indexed by the disjoint union of the J_i, for $i \in I$, is strictly epimorphic. Since $\operatorname{card} J \leq \Pi^2 = \Pi$, the conclusion follows. Notice that we have used only the existence of $\varinjlim_I X_i$, and not the fact that I is large with respect to Π_0, nor even that it is filtered.

It remains to prove (c). First note that for every $X \in \operatorname{ob} E$, the family of arrows $X_i \to X$ with source in $\operatorname{ob} C$ is strictly epimorphic, because C is generating by strict epimorphisms, and is plainly small. Thus there is a cardinal $\Pi \geq \Pi_0$ such that $X \in \operatorname{ob} \mathrm{Filt}^{\Pi}(E)$. It remains to prove that, if $\Pi' \geq \Pi \geq \Pi_0$ and $X \in \operatorname{ob} \mathrm{Filt}^{\Pi'}(E)$, then there is an isomorphism

$$X \simeq \varinjlim_{I} X_i,$$

where I is large with respect to Π, $\operatorname{card} I \leq (\Pi')^{\Pi}$, and the X_i lie in $\operatorname{ob} \mathrm{Filt}^{\Pi}(E)$.

Since C is generating by strict epimorphisms, one has

$$X \simeq \varinjlim_{C_{/X}} T. \tag{$*$}$$

Furthermore, by successively adjoining to C sums and cokernels of double arrows in C, and then doing the same for the category so obtained, and so on, we are reduced to the case where C

is stable under sums of two objects in E and under cokernels of double arrows in E, without destroying the hypothesis $\Pi_0 \geq \operatorname{card} \operatorname{Fl} C$. We may also suppose that, if E contains a fixed initial object $\varnothing_E$, then $\varnothing_E \in \operatorname{ob} C$. This implies that the category $C_{/X}$ is stable under sums of two objects and under cokernels of double arrows. It is then filtered: it is non-empty, since otherwise the relation $(*)$ would show that X is an initial object, so that $C_{/X}$ would contain the arrow $\varnothing_E \to X$, a contradiction.

Let I be the set, ordered by inclusion, of full subcategories i of $C_{/X}$ which are filtered and such that $\operatorname{card} \operatorname{ob} i \leq \Pi$. For every $i \in I$, let $X_i \in \operatorname{ob} E$ be the inductive limit of the composite functor

$$i \longrightarrow C_{/X} \longrightarrow E,$$

which exists by hypothesis. Then one plainly has

$$\varinjlim_{I} X_i \simeq \varinjlim_{C_{/X}} T \simeq X.$$

On the other hand, we have already noted that

$$\operatorname{card} \operatorname{ob} C_{/X} \leq (\Pi')^{\Pi_0}.$$

It follows that I is large with respect to Π and that

$$\operatorname{card} I \leq ((\Pi')^{\Pi_0})^{\Pi} = (\Pi')^{\Pi\Pi_0} = (\Pi')^{\Pi}$$

by the following lemma, whose proof is left to the reader; in applying it one takes $J = C_{/X}$ and $c = (\Pi')^{\Pi_0}$.

Lemma 9.13.1. Let J be a filtered category, and let c and Π be two cardinals such that

$$c \geq \operatorname{Sup}(\operatorname{card} \operatorname{Fl} J, \Pi)$$

and such that $\operatorname{card} \operatorname{Hom}(j, j') \leq \Pi_0$ for every pair of objects j, j' of J. Let I be the set of full filtered subcategories i of J such that $\operatorname{card} \operatorname{ob} i \leq \Pi$. Then, ordered by inclusion, I is large with respect to Π, and one has $\operatorname{card} I \leq c^{\Pi}$.

This completes the proof of Proposition 9.13.

Exposé I. Presheaves

Section 9, Remark 9.13.2 through Corollary 9.26

A. Grothendieck and J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 9 from Remark 9.13.2 through Corollary 9.26. The terminology follows the previous batches: "large with respect to Π" translates *grand devant* Π; "accessible" is used in the sense of 9.2–9.3; and $\operatorname{card} \operatorname{Fl} C$ denotes the cardinal of the set of arrows of a category C. A few evident typographical slips in the source have been corrected where the formulas force the correction, notably in some subscripts in 9.14, 9.21, and 9.24.

9 Accessible functors, cardinal filtrations, and construction of small generating subcategories

9.13.2

Remark 9.13.2. A slight additional effort should make it possible, in Proposition 9.13, to replace the hypothesis that E is stable under small filtered inductive limits by the hypothesis that E satisfies L (9.1), provided that E is stable under small sums, or that in E every epimorphic *family* is universally epimorphic. In the proof of c), one must then choose Π_0 so that E satisfies L_{Π_0}, and restrict to those full subcategories i of $C_{/X}$ which are not only filtered, but whose preordered set $\operatorname{ob} i$ is large with respect to Π_0. Using Section 8 and the hypothesis that E satisfies L_{Π_0}, one then finds that the X_i exist; one should conclude by means of a suitable variant of Lemma 9.13.1, which the editor has not checked.

Proposition 9.14. Let $f : E \to F$ be an accessible functor (9.2) between two categories endowed with cardinal filtrations

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0} \qquad \text{and} \qquad (\mathrm{Filt}^{\Pi}(F))_{\Pi \geq \Pi'_0}.$$

Then there exists a cardinal $\Pi_1 \geq \operatorname{Sup}(\Pi_0, \Pi'_0)$ such that, for every cardinal $\Pi \geq \Pi_1$, one has

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi^{\Pi_1}}(F). \tag{9.14.1}$$

In particular, if $\Pi = 2^c$ with $c \geq \Pi_1$, whence

$$\Pi^{\Pi_1} = 2^{c\Pi_1} = 2^c = \Pi,$$

then

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi}(F). \tag{9.14.2}$$

Let us immediately record the following consequence.

Corollary 9.15. Let E be a category endowed with two cardinal filtrations

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0} \qquad \text{and} \qquad (\mathrm{Filt}'^{\Pi}(E))_{\Pi \geq \Pi'_0}.$$

Then there exists a cardinal $\Pi_1 \geq \mathrm{Sup}(\Pi_0, \Pi_0')$ such that, for every cardinal $c \geq \Pi_1$, if $\Pi = 2^c$, then

$$\mathrm{Filt}^{\Pi}(E) = \mathrm{Filt}'^{\Pi}(E).$$

Proof. *Proof of Proposition 9.14.* Let c be a cardinal such that

$$c \geq \mathrm{Sup}(\Pi_0, \Pi_0')$$

and such that f is c-admissible. Let also $\Pi_1 \geq c$ be such that

$$f(\mathrm{Filt}^c(E)) \subset \mathrm{Filt}^{\Pi_1}(F). \tag{$*$}$$

Such a Π_1 exists because $\mathrm{Filt}^c(E)$ is equivalent to a small category (9.12 a)); hence $f(\mathrm{Filt}^c(E))$ is also essentially small, and one may apply 9.12.1 to the objects of the latter category to find one $\mathrm{Filt}^{\Pi}(F)$ containing all of them, taking into account that the $\mathrm{Filt}^{\Pi}(F)$ are full subcategories.

Let $\Pi \geq \Pi_1$, and let $X \in \mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$. We prove that

$$f(X) \in \mathrm{Filt}^{\Pi^{\Pi_1}}(F).$$

Write

$$X = \varinjlim_I X_i,$$

where $X_i \in \mathrm{ob}\,\mathrm{Filt}^c(E)$, where I is large with respect to c, and where

$$\mathrm{card}\, I \leq \Pi^c \leq \Pi^{\Pi_1}$$

(9.12 c)). Since f is c-admissible and I is large with respect to c, one has

$$f(X) \simeq \varinjlim_I f(X_i).$$

Moreover $\mathrm{card}\, I \leq \Pi^{\Pi_1}$, and $f(X_i) \in \mathrm{ob}\,\mathrm{Filt}^{\Pi_1}(F) \subset \mathrm{ob}\,\mathrm{Filt}^{\Pi^{\Pi_1}}(F)$ by $(*)$. Therefore $f(X) \in \mathrm{Filt}^{\Pi^{\Pi_1}}(F)$ by 9.12 b). $\square$

9.15.1

The notion of a cardinal filtration (9.12) is of interest only when the objects of E are accessible. Let us point out that Proposition 9.11 implies that this condition is satisfied when, in addition to the hypotheses of Proposition 9.13, one assumes that the functor Ker on the category of double arrows of E is accessible. We also record the following.

Proposition 9.16. Let E be a category endowed with a cardinal filtration

$$(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0}.$$

Suppose that the elements of $\mathrm{Filt}^{\Pi_0}(E)$ are accessible objects of E. Then there exists a cardinal $\Pi_1 \geq \Pi_0$ in $\mathcal{U}$ such that the objects of $\mathrm{Filt}^{\Pi_0}(E)$ are Π_1-accessible. If Π_1 is chosen in this way, then for every cardinal $\Pi \geq \Pi_1$ one has, with the notation of 9.3.1,

$$E_{\Pi} \subset \mathrm{Filt}^{\Pi}(E) \subset E_{\Pi^{\Pi_0}}. \tag{9.16.1}$$

The existence of Π_1 follows immediately from the fact that $\mathrm{Filt}^{\Pi_0}(E)$ is equivalent to a small category (9.12 a)). Let $\Pi \geq \Pi_1$. If X belongs to $\mathrm{Filt}^{\Pi}(E)$, write

$$X \simeq \varinjlim_I X_i$$

with $\operatorname{card} I \leq \Pi^{\Pi_0}$ and $X_i \in \operatorname{ob}\mathrm{Filt}^{\Pi_0}(E)$ for every i (9.12 c)). It then follows from Corollary 9.9 that X is Π^{Π_0}-accessible; this gives the second inclusion in (9.16.1).

Conversely suppose that X is Π-accessible, and write

$$X \simeq \varinjlim_I X_i,$$

where I is large with respect to Π and the X_i lie in $\operatorname{ob}\mathrm{Filt}^{\Pi}(E)$ (9.12 c)). By the hypothesis on X, the given isomorphism $X \xrightarrow{\sim} \varinjlim_I X_i$ factors through one of the X_i; hence X is isomorphic to a direct factor of this X_i. We conclude that $X \in \operatorname{ob}\mathrm{Filt}^{\Pi}(E)$, and hence obtain the first inclusion in (9.16.1), by the following lemma.

Lemma 9.16.2. Every object of E which is a direct factor of an object of $\mathrm{Filt}^{\Pi}(E)$ belongs to $\mathrm{Filt}^{\Pi}(E)$.

Indeed, if X is the image of an idempotent p on an object Y of E (i.e.an endomorphism p such that $p^2 = p$), and if I is a filtered ordered set, then X is the inductive limit of the filtered inductive system $(Y_i)_{i\in I}$ defined by $Y_i = Y$ for every i, and with transition map $p : Y_i \to Y_j$ whenever $i < j$. Taking I large with respect to Π_0 and with $\operatorname{card} I \leq \Pi$, we see that if $Y \in \mathrm{Filt}^{\Pi}(E)$, then the same is true of X by 9.12 b).

Corollary 9.17. Under the hypotheses of Proposition 9.16, for every cardinal $c \geq \Pi_1$, if $\Pi = 2^c$, then $\mathrm{Filt}^{\Pi}(E)$ is identical with the strictly full subcategory E_Π of E formed by the Π-accessible objects.

Corollary 9.18. Let E be a category satisfying condition L_Π (9.1), where Π is a cardinal, and let C be a full subcategory of E. Denote by $\mathrm{Ind}(C)_\Pi$ the full subcategory of the category $\mathrm{Ind}(C)$ of ind-objects of C (8.2) formed by ind-objects of the form $(X_i)_{i\in I}$, where I is an ordered set large with respect to Π. Consider the canonical functor

$$(X_i)_{i\in I} \longmapsto \varinjlim_I X_i : \mathrm{Ind}(C)_\Pi \longrightarrow E. \tag{9.18.1}$$

a) This functor is fully faithful if and only if every object X of C is a Π-accessible object (9.3) of E.

b) Assume the hypotheses of Corollary 9.17; in particular $\Pi = 2^c$, and take $C = \mathrm{Filt}^{\Pi}(E)$. Then the functor (9.18.1) is an equivalence of categories.

Assertion a) is an immediate generalization of 8.7.5 a), and is proved in the same way. Assertion b) follows from Corollary 9.17 and from condition 9.12 c) in the definition of cardinal filtrations, which implies that the functor under consideration is essentially surjective.

Corollary 9.19. Under the hypotheses of Corollary 9.17, let $C = \mathrm{Filt}^{\Pi}(E)$, let F be a $\mathcal{U}$-category, and consider the functor

$$\mathrm{Hom}(E, F)_\Pi \longrightarrow \mathrm{Hom}(C, F) \tag{9.19.1}$$

induced by "restriction to C", $f \mapsto f|_C$, where the source of (9.19.1) is the category of Π-accessible functors from E to F (9.2). This functor is fully faithful. If F satisfies condition L_Π (9.1), then the functor (9.19.1) is an equivalence of categories.

The first assertion is proved as in Proposition 7.8, using Corollary 9.18 b). For the second, one constructs a quasi-inverse to (9.19.1) by associating to a functor $f : C \to F$ the functor

$$(X_i)_{i\in I} \longmapsto \varinjlim_i f(X_i)$$

from $\mathrm{Ind}(E_\Pi)$ to F, and then using Corollary 9.18 b) to obtain a functor $\bar{f} : E \to F$. It remains only to show that this last functor is Π-accessible. This is proved as in the analogous assertion 8.7.3.

Corollary 9.20. Let E be a category admitting a cardinal filtration and such that every object of E is accessible (cf.Proposition 9.16), and let F be a category. Then:

a) The category $\mathrm{Hom}(E, F)_{\mathrm{acc}}$ of accessible functors from E to F (9.2) is a $\mathcal{U}$-category. (N.B. the given categories E, F are assumed to be $\mathcal{U}$-categories; see 9.0.)

b) Suppose that F is stable under small filtered inductive limits. For every full subcategory C of E equivalent to a small category, with inclusion functor $i : C \to E$, consider the corresponding functor
$$i_! : \mathrm{Hom}(C, F) \longrightarrow \mathrm{Hom}(E, F)$$
(5.1). A functor $f : E \to F$ is accessible if and only if there exists a small subcategory C of E such that f lies in the essential image of the preceding functor $i_!$.

Proof.

a) It is enough to prove that, for every cardinal Π such that E satisfies L_Π, the full subcategory $\mathrm{Hom}(E, F)_\Pi$ of $\mathrm{Hom}(E, F)_{\mathrm{acc}}$ is a $\mathcal{U}$-category. It plainly suffices to check this for cardinals of the form 2^c, with c sufficiently large. But then this follows from Corollary 9.19, since $C = \mathrm{Filt}^\Pi(E)$ is essentially small and therefore $\mathrm{Hom}(C, F)$ is plainly a $\mathcal{U}$-category.

b) By transitivity in the formation of the functors $i_!$, it suffices in the statement to restrict to subcategories C of the form $\mathrm{Filt}^\Pi(E)$, where Π is as in Corollary 9.17. One then sees easily that the composite functor

$$\mathrm{Hom}(\mathrm{Filt}^\Pi(E), F) \longrightarrow \mathrm{Hom}(E, F)_\Pi \longrightarrow \mathrm{Hom}(E, F),$$

where the first arrow is quasi-inverse to (9.19.1) and the second is the inclusion, is none other than the functor $i_!$, up to isomorphism. Thus b) follows from Corollary 9.19.

Exercise 9.20.1. This exercise uses the notions of site and topos, developed in Exposés II and IV. Let E be a $\mathcal{U}$-topos, let $\mathcal{V}$ be a universe such that $\mathcal{U} \in \mathcal{V}$, let C be a small full generating subcategory of the $\mathcal{U}$-topos E, and let $\Pi_0 \in \mathcal{U}$ be an infinite cardinal such that $\Pi_0 \geq \mathrm{card}\,\mathrm{Fl}\,C$. For every cardinal $\Pi \geq \Pi_0$, $\Pi \in \mathcal{U}$, let $\mathrm{Filt}^\Pi(E)$ be the strictly full subcategory of E formed by objects X for which there exists a strict epimorphic family $(X_i \to X)_{i\in I}$ with target X, such that $\mathrm{card}\, I \leq \Pi$ and $X_i \in \mathrm{Ob}\, C$ for every $i \in I$.

a) Show that $(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_0}$ is a cardinal filtration of the $\mathcal{U}$-category E, and that one may choose Π_0 so that, for every $\Pi \geq \Pi_0$, one has inclusions

$$E_{\Pi} \subset \mathrm{Filt}^{\Pi}(E) \subset E_{\Pi^{\Pi_0}},$$

where for every cardinal c, E_c denotes the strictly full subcategory of E formed by the c-accessible objects.

b) Choose a cardinal $\Pi \geq \Pi_0$ such that $\Pi = \Pi^{\Pi_0}$, for example a cardinal of the form 2^c with $c > \Pi_0$, and put $C = \mathrm{Filt}^{\Pi}(E)$. Show that one has an equivalence of categories

$$\mathrm{Ind}(C)_{\Pi} \xrightarrow{\sim} E$$

(with the notation $\mathrm{Ind}(C)_{\Pi}$ of Corollary 9.18).

c) Keep the notation of b), let $\mathcal{V}$ be a universe such that $\mathcal{U} \subset \mathcal{V}$, and let $\widetilde{E}_{\mathcal{V}}$ be the $\mathcal{V}$-topos of $\mathcal{V}$-sheaves on the site E endowed with its canonical topology. Denote by $\mathrm{Ind}(C, \mathcal{V})_{\Pi}$ the category of $\mathcal{V}$-Ind-objects of C indexed by preordered sets of indices which are *large with respect to* Π. Show that one has an equivalence of categories

$$\mathrm{Ind}(C, \mathcal{V})_{\Pi} \xrightarrow{\sim} \widetilde{E}_{\mathcal{V}}.$$

d) Let $c \in \mathcal{V}$ be a cardinal, and let $\mathrm{Ind}(E, \mathcal{V})'_c$ be the full subcategory of $\mathrm{Ind}(E, \mathcal{V})$ formed by $\mathcal{V}$-ind-objects of E indexed by a preordered set which is large with respect to every cardinal $< c$. Taking $c = \mathrm{card}\,\mathcal{U}$, show that one has an equivalence of categories

$$\mathrm{Ind}(E, \mathcal{V})'_c \xrightarrow{\sim} \widetilde{E}_{\mathcal{V}}.$$

9.21

This section develops the technical preliminaries for the proof of Theorem 9.22 below, which is the main result of Section 9. Let

$$p : E \longrightarrow B \tag{9.21.1}$$

be a fibred functor, where B is a small category, and where the inverse image functors

$$f^* : E_{\beta} \longrightarrow E_{\alpha}$$

associated with arrows $f : \alpha \to \beta$ of B are accessible (9.2). In particular, the fibre categories E_{α} ($\alpha \in \mathrm{ob}\, B$) satisfy condition L (9.1). Since B is small, there is a cardinal $\Pi'_0 \in \mathcal{U}$ such that all the categories E_{α} satisfy $L_{\Pi'_0}$. Let $\Pi_0 \in \mathcal{U}$ be an infinite cardinal $> \Pi'_0$, so that

$$\Pi_0 > \Pi'_0, \qquad E_{\alpha} \text{ satisfies } L_{\Pi'_0} \text{ for all } \alpha \in \mathrm{ob}\, B. \tag{9.21.2}$$

Moreover, the hypotheses just made immediately imply the existence of a cardinal $c \in \mathcal{U}$ satisfying:

$$\begin{cases} \text{a)} & c \text{ is infinite,} \\ \text{b)} & c \geq \mathrm{card}\, \mathrm{Fl}\, B, \\ \text{c)} & \text{for every arrow } f : \alpha \to \beta \text{ of } B,\ f^* : E_{\beta} \to E_{\alpha} \text{ is } c\text{-accessible.} \end{cases} \tag{9.21.3}$$

Suppose also that for each $\alpha \in \operatorname{ob} B$ one can find a full subcategory C_α of E_α satisfying:

$$\begin{cases} \text{a)} & \text{For every } X \in \operatorname{ob} C_\alpha,\ X \text{ is } c\text{-accessible in } E_\alpha. \\ \text{b)} & \text{For every } X \in \operatorname{ob} E_\alpha,\ \text{there is an isomorphism } X \simeq \varinjlim_I X_i \\ & \text{in } E_\alpha,\ \text{where the } X_i \text{ lie in } C_\alpha \text{ and } I \text{ is large with respect to } c. \end{cases} \tag{9.21.4}$$

Let Π be a cardinal such that

$$\Pi \geq \Pi_0, \qquad \text{and hence} \qquad \Pi > \Pi_0', \tag{9.21.5}$$

and for every $\alpha \in \operatorname{ob} B$ let

$$C_\alpha^\Pi \subset E_\alpha \tag{9.21.6}$$

be the full subcategory formed by objects which can be represented in the form $\varinjlim_I X_i$, where I is an ordered set large with respect to Π_0', $\operatorname{card} I \leq \Pi$, and the X_i lie in C_α. It is then clear, by 7.5.2, that C_α^Π is essentially small, i.e.equivalent to a small category.

Let

$$F = \operatorname{Hom}_B(B, E) \tag{9.21.7}$$

be the category of sections of E over B, and let

$$F^\Pi \subset F \tag{9.21.8}$$

be the strictly full subcategory of F formed by sections $X : \alpha \mapsto X(\alpha)$ such that, for every $\alpha \in \operatorname{ob} B$,

$$X(\alpha) \in C_\alpha^\Pi.$$

Since the C_α^Π are essentially small, it is clear that the same is true of F^Π. We shall show that this category is generating, and more precisely:

Lemma 9.21.9. Under the preceding hypotheses and with the preceding notation, the following hold.

(i) Every object of F is accessible (9.3). If d is a cardinal $\geq \Pi_0'$, and if $\alpha \mapsto X(\alpha)$ is an element of F such that, for every α, $X(\alpha)$ is d-accessible in E_α, then X is d-accessible in F.

(ii) Suppose that $\Pi \leq c$, or that Π_0' is finite (i.e.that the E_α are stable under small filtered inductive limits). Then every object X of F is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i lie in F^Π and I is large with respect to Π.

To prove (i), note first that by (9.21.4) and Corollary 9.9, every object of E_α is accessible. On the other hand, F satisfies condition $L_{\Pi_0'}$, by the following lemma.

Lemma 9.21.10. Let $p : E \to B$ be a fibred functor, let $F = \operatorname{Hom}_B(B, E)$, let I be a category, and let $i \mapsto X_i$ be a functor from I to F. In order that $\varinjlim_I X_i$ be representable in F, it is enough that, for every $\alpha \in \operatorname{ob} B$, $\varinjlim_I X_i(\alpha)$ be representable in the fibre category E_α. When this is the case, $\varinjlim_I X_i$ is computed "argument by argument". In particular, if the fibre categories satisfy condition $L_{\Pi_0'}$, where Π_0' is a fixed cardinal, then so does F.

We leave the easy proof of Lemma 9.21.10 to the reader, merely noting that it is convenient to use the following immediate result.

Corollary 9.21.10.1. With the hypotheses and notation of Lemma 9.21.10 for $p : E \to B$ and for I, for every $\alpha \in \operatorname{ob} B$, the inclusion functor $E_\alpha \to E$ commutes with inductive limits of type I.

Returning to the general hypotheses of Lemma 9.21.9, let X be an object of F. Since for every $\alpha \in \operatorname{ob} B$, $X(\alpha)$ is accessible in E_α, there exists a cardinal $d \in \mathcal{U}$ such that for every α, $X(\alpha)$ is d-accessible in E_α. We may choose $d \geq \Pi'_0$, so that F satisfies L_d by Lemma 9.21.10. Using Lemma 9.21.10 again to compute inductive limits $\varinjlim_I Y_i$ in F, with I large with respect to d, we immediately see that X is d-accessible. This proves (i).

We now prove Lemma 9.21.9(ii) in several steps, namely 9.21.11 through 9.21.16.

Let

$$X : \alpha \longmapsto X(\alpha)$$

be an object of F. By (9.21.4 b)), for every $\alpha \in \operatorname{ob} B$ one can find an ordered set I_α large with respect to c, and an isomorphism

$$X(\alpha) = \varinjlim_{i \in I_\alpha} X(\alpha)_i,$$

where $i \mapsto X(\alpha)_i$ is an inductive system of type I_α in E_α, with $X(\alpha)_i$ lying in C_α. Consider the product ordered set

$$I = \prod_\alpha I_\alpha.$$

It is clear that it is large with respect to c, and that the preceding inductive systems give rise to inductive systems in the categories E,

$$i \longmapsto X(\alpha)_i \in \operatorname{ob} C_\alpha, \qquad i \in \operatorname{ob} I, \tag{9.21.11}$$

indexed by the same ordered set I, and to isomorphisms

$$X(\alpha) \xrightarrow{\sim} \varinjlim_I X(\alpha)_i \quad \text{in } E_\alpha, \tag{9.21.12}$$

for all $\alpha \in \operatorname{ob} B$. In what follows we fix data (9.21.11) and (9.21.12).

Lemma 9.21.13. Under the preceding hypotheses, one can find a map $\varphi : I \to I$ such that $\varphi(i) \geq i$ for every $i \in I$, and a map

$$(i, f) \longmapsto \lambda(i, f)$$

from $I \times \operatorname{Fl} B$ to $\operatorname{Fl} E$, satisfying the following conditions.

a) For $i \in I$ and $f : \alpha \to \beta$ in $\operatorname{Fl} B$, $\lambda(i, f)$ is an f-morphism from $X(\alpha)_i$ to $X(\beta)_{\varphi(i)}$, making commutative the diagram

$$\begin{tikzcd}
X(\alpha) \arrow[r, "X(f)"] & X(\beta) \\
 & X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uu] \arrow[ur, "\lambda(i,f)"'] &
\end{tikzcd}$$

$$\alpha \xrightarrow[f]{} \beta.$$

Here the vertical arrows are the canonical morphisms deduced from (9.21.12).

b) For every $i \in I$ and every $f : \alpha \to \beta$ in $\mathrm{Fl}\, B$, one has commutativity in

$$\begin{tikzcd}
 & X(\beta)_{\varphi^2(i)} \\
X(\alpha)_{\varphi(i)} \arrow[ur, "\lambda(\varphi(i),f)"] & X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[u] \arrow[ur, "\lambda(i,f)"'] & \\
\alpha \arrow[r, "f"'] & \beta.
\end{tikzcd}$$

c) For every pair of consecutive arrows

$$\alpha \xrightarrow{f} \beta \xrightarrow{g} \gamma$$

of B, one has commutativity in the following diagram:

$$\begin{tikzcd}
X(\alpha) \arrow[r, "X(f)"] & X(\beta) \arrow[r, "X(g)"] & X(\gamma) \\
 & & X(\gamma)_{\varphi^2(i)} \arrow[u] \\
 & X(\beta)_{\varphi(i)} \arrow[uu] \arrow[ur, "\lambda(\varphi(i),g)"'] & X(\gamma)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uuu] \arrow[ur, "\lambda(i,f)"] \arrow[urr, "\lambda(i,gf)"'] & & \\
\alpha \arrow[r, "f"'] & \beta \arrow[r, "g"'] & \gamma.
\end{tikzcd}$$

Again, the vertical arrows are deduced from (9.21.12).

Notice also that the commutativity of the two upper trapezia in b) is already contained in a); thus b) is really asserting the commutativity of the lower triangle of the diagram considered.

We prove Lemma 9.21.13. Let $i \in I$, and let $f : \alpha \to \beta$ be an arrow of B. Consider the composite

$$X(\alpha)_i \longrightarrow X(\alpha) \xrightarrow{X(f)} X(\beta) \simeq \varinjlim_j X(\beta)_j .$$

It defines a morphism in E_α

$$X(\alpha)_i \longrightarrow f^*(X(\beta)) \simeq f^*\left(\varinjlim_j X(\beta)_j\right) \simeq \varinjlim_j f^*(X(\beta)_j),$$

where the last equality follows from the fact that f^* is c-accessible (9.21.3 c)) and that I is large with respect to c. By (9.21.4 a)), since $X(\alpha)_i$ lies in C_α, the morphism under consideration factors through one of the $f^*(X(\beta)_j)$; here j *a priori* depends on i and on $f \in \mathrm{Fl}\, B$. But by (9.21.3 b)), and since I is large with respect to c, we may choose j independently of f; write

$j = \varphi(i)$. Since I is filtered, we may suppose $\varphi(i) \geq i$. We thus obtain, for $i \in I$ and $f \in \mathrm{Fl}\, B$, a morphism

$$X(\alpha)_i \longrightarrow f^*(X(\beta)_{\varphi(i)}),$$

or equivalently an f-morphism

$$\lambda(i, f) : X(\alpha)_i \longrightarrow X(\beta)_{\varphi(i)}$$

which, by construction, makes the diagram in a) commute.

Next fix i, f, g and consider the lower triangle of the diagram in 9.21.13 c). It is not clear that this triangle commutes, but the two composites

$$X(\alpha)_i \rightrightarrows X(\gamma)_{\varphi^2(i)}$$

become equal after composition with $X(\gamma)_{\varphi^2(i)} \to X(\gamma)$, as follows from the commutativity of the two upper trapezia and of the outer trapezium; these are respectively the diagrams $D(f)$, $D(g)$, and $D(gf)$. Since $X(\alpha)_i$ is c-accessible (9.21.4 a)) and

$$X(\gamma) \simeq \varinjlim_{j \in I} X(\gamma)_j$$

with I large with respect to c, one can find an element $j \geq \varphi^2(i)$ of I such that the two arrows just considered become equal after composition with

$$X(\gamma)_{\varphi^2(i)} \longrightarrow X(\gamma)_j.$$

A priori j depends on i, f, g. But for fixed i, the set of possible pairs f, g has cardinal $\leq c^2 = c$, by (9.21.3 a) and b)); since I is large with respect to c, we may choose j independently of f and g. Write $j = \varphi'(i) \geq \varphi^2(i) \geq \varphi(i)$. Define, for every $i \in I$ and every $f : \alpha \to \beta$, $\lambda'(i, f)$ to be the composite f-morphism

$$X(\alpha)_i \longrightarrow X(\beta)_{\varphi(i)} \longrightarrow X(\beta)_{\varphi'(i)},$$

where the second arrow is the transition morphism. It is then immediate from the construction that (φ', λ') satisfies conditions 9.21.13 a) and c). Proceeding in the same way for condition b), one sees that φ' can be chosen so that this condition is also satisfied. This completes the proof of Lemma 9.21.13.

9.21.14

Let

$$J \subset I$$

be a subset of I satisfying the following conditions:

a) J is filtered;

b) for every $j \in J$, one has $\varphi(j) \in J$;

c) for every $\alpha \in \mathrm{ob}\, B$, the object
$$X_J(\alpha) = \varinjlim_{i \in J} X(\alpha)_i$$
is representable in E_α.

Conditions a) and c) are satisfied in particular if J is large with respect to Π_0' (9.21.2). It then follows from Corollary 9.21.10.1 that, for every $\alpha \in \operatorname{ob} B$, $X_J(\alpha)$ is the inductive limit $\varinjlim_{i\in J} X(\alpha)_i$ in E. Now for an arrow $f : \alpha \to \beta$ of B, the arrows $\lambda(i, f)$, with i varying in J, define by 9.21.13 b) a morphism of ind-objects from $(X(\alpha)_i)_{i\in J}$ to $(X(\beta)_i)_{i\in J}$. Hence they induce a homomorphism

$$X_J(f) : X_J(\alpha) \longrightarrow X_J(\beta)$$

on inductive limits, which is visibly an f-homomorphism. Using 9.21.13 c), one finds the transitivity relations

$$X_J(gf) = X_J(g)X_J(f),$$

so that a section $X_J \in \operatorname{ob} F$ of E over B has been defined. Finally, 9.21.13 a) shows that the canonical homomorphisms

$$X_J(\alpha) \longrightarrow X(\alpha), \qquad \alpha \in \operatorname{ob} B,$$

are functorial in α; thus we have a canonical homomorphism

$$u_J : X_J \longrightarrow X.$$

Moreover, if

$$J' \supset J$$

is another subset of I satisfying conditions a), b), c) above, one obtains a canonical homomorphism

$$u_{J',J} : X_J \longrightarrow X_{J'},$$

so that the X_J form an inductive system in F, parametrized by the set, ordered by inclusion, of subsets J of I satisfying the conditions under consideration. Finally, the homomorphisms u_J above define a homomorphism from this inductive system to the constant inductive system defined by X.

9.21.15

Let K be a set of subsets J of I satisfying conditions a), b), c) of 9.21.14, and suppose that K is filtered and has union I. Then it is clear that

$$\varinjlim_{J\in K} X_J \xrightarrow{\sim} X,$$

using Corollary 9.21.10.1 to reduce the verification to an isomorphism argument by argument.

For example, take K to be the set of all subsets J of I which are large with respect to Π_0', stable under φ, and such that $\operatorname{card} J \leq \Pi$. Then, by the definition (9.21.6) of C_α^Π, one has

$$X_J(\alpha) \in \operatorname{ob} C_\alpha^\Pi$$

for every $\alpha \in \operatorname{ob} B$, and hence $X_J \in \operatorname{ob} F^\Pi$. Therefore Lemma 9.21.9(ii) will be proved if we show that K is large with respect to Π (hence filtered) and has union I. For this it plainly suffices to prove that every subset S of I with $\operatorname{card} S \leq \Pi$ is contained in some $J \in K$. This follows from the preliminary hypothesis stated in Lemma 9.21.9(ii), and from the following lemma.

Lemma 9.21.16. Let I be an ordered set, let Π be an infinite cardinal, and let $\varphi : I \to I$ be a map.

a) Suppose that I is filtered. Then for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a filtered subset $J \supset S$ of I such that $\varphi(J) \subset J$ and $\operatorname{card} J \leq \Pi$.

b) Let Π'_0 be a cardinal $< \Pi$, and suppose that I is large with respect to Π. Then for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a subset $J \supset S$ of I which is large with respect to Π'_0, such that $\varphi(J) \subset J$ and $\operatorname{card} J \leq \Pi$.

We prove b); the proof of a) is analogous and simpler. Let P be the set of subsets of I of cardinal $\leq \Pi$, and let $T \mapsto i_T : P \to I$ be a map such that, for $T \in P$, i_T is an upper bound for T in I. The existence of this map is simply the assertion that I is large with respect to Π. Let A be a well-ordered set such that $\operatorname{card} A = \Pi$ and such that, for every $a \in A$, the set of $a' \leq a$ has cardinal $< \Pi$. In particular, since $\Pi'_0 < \Pi$, it follows that A is large with respect to Π'_0.

Define, by transfinite recursion, maps

$$S \longmapsto S_a : P \to P \qquad (a \in A)$$

as follows:

$$S_0 = S, \qquad S_{a+1} = S_a \cup \varphi(S_a) \cup \{i_{S_a}\}, \qquad S_a = \bigcup_{a' < a} S_{a'} \quad \text{if } a \text{ is a limit ordinal.}$$

Then put, for $S \in P$,

$$J(S) = \bigcup_{a \in A} S_a.$$

It is clear that $J(S) \supset S$, that $J(S)$ is stable under φ, that $\operatorname{card} J(S) \leq \Pi$ (since $\operatorname{card} A = \Pi$), and finally that $J(S)$ is large with respect to Π'_0 (using that A is large with respect to Π'_0). This proves Lemma 9.21.16 and completes the proof of Lemma 9.21.9.

We also note the following variant of Lemma 9.21.9.

Lemma 9.21.17. The notation is that of Lemma 9.21.9. Denote by F' the full subcategory

$$F' = \operatorname{Hom}_B^{\text{cart}}(B, E)$$

of $F = \operatorname{Hom}_B(B, E)$ formed by the cartesian sections of E over B. Then:

(i) Every object of F' is accessible. More precisely, let $\alpha \mapsto X(\alpha)$ be an element of F', and let d be a cardinal such that $d \geq \Pi'_0$, $d \geq c$, and such that for every $\alpha \in \operatorname{ob} B$, $X(\alpha)$ is a d-accessible object of E_α. Then X is a d-accessible object of F'.

(ii) Suppose that $\Pi \leq c$ and that the functors $f^* : E_\beta \to E_\alpha$ are Π'_0-accessible, or else that the categories E_α are stable under small filtered inductive limits and that the functors $f^* : E_\beta \to E_\alpha$ commute with these inductive limits. Then every object X of F' is isomorphic to an object of the form $\varinjlim_I X_i$, where for every $i \in I$, X_i is an object of the full subcategory

$$F'^{\Pi} = F' \cap F^{\Pi}$$

of F', and where I is large with respect to Π.

The proof being entirely analogous to that of Lemma 9.21.9, we merely indicate where modifications of that proof are necessary. First note the following.

Lemma 9.21.18. The statement of Lemma 9.21.10 remains valid when one replaces $F = \mathrm{Hom}_B(B, E)$ by $F' = \mathrm{Hom}_B^{\mathrm{cart}}(B, E)$, provided that one assumes that the inverse image functors

$$f^* : E_\beta \longrightarrow E_\alpha$$

commute with inductive limits of type I.

Thus, under the hypotheses of Lemma 9.21.17, if d is a cardinal such that $d \geq c$ and $d \geq \Pi_0'$, it follows from what precedes and from hypothesis (9.21.3 c)) that for every ordered set I large with respect to d, inductive limits of type I are representable in F' and are computed argument by argument. Lemma 9.21.17(i) follows by an immediate argument.

To prove (ii), one must supplement Lemma 9.21.13.

Lemma 9.21.19. Under the hypotheses of Lemma 9.21.13, suppose that $X \in \mathrm{ob}\, F'$, i.e.that X is a cartesian section of E over B. Then the conclusion can be strengthened as follows: there exists a function μ which assigns, to every $i \in I$ and every arrow $f : \alpha \to \beta$ of B, a morphism

$$\mu(i, f) : f^* X(\beta)_i \longrightarrow X(\alpha)_{\varphi(i)}$$

in E_α, in such a way that:

d) For every $i \in I$ and every arrow $f : \alpha \to \beta$ of B, the two following diagrams commute:

$$\begin{tikzcd}
& f^*X(\beta)_{\varphi^2(i)} \\
X(\alpha)_{\varphi(i)} \arrow[ur, "\lambda(\varphi(i),f)"] & \\
& f^*X(\beta)_i \arrow[ul, "\mu(i,f)"'] \arrow[uu]
\end{tikzcd}
\qquad
\begin{tikzcd}
X(\alpha)_{\varphi^2(i)} & \\
& f^*X(\beta)_{\varphi(i)} \arrow[ul, "\mu(\varphi(i),f)"'] \\
X(\alpha)_i \arrow[uu] \arrow[ur, "\lambda(i,f)"'] &
\end{tikzcd}$$

where the unlabeled arrows are the evident transition maps.

To prove this, one proceeds as in the proof of 9.21.13, by considering the composite morphism

$$f^*(X(\beta)_i) \longrightarrow f^*(X(\beta)) \xrightarrow{X(f)^{-1}} X(\alpha),$$

where, as already in the notation for condition d), an f-morphism $R \to S$ of E is identified with the corresponding morphism $R \to f^*(S)$ in E_α. Since

$$X(\alpha) = \varinjlim_j X(\alpha)_j$$

and I is large with respect to c, the preceding morphism can be factored through one of the $X(\alpha)_j$. Here j *a priori* depends on i and on f, but can be chosen independently of f; write it $\psi(i)$. Enlarging the function φ of Lemma 9.21.13 if necessary, we may suppose that $\psi = \varphi$. Proceeding as for conditions b) and c) of Lemma 9.21.13, we see that, after enlarging φ again if necessary, the two diagrams of Lemma 9.21.19 d) may be assumed commutative.

9.21.20

With φ chosen as in Lemma 9.21.19, resume the argument of 9.21.14. One must, however, assume that J satisfies, besides conditions a), b), c) stated there, the following condition:

d) For every arrow $f : \alpha \to \beta$ in B, the functor $f^* : E_\beta \to E_\alpha$ commutes with inductive limits of type J.

I claim that, under this additional condition, the section X_J of E over B is cartesian; that is, for every arrow $f : \alpha \to \beta$ of B, the morphism

$$X_J(\alpha) \longrightarrow f^* X_J(\beta) \tag{$*$}$$

is an isomorphism. Indeed, by condition d) above, the target of this morphism is identified with

$$\varinjlim_{i \in J} f^*(X(\beta)_i),$$

and the morphism itself is obtained by passing to $\varinjlim_J$ from the morphism of ind-objects in E_α

$$(X(\alpha)_i)_{i \in J} \longrightarrow (f^* X(\beta)_i)_{i \in J},$$

deduced from the morphisms

$$\lambda(i, f) : X(\alpha)_i \longrightarrow f^*(X(\beta)_{\varphi(i)})$$

for $i \in J$. But the conditions made explicit in Lemma 9.21.19 ensure that this morphism of ind-objects is in fact an isomorphism, with inverse obtained from the morphism deduced from the system of the $\mu(i, f)$, $i \in J$. This proves that $(*)$ is an isomorphism, hence that $X_J \in \operatorname{ob} F'$.

9.21.21

We now assume that the functors $f^* : E_\beta \to E_\alpha$ are Π_0'-*accessible*. Then the conditions on J considered in 9.21.20 are satisfied if J is large with respect to Π_0', stable under φ, and has $\operatorname{card} J \leq \Pi$. The proof of Lemma 9.21.17 is then completed as in 9.21.15.

Theorem 9.22. *Let $p : E \to B$ be a fibred functor, where B is a category essentially equivalent to a small category, and where for every arrow $f : \alpha \to \beta$ of B, the inverse image functor*

$$f^* : E_\beta \longrightarrow E_\alpha$$

is accessible (9.2). Suppose further that, for every $\alpha \in \operatorname{ob} B$, the fibre category E_α admits a cardinal filtration (9.12), and that every object of E_α is accessible (9.3). Then each of the categories

$$F = \operatorname{Hom}_B(B, E), \qquad F' = \operatorname{Hom}_B^{\text{cart}}(B, E)$$

admits a small full subcategory which is generating by strict epimorphisms (7.1), and all of its objects are accessible.

It is immediate that the statement is not essentially changed when B is replaced by a full subcategory B_0 such that the inclusion functor $B_0 \to B$ is an equivalence, and E is replaced by $E_0 = E \times_B B_0$. Thus we may suppose that B is a small category.

For every $\alpha \in \operatorname{ob} B$, let

$$(\operatorname{Filt}^{\Pi}(E_\alpha))_{\Pi \geq \Pi_\alpha}$$

be a cardinal filtration of E_α. Let $c_0 \in \mathcal{U}$ be a cardinal satisfying:

$$\begin{cases} c_0 \geq \operatorname{Sup}_{\alpha \in \operatorname{ob} B} \Pi_\alpha, \qquad c_0 \geq \operatorname{card} \operatorname{Fl} B; \\ \text{for every arrow } f : \alpha \to \beta \text{ of } B,\ f^* : E_\beta \to E_\alpha \text{ is } c_0\text{-accessible;} \\ \text{for every } \alpha \in \operatorname{ob} B,\ \text{the objects of } \operatorname{Filt}^{\Pi_\alpha}(E_\alpha) \text{ are } c_0\text{-accessible.} \end{cases} \tag{9.22.1}$$

Put

$$c = 2^{c_0},$$

so that $c > c_0$, and for every $\alpha \in \operatorname{ob} B$ put

$$C_\alpha = \operatorname{Filt}^c(E_\alpha).$$

I claim that the preliminary hypotheses of Lemmas 9.21.9 and 9.21.17 are satisfied, taking $\Pi = \Pi_0 = c$. This is clear for (9.21.2) and (9.21.3). Condition (9.21.4 a)) follows from Corollary 9.17, and (9.21.4 b)) follows from condition 9.12 c) for a cardinal filtration. Notice also that condition 9.12 b) in the definition of a cardinal filtration implies that, for every $\alpha \in \operatorname{ob} B$, one has

$$C_\alpha^\Pi = C_\alpha = \operatorname{Filt}^c(E_\alpha),$$

where C_α^Π is defined in (9.21.6). Theorem 9.22 therefore follows from Lemmas 9.21.9 and 9.21.17. More precisely, we have proved:

Corollary 9.23. Under the hypotheses of Theorem 9.22, if $c_0 \in \mathcal{U}$ is a cardinal satisfying (9.22.1), and if $c = 2^{c_0}$, then the subcategory F^c of F (respectively F'^c of F') formed by objects X such that $X(\alpha) \in \operatorname{ob} \operatorname{Filt}^c(E_\alpha)$ for every $\alpha \in \operatorname{ob} B$ is essentially small and generating by strict epimorphisms. More precisely, every object X of F (respectively of F') is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i lie in F^c (respectively in F'^c), and where I is an ordered set large with respect to c.

Under slightly stronger hypotheses in Theorem 9.22, one can also make Theorem 9.22 and Corollary 9.23 considerably more precise.

Corollary 9.24. Under the hypotheses of Theorem 9.22, suppose that each of the categories E_α ($\alpha \in \operatorname{ob} B$) is stable under small filtered inductive limits, and that for every $\Pi \geq \Pi_\alpha$, $\operatorname{Filt}^\Pi(E_\alpha)$ is stable under filtered inductive limits indexed by filtered ordered sets I such that $\operatorname{card} I \leq \Pi$; this is a slight strengthening of condition 9.12 b) for cardinal filtrations. In the case where F' is considered, suppose moreover that, for every arrow $f : \alpha \to \beta$ of B, the functor

$$f^* : E_\beta \to E_\alpha$$

commutes with small filtered inductive limits. Finally let $c_0 \in \mathcal{U}$ be a cardinal satisfying (9.22.1), and for every cardinal $\Pi \geq c = 2^{c_0}$ consider the strictly full subcategory F^Π (respectively F'^Π) of F (respectively of F') formed by the X such that

$$X(\alpha) \in \operatorname{Filt}^\Pi(E_\alpha)$$

for every $\alpha \in \operatorname{ob} B$. Then the subcategories under consideration define a cardinal filtration (9.12) of F (respectively of F').

One must verify conditions a), b), c) of 9.12. Conditions a) and b) follow from the analogous conditions for the given cardinal filtrations of the E_α, and from Lemma 9.21.10 (respectively Lemma 9.21.18, taking into account the second condition in (9.22.1)). It remains to prove c). The first part of c) follows at once from Lemma 9.21.9 (respectively Lemma 9.21.17), by taking $\Pi'_0 = 0$ and $\Pi_0 = c$.

It remains to prove that if $\Pi' \geq \Pi \geq c$ are two cardinals, then for every X in $F^{\Pi'}$ (respectively $F'^{\Pi'}$) one has

$$X \simeq \varinjlim_{i \in K} X_i,$$

where the X_i lie in F^{Π} (respectively in F'^{Π}), where K is large with respect to Π, and where $\operatorname{card} K \leq \Pi'^{\Pi}$. To this end, resume the proofs of Lemma 9.21.9(ii) and Lemma 9.21.17(ii). We may suppose that each I_α has cardinal $\leq \Pi'$ by condition 9.12 c) on the cardinal filtration of E_α. Hence

$$\operatorname{card} I \leq (\Pi'^{\Pi})^{\operatorname{card} \operatorname{ob} B} \leq (\Pi'^{\Pi})^c = \Pi'^{\Pi c} = \Pi'^{\Pi}.$$

The indexing set K used in 9.21.15, respectively 9.21.21, is the set of subsets J of I which are filtered (i.e.large with respect to $\Pi'_0 = 0$), stable under φ, and such that $\operatorname{card} J \leq \Pi$. It is then immediate that

$$\operatorname{card} K \leq (\operatorname{card} I)^{\Pi} \leq (\Pi'^{\Pi})^{\Pi} = \Pi'^{\Pi^2} = \Pi'^{\Pi},$$

which completes the proof of Corollary 9.24.

To finish, it is appropriate to give a statement freed from the somewhat technical hypotheses of Theorem 9.22, by replacing them, for the E_α, by the conjunction of the hypotheses occurring in Proposition 9.11, which ensure accessibility of the objects of the E_α, and in Proposition 9.13, which ensure the existence of a cardinal filtration in E_α.

Corollary 9.25. Let $p : E \to B$ be a fibred functor, where B is a category equivalent to a small category, and where for every arrow $f : \alpha \to \beta$ of B, the inverse image functor

$$f^* : E_\beta \to E_\alpha$$

is accessible (9.2). Suppose further that, for every $\alpha \in \operatorname{ob} B$, the fibre category E_α satisfies the following conditions:

a) E_α admits a small full subcategory generating by strict epimorphisms (7.1).

b) E_α is stable under fibre products, under kernels of double arrows, under sums of two objects, under cokernels of double arrows, and finally under small filtered inductive limits.[1]

c) The functor $\operatorname{Ker}(u, v)$ on the category of double arrows of E_α is accessible; for example, it commutes with small filtered inductive limits.

d) Every strict epimorphism of E_α (10.2) is a universal strict epimorphism.

Under these hypotheses, each of the categories

$$F = \operatorname{Hom}_B(B, E), \qquad F' = \operatorname{Hom}_B^{\mathrm{cart}}(B, E)$$

admits a small subcategory generating by strict epimorphisms, and all of its objects are accessible (9.3).

[1]This last condition is probably unnecessary; cf.9.13.2.

Moreover, Corollary 9.24 gives:

Corollary 9.26. Under the hypotheses of 9.25, and in the case where one considers $F' = \mathrm{Hom}^{\mathrm{cart}}_B(B, E)$, suppose that the functors

$$f^* : E_\beta \to E_\alpha$$

commute with small filtered inductive limits. Then F (respectively F') admits a cardinal filtration, which can be made explicit by the procedure of Corollary 9.24.

Continuation anchor. The next part of Exposé I is Section 10, the glossary of terms used in the preceding sections.

Exposé I. Presheaves

Section 10 and Appendix II: Glossary; Universes

A. Grothendieck and J.-L. Verdier; Appendix attributed to N. Bourbaki

Translator's status note. This is a working English translation of SGA 4, Exposé I, Section 10 and the appended note "Universes." The range corresponds to source lines 7424–8921 of the Orgogozo–Laszlo TeX snapshot. The translation keeps the period terminology where it is still mathematically useful: *strict epimorphism*, *effective equivalence relation*, *direct factor*, and *base-changeable* for French *quarrable*. A few evident typographical slips in the source have been silently normalized where the formulas force the correction.

10 Glossary

For the convenience of the reader, we collect here the definitions of several terms used in the preceding sections. Throughout this section, C denotes a category.

10.1 Cartesian, cocartesian

A diagram in C

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ C \arrow[r] & D \end{tikzcd}$$

is called *cartesian* if it is commutative and if the canonical morphism from A to the fiber product $C \times_D B$ is an isomorphism. It is called *cocartesian* if the corresponding diagram in the opposite category C° is cartesian.

10.2 Epimorphism, strict epimorphism

See 10.3.

10.3 Epimorphic family, strictly epimorphic family

A family

$$f_i : A_i \longrightarrow B, \qquad i \in I,$$

of arrows with common target in C is called an *epimorphic family* if, for every object T of C, the map induced by the f_i,

$$\mathrm{Hom}_C(B,T) \longrightarrow \prod_i \mathrm{Hom}_C(A_i,T), \tag{10.3.1}$$

is injective.

The family is called *strictly epimorphic* if the image of (10.3.1) consists of those families (g_i) such that, for every object R of C, every pair of indices $i, j \in I$, and every pair of arrows $u : R \to A_i$, $v : R \to A_j$ satisfying $f_i u = f_j v$, one also has $g_i u = g_j v$. When the fiber products

$A_i \times_B A_j$ are representable for $i, j \in I$, this is equivalent to saying that the image of (10.3.1) consists of those (g_i) such that, for all i, j,

$$g_i \,\mathrm{pr}_1 = g_j \,\mathrm{pr}_2,$$

where $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections from $A_i \times_B A_j$. The family $(f_i)_{i\in I}$ is called an *effective epimorphic family* if this condition is satisfied, that is, if it is strictly epimorphic and the fiber products $A_i \times_B A_j$ are representable.

The family $(f_i)_{i\in I}$ is called *universally epimorphic* (respectively, *universally effectively epimorphic*) if the morphisms f_i are base-changeable (10.7) and if, for every arrow $B' \to B$, the family of arrows obtained from (f_i) by base change along $B' \to B$ is epimorphic (respectively, effective epimorphic).

The notion of a *universally strictly epimorphic* family will be defined in II 2.5–2.6. Note that when the morphisms f_i are base-changeable, for example if fiber products are representable in C, this notion agrees with that of a universally effectively epimorphic family, using II 2.4. In general, between the six variants of epimorphicity considered here, one has the following logical implications:

universally effective epimorphic ⇒ effective epimorphic ⇒ strictly epimorphic ⇒ epimorphic,
universally effective epimorphic ⇒ universally strictly epimorphic ⇒ strictly epimorphic,
universally strictly epimorphic ⇒ universally epimorphic ⇒ epimorphic.

A morphism $f : A \to B$ of C is called an *epimorphism* (respectively, a strict epimorphism, effective epimorphism, universal epimorphism, universally effective epimorphism, universally strict epimorphism) if the one-element family consisting of f is epimorphic (respectively, ...).

10.4 Monomorphic family, strictly monomorphic family

A family of arrows $f_i : A_i \to B$ of C is called *monomorphic* (respectively, strictly monomorphic, ...) if, as a family of arrows in the opposite category C°, it is epimorphic (respectively, strictly epimorphic, ...). An arrow $f : A \to B$ of C is called a *monomorphism* (respectively, a strict monomorphism, ...) if the one-element family consisting of f is monomorphic (respectively, strictly monomorphic, ...), that is, if f, viewed as an arrow of C°, is an epimorphism (respectively, a strict epimorphism, ...).

10.5 Monomorphism, strict monomorphism

See 10.4.

10.6 Projector, image of a projector, direct factor of an object

Let $f : X \to X$ be an endomorphism of an object of C. We say that f is a *projector* if $f^2 = f$. Then the pair (f, id_X) admits a representable cokernel X' if and only if it admits a representable kernel X''; and when this is so, there is a unique isomorphism $u : X' \to X''$ in C such that $iup = f$, where $p : X \to X'$ and $i : X'' \to X$ are the canonical morphisms. The representability of the kernel, or of the cokernel, is also equivalent to the existence of morphisms i, p and of an isomorphism u such that $iup = f$ and $pi = u^{-1}$. In particular, this property is preserved by any functor.

One then usually identifies X' and X'', and calls it the *image of the projector* f. We say that the projector f *admits an image* if $\mathrm{Ker}(f, \mathrm{id}_X)$ is representable, equivalently if $\mathrm{Coker}(f, \mathrm{id}_X)$ is representable. A subobject (respectively, a quotient) Y of X is called a *direct factor* of X if one can find a projector f in X which factors through Y, and which admits Y as image as a subobject (respectively, as a quotient) of X. By abuse of language, one sometimes says that an object of C is a *direct factor of* X if it is isomorphic to the image of a projector in X.

10.7 Base-changeable

An arrow $f : A \to B$ of C is called *base-changeable* (French: *quarrable*) if, for every arrow $B' \to B$ of C, the fiber product $A \times_B B'$ is representable in C.

10.8 Quotient, strict quotient

Let A be an object of C. Two epimorphisms $f : A \to B$, $f' : A \to B'$ with source A are called *equivalent* if there exists an isomorphism $u : B \xrightarrow{\sim} B'$ such that $uf = f'$. This gives an equivalence relation on the set of epimorphisms with source A; its classes are called the *quotients*, or *quotient objects*, of A. For every quotient of A one usually chooses an element of this class, say $f : A \to B$, and one often speaks, by abuse of language, of the quotient B of A, or of the quotient $f : A \to B$ of A. We say that B is a *strict quotient* (respectively, an effective quotient, universal quotient, $\ldots$) if the morphism $f : A \to B$ is a strict epimorphism (respectively, an effective epimorphism, universal epimorphism, $\ldots$) in the sense of 10.3.

10.9 Equivalence relations

Let F and G be two presheaves of sets on C. A diagram $p_1, p_2 : F \rightrightarrows G$ is called an *equivalence relation on* G if, for every object A of C, the map

$$(p_1(A), p_2(A)) : F(A) \longrightarrow G(A) \times G(A)$$

induces a bijection from $F(A)$ onto the graph of an equivalence relation on the set $G(A)$.

A diagram $p_1, p_2 : B \rightrightarrows C$ in C is called an equivalence relation on C if the corresponding diagram of presheaves is an equivalence relation. When finite products and fiber products are representable in C, a functor $\Phi : C \to C'$ commuting with finite products and fiber products carries equivalence relations on C to equivalence relations on $\Phi(C)$.

Let $\Pi : C \to D$ be a morphism of C such that the fiber product $C \times_D C$ is representable. Then the diagram

$$\mathrm{pr}_1, \mathrm{pr}_2 : C \times_D C \rightrightarrows C$$

is an equivalence relation on C.

10.10 Effective equivalence relation, universally effective equivalence relation

An equivalence relation $p_1, p_2 : R \rightrightarrows A$ on an object A of C is called an *effective equivalence relation* if there exists a morphism $\pi : A \to B$ such that the square

$$\begin{tikzcd} R \arrow[r, "p_2"] \arrow[d, "p_1"'] & A \arrow[d, "\pi"] \\ A \arrow[r] & B \end{tikzcd}$$

is cartesian and cocartesian. The morphism π is the cokernel of the pair (p_1, p_2), and is therefore determined up to unique isomorphism. The morphism π is then an effective epimorphism; if it is a universally effective epimorphism in the sense of 10.3, one says that the *equivalence relation* is *universally effective.*

10.11 Subobject, strict subobject

These are the notions dual to those of quotient, strict quotient, and so on; see 10.8.

References

[1] B. Mitchell, *Theory of Categories*, Academic Press, 1965.

II. Appendix: Universes (by N. Bourbaki*)

*We reproduce here, with his permission, secret papers of N. Bourbaki. The references in this text are to his learned work.

1 Definition and first properties of universes

Definition 1. A set U is called a *universe* if it satisfies the following conditions:

(U.I) if $x \in U$ and $y \in x$, then $y \in U$;

(U.II) if $x, y \in U$, then $\{x, y\} \in U$;

(U.III) if $x \in U$, then $\mathcal{P}(x) \in U$;

(U.IV) if $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then the union $\bigcup_{\alpha \in I} x_\alpha$ belongs to U;

(U.?) if $x, y \in U$, then the pair $(x, y) \in U$.

N.B. Since, I believe, it has been decided for future editions to define the ordered pair à la Kuratowski by

$$(x, y) = \{\{x, y\}, \{x\}\},$$

condition (U.?) is useless, because it follows from (U.II).

Examples.

1) The empty set is a universe, denoted U_0.

2) Consider the nonempty finite words formed with the four symbols "{", "}", ",", and "$\emptyset$" (cf. Alg. I). Define, by induction on the length n of such a word, the notion of a *meaningful* word:

   (a) for $n = 1$, only the word $\emptyset$ is meaningful;

   (b) for a word A of length n to be meaningful, it is necessary and sufficient that there exist p distinct meaningful words $A_1, \ldots, A_p$, with $p \geq 1$ and of lengths $< n$, such that

$$A = \{A_1, A_2, \ldots, A_p\}.$$

For example, $\emptyset$, $\{\emptyset\}$, and

$$\{\{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$$

are meaningful words. It is clear, by induction on the length, that every meaningful word denotes a term of set theory; for example, if $A = \{A_1, A_2, \ldots, A_p\}$, then A denotes the set whose elements are $A_1, A_2, \ldots, A_p$. These terms are of course *finite sets*. Let U_1 be the set of the sets thus obtained. One easily checks that U_1 satisfies conditions (U.I), (U.II), (U.III), and (U.IV) of Definition 1, though not that idiotic (U.?), which is not serious if one is willing to de-cannulate the pair. Thus U_1 is a *universe*. Notice that the elements of U_1 are finite and that U_1 is countable.

In the statements that follow, U denotes a universe.

Proposition 1. If $x \in U$ and $y \subset x$, then $y \in U$.

Indeed $\mathcal{P}(x) \in U$ by (U.III), whence $y \in \mathcal{P}(x)$ and $y \in U$ by (U.I).

Corollary. If $x \in U$, every quotient set y of x is an element of U.

Indeed y is a subset of $\mathcal{P}(x)$. Hence $y \in U$ by (U.III) and Proposition 1.

Proposition 2. If $x \in U$, then $\{x\} \in U$.

This follows from (U.II), applied with $y = x$.

Proposition 3. Every pair, triple, and quadruple of elements of U is an element of U.

For pairs this is true by the preceding N.B. or by (U.?). The case of triples follows since $(x, y, z) = ((x, y), z)$, and the case of quadruples then follows since $(x, y, z, t) = ((x, y, z), t)$.

Proposition 4. If $X, Y \in U$, then $X \times Y \in U$.

Indeed, for $x \in X$, the set $\{x\} \times Y$ is the union of the family $\{(x, y)\}_{y \in Y}$; since $\{(x, y)\} \in U$ by (U.I), (U.II), and Proposition 3, we have $\{x\} \times Y \in U$ by (U.IV). Finally, $X \times Y$ is the union of the family $(\{x\} \times Y)_{x \in X}$, so it is again an element of U by (U.IV).

Corollary 1. If $X, Y, Z, \ldots$ are elements of U, then all sets in the scale built on $X, Y, Z, \ldots$ are elements of U (cf. Chapter IV, §).

This follows from successive applications of Proposition 4 and (U.III).

Corollary 2. If $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then the sum set $\sum_{\alpha \in I} x_\alpha$ is an element of U.

Indeed this sum set is a subset of the product $(\bigcup_{\alpha \in I} x_\alpha) \times I$, whose two factors are elements of U, by (U.IV) and the hypothesis. One then applies Propositions 4 and 1.

Proposition 5. If X and Y are elements of U, then every correspondence between X and Y, in particular every map from X to Y, is an element of U.

Indeed such a correspondence C is a triple (X, Y, Γ), where Γ is a subset of $X \times Y$, the graph of C (Chapter II, §). We have $\Gamma \in U$ by Propositions 4 and 1. Hence $C \in U$ by Proposition 3.

Proposition 6. If $X, Y \in U$, then every set Z of correspondences between X and Y, in particular every set of maps from X to Y, is an element of U.

Indeed Z is a subset of $\{X\} \times \{Y\} \times \mathcal{P}(X \times Y)$. This product is an element of U, by Proposition 2 and the corollary to Proposition 4. Therefore $Z \in U$ by Proposition 1.

Corollary. If $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then $\prod_{\alpha \in I} x_\alpha \in U$.

Indeed this product is a set of maps from I to $\bigcup_{\alpha \in I} x_\alpha$, and this union is an element of U by (U.IV).

Proposition 7. If X is a subset of U whose cardinal is at most that of an element of U, then X is an element of U.

Let I be an element of U such that $\operatorname{card}(X) \leq \operatorname{card}(I)$. There is a surjection $i \mapsto x_i$ from a subset I' of I onto X. Then X is the union of the family $(\{x_i\})_{i \in I'}$. This union is an element of U by Proposition 1, Proposition 2, and (U.IV).

Corollary. If U is nonempty, every finite subset of U is an element of U, and U has elements of arbitrarily large finite cardinality.

On the other hand, if $U = \emptyset$, then $\emptyset$ is a finite subset of $\emptyset$ but not an element of $\emptyset$.

Indeed, if U is nonempty, Proposition 2 shows that a one-element set x_0 belongs to U. By induction on n, put $x_{n+1} = \mathcal{P}(x_n)$. Then $x_n \in U$ by (U.III), and $\mathrm{card}(x_{n+1}) = 2^{\mathrm{card}(x_n)}$, so U has elements of arbitrarily large finite cardinality.

Remark. It follows from Proposition 2 and the corollary to Proposition 7 that every nonempty universe U contains the universe U_1 of Example 2; indeed $\emptyset \in U$ by Proposition 1. Thus U_1 is the intersection of all nonempty universes. More generally:

Proposition 8. If $(U_\lambda)_{\lambda \in L}$ is a nonempty family of universes, then $U = \bigcap_{\lambda \in L} U_\lambda$ is a universe.

This follows immediately from Definition 1.

2 Universes and species of structures

Let $\mathcal{E}$ be a species of structure. To fix ideas and lighten the exposition, suppose that every structure of species $\mathcal{E}$ is defined on one underlying set. Let (X, S) be a structure of species $\mathcal{E}$, where X is the underlying set and S the structure, and let U be a universe. If $X \in U$, then all the constituent objects of the structure S on X are elements of U, by Corollary 1 to Proposition 4 of Section 1 and by (U.I); hence the structure (X, S) is an element of U.

Now suppose that $\mathcal{E}$ is a species of structure with *morphisms*. If X and X' are elements of U endowed with structures of species $\mathcal{E}$, then the set of morphisms from X to X' is again an element of U, by Proposition 6 of Section 1.

Consider the category $(U\text{-}\mathcal{E})$ defined in Section 1, no. 2, Example c). Since the objects and arrows of $(U\text{-}\mathcal{E})$ are elements of U, $(U\text{-}\mathcal{E})$ is a pair of subsets of U, endowed with a quadruple of maps. It is not, in general, an element of U. The notation $X \in (U\text{-}\mathcal{E})$ will mean that X is an element of U endowed with a structure of species $\mathcal{E}$.

The category $(U\text{-}\mathcal{E})$ is stable under many operations:

a) If $X \in (U\text{-}\mathcal{E})$, and if X' is a subset of X admitting an *induced structure*, then $X' \in (U\text{-}\mathcal{E})$. This follows from Proposition 1 of Section 1.

b) If $X \in (U\text{-}\mathcal{E})$, and if X'' is a quotient set of X admitting a *quotient structure*, then $X'' \in (U\text{-}\mathcal{E})$, by the corollary to Proposition 1 of Section 1.

c) If $X, Y \in (U\text{-}\mathcal{E})$, and if $X \times Y$ admits a *product structure*, then $X \times Y \in (U\text{-}\mathcal{E})$, by Proposition 4 of Section 1. More generally, if $(X_i)_{i \in I}$ is a family of elements of $(U\text{-}\mathcal{E})$, if one has $I \in U$, and if $\prod_{i \in I} X_i$ admits a *product structure*, then $\prod_{i \in I} X_i \in (U\text{-}\mathcal{E})$, by the corollary to Proposition 6. The analogous assertions hold for *sum structures*, by Corollary 2 to Proposition 4.

d) Let I be a preordered set, and let $(X_i, f_{ij})_{i,j \in I}$ be a *projective* (respectively, *inductive*) system of sets endowed with structures of species $\mathcal{E}$ and morphisms. Let L be the limit of this system, in the sense of Chapter III. If $X_i \in U$ for all i, if $I \in U$, and if L admits a projective (respectively, inductive) limit structure, then $L \in (U\text{-}\mathcal{E})$: indeed L is a subset of $\prod_{i \in I} X_i$ (respectively, a quotient set of $\sum_{i \in I} X_i$), and is therefore an element of U by Section 1.

Here are a few more particular examples:

* 1) Let $X, Y \in (U\text{-}\mathbf{Top})$ be two *locally compact* spaces. Then the set $\mathcal{C}(X, Y)$ of continuous maps from X to Y, endowed with the compact-open topology (General Topology, X), is an element of $(U\text{-}\mathbf{Top})$. Indeed $\mathcal{C}(X, Y) \in U$ by Proposition 6 of Section 1.

* 2) Let $X, Y \in (U\text{-}\mathbf{Ab})$. Then the group $\mathrm{Hom}_{\mathbb{Z}}(X, Y)$ is an element of $(U\text{-}\mathbf{Ab})$ by Proposition 6 of Section 1. If, moreover, $\mathbb{Z} \in U$, then the tensor product $X \otimes_{\mathbb{Z}} Y$ is an element of $(U\text{-}\mathbf{Ab})$. Indeed, this tensor product is a quotient of the free abelian group $\mathbb{Z}^{(X \times Y)}$, whose underlying set is a subset of $\mathbb{Z}^{X \times Y}$, and the latter is an element of U by Propositions 4 and 6 of Section 1.

* 3) Let $X \in (U\text{-}\mathbf{Unif})$ be a uniform space. Then its *completion* $\widehat{X}$ is an element of $(U\text{-}\mathbf{Unif})$. Indeed $\widehat{X}$ is a set of equivalence classes of Cauchy filters on X; but a filter on X is an element of $\mathcal{P}(\mathcal{P}(X))$, and hence an equivalence class of filters is an element of $\mathcal{P}(\mathcal{P}(\mathcal{P}(X)))$. Thus $\widehat{X}$ is an element of $\mathcal{P}(\mathcal{P}(\mathcal{P}(\mathcal{P}(X))))$, and therefore an element of U by (U.III) and (U.I).

N.B. Moral: Bourbaki must take care to "canonify" his constructions. For example, in constructions in which one adjoins an element ∞, such as the Alexandroff compactification or the projective field, it will be useful to take $\infty = \emptyset$, because $\emptyset$ is an element of every nonempty universe, and to form the sum set of $\{\emptyset\}$ and the given set. Of course one should also "canonify" the two-element set used to construct sum sets; $\{\emptyset, \{\emptyset\}\}$ seems to me a good candidate, since it belongs to every nonempty universe.

3 Universes and categories

Let U be a universe and C a category. We write $C \in U$ if $\mathrm{Ob}(C) \in U$ and $\mathrm{Fl}(C) \in U$. This notation is justified by the fact that C is the sextuple consisting of $\mathrm{Ob}(C)$, $\mathrm{Fl}(C)$, and the four structural maps. Thus if $\mathrm{Ob}(C) \in U$ and $\mathrm{Fl}(C) \in U$, these four maps are elements of U, by Corollary 1 to Proposition 4 of Section 1, and so is the sextuple C, by Proposition 3 of Section 1, *cum grano salis*. This is moreover a special case of Section 2 if one considers the species of structure "category". With the notation of Section 2, the relation $C \in U$ is also written $C \in (U\text{-}\mathbf{Cat})$.

Notice that a category such as $(U\text{-}\mathbf{Set})$, $(U\text{-}\mathbf{Ord})$, $(U\text{-}\mathbf{Top})$, or $(U\text{-}\mathbf{Gr})$ is not, in general, an element of U.

Let C, D be two categories and U a universe such that $C, D \in U$. If C' is a subcategory of C, then $C' \in U$. The product category $C \times D$, the sum category of C and D, and the opposite categories C° and D° are also elements of U, by Section 2. The functor category $E = \mathrm{Hom}(C, D)$ is also an element of U: indeed, $\mathrm{Ob}(E)$ is a set of pairs of maps $\mathrm{Ob}(C) \to \mathrm{Ob}(D)$, $\mathrm{Fl}(C) \to \mathrm{Fl}(D)$, whence $\mathrm{Ob}(E) \in U$ by Section 1. As for $\mathrm{Fl}(E)$, it is a set of functorial morphisms, that is, of maps $\mathrm{Ob}(C) \to \mathrm{Fl}(D)$; hence $\mathrm{Fl}(E) \in U$ by Proposition 6 of Section 1.

Proposition 9. Let $\mathcal{E}$ be a species of structure with morphisms, and let U', U be two universes such that $U' \subset U$. Then $(U'\text{-}\mathcal{E})$ is a full subcategory of $(U\text{-}\mathcal{E})$.

Indeed, if x and y are objects of $(U'\text{-}\mathcal{E})$, then by definition they are objects of $(U\text{-}\mathcal{E})$. As for $\mathrm{Hom}(x, y)$, it is the same set in the two categories; see Section 1, no. 2, Example c).

N.B. The hypothesis that U and U' are universes is useless, so Proposition 9 would advantageously belong back in Section 1, no. 4. But the Tribe asked that it be placed here.

Remark. It may happen that the axioms of the species of structure $\mathcal{E}$ imply that the cardinals of the sets endowed with structures of species $\mathcal{E}$ are *bounded* by a fixed cardinal, say $\mathfrak{c}$; for example, the species of finite groups, finitely generated groups, finitely generated modules over a fixed ring A, or algebras of finite type over A. Suppose then that there exists an element z of U' such that $\mathfrak{c} \leq \operatorname{card}(z)$. Then, for every set x endowed with a structure of species $\mathcal{E}$, there exists an element x' of U' equipotent to x, for instance a subset of z. Endow x' with the structure transported from that of x; one obtains an element of $(U'\text{-}\mathcal{E})$. It follows that the inclusion functor from $(U'\text{-}\mathcal{E})$ into $(U\text{-}\mathcal{E})$ is then essentially surjective (Section 4, no. 2, Definition 2), and is therefore an equivalence of categories (Section 4, no. 3, Theorem 1).

4 The axiom of universes

The very agreeable stability results of Section 2 are of interest only if they can be applied to something other than the two small universes U_0 and U_1 described in Section 1. We will therefore add to the axioms of set theory the following axiom:

(A.6) *For every set x, there exists a universe U such that $x \in U$.*

This axiom implies that, if $(x_\alpha)_{\alpha \in I}$ is a *family* of sets, then there exists a universe U such that $x_\alpha \in U$ for every $\alpha \in I$. Indeed, it suffices to apply (A.6) to $x = \bigcup_{\alpha \in I} x_\alpha$, and then to apply Proposition 1 of Section 1.

In particular, given a category C, there exists a universe U such that $C \in U$ in the sense of Section 3: apply the preceding paragraph to the family $(\mathrm{Ob}(C), \mathrm{Fl}(C))$. This applies to categories of the form $(V\text{-}\mathcal{E})$, where $\mathcal{E}$ is a species of structure with morphisms and V is a set, for example a universe; in general $V \neq U$.

For example, if V is a universe, then $\mathrm{Ob}(V\text{-}\mathbf{Set}) = V$ and $\mathrm{Fl}(V\text{-}\mathbf{Set}) \subset V$, by Proposition 5 of Section 1. The universes U such that $(V\text{-}\mathbf{Set}) \in U$ are therefore those such that $V \in U$. But the relation $V \in V$ is impossible for a universe: indeed, for every subset A of V, one would have $A \in V$, by Proposition 1 of Section 1, whence $\operatorname{card}(\mathcal{P}(V)) \leq \operatorname{card}(V)$, which is impossible.

5 Universes and strongly inaccessible cardinals

Let U be a universe. By condition (U.I) of Definition 1, every element x of U is a subset of U; hence $\operatorname{card}(x) \leq \operatorname{card}(U)$. Since the cardinals of subsets of U form a well-ordered set, the cardinal

$$\mathfrak{c}(U) = \sup_{x \in U} \operatorname{card}(x) \tag{1}$$

exists. Notice that, for every cardinal $\mathfrak{c} < \mathfrak{c}(U)$, there exists an element x of U such that $\operatorname{card}(x) = \mathfrak{c}$. Indeed, by definition there exists $y \in U$ such that $\mathfrak{c} \leq \operatorname{card}(y) \leq \mathfrak{c}(U)$, and one takes for x a suitable subset of y. Conversely, if $x \in U$, then $\operatorname{card}(x) < \mathfrak{c}(U)$: indeed $\mathcal{P}(x) \in U$ by (U.III), whence $2^{\operatorname{card}(x)} \leq \mathfrak{c}(U)$. Thus the cardinal $\mathfrak{c}(U)$ has the following properties:

1) If $\mathfrak{c}$ is a cardinal $< \mathfrak{c}(U)$, then $2^{\mathfrak{c}} < \mathfrak{c}(U)$. Indeed, if x is an element of U of cardinal $\mathfrak{c}$, then $\mathcal{P}(x) \in U$ by (U.III), whence $2^{\mathfrak{c}} < \mathfrak{c}(U)$.

2) If $(\mathfrak{c}_\lambda)_{\lambda \in I}$ is a family of cardinals such that $\mathfrak{c}_\lambda < \mathfrak{c}(U)$ for every $\lambda \in I$, and if $\operatorname{card}(I) < \mathfrak{c}(U)$, then the sum cardinal $\sum_{\lambda \in I} \mathfrak{c}_\lambda$ is $< \mathfrak{c}(U)$. Indeed, let $x_\lambda \in U$ be such that $\operatorname{card}(x_\lambda) = \mathfrak{c}_\lambda$. Replacing I, if necessary, by an equipotent index set, we may suppose that $I \in U$. Then the sum set of the x_λ is an element of U, by Corollary 2 to Proposition 4 of Section 1, proving the assertion.

We make the following definition.

Definition 2. A cardinal $\mathfrak{d}$ is called *strongly inaccessible* if:

(FI.1) if $\mathfrak{c}$ is a cardinal such that $\mathfrak{c} < \mathfrak{d}$, then $2^{\mathfrak{c}} < \mathfrak{d}$;

(FI.2) if $(\mathfrak{c}_\lambda)_{\lambda \in I}$ is a family of cardinals such that $\mathfrak{c}_\lambda < \mathfrak{d}$ for all $\lambda \in I$, and if $\operatorname{card}(I) < \mathfrak{d}$, then $\sum_{\lambda \in I} \mathfrak{c}_\lambda < \mathfrak{d}$.

Examples. The cardinal 0 and the countably infinite cardinal are strongly inaccessible. No nonzero finite cardinal is strongly inaccessible. Thus we have just proved that the universe axiom (A.6) implies:

(A'.6) *Every cardinal is strictly dominated by a strongly inaccessible cardinal.*

Conversely:

Theorem 1. The assertion (A'.6) implies the axiom of universes (A.6).

Indeed, let A be a set. We must construct a universe U of which A is an element. Define by recursion a sequence $(A_n)_{n \geq 0}$ of sets by

(2) $A_0 = A$, and A_{n+1} is the union of the elements of A_n, that is, the set of elements of elements of A_n.

Put $B = \bigcup_{n=0}^{\infty} A_n$. By (A'.6), choose a strongly inaccessible cardinal $\mathfrak{c}$ such that $\operatorname{card}(B) < \mathfrak{c}$.

There exists a well-ordered set I such that $\operatorname{card}(I) = \mathfrak{c}$. Replacing I by its least segment of cardinal $\mathfrak{c}$, we may suppose that every proper segment of I has cardinal $< \mathfrak{c}$. We denote by $\mathcal{E}$ the least element of I. It follows from the hypothesis on segments that I has no largest element, and hence every element of I has a successor; for $\alpha \in I$, we denote the successor of α by $s(\alpha)$.

Now define, by transfinite recursion, a family $(B_\alpha)_{\alpha \in I}$ of sets by

$$\begin{cases} B_{\mathcal{E}} = B, \\ B_{s(\alpha)} = B_\alpha \cup \mathcal{P}(B_\alpha), \\ B_\alpha = \bigcup\limits_{\beta < \alpha} B_\beta \quad \text{if } \alpha \text{ has no predecessor.} \end{cases} \tag{3}$$

Put $U = \bigcup_{\alpha \in I} B_\alpha$. We will show that U is the required universe.

First $A \in U$, because $A = A_0$ is a subset of $B = B_{\mathcal{E}}$, and hence an element of $\mathcal{P}(B_{\mathcal{E}}) \subset B_{s(\mathcal{E})}$.

Next note that, for every $\alpha \in I$,

$$\operatorname{card}(B_\alpha) < \mathfrak{c}. \tag{4}$$

This is proved by transfinite induction on α. It is true for $\alpha = \mathcal{E}$ by hypothesis. From (4) one obtains

$$\operatorname{card}(B_{s(\alpha)}) \leq \operatorname{card}(B_\alpha) + 2^{\operatorname{card}(B_\alpha)} < \mathfrak{c} + \mathfrak{c} = \mathfrak{c},$$

using (FI.1) and the fact that $\mathfrak{c}$ is infinite. Finally, if α has no predecessor, the set I' of $\beta < \alpha$ has cardinal $< \mathfrak{c}$ by construction. Thus, if $\operatorname{card}(B_\beta) < \mathfrak{c}$ for all $\beta < \alpha$, then

$$\operatorname{card}(B_\alpha) = \operatorname{card}\Big(\bigcup_{\beta \in I'} B_\beta\Big) \leq \sum_{\beta \in I'} \operatorname{card}(B_\beta) < \mathfrak{c}$$

by (FI.2). This proves (4).

We now show that

$$\operatorname{card}(x) < \mathfrak{c} \quad \text{for every } x \in U. \tag{5}$$

It is enough to show, for every $\alpha \in I$, that $\operatorname{card}(x) < \mathfrak{c}$ for all $x \in B_\alpha$. Again proceed by transfinite induction on α. For $\alpha = \mathcal{E}$, if $x \in B_{\mathcal{E}} = B$, then $x \in A_n$ for some $n \geq 0$; hence $x \subset A_{n+1}$, so $x \subset B$ and $\operatorname{card}(x) \leq \operatorname{card}(B) < \mathfrak{c}$. The case where α has no predecessor is evident. Finally, if $x \in B_\alpha$ and $\alpha = s(\beta)$, then either $x \in B_\beta$, and the assertion follows by induction, or $x \in \mathcal{P}(B_\beta)$, whence $x \subset B_\beta$, and the assertion follows from (4).

We can now prove that U is indeed a universe:

(U.I) Let $x \in U$ and $y \in x$. We must show that $y \in U$. In other words, we must prove, for every $\alpha \in I$, the relation

$$x \in B_\alpha \text{ and } y \in x \Longrightarrow y \in B_\alpha.$$

Proceed once more by transfinite induction on α. It is true for $\alpha = \mathcal{E}$, since $x \in B_{\mathcal{E}} = B$ implies $x \in A_n$ for some n, and so, if $y \in x$, then $y \in A_{n+1}$, whence $y \in B = B_{\mathcal{E}}$. The passage to an element with no predecessor is evident. Finally pass from α to $s(\alpha)$. If $x \in B_{s(\alpha)}$ and $y \in x$, then either $x \in B_\alpha$, so $y \in B_\alpha \subset B_{s(\alpha)}$ by induction, or $x \in \mathcal{P}(B_\alpha)$, and again $y \in B_\alpha \subset B_{s(\alpha)}$.

(U.II) Let $x, y \in U$. Since the family $(B_\alpha)_{\alpha \in I}$ is increasing by (3), there exists $\alpha \in I$ such that $x, y \in B_\alpha$. Then $\{x, y\} \in \mathcal{P}(B_\alpha) \subset B_{s(\alpha)} \subset U$.

(U.III) Let $x \in U$. There exists $\alpha \in I$ such that $x \in B_\alpha$. It suffices to show, for every $\alpha \in I$, that

$$x \in B_\alpha \Longrightarrow \mathcal{P}(x) \in B_{s(s(\alpha))}.$$

Again proceed by transfinite induction on α. If $\alpha = \mathcal{E}$, then $x \in A_n$ for some n, whence every subset y of x is contained in $A_{n+1} \subset B = B_{\mathcal{E}}$. Thus $y \in \mathcal{P}(B_{\mathcal{E}}) \subset B_{s(\mathcal{E})}$ for every $y \in \mathcal{P}(x)$, and hence $\mathcal{P}(x) \subset B_{s(\mathcal{E})}$ and $\mathcal{P}(x) \in \mathcal{P}(B_{s(\mathcal{E})}) \subset B_{s(s(\mathcal{E}))}$. The passage to an element α with no predecessor is immediate, because $\beta < \alpha$ then implies $s(s(\beta)) < \alpha$. Finally pass from α to $s(\alpha)$. Let $x \in B_{s(\alpha)}$. If $x \in B_\alpha$, then $\mathcal{P}(x) \in B_{s(s(\alpha))} \subset B_{s(s(s(\alpha)))}$ by induction. If $x \in \mathcal{P}(B_\alpha)$, then $x \subset B_\alpha$, so $\mathcal{P}(x) \subset \mathcal{P}(B_\alpha)$ and $\mathcal{P}(x) \in \mathcal{P}(\mathcal{P}(B_\alpha)) \in B_{s(s(s(\alpha)))}$.

(U.IV) Let $(x_\lambda)_{\lambda \in K}$ be a family of elements of U such that $K \in U$. We must show that $x = \bigcup_{\lambda \in K} x_\lambda$ is an element of U. For every $\lambda \in K$, choose $\alpha(\lambda) \in I$ such that $x_\lambda \in B_{\alpha(\lambda)}$. We show that the set $\alpha(K)$ of the $\alpha(\lambda)$ is bounded in I. If it were not, then

$$I = \bigcup_{\lambda \in K} [\mathcal{E}, \alpha(\lambda)],$$

contradicting (FI.2), since $\mathrm{card}(K) < \mathfrak{c}$ by (5) and $\mathrm{card}([\mathcal{E}, \alpha(\lambda)]) < \mathfrak{c}$ for every $\lambda \in K$. Let β be an upper bound for $\alpha(K)$. Then $x_\lambda \in B_\beta$ for every $\lambda \in K$, because the family $(B_\alpha)_{\alpha \in I}$ is increasing. By what was proved in the verification of (U.I), each x_λ is a subset of B_β. Hence $x = \bigcup_{\lambda \in K} x_\lambda \subset B_\beta$. By (3), it follows that $x \in B_{s(\beta)}$, whence $x \in U$. □

Definition 3. Let x, y be two sets and let n be an integer ≥ 0. We say that y is a *component of order* n of x if there exists a sequence $(x_j)_{j=0,\ldots,n}$ such that $x_0 = x$, $x_n = y$, and $x_{j+1} \in x_j$ for $j = 0, \ldots, n-1$.

Thus x is the only component of order 0 of x. The components of order 1, respectively 2, of x are the elements of x, respectively the elements of elements of x. We say that y is a *component* of x if there exists $n \geq 0$ such that y is a component of order n of x. The relation "y is a component of x" is a preorder relation. By the selection-union scheme (Chapter II, . . .), the relation "y is a component of x" is collectivizing with respect to y, so the components of x form a set.

Definition 4. Let $\mathfrak{c}$ be a cardinal. A set x is said to be of *type* $\mathfrak{c}$ (respectively, of *strict type* $\mathfrak{c}$, of *finite type*) if all components of x have cardinal $\leq \mathfrak{c}$ (respectively, $< \mathfrak{c}$, finite).

Examples. The elements of the universe U_1 of Section 1, Example 2, are all of finite type. If $\mathfrak{c}$ is a cardinal and x is of type $\mathfrak{c}$ (respectively, strict type $\mathfrak{c}$, finite type), then every component of x and every subset of x are of type $\mathfrak{c}$ (respectively, strict type $\mathfrak{c}$, finite type). Similarly, $\mathcal{P}(x)$ is of type $2^{\mathfrak{c}}$ (respectively, of strict type $2^{\mathfrak{c}}$, of finite type). If x is of type $\mathfrak{c}$ and $\mathfrak{c} \leq \mathfrak{c}'$, then x is of type $\mathfrak{c}'$.

Lemma. Let $\mathfrak{c}$ be a noncountable strongly inaccessible cardinal, and let X be a set of strict type $\mathfrak{c}$. Then there exists a cardinal $\mathfrak{d} < \mathfrak{c}$ such that X is of type $\mathfrak{d}$.

Indeed, for every integer n, let $\mathfrak{d}_n$ be the cardinal of the set X_n of components of order n of X. We have $\mathfrak{d}_0 = 1$, $\mathfrak{d}_1 = \mathrm{card}(X) < \mathfrak{c}$, and, since $X_n = \bigcup_{y \in X_{n-1}} y$, it follows by induction on n and by use of (FI.2) that $\mathfrak{d}_n < \mathfrak{c}$ for every n. Then the $\mathfrak{d}_n$ are bounded above by the cardinal $\mathfrak{d} = \sum_{j \geq 0} \mathfrak{d}_j$, which is $< \mathfrak{c}$ by (FI.2) and the noncountability of $\mathfrak{c}$. If Y is a component of order n of X, then $\mathrm{card}(Y) \leq \mathrm{card}(X_{n+1}) \leq \mathfrak{d}$. □

Proposition 10. Let U be a universe and let $\mathfrak{c}$ be a strongly inaccessible cardinal. Then the set U' of those $x \in U$ which are of strict type $\mathfrak{c}$ is a universe.

Indeed, if $x, y \in U'$ and $z \in x$, then obviously $z \in U'$ and $\{x, y\} \in U'$. If $x \in U'$, then $\mathrm{card}(x) < \mathfrak{c}$, whence $\mathrm{card}(\mathcal{P}(x)) < \mathfrak{c}$ by (FI.1). Since a component of $\mathcal{P}(x)$ is either $\mathcal{P}(x)$, or a subset of x, or a component of x, one has $\mathcal{P}(x) \in U'$. Finally, if $(x_\alpha)_{\alpha \in I}$ is a family of elements of U' such that $I \in U'$, then $\mathrm{card}(\bigcup_\alpha x_\alpha) < \mathfrak{c}$ by (FI.2); since every component of order > 0 of $\bigcup_\alpha x_\alpha$ is a component of some x_α, we do indeed have $\bigcup_\alpha x_\alpha \in U'$. □

Remark on the cardinal $\mathfrak{c}(U)$. Let U be a universe and let $\mathfrak{c}(U)$ be the cardinal defined by (1), that is,

$$\mathfrak{c}(U) = \sup_{x \in U} \mathrm{card}(x).$$

Obviously $\mathfrak{c}(U) \leq \mathrm{card}(U)$, because, recall, $x \in U$ implies $x \subset U$. But the equality $\mathfrak{c}(U) = \mathrm{card}(U)$ is not always true. For example, let $(x_\alpha)_{\alpha \in I}$ be a noncountable family of symbols with the axioms

$$\text{“}x_\alpha = \{x_\alpha\} \text{ for every } \alpha \in I\text{”} \quad \text{and} \quad \text{“}x_\alpha \neq x_\beta \text{ if } \alpha \neq \beta\text{”}.$$

In other words, x_α is a one-element set, namely itself. As in Example 2 of Section 1, form the "meaningful words" made from the symbols $\emptyset$, x_α, braces, and so on. Each denotes a *finite* set. Let U be the set of these. One easily checks that the conditions of Definition 1 are satisfied, so U is a universe. Since meaningful words are finite sequences of elements of a set of cardinal $\operatorname{card}(I)+4=\operatorname{card}(I)$, we have $\operatorname{card}(U)=\operatorname{card}(I)$; in fact $\operatorname{card}(U)\geq\operatorname{card}(I)$ would suffice, and this is evident. On the other hand, $\mathfrak{c}(U)$ is the countable cardinal, whence $\mathfrak{c}(U)<\operatorname{card}(U)$.

The strict inequality $\mathfrak{c}(U)<\operatorname{card}(U)$ is due to the fact that here we have introduced sets x_α such that $x_\alpha\in x_\alpha$. If one forbids horrors of this kind, one obtains $\operatorname{card}(U)=\mathfrak{c}(U)$ for every universe U, as well as other very pretty results "which cannot serve any purpose." This is what we shall do in the next section.

6 Artinian sets and artinian universes

Definition 5. A set x is called *artinian* if there exists no infinite sequence $(x_n)_{n\geq 0}$ such that $x_0=x$ and $x_{n+1}\in x_n$ for every $n\geq 0$.

Examples. The sets $\emptyset$, $\{\emptyset\}$, $\{\emptyset,\{\emptyset\}\}$, and more generally the elements of the universe U_1 of Section 1, are artinian.

In pictorial terms, if x is artinian and one takes an element x_1 of x, then an element x_2 of x_1, and so on, the process must stop, and one arrives at a component x_n of x which is *empty.* In other words, artinian sets are "built out of $\emptyset$." This will be made precise later (cf. (*) in the proof of Theorem 2).

Artinian sets evidently enjoy the following properties:

(AR.I) If x is artinian, every subset of x and every component of x is artinian. For x to be artinian, it is necessary and sufficient that every element of x be artinian.

(AR.II) If x and y are artinian, then $\{x,y\}$ is artinian.

(AR.III) If x is artinian, then $\mathcal{P}(x)$ is artinian.

(AR.IV) Every union of artinian sets is an artinian set.

These properties immediately show:

Proposition 11. If U is a universe, the set of artinian elements of U is a universe, necessarily artinian.

Corollary. If x is an artinian set, there exists an artinian universe U such that $x\in U$.

Indeed, by axiom (A.6), x is an element of a universe V; take for U the set of artinian elements of V.

The following proposition is still less useful than the rest of the section:

Proposition 12. Let A be an artinian set. Then:

a) for every $x\in A$, one has $x\notin x$;

b) if $x,y\in A$, one cannot have both $x\in y$ and $y\in x$;

c) for every $x \in A$, the relation "y is a component of z" between components y, z of x is an order relation;

d) for every nonempty element x of A, there exists $y \in x$ such that $x \cap y = \emptyset$.

Indeed, the negation of a), respectively of b), entails the existence of an infinite sequence $(x_n)_{n\geq 0}$ contradicting Definition 5, namely $(x, x, x, \ldots)$, respectively $(x, y, x, y, x, y, \ldots)$. The negation of c) means that there exist $x \in A$, distinct components y, z of x, and membership chains

$$y \in y_1 \in \cdots \in y_q \in z, \qquad z \in z_1 \in \cdots \in z_r \in y;$$

then, as in b), one obtains an infinite sequence $(x_n)_{n\geq 0}$ contradicting Definition 5. Finally, if d) is false, there exists a nonempty element x of A such that $x \cap y \neq \emptyset$ for every $y \in x$. Put $x_0 = x$, and choose $x_1 \in x$. Since $x \cap x_1 \neq \emptyset$, choose $x_2 \in x \cap x_1$, and so on. More formally, define by recursion an infinite sequence $(x_n)_{n\geq 1}$ of elements of x by choosing $x_1 \in x$ and $x_{n+1} \in x_n \cap x$ for $n \geq 1$. Then the sequence $(x_n)_{n\geq 0}$ contradicts Definition 5.

Remark. Let B be a set. For B to be artinian, it is necessary and sufficient that every set A of sets of components of B satisfy condition d) of Proposition 12. Necessity follows from (AR.I) and Proposition 12. Conversely, if B is not artinian, there exists an infinite sequence $(x_n)_{n\geq 0}$ with $x_0 = B$ and $x_{n+1} \in x_n$ for every $n \geq 0$. Take for A the singleton consisting of the set X of the x_n. Thus A contains a nonempty element X such that, for every element y of X, one has $y \cap X \neq \emptyset$: indeed $y = x_n$ for some n, and $x_{n+1} \in y \cap X$.

Theorem 2. Let $\mathfrak{c}$ be an infinite cardinal. Then:

a) The relation "x is an artinian set of type $\mathfrak{c}$" (respectively, of strict type $\mathfrak{c}$) is collectivizing with respect to x. The set $A_{\mathfrak{c}}$ of artinian sets of type $\mathfrak{c}$ has cardinal $2^{\mathfrak{c}}$.

b) If $\mathfrak{c}$ is strongly inaccessible, the set $U_{\mathfrak{c}}$ of artinian sets of strict type $\mathfrak{c}$ is a universe of cardinal $\mathfrak{c}$; the cardinal $\mathfrak{c}(U_{\mathfrak{c}}) = \sup_{x\in U_{\mathfrak{c}}} \operatorname{card}(x)$ is $\mathfrak{c}$.

c) If a universe U has an element of cardinal $\mathfrak{c}$, then every artinian set of type $\mathfrak{c}$ belongs to U, that is, $A_{\mathfrak{c}} \subset U$.

c') If a universe U is nonempty, then every artinian set of finite type is an element of U.

Before proving Theorem 2, let us derive a few illuminating corollaries.

Corollary 1. If a universe U is artinian, then $\operatorname{card}(U)$ is strongly inaccessible, and U is the set of artinian sets of strict type $\operatorname{card}(U)$.

Indeed put $\mathfrak{c}(U) = \sup_{x\in U} \operatorname{card}(x)$. This is a strongly inaccessible cardinal, by the beginning of Section 5. First suppose it is noncountable. Then, for every infinite cardinal $\mathfrak{c} < \mathfrak{c}(U)$, every artinian set of type $\mathfrak{c}$ is an element of U by c); hence every artinian set of strict type $\mathfrak{c}(U)$ is an element of U, by the lemma of Section 5. This last assertion remains true if $\mathfrak{c}(U)$ is countable, by c'). Conversely, by (U.I), every set of U is of strict type $\mathfrak{c}(U)$. Thus U is the universe $U_{\mathfrak{c}(U)}$ of b), whence $\mathfrak{c}(U) = \operatorname{card}(U)$ by b).

It follows from Corollary 1 that an artinian universe is uniquely determined by its cardinal, which is strongly inaccessible. Thus one has a "one-to-one correspondence" between artinian universes and strongly inaccessible cardinals. In particular:

Corollary 2. The inclusion relation $U \subset U'$ between artinian universes is a well-ordering relation.

Indeed, the relation $\mathfrak{c} \leq \mathfrak{c}'$ between cardinals is a well-ordering relation (Chapter III), and, with the notation of Theorem 2 b), the relations $\mathfrak{c} \leq \mathfrak{c}'$ and $U_{\mathfrak{c}} \subset U_{\mathfrak{c}'}$ are equivalent.

Note that Theorem 2 b) gives a second proof of Theorem 1.

We now prove Theorem 2. Given a set A, call a *chain* of A any finite sequence $(x_j)_{j=0,\ldots,n}$ such that $x_0 = A$ and $x_{j+1} \in x_j$ for $j = 0, \ldots, n-1$. The chains of A form a set, by the selection-union scheme; denote it by $G(A)$. Given a chain $X = (x_j)_{j=0,\ldots,n}$, the chains of the form $(x_i)_{i=0,\ldots,q}$, with $q \leq n$, are said to be smaller than X. This gives an ordered-set structure on $G(A)$. For A to be artinian, it is necessary and sufficient that $G(A)$ be a noetherian ordered set, in the sense of the equivalent conditions of Chapter III, §6, no. 5.

We shall show that:

> (*) *If A is artinian, then A is uniquely determined by the isomorphism class of the ordered set $G(A)$.*

Indeed, given an ordered set G and an element $g \in G$, write $S(g)$ for the set of $g' \in G$ such that $g < g'$ and such that $g \leq h \leq g'$ implies $h = g$ or $h = g'$. In other words, $S(g)$ is the set of immediate successors of g. Consider the map θ which sends a chain $X = (x_0, \ldots, x_n)$ of A to the set x_n. Then, for every $X \in G(A)$,

$$\theta(X) = \{\theta(X') \mid X' \in S(X)\}.$$

Since A is the image under θ of the least element $X_0 = (A)$ of $G(A)$, it will be enough to show that θ is uniquely determined by the isomorphism class of the ordered set $G(A)$. This follows from the following lemma.

Lemma 1. Let G be a noetherian ordered set, and let φ be a map on G such that, for every $g \in G$,

$$\varphi(g) = \{\varphi(g') \mid g' \in S(g)\}. \tag{1}$$

Then φ is uniquely determined. Moreover, if U is a universe containing an element equipotent to G, then φ takes its values in U.

The hypothesis implies that $\varphi(g) = \emptyset$ if g is a maximal element of G. Indeed, let φ and φ' be two maps satisfying (1). If $\varphi \neq \varphi'$, the set of $g \in G$ such that $\varphi(g) \neq \varphi'(g)$ is nonempty, hence has a maximal element h, since G is noetherian. Then $\varphi(g) = \varphi'(g)$ for every $g > h$, in particular for every successor $g \in S(h)$, whence $\varphi(h) = \varphi'(h)$ by (1), a contradiction. Thus $\varphi = \varphi'$.

Now let U be a universe containing an element x equipotent to G. We show that φ takes its values in U. Otherwise, let h be maximal among the $g \in G$ such that $\varphi(g) \notin U$. We have $\varphi(g') \in U$ for every $g' \in S(h)$, so

$$\varphi(h) = \{\varphi(g') \mid g' \in S(h)\}$$

is a subset of U. Since its cardinal is less than or equal to $\mathrm{card}(G)$, hence to the cardinal of an element of U, one has $\varphi(h) \in U$ by Proposition 7 of Section 1. This contradiction shows that φ takes its values in U.

We now prove Theorem 2 a). It is enough to treat the non-respective assertion, since the other follows at once. Let $\mathfrak{c}$ be an infinite cardinal. If A is a set of type $\mathfrak{c}$, then

$$\operatorname{card}(G(A)) \leq \mathfrak{c}. \tag{2}$$

Indeed, if A_n denotes the set of components of order n of A, then $\operatorname{card}(A_1) \leq \mathfrak{c}$ and $\operatorname{card}(A_{n+1}) \leq \mathfrak{c} \cdot \operatorname{card}(A_n)$. Hence $\operatorname{card}(A_n) \leq \mathfrak{c}^n$ by induction; but $\mathfrak{c}^n = \mathfrak{c}$, because $\mathfrak{c}$ is infinite (Chapter III). Therefore

$$\operatorname{card}\Big(\bigcup_{n=0}^{\infty} A_n\Big) \leq \operatorname{card}(\mathbb{N}) \cdot \mathfrak{c} = \mathfrak{c}.$$

Since $G(A)$ is a set of finite sequences of elements of $\bigcup_n A_n$, inequality (2) follows.

Now let E be a set of cardinal $\mathfrak{c}$. Giving an order structure on a subset E' of E is equivalent to giving the subset of $E \times E$ consisting of the pairs (x, y) such that $x \in E'$, $y \in E'$, and $x \leq y$. Therefore, by (2), the isomorphism classes of ordered sets $G(A)$, where A is of type $\mathfrak{c}$, form a set $\mathfrak{F}_\mathfrak{c}$, and

$$\operatorname{card}(\mathfrak{F}_\mathfrak{c}) \leq \operatorname{card}(\mathcal{P}(E \times E)) = 2^\mathfrak{c}.$$

Let $\mathfrak{F}'_\mathfrak{c}$ be the subset of $\mathfrak{F}_\mathfrak{c}$ formed by the classes of noetherian ordered sets having a least element. If, to each $G \in \mathfrak{F}'_\mathfrak{c}$, we associate the value of the function φ of Lemma 1 at the least element of G, we obtain a map θ on $\mathfrak{F}'_\mathfrak{c}$ whose image contains *all* artinian sets of type $\mathfrak{c}$. These therefore form a set $A_\mathfrak{c}$, and

$$\operatorname{card}(A_\mathfrak{c}) \leq \operatorname{card}(\mathfrak{F}'_\mathfrak{c}) \leq \operatorname{card}(\mathfrak{F}_\mathfrak{c}) \leq 2^\mathfrak{c}.$$

It remains to see that $\operatorname{card}(A_\mathfrak{c}) = 2^\mathfrak{c}$. For this it is enough to see that there exists an artinian set B of type $\mathfrak{c}$ and cardinal $\mathfrak{c}$, since the subsets of B will then be elements of $A_\mathfrak{c}$. This existence follows from the following lemma.

Lemma 2. For every cardinal $\mathfrak{c}$, there exists an artinian set B of type $\mathfrak{c}$ and cardinal $\mathfrak{c}$.

Proceed by transfinite induction on $\mathfrak{c}$. For $\mathfrak{c} = 0$ there is no choice: take $B = \emptyset$. If $\mathfrak{c}$ has a predecessor $\mathfrak{c}'$, let B' be an artinian set of type $\mathfrak{c}'$ and of cardinal $\mathfrak{c}'$. Then $\mathfrak{c} \leq 2^{\mathfrak{c}'}$, so there exists a subset B of $\mathcal{P}(B')$ of cardinal $\mathfrak{c}$. The elements of B are subsets of B', and hence have cardinals $\leq \mathfrak{c}' \leq \mathfrak{c}$; the components of higher order of B are components of B', and hence also have cardinals $\leq \mathfrak{c}' \leq \mathfrak{c}$. Finally, if $\mathfrak{c}$ has no predecessor, choose, for every cardinal $\mathfrak{c}_\lambda < \mathfrak{c}$, an artinian set B_λ of type $\mathfrak{c}_\lambda$ and cardinal $\mathfrak{c}_\lambda$. Then $B = \bigcup_\lambda B_\lambda$ has the required property. This proves Lemma 2 and completes the proof of part a).

We turn to b). Let $\mathfrak{c}$ be a strongly inaccessible cardinal. We already know, by a), that the artinian sets of strict type $\mathfrak{c}$ form a set $U_\mathfrak{c}$. The fact that $U_\mathfrak{c}$ is a universe follows immediately from properties (AR.I)–(AR.IV) of artinian sets, at the beginning of this section, and from cardinal estimates analogous to those of Proposition 10 of Section 5. The relation

$$\sup_{x \in U_\mathfrak{c}} \operatorname{card}(x) = \mathfrak{c}$$

follows from Lemma 2 applied to cardinals $< \mathfrak{c}$. Finally, to prove $\operatorname{card}(U_\mathfrak{c}) = \mathfrak{c}$, first suppose that $\mathfrak{c}$ is noncountable. By the lemma of Section 5, $U_\mathfrak{c}$ is the union

$$\bigcup_{\mathfrak{d} < \mathfrak{c}} A_\mathfrak{d},$$

where $A_{\mathfrak{d}}$ denotes the set of artinian sets of type $\mathfrak{d}$. But $\mathrm{card}(A_{\mathfrak{d}}) = 2^{\mathfrak{d}} < \mathfrak{c}$ by a); on the other hand, the set of cardinals $\mathfrak{d} < \mathfrak{c}$ has cardinal $\leq \mathfrak{c}$ (Chapter III). Hence

$$\mathrm{card}(U_{\mathfrak{c}}) \leq \mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c},$$

and also $\mathrm{card}(U_{\mathfrak{c}}) \geq \mathfrak{c}$, since $\mathrm{card}(U_{\mathfrak{c}}) \geq \mathrm{card}(A_{\mathfrak{d}}) = 2^{\mathfrak{d}}$ for every $\mathfrak{d} < \mathfrak{c}$.

The case $\mathfrak{c} = 0$ is trivial. It remains to treat the case where $\mathfrak{c}$ is the countably infinite cardinal. In this case $U_{\mathfrak{c}}$ is the set of artinian sets of finite type, that is, sets which are finite, as are all their components; and one uses a pretty combinatorial result.

Lemma 3 (D. König?). Consider two infinite sequences $(E_n)_{n\geq 1}$, $(f_n)_{n\geq 1}$ of finite sets E_n and maps $f_n : E_{n+1} \to E_n$. If there exists no infinite sequence $(x_n)_{n\geq 1}$ such that $x_n \in E_n$ and $f_n(x_{n+1}) = x_n$ for every n, then E_n is empty for all sufficiently large n.

In other words, if all sequences (x_n) such that $f_n(x_{n+1}) = x_n$ are finite, then their lengths are bounded. This can be expressed in terms of projective limits: a projective limit of nonempty finite sets is nonempty; see General Topology, Chapter I, 2nd ed., §9, no. 6, Proposition 8, 2°.

Argue by contradiction. If there exist nonempty E_n for arbitrarily large n, then no E_n is empty, because $E_n = \emptyset$ implies $E_{n+1} = \emptyset$, given the existence of $f_n : E_{n+1} \to E_n$. Call a finite sequence $(x_j)_{1\leq j\leq n}$ *coherent* if $f_j(x_{j+1}) = x_j$ for $j = 1, \ldots, n-1$. We prove by induction on n the existence of a coherent sequence $(a_1, \ldots, a_n)$, with $a_i \in E_i$, which, for every $q \geq n$, can be extended to a coherent sequence $(a_1, \ldots, a_n, x_{n+1}, \ldots, x_q)$ of length q. This is evident for $n = 0$, since no E_q is empty. Passing from n to $n+1$, if, for every $x \in f_n^{-1}(\{a_n\}) \subset E_{n+1}$, all coherent sequences extending $(a_1, \ldots, a_n, x)$ had lengths bounded by an integer $q(x)$, then all coherent sequences extending $(a_1, \ldots, a_n)$ would have bounded lengths, by $\sup_x q(x)$, since E_{n+1} is finite. Hence there exists $a_{n+1} \in f_n^{-1}(\{a_n\})$ such that the coherent sequence $(a_1, \ldots, a_n, a_{n+1})$ has extensions of arbitrary length. This gives an infinite sequence $(a_n)_{n\geq 1}$, contrary to the hypothesis.

It follows from Lemma 3 that if A is an artinian set of finite type, then the ordered set $G(A)$ of its chains is *finite*. Indeed, take E_n to be the set of chains

$$x_n \in x_{n-1} \in \cdots \in x_1 \in A$$

with $n+1$ terms. This is finite because the components of A of order $\leq n+1$ are finite in number and are themselves finite. Take f_n to be the map which deletes the last term:

$$(x_{n+1} \in x_n \in \cdots \in x_1 \in A) \longmapsto (x_n \in x_{n-1} \in \cdots \in x_1 \in A).$$

The set of isomorphism classes of finite ordered sets is countable: indeed, giving an order structure on a finite subset of $\mathbb{N}$ is equivalent to giving its graph, a finite subset of $\mathbb{N} \times \mathbb{N}$, and the set of finite subsets of a countable set is countable (Chapter III). It follows from (*) and Lemma 1 that the set $\mathfrak{a}$ of artinian sets of finite type is countable. It is infinite by Lemma 2, or more simply by using the sequence $(z_n)_{n\geq 0}$ defined by $z_0 = \emptyset$, $z_{n+1} = \{z_n\}$. This completes the proof of b).

Remark. We have just proved that, if A is an artinian set of finite type, then it has only finitely many components. Thus there exists an integer n such that A is of type n.

We now prove c). Let $\mathfrak{c}$ be an infinite cardinal, let U be a universe having an element of cardinal $\mathfrak{c}$, and let A be an artinian set of type $\mathfrak{c}$. The ordered set $G(A)$ has cardinal $\leq \mathfrak{c}$, by formula (2). The second assertion of Lemma 1, applied to $G(A)$, shows that the map

$$(x_0, \ldots, x_n) \rightsquigarrow x_n$$

from $G(A)$ takes its values in U. In other words, all components of A are elements of U. This proves c). The proof of c') is analogous: if A is an artinian set of finite type, then we have just seen that $G(A)$ is finite; since U is nonempty, it contains an element equipotent to $G(A)$, by the corollary to Proposition 7 of Section 1. □

7 Vague metamathematical remarks

a) The axiom "*every set is artinian*" is *inoffensive*. Indeed, if one has a model M of set theory, the set M' of artinian elements of M is also a model (cf. Proposition 11).

b) The universe axiom (A.6), and the equivalent axiom (A'.6) on strongly inaccessible cardinals, is *independent* of the rest of set theory. Indeed, let $\mathfrak{c}$ be the first noncountable strongly inaccessible cardinal. The universe $U_\mathfrak{c}$ of artinian sets of strict type $\mathfrak{c}$ (Theorem 2 b)) is a model of set theory without (A.6): one calls the elements of $U_\mathfrak{c}$ "sets," the "membership relation" is the restriction to $U_\mathfrak{c}$ of the ordinary one, and so on. The "universes" of the model are therefore the ordinary universes which are elements of $U_\mathfrak{c}$. But we have seen that the only universes which are elements of $U_\mathfrak{c}$ are the two pequeños $U_0 = \emptyset$ and U_1. Thus U_1 is a "set" which is an element of no "universe." Hence we have a model of set theory in which (A.6) is false.

c) Bourbaki was too cautious in merely "presuming" that the *axiom of infinity* (A.5) is independent of the preceding axioms and schemes. It is indeed independent, because the countable universe U_1 of artinian sets of finite type is a model in which (A.5) is false and in which the preceding axioms and schemes are true.

d) It would be very interesting to prove that the universe axiom (A.6) is inoffensive. This seems difficult, and is even unprovable, says Paul Cohen.

The adjective "vague" in the title means that, in constructing models, one did not take the trouble to cannulate the symbol τ so that it does not spill out. The key to this, if one considers a model M, is to replace the ordinary

$$\tau_x(R(x))$$

by

$$\tau_x(R(x) \text{ and } x \in M).$$

One still has to check that this really transforms ordinary quantifiers into the quantifiers formerly called "typical":

$$(\forall x \in M) \qquad (\exists x \in M).$$

Exercises.

1) Let $n \geq 1$ be an integer. Show that the set of artinian sets of type n is infinite. The sets $z_0 = \emptyset$, $z_1 = \{\emptyset\}$, and $z_{q+1} = \{z_q\}$ are of type 1.

2) Call the *height* of a set A the least upper bound, finite or infinite, of the integers n such that there exists a sequence

$$x_n \in x_{n-1} \in \cdots \in x_0 = A.$$

Show that the sets of height $\leq n$ are finite and form a finite set whose cardinal p_n is calculated from $p_0 = 1$, $p_{n+1} = 2^{p_n}$. Proceed by induction on n, noting that the elements of a set A of height $\leq n$ are sets of height $\leq n-1$.

3) Let G be a noetherian ordered set having a least element g_0, and such that, for every $h \in G$, the set of $g \leq h$ is finite and totally ordered.

   a) Show that every element $h \neq g_0$ of G has a unique predecessor.

   b) Let φ be the map on G defined in Lemma 1, that is, $\varphi(g)$ is the set of the $\varphi(g')$ where g' runs over the successors of g. Put $A = \varphi(g_0)$. Show that A is an artinian set. For $g \in G$, let
   $$g_0 < g_1 < \cdots < g_n = g$$
   be the sequence of elements $\leq g$, and put
   $$f(g) = (\varphi(g_0), \varphi(g_1), \ldots, \varphi(g_n)).$$
   Show that f is an increasing surjective map from G onto the ordered set $G(A)$ of the text. Show that, if f is *injective*, it is an isomorphism from G onto $G(A)$.

   c) For $g \in G$, let M_g be the set of majorants of g. Suppose that, for every pair of distinct elements g, g' having the same predecessor p, the ordered sets M_g and $M_{g'}$ are not isomorphic. Show that the map f of b) is then an *isomorphism* from G onto $G(A)$. Hint: if $f(g) = f(g')$ with $g \neq g'$, and if
   $$g_0 < g_1 < \cdots < g_n = g, \qquad g'_0 < g'_1 < \cdots < g'_{n'} = g'$$
   are the sequences of elements $\leq g$ and $\leq g'$, show that $n = n'$, and that one may suppose that g and g' have the same predecessor p. Then consider the set of $p \in G$ such that there exist two distinct successors g, g' of p with $f(g) = f(g')$; take a maximal element q of this set and two distinct successors h, h' of q such that $f(h) = f(h')$. Observe that the restrictions of f to M_h and to $M_{h'}$ are injective, and hence, by b), are isomorphisms from M_h to $G(\varphi(h))$ and from $M_{h'}$ to $G(\varphi(h'))$. Deduce from the non-isomorphism hypothesis on M_h and $M_{h'}$ that $\varphi(h) \neq \varphi(h')$, contradicting $f(h) = f(h')$.

N.B. Exercise 3 gives very precise information on how artinian sets are "made." We already knew that such a set A is determined by the isomorphism class of the ordered set $G(A)$ (Lemma 1). We now know how to *characterize* the ordered sets G isomorphic to some $G(A)$:

- α) G is noetherian and has a least element;
- β) for every $g \in G$, the set of $h \leq g$ is totally ordered and finite, whence the existence and uniqueness of the predecessor of g;
- γ) if g, g' are distinct elements having the same predecessor, then the set M_g of majorants of g and the set $M_{g'}$ of majorants of g' are not isomorphic.

4) Let $(A_n)_{n\geq 0}$ be the sequence of finite ordinals defined by $A_0 = \emptyset$, $A_{n+1} = A_n \cup \{A_n\}$. Show that the ordered set $G(A_n)$ has 2^n elements, and that the number of its elements of height q, in the sense of Exercise 2, is $\binom{n}{q}$.

References

[1] B. Mitchell, *Theory of Categories*, Academic Press, 1965.

Exposé II. Topologies and Sheaves

Opening; Sections 1–2.7

J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé II, from the opening overview through Remarks 2.7. The range corresponds to source lines 23–421 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current English usage where this is harmless: *sieve* for *crible*, *covering family*, *pretopology*, and *base-changeable* for French *quarrable*. References to Exposé I are kept in their original numerical form.

After the definition of topologies and pretopologies (no. 1) and of sheaves of sets (no. 2), no. 3 turns to the central theorem of this exposé (3.4): the existence of the sheaf associated to a presheaf. In order to cover all cases encountered in practice, the existence of this functor is proved for presheaves on a $\mathcal{U}$-site (3.0.2). Nos. 4 and 5 draw the consequences of this theorem for the behavior of inductive and projective limits in categories of sheaves. In no. 6 one defines and studies sheaves with values in arbitrary categories, and then immediately turns attention to sheaves of rings, groups, modules, etc.

1 Topologies, covering families, pretopologies

Definition 1.1. A *topology* on a category C consists, for each object X of C, of a set $J(X)$ of sieves on X, subject to the following axioms:

T 1) *Stability under base change.* For every object X of C, every sieve $R \in J(X)$, and every morphism $f : Y \to X$ with $Y \in \mathrm{Ob}(C)$, the sieve $R \times_X Y$ on Y belongs to $J(Y)$.

T 2) *Local character.* If R and R' are two sieves on X, if $R \in J(X)$, and if for every $Y \in \mathrm{Ob}(C)$ and every morphism $Y \to R$ the sieve $R' \times_X Y$ belongs to $J(Y)$, then R' belongs to $J(X)$.

T 3) For every object X of C, the maximal sieve X belongs to $J(X)$.

1.1.1

The sieves belonging to $J(X)$ will be called the *sieves covering* X, or also the *refinements of* X. From axioms (T1), (T2), and (T3), one immediately deduces that the set of sieves covering X is stable under finite intersections, and that every sieve containing a covering sieve is again a covering sieve. Thus $J(X)$, ordered by inclusion, is cofiltered (I 8).

1.1.2

Let C be a category, and let T and T' be two topologies on C. The topology T is said to be *finer than* the topology T' if, for every object X of C, every refinement of X for the topology T' is a refinement of X for the topology T. In this way one defines an order structure on the set of topologies.

1.1.3

Let $(T_i)_{i\in I}$ be a family of topologies on C. For each object X of C, the sets of sieves on X which are covering for all the topologies T_i satisfy axioms (T1), (T2), and (T3), and therefore define a topology: the *intersection topology* of the T_i, i.e. the greatest lower bound of the T_i. It is the finest of the topologies which is less fine than all the topologies T_i. It follows that the family $(T_i)_{i\in I}$ has a *least upper bound*: the intersection topology of the topologies finer than each of the T_i.

1.1.4

In particular, the assignment $J(X) =$ the set of all sieves on X is a topology finer than every topology on C; we shall call it the *discrete topology.*

There is a topology less fine than every topology on C: the topology for which $J(X) = \{X\}$ for every X of C. This topology is called the *coarse* or *chaotic* topology.

1.1.5

A category C endowed with a topology is called a *site*. The category C is called the *underlying category* of the site.

Definition 1.2. Let C be a site and let X be an object of C. A family of morphisms $(f_\alpha : X_\alpha \to X)$, $\alpha \in A$, is called *covering* if the sieve generated by the family f_α (I 4.3.3) is a sieve covering X.

1.1.6

Let C be a category. Suppose that for every object X of C we are given a set of families of morphisms with target X. Then there exists a topology T which is the least fine among the topologies for which the given families are covering, namely the intersection (1.1.3) of all topologies in question. We call this topology the *topology generated by the given sets of families of morphisms*. In general it is awkward to describe all covering sieves of this topology. The situation is more agreeable in the following case.

Definition 1.3. Let C be a category. A *pretopology* on C consists, for each object X of C, of a set $\mathrm{Cov}(X)$ of families of morphisms with target X, subject to the following axioms:

PT 0)

For every object X of C, the morphisms occurring in the families of morphisms of $\mathrm{Cov}(X)$ are base-changeable. Recall that a morphism $Y \to X$ is called base-changeable if for every morphism $Z \to X$, the fiber product $Z \times_X Y$ is representable.

PT 1)

For every object X of C, every family $(X_\alpha \to X)_{\alpha\in A}$ belonging to $\mathrm{Cov}(X)$, and every morphism $Y \to X$ with $Y \in \mathrm{Ob}(C)$, the family

$$(X_\alpha \times_X Y \to Y)_{\alpha\in A}$$

belongs to $\mathrm{Cov}(Y)$. This is stability under base change.

PT 2)

If $(X_\alpha \to X)_{\alpha\in A}$ belongs to $\mathrm{Cov}(X)$, and if for each $\alpha \in A$ the family

$$(X_{\beta_\alpha} \to X_\alpha)_{\beta_\alpha \in B_\alpha}$$

belongs to $\mathrm{Cov}(X_\alpha)$, then the family $(X_\gamma \to X)_{\gamma\in\coprod_{\alpha\in A} B_\alpha}$, where for $\gamma = (\alpha, \beta_\alpha)$ the morphism $X_\gamma \to X$ is the composite

$$X_\gamma = X_{\beta_\alpha} \to X_\alpha \to X,$$

belongs to $\mathrm{Cov}(X)$. This is stability under composition.

PT 3)

The family $X \xrightarrow{\mathrm{id}_X} X$ belongs to $\mathrm{Cov}(X)$.

1.3.1

Definition 1.1.6 allows us, for a given *pretopology* on C, to consider the *topology* on C generated by this pretopology. Note that if C is a category in which fiber products are representable, then every topology T on C can be defined by a pretopology, namely the one for which $\mathrm{Cov}(X)$ consists of *all* families covering X for the topology T.

Proposition 1.4. Let C be a category, let E be a pretopology on C, let T be the topology defined by the pretopology E (1.1.6), and let X be an object of C. Denote by $J_E(X)$ the set of sieves on X generated by the families of the pretopology, and by $J_T(X)$ the set of refinements of X for the topology T. Then $J_E(X)$ is cofinal in $J_T(X)$. In other words, for a sieve R on X to belong to $J_T(X)$, it is necessary and sufficient that there exist a sieve $R' \in J_E(X)$ such that $R' \subset R$.

Proof. For each object X of C, let $J'(X)$ be the set of sieves on X which contain a sieve of $J_E(X)$. Evidently $J'(X) \subset J_T(X)$. To show that $J'(X) = J_T(X)$, it is enough to show that the data of the $J'(X)$ define a topology on C. Now the $J'(X)$ plainly satisfy axioms (T1) and (T3). It remains to verify (T2). For this it is enough to prove that, if R' is a subsieve of $R \in J_E(X)$ such that for every $Y \to R$ the sieve $R' \times_X Y$ on Y belongs to $J'(Y)$, then R' belongs to $J'(X)$. The sieve R is generated by a family $(X_\alpha \to X)$ belonging to $\mathrm{Cov}(X)$, and for every α the sieve $R' \times_X X_\alpha$ contains a sieve generated by a family $(X_{\beta_\alpha} \to X_\alpha)$ belonging to $\mathrm{Cov}(X_\alpha)$. It follows that R' contains a sieve generated by the family $(X_{\beta_\alpha} \to X)$. Hence, by axiom (PT2), R' contains a sieve of $J_E(X)$, and therefore belongs to $J'(X)$. □

2 Sheaves of sets

Definition 2.1. Let C be a site whose underlying category is a $\mathcal{U}$-category. A presheaf F with values in $\mathcal{U}$-**Set** is called *separated* (respectively, a *sheaf*) if, for every sieve R covering an object X of C, the map

$$\mathrm{Hom}_{\widehat{C}}(X, F) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F)$$

is injective (respectively, bijective). The full subcategory of $\widehat{C}$ whose objects are the sheaves is called the category of sheaves of sets on C, and is most often denoted $\widetilde{C}$. When no ambiguity can result, we shall simply say the category of sheaves on C; conversely, we shall sometimes be more precise and say the category of sheaves with values in $\mathcal{U}$-**Set**, or the category of $\mathcal{U}$-sheaves.

Proposition 2.2. Let C be a $\mathcal{U}$-category, and let $\mathcal{F} = (F_i)_{i\in I}$ be a family of $\mathcal{U}$-presheaves on C. For each object X of C, denote by $J_{\mathcal{F}}(X)$ the set of sieves $R \to X$ such that, for every morphism $Y \to X$ of C with target X, the sieve $R \times_X Y$ has the following property: the map

$$\mathrm{Hom}_{\widehat{C}}(Y, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R \times_X Y, F_i)$$

is bijective (respectively, injective) for every $i \in I$. Then the sets $J_{\mathcal{F}}(X)$ define a topology on C, which is the finest topology for which each of the F_i is a sheaf (respectively, a separated presheaf).

Proof. The $J_{\mathcal{F}}(X)$ plainly satisfy axioms (T1) and (T3). It remains to show that they satisfy (T2). For this it is enough to show that:

1) if $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X, with $R \in J_{\mathcal{F}}(X)$, and such that for every $Y \to R$, where Y is an object of C, the sieve $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$, then R' belongs to $J_{\mathcal{F}}(X)$;

2) if $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X such that $R' \in J_{\mathcal{F}}(X)$, then R belongs to $J_{\mathcal{F}}(X)$.

In case 2), note that for every $Y \to R$, where Y is an object of C, the sieve $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$. In cases 1) and 2), the sieve R is the inductive limit of the objects Y of C lying over R (I 3.4). Inductive limits in $\widehat{C}$ are universal (I 3.3). Consequently, in cases 1) and 2), R' is the inductive limit of the $R' \times_X Y$. But in cases 1) and 2), $R' \times_X Y$ belongs to $J_{\mathcal{F}}(Y)$. Passing to the inductive limit over the objects Y of C lying over R, one deduces that the map

$$\mathrm{Hom}_{\widehat{C}}(R, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R', F_i)$$

is bijective (respectively, injective) for every $i \in I$. It follows that the maps

$$\mathrm{Hom}_{\widehat{C}}(X, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R', F_i), \qquad \mathrm{Hom}_{\widehat{C}}(X, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F_i)$$

are, in cases 1) and 2), bijections (respectively, injections). Moreover, hypotheses 1) and 2) are visibly stable under arbitrary base change $Y \to X$, with Y an object of C. Hence, for every object Y of C lying over X, and every $i \in I$, the maps

$$\begin{gathered}\mathrm{Hom}_{\widehat{C}}(Y, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R \times_X Y, F_i),\\ \mathrm{Hom}_{\widehat{C}}(Y, F_i) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R' \times_X Y, F_i)\end{gathered}$$

are in both cases bijections (respectively, injections). This shows that R' and R belong to $J_{\mathcal{F}}(X)$. □

Corollary 2.3. Let C be a $\mathcal{U}$-category, and for every object X of C let $K(X)$ be a set of sieves on X. Assume that the $K(X)$ satisfy axiom (T1) of (1.1). For a presheaf F to be a sheaf (respectively, a separated presheaf) for the topology generated (1.1.6) by the $K(X)$, it is necessary and sufficient that, for every object X of C and every sieve $R \in K(X)$, the map

$$\mathrm{Hom}_{\widehat{C}}(X, F) \longrightarrow \mathrm{Hom}_{\widehat{C}}(R, F)$$

be bijective (respectively, injective).

Corollary 2.4. In particular, let C be a $\mathcal{U}$-category endowed with a pretopology. For a presheaf F to be a sheaf (respectively, a separated presheaf), it is necessary and sufficient that, for every object X of C and every family $(X_\alpha \to X)$ belonging to $\mathrm{Cov}(X)$, the diagram of sets

$$F(X) \longrightarrow \prod_{\alpha \in A} F(X_\alpha) \rightrightarrows \prod_{(\alpha,\beta)\in A\times A} F(X_\alpha \times_X X_\beta)$$

be exact (respectively, that the map

$$F(X) \longrightarrow \prod_{\alpha\in A} F(X_\alpha)$$

be injective).

Proof. Apply 2.3, then I 3.5 and I 2.12.

With Corollary 2.4 one recovers the definition given in [1].

Definition 2.5. Let C be a $\mathcal{U}$-category. The *canonical topology* of C is the finest topology for which the representable functors are sheaves (2.2). A sieve covering X for the canonical topology will be called a *universally strict epimorphic sieve.* A covering family for the canonical topology will be called a *universally strict epimorphic family.* When, moreover, the morphisms of the covering family are base-changeable, the family will be called *universally effective epimorphic* [2].

Remark 2.5.1. For almost all the sites that have had to be used up to now, the topology is *less fine* than the canonical topology; in other words, the representable functors on C are sheaves, i.e. the covering families of C are universally strict epimorphic. The only exception to this rule is the site $\widehat{C}$ studied in no. 5, whose topology is *finer*, and most often strictly finer, than the canonical topology.

Proposition 2.6. For a sieve $R \hookrightarrow X$ to be universally strict epimorphic, it is necessary and sufficient that, for every object $Y \to X$ of C lying over X, and for every object Z of C, the map

$$\mathrm{Hom}_C(Y,Z) \longrightarrow \varprojlim_{C/(Y\times_X R)} \mathrm{Hom}_C(\,\cdot\,,Z)$$

be a bijection.

Proof. Immediate by applying 2.2 and then I 5.3. □

Remarks 2.7.

1) Proposition 2.6 gives a characterization of universally strict epimorphic sieves of a category C which is independent of the universe in which the presheaves take their values, under the sole condition that the Hom-sets $\mathrm{Hom}(X,Y)$ of C belong to this universe. Thus it allows one to define the canonical topology for any category C, without needing to specify the universes.

2) Let C be a site whose underlying category is a $\mathcal{U}$-category, let F be a $\mathcal{U}$-presheaf, and let $\mathcal{V}$ be a universe containing $\mathcal{U}$. The underlying category of C is a $\mathcal{V}$-category, and F may be regarded as a $\mathcal{V}$-presheaf. The $\mathcal{U}$-presheaf F is a sheaf (respectively, a separated presheaf) if and only if the $\mathcal{V}$-presheaf F is a sheaf (respectively, a separated presheaf). In other

words, the condition for a presheaf F to be a sheaf (respectively, a separated presheaf) does not depend, in the sense just specified, on the universe in which the presheaf F takes its values. In particular, let C be a site and let F be a $\mathcal{U}$-presheaf; one says that F is a sheaf (respectively, a separated presheaf) if there exists a universe $\mathcal{V}$ containing $\mathcal{U}$ such that the category C is a $\mathcal{V}$-category and such that F is a $\mathcal{V}$-sheaf (respectively, a separated $\mathcal{V}$-presheaf). This property does not depend on the universe $\mathcal{V}$.

Exposé II. Topologies and Sheaves

Sections 3–6.9

J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé II, Sections 3–6.9. It continues immediately after Remarks 2.7 and completes Exposé II. The range corresponds to source lines 422–2114 of the Orgogozo–Laszlo TeX snapshot. I write $\widehat{C}$ for the category of presheaves on C, and $\widetilde{C}$ for the category of sheaves on C. Terminology is normalized to current English usage: *sieve*, *covering morphism*, *bicovering morphism*, *associated sheaf* or *sheafification*, and *base-changeable* for French *quarrable*.

3 The sheaf associated to a presheaf

Definition 3.0.1. Let C be a site. A *topologically generating family* of C, or simply a *generating family* when no confusion can result, is a set G of objects of C such that every object of C is the target of a covering family of morphisms of C (1.2) whose sources are elements of G.

Definition 3.0.2. Let $\mathcal{U}$ be a universe. A $\mathcal{U}$*-site* is a site C whose underlying category is a $\mathcal{U}$-category (Exposé I, 1.1), and which has a small topologically generating family. If C is a $\mathcal{U}$-category, a $\mathcal{U}$*-topology* on C is a topology on C making C into a $\mathcal{U}$-site. A site C is said to be $\mathcal{U}$*-small*, or, by abuse of language, *small*, if the underlying category of C is small (Exposé I, 1.0).

3.0.3

It follows immediately from the definitions that every topology finer than a $\mathcal{U}$-topology is again a $\mathcal{U}$-topology, and that a small site is a $\mathcal{U}$-site.

Proposition 3.0.4. Let C be a $\mathcal{U}$-site and let G be a small topologically generating family of C. For $X \in \mathrm{Ob}(C)$, let $J_G(X)$ denote the set of covering sieves on X generated by a family of morphisms

$$(Y_\alpha \xrightarrow{u_\alpha} X)_{\alpha\in A}, \qquad Y_\alpha \in G.$$

Then:

1) the set $J_G(X)$ is small;

2) $J_G(X)$ is cofinal in the set $J(X)$ of all covering sieves on X, ordered by inclusion;

3) for every $R \in J_G(X)$, there is a small epimorphic family $(u_\alpha : Y \to R)_{\alpha\in A}$ with $Y \in G$ (Exposé I, 10.2).

Proof. For (1), put

$$A(X) = \coprod_{Y\in G} \mathrm{Hom}(Y, X).$$

The set $A(X)$ is small (Exposé I, 0), and $\operatorname{card} J_G(X) \le 2^{\operatorname{card} A(X)}$. Hence $J_G(X)$ is small.

For (2), let $R \in J(X)$. Put
$$A(R) = \coprod_{Y \in G} \mathrm{Hom}(Y, R),$$
and let R' be the sieve on X generated by the family
$$Y \xrightarrow{u} R \hookrightarrow X, \qquad u \in A(R).$$
We have $R' \subset R$, and it remains only to prove that R' is covering. By axiom (T2) of 1.1, it is enough to show that, for every morphism $Z \to R$ with $Z \in \mathrm{Ob}(C)$, the sieve $R' \times_X Z$ on Z is covering. But this sieve contains the sieve generated by the family of all morphisms $Y \to Z$ with $Y \in G$, which is covering by the hypothesis on G. Hence $R' \times_X Z$ is covering, again by axiom (T2).

For (3), let $R \in J_G(X)$. The family
$$(Y \xrightarrow{u} R), \qquad u \in A(R) = \coprod_{Y \in G} \mathrm{Hom}(Y, R),$$
is epimorphic by construction. For every $Y \in \mathrm{Ob}(C)$, $\mathrm{Hom}(Y, R)$ is contained in $\mathrm{Hom}(Y, X)$, which is small. Hence $A(R)$ is small. $\square$

3.0.5

Let C be a $\mathcal{U}$-site, let $\mathcal{V}$ be a universe such that $C \in \mathcal{V}$ and $\mathcal{U} \subset \mathcal{V}$, and let G be a $\mathcal{U}$-small topologically generating family of C. The category $\widehat{C}$ of presheaves of $\mathcal{U}$-sets on C is a $\mathcal{V}$-category (Exposé I, 1.1.1). Let X be an object of C. The set $J(X)$ of covering sieves on X, ordered by inclusion, is $\mathcal{V}$-small and cofiltered (1.1.1). For every $\mathcal{U}$-presheaf F, the inductive limit
$$\varinjlim_{R \in J(X)} \mathrm{Hom}_{\widehat{C}}(R, F)$$
is therefore representable by an element of $\mathcal{V}$ (Exposé I, 2.4.1). Moreover, by 3.0.4(3), for every $R \in J_G(X)$, the set $\mathrm{Hom}_{\widehat{C}}(R, F)$ is $\mathcal{U}$-small; since $J_G(X)$ is a $\mathcal{U}$-small cofinal subset of $J(X)$ (loc. cit.), it follows from Exposé I, 2.4.2 that
$$\varinjlim_{R \in J(X)} \mathrm{Hom}_{\widehat{C}}(R, F)$$
is $\mathcal{U}$-small. We choose, for each F and X, an element of $\mathcal{U}$ representing this inductive limit, and put
$$LF(X) = \varinjlim_{R \in J(X)} \mathrm{Hom}_{\widehat{C}}(R, F).$$
If $g : Y \to X$ is a morphism of C, the base-change functor $g^* : J(X) \to J(Y)$ defines a map
$$LF(g) : LF(X) \longrightarrow LF(Y),$$
making $X \mapsto LF(X)$ into a presheaf on C.

Since the morphism $\mathrm{id}_X : X \to X$ is an element of $J(X)$, for every object X of C one has a map
$$\ell(F)(X) : F(X) \longrightarrow LF(X),$$

and these maps plainly define a morphism of presheaves

$$\ell(F) : F \longrightarrow LF.$$

It is also clear that $F \mapsto LF$ is functorial in F, and that the morphisms $\ell(F)$ define a natural transformation

$$\ell : \mathrm{Id} \longrightarrow L.$$

Finally, let $R \hookrightarrow X$ be a refinement of X. The definition of $LF(X)$ and Exposé I, 1.4 give a map

$$Z_R : \mathrm{Hom}_{\widehat{C}}(R, F) \longrightarrow \mathrm{Hom}_{\widehat{C}}(X, LF).$$

For every morphism $g : Y \to X$ of C, the definition of LF shows that the square

$$\begin{array}{ccc} \mathrm{Hom}_{\widehat{C}}(R, F) & \xrightarrow{Z_R} & \mathrm{Hom}_{\widehat{C}}(X, LF) \\ \downarrow & & \downarrow \\ \mathrm{Hom}_{\widehat{C}}(R \times_X Y, F) & \xrightarrow{Z_{R\times_X Y}} & \mathrm{Hom}_{\widehat{C}}(Y, LF) \end{array} \tag{*}$$

is commutative, where the vertical arrows are the evident ones.

We now copy the standard argument of Demazure.

Lemma 3.1.

1) For every refinement $i_R : R \hookrightarrow X$ and every $u : R \to F$, the diagram

$$\begin{array}{ccc} R & \xrightarrow{i_R} & X \\ u \downarrow & & \downarrow Z_R(u) \\ F & \xrightarrow{\ell(F)} & LF \end{array} \tag{**}$$

is commutative.

2) For every morphism $v : X \to LF$, there exists a refinement R of X and a morphism $u : R \to F$ such that $Z_R(u) = v$.

3) Let Y be an object of C, and let $u, v : Y \rightrightarrows F$ be two morphisms such that $\ell(F)\circ u = \ell(F)\circ v$. Then the equalizer, or kernel, of the pair (u, v) is a refinement of Y.

4) Let R and R' be two refinements of X, and let $u : R \to F$ and $u' : R' \to F$ be two morphisms. Then $Z_R(u) = Z_{R'}(u')$ if and only if u and u' coincide on some refinement

$$R'' \hookrightarrow R \times_X R'.$$

Proof. Only assertion (1) is nontrivial. We must prove

$$Z_R(u) \circ i_R = \ell(F) \circ u.$$

By Exposé I, 3.4 it is enough to prove equality after composition with every morphism $g : Y \to R$, where Y is an object of C. Let $f = i_R \circ g$, and consider $R \times_X Y$. The canonical monomorphism $R \times_X Y \to Y$ has a section, hence is an isomorphism by 1.3(3). By definition of $\ell(F)$, the morphism $Z_{R\times_X Y}(u \circ g')$ is $\ell(F) \circ u \circ g$. On the other hand, the commutativity of $(*)$ gives

$$Z_{R\times_X Y}(u \circ g') = Z_R(u) \circ f.$$

Thus $\ell(F) \circ u \circ g = Z_R(u) \circ i_R \circ g$, as required. □

Proposition 3.2.

1) The functor L is left exact (Exposé I, 2.3.2).

2) For every presheaf F, LF is a separated presheaf.

3) A presheaf F is separated if and only if the morphism $\ell(F) : F \to LF$ is a monomorphism. In that case LF is a sheaf.

4) The following properties are equivalent:

   (i) $\ell(F) : F \to LF$ is an isomorphism;

   (ii) F is a sheaf.

Proof. For (1), it is enough, by Exposé I, 3.1, to prove that for every object X of C, the functor $F \mapsto LF(X)$ commutes with finite projective limits. But for every $R \hookrightarrow X$ in $J(X)$, the functor

$$F \longmapsto \mathrm{Hom}_{\widehat{C}}(R, F)$$

commutes with projective limits, by the definition of projective limit; and the filtered inductive limit $\varinjlim_{J(X)}$ commutes with finite projective limits because $J(X)$ is a cofiltered set (Exposé I, 2.7).

For (2), let X be an object of C, and let $f, g : X \rightrightarrows LF$ be two morphisms which coincide on a refinement $R \hookrightarrow X$. By Lemma 3.1(2), there is a covering sieve $R' \hookrightarrow X$, which we may suppose contained in R, and two morphisms $u, v : R' \rightrightarrows F$ such that $Z_{R'}(u) = f$ and $Z_{R'}(v) = g$. By Lemma 3.1(1), $\ell(F) \circ u = \ell(F) \circ v$. Hence, by Lemma 3.1(4), u and v coincide on a refinement $R'' \hookrightarrow R'$. If w denotes the restriction of u to R'', then

$$Z_{R''}(w) = Z_{R'}(u) = Z_{R'}(v),$$

and consequently $f = g$. Thus LF is separated.

For (3), if F is separated, then $\ell(F)$ is a monomorphism, because a filtered inductive limit of monomorphisms is a monomorphism. Conversely, if $\ell(F)$ is a monomorphism, then F is a subpresheaf of a separated presheaf, hence is separated. We now prove that LF is then a sheaf. Let $i : R \hookrightarrow X$ be a covering sieve of an object X of C, and let $u : R \to LF$ be a morphism. It is enough to prove that u factors through X. Put

$$R' = F \times_{LF} R, \qquad v = Z_{R'}(\mathrm{pr}_1).$$

To prove $u = v \circ i$, it is enough, by Exposé I, 3.4, to test after every morphism $m : Y \to R$ with Y an object of C. Put $R'' = Y \times_R R'$, with projections p_1, p_2. The projection $R'' \to Y$ is a monomorphism making R'' into a covering sieve of Y by Lemma 3.1(2). We have

$$v \circ i \circ p_2 = \ell(F) \circ \mathrm{pr}_1$$

by Lemma 3.1(1), and hence

$$v \circ i \circ p_2 = u \circ m \circ p_2.$$

Pulling back along p_1 gives

$$v \circ i \circ m \circ p_2 = u \circ m \circ p_2.$$

Since LF is separated, $v \circ i \circ m = u \circ m$. This proves the sheaf condition.

Assertion (4) is clear from (3). $\square$

Remark 3.3. Let $J'(X)$ be a cofinal subset of $J(X)$. Then

$$LF(X) = \varinjlim_{R \in J'(X)} \mathrm{Hom}_{\widehat{C}}(R, F).$$

In particular, if the topology of C is defined by a pretopology $X \mapsto \mathrm{Cov}(X)$ (1.1.3), then the functor L can be described by means of the covering families of $\mathrm{Cov}(X)$ (Exposé I, 2.12 and 3.5). Making the formulae explicit, one recovers the usual construction.

Theorem 3.4. Let C be a $\mathcal{U}$-site. The inclusion functor

$$i : \widetilde{C} \hookrightarrow \widehat{C}$$

from sheaves to presheaves has a left adjoint

$$\underline{a} : \widehat{C} \longrightarrow \widetilde{C},$$

which is left exact (Exposé I, 2.3.2). The functor $i \circ \underline{a}$ is canonically isomorphic to $L \circ L$ (cf. 3.0.5). For every presheaf F, the adjunction morphism

$$F \longrightarrow i\,\underline{a}(F)$$

is, through the preceding isomorphism, the composite

$$F \xrightarrow{\ell(F)} LF \xrightarrow{\ell(LF)} L(LF).$$

Definition 3.5. The sheaf $\underline{a}F$ is called the *sheaf associated to the presheaf* F, or the *sheafification* of F.

Theorem 3.4 follows immediately from Proposition 3.2.

Proposition 3.6. Let C be a $\mathcal{U}$-site and let $\mathcal{V} \supset \mathcal{U}$ be a universe. Denote by

$$\widehat{C}_{\mathcal{U}}, \quad \widetilde{C}_{\mathcal{U}}$$

the categories of $\mathcal{U}$-presheaves and $\mathcal{U}$-sheaves, and by

$$\widehat{C}_{\mathcal{V}}, \quad \widetilde{C}_{\mathcal{V}}$$

the categories of $\mathcal{V}$-presheaves and $\mathcal{V}$-sheaves. Let

$$\underline{a}_{\mathcal{U}} : \widehat{C}_{\mathcal{U}} \to \widetilde{C}_{\mathcal{U}}, \qquad \underline{a}_{\mathcal{V}} : \widehat{C}_{\mathcal{V}} \to \widetilde{C}_{\mathcal{V}}$$

be the corresponding associated-sheaf functors. Then the diagram

$$\begin{array}{ccc} \widehat{C}_{\mathcal{U}} & \xrightarrow{\underline{a}_{\mathcal{U}}} & \widetilde{C}_{\mathcal{U}} \\ \downarrow & & \downarrow \\ \widehat{C}_{\mathcal{V}} & \xrightarrow{\underline{a}_{\mathcal{V}}} & \widetilde{C}_{\mathcal{V}} \end{array}$$

where the vertical functors are the canonical inclusions, commutes up to canonical isomorphism.

Proof. This follows from the construction of the functors $\underline{a}_{\mathcal{U}}$ and $\underline{a}_{\mathcal{V}}$ (3.4 and 3.0.5). $\square$

4 Exactness properties of the category of sheaves

The exactness properties of the category of sheaves are deduced from those of the category of presheaves by means of Theorem 3.4. This section makes that philosophy explicit in a number of typical statements which are among the most useful.

Theorem 4.1. Let C be a $\mathcal{U}$-site, let $\widetilde{C}$ be the category of sheaves, let

$$\underline{a} : \widehat{C} \longrightarrow \widetilde{C}$$

be the associated-sheaf functor, and let

$$i : \widetilde{C} \longrightarrow \widehat{C}$$

be the inclusion functor.

1) The functor $\underline{a}$ commutes with inductive limits and is exact.

2) $\mathcal{U}$-inductive limits in $\widetilde{C}$ are representable. For every small category I and every functor $E : I \to \widetilde{C}$, the canonical morphism

$$\varinjlim_{I} E \longrightarrow \underline{a}\left(\varinjlim_{I} iE\right)$$

is an isomorphism.

3) $\mathcal{U}$-projective limits in $\widetilde{C}$ are representable. For every object X of C, the functor $F \mapsto F(X)$ on $\widetilde{C}$ commutes with projective limits; equivalently, the inclusion $i : \widetilde{C} \to \widehat{C}$ commutes with projective limits.

Proof. These properties follow essentially from Theorem 3.4 and Exposé I, 2.11. □

Thus, in the category of sheaves, products indexed by an element of $\mathcal{U}$, fiber products, sums indexed by an element of $\mathcal{U}$, amalgamated sums, kernels, cokernels, images, and coimages are representable.

Corollary 4.1.1. Let C be a $\mathcal{U}$-site and let F be a sheaf of sets on C. The canonical homomorphism

$$\varinjlim_{C/F} \underline{a}X \longrightarrow F$$

is an isomorphism.

Proof. This follows from Exposé I, 3.4 and the fact that $\underline{a}$ commutes with inductive limits. □

Proposition 4.2. Every morphism of $\widetilde{C}$ which is both an epimorphism and a monomorphism is an isomorphism.

Proof. Let $u : G \to H$ be a morphism of $\widetilde{C}$ which is both an epimorphism and a monomorphism. First note that u is a monomorphism of presheaves (4.1(1)). Form the amalgamated sum

$H \amalg_G H = K$ and the two canonical morphisms $i_1, i_2 : H \rightrightarrows K$ in the category of presheaves. Since u is a monomorphism of presheaves, the square

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ u \downarrow & & \downarrow i_2 \\ H & \xrightarrow{i_1} & K \end{array}$$

is immediately verified to be both cartesian and cocartesian (Exposé I, 10.1). Applying the associated-sheaf functor, we get a cartesian and cocartesian square in the category of sheaves:

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ u \downarrow & & \downarrow \underline{a}i_2 \\ H & \xrightarrow{\underline{a}i_1} & \underline{a}K. \end{array} \tag{*}$$

Since u is an epimorphism of sheaves, $\underline{a}i_1$ is an isomorphism; since the square $(*)$ is cartesian, u is an isomorphism. ☐

Proposition 4.3.

1) Representable inductive limits in $\widetilde{C}$ are universal (Exposé I, 2.5).

2) Every epimorphic family of morphisms is an effective universal epimorphic family (2.6).

3) Every equivalence relation is effective and universal (Exposé I, 10.6).

4) $\mathcal{U}$-filtered inductive limits commute with finite projective limits (Exposé I, 2.6).

Proof. Assertions (1)–(4) hold in the category of sets and hence in the category of presheaves of sets (Exposé I, 3.1). Assertions (1) and (4) then follow at once from the corresponding assertions for presheaves and from Theorem 4.1.

We prove (3). Let $p_1, p_2 : R \rightrightarrows X$ be an equivalence relation in $\widetilde{C}$. It is also an equivalence relation of presheaves (4.1.1). Hence there exists a morphism of presheaves $u : X \to Y$ such that

$$\begin{array}{ccc} R & \xrightarrow{p_2} & X \\ p_1 \downarrow & & \downarrow u \\ X & \xrightarrow{u} & Y \end{array}$$

is both cartesian and cocartesian in the category of presheaves. Applying the associated-sheaf functor, one obtains a cartesian and cocartesian square in the category of sheaves (4.1(1)). Thus $R \rightrightarrows X$ is an effective equivalence relation. Since every equivalence relation is effective by the preceding argument, every equivalence relation is universally effective.

We prove (2). Let $(u_i : X_i \to X)_{i \in I}$ be an epimorphic family in $\widetilde{C}$. By 2.6 and Exposé I, 2.12, it is enough to prove:

a) for every morphism of sheaves $v : Y \to X$, the family

$$\mathrm{pr}_{2,i} : X_i \times_X Y \to Y, \qquad i \in I,$$

is epimorphic in $\widetilde{C}$;

b) for every sheaf Z, the diagram of sets

$$\mathrm{Hom}(X,Z) \longrightarrow \prod_{i\in I}\mathrm{Hom}(X_i,Z) \rightrightarrows \prod_{(i,k)\in I\times I}\mathrm{Hom}(X_i\times_X X_k, Z)$$

is exact.

Let X' be the union of the images of the morphisms u_i, in the sense of presheaves, and let $j : X' \hookrightarrow X$ be the canonical injection. For every i, the morphism $u_i : X_i \to X$ factors as $X_i \xrightarrow{u_i'} X' \xrightarrow{j} X$, and the family $u_i' : X_i \to X'$ is epimorphic in $\widehat{C}$. Since j is a monomorphism, the morphism $\underline{a}j : \underline{a}X' \to X$ is a monomorphism of sheaves (4.1). Since $u_i = \underline{a}j\,\underline{a}u_i'$ for each i, $\underline{a}j$ is also an epimorphism of sheaves. Hence $\underline{a}j$ is an isomorphism by Proposition 4.2.

Let $v : Y \to X$ be a morphism of sheaves. After base change we obtain a morphism $j_v : X' \times_X Y \to Y$ and morphisms $u_{i,v}' : X_i \times_X Y \to X' \times_X Y$. Since $\underline{a}$ commutes with fiber products, $\underline{a}j_v$ is an isomorphism. Since epimorphic families of presheaves remain epimorphic after base change, the family $u_{i,v}'$ is epimorphic in $\widehat{C}$. As $\underline{a}$ commutes with inductive limits, the family

$$\underline{a}u_{i,v}' : X_i \times_X Y \to \underline{a}(X' \times_X Y)$$

is epimorphic in $\widetilde{C}$. This proves (a).

For (b), let Z be a sheaf. Since $\underline{a}j$ is an isomorphism, the map

$$\mathrm{Hom}(X,Z) \longrightarrow \mathrm{Hom}(X',Z)$$

is a bijection. Since the u_i' form an epimorphic family in $\widehat{C}$, the diagram

$$\mathrm{Hom}(X',Z) \longrightarrow \prod_{i\in I}\mathrm{Hom}(X_i,Z) \rightrightarrows \prod_{(i,k)\in I\times I}\mathrm{Hom}(X_i\times_{X'} X_k, Z)$$

is exact. Finally, since X' is a subpresheaf of X, the presheaf $X_i \times_{X'} X_k$ is canonically isomorphic to $X_i \times_X X_k$. This gives (b). $\square$

Remarks 4.3.1.

1) Proposition 4.3(2) formally gives a second proof of Proposition 4.2. In any category, a morphism which is both an effective universal epimorphism and a monomorphism is plainly an isomorphism.

2) From the preceding propositions it follows that every morphism in the category of sheaves factors uniquely as an effective epimorphism followed by an effective monomorphism. For non-additive categories, this property is the natural analogue of axiom (AB2) for abelian categories, namely coim $\simeq$ im [Tohoku].

4.4.0

Let C be a $\mathcal{U}$-site. The functor $h : C \to \widehat{C}$ (Exposé I, 1.3.1), composed with the associated-sheaf functor, gives a functor

$$\epsilon_C : C \longrightarrow \widetilde{C},$$

called the *canonical functor* from C to $\widetilde{C}$. It will be used constantly below. The functor ϵ_C commutes with finite projective limits. When the topology of C is less fine than the canonical topology, ϵ_C is fully faithful and commutes with projective limits; in that case it is, moreover, defined even if C need not have a small topologically generating family. We shall not study in detail the behavior of ϵ_C with respect to inductive limits (cf. SGA 3, IV). We shall, however, need the following propositions.

Theorem 4.4. Let C be a $\mathcal{U}$-site and let $(u_i : X_i \to X)_{i \in I}$ be a family of morphisms of C with target X. The following conditions are equivalent:[1]

(i) The family
$$(\epsilon_C u_i : \epsilon_C X_i \to \epsilon_C X)_{i \in I}$$
is an epimorphic family in $\widetilde{C}$ (Exposé I, 10.2).

(ii) The family $(u_i : X_i \to X)_{i \in I}$ is a covering family of C (1.2).

Proof. (ii) $\Rightarrow$ (i). Let $R \hookrightarrow X$ be the sieve generated by the u_i (Exposé I, 4.3.3), and let $u_i' : X_i \to R$ be the morphisms induced by the u_i. The family u_i' is epimorphic in $\widehat{C}$, and R is a covering sieve on X. Hence, for every sheaf F, the map
$$\mathrm{Hom}(X, F) \longrightarrow \mathrm{Hom}(R, F)$$
is bijective, while
$$\mathrm{Hom}(R, F) \longrightarrow \prod_i \mathrm{Hom}(X_i, F)$$
is injective. Thus
$$\mathrm{Hom}(X, F) \longrightarrow \prod_i \mathrm{Hom}(X_i, F)$$
is injective, and consequently
$$\mathrm{Hom}(\underline{a}X, F) \longrightarrow \prod_i \mathrm{Hom}(\underline{a}X_i, F)$$
is injective. This is exactly (i).

(i) $\Rightarrow$ (ii). With the notation just introduced, let $J_R : R \hookrightarrow X$ be the canonical inclusion of the sieve generated by the u_i. From (i) it follows that the morphism of sheaves
$$\underline{a}J_R : \underline{a}R \longrightarrow \underline{a}X$$
is both an epimorphism and a monomorphism. Hence it is an isomorphism by Proposition 4.2. With the notation of Section 3, we have a commutative diagram
$$\begin{array}{ccccc} X & \xrightarrow{\ell(X)} & LX & \xrightarrow{\ell(LX)} & \underline{a}X \\ \uparrow J_R & & \uparrow LJ_R & & \uparrow \wr \\ R & \xrightarrow{\ell(R)} & LR & \xrightarrow{\ell(LR)} & \underline{a}R. \end{array}$$

[1] When the site C need not have a small topologically generating family and when the topology of C is less fine than the canonical topology, one has (ii) $\Rightarrow$ (i); see the proof of 4.4.

By Lemma 3.1(2), there exists a covering sieve $J_{S_1} : S_1 \to X$ and a morphism $u_1 : S_1 \to LR$ such that

$$\ell(LX) \circ \ell(X) \circ J_{S_1} = \underline{a}J_R \circ \ell(LR) \circ u_1 = \ell(LX) \circ LJ_R \circ u_1. \tag{4.6.1}$$

Put $m = \ell(X) \circ J_{S_1}$ and $n = LJ_R \circ u_1$. These are morphisms $S_1 \to LX$. Let $S_2 \hookrightarrow S_1$ be the equalizer of the pair (m, n). For every object Y of C and every morphism $\alpha : Y \to S_1$, equation (4.6.1) gives

$$\ell(LX)m\alpha = \ell(LX)n\alpha.$$

By Lemma 3.1(3), the equalizer $S_2 \times_{S_1} Y$ of $m\alpha$ and $n\alpha$ is a covering sieve on Y. Hence, by axiom (T2), $J_{S_2} : S_2 \hookrightarrow X$ is a covering sieve on X. Thus we have a morphism $u_2 : S_2 \to LR$ and a commutative diagram

$$\begin{array}{ccc} S_2 & \xrightarrow{J_{S_2}} & X \\ {\scriptstyle u_2}\downarrow & & \downarrow{\scriptstyle \ell(X)} \\ LR & \xrightarrow{LJ_R} & LX \end{array} \tag{4.6.2}$$

with $R \hookrightarrow X$ inserted in the evident way through J_R.

Let Y be an object of C and let $\beta : Y \to S_2$ be a morphism. By Lemma 3.1(1), there exists a covering sieve $J_{Q_1} : Q_1 \hookrightarrow Y$ and a morphism $v_1 : Q_1 \to R$ such that

$$u_2\beta J_{Q_1} = \ell(R)v_1.$$

Put

$$f = J_{S_2}\beta J_{Q_1}, \qquad g = J_R v_1.$$

Let Q_2 be the equalizer of the pair $f, g : Q_1 \rightrightarrows X$, and let $J_{Q_2} : Q_2 \hookrightarrow Y$ be the canonical inclusion. For every object Z of C and every morphism $\gamma : Z \to Q_1$, the commutativity of (4.6.2) gives

$$\ell(X)f\gamma = LJ_R\, u_2\beta J_{Q_1}\gamma = LJ_R\, \ell(R)v_1\gamma = \ell(X)J_R v_1\gamma = \ell(X)g\gamma.$$

By Lemma 3.1(3), the equalizer $Q_2 \times_{Q_1} Z$ of $f\gamma$ and $g\gamma$ is a covering sieve on Z. By axiom (T2), $J_{Q_2} : Q_2 \hookrightarrow Y$ is a covering sieve on Y. Hence, for every morphism $\beta : Y \to S_2$, there exists a covering sieve $Q_2 \hookrightarrow Y$ and a morphism $v_2 : Q_2 \to R$ making

$$\begin{array}{ccccc} Q_2 & \xrightarrow{v_2} & & & R \\ \downarrow & & & & \downarrow J_R \\ Y & \xrightarrow{\beta} & S_2 & \xrightarrow{J_{S_2}} & X \end{array}$$

commutative.

Now let $S_2 \cap R$ denote the fiber product of S_2 and R over X. The sieve

$$(S_2 \cap R) \times_{S_2} Y$$

on Y contains Q_2; hence it is covering. Axiom (T2) implies that $S_2 \cap R$ is a covering sieve on X. Since R contains $S_2 \cap R$, R is itself a covering sieve on X. □

Corollary 4.4.4. Let T be the topology of the $\mathcal{U}$-site C. Let T' (respectively T'') be the finest topology on C among those for which the objects of $\widetilde{C}$ are sheaves (respectively separated presheaves) (2.2). Then

$$T = T' = T''.$$

Indeed, trivially $T \leq T' \leq T''$, while Theorem 4.4 and Proposition 2.2 imply $T'' \leq T$. $\square$

Definition 4.5. An *initial object* of a category A is an object ϕ_A representing the empty inductive limit, i.e. such that for every $X \in \mathrm{Ob}(A)$ there is exactly one arrow $\phi_A \to X$. An object ϕ_A is called *strict initial* if it is initial and every morphism with target ϕ_A is an isomorphism.

Let $(S_i)_{i\in I}$ be a family of objects of a category A. Suppose that the sum $S = \coprod_{i\in I} S_i$ is representable. The sum S is said to be *disjoint* if the structural morphisms $S_i \to S$ are base-changeable monomorphisms and, for every pair i, j with $i \neq j$, the fiber product $S_i \times_S S_j$ is an initial object of A. The sum is said to be *universally disjoint* if it is disjoint and remains a disjoint sum after every base change $T \to S$. It follows that, for every $i \neq j$, the objects $S_i \times_S S_j$ are strict initial objects of A.

Example 4.5.1. In the category of sets, direct sums are disjoint and universal. Hence the same is true in every category of presheaves of sets (Exposé I, 3.1), and therefore in every category of sheaves of sets on a $\mathcal{U}$-site (4.1). In particular, the initial object of $\widetilde{C}$ is strict.

Proposition 4.6. Let C be a $\mathcal{U}$-category and let $(s_i : X_i \to X)_{i\in I}$ be a family of morphisms of C. For every $\mathcal{U}$-topology $\mathcal{T}$ on C (3.0.2), let $\widetilde{C}_{\mathcal{T}}$ be the corresponding category of sheaves, and let

$$\epsilon_{\mathcal{T}} : C \to \widetilde{C}_{\mathcal{T}}$$

be the corresponding canonical functor.

1) Let $\mathcal{T}$ be a $\mathcal{U}$-topology such that

$$\coprod_{i\in I} \epsilon_{\mathcal{T}}(X_i) \xrightarrow{(\epsilon_{\mathcal{T}}(s_i))} \epsilon_{\mathcal{T}}(X)$$

is an isomorphism. Then, for every topology $\mathcal{T}'$ finer than $\mathcal{T}$, the morphism

$$\coprod_{i\in I} \epsilon_{\mathcal{T}'}(X_i) \xrightarrow{(\epsilon_{\mathcal{T}'}(s_i))} \epsilon_{\mathcal{T}'}(X)$$

is an isomorphism.

2) Let $\mathcal{T}$ be a $\mathcal{U}$-topology on C. The following properties (i) and (ii) are equivalent:

(i) The following three conditions hold:

(a) The family $(s_i : X_i \to X)_{i\in I}$ is covering for $\mathcal{T}$.

(b) For every $i \in I$, the diagonal morphism of presheaves

$$\Delta_i : X_i \hookrightarrow X_i \times_X X_i$$

is transformed by the associated-sheaf functor for $\mathcal{T}$ into an isomorphism; this is automatically the case if the s_i are monomorphisms.

(c) For every pair $i \neq j$, the presheaf $X_i \times_X X_j$ is transformed by the associated-sheaf functor for $\mathcal{T}$ into the initial object of $\widetilde{C}_{\mathcal{T}}$.

(ii) $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$.

Proof. For (1), a sheaf for $\mathcal{T}'$ is a sheaf for $\mathcal{T}$. Hence, for every sheaf F for $\mathcal{T}'$, the morphism

$$\operatorname{Hom}_{\widehat{C}}(X, F) \longrightarrow \prod_{i \in I} \operatorname{Hom}_{\widehat{C}}(X_i, F)$$

is an isomorphism. This implies the assertion.

For (2), by definition property (ii) holds if and only if the morphism of presheaves

$$\Phi = (h(s_i))_{i \in I} : \coprod_i h(X_i) \longrightarrow h(X)$$

is transformed by the associated-sheaf functor into an isomorphism; equivalently, by Proposition 4.2, $\underline{a}(\Phi)$ is both an epimorphism and a monomorphism of sheaves. By Theorem 4.4, $\underline{a}(\Phi)$ is an epimorphism if and only if (a) holds. The diagonal

$$\Delta : \coprod_i h(X_i) \longrightarrow \left(\coprod_i h(X_i)\right) \times_{h(X)} \left(\coprod_i h(X_i)\right)$$

is the direct sum of morphisms $\Delta_{i,j}$, indexed by $I \times I$, defined as follows:

- if $i = j$, $\Delta_{i,i}$ is the diagonal

$$h(X_i) \to h(X_i) \times_{h(X)} h(X_i);$$

- if $i \neq j$, $\Delta_{i,j}$ is the morphism

$$\phi_{\widehat{C}} \to h(X_i) \times_{h(X)} h(X_j),$$

  where $\phi_{\widehat{C}}$ denotes the initial object of $\widehat{C}$.

The morphism $\underline{a}(\Phi)$ is a monomorphism if and only if $\underline{a}(\Delta)$ is an isomorphism; and by Theorem 4.1, $\underline{a}(\Delta)$ is an isomorphism if and only if each $\underline{a}(\Delta_{i,j})$ is an isomorphism, i.e. if and only if (b) and (c) hold. ☐

Corollary 4.6.1. Let C be a $\mathcal{U}$-category and let X be an object of C.

1) Let $\mathcal{T}$ be a $\mathcal{U}$-topology on C. The object X of C is transformed by the associated-sheaf functor for $\mathcal{T}$ into the initial object of $\widetilde{C}_{\mathcal{T}}$ if and only if the empty sieve covers X.

2) When X is a strict initial object of C, the sheaf associated to X, for every $\mathcal{U}$-topology finer than the canonical topology, is an initial object.

Proof. For (1), take the indexing set I in Proposition 4.6(2) to be empty. For (2), by (1) and Proposition 4.6(1), it is enough to prove that the empty sieve covers X for the canonical topology; equivalently, by 2.6, it is enough to prove that every object $Y \to X$ over X is initial, which follows from the definition of strict initial object. ☐

Corollary 4.6.2. Let C be a $\mathcal{U}$-site, and let $(s_i : X_i \to X)_{i \in I}$ be a family of base-changeable morphisms in C, all with target X, satisfying:

α the family s_i is covering;

β each $s_i : X_i \to X$ is a monomorphism;

γ for each pair $i \neq j$, the object $X_i \times_X X_j$ is covered by the empty sieve, i.e. for every sheaf F on C, the set $F(X_i \times_X X_j)$ is a singleton.

Then $\epsilon_C(X)$ is the sum of the $\epsilon_C(X_i)$, $i \in I$.

Proof. This follows immediately from Proposition 4.6(2) and Corollary 4.6.1(1). $\square$

Corollary 4.6.3. Let C be a $\mathcal{U}$-category and let $(s_i : X_i \to X)_{i \in I}$ be a family of base-changeable morphisms with the same target. Suppose that the canonical topology of C is a $\mathcal{U}$-topology. The following conditions are equivalent:

(i) There exists a $\mathcal{U}$-topology $\mathcal{T}$ on C, less fine than the canonical topology, such that $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$, $i \in I$.

(ii) The object X is the disjoint universal sum, in C, of the X_i (4.5).

Proof. (ii) $\Rightarrow$ (i). Since X is the universal sum of the X_i, the family s_i is covering for the canonical topology of C (2.6). Thus condition α of Corollary 4.6.2 holds for the canonical topology. Condition β is immediate, and condition γ follows from Proposition 4.6(1),(2). Hence (i) holds.

(i) $\Rightarrow$ (ii). Let $\mathcal{T}$ be a topology on C, less fine than the canonical topology, such that $\epsilon_{\mathcal{T}}(X)$ is the sum of the $\epsilon_{\mathcal{T}}(X_i)$. For every sheaf F for $\mathcal{T}$, the canonical map

$$\mathrm{Hom}(X, F) \to \prod_i \mathrm{Hom}(X_i, F)$$

is a bijection. Since every representable presheaf is a sheaf, it follows at once that X is the sum in C of the X_i. Applying the same argument when I is empty, we see that if an object of C is transformed into the initial object of $\widetilde{C}_{\mathcal{T}}$, then it is an initial object of C. The functor $\epsilon_{\mathcal{T}} : C \to \widetilde{C}_{\mathcal{T}}$ is fully faithful and commutes with finite projective limits. Hence condition (b) of Corollary 4.6.2 implies that the s_i are monomorphisms in C, and condition (c), by what has just been proved, implies that $X_i \times_X X_j$ is an initial object of C for $i \neq j$. Thus X is a disjoint sum of the X_i. Since $\epsilon_{\mathcal{T}}$ commutes with fiber products, this property is stable under every base change $Y \to X$, and therefore X is the disjoint universal sum of the X_i. $\square$

Corollary 4.6.4. Let C be a $\mathcal{U}$-category and let X be an object of C. Suppose that the canonical topology on C is a $\mathcal{U}$-topology. The following properties are equivalent:

(i) There exists on C a topology less fine than the canonical topology such that $\epsilon_{\mathcal{T}}(X)$ is an initial object of $\widetilde{C}_{\mathcal{T}}$.

(ii) X is a strict initial object of C.

Proof. Take I to be empty in Corollary 4.6.3. $\square$

Proposition 4.7. Let C be a $\mathcal{U}$-category, let $\widetilde{C}$ be the category of sheaves on C for the canonical topology, let

$$\epsilon_C : C \to \widetilde{C}$$

be the canonical functor (4.4.0), and let $R \rightrightarrows X$ be an equivalence relation in C admitting a cokernel Y in C. Let $\pi : X \to Y$ be the canonical morphism and suppose it is base-changeable. Consider:

(i) $\epsilon_C(Y)$ is the quotient of the equivalence relation

$$\epsilon_C(R) \rightrightarrows \epsilon_C(X).$$

(ii) The equivalence relation $R \rightrightarrows X$ is effective and universal (cf. no. 7).

Then (ii) implies (i). If C has a $\mathcal{U}$-topology less fine than the canonical topology, then (i) implies (ii).

Proof. (ii) $\Rightarrow$ (i). The canonical morphism $R \to X \times_Y X$ is an isomorphism. Let $S \to Y$ be the sieve generated by $\pi : X \to Y$. For every presheaf F, the diagram

$$\mathrm{Hom}(S, F) \longrightarrow \mathrm{Hom}(X, F) \rightrightarrows \mathrm{Hom}(R, F)$$

is exact (Exposé I, 2.12). It is therefore enough to show that, for every sheaf F for the canonical topology, the map

$$\mathrm{Hom}(Y, F) \longrightarrow \mathrm{Hom}(S, F)$$

is a bijection; equivalently, that S is covering for the canonical topology, or that the sieve generated by $\pi : X \to Y$ is covering. This follows from Definition 2.5.

(i) $\Rightarrow$ (ii). The functor $\epsilon_C : C \to \widetilde{C}$ commutes with finite projective limits and is fully faithful. Since the canonical morphism

$$\epsilon_C(R) \to \epsilon_C(X) \times_{\epsilon_C(Y)} \epsilon_C(X)$$

is an isomorphism (4.3), the canonical morphism $R \to X \times_Y X$ is an isomorphism. The morphism $\epsilon_C(X) \to \epsilon_C(Y)$ is an epimorphism; hence, by Theorem 4.4, $\pi : X \to Y$ is a covering morphism for the canonical topology. □

Proposition 4.8. Let C be a $\mathcal{U}$-site. The category of sheaves on C has the following properties:

a) finite projective limits are representable;

b) direct sums indexed by an element of $\mathcal{U}$ are representable, and are disjoint and universal (4.5);

c) equivalence relations are effective and universal.

These facts were proved in Theorem 4.1(2),(3), Proposition 4.3(3), and Example 4.5.1.

These properties have been singled out because they will later make it possible to characterize the $\mathcal{U}$-categories equivalent to categories of sheaves on categories belonging to $\mathcal{U}$ (Exposé IV, 1.2).

Remark. Recall that property (a) is equivalent to the following: there exists a final object, and fiber products are representable (Exposé I, 2.3.1).

4.9

Let C be a site whose underlying category is a $\mathcal{U}$-category. Then the category of $\mathcal{U}$-sheaves on C satisfies the conditions considered in Exposé I, 7.3, either (i) or (ii); hence, by loc. cit., the various variants considered in Exposé I, 7.1 for the notion of a *generating family* in C coincide. With this understood:

Proposition 4.10. With the preceding notation, consider the canonical functor

$$\epsilon : C \to \widetilde{C}$$

(4.4.0), and let $G \subset \mathrm{Ob}(C)$ be a topologically generating family in C (3.0.1). Then the family $(\epsilon(X))_{X \in G}$ of objects of $\widetilde{C}$ is a generating family. In particular, the family $(\epsilon(X))_{X \in \mathrm{Ob}(C)}$ is generating.

We must prove that every morphism $u : F \to F'$ in $\widetilde{C}$ such that

$$\mathrm{Hom}(\epsilon(X), F) \longrightarrow \mathrm{Hom}(\epsilon(X), F') \tag{4.10.1}$$

is a bijection for every $X \in G$, is an isomorphism. The preceding map is identified with

$$u(X) : F(X) \longrightarrow F'(X), \tag{4.10.2}$$

and it remains to show that, if this map is bijective for every $X \in G$, then it is bijective for every $X \in \mathrm{Ob}(C)$. Let $(X_i \to X)_{i \in I}$ be a covering family of X by objects of G. For every pair (i, j) in $I \times I$, consider the set of all morphisms

$$X_{ijk} \to X_i \times_X X_j$$

in $\widehat{C}$, with $X_{ijk} \in G$, where k ranges over an index set $I_{i,j}$. We obtain a morphism of exact diagrams of sets

$$\begin{array}{ccccc} F(X) & \to & \prod\limits_i F(X_i) & \rightrightarrows & \prod\limits_{i,j,k} F(X_{ijk}) \\ \downarrow & & \downarrow \wr & & \downarrow \wr \\ F'(X) & \to & \prod\limits_i F'(X_i) & \rightrightarrows & \prod\limits_{i,j,k} F'(X_{ijk}), \end{array}$$

where the last two vertical arrows are bijections by hypothesis. Hence the first vertical arrow is a bijection as well. $\square$

Corollary 4.11. Suppose that C is a $\mathcal{U}$-site (3.0.2). Then $\widetilde{C}$ is a $\mathcal{U}$-category and has a $\mathcal{U}$-small generating family.

Indeed, by hypothesis one may take in Proposition 4.10 a small generating family G, which proves the existence of a small generating family. On the other hand, if $F, F' \in \mathrm{Ob}(\widetilde{C})$, a homomorphism $F \to F'$ is known once the maps (4.10.1), or equivalently (4.10.2), are known for every $X \in G$ (Exposé I, 7.1.1). Hence the map

$$\mathrm{Hom}(F, F') \longrightarrow \prod_{X \in G} \mathrm{Hom}(F(X), F'(X))$$

is injective. Since the right-hand side is $\mathcal{U}$-small, the left-hand side is $\mathcal{U}$-small, which proves that $\widetilde{C}$ is a $\mathcal{U}$-category.

Corollary 4.12. Let C be a $\mathcal{U}$-site. Then for every object X of $\widetilde{C}$, both the set of subobjects of X and the set of quotient objects of X are $\mathcal{U}$-small.

This follows from Exposé I, 7.4 and 7.5, respectively, which apply by Corollary 4.11.

5 Extension of a topology from C to $\widehat{C}$

Proposition 5.1. Let C be a $\mathcal{U}$-site and let $f : H \to K$ be a morphism of $\widehat{C}$. The following conditions are equivalent:

(i) For every $X \to K$, with $X \in \mathrm{Ob}(C)$, the corresponding morphism

$$H \times_K X \to X$$

has as image a covering sieve of X.

(ii) The morphism of associated sheaves

$$\underline{a}(f) : \underline{a}H \to \underline{a}K$$

is an epimorphism in $\widetilde{C}$.

(ii bis) For every sheaf F on C, the map

$$\mathrm{Hom}(K, F) \to \mathrm{Hom}(H, F)$$

deduced from f is injective.

Proof. It is clear that (ii) is equivalent to (ii bis).

We prove (i) $\Rightarrow$ (ii). By Theorem 4.4, (i) means that

$$\underline{a}(H \times_K X) \to \underline{a}(X)$$

is an epimorphism. Since

$$\underline{a}(H \times_K X) \simeq \underline{a}(H) \times_{\underline{a}(K)} \underline{a}(X),$$

because $\underline{a}$ commutes with finite projective limits (4.1), the morphism in question is obtained from

$$\underline{a}(f) : \underline{a}(H) \to \underline{a}(K)$$

by base change. As epimorphisms in $\widetilde{C}$ are universal, this shows that (ii) implies (i) after base change; conversely, since the family of all $X \to K$ is epimorphic and the corresponding family $\underline{a}(X) \to \underline{a}(K)$ is epimorphic in $\widetilde{C}$ (4.1), to verify that $\underline{a}(f)$ is an epimorphism it is enough to check it after every such base change. This proves (i) $\Rightarrow$ (ii), and the equivalence follows. $\square$

Definition 5.2.

1) A morphism $f : H \to K$ satisfying the equivalent conditions of Proposition 5.1 is called a *covering morphism*. A family of morphisms $(f_i : H_i \to K)_{i \in I}$ with common target is called *covering* if the corresponding morphism

$$f : \coprod_i H_i \to K$$

is covering.

2) A morphism $f : H \to K$ is called *bicovering* if it is covering and if its diagonal morphism

$$H \to H \times_K H$$

is covering. A family $(f_i : H_i \to K)_{i \in I}$ with common target is called *bicovering* if the corresponding morphism $\coprod_i H_i \to K$ is bicovering.

By condition (i) of Proposition 5.1, to say that a family $(f_i : H_i \to K)_{i \in I}$ is covering means that, for every morphism $X \to K$, with $X \in \mathrm{Ob}(C)$, the family

$$H_i \times_K X \to X, \qquad i \in I,$$

has as image a covering sieve of X; or equivalently, by condition (ii), that the family of morphisms

$$\underline{a}(f_i) : \underline{a}H_i \to \underline{a}K$$

in $\widetilde{C}$ is epimorphic, taking into account that $\underline{a}$ commutes with direct sums (4.1).[2]

Proposition 5.3. Let C be a $\mathcal{U}$-site and let $f : H \to K$ be a morphism of $\widehat{C}$. The following conditions are equivalent:

(i) f is bicovering (5.2).

(i bis) f is covering, and for every object X of C and every pair of morphisms $u, v : X \rightrightarrows H$ such that $fu = fv$, the equalizer of (u, v) is a covering sieve of X.

(ii) $\underline{a}(f) : \underline{a}H \to \underline{a}K$ is an isomorphism in $\widetilde{C}$.

(ii bis) For every sheaf F on C, the map

$$\mathrm{Hom}(K, F) \to \mathrm{Hom}(H, F)$$

is a bijection.

Proof. The equivalence (i) $\Leftrightarrow$ (i bis) follows from condition (i) of Proposition 5.1, applied to the diagonal morphism $H \to H \times_K H$. The equivalence (ii) $\Leftrightarrow$ (ii bis) is trivial.

(i) $\Rightarrow$ (ii). The morphism

$$\underline{a}(f) : \underline{a}H \to \underline{a}K$$

is an epimorphism by Proposition 5.1. Since $\underline{a}$ commutes with fiber products (4.1), the diagonal

$$\underline{a}H \to \underline{a}H \times_{\underline{a}K} \underline{a}H$$

is an epimorphism by Proposition 5.1. A diagonal morphism is always a monomorphism; hence it is an isomorphism by Proposition 4.2. Thus $\underline{a}(f)$ is a monomorphism. It is therefore an isomorphism by Proposition 4.2.

(ii) $\Rightarrow$ (i). Since $\underline{a}$ commutes with fiber products, both f and the diagonal $H \to H \times_K H$ are transformed by $\underline{a}$ into isomorphisms. In particular they are transformed into epimorphisms. Hence they are covering. $\square$

[2] The property of being covering (respectively bicovering), for a morphism or family of morphisms of $\widehat{C}$, is stable under base change.

5.3.1

It follows from Proposition 5.3 and from the fact that $\underline{a}$ commutes with direct sums that a family

$$(f_i : H_i \to K)_{i \in I}$$

of morphisms of $\widehat{C}$ is bicovering if and only if the f_i induce, on associated sheaves, an isomorphism

$$\coprod_i \underline{a}H_i \xrightarrow{\sim} \underline{a}K,$$

or equivalently if and only if, for every sheaf F, the map

$$\operatorname{Hom}(K, F) \longrightarrow \prod_i \operatorname{Hom}(H_i, F)$$

is bijective.

Proposition 5.4. Let C be a $\mathcal{U}$-site. There exists on $\widehat{C}$ a topology, clearly unique, such that a family $H_i \to K$ of arrows of $\widehat{C}$ with common target is covering for this topology if and only if it is covering in the sense of Definition 5.2. It is also the least fine topology on $\widehat{C}$ among the topologies T satisfying:

a) T is finer than the canonical topology of $\widehat{C}$, i.e. every epimorphic family in $\widehat{C}$ is covering for T;

b) every covering family in C is covering in $\widehat{C}$.

Proof. We only give indications. It is left to the reader to show that the families covering in the sense of Definition 5.2 are the covering families of a topology T_C on $\widehat{C}$; one uses Theorem 4.1. The covering families for the canonical topology on $\widehat{C}$ are the epimorphic families of $\widehat{C}$ (2.6 and Exposé I, 3.1). Since $\underline{a}$ commutes with inductive limits (4.1), the topology T_C is finer than the canonical topology of $\widehat{C}$. Moreover, the covering families of C are covering families for T_C (5.1).

Let T' denote the least fine topology on $\widehat{C}$ having properties (a) and (b). Then T_C is finer than T'. Let

$$(f_i : H_i \to K)_{i \in I}$$

be a T_C-covering family. We show that it is a covering family for T'. Let $s_i : H_i \to H = \coprod_i H_i$ be the canonical monomorphisms, and let $f = (f_i) : H \to K$ be the morphism defined by the f_i. The family s_i is covering for T'. Thus, by axiom (T2), to prove that (f_i) is covering for T', it is enough to prove that the morphism $f : H \to K$ is covering for T'.

There exists an epimorphic family in $\widehat{C}$

$$u_\lambda : X_\lambda \to K, \qquad \lambda \in \Lambda, \quad X_\lambda \in \operatorname{Ob}(C)$$

(Exposé I, 3.4). By axiom (T2), to prove that $f : H \to K$ is covering for T', it is enough to prove that, for every λ, the projection

$$H \times_K X_\lambda \to X_\lambda$$

is covering for T'. Let

$$v_j : Y_j \to H \times_K X_\lambda, \qquad j \in J, \quad Y_j \in \operatorname{Ob}(C),$$

be an epimorphic family in $\widehat{C}$. The family $(\mathrm{pr}_2 \circ v_j)_{j\in J}$ is covering for T_C. Hence it is a covering family in C by Proposition 5.1, and therefore is covering for T'. Consequently the sieve generated by $H \times_K X_\lambda \to X_\lambda$ contains a T'-covering sieve; it is therefore covering for T'. This proves the proposition. □

Remark 5.4.1. The proof of Proposition 5.4 actually shows that every topology T' on $\widehat{C}$ which is finer than the canonical topology of $\widehat{C}$, is the least fine of the topologies T on $\widehat{C}$ satisfying:

a) T is finer than the canonical topology of $\widehat{C}$;

b) every T'-covering family of the form

$$u_i : X_i \to X, \qquad i \in I,$$

where X and the X_i are objects of C, is covering for T.

In particular, every topology T' on $\widehat{C}$ finer than the canonical topology is uniquely determined by the families $u_i : X_i \to X$, $i \in I$, with X_i and X objects of C, which are covering for T'.

Remark 5.4.2. One can easily show that, for every topology T' on $\widehat{C}$ finer than the canonical topology, the set of families of morphisms

$$(X_i \to X)_{i\in I}$$

with common target in C, which are covering for T', is the set of covering families of a topology on C. Thus Proposition 5.4 and Remark 5.4.1 establish a one-to-one correspondence between topologies on C and topologies on $\widehat{C}$ finer than the canonical topology.

5.5.0

Let C be a small category. Let $\mathcal{C}$ be the set of strictly full subcategories of $\widehat{C}$ (strictly full means that every object isomorphic to an object of the subcategory is itself an object of the subcategory) whose inclusion functor admits a left adjoint commuting with finite projective limits. Also let $\mathcal{T}$ denote the set of topologies on C. Theorem 3.4 defines a map

$$\Phi : \mathcal{T} \longrightarrow \mathcal{C}.$$

We now define a map in the other direction. For this, to every element

$$i' : C' \hookrightarrow \widehat{C}, \qquad \underline{a}' \dashv i',$$

with $\underline{a}'$ left adjoint to i', we associate a topology T_e on C. For every object X of C, $J_e(X)$ is defined to be the set of subobjects of X whose inclusion morphism is transformed by $\underline{a}'$ into an isomorphism. Using the hypotheses on $\underline{a}'$, one immediately checks that this defines a topology T_e on C. We have therefore defined a map

$$\Psi : \mathcal{C} \longrightarrow \mathcal{T}.$$

We then have the following result, due to J. Giraud.

Theorem 5.5. The map Φ is a bijection, and Ψ is its inverse.

Proof. The composite $\Psi \circ \Phi$ is the identity. Indeed, this follows immediately from Proposition 5.1.

The composite $\Phi \circ \Psi$ is the identity. Let

$$i' : C' \hookrightarrow \widehat{C}, \qquad \underline{a}' \dashv i',$$

be an element of $\mathcal{C}$, let T_e be the topology corresponding to it under Ψ, and let C_e be the category of sheaves for T_e. Using the definition of the topology T_e and the definition of bicovering morphisms (5.2), one proves the equivalence of the following assertions:

(i) a morphism u of $\widehat{C}$ is bicovering for the topology T_e;

(ii) the morphism u of $\widehat{C}$ is transformed by $\underline{a}'$ into an isomorphism.

It is clear that C' is a full subcategory of C_e. It remains only to prove that every sheaf F for the topology T_e is an object of C'. By the equivalence above, the morphism

$$F \longrightarrow i'\underline{a}'(F)$$

is bicovering; since its source and target are sheaves, Proposition 5.3(ii) implies that it is an isomorphism. □

6 Sheaves with values in a category

6.0

Let C and D be two categories. A contravariant functor

$$F : C^\circ \to D$$

is called a *presheaf on C with values in D*.

Definition 6.1. Let C be a site and let D be a category. A presheaf $F : C^\circ \to D$ is called a *sheaf on C with values in D*, or more briefly a *D-valued sheaf*, if for every object S of D, the presheaf of sets

$$X \longmapsto \mathrm{Hom}_D(S, F(X)), \qquad X \in \mathrm{Ob}(C),$$

is a sheaf. The full subcategory of $\mathrm{Hom}(C^\circ, D)$ consisting of sheaves on C with values in D will be denoted

$$\mathrm{Hom}^\sim(C^\circ, D).$$

Remarks 6.2.

1) Condition 6.1 means that for every object X of C, every covering sieve R of X, and every object S of D, one has an isomorphism

$$\mathrm{Hom}_D(S, F(X)) \xrightarrow{\sim} \varprojlim_{Y \to R} \mathrm{Hom}_D(S, F(Y))$$

(Exposé I, 3.5 and 2.1). When D is the category of $\mathcal{U}$-sets, this recovers the definition of sheaves of sets.

2) Let D' be a $\mathcal{U}$-category and let $G : D \to D'$ be a functor commuting with projective limits. By composition, G sends sheaves with values in D to sheaves with values in D'.

6.3.0

Let γ be a species of algebraic structure defined by finite projective limits (Exposé I, 2.9), and let $\mathcal{U}$-γ be the category of γ-sets which belong to $\mathcal{U}$. Denote by

$$\mathrm{For} : \mathcal{U}\text{-}\gamma \to \mathcal{U}\text{-}\mathbf{Set}$$

the forgetful functor to the underlying set. The functor For commutes with projective limits and therefore, by the preceding remark, defines by composition functors

$$\widehat{\mathrm{For}} : \mathrm{Hom}(C^\circ,\mathcal{U}\text{-}\gamma) \longrightarrow \widehat{C},$$

and

$$\widetilde{\mathrm{For}} : \mathrm{Hom}^\sim(C^\circ,\mathcal{U}\text{-}\gamma) \longrightarrow \widetilde{C}.$$

The latter is called the *underlying sheaf of sets* functor. In fact, it factors canonically through the category γ-$\widetilde{C}$ of γ-objects of $\widetilde{C}$. Denoting the resulting functor again by

$$\widetilde{\mathrm{For}} : \mathrm{Hom}^\sim(C^\circ,\mathcal{U}\text{-}\gamma) \longrightarrow \gamma\text{-}\widetilde{C},$$

one may state:

Proposition 6.3.1. The functor $\widetilde{\mathrm{For}}$ establishes equivalences of categories between:

1) sheaves on C with values in $\mathcal{U}$-γ;

2) γ-objects of the category of sheaves of sets;

3) γ-objects of the category of presheaves of sets whose underlying presheaf is a sheaf.

Proof. The proof uses essentially Exposé I, 3.2 and the fact that a finite projective limit of sheaves, computed in the category of presheaves, is again a sheaf (4.1). □

6.3.2

The preceding proposition justifies the abuse of language by which one identifies a sheaf with values in $\mathcal{U}$-γ with the corresponding γ-object in $\widetilde{C}$. We shall henceforth use this abuse of language systematically.

We shall use the classical terminology: *sheaves of groups*, *sheaves of commutative groups*, which we shall most often call *abelian sheaves*, *sheaves of rings*, *sheaves of A-modules*, and so on.

6.3.3

We denote by

$$\widetilde{C}_\gamma \quad (\text{respectively } \widehat{C}_\gamma)$$

the category of γ-objects of $\widetilde{C}$ (respectively of $\widehat{C}$). When γ is the species of structure "A-module", we also write $\widetilde{C}_A$, or simply $\widetilde{C}_{\mathrm{ab}}$ when $A = \mathbb{Z}$.

Assume that C is a $\mathcal{U}$-site. The associated-sheaf functor is left exact (4.1), and therefore:

Proposition 6.4. The inclusion functor

$$i_\gamma : \widetilde{C}_\gamma \to \widehat{C}_\gamma$$

has a left adjoint

$$a_\gamma : \widehat{C}_\gamma \to \widetilde{C}_\gamma$$

which is left exact; in other words, there is an isomorphism

$$\mathrm{Hom}_{\widetilde{C}_\gamma}(a_\gamma(\,\cdot\,),\,\cdot\,) \xrightarrow{\sim} \mathrm{Hom}_{\widehat{C}_\gamma}(\,\cdot\,, i_\gamma(\,\cdot\,)).$$

Let X be an object of $\widehat{C}_\gamma$. The underlying sheaf of sets of $a_\gamma(X)$ is canonically isomorphic to the sheaf of sets associated to the underlying presheaf of sets of X. The morphism of underlying presheaves of sets associated to the adjunction morphism

$$\mathrm{id} \to i_\gamma a_\gamma$$

is identified with the adjunction morphism applied to the underlying presheaf of sets.

Proposition 6.5. Suppose that the forgetful functor

$$\mathrm{For} : \gamma\text{-}\mathcal{U} \to \mathcal{U}\text{-}\mathbf{Set}$$

has a left adjoint[3]

$$\mathrm{Lib} : \mathcal{U}\text{-}\mathbf{Set} \to \gamma\text{-}\mathcal{U},$$

i.e.

$$\mathrm{Hom}_{\mathcal{U}\text{-}\mathbf{Set}}(\,\cdot\,, \mathrm{For}(\,\cdot\,)) \xrightarrow{\sim} \mathrm{Hom}_{\gamma\text{-}\mathcal{U}}(\mathrm{Lib}(\,\cdot\,),\,\cdot\,).$$

Then the functor

$$\mathrm{For}^{\sim} : \widetilde{C}_\gamma \to \widetilde{C}$$

(respectively

$$\mathrm{For}^{\frown} : \widehat{C}_\gamma \to \widehat{C})$$

has a left adjoint

$$\mathrm{Lib}^{\sim} \quad (\text{respectively } \mathrm{Lib}^{\frown}),$$

and there is a canonical isomorphism

$$\mathrm{Lib}^{\sim} = a_\gamma \,\mathrm{Lib}^{\frown}\, i.$$

Proof. The proof is formal and is left to the reader. □

Corollary 6.6. Let $(X_i)_{i\in I}$ be a generating family of $\widetilde{C}$ (4.9). Then the family

$$\mathrm{Lib}^{\sim}(X_i)$$

is a generating family of $\widetilde{C}_\gamma$.

Proof. The proof is formal once one has observed that the functor $\mathrm{For}^{\sim}$ commutes with projective limits and is conservative: $\mathrm{For}^{\sim}(u)$ is an isomorphism if and only if u is an isomorphism. □

[3]One proves that such a left adjoint always exists in the usual examples: free group, free abelian group, free A-module, and so on.

Proposition 6.7. Let A be a $\mathcal{U}$-sheaf of rings, or a small ring. Let

$$\widetilde{C}_A \quad \text{(respectively } \widehat{C}_A\text{)}$$

be the category of sheaves (respectively presheaves) of unital A-modules (6.3.3) on a $\mathcal{U}$-site C. Then $\widetilde{C}_A$ is a $\mathcal{U}$-abelian category satisfying the axioms (AB5), "existence of left exact filtered inductive limits", and (AB3)*, "existence of infinite products", of TOHOKU. It has a family of generators indexed by an element of $\mathcal{U}$.

Proof. It is clear that $\widehat{C}_A$ is an abelian category satisfying axioms (AB3), (AB4), and (AB5) (Exposé I, 2.8 and 3.3). From Proposition 6.4 it follows that $\widetilde{C}_A$ is an additive category in which kernels and cokernels are representable.

More explicitly, let i_A and a_A be the inclusion and associated-sheaf functors, and let $u : X \to Y$ be a morphism of $\widetilde{C}_A$. The functor a_A is left exact and commutes with inductive limits. The functor i_A commutes with projective limits. Hence the functor

$$i_A a_A : \widehat{C}_A \to \widehat{C}_A$$

is additive and left exact. We get canonical isomorphisms

$$i_A(\mathrm{Ker}(u)) \xrightarrow{\sim} \mathrm{Ker}(i_A(u)), \tag{*}$$

and

$$\mathrm{coker}(u) \xrightarrow{\sim} a_A\ \mathrm{coker}(i_A(u)). \tag{**}$$

It follows that there is a canonical isomorphism

$$\mathrm{coim}(u) \xrightarrow{\sim} a_A\ \mathrm{coim}(i_A(u)). \tag{***}$$

We prove that the canonical morphism $\mathrm{coim}(u) \to \mathrm{im}(u)$ is an isomorphism. By $(*)$, it is enough to prove exactness of the sequence

$$0 \longrightarrow i_A(\mathrm{coim}(u)) \longrightarrow i_A(Y) \longrightarrow i_A(\mathrm{coker}(u)).$$

Using the isomorphisms $(**)$ and $(***)$, and observing that $a_A i_A(Y)$ is isomorphic to Y, this sequence is identified with

$$0 \longrightarrow i_A a_A(\mathrm{coim}(i_A(u))) \longrightarrow i_A a_A i_A(Y) \longrightarrow i_A a_A(\mathrm{coker}(i_A(u))),$$

which is simply the image under the left exact functor $i_A a_A$ of the exact sequence

$$0 \longrightarrow \mathrm{coim}(i_A(u)) \longrightarrow i_A(Y) \longrightarrow \mathrm{coker}(i_A(u)).$$

The category $\widetilde{C}_A$ is therefore abelian.

The functor

$$a_A : \widehat{C}_A \to \widetilde{C}_A$$

is left exact and commutes with arbitrary inductive limits. Hence it is additive, exact, and commutes with inductive limits. Since $\widehat{C}_A$ satisfies axiom (AB5), so does $\widetilde{C}_A$. Finally, it is clear that in $\widetilde{C}_A$, products indexed by an element of $\mathcal{U}$ are representable; hence axiom (AB3)* holds. This completes the proof of the first assertion.

For every ring A which is an element of $\mathcal{U}$, the functor

$$\text{For} : \mathcal{U}\text{-}A\text{-Mod} \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

has a left adjoint X_A, the free A-module generated by X. It remains only to apply Proposition 4.10 and Corollary 6.6. $\square$

Notation 6.8. Let H be a sheaf of sets. The sheaf of A-modules associated to the presheaf (cf. notation 6.5)

$$X \longmapsto \mathrm{Lib}_A\, H(X), \qquad X \in \mathrm{Ob}(C),$$

is denoted A_H. When $H = \underline{a}(X)$, the sheaf associated to the presheaf represented by X, one sometimes simply writes A_X, by abuse of notation. The proof of Proposition 6.7 shows that the family of sheaves of A-modules A_X, $X \in \mathrm{Ob}(C)$, is a generating family, indexed by an element of $\mathcal{U}$, for the category of sheaves of A-modules.

Remark 6.9. By Tohoku, Proposition 6.7 shows that when A is an element of $\mathcal{U}$, the category $\widetilde{C}_A$ has enough injectives. It is known, moreover, that infinite products are not necessarily exact in $\widetilde{C}_A$, and therefore that the category $\widetilde{C}_A$ does not in general have enough projectives [Roos].

References

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[3] A. Grothendieck, *Sur quelques points d'algèbre homologique*, Tohoku Math. Journal.

[4] J.-E. Roos, *C. R.*

Exposé III. Functoriality of Categories of Sheaves

Complete draft translation

J.-L. Verdier

Translator's status note. This is a working English translation of SGA 4, Exposé III. The range corresponds to source lines 23–1551 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current English usage where this is harmless: *continuous functor*, *cocontinuous functor*, *induced topology*, *comparison lemma*, *sieve*, *bicovering morphism*, *associated sheaf*, and *base-changeable* for French *quarrable*. Numbering follows the original exposé. Cross-references to other exposés are kept numerically rather than converted into active links.

In Exposé I, 5, the behavior of categories of presheaves with respect to functors between the categories of arguments was studied. In this exposé this study is extended to the case of sites and categories of sheaves (nos. 1 and 2). After introducing the induced topology (no. 3), we turn in no. 4 to the comparison lemma, which will play an important role in Exposé IV. In no. 5, for the patient reader, certain commutative diagrams related to localized categories of the type C/X are studied. In Theorem I, 5.1 smallness hypotheses are imposed on the categories under consideration. These hypotheses are not, in general, satisfied by the sites encountered in practice ($\mathcal{U}$-sites). This forces a few contortions (4.2, 4.3, 4.4).

1 Continuous functors

Definition 1.1. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a functor between the underlying categories. We say that u is *continuous* if, for every sheaf of sets F on C', the presheaf

$$X \longmapsto F(u(X))$$

on C is a sheaf.

This notion of continuity depends a priori on the universe $\mathcal{U}$ for which the two sites are $\mathcal{U}$-sites. Proposition 1.5 shows that it does not depend on it.

1.1.1

Let

$$i : \widetilde{C} \to \widehat{C} \qquad \text{(respectively } i' : \widetilde{C'} \to \widehat{C'}\text{)}$$

denote the canonical inclusion functor from sheaves to presheaves. By Definition 1.1, the functor u is continuous if and only if there exists a functor

$$u_s : \widetilde{C'} \to \widetilde{C}$$

such that

$$iu_s = \widehat{u}^* i', \qquad \widehat{u}^* F = F \circ u$$

(Exposé I, 5.0).

Proposition 1.2. Let C be a small site, let C' be a $\mathcal{U}$-site, and let $u : C \to C'$ be a functor between the underlying categories. The following properties are equivalent:

i) The functor u is continuous.

ii) For every object X of C and every sieve $R \hookrightarrow X$ covering X, the morphism

$$u_! R \hookrightarrow u(X)$$

is bicovering in $\widehat{C'}$ (Exposé II, 5.2 and Exposé I, 5.1).[1]

iii) For every bicovering family $H_i \to K$, $i \in I$, in $\widehat{C}$, the family

$$u_! H_i \to u_! K, \qquad i \in I,$$

is bicovering in $\widehat{C'}$.

iv) There exists a functor

$$u^s : \widetilde{C} \to \widetilde{C'}$$

commuting with inductive limits and "extending u", that is, such that the diagram of canonical functors

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes up to isomorphism.

Moreover, when C is no longer necessarily a small site but is a $\mathcal{U}$-site, one still has i) $\Leftrightarrow$ iv), and the functor u^s in iv) is necessarily left adjoint to $u_s : \widetilde{C'} \to \widetilde{C}$, and is therefore determined up to a unique isomorphism.

Proof. The proof of i) $\Leftrightarrow$ iv) when C is a $\mathcal{U}$-site will be given in 4.2. Suppose that C is small. Clearly iii) $\Rightarrow$ ii).

i) $\Rightarrow$ iii). Let $H_i \to K$, $i \in I$, be a bicovering family of $\widehat{C}$. For every sheaf F on C', the presheaf $u^* F$ (Exposé I, 5.0) is a sheaf on C. Hence (Exposé II, 5.3) the canonical map

$$\mathrm{Hom}(K, u^*F) \longrightarrow \prod_i \mathrm{Hom}(H_i, u^*F)$$

is bijective. By adjunction (Exposé I, 5.1), it follows that the canonical map

$$\mathrm{Hom}(u_! K, F) \longrightarrow \prod_i \mathrm{Hom}(u_! H_i, F)$$

is bijective. Therefore (Exposé II, 5.3) the family $u_! H_i \to u_! K$, $i \in I$, is bicovering.

ii) $\Rightarrow$ i). Let X be an object of C, let $R \hookrightarrow X$ be a covering sieve, and let F be a sheaf on C'. By Exposé II, 5.3, the map

$$\mathrm{Hom}(u_! X, F) \longrightarrow \mathrm{Hom}(u_! R, F)$$

is bijective. By adjunction (Exposé I, 5.1), the map

$$\mathrm{Hom}(X, u^*F) \longrightarrow \mathrm{Hom}(R, u^*F)$$

[1] "Covering" instead of "bicovering" was not necessarily sufficient; see Example 1.9.3 below.

is bijective. Therefore u^*F is a sheaf.

i) $\Rightarrow$ iv). Use the usual notation $\underline{a}$ and i (respectively $\underline{a}'$ and i') for the associated-sheaf functor and the inclusion into presheaves. For every sheaf G on C, put

$$u^s G = \underline{a}' u_! i G.$$

For every sheaf F on C', one has the sequence of natural isomorphisms

$$\begin{aligned}\operatorname{Hom}(G, u_s F) &\simeq \operatorname{Hom}(iG, iu_s F) \simeq \operatorname{Hom}(iG, u^* i' F)\\ &\simeq \operatorname{Hom}(u_! iG, i'F) \simeq \operatorname{Hom}(\underline{a}' u_! iG, F).\end{aligned}$$

The first isomorphism comes from the full faithfulness of i, the second from the definition of u_s, and the third and fourth from adjunction. Thus u^s is left adjoint to u_s. In particular u^s commutes with inductive limits. For every presheaf K on C and every sheaf F on C', one has natural isomorphisms

$$\operatorname{Hom}(u^s \underline{a} K, F) \simeq \operatorname{Hom}(\underline{a} K, u_s F) \simeq \operatorname{Hom}(K, iu_s F) \simeq \operatorname{Hom}(K, u^* i' F) \simeq \operatorname{Hom}(u_! K, i' F) \simeq \operatorname{Hom}(\underline{a}' u_! K, F),$$

whence a functorial isomorphism

$$u^s \underline{a} K \simeq \underline{a}' u_! K.$$

In particular, using this isomorphism for representable K, and taking Exposé I, 5.4(3) into account, one obtains a diagram, commutative up to isomorphism:

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}. \end{tikzcd}$$

iv) $\Rightarrow$ ii). Let $h : C \to \widehat{C}$ (respectively $h' : C' \to \widehat{C'}$) be the functor which sends an object to the presheaf it represents. Consider the diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "h"'] & C' \arrow[d, "h'"] \\ \widehat{C} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & \widehat{C'} \arrow[d, "\underline{a}'"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}. \end{tikzcd}$$

The upper square is commutative (Remark 1.4(3)), and $\underline{a}h = \epsilon_C$, $\underline{a}'h' = \epsilon_{C'}$. For every object K of $\widehat{C}$, there is a canonical isomorphism (Exposé I, 3.4)

$$\varinjlim_{(h(X)\to K)\in \operatorname{Ob}(C/K)} h(X) \xrightarrow{\sim} K.$$

Since $u_!$, u^s, $\underline{a}$, and $\underline{a}'$ commute with inductive limits, this gives isomorphisms

$$\varinjlim_{(h(X)\to K)} u^s \underline{a} h(X) \xrightarrow{\sim} u^s \underline{a} K, \qquad \varinjlim_{(h(X)\to K)} \underline{a}' u_! h(X) \xrightarrow{\sim} \underline{a}' u_! K.$$

We have $u_!h = h'u$, and, by hypothesis, an isomorphism $\underline{a}'h'u \simeq u^s\underline{a}h$. Hence there is an isomorphism

$$u^s\underline{a}K \simeq \underline{a}'u_!K,$$

which is immediately checked to be functorial in K. Thus the diagram

$$\begin{tikzcd}
\widehat{C} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & \widehat{C'} \arrow[d, "\underline{a}'"] \\
\widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}
\end{tikzcd}$$

commutes up to isomorphism. Since $\underline{a}i$ is isomorphic to the identity functor, we have $u^s \simeq \underline{a}'u_!i$, whence the uniqueness of u^s. Let $v : H \to K$ be a bicovering morphism of $\widehat{C}$. Then $\underline{a}(v)$ is an isomorphism (Exposé II, 5.3), and therefore $\underline{a}'u_!(v)$, isomorphic to $u^s\underline{a}(v)$, is an isomorphism. Thus $u_!(v)$ is sent by $\underline{a}'$ to an isomorphism; hence $u_!(v)$ is bicovering (Exposé II, 5.3), which proves ii).

The supplementary assertion, in the case where C is small, has been proved along the way. □

1.2.1

The functor $u^s : \widetilde{C} \to \widetilde{C'}$ of Proposition 1.2(iv) can often be interpreted as an "inverse image" functor for a "morphism of topoi" $\widetilde{C'} \to \widetilde{C}$; cf. Exposé IV, 4.9. Its properties are summarized in the following proposition.

Proposition 1.3. Let $u : C \to C'$ be a continuous functor from a small site C to a $\mathcal{U}$-site C'.

1) The functor u^s is left adjoint to u_s.
2) There is a canonical isomorphism $u^s \simeq \underline{a}'u_!i$.
3) There is a canonical isomorphism $u^s\underline{a} \simeq \underline{a}'u_!$.
4) The functor u^s commutes with inductive limits.
5) When $u_!$ is left exact (cf. Exposé I, 5.4(4)), so is u^s. More generally, u^s commutes with all types of finite projective limits with which $u_!$ commutes.

Proof. Assertions (1)–(4) were proved in the course of the proof of Proposition 1.2. Assertion (5) follows from the isomorphism in (2), using the fact that i commutes with projective limits and that $\underline{a}'$ commutes with finite projective limits (Exposé II, 4.1). □

Remark 1.4. Combining Exposé I, 5.4(3) and Proposition 1.3(3), one obtains a diagram

$$\begin{tikzcd}
C \arrow[r, "u"] \arrow[d, "h"'] & C' \arrow[d, "h'"] \\
\widehat{C} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & \widehat{C'} \arrow[d, "\underline{a}'"] \\
\widetilde{C} \arrow[r] & \widetilde{C'},
\end{tikzcd}$$

in which the upper square is commutative and the lower square is commutative up to isomorphism. But the functors $\underline{a}$, $\underline{a}'$, $u_!$, and u^s are defined only up to isomorphism. The reader may check that the functors $\underline{a}$, $\underline{a}'$, and u^s can be chosen so that:

1) the composite functors $\underline{a}h$ and $\underline{a}'h'$ are injective on sets of objects;
2) the diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\underline{a}h"'] & C' \arrow[d, "\underline{a}'h'"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes.

More precisely, one can choose $\underline{a}$ and $\underline{a}'$ so as to satisfy condition (1). Once $\underline{a}$ and $\underline{a}'$ have been chosen, one can choose u^s so as to satisfy condition (2).

Proposition 1.5. Let $\mathcal{U} \subset \mathcal{V}$ be two universes, let C and C' be two $\mathcal{U}$-sites, and let $u : C \to C'$ be a functor between the underlying categories. Then u is continuous relative to $\mathcal{U}$ if and only if it is continuous relative to $\mathcal{V}$. Moreover, when u is continuous, let $u^s_{\mathcal{U}}$ (respectively $u^s_{\mathcal{V}}$) denote the functor between categories of $\mathcal{U}$-sheaves (respectively $\mathcal{V}$-sheaves) introduced in Proposition 1.2(iv). Then the diagram

$$\begin{tikzcd} \widetilde{C}_{\mathcal{U}} \arrow[r, "u^s_{\mathcal{U}}"] \arrow[d] & \widetilde{C'}_{\mathcal{U}} \arrow[d] \\ \widetilde{C}_{\mathcal{V}} \arrow[r, "u^s_{\mathcal{V}}"] & \widetilde{C'}_{\mathcal{V}} \end{tikzcd} \tag{1.5.1}$$

where the vertical functors are the canonical inclusion functors, is commutative up to canonical isomorphism.

Proof. We give the proof only in the case where C is $\mathcal{U}$-small. The general case is proved in 4.3. It is clear that if u^* carries every $\mathcal{V}$-sheaf on C' to a $\mathcal{V}$-sheaf on C, then it carries every $\mathcal{U}$-sheaf on C' to a $\mathcal{U}$-sheaf on C.

Suppose conversely that u^* carries every $\mathcal{U}$-sheaf on C' to a $\mathcal{U}$-sheaf on C. Denote by

$$\widehat{C}_{\mathcal{U}} \hookrightarrow \widehat{C}_{\mathcal{V}}, \qquad \widehat{C'}_{\mathcal{U}} \hookrightarrow \widehat{C'}_{\mathcal{V}}$$

the categories of $\mathcal{U}$-presheaves and $\mathcal{V}$-presheaves, and by $u_{\mathcal{U}!}$ and $u_{\mathcal{V}!}$ the "inverse image" functors for $\mathcal{U}$-presheaves and $\mathcal{V}$-presheaves respectively. There is a commutative diagram (cf. the explicit construction of $u_!$ in Exposé I, 5.1)

$$\begin{tikzcd} \widehat{C}_{\mathcal{U}} \arrow[r, "u_{\mathcal{U}!}"] \arrow[d, hook] & \widehat{C'}_{\mathcal{U}} \arrow[d, hook] \\ \widehat{C}_{\mathcal{V}} \arrow[r, "u_{\mathcal{V}!}"] & \widehat{C'}_{\mathcal{V}}. \end{tikzcd} \tag{*}$$

The inclusion functors from $\mathcal{U}$-presheaves to $\mathcal{V}$-presheaves are fully faithful and commute with projective limits. It follows from Exposé II, 5.1(i) that a bicovering morphism of $\mathcal{U}$-presheaves is a bicovering morphism of $\mathcal{V}$-presheaves. Let $R \hookrightarrow X$ be a sieve covering an object X of C. By Proposition 1.2(ii), $u_{\mathcal{U}!}R \to u_{\mathcal{U}!}X$ is bicovering. Using the commutativity of $(*)$ and the preceding observation, it follows that $u_{\mathcal{V}!}R \to u_{\mathcal{V}!}X$ is bicovering. Hence (Proposition 1.2) u^* carries $\mathcal{V}$-sheaves on C' to $\mathcal{V}$-sheaves on C. The commutativity of (1.5.1) follows from the commutativity of $(*)$, from Exposé II, 3.6, and from Proposition 1.3(2). □

Proposition 1.6. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a functor between the underlying categories. Consider the conditions:

i) u is continuous (cf. Definition 1.1 and Proposition 1.2).

ii) For every covering family $(X_i \to X)_{i \in I}$ of C, the family $(uX_i \to uX)_{i\in I}$ of C' is covering.

There is an implication i) ⇒ ii).

Assume that the topology of C can be defined by a pretopology T (Exposé II, 1.3) and that u commutes with fiber products.[2] Then i) and ii) are equivalent, and are equivalent to the following condition:

iii) The functor u carries the covering families of T to covering families of C'.

In particular, if fiber products are representable in C and if u commutes with fiber products, then i) and ii) are equivalent.

Proof. Let us prove i) ⇒ ii). Let $(X_i \to X)_{i\in I}$ be a covering family of C. For every sheaf F on C', u^*F is a sheaf. Hence the map

$$\mathrm{Hom}(X, u^*F) \longrightarrow \prod_i \mathrm{Hom}(X_i, u^*F)$$

is injective (Exposé II, 5.1). Thus

$$\mathrm{Hom}(u_!X, F) \longrightarrow \prod_i \mathrm{Hom}(u_!X_i, F)$$

is injective. Hence the family $u_!X_i \to u_!X$, $i \in I$, is covering in C' (Exposé II, 5.1).

Every covering family of the pretopology T is a covering family of C, whence ii) ⇒ iii). Let us prove iii) ⇒ i). Let X be an object of C, and let $(X_i \to X)_{i\in I}$ be a covering family in $\mathrm{Cov}(X)$ (Exposé II, 1.3). The morphisms $X_i \to X$ are base-changeable and u commutes with fiber products. Thus the fiber products $u(X_i) \times_{u(X)} u(X_j)$ are representable and canonically isomorphic to $u(X_i \times_X X_j)$. Let F be a sheaf on C'. Then the diagram of sets

$$F(u(X)) \longrightarrow \prod_i F(u(X_i)) \rightrightarrows \prod_{i,j} F(u(X_i \times_X X_j))$$

is exact (Exposé II, 2.1; Exposé I, 3.5 and 2.12). Hence the diagram

$$u^*F(X) \longrightarrow \prod_i u^*F(X_i) \rightrightarrows \prod_{i,j} u^*F(X_i \times_X X_j)$$

[2] In fact, it is enough that u commute with the fiber products occurring in base changes of morphisms coming from covering families for T.

is exact. Thus u^*F is a sheaf (Exposé II, 2.4), and i) follows. The last assertion follows from the preceding argument and from the fact that, when fiber products are representable in C, all covering families of C[3] define a pretopology on C whose associated topology is the topology of C. $\square$

Proposition 1.7. Let C be a small site, let C' be a $\mathcal{U}$-site, and let $u : C \to C'$ be a continuous functor. Let γ be an algebraic structure defined by finite projective limits, such that in the category of γ-objects of $\mathcal{U}$-**Set**, $\mathcal{U}$-inductive limits are representable.[4] We use the notation of Proposition II, 6.4. The functor u_s commutes with projective limits and therefore defines a functor

$$u_s^\gamma : \widetilde{C}_\gamma \to \widetilde{C'}_\gamma.$$

The functor u_s^γ admits a left adjoint u_γ^s, with the following properties:

1) There is a canonical isomorphism

$$u_\gamma^s \xrightarrow{\sim} \underline{a}'_\gamma u_{\gamma!} i_\gamma,$$

and u_γ^s commutes with all finite projective limits with which $u_{\gamma!}$ commutes.

2) There is a canonical isomorphism

$$u_\gamma^s \underline{a}_\gamma \xrightarrow{\sim} \underline{a}'_\gamma u'_{\gamma!}.$$

3) The functor u_γ^s commutes with inductive limits.

4) If u^s is left exact, then the diagram (notation of Exposé II, 6.3.0)

$$\begin{tikzcd} \widetilde{C}_\gamma \arrow[r, "u_\gamma^s"] \arrow[d, "j^\sim"'] & \widetilde{C'}_\gamma \arrow[d, "j'^\sim"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes up to isomorphism, and u_γ^s is exact.

5) Suppose that $j^\sim$ (respectively $j'^\sim$) has a left adjoint $\mathrm{Lib}^\sim$ (respectively $\mathrm{Lib}'^\sim$) (Exposé II, 6.5). There is a canonical isomorphism

$$u_\gamma^s \,\mathrm{Lib}^\sim \xrightarrow{\sim} \mathrm{Lib}'^\sim u^s.$$

Moreover, when C is not necessarily a small site but is a $\mathcal{U}$-site, the functor u_s^γ admits a left adjoint u_γ^s. Finally, if $\mathcal{V}$ is a universe containing $\mathcal{U}$, and if $u_{\gamma\mathcal{U}}^s$ (respectively $u_{\gamma\mathcal{V}}^s$) denotes the left adjoint of u_s relative to $\mathcal{U}$ (respectively $\mathcal{V}$), the following diagram, analogous to (1.5.1),

$$\begin{tikzcd} \widetilde{C}_{\gamma\mathcal{U}} \arrow[r, "u_{\gamma\mathcal{U}}^s"] \arrow[d, hook] & \widetilde{C'}_{\gamma\mathcal{U}} \arrow[d, hook] \\ \widetilde{C}_{\gamma\mathcal{V}} \arrow[r, "u_{\gamma\mathcal{V}}^s"] & \widetilde{C'}_{\gamma\mathcal{V}} \end{tikzcd} \tag{1.7.1}$$

[3] Indexed by sets belonging to a suitable universe.

[4] In fact one can show that this condition is always satisfied.

commutes up to canonical isomorphism.

Proof. The proof in the case where C is a small site is left to the reader. It will be completed in 4.3 in the case where C is a $\mathcal{U}$-site. $\square$

Notation 1.8. From now on we shall use the notation u_s for the functor u_s^γ. This notation causes no confusion, because the functor u_s defined on sheaves of sets commutes with finite projective limits.

Similarly, when u^s is left exact, we shall use the notation u^s for the functor u^s_γ. This abuse of notation is justified by Proposition 1.7(4).

Example 1.9.1. Let X be a small topological space. Let $\mathrm{Open}(X)$ denote the category of open subsets of X, whose objects are the open subsets of X and whose morphisms are inclusions, endowed with the canonical topology. A family $(U_i \subset U)_{i\in I}$ is covering if and only if the open subsets U_i cover U (Exposé II, 2.6). Thus sheaves of sets on $\mathrm{Open}(X)$ are sheaves of sets in the sense of Godement [TF]. Let $f : X \to Y$ be a continuous map. It gives a functor

$$f^{-1} : \mathrm{Open}(Y) \to \mathrm{Open}(X),$$

which sends an open set U of Y to its inverse image $f^{-1}(U)$. The functor f^{-1} is continuous (Proposition 1.6). The functor

$$f_s^{-1} : \widetilde{\mathrm{Open}(X)} \to \widetilde{\mathrm{Open}(Y)}$$

is the direct image functor for sheaves in the sense of [TF]. The functor

$$(f^{-1})^s : \widetilde{\mathrm{Open}(Y)} \to \widetilde{\mathrm{Open}(X)}$$

is the inverse image functor for sheaves in the sense of [TF]. The functor $(f^{-1})^s$ commutes with finite projective limits, by Proposition 1.3(5).

Example 1.9.2. Let X be a topological space and let $Y \hookrightarrow X$ be an open subspace. Every open subset of Y is an open subset of X, hence there is a functor

$$\Phi : \mathrm{Open}(Y) \to \mathrm{Open}(X).$$

The functor Φ is continuous (Proposition 1.6). The functor $\Phi_s : \widetilde{\mathrm{Open}(X)} \to \widetilde{\mathrm{Open}(Y)}$ is the "restriction to Y" functor. The functor

$$\Phi^s_{\mathbf{Ab}} : \widetilde{\mathrm{Open}(Y)}_{\mathbf{Ab}} \to \widetilde{\mathrm{Open}(X)}_{\mathbf{Ab}}$$

(Exposé II, 6.3.3; Exposé I, 1.7) is the "extension by zero outside Y" functor introduced in [TF]. The functor

$$\Phi^s : \widetilde{\mathrm{Open}(Y)} \to \widetilde{\mathrm{Open}(X)}$$

(Proposition 1.3) is analogous to extension by zero: for every sheaf H on Y and every point $y \in Y$, the stalk of $\Phi^s H$ at y is canonically isomorphic to the stalk of H at y; for every point $x \in X$ with $x \notin Y$, the stalk of $\Phi^s H$ at x is empty. The functor Φ^s is called the "extension by the empty set outside Y" functor. It commutes with fiber products and with finite products over a nonempty set of indices, but it does not carry the final object of $\widetilde{\mathrm{Open}(Y)}$ to the final object of $\widetilde{\mathrm{Open}(X)}$.

Example 1.9.3. Let C and C' be two small sites, and let $u : C \to C'$ be a functor between the underlying categories which carries every covering family of C to a covering family of C'. The functor u is not continuous in general (cf. however Proposition 1.6). Here is a counterexample. Let S be an infinite set belonging to $\mathcal{U}$, and let C be the full subcategory of the category of sets whose objects are the finite nonempty subsets of S. Endow C with the canonical topology. Notice that the covering families of C are the surjective families of maps, and that covering families are nonempty. Notice also that, up to isomorphism, there are only two constant sheaves on C: the sheaf with value the empty set and the sheaf with value a one-point set. Put $C = C'$, and let $u : C \to C'$ be a constant functor. The functor u carries covering families of C to covering families of C', but u is not continuous. Indeed, since the representable presheaves of C' are sheaves, for every object X of C' there exists a sheaf F on C' such that the cardinality of $F(X)$ is at least 2; consequently the presheaf u^*F is not a sheaf.

2 Cocontinuous functors

Definition 2.1. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a functor between the underlying categories. We say that u is *cocontinuous* if it has the following property:

(COC) For every object Y of C, and every sieve $R \hookrightarrow u(Y)$ covering $u(Y)$, the sieve of Y generated by the arrows $Z \to Y$ such that $u(Z) \to u(Y)$ factors through R, covers Y.

Proposition 2.2. Let C be a small site, let C' be a $\mathcal{U}$-site, and let $u : C \to C'$ be a functor between the underlying categories. Let

$$\widehat{u}_* : \widehat{C} \to \widehat{C'}$$

denote the functor right adjoint to $F \mapsto F \circ u = \widehat{u}^*F$ (Exposé I, 5.1). The functor u is cocontinuous if and only if, for every sheaf G on C, $\widehat{u}_*G$ is a sheaf on C'.

Proof. Sufficiency. Suppose that for every sheaf G on C, the presheaf $\widehat{u}_*G$ is a sheaf. Let Y be an object of C, and let $R \hookrightarrow u(Y)$ be a covering sieve. We have an isomorphism

$$\mathrm{Hom}(u(Y), u_*G) \xrightarrow{\sim} \mathrm{Hom}(R, u_*G),$$

whence, by adjunction (Exposé I, 5.1), an isomorphism

$$\mathrm{Hom}(\widehat{u}^*u(Y), G) \xrightarrow{\sim} \mathrm{Hom}(\widehat{u}^*R, G).$$

It follows (Exposé II, 5.3) that the morphism

$$\widehat{u}^*R \to \widehat{u}^*u(Y)$$

is bicovering. Since $\widehat{u}^*$ commutes with projective limits, this morphism is a monomorphism, and therefore is a covering monomorphism. On the other hand, there is a canonical morphism

$$Y \xrightarrow{p} \widehat{u}^*u(Y)$$

deduced from the adjunction morphism $\mathrm{id}_C \to \widehat{u}^* u_!$ (Exposé I, 5.4(3)). By base change along p, we obtain a covering sieve of Y:

$$\widehat{u}^* R \times_{\widehat{u}^* u(Y)} Y \to Y.$$

The arrows $Z \to Y$ which factor through this sieve are precisely those described in property (COC).

Necessity. Let S be an object of C' and let $R \hookrightarrow S$ be a sieve covering S. We must prove that, for every sheaf G on C, the map

$$\mathrm{Hom}(S, u_* G) \xrightarrow{\sim} \mathrm{Hom}(R, u_* G)$$

is an isomorphism. Using Proposition II, 5.3 and the adjunction isomorphisms, it is enough to prove that the monomorphism

$$\widehat{u}^* R \to \widehat{u}^* S$$

is covering. For this, it is enough to prove (Exposé II, 5.1) that, for every base change $Y \to \widehat{u}^* S$, where Y is an object of C, the sieve

$$\widehat{u}^* R \times_{\widehat{u}^* S} Y$$

is covering. By the properties of adjoint functors and Exposé I, 5.4(3), the morphism $Y \to \widehat{u}^* S$ factors as

$$Y \xrightarrow{p} \widehat{u}^* u(Y) \to \widehat{u}^* S.$$

Thus the sieve $\widehat{u}^* R \times_{\widehat{u}^* S} Y \hookrightarrow Y$ is obtained by base change along p from the monomorphism

$$\widehat{u}^* R \times_{\widehat{u}^* S} \widehat{u}^* u(Y) \hookrightarrow \widehat{u}^* u(Y).$$

Since $\widehat{u}^*$ commutes with projective limits, this monomorphism is the image under $\widehat{u}^*$ of the covering sieve

$$R \times_S u(Y) \hookrightarrow u(Y).$$

The hypothesis (COC) therefore implies that

$$\widehat{u}^* R \times_{\widehat{u}^* S} Y \hookrightarrow Y$$

is covering. □

Proposition 2.3. Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a cocontinuous functor. Let

$$u^* : \widetilde{C'} \to \widetilde{C}$$

be the functor $u^* = \underline{a}\widehat{u}^* i'$, where $\underline{a}$ denotes the associated-sheaf functor for $\widehat{C}$, $\widehat{u}^*$ the functor $F \mapsto F \circ u$, and i' the inclusion functor $\widetilde{C'} \to \widehat{C'}$.

1) The functor u^* commutes with inductive limits and is exact.

2) There is a canonical isomorphism

$$u^* \underline{a}' \simeq \underline{a}\widehat{u}^*,$$

where $\underline{a}'$ denotes the associated-sheaf functor for $\widehat{C'}$.

3) The functor u^* has a right adjoint $u_* : \widetilde{C} \to \widetilde{C'}$. When C is small, the diagram

$$\begin{tikzcd} \widetilde{C} \arrow[r, "u_*"] \arrow[d, hook, "i"'] & \widetilde{C'} \arrow[d, hook, "i'"] \\ \widehat{C} \arrow[r, "\widehat{u}_*"] & \widehat{C'} \end{tikzcd} \tag{2.3.1}$$

in which the lower functor is right adjoint to $\widehat{u}^*$ (Exposé I, 5.1), is commutative up to canonical isomorphism.

4) Let $\mathcal{V} \supset \mathcal{U}$ be a universe. Let

$$u_{*\mathcal{U}} : \widetilde{C}_{\mathcal{U}} \to \widetilde{C'}_{\mathcal{U}} \quad \text{(respectively } u_{*\mathcal{V}} : \widetilde{C}_{\mathcal{V}} \to \widetilde{C'}_{\mathcal{V}})$$

denote the right adjoint relative to the universe $\mathcal{U}$ (respectively $\mathcal{V}$) whose existence is asserted in (3). Then the diagram

$$\begin{tikzcd} \widetilde{C}_{\mathcal{U}} \arrow[r, "u_{*\mathcal{U}}"] \arrow[d, hook] & \widetilde{C'}_{\mathcal{U}} \arrow[d, hook] \\ \widetilde{C}_{\mathcal{V}} \arrow[r, "u_{*\mathcal{V}}"] & \widetilde{C'}_{\mathcal{V}} \end{tikzcd} \tag{2.3.2}$$

commutes up to isomorphism.

Proof. Assertions (1) and (2) follow immediately from Exposé II, 3.4. Assertion (3), when C is small, follows from Proposition 2.2. When C is a $\mathcal{U}$-site, it will be proved in 4.4. Assertion (4), when C is small, follows from the analogous commutativity assertion when the topologies on C and C' are chaotic, and from Exposé I, 3.5. When the topologies on C and C' are chaotic, the commutativity of (2.3.2) is immediate from the explicit description of u_* (Exposé I, 5.1). □

2.4

We shall not develop here the considerations concerning γ-objects of categories of sheaves. It will be enough for us to observe that the functors u^* and u_* introduced in this number always commute with finite projective limits, unlike what happened in the preceding number for the functor u^s. Consequently they extend naturally to functors defined on γ-objects; these extensions are adjoint to one another and commute with the "underlying sheaf of sets" functors. We shall make the abuses of notation indicated in 1.8, writing u^* and u_* for the extensions to γ-objects.

Proposition 2.5. Let C and C' be two $\mathcal{U}$-sites and let

$$\begin{tikzcd} C \arrow[r, shift left, "v"] & C' \arrow[l, shift left, "u"] \end{tikzcd}$$

be a pair of functors, where v is left adjoint to u. The following properties are equivalent:

i) The functor u is continuous.

ii) The functor v is cocontinuous.

Moreover, under these equivalent conditions, there are canonical isomorphisms $v_* \simeq u_s$ and $v^* \simeq u^s$. In particular, u^s commutes with finite projective limits.

Proof. The property of a functor being continuous or cocontinuous does not depend on the universe $\mathcal{U}$ for which C and C' are $\mathcal{U}$-sites; this is immediate for cocontinuous functors, and follows from Proposition 1.5 for continuous functors. Thus, after enlarging the universe if necessary, we may suppose that C and C' are small. The proposition then follows from Proposition 2.2 and Exposé I, 5.6. □

Proposition 2.6. Let $u : C \to C'$ be a continuous and cocontinuous functor between two $\mathcal{U}$-sites. Then the functor

$$u^* : \widetilde{C'} \to \widetilde{C}$$

(Proposition 2.3) commutes with $\mathcal{U}$-inductive and projective limits. Let

$$u_! : \widetilde{C} \to \widetilde{C'}, \qquad u_* : \widetilde{C} \to \widetilde{C'}$$

denote the left and right adjoints of u^*, respectively. The functor $u_!$ is fully faithful if and only if u_* is fully faithful. When u is fully faithful, $u_!$ is fully faithful; the converse holds when the topologies of C and C' are less fine than the canonical topology.

The functor u^* has a right adjoint u_* (Proposition 2.3) and a left adjoint $u_!$ (Proposition 1.2). Hence u^* commutes with inductive and projective limits (Exposé I, 2.11). The fact that $u_!$ is fully faithful if and only if u_* is fully faithful is a general property of adjoint functors (Exposé I, 5.7(1)). When u is fully faithful, the functor

$$\widehat{u}_* : \widehat{C}_{\mathcal{V}} \to \widehat{C'}_{\mathcal{V}},$$

relative to $\mathcal{V}$-presheaves for a sufficiently large universe $\mathcal{V}$, is fully faithful (Exposé I, 5.7). Therefore $u_* : \widetilde{C} \to \widetilde{C'}$ is fully faithful by the commutativity of (2.3.1) and (2.3.2), and hence $u_!$ is fully faithful by what precedes. The converse follows from the existence of the commutative diagram in Proposition 1.2(iv), taking into account the fact that the functors ϵ_C and $\epsilon_{C'}$ are fully faithful when the topologies are less fine than the canonical topology. □

Example 2.7. With the notation of 1.9.2, the functor

$$\Phi : \mathrm{Open}(Y) \to \mathrm{Open}(X)$$

is continuous (1.9.2) and cocontinuous (Proposition 2.2). Moreover, let $i : Y \to X$ be the canonical injection and let $i^{-1} : \mathrm{Open}(X) \to \mathrm{Open}(Y)$ be the functor it defines (1.9.1). The functor i^{-1} is right adjoint to Φ. Thus there is a sequence of three adjoint functors

$$\Phi^s, \quad \Phi_s = \Phi^*, \quad \Phi_*,$$

and canonical isomorphisms

$$\Phi^* \simeq (i^{-1})^s, \qquad \Phi_* \simeq i_s^{-1}.$$

3 Induced topology

3.1

Let C' be a site, let C be a category, and let $u : C \to C'$ be a functor. For every universe $\mathcal{U}$ such that C' is a $\mathcal{U}$-site and C is a $\mathcal{U}$-small category, denote by $\mathcal{T}_{\mathcal{U}}$ the finest among the topologies T on C which make u continuous (Definition 1.1). Such a topology exists by Exposé II, 2.2. The topology $\mathcal{T}_{\mathcal{U}}$ does not depend on $\mathcal{U}$. Indeed, if $\mathcal{V} \supset \mathcal{U}$ is a universe, one has $\mathcal{T}_{\mathcal{U}} = \mathcal{T}_{\mathcal{V}}$ (Definition 1.1 and Proposition 1.5). The topology $\mathcal{T}_{\mathcal{U}}$ is called the *topology induced on C by the topology of C' by means of the functor u.*[5]

Proposition 3.2. Let C be a small category, let C' be a $\mathcal{U}$-site, let $u : C \to C'$ be a functor, and let $\mathcal{T}$ be the topology on C induced by u. Let X be an object of C and let $R \to X$ be a sieve on X. The following properties are equivalent:

i) The sieve $R \hookrightarrow X$ is covering for $\mathcal{T}$.

ii) For every base change $Y \to X$, where Y is an object of C, the morphism

$$u_!(R \times_X Y) \to u(Y)$$

is bicovering in $\widehat{C'}$ (Exposé II, 5.2).

Proof. i) $\Rightarrow$ ii) follows from axiom (T1) for topologies and from Proposition 1.2.

ii) $\Rightarrow$ i). For every sheaf F on C', the map

$$\operatorname{Hom}(Y, u^*F) \to \operatorname{Hom}(R \times_X Y, u^*F)$$

is isomorphic, by adjunction (Exposé I, 5.1), to the map

$$\operatorname{Hom}(u(Y), F) \to \operatorname{Hom}(u_!(R \times_X Y), F),$$

which is bijective (Exposé II, 5.3). Hence, by Exposé II, 2.2, the sieve $R \hookrightarrow X$ is covering for $\mathcal{T}$. $\square$

Corollary 3.3. Let C' be a site, let C be a category, let $u : C \to C'$ be a functor, and let $\mathcal{T}$ be the topology on C induced by the topology of C'. Let $(X_i \to X)_{i \in I}$ be a family of base-changeable morphisms of C, and suppose that u commutes with fiber products.[6] The following properties are equivalent:

i) The family $(X_i \to X)_{i \in I}$ is covering for $\mathcal{T}$.

ii) The family $(u(X_i) \to u(X))_{i \in I}$ is covering.

Proof. i) $\Rightarrow$ ii) follows from Proposition 1.6.

ii) $\Rightarrow$ i). Let $\mathcal{U}$ be a universe such that C is $\mathcal{U}$-small and C' is a $\mathcal{U}$-site. Let $R \hookrightarrow X$ be the sieve generated by the $X_i \to X$. The presheaf R is the cokernel of the pair of arrows (Exposé I, 2.12 and 3.5)

$$\coprod_{i,j} X_i \times_X X_j \rightrightarrows \coprod_i X_i,$$

[5] When no confusion can result, this topology is called simply the *topology induced on C by the topology of C'*.

[6] In fact, it is enough that u commute with fiber products of the form $X_i \times_X X'$.

where the coproduct is taken in $\widehat{C}$. Since $u_!$ commutes with inductive limits (Exposé I, 5.4), the presheaf $u_!R$ is the cokernel of

$$\coprod_{i,j} u(X_i \times_X X_j) \rightrightarrows \coprod_i u(X_i).$$

Since u commutes with fiber products, $u_!R$ is the cokernel of

$$\coprod_{i,j} u(X_i) \times_{u(X)} u(X_j) \rightrightarrows \coprod_i u(X_i),$$

and hence $u_!R \to u(X)$ is the sieve on $u(X)$ generated by the $u(X_i) \to u(X)$, $i \in I$. Using once again that u commutes with fiber products, one shows that, for every base change $Y \to X$ with Y an object of C, the morphism

$$u_!(R \times_X Y) \to u(Y)$$

is the sieve on $u(Y)$ generated by the morphisms

$$u(X_i) \times_{u(X)} u(Y) \to u(Y), \qquad i \in I.$$

Since $(u(X_i) \to u(X))_{i\in I}$ is covering, the family $u(X_i) \times_{u(X)} u(Y) \to u(Y)$, $i \in I$, is covering. Thus $u_!(R \times_X Y) \to u(Y)$ is a covering sieve, and therefore by Proposition 3.2 the sieve $R \to X$ is covering for $\mathcal{T}$. $\square$

Corollary 3.4. Let C' be a $\mathcal{U}$-site, let C be a full subcategory of C', and let $u : C \to C'$ be the inclusion functor. Suppose that fiber products are representable in C and that u commutes with fiber products. The following conditions are equivalent:

- i) a) For every object X of C, every covering family $(Y_j \to X)_{j\in J}$ of C' is dominated by a covering family $(X_i \to X)_{i\in I}$, where the X_i are objects of C.
  - b) There exists a small set G of objects of C such that every object of C is the target of a covering family in C' of morphisms whose sources belong to G.
- ii) The topology induced by the topology of C' is a $\mathcal{U}$-topology (Exposé I, 3.0.2), and u is continuous and cocontinuous for this topology.

Proof. This follows immediately from Corollary 3.3 and Definition 2.1. $\square$

Proposition 3.5. Let C be a $\mathcal{U}$-site and let $\epsilon_C : C \to \widetilde{C}$ be the canonical functor (Exposé II, 4.4.0). Endow $\widetilde{C}$ with the canonical topology. Then the topology of the site C is the topology induced by the topology of $\widetilde{C}$.

Proof. The covering families of $\widetilde{C}$ for the canonical topology are the universal effective epimorphic families (Exposé II, 2.5), that is (Exposé II, 4.3), the epimorphic families. Let T be the topology of the site C, and let $\mathcal{T}_{\mathcal{U}}$ be the finest topology T' on C such that, for every sheaf F on $\widetilde{C}$, $F \circ \epsilon_C$ is a sheaf for T'. It is enough to show that $T = \mathcal{T}_{\mathcal{U}}$; then $\mathcal{T}_{\mathcal{U}}$ is a $\mathcal{U}$-topology, and hence $\mathcal{T}_{\mathcal{U}}$ is the induced topology.

A) The topology T is finer than $\mathcal{T}_{\mathcal{U}}$. Let $(X_i \to X)_{i\in I}$ be a covering family for $\mathcal{T}_{\mathcal{U}}$. Then for every sheaf F on $\widetilde{C}$, the map

$$F(\epsilon_C(X)) \to \prod_i F(\epsilon_C(X_i))$$

is injective. Hence (Exposé II, 5.2) the family $(\epsilon_C X_i \to \epsilon_C X)_{i\in I}$ is covering for the canonical topology of $\widetilde{C}$. Therefore (Exposé II, 5.2) $(X_i \to X)_{i\in I}$ is covering for T.

B) The topology T is less fine than $\mathcal{T}_{\mathcal{U}}$. It is enough to show that $\epsilon_C : C \to \widetilde{C}$ is continuous (Definition 1.1). But it will be proved in Exposé IV, no. 1, that every sheaf on $\widetilde{C}$ is representable; hence, for every sheaf F on $\widetilde{C}$, $F \circ \epsilon_C$ is a sheaf on C. We shall not use Proposition 3.5 before Exposé IV, 1.

We record two results which will be used in Exposé VI, 7. □

Proposition 3.6. Let $(C_i)_{i\in I}$ be a family of sites, let C be a category, and for each $i \in I$, let $u_i : C_i \to C$ be a functor. Let $\mathcal{U}$ be a universe such that the categories C_i and C are $\mathcal{U}$-small. There exists a topology $\mathcal{T}_{\mathcal{U}}$ on C which is the least fine among the topologies for which the u_i are continuous. The topology $\mathcal{T}_{\mathcal{U}}$ does not depend on the universe $\mathcal{U}$ for which the categories considered are small.

The last assertion of Proposition 3.6 follows from Proposition 1.5.

Let T be a topology on C. The functors u_i are continuous if and only if, for every $i \in I$ and every covering sieve $R \hookrightarrow X$ of an object X of C_i, the morphism

$$u_{i!}(R) \to u_i(X)$$

of $\widehat{C}$ is bicovering (Proposition 1.2(ii)). Thus Proposition 3.6 is a consequence of the following lemma.

Lemma 3.6.1. Let C be a small category and let $(u_i : F_i \to G_i)_{i\in I}$ be a family of arrows of $\widehat{C}$. Then there exists on C a least fine topology among those which make the morphisms u_i covering (respectively bicovering) (Exposé II, 5.2).

To say that $u : F \to G$ is covering for a given topology T means that, for every arrow $X \to G$, with $X \in \operatorname{Ob} C$, the corresponding arrow $F \times_G X \to X$ is covering; equivalently, the family of arrows $X' \to X$ of C which factor through that arrow is covering. The existence of a least fine topology among those for which the $u_i : F_i \to G_i$ are covering follows from Exposé I, 1.1.6, proving the non-parenthesized assertion of Lemma 3.6.1. The parenthesized assertion follows by recalling that a morphism $u : F \to G$ is bicovering if and only if the morphisms $u : F \to G$ and $\Delta_u : F \to F \times_G F$ are covering.

Proposition 3.7. Let $(C_i)_{i\in I}$ be a family of sites, let C be a category, and for each $i \in I$, let $u_i : C_i \to C$ be a functor. Let $\mathcal{U}$ be a universe such that the categories C_i and C are $\mathcal{U}$-small. There exists a topology $\mathfrak{T}_{\mathcal{U}}$ which is the finest topology for which the u_i are cocontinuous. The topology $\mathfrak{T}_{\mathcal{U}}$ does not depend on the universe $\mathcal{U}$ for which the categories considered are small.

Let $\mathcal{U}$ be a universe for which the categories considered are small. The functors u_i are cocontinuous for a topology $\mathfrak{T}$ on C if and only if, for every $i \in I$ and every sheaf F on C_i, the presheaf $\widehat{u}_{i*}F$ is a sheaf for $\mathfrak{T}$ (Proposition 2.2). It follows that $\mathfrak{T}_{\mathcal{U}}$ is the finest topology for which the presheaves $\widehat{u}_{i*}F$, $i \in I$, $F \in \operatorname{Ob}(C_i)$, are sheaves (Exposé II, 2.2). The last assertion follows from Proposition 2.2.

4 The comparison lemma

Theorem 4.1 (comparison lemma). Let C be a small category, let C' be a site whose underlying category is a $\mathcal{U}$-category, and let $u : C \to C'$ be a fully faithful functor. Endow C with the topology induced by u (3.1). Consider the properties:

i) Every object of C' can be covered by objects coming from C.

ii) The functor $F \mapsto F \circ u$ induces an equivalence from the category of sheaves on C' to the category of sheaves on C.

One always has i) $\Rightarrow$ ii). When C' is a $\mathcal{U}$-site and the topology of C' is less fine than the canonical topology, one has ii) $\Rightarrow$ i).

4.1.1

We first prove i) $\Rightarrow$ ii). The proof is in two steps.

First step. For every presheaf H of $\widehat{C'}$, the adjunction morphism

$$\varphi : u_! u^* H \to H$$

(Exposé I, 5.1) is bicovering (Exposé II, 5.3), and the functor $u_s : \widetilde{C'} \to \widetilde{C}$ (1.1.1) is fully faithful.

Let C/H be the small category whose objects are pairs consisting of an object Y of C and a morphism $u(Y) \to H$, and whose morphisms are the commutative diagrams

$$\begin{tikzcd} u(Y) \arrow[rr, "u(m)"] \arrow[dr] & & u(Y') \arrow[dl] \\ & H & \end{tikzcd} .$$

We have

$$u^* H = \varinjlim_{Y \in \mathrm{Ob}(C/H)} Y$$

(Exposé I, 3.4), and hence (Exposé I, 5.4)

$$u_! u^* H = \varinjlim_{Y \in \mathrm{Ob}(C/H)} u(Y).$$

The adjunction morphism is the evident morphism coming from this description of $u_! u^* H$ as an inductive limit. It has the following property:

($*$) For every object Y of C, every morphism $m : u(Y) \to H$ factors uniquely as the canonical morphism $\alpha(m) : u(Y) \to u_! u^* H$ followed by φ.

It follows immediately from (i) that φ is covering (Exposé II, 5.1). Let us prove that it is bicovering. Let $p, q : Z \rightrightarrows u_! u^* H$ be two morphisms from an object Z of C' such that $\varphi p = \varphi q$. For every object Y of C and every morphism $n : u(Y) \to Z$, one has $\varphi p n = \varphi q n$. Property ($*$) then implies $pn = qn$, and since the $u(Y)$ cover Z, the kernel of (p, q) is a covering sieve of Z. Thus φ is bicovering (Exposé II, 5.3). For every sheaf H on C', $u^* H$ is a sheaf on C, denoted $u_s H$ (1.1.1). Moreover

$$u^s u_s H = \underline{a}' u_! u_s H$$

(Proposition 1.3), and the adjunction morphism $u^s u_s H \to H$ is obtained by applying the associated-sheaf functor to $\varphi : u_! u^* H \to H$. Therefore (Exposé II, 5.3) the adjunction morphism $u^s u_s H \to H$ is an isomorphism. Hence $u_s : \widetilde{C'} \to \widetilde{C}$ is fully faithful.

Second step. The functor u is cocontinuous and the functor $u^s : \widetilde{C} \to \widetilde{C'}$ (Proposition 1.2) is fully faithful. Therefore $u_s : \widetilde{C'} \to \widetilde{C}$ is an equivalence.

Let Y be an object of C, and let $i : R \hookrightarrow u(Y)$ be a covering sieve. Since u^* commutes with projective limits (Exposé I, 5.5), the morphism

$$u^*(i) : u^*(R) \hookrightarrow u^*u(Y)$$

is a monomorphism. Since u is fully faithful, $u^*u(Y) \simeq Y$, and hence this gives a sieve $u^*(R) \to Y$ on Y. To show that u is cocontinuous, it is enough to show that $u^*(i) : u^*(R) \hookrightarrow Y$ is a covering sieve for the induced topology on C (Definition 2.1). For this it is enough to prove (Proposition 3.2) that, for every base change $m : X \to Y$, the morphism

$$u_!(u^*(R) \times_Y X) \to u(X)$$

is bicovering. Since u^* commutes with projective limits, one has

$$u^*(R) \times_Y X = u^*(R \times_{u(Y)} u(X)),$$

and $R \times_{u(Y)} u(X)$ is a covering sieve of $u(X)$. It is therefore enough to prove that, for every object Y of C and every covering sieve $i : R \hookrightarrow u(Y)$, the morphism in $\widehat{C'}$

$$u_!u^*(R) \xrightarrow{u_!u^*(i)} u(Y)$$

is bicovering. But this morphism factors as the adjunction morphism $u_!u^*(R) \to R$, which is bicovering by the first step, followed by the monomorphism $R \hookrightarrow u(Y)$, which is covering. Hence it is bicovering (Exposé II, 5.3). This proves that u is cocontinuous. Since u is continuous and fully faithful, Proposition 2.6 (used in the case where C is small) implies that u^s (denoted $u_!$ in Proposition 2.6) is fully faithful. Since u^s and u_s are adjoint to one another and both are fully faithful, they are quasi-inverse functors; hence u_s is an equivalence.

4.1.2

We now prove ii) $\Rightarrow$ i). The objects Y of C form a generating family of $\widetilde{C}$ (Exposé II, 4.10). Therefore, for every object X of C', there exists an epimorphic family in $\widetilde{C}$

$$v_i : Y_i \to u_sX, \qquad i \in I.$$

We have $u_su(Y_i) = Y_i$, and, since u_s is an equivalence of categories, $v_i = u_s(w_i)$, where the $w_i : u(Y_i) \to X$ form an epimorphic family of $\widetilde{C'}$. It follows from Exposé II, 5.1, that the family $w_i : u(Y_i) \to X$, $i \in I$, is covering.

4.1 End of the proof of Proposition 1.2

Let G be a full subcategory of C whose objects form a small topologically generating family of C (Exposé II, 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \to C$ (3.1). The functor $(u \circ i)_s = i_s \circ u_s$ (1.1.1) has a left adjoint by the first part of the proof of Proposition 1.2. Since i_s is an equivalence of categories (Theorem 4.1), the functor u_s has a left adjoint u^s, and there is a functorial isomorphism

$$u^s \simeq (u \circ i)^s \circ i_s.$$

Let us prove that the diagram

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd} \tag{4.2.1}$$

commutes up to isomorphism. For every sheaf H on C' and every $X \in \operatorname{Ob} C$,

$$\operatorname{Hom}_{\widetilde{C'}}(\epsilon_{C'} uX, H) \simeq u_s H(X).$$

Moreover,

$$\operatorname{Hom}_{\widetilde{C'}}(u^s \epsilon_C X, H) \simeq \operatorname{Hom}_{\widetilde{C}}(\epsilon_C X, u_s H) \simeq u_s H(X),$$

by adjunction and by the definition of ϵ_C. Hence $\epsilon_{C'} uX \simeq u^s \epsilon_C X$ for every $X \in \operatorname{Ob} C$. This proves i) $\Rightarrow$ iv). Let us prove iv) $\Rightarrow$ i). There is a commutative diagram

$$\begin{tikzcd} G \arrow[r, "i"] \arrow[d, "\epsilon_G"'] & C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"'] \\ \widetilde{G} \arrow[r, "i^s"] & \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}, \end{tikzcd} \tag{4.2.2}$$

and therefore, by the first part of the proof, the functor $u \circ i : G \to C'$ is continuous. From Theorem 4.1 and Definition 1.1 it follows immediately that u is continuous, hence iv) $\Rightarrow$ i). One also obtains the uniqueness of u^s, knowing from the first part of the proof the uniqueness when C is small.

4.2 End of the proof of Propositions 1.5 and 1.7

Let $u : C \to C'$ be a functor between two $\mathcal{U}$-sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (Exposé II, 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \to C$. It follows immediately from Theorem 4.1 and Definition 1.1 that u is continuous if and only if $u \circ i$ is continuous. From this observation and from the first part of the proof of Proposition 1.5 the general case follows. This observation also reduces the proof of Proposition 1.7 to the case where C is small.

4.3 End of the proof of Proposition 2.3

Let $u : C \to C'$ be a functor between two $\mathcal{U}$-sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (Exposé II, 3.0.1), endowed with the topology induced by the inclusion functor $i : G \to C$. It follows from the proof of Theorem 4.1 (second step in 4.1.1) that i is cocontinuous. Therefore, when u is cocontinuous, the functor $u \circ i : G \to C'$ is cocontinuous. Moreover,

$$i^* : \widetilde{C} \to \widetilde{G}$$

is an equivalence of categories (Theorem 4.1). Thus the functor $u^* : \widetilde{C'} \to \widetilde{C}$ has a right adjoint. This proves assertion (3). To prove assertion (4), it is enough to observe that

$$(u \circ i)_* \simeq u_* \circ i_*$$

and that i_* is an equivalence (Theorem 4.1). The commutativity of (2.3.2) follows from the commutativity of these diagrams when C is small.

5 Localization

5.1

Let now C be a $\mathcal{U}$-site and let X be an object of $\widehat{C}$. Unless explicitly stated otherwise, the category C/X will be endowed with the topology $\mathcal{T}$ induced by the functor

$$j_X : C/X \to C$$

(3.1). The notation C/X will denote the category C/X endowed with the topology $\mathcal{T}$. Proposition I, 5.11 shows that the functor $j_{X!}$ commutes with fiber products and therefore carries every monomorphism to a monomorphism. In particular, for every object $(Y \to X)$ of C/X, the functor $j_{X!}$ gives a bijective correspondence between the sieves, in the category C/X, of the object $(Y \to X)$, and the sieves, in C, of the object Y.

Proposition 5.2. Let C be a $\mathcal{U}$-site, let X be an object of $\widehat{C}$, and let $j_X : C/X \to C$ be the corresponding continuous functor.

1) A sieve R of an object $(Z \to X)$ is covering in C/X if and only if the sieve $j_{X!}(R) \hookrightarrow Z$ is covering in C.

2) The functor $j_X : C/X \to C$ is cocontinuous and continuous.

3) Let $m : Y \to X$ be a morphism of $\widehat{C}$. Then there is a commutative diagram

$$\begin{tikzcd} C/Y \arrow[rr, "j_m"] \arrow[dr, "j_Y"'] & & C/X \arrow[dl, "j_X"] \\ & C & \end{tikzcd} .$$

The topology induced by j_Y on C/Y is equal to the topology induced by j_m on C/Y.

4) The topology induced by $j_X : C/X \to C$ is a $\mathcal{U}$-topology (Exposé II, 3.0.2).

Proof.

1) If the sieve R of $(Z \to X)$ is covering, the sieve $j_{X!}(R) \hookrightarrow Z$ is covering (Proposition 1.6). Conversely, if $j_{X!}(R) \to Z$ is covering in C, the same is true for every sieve obtained by base change in C/X. Hence R is covering (Proposition 3.2).

2) This follows immediately from (1) by applying Definition 2.1.

3) This follows immediately from the description of covering sieves given in (1).

4) Let $(G_i)_{i\in I}$ be a small topologically generating family of C. One checks immediately that the small family of pairs

$$(u : G_i \to X), \qquad u \in \coprod_{i\in I} \mathrm{Hom}_{\widehat{C}}(G_i, X),$$

is topologically generating in C/X.

$\square$

Terminology and notation 5.3. By the preceding proposition, the functor j_X is both continuous and cocontinuous. It therefore defines a sequence of three adjoint functors (4.3.2) between categories of sheaves of sets (Propositions 1.3 and 2.3):

$$j_X^s : (C/X)^\sim \to \widetilde{C}, \qquad j_X^* = j_{X,s} : \widetilde{C} \to (C/X)^\sim, \qquad j_{X*} : (C/X)^\sim \to \widetilde{C}.$$

In the special situation of Proposition 5.2, we shall use the following terminology and notation:

1) The functor j_{X*} will be called the *direct image functor*.

2) The functor $j_{X,s} = j_X^*$ will be denoted j_X^* and called the *restriction functor to C/X*.

3) The functor j_X^s on sheaves of sets will be called the "extension by the empty set to the category C" functor and will be denoted $j_{X!}$ (cf. 2.9.2).

Thus there is a sequence of three adjoint functors between $(C/X)^\sim$ and $\widetilde{C}$:

$$j_{X!}, \quad j_X^*, \quad j_{X*}.$$

Proposition 5.4. The functor $j_{X!} : (C/X)^\sim \to \widetilde{C}$ factors through the category $\widetilde{C}/\underline{a}X$, where $\underline{a}$ is the associated-sheaf functor:

$$(C/X)^\sim \xrightarrow{e_X^\sim} \widetilde{C}/\underline{a}X \to \widetilde{C}.$$

The functor

$$e_X^\sim : (C/X)^\sim \to \widetilde{C}/\underline{a}X$$

is an equivalence of categories. The restriction functor to C/X, composed with the equivalence $e_X^\sim$, is isomorphic to the "base change by $\underline{a}X \to \epsilon$" functor

$$F \longmapsto (F \times \underline{a}X \xrightarrow{\mathrm{pr}_2} \underline{a}X),$$

where ϵ is the final object of $\widetilde{C}$.

Proof. The image under $j_{X!}$ of the final object of $(C/X)^\sim$ is the object $\underline{a}X$, whence the factorization. To show that $e_X^\sim$ is an equivalence, we restrict ourselves to a few indications. By Exposé I, 5.11, a presheaf on C/X is defined by a presheaf F on C endowed with a morphism $F \to X$. One then proves that the following two properties are equivalent:

i) The presheaf on C/X defined by $F \to X$ is a sheaf.

ii) The following diagram is cartesian, where i and $\underline{a}$ denote the inclusion into presheaves and the associated-sheaf functor:

$$\begin{tikzcd}
F \arrow[r] \arrow[d] & i\underline{a}F \arrow[d] \\
X \arrow[r] & i\underline{a}X.
\end{tikzcd}$$

It follows immediately that $e^{\sim}_X$ is an equivalence. The last assertion is trivial. □

Proposition 5.5.

1) Let C be a $\mathcal{U}$-site and let X be an object of $\widehat{C}$. The following diagram of categories and functors is commutative up to canonical isomorphisms:

$$\begin{tikzcd}
C/X \arrow[r, "h_X"] \arrow[d] & \widehat{(C/X)} \arrow[r, "\underline{a}_X"] \arrow[d, "\widehat{e}_X"] & (C/X)^{\sim} \arrow[rr, "i_X"] \arrow[d, "e^{\sim}_X"] & & \widehat{(C/X)} \arrow[d, "\widehat{e}_X/X"] \\
C/X \arrow[r, "h/X"] & \widehat{C}/X \arrow[r, "\underline{a}/X"] & \widetilde{C}/\underline{a}X \arrow[r, "i/\underline{a}X"] & \widehat{C}/i\underline{a}X \arrow[r, "\pi"] & \widehat{C}/X.
\end{tikzcd}$$

The notation $\underline{a}_X$ and i_X denotes the associated-sheaf functor and the inclusion into presheaves for the site C/X. The notation $\underline{a}/X$ denotes the natural extension of $\underline{a}$, the associated-sheaf functor for the site C, to the category of arrows with target X; similarly for $i/\underline{a}X$. Finally, π denotes the base-change functor by the canonical arrow $X \to i\underline{a}X$.

2) Suppose in addition that $m : Y \to X$ is a morphism of $\widehat{C}$. The following diagram is commutative up to canonical isomorphism:

$$\begin{tikzcd}
\widetilde{C}/\underline{a}X/\underline{a}Y \arrow[d, equal] & (C/X)^{\sim}/\underline{a}_X Y \arrow[l, "f"'] & \\
 & & (C/X/Y)^{\sim} \arrow[ul, "g"'] \arrow[dl, "e^{\sim}_m"] \\
\widetilde{C}/\underline{a}Y & (C/Y)^{\sim} \arrow[l, "e^{\sim}_Y"'] & .
\end{tikzcd}$$

The arrow f is the natural extension of the equivalence $e^{\sim}_X$ to the category of arrows with target $\underline{a}_X Y$, and therefore is an equivalence of categories. The arrow g is none other than the equivalence of Proposition 5.4 applied to the situation $(C/X)/[m]$.

3) Let

$$j_{\underline{a}X!} : \widetilde{C}/\underline{a}X \to \widetilde{C}$$

denote the functor "forget $\underline{a}X$", and let

$$j^*_{\underline{a}X} : \widetilde{C} \to \widetilde{C}/\underline{a}X$$

denote the functor "product with $\underline{a}X$". The following diagrams are commutative up to canonical isomorphism:

$$\begin{tikzcd} (C/X)^\sim \arrow[r, "j_{X!}"] \arrow[d] & \widetilde{C} \arrow[d, equal] \\ \widetilde{C}/\underline{a}X \arrow[r, "j_{\underline{a}X!}"] & \widetilde{C} \end{tikzcd} \qquad \begin{tikzcd} \widetilde{C} \arrow[r, "j_X^*"] \arrow[d, equal] & (C/X)^\sim \arrow[d, "e_X^\sim"] \\ \widetilde{C} \arrow[r, "j_{\underline{a}X}^*"] & \widetilde{C}/\underline{a}X. \end{tikzcd}$$

Proof. The proof of these assertions is immediate from the definition of the equivalences $e_X^\sim$ (Proposition 5.4) and $\widehat{e}_X$ (Exposé I, 5.11). □

References

[1] R. Godement, *Théorie des faisceaux*, Hermann, 1958, Act. Scient. Ind. no. 1252, Paris.

Exposé IV. Topoi

Draft translation, Sections 0–8.8

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Translator's status note. This is a working English translation of SGA 4, Exposé IV, from the introduction through Exercise 8.8. The cumulative range corresponds to source lines 1–5253 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current mathematical English where this is harmless. Numbering follows the source.

0. Introduction

0.1

We have seen in Exposé II various exactness properties of categories of the form

$$\widetilde{\mathcal{C}} = \text{the category of sheaves of sets on } \mathcal{C},$$

where $\mathcal{C}$ is a small site. These properties may be expressed by saying that, in many respects, these categories, which we shall call *topoi*, inherit the familiar properties of the category (**Set**) of small sets. On the other hand, experience has shown that one should often regard various situations in mathematics *mainly as a technical means for constructing the corresponding categories of sheaves* of sets, i.e. the corresponding *topoi*. It appears that all the genuinely important notions attached to a site–for instance its cohomological invariants, studied in Exposé V, various other "topological" invariants, such as the homotopy invariants recently studied by M. Artin and B. Mazur, and the notions studied in J. Giraud's book on non-commutative cohomology–are in fact expressed directly in terms of the associated topos.

From this viewpoint, it is appropriate to regard two sites as essentially equivalent when their associated topoi are equivalent categories, and to regard the datum of a site, at least in the practically most important case where its topology is no finer than the canonical topology, as equivalent to the datum of a topos E, namely the associated topos of sheaves of sets on the site, together with a generating family of objects of E (cf. Exposé II, 4.9 and 1.2.1 below). This viewpoint is analogous to the one in which one attaches a group to a system of generators and a system of relations among these generators, and focuses on the structure of the group rather than on the generators and relations by which it was produced, these being regarded as auxiliary data of the situation. Moreover, the comparison lemma, Exposé III, 4.1, gives numerous examples of pairs of sites $\mathcal{C}, \mathcal{C}'$ which are not isomorphic, and not even equivalent as categories, but give rise to equivalent topoi; hence $\mathcal{C}$ and $\mathcal{C}'$ should be regarded as essentially equivalent.

0.2

In this exposé we give a characterization of topoi by simple exactness properties, due to J. Giraud. We study the natural notion of morphism of topoi, inspired by the notion of a continuous map from one topological space to another, and we develop in the setting of topoi some familiar constructions from ordinary sheaf theory, such as sheaves $\mathcal{H}om$, sheaf tensor products, and supports. Finally, following M. Artin, we show how a topos can be reconstructed from an "open" part of it, the complementary "closed" part, and a certain left exact functor relating them, which, up to minor qualifications, may in fact be chosen arbitrarily.

0.3

This gives a gluing procedure for topoi which, when applied to topoi coming from ordinary topological spaces, will generally produce a topos no longer of that same type. This is a first indication of the remarkable stability of the notion of topos under various natural constructions, a stability lacking in the notion of topological space, from which the notion of topos is inspired. As a second remarkable example, let us also mention the classifying topos associated with a group object of a topos (cf. [1] or 5.9 below), inspired by the classical notion of classifying space of a topological group, and the notion of a "modular topos" associated with various moduli problems in algebraic geometry or analytic geometry [10], [13].

Other topoi, such as the *étale topos* of a scheme, studied systematically in this seminar from Exposé VII onward, or the *crystalline topos* of a relative scheme [6], arise naturally when one wants to develop usable cohomology theories for abstract algebraic varieties, and more generally for schemes, replacing the classical Betti cohomology of algebraic varieties over the complex numbers.

0.4

Thus one may say that the notion of topos, a natural derivative of the *sheaf-theoretic point of view* in topology, is itself a substantial enlargement of the notion of topological space.[1] It embraces many situations which previously were not regarded as belonging to topological intuition. The characteristic feature of such situations is that one has a notion of "localization"; this is formalized precisely by the notion of site, and ultimately by that of topos, via the topos associated with the site. As the term "topos" itself is meant to suggest, it seems reasonable and legitimate to the authors of this seminar to regard topology as the study of *topoi*, and not merely of topological spaces.

0.5

It has seemed useful to include in this general exposé on topoi a fairly large number of examples, many of which have only distant connections with the original aim of this seminar, namely the study of étale cohomology. The hurried reader, interested exclusively in étale cohomology, may of course omit these examples without loss, and likewise most of the present exposé, referring back to it only as needed.

1 Definition and characterization of topoi

Definition 1.1. A $\mathcal{U}$-*topos*, or simply a topos when no confusion can arise, is a category E such that there exists a site $C \in \mathcal{U}$ for which E is equivalent to the category $\widetilde{C}$ of $\mathcal{U}$-sheaves of sets on C.

1.1.1

Let E be a $\mathcal{U}$-topos. We shall always regard E as endowed with its canonical topology (Exposé II, 2.5), making it a site, and even, by 1.1.2(d) below, a $\mathcal{U}$-site (Exposé II, 3.0.2). Unless explicitly stated otherwise, we shall consider no topology on E other than the one just described.

[1] Cf. [9], or 4.1 and 4.2 below, for the precise relation between topoi and topological spaces.

1.1.2

We saw in Exposé II, 4.8 and 4.11, that a $\mathcal{U}$-topos E is a $\mathcal{U}$-category satisfying the following conditions:

a) finite projective limits are representable in E;

b) coproducts indexed by an element of $\mathcal{U}$ are representable in E, and are disjoint and universal;

c) equivalence relations in E are effective and universal;

d) E admits a generating family indexed by an element of $\mathcal{U}$.

In fact, we shall see that these intrinsic properties *characterize* $\mathcal{U}$-topoi.

Theorem 1.2 (J. Giraud). Let E be a $\mathcal{U}$-category. The following properties are equivalent:

i) E is a $\mathcal{U}$-topos.

ii) E satisfies conditions a), b), c), and d) of 1.1.2.

iii) The $\mathcal{U}$-sheaves on E for the canonical topology are representable, and E has a small generating family, namely condition 1.1.2(d).

iv) There exist a category $C \in \mathcal{U}$ and a fully faithful functor $i : E \to \widehat{C}$, where $\widehat{C}$ denotes the category of $\mathcal{U}$-presheaves on C, which admits a left adjoint a that is left exact.

i′ There exists a site $C \in \mathcal{U}$, in which projective limits are representable and whose topology is no finer than its canonical topology, such that E is equivalent to the category $\widetilde{C}$ of $\mathcal{U}$-sheaves of sets on C.

Moreover:

Corollary 1.2.1. Let E be a $\mathcal{U}$-topos, and let C be a full subcategory of E, endowed with the topology induced from that of E. Consider the functor

$$E \longrightarrow \widetilde{C}$$

which sends $X \in \operatorname{Ob} E$ to the restriction to C of the functor represented by X. This functor is an equivalence of categories if and only if $\operatorname{Ob} C$ is a generating family of E.

The equivalence i) $\Leftrightarrow$ iv) follows at once from Exposé II, 5.5, and we have already recalled that i) $\Rightarrow$ ii). Since i′ $\Rightarrow$ i is trivial, it remains to prove ii) $\Rightarrow$ iii) and iii) $\Rightarrow$ i′, which is done in 1.2.4 and 1.2.3 below. The corollary is then obtained by noting that the functor under consideration factors as

$$E \longrightarrow \widetilde{E} \longrightarrow \widetilde{C}.$$

Since $E \to \widetilde{E}$ is an equivalence by (iii), the question reduces to determining when $\widetilde{E} \to \widetilde{C}$ is an equivalence. The comparison lemma of Exposé III, 4.1, then gives the conclusion; cf. the proof in 1.2.3 below.

1.2.3. Proof of iii) $\Rightarrow$ i')

Let $\mathfrak{X} = (X_i)_{i\in I}$, $I \in \mathcal{U}$, be a small generating family of E. Since E is a $\mathcal{U}$-category, the set of isomorphism classes of finite diagrams in E whose objects are among the X_i is $\mathcal{U}$-small. Hence the smallest set $\mathfrak{X}'$ of objects of E, containing the finite projective limits of objects of $\mathfrak{X}$, is a countable union of small sets and is therefore small.[2] Replacing $\mathfrak{X}$ successively by $\mathfrak{X}^{(n+1)} = (\mathfrak{X}^{(n)})'$ and then by $\bigcup_n \mathfrak{X}^{(n)}$, we may assume, after enlarging the generating family, that $\mathfrak{X}$ is stable under finite projective limits.

Let $\mathcal{V}$ be a universe containing $\mathcal{U}$ such that E is $\mathcal{V}$-small. For each object H of E, write

$$I(H) = \coprod_{i\in I} \mathrm{Hom}(X_i, H).$$

Let H be an object of E, and let $H' \hookrightarrow H$ be the subsheaf of H, for the canonical topology, which is the "union" of the images of the morphisms $u : X_i \to H$, $(u,i) \in I(H)$. Since H' is a subsheaf of a $\mathcal{U}$-sheaf, H' is a $\mathcal{U}$-sheaf. It is therefore representable. Since $\mathfrak{X}$ is generating and, for every $i \in I$, the map $\mathrm{Hom}(X_i, H') \to \mathrm{Hom}(X_i, H)$ is bijective, the monomorphism $H' \hookrightarrow H$ is an isomorphism. Thus the family $(u : X_i \to H)$, $(u,i) \in I(H)$, is covering for the canonical topology of E. Hence every object of E can be covered, for the canonical topology of E, by objects of $\mathfrak{X}$.

Let C be the full subcategory of E defined by the objects of $\mathfrak{X}$, and let $u : C \hookrightarrow E$ be the inclusion. The topology induced on C by the canonical topology of E is no finer than the canonical topology of C; when $\mathcal{U}$ contains an element of infinite cardinality, finite projective limits are representable in C. It follows from Exposé III, 5.1, that the functor $F \mapsto F \circ u$ is an equivalence from E to the category of $\mathcal{U}$-sheaves on C for the induced topology. ☐

1.2.4. Proof of ii) $\Rightarrow$ iii)

The proof has four steps. Let $\mathcal{V}$ be a universe containing $\mathcal{U}$, such that E is an element of $\mathcal{V}$. Let $\widetilde{E}$ be the category of $\mathcal{V}$-sheaves on E for the canonical topology, and let $J_E : E \to \widetilde{E}$ be the canonical functor.

1.2.4.1. Let $(g_i : G_i \to H)$, $i \in I \in \mathcal{U}$, be an epimorphic family in $\widetilde{E}$. If the G_i and the fiber products $G_i \times_H G_j$ are representable, then the sheaf H is representable.

Indeed, the coproduct $\coprod_{i\in I} G_i$ is representable in $\widetilde{E}$, by a representable sheaf G. The same holds for $K = \coprod_{(i,j)\in I\times I}(G_i \times_H G_j)$. Moreover, the diagram $K \rightrightarrows G \to H$ is exact, and K is the fiber square of G over H. The latter property holds in the category of presheaves, hence in the category of sheaves. It follows from condition c) and Exposé II, 4.7, that H is representable.

1.2.4.2. Let X_α, $\alpha \in A \in \mathcal{U}$, be a generating family of E. For every sheaf H, let

$$I(H) = \coprod_{\alpha\in A} \mathrm{Hom}_E(X_\alpha, H).$$

The family $(u : X_\alpha \to H, (u,\alpha) \in I(H))$ is epimorphic in $\widetilde{E}$. When H is a $\mathcal{U}$-sheaf, $I(H)$ is an element of $\mathcal{U}$.

[2] We are reasoning here with classes of objects of E up to isomorphism.

Indeed, every sheaf H is the target of an epimorphic family of morphisms whose sources are representable sheaves; therefore it suffices to prove the first assertion when H is representable. Let G be the image of the family $(u : X_\alpha \to H, (u, \alpha) \in I(H))$. The morphism $G \to H$ is a monomorphism. We are then in the situation of the first step, because $X_\alpha \times_G X_\beta$ is isomorphic to $X_\alpha \times_H X_\beta$, which is representable, and because $I(H)$ is an element of $\mathcal{U}$. Hence G is representable. Since the X_α form a generating family, $G \to H$ is an isomorphism. The final assertion is immediate from the definition of $\mathcal{U}$-sheaves.

1.2.4.3. Every subsheaf of a representable sheaf is representable.

Indeed, let $G \to H$ be a subsheaf of a representable sheaf. Then G is a $\mathcal{U}$-sheaf. The family $(u : X_\alpha \to G, (u, \alpha) \in I(G))$ is therefore epimorphic in $\widetilde{E}$ and indexed by an element of $\mathcal{U}$. Moreover, the fiber products $X_\alpha \times_G X_\beta$ are isomorphic to the fiber products $X_\alpha \times_H X_\beta$, which are representable. Thus we are in the situation of the first step, and G is representable.

1.2.4.4. Every $\mathcal{U}$-sheaf is representable.

Indeed, by the first and second steps, it suffices to show that the fiber products $X_\alpha \times_H X_\beta$ are representable. But these fiber products are subobjects of the products $X_\alpha \times X_\beta$. The conclusion follows from the third step. This completes the proof of Theorem 1.2.

Remark 1.3. Of course, for a given $\mathcal{U}$-topos E, there is generally no preferred way to represent it, up to equivalence, as $\widetilde{C}$ with C a *small* site; or, what is essentially the same by 1.2.1 when one restricts to C whose topology is no finer than the canonical topology, there is no preferred *small* generating family in E. Once one drops the requirement that C be small, however, there is, by 1.2(iii), a completely canonical choice of a $\mathcal{U}$-site C such that E is equivalent to $\widetilde{C}$, namely E itself. This is one technical reason why, in practice, it is inconvenient to work only with small sites: indeed, the most important sites of all, namely $\mathcal{U}$-topoi, are not small. Moreover, the sites generating topoi which arise in many questions in algebraic geometry, and even in topology (cf. 2.5), are also not small; for example, the étale site of a scheme.

Proposition 1.4. Let E be a $\mathcal{U}$-topos, E' a category, not necessarily a $\mathcal{U}$-category, and F a presheaf on E with values in E'. Then F is a sheaf with values in E' if and only if F transforms $\mathcal{U}$-inductive limits in E into projective limits in E'.

Let $\mathcal{V}$ be a universe containing $\mathcal{U}$ and such that E' is a $\mathcal{V}$-category. By composing F with the functors $\mathrm{Hom}(X', -) : E' \to \mathcal{V}$-**Set** defined by the objects X' of E', we reduce to the case $E' = \mathcal{V}$-**Set**. Suppose first that F transforms $\mathcal{U}$-inductive limits into projective limits. Then it follows immediately from the definitions that F is a sheaf, since a covering family $X_i \to X$ in E, by definition of the canonical topology on E, allows X to be regarded as the inductive limit of the usual diagram $X_i \times_X X_j \rightrightarrows X_i$.

Conversely, suppose that F is a sheaf, and prove that it transforms $\mathcal{U}$-inductive limits into projective limits. If $\mathcal{V} \subset \mathcal{U}$, we may assume $\mathcal{V} = \mathcal{U}$, and it suffices to apply criterion 1.2(iii). In the general case a little more work is needed. Let C be a full subcategory of E generated by a generating family indexed by an element of $\mathcal{U}$, and let $u : C \to E$ be the inclusion. Endow C with the topology induced by the canonical topology of E. Let $\widetilde{C}_{\mathcal{U}}$ and $\widetilde{C}_{\mathcal{V}}$ denote the categories of sheaves on C, and let $\widetilde{E}_{\mathcal{V}}$ denote the category of $\mathcal{V}$-sheaves on E for the canonical topology. Let $J_E : E \to \widetilde{E}_{\mathcal{V}}$ be the canonical functor, and let

$$i_{\mathcal{U},\mathcal{V}} : \widetilde{C}_{\mathcal{U}} \longrightarrow \widetilde{C}_{\mathcal{V}}$$

be the inclusion. The diagram

$$\begin{tikzcd}
\widetilde{C}_{\mathcal{U}} \arrow[r, "i_{\mathcal{U},\mathcal{V}}"] & \widetilde{C}_{\mathcal{V}} \\
E \arrow[u] \arrow[r, "J_E"] & \widetilde{E}_{\mathcal{V}} \arrow[u]
\end{tikzcd}$$

where the vertical arrows are induced by $F \mapsto F \circ u$, is commutative. By the explicit construction of the associated-sheaf functor and by Exposé II, 4.1, the functor $i_{\mathcal{U},\mathcal{V}}$ commutes with $\mathcal{U}$-inductive limits. Moreover, by Exposé III, 5.1, and Corollary 1.5, the vertical arrows in the diagram are equivalences of categories. Hence $J_E : E \to \widetilde{E}_{\mathcal{V}}$ commutes with $\mathcal{U}$-inductive limits. For every object X of E, one has

$$F(X) \simeq \mathrm{Hom}_{\widetilde{E}_{\mathcal{V}}}(J_E(X), F).$$

Therefore F transforms $\mathcal{U}$-inductive limits of E into projective limits.

Corollary 1.5. Let E be a $\mathcal{U}$-topos, E' a $\mathcal{U}$-category, and $f : E \to E'$ a functor. Then f admits a right adjoint if and only if f commutes with $\mathcal{U}$-inductive limits.

This is an immediate consequence of 1.4 in the case $E' = \mathcal{U}$-**Set**.

Corollary 1.6. Let E, E' be two $\mathcal{U}$-topoi and let $f : E \to E'$ be a functor. The following conditions are equivalent:

i) f commutes with $\mathcal{U}$-inductive limits;

ii) f admits a right adjoint;

iii) f is continuous in the sense of Exposé III, 1.1.

The equivalence i) $\Leftrightarrow$ ii) was proved in 1.5. To prove i) $\Leftrightarrow$ iii), apply the definition of continuous functors, choosing a universe $\mathcal{V}$ such that $\mathcal{U} \in \mathcal{V}$, so that E and E' are $\mathcal{V}$-small sites. One must express that, for every $\mathcal{V}$-sheaf F on E', the composite $F \circ f$ is a sheaf on E, that is, by 1.4, that it transforms $\mathcal{U}$-inductive limits into projective limits. This follows from (i), since by 1.4 the functor F itself transforms $\mathcal{U}$-inductive limits into projective limits. It is also necessary, as is seen by taking F representable.

Corollary 1.7. With the notation of 1.6, f is the inverse-image functor u^* of a geometric morphism $u : E' \to E$ if and only if f is left exact and sends epimorphic families to epimorphic families.

Necessity is trivial from the definition. Sufficiency follows from Exposé III, 1.6, which implies that f is continuous under the stated conditions, hence commutes with $\mathcal{U}$-inductive limits by 1.6.

Remark 1.8. With the notation of 1.5, one can show that f admits a left adjoint if and only if it commutes with $\mathcal{U}$-projective limits. Equivalently, a covariant functor $E \to (\mathcal{U}\text{-}\mathbf{Set})$ is representable if and only if it commutes with $\mathcal{U}$-projective limits.[3] We indicate only the principle of the proof, which has two steps:

[3] A more general statement is found in Exposé I, 8.12.8 and 8.12.9; the sketch below corresponds to the proof given there.

a) Standard arguments show that, if F commutes with projective limits, it is pro-representable by a strict projective system $(T_i)_{i\in I}$, where I is a filtered ordered set, not necessarily small. One may assume that, if $i > j$, then $T_i \to T_j$ is not an isomorphism; under this hypothesis, F is representable if and only if I is small, which in fact implies that the projective system is essentially constant.

b) To prove that I is small, knowing that for every object X of E the set $F(X) = \varinjlim_i \operatorname{Hom}(T_i, X)$ is small, it suffices to have a *small cogenerating family* $(X_j)_{j\in J}$, that is, a generating family for the opposite category E°. One shows that such a small cogenerating family always exists in a $\mathcal{U}$-topos E.

c) To prove this last point, standard arguments show that every object X of E admits a monomorphism into an "injective" object; then, for every generating family (L_α) of E, if one embeds each L_α in this way into an injective object I_α, the family (I_α) is cogenerating.

2 Examples of topoi

2.0

In this section we have gathered a fairly large number of typical examples of topoi which the reader will already have met elsewhere, and which are meant to ease access to the "yoga" of topoi. For other examples, drawn from algebraic geometry, of topologies on sites giving rise to topoi, the reader may consult SGA 3, IV, 6, and, for the étale topos, Exposé VII of the present seminar. Since later references to this section will mostly concern notation or terminology, we leave the verification of the statements accompanying these examples as exercises for the reader's general instruction. All the examples in this section will be made more precise in Section 4, where we examine their functorial dependence on the data, and in later sections as illustrations of the general notions concerning topoi.

2.1. Topos associated with a topological space

Let X be a small topological space, and let $\operatorname{Open}(X)$ be the category of open subsets of X, endowed with the canonical topology. We denote by $\operatorname{Top}(X)$ the topos of $\mathcal{U}$-sheaves on $\operatorname{Open}(X)$. This topos is equivalent to the category of spaces *étalés* over X, by associating to such a space X' over X the sheaf

$$U \longmapsto \Gamma(X'/U) = \operatorname{Hom}_X(U, X')$$

on $\operatorname{Open}(X)$. By abuse of language we shall sometimes denote by the same letter X the topos $\operatorname{Top}(X)$ defined by the topological space X.

It is clearly this example which served chiefly as a guide and intuitive support for the development of the theory of topoi. One should nevertheless be careful that the topoi deduced from topological spaces are of a very special nature, notably because they have been described by sites $C = \operatorname{Open}(X)$ in which *all morphisms are monomorphisms*; the underlying category is therefore just a preordered set. It follows in particular that the sheaves represented by the objects of C are subsheaves of the final sheaf, and hence that *subsheaves of the final sheaf form a generating family* for the topos in question. This property is not shared by most of the topoi arising naturally in algebraic geometry or algebra, cf. the examples below. Up to minor qualifications, it characterizes topoi of the form $\mathrm{Top}(X)$ (cf. 7.1.9 below).

One checks easily that the map $\mathrm{Open}(X) \to \mathrm{Top}(X)$, which sends an open set of X to the sheaf it represents, is a bijection from $\mathrm{Open}(X)$ onto the set of subobjects of the final object of $\mathrm{Top}(X)$, and that this bijection is even an isomorphism for the natural order structures. This suggests that it should be possible to reconstruct the topological space X, up to homeomorphism, from $\mathrm{Top}(X)$ up to equivalence. We shall see below (4.2) that this is indeed so, subject to a slight restriction on X.

2.2. The punctual or final topos, and the empty or initial topos

When X is a topological space reduced to a single point, the functor

$$F \longmapsto F(X) : \mathrm{Top}(X) \longrightarrow \mathcal{U}\text{-}\mathbf{Set}$$

is an equivalence of categories. In particular, the category $\mathcal{U}$-**Set** is a $\mathcal{U}$-topos. We saw in Exposé II that this $\mathcal{U}$-topos is typical from the point of view of exactness properties, in that the verification of many properties of general topoi, especially exactness properties, reduces to this particular topos. The interpretation given here in terms of the one-point topological space justifies the abuse of language by which a topos equivalent to $\mathcal{U}$-**Set** is called a *punctual topos*, although as a category it is not at all equivalent to the one-object category. This terminology corresponds to the correct geometric intuition for the role played by these topoi. Also by abuse of language, a punctual topos is sometimes called a *final topos*; we shall say “the final topos” for $\mathcal{U}$-**Set**.

When X is the empty topological space, so that $\mathrm{Open}(X)$ is the one-object category, a presheaf F on $\mathrm{Open}(X)$ is a sheaf if and only if its value at the unique object $\varnothing$ of $\mathrm{Open}(X)$ is a singleton. It follows that $\mathrm{Top}(\varnothing)$ is isomorphic to the category of singleton $\mathcal{U}$-sets, a category equivalent to the one-object category. Thus the one-object category, and any $\mathcal{U}$-category equivalent to it, is a $\mathcal{U}$-topos. It is sometimes called, by abuse of language, the *empty topos* or *initial topos*; note that it is not equivalent to the empty category.

2.3. Topos associated with a space with operators

Let X be a topological space and let G be a discrete group acting on X by homeomorphisms. In [Toh., 5.1] the category of *G-sheaves on X* was defined; we shall also say sheaves on (X, G). These are sheaves of sets on X endowed with actions of G compatible with the action of G on X. Giraud's criterion 1.2(iii) immediately shows that this category is a topos, denoted $\mathrm{Top}(X, G)$. When G is the trivial group, this gives the example of 2.1; when X is a point, this gives the topos of sets with a left action of G, or *G-sets*, also called the *classifying topos of the discrete group G*, and denoted B_G. One checks easily that the only subobjects of the final object e of B_G are e and the empty sheaf $\varnothing$. In particular, if G is not the trivial group, subobjects of the final object of B_G do not form a generating family for B_G. Thus B_G is then not equivalent to a topos of the form $\mathrm{Top}(X)$ considered in 2.1.

The notion of G-sheaf was introduced in loc. cit. to develop the cohomological theory of abelian G-sheaves. Interpreting these as the abelian sheaves of the topos (X, G), that theory is included in the theory of Exposé V, developed for general topoi.

More generally, one may wish to attach an appropriate topos to a topological space X endowed with a topological group G, not necessarily discrete, of automorphisms, giving rise to an adequate cohomological theory. The same question arises in the contexts of differentiable manifolds, real or complex analytic manifolds or spaces, and schemes. This is indeed possible; cf. 2.5 below.

2.4. Classifying topos of a group object

Let E be a topos and let G be a group object of E. Let (E, G) be the category of objects of E on which G acts. Giraud's criterion immediately shows that this is a topos. It is called the *classifying topos* of the group object G, and is denoted B_G. When E is the punctual topos, i.e. when G is an ordinary group, one recovers the classifying topos of 2.3.

The terminology is justified by the fact that B_G plays a universal role in the classification of torsors, or principal homogeneous bundles, under G, and more generally under $G_{E'} = f^*(G)$, where E' is a topos "over E", i.e. endowed with a morphism $f : E' \to E$ (cf. 3.1 below). This role, made explicit in [3, Chap. V] or in 5.9 below, shows that B_G plays in the context of topoi the same role as the classical classifying spaces of topological groups in the homotopy theory of topological spaces. The latter may be regarded (cf. 2.5) as a weakened version of the former, obtained by retaining only the "homotopy type" of the classifying topos, in a suitable sense which need not be made precise here.

2.5. The big site and big topos of a topological space. Classifying topos of a topological group

Let $\mathcal{U}$-**Top**, or simply (**Top**), be the category of topological spaces belonging to $\mathcal{U}$. Finite projective limits are representable in (**Top**). Consider on (**Top**) the pretopology for which $\mathrm{Cov}(X)$ is the set of surjective families of open immersions $u_i : X_i \to X$. We shall regard (**Top**) as a site via the topology generated by this pretopology. For every object X of (**Top**), consider the category

$$(\mathbf{Top})_{/X}$$

of objects of (**Top**) over X, i.e. topological spaces over X, as a site by means of the topology induced from that of (**Top**) by the forgetful functor $(\mathbf{Top})_{/X} \to (\mathbf{Top})$. This site is called the *big site* associated with X. It is not an element of $\mathcal{U}$; nor is it a $\mathcal{U}$-site in the sense of Exposé II, 3.0.2, so precautions are needed when applying the usual results to it. To avoid this inconvenience, one may choose a universe $\mathcal{V}$ such that $\mathcal{U} \in \mathcal{V}$, so that $(\mathbf{Top})_{/X}$ becomes a $\mathcal{V}$-site, and work with the associated $\mathcal{V}$-topos, which may be denoted $\mathrm{TOP}(X)$ and will be called the *big topos of* X. If one is reluctant to enlarge $\mathcal{U}$, one may choose a cardinal c bounding the cardinalities of X and of all topological spaces one expects to use in the arguments–most often $\sup(\operatorname{card} X, \operatorname{card} \mathbb{R})$ will suffice–and replace $(\mathbf{Top})_{/X}$ by the subcategory $(\mathbf{Top})'_{/X}$ formed by those X' over X with $\operatorname{card} X' \leq c$, endowed with the induced topology, and denote the sheaf topos on that site by $\mathrm{TOP}(X)$. To fix ideas, suppose the first definition has been adopted.

The advantage of the big topos of X over the small one is that the site defining it contains $(\mathbf{Top})_{/X}$ as a full subcategory. Since the topology of this site is manifestly no finer than the canonical topology, the canonical functor from $(\mathbf{Top})_{/X}$ to $\mathrm{TOP}(X)$, sending a space X' over X to the sheaf it represents, is *fully faithful.* Thus *a space X' over X is known up to X-isomorphism once the sheaf it defines is known*; the notion of sheaf on the big site of X may therefore be regarded as a *generalization* of the notion of topological space over X, in terms of which all constructions of sheaf theory acquire meaning for topological spaces over X.

Thus, when G is a group object of the category $(\mathbf{Top})_{/X}$ of topological spaces over X, one can associate to it the classifying topos B_G (2.4), hence classifying cohomology groups, classifying homotopy groups, and so on, defined as the corresponding invariants of the $\mathcal{V}$-topos B_G. In particular, when X is a point, G is an ordinary topological group. Under the usual local hypotheses ensuring that the singular cohomology of the Cartesian powers G^n agrees with sheaf cohomology, say for constant coefficients–for example when G is locally contractible–the cohomology of the classifying topos of G is canonically isomorphic to that of the classifying space of G in the sense of topologists.

The introduction of classifying topoi through big sites has, compared with classifying spaces, the advantage of giving a richer theory, since it provides useful cohomological invariants for coefficients more general than constant or locally constant coefficients. Moreover, the definition considered here adapts in the evident way to the usual other contexts: differentiable manifolds, real or complex analytic manifolds or spaces, and schemes. This point of view makes it possible, in particular, to connect the traditional and the "arithmetic" approaches to characteristic classes by considering the classical groups as coming from schemes defined over the ring of integers; cf. Section 7 for indications in this direction. Similarly, J. Giraud's general results [3] on the classification of extensions of group objects, developed in the very general and flexible setting of topoi, may, through big topoi, specialize to results on the classification of extensions of topological groups, or of real or complex Lie groups, which seem to have been known to topologists essentially only in the case of extensions with abelian kernel [11].

2.6. Topoi of the form $\widehat{C}$

Let C be a small category. The category $\widehat{C}$ of $\mathcal{U}$-presheaves on C is plainly a $\mathcal{U}$-topos, since it is of the form $\widetilde{C}$ when C is endowed with the chaotic topology. We shall give below some details on the relations between C and $\widehat{C}$. Let us note here only that a topos E equivalent to one of the form $\widehat{C}$ is rather special, since *it admits a small generating family* (X_i) *formed of connected projective objects*, i.e. objects X such that the functor $Y \mapsto \mathrm{Hom}(X, Y)$ sends epimorphisms to epimorphisms and coproducts to coproducts. It suffices to take in $\widehat{C}$ the generating family formed by the functors represented by the objects $X \in \mathrm{Ob}\, C$. Note also that if, in a topos E, one has a covering family $X_i \to X$ with the X_i projective and connected, then every other covering family of X is refined by this one. Hence in a topos E of the form $\widehat{C}$, every object X admits a covering family refining all the others. A topos of the form $\mathrm{Top}(X)$, with X a topological space whose points are closed, has this property only when X is discrete.

When the category C has a single object, C is identified with a monoid G. A presheaf on C is then a set on which G acts on the *right*, since it is a functor $G^\circ \to \mathbf{Set}$, and $\widehat{C}$ is the topos of sets with a right monoid of operators. Taking 2.3 into account, this may also be denoted B_{G°: it is the *topos of sets with monoid of operators* G°, where G° is the opposite monoid. When G is a group, using the isomorphism $g \mapsto g^{-1}$ from G to G°, one recovers the classifying topos B_G of 2.3.

2.7. Classifying topos of a pro-group

2.7.1

Let $\mathcal{G} = (G_i)_{i \in I}$ be a projective system of groups, with $G_i, I \in \mathcal{U}$. Suppose the projective system is *strict*, i.e. the transition morphisms $G_j \to G_i$ are surjective. If E is a set, an *action of* $\mathcal{G}$ *on*

E (say on the left) is the following structure: a family $(E_i)_{i\in I}$ of subsets of E, with union E; and, for every $i \in I$, an action of the group G_i on the set E_i. These data are required to satisfy the following condition: for $j \geq i$, E_i is the subset of E_j formed by the elements fixed under the kernel of $G_j \to G_i$. The $\mathcal{U}$-sets endowed with an action of $\mathcal{G}$ form a category in the evident way. Giraud's criterion immediately shows that this category is a $\mathcal{U}$-topos. It is denoted $B_{\mathcal{G}}$ and called the *classifying topos of* $\mathcal{G}$. When I has an initial object i_0, and $G = G_{i_0}$, this recovers the classifying topos of 2.3.

2.7.2

Another important example is the case where the groups G_i are finite, so that

$$G = \varprojlim G_i$$

is a compact totally disconnected topological group, or *profinite group*. An action of $\mathcal{G}$ on E is then the same as a continuous action of G on E, or equivalently an action such that the stabilizer of every point of E is an open subgroup of G. The classifying topos $B_{\mathcal{G}}$ will also be denoted B_G, where of course G is considered with its profinite topology.

2.7.3

Using the remarks of 2.6, it is easy to check that the topos $B_{\mathcal{G}}$ defined by a strict projective system $\mathcal{G} = (G_i)_{i\in I}$ of groups is equivalent to a topos of the form $\widehat{C}$ only if this projective system is essentially constant. In the case of a profinite group, this means that the group is actually finite.

2.7.4

The following geometric interpretation of the classifying topos B_G of a discrete group G is useful for giving the correct geometric intuition for these topoi; cf. also 4.5, 5.8, 5.9, and 7.2 below. Let X be a connected, locally connected, and locally simply connected topological space, let x be a point of X, and let G be its fundamental group at x. It is known that, up to isomorphism, every discrete group G can be obtained in this way. Galois theory of coverings of X gives an equivalence between the category B_G of G-sets and the category of *étale coverings* of X, that is, of spaces X' over X which are locally X-isomorphic to X-spaces of the form $X \times I$, where I is a discrete space. This latter category may also be interpreted as the category of locally constant sheaves on X, i.e. the category of locally constant objects of the topos $\mathrm{Top}(X)$.

When G is a profinite group, there is an analogous geometric interpretation of B_G as the category of X-schemes which are sums of finite étale coverings of a connected scheme X, endowed with a geometric point x and an isomorphism $G \simeq \pi_1(X, x)$. It is again known that every profinite group can occur as the fundamental group of a suitable connected scheme, the spectrum of a field if desired. Finally, pro-groups which are not necessarily profinite and not necessarily essentially constant also occur in the classification of coverings of connected and locally connected spaces which are not locally simply connected, or in the classification of étale coverings, not necessarily finite or ind-finite, of non-normal connected schemes. For this last case, cf. SGA 3, X, 6.

Exercise 2.7.5. Define, for a topos E, the notions of connectedness, local connectedness, simple connectedness, and local simple connectedness. Define the notions of constant and locally constant object of E (cf. IX, 2.0). Let $f : E' \to E$ be a geometric morphism, with E' simply connected (for example E' the punctual topos), and E connected and locally connected. Define a strict pro-group

$$\pi_1(E, f) = (G_i)_{i\in I} = \mathcal{G},$$

called the *fundamental pro-group of E at f*, and an equivalence of categories from $B_{\mathcal{G}}$ to the category of locally constant objects of E. Show that when E is locally simply connected, $\pi_1(E, f)$ is essentially constant and therefore identifies with an ordinary discrete group $\pi_1(E, f)$, called the *fundamental group of E at f*. Show that every strict pro-group $\mathcal{G}$ is isomorphic, as a pro-group, to the fundamental pro-group of a suitable connected and locally connected topos E, for a suitable f, with E' the punctual topos: take $E = B_{\mathcal{G}}$, and $f : E' \to E$ defined by the forgetful functor $f^* : E \to E' = \mathbf{Set}$. When $\mathcal{G}$ is essentially constant, i.e. isomorphic as a pro-group to an ordinary discrete group, prove that one may take E above to be locally simply connected, again taking $E = B_G$.

2.8. Example of a false topos

Let $\mathcal{G} = (G_i)_{i\in I}$ be a strict pro-group, where I is a filtered ordered set and where $i > j$ implies that $G_i \to G_j$ is not an isomorphism. Suppose that $\mathrm{card}(I) \notin \mathcal{U}$. Consider the category of sets $E \in \mathcal{U}$ on which $\mathcal{G}$ acts on the left, in the sense of 2.7.1. This is a $\mathcal{U}$-category, and, as in 2.7.1, it satisfies conditions a), b), c) of 1.1.2. However, it is not a $\mathcal{U}$-topos, since it has no generating family that is $\mathcal{U}$-small. Likewise, it is not a $\mathcal{V}$-topos for any universe $\mathcal{V}$.

3 Geometric morphisms of topoi

Definition 3.1. Let E and E' be two $\mathcal{U}$-topoi. A *geometric morphism* from E to E', or sometimes, by abuse of language, a continuous map from E to E', is a triple $u = (u_*, u^*, \varphi)$, formed of functors

$$u_* : E \longrightarrow E', \qquad u^* : E' \longrightarrow E,$$

and an adjunction isomorphism of bifunctors, for $X' \in \mathrm{Ob}\, E'$ and $Y \in \mathrm{Ob}\, E$,

$$\varphi : \mathrm{Hom}_E(u^*X', Y) \xrightarrow{\sim} \mathrm{Hom}_{E'}(X', u_*Y),$$

where u^* is moreover required to be left exact, i.e. to commute with finite projective limits. The functor u_* is called the direct-image functor of the geometric morphism u, the functor u^* is called the inverse-image functor of u, and φ is called the adjunction isomorphism for u.

3.1.1

Unless explicitly stated otherwise, for a geometric morphism $u : E \to E'$ we shall denote the corresponding direct-image and inverse-image functors by u_* and u^*.[4] Since u_* is right adjoint to u^*, and u^* is left adjoint to u_* by the adjunction isomorphism φ, either of the two functors u_*, u^* determines the other up to unique isomorphism, by the usual formalism of adjoint functors. In practice, depending on the case, it may be more convenient to define a geometric morphism

[4] Sometimes one should also write u^{-1} instead of u^*, cf. 13.2.3.

$u : E \to E'$ either by giving $u^* : E' \to E$, or by giving $u_* : E \to E'$. In the first case one must simply verify that the given functor u^* admits a right adjoint and is left exact. In the second, one must verify that the given functor u_* admits a left adjoint which is left exact. In either case, from the partial datum and from a *choice* of adjoint functor and adjunction isomorphism, one obtains a geometric morphism $u : E \to E'$, which is unique up to unique isomorphism in terms of the datum u^*, respectively u_*, in an evident sense made entirely explicit below in 3.2.1.

3.1.2

If $u : E \to E'$ is a geometric morphism, it follows from the properties of adjoint functors that the direct image $u_* : E \to E'$ commutes with projective limits, while $u^* : E' \to E$ commutes with inductive limits.[5] Since u^* is also assumed left exact, it follows in particular that u^* is *exact*. Thus, in the pair (u_*, u^*), it is the *inverse-image functor* u^* which has the strongest exactness properties.

These properties ensure that, for any kind of algebraic structure Σ whose data can be described in terms of arrows between underlying sets and sets obtained from them by repeated application of finite projective limits and arbitrary inductive limits, and for any "object of E' endowed with a structure of kind Σ", its image under u^* is endowed with the same structures. Rather than undertake the unattractive task of giving a precise formal meaning to this statement and justifying it formally, we advise the reader to spell it out and verify it for structures such as group, ring, module over a ring, comodule over a co-ring, bialgebra over a ring, and torsor under a group. In these examples, the first three kinds of structure are defined exclusively in terms of finite projective limits, while the others implicitly involve constructions using inductive limits as well. Moreover, u^* "commutes" with all the usual functorial operations made in terms of such structures, more precisely with all operations expressible in terms of inductive limits and finite projective limits: construction of free objects, such as free groups or free modules generated by an object, tensor products, cf. Section 12 below, and so on.

The direct image $u_* : E \to E'$, which commutes with projective limits, consequently respects every algebraic structure on an object, or a family of objects, of E which is definable solely in terms of projective limits, such as group, ring, or module structures among the examples just mentioned. On the other hand, u_* is generally not right exact, i.e. it does not generally commute with finite inductive limits, and it does not in general even send epimorphisms to epimorphisms. This failure of exactness of u_* is precisely the source of cohomological phenomena, to be studied from the standpoint of commutative homological algebra in the following exposé. Consequently, u_* does not generally extend to objects such as comodules, bialgebras, or torsors under a group, and does not in general commute with operations such as the free module generated by an object, tensor products of modules, and so on.

3.1.3

In practice, when one has a functor $f : E \to F$ from one $\mathcal{U}$-topos to another, one should make explicit all its exactness properties, including the possible existence of left or right adjoints, in order to understand the "geometric nature" of f. Such an understanding is usually an indispensable guide to the correct geometric intuition. Thus, if f commutes with arbitrary

[5] Moreover, by 1.5 and 1.8, for a given functor $u_* : E \to E'$, respectively $u^* : E' \to E$, this functor admits a left, respectively right, adjoint if and only if it commutes with $\mathcal{U}$-projective, respectively $\mathcal{U}$-inductive, limits.

inductive limits and finite projective limits, it should be written

$$f = u^*,$$

where

$$u : F \longrightarrow E$$

is a geometric morphism; in other words, f should be interpreted as an inverse-image functor for a continuous map of topoi. If f commutes with arbitrary projective limits, so that it has a left adjoint, and if that left adjoint, which a priori commutes with arbitrary inductive limits, also commutes with finite projective limits, then f should be written

$$f = v_*,$$

where

$$v : E \longrightarrow F$$

is a geometric morphism. In some cases f may satisfy both of these conditions; cf. 4.10 for an example. In that case one should introduce simultaneously the geometric morphisms

$$u : F \longrightarrow E, \qquad v : E \longrightarrow F,$$

which give rise to a sequence of three adjoint functors

$$v^*, \qquad v_* = u^* = f, \qquad u_*.$$

One must then carefully distinguish the geometric morphisms u and v, under penalty of losing the geometric intuition of the situation.

In this connection, note that when between two topoi E, F one has a sequence of three adjoint functors

$$e, \quad f, \quad g \qquad (e, g : F \rightrightarrows E,\ f : E \to F),$$

then f commutes with arbitrary inductive and projective limits, and hence can always be written as u^*, where $u : F \to E$ is a geometric morphism. Then g is u_*. On the other hand, of course, e can be written as v^*, and then f as v_*, only if e also commutes with finite projective limits. This will certainly be the case if e is itself the right adjoint of a fourth functor d. Without such a condition, one often writes

$$e = u_!$$

for the left adjoint of an inverse-image functor $f = u^*$, when such a left adjoint exists; this notation is suggested by the example of an open immersion $u : X \to Y$ of topological spaces. Similarly, if e is of the form v^*, i.e. if f is of the form v_*, one sometimes writes $g = v^!$ for the right adjoint of a direct-image functor v_*, when such an adjoint exists.[6]

[6] Cf. below, 14.4, in the case of abelian sheaves.

3.2

Let E, E' be two $\mathcal{U}$-topoi, and let

$$u = (u_*, u^*, \varphi), \qquad v = (v_*, v^*, \Psi) : E \rightrightarrows E'$$

be two geometric morphisms from E to E'. A *morphism from u to v* is a morphism from u_* to v_* in the category $\mathcal{H}om(E, E')$ of functors from E to E'. Morphisms of geometric morphisms compose in the evident way, thereby defining a category, in fact a $\mathcal{U}$-category, denoted

$$\mathcal{H}om_{\mathrm{top}}(E, E'),$$

called the *category of geometric morphisms*, or of continuous maps, from E to E'. There is an evident functor

$$u \longmapsto u_* : \mathcal{H}om_{\mathrm{top}}(E, E') \longrightarrow \mathcal{H}om(E, E'),$$

which is fully faithful by the definition of arrows in the first category, though it is not injective on objects.

3.2.1

If u and v are as above, the theory of adjoint functors gives a canonical bijection

$$\mathrm{Hom}(u_*, v_*) \xrightarrow{\sim} \mathrm{Hom}(v^*, u^*).$$

In particular, one obtains a contravariant functor on $\mathcal{H}om_{\mathrm{top}}(E, E')$:

$$u \longmapsto u^* : \mathcal{H}om_{\mathrm{top}}(E, E')^\circ \longrightarrow \mathcal{H}om(E, E').$$

In accordance with the geometric intuition, according to which the direct-image functor u_* goes in the same direction as the continuous map giving rise to it, the direction of arrows between geometric morphisms should therefore be defined in terms of direct-image functors, not in terms of inverse-image functors, even though the latter, as we have seen, have the characteristic exactness properties of the notion of geometric morphism.

3.2.2

Having defined the category $\mathcal{H}om_{\mathrm{top}}(E, E')$ of morphisms from the topos E to the topos E', the notion of *isomorphism* between two morphisms u, v from E to E' is also defined. In practice, one usually does not distinguish essentially between two isomorphic geometric morphisms, at least when a *canonical* isomorphism between them is available, just as one often does not distinguish between two objects of a category when one has chosen a canonical isomorphism between them. In this connection, let us point out that when one deals with diagrams of geometric morphisms and commutativity questions for such diagrams, a notion meaningful by 3.3, one usually means commutativity only up to “canonical” isomorphism. By abuse of language, such diagrams are then treated as genuinely commutative.

One might try to justify this abuse by introducing the set $\mathrm{Hom}_{\mathrm{top}}(E, E')/\,\mathrm{Isom}$ of morphisms from E to E' up to isomorphism, and by calling such an isomorphism class a morphism, instead of following Definition 3.1. But this meets the same serious drawbacks that arise whenever one tries to identify two isomorphic objects of a category without having a canonical isomorphism between them. Experience shows that such a viewpoint is impracticable, and that one must retain the fine notion of 3.1, together with the notion of morphism between geometric morphisms, even if this sometimes forces one to fight with compatibilities among canonical isomorphisms.[7]

3.3.1

Let E, E', E'' be three $\mathcal{U}$-topoi, and consider geometric morphisms

$$u : E \longrightarrow E', \qquad v : E' \longrightarrow E''.$$

The theory of adjoint functors gives an adjunction isomorphism between the composite functors $v_* u_*$ and $u^* v^*$, in terms of the adjunction isomorphisms for the pairs (u_*, u^*) and (v_*, v^*). On the other hand, $u^* v^*$ is left exact as a composite of left exact functors. Thus one obtains a morphism from E to E'', called the *composite of the morphisms u and v*, and denoted vu:

$$vu : E \longrightarrow E''.$$

One then verifies trivially that composition of morphisms is associative, and that for every $\mathcal{U}$-topos E there is a morphism from E to itself which is a two-sided unit for composition, namely $(\mathrm{id}_E, \mathrm{id}_E, \varphi)$, where φ is the evident adjunction isomorphism from id_E to itself. If $\mathcal{V}$ is a universe such that $\mathcal{U} \in \mathcal{V}$, this defines a category

$$(\mathcal{V}\text{-}\mathcal{U}\text{- Top}),$$

whose objects are the $\mathcal{U}$-topoi belonging to $\mathcal{V}$, whose arrows are the morphisms between such $\mathcal{U}$-topoi, and whose composition is the one just described.

3.3.2

In fact, the composition map

$$\mathrm{Hom}_{\mathrm{top}}(E, E') \times \mathrm{Hom}_{\mathrm{top}}(E', E'') \longrightarrow \mathrm{Hom}_{\mathrm{top}}(E, E'')$$

is the map on objects induced by a "composition functor for morphisms"

$$\mathcal{H}om_{\mathrm{top}}(E, E') \times \mathcal{H}om_{\mathrm{top}}(E', E'') \longrightarrow \mathcal{H}om_{\mathrm{top}}(E, E''),$$

whose effect on arrows is the usual convolution product for morphisms between functors, here direct-image functors. These composition functors satisfy a strict associativity property for four topoi E, E', E'', E''', refining the associativity of composition of geometric morphisms. In language which is becoming familiar, one may also say that $\mathcal{U}$-topoi are the objects, or 0-arrows, of a *2-category*, whose 1-arrows are geometric morphisms and whose 2-arrows are morphisms of geometric morphisms.

It is this fact–that $\mathcal{U}$-topoi belonging to a universe $\mathcal{V}$ form a 2-category, and not merely an ordinary category as ordinary topological spaces do–which, from the technical point of view, is the most important difference between the theory of topoi and that of topological spaces. It is the source of certain technical complications already mentioned, but also of essentially new facts compared with traditional topology.

[7] For examples of such battles, apparently victorious, we refer to M. Hakim's book on relative schemes [9].

3.4

The fact that $\mathcal{U}$-topoi belonging to a universe $\mathcal{V}$ form a 2-category makes it possible, in particular, to define the notion of *equivalence of two $\mathcal{U}$-topoi E, E'*: we say that E and E' are *equivalent* if there exist geometric morphisms $u : E \to E'$ and $v : E' \to E$ such that the composites uv and vu are isomorphic to the identity morphisms of E and of E', respectively. Then u and v are said to be *quasi-inverse equivalences* of each other.

One sees at once that a geometric morphism $u : E \to E'$ is an equivalence if and only if u_* is an equivalence, or equivalently if u^* is an equivalence. Use the fact that a functor $f : E \to E'$ between two topoi which is an equivalence is both of the form u_* and of the form v^*, for geometric morphisms u and v. Also, E and E' are equivalent in the sense of the preceding paragraph if and only if they are equivalent as categories, i.e. as objects of the 2-category ($\mathcal{V}$-**Cat**). As expected, this shows that the notions of equivalence introduced here do not depend on the choice of the universe $\mathcal{V}$ in 3.3.1.

3.4.1

In practice, one usually does not distinguish essentially between equivalent $\mathcal{U}$-topoi, just as one often does not distinguish between equivalent categories, provided however that one has an explicit equivalence between them, or at least an equivalence defined up to unique isomorphism. This notion of equivalence of topoi replaces here the traditional notion of homeomorphism between topological spaces. See the example in 4.2 below for the precise relation between these two notions.

4 Examples of geometric morphisms of topoi

We return here to the examples of Section 2, using the notion of geometric morphism. The comments of 2.0 apply equally to this section. The universe $\mathcal{U}$ will generally be left implicit.

4.1. The topos $\mathrm{Top}(X)$ for variable topological space X

4.1.1

Let

$$f : X \longrightarrow Y$$

be a continuous map of topological spaces. We shall associate to it canonically a geometric morphism

$$\mathrm{Top}(f) \text{ or } f : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(Y),$$

with the notation of 2.1. When $\mathrm{Top}(X)$ is defined as $\widetilde{\mathrm{Open}(X)}$, the most convenient description of $\mathrm{Top}(f)$ is by the direct-image functor for sheaves

$$f_* : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(Y),$$

defined by the formula

$$f_*(F) = F \circ f^{-1},$$

where

$$f^{-1} : \mathrm{Open}(Y) \longrightarrow \mathrm{Open}(X)$$

is the evident functor $U \mapsto f^{-1}(U)$. We have already noted that this functor is continuous and left exact, and thus defines the above functor f_*, which has a right adjoint f^* that is left exact (Exposé III, 1.9.1). Strictly speaking, the geometric morphism $\mathrm{Top}(f)$ depends on the choice of the right adjoint f^* of f_*, and is therefore defined only up to canonical isomorphism. We shall henceforth refrain from explicitly signaling phenomena of this sort.

When one adopts the viewpoint of *étalé spaces* to define $\mathrm{Top}(X)$, it is the inverse-image functor

$$f^* : \mathrm{Top}(Y) \longrightarrow \mathrm{Top}(X)$$

which is most convenient for defining $\mathrm{Top}(f)$, by simply putting

$$f^*(Y') = X \times_Y Y'$$

for every étale space Y' over Y. It is clear that the fiber product is an étale space over X, and that the functor f^* thus obtained is left exact and commutes with arbitrary inductive limits, hence defines a geometric morphism $\mathrm{Top}(f)$. For the compatibility of the two definitions we refer to [TF].

For composable continuous maps

$$X \xrightarrow{f} Y \xrightarrow{g} Z,$$

there are transitivity isomorphisms

$$\mathrm{Top}(gf) \simeq \mathrm{Top}(g)\,\mathrm{Top}(f)$$

of geometric morphisms. For three composable continuous maps f, g, h, these *transitivity isomorphisms* satisfy a compatibility relation which we do not write out here, and which is precisely the one considered in SGA 1, VI, 7.4(B). This may be expressed by saying that, for X varying in the category (**Top**),

$$X \longmapsto \mathrm{Top}(X)$$

is a "pseudo-functor"

$$(\mathcal{U}\text{-}\mathbf{Top}) \longrightarrow (\mathcal{V}\text{-}\mathcal{U}\text{- Top}), \tag{4.1.1.1}$$

or, in the terminology of 2-categories, that one has a non-strict functor of 2-categories. In practice we shall usually allow the abuse of language that identifies $\mathrm{Top}(gf)$ and $\mathrm{Top}(g)\,\mathrm{Top}(f)$, reasoning as though (4.1.1.1) were a genuine functor of ordinary categories. We shall allow the analogous abuses in the other examples below.

4.1.2

The preceding considerations extend immediately to topoi associated with topological spaces with groups of operators (2.3). If $f = (f^{\mathrm{sp}}, f^{\mathrm{gr}})$,

$$f : (X, G) \longrightarrow (Y, H)$$

is a morphism of spaces with operators, where

$$f^{\mathrm{sp}} : X \to Y, \qquad f^{\mathrm{gr}} : G \to H$$

are respectively a continuous map and a group homomorphism, compatible in the evident sense, then one associates to it a geometric morphism

$$\mathrm{Top}(f) \text{ or } f : \mathrm{Top}(X, G) \longrightarrow \mathrm{Top}(Y, H),$$

whose definition is left to the reader. When G and H are the trivial groups, this gives the definition of 4.1.1. When instead X and Y are points, the inverse-image functor f^* is the "restriction of the group of operators"

$$f^* : B_H \longrightarrow B_G,$$

which sends an H-set to the G-set it defines via $f : G \to H$. We shall encounter this example again in other forms in 4.5 and 4.6.1.

4.1.3

Given a continuous map $f : X \to Y$ of topological spaces, one also associates to it a morphism of the corresponding big topoi (2.5),

$$\mathrm{TOP}(f) \text{ or } f : \mathrm{TOP}(X) \longrightarrow \mathrm{TOP}(Y),$$

most conveniently defined by the inverse-image functor

$$f^* : \mathrm{TOP}(Y) \longrightarrow \mathrm{TOP}(X),$$

which is just the *restriction functor*. This morphism $\mathrm{TOP}(f)$ is a special case of the so-called "inclusion" morphism for an induced topos, which will be studied in Section 5. Thus, with the same reservations as in 4.1.1, the topos $\mathrm{TOP}(X)$ may be regarded as a functor in X, for X varying in (**Top**).

4.2. Faithfulness properties of $X \mapsto \mathrm{Top}(X)$

We shall specify to what extent a topological space X can be reconstructed from the topos $\mathrm{Top}(X)$. For this purpose, for two spaces X and Y, one must describe the category of morphisms from $\mathrm{Top}(X)$ to $\mathrm{Top}(Y)$, in order to state the faithfulness properties of the "functor" $X \mapsto \mathrm{Top}(X)$. We merely state the results obtained, referring the reader to [9] for details. The reader who wants to verify them directly may consult Exercise 7.8.

4.2.1

Recall that a topological space X is called *sober* if every irreducible closed subset of X has exactly one generic point. Almost all spaces used in practice are sober; this is true in particular for Hausdorff spaces, more generally for spaces all of whose points are closed, and for the underlying space of a scheme. With any topological space X one associates a sober topological space X_{sob} and a continuous map

$$\varphi : X \longrightarrow X_{\mathrm{sob}} \tag{4.2.1.1}$$

which is *universal* for continuous maps from X to sober spaces. Equivalently, one constructs a left adjoint $X \mapsto X_{\mathrm{sob}}$ to the inclusion functor from sober spaces to all topological spaces. Explicitly, the points of X_{sob} are the irreducible closed subsets of X, and the open subsets are those of the form U', where U is an open subset of X and $U' \subset X_{\mathrm{sob}}$ is the set of irreducible closed subsets of X meeting U. The map (4.2.1.1) sends $x \in X$ to the closure of $\{x\}$. The space X is sober if and only if the preceding map is bijective, hence a homeomorphism.

The functor

$$\varphi^{-1} : \mathrm{Open}(X_{\mathrm{sob}}) \longrightarrow \mathrm{Open}(X)$$

induced by φ is an isomorphism. It follows that the geometric morphism

$$\mathrm{Top}(\varphi) : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(X_{\mathrm{sob}})$$

defined by φ is also an isomorphism. This explains a priori why X_{sob} must enter into the question of reconstructing X from $\mathrm{Top}(X)$: the latter depends only on X_{sob}, up to isomorphism, so the question can have an affirmative answer only if X is sober. We shall specify below (7.1) how X_{sob} can in fact be reconstructed in terms of $\mathrm{Top}(X)$, by interpreting its points as "points" of the topos $\mathrm{Top}(X)$, or equivalently as fiber functors.

4.2.2

On every topological space one should introduce the order relation $\leq$ defined by

$$x \leq y \quad \Longleftrightarrow \quad \overline{\{x\}} \subset \overline{\{y\}} \quad \Longleftrightarrow \quad x \in \overline{\{y\}},$$

which is also expressed by saying that x is a *specialization* of y, or that y is a *generalization* of x. For a space of the form X_{sob}, this is just the inclusion relation among irreducible closed subsets of X.

This being so, on the set of maps from a space X to a space Y one introduces the so-called specialization order induced from that of Y, namely

$$f \leq g \quad \Longleftrightarrow \quad f(x) \leq g(x) \text{ for all } x \in X.$$

With these conventions one has the following result.

4.2.3

Let X, Y be topological spaces, with Y sober, and let $f, g : X \to Y$ be continuous maps. Then there is at most one morphism from $\mathrm{Top}(f)$ to $\mathrm{Top}(g)$, and such a morphism exists if and only if g is a specialization of f. Finally, every geometric morphism $\mathrm{Top}(X) \to \mathrm{Top}(Y)$ is isomorphic to a morphism of the form $\mathrm{Top}(f)$, where $f : X \to Y$ is a continuous map, uniquely determined by the first assertion.

If $\mathrm{cat}(I)$ denotes the category associated with an ordered set I, defined by declaring that for $i \geq j$ there is exactly one arrow from i to j, the preceding result may be summarized by saying that there is a canonical equivalence of categories

$$\mathrm{cat}(\mathrm{Hom}_{\mathbf{Top}}(X, Y)) \xrightarrow{\approx} \mathcal{H}om_{\mathrm{top}}(\mathrm{Top}(X), \mathrm{Top}(Y)).$$

Formally, these results imply:

Corollary 4.2.4.

a) Let $f : X \to Y$ be a continuous map. Then $\mathrm{Top}(f) : \mathrm{Top}(X) \to \mathrm{Top}(Y)$ is an equivalence of topoi if and only if $f_{\mathrm{sob}} : X_{\mathrm{sob}} \to Y_{\mathrm{sob}}$ is a homeomorphism. Thus, when X and Y are sober, this is equivalent to f being a homeomorphism.

b) Let X, Y be topological spaces. The topoi $\mathrm{Top}(X)$ and $\mathrm{Top}(Y)$ are equivalent if and only if X_{sob} and Y_{sob} are homeomorphic. Thus, if X and Y are sober, this is equivalent to X and Y being homeomorphic.

4.3. Morphisms to the final topos: constant objects of a topos, section functors

Let P, for "point", denote the standard final topos, i.e. $P = \mathbf{Set}$ (2.2). Let E be any topos. We shall see that *up to unique isomorphism, there exists a unique geometric morphism*

$$f : E \longrightarrow P,$$

more precisely that $\mathcal{H}om_{\mathrm{top}}(E, P)$ is equivalent to the one-object category. Thus, between any two objects of this category there exists a unique arrow, and it is an isomorphism. This justifies, to some extent, the name "final topos".

For this, recall from 3.2.1 that $\mathcal{H}om_{\mathrm{top}}(E, P)$ is equivalent to the full subcategory of $\mathcal{H}om(P, E)^\circ$ formed by the functors

$$f^* : P = \mathbf{Set} \longrightarrow E$$

which commute with inductive limits and are left exact. Let e be a one-point set. Then every set X can be written canonically as the sum of X copies of e. It follows that the category of functors $g : \mathbf{Set} \to E$ which commute with inductive limits is equivalent to E, by associating to g the object $g(e)$ of E. Conversely, g is reconstructed from $T = g(e)$, up to unique isomorphism, by

$$g(I) = T \times I,$$

where $T \times I = \coprod_{i \in I} T_i$ with $T_i = T$ for all $i \in I$. The functor in I thus defined by T indeed commutes with inductive limits, since it is plainly left adjoint to the functor $X \mapsto \mathrm{Hom}(T, X)$ from E to $\mathbf{Set}$. For g to be left exact, it is plainly necessary that $g(e) = T$ be a final object of E, since e is a final object of $\mathbf{Set}$. It follows easily from the fact that coproducts in E are universal that this condition is also sufficient. Thus the category of inverse-image functors f^* is equivalent to the category of final objects of E, and this is itself plainly equivalent to the final category.

The preceding discussion shows that choosing a morphism $f : E \to P$ is essentially the same as choosing a final object e_E of E. In terms of this object, one has canonical isomorphisms of functors

$$f^*(I) \simeq e_E \times I = \text{the sum of } I \text{ copies of } e_E \qquad (I \in \mathrm{Ob}\,\mathbf{Set}), \tag{4.3.2}$$

and

$$f_*(X) = \mathrm{Hom}(e_E, X) \qquad (X \in \mathrm{Ob}\, E). \tag{4.3.3}$$

4.3.4

The preceding two functors will play an important role below. For a set I, the object $f^*(I) = e_E \times I$ of (4.3.2) is called the *constant object of value* I in E, or, when E is realized as a category $\widetilde{C}$ by means of a site C, the *constant sheaf of value* I *on* C. It will often be denoted I_E, or I_C when E is defined by the site C. The fact that $I \mapsto I_E$ is the inverse-image functor of a geometric morphism specifies its exactness properties; in particular, it respects all usual kinds of algebraic structure, sending a group to a group object of E, and so on (3.1.2). For example, if G

is a group object of E, we shall say that it is a *constant group object*, or, as the case may be, a *constant sheaf of groups*, if it is isomorphic to a group of the form $G_{0,E}$, where G_0 is an ordinary group. The same terminology applies to every other kind of "algebraic" structure, in the sense made more or less precise in 3.1.2.

4.3.5

One should be careful that the functor $I \mapsto I_E$ is not necessarily fully faithful, nor even faithful: take for E the empty topos. Hence a constant object of E does not generally determine, up to unique isomorphism, the set I from which it comes. The topos E is said to be 0-*acyclic*, or *connected and non-empty*, if the functor $I \mapsto I_E$ is fully faithful. By the general properties of adjoint functors, this is equivalent to saying that the adjunction morphism

$$I \longrightarrow f_* f^*(I) = f_*(I_E) = \mathrm{Hom}(e_E, I_E)$$

is an isomorphism, so that the functor (4.3.3) recovers the "value" of a constant object of E. One checks easily that this holds if and only if e_E is not the initial object $\varnothing_E$ of E, i.e. E is not an empty topos, which expresses the faithfulness of $I \mapsto I_E$,[8] and e_E is a connected object of E, i.e. is not a sum of two objects of E which are not empty, meaning not initial objects of E.

4.3.6

The functor (4.3.3) is also often called the *section functor* and denoted Γ_E, $\Gamma(E, -)$, or simply Γ:

$$\mathrm{Hom}(e_E, X) = \Gamma_E(X) = \Gamma(E, X) = \Gamma(X). \tag{4.3.6.1}$$

It commutes with arbitrary projective limits, but is not generally right exact; its derived functors on abelian group objects will be studied in the next exposé.

4.4. Morphisms from the empty topos

Let $\varnothing_{\mathrm{top}}$ be an empty topos, corresponding to a category of sheaves Φ equivalent to the final category (2.2). Let E be a topos. The category of functors from E to Φ is evidently equivalent to the one-object category, and every such functor commutes with inductive and projective limits. Thus it may be regarded as an inverse-image functor for a geometric morphism $\varnothing_{\mathrm{top}} \to E$. It follows that $\mathcal{H}om_{\mathrm{top}}(\varnothing_{\mathrm{top}}, E)$ is equivalent to the one-object category, and in particular that *there exists, up to unique isomorphism, a unique geometric morphism*

$$\varnothing_{\mathrm{top}} \longrightarrow E. \tag{4.4.1}$$

This justifies, to some extent, the terminology "initial topos" introduced in 2.2.

One may also determine the geometric morphisms

$$E \longrightarrow \varnothing_{\mathrm{top}}. \tag{4.4.2}$$

One checks at once that such a morphism exists if and only if the initial object of E is also final, i.e. if and only if E itself is an empty topos, and that in this case $\mathcal{H}om_{\mathrm{top}}(E, \varnothing_{\mathrm{top}})$ is again equivalent to the one-object category. The unique morphism (4.4.2), modulo isomorphism, is then an equivalence of topoi.

[8] Or equivalently the fact that this functor is conservative.

4.5. The classifying topos B_G for variable group object G

4.5.1

Let E be a topos and let
$$f : G \longrightarrow H$$
be a morphism of group objects in E. It induces a “restriction of the group of operators” functor
$$f^* : B_H \longrightarrow B_G,$$
with the notation of 2.4. It is trivial that this functor commutes with inductive limits and projective limits; a fortiori it may be interpreted as the inverse-image functor associated with a geometric morphism
$$B_f \text{ or } f : B_G \longrightarrow B_H.$$
The corresponding direct-image functor
$$f_* : B_G \longrightarrow B_H$$
is described easily by the formula
$$f_*(X) = \mathcal{H}om_G(H_s, X),$$
where X is an object of E with a left action of G, where H_s is H endowed with the left action of G induced by f, and where $\mathcal{H}om_G$ denotes the evident subobject of the $\mathcal{H}om$-object defined below in 10.2. The group object H acts on the left on $\mathcal{H}om_G(H_s, X)$ through the right actions of H on H_s by right translations.

Since the inverse-image functor f^* commutes with arbitrary projective limits, it is itself the right adjoint of a functor
$$f_! : B_G \longrightarrow B_H,$$
so that there is a sequence of three adjoint functors as in 3.1.3:
$$f_!, \quad f^*, \quad f_*.$$
The functor $f_!$ is described by the formula
$$f_!(X) = H \overset{G}{\times} X,$$
where the right-hand side denotes the contracted product, deduced from the action of G on X on the left and on H on the right via right translations and f, defined as the quotient of $H \times X$ by the action of G given by
$$g \circ (h, x) = (hg^{-1}, gx).$$

The functor $f_!$, being a left adjoint, plainly commutes with inductive limits, but it is generally not left exact, and hence cannot in turn be regarded as an inverse-image functor for a geometric morphism $B_H \to B_G$. In fact, it is easy to check that it can be left exact only if $f : G \to H$ is an isomorphism. Similarly, the functor f_*, which commutes with projective limits, is generally not right exact, and a fortiori does not generally admit a right adjoint. At least when E is the punctual topos, i.e. when G and H are ordinary groups, f_* is right exact only if f is an isomorphism.

When there is a second homomorphism of group objects $g : H \to K$, one obtains as in 4.1.1 a transitivity property up to canonical isomorphism. Thus, subject to the usual reservation, the classifying topos B_G may be regarded as depending *functorially* on the group object G. We leave to the reader the generalization of this functorial behavior to the case where both G and the topos E vary simultaneously.

4.5.2. The topos $B_{\mathcal{G}}$ for a variable pro-group $\mathcal{G}$

We leave to the reader the task of specifying the covariant character of the topos $B_{\mathcal{G}}$ (2.7) with respect to $\mathcal{G}$, by imitating the exposition given in 4.5.1. One should be careful, however, that in the case of a morphism $f : \mathcal{G} \to \mathcal{H}$ of pro-groups which are not essentially constant, the corresponding geometric morphism $B_{\mathcal{G}} \to B_{\mathcal{H}}$ does not generally allow the definition of a functor $f_!$, whose right adjoint would be the restriction functor f^* for the pro-group of operators. Assuming, for simplicity, that $\mathcal{G}$ and $\mathcal{H}$ are defined by profinite groups G and H, one sees easily that $f_!$ exists if and only if the image of $f : G \to H$ has finite index in H, and in that case $f_!$ is given by the same formula as in 4.5.

4.6. The topos $\widehat{C}$ for variable category C

4.6.1

Let

$$f : C \longrightarrow C'$$

be a functor from a category $C \in \mathcal{U}$ to another category C'. It gives a functor

$$f^* : \widehat{C'} \longrightarrow \widehat{C}, \qquad f^*(F) = F \circ f.$$

It is trivial that this functor commutes with projective and inductive limits; a fortiori it may be regarded as the inverse-image functor for a geometric morphism

$$\widehat{f} \text{ or } f : \widehat{C} \longrightarrow \widehat{C'}.$$

The corresponding direct-image functor

$$f_* : \widehat{C} \longrightarrow \widehat{C'}$$

is just the functor also denoted f_* in Exposé I, 5.1. Moreover, as expected from the fact that f^* also commutes with arbitrary projective limits, f^* also admits a left adjoint

$$f_! : \widehat{C} \longrightarrow \widehat{C'},$$

which was also denoted $f_!$ in Exposé I, 5.1. Thus one obtains a sequence of three adjoint functors

$$f_!, \quad f^*, \quad f_*,$$

the first being an extension of $f : C \to C'$ to categories of presheaves, for the usual embeddings of C, C' into $\widehat{C}, \widehat{C'}$. Note that $f_!$ is generally not left exact, nor is f_* generally right exact; this removes any ambiguity about the variance direction of the topos $\widehat{C}$ associated with the variable category C: it is *covariant* in C, subject to the usual reservation concerning transitivity isomorphisms. When the sets of objects of C and C' are each a singleton, so that C and C' identify with monoids G and G', the corresponding topoi are the classifying topoi B_{G° and $B_{G'^\circ}$, and one recovers the variance for variable G already encountered from other viewpoints in 4.1.2 and 4.5.

4.6.2

The 2-functorial dependence of the topos $\widehat{C}$ on C may be made precise by introducing, for two categories $C, C' \in \mathcal{U}$, a canonical functor

$$\mathcal{H}om(C, C') \longrightarrow \mathcal{H}om_{\mathrm{top}}(\widehat{C}, \widehat{C'}). \tag{4.6.2.1}$$

It remains to define this functor on arrows. For this, note that $f \mapsto f^*$ identifies, up to equivalence of categories, the target with a full subcategory of $\mathcal{H}om(\widehat{C'}, \widehat{C})^\circ$. If $f, g : C \to C'$ are two functors, any morphism $u : f \to g$ of functors defines a morphism $F \circ g \to F \circ f$ of functors in $F \in \mathrm{Ob}\,\widehat{C'}$, because F is a contravariant functor. This is a morphism $g^* \to f^*$, hence defines a morphism $\widehat{f} \to \widehat{g}$, as asserted.

When C is the one-object category, $\widehat{C}$ is the punctual topos P, and (4.6.2.1) is interpreted as a natural functor

$$C' \longrightarrow \mathcal{H}om_{\mathrm{top}}(P, \widehat{C'}) \stackrel{\mathrm{def}}{=} \mathrm{Pt}(\widehat{C'}), \tag{4.6.2.2}$$

from C' to the *category of points* of $\widehat{C'}$, which will be studied in Section 6. This functor is not necessarily an equivalence of categories (7.3.3); a fortiori (4.6.2.1) is not necessarily an equivalence of categories.

On the other hand, the functor (4.6.2.1) is always *fully faithful*. To see this, note that if $f, g : E \to E'$ are two geometric morphisms such that $f_!$ and $g_!$ are defined (3.1.3), then the theory of adjoint functors gives canonical isomorphisms

$$\mathrm{Hom}(f_!, g_!) \simeq \mathrm{Hom}(g^*, f^*) \simeq \mathrm{Hom}(f_*, g_*) \stackrel{\mathrm{def}}{=} \mathrm{Hom}(f, g).$$

Applying this to functors of the form $\widehat{f}, \widehat{g}$ associated with $f, g : C \to C'$, one obtains the result, because the natural extension map $\mathrm{Hom}(f, g) \to \mathrm{Hom}(f_!, g_!)$ is bijective (Exposé I, 7.8).

4.6.3

One may ask when the functor (4.6.2.1) is an equivalence of categories, i.e. when it is essentially surjective. This is a special case of determining all geometric morphisms $\widehat{C} \to \widehat{C'}$. More generally, if C is a category in $\mathcal{U}$ and E is a topos, one may try to determine the geometric morphisms

$$f : \widehat{C} \longrightarrow E.$$

The category of these morphisms is equivalent to the opposite of the category of functors $f^* : E \to \widehat{C}$ commuting with arbitrary inductive limits and finite projective limits. Interpreting functors $E \to \widehat{C}$ as functors $E \times C^\circ \to \mathcal{U}\text{-}\mathbf{Set} = \mathbf{Set}$, or equivalently as functors $F : C^\circ \to \mathcal{H}om(E, \mathbf{Set})$, the exactness condition in question is expressed by requiring that, for every object X of C, the functor $F(X) : E \to \mathbf{Set}$ commute with arbitrary inductive limits and finite projective limits; as we shall say in Section 6, $F(X)$ is a *fiber functor* on E. Equivalently, $F(X)$ is the inverse-image functor of a geometric morphism $P \to E$. Thus one finally obtains a canonical equivalence of categories

$$\mathcal{H}om_{\mathrm{top}}(\widehat{C}, E) \xrightarrow{\approx} \mathcal{H}om(C, \mathrm{Pt}(E)), \tag{4.6.3.1}$$

where, for brevity, $\mathrm{Pt}(E) = \mathcal{H}om_{\mathrm{top}}(P, E)$ denotes the category of points of the topos E.

When E is of the form $\widehat{C'}$, it follows at once from the definitions that the composite of (4.6.2.1) and the equivalence (4.6.3.1) is the functor

$$\mathcal{H}om(C, C') \longrightarrow \mathcal{H}om(C, \mathrm{Pt}(\widehat{C'})) \tag{4.6.3.2}$$

defined by $F \mapsto i \circ F$, where $i : C' \to \mathrm{Pt}(\widehat{C'})$ is the canonical embedding (4.6.2.2). It follows immediately that *(4.6.2.1) is an equivalence if and only if C is empty or (4.6.2.2) is essentially surjective*, and hence an equivalence of categories. This last condition on C' is satisfied in some interesting cases, notably when C' is the one-object category defined by a group G (7.2.5).

Remark 4.6.4. The fact that a topos $\widehat{C}$ has been attached to an arbitrary category C suggests that a category C has invariants of topological nature, such as cohomology and homotopy groups, just as a topos does. The cohomology groups of $\widehat{C}$ with coefficients in an abelian group object F, in the general sense studied in Exposé V, are just the values of the derived functors $\varprojlim^{(n)}$ of $\varprojlim$, already familiar to topologists. J.-L. Verdier and, independently, D. G. Quillen verified that, when one restricts to constant coefficients, or more generally to locally constant coefficients, these cohomology groups identify with the cohomology groups of the semi-simplicial set $\mathrm{Ner}(C)$ canonically associated with C [5, no. 212, Prop. 4.1], and also that, up to isomorphism in the homotopy category of [2], every semi-simplicial set can be obtained from a category C. With a suitable notion of homotopy type for topoi, which we do not specify here, one may say that the semi-simplicial homotopy types of topologists are precisely the homotopy types of topoi of the special form $\widehat{C}$, and more generally of topoi E in which every object admits a cover refining all others (2.6).[9] In the absence of this condition on E, its homotopy type can at best be expressed by a suitable projective system of semi-simplicial sets [1].

4.7. The topos $\widetilde{C}$ for variable site C (cocontinuous functors)

Let

$$f : C \longrightarrow C'$$

be a cocontinuous functor between sites in $\mathcal{U}$, i.e. a functor such that the functor $\widehat{f}_* : \widehat{C} \to \widehat{C'}$ sends $\widetilde{C}$ into $\widetilde{C'}$, or equivalently induces a functor

$$\widetilde{f}_* : \widetilde{C} \longrightarrow \widetilde{C'} \tag{4.7.1}$$

making the diagram

$$\begin{tikzcd} \widetilde{C} \arrow[r, "\widetilde{f}_*"] \arrow[d, "i"'] & \widetilde{C'} \arrow[d, "i'"] \\ \widehat{C} \arrow[r, "\widehat{f}_*"] & \widehat{C'} \end{tikzcd} \tag{4.7.2}$$

commute, where i, i' are the inclusion functors. We then saw in Exposé III, 2.3, that $\widetilde{f}_*$ has a left adjoint $\widetilde{f}^*$, and that this left adjoint is left exact. In other words, $\widetilde{f}_*$ is the direct-image functor associated with a geometric morphism

$$\widetilde{f} \text{ or } f : \widetilde{C} \longrightarrow \widetilde{C'},$$

[9] And still more generally of topoi which are "locally ∞-connected" in an evident sense which we leave to the reader to specify.

whose inverse-image functor is, of course, $\widetilde{f}^*$. Taking left adjoints of the functors involved, the commutative diagram (4.7.2) also gives a diagram commutative up to isomorphism

$$\begin{tikzcd} \widetilde{C} & \widetilde{C'} \arrow[l, "\widetilde{f}^*"'] \\ \widehat{C} \arrow[u, "\underline{a}"] & \widehat{C'} \arrow[l, "\widehat{f}^*"'] \arrow[u, "\underline{a}'"'] \end{tikzcd} \tag{4.7.3}$$

where $\underline{a}$ and $\underline{a}'$ are the associated-sheaf functors. This immediately recovers the formula

$$\widetilde{f}^* = \underline{a}\,\widehat{f}^* i'.$$

The transitivity property for geometric morphisms $\widehat{f} : \widehat{C} \to \widehat{C'}$ implies the same property for the geometric morphisms $\widetilde{f} : \widetilde{C} \to \widetilde{C'}$ associated with cocontinuous functors. Thus the topos $\widetilde{C}$ varies functorially, in a *covariant* way, with C, when cocontinuous functors are taken as the "morphisms" of sites.

In all cases encountered so far, the cocontinuous functor f used is also continuous, that is, it extends to a functor

$$\widetilde{f}_! : \widetilde{C} \longrightarrow \widetilde{C'}$$

commuting with inductive limits and left adjoint to $\widetilde{f}^*$, so that there is a sequence of three adjoint functors

$$\widetilde{f}_!, \quad \widetilde{f}^*, \quad \widetilde{f}_*.$$

One should note that $\widetilde{f}_!$ is generally not left exact, nor is $\widetilde{f}_*$ generally right exact; this removes any ambiguity about the variance direction of the topos $\widetilde{C}$ when C varies by functors assumed only to be cocontinuous, or even continuous and cocontinuous.

Remark 4.7.4. Given a geometric morphism

$$F : E \longrightarrow E',$$

there exists a functor $F_!$ left adjoint to F^* if and only if one can realize E and E', up to equivalence, as $\widetilde{C}$ and $\widetilde{C'}$ for two sites $C, C' \in \mathcal{U}$, and find a *continuous and cocontinuous* functor $f : C \to C'$ such that F identifies with $\widetilde{f}$. This is plainly sufficient. For necessity, it suffices to take for C and C' small full generating subcategories of E and E', respectively, such that

$$F_!(\operatorname{Ob} C) \subset \operatorname{Ob} C',$$

endow them with the topologies induced from those of E and E', and take f to be the functor induced by $F_!$ (cf. 1.2.1). Recall (1.8) that existence of $F_!$ also means that F^* commutes with projective limits, or equivalently here, since this functor is left exact, that it commutes with products.

4.7.5

If one asks which geometric morphisms $F : E \to E'$ can be realized by a cocontinuous, not necessarily continuous, functor of sites, one obtains similarly the following answer: the set of objects X of E such that the functor

$$X' \longmapsto \mathrm{Hom}(X, F^*(X'))$$

from E' to **Set** commutes with projective limits, or equivalently with products, must be a *generating family of E*.[10]

4.8. The geometric morphism $\widetilde{C} \to \widehat{C}$ for a site C

Let C be a small site, with associated topoi $\widetilde{C}$ and $\widehat{C}$, the latter independent of the topology put on C. In Exposé II, 3.4, the associated-sheaf functor

$$\underline{a} : \widehat{C} \longrightarrow \widetilde{C}$$

was defined and shown to be left exact and to commute with inductive limits. It is therefore the inverse-image functor associated with a geometric morphism

$$p : \widetilde{C} \longrightarrow \widehat{C}, \qquad p^* = \underline{a},$$

whose direct-image functor is the canonical inclusion

$$i = p_* : \widetilde{C} \longrightarrow \widehat{C}.$$

It is well known that this latter functor is generally not right exact–its derived functors on abelian objects give rise to the presheaves of cohomology $\mathcal{H}^n(F)$ of Exposé V, 2–and that $\underline{a}$ generally does not commute with arbitrary projective limits. This removes any ambiguity about the direction of the "natural" geometric morphism between $\widehat{C}$ and $\widetilde{C}$.

When one has a cocontinuous functor

$$f : C \longrightarrow C'$$

of sites, one deduces a diagram of geometric morphisms

$$\begin{tikzcd} \widetilde{C} \arrow[r, "\widetilde{f}"] \arrow[d, "p"'] & \widetilde{C'} \arrow[d, "p'"] \\ \widehat{C} \arrow[r, "\widehat{f}"] & \widehat{C'} \end{tikzcd}$$

commutative up to canonical isomorphism. This is precisely the content of the commutativity of the diagram of functors (4.7.2). Thus the geometric morphism $p : \widetilde{C} \to \widehat{C}$ is functorial in C, when C varies by *cocontinuous* functors between sites.

[10] Editorial note in the source: this statement does not appear correct. The expressed condition is equivalent to F^* commuting with arbitrary products, or equivalently to F^* having a left adjoint $F_!$. For any site C, the identity functor $F = \mathrm{id} : C \to C$ is cocontinuous when the codomain is given the trivial topology. Then $F^* : \widehat{C} \to \widetilde{C}$ is the associated-sheaf functor, and it does not in general commute with arbitrary products.

4.9. Effect of a continuous functor of sites. Morphisms of sites

4.9.1

Let
$$f : C \longrightarrow C'$$
be a continuous functor from C to C', i.e. such that the functor $\widehat{f}^* : \widehat{C'} \to \widehat{C}$ sends $\widetilde{C'}$ into $\widetilde{C}$, and hence induces a functor
$$f_s : \widetilde{C'} \longrightarrow \widetilde{C}.$$
We saw in Exposé III, 1.2, that f_s has a left adjoint
$$f^s : \widetilde{C} \longrightarrow \widetilde{C'},$$
which moreover extends f in an evident sense. One should be careful that in general f^s is not left exact, even if f is also cocontinuous, nor does f_s commute with inductive limits. Thus the datum of f does not, without further hypotheses, describe a geometric morphism in either direction between $\widetilde{C}$ and $\widetilde{C'}$. The case where f is cocontinuous, i.e. where f_s commutes with inductive limits and can be regarded as an inverse-image functor for a geometric morphism $\widetilde{f} : \widetilde{C} \to \widetilde{C'}$, was examined in 4.7. We now examine the case where f^s is left exact, hence may be regarded as the inverse-image functor for a geometric morphism in the *opposite* direction:
$$\mathrm{Top}(f) = g : \widetilde{C'} \longrightarrow \widetilde{C}. \tag{4.9.1.1}$$
Notice that *we have deliberately avoided denoting this morphism by $\widetilde{f}$ or f, to prevent confusion with the situation of 4.7, following the general recommendations of 3.1.3.* One sometimes says that the functor $f : C \to C'$ is a *morphism of sites from C' to C*–note: *not from C to C'*–if it is continuous and if the functor f^s is left exact. Equivalently, there exists a geometric morphism (4.9.1.1) whose corresponding inverse-image functor
$$g^* : \widetilde{C} \longrightarrow \widetilde{C'}$$
extends the functor f, i.e. makes the diagram of functors
$$\begin{tikzcd} C \arrow[r, "f"] \arrow[d, "\epsilon"'] & C' \arrow[d, "\epsilon'"] \\ \widetilde{C} \arrow[r, "g^*"] & \widetilde{C'} \end{tikzcd}$$
commute up to isomorphism, where ϵ, ϵ' are the canonical functors of Exposé II, 4.4.0.

4.9.2

In practice, one recognizes that a continuous functor $f : C \to C'$ is a morphism of sites from C' to C by the fact that finite projective limits are representable in C and that f commutes with them (Exposé III, 1.3(5)). Under the stated condition on C, almost always satisfied in practice, and assuming moreover that the topology of C' is no finer than the canonical topology, which is also almost always satisfied, this sufficient condition–that f be left exact–is also necessary.

4.9.3

In the spirit of the preceding discussion, if C and C' are two $\mathcal{U}$-sites, one should define the category of morphisms of sites $\mathcal{M}or_{\mathrm{site}}(C', C)$ from C to C' as the full subcategory of the *opposite* category $\mathcal{H}om(C, C')^{\circ}$ of the category of functors from C to C', formed by the functors which are willing to be morphisms of sites, from C' to C. In this way one obtains a canonical functor, defined up to unique isomorphism,

$$\mathcal{M}or_{\mathrm{site}}(C', C) \longrightarrow \mathcal{H}om_{\mathrm{top}}(\widetilde{C'}, \widetilde{C}).$$

Proposition 4.9.4. Let E be a $\mathcal{U}$-topos and C a $\mathcal{U}$-site. The functor $f \mapsto f^*|_C = f^* \circ \epsilon_C$, which associates with every geometric morphism $f : E \to \widetilde{C}$ the restriction to C of its inverse-image functor $f^* : \widetilde{C} \to E$, induces an equivalence of categories

$$\mathcal{H}om_{\mathrm{top}}(E, \widetilde{C}) \xrightarrow{\approx} \mathcal{M}or_{\mathrm{site}}(E, C) \hookrightarrow \mathcal{H}om(C, E)^{\circ}.$$

When finite projective limits are representable in C, the corresponding fully faithful functor

$$\mathcal{H}om_{\mathrm{top}}(E, \widetilde{C}) \longrightarrow \mathcal{H}om(C, E)^{\circ}$$

has as essential image the set of functors $g : C \to E$ which are left exact and continuous, or equivalently left exact and send covering families to covering families.

The last assertion follows from the first by 4.9.2. On the other hand, Exposé III, 1.2(iv), together with Theorem 1.2(iii) above, implies that $G \mapsto G \circ \epsilon$ induces an equivalence between the category of continuous functors from $\widetilde{C}$ to E and the category of continuous functors from C to E; a quasi-inverse is given by $g \mapsto g^s$, with the notation of loc. cit.. By definition, g is a morphism of sites if and only if g^s is the inverse-image functor associated with a geometric morphism $E \to \widetilde{C}$. The conclusion follows from 3.2.1, and the “or equivalently” from Exposé III, 1.6.

The main point to retain from 4.9.4 is that, when finite projective limits are representable in C, giving a geometric morphism $f : E \to \widetilde{C}$ is “the same as” giving a functor $g : C \to E$ which is left exact and sends covering families to covering families.

4.9.5

Using 4.9.1 and 4.9.3, one sees as usual that for a variable $\mathcal{U}$-site C, using the notion of morphism of sites and of morphism of morphisms of sites just described, the topos $\widetilde{C}$ depends functorially, or more exactly 2-functorially, on the site C. Notice that, with the terminology introduced, $\widetilde{C}$ depends *covariantly* on the site C.

4.9.6

It is immediate that, contrary to what happens for cocontinuous functors (cf. 4.7.4), *every* geometric morphism $F : E \to E'$ can be realized by a morphism of sites $f : C \to C'$, that is, by a continuous functor $f : C' \to C$ satisfying the above condition. It suffices to choose small full generating subcategories C and C' of E and E', respectively, endowed with the topologies induced by those of E and E', such that

$$F^*(\mathrm{Ob}\, C') \subset \mathrm{Ob}\, C,$$

and to take f to be the functor induced by F^*. This explains why most geometric morphisms of topoi encountered in practice are in fact described by morphisms of sites, rather than by cocontinuous functors as in 4.7.

4.10. Relations between the small and big topoi associated with a topological space X

We resume the notation of 2.5. In particular, $\mathcal{V}$ is a universe such that $\mathcal{U} \in \mathcal{V}$. We depart from the convention of 2.1 by denoting by $\mathrm{Top}(X)$ the topos of $\mathcal{V}$-sheaves, not $\mathcal{U}$-sheaves, on X. Thus we shall reason with $\mathcal{V}$-topoi rather than $\mathcal{U}$-topoi. It would be possible to keep $\mathcal{U}$-topoi by adopting the appropriate convention in the definition of $\mathrm{TOP}(X)$, so that it is a $\mathcal{U}$-topos; cf. 2.5. With this understood, we shall define two geometric morphisms

$$\begin{cases} f : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X), \\ g : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X), \qquad gf \simeq \mathrm{id}_{\mathrm{Top}(X)}, \end{cases} \tag{4.10.1}$$

giving rise to a sequence of three adjoint functors

$$g^* = f_!, \qquad g_* = f^*, \qquad g^! = f_*, \tag{4.10.1.1}$$

with the notation of 3.1.3. We shall define the first two functors in this sequence successively; the third is then defined as the right adjoint of the second.

4.10.2

The functor

$$g_* = f^* : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X)$$

is simply the *restriction functor* from $(\mathbf{Top})_{/X}$ to the site $\mathrm{Open}(X)$ of open subsets of X. It indeed sends sheaves to sheaves, as follows immediately from the definition. Denoting sheaves on the big site of X by underlined letters, we denote, for such a sheaf $\underline{F}$, by $\underline{F}_X$ its restriction to the site $\mathrm{Open}(X)$. Thus we obtain a functor

$$\mathrm{Res} : \underline{F} \longmapsto \underline{F}_X : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X). \tag{4.10.2.1}$$

It is clear that this functor commutes with $\mathcal{V}$-projective limits, since these are computed argument by argument; one could also invoke the existence of the left adjoint constructed in 4.10.4 below. I claim that it also commutes with $\mathcal{V}$-inductive limits. To see this, we give a very convenient interpretation of big sheaves on X, i.e. objects of $\mathrm{TOP}(X)$, in terms of ordinary sheaves, meaning $\mathcal{V}$-sheaves, on topological spaces X' over X.

4.10.3

For a big sheaf $\underline{F}$ on X, and for every space X' over X, with $X' \in \mathcal{U}$ understood, one defines in the evident way, as in 4.10.2, a sheaf $\underline{F}_{X'}$, the restriction of $\underline{F}$ to X'. If

$$u : X'' \longrightarrow X'$$

is a morphism in $(\mathbf{Top})_{/X}$, one defines in the evident way a morphism $u_*(\underline{F}_{X''}) \to \underline{F}_{X'}$, or equivalently a *transition morphism*

$$\varphi_u : u^*(\underline{F}_{X'}) \longrightarrow \underline{F}_{X''}. \tag{4.10.3.1}$$

As u varies, these morphisms satisfy an evident transitivity condition for a composite

$$X''' \xrightarrow{u} X'' \xrightarrow{v} X'$$

of morphisms in $(\mathbf{Top})_{/X}$, which we leave the reader to spell out. This gives a natural functor from the category $\mathrm{TOP}(X)$ of big sheaves on X to the category of systems

$$(\underline{F}_{X'}) \quad (X' \in \mathrm{Ob}(\mathbf{Top})_{/X}), \qquad (\varphi_u) \quad (u \in \mathrm{Fl}(\mathbf{Top})_{/X}),$$

satisfying the transitivity condition just described and, moreover, such that for every morphism $u : X'' \to X'$ which is an *open immersion*, or more generally an étale map, the transition morphism φ_u is an isomorphism. Since the functors $u^* : \mathrm{Top}(X') \to \mathrm{Top}(X'')$ used in the description of the φ_u commute with $\mathcal{V}$-inductive limits, it follows at once that, in the preceding description of objects of $\mathrm{TOP}(X)$ in terms of small sheaves $F_{X'}$, $\mathcal{V}$-inductive limits are computed argument by argument. In other words, the functors $\underline{F} \mapsto \underline{F}_{X'}$ commute with $\mathcal{V}$-inductive limits.

In particular, this is true of the functor $\underline{F} \mapsto \underline{F}_X$ considered in 4.10.2. Hence it has a right adjoint (1.5). It remains to construct a left adjoint for it, whose existence also follows a priori from 1.8, and to verify that this left adjoint is left exact. This will complete the definition of the announced geometric morphisms (4.10.1) in terms of the functors (4.10.1.1).

4.10.4

The functor

$$g^* = f_! : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X)$$

is obtained by associating with every small sheaf F on X the system of its inverse images $(F_{X'})$, for $X' \in \mathrm{Ob}(\mathbf{Top})_{/X}$, by the structural morphisms $X' \to X$; the transition morphism φ_u for $u : X'' \to X'$ is the transitivity isomorphism for inverse images of sheaves (4.1.1). One sees at once that this gives a fully faithful functor

$$\mathrm{Prol} : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X),$$

whose essential image consists of the big sheaves $\mathcal{F}$ on X for which all the transition morphisms φ_u of (4.10.3.1) are isomorphisms. We leave to the reader the definition of an adjunction isomorphism between this canonical prolongation functor and the restriction functor of 4.10.2, proving that the latter is right adjoint to the former. This is immediate in terms of the description 4.10.3 of the category $\mathrm{TOP}(X)$.

Remarks 4.10.5.

a) The construction of 4.10.4 shows at the same time that $g^* = f_!$ is fully faithful, or equivalently that $g_* = f^*$ is a localization functor, or again that $g^! = f_*$ is fully faithful. The big sheaves on X which belong to the essential image of $g^* = f_! = \mathrm{Prol}$ deserve the name of big *étale sheaves* on X, since they form a category equivalent to that of ordinary sheaves on X, or equivalently to that of étale spaces over X. It is also easy to see that if X' is a topological space over X, then the big sheaf on X represented by X' is étale in the preceding sense if and only if X' is an étale space over X.

b) The full faithfulness of f_* may also be expressed by saying that the adjunction morphism $f^*f_* \to \mathrm{id}_{\mathrm{Top}(X)}$ is an isomorphism, i.e. that $g_*f_* \simeq \mathrm{id}$, i.e. that $gf \simeq \mathrm{id}$. Thus g makes $\mathrm{TOP}(X)$ a topos over $\mathrm{Top}(X)$, admitting a section f over $\mathrm{Top}(X)$.

c) The fact that $g^* = f_!$ is fully faithful, exact, and commutes with $\mathcal{V}$-inductive limits partially justifies the very convenient viewpoint, due to J. Giraud, that in practically all questions of sheaf theory on X, it is harmless to replace ordinary, or small, sheaves by the associated big sheaves. This is especially true for cohomological questions, since g_* is exact, so the functors $R^i g_*$ vanish for $i > 0$. Therefore, for every big sheaf $\mathcal{F}$ on X, there are canonical isomorphisms

$$H^i(\mathrm{TOP}(X), \mathcal{F}) \simeq H^i(\mathrm{Top}(X), \mathcal{F}_X) = H^i(X, F).$$

Applying this to a sheaf of the form $\mathrm{Prol}(F)$, where F is a small sheaf on X, and using $\mathrm{Prol}(F)_X \simeq F$, one obtains a canonical isomorphism

$$H^i(X, F) = H^i(\mathrm{TOP}(X), \mathrm{Prol}(F)).$$

Thus *the cohomological invariants of X, computed via the small or the big topos of X, are essentially identical.* The same result also holds in non-commutative cohomology.

Exercise 4.10.6. Let S be a $\mathcal{U}$-site whose topology is no finer than the canonical topology, and let $\mathcal{M}$ be a subset of $\mathrm{Fl}\, S$ satisfying the following conditions:

a) The morphisms in $\mathcal{M}$ are base-changeable, and $\mathcal{M}$ is stable under base change.

b) $\mathcal{M}$ contains the identity arrows and is stable under composition.

c) If a morphism $u : X \to Y$ admits a covering family $Y_i \to Y$ such that the morphisms $X \times_Y Y_i \to Y_i$ belong to $\mathcal{M}$, then u belongs to $\mathcal{M}$.

d) For every $X \in \mathrm{Ob}\, S$, every covering family of X is refined by a covering family $(f_i : X_i \to X)$ with $f_i \in \mathcal{M}$.

For every $X \in \mathrm{Ob}\, S$, consider the site $S(X)$, the "small site of X", whose underlying category is the full subcategory of $S_{/X}$ formed by the objects whose structural morphism is in $\mathcal{M}$, endowed with the topology induced from that of S.

1) For every arrow $u : X \to Y$ of S, show that base change by u from $S(Y)$ to $S(X)$ is a continuous functor, giving a functor commuting with inductive limits

$$S(u)^* : S(Y)^\sim \longrightarrow S(X)^\sim.$$

2) Define an equivalence between the topos $S^\sim$ and the category of systems

$$(F_X) \quad (X \in \mathrm{Ob}\, S), \qquad (\varphi_u) \quad (u \in \mathrm{Fl}\, S)$$

formed of objects $F_X \in \mathrm{Ob}\, S(X)^\sim$, and, for every arrow $u : X \to Y$ in S, a morphism $\varphi_u : S(u)^*(F_Y) \to F_X$, subject to a transitivity condition for a composite $v \circ u$ of arrows of S, and to the condition that $u \in \mathcal{M}$ implies that φ_u is an isomorphism.

3) Define the restriction and prolongation functors

$$\mathrm{Res}_X : (S_{/X})^\sim \longrightarrow S(X)^\sim, \qquad \mathrm{Prol}_X : S(X)^\sim \longrightarrow (S_{/X})^\sim.$$

Show that Res_X commutes with small inductive and projective limits and that Prol_X is fully faithful, its essential image consisting of the sheaves F such that φ_u is an isomorphism for every arrow u of $S_{/X}$.

4) Define an adjunction morphism making Res_X right adjoint to Prol_X. Deduce that there exists a geometric morphism
$$f : S(X)^\sim \longrightarrow (S_{/X})^\sim$$
which makes $S(X)^\sim$ a subtopos of $(S_{/X})^\sim$, and such that
$$f_! = \mathrm{Prol}_X , \qquad f^* = \mathrm{Res}_X \,.$$
Show that Prol_X sends abelian sheaves to abelian sheaves.

5) Show that if $S(X)$ has fiber products, then Prol_X is exact. Deduce that there then exists a geometric morphism
$$g : (S_{/X})^\sim \longrightarrow S(X)^\sim$$
which is a left retraction of f, i.e. such that
$$g^* = \mathrm{Prol}_X , \qquad g_* = \mathrm{Res}_X \,.$$
For an example where $S(X)$ does not have fiber products, take S to be the category of schemes with the étale topology, take $\mathcal{M}$ to be the class of smooth morphisms, and take X to be a noetherian scheme of dimension > 0.

6) Show that for every abelian sheaf F on $(S_{/X})^\sim$ there is an isomorphism
$$H^q(X, F) \simeq H^q(X, \mathrm{Res}_X F) \qquad \forall q.$$
Show, using for instance hypercoverings, that for every abelian sheaf G on $S(X)^\sim$ there is an isomorphism
$$H^q(X, G) \simeq H^q(X, \mathrm{Prol}_X G) \qquad \forall q.$$

7) Buy the expositor a chocolate medal.

5 Induced topoi

5.1

Let E be a topos and let X be an object of E. Then the category $E_{/X}$ of objects of E over X is a topos. This follows immediately, for instance, from Giraud's criterion, 1.2(ii). Equally, using 1.2.1, one may write E as a category of sheaves $C^\sim$, where C is a generating subcategory of E, chosen so that $X \in \mathrm{Ob}\, C$. Then
$$E_{/X} = C^\sim_{/X}$$
is equivalent to $(C_{/X})^\sim$, by Exposé III, 5.4, and hence is a topos.

The topos $E_{/X}$ is called the *topos induced on the object* X of E.

5.2

We now define a canonical geometric morphism

$$j_X : E_{/X} \longrightarrow E, \tag{5.2.1}$$

called the *inclusion morphism* of the induced topos $E_{/X}$ into the ambient topos E, or better, in view of 5.7, the *localization morphism of E at X*. This morphism corresponds to a sequence of three adjoint functors

$$j_{X!} \quad j_X^* \quad j_{X*}, \tag{5.2.2}$$

which may be described as follows.

First, $j_{X!} : E_{/X} \to E$ is the functor which forgets the structural arrow to X. Second,

$$j_X^* : E \longrightarrow E_{/X}$$

is given by

$$j_X^*(Z) = (X \times Z, \mathrm{pr}_1),$$

where $\mathrm{pr}_1 : X \times Z \to X$ is the first projection. It is also the base-change functor for the morphism $X \to e_E$, where e_E is the final object of E. It is immediate from the definition of base change that j_X^* is right adjoint to $j_{X!}$. In particular it commutes with projective limits. It also commutes with inductive limits, by the special exactness properties of topoi (Exposé II, 4).

It follows from the latter fact, by 1.6, that j_X^* admits a right adjoint

$$j_{X*} : E_{/X} \longrightarrow E.$$

For an object X' over X, the object $j_{X*}(X')$ is also sometimes denoted by one of

$$\prod_{X/e_E} (X'/X), \qquad \mathcal{H}om_{X/e_E}(X, X'), \qquad \mathrm{Res}_{X/e_E}(X'/X),$$

the last notation being the “Weil restriction”; one of these notations is no doubt already familiar to the learned reader of our modest work.

5.3

One should note that the functor $j_{X!}$ commutes with fiber products and sends monomorphisms to monomorphisms, but is not in general left exact. It is left exact only when X is a final object of E, since $X = j_{X!}(X, \mathrm{id}_X)$ and (X, id_X) is final in $E_{/X}$. Equivalently, this is the case precisely when j_X is in fact an equivalence of topoi, and hence when all three functors in (5.2.2) are equivalences.

Similarly, j_{X*} is not in general right exact and need not send epimorphisms to epimorphisms. These observations remove any possible ambiguity about the direction of the geometric morphism relating $E_{/X}$ to E.

5.4

The functor j_X^* is often called the *localization functor*, or the *restriction functor*. The latter terminology is justified as follows. Identify objects of E, respectively of $E_{/X}$, with sheaves on E, respectively on $E_{/X}$, by 1.2(iii). Under this identification, the adjunction between $j_{X!}$ and j_X^*

says that $j_X^*(F)$ is the restriction to $E_{/X}$ of the sheaf F on E, where "restriction" is understood in the generalized sense: composition with the "inclusion" functor $j_{X!} : E_{/X} \to E$. In familiar notation we shall therefore also write, interchangeably,

$$j_X^*(F) = F|X = F_X = \text{ the restriction of } F \text{ to } X. \tag{5.4.1}$$

Similarly, when $E = C^{\sim}$, with C a small site, and $X = \varepsilon(S)$ for $S \in \mathrm{Ob}\, C$, so that the equivalence recalled in 5.1

$$E_{/X} = C^{\sim}_{/\varepsilon(S)} \simeq (C_{/S})^{\sim}$$

holds, where $C_{/S}$ carries the topology induced from that of C, the functor j_X^* is simply the restriction of a variable sheaf F on C to the category $C_{/S}$.

These considerations already indicate the important role played by induced topoi and localization morphisms in all questions where one reasons by localization on E, and hence in practically every question involving sites or topoi.

5.5

Let

$$f : X \longrightarrow Y$$

be an arrow of E, so that X, more precisely (X, f), may be regarded as an object of $E_{/Y}$. There is an evident canonical isomorphism

$$(E_{/Y})_{/X} \xrightarrow{\sim} E_{/X}. \tag{5.5.1}$$

This is the transitivity of induced topoi. Applying the construction of 5.1 to $E_{/Y}$ instead of E, one obtains a canonical geometric morphism

$$\mathrm{loc}(f), \text{ or simply } f : E_{/X} \longrightarrow E_{/Y}, \tag{5.5.2}$$

also called the *localization morphism associated with* f. When Y is the final object of E, this recovers (5.2.1). The localization morphism for f is associated with three adjoint functors

$$f_! \quad f^* \quad f_*, \tag{5.5.3}$$

which are interpreted respectively as the forgetful functor, the restriction functor, and a functor denoted, according to preference and with X' as argument, by

$$\prod_{X/Y} (X'/X), \qquad \mathcal{H}om_{X/Y}(X, X'), \qquad \mathrm{Res}_{X/Y}(X'/X).$$

5.6

The transitivity of forgetful functors implies that, for two composable morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

in E, there is a canonical isomorphism of geometric morphisms

$$\mathrm{loc}(gf) \simeq \mathrm{loc}(g)\, \mathrm{loc}(f) : E_{/X} \longrightarrow E_{/Y} \longrightarrow E_{/Z}.$$

For three composable morphisms, the usual compatibility relation holds for these transitivity isomorphisms. Thus, with the usual abuse of language, one may regard

$$X \longmapsto E_{/X}$$

as a covariant functor from E to the category of topoi. More precisely, it is a 2-functor, not necessarily strict, from E to the 2-category of topoi. Before continuing the general theory of induced topoi in 5.10, we give some instructive examples.

5.7

Let X be a topological space, hence giving a topos $\mathrm{Top}(X)$. Let X' be an object of $\mathrm{Top}(X)$, which it is convenient to interpret as an *étale space* over X, say $p : X' \to X$. The category $\mathrm{Top}(X)_{/X'}$ identifies with the category of étale spaces X'' over X, endowed with an X-morphism $X'' \to X'$. Such a morphism makes X'' an étale space over X', and in this way one obtains a canonical equivalence of topoi

$$\mathrm{Top}(X)_{/X'} \simeq \mathrm{Top}(X').$$

The localization morphism is therefore a geometric morphism $\mathrm{Top}(X') \to \mathrm{Top}(X)$, and one immediately checks that this is precisely the morphism

$$\mathrm{Top}(p) : \mathrm{Top}(X') \longrightarrow \mathrm{Top}(X)$$

associated with the structural continuous map $p : X' \to X$. Intuitively, the localization morphism is just the translation, into topos language, of the étale-space morphism $p : X' \to X$. This both justifies the term "localization morphism" and calls for caution with the term "inclusion morphism", which seems appropriate mainly when p is an open immersion. It is therefore prudent, in 5.1, to reserve "inclusion morphism" for the case where $X \to e_E$ is a monomorphism, i.e. where X identifies with a subobject of the final object of E.

5.8

Let E be a topos, let G be a group object of E, let $H \subset G$ be a subgroup, and put

$$X = G/H.$$

We regard this quotient homogeneous space as an object of E with a left action of G, i.e. as an object of the classifying topos B_G. We determine the induced topos

$$(B_G)_{/X} \qquad (X = G/H)$$

by defining an equivalence of topoi

$$c : B_H \xrightarrow{\sim} (B_G)_{/X} \tag{5.8.1}$$

such that the diagram

$$\begin{tikzcd} B_H \arrow[r, hook, "c"] & (B_G)_{/X} \arrow[r, "j_X"] \arrow[d, phantom, "B_i"] & B_G \\ & & \end{tikzcd} \tag{5.8.2}$$

commutes up to canonical isomorphism. Here j_X is the localization morphism in B_G, and

$$B_i : B_H \longrightarrow B_G$$

is the morphism induced by the inclusion $i : H \to G$.

To define (5.8.1), we only describe the inverse-image and direct-image functors c^* and c_*, leaving to the reader the verification that they are quasi-inverse equivalences and give rise to the commutative diagram (5.8.2). Let e be the subobject of E underlying X, namely the image of the section of X induced, over a chosen final object e_E, by the unit section of G. If X' is an object of B_G over X, define

$$c^*(X') = X' \times_X e.$$

This object is stable under the left action of H, obtained by restricting the given action of G on X'. This gives a functor

$$c^* : (B_G)_{/X} \longrightarrow B_H.$$

Conversely, if X' is an object of B_H, the morphism $X' \to e_H$, where e_H is the final object of B_H, gives after applying $i_!$ a morphism in B_G

$$i_!(X') \longrightarrow i_!(e_H) \simeq X.$$

Thus $i_!(X')$ is an object of $(B_G)_{/X}$, and the desired functor is

$$c_* : X' \longmapsto (i_!(X') \to X) : B_H \longrightarrow (B_G)_{/X}.$$

5.8.3

It follows from (5.8.2) that if G is a group object of a topos and H is a subgroup, then the functor which restricts the operators from G to H may be interpreted as a localization functor. In particular, taking H to be the unit subgroup, the functor which forgets the action of G is a localization functor.

More generally, if (X, G) is an object of E with an action of G, i.e. an object of B_G, then the functor

$$(B_G)_{/(X,G)} \longrightarrow E_{/X}$$

which forgets the action of G may be interpreted as a localization functor, relative to a suitable object of $(B_G)_{/(X,G)}$ covering the final object. One takes

$$E_{G,(X,G)} = E_G \times (X, G),$$

where $E_G = G_s = G$ carries the left-translation action of G. This gives a cartesian diagram

$$\begin{tikzcd} E_G \arrow[d] & Z = E_{G,(X,G)} \arrow[l] \arrow[d] \\ e_G & (X,G) \arrow[l] \end{tikzcd} \tag{5.8.3.1}$$

and, by transitivity of induced topoi, the induced topos over Z identifies with $E_{/X}$. The resulting diagram of topoi

$$\begin{tikzcd}
(B_G)_{/E_G} \simeq E \arrow[d, "j"'] & (B_G)_{/Z} \simeq E_{/X} \arrow[l, "i"'] \arrow[d, "j'"] \\
B_G & (B_G)_{/(X,G)} \arrow[l, "f"']
\end{tikzcd} \tag{5.8.3.2}$$

commutes up to canonical isomorphism. The inverse-image functors associated with the vertical arrows are the functors forgetting the action of G, while i^* identifies with the localization functor $E \to E_{/X}$. This diagram is in fact 2-cartesian in the sense of Proposition 5.11 below.

5.8.4

The reader may extend the preceding considerations to a pro-group $\mathcal{G} = (G_i)$, in the special case where H is defined as the kernel of $\mathcal{G} \to G_{i_0}$. For profinite groups this means that one restricts to open subgroups H of G, equivalently closed subgroups of finite index.

Exercise 5.9. Let E be a topos, let G be a group object of E, let B_G be the classifying topos of G, and let

$$\pi : B_G \longrightarrow E$$

be the morphism induced by the homomorphism from G to the unit group of E, using $B_e \simeq E$.

a) Let $E_G = G_s$ be the object of B_G whose underlying object of E is G, with G acting by left translations. Regard $\pi^*(G)$ as a group object of B_G, namely the object G of E with the trivial action of G. Show that right translations of G on G_s define a morphism in B_G

$$E_G \times \pi^*(G) \longrightarrow E_G.$$

Show that this makes E_G an object of B_G with a right action of the B_G-group $\pi^*(G)$, and that this action makes E_G a torsor under $\pi^*(G)$.

b) If $q : E' \to E$ is a topos over E, and B is any topos over E, define the category $\mathcal{H}om_{\mathrm{top},E}(E', B)$ of geometric morphisms $E' \to B$ compatible with the structural morphisms to E, up to a specified isomorphism. For $B = B_G$, show that the rule $f \mapsto f^*(E_G)$ defines an equivalence of categories

$$\mathcal{H}om_{\mathrm{top},E}(E', B_G) \longrightarrow \mathrm{Tors}(E', q^*(G)),$$

where the right-hand side is the category of right $q^*(G)$-torsors on E'.

c) Let $H \subset G$ be a subgroup. The quotient

$$X = E_G/\pi^*(H)$$

is the object G/H with its usual left action of G. Passing to induced topoi in the diagram $E_G \to X \leftarrow e_G$, show that the resulting diagram is equivalent to the diagram attached by 4.5 to the group homomorphisms $e \to H \to G$. Thus the classifying topos B_H is to be interpreted intuitively as a homogeneous space over B_G, with group $\pi^*(G)$, associated with the universal torsor E_G over B_G.

d) Revisit the example of 5.7 when X' is an étale covering of X, locally trivial in the sense of 2.7.4, and interpret it in terms of the present exercise.

5.10. Inverse images of induced topoi

Let

$$f : E' \longrightarrow E$$

be a morphism of $\mathcal{U}$-topoi, let X be an object of E, and put $X' = f^*(X)$. We define a diagram of geometric morphisms

$$\begin{tikzcd} E_{/X} \arrow[d, "j_X"'] & E'_{/X'} \arrow[l, "f_{/X}"'] \arrow[d, "j_{X'}"] \\ E & E' \arrow[l, "f"'] \end{tikzcd} \tag{5.10.1}$$

where j_X and $j_{X'}$ are the localization morphisms and where $f_{/X}$ is defined below. The diagram is commutative up to canonical isomorphism.

We define $f_{/X}$ by its inverse-image functor

$$(f_{/X})^* : E_{/X} \longrightarrow E'_{/X'}, \tag{5.10.2}$$

namely the functor induced by f^* in the evident sense. Since the forgetful functors $j_{X!}$ and $j_{X'!}$ commute with inductive limits and fiber products and are conservative, it follows at once that $(f_{/X})^*$ commutes with inductive limits and fiber products, just as f^* does. It also sends the final object X to the final object X'. Hence it is left exact and is associated to a geometric morphism

$$f_{/X} : E'_{/X'} \longrightarrow E_{/X}, \tag{5.10.3}$$

defined up to unique isomorphism. The diagram obtained from (5.10.1) by passing to inverse-image functors is commutative up to canonical isomorphism by construction and because f^* commutes with products $X \times Y$. Thus (5.10.1) itself is commutative, with the unique isomorphism inducing the canonical isomorphism on inverse-image functors.

5.10.4

When $f : E' \to E$ is associated with a continuous map $Y' \to Y$ of topological spaces, so that X is an étale space over Y and $X' = X \times_Y Y'$, the identifications $\mathrm{Top}(Y)_{/X} \simeq \mathrm{Top}(X)$ and $\mathrm{Top}(Y')_{/X'} \simeq \mathrm{Top}(X')$ identify $f_{/X}$ with the morphism $\mathrm{Top}(X') \to \mathrm{Top}(X)$ induced by the canonical continuous map $X' \to X$. Thus $E'_{/X'}$ plays the role of the fiber product of $E_{/X}$ and E' over E. The following result makes this intuition precise.

Proposition 5.11. With the notation of 5.10, the diagram (5.10.1), together with the compatibility isomorphism

$$\alpha : f \circ j_{X'} \xrightarrow{\sim} j_X \circ f_{/X},$$

is 2-cartesian. More explicitly, for every $\mathcal{U}$-topos F, associating to a geometric morphism $g : F \to E'_{/X'}$ the triple

$$(f_{/X} g,\ j_{X'} g,\ \alpha_g)$$

defines an equivalence from $\mathcal{H}om_{\mathrm{top}}(F, E'_{/X'})$ to the category of triples (g_1, g_2, β), where $g_1 : F \to E_{/X}$, $g_2 : F \to E'$, and $\beta : f \circ g_2 \xrightarrow{\sim} j_X \circ g_1$ is an isomorphism of geometric morphisms to E.

5.11.1

We record the elementary categorical calculation behind the preceding statement. Suppose given categories E, E', an object X of E such that $A \mapsto A \times X$ is defined, and a left exact functor

$$\varphi : E_{/X} \longrightarrow E'. \tag{*}$$

Since $E_{/X}$ has final object (X, id_X), so does E', with final object $e' \simeq \varphi(X)$. Choose e'. To φ we associate the pair

$$(\Psi, u), \qquad \Psi = \varphi \circ j : E \to E', \qquad u \in \mathrm{Hom}(e', \Psi(X)), \tag{**}$$

where $j(A) = A \times X$. The element u is obtained by applying φ to the diagonal section $\delta_X : X \to X \times X$ over the final object X of $E_{/X}$:

$$u = \varphi(\delta_X) : e' = \varphi(X) \longrightarrow \varphi(X \times X) = \Psi(X).$$

Conversely, the pair (Ψ, u) recovers φ up to unique isomorphism. For an object $p : X' \to X$ of $E_{/X}$, the graph morphism $\Gamma_p = (\mathrm{id}_{X'}, p)$ gives a cartesian square in $E_{/X}$. Applying φ yields the canonical formula

$$\varphi(X', p) \simeq \Psi(X') \times_{\Psi(X)} e'. \tag{5.11.1.1}$$

This is functorial in (X', p). The functor Ψ is left exact, and more generally commutes with any type of projective limits with which φ commutes.

Conversely, when finite projective limits are representable in E and E', a pair (Ψ, u) with Ψ left exact defines φ by (5.11.1.1). This φ is left exact and has the corresponding limit-commutation properties. Moreover one recovers Ψ by the canonical isomorphism

$$\Psi(X') \simeq \varphi(j(X')), \tag{5.11.1.2}$$

and under this isomorphism $u : e' \to \Psi(X)$ identifies with $\varphi(\delta_X)$.

Lemma 5.11.2. Let E and E' be categories in which finite projective limits are representable, let X be an object of E, and let $\mathrm{Lex}(E, E')$, respectively $\mathrm{Lex}(E_{/X}, E')$, denote the category of left exact functors. Then $\mathrm{Lex}(E_{/X}, E')$ is equivalent to the category whose objects are pairs (Ψ, u), with $\Psi \in \mathrm{Ob}\,\mathrm{Lex}(E, E')$ and $u : e' \to \Psi(X)$, where e' is the final object of E'. The construction and a quasi-inverse are as described above. If φ and (Ψ, u) correspond, then φ commutes with a given type of projective limits if and only if Ψ does. If in E and E' base-change functors commute with a given type of inductive limits, then φ commutes with inductive limits of that type if and only if Ψ does.

We leave to the reader the explicit categorical structure on the category of pairs (Ψ, u). If $\Gamma : \mathrm{Lex}(E, E') \to \mathbf{Set}$ is given by $\Gamma(\Psi) = \mathrm{Hom}(e', \Psi(X))$, then u may be viewed as an element of $\Gamma(\Psi)$, which explains the notation $\mathrm{Lex}(E, E')_{/\Gamma}$. The remaining functorial details and the assertions on inductive limits follow at once from the explicit formulas (5.11.1.1) and (5.11.1.2).

Proposition 5.12. Let E and E' be $\mathcal{U}$-topoi and let X be an object of E. On $\mathcal{H}om_{\mathrm{top}}(E', E)$ consider the contravariant functor

$$\gamma : f \longmapsto \Gamma(E', f^*(X)) = \mathrm{Hom}(e', f^*(X)).$$

Then there is an equivalence of categories

$$\mathcal{H}om_{\mathrm{top}}(E', E_{/X}) \xrightarrow{\sim} \mathcal{H}om_{\mathrm{top}}(E', E)_{/\gamma}. \tag{5.12.1}$$

It sends a geometric morphism $h : E' \to E_{/X}$ to the pair (f, u), where $f = j_X h : E' \to E$, and where

$$u : e' = h^*(X, \mathrm{id}_X) \longrightarrow f^*(X) = h^*(j_X^* X) = h^*(X \times X)$$

is induced by the diagonal $\delta_X : X \to X \times X$.

5.12.2

Using 5.12, the proof of Proposition 5.11 becomes almost tautological. To give a geometric morphism $g : F \to E'_{/X'}$ amounts to giving a morphism $g_2 : F \to E'$ together with a section of $g_2^*(X')$. But $g_2^*(X') = g_2^* f^*(X) = (fg_2)^*(X)$, so such a section is equivalently a lifting of $fg_2 : F \to E$ to a morphism $g_1 : F \to E_{/X}$. This is exactly the data of a triple (g_1, g_2, α) as in 5.11.

Remark 5.13. For any diagram of topoi

$$\begin{tikzcd} E_1 \arrow[d] & \\ E & E_2 \arrow[l] \end{tikzcd}$$

there exists a 2-fiber product in the 2-category of topoi, independent up to equivalence of the chosen enlargement of universes. For other natural examples of fiber products of topoi, besides Proposition 5.11, see the final chapter of Giraud's book.

Exercise 5.14.

a) If F and G are topoi over a topos E, define $\mathcal{H}om_{\mathrm{top},E}(F, G)$ to be the category of pairs (f, α), where $f : F \to G$ is a geometric morphism and $\alpha : p \xrightarrow{\sim} qf$ is an isomorphism of the structural morphisms to E. Define composition functors $\mathcal{H}om_{\mathrm{top},E}(F, G) \times \mathcal{H}om_{\mathrm{top},E}(G, H) \to \mathcal{H}om_{\mathrm{top},E}(F, H)$ and associativity isomorphisms.

b) If X, Y are objects of E, define a natural functor from the discrete category $\mathrm{Hom}(X, Y)$ to $\mathcal{H}om_{\mathrm{top},E}(E_{/X}, E_{/Y})$, and check compatibility with composition in E and the compositions of part a).

c) Prove that the functor of b) is an equivalence. In particular, an object X of a topos E is recovered, up to unique isomorphism, from the induced topos $E_{/X}$ considered as a topos over E, up to E-equivalence. This result is due to P. Deligne; the case where E is the punctual topos was previously treated by Mme Hakim.

6 Points of a topos and fiber functors

6.0

Let P be the standard punctual $\mathcal{U}$-topos

$$P = (\mathcal{U}\text{-}\mathbf{Set}).$$

Because the symbol P evokes a geometric object of the nature of a point, whereas ($\mathcal{U}$-**Set**) evokes a large category, to be interpreted as the category of sheaves on P, it is often preferable, according to context, to use one notation or the other exclusively, although strictly speaking they denote one and the same object.

Definition 6.1. Let E be a $\mathcal{U}$-topos. A *point* of E is a geometric morphism $P \to E$ from the punctual topos to E. The *category of points* of E, denoted $\mathrm{Pt}(E)$ or $\mathrm{Pt}(E)$, is the category $\mathcal{H}om_{\mathrm{top}}(P, E)$. If $p : P \to E$ is a point of E, with inverse-image functor $p^* : E \to (\mathcal{U}\text{-}\mathbf{Set})$, and if F is an object of E, the set $p^*(F)$ is called the *fiber of F at p* and is denoted F_p.

6.1.1

If F is a group object, ring object, and so on, in E, then F_p is respectively a group, a ring, and so on.

6.1.2

By 3.2.1, the category of points of E is equivalent, via $p \mapsto p^*$, to the opposite of the category of functors

$$\varphi : E \longrightarrow (\mathcal{U}\text{-}\mathbf{Set})$$

which commute with $\mathcal{U}$-inductive limits and are left exact. Thus giving a point of E is essentially the same thing as giving such a functor φ.

Definition 6.2. A *fiber functor* of the $\mathcal{U}$-topos E is a functor

$$\varphi : E \longrightarrow (\mathcal{U}\text{-}\mathbf{Set})$$

which commutes with $\mathcal{U}$-inductive limits and is left exact. The *category of fiber functors* on E, denoted $\mathrm{Fib}(E)$, is the full subcategory of $\mathcal{H}om(E, (\mathcal{U}\text{-}\mathbf{Set}))$ formed by the fiber functors.

6.2.1

The category $\mathrm{Fib}(E)$ of fiber functors on E is equivalent to the opposite of the category $\mathrm{Pt}(E)$ of points of E, via the functor $p \mapsto p^*$:

$$\mathrm{Pt}(E) \xrightarrow{\sim} \mathrm{Fib}(E)^\circ. \tag{6.2.1.1}$$

Thus a functor $\varphi : E \to (\mathcal{U}\text{-}\mathbf{Set})$ is a fiber functor if and only if it is the fiber functor $F \mapsto F_p$ associated with a point p of E. Equivalently, by 1.7, φ is left exact and sends covering families to surjective families. This last condition may also be expressed by saying that φ commutes with $\mathcal{U}$-small sums and sends epimorphisms to epimorphisms.

6.3

Let $C \in \mathcal{U}$ be a site. A *point of the site* C is a point of the topos $C^\sim$, and the *category of points of the site* C is

$$\mathrm{Pt}(C) = \mathrm{Pt}(C^\sim).$$

Consider the functor

$$\varphi \longmapsto \varphi|C = \varphi \circ \varepsilon_C : \mathrm{Fib}(C^\sim) \longrightarrow \mathcal{H}om(C, (\mathcal{U}\text{-}\mathbf{Set})). \tag{6.3.1}$$

It is fully faithful, with essential image the full subcategory of site morphisms $(\mathcal{U}\text{-}\mathbf{Set}) \to C$ in the sense of 4.9.4; denote this subcategory also by $\mathrm{Fib}(C)$. A functor $\Psi : C \to (\mathcal{U}\text{-}\mathbf{Set})$ is called a *fiber functor on the site* C if it belongs to this essential image. Such a functor is continuous, hence sends covering families to covering families, i.e. to surjective families, and is left exact. When finite projective limits are representable in C, as in nearly all practical cases, these two properties characterize the fiber functors on C: they are precisely the left exact functors $C \to (\mathcal{U}\text{-}\mathbf{Set})$ which send covering families to surjective families.

The functor $\varphi \mapsto \varphi|C$ is an equivalence

$$\mathrm{Fib}(C^\sim) \xrightarrow{\sim} \mathrm{Fib}(C),$$

so it is essentially the same thing to give a fiber functor on the topos $C^\sim$, or a point of it, as to give a fiber functor on the site C. If Ψ is a fiber functor on C, and φ is the corresponding fiber functor on $C^\sim$, then φ is recovered by

$$\varphi(F) \simeq \varinjlim_{C_{/F}} \Psi(X), \tag{6.3.2}$$

where $C_{/F}$ is the category of objects X of C equipped with a morphism $X \to F$ in $\widehat{C}$, equivalently with an element of $F(X)$. This follows immediately from the canonical formula

$$F \simeq \varinjlim_{C_{/F}} \varepsilon_C(X).$$

More generally, for a presheaf G on C, one has a canonical isomorphism

$$\varphi(aG) \simeq \varinjlim_{C_{/G}} \Psi(X), \tag{6.3.3}$$

where aG denotes the sheaf associated with G.

6.4.0

We refer to Exposé I, 6.1 for the notion of a conservative family of functors

$$\varphi_i : E \longrightarrow E_i \qquad (i \in I).$$

When E and the E_i are topoi and the φ_i are inverse-image functors f_i^* of geometric morphisms $f_i : E_i \to E$, the exactness properties required in Exposé I, 6.2, 6.3 and 6.4 are satisfied, apart from the harmless finiteness restriction there indicated. In this setting it is equivalent to say that $(f_i^*)_{i \in I}$ is conservative, conservative for monomorphisms, conservative for epimorphisms, or faithful.

When the family (f_i^*) is conservative we shall also say that the family of geometric morphisms (f_i) is conservative. In particular:

Definition 6.4.1. Let E be a $\mathcal{U}$-topos and let $(p_i : P \to E)_{i \in I}$ be a family of points of E. This family is *conservative* if the associated family of fiber functors $F \mapsto F_{p_i}$ from E to ($\mathcal{U}$-**Set**) is conservative. The topos E is said to have *enough points* if it admits a conservative family of points, equivalently if the family of all its points is conservative.

6.4.2

All topoi used up to this point have enough points. It is nevertheless possible, deliberately, to construct topoi which do not have enough points; such a topos is necessarily non-empty in the geometric sense of 2.2. A topos with enough points admits a conservative family of points indexed by a $\mathcal{U}$-small set (Proposition 6.5). However, the set of isomorphism classes of points of a $\mathcal{U}$-topos need not itself be $\mathcal{U}$-small, although it is so in many practical cases. Finally, for an important existence theorem of P. Deligne giving enough points and covering all cases occurring in algebraic geometry, see Exposé VI, 9.

6.4.3

If $(p_i)_{i \in I}$ is a conservative family of points of E, then two arrows $u, v : F \rightrightarrows G$ in E are equal if and only if $u_{p_i} = v_{p_i}$ for all i. A morphism $u : F \to G$ is a monomorphism, respectively an epimorphism, if and only if each map $u_{p_i} : F_{p_i} \to G_{p_i}$ is injective, respectively surjective. An object F is initial, respectively final, if and only if every F_{p_i} is empty, respectively a singleton. Finally, F is a subobject of the final object e_E if and only if each F_{p_i} has at most one element.

Proposition 6.5.

a) Let C be a $\mathcal{U}$-site and let $(\varphi_i)_{i \in I}$ be a family of fiber functors on C. The corresponding family of points of $E = C^{\sim}$ is conservative if and only if, for every family $(X_j \to X)_{j \in J}$ in C, the following implication holds: if for every $i \in I$ the family $(\varphi_i(X_j) \to \varphi_i(X))_{j \in J}$ is surjective, then $(X_j \to X)_{j \in J}$ is covering.

b) If a $\mathcal{U}$-topos E has enough points, then it admits a $\mathcal{U}$-small conservative family of points.

Assertion b) is a special case of Exposé I, 7.7. For a), apply that criterion to the generating family of E consisting of the $\varepsilon(X)$, $X \in \operatorname{Ob} C$. Recall that $(X_j \to X)$ is covering if and only if $(\varepsilon(X_j) \to \varepsilon(X))$ is epimorphic. If the family of points is conservative, this is equivalent to surjectivity on all fibers, i.e. to surjectivity of all the families $(\varphi_i(X_j) \to \varphi_i(X))$. This proves necessity. Conversely, the same criterion reduces sufficiency to checking that a monomorphism $F \to \varepsilon(X)$ which is an isomorphism on all fibers is an isomorphism. Choose a covering family $\varepsilon(X_j) \to F$, refined so that the composites $\varepsilon(X_j) \to \varepsilon(X)$ come from morphisms $X_j \to X$. The fiberwise hypothesis and the assumption of a) imply that $(X_j \to X)$ is covering. Hence $(\varepsilon(X_j) \to \varepsilon(X))$ is epimorphic, and therefore $F \to \varepsilon(X)$ is epimorphic.

Corollary 6.5.1. Let C be a category. A topology on C making C into a $\mathcal{U}$-site with enough fiber functors, equivalently such that the associated topos has enough points, is completely determined by the full subcategory $\Phi \subset \mathcal{H}om(C, \mathbf{Set})$ formed by the fiber functors on C.

Indeed, Proposition 6.5(a) describes the covering families in terms of the family of fiber functors. Below, in 6.8.5, we shall see that Φ is contained in the category of pro-representable functors on C, so that $\Phi^\circ \simeq \mathrm{Pt}(C^\sim)$ identifies, up to equivalence, with a full subcategory of $\mathrm{Pro}(C)$. Compare this corollary with Giraud's theorem, Exposé II, 5.5, which establishes a bijection between topologies on C and certain full subcategories of $\mathcal{H}om(C^\circ, \mathbf{Set})$.

Exercise 6.5.2. Let C be a category equivalent to a $\mathcal{U}$-small category, and let P be a subcategory of $\mathrm{Pro}(C)$. For $p \in P$, write $X \mapsto X_p$ for the functor pro-represented by p. A family $(X_i \to X)_{i\in I}$ in C is called P-covering if for every $p \in P$ the family $(X_{ip} \to X_p)_{i\in I}$ is surjective. Show that there exists a topology T_P on C for which the covering families are precisely the P-covering families; that T_P is the finest topology on C for which all functors $X \mapsto X_p$, $p \in P$, are fiber functors; and that T_P gives C enough fiber functors. For any topology T on C, show that among the topologies finer than T with enough fiber functors there is a least one, namely T_P, where P is the strictly full subcategory of $\mathrm{Pro}(C)$ equivalent to $\mathrm{Pt}(C^\sim)$ described in 6.8.5.

Problem 6.5.3. Characterize the strictly full subcategories P of $\mathrm{Pro}(C)$, stable under filtered projective limits and satisfying the appropriate further conditions, which arise from a topology on C as the essential image of $\mathrm{Pt}(C^\sim)$ in $\mathrm{Pro}(C)$.

6.6

Let

$$f : E \longrightarrow E'$$

be a morphism of $\mathcal{U}$-topoi. It induces a canonical functor

$$\mathrm{Pt}(f) = (p \longmapsto f \circ p) : \mathrm{Pt}(E) \longrightarrow \mathrm{Pt}(E').$$

For two composable geometric morphisms

$$E \xrightarrow{f} E' \xrightarrow{g} E'',$$

one has, tautologically,

$$\mathrm{Pt}(gf) = \mathrm{Pt}(g) \circ \mathrm{Pt}(f).$$

Thus the dependence of $\mathrm{Pt}(E)$ on E is genuinely functorial: if $\mathcal{V}$ is a universe with $\mathcal{U} \in \mathcal{V}$, one obtains an honest functor from the category of $\mathcal{U}$-topoi which are elements of $\mathcal{V}$ to the category of categories which are elements of $\mathcal{V}$.

6.7

Let E be a $\mathcal{U}$-topos. Every object F of E defines a contravariant functor

$$p \longmapsto F_p = p^*(F) : \mathrm{Pt}(E)^\circ \longrightarrow (\mathcal{U}\text{-}\mathbf{Set}),$$

and hence, for variable F, a canonical functor

$$E \longrightarrow \widehat{\mathrm{Pt}(E)} = \mathcal{H}om(\mathrm{Pt}(E)^\circ, (\mathcal{U}\text{-}\mathbf{Set})). \tag{6.7.1}$$

By definition, this functor is faithful if and only if E has enough points. It has a certain formal analogy with a Fourier transform.

Let X be an object of E, and let $\widehat{X}$ be the corresponding presheaf on $C = \mathrm{Pt}(E)$. The category

$$C_{/\widehat{X}} = \mathrm{Pt}(E)_{/\widehat{X}}$$

is the category of pairs (p, ξ), where p is a point of E and $\xi \in X_p$. There is a canonical equivalence of categories

$$\mathrm{Pt}(E_{/X}) \xrightarrow{\sim} \mathrm{Pt}(E)_{/\widehat{X}}. \tag{6.7.2}$$

Composed with $\mathrm{Pt}(E)_{/\widehat{X}} \to \mathrm{Pt}(E)$, it is just the functor on points induced by the localization morphism $j_X : E_{/X} \to E$. This is the special case of Proposition 5.12 obtained by taking $E' = P = (\mathbf{Set})$.

Corollary 6.7.3. Let E be a topos. If E has enough points, then so does every induced topos $E_{/X}$. Conversely, if $(X_i)_{i \in I}$ is a family of objects of E covering the final object, and if each $E_{/X_i}$ has enough points, then E has enough points.

For the first assertion, let $f : X' \to X''$ be a morphism in $E_{/X}$ which is not an isomorphism. Since E has enough points, there is a point p of E such that $f_p : X'_p \to X''_p$ is not an isomorphism. Then for some $q \in X_p$, the map induced on the fibers over q is not an isomorphism. Under (6.7.2), this element q is a point of $E_{/X}$, and the map just described is precisely the map on fibers at that point.

Conversely, suppose the X_i cover the final object of E, and that each $E_{/X_i}$ has enough points. If $f : X' \to X''$ in E is not an isomorphism, then for some i, its restriction $f_i : X'_i \to X''_i$ over X_i is not an isomorphism. A point p_i of $E_{/X_i}$ detects this failure. Its image p in E then detects the failure of f to be an isomorphism. Thus E has enough points.

Remark 6.7.4. The same argument shows that if (p_α) is a conservative family of points of E, then the family of points of $E_{/X}$ lying above one of the p_α is conservative. Likewise, if the X_i cover the final object of E, and P_i is a conservative set of points of $E_{/X_i}$, then the set of their images in E under the localization morphisms $E_{/X_i} \to E$ is conservative.

6.8

Let E be a topos and let p be a point of E. A *neighborhood of p in E* is a pair (X, u), where $X \in \mathrm{Ob}\, E$ and $u \in X_p$. By (6.7.2), one may also interpret u as a lifting of p to a point of the induced topos $E_{/X}$. If φ_p is the fiber functor associated with p, a neighborhood of p is also an object of the category $E_{/\varphi_p}$. This category of pairs (X, u) is called the *category of neighborhoods* of p and is denoted $\mathrm{Neigh}(p)$. Since φ_p is left exact and finite projective limits are representable in E, finite projective limits are representable in $\mathrm{Neigh}(p)$, and the canonical functor $\mathrm{Neigh}(p) \to E$ commutes with them. Hence $\mathrm{Neigh}(p)^\circ$ is filtered. Moreover, for every $F \in \mathrm{Ob}\, E$, there is a canonical isomorphism

$$F_p = \varphi_p(F) \simeq \varinjlim_{\mathrm{Neigh}(p)^\circ} F(X). \tag{6.8.1}$$

In other words, the fiber functor at p is the filtered inductive limit of the functors represented in E by the neighborhoods of p. It is in fact ind-representable, equivalently representable by a pro-object $(X_i)_{i \in I}$ of E, with I a filtered ordered set in $\mathcal{U}$. To see this, choose a small full generating subcategory $C \subset E$. The full subcategory of $\mathrm{Neigh}(p)$ formed by the neighborhoods (X, u) with $X \in C$ is cofinal: every neighborhood (X, u) is refined by one whose object lies in C, using a covering family from C and the surjectivity on fibers.

6.8.2

Let C be a $\mathcal{U}$-site and let p be a point of C, i.e. of $C^\sim$. By abuse of notation write again φ_p for its restriction to C, and write $X_p = \varphi_p(X) = \varepsilon(X)_p$. A *neighborhood of p in the site C* is a pair (X, u), with $X \in \operatorname{Ob} C$ and $u \in X_p$. These neighborhoods form the category $C_{/\varphi_p}$, also denoted $\operatorname{Neigh}_C(p)$. Then $\operatorname{Neigh}_C(p)^\circ$ is filtered, admits a cofinal $\mathcal{U}$-small full subcategory, and for every sheaf F on C there is a functorial isomorphism

$$F_p \simeq \varinjlim_{\operatorname{Neigh}_C(p)^\circ} F(X). \tag{6.8.3}$$

Indeed, if $C' \subset C$ is a topologically generating full subcategory, then $\operatorname{Neigh}_{C'}(p)^\circ$ is cofinal in $\operatorname{Neigh}_C(p)^\circ$. Thus one reduces to the case $C \in \mathcal{U}$. Then $\widehat{C}$ is a topos and the associated-sheaf functor $a : \widehat{C} \to C^\sim$ allows one to apply (6.8.1) to $\widehat{C}$, with generating full subcategory C. This gives, more generally, for every presheaf P on C,

$$(aP)_p \simeq \varinjlim_{\operatorname{Neigh}_C(p)^\circ} P(X). \tag{6.8.4}$$

The general large-site case follows after enlargement of universes and the compatibility of associated sheaves with such enlargement.

6.8.5

To each point p of $C^\sim$ one has therefore associated a pro-object of the site C, defined by the canonical functor $\operatorname{Neigh}_C(p) \to C$. For variable p this gives a functor

$$\operatorname{Pt}(C^\sim) \longrightarrow \operatorname{Pro}(C). \tag{6.8.5.1}$$

It is fully faithful. Indeed, via associated fiber functors it is the composite $\operatorname{Fib}(C) \to \operatorname{Fib}(C^\sim) \to \operatorname{Pro}(C)^\circ$. The first functor is an equivalence by 4.9.4, recalled for fiber functors in 6.3, and the second is fully faithful by the general formalism of pro-objects.

6.8.6

If C is $\mathcal{U}$-small and carries the chaotic topology, so that $C^\sim = \widehat{C}$, then the preceding functor is an equivalence

$$\operatorname{Pt}(\widehat{C}) \xrightarrow{\sim} \operatorname{Pro}(C). \tag{6.8.6.1}$$

Essential surjectivity follows because the functors $P \mapsto P(X)$ on $\widehat{C}$ are fiber functors, and a filtered inductive limit of fiber functors on a topos is again a fiber functor. Under the embedding $C \subset \operatorname{Pro}(C)$, the induced functor $C \to \operatorname{Pt}(\widehat{C})$ is the canonical one already considered in 4.6.2.2; a quasi-inverse to (6.8.6.1) is the canonical extension of that functor to pro-objects.

6.8.7

Let $(X_i)_{i \in I}$ be a pro-object of the $\mathcal{U}$-site C. It defines on $C^\sim$ the functor

$$\varphi(F) = \varinjlim_I F(X_i). \tag{6.8.7.1}$$

In view of 6.8.5 it is natural to ask when this is a fiber functor. The following conditions are equivalent:

(i) φ is a fiber functor on $C^{\sim}$.

(ii) Its restriction to C is a fiber functor on the site C.

(iii) For every covering family $(Y_\alpha \to Y)$ in C, every $i_0 \in I$, and every morphism $X_{i_0} \to Y$, there exist $i \geq i_0$, an index α, and a morphism $X_i \to Y_\alpha$ making the evident square

$$\begin{tikzcd} X_i \arrow[r] \arrow[d] & Y_\alpha \arrow[d] \\ X_{i_0} \arrow[r] & Y \end{tikzcd}$$

commute.

Condition (iii) simply expresses that $\varphi|C$ sends covering families to covering families. The implications (i)$\Rightarrow$(ii)$\Rightarrow$(iii) are immediate. Conversely, φ is manifestly left exact; by 4.9.4 it remains to show that it sends covering families of sheaves to covering families. Given a covering family $F_\alpha \to F$, an index i_0, and an element $x_0 \in F(X_{i_0})$, choose a covering family $Z_\gamma \to X_{i_0}$ over which x_0 lifts to some $F_\alpha(Z_\gamma)$. By (iii), for some $i \geq i_0$ there is an X_{i_0}-morphism $X_i \to Z_\gamma$, so the image of x_0 in $F(X_i)$ lifts to some $F_\alpha(X_i)$. This proves the claim.

Exercise 6.9. Let C be a site in $\mathcal{U}$, and let φ be a fiber functor on C, pro-represented by a pro-object $(X_i)_{i\in I}$ of $\mathrm{Pro}(C)$. For every index i, every object Y, every morphism $u : X_i \to Y$, and every covering sieve R on Y, choose an index $j \geq i$ such that $X_j \to X_i \xrightarrow{u} Y$ factors through R. For fixed i, let $J(i)$ be the set formed by i and all such j; define $J(I')$ similarly for subsets $I' \subset I$. For a finite subset I', choose an upper bound $m(I')$, with $m(\{i\}) = i$; let $M(I')$ be the union of these upper bounds over all finite subsets of I'. Let $P(I')$ be the union of the iterates $(MJ)^n(I')$.

a) Show that $P(I')$ is filtered for the induced order, and that I is the increasing filtered union of the $P(I')$ as I' ranges over finite subsets of I.

b) Show that there is a cardinal $c \in \mathcal{U}$, depending only on $a = \operatorname{card}\mathrm{Fl}(C)$, such that $\operatorname{card} P(I') \leq c$ for every finite I'. If a is infinite, one may take $c = 2^a$.

c) Using 6.8.7, show that for every I', the pro-object $(X_i)_{i\in P(I')}$ pro-represents a fiber functor $\varphi_{I'}$, and that φ is the filtered inductive limit of the $\varphi_{I'}$ for finite I'.

d) Deduce that there is a small full subcategory Φ of $\mathrm{Fib}(C)$ such that every object of $\mathrm{Fib}(C)$ is a filtered inductive limit of objects of Φ. If a is infinite, Φ may be chosen of cardinal $\leq 2^{2^a}$; if a is finite, one may of course take $\Phi = C$.

Exercise 6.10. Let C be a $\mathcal{U}$-site, let D be a category in which small filtered inductive limits are representable, and let $\mathrm{Sh}(C, D)$ be the category of sheaves on C with values in D. Extending (6.8.3), define a fiber functor

$$F \longmapsto F_p : \mathrm{Sh}(C, D) \longrightarrow D. \tag{6.10.1}$$

If finite projective limits are representable in D and commute with filtered inductive limits, then this functor is left exact.

7 Examples of fiber functors and points of topoi

7.1. The case of $\mathrm{Top}(X)$, for a topological space X

Let x be a point of X. Regard the one-point set $\{x\}$ as a topological space, and consider the inclusion

$$i_x : \{x\} \longrightarrow X.$$

By 4.1.1 it defines a geometric morphism, defined up to unique isomorphism,

$$\mathrm{Top}(i_x) : \mathrm{Top}(\{x\}) \longrightarrow \mathrm{Top}(X). \tag{7.1.1}$$

On the other hand, one may identify P with $\mathrm{Top}(\{x\})$ (2.2), and hence obtains a point p_x of $\mathrm{Top}(X)$:

$$p_x : P \longrightarrow \mathrm{Top}(X), \tag{7.1.2}$$

defined by x up to unique isomorphism. The associated fiber functor is given by the familiar formula [TF], which follows immediately from I, 5.1:

$$p_x^*(F) = F_x = F_{p_x} = \varinjlim_{U \ni x} F(U), \tag{7.1.3}$$

the limit being taken over the open neighborhoods U of x in X.

Since $\mathrm{Top}(X)$ is known to be isomorphic to $\mathrm{Top}(X_{\mathrm{sob}})$ (4.2.1), one sees more generally that every point of X_{sob}, i.e. every irreducible closed subset Z of X, defines a point of $\mathrm{Top}(X)$, or equivalently a fiber functor on $\mathrm{Top}(X)$, still denoted $F \mapsto F_Z$. Returning to its definition by the formula (7.1.3) on X_{sob}, one obtains

$$F_Z = \varinjlim_{U \cap Z \neq \varnothing} F(U), \tag{7.1.4}$$

the inductive limit being taken over the open subsets U of X which meet Z.

7.1.5

It is well known that the fiber functors $F \mapsto F_x$, $x \in X$, already form a conservative family. This is especially clear from the interpretation of sheaves on X in terms of étale spaces: the fiber of F at x is then just its fiber in the sense of spaces fibered over X. Notice that the index set of this conservative family of fiber functors is X, and hence is $\mathcal{U}$-small.

7.1.6

By 4.2.3, the category $\mathrm{Pt}(\mathrm{Top}(X))$ is equivalent to the category associated with the set X_{sob} ordered by the specialization relation. In other words, every fiber functor on $\mathrm{Top}(X)$ is isomorphic to a fiber functor $F \mapsto F_Z$, where Z is a uniquely determined element of X_{sob}, i.e. a uniquely determined irreducible closed subset of X. Moreover, if $Z, Z' \in X_{\mathrm{sob}}$, then $\mathrm{Hom}(F_Z, F_{Z'})$ is either empty or reduced to one element; the latter case occurs if and only if Z is a specialization of Z' in X_{sob}, i.e. if and only if $Z \subset Z'$ as a subset of X. When X itself is sober, one may replace X_{sob} by X in these statements.

Observe that the automorphism group of a fiber functor of $\mathrm{Top}(X)$, opposite to the automorphism group of the corresponding point of $\mathrm{Top}(X)$, is always the trivial group. More generally, 4.2.3 says the same thing for the automorphism group of any geometric morphism associated with topological spaces. This is a very special phenomenon of the present case; see the example 7.2, as well as VIII, 7. In the case of topoi associated with "moduli problems" (cf. for instance [10]), the automorphism groups of the fiber functors have, moreover, a remarkable interpretation: they are the automorphism groups of the algebraic structures over algebraically closed fields which one proposes to classify.

7.1.7

We have just seen how one can reconstruct a sober space X, or the sober space X_{sob} associated with an arbitrary topological space, at least as an ordered set under the specialization relation, from the topos $\mathrm{Top}(X)$, which it is enough to know up to equivalence, as the set of isomorphism classes of "points" of $\mathrm{Top}(X)$. According to 2.1, one also reconstructs the topology of X, i.e. the family of its open subsets, as follows. For every subobject U of the final object e_E of E, let $\mathrm{Pt}(U)$ be the set of $x \in X$ such that $U_x \neq \varnothing$. By (6.7.2), this also identifies with the set of isomorphism classes of points of the induced topos $E_{/U}$. Then $U \mapsto \mathrm{Pt}(U)$ is an isomorphism of ordered sets from the set of subobjects of e_E onto the set of open subsets of X.

7.1.8

The determination 7.1.6 of the category of points of the topos $\mathrm{Top}(X)$ defined by a topological space X leads to the following terminology for points p, p' of an arbitrary topos E: one says that p is a *specialization* of p', or that p' is a *generization* of p, if there is a morphism from p' to p, in the sense of the category $\mathrm{Pt}(E)$ of 6.1. These relations are still transitive, but by 6.8.6 one easily finds examples in which p and p' are each a specialization of the other without being isomorphic, much less equal. The category of generizations of a point p of the topos E, meaning the category $\mathrm{Pt}(E)_{/p}$, plays in some respects a role analogous to localization of a scheme at a point. This role will be made precise in Chapter VI, 8.4, with the construction of the topos localized at p; its category of points is canonically equivalent to the category of generizations of p.

Exercise 7.1.9. Let E be a $\mathcal{U}$-topos. Prove that E is equivalent to a topos of the form $\mathrm{Top}(X)$, where X is a topological space belonging to $\mathcal{U}$, if and only if it satisfies the following two conditions:

a) The family of subobjects of the final object e_E is generating for the topos E.

b) E has enough points (6.5). This condition is not superfluous; cf. 7.4.

Compare with 7.8 c).

Exercise 7.1.10. Let X be a topological space endowed with a group G of operators, with $X, G \in \mathcal{U}$, and let $E = \mathrm{Top}(X, G)$ (2.3).

a) Show that for every object (X', G) of E, the induced topos $E_{/(X',G)}$ is canonically equivalent to $\mathrm{Top}(X', G)$ (compare 5.7).

b) When G acts properly and freely on X' (Bourbaki, *General Topology*, Chapter III, §4), show that the morphism of spaces with operators $(X', G) \to (X'/G, e)$ induces an equivalence of topoi

$$\mathrm{Top}(X', G) \longrightarrow \mathrm{Top}(X'/G).$$

c) Deduce from b) that there is an object Z of E such that the induced topos $E_{/Z}$ is equivalent to $\mathrm{Top}(X)$, the localization functor $E \to E_{/Z}$ being isomorphic to the functor "forget the operations of G". Imitate the argument of 5.8.3.

d) Deduce from c) that every fiber functor on E is induced, via the functor "forget the operations of G", by a fiber functor on $\mathrm{Top}(X)$, and is therefore definable by a point of X_{sob}.

e) Determine the structure of the category $\mathrm{Pt}(E)$, up to equivalence, in terms of the space with operators (X_{sob}, G). Conclude in particular that two points of X_{sob} define isomorphic fiber functors on E if and only if they are conjugate under the action of G.

Exercise 7.1.11. Let $X \in \mathcal{U}$ be a topological space. Prove that the $\mathcal{V}$-topos $\mathrm{TOP}(X)$ (2.5) has enough points. More precisely, for every object X' of $\mathrm{TOP}(X)$, and every $x' \in X'$, define a fiber functor $F \mapsto F_{X',x'}$ on $\mathrm{TOP}(X)$, depending only on the germ of the space (X', x') over X, and prove that this family of fiber functors is conservative and indexed by a $\mathcal{V}$-small set. Using a suitable filtered projective system $(X'_i, x'_i)_{i \in I}$, with I not $\mathcal{U}$-small, show that one does not obtain in this way all the fiber functors of the $\mathcal{V}$-topos $\mathrm{TOP}(X)$. Show that the set of isomorphism classes of such fiber functors does not have cardinal belonging to $\mathcal{V}$.

7.2. Points of a classifying topos B_G

Let E be a $\mathcal{U}$-topos and let G be a group object of E, whence a classifying topos B_G (2.4). If e is the trivial group object, the morphisms $e \to G \to e$ define, by 4.5, geometric morphisms, using $B_e \simeq E$,

$$E \longrightarrow B_G \longrightarrow E, \tag{7.2.1}$$

whose composite is isomorphic to id_E. Passing to categories of points gives functors

$$\mathrm{Pt}(E) \longrightarrow \mathrm{Pt}(B_G) \longrightarrow \mathrm{Pt}(E), \tag{7.2.2}$$

whose composite is isomorphic to the identity. Use the fact that the first geometric morphism in (7.2.1) identifies with a localization morphism relative to the object $E_G = G$ of B_G (5.8). By (6.7.2), the first functor in (7.2.2) then identifies with the natural "inclusion" functor

$$\mathrm{Pt}(B_G)_{/\widehat{E}_G} \longrightarrow \mathrm{Pt}(B_G), \tag{7.2.3}$$

where $\widehat{E}_G$ denotes the presheaf $p' \mapsto (E_G)_{p'}$ on $\mathrm{Pt}(B_G)$. Since E_G plainly covers the final object of B_G, its fibers $(E_G)_p$ are non-empty. It follows that (7.2.3) is essentially surjective. Equivalently, *every fiber functor on B_G is induced, up to isomorphism, by a fiber functor on E, via the functor $B_G \to E$ which forgets the operations of G.* In particular, *if E has enough points, respectively a small conservative family of points, then so does B_G.*

Remark 7.2.4. One can go further and determine, up to equivalence, the structure of the category $C' = \mathrm{Pt}(B_G)$ in terms of the category $C = \mathrm{Pt}(E)$ and the presheaf of groups

$$\widehat{G} : p \longmapsto G_p : C^\circ \longrightarrow (\mathrm{Groups})$$

on it. Define a new category C_1, with the same objects as C, and such that for two objects p, q of C one has

$$\operatorname{Hom}_{C_1}(p,q) = \operatorname{Hom}_C(p,q) \times \widehat{G}(p),$$

composition in C_1 being induced by composition in C, and by the group law of the presheaf $\widehat{G}$. The datum of a functor $\varphi_1 : C_1 \to D$, for an arbitrary category D, is then equivalent to the datum of a functor $\varphi : C \to D$ endowed with an action of the presheaf $\widehat{G}$ on it, meaning, for every $p \in \operatorname{ob} C$, an action of $\widehat{G}(p)$ on $\varphi(p)$, satisfying the evident functoriality condition as p varies. Using this observation, one defines a canonical functor

$$\varphi_1 : C_1 \longrightarrow C', \tag{7.2.4.1}$$

corresponding to the functor $\varphi : C \to C'$ of (7.2.2), which associates to every point p of E the point $\varphi(p)$ induced on B_G, with the natural actions of $\widehat{G}(p) = G_p$ on it, coming from the actions of G_p on the X_p as X ranges over B_G. We leave to the reader the proof that (7.2.4.1) is an equivalence of categories, using the structure (7.2.3) of the first functor in (7.2.2), and observing that the natural inclusion functor $C \to C_1$ has an analogous structure relative to the inverse image P_1 of the presheaf $P' = \widehat{E}_G$ on C'.

7.2.5

One concludes in particular that the *isomorphism classes of points of B_G correspond exactly to the isomorphism classes of points of E: every point of the classifying topos B_G is induced, up to non-unique isomorphism, by a point of E.* In particular, when E is the punctual topos P, so that G is an ordinary group, the category $\operatorname{Pt}(B_G)$ is a connected groupoid with fundamental group G: every fiber functor on B_G is isomorphic, non-canonically, to the functor ω which forgets the operations of G, and the monoid of endomorphisms of this functor is the group G. Thus every endomorphism of ω is an automorphism.

Exercise 7.2.6.

a) Let (Tors) be the category of pairs $(\Gamma, P) \in \mathcal{U}$, where Γ is a group and P is a right torsor under Γ. The functor

$$\text{(Tors)} \longrightarrow \text{(Groups)}, \quad (\Gamma, P) \longmapsto \Gamma \tag{7.2.6.1}$$

is a cofibered functor. With the notation of 7.2.4, consider the functor

$$\widehat{G} : C^\circ \longrightarrow \text{(Groups)},$$

and let D' be the cofibered category over C° inverse image of the cofibered category (7.2.6.1). Let D be the corresponding fibered category over C, with the same fiber categories as D', so that for $p \in \operatorname{ob} C$,

$$D_p = \text{the category of right } G_p\text{-torsors.}$$

Construct canonical functors

$$D \longrightarrow C' = \operatorname{Pt}(B_G) \quad \text{and} \quad C' \longrightarrow D, \tag{7.2.6.2}$$

quasi-inverse to one another. In particular, conclude that the category of points of B_G lying above a given point p of E is canonically equivalent to the category of right torsors under the group G_p.

b) Let $\mathcal{G} = (G_i)_{i\in I}$ be a strict projective system of group objects of the $\mathcal{U}$-topos E, where $I \in \mathcal{U}$ is a filtered preordered set, whence a classifying topos $B_{\mathcal{G}}$, imitating the definition of 2.7.1. Determine the category of points of $B_{\mathcal{G}}$, imitating a). In other words, replace the categories (Tors) and (Groups) by the categories of projective systems indexed by I in these categories. Conclude that if I admits a countable cofinal subset, then every fiber functor on $B_{\mathcal{G}}$ is isomorphic to a functor induced by a fiber functor on E, via the functor which forgets the operations of G; the latter is determined up to non-unique isomorphism.

c) Take E to be the punctual topos, so that $\mathcal{G}$ is an ordinary pro-group. Show that the conclusion of b) is also valid if I is arbitrary but $\mathcal{G}$ is profinite, i.e. the G_i are finite: every fiber functor is isomorphic to the functor which forgets the operations of G.

d) Still in the case where E is the punctual topos, give an example of a fiber functor on $B_{\mathcal{G}}$, for a suitable projective system $\mathcal{G} = (G_i)_{i\in I}$, which is not isomorphic to the functor which forgets the operations of G.

e) Regard $\mathcal{G} = (G_i)$ as a group object of the topos $\widehat{I}$, and let $\mathcal{T}$ be a *gerbe* on I with *band* G [3]. Show how one may "twist" the classifying topos B_G by means of the gerbe T, obtaining a topos $B^T_{\mathcal{G}}$, an inductive limit of subcategories $B^T(i)$ equivalent, non-canonically, to the classifying topoi B_{G_i}. Show that the topos $B^T_{\mathcal{G}}$ admits a fiber functor if and only if the gerbe T is neutral, by establishing an equivalence of categories between the category of fiber functors of $B^T_{\mathcal{G}}$ and the category of sections of T over I. Conclude, if the G_i are commutative, so that gerbes on I banded by $\mathcal{G}$ are classified by

$$\varprojlim_I{}^{(2)} G_i = H^2(I, \mathcal{G})$$

(loc. cit.), that there is an example of a topos, non-empty in the sense of 2.2, of the form $B^T_{\mathcal{G}}$ which has no points. Take an example where $\varprojlim_I{}^{(2)} G_i \neq 0$.

7.3. Points of the topoi $\widehat{C}$; examples of $\mathcal{U}$-topoi $\widehat{C}$ whose category of points is not equivalent to a small category

Let E be a $\mathcal{U}$-topos and let $(\varphi_i)_{i\in I}$ be a filtered family of fiber functors on E. It follows from the exactness properties of the functors $\varinjlim$ (I, 2.8) that $\varphi = \varinjlim \varphi_i$ is again a fiber functor: *every filtered inductive limit of fiber functors is a fiber functor*. Consequently the category $\mathrm{Fib}(E)$ admits $\mathcal{U}$-small filtered inductive limits, and the inclusion functor $\mathrm{Fib}(E) \to \mathcal{H}om(E, (\mathcal{U}\text{-}\mathbf{Set}))$ commutes with them. Equivalently, the category $\mathrm{Pt}(E)$ admits $\mathcal{U}$-small filtered projective limits. This fact was already used in Exercises 7.1.10 and 7.1.11 to give examples of large categories of fiber functors.

One can construct much simpler examples, with topoi of the form $E = \widehat{C}$, $C \in \mathcal{U}$, using (6.8.6.1). This immediately gives examples where $\mathrm{Fib}(\widehat{C})$ is not equivalent to a category belonging to $\mathcal{U}$, i.e. where the cardinality of the set of isomorphism classes of objects of this $\mathcal{U}$-category is not in $\mathcal{U}$. This means that the category $\mathrm{Pro}(C)$ is not equivalent to a category in $\mathcal{U}$. It suffices, for example, to take $D = C^\circ$ to be the category of finite sets of the form $[0, n]$, where $n \geq 0$ is an integer. Then $\mathrm{Ind}(D) \simeq \mathrm{Pro}(C)^\circ$ is equivalent to the category of non-empty sets. In this case the topos E is the familiar category of *cosimplicial sets*. One could also take for C the category of non-empty finite sets; then the category of points of the topos $\widehat{C}$ of simplicial sets is equivalent to the category of profinite sets, or equivalently to the category of totally disconnected compact spaces.

7.4. Non-empty topoi without points

The following example is due to P. Deligne. For another example, cf. 7.2.6 e). Take a compact space K endowed with a measure μ, and the ordered set U of measurable subsets of K, modulo sets of measure zero. Make U into a category $\underline{U}$ with $\operatorname{ob}\underline{U} = U$, whose morphisms are the inclusion morphisms among elements of U. Make $\underline{U}$ into a site by taking the pretopology in which $\operatorname{Cov}(E)$, for $E \in U$, consists of countable families of elements E_i of U, bounded above by E, such that E is the union of the E_i, modulo a set of measure zero. One obtains a topos $\operatorname{Top}(\mu) = \underline{U}^{\sim}$, which admits the set of subobjects of the final object as a generating family. This topos seems to have escaped the attention of probabilists. It is the empty topos (2.2) if and only if $\mu = 0$. On the other hand, the category of points of this topos is equivalent to the discrete category defined by the set of points $x \in K$ such that $\mu(\{x\}) \neq 0$. The proof is left to the reader. Thus it is empty when K has no such points, for example when K is the unit interval on the real line, with the measure induced by Lebesgue measure.

Exercise 7.5 (Karoubian categories and geometric morphisms).

a) Let C be a category, X an object of C, and p an endomorphism of X. One says that p is a *projector* if $p^2 = p$. Prove that, if p is a projector, then $\operatorname{Ker}(\mathrm{id}_X, p)$ is representable if and only if $\operatorname{Coker}(\mathrm{id}_X, p)$ is representable, and the two objects of C so obtained are canonically isomorphic. One then says that the projector p *admits an image*, and $\operatorname{Ker}(\mathrm{id}_X, p) \simeq \operatorname{Coker}(\mathrm{id}_X, p)$ is called the *image of the projector* p. Depending on the context, it is identified with a subobject or with a quotient object of C. An object isomorphic to $\operatorname{Im} p$, for a suitable projector p of X, is called a *direct summand of the object* X *of* C. One says that C is a *category with direct summands*, or a *Karoubian category*, if every projector in an object of C admits an image. If $F : C \to C'$ is a functor which commutes with kernels or cokernels, then F carries a projector admitting an image to a projector admitting an image.[11]

b) Show that for every category C one can find a functor $\varphi : C \to \operatorname{Kar}(C)$ from C into a Karoubian category, determined up to equivalence, such that for every Karoubian category C', the functor

$$f \longmapsto f \circ \varphi : \mathcal{H}om(\operatorname{Kar}(C), C') \longrightarrow \mathcal{H}om(C, C')$$

is an equivalence of categories. We call $\operatorname{Kar}(C)$ the *Karoubi envelope* of C. Hint: take $\operatorname{ob}\operatorname{Kar}(C)$ to be the set of pairs (X, p), where $X \in \operatorname{ob} C$ and p is a projector of X, and take $\operatorname{Hom}((X, p), (Y, q))$ to be the subset of $\operatorname{Hom}(X, Y)$ formed by those $f : X \to Y$ such that $f = qfp$. Show that φ is fully faithful.

c) Let $F : E \to E'$ be a functor from one category to another. Show that if F commutes with a certain type of inductive or projective limits, then the same holds for every direct summand of F. In particular, if E and E' are $\mathcal{U}$-topoi and if F is an inverse image functor for a geometric morphism $E' \to E$, then the same holds for every direct summand of F.

[11] In fact no condition on F is needed here: all functors preserve projectors which admit images.

Hence the category $\mathcal{H}om_{\mathrm{top}}(E', E)$ is Karoubian, and in particular the category $\mathrm{Pt}(E)$ is Karoubian.

d) Show that for every $\mathcal{U}$-category C, the category $\mathrm{Ind}(C)$ is Karoubian, where $\mathrm{Ind}(C)$ is formed using inductive systems in C indexed by a filtered preordered set $I \in \mathcal{U}$. If $C \in \mathcal{U}$, use for instance the fact (7.3) that $\mathrm{Ind}(C)$ is equivalent to the category $\mathrm{Fib}(\widehat{(C^\circ)}) = \mathrm{Pt}(\widehat{(C^\circ)})^\circ$. Deduce another construction of $\mathrm{Kar}(C)$, as the full subcategory of $\mathrm{Ind}(C)$ formed by the images in $\mathrm{Ind}(C)$ of projectors of objects X of C.

Exercise 7.6 (Essential geometric morphisms; essential points).

a) Let $f : E \to F$ be a geometric morphism. Show that the following conditions are equivalent:

   (i) $f_!$ exists, i.e. f^* admits a left adjoint.

   (ii) f^* commutes with $\mathcal{U}$-small projective limits.

   (iii) f^* commutes with $\mathcal{U}$-small products.

   One then says that f is an *essential morphism* of topoi from E to F. Show that if f satisfies these conditions, then the same holds for every direct summand of f. Use 1.8 and 7.5 c).

b) Let E be a topos. An *essential point* of E is a point $p : P \to E$ such that $p_!$ exists. Show that if $E = \mathrm{Top}(X)$, with X a topological space, and if p is the point of E defined by $x \in X$, then p is essential if and only if x admits a smallest open neighborhood U. If the points of X are closed, this means that x is an isolated point of X. For a geometric morphism $f : E \to F$ to be essential, it is necessary that f carry essential points to essential points; this condition is also sufficient when E has enough essential points, meaning when these points form a conservative family. Cf. d).

c) A point p of the topos E is essential if and only if the associated fiber functor is represented by an object X of E. For the covariant functor $\varphi : E \to (\mathbf{Set})$ represented by a given object X of E to be a fiber functor, i.e. to define a necessarily essential point of E, it is necessary and sufficient that X be non-empty connected (4.3.5) and projective, i.e. that φ carry epimorphisms to epimorphisms. Hint: first show that φ commutes with sums if and only if X is non-empty connected, then use the criterion 4.9.4. Conclude an equivalence between the category $\mathrm{Pt}_{\mathrm{ess}}(E)$ and the full subcategory P of E formed by the non-empty connected projective objects.

d) Show that the topology on P induced by that of E is the chaotic topology. Deduce the equivalence of the following conditions on E, due to J. E. Roos [12 c), Proposition 1]: (i) the family of essential points of E is conservative; (ii) the full subcategory P of E formed by non-empty connected projective objects is generating; (iii) E is equivalent to a topos of the form $\widehat{C}$, where C is a category equivalent to a category in $\mathcal{U}$, or, equivalently, one may take $C \in \mathcal{U}$. Use c) and the fact that, if P is generating, the natural functor $E \to P^\sim$ is an equivalence of categories.

e) The category of essential points of a topos E is Karoubian (cf. 7.5 c)). Let C be a category equivalent to a category in $\mathcal{U}$. Prove that the canonical fully faithful functor (4.6.2)

$$C \longrightarrow \mathrm{Pt}(\widehat{C})$$

factors through a functor

$$C \longrightarrow \mathrm{Pt}_{\mathrm{ess}}(\widehat{C}),$$

which makes $\mathrm{Pt}_{\mathrm{ess}}(\widehat{C})$ a *Karoubi envelope* of C (7.5 b)). In particular, this functor is an equivalence of categories if and only if C is Karoubian. Hint: using c), prove that every isolated point of $\widehat{C}$ is isomorphic to a direct summand of an $X \in \mathrm{ob}\, C$.

f) Let C and C' be two categories equivalent to categories in $\mathcal{U}$. Prove that the canonical fully faithful functor (4.6.2)

$$\mathcal{H}om(C, C') \longrightarrow \mathcal{H}om_{\mathrm{top}}(\widehat{C}, \widehat{C'})$$

takes its values in the full subcategory $\mathcal{H}om_{\mathrm{top,ess}}(\widehat{C}, \widehat{C'})$ of essential geometric morphisms, and that the induced functor

$$\mathcal{H}om(C, C') \longrightarrow \mathcal{H}om_{\mathrm{top,ess}}(\widehat{C}, \widehat{C'})$$

is an equivalence of categories if and only if C is empty or C' is Karoubian. Use g) below.

g) Let C be a category equivalent to a category in $\mathcal{U}$, and let E be a topos. Define an equivalence between the category $\mathcal{H}om_{\mathrm{top,ess}}(\widehat{C}, E)$ of essential geometric morphisms $\widehat{C} \to E$, and the category $\mathcal{H}om(C, \mathrm{Pt}_{\mathrm{ess}}(E))$. Use b) and the fact that $\widehat{C}$ has enough essential points.

h) Let C be a finite category. Prove that every point of $\widehat{C}$ is essential, and that $\mathrm{Pt}(\widehat{C})$ is equivalent to a finite category. Noting that the Karoubi envelope of C is finite and using (6.8.6.1), reduce to proving that if C is a finite Karoubian category, then the canonical functor $C \to \mathrm{Pro}(C)$ is an equivalence of categories. Using b), conclude that *every* geometric morphism $\widehat{C'} \to \widehat{C}$ is essential, and that $\mathcal{H}om(C, C') \to \mathrm{Hom}(\widehat{C}, \widehat{C'})$ is an equivalence if and only if C is empty or C' is Karoubian.

i) Let X be a topological space, and let $f : \mathrm{Top}(X) \to P$ be the geometric morphism induced by the continuous map from X to the one-point space. Show that f is essential if and only if X is *locally connected*, i.e. satisfies the following condition: for every $x \in X$ and every open neighborhood U of x, the connected component of x in U is a neighborhood of x; equivalently, for every open subset U of X, the connected components of U are open. Cf. 8.7 l) for a generalization.

Exercise 7.7 (Unusual points of a classifying topos).

a) Let G be a monoid. The monoid G contains an element g such that $g \neq e$ and $g^2 = g$ if and only if the classifying topos B_G admits an essential point not isomorphic to the trivial point, the one corresponding to the fiber functor which forgets the operations of G. Use 7.6 e).

b) Let G be the additive monoid of integers ≥ 0. Show that every essential point of the classifying topos B_G is isomorphic to the trivial point. Construct a point of B_G which is not isomorphic to the trivial point. Show that the category $\mathrm{Pt}(B_G)$ is not equivalent to a category in $\mathcal{U}$.

Exercise 7.8 (Topology on $\mathrm{Point}(E)$**, and topoi associated with ordered sets).**

a) Let E be a topos. Denote by $\mathrm{Point}(E)$ the set of isomorphism classes of points of E. For every open U of E, let U' be the set of classes of points p of E such that $U_p \neq \varnothing$. Show that $U \mapsto U'$ is an increasing map from the ordered set $\mathrm{Open}(E)$ of open subobjects of E to the set of subsets of $X = \mathrm{Point}(E)$, and that this map commutes with finite infima and arbitrary suprema. Deduce that the subsets of X of the form U' define a topology on X, called the *canonical topology on the set of classes of points of the topos* E. Show that if E has enough points, the map $U \mapsto U'$ is an isomorphism from the ordered set $\mathrm{Open}(E)$ to the ordered set $\mathrm{Open}(X)$.

b) Let $\mathcal{O} \in \mathcal{U}$ be an ordered set admitting finite infima and arbitrary suprema; it therefore defines a category $\mathrm{cat}(\mathcal{O})$ having finite products and arbitrary sums. We say that a family of morphisms $U_i \to U$ with common target U is covering if U is the supremum of the U_i. Assuming that sums in $\mathrm{cat}(\mathcal{O})$ are universal, i.e. that arbitrary suprema in $\mathcal{O}$ commute with finite infima, this defines a topology on $\mathrm{cat}(\mathcal{O})$, making it a site. The resulting topos $\mathrm{cat}(\mathcal{O})^\sim$ is also denoted simply $\mathcal{O}^\sim$. Show that $\mathcal{O}$ is canonically isomorphic to $\mathrm{Open}(\mathcal{O}^\sim)$.

Show that the category $\mathcal{H}om_{\mathrm{top}}(E, \mathcal{O}^\sim)$ is equivalent to the category associated with the ordered set of morphisms of ordered sets $\mathcal{O} \to \mathrm{Open}(E)$ which commute with finite infima and arbitrary suprema. Use the criterion 4.9.4. If $\mathcal{O}'$ is an ordered set satisfying the same conditions as $\mathcal{O}$, conclude that $\mathcal{H}om_{\mathrm{top}}(\mathcal{O}^\sim, \mathcal{O}'^\sim)$ is equivalent to the category associated with the ordered set of all morphisms of ordered sets $\mathcal{O} \to \mathcal{O}'$ which commute with finite infima and arbitrary suprema.

c) Show that for the topos E to be equivalent to a topos of the form $\mathcal{O}^\sim$, with $\mathcal{O}$ as in b), it is necessary and sufficient that the family of opens of E be generating in E. For it to be equivalent to $\mathrm{Top}(X)$ for $X \in \mathcal{U}$, it is necessary and sufficient that it satisfy the preceding condition and have enough points. Assuming the first condition holds, express the second in terms of the structure of the ordered set $\mathcal{O} = \mathrm{Open}(E)$, using the last assertion of b).

d) Define a canonical geometric morphism

$$E \longrightarrow \mathrm{Open}(E)^\sim$$

which is universal, up to equivalence, for morphisms from E to topoi generated by their opens (cf. c)).

e) Let X' be a small subset of $X = \mathrm{Point}(E)$, endowed with the topology induced from that of X. Define a canonical geometric morphism

$$\mathrm{Open}(E)^\sim \longrightarrow \mathrm{Top}(X')$$

which is an equivalence when the family X' of points of E is conservative. In that case, deduce a canonical geometric morphism

$$E \longrightarrow \mathrm{Top}(X').$$

Give a universal characterization, up to equivalence, of this geometric morphism for morphisms from the topos E to topoi of the form $\mathrm{Top}(Y)$. Notice in this connection that, up to equivalence, $\mathrm{Top}(X')$ does not depend on the small conservative family of fiber functors chosen; equivalently, by 4.2, X'_{sob} does not depend, up to homeomorphism, on the choice of such a family.

f) Suppose that the category $\mathrm{Pt}(E)$ is not equivalent to a small category (cf. 7.3), and that E admits a small conservative family of fiber functors. Show that if such a family is chosen sufficiently large, the subset X' of $\mathrm{Pt}(E)$ which it defines is not a sober topological space for the induced topology.

g) For the relations with Exercise 7.6, see Section 8.

8 Localization. Opens of a topos

8.1

Let C be a category, and let $T(X)$ be a relation involving an object X of C. One says that the relation $T(X)$ is *stable under base change*, in X, if for every morphism $X' \to X$ of C, the relation $T(X)$ implies $T(X')$. This is equivalent to saying that the full subcategory R of C, formed by the objects X of C for which $T(X)$ holds, is a *sieve*.

Remark 8.1.1. Suppose that C is a $\mathcal{U}$-category, so that C may be identified with a full subcategory of $\widehat{C}$ (I, 1.4). It is sometimes convenient to extend the definition of the relation T on $\mathrm{ob}\, C$ to a relation $\widehat{T}$ concerning an object F of $\widehat{C}$, by defining $\widehat{T}(F)$ to mean: for every arrow $X \to F$ in $\widehat{C}$, with $X \in \mathrm{ob}\, C$, one has $T(X)$. The relation $\widehat{T}(F)$ is clearly still stable under base change in F. Still denoting by R the subobject of the final object e of $\widehat{C}$ defined by the sieve R of C (I, 4.2.1), the relation $\widehat{T}(F)$ may also be expressed by $\mathrm{Hom}_{\widehat{C}}(F, R) \neq \varnothing$, meaning that the unique morphism $F \to e$ factors through the subobject R of e.

8.2

Now suppose that C is a *site*. A relation $T(X)$, in one argument $X \in \mathrm{ob}\, C$, is said to be *stable under descent* if, for every covering family $(X_i \to X)_{i \in I}$ of an object X of C, the relation "$T(X_i)$ for all $i \in I$" implies $T(X)$. One says that T is a *relation of local nature* if it is stable under base change (8.1) and stable under descent. When C is a $\mathcal{U}$-category, and T is already assumed stable under base change, i.e. definable by a sieve R of C, which is identified with a subpresheaf of the final presheaf on C, one sees immediately that T is of local nature if and only if R is a *sheaf*. Thus *relations of local nature in one argument $X \in \mathrm{ob}\, C$ correspond exactly to subsheaves of the final sheaf of C, i.e. to subobjects of the final object of the topos $C^\sim$.* They therefore essentially depend only on the topos $C^\sim$ defined by the site C, as do all important notions associated with a site. In terms of the subsheaf R of the final sheaf, the relation $T(X)$ is expressed as $\mathrm{Hom}(\varepsilon(X), R) \neq \varnothing$. It extends canonically to the relation $T^\sim(F) : \mathrm{Hom}(F, R) \neq \varnothing$ on the topos $C^\sim$.

Remark 8.2.1. With the notation introduced in 8.1.1, for a presheaf F on C, since $\mathrm{Hom}(F, R) \simeq \mathrm{Hom}(a(F), R)$, the relation $\widehat{T}(F)$ is equivalent to the relation $\widehat{T}(a(F))$.

Definition 8.3. An *open* of a $\mathcal{U}$-topos E is a subobject of the final object e_E of E. If X is an object of E, an *open of* X sometimes means an open of the induced topos $E_{/X}$, i.e. a subobject of X.

An *open* of a $\mathcal{U}$-site C is an open of the associated $\mathcal{U}$-topos $C^\sim$.

Notice that, since final objects of E are canonically isomorphic, the ambiguity introduced in 8.3 by the choice of e_E is harmless. One may also, more intrinsically but less conveniently in practice, define the opens of E to be the full subcategories R of E such that the relation $F \in \operatorname{ob} R$, for an object F of E, is a relation of local nature. Likewise, the opens of a $\mathcal{U}$-site C correspond bijectively to full subcategories of C such that the relation $F \in \operatorname{ob} R$, for an object of C, is of local nature. Thus this notion is essentially independent of the chosen universe $\mathcal{U}$. Finally, by 8.2, a relation of local nature on a topos, or on a site, is essentially the same thing as an open of that topos, or of that site.

8.4. Examples of opens

8.4.1

Let X be a topological space, and interpret $\operatorname{Top}(X)$ (2.1) as the category of étale spaces over X. Since monomorphisms in $\operatorname{Top}(X)$ are plainly the injective maps, it follows that the opens of the topos $\operatorname{Top}(X)$ identify with the open subsets of the space X. More generally, the opens of a topos $\operatorname{Top}(X, G)$ (2.4) identify with the open subsets of X which are invariant under the action of G.

8.4.2

The opens of a $\mathcal{U}$-topos E form a $\mathcal{U}$-small ordered set under inclusion, admitting arbitrary suprema and finite infima. It has the initial object, or empty object, $\varnothing_E$, as least element, and the final object e_E of E as greatest element. These two elements are identical only if E is an empty topos (2.2).

8.4.3

Let G be a monoid object of E, whence a topos B_G (2.4, 2.6), the category of objects of E on which G acts on the left. The final object e_G of B_G is the final object e_E of E, with trivial action of G. It follows that the ordered set of opens of B_G is canonically isomorphic to the category of opens of E. In particular, if E is the punctual topos, so that G is an ordinary monoid, the set of opens of B_G consists exactly of two elements, namely $\varnothing_{B_G}$ and e_G.

8.4.4

If E is a topos of the form $\widehat{C}$, where C is a $\mathcal{U}$-category, then by 8.1 the ordered set of opens of E is isomorphic to the ordered set of sieves of C.

8.4.5

Let X be a sober non-discrete topological space. Show that $\operatorname{TOP}(X)$ (2.5) has opens which do not come from opens of X, i.e. from opens of $\operatorname{Top}(X)$, by inverse image under the canonical morphism (4.10.1) $\operatorname{TOP}(X) \to \operatorname{Top}(X)$.

8.5. Examples of relations of local nature

For every object X of the topos E, denote by $F \mapsto F_X$ the *localization functor* $j_X^* : E \to E_{/X}$, $Y \mapsto Y \times X$ (5.2).

8.5.1

Let $u : F \to G$ be a morphism of E. The relation in the argument $X \in \operatorname{ob} E$:

$u_X : F_X \to G_X$ is a monomorphism, respectively an epimorphism, respectively an isomorphism,

is a relation of local nature. In particular, if G is a group object of E, the relation "G_X is the trivial group over X" is of local nature; the open of E which corresponds to it is called the *cosupport of the group object G.*

8.5.2

Let $u, v : F \rightrightarrows G$ be two morphisms of E. The relation in the argument $X \in \operatorname{ob} E$

$$u_X = v_X : F_X \longrightarrow G_X$$

is a relation of local nature. In particular, if G is a group object of E, and u is a *section* of G (4.3.6), the relation in X

the section u_X of the group object G_X of $E_{/X}$ is zero

is of local nature; the open of E which corresponds to it is called the *cosupport of the section u of the group object G.*

Remark 8.5.3. When $E = \operatorname{Top}(X)$, the cosupport of a sheaf of groups G, respectively of a section of such a sheaf G, is just the open subset of X complementary to the *support of G*, respectively the *support of u*, in the classical terminology [TF].

8.5.4

Let $P(f)$ be a relation in the argument $f \in \operatorname{Fl} C$, where C is a site. When fiber products are representable in C, one says that the relation $P(f)$ is *stable under base change*, respectively *stable under descent*, respectively *of local nature*, *on the target*, or *on the base*, or "downstairs", if for every morphism $f : X \to Y$ of C, the relation in the argument $Y' \in \operatorname{ob} C_{/Y}$:

the morphism $f' : X' = X \times_Y Y' \to Y'$, obtained from f by base change along the structural morphism $Y' \to Y$, satisfies $P(f')$,

is stable under base change, respectively stable under descent, respectively is of local nature. When the topology of C is defined by a pretopology, one says that the relation $P(f)$ is *of local nature on the source*, or "upstairs", if for every morphism $f : X \to Y$ of C and every family $(g_i : X_i \to X)_{i \in I}$ in $\operatorname{Cov}(X)$, the relation $P(f)$ is equivalent to the relation "$P(fg_i)$ for all $i \in I$". When $\operatorname{Cov}(X)$ consists of all covering families of C, this condition also says that the relation in the object X' of the induced site $C_{/X}$,

$P(fg)$, where $g : X' \to X$ is the structural morphism,

is of local nature. Notice that if $P(f)$ is of local nature upstairs, then it is a fortiori of local nature downstairs.

8.5.5

For example, take $C = (\mathbf{Sch})$, the category of schemes belonging to $\mathcal{U}$, with the faithfully flat quasi-compact topology (SGA 3, IV, 6.3) or a less fine topology. Then each of the following properties of a morphism is of local nature on the target:

> surjective, radicial, universally open, universally closed, proper, quasi-compact, quasi-compact and dominant, universal homeomorphism, separated, quasi-separated, locally of finite type, locally of finite presentation, of finite type, of finite presentation, an isomorphism, a monomorphism, an open immersion, a closed immersion, a quasi-compact immersion, affine, quasi-affine, finite, quasi-finite, integral, flat, faithfully flat, unramified, smooth, étale.

See EGA IV, 2.6.4, 2.7.1, and EGA IV, 17.7.1. On the other hand, endow $(\mathbf{Sch})$ with the pretopology for which, for every scheme $X \in \mathcal{U}$, $\mathrm{Cov}(X)$ consists of families of morphisms $(f_i : X_i \to X)_{i \in I}$ which are surjective and such that the f_i are flat and locally of finite presentation. Then each of the following properties is *of local nature on the source*:

locally of finite type, locally of finite presentation, of finite type, flat, unramified, smooth, étale.

See EGA IV, 17.7.5, 17.7.7.

Exercise 8.6. Let $f : E \to F$ be a morphism of $\mathcal{U}$-topoi. Consider the relation in the argument $X \in \mathrm{ob}\, F$: the induced morphism $E_{/f^*(X)} \to F_{/X}$ is an equivalence of topoi. Prove that this relation is of local nature.

Exercise 8.7 (Partitions of a topos; sums of topoi). Let E be a $\mathcal{U}$-topos. A family $(e_i)_{i \in I}$ of opens of E, i.e. of subobjects of the final object e of E, is called a *partition of* e, or of the topos E, if the canonical morphism $\coprod_{i \in I} e_i \to e$ is an isomorphism; equivalently, by II, 4.6.2, e is the supremum of the e_i and $i \neq j$ implies $e_i \cap e_j = \varnothing_E$.

a) Show that a family $(e_i)_{i \in I}$ of objects of E is a partition of E if and only if the functor

$$E \longrightarrow \prod_i E_{/e_i} \tag{8.7.1}$$

defined by the localization functors $E \to E_{/e_i}$ is an equivalence of categories.

b) Conversely, let $(E_i)_{i \in I}$ be a family of $\mathcal{U}$-topoi, with $I \in \mathcal{U}$. Prove that $E = \prod_{i \in I} E_i$ is a $\mathcal{U}$-topos, called the *sum topos* of the family $(E_i)_{i \in I}$. Define a partition $(e_i)_{i \in I}$ of E, and equivalences of categories $E_{/e_i} \simeq E_i$, in such a way that the projection functors $E \to E_i$ identify with the localization functors $E \to E_{/e_i}$.

c) Let E be the sum topos of the family $(E_i)_{i \in I}$. Show that for every $i \in I$ the projection functor $E \to E_i$ has the form u_i^*, where

$$u_i : E_i \longrightarrow E \tag{8.7.2}$$

is a geometric morphism. Prove that for every $\mathcal{U}$-topos F, the functor $v \mapsto (v \circ u_i)_{i \in I}$

$$\mathcal{H}om_{\mathrm{top}}(E, F) \longrightarrow \prod_i \mathcal{H}om_{\mathrm{top}}(E_i, F) \tag{8.7.3}$$

is an equivalence of categories.

d) Let $(e_i)_{i\in I}$ be a partition of the topos E. For every geometric morphism $g : F \to E$, the family $(f_i)_{i\in I}$, with $f_i = g^*(e_i)$, is a partition of F, and the induced morphisms $g_i : E_{/e_i} \to F_{/f_i}$ allow one to reconstruct g, up to unique isomorphism; the functor g^* identifies with the Cartesian product of the functors g_i^*, using the equivalences of type 8.7.1. Deduce a complete description of the category $\mathcal{H}om_{\mathrm{top}}(F, E)$ in terms of categories of the form $\mathcal{H}om_{\mathrm{top}}(F', E_i)$, where F' is a topos of the form $F_{/f}$, with f a direct summand of the final object of F, and $E_i = E_{/e_i}$.

e) In particular, if F is non-empty connected (cf. 4.3.5), i.e. if for every partition $(f_i)_{i\in I}$ of F there is exactly one $i \in I$ with $f_i \neq \varnothing_F$, prove that the canonical functor

$$\coprod_{i\in I} \mathcal{H}om_{\mathrm{top}}(F, E_i) \longrightarrow \mathcal{H}om_{\mathrm{top}}(F, E) \tag{8.7.4}$$

induced by the geometric morphisms (8.7.2) is an equivalence of categories. More particularly, the family of geometric morphisms (8.7.2) induces an equivalence of categories

$$\coprod_{i\in I} \mathrm{Pt}(E_i) \xrightarrow{\sim} \mathrm{Pt}(E).$$

f) A subset I of the set of opens of E is called a *reduced partition* of E if the identical family $(e_i)_{i\in I}$, indexed by I, is a partition and if each $e_i \neq \varnothing_E$. Show that the refinement relation (I, 4.3.2) among reduced partitions makes the set P of reduced partitions into a $\mathcal{U}$-small ordered set such that the least upper bound of two elements of P exists. Thus one obtains a strict projective system $(I_p)_{p\in P}$, regarded as a pro-set and denoted $\pi_0(E)$. It is called the *pro-π_0 of the topos E*.

g) Prove that $\pi_0(E)$ pro-represents the functor

$$T \longmapsto f_* f^*(T) = \Gamma(E, T_E) : (\mathcal{U}\text{-}\mathbf{Set}) \longrightarrow (\mathcal{U}\text{-}\mathbf{Set})$$

associated with the canonical morphism $f : E \to P$ from E to the punctual topos P (4.3). In particular, this functor is representable if and only if $\pi_0(E)$ is essentially constant, i.e. isomorphic, as a pro-set, to an ordinary set, again denoted $\pi_0(E)$. The pro-set $\pi_0(E)$ is reduced to one point if and only if E is non-empty connected. It is empty if and only if E is the empty topos (2.2).

h) Suppose that E is quasi-compact, i.e. every covering of its final object e admits a finite subcovering. Show that $\pi_0(E)$ is then a profinite set, and may therefore be identified, via the well-known equivalence between pro-objects of the category of finite sets and totally disconnected compact spaces, with a totally disconnected compact space, also denoted $\pi_0(E)$.

i) Let X be a topological space, and let $\pi_0(X)$ be the set of connected components of X, regarded as a constant pro-set. Define a canonical morphism of pro-sets

$$\pi_0(X) \longrightarrow \pi_0(\mathrm{Top}(X)), \tag{8.7.5}$$

or equivalently a canonical map

$$\pi_0(X) \longrightarrow \varprojlim \pi_0(\mathrm{Top}(X)). \tag{8.7.6}$$

Show that the first morphism is an isomorphism if and only if X is homeomorphic to a sum space of connected spaces. This is the case in particular if X is locally connected.

j) Suppose that X is a quasi-compact scheme. Prove that (8.7.6) is bijective.

k) Establish the functorial character in X of the morphisms (8.7.5) and (8.7.6), beginning by specifying the meaning of this expression.

l) Compare 7.6 i). Show that the following two conditions on the $\mathcal{U}$-topos E are equivalent: (i) the canonical morphism from E to the punctual topos is essential (Exercise 7.6 a)), i.e. the functor $I \mapsto I_E$ from ($\mathcal{U}$-**Set**) to E commutes with products indexed by sets belonging to $\mathcal{U}$; (ii) for every object X of E, the induced topos $E_{/X}$ has a $\pi_0(E_{/X})$ which is an ordinary set, i.e. X is isomorphic to the sum of a family of connected objects of E. One then says that E is a *locally connected topos* (compare IV, 2.7.5).

Exercise 8.8 (Dominant geometric morphisms).

a) Let $f : E' \to E$ be a geometric morphism. Show that the following conditions are equivalent:

   (i) $f_*(\varnothing_{E'}) = \varnothing_E$, where $\varnothing_E$ and $\varnothing_{E'}$ denote the initial objects of E and E'.

   (ii) For every object X of E which is non-empty, i.e. not isomorphic to $\varnothing_E$, the object $f^*(X)$ is non-empty in E'.

   (iii) As in (ii), with X a subobject of the final object e_E of E, i.e. an open of the topos E.

   One then says that the geometric morphism f is *dominant*.

b) Let $f : X' \to X$ be a continuous map of topological spaces, whence a geometric morphism $\mathrm{Top}(f) : \mathrm{Top}(X') \to \mathrm{Top}(X)$ (4.1). Show that $\mathrm{Top}(f)$ is dominant if and only if f is dominant, i.e. $f(X')$ is dense in X.

c) Show that if $f : E' \to E$ is a conservative geometric morphism (6.4.0), i.e. if f^* is faithful, then f is dominant. Show that the converse is not necessarily true, even for a localization morphism $E_{/U} \to E$, where U is an open of the topos E. In this case, show that f is conservative only if U is the final object of E, hence only if f is an equivalence of topoi, and use the example b).

d) Let $f : X' \to X$ be an arrow in a topos E. One says that f is a *dominant morphism* in the topos E if the corresponding morphism of induced topoi $E_{/X'} \to E_{/X}$ (5.5.2) is dominant. Show that this property depends only on the image $f(X')$ of X' in X, as an element of the ordered set $\mathrm{Open}(X)$ of subobjects of X. Show that the morphism $E_{/X'} \to E_{/X}$ is conservative if and only if f is covering.

Continuation anchor. The next untranslated point is SGA 4, Exposé IV, Section 9, “Subtopoi and gluing of topoi” (French source line 5254 of `04/04.tex`).

SGA 4, Exposé IV. Topoi

Section 9. Subtopoi and gluing of topoi

A. Grothendieck and J.-L. Verdier

9. Subtopoi and gluing of topoi

9.1. Subtopoi and embeddings of topoi

Definition 9.1.1.

a) A morphism of topoi $f : F \to E$ is called an *embedding* if the functor $f_* : F \to E$ is fully faithful, or equivalently if the adjunction morphism $f^* f_* \to \mathrm{id}_F$ is an isomorphism.

b) A *subtopos* of a topos E is a strictly full subcategory E' of E which is a topos and for which the inclusion functor $\alpha : E' \to E$ is of the form i_*, where $i : E' \to E$ is a morphism of topoi; equivalently, α has a left adjoint i^* which is left exact. The morphism i, determined up to unique isomorphism, is then called the *inclusion morphism* of the subtopos E' into E.

Remark 9.1.2.

a) The inclusion morphism of a subtopos is an embedding. A strictly full subcategory $E' \subset E$ is a subtopos if and only if it is the essential image of the direct-image functor f_* attached to a morphism of topoi $f : F \to E$ which is an embedding.

b) A morphism $f : F \to E$ of topoi is an embedding if and only if it is isomorphic to a composite

$$F \xrightarrow{g} E' \xrightarrow{i} E,$$

where g is an equivalence of topoi and i is the inclusion of a subtopos E' of E. The subtopos E', namely the essential image of f_*, is uniquely determined, and the factorization is unique up to unique isomorphism. In practice one identifies F, by means of g, with the corresponding subtopos of E.

c) An equivalence of topoi $u : E \xrightarrow{\sim} E_1$ gives, in the evident way, a bijection between the subtopoi of E and the subtopoi of E_1. Thus for the study of subtopoi one may reduce to the case $E = \widetilde{C}$, where C is a small site. In that case a strictly full subcategory $E' \subset E$ is a subtopos if and only if its inclusion has a left exact left adjoint. Moreover the subtopoi so obtained correspond bijectively to the topologies T' on C finer than the given topology T: to T' one associates the strictly full subcategory $(C, T')^\sim$ of $C^\sim$ consisting of the sheaves for T'. Consequently, for any $\mathcal{U}$-topos E, a strictly full subcategory $E' \subset E$ is a subtopos if and only if its inclusion has a left exact left adjoint; the ordered set $\mathfrak{E}(E)$ of subtopoi of E is $\mathcal{U}$-small, and every subset of it admits a supremum and an infimum.

d) If E' and E'' are two subtopoi of E, their intersection is again a subtopos. In the case $E = C^\sim$, let T' and T'' be the corresponding topologies on C, both finer than T. Then $E' \cap E''$ is the category of sheaves for the topology T''' which is the supremum of T' and T''. More generally, if T' is the supremum of a family (T_i) of topologies, then the sheaves for T' form the intersection of the corresponding categories of sheaves.

Proposition 9.1.3. For every family $(E_i)_{i\in I}$ of subtopoi of a topos E, the intersection

$$\bigcap_{i\in I} E_i$$

is a subtopos of E.

Proposition 9.1.4. Let $i : E' \to E$ be an embedding of topoi, and let F be a topos. The functor

$$f \longmapsto i \circ f : \mathcal{H}om_{\mathrm{top}}(F, E') \longrightarrow \mathcal{H}om_{\mathrm{top}}(F, E) \tag{9.1.4.1}$$

is fully faithful. Its essential image consists of those morphisms $g : F \to E$ for which the essential image of g_* is contained in that of i_*.

Proof. Consider the evident commutative square

$$\begin{tikzcd}
\mathcal{H}om_{\mathrm{top}}(F, E') \arrow[r] \arrow[d, hook] & \mathcal{H}om_{\mathrm{top}}(F, E) \arrow[d, hook] \\
\mathcal{H}om(F, E') \arrow[r] & \mathcal{H}om(F, E).
\end{tikzcd}$$

The vertical functors send a morphism of topoi to its direct image and are fully faithful by definition of $\mathcal{H}om_{\mathrm{top}}$. Since i_* is fully faithful, the lower horizontal functor is fully faithful, and so is the upper one. The characterization of the image follows from $(if)_* = i_*f_*$. Conversely, if $g_* = i_*f_*$ for a functor $f_* : F \to E'$, then f_* has the left adjoint g^*i_*, which is left exact; hence f_* is the direct image of a morphism of topoi $f : F \to E'$.

Corollary 9.1.5. If $f : E' \to E$ is an embedding of topoi, then the induced functor on categories of points

$$\mathrm{Point}(f) : \mathrm{Pt}(E') \longrightarrow \mathrm{Pt}(E) \tag{9.1.5.1}$$

is fully faithful.

Exercise 9.1.6. Embeddings of topoi and 2-monomorphisms. Let $i : E' \to E$ be a morphism of topoi.

a) If F and G are topoi over E', define the canonical functor

$$f \longmapsto f * i : \mathcal{H}om_{\mathrm{top},E'}(F, G) \longrightarrow \mathcal{H}om_{\mathrm{top},E}(F, G). \tag{9.1.6.1}$$

b) Suppose that i, as a 1-arrow of the 2-category of topoi, is a 2-monomorphism, that is, for every topos F belonging to a sufficiently large universe the functor (9.1.4.1) is fully faithful. Show that for every pair of topoi F, G the functor (9.1.6.1) is fully faithful.

c) Show that i is an embedding if and only if i is a 2-monomorphism.

d) Let $g : F \to E$ be a morphism of topoi and assume the 2-fiber product

$$F' = F \times^2_E E'$$

exists. Let $j : F' \to F$ be the projection. If i is an embedding, show that j is an embedding. When E' is a subtopos of E, the essential image of j_* is called the *inverse-image subtopos of* F of E' by g.

e) With the preceding notation, let X be an object of F. Show that if X belongs to the inverse-image subtopos F', then for every object U of F, writing $g_U : F_{/U} \to E$ for the induced morphism and $X|U = (X \times U \to U)$, one has

$$g_{U*}(X|U) \in \operatorname{Ob} E'.$$

Ask conversely whether this condition is sufficient, or equivalently whether the strictly full subcategory so obtained is a subtopos of F.

f) If $X \in E$, put $X' = i^*X$ and let $i_{/X} : E'_{/X'} \to E_{/X}$ be the induced morphism of topoi. Show that if i is an embedding, then $i_{/X}$ is an embedding. Conversely, if $(X_\alpha)_\alpha$ is a covering family of the final object and every $i_{/X_\alpha}$ is an embedding, then i is an embedding.

Exercise 9.1.7. Subtopoi and multiplicative sets of arrows.

a) Let $i : E' \to E$ be a morphism of topoi and let S be the set of arrows u of E such that $i^*(u)$ is an isomorphism. Let

$$\rho : E[S^{-1}] \longrightarrow E' \tag{9.1.7.1}$$

be the canonical functor induced by i^*. Show that i is an embedding if and only if ρ is an equivalence. In this case S admits both a left and a right calculus of fractions. The essential image of i_* consists of the objects $X \in E$ such that, for every $u : Y \to Y'$ in S, the map

$$\operatorname{Hom}(Y', X) \longrightarrow \operatorname{Hom}(Y, X)$$

is bijective.

b) Assume now that i is an embedding. For any topos F, show that the functor

$$f^* \longmapsto f^*i^* : \mathcal{H}om(E', F) \longrightarrow \mathcal{H}om(E, F)$$

induces an equivalence between inverse-image functors arising from morphisms $F \to E'$ and inverse-image functors $g^* : E \to F$ arising from morphisms $F \to E$ which carry every arrow of S to an isomorphism.

c) Let $S \subset \operatorname{Fl}(E)$. Show that S comes from a subtopos by the preceding construction if and only if it satisfies:

ST1) S contains the isomorphisms and is stable under composition and base change.

ST2) If a composite vu lies in S, then $u \in S$ if and only if $v \in S$.

ST3) If $u : X \to Y$ becomes an element of S after base change by a covering family $Y_i \to Y$, then $u \in S$.

This gives a bijection between subtopoi of E and subsets $S \subset \operatorname{Fl}(E)$ satisfying ST1–ST3. Given S, associate to it the topology T' finer than the canonical topology for which $(X_i \to X)$ is covering precisely when the image of $\coprod X_i \to X$ is included by a monomorphism belonging to S.

d) S is determined by its monomorphisms. The assignment $S \mapsto S_0$, where S_0 is the subset of monomorphisms in S, gives a bijection between subsets of $\operatorname{Fl}(E)$ satisfying ST1–ST3 and subsets of monomorphisms satisfying the same conditions.

e) If $(E'_i)_{i\in I}$ is a family of subtopoi and S_i are the corresponding subsets of $\mathrm{Fl}(E)$, then $\bigcap_i S_i$ satisfies ST1–ST3 and corresponds to the subtopos generated by the E'_i, that is, to their supremum.

f) Given $A \subset \mathrm{Fl}(E)$, there is a smallest subset $S \supset A$ satisfying ST1–ST3. It is the union of subsets A_n, $n \geq 0$, obtained inductively by adjoining isomorphisms, composites, base changes, factors of composites whose other factor and composite are already present, small sums, and arrows which become present after an epimorphic base change.

g) With the notation of e), if $A = \bigcup_i S_i$ and S is obtained from A as in f), then the corresponding subtopos is the intersection, or infimum, of the E'_i.

Exercise 9.1.7.2. Canonical factorization of a morphism of topoi.

a) Let $f : F \to E$ be a morphism of topoi. Show that, up to isomorphism, f factors as

$$F \xrightarrow{f'} E' \xrightarrow{i} E, \tag{9.1.7.3}$$

where f' is conservative and i is an embedding; this factorization is unique up to equivalence. Introduce $S_f \subset \mathrm{Fl}(E)$, the arrows u such that f^*u is an isomorphism, and take for E' the subtopos associated with S_f.

b) The subtopos defined by i is called the *image subtopos* of f. It is the smallest subtopos through which f factors, or equivalently the smallest subtopos containing the essential image of f_*. The morphism f is an embedding if and only if it induces an equivalence of F with its image; it is conservative if and only if its image is all of E.

c) Suppose $f = \mathrm{Top}(f_0)$ comes from a continuous map $f_0 : Y \to X$ of sober spaces. Let $Y \to X' = f_0(Y) \to X$ be the usual factorization. The canonical factorization of f is the corresponding factorization

$$\mathrm{Top}(Y) \longrightarrow \mathrm{Top}(X') \hookrightarrow \mathrm{Top}(X).$$

If Z is the smallest sober subspace of X containing $f_0(Y)$, then f is conservative if and only if $Z = X$, that is, if $f_0(Y)$ is very dense in X. Thus conservative morphisms of topoi generalize surjective continuous maps.

d) Give a 2-cartesian square of topoi

$$\begin{tikzcd} F' \arrow[r] \arrow[d] & F \arrow[d, "f"] \\ E' \arrow[r] & E \end{tikzcd}$$

with f conservative but f' not conservative, for instance E' a punctual topos and F' the empty topos.

e) For a family $(E_i)_{i\in I}$ of subtopoi of E, prove that $E = \bigvee_i E_i$ if and only if the family of embedding morphisms $E_i \to E$ is conservative.

Exercise 9.1.8. Subtopoi and points; the case of topological spaces. Let E be a topos.

a) Let Z be a full subcategory of $\mathrm{Pt}(E)$, or the corresponding subset of isomorphism classes of points. Define E_Z to be the intersection of all subtopoi E' of E through which every point of Z factors. Show that E_Z is the smallest such subtopos.

b) If (Z_i) is a family of such full subcategories, show that $E_{\bigcup_i Z_i}$ is the subtopos generated by the E_{Z_i}.

c) If $E' \subset E$, describe the full subcategory of points of E coming from E', and show that $E' \subset E_Z$ exactly when the points of Z factor through E'.

d) Let $E = \mathrm{Top}(X)$ for a sober space X. For a sober subspace $X' \subset X$, the associated subtopos $E_{X'}$ is the essential image of $\mathrm{Top}(X') \to \mathrm{Top}(X)$. Conversely, every subtopos with enough points is of this form.

e) If (X'_i) is a family of sober subspaces of X, there is a smallest sober subspace X' containing them; it is the union when the family is finite or when X is separated. The subtopos $E_{X'}$ is generated by the $E_{X'_i}$.

f) Give an example of a subtopos of a topos of the form $\mathrm{Top}(X)$ which is nonempty but has no points.

Exercise 9.1.9. Topoi defined by ordered sets. Use the dictionary between topoi generated by their open subobjects and ordered sets admitting arbitrary suprema, finite infima, and distributivity of finite infima over arbitrary suprema.

a) If $f : E' \to E$ is an embedding and (X_i) is a generating family of E, then (f^*X_i) is a generating family of E'. Thus a subtopos of a topos generated by its opens is again generated by its opens.

b) Let $f : E' \to E$ be a morphism of such topoi, corresponding to a morphism $g : \mathcal{O} \to \mathcal{O}'$ of ordered sets. Show that f is an embedding if and only if g is surjective and also surjective on the graphs of the order relations; equivalently, the order on $\mathcal{O}'$ is the quotient of the order on $\mathcal{O}$.

c) If $E = \mathcal{O}^\sim$, subtopoi correspond to equivalence relations R on $\mathcal{O}$ compatible with finite infima and arbitrary suprema. Under this correspondence, the subtopos generated by a family corresponds to the intersection of the corresponding equivalence relations.

Exercise 9.1.10. Intersections of subspaces, point-free topoi, and nonexistence of complements. Let X be a topological space and $E = \mathrm{Top}(X) = \mathrm{Open}(X)^\sim$.

a) Define R on $\mathrm{Open}(X)$ by $(U, U') \in R$ if $U \cap U'$ is dense in both U and U'. Show that R satisfies the conditions of 9.1.9 and defines a subtopos E'. A point x of X comes from E' if and only if it is *thick*, that is, belongs to the closure of the interior of $\overline{\{x\}}$. The subtopos E' is empty only when X is empty.

b) If X is sober, $\mathrm{Pt}(E')$ is equivalent to the ordered set of thick points. If the points of X are closed, the thick points are precisely the isolated points; hence a nonempty separated space without isolated points gives a nonempty subtopos without points.

c) If $X' \subset X$ and $E_{X'}$ is as in 9.1.8, then for dense X', $E_{X'}$ contains the preceding E'. If X is sober and X', X'' are sober subspaces, then $X' \cap X''$ is sober and

$$E_{X' \cap X''} \subset E_{X'} \cap E_{X''},$$

and the inclusion may be strict. It is an equality exactly when the right-hand side has enough points; this holds if one of X', X'' is locally closed.

d) Let $X'' = X - X'$. If X'' is a union of locally closed subsets, then the subtopoi disjoint from $E_{X'}$ have a largest element, the *weak complement* of $E_{X'}$, precisely when $E_{X'} \cap E_{X''}$ is empty. If this fails, the infimum in the ordered set of subtopoi is not distributive over arbitrary suprema.

Exercise 9.1.11. Subtopoi of locally noetherian topoi.

a) If $f : E' \to E$ is an embedding and X is a prenoetherian object of E, then f^*X is prenoetherian. Hence if E has a generating family of prenoetherian objects, or is noetherian or locally noetherian, then E' has the same property.

b) If E is locally noetherian, every subtopos of E is of the form E_Z for a suitable full subcategory Z of $\mathrm{Pt}(E)$. In particular, if $E = \mathrm{Top}(X)$ with X sober and locally noetherian, then $Z \mapsto E_Z$ identifies sober subspaces of X with subtopoi of E, and intersections of sober subspaces correspond to intersections of subtopoi.

c) For X sober locally noetherian and $X' \subset X$, with complement X'', the following are equivalent: X' is constructible; both X' and X'' are sober; $E_{X'}$ has a complementary subtopos; and $E_{X'} \cap E_{X''}$ is empty.

d) For X sober noetherian, the following are equivalent: every sober subspace is constructible; it is enough to test $X - \{x\}$ for maximal x; X is finite; every subtopos of $\mathrm{Top}(X)$ has a complement; and infima distribute over arbitrary suprema.

e) If X is locally noetherian, then in the ordered set of subtopoi of $\mathrm{Top}(X)$ finite suprema distribute over binary infima.

f) For every sober subspace Z of a sober locally noetherian X, define a topology T_Z on $\mathrm{Open}(X)$ by declaring $(U_i \to U)$ covering when $U \cap Z = \bigcup_i (U_i \cap Z)$. Then $Z \mapsto T_Z$ is an order-reversing bijection from sober subspaces of X to topologies on $\mathrm{Open}(X)$ finer than the canonical topology.

g) For a sober topological space X, a subset is sober if and only if its complement is a union of locally closed subsets. If X is locally noetherian this is the same as proconstructibility. The constructible topology X_{cons} identifies the ordered set of subtopoi of $\mathrm{Top}(X)$ with the closed subsets of X_{cons}; complemented subtopoi correspond to clopen subsets.

h) If E is locally noetherian and has the property stated in 9.1.14 b), then the essential images of $\mathrm{Pt}(E') \to \mathrm{Pt}(E)$, with E' running over all subtopoi, are the closed subsets of a topology T_{cons} on the set of isomorphism classes of points, and the ordered set of subtopoi is identified with these closed subsets.

Exercise 9.1.12. Finite topoi. A topos E is called *finite* if it is equivalent to $\widehat{C}$ for a finite category C.

a) There are quasi-inverse 2-equivalences between Karoubian categories equivalent to finite categories and finite topoi:
$$C \longmapsto \widehat{C}, \qquad E \longmapsto \mathrm{Pt}(E).$$

b) A finite topos is noetherian.

c) If $E \simeq \widehat{C}$, with C Karoubian and finite up to equivalence, then subtopoi of E correspond to strictly full Karoubian subcategories $C' \subset C$. The subtopos is the essential image of $\widehat{C'} \to \widehat{C}$.

d) The finite set $X = \mathrm{Point}(E)$ of isomorphism classes of points carries a canonical sober topology, for which subtopoi of E correspond to closed subsets of X. Thus the ordered set of subtopoi is finite.

e) For a category C equivalent to a finite category and a full subcategory C', define a topology $T_{C'}$ by declaring $(X_i \to X)$ covering when, for every $Y \in C'$, the maps $\mathrm{Hom}(Y, X_i) \to \mathrm{Hom}(Y, X)$ are jointly surjective. This gives an order-reversing bijection from strictly full Karoubian subcategories of C to topologies on C.

f) If $f : C' \to C$ and C is equivalent to a finite category, then $\widehat{f} : \widehat{C'} \to \widehat{C}$ is conservative if and only if every object of C is a direct factor of an object in the image of f.

g) For the category with objects e, a and arrows $f : a \to e$, $g : e \to a$, $p = gf$, $p^2 = p$, show that $\widehat{C}$ has exactly one proper nonempty subtopos, namely $\widehat{\{e\}}$. Thus any two nonempty subtopoi meet nontrivially.

Exercise 9.1.13. Complementary subtopoi. Two subtopoi E', E'' of E are called *complementary* if
$$E' \cap E'' = \mathrm{Top}(\varnothing), \qquad E' \vee E'' = E.$$
Assume binary infima distribute over finite suprema in the ordered set of subtopoi.

a) If E' and E'' are complementary, then E'' is the largest subtopos disjoint from E'.

b) The following are equivalent: every weak complement is a complement; every proper subtopos has a nonempty disjoint subtopos; every maximal subtopos disjoint from a given one is a complement. These conditions may fail even for finite topoi, but hold for $\mathrm{Top}(X)$ with X locally noetherian.

c) Complemented subtopoi are stable under finite infima, finite suprema, and complements; complementation interchanges infima and suprema.

d) State the dual assertions.

9.1.14. Open questions. Let E be a topos.

a) In the ordered set of subtopoi of E, is the infimum of two elements distributive over finite suprema?

b) If $E' \vee E'' = E$, is every point of E isomorphic to the image of a point of E' or of E''?

c) If E' and E'' are subtopoi of a topos F whose supremum is F, equivalently if

$$E' \amalg E'' \longrightarrow F \tag{9.1.14.1}$$

is conservative, is this morphism universally conservative, that is, still conservative after every 2-base change?

d) In the ordered set Σ of subtopoi of E, is the infimum of two elements distributive over arbitrary suprema?

e) If F is noetherian and $(E_i)_{i \in I}$ is a family of subtopoi with intersection $\mathrm{Top}(\varnothing)$, does a finite subfamily already have empty intersection?

Positive answers to a) and d) were later supplied in the literature cited in the French edition.

9.2. The case of open subtopoi

Let U be an open of E, i.e. a subobject of the final object e. The localization morphism

$$j : E_{/U} \longrightarrow E \tag{9.2.2}$$

is an embedding. An embedding is called an *open embedding* if it is isomorphic to one so obtained; the corresponding subtopos is an *open subtopos*. The open U attached to an open embedding $j : E' \to E$ is uniquely determined as the subobject $j_!(e') \subset e$, where e' is the final object of E'. Thus opens of E are in bijection with open subtopoi of E.

Remark 9.2.3. The open subtopos defined by U is the essential image of $j_* : E_{/U} \to E$. It must not be confused with the essential image of $j_!$, namely the objects whose structural morphism to e factors through U. These two subcategories determine one another and are canonically equivalent, but the former is the subtopos in the sense fixed here.

Proposition 9.2.4. Let $j : \mathcal{U} \to E$ be an open embedding. Then $j_!$, left adjoint to j^*, exists; hence there is a chain of adjoint functors

$$j_!, \quad j^*, \quad j_*.$$

Moreover:

a) $j_!$ and j_* are fully faithful.

b) $j_!$ commutes with fiber products, with products indexed by nonempty small sets, and with projective limits over small cofiltered categories.

c) For every $X \in E$, the adjunction morphism $j_! j^* X \to X$ is a monomorphism.

Proof. For the localization defined by an open U, the functor $j_!$ is the forgetful functor. The assertions follow from the fact that $U \to e$ is a monomorphism and monomorphisms are stable under base change.

Remark 9.2.4.1. There are embeddings $j : E' \to E$ for which $j_!$ exists, but $j_!$ does not commute with binary products or cofiltered projective limits.

Let $g : E' \to E$ be any morphism of topoi and put $U' = g^*U$. The square

$$\begin{tikzcd} E'_{/U'} \arrow[r, "j'"] \arrow[d, "g_U"'] & E' \arrow[d, "g"] \\ E_{/U} \arrow[r, "j"] & E \end{tikzcd}$$

is 2-cartesian. Thus inverse images of open subtopoi are open subtopoi, and open embeddings are stable under 2-base change. Moreover, g factors through $E_{/U}$ if and only if $U' = e_{E'}$.

For two opens U, V of E, the 2-fiber product of $E_{/U}$ and $E_{/V}$ over E is $E_{/U\cap V}$. Hence finite intersections of open subtopoi are open. The assignment $U \mapsto E_{/U}$ also commutes with arbitrary suprema.

9.3. Construction of the closed subtopos complementary to an open subtopos

Let T be a topological space, $U \subset T$ an open, and $Y = T - U$ the closed complement. Then $\mathrm{Top}(Y)$ may be regarded as the subtopos of $\mathrm{Top}(T)$ consisting of objects whose restriction to U is the final object. This description makes sense for any topos E and open $U \subset e$, and it defines the closed subtopos complementary to U.

Let

$$j : \mathcal{U} = E_{/U} \longrightarrow E \tag{9.3.2.1}$$

be the localization. For $X \in E$, define

$$X_{\complement U} = U \coprod_{U \times X} X, \tag{9.3.2.2}$$

where the amalgamation is along the projections $U \times X \to U$ and $U \times X \to X$. Thus

$$\begin{tikzcd} U \times X \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] & X \arrow[d] \\ U \arrow[r] & X_{\complement U} \end{tikzcd} \tag{9.3.2.3}$$

is cocartesian.

Proposition 9.3.3. With the preceding notation:

a) For $X \in E$, the following are equivalent:

   (i) $X \simeq Y_{\complement U}$ for some $Y \in E$;
   (ii) $X \to X_{\complement U}$ is an isomorphism;
   (iii) $X \times U \to U$ is an isomorphism, equivalently j^*X is final in $E_{/U}$.

b) For a morphism $m : X \to X'$, the following are equivalent:

(i') the square

$$\begin{tikzcd} X \arrow[r, "m"] \arrow[d] & X' \arrow[d] \\ X_{\complement U} \arrow[r] & X'_{\complement U} \end{tikzcd}$$

is cartesian;

(ii') $m \times \mathrm{id}_U : X \times U \to X' \times U$ is an isomorphism, equivalently j^*m is an isomorphism.

c) The functor $X \mapsto X_{\complement U}$ is left exact, sends epimorphic families to epimorphic families, and commutes with filtered inductive limits and pushouts.

Proof. First prove the assertion when $E = \widehat{C}$ and reduce, by the usual arguments, to the punctual topos **Set**, where it is immediate. The general case follows by the standard sheafification procedure.

Proposition 9.3.4. Let $\mathcal{F}$ be the strictly full subcategory of E consisting of objects X such that j^*X is final in $\mathcal{U}$. Then $\mathcal{F}$ is a subtopos of E. Its inclusion $i : \mathcal{F} \to E$ has inverse-image functor

$$i^*X = X_{\complement U}.$$

The adjunction morphism $X \to i_*i^*X$ is the canonical morphism $X \to X_{\complement U}$.

Proof. Let $u(X) : X \to X_{\complement U}$. This morphism is functorial in X, and $u(X_{\complement U})$ is an isomorphism. These properties formally give an adjunction $i^* \dashv i_*$, with $i^*(X) = X_{\complement U}$. Since i^* is left exact, the general criterion for a full subcategory to be a topos applies.

The subtopos $\mathcal{F}$ is called the *closed subtopos of E complementary to the open U*. Its inclusion $i : \mathcal{F} \to E$ is a *closed embedding*. Conversely a subtopos is called closed if it is obtained in this way. The complementary open is recovered as $i_*(\varnothing_{\mathcal{F}}) \subset e_E$. For a group object G on E, and for a section $s \in \mathrm{Hom}(e_E, G)$, the *support* of G or s is the closed subtopos complementary to its cosupport.

With $j : \mathcal{U} \to E$ and $i : \mathcal{F} \to E$, we have the two adjoint strings

$$\begin{tikzcd} \mathcal{U} \arrow[r, shift left=2, "j_!"] \arrow[r, shift right=2, "j_*"'] & E \arrow[l, "j^*"'] \arrow[r, shift left, "i^*"] & \mathcal{F}. \arrow[l, shift left, "i_*"] \end{tikzcd} \tag{9.3.6.2}$$

The functor

$$\rho = i^*j_* : \mathcal{U} \longrightarrow \mathcal{F} \tag{9.3.6.3}$$

is called the *gluing functor* relative to U. More generally, a functor $\rho : \mathcal{U} \to \mathcal{F}$ between topoi is called a gluing functor if it is left exact and accessible, equivalently if it admits a pro-adjoint.

9.4. First properties of the closed subtopos $\mathcal{F}$

Proposition 9.4.1. With the preceding notation:

a) $i_* : \mathcal{F} \to E$ sends epimorphic families to epimorphic families and commutes with pushouts and filtered inductive limits.

b) If $X \in E$ and $U \times X \to U$ is an epimorphism, then $X \to i_* i^* X$ is an epimorphism.

c) The pair of functors (i^*, j^*) from E to $\mathcal{F}$ and $\mathcal{U}$ is conservative.

d) The gluing functor $\rho = i^* j_* : \mathcal{U} \to \mathcal{F}$ is left exact and accessible; equivalently it admits a pro-adjoint

$$\sigma : \mathrm{Pro}(\mathcal{F}) \longrightarrow \mathrm{Pro}(\mathcal{U}).$$

Proof. The assertions follow directly from 9.3.3 and the construction of $X_{\complement U}$. For d), j_* is a direct image functor and is accessible, while i^* commutes with inductive limits; the composite is therefore accessible.

Proposition 9.4.2. For any topos E', the functor

$$h \longmapsto i \circ h : \mathrm{Hom}_{\mathrm{top}}(E', \mathcal{F}) \longrightarrow \mathrm{Hom}_{\mathrm{top}}(E', E)$$

is fully faithful. Its essential image consists of those morphisms $g : E' \to E$ such that $g^*(U)$ is an initial object of E'.

Proof. Full faithfulness is 9.1.4. The criterion follows by writing $U' = g^*U$ and considering the square

$$\begin{tikzcd} E'_{/U'} \arrow[r, "j'"] \arrow[d, "g_U"'] & E' \arrow[d, "g"] \\ E_{/U} \arrow[r, "j"] & E. \end{tikzcd}$$

The condition that $g_*(X')$ lie in $\mathcal{F}$ for all X' is equivalent to $g_{U*}(Y')$ being final for every $Y' \in E'_{/U'}$, and by adjunction this is equivalent to U' being initial in E'.

Corollary 9.4.3. Let $g : E' \to E$ be a morphism of topoi, let $U' = g^*U$, and let $\mathcal{F}'$ be the closed subtopos of E' complementary to U'. Then $g \circ i' : \mathcal{F}' \to E$ factors, up to isomorphism, through a morphism $g_{\mathcal{F}} : \mathcal{F}' \to \mathcal{F}$, and the square

$$\begin{tikzcd} \mathcal{F}' \arrow[r, "i'"] \arrow[d, "g_{\mathcal{F}}"'] & E' \arrow[d, "g"] \\ \mathcal{F} \arrow[r, "i"] & E \end{tikzcd} \tag{9.4.3.1}$$

is 2-cartesian.

Thus inverse images of closed subtopoi are closed subtopoi.

Corollary 9.4.4. Let $\mathcal{U}$ be an open subtopos of E and $\mathcal{F}$ its closed complement.

a)

$$\mathcal{U} \cap \mathcal{F} \simeq \mathrm{Top}(\varnothing), \qquad \mathcal{U} \vee \mathcal{F} = E.$$

Thus $\mathcal{U}$ and $\mathcal{F}$ are complementary subtopoi.

b) $\mathcal{U}$ is the largest subtopos E' with $E' \cap \mathcal{F} \simeq \mathrm{Top}(\varnothing)$, and $\mathcal{F}$ is the largest subtopos E' with $\mathcal{U} \cap E' \simeq \mathrm{Top}(\varnothing)$.

Proof. The first assertion reduces to the criteria 9.2.5 and 9.4.2 for a morphism to factor through the open or through the closed complement. The equality $\mathcal{U} \vee \mathcal{F} = E$ is equivalent to conservativity of (i^*, j^*). The maximality statements follow by applying 9.4.3 to a subtopos $E' \subset E$.

Corollary 9.4.5. If $(\mathcal{F}_i)_{i \in I}$ is a family of closed subtopoi of E, and U_i is the complementary open of $\mathcal{F}_i$, then $\bigcap_i \mathcal{F}_i$ is closed and its complementary open is $U = \mathrm{Sup}_i\, U_i$.

Corollary 9.4.6. If the family in 9.4.5 is finite, then the supremum $\bigvee_i \mathcal{F}_i$ is closed and its complementary open is $\bigcup_i U_i$.

Exercise 9.4.7. Gluing subtopoi. Let E' be a subtopos of E, and let $(S_i)_{i \in I}$ be a covering family of the final object of E. Write $E_i = E_{/S_i}$ and E_i' for the inverse-image subtopos of E' in E_i. Show that an object $X \in E$ belongs to E' if and only if every restriction $X|S_i$ belongs to E_i'. Conversely, if subtopoi $E_i' \subset E_i$ are compatible on overlaps $E_{/S_i \times S_j}$, construct a unique subtopos $E' \subset E$ inducing them.

Exercise 9.4.8. Exterior, closure, interior, and boundary. For a subtopos $E' \subset E$, define its exterior as the largest open subtopos disjoint from E', its closure as the closed complement of this exterior, its interior as the largest open subtopos contained in E', and its boundary as the closed complement of the union of the exterior and the interior. Show the expected formal properties, including compatibility with induced topoi. The exterior of E' is empty exactly when the embedding $E' \to E$ is dominant.

Exercise 9.4.9. Locally closed subtopoi. A subtopos F of E is *locally closed* if it is the intersection of an open subtopos and a closed subtopos.

a) Finite intersections of locally closed subtopoi are locally closed, and inverse images of locally closed subtopoi are locally closed.

b) For $E = \mathrm{Top}(X)$, locally closed subsets $X' \subset X$ give locally closed subtopoi $T(X')$; conversely, under sobriety assumptions, every locally closed subtopos comes in this way.

c) A subtopos $F \subset E$ is locally closed if and only if it is open in its closure. This is a local property on E.

d) Among all expressions $F = \mathcal{U} \cap \mathcal{F}$ with $\mathcal{U}$ open and $\mathcal{F}$ closed, there is a canonical one with $\mathcal{U}$ largest and $\mathcal{F}$ smallest, namely $\mathcal{F} = \overline{F}$ and $\mathcal{U}$ the largest open inducing F on $\mathcal{F}$.

Exercise 9.4.10. Develop the notions of constructible, locally constructible, quasi-constructible, and locally quasi-constructible subtopoi on the model of the analogous notions for topological spaces. For $E = \mathrm{Top}(X)$, these notions recover the corresponding classes of subsets of X.

9.5. The gluing theorem

Let $f : A \to B$ be a functor. Denote by (B, A, f) the category whose objects are triples (X, Y, u), with $X \in B$, $Y \in A$, and $u : X \to f(Y)$; a morphism $(X, Y, u) \to (X', Y', u')$ is a pair (m, m'), with $m : X \to X'$, $m' : Y \to Y'$, such that

$$\begin{tikzcd}
X \arrow[r, "u"] \arrow[d, "m"'] & f(Y) \arrow[d, "f(m')"] \\
X' \arrow[r, "u'"] & f(Y')
\end{tikzcd}$$

commutes. This construction is functorial in A, B, f; in particular a natural transformation $f \to f'$ gives a functor $(B, A, f) \to (B, A, f')$, an isomorphism if the transformation is one.

Return to E, an open subtopos $\mathcal{U}$, its closed complement $\mathcal{F}$, and $\rho = i^* j_* : \mathcal{U} \to \mathcal{F}$. To $X \in E$ attach

$$(i^*X,\ j^*X,\ h(X) : i^*X \to \rho j^*X),$$

where $h(X)$ is obtained by applying i^* to $X \to j_* j^* X$. This defines a functor

$$\Phi : E \longrightarrow (\mathcal{F}, \mathcal{U}, \rho). \tag{9.5.3.1}$$

Theorem 9.5.4.

a) Let E be a topos, $\mathcal{U}$ an open subtopos, and $\mathcal{F}$ its closed complement. Then $\rho = i^* j_* : \mathcal{U} \to \mathcal{F}$ is left exact and accessible, and

$$\Phi : E \longrightarrow (\mathcal{F}, \mathcal{U}, \rho)$$

is an equivalence of categories.

b) Conversely, if $\mathcal{U}$, $\mathcal{F}$ are topoi and $\rho : \mathcal{U} \to \mathcal{F}$ is a gluing functor, then $E = (\mathcal{F}, \mathcal{U}, \rho)$ is a topos. The object

$$U = (\varnothing_{\mathcal{F}}, e_{\mathcal{U}}, \varnothing_{\mathcal{F}} \to \rho(e_{\mathcal{U}}))$$

defines an open subtopos equivalent to $\mathcal{U}$, and the functor

$$Y \longmapsto (Y, e_{\mathcal{U}}, Y \to \rho(e_{\mathcal{U}}))$$

identifies $\mathcal{F}$ with the closed complement. The resulting gluing functor is compatible with the given ρ.

Definition 9.5.5. For topoi $\mathcal{U}, \mathcal{F}$ and a gluing functor $\rho : \mathcal{U} \to \mathcal{F}$, the topos $(\mathcal{F}, \mathcal{U}, \rho)$ is called the topos obtained by *gluing* $\mathcal{U}$ and $\mathcal{F}$ by means of ρ.

Proof. Proof of 9.5.4 a). The pair (i^*, j^*) is conservative, so Φ is faithful. For every $X \in E$, the square

$$\begin{tikzcd}
X \arrow[r] \arrow[d] & j_* j^* X \arrow[d] \\
i_* i^* X \arrow[r] & i_* i^* j_* j^* X
\end{tikzcd} \tag{*}$$

is cartesian, by 9.3.3. Given a morphism between $\Phi(X)$ and $\Phi(Y)$, the universal property of this cartesian square produces a unique $X \to Y$, showing full faithfulness. Conversely, an object $(Y, X, u : Y \to i^*j_*X)$ of $(\mathcal{F}, \mathcal{U}, i^*j_*)$ is obtained from the fiber product

$$\begin{tikzcd} W \arrow[r] \arrow[d] & j_*X \arrow[d] \\ i_*Y \arrow[r, "i_*u"] & i_*i^*j_*X. \end{tikzcd}$$

Then $\Phi(W) \simeq (Y, X, u)$, so Φ is essentially surjective.

Proof. Proof of 9.5.4 b). It remains chiefly to see that $(\mathcal{F}, \mathcal{U}, \rho)$ is a topos. Finite projective limits are computed componentwise, since ρ is left exact. Sums are constructed by

$$\coprod_i (Y_i, X_i, u_i) = \left(\coprod_i Y_i,\ \coprod_i X_i,\ \coprod_i Y_i \to \rho(\coprod_i X_i) \right),$$

using the canonical map $\coprod_i \rho(X_i) \to \rho(\coprod_i X_i)$. Equivalence relations are effective and universal by applying the corresponding assertions in $\mathcal{F}$ and $\mathcal{U}$. Finally, accessibility of ρ, or equivalently the existence of a pro-adjoint $\sigma : \mathrm{Pro}(\mathcal{F}) \to \mathrm{Pro}(\mathcal{U})$, supplies a small generating family: take generators C' of $\mathcal{U}$, C'' of $\mathcal{F}$, represent each $\sigma(X'')$ by a small filtered projective system, and use the objects

$$(\varnothing_{\mathcal{F}}, X', \varnothing_{\mathcal{F}} \to \rho X') \quad \text{and} \quad (X'', \sigma(X'')_i, X'' \to \rho(\sigma(X'')_i)).$$

For $E = (\mathcal{F}, \mathcal{U}, \rho)$, the structural functors are explicitly:

$$i^*(X, Y, u) = X, \qquad i_*(X) = (X, e_{\mathcal{U}}, X \to \rho(e_{\mathcal{U}})),$$

$$j^*(X, Y, u) = Y, \qquad j_!(Y) = (\varnothing_{\mathcal{F}}, Y, \varnothing_{\mathcal{F}} \to \rho(Y)),$$

$$j_*(Y) = (\rho(Y), Y, \mathrm{id}_{\rho(Y)}).$$

The adjunction morphisms are the evident morphisms represented by the arrows $u : X \to \rho(Y)$ and by the maps to $\rho(e_{\mathcal{U}})$.

Exercise 9.5.9. Exactness properties of a glued topos. Let $E = (\mathcal{F}, \mathcal{U}, \rho)$.

a) Show that $j_!$ preserves all inductive limits and j^* preserves both all inductive limits and finite projective limits. Show that i^* preserves inductive limits, and i_* preserves filtered inductive limits and pushouts. If ρ preserves a given class of projective limits, then j_* preserves it.

b) Give an example where $j_!$ is not exact on the right for arbitrary embeddings, and one where $j_!$ fails to preserve products.

Exercise 9.5.10. Let $E = (\mathcal{F}, \mathcal{U}, \rho)$, and let A be an object of E corresponding to (B, C, u). Describe the induced topos $E_{/A}$ as a glued topos built from $\mathcal{F}_{/B}$ and $\mathcal{U}_{/C}$, and identify its gluing functor.

Exercise 9.5.11. Morphisms from a glued topos. Let F be a category or a topos.

a) Describe $\mathcal{H}om(F, E)$ as a category glued from $\mathcal{H}om(F, \mathcal{U})$ and $\mathcal{H}om(F, \mathcal{F})$, with gluing functor induced by ρ.

b) A functor $f : F \to E$ commutes with a given class of inductive limits if and only if j^*f and i^*f do; for projective limits also require ρ to commute with them.

c) If F is a topos, then f is an inverse-image functor of a morphism $E \to F$ if and only if j^*f and i^*f are. Hence $\mathcal{H}om_{\mathrm{top}}(E, F)$ is equivalent to the category of triples (f', f'', u), where $f' \in \mathcal{H}om_{\mathrm{top}}(\mathcal{U}, F)$, $f'' \in \mathcal{H}om_{\mathrm{top}}(\mathcal{F}, F)$, and $u : f''^* \to \rho\, f'^*$ is a morphism of functors $F \to \mathcal{F}$.

d) Spell out in what sense this gives a characterization of E, up to equivalence in the 2-category of topoi, by the triple $(\mathcal{U}, \mathcal{F}, \rho)$.

Exercise 9.5.12. Gluing of topological spaces.

a) Let X be a topological space and $\rho : \mathrm{Top}(X) \to \mathbf{Set}$ a gluing functor, represented by a pro-object $A \in \mathrm{Pro}(\mathrm{Top}(X))$. Show that the glued topos is of the form $\mathrm{Top}(Y)$ if and only if A is isomorphic to an object of $\mathrm{Pro}(\mathrm{Open}(X))$. If $X \neq \varnothing$, there are choices of ρ for which the glued topos is not of the form $\mathrm{Top}(Y)$.

b) More generally, for $\rho : \mathrm{Top}(X) \to \mathrm{Top}(Y)$, use the pro-adjoint σ to obtain a map

$$\phi : X \longrightarrow \mathrm{Ob}\,\mathrm{Pro}(\mathrm{Top}(Y)).$$

If the glued topos is $\mathrm{Top}(Z)$, then ϕ takes values in the essential image of $\mathrm{Pro}(\mathrm{Open}(Y))$.

Problem. Does the converse in 9.5.12 b) hold?

9.6. The gluing sheaf

The data of a topos E with an open U is, by 9.5.4, essentially the same as the data of two topoi $\mathcal{U}$, $\mathcal{F}$ and a gluing functor

$$\rho : \mathcal{U} \longrightarrow \mathcal{F}, \tag{9.6.1}$$

or equivalently a pro-adjoint

$$\sigma : \mathrm{Pro}(\mathcal{F}) \longrightarrow \mathrm{Pro}(\mathcal{U}). \tag{9.6.2}$$

The functor σ is determined by its restriction

$$\sigma_\circ : \mathcal{F} \longrightarrow \mathrm{Pro}(\mathcal{U}), \tag{9.6.3}$$

extended to filtered projective limits. Conversely, $\sigma_\circ$ arises from a pro-adjoint exactly when, for every $X \in \mathcal{U}$, the functor

$$Y \longmapsto \mathrm{Hom}_{\mathrm{Pro}(\mathcal{U})}(\sigma_\circ(Y), X)$$

on $\mathcal{F}^\circ$ is representable, equivalently a sheaf for the canonical topology. Thus the opposite functor

$$\sigma_\circ^\circ : \mathcal{F}^\circ \longrightarrow \mathrm{Pro}(\mathcal{U})^\circ \subset \check{\mathcal{U}} = \mathcal{H}om(\mathcal{U}, \mathbf{Set}) \tag{9.6.4}$$

is a sheaf on $\mathcal{F}$ with values in $\mathrm{Pro}(\mathcal{U})^\circ$. The data of a gluing functor is therefore equivalent to such a sheaf. This sheaf is called the *gluing sheaf* attached to ρ, or to the open U of E.

Exercise 9.6.5. Refine 9.5.4 to a statement of 2-equivalence between the 2-category of pairs (E, U), where E is a topos and U an open, and the 2-category of triples $(\mathcal{U}, \mathcal{F}, \rho)$ with ρ a gluing functor. Give the corresponding variant where ρ is replaced by the gluing sheaf ϕ on $\mathcal{F}$ with values in $\mathrm{Pro}(\mathcal{U})^\circ$.

9.7. Points of a glued topos

Let C be a category, and let $D' \to C$, $D'' \to C$ be fully faithful functors with essential images C', C'', such that the objects of C are partitioned by C' and C'', and such that

$$\mathrm{Hom}(X'', X') = \varnothing \qquad (X' \in C',\ X'' \in C'').$$

Then C is reconstructed from C', C'', and the functor

$$(X', X'') \longmapsto \mathrm{Hom}(X', X'') : C'^{\circ} \times C'' \to \mathbf{Set}.$$

This applies to the points of a glued topos.

Proposition 9.7.2. Let E be a topos, U an open, $\mathcal{U} = E_{/U}$, and $\mathcal{F}$ the closed complement. The inclusion morphisms $j : \mathcal{U} \to E$, $i : \mathcal{F} \to E$ induce

$$u : \mathrm{Pt}(\mathcal{U}) \longrightarrow \mathrm{Pt}(E), \qquad v : \mathrm{Pt}(\mathcal{F}) \longrightarrow \mathrm{Pt}(E). \tag{9.7.2.1}$$

Then:

a) u and v are fully faithful, and every point of E belongs to exactly one of their essential images.

b) If p comes from $\mathcal{U}$ and q comes from $\mathcal{F}$, then

$$\mathrm{Hom}(q, p) = \varnothing. \tag{9.7.2.2}$$

c) For $p \in \mathrm{Pt}(\mathcal{U})$ and $q \in \mathrm{Pt}(\mathcal{F})$, there are functorial isomorphisms

$$\mathrm{Hom}(u(p), v(q)) \simeq \mathrm{Hom}(\mathrm{Pro}(\rho)(p), q) \simeq \mathrm{Hom}(p, \sigma(q)), \tag{9.7.2.3}$$

where $\sigma : \mathrm{Pro}(\mathcal{F}) \to \mathrm{Pro}(\mathcal{U})$ is pro-adjoint to ρ.

Proof. Full faithfulness is 9.1.4. A point factors through $\mathcal{U}$ or $\mathcal{F}$ according as p^*U is final or initial in the punctual topos; these are the only two possibilities. For c), use the commutative diagram

$$\begin{tikzcd}
\mathrm{Pt}(\mathcal{U}) \arrow[r, "u"] \arrow[d, hook] & \mathrm{Pt}(E) \arrow[d, hook] & \mathrm{Pt}(\mathcal{F}) \arrow[l, "v"'] \arrow[d, hook] \\
\mathrm{Pro}(\mathcal{U}) \arrow[r, "\mathrm{Pro}(j_*)"] & \mathrm{Pro}(E) & \mathrm{Pro}(\mathcal{F}) \arrow[l, "\mathrm{Pro}(i_*)"']
\end{tikzcd}$$

and the adjunction $\mathrm{Pro}(i^*) \dashv \mathrm{Pro}(i_*)$.

Lemma 9.7.2.4. For a morphism of topoi $f : F \to E$, the diagram

$$\begin{tikzcd} \mathrm{Pt}(F) \arrow[r, "p \mapsto f\circ p"] \arrow[d] & \mathrm{Pt}(E) \arrow[d] \\ \mathrm{Pro}(F) \arrow[r, "\mathrm{Pro}(f_*)"] & \mathrm{Pro}(E) \end{tikzcd} \tag{9.7.2.4.1}$$

commutes up to canonical isomorphism.

Corollary 9.7.3. If E has enough points, then every closed subtopos $\mathcal{F}$ of E has enough points. More precisely, if $(p_i)_{i\in I}$ is a conservative family of points of E, then the subfamily consisting of those p_i which come from $\mathcal{F}$ is conservative on $\mathcal{F}$.

Remark 9.7.4. If E has enough points, an open U is determined by the full subcategory of $\mathrm{Pt}(E)$ consisting of points factoring through $E_{/U}$. Therefore a closed subtopos is also determined by the essential image of its category of points in $\mathrm{Pt}(E)$; it even suffices to know this intersection with any conservative family of points.

Exercise 9.7.5. For a subtopos $F \subset E$, let $P(F) \subset \mathrm{Point}(E)$ be the set of isomorphism classes of points factoring through F. If F is open, closed, or locally closed, then $P(F)$ is respectively open, closed, or locally closed in $\mathrm{Point}(E)$. If E has enough points, this gives bijections between open, closed, locally closed subtopoi of E and open, closed, locally closed subsets of $\mathrm{Point}(E)$. Generalize to any subspace of $\mathrm{Point}(E)$ coming from a conservative family of points.

9.8. Complements on certain topoi attached to topological spaces

The following exercises are meant to accustom the reader to the point of view of topoi in classical-looking situations.

Exercise 9.8.1. Descent of sheaves on a topological space. Let $f : X \to Y$ be a continuous map and $f^* : \mathrm{Top}(Y) \to \mathrm{Top}(X)$.

a) The following are equivalent: f^* is faithful; f^* is conservative; and $f(X)$ is very dense in Y. It is enough that f, or f_{sob}, be surjective. Give an example where f^* is faithful but f_{sob} is not surjective.

b) Let $R = X \times_Y X \rightrightarrows X$. Show that a sheaf on Y is the same as a sheaf on X with descent datum relative to this equivalence relation exactly when f^* is fully faithful. Relate this to very density.

c) Study when the usual descent category is effective for open coverings, for surjective local homeomorphisms, and more generally for maps satisfying suitable local section conditions.

Exercise 9.8.2. Étendues and spaces with operators.

a) Define an *étendue* as a topos locally equivalent to the topos of a topological space. Show that if X is a space with a discrete group G of operators, the topos $\mathrm{Top}(X/G)$ is an étendue.

b) For $x \in X$, identify the monoid of endomorphisms of the corresponding point of $\mathrm{Top}(X, G)$ with the stabilizer group G_x. Thus points of an étendue may have nontrivial automorphism groups.

c) Determine $\mathrm{Pt}(T)$ for the étendue defined by an étale pre-equivalence relation on X, and show that every endomorphism of a point is an automorphism.

d) A topos T is *pointwise rigid*, or simply *rigid*, if between any two of its points there is at most one morphism. Show that for an étale pre-equivalence relation R on X, the quotient étendue $\mathrm{Top}(X)/R$ is rigid if and only if R is an equivalence relation.

e) Let G act on X, let $H \triangleleft G$ act properly and freely, put $X' = X/H$, $G' = G/H$. Show that $\mathrm{Top}(X, G) \to \mathrm{Top}(X', G')$ is an equivalence of topoi. Identify opens of $\mathrm{Top}(X/G)$ with G-stable opens of X, and express connectedness and π_0 in terms of the G-action.

f) If X is locally connected and locally simply connected and $T = \mathrm{Top}(X, G)$ is connected, show that there is a realization $\mathrm{Top}(X', G') \simeq T$ with X' connected and simply connected, unique up to nonunique isomorphism. This is equivalent to choosing a universal covering of the final object of T, and G' is the corresponding fundamental group.

g) Characterize the connected, locally connected, locally simply connected étendues which arise from such spaces with operators. Give an example of a connected simply connected étendue locally isomorphic to $\mathbb{R}$ which is not a topological space, for instance from the universal covering of the line with doubled origin.

h) Define differentiable, real analytic, and complex analytic étendues as ringed topoi locally of the corresponding form. Give their construction by spaces with operators and examples which are not ringed topoi of ordinary spaces.

Exercise 9.8.3. Topoi associated with local equivalence relations and foliations. This exercise, in the French text, was explicitly marked as provisional.

a) For every open $U \subset X$, let $\mathrm{Quot}(U)$ be the set of equivalence relations on U. These form a presheaf, and its associated sheaf is denoted Q_X. A section r of Q_X is a *local equivalence relation* on X. The order by fineness on equivalence relations makes Q_X a sheaf of ordered sets.

b) For a global equivalence relation R, let $\mathrm{loc}(R)$ be the associated section. For a local relation r, define $\mathrm{glob}(r)$ as the supremum of all global equivalence relations whose localization is no finer than r. Describe $\mathrm{glob}(r)$ using local representatives R_x and restrictions to smaller neighborhoods.

c) Define r to be precoherent, coherent, and globally coherent by the conditions $\mathrm{glob}(r) \in E(r)$, stability under restriction, and $r = \mathrm{locglob}(r)$. Define the corresponding notions for global equivalence relations. Show that globally coherent local relations correspond bijectively to coherent global relations.

d) Give local connectedness criteria ensuring coherence, and discuss examples with connected locally connected fibers which are not locally coherent.

e) For a continuous map $f : X' \to X$, define the inverse image of a local equivalence relation. A map is a *fiber map* for r if the inverse image of r is the coarse local equivalence relation. Show that $X' \mapsto \mathrm{Hom}_{\mathrm{fib},r}(X', X)$ is contravariantly functorial.

f) If r is coherent, show that the preceding functor is represented by a space X^r over X. The universal fiber map $X^r \to X$ is bijective and locally a homeomorphism onto its image; connected components of X^r are open and correspond to fibers of $R = \mathrm{glob}(r)$.

g) Define open and strictly open local equivalence relations. If r is open, give sufficient, and in the locally closed-fiber case necessary, conditions for strict openness.

h) For a sheaf F on X, define $Q(F/X)$, whose sections describe equivalence relations on F compatible with equivalence relations on the base. An r-structure on F is a section of $Q(F/X)$ above r. Define the category $\mathrm{Top}(X/r)$ of r-sheaves and the faithful conservative forgetful functor $\mathrm{Top}(X/r) \to \mathrm{Top}(X)$. Prove finite limits, finite colimits, and a small generating family in the appropriate cases.

i) If $f : X \to Y$ is compatible with $R = \mathrm{glob}(r)$, define

$$f_r^* : \mathrm{Top}(Y) \longrightarrow \mathrm{Top}(X/r).$$

If r is globally coherent and strictly open and $f : X \to X/R$ is the quotient map, then f_r^* is an equivalence.

j) If r is strictly open, conclude that $\mathrm{Top}(X/r)$ is a topos and that the forgetful functor is the inverse image of a morphism

$$p : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(X/r).$$

k) Establish functoriality of $\mathrm{Top}(X/r)$ in (X, r). If $X' \to X$ is étale and r' is the inverse image of r, then $\mathrm{Top}(X'/r') \to \mathrm{Top}(X/r)$ is a localization.

l) Show that $\mathrm{Top}(X/r)$ is a rigid étendue. If X is sober, the set of isomorphism classes of points is homeomorphic to $X/\mathrm{glob}(r)$, and there is at most one morphism between two points.

m) Study the canonical morphism $\mathrm{Top}(X) \to \mathrm{Top}(X/r)$; over an object U for which the induced topos is an ordinary space, the corresponding map of spaces has fibers homeomorphic to components of X^r.

n) For a differentiable manifold X with a foliation, define the associated strictly open local equivalence relation, and put a ring on $T = \mathrm{Top}(X/r)$ making it a differentiable étendue. Identify locally free modules on T with locally free modules on X equipped with a connection along the foliation. Give real and complex analytic variants.

Continuation anchor. Exposé IV continues with Section 10, "Sheaves of morphisms", which was redelivered separately in Batch 015 and incorporated again in Batch 016.

SGA 4, Exposé IV. Topoi

Draft translation, Sections 10–14

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Translator's status note. This is a fresh-only working package. It replaces the undownloadable cumulative Batch 015 fresh content and continues through the end of Exposé IV. The translated range corresponds to source lines 8246–9852 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current mathematical English: *geometric morphism* for *morphisme de topos*, *ringed topos*, *locally ringed topos*, *inverse/direct image for modules*, *extension by zero*, *underlying sheaf of sets/abelian groups*, *internal Hom*, and *gluing*. Numbering follows the source.

10 Sheaves of morphisms

Proposition 10.1. Let E be a topos, and let X and Y be two objects of E. The functor

$$Z \longmapsto \mathrm{Hom}_E(Z \times X, Y) \simeq \mathrm{Hom}_{E_{/Z}}(X_Z, Y_Z)$$

is representable.

Indeed, inductive limits are universal in E (II, 4.3). Hence the functor $Z \mapsto Z \times X$ commutes with inductive limits. It follows that $Z \mapsto \mathrm{Hom}_E(Z \times X, Y)$ carries inductive limits in the argument Z into projective limits. It is therefore representable (IV, 1.4 and 1.2, equivalently 1.2.1).

10.2

The object representing the functor $Z \mapsto \mathrm{Hom}_E(Z \times X, Y)$ is denoted

$$\mathscr{H}om_E(X, Y),$$

or simply $\mathscr{H}om(X, Y)$, and is called the *sheaf of morphisms from X to Y*. It is bifunctorial in X and Y. Thus one has a trifunctorial isomorphism

$$\mathrm{Hom}_E(Z, \mathscr{H}om_E(X, Y)) \simeq \mathrm{Hom}_E(Z \times X, Y). \tag{10.2.1}$$

It follows from (10.2.1) that the bifunctor $(X, Y) \mapsto \mathscr{H}om(X, Y)$ sends inductive limits in the argument X (respectively projective limits in the argument Y) to projective limits in E.

Proposition 10.3. Let $v : E \to E'$ be a geometric morphism. For a variable object X of E and a variable object Y of E', there is a bifunctorial isomorphism

$$v_*\mathscr{H}om_E(v^*Y, X) \simeq \mathscr{H}om_{E'}(Y, v_*X). \tag{10.3.1}$$

Proof. For every object Z of E' there is a chain of isomorphisms, functorial in all arguments:

$$\begin{aligned}
\mathrm{Hom}_{E'}(Z, v_*\mathscr{H}om_E(v^*Y, X)) &\simeq \mathrm{Hom}_E(v^*Z, \mathscr{H}om_E(v^*Y, X)) && \text{(adjunction)}\\
&\simeq \mathrm{Hom}_E(v^*Z \times v^*Y, X) && \text{by (10.2.1)}\\
&\simeq \mathrm{Hom}_E(v^*(Z \times Y), X) && \text{since } v^* \text{ is left exact}\\
&\simeq \mathrm{Hom}_{E'}(Z \times Y, v_*X) && \text{(adjunction)}\\
&\simeq \mathrm{Hom}_{E'}(Z, \mathscr{H}om_{E'}(Y, v_*X)) && \text{by (10.2.1).}
\end{aligned}$$

Thus the two sides of (10.3.1) represent isomorphic functors.

Corollary 10.4. Let C be a $\mathcal{U}$-site, X a $\mathcal{U}$-presheaf on C, Y a $\mathcal{U}$-sheaf on C, and $\mathcal{V}$ a universe containing $\mathcal{U}$ such that C is $\mathcal{V}$-small. Let $\widehat{C}_{\mathcal{V}}$ be the topos of $\mathcal{V}$-presheaves on C, and let $C^{\sim}_{\mathcal{U}}$ be the topos of $\mathcal{U}$-sheaves on C. Then the presheaf

$$\mathscr{H}om_{\widehat{C}_{\mathcal{V}}}(X, Y)$$

is a $\mathcal{U}$-sheaf. If $X \to \underline{a}X$ is the canonical morphism from X to its associated sheaf, the corresponding morphism

$$\mathscr{H}om_{\widehat{C}_{\mathcal{V}}}(\underline{a}X, Y) \longrightarrow \mathscr{H}om_{\widehat{C}_{\mathcal{V}}}(X, Y)$$

is an isomorphism. Moreover there is a canonical isomorphism

$$\mathscr{H}om_{\widehat{C}_{\mathcal{V}}}(\underline{a}X, Y) \simeq \mathscr{H}om_{C^{\sim}_{\mathcal{U}}}(\underline{a}X, Y).$$

10.4.1

First suppose that C is a small site and that $\mathcal{V} = \mathcal{U}$. There is then a geometric morphism

$$C^{\sim}_{\mathcal{U}} \longrightarrow \widehat{C}_{\mathcal{U}}$$

(IV, 4.8), and (10.3.1) gives the assertions in this case. To pass from this to the general case, first observe that the associated-sheaf functor is independent of the universe (II, 3.6), and that the topos of $\mathcal{V}$-sheaves on C is equivalent to the topos of $\mathcal{V}$-sheaves on $C^{\sim}_{\mathcal{U}}$ (III, 4 and IV, 1). It remains only to prove the following lemma.

Lemma 10.4.2. Let E be a $\mathcal{U}$-topos, let $\mathcal{V}$ be a universe containing $\mathcal{U}$, let $E^{\sim}_{\mathcal{V}}$ be the topos of $\mathcal{V}$-sheaves on E, and let X, Y be two objects of E. There is a canonical isomorphism

$$\mathscr{H}om_E(X, Y) \simeq \mathscr{H}om_{E^{\sim}_{\mathcal{V}}}(X, Y).$$

10.4.3

The functor $\epsilon_E : E \to E^{\sim}_{\mathcal{V}}$ is fully faithful and left exact. Therefore, for every object Z of E, there is an isomorphism

$$\mathrm{Hom}_{E^{\sim}_{\mathcal{V}}}(\epsilon_E Z, \epsilon_E \mathscr{H}om_E(X, Y)) \simeq \mathrm{Hom}_{E^{\sim}_{\mathcal{V}}}(\epsilon_E Z \times \epsilon_E X, \epsilon_E Y).$$

But every object of $E^{\sim}_{\mathcal{V}}$ is an inductive limit of objects coming from E; by (10.2.1) this proves the assertion.

10.5

Let $v : E \to E'$ be a geometric morphism. We shall define four bifunctorial morphisms:

$$\begin{aligned}
&\Phi_v : v^* \mathscr{H}om(X, Y) \longrightarrow \mathscr{H}om(v^* X, v^* Y), && X, Y \in \mathrm{ob}\, E',\\
&\Psi_v : v_* \mathscr{H}om(X, Y) \longrightarrow \mathscr{H}om(v_* X, v_* Y), && X, Y \in \mathrm{ob}\, E,\\
&\Xi_v : \mathscr{H}om(v_! X, Y) \longrightarrow v_* \mathscr{H}om(X, v^* Y), && X \in \mathrm{ob}\, E,\ Y \in \mathrm{ob}\, E',\\
&\Lambda_v : v_!(X \times v^* Y) \longrightarrow v_!(X) \times Y, && X \in \mathrm{ob}\, E,\ Y \in \mathrm{ob}\, E'.
\end{aligned}$$

In the definitions of Ξ_v and Λ_v one assumes that the inverse-image functor v^* has a left adjoint $v_!$.

10.5.1. Definition of Ψ_v

Let X be an object of E. The adjunction morphism $v^*v_*X \to X$ gives, for every object Z of E', a bifunctorial morphism

$$v^*(Z \times v_*X) \longrightarrow v^*Z \times X.$$

Thus, for every object Y of E, one has a trifunctorial morphism

$$\mathrm{Hom}_E(v^*Z \times X, Y) \longrightarrow \mathrm{Hom}_E(v^*(Z \times v_*X), Y).$$

By adjunction this gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z, v_*\mathscr{H}om(X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, \mathscr{H}om(v_*X, v_*Y)),$$

and hence the morphism Ψ_v.

10.5.2. Definition of Λ_v

For every object X of E there is an adjunction morphism $X \to v^*v_!X$. Therefore, for every object Y of E', one obtains a bifunctorial morphism

$$X \times v^*Y \longrightarrow v^*v_!(X) \times v^*Y \simeq v^*(v_!(X) \times Y),$$

and hence, by adjunction, the morphism Λ_v.

10.5.3. Definition of Ξ_v

Let X be an object of E, and let Z, Y be objects of E'. The morphism

$$\Lambda_v : v_!(X \times v^*Z) \to v_!(X) \times Z$$

gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(v_!(X) \times Z, Y) \longrightarrow \mathrm{Hom}_{E'}(v_!(X \times v^*Z), Y).$$

By adjunction this gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z, \mathscr{H}om(v_!X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, v_*\mathscr{H}om(X, v^*Y)),$$

and hence the morphism Ξ_v.

10.5.4. Definition of Φ_v

The trifunctional morphism of 10.5.3,

$$\mathrm{Hom}_{E'}(v_!Z \times X, Y) \longrightarrow \mathrm{Hom}_{E'}(v_!(Z \times v^*X), Y),$$

where Z is an object of E and X, Y are objects of E', gives by adjunction a trifunctional morphism

$$\mathrm{Hom}_E(Z, v^*\mathscr{H}om(X, Y)) \longrightarrow \mathrm{Hom}_E(Z, \mathscr{H}om(v^*X, v^*Y)),$$

and hence one definition of Φ_v.

There is a second way to define Φ_v. Let X, Y, Z be three objects of E'. The functor $v^* : E' \to E$ gives a trifunctional morphism

$$\mathrm{Hom}_{E'}(Z \times X, Y) \longrightarrow \mathrm{Hom}_E(v^*Z \times v^*X, v^*Y).$$

By adjunction this yields

$$\mathrm{Hom}_{E'}(Z, \mathscr{H}om(X,Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, v_*\mathscr{H}om(v^*X, v^*Y)).$$

Thus one obtains a bifunctorial morphism

$$\mathscr{H}om(X,Y) \longrightarrow v_*\mathscr{H}om(v^*X, v^*Y),$$

and, again by adjunction, a morphism which the reader may verify is the same as the morphism Φ_v defined above.

Proposition 10.6. Let $v : E \to E'$ be a geometric morphism.

1) The morphism Ψ_v of 10.5.1 is an isomorphism for every object Y of E if and only if the adjunction morphism $v^*v_*X \to X$ is an isomorphism. In particular, Ψ_v is an isomorphism for every pair (X, Y) if and only if $v_* : E \to E'$ is fully faithful, i.e. if and only if v is an embedding of topoi.

2) The following conditions are equivalent:

   (i) for every pair (X, Y) of objects of E', the morphism Φ_v of 10.5.4 is an isomorphism;

   (ii) the functor v^* has a left adjoint $v_!$ and, for every object X of E and every object Y of E', the morphism Ξ_v of 10.5.3 is an isomorphism;

   (iii) the functor v^* has a left adjoint $v_!$ and, for every object X of E and every object Y of E', the morphism Λ_v of 10.5.2 is an isomorphism.

10.6.1

The first assertion follows immediately from 10.5.1. It also follows immediately from 10.5.2, 10.5.3, and 10.5.4 that (iii) is equivalent to (ii), and that these are equivalent to (i) together with the existence of a left adjoint $v_!$ to v^*. It remains to prove that (i) implies that v^* has a left adjoint, i.e. by IV, 1.8, that (i) implies that v^* commutes with small products. Let e' be the final object of E' and let I be a small set. Since v^* is left exact, $v^*(e')$ is final in E; since v^* commutes with inductive limits, $v^*(\coprod_I e') \simeq \coprod_I v^*(e')$. For every object Y of E' one has

$$\mathscr{H}om(\coprod_I e', Y) \simeq \prod_I \mathscr{H}om(e', Y) \simeq \prod_I Y$$

and

$$\mathscr{H}om(\coprod_I v^*(e'), v^*Y) \simeq \prod_I \mathscr{H}om(v^*(e'), v^*Y) \simeq \prod_I v^*Y.$$

The functorial morphism Φ_v therefore induces an isomorphism

$$v^* \prod_I Y \simeq \prod_I v^*Y,$$

and one checks that this is the canonical isomorphism.

Corollary 10.7. Let E be a topos, let Z be an object of E, let $j_Z : E_{/Z} \to E$ be the localization morphism, and let X, Y be two objects of E.

1) There is a bifunctorial isomorphism

$$j_Z^* \mathscr{H}om(X, Y) \simeq \mathscr{H}om(j_Z^* X, j_Z^* Y).$$

2) If X' is an object of $E_{/Z}$, there is a bifunctorial isomorphism

$$\mathscr{H}om_E(j_{Z!} X', Y) \simeq j_{Z*} \mathscr{H}om_{E_{/Z}}(X', j_Z^* Y).$$

3) If Y' is an object of $E_{/Z}$ and Z is an open of E, then there is a bifunctorial isomorphism

$$j_{Z*} \mathscr{H}om(X', Y') \simeq \mathscr{H}om(j_{Z*} X', j_{Z*} Y').$$

Assertions 1) and 2) are proved by observing that Λ_{j_Z} is an isomorphism. For 3) it is enough to observe that j_{Z*} is fully faithful, because $j_{Z!}$ is fully faithful (I, 5.7 a)).

Corollary 10.8. Let E be a topos, let X, Y, Z be three objects of E, and let $j_Z : E_{/Z} \to E$ be the localization morphism.

1) There is a canonical isomorphism

$$j_{Z*} j_Z^* Y \simeq \mathscr{H}om(Z, Y).$$

2) There is a canonical isomorphism

$$\mathrm{Hom}_E(Z, \mathscr{H}om(X, Y)) \simeq \mathrm{Hom}_{E_{/Z}}(j_Z^* X, j_Z^* Y).$$

Let e_Z be the final object of $E_{/Z}$, i.e. the object $\mathrm{id}_Z : Z \to Z$. From (10.2.1) one has a canonical isomorphism

$$\mathscr{H}om_{E_{/Z}}(e_Z, j_Z^* Y) \simeq j_Z^* Y.$$

By Corollary 10.7, 2), this gives an isomorphism

$$j_{Z*} j_Z^* Y \simeq \mathscr{H}om(j_{Z!} e_Z, Y).$$

Since $j_{Z!} e_Z = Z$, this proves 1). To prove 2), Corollary 10.7 gives a canonical isomorphism

$$\mathrm{Hom}_{E_{/Z}}(e_Z, j_Z^* \mathscr{H}om(X, Y)) \simeq \mathrm{Hom}_{E_{/Z}}(j_Z^* X, j_Z^* Y).$$

Adjunction on the first member then gives the asserted isomorphism.

11 Ringed topoi, localization in ringed topoi

11.1.1

Let $\mathcal{U}$ be a universe. A *$\mathcal{U}$-ringed topos* is a pair (E, A), where E is a $\mathcal{U}$-topos and A is an object endowed with a ring structure. A *$\mathcal{U}$-ringed site* is a pair (C, A), where C is a $\mathcal{U}$-site and A is a $\mathcal{U}$-sheaf of rings on C. The universe will not be mentioned when there is no risk of ambiguity. A ringed site (C, A) has associated ringed topos $(C^\sim, A)$. A ringed topos (E, A) has associated ringed site consisting of the site E and the sheaf of rings represented by A.

11.1.2

Let (E, A) be a ringed topos. Write ${}_AE$ (respectively E_A) for the category of sheaves of unital left (respectively right) A-modules; we shall also write "A-modules". The category ${}_AE$ (respectively E_A) is abelian (II, 6.7). If M and N are two sheaves of left (respectively right) A-modules, the abelian group of A-linear morphisms from M to N is denoted $\mathrm{Hom}_A(M, N)$.

11.1.3

Let A, B, C be three rings in a topos E, let M be a sheaf of A-B-bimodules, and let N be a sheaf of A-C-bimodules on the left. Let e be a final object of E and put $\mathrm{Hom}(e, B) = \Gamma(B)$ and $\mathrm{Hom}(e, C) = \Gamma(C)$. The A-B-bimodule structure on M gives a ring homomorphism from $\Gamma(B)^\circ$ to the endomorphism ring of the A-module M, and similarly the A-C-bimodule structure on N gives a ring homomorphism from $\Gamma(C)^\circ$ to the endomorphism ring of the A-module N. By functoriality one obtains a $\Gamma(B)$-$\Gamma(C)$-bimodule structure on the abelian group $\mathrm{Hom}_A(M, N)$. In particular, when A is a sheaf of commutative rings, $\mathrm{Hom}_A(M, N)$ is canonically a $\Gamma(A)$-module.

11.1.4

Let E be a topos, let A and B be two rings in E, let $u : A \to B$ be a morphism of sheaves of rings, and let M be a B-module, say on the left. The sheaf M may be regarded as a sheaf of A-modules via u: for every object X of E, the $A(X)$-module structure on $M(X)$ is deduced from its $B(X)$-module structure and from $u(X) : A(X) \to B(X)$ by restriction of scalars. Thus one obtains a functor

$$\mathrm{Res}(u) : {}_BE \longrightarrow {}_AE,$$

called *restriction of scalars by u*.

11.1.5

In particular, when $A = \mathbb{Z}$ is the constant sheaf $\underline{\mathbb{Z}}$ (the sheaf associated with the constant presheaf $\mathbb{Z}$), and when $u : \underline{\mathbb{Z}} \to B$ is the unique canonical morphism, one obtains a functor from ${}_BE$ to the category ${}_{\underline{\mathbb{Z}}}E$, which is none other than the category of sheaves of abelian groups, denoted $E_{\mathbf{Ab}}$. This functor is called the *underlying abelian sheaf functor*.

Proposition 11.1.6. Restriction of scalars by u commutes with inductive and projective limits. It is conservative.

The proposition is true for the ordinary restriction-of-scalars functor for modules, i.e. when E is the punctual topos. It is therefore true when E is the topos of presheaves on a small site. The description of inductive and projective limits by means of the associated-sheaf functor (II, 6.4) then implies the proposition in the general case.

11.2.1

Let (E, A) be a ringed topos, let X be an object of E, and let $j_X : E_{/X} \to E$ be the localization morphism. By III, 1.7, or IV, 3.1.2, the sheaf j_X^*A has a canonical ring structure. It is usually denoted $A|X$, or abusively again by A. Unless otherwise stated, the topos $E_{/X}$ will be ringed by $A|X$. The sheaf $j_{X*}j_X^*A$ has a canonical ring structure, and the adjunction morphism

$$A \longrightarrow j_{X*}j_X^*A$$

is a morphism of sheaves of rings.

11.2.2

Let in addition M be an A-module, say on the left. The sheaf j_X^*M has an $A|X$-module structure (III, 1.7, or IV, 3.1.2). This gives a functor

$$j_X^* : {}_A E \longrightarrow {}_{A|X}E_{/X},$$

called restriction to $E_{/X}$, or simply restriction to X. The functor j_X^* commutes with inductive and projective limits, hence in particular is exact. If N is an $A|X$-module, then $j_{X*}N$ is a sheaf of $j_{X*}(A|X)$-modules (III, 1.7 or IV, 3.1.2), and therefore, by restriction of scalars through the adjunction morphism $A \to j_{X*}(A|X)$ (11.1.4), it is an A-module, again denoted $j_{X*}N$. We have thus defined a functor

$$j_{X*} : {}_{A|X}E_{/X} \longrightarrow {}_A E,$$

which commutes with projective limits (11.1.6 and III, 1.7). For every A-module M, the adjunction morphism $M \to j_{X*}j_X^*M$ is a morphism of A-modules. For every $A|X$-module N and every A-module M, this adjunction morphism defines a morphism, bifunctorial in M and N,

$$\mathrm{Hom}_{A|X}(j_X^*M, N) \longrightarrow \mathrm{Hom}_A(M, j_{X*}N). \tag{11.2.2.1}$$

This morphism is an isomorphism; equivalently, the functors j_X^* and j_{X*} for A-modules are adjoint (III, 1.7).

11.2.3

The functors j_X^* and j_{X*} for modules commute with the underlying-set functor (III, 1.7, 4). They therefore commute with the restriction-of-scalars functors, and in particular with the underlying abelian sheaf functor.

Proposition 11.3.1. Let (E, A) be a ringed topos and let X be an object of E. The functor

$$j_X^* : {}_A E \longrightarrow {}_{A|X}E_{/X}$$

has a left adjoint

$$j_{X!} : {}_{A|X}E_{/X} \longrightarrow {}_A E,$$

called *extension by zero*. The extension-by-zero functor is exact and faithful and commutes with inductive limits. The functors $j_{X!}$ for modules commute with restriction of scalars and, in particular, with the underlying abelian sheaf functor.

The existence of $j_{X!}$ follows from III, 1.7. Since $j_{X!}$ is a left adjoint, it commutes with inductive limits (I, 2). To prove the other assertions, first suppose that E is the topos of presheaves of sets on a small category C containing the object X. Then $E_{/X}$ is equivalent to the category of presheaves on $C_{/X}$, and, modulo this equivalence, the functor $j_X^* : E \to E_{/X}$ is just composition with the forgetful functor $C_{/X} \to C$ (I, 5.11). It follows immediately from the explicit construction of $j_{X!}$ (I, 5.1) that, for every $A|X$-module N and every object Y of C, one has

$$j_{X!}N(Y) = \bigoplus_{u \in \mathrm{Hom}_C(Y,X)} N(u).$$

This gives the exactness of $j_{X!}$ and the fact that extension by zero commutes with restriction of scalars in this case. In the general case one may suppose that E is the topos of sheaves on a small site C and that X comes from an object of C (IV, 1.2.1). Then $E_{/X}$ is equivalent to the topos of sheaves on $C_{/X}$ with its induced topology (III, 5.4), and the geometric morphism $j_X : E_{/X} \to E$ comes from the forgetful functor $C_{/X} \to C$, which is continuous and cocontinuous (III, 5.2). It follows from III, 1.7 that extension by zero for sheaves is obtained by composing extension by zero for presheaves with the associated-sheaf functor. This proves exactness and commutation with restriction of scalars.

To prove faithfulness, it is equivalent to show that, for every $A|X$-module N, the adjunction morphism

$$\mathrm{ad}_N : N \longrightarrow j_X^* j_{X!} N$$

is a monomorphism. Let $j_{\widehat{X}!}$ denote extension by zero for presheaves and let $\underline{a}$ denote the associated-sheaf functor. There is a commutative diagram

$$\begin{tikzcd}
N \arrow[r, "\mathrm{ad}_N"] \arrow[dr, "\mathrm{ad}_{\widehat{N}}"'] & j_X^* \underline{a}\, j_{\widehat{X}!} N \\
& \underline{a}\, j_X^* j_{\widehat{X}!} N \arrow[u]
\end{tikzcd}$$

where $\mathrm{ad}_{\widehat{N}}$ is the adjunction morphism for presheaves. The morphism $\mathrm{ad}_{\widehat{N}}$ is a monomorphism by what has just been proved, hence so is $\underline{a}(\mathrm{ad}_{\widehat{N}})$. Moreover, since the forgetful functor $C_{/X} \to C$ is continuous and cocontinuous, the canonical morphism $\underline{a}\, j_X^* \to j_X^* \underline{a}$ is an isomorphism (III, 2.3). Hence ad_N is a monomorphism.

Remark 11.3.2. Let N be an $A|X$-module. The underlying sheaf of sets of $j_{X!}N$ is not, in general, isomorphic to the sheaf obtained by extending the underlying sheaf of sets of N by the empty set (III, 5.3 and IV, 5.2). One should therefore take care not to confuse the functor *extension by zero*, denoted

$$j_{X!} : {}_{A|X}E_{/X} \to {}_A E$$

in 11.3.1, with the functor extension by the empty set, also denoted $j_{X!} : E_{/X} \to E$ in III, 5.3 and IV, 5.2. In most cases encountered in practice, the abuse of notation just mentioned causes no confusion. When confusion is nevertheless possible, we shall write $j_{X!}^{\mathrm{ab}}$ and $j_{X!}^{\mathbf{Set}}$ for extension by zero and extension by the empty set, respectively.

Proposition 11.3.3. Let (E, A) be a ringed topos and let X be an object of E. The A-module $j_{X!}(A|X)$, usually denoted A_X or $A_{X,E}$, is the free A-module generated by X (II, 6.5): for every A-module M there is a canonical isomorphism, functorial in M,

$$\mathrm{Hom}_E(X, M) \xleftarrow{\sim} \mathrm{Hom}_A(A_X, M).$$

If $(X_i)_{i\in I}$ is a topologically generating family of E (II, 3.0.1), then $(A_{X_i})_{i\in I}$ is a generating family of the category of A-modules.

Let e_X be the final object of $E_{/X}$, namely $X \xrightarrow{\mathrm{id}} X$. There is an isomorphism

$$\mathrm{Hom}_{A|X}(A|X, j_X^* M) \xrightarrow{\sim} \mathrm{Hom}_{E_{/X}}(e_X, j_X^* M)$$

induced by the unit section $e_X \to A|X$. By adjunction this gives an isomorphism

$$\mathrm{Hom}_A(j_{X!}^{\mathrm{ab}}(A|X), M) \xrightarrow{\sim} \mathrm{Hom}_E(j_{X!}^{\mathbf{Set}}(e_X), M).$$

Since $j_{X!}^{\mathbf{Set}}(e_X) = X$, one obtains the announced isomorphism. The final assertion follows from II, 6.6.

Remark 11.3.4. Let A and B be two sheaves of rings on a topos E, let X be an object of E, and let N be an $A|X$-$B|X$-bimodule. The abelian sheaf obtained by extending N by zero has a canonical A-B-bimodule structure, as follows immediately from its explicit description (11.3.1). In particular the free A-module A_X generated by X is an A-bimodule.

Exercise 11.3.5. Let E be a topos, let X be an object of E, let $x : P \to E$ be a point of E, and let $(x_i)_{i \in I}$ be the family of points of $E_{/X}$ over x (in bijective correspondence with the fiber X_x, by (6.7.2)). Show that for every abelian sheaf M on $E_{/X}$, the fiber $(j_{X!}M)_x$ is canonically isomorphic to $\bigoplus_i M_{x_i}$.

12 Operations on modules

Proposition 12.1. Let (E, A) be a ringed topos, and let M, N be two left (respectively right) A-modules. The functor on E which sends an object X of E to the abelian group

$$\mathrm{Hom}_A(M, \mathscr{H}om_E(X, N))$$

is representable by an abelian sheaf denoted $\mathscr{H}om_A(M, N)$, or sometimes $\mathscr{H}om(M, N)$ when no confusion can result. For every object X of E there is a canonical isomorphism

$$\mathscr{H}om_A(M, N)(X) \simeq \mathrm{Hom}_{A|X}(j_X^* M, j_X^* N). \tag{12.1.1}$$

There is a canonical isomorphism $\mathscr{H}om_E(X, N) = j_{X*} j_X^* N$ (10.8); hence $\mathscr{H}om_E(X, N)$ is functorially in X a left (respectively right) A-module. The functor $X \mapsto \mathscr{H}om_E(X, N)$ transforms inductive limits in X into projective limits of A-modules (10.2 and 11.2.3), and therefore the functor

$$X \longmapsto \mathrm{Hom}_{\mathbb{Z}}(M, \mathscr{H}om_E(X, N))$$

transforms inductive limits in X into projective limits of abelian groups, hence into projective limits of the underlying sets. It is therefore representable (IV, 1.4 and 1.2), and the representing object has an abelian sheaf structure. By definition, for every object X of E there is a canonical isomorphism

$$\mathscr{H}om_A(M, N)(X) = \mathrm{Hom}(X, \mathscr{H}om_A(M, N)) \simeq \mathrm{Hom}_A(M, \mathscr{H}om(X, N)). \tag{12.1.2}$$

The isomorphism (12.1.1) follows from (12.1.2), the isomorphism $\mathscr{H}om(X, N) \simeq j_{X*} j_X^* N$ (10.8), and the adjunction formulas.

12.2

The abelian sheaf $\mathscr{H}om_A(M,N)$ is called the *sheaf of morphisms of A-modules from M to N*. It is bifunctorial in M and N. It follows from its definition and from 10.2 that it sends projective limits in the argument N (respectively inductive limits in the argument M) to projective limits of abelian sheaves. In particular it is left exact in both arguments.

Proposition 12.3. Let (E,A) be a ringed topos, let M,N be two right (respectively left) A-modules, and let X be an object of E.

a) There are canonical isomorphisms

$$\mathscr{H}om_A(M,N)(X) \simeq \mathrm{Hom}_A(M, j_{X*}j_X^*N) \simeq \mathrm{Hom}_{A|X}(j_X^*M, j_X^*N) \simeq \mathrm{Hom}_A(j_{X!}j_X^*M, N).$$

b) There is a canonical isomorphism

$$\Phi_X : j_X^*\mathscr{H}om_A(M,N) \simeq \mathscr{H}om_{A|X}(j_X^*M, j_X^*N).$$

Let moreover P be a right (respectively left) $A|X$-module.

c) There is a canonical isomorphism

$$j_{X*}\mathscr{H}om_{A|X}(j_X^*M, P) \simeq \mathscr{H}om_A(M, j_{X*}P).$$

d) There is a canonical isomorphism

$$\Xi_X : \mathscr{H}om_A(j_{X!}P, N) \simeq j_{X*}\mathscr{H}om_{A|X}(P, j_X^*N).$$

The isomorphisms in a) follow from (12.1.2), from $\mathscr{H}om_E(X,N) \simeq j_{X*}j_X^*N$ (10.8), and from the adjunction formulas.

For b), for every object Y of $E_{/X}$ one has a chain of isomorphisms

$$\begin{aligned}
\mathrm{Hom}(Y, j_X^*\mathscr{H}om_A(M,N)) &\simeq \mathrm{Hom}(j_{X!}Y, \mathscr{H}om_A(M,N)) \\
&\simeq \mathrm{Hom}_A(M, \mathscr{H}om_E(j_{X!}Y, N)) \\
&\simeq \mathrm{Hom}_A(M, j_{X*}\mathscr{H}om_{E_{/X}}(Y, j_X^*N)) \\
&\simeq \mathrm{Hom}_{A|X}(j_X^*M, \mathscr{H}om_{E_{/X}}(Y, j_X^*N)) \\
&\simeq \mathrm{Hom}(Y, \mathscr{H}om_{A|X}(j_X^*M, j_X^*N)).
\end{aligned}$$

For c), for every object Y of E one has

$$\begin{aligned}
\mathrm{Hom}(Y, j_{X*}\mathscr{H}om_{A|X}(j_X^*M, P)) &\simeq \mathrm{Hom}(j_X^*Y, \mathscr{H}om_{A|X}(j_X^*M, P)) \\
&\simeq \mathrm{Hom}_{A|X}(j_X^*M, \mathscr{H}om_{E_{/X}}(j_X^*Y, P)) \\
&\simeq \mathrm{Hom}_A(M, j_{X*}\mathscr{H}om_{E_{/X}}(j_X^*Y, P)) \\
&\simeq \mathrm{Hom}_A(M, \mathscr{H}om_E(Y, j_{X*}P)) \\
&\simeq \mathrm{Hom}(Y, \mathscr{H}om_A(M, j_{X*}P)).
\end{aligned}$$

For d), for every object Y of E one has

$$\begin{aligned}\mathrm{Hom}(Y, \mathscr{H}om_A(j_{X!}P, N)) &\simeq \mathrm{Hom}_A(j_{X!}P, \mathscr{H}om_E(Y, N)) \\ &\simeq \mathrm{Hom}_{A|X}(P, j_X^* \mathscr{H}om_E(Y, N)) \\ &\simeq \mathrm{Hom}_{A|X}(P, \mathscr{H}om_{E_{/X}}(j_X^* Y, j_X^* N)) \\ &\simeq \mathrm{Hom}(j_X^* Y, \mathscr{H}om_{A|X}(P, j_X^* N)) \\ &\simeq \mathrm{Hom}(Y, j_{X*} \mathscr{H}om_{A|X}(P, j_X^* N)).\end{aligned}$$

Corollary 12.4. Let C be a $\mathcal{U}$-site, let A' be a $\mathcal{U}$-presheaf of rings on C, let M be a $\mathcal{U}$-presheaf of left A'-modules, and let N be a $\mathcal{U}$-sheaf of left A'-modules. Let A denote the sheaf associated with A', so that N and $\underline{a}M$ are A-modules. Then the presheaf

$$X \longmapsto \mathrm{Hom}_{A'|X}(j_X^* M, j_X^* N)$$

is a $\mathcal{U}$-sheaf of abelian groups, canonically isomorphic to $\mathscr{H}om_A(\underline{a}M, N)$.

Indeed, by III, 5.5 and III, 2.3, localization to $C_{/X}$ commutes with associated-sheaf functors (for C and for $C_{/X}$). Hence one has isomorphisms, functorial in the variable object X of C,

$$\begin{aligned}\mathrm{Hom}_{A'|X}(j_X^* M, j_X^* N) &\simeq \mathrm{Hom}_{A|X}(\underline{a}\, j_X^* M, j_X^* N) \\ &\simeq \mathrm{Hom}_{A|X}(j_X^* \underline{a} M, j_X^* N) \\ &\simeq \mathscr{H}om_A(\underline{a}M, N)(X).\end{aligned}$$

Corollary 12.5. Let E be a topos, let A, B, C be three sheaves of rings on E, let M be an A-B-bimodule, and let N be an A-C-bimodule. The abelian sheaf $\mathscr{H}om_A(M, N)$ has a canonical B-C-bimodule structure. In particular, when A is a sheaf of commutative rings and when M, N are A-modules, $\mathscr{H}om_A(M, N)$ has a canonical A-module structure. The canonical isomorphisms b), c), d) of 12.3 are isomorphisms of bimodules.

We only give indications. For every object X of E, one has

$$\mathscr{H}om_A(M, N)(X) \simeq \mathrm{Hom}_{A|X}(j_X^* M, j_X^* N).$$

Thus the abelian group $\mathscr{H}om_A(M, N)(X)$ has a canonical $B(X)$-$C(X)$-bimodule structure (11.1.3), which one checks is functorial in X. The other assertions are left to the reader.

Corollary 12.6. Let (E, A) be a ringed topos, let X be an object of E, and let N be an A-module, say on the left. There are canonical isomorphisms of A-modules

$$\mathscr{H}om_E(X, N) \simeq j_{X*} j_X^* N \simeq \mathscr{H}om_A(A_X, N),$$

where the A-module structure on $\mathscr{H}om_A(A_X, N)$ comes from the bimodule structure on A_X (11.3.4).

The first isomorphism follows from 10.7, and the second from the isomorphism of A-modules $N \simeq \mathscr{H}om_A(A, N)$ and from 12.3.

Proposition 12.7. Let (E, A) be a ringed topos, let M be a right A-module and N a left A-module. The functor which sends an abelian sheaf P to the abelian group

$$\mathrm{Hom}_A(M, \mathscr{H}om_{\underline{\mathbb{Z}}}(N, P))$$

is representable by an abelian sheaf denoted $M \otimes_A N$ and called the tensor product of M and N over A.

There is a canonical injection of $\mathrm{Hom}_A(M, \mathscr{H}om_{\underline{\mathbb{Z}}}(N,P))$ into

$$\mathrm{Hom}(M, \mathscr{H}om(N,P)) \simeq \mathrm{Hom}(M \times N, P).$$

Returning to the definitions, one immediately sees that a morphism f of sheaves of sets from $M \times N$ to P comes from an element of $\mathrm{Hom}_A(M, \mathscr{H}om_{\mathbb{Z}}(N,P))$ if and only if, for every object X of E, the map

$$f(X) : M(X) \times N(X) \longrightarrow P(X)$$

is $A(X)$-bilinear. Call such morphisms A-bilinear morphisms. It remains to represent the functor of A-bilinear morphisms from $M \times N$ to P. One proceeds as in the ordinary case, i.e. in the punctual topos. Consider the eight morphisms

$$\begin{aligned}
&M \times M \times N \longrightarrow M \times N && (1 \le i \le 3),\\
&M \times N \times N \longrightarrow M \times N && (4 \le i \le 6),\\
&M \times A \times N \longrightarrow M \times N && (7 \le i \le 8),
\end{aligned}$$

defined by

$$\begin{aligned}
&(1) \quad (m_1, m_2, n) \longmapsto (m_1 + m_2, n), && (2) \quad (m_1, m_2, n) \longmapsto (m_1, n),\\
&(3) \quad (m_1, m_2, n) \longmapsto (m_2, n), && (4) \quad (m, n_1, n_2) \longmapsto (m, n_1 + n_2),\\
&(5) \quad (m, n_1, n_2) \longmapsto (m, n_1), && (6) \quad (m, n_1, n_2) \longmapsto (m, n_2),\\
&(7) \quad (m, a, n) \longmapsto (ma, n), && (8) \quad (m, a, n) \longmapsto (m, an).
\end{aligned}$$

Here the formula (i) describes the map $(i)(X)$ for variable objects X of E. Let L denote the "free abelian sheaf generated by" functor. It is clear that the largest quotient, in the category of abelian sheaves, of $L(M \times N)$ which equalizes $L(1)$ with $L(2) + L(3)$, $L(4)$ with $L(5) + L(6)$, and $L(7)$ with $L(8)$ represents the functor of A-bilinear morphisms from $M \times N$ to P.

12.8

The functor

$$P \longmapsto \mathrm{Hom}_A(N, \mathscr{H}om_{\underline{\mathbb{Z}}}(M,P))$$

is also canonically isomorphic to the functor of A-bilinear maps from $M \times N$ to P. Hence $M \otimes_A N$ also represents the functor $\mathrm{Hom}_A(N, \mathscr{H}om_{\underline{\mathbb{Z}}}(M,P))$. Thus one has isomorphisms, functorial in all arguments,

$$\mathrm{Hom}_{\underline{\mathbb{Z}}}(M \otimes_A N, P) \simeq \mathrm{Hom}_A(M, \mathscr{H}om_{\underline{\mathbb{Z}}}(N,P)), \tag{12.8.1}$$

$$\mathrm{Hom}_{\underline{\mathbb{Z}}}(M \otimes_A N, P) \simeq \mathrm{Hom}_A(N, \mathscr{H}om_{\underline{\mathbb{Z}}}(M,P)). \tag{12.8.2}$$

12.9

It follows from 12.8, or from (12.8.1) and 12.2, that the functor $\otimes_A$ commutes with inductive limits in both arguments.

Proposition 12.10. Let C be a $\mathcal{U}$-site, let A' be a presheaf of rings on C, let M' (respectively N') be a right (respectively left) A'-module, and let A, M, N be the sheaves associated with A', M', N', respectively. The sheaf associated with the presheaf

$$X \longmapsto M'(X) \otimes_{A'(X)} N'(X), \qquad X \in \operatorname{ob} C,$$

is canonically isomorphic to the sheaf $M \otimes_A N$.

The presheaf $X \mapsto M'(X) \otimes_{A'(X)} N'(X)$ may be constructed from the presheaves M', N', and A' by the operations indicated in the proof of 12.7. Since the associated-sheaf functor commutes with finite projective and inductive limits and with the free-abelian-object functor, formation of the tensor product commutes with the associated-sheaf functor.

Proposition 12.11. Let (E, A) be a ringed topos, let M be a right A-module, let N be a left A-module, and let X be an object of E.

a) There is a canonical isomorphism

$$j_X^*(M \otimes_A N) \simeq (j_X^* M) \otimes_{A|X} (j_X^* N).$$

Let moreover P be a right $A|X$-module and Q a left $A|X$-module.

b) There are canonical isomorphisms, the projection formulas,

$$j_{X!}(P \otimes_{A|X} j_X^* N) \simeq (j_{X!} P) \otimes_A N,$$
$$j_{X!}(j_X^* M \otimes_{A|X} Q) \simeq M \otimes_A (j_{X!} Q).$$

For a), for every abelian sheaf R on $E_{/X}$ there is a chain of isomorphisms, functorial in R,

$$\begin{aligned}
\operatorname{Hom}_{\underline{\mathbb{Z}}}(j_X^*(M \otimes_A N), R) &\simeq \operatorname{Hom}_{\underline{\mathbb{Z}}}(M \otimes_A N, j_{X*} R) \\
&\simeq \operatorname{Hom}_A(M, \mathscr{H}\!om_{\underline{\mathbb{Z}}}(N, j_{X*} R)) \\
&\simeq \operatorname{Hom}_A(M, j_{X*} \mathscr{H}\!om_{\underline{\mathbb{Z}}}(j_X^* N, R)) \\
&\simeq \operatorname{Hom}_{A|X}(j_X^* M, \mathscr{H}\!om_{\underline{\mathbb{Z}}}(j_X^* N, R)) \\
&\simeq \operatorname{Hom}_{\underline{\mathbb{Z}}}(j_X^* M \otimes_{A|X} j_X^* N, R).
\end{aligned}$$

For the first isomorphism in b), for every abelian sheaf R there is a functorial chain

$$\begin{aligned}
\operatorname{Hom}_{\underline{\mathbb{Z}}}(j_{X!}(P \otimes_{A|X} j_X^* N), R) &\simeq \operatorname{Hom}_{\underline{\mathbb{Z}}}(P \otimes_{A|X} j_X^* N, j_X^* R) \\
&\simeq \operatorname{Hom}_{A|X}(P, \mathscr{H}\!om_{\underline{\mathbb{Z}}}(j_X^* N, j_X^* R)) \\
&\simeq \operatorname{Hom}_{A|X}(P, j_X^* \mathscr{H}\!om_{\underline{\mathbb{Z}}}(N, R)) \\
&\simeq \operatorname{Hom}_A(j_{X!} P, \mathscr{H}\!om_{\underline{\mathbb{Z}}}(N, R)) \\
&\simeq \operatorname{Hom}_{\underline{\mathbb{Z}}}(j_{X!} P \otimes_A N, R).
\end{aligned}$$

The second isomorphism is obtained in the same way using (12.8.2).

Corollary 12.12. Let E be a topos, let A, B, C be three sheaves of rings on E, let M be a B-A-bimodule, and let N be an A-C-bimodule. The abelian sheaf $M \otimes_A N$ has a canonical B-C-bimodule structure. In particular, when A is a sheaf of commutative rings and M, N

are A-modules, $M \otimes_A N$ has a canonical A-module structure. The isomorphisms of 12.9 are isomorphisms of bimodules.

For every object X of E, $j_X^* M$ is a right $A|X$-module on which the ring $B(X)$ acts on the left, and $j_X^* N$ is a left $A|X$-module on which $C(X)$ acts on the right. By functoriality, the abelian sheaf

$$j_X^* M \otimes_{A|X} j_X^* N \simeq j_X^*(M \otimes_A N)$$

has the structure of a $B(X)$-$C(X)$-object, functorially in X. Hence $M \otimes_A N$ has a B-C-bimodule structure. This $B(X)$-$C(X)$-structure on $j_X^*(M \otimes_A N)$ is reflected in the evident way on the functor of A-bilinear morphisms. One then checks that when A is commutative and M, N are A-modules, the right and left A-module structures so obtained are equal; hence $M \otimes_A N$ is an A-module. The last assertion is left to the reader.

Corollary 12.13. *Let (E, A) be a ringed topos, let M be a left (respectively right) A-module, and let X be an object of E. With the notation A_X of 11.3.3, there is a canonical isomorphism*

$$A_X \otimes_A M \simeq j_{X!} j_X^* M \qquad (\textit{respectively } M \otimes_A A_X \simeq j_{X!} j_X^* M).$$

This follows from the projection formulas (12.11).

Proposition 12.14. *Let E be a topos, let A and B be two sheaves of rings on E, let M be a right A-module, let N be an A-B-bimodule, and let P be a right B-module. There is a canonical isomorphism*

$$\mathrm{Hom}_B(M \otimes_A N, P) \simeq \mathrm{Hom}_A(M, \mathscr{H}om_B(N, P)).$$

From (12.8.1) one obtains a canonical A-bilinear morphism

$$\mathscr{H}om_{\mathbb{Z}}(N, P) \times N \longrightarrow P.$$

Restricting to the subsheaf $\mathscr{H}om_B(N, P) \times N$, this gives an A-bilinear morphism

$$\mathscr{H}om_B(N, P) \times N \longrightarrow P,$$

which is immediately checked to be B-linear in the second factor. Thus one has a canonical morphism of B-modules

$$\mathscr{H}om_B(N, P) \otimes_A N \longrightarrow P,$$

and hence a canonical map, functorial in M,

$$\mathrm{Hom}_A(M, \mathscr{H}om_B(N, P)) \longrightarrow \mathrm{Hom}_B(M \otimes_A N, P). \tag{12.14.1}$$

We prove that this morphism of functors is an isomorphism. Since both sides send inductive limits in M to projective limits, it is enough to prove that (12.14.1) is an isomorphism for M running through a generating family of the category E_A. It is therefore enough to treat the case $M = A_X$, where X is an object of E. The verification is then immediate.

13 Morphisms of ringed topoi

Definition 13.1. Let (E, A) and (E', A') be two ringed topoi. A morphism of ringed topoi

$$u : (E, A) \longrightarrow (E', A')$$

is a pair (m, θ), where $m : E \to E'$ is a geometric morphism and $\theta : m^* A' \to A$ is a morphism of rings.

13.1.1

Since $m^* : E' \to E$ is left adjoint to $m_* : E \to E'$, giving a morphism of ringed topoi $(m, \theta) : (E, A) \to (E', A')$ is equivalent to giving a geometric morphism $m : E \to E'$ and a morphism of sheaves of rings

$$\Theta : A' \longrightarrow m_* A.$$

13.2

With a morphism of ringed topoi $u = (m, \theta) : (E, A) \to (E', A')$ one associates two important functors between the corresponding categories of modules.

13.2.1. The direct-image functor for modules

Let M be a left (respectively right) A-module. The object $m_* M$ has a canonical $m_* A$-module structure; by restriction of scalars along the canonical morphism $\Theta : A' \to m_* A$, it becomes an A'-module, denoted $u_* M$ and called the *direct image of M by u*.

13.2.2. The inverse-image functor for modules

Let N be a left (respectively right) A'-module. The object $m^* N$ of E has a canonical left (respectively right) $m^* A'$-module structure. The left A-module

$$A \otimes_{m^* A'} m^* N$$

(respectively the right A-module $m^* N \otimes_{m^* A'} A$), where A is endowed with the $m^* A'$-module structure defined by $\theta : m^* A' \to A$, is denoted $u^* N$ and is called the *inverse image of the module N by the morphism of ringed topoi u.*

13.2.3

For every A-module M, the underlying sheaf of sets and the underlying abelian sheaf of $u_* M$ are both $m_* M$. Thus the notation u_* is most often also used for the direct image m_* of sheaves of sets. On the other hand, for an A'-module N, the underlying sheaf of sets or abelian sheaf of $u^* N$ is in general neither equal nor isomorphic to $m^* N$ (except when θ is an isomorphism). One must therefore distinguish inverse image for modules from inverse image for sheaves of sets or abelian sheaves. The notation $u^{-1} : E' \to E$ is most often used for the functor m^*, the inverse image for sheaves of sets. The functor u^{-1} is called the *set-theoretic inverse image* by the morphism of ringed topoi u, as opposed to the inverse image u^* "in the sense of modules".

Definition 13.3. Let (C, A) and (C', A') be two $\mathcal{U}$-ringed sites. A morphism of ringed sites

$$u : (C, A) \longrightarrow (C', A')$$

is a pair (m, θ), where m is a morphism from the site C to the site C' (IV, 4.9) and $\theta : m^* A' \to A$ is a morphism of rings.

13.3.1

A morphism of ringed topoi is a morphism of $\mathcal{U}$-ringed sites. A morphism of ringed sites gives rise to a morphism between the corresponding ringed topoi (11.1.1 and 4.9.1).

Proposition 13.4. Let $u : (E, A) \to (E', A')$ be a morphism of ringed topoi.

a) For a variable left (respectively right) A-module M, and for a variable left (respectively right) A'-module N, there are canonical bifunctorial isomorphisms, called adjunction isomorphisms,

$$\operatorname{Hom}_A(u^*N, M) \simeq \operatorname{Hom}_{A'}(N, u_*M), \tag{13.4.1}$$
$$u_*\mathscr{H}om_A(u^*N, M) \simeq \mathscr{H}om_{A'}(N, u_*M). \tag{13.4.2}$$

b) For a variable object X of E', there is a canonical isomorphism, functorial in X,

$$u^*A'_X \longrightarrow A_{u^{-1}(X)}, \tag{13.4.3}$$

where A'_X and $A_{u^{-1}(X)}$ denote the free modules generated by the indicated objects (11.3.3).

c) If A' is commutative and if the canonical morphism $u^{-1}A' \to A$ is central (respectively if the canonical morphism $u^{-1}A' \to A$ is an isomorphism), then, for a variable right A'-module M and a variable left A'-module N, there is a canonical bifunctorial isomorphism

$$u^*M \otimes_A u^*N \simeq u^*(M \otimes_{A'} N), \tag{13.4.4}$$
$$u^*M \otimes_A u^*N \simeq u^{-1}(M \otimes_{A'} N) \qquad (\textit{respectively}). \tag{13.4.5}$$

First there is a canonical isomorphism

$$\operatorname{Hom}_A(u^*N, M) \simeq \operatorname{Hom}_{u^{-1}A'}(u^{-1}N, M)$$

(12.14), and then an isomorphism

$$\operatorname{Hom}_{u^{-1}A'}(u^{-1}N, M) \simeq \operatorname{Hom}_{A'}(N, u_*M)$$

(III, 1.7); this proves (13.4.1). To exhibit (13.4.2), for every object X of E' one has functorial isomorphisms

$$\begin{aligned}
\operatorname{Hom}_{E'}(X, u_*\mathscr{H}om_A(u^*N, M)) &\simeq \operatorname{Hom}_E(u^{-1}X, \mathscr{H}om_A(u^*N, M)) \\
&\simeq \operatorname{Hom}_A(u^*N, \mathscr{H}om_E(u^{-1}X, M)) \\
&\simeq \operatorname{Hom}_{A'}(N, u_*\mathscr{H}om_E(u^{-1}X, M)) \\
&\simeq \operatorname{Hom}_{A'}(N, \mathscr{H}om_{E'}(X, u_*M)) \\
&\simeq \operatorname{Hom}_{E'}(X, \mathscr{H}om_{A'}(N, u_*M)).
\end{aligned}$$

Formula (13.4.5) is obtained from the corresponding adjunction formula and the definition of the tensor product (12.7). Formula (13.4.4) follows from (13.4.5), using commutativity of the tensor product when the base ring is commutative (12.8). Finally, to prove (13.4.3), consider the chain of isomorphisms

$$\begin{aligned}
\operatorname{Hom}_A(u^*A'_X, M) &\simeq \operatorname{Hom}_{A'}(A'_X, u_*M) && \text{by (13.4.1),}\\
&\simeq \operatorname{Hom}_{E'}(X, u_*M) && \text{by (11.3.3),}\\
&\simeq \operatorname{Hom}_E(u^{-1}X, M) && \text{by adjunction,}\\
&\simeq \operatorname{Hom}_A(A_{u^{-1}X}, M) && \text{by (11.3.3).}
\end{aligned}$$

Corollary 13.5. Let (E, A) be a ringed topos, let $x : P \to E$ be a point of E, and let $F \mapsto F_x$ be the associated fiber functor. The fiber functor at x sends the tensor product of A-modules to the ordinary tensor product of A_x-modules.

The tensor product in P is the ordinary tensor product of modules, since P is the punctual topos, namely the category of sets. The assertion follows from (13.4.4).

Corollary 13.6. Let $u : (E, A) \to (E', A')$ be a morphism of ringed topoi. The direct-image functor u_* for right or left modules commutes with projective limits and in particular is left exact. The inverse-image functor u^* for right or left modules commutes with inductive limits and in particular is right exact.

This follows from formula (13.4.1) (I, 2.11).

13.7

Note that the inverse-image functor u^* for modules is not, in general, exact, whereas the set-theoretic inverse-image functor u^{-1} is exact. One can nevertheless assert the exactness of u^* when the canonical morphism $u^{-1}A' \to A$ is flat on the right or on the left (V, 1.8). This is in particular the case when the canonical morphism $u^{-1}A' \to A$ is an isomorphism.

13.8

Let C be a $\mathcal{U}$-site, let A' be a presheaf of rings on C, and let A denote the associated sheaf. The category of presheaves of A'-modules which are sheaves is equivalent to the category of A-modules. One sometimes writes $\operatorname{Hom}_{A'}(M, N)$ for the group $\operatorname{Hom}_A(M, N)$ (11.1.2). Similarly, and abusively, one writes $\mathscr{H}om_{A'}(M, N)$ and $M \otimes_{A'} N$ for the sheaves $\mathscr{H}om_A(M, N)$ and $M \otimes_A N$. When $A' = k_E$ is the constant presheaf defined by an ordinary ring k, one also writes $\mathscr{H}om_k$ and $\otimes_k$ instead of $\mathscr{H}om_{k_E}$ and $\otimes_{k_E}$.

Exercise 13.9. Locally ringed topoi. Let $(E, \mathcal{O}_E)$ be a commutatively ringed topos. For $X \in \operatorname{ob} E$ and $f \in \mathcal{O}_E(X)$, let X_f be the largest subobject of X on which f is invertible.

a) Show that for $X' \in \operatorname{ob} E_{/X}$, the section $f_{X'}$ is invertible if and only if the structural morphism $X' \to X$ factors through X_f, and that for $f, g \in \mathcal{O}_E(X)$ one has

$$X_{fg} = X_f \cap X_g.$$

b) Show that the following conditions (i)–(iii) are equivalent:

(i) for $X \in \operatorname{ob} E$ and $f, g \in \mathcal{O}_E(X)$,

$$X_{f+g} \subset \operatorname{Sup}(X_f, X_g);$$

(ii) for $X \in \operatorname{ob} E$ and $f, g \in \mathcal{O}_E(X)$ such that $f+g$ is invertible, one has $X = \operatorname{Sup}(X_f, X_g)$;

(iii) for $X \in \operatorname{ob} E$ and $f \in \mathcal{O}_E(X)$, one has

$$X = \operatorname{Sup}(X_f, X_{1-f}).$$

(iv) Show that these conditions imply the following condition, and are equivalent to it if E has enough points: for every point p of E, the fiber ring $\mathcal{O}_{E,p}$ is a *local ring*.

When the equivalent conditions (i)–(iii) are satisfied, one says that $(E, \mathcal{O}_E)$ is a *locally ringed topos*; similarly a ringed site is called a *locally ringed site* if the corresponding ringed topos is locally ringed.

c) Let $(E, \mathcal{O}_E)$ and $(E', \mathcal{O}_{E'})$ be two locally ringed topoi, and let f be a morphism of ringed topoi from the first to the second. Show that condition (i) below implies condition (ii), and is equivalent to it if E has enough points:

(i) for every $X' \in \operatorname{ob} E'$ and every $s' \in \mathcal{O}_{E'}(X')$, putting $X = f^{-1}(X')$ and $s = f^*(s')$, one has

$$X'_{s'} = f^{-1}(X_s);$$

(ii) for every point p of E, putting $p' = f(p)$, the natural homomorphism on fibers

$$\mathcal{O}_{E',p'} \longrightarrow \mathcal{O}_{E,p}$$

is a *local* homomorphism of local rings.

When condition (i) is satisfied, one says that f is a *morphism of locally ringed topoi*, or, if confusion is possible, an *admissible morphism of locally ringed topoi.* Denote by $\mathcal{H}om_{\text{top,loc.ann}}(E, E')$ the full subcategory of the category $\mathcal{H}om_{\text{top,ann}}(E, E')$ of all morphisms of ringed topoi from E to E' defined by the admissible morphisms.

d) Suppose that $(E, \mathcal{O}_E)$ is the ringed topos associated with a ringed topological space $(X, \mathcal{O}_X)$ (IV, 2.1). Show that $(E, \mathcal{O}_E)$ is locally ringed if and only if $(X, \mathcal{O}_X)$ is locally ringed, i.e. if and only if for every $x \in X$ the fiber $\mathcal{O}_{X,x}$ is a local ring. In criterion (iv) of b), it is enough to take p in a conservative family of points of E. Let $f : (X, \mathcal{O}_X) \to (X', \mathcal{O}_{X'})$ be a morphism of ringed spaces, where $\mathcal{O}_X$ and $\mathcal{O}_{X'}$ are sheaves of local rings. Show that the corresponding morphism of ringed topoi

$$\operatorname{Top}(X, \mathcal{O}_X) \longrightarrow \operatorname{Top}(X', \mathcal{O}_{X'})$$

is admissible if and only if the same is true of f, i.e. if and only if for every $x \in X$ the homomorphism $\mathcal{O}_{X',f(x)} \to \mathcal{O}_{X,x}$ is a *local* homomorphism of local rings.

e) Let $(E, \mathcal{O}_E)$ and $(E', \mathcal{O}_{E'})$ be two locally ringed topoi, suppose that E has enough points and that $(E', \mathcal{O}_{E'})$ is equivalent to the ringed topos defined by a *scheme* $(X', \mathcal{O}_{X'})$. Prove that the category $\mathcal{H}om_{\text{top,loc.ann}}(E, E')$ is equivalent to a *discrete* category, i.e. it is a rigid groupoid: every morphism is an isomorphism and every automorphism group is trivial.

Hint: reduce to the case where $(E, \mathcal{O}_E)$ is the punctual topos ringed by a field. Recall, by contrast, that $\mathcal{H}om_{\text{top}}(E, E')$ is not in general equivalent to a discrete category, even if E and E' are topoi defined by schemes, and even if E is the punctual topos (cf. 4.2.3).

f) Let E be a locally ringed topos, and let P be the ringed topos defined by the ringed space $\operatorname{Spec} k$, where k is a field. Show that admissible morphisms of locally ringed topoi $P \to E$ correspond to pairs (p, u) consisting of a point p of E and an injection of $k(p)$ into k, where $k(p)$ is the residue field of the local ring $\mathcal{O}_{E,p}$. Generalize this to a statement exhibiting admissible morphisms from a punctual locally ringed topos into E (generalizing EGA I, 2.4.4).

A *geometric point of a locally ringed topos* E means any admissible morphism into E from the locally ringed topos defined by a ringed space of the form $\operatorname{Spec}(k)$, where k is an *algebraically closed* field. Define the *category of geometric points of* E (understood: corresponding to fields $k \in \mathcal{U}$), denoted $\mathrm{Pt}_{\mathrm{geom}}(E)$, and a canonical functor

$$\mathrm{Pt}_{\mathrm{geom}}(E) \longrightarrow \mathrm{Pt}(E).$$

Show that this functor is faithful if and only if E has no point, i.e. if and only if $\mathrm{Pt}(E)$ is the empty category.

g) Let E be a $\mathcal{U}$-topos and let $\mathcal{V}$ be a universe such that $\mathcal{U} \in \mathcal{V}$. Define a 2-category $(\mathcal{V}\text{-}\mathcal{U}\text{-}\mathrm{Top})_{/E}$ in terms of the object E of the 2-category $(\mathcal{V}\text{-}\mathcal{U}\text{-}\mathrm{Top})$, following the model of 5.14 a). When E is ringed by a ring A, define similarly a 2-category $(\mathcal{V}\text{-}\mathcal{U}\text{-}\mathrm{Topan})_{/E}$; and when E is locally ringed, using the notion of admissible morphism, define a 2-category $(\mathcal{V}\text{-}\mathcal{U}\text{-}\mathrm{Topanloc})_{/E}$, admitting $\mathrm{Pt}_{\mathrm{geom}}(E)$ as a full subcategory.

14 Modules on a topos defined by gluing

14.1

Let (E, A) be a ringed topos, let U be an open subtopos of E, let F be its complementary closed subtopos, and let

$$j : U \to E, \qquad i : F \to E$$

be the canonical morphisms (9.3). The topos U is ringed by j^*A (11.2.1), and the morphism i, completed in the evident way, becomes a morphism of ringed topoi. In this section, the topos F will be ringed by the sheaf i^*A, denoted $A|F$, and the morphism j, completed in the evident way, is then a morphism of ringed topoi.

14.2

Thus one has two morphisms of ringed topoi

$$j : (U, A|U) \longrightarrow (E, A), \qquad i : (F, A|F) \longrightarrow (E, A),$$

and hence five functors between the corresponding module categories, written here for left modules:

$$\begin{tikzcd} {}_{A|U}U \arrow[r, shift left=3, "j_!"] \arrow[r, shift right=3, "j_*"'] & {}_{A}E \arrow[l, "j^*"'] \arrow[r, shift left=1.5, "i^*"] & {}_{A|F}F \arrow[l, shift left=1.5, "i_*"] \end{tikzcd} \qquad (14.2.1)$$

Each functor in (14.2.1) is left adjoint to the functor displayed below it.

14.3

We know that the functor

$$X \longmapsto (j^*X, i^*X; i^*X \to i^*j_*j^*X)$$

is an equivalence from E to the gluing topos (U, F, i^*j_*) (9.5.4). This equivalence induces an equivalence between the corresponding module categories. Hence the functor

$$P \longmapsto (j^*P, i^*P; i^*P \to i^*j_*j^*P)$$

is an equivalence, denoted Φ, from ${}_AE$ to the category $({}_{A|U}U, {}_{A|F}F, i^*j_*)$. The functors of (14.2.1), composed with Φ or Φ^{-1}, are then the following functors:

$$\begin{aligned}
\Phi \circ j_* &: M \longmapsto (M, i^*j_*M;\ i^*j_*M \xrightarrow{\mathrm{id}} i^*j_*M),\\
j^* \circ \Phi^{-1} &: (M, N;\ N \to i^*j_*M) \longmapsto M,\\
\Phi \circ j_! &: M \longmapsto (M, 0;\ 0 \to i^*j_*M),\\
\Phi \circ i_* &: N \longmapsto (0, N;\ N \to 0),\\
i^* \circ \Phi^{-1} &: (M, N;\ N \to i^*j_*M) \longmapsto N.
\end{aligned}$$

As an exercise, the reader may spell out the various adjunction morphisms.

14.4

Denote by

$$i^! : {}_AE \longrightarrow {}_{A|F}F \tag{14.4.1}$$

the functor defined by the formula

$$i^! \circ \Phi^{-1}(M, N; N \xrightarrow{u} i^*j_*M) = \mathrm{Ker}(u). \tag{14.4.2}$$

Proposition 14.5. The functor $i^!$ is right adjoint to i_*. The adjunction morphism $i_*i^! \to \mathrm{id}$ is a monomorphism. The functors $i^!$, for variable rings, commute with restriction-of-scalars functors.

It is clear that every morphism

$$(0, N; 0) \longrightarrow (M, N; N \xrightarrow{u} i^*j_*M)$$

factors uniquely through $(0, \mathrm{Ker}(u); 0)$, proving the adjunction property. The other assertions are trivial.

Proposition 14.6. For every object P of ${}_AE$ there are exact sequences, functorial in P,

$$0 \longrightarrow j_!j^*P \longrightarrow P \longrightarrow i_*i^*P \longrightarrow 0, \tag{14.6.1}$$

$$0 \longrightarrow i_*i^!P \longrightarrow P \longrightarrow j_*j^*P, \tag{14.6.2}$$

where the non-trivial arrows are the adjunction morphisms.

First observe that a sequence

$$(M', N'; u') \longrightarrow (M, N; u) \longrightarrow (M'', N''; u'')$$

in $({}_{A|U}U, {}_{A|F}F; i^*j_*)$ is exact if and only if the corresponding sequences

$$M' \to M \to M'', \qquad N' \to N \to N''$$

are exact. The sequences (14.6.1) and (14.6.2) are transformed by Φ into sequences of the form

$$\begin{aligned}
&0 \longrightarrow (M, 0; 0) \longrightarrow (M, N; u) \longrightarrow (0, N; 0) \longrightarrow 0,\\
&0 \longrightarrow (0, \mathrm{Ker}(u); 0) \longrightarrow (M, N; u) \longrightarrow (M, i^*j_*M; \mathrm{id}).
\end{aligned}$$

The exactness of these sequences is immediate.

14.7

The exact sequence (14.6.2) gives a new interpretation of the functor $i_*i^!$. Let X be an object of E. From (14.6.2) one obtains an exact sequence of abelian groups

$$0 \longrightarrow \mathrm{Hom}_E(X, i_*i^!P) \longrightarrow \mathrm{Hom}_E(X, P) \longrightarrow \mathrm{Hom}_E(X, j_*j^*P).$$

Write again U for the open of E, the final object of the open subtopos U. The adjunction properties of j_*, j^*, and $j_!^{\mathbf{Set}}$ imply that $\mathrm{Hom}_E(X, j_*j^*P)$ is canonically isomorphic to $\mathrm{Hom}_E(X \times U, P)$, and that the morphism

$$\mathrm{Hom}_E(X, P) \longrightarrow \mathrm{Hom}_E(X, j_*j^*P)$$

is the morphism defined by the monomorphism $X \times U \to X$ (IV, 5). In other words:

Proposition 14.8. The sections of $i_*i^!P$ on $E_{/X}$ are the sections of P on $E_{/X}$ whose support is contained in F (notation of 9.3.5). Equivalently, $i_*i^!P$ is the largest subsheaf of P with support contained in F.

14.9

By abuse of language, one sometimes says that $i_*i^!P$ is the submodule of P defined by the sections of P with support in F. By 8.5.3, this terminology merely extends to general topoi the terminology used for topoi of sheaves on topological spaces.

Proposition 14.10.

1) The functor $P \mapsto i_*i^*P$ is isomorphic to the functor

$$P \mapsto i_*i^*A \otimes_A P.$$

2) The functor $P \mapsto i_*i^!P$ is isomorphic to the functor

$$P \mapsto \mathscr{H}om_A(i_*i^*A, P).$$

The exact sequence (14.6.1), in the special case where P is the A-module A, reads

$$0 \longrightarrow A_U \longrightarrow A \longrightarrow i_*i^*A \longrightarrow 0, \tag{14.10.1}$$

where A_U is the free A-module generated by the open U corresponding to the open subtopos U (9 and 11.3.3). For every A-module P, the A-modules $j_!j^*P$ and j_*j^*P are canonically isomorphic to

$A_U \otimes_A P$ and $\mathscr{H}om_A(A_U, P)$ respectively (12.3 and 12.6). Moreover, modulo these isomorphisms, the canonical morphisms $j_! j^* P \to P$ and $P \to j_* j^* P$ come from the monomorphism $A_U \to A$. From the exact sequence (14.10.1) one therefore obtains two exact sequences (12.2 and 12.11)

$$\begin{gathered} j_! j^* P \longrightarrow P \longrightarrow i_* i^* A \otimes_A P \longrightarrow 0, \\ 0 \longrightarrow \mathscr{H}om_A(i_* i^* A, P) \longrightarrow P \longrightarrow j_* j^* P. \end{gathered}$$

Comparison with (14.6.1) and (14.6.2) gives the asserted isomorphisms.

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SGA 4, Exposé V. Cohomology in Topoi

Draft translation, Introduction and Section 0

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Translator's status note. This fresh-only continuation starts Exposé V. The translated range corresponds to source lines 57–369 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current mathematical English: *topos*, *ringed topos*, *flat module*, *spectral sequence*, *derived category*, *sheaf cohomology*, *Čech cohomology*, *hypercovering*, *abelian category*, *injective object*, *generator*, *exact filtered inductive limits*, and *cofinal/cogenerating family* in the usual modern sense. Numbering follows the source.

Introduction

In this exposé we present the elementary commutative cohomological invariants of topoi. In no. 1, flat modules and flat morphisms of ringed topoi are studied. The proofs are given using the hypothesis, most often satisfied in practice, that the topoi have enough points (IV, 6). These proofs are taken up again in the general case in Appendix no. 8, where Deligne, using the technique of local inductive limits, also generalizes to the case of topoi D. Lazard's theorem on the structure of flat modules. The theorems of this exposé are existence theorems for spectral sequences relating the various cohomological invariants (nos. 3, 5, 6). It is known that, even for topological spaces, Čech cohomology does not in general coincide with sheaf cohomology [11]. In Appendix no. 7 a modified Čech calculation is introduced which makes it possible to obtain, by means of coverings, sheaf cohomology in an arbitrary topos. In that appendix one is led to use simplicial coverings, or hypercoverings, whose homotopy invariants were studied in [1] (cf. also [17]).

The cohomological invariants introduced here are elementary in the sense that derived categories are not used [12]. The reader familiar with that language will immediately translate the various statements of this exposé and will then be able to generalize them to complexes and to hypercohomology. The language of derived categories is, moreover, used later in this seminar.

We restrict ourselves here to commutative cohomology. For non-commutative H^1, used in this seminar, and for non-commutative H^2, we refer to [9]. The functors that are derived are additive; multiplicative structures are not discussed (cf. [7]).

0 Generalities on abelian categories

In this section we recall a few lemmas, most of which may be found in [11].

Proposition 0.1. Let $\mathfrak{A}$ be an abelian category having a generator. The following conditions are equivalent:

(i) The category $\mathfrak{A}$ satisfies axiom $AB5$): small direct sums are representable, and if $(X_i)_{i\in I}$ is a small increasing filtered family of subobjects of an object X of $\mathfrak{A}$, and if Y is a subobject of X, then

$$\left(\sup_i X_i\right) \cap Y = \sup_i (X_i \cap Y).$$

(ii) Small pseudo-filtered inductive limits *(I, 2.7)* are representable and commute with finite projective limits.

Moreover, if these conditions are satisfied, small filtered inductive limits are universal *(I, 2.6)*.

Proof. It is clear that (ii) implies (i). To show that (i) implies (ii), and to prove the supplementary assertion, it is enough to use the fact that $\mathfrak{A}$ is a full subcategory of a category $\mathcal{M}$ of modules over a suitable ring, in such a way that the inclusion functor

$$u : \mathfrak{A} \longrightarrow \mathcal{M}$$

has an exact left adjoint v [5]. The verification is then trivial.

0.1.1

It is known [11] that an abelian category $\mathfrak{A}$ having a generator and satisfying axiom $AB5$) has enough injectives, i.e. every object embeds into an injective object. Moreover, by the result already cited [5], small products are representable in $\mathfrak{A}$ (axiom $AB3$)*).

Proposition 0.2. Let $\mathfrak{A}$ and $\mathfrak{B}$ be two abelian categories, and let

$$\mathfrak{A} \underset{u}{\overset{v}{\rightleftarrows}} \mathfrak{B}$$

be two adjoint functors, with u left adjoint to v. Consider the two properties:

(i) the functor u is exact;

(ii) the functor v carries injective objects of $\mathfrak{B}$ to injective objects of $\mathfrak{A}$.

1) One always has the implication (i) $\Rightarrow$ (ii). If, moreover, every non-zero object of $\mathfrak{B}$ is the source of a non-zero morphism into an injective object, then (ii) $\Leftrightarrow$ (i).

2) Suppose that:

   a) the category $\mathfrak{B}$ has enough injectives;
   b) one of the two equivalent conditions (i) and (ii) above is satisfied;
   c) the functor u is faithful.

   Then the category $\mathfrak{A}$ has enough injectives.

Proof. The proof is left to the reader.

Remark 0.2.1. The category of abelian groups has enough injectives. Applying Proposition 0.2, one deduces that every category of unital modules over a unital ring has enough injectives. Applying then the result of [5] used in the proof of Proposition 0.1, together with Proposition 0.2, one deduces that every abelian category having a generator and exact small filtered inductive limits has enough injectives. This gives a new proof of that fact.

Proposition 0.3. Let $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$ be three abelian categories, and let

$$u : \mathfrak{A} \longrightarrow \mathfrak{B}, \qquad v : \mathfrak{B} \longrightarrow \mathfrak{C}$$

be two additive left exact functors. Suppose that $\mathfrak{A}$ and $\mathfrak{B}$ have enough injective objects. The following two properties are equivalent:

(i) There exists a spectral functor whose $E_2^{p,q}$-term is

$$R^p v\, R^q u$$

and which abuts to $R^{p+q}(vu)$, suitably filtered.

(ii) The functor u sends injective objects of $\mathfrak{A}$ to objects that are acyclic for the functor v.

Proof. The implication (i) $\Rightarrow$ (ii) is trivial, since it is enough to apply the spectral functor to an injective object. The implication (ii) $\Rightarrow$ (i) is proved in [11].

Proposition 0.4. Let $\mathfrak{A}$ and $\mathfrak{B}$ be two abelian categories, and let

$$u : \mathfrak{A} \longrightarrow \mathfrak{B}$$

be an additive left exact functor. Let M be a subset of the set of objects of $\mathfrak{A}$ having the following properties:

1) Every object of $\mathfrak{A}$ embeds into an element of M.

2) If $X \oplus Y$ belongs to M, then X belongs to M.

3) If

$$0 \longrightarrow X' \longrightarrow X \longrightarrow X'' \longrightarrow 0$$

is an exact sequence and if X' and X belong to M, then X'' belongs to M and the sequence

$$0 \longrightarrow u(X') \longrightarrow u(X) \longrightarrow u(X'') \longrightarrow 0$$

is exact. The zero objects belong to M.

Then every injective object belongs to M, and the objects of M are acyclic for u, i.e. for every $q \neq 0$ and every object X of M one has $R^q u(X) = 0$. In particular, resolutions by objects of M allow one to compute the derived functors of u.

For the proof, see [11], 3.3.1.

Proposition 0.5. Let $\mathcal{U} \subset \mathcal{V}$ be two universes. Let $\mathfrak{A}$ (respectively $\mathfrak{B}$) be a $\mathcal{U}$-category (respectively a $\mathcal{V}$-category) which is abelian, satisfies axiom $AB5$) relative to $\mathcal{U}$ (respectively to $\mathcal{V}$), and has a $\mathcal{U}$-small (respectively $\mathcal{V}$-small) generating family. Let

$$\epsilon : \mathfrak{A} \longrightarrow \mathfrak{B}$$

be an exact fully faithful functor. The following conditions are equivalent:

1) There exists a generating family $(X_i)_{i \in I}$ of $\mathfrak{A}$ such that the family $(\epsilon(X_i))_{i \in I}$ is generating in $\mathfrak{B}$.

1′) Every object of $\mathfrak{B}$ is isomorphic to a quotient of an object of the form

$$\bigoplus_{\alpha \in A} \epsilon(Y_\alpha), \qquad A \in \mathcal{V}.$$

Under these conditions, one has the following property:

2) ϵ sends $\mathcal{U}$-small products to products, and hence commutes with $\mathcal{U}$-small projective limits.

Moreover, under the equivalent conditions 1) or 1′), the following conditions are equivalent:

a) For every object Y of $\mathfrak{A}$, every subobject of $\epsilon(Y)$ is isomorphic to the image under ϵ of a subobject of Y.

a′) There exists a generating family $(X_i)_{i\in I}$ of $\mathfrak{A}$ such that the family $(\epsilon(X_i))_{i\in I}$ is generating in $\mathfrak{B}$, and such that, for every $i \in I$, every subobject of $\epsilon(X_i)$ is isomorphic to the image under ϵ of a subobject of X_i.

b) Every object of $\mathfrak{B}$ is isomorphic to a subobject of an object of the form

$$\prod_{\alpha\in A} \epsilon(Y_\alpha), \qquad A \in \mathcal{V}.$$

c) ϵ commutes with $\mathcal{U}$-small direct sums, and hence commutes with $\mathcal{U}$-small inductive limits.

Finally, under conditions 1) and a′), one has:

d) ϵ sends injective objects to injective objects.

Remark 0.5.1. When in Proposition 0.5 one has $\mathcal{U} = \mathcal{V}$, conditions 1) and a′) imply that ϵ is an equivalence of categories, since one then has b), 2), and a).

0.5.2

We shall give only indications for the proof. The implications 1) $\Leftrightarrow$ 1′) and 1) $\Rightarrow$ 2) are left to the reader. It is clear that a) implies a′). Let us show that a′) implies d). Let M be an injective object of $\mathfrak{A}$. For every $i \in I$ and every subobject $Y \hookrightarrow X_i$ of X_i, the homomorphism

$$\mathrm{Hom}(X_i, M) \longrightarrow \mathrm{Hom}(Y, M)$$

is surjective. Hence, by a′) and by the full faithfulness of ϵ, for every subobject $U \hookrightarrow \epsilon(X_i)$ the homomorphism

$$\mathrm{Hom}(\epsilon(X_i), \epsilon(M)) \longrightarrow \mathrm{Hom}(U, \epsilon(M))$$

is surjective. Since the $\epsilon(X_i)$ form a generating family of $\mathfrak{B}$, the object $\epsilon(M)$ is injective [11].

Let us show that a′) implies b). Enlarging the family of the X_i if necessary, one may suppose that for every $i \in I$ every quotient of X_i is isomorphic to some X_j. Let, for every $i \in I$, $X_i \hookrightarrow M_i$ be a monomorphism into an injective object. The family $(\epsilon(X_i))_{i\in I}$ is stable under quotients, and for $i \in I$ the morphism

$$\epsilon(X_i) \hookrightarrow \epsilon(M_i)$$

is, by d), a monomorphism into an injective object. One then checks immediately that, since the family $\epsilon(X_i)$ is generating, the family $(\epsilon(M_i))_{i\in I}$ is cogenerating; hence b).

Let us show that b) implies c). Let $(Z_\alpha)_{\alpha\in A}$ be a $\mathcal{U}$-small family of objects of $\mathfrak{A}$, and let us prove that the canonical morphism

$$\bigoplus_{\alpha\in A} \epsilon(Z_\alpha) \longrightarrow \epsilon\left(\bigoplus_{\alpha\in A} Z_\alpha\right)$$

is an isomorphism. Since the family of objects $\epsilon(Y)$ is cogenerating, it is enough to show that for every object Y of $\mathfrak{A}$ the homomorphism

$$\operatorname{Hom}\left(\epsilon\left(\bigoplus_{\alpha} Z_\alpha\right), \epsilon(Y)\right) \longrightarrow \operatorname{Hom}\left(\bigoplus_{\alpha} \epsilon(Z_\alpha), \epsilon(Y)\right)$$

is an isomorphism, which follows from the full faithfulness of ϵ.

It remains to show that c) implies a). Let Z be a subobject of $\epsilon(Y)$. By $1'$), there exists a $\mathcal{U}$-small family Y_α of objects of $\mathfrak{A}$ and an epimorphism from $\bigoplus_\alpha \epsilon(Y_\alpha)$ onto Z. Since ϵ commutes with $\mathcal{U}$-small direct sums, there is therefore an epimorphism

$$\epsilon(Y') \longrightarrow Z.$$

Let $u : \epsilon(Y') \to Z \to \epsilon(Y)$ be the composite morphism. One has $Z = \operatorname{Im}(u)$. Since ϵ is fully faithful, one has $u = \epsilon(v)$, and consequently

$$Z = \operatorname{Im}(\epsilon(v)) = \epsilon(\operatorname{Im}(v)).$$

Exercise 0.5.2. Let $\mathcal{S}ex_{\mathcal{V}}(\mathfrak{A})$ be the category of contravariant functors from $\mathfrak{A}$ to the category of $\mathcal{V}$-abelian groups which commute with $\mathcal{U}$-small inductive limits. Show that under conditions 1) and a$'$) the canonical functor from $\mathfrak{B}$ to $\mathcal{S}ex_{\mathcal{V}}(\mathfrak{A})$ is an equivalence.

SGA 4, Exposé V. Cohomology in Topoi

Draft translation, Sections 1–4

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Translator's status note. This fresh-only continuation covers Exposé V, Sections 1–4, corresponding to source lines 370–1946 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current mathematical English: *ringed topos*, *flat module*, *flat morphism of ringed topoi*, *point/fiber functor*, *localization*, *presheaf topos*, *Cech cohomology*, *sieve*, *covering family*, and *change of universe*. Numbering follows the source.

1 Flat modules

Definition 1.1. Let (E, A) be a ringed topos *(IV, 11.1.1)*. A right A-module (respectively a left A-module) M is called *flat* if the functor

$$M \otimes_A - \qquad (\textit{respectively } - \otimes_A M)$$

from the category of left A-modules (respectively right A-modules) to the category of abelian sheaves on E is exact.

Proposition 1.2. Let M be a B-A-bimodule.

1) The following properties are equivalent:

   (i) M is left A-flat.

   (ii) For every injective B-module I, the left A-module $\mathcal{H}om_B(M, I)$ is injective.

2) A module M which is a pseudo-filtered inductive limit *(I, 2.7.1)* of flat modules is flat.

3) Finally, if

   $$M_\bullet = \cdots \longrightarrow M_{i+1} \longrightarrow M_i \longrightarrow \cdots$$

   is an acyclic complex of flat modules, with $M_i = 0$ for $i < i_0$, then for every module F the complex

   $$M_\bullet \otimes_A F = \cdots \longrightarrow M_{i+1} \otimes_A F \longrightarrow M_i \otimes_A F \longrightarrow \cdots$$

   is acyclic.

Proof. By (IV, 12.12) one has an adjunction isomorphism

$$\mathrm{Hom}_B(M \otimes_A -, -) \xrightarrow{\sim} \mathrm{Hom}_A(-, \mathcal{H}om_B(M, -)).$$

It is then enough to apply 0.2 to obtain the equivalence (i) $\Leftrightarrow$ (ii). This adjunction isomorphism also shows that tensor product commutes with inductive limits. The fact that pseudo-filtered inductive limits are exact (0.1) gives the second assertion. To show that the complex $M_\bullet \otimes_A F$ is acyclic, it is enough to show that, for every injective abelian sheaf I, the complex

$$\mathrm{Hom}^\bullet_{\mathbb{Z}}(M_\bullet \otimes_A F, I)$$

is acyclic. By the adjunction formulae this complex is isomorphic to

$$\mathrm{Hom}_A^\bullet(F, \mathcal{H}om_{\mathbb{Z}}(M_\bullet, I)).$$

But, by the equivalence (i) $\Leftrightarrow$ (ii), the complex of sheaves

$$\mathcal{H}om_{\mathbb{Z}}(M_\bullet, I)$$

is an acyclic complex whose objects are injective. This proves the assertion.

Proposition 1.3.1. Let (E, A) be a ringed topos, let H be an object of E, and let M be a flat $A_{/H}$-module. Then $j_{H!}M$ is a flat A-module. In particular A_H is flat on both the right and the left.

Suppose, to fix ideas, that M is a right $A_{/H}$-module. For every left A-module P there is a canonical isomorphism (IV, 12)

$$P \otimes_A j_{H!}M \simeq j_{H!}(P \otimes_{A_{/H}} M).$$

The functors $j_{H!}$ and j_H^* are exact (IV, 11.3.1 and 11.12.2), and by hypothesis the functor $- \otimes_{A_{/H}} M$ is exact. Hence the functor

$$P \longmapsto P \otimes_A j_{H!}M$$

is exact, and $j_{H!}M$ is flat.

Proposition 1.3.2. *Projection formula for closed immersions.* Let (E, A) be a ringed topos, let $i : F \to E$ be a closed subtopos of E, and put $i^*A = A_{/F}$. For every right $A_{/F}$-module M and every left A-module P, there is a functorial isomorphism

$$i_*(M \otimes_{A_{/F}} i^*P) \simeq (i_*M) \otimes_A P.$$

Let U be the open complement of F and let $j : U \to E$ be the canonical immersion. One has

$$j^*(i_*M \otimes_A P) \simeq 0 \otimes_{A_{/U}} j^*P$$

(IV, 12); hence $i_*M \otimes_A P$ is supported on F. Therefore the adjunction morphism

$$i_*M \otimes_A P \longrightarrow i_*i^*(i_*M \otimes_A P)$$

is an isomorphism (IV, 14). Moreover,

$$i^*(i_*M \otimes_A P) \simeq i^*i_*M \otimes_{A_{/F}} i^*P$$

(IV, 12), and since $i : F \to E$ is a closed immersion, $i^*i_*M \simeq M$ (IV, 14). The announced isomorphism follows.

Corollary 1.3.3. For every flat $A_{/F}$-module M, the module i_*M is flat.

This follows from 1.3.2 and (IV, 14), since the functor

$$P \longmapsto (i_*M) \otimes_A P$$

is exact.

1.4

Let (E, A) be a ringed topos, let $x : P \to E$ be a point of E (IV, 6.1), and let $\mathrm{Neigh}(x)$ denote the category of neighborhoods of x (IV, 6.8). To every object V of $\mathrm{Neigh}(x)$ there corresponds an object of E, still denoted V, and a point $x_V : P \to E/V$ of E/V. Moreover, to every morphism $u : V \to W$ in $\mathrm{Neigh}(x)$ there corresponds an essentially commutative diagram of morphisms of topoi (IV, 6.7)

$$\begin{tikzcd} P \arrow[r, "x_V"] \arrow[dr, "x_W"'] & E/V \arrow[d, "j_u"] \\ & E/W. \end{tikzcd} \tag{1.4.1}$$

Let N be an A_x-module (respectively a set). From (1.4.1) one obtains a diagram of A_x-modules (respectively sets):

$$\begin{tikzcd} x_W^* x_{W*} N \arrow[r, "\mathrm{ad}_W"] \arrow[d, "\phi(u)"'] & N \\ x_V^* x_{V*} N \arrow[ur, "\mathrm{ad}_V"'] & \end{tikzcd} \tag{1.4.2}$$

where ad_W and ad_V are the adjunction morphisms. One checks immediately that $\phi(uv) = \phi(v)\phi(u)$. Thus one has a functor from the filtered category $\mathrm{Neigh}(x)^{\mathrm{op}}$ to the category of A_x-modules (respectively sets), and a homomorphism

$$\Lambda(N) : \varinjlim_{V \in \mathrm{Neigh}(x)^{\mathrm{op}}} x_V^* x_{V*} N \longrightarrow N. \tag{1.4.3}$$

Proposition 1.5. For every A_x-module (respectively set) N, the map $\Lambda(N)$ is an isomorphism.

This proposition is a special case of a proposition due to Deligne (8.2.6). We give a direct proof. Passing to the underlying sets, it is enough to prove the proposition when N is a set (I, 2.8). The cofiltered category $\mathrm{Fl}(\mathrm{Neigh}(x))$ of arrows of $\mathrm{Neigh}(x)$ is fibered over $\mathrm{Neigh}(x)$ by the functor

$$p : \mathrm{Fl}(\mathrm{Neigh}(x)) \longrightarrow \mathrm{Neigh}(x)$$

which sends an arrow to its target. Let D be a category and let

$$F : \mathrm{Fl}(\mathrm{Neigh}(x)) \longrightarrow D$$

be a pro-object (I, 8.10). For every object V of $\mathrm{Neigh}(x)$, denote by F_V the pro-object obtained by restricting F to the fiber category $\mathrm{Neigh}(x)/V$. To every morphism $m : U \to V$ in $\mathrm{Neigh}(x)$, base change by m associates a morphism of pro-objects $F_U \to F_V$, and the canonical morphisms of pro-objects $F \to F_V$ determine a morphism

$$F \longrightarrow \varprojlim_{V \in \mathrm{Neigh}(x)^{\mathrm{op}}} F_V \tag{1.5.1}$$

which is immediately checked to be an isomorphism. Apply this remark to the pro-object of pointed sets

$$(U, V, m) \longmapsto x_V^*(m) = F(U, V, m). \tag{1.5.2}$$

One obtains an isomorphism of pro-objects

$$F \xrightarrow{\sim} \varprojlim_{V \in \mathrm{Neigh}(x)} \varprojlim_{\mathrm{Neigh}(x)/V} x_V^*(m), \tag{1.5.3}$$

and hence, for every set N, a bijection

$$\varinjlim_{\mathrm{Fl}(\mathrm{Neigh}(x))^{\mathrm{op}}} \mathrm{Hom}(x_V^*(m), N) \simeq \varinjlim_{\mathrm{Neigh}(x)^{\mathrm{op}}} \varinjlim_{(\mathrm{Neigh}(x)/V)^{\mathrm{op}}} \mathrm{Hom}(x_V^*(m), N). \tag{1.5.4}$$

But for fixed V,

$$\varinjlim_{(\mathrm{Neigh}(x)/V)^{\mathrm{op}}} \mathrm{Hom}(x_V^*(m), N)$$

identifies with $x_V^* x_{V*} N$. Indeed,

$$x_V^* x_{V*} N \simeq \varinjlim_{\mathrm{Neigh}(x_V)^{\mathrm{op}}} \mathrm{Hom}_{E/V}(W, x_{V*} N)$$

by IV, 6.8.1, and therefore by adjunction

$$x_V^* x_{V*} N \simeq \varinjlim_{\mathrm{Neigh}(x_V)^{\mathrm{op}}} \mathrm{Hom}(x_V^*(w), N).$$

Moreover, the category $\mathrm{Neigh}(x_V)$ is equivalent to $\mathrm{Neigh}(x)/V$ (IV, 6.7.2). Finally, the canonical transition maps appearing in (1.5.4) identify with the canonical maps

$$x_U^* x_{U*} N \longrightarrow x_V^* x_{V*} N$$

of (1.4.3), as the reader will readily verify. Thus one has a bijection

$$\varinjlim_{\mathrm{Fl}(\mathrm{Neigh}(x))^{\mathrm{op}}} \mathrm{Hom}(x_V^*(m), N) \simeq \varinjlim_{\mathrm{Neigh}(x)^{\mathrm{op}}} x_V^* x_{V*} N. \tag{1.5.5}$$

Composing $\Lambda(N)$ with this bijection, one obtains the map induced by the maps

$$\mathrm{Hom}(x_V^*(m), N) \longrightarrow N$$

which send a map $r : x_V^*(m) \to N$ to the image under r of the marked point of $x_V^*(m)$. It remains only to observe that every object (U, V, m) of $\mathrm{Fl}(\mathrm{Neigh}(x))$ is bounded below by (U, U, id_U) and that $x_U^*(\mathrm{id}_U)$ consists of a single element; equivalently, the canonical morphism from the constant pro-object reduced to one element to F is an isomorphism of pro-objects.

Proposition 1.6. Let (E, A) be a ringed topos and let M be an A-module.

1) If M is flat, then for every point $x : P \to E$ of E, the A_x-module M_x is flat.

2) Let $(x_i)_{i \in I}$ be a conservative family of points of E such that, for every $i \in I$, M_{x_i} is a flat A_{x_i}-module.

Then M is flat.

Lemma 1.6.1. Let M be a flat module and let V be an object of E. The A/V-module $j_V^* M$ is flat.

We must show that the functor

$$P \longmapsto P \otimes_{A/V} j_V^* M$$

is exact. Since extension by zero $j_{V!}$ is exact and faithful (IV, 11.3.1), it is enough to show that

$$P \longmapsto j_{V!}(P \otimes_{A/V} j_V^* M)$$

is exact. There is a functorial isomorphism

$$j_{V!}(P \otimes_{A/V} j_V^* M) \simeq j_{V!}(P) \otimes_A M$$

(IV, 12.11), and hence the functor $P \mapsto j_{V!}(P) \otimes_A M$ is exact (IV, 11.3.1).

1.6.2

Let us prove 1). Suppose, to fix ideas, that M is a flat left A-module, and let $x : P \to E$ be a point of E. We show that the functor

$$N \longmapsto N \otimes_{A_x} M_x$$

from the category of right A_x-modules to the category of abelian groups is exact. It suffices to show that this functor is left exact. With the notation of 1.4, let V be a neighborhood of x and let $j_V : E/V \to E$ be the localization morphism. For every A_x-module N one has

$$x_V^* x_{V*} N \otimes_{A_x} M_x \simeq x_V^*(x_{V*} N \otimes_{A/V} j_V^* M)$$

(IV, 12.11). Hence the functor

$$N \longmapsto x_V^* x_{V*} N \otimes_{A_x} M_x$$

is left exact. Therefore

$$N \longmapsto N \otimes_{A_x} M_x,$$

being a filtered inductive limit of left exact functors (1.5), is left exact.

Let us prove 2). Let

$$0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0$$

be an exact sequence of A-modules. We must show that

$$0 \longrightarrow P' \otimes_A M \longrightarrow P \otimes_A M \longrightarrow P'' \otimes_A M \longrightarrow 0$$

is exact. It is enough to check exactness on all fibers at the points x_i (IV, 6). The sequence of fibers at x_i is

$$0 \longrightarrow P'_{x_i} \otimes_{A_{x_i}} M_{x_i} \longrightarrow P_{x_i} \otimes_{A_{x_i}} M_{x_i} \longrightarrow P''_{x_i} \otimes_{A_{x_i}} M_{x_i} \longrightarrow 0,$$

and this sequence is exact because M_{x_i} is flat.

Proposition 1.7. Let (E, A) be a ringed topos, let $u : E' \to E$ be a morphism of topoi, and put $A' = u^* A$. Let M be a right A'-module (respectively a left A'-module). The following conditions are equivalent:

(i) The functor

$$N \longmapsto M \otimes_{A'} u^*N \qquad (\textit{respectively } N \longmapsto u^*N \otimes_{A'} M)$$

from the category of left A-modules (respectively right A-modules) to the category of abelian sheaves on E' is exact.

(ii) M is a flat A'-module.

It is clear that (ii) implies (i). We prove that (i) implies (ii). We give the proof only in the case where E' has a conservative family of points $(x_i)_{i\in I}$. The general case is treated in 8.2.7. In this special case, which covers most applications, one is reduced to the case where E' is the punctual topos and where the morphism u, now denoted by x, is a point of E (1.6 and IV, 11.3.1). We restrict ourselves to the case where M is a right A'-module. The case where M is a left A'-module is reduced to this one by passing to opposite rings.

Let V be a neighborhood of x and let V_x be its fiber at x. With the notation of 1.4, one has an essentially commutative diagram of morphisms of topoi

$$\begin{tikzcd} P \arrow[r, "j"] \arrow[dr, "x_V"'] & P/V_x \arrow[r, "j_{V_x}"] \arrow[d, "x/V"] & P \arrow[d, "x"] \\ & E/V \arrow[r] & E, \end{tikzcd}$$

where x/V is the morphism induced from x by localization on V (IV, 5.10), and j is the morphism induced by the marked point of V_x. We show that the functor

$$N' \longmapsto M \otimes_{A'} x_V^* N'$$

is exact. One has $x_V^* = j^*(x/V)^*$ and $\mathrm{id} = j^* j_{V_x}^*$, and hence a canonical isomorphism

$$M \otimes_{A'} x_V^* N' \simeq j^*\big(j_{V_x}^* M \otimes_{A'/V_x} (x/V)^* N'\big).$$

Since j^* is exact and since $j_{V_x!}$ is exact and faithful (IV, 11.3.1), it is enough to show that the functor

$$N' \longmapsto j_{V_x!}\big(j_{V_x}^* M \otimes (x/V)^* N'\big)$$

is exact. By (IV, 12.11), it is enough to show that

$$N' \longmapsto M \otimes_{A'} x^* j_{V!} N'$$

is exact. But by the definition of the inverse image of an induced topos (IV, 5.10.2), $j_{V_x!}(x/V)^*N'$ is equal to $x^* j_{V!} N'$, and since $j_{V!}$ is exact (IV, 11.3.1), exactness follows from the hypothesis.

For every A'-module Q one has a functorial isomorphism (1.5)

$$Q \simeq \varinjlim_{\mathrm{Neigh}(x)^{\mathrm{op}}} x_V^* x_{V*} Q.$$

By what precedes, the functor

$$Q \longmapsto M \otimes_{A'} Q$$

is a filtered inductive limit of left exact functors. It is therefore left exact, and M is flat.

Corollary 1.7.1. Let $u : (E', A') \to (E, A)$ be a morphism of ringed topoi, and let M be a flat A-module. Then u^*M is a flat A'-module.

This follows from the criterion in 1.7 and the formula (IV, 13.4.5).

Corollary 1.7.2. Let $u : (E', A') \to (E, A)$ be a morphism of ringed topoi. The following conditions are equivalent:

(i) the right $u^{-1}(A)$-module (respectively left $u^{-1}(A)$-module) A' is flat;

(ii) the functor u^* from the category of left A-modules (respectively right A-modules) to the category of left A'-modules (respectively right A'-modules) is exact.

The equivalence follows from 1.7 and from the definition of u^* (IV, 13.2.1).

Definition 1.8. A morphism of ringed topoi satisfying the equivalent properties of 1.7.2 is called a *left flat* (respectively *right flat*) morphism of topoi.

Proposition 1.9. Let $\mathcal{U} \subset \mathcal{V}$ be two univers, let C be a $\mathcal{U}$-site, let $C^\sim_{\mathcal{U}}$ and $C^\sim_{\mathcal{V}}$ be the categories of $\mathcal{U}$- and $\mathcal{V}$-sheaves of sets, respectively, let

$$\epsilon : C^\sim_{\mathcal{U}} \longrightarrow C^\sim_{\mathcal{V}}$$

be the canonical inclusion functor, and let A be a $\mathcal{U}$-sheaf of rings on C.

1) The functor ϵ commutes with $\mathcal{U}$-small inductive and projective limits of modules. On categories of modules, ϵ is conservative and fully faithful.

2) For every object H of $C^\sim_{\mathcal{U}}$, there is a canonical isomorphism

$$\epsilon(A_H) \simeq \epsilon(A)_{\epsilon(H)}.$$

3) Every injective left or right A-module is carried by ϵ to an injective $\epsilon(A)$-module.

4) Every flat right or left A-module is carried by ϵ to a flat $\epsilon(A)$-module.

5) The functor ϵ commutes with formation of the tensor product and of the sheaf of homomorphisms.

Formation of the associated sheaf does not depend on the universe (II, 3.6). It then follows from the construction of inductive and projective limits in categories of sheaves (II, 4.1 and 6.4) that ϵ commutes with $\mathcal{U}$-small projective limits of sets or modules, giving 1). To prove 2), by taking a $\mathcal{U}$-small generating subcategory of $C^\sim_{\mathcal{U}}$, one is reduced to the case where C is $\mathcal{U}$-small (III, 4.1). It then follows from (II, 6.5) that the free A-module generated by H is obtained by first forming the presheaf of free A-modules generated by H, a construction which commutes with enlargement of universes, and then taking the associated sheaf, an operation which also commutes with enlargement of universes (II, 3.6).

To prove 3), it is enough, by 0.5, to show that every $\mathcal{V}$-subsheaf of a $\mathcal{U}$-sheaf is a $\mathcal{U}$-sheaf; this is clear. To prove 4), let M be a flat A-module. It is enough to show that, for every injective $\epsilon(A)$-module J, the abelian sheaf

$$\mathcal{H}om_{\epsilon(A)}(\epsilon(M), J)$$

is injective (1.2). By 0.5, J is a subobject, hence a direct factor, of an object of the form

$$\prod_{\alpha} \epsilon(I_\alpha),$$

where the I_α are injective. We may therefore suppose that $J = \epsilon(I)$ with I injective. If the homomorphism

$$\epsilon(\mathcal{H}om_A(M, I)) \longrightarrow \mathcal{H}om_{\epsilon(A)}(\epsilon(M), \epsilon(I))$$

is an isomorphism, then $\mathcal{H}om_{\epsilon(A)}(\epsilon(M), \epsilon(I))$ is injective by 3). It remains to prove 5).

The assertion that formation of the tensor product commutes with ϵ follows from IV, 12.6 and from the fact that formation of the associated sheaf commutes with enlargement of the universe (II, 3.6). For every object X of $C^{\sim}_{\mathcal{U}}$ and every pair of A-modules M, N on $C^{\sim}_{\mathcal{U}}$, one has

$$\mathrm{Hom}_{C^{\sim}_{\mathcal{V}}}\big(\epsilon(X), \epsilon(\mathcal{H}om_A(M, N))\big) \simeq \mathrm{Hom}_{C^{\sim}_{\mathcal{U}}}\big(X, \mathcal{H}om_A(M, N)\big) \simeq \mathrm{Hom}_A(\mathbb{Z}_X \otimes_{\mathbb{Z}} M, N)$$

by full faithfulness of ϵ, and then

$$\mathrm{Hom}_A(\mathbb{Z}_X \otimes_{\mathbb{Z}} M, N) \simeq \mathrm{Hom}_{\epsilon(A)}\big(\epsilon(\mathbb{Z}_X \otimes_{\mathbb{Z}} M), \epsilon(N)\big) \simeq \mathrm{Hom}_{\epsilon(A)}\big(\epsilon(\mathbb{Z}_X) \otimes_{\mathbb{Z}} \epsilon(M), \epsilon(N)\big)$$

by full faithfulness of ϵ and by what was just proved. Using 2) and IV, 12.14, one obtains finally an isomorphism

$$\mathrm{Hom}_{C^{\sim}_{\mathcal{V}}}\big(\epsilon(X), \epsilon(\mathcal{H}om_A(M, N))\big) \simeq \mathrm{Hom}_{C^{\sim}_{\mathcal{V}}}\big(\epsilon(X), \mathcal{H}om_{\epsilon(A)}(\epsilon(M), \epsilon(N))\big),$$

and hence the announced isomorphism.

1.10

Let E be a topos and let

$$\mathcal{U} = (U_i \to X)_{i \in I}$$

be a small family of morphisms with common target. For the finite set $\Delta_n = \{0, \dots, n\}$, put

$$S_n(\mathcal{U}) = \coprod_{f : \Delta_n \to I} \prod_{a \in \Delta_n, X} U_{f(a)},$$

the coproduct being taken over the set of maps $\Delta_n \to I$. Let m, n be integers and let $g : \Delta_m \to \Delta_n$ be a map. Define a morphism

$$s(g) : S_n(\mathcal{U}) \longrightarrow S_m(\mathcal{U}) \tag{1.10.1}$$

as follows: for every $f : \Delta_n \to I$, the restriction of $s(g)$ to

$$\prod_{a \in \Delta_n, X} U_{f(a)}$$

is the composite of the canonical inclusion

$$\prod_{b \in \Delta_m, X} U_{f(g(b))} \hookrightarrow S_m(\mathcal{U})$$

and the unique morphism

$$s_f(g) : \prod_{a \in \Delta_n, X} U_{f(a)} \longrightarrow \prod_{b \in \Delta_m, X} U_{f(g(b))}$$

such that, for every $b \in \Delta_m$,

$$\mathrm{pr}_{f(g(b))}\, s_f(g) = \mathrm{pr}_{f(g(b))}\,.$$

Here pr_a denotes the a-th projection. Thus one obtains a contravariant functor $\Delta_n \mapsto S_n$ from the category of finite sets to E, in other words a semi-simplicial complex $S_\bullet(\mathcal{U})$ of objects of E. Note that this complex is canonically augmented toward X.

Every functor from E to a category C carries $S_\bullet(\mathcal{U})$ to a simplicial object of C. In particular, if A is a ring of E, the functor "free A-module generated by" carries $S_\bullet(\mathcal{U})$ to a simplicial complex of A-bimodules, augmented toward A_X, denoted $A_\bullet(\mathcal{U})$. One has

$$A_n(\mathcal{U}) = \bigoplus_{f:\Delta_n \to I} A_{\prod_{a\in\Delta_n,\, X} U_{f(a)}}. \tag{1.10.2}$$

This complex will often be denoted

$$\cdots \quad \overset{s_0}{\underset{s_2}{\overset{\xrightarrow{\quad s_1 \quad}}{\rightrightarrows}}} \quad \bigoplus_{i,j} A_{U_i\times_X U_j} \quad \overset{s_0}{\underset{s_1}{\rightrightarrows}} \quad \bigoplus_i A_{U_i} \quad \longrightarrow \quad A_X. \tag{1.10.3}$$

In (1.10.3), except for the last arrow, which is the augmentation, the arrows represent the face operators of the simplicial complex $A_\bullet(\mathcal{U})$; that is, the morphisms corresponding to the increasing injective maps $\Delta_n \to \Delta_{n+1}$, where s_i omits the integer i. To $A_\bullet(\mathcal{U})$ one associates a differential complex augmented toward A_X:

$$\cdots \xrightarrow{d} \bigoplus_{i,j} A_{U_i\times_X U_j} \xrightarrow{d} \bigoplus_i A_{U_i} \longrightarrow A_X, \tag{1.10.4}$$

with

$$d = \sum_i (-1)^i s_i. \tag{1.10.5}$$

Proposition 1.11. If the family $\mathcal{U} = (U_i \to X)_{i\in I}$ is epimorphic, the differential complex (1.10.4) is acyclic and gives a resolution of A_X.

Let $\mathbb{Z}$ denote the constant sheaf with values in $\mathbb{Z}$. By the definition of the functor "free A-module generated by", for every object H of E one has

$$A_H \simeq \mathbb{Z}_H \otimes_{\mathbb{Z}} A,$$

and therefore

$$A_\bullet(\mathcal{U}) \simeq \mathbb{Z}_\bullet(\mathcal{U}) \otimes_{\mathbb{Z}} A.$$

Since the components of $\mathbb{Z}_\bullet(\mathcal{U})$ are flat $\mathbb{Z}$-modules (1.3.1), it is enough to prove the proposition for $A = \mathbb{Z}$.

Suppose first that E is the topos of sets. Then the augmented complex $S_\bullet(\mathcal{U})$ is a direct sum of augmented complexes of the form

$$\cdots \ \overset{\longrightarrow}{\underset{\longrightarrow}{\rightrightarrows}}\ S\times S\times S \ \overset{\longrightarrow}{\rightrightarrows}\ S\times S \rightrightarrows S \longrightarrow \{e\},$$

where $\{e\}$ is a one-point set. Each of these augmented complexes is homotopically trivial. Thus $S_\bullet(\mathcal{U})$ is a homotopically trivial augmented complex, and its homology is therefore trivial. This proves the proposition in this case.

Now let $p : \mathbf{Set} \to E$ be a point of E. Since formation of the complex $\mathbb{Z}_\bullet(\mathcal{U})$ commutes with inverse-image functors of morphisms of topoi,

$$p^*(\mathbb{Z}_\bullet(\mathcal{U})) \simeq \mathbb{Z}_\bullet(p^*(\mathcal{U}))$$

is a resolution of $\mathbb{Z}_{p^*X} \simeq p^*(\mathbb{Z}_X)$. This proves the proposition when E has enough fiber functors (IV, 4.6). This is the case in particular when E is a presheaf topos $\widehat{C}$, since for every object X of C, $\Gamma(X, -)$ is a fiber functor. In the general case, E is equivalent to a topos of sheaves on a small site C (IV, 1), and the epimorphic family

$$\mathcal{U} = (U_i \to X)_{i \in I}$$

is the image, under the associated-sheaf functor, of an epimorphic family

$$\mathcal{U}' = (U'_i \to X')_{i \in I}.$$

Consequently

$$\mathbb{Z}_\bullet(\mathcal{U}) \simeq \underline{a}\,\mathbb{Z}_\bullet(\mathcal{U}')$$

is a resolution of $\underline{a}\,\mathbb{Z}_{X'} \simeq \mathbb{Z}_X$.

2 Čech cohomology. Cohomological notation

2.1. General notation

2.1.1

Let (E, A) be a ringed topos and let M, N be two A-modules, left modules to fix ideas. We denote by

$$\operatorname{Ext}^q_A(E; M, N)$$

the value at N of the q-th right derived functor of the functor $\operatorname{Hom}_A(M, -)$ [3]. The functors $\operatorname{Ext}^q_A(E; M, N)$, $q \in \mathbb{N}$, form a δ-functor in the variable N. They also form a contravariant δ-functor in the variable M. By definition,

$$\operatorname{Ext}^0_A(E; M, N) = \operatorname{Hom}_A(M, N).$$

2.1.2

Let X be an object of E. When $M = A_X$, the free A-module generated by X (IV, 12), we put

$$\operatorname{Ext}^q_A(E; A_X, N) = H^q(X, N).$$

Notice that in this new notation the ring A no longer appears. This cannot cause confusion, since we shall show (3.5) that formation of the $H^q(X, -)$ commutes with restriction of scalars. The functor $H^q(X, -)$ is the q-th right derived functor of

$$\operatorname{Hom}_A(A_X, -) = \operatorname{Hom}_E(X, -),$$

also denoted $\Gamma(X, -)$. When $M = A$, we put

$$\operatorname{Ext}^q_A(E; A, N) = H^q(E, N).$$

2.2. Localization

Let X be an object of E and let

$$j : E_{/X} \longrightarrow E$$

be the localization morphism (IV, 8). The functor j_X^* on modules is exact (4.11) and admits an exact left adjoint $j_{X!}$ (4.11). Consequently it carries injective modules to injective modules. Thus, for a variable A-module N on E and a variable $A|X$-module M on $E_{/X}$, one has functorial isomorphisms

$$\mathrm{Ext}^q_{A|X}(E_{/X}; M, j_X^* N) \simeq \mathrm{Ext}^q_A(E; j_{X!}M, N). \tag{2.2.1}$$

In particular, one has canonical isomorphisms

$$H^q(E_{/X}, j_X^* N) \simeq H^q(X, N).$$

For every object X of E and every pair of A-modules M, N, set

$$\mathrm{Ext}^q_A(X; M, N) = \mathrm{Ext}^q_{A|X}(E_{/X}; M|X, N|X). \tag{2.2.2}$$

By what precedes, the functors $\mathrm{Ext}^q_A(X; M, -)$ are the derived functors of

$$N \longmapsto \mathrm{Hom}_{A|X}(M|X, N|X).$$

The functors $(M, N) \mapsto \mathrm{Ext}^q_A(X; M, N)$, $q > 0$, form a δ-functor with respect to each variable.

2.3. The case of presheaf topoi. Cohomology of a covering

2.3.1

Let C be a small category endowed with a presheaf of rings A, let $\widehat{C}$ be the topos of presheaves on C, and let X be a representable object of $\widehat{C}$. The functor which sends an A-module M to the group

$$\Gamma(X, M) = M(X)$$

is exact (I, 3). Hence $H^q(X, M) = 0$ for all $q > 0$ and every A-module M; equivalently, A_X is a projective A-module.

2.3.2

Let S be a presheaf on C. For every A-module M there is a canonical isomorphism (I, 2)

$$\Gamma(S, M) \simeq \varprojlim_{C/S} M|S.$$

Moreover, for every injective A-module M, the $A|S$-module $M|S$ is injective (2.2). Therefore the groups $H^q(S, M)$ are the values at $M|S$ of the right derived functors of the functor $\varprojlim_{C/S}$. Denoting these derived functors by $\varprojlim^q_{C/S}$, one has canonical isomorphisms

$$H^q(S, M) \simeq \varprojlim_{C/S}{}^{q}\, M|S.$$

In particular, one has canonical isomorphisms

$$H^q(\widehat{C}, M) \simeq \varprojlim_{C}{}^{q}\, M.$$

2.3.3

Let X be an object of C and let

$$\mathcal{U} = (U_i \to X)_{i \in I}$$

be a family of morphisms of C such that, for every $i \in I$, the morphism $U_i \to X$ is squareable (I, 10). Let $A_\bullet$ be the simplicial complex studied in 1.10:

$$A_\bullet = \cdots \; \substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow} \coprod_{(i,j) \in I \times I} A_{U_i \times_X U_j} \; \substack{\longrightarrow\\ \longrightarrow} \coprod_{i \in I} A_{U_i}.$$

For every A-module M, put

$$C^\bullet(\mathcal{U}, M) = \operatorname{Hom}_A(A_\bullet, M) :$$

$$C^\bullet(\mathcal{U}, M) : \prod_{i \in I} M(U_i) \; \substack{\longrightarrow\\ \longrightarrow} \prod_{(i,j) \in I \times I} M(U_i \times_X U_j) \; \substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow} \cdots . \tag{2.3.3.1}$$

Set

$$H^q(\mathcal{U}, M) = H^q(C^\bullet(\mathcal{U}, M)).$$

Proposition 2.3.4.

1) With the notation of 2.3.3, let $R \hookrightarrow X$ be the sieve generated by the family $(U_i \to X)$, $i \in I$. There is a canonical isomorphism

$$H^q(\mathcal{U}, M) \simeq H^q(R, M).$$

2) The functors $H^q(\mathcal{U}, -)$ commute with restrictions of scalars.

Since R is a subobject of X in $\widehat{C}$, the fiber products

$$U_{i_1} \times_R U_{i_2} \times_R \cdots \times_R U_{i_n} \quad \text{and} \quad U_{i_1} \times_X U_{i_2} \times_X \cdots \times_X U_{i_n}$$

are canonically isomorphic. It follows from 1.11 that the complex $A_{\mathcal{U},\bullet}$ is a resolution of A_R, and from 2.3.1 that the components of $A_{\mathcal{U},\bullet}$ are projective modules. The cohomology groups of $C^\bullet(\mathcal{U}, M)$ are therefore canonically isomorphic to

$$\operatorname{Ext}^q_A(\widehat{C}; A_R, M),$$

which gives the isomorphism. Assertion 2) follows immediately from the description (2.3.3.1) of $C^\bullet(\mathcal{U}, M)$.

Corollary 2.3.5. Let

$$\mathcal{U} = (U_i \to X, \; i \in I), \qquad \mathcal{U}' = (U'_j \to X, \; j \in J)$$

be two families of morphisms with target X. Let

$$\phi = (\phi : I \to J, \; m_i : U_i \to U'_{\phi(i)})$$

and

$$\phi' = (\phi' : I \to J,\ m'_i : U_i \to U'_{\phi'(i)})$$

be two morphisms, over X, from $\mathcal{U}$ to $\mathcal{U}'$. The morphisms ϕ and ϕ' induce morphisms

$$\phi^q, \phi'^q : H^q(\mathcal{U}, M) \longrightarrow H^q(\mathcal{U}', M).$$

These morphisms are equal. In particular, if the families $\mathcal{U}$ and $\mathcal{U}'$ are equivalent (i.e., if there exists a morphism from $\mathcal{U}$ to $\mathcal{U}'$ and a morphism from $\mathcal{U}'$ to $\mathcal{U}$), then the A-modules $H^q(\mathcal{U}, M)$ and $H^q(\mathcal{U}', M)$ are canonically isomorphic.

Proof. Let R and R' be the sieves of X generated by the families $\mathcal{U}$ and $\mathcal{U}'$. The morphisms ϕ and ϕ' define the same morphism from R to R' and hence induce two homotopic morphisms between the projective resolutions $A_{\mathcal{U},\bullet}$ and $A_{\mathcal{U}',\bullet}$.

Exercise 2.3.6 (standard resolution).

a) Let C be a small category. For every integer $n > 0$, let $\mathrm{Fl}^n(C)$ denote the set of sequences of morphisms in C,

$$(u_1, \ldots, u_n),$$

such that, for every i, the target of u_i is equal to the source of u_{i+1}, so that u_i and u_{i+1} are composable. Define a semi-simplicial set

$$ES(C) = \mathrm{Ob}\, C \;\leftleftarrows\; \mathrm{Fl}^1(C) \;\overset{\longleftarrow}{\leftleftarrows}\; \mathrm{Fl}^2(C) \;\overset{\leftleftarrows}{\leftleftarrows}\; \mathrm{Fl}^3(C) \qquad \cdots,$$

whose face operators $s_i : \mathrm{Fl}^n(C) \to \mathrm{Fl}^{n-1}(C)$ are

$$\begin{aligned} s_0(u_1, \ldots, u_n) &= (u_2, \ldots, u_n), \\ s_i(u_1, \ldots, u_n) &= (u_1, \ldots, u_{i+1}u_i, \ldots, u_n), \qquad 0 < i < n, \\ s_n(u_1, \ldots, u_n) &= (u_1, \ldots, u_{n-1}). \end{aligned}$$

b) Show that if C has an initial object or a final object, the complex $ES(C)$ is homotopically trivial.

c) For every object X of C, let ${}_{X\backslash}C$ denote the category of arrows with source X. Define a semi-simplicial presheaf $ES(C)$ whose value at every object X of C is $ES({}_{X\backslash}C)$. Let $\mathbb{Z}_{ES(C)}$ be the free abelian semi-simplicial presheaf generated by $ES(C)$. It is equipped with a canonical augmentation

$$\mathbb{Z}_{ES(C)} \longrightarrow \mathbb{Z}$$

to the constant presheaf with value $\mathbb{Z}$. Show that, on passing to the associated differential complex, $\mathbb{Z}_{ES(C)}$ is a resolution of $\mathbb{Z}$ and that the components of this complex are projective abelian presheaves.

d) For every abelian presheaf M, put

$$ST^\bullet(M) = \mathrm{Hom}(\mathbb{Z}_{ES(C)}, M).$$

Describe explicitly the components of $ST^\bullet(M)$. Show that, for every integer $q \geq 0$,

$$H^q(ST^\bullet(M)) = H^q(\widehat{C}, M).$$

e) Show that, for every sheaf S on C, the functors $H^q(S,-)$ commute with restrictions of scalars.

f) Observe that, when C is a group, $\mathbb{Z}_{ES(C)}$ is the standard resolution of the trivial module $\mathbb{Z}$ [1].

g) Show that, for every abelian presheaf M on C, $\mathbb{Z}_{ES(C)} \otimes M$ is a left resolution of M. Define the functor $\varprojlim_C$ on presheaves. Show that it is right exact. Denote its left derived functors by $H_q(C,M)$. Show that the components of the complex $\mathbb{Z}_{ES(C)} \otimes M$ are acyclic for $\varprojlim_C$. Deduce, writing

$$ST_\bullet(M) = \varprojlim_C(\mathbb{Z}_{ES(C)} \otimes M),$$

canonical isomorphisms

$$H_q(ST_\bullet(M)) \simeq H_q(C,M).$$

h) Let $u : \widehat{C} \to \mathbf{Set}$ be the unique morphism from the topos $\widehat{C}$ to the punctual topos. Let

$$u_! : \widehat{C}_{\mathbb{Z}} \longrightarrow \mathbf{Ab}$$

be the left adjoint of $u^* : \mathbf{Ab} \to \widehat{C}_{\mathbb{Z}}$. Show that, for every abelian presheaf M,

$$u_! M = \varinjlim_C M.$$

i) From now on denote by $ST^\bullet(C,M)$ and $ST_\bullet(C,M)$ the complexes previously denoted $ST^\bullet(M)$ and $ST_\bullet(M)$. A presheaf M on C is called *locally constant* if it sends all morphisms of C to isomorphisms. Associate to every locally constant presheaf M a locally constant presheaf $\check{M}$ on the opposite category C^{op}, such that for every morphism u of C,

$$\check{M}(u) = M(u)^{-1}.$$

Find canonical isomorphisms between the complexes $ST^\bullet(C,M)$ and $ST^\bullet(C^{\mathrm{op}},\check{M})$, and between $ST_\bullet(C,M)$ and $ST_\bullet(C^{\mathrm{op}},\check{M})$.

j) Let C be a small category having an initial object (respectively a small filtered category). Show that, for every constant presheaf M,

$$H^q(\widehat{C},M) = 0 \quad (q > 0)$$

(respectively $H_q(C,M) = 0$ for $q > 0$).

2.4. The case of small sites. Čech cohomology

2.4.1

Let (C,A) be a ringed $\mathcal{U}$-site, let $C^\sim$ be the topos of sheaves on C, and let

$$\epsilon : C \longrightarrow C^\sim$$

be the canonical functor which assigns to an object of C the sheaf associated with the presheaf represented by that object. By abuse of notation, for every object X of C and every sheaf of A-modules M we write

$$H^q(X, M) = H^q(\epsilon(X), M)$$

(cf. 2.3.1). Recall that when the topology of C is no finer than the canonical topology, as is always the case in practice, the functor ϵ is fully faithful and allows one to identify C with its image under ϵ.

2.4.2

Let

$$\mathcal{H}^0 : C_A^\sim \longrightarrow \widehat{C}_A$$

denote the inclusion functor from sheaves of A-modules into presheaves of A-modules. Thus, for every sheaf of A-modules M,

$$\mathcal{H}^0(M)(X) = H^0(X, M) = M(X) \tag{2.4.2.1}$$

for every object X of C. The functor $\mathcal{H}^0$ is left exact. Its right derived functors are denoted $\mathcal{H}^q$. Since, for every object X of C, the functor "sections over X" is exact in the category of presheaves, one has

$$\mathcal{H}^q(M)(X) = H^q(X, M) \tag{2.4.2.2}$$

for every object X of C and every sheaf of A-modules M. Thus the presheaf $\mathcal{H}^q(M)$ is simply the presheaf

$$X \longmapsto H^q(X, M).$$

2.4.3

Assume that (C, A) is a *small* ringed site, so that $C^\sim$ is a *topos* to which the results of 2.3 apply. Let X be an object of C and let $R \hookrightarrow X$ be a covering sieve. For every *presheaf* G of A-modules, the groups $H^q(R, G)$, computed in the topos $\widehat{C}$, are called the *Čech cohomology groups of the presheaf* G *relative to the covering sieve* R. When $R \hookrightarrow X$ is the sieve generated by a covering family

$$\mathcal{U} = (U_i \to X)_{i \in I}$$

of squareable morphisms, these groups may be computed by means of the complex $C^\bullet(\mathcal{U}, G)$ (2.3.3), called the *Čech complex of* G *relative to the covering family* $\mathcal{U}$, or relative to the covering $\mathcal{U}$. The groups

$$H^q(\mathcal{U}, G) = H^q(R, G)$$

(2.3.4) are then called the *Čech cohomology groups of* G *relative to the covering family* $\mathcal{U}$.

2.4.4

Let M be a sheaf of A-modules on C. The groups

$$H^q(\mathcal{U}, \mathcal{H}^0(M))$$

are most often denoted, abusively, by $H^q(\mathcal{U}, M)$ and called the Čech cohomology groups of the sheaf M relative to the covering family $\mathcal{U}$.

2.4.5

Let
$$\check{\mathcal{H}}^0 : \widehat{C}_A \longrightarrow \widehat{C}_A$$
be the natural extension to presheaves of A-modules of the functor L described in II. Thus, by definition, for a presheaf G and an object X of C,
$$\check{\mathcal{H}}^0(G)(X) = \varinjlim_{R \hookrightarrow X} G(R), \tag{2.4.5.1}$$
where the inductive limit is taken over the covering sieves of X. It follows from (2.4.5.1) that $\check{\mathcal{H}}^0$ is left exact. Its right derived functors are denoted $\check{\mathcal{H}}^q$. Since the functors "sections over X" and "filtered inductive limit" are exact, (2.4.5.1) gives
$$\check{\mathcal{H}}^q(G)(X) = \varinjlim_{R \hookrightarrow X} H^q(R, G), \tag{2.4.5.2}$$
where the limit is taken over the covering sieves of X.

The presheaves $\check{\mathcal{H}}^q$ are called the *Čech cohomology presheaves*. For every object X of C, put
$$\check{H}^q(X, G) = \check{\mathcal{H}}^q(G)(X). \tag{2.4.5.3}$$
The groups $\check{H}^q(X, G)$ are called the *Čech cohomology groups*. When the topology of C is defined by a pretopology, as is most often the case in practice, one has, taking 2.3.4 into account,
$$\check{H}^q(X, G) \simeq \varinjlim_{\mathcal{U}} H^q(\mathcal{U}, G), \tag{2.4.5.4}$$
the inductive limit being taken over the squareable covering families, preordered by the natural order relation on the corresponding sieves (2.3.5).

2.4.6

Let M be a sheaf of A-modules. We set, abusively,
$$\check{H}^q(X, M) = \check{H}^q(X, \mathcal{H}^0(M)), \qquad \check{\mathcal{H}}^q(M) = \check{\mathcal{H}}^q(\mathcal{H}^0(M)). \tag{2.4.6.1}$$
The groups $\check{H}^q(X, M)$ are called the Čech cohomology groups of the sheaf M. Note that, although the functors $\check{H}^q$ are derived functors on the category of presheaves, they do not in general form a δ-functor on the category of sheaves.

2.5. Change of universe. Čech cohomology for $\mathcal{U}$-sites

2.5.1

Let (C, A) be a ringed $\mathcal{U}$-site and let $\mathcal{V}$ be a universe containing $\mathcal{U}$. The site (C, A) is then a $\mathcal{V}$-site, and one has a $\mathcal{U}$-topos $C^{\sim}_{\mathcal{U}}$, a $\mathcal{V}$-topos $C^{\sim}_{\mathcal{V}}$, and a canonical inclusion functor
$$\epsilon : C^{\sim}_{\mathcal{U}} \hookrightarrow C^{\sim}_{\mathcal{V}}.$$
The functor ϵ is exact and fully faithful on categories of modules, and carries injective modules to injective modules (1.9). Hence, for every pair of $\mathcal{U}$-sheaves of A-modules, there are canonical isomorphisms
$$\operatorname{Ext}^q_A(C^{\sim}_{\mathcal{U}}; M, N) \simeq \operatorname{Ext}^q_{\epsilon A}(C^{\sim}_{\mathcal{V}}; \epsilon M, \epsilon N), \qquad q \geq 0. \tag{2.5.1.1}$$

In particular, for every $\mathcal{U}$-sheaf of sets R on C, there are canonical isomorphisms (2.1.1)

$$H^q(R, M) \simeq H^q(\epsilon R, \epsilon M), \qquad q \geq 0, \tag{2.5.1.2}$$

and still more particularly, for every object X of C, one has canonical isomorphisms (2.4.1)

$$H^q(X, M) \simeq H^q(X, \epsilon M). \tag{2.5.1.3}$$

In loose terms, sheaf cohomology therefore does not depend on the choice of universe, and one may always enlarge the universe, when needed for a proof or a construction, in order to compute the cohomology of a sheaf.

2.5.2

Let (C, A) be a ringed $\mathcal{U}$-site and let $\mathcal{V}$ be a universe containing $\mathcal{U}$. Denote by $C^{\sim}_A$ the category of $\mathcal{U}$-sheaves, and let

$$\widehat{\epsilon} : \widehat{C}_{A,\mathcal{U}} \longrightarrow \widehat{C}_{A,\mathcal{V}}$$

be the inclusion functor from $\mathcal{U}$-presheaves into $\mathcal{V}$-presheaves of A-modules. The functor $\widehat{\epsilon}$ is exact; hence the derived functors of

$$\widehat{\epsilon}\mathcal{H}^0 : C^{\sim}_A \longrightarrow \widehat{C}_{A,\mathcal{V}}$$

are the functors $\widehat{\epsilon}\mathcal{H}^q$, $q \geq 0$. By abuse of notation, we again write

$$\mathcal{H}^q : C^{\sim}_A \longrightarrow \widehat{C}_{A,\mathcal{V}}, \qquad q \geq 0,$$

for the functors $\widehat{\epsilon}\mathcal{H}^q$.

When C is $\mathcal{V}$-small, this enlargement of the universe has the following advantage. The category $\widehat{C}_{\mathcal{U}}$ is not in general a $\mathcal{U}$-topos, and the $\mathcal{U}$-presheaves of A-modules are not necessarily submodules of injective $\mathcal{U}$-presheaves. By contrast, the category of $\mathcal{V}$-presheaves is a topos, a $\mathcal{V}$-topos, and therefore every $\mathcal{V}$-presheaf of A-modules is a subobject of a $\mathcal{V}$-injective presheaf (0.1.1). Thus, for every $\mathcal{V}$-presheaf of sets R and, in particular, when R is a $\mathcal{U}$-presheaf, and for every $\mathcal{U}$-sheaf of A-modules M, the groups

$$H^q(R, \mathcal{H}^q(M))$$

are defined by 2.1.1, and it follows from (2.5.1.2) that these groups do not depend on the universe $\mathcal{V}$ considered. Similarly, for every pair of integers $p, q \geq 0$, the presheaves

$$\check{\mathcal{H}}^p(\mathcal{H}^q(M))$$

are defined by 2.4.5, and it follows from (2.5.1.2) and (2.4.5.4) that these presheaves do not depend on the universe $\mathcal{V}$ used to define them.

3 The Cartan–Leray spectral sequence relative to a covering

Proposition 3.1. Let (C, A) be a ringed $\mathcal{U}$-site and let $\mathcal{V}$ be a universe containing $\mathcal{U}$. The functor

$$\mathcal{H}^0 : C^{\sim}_A \longrightarrow \widehat{C}_{A,\mathcal{V}}$$

of 2.5.2 carries injective A-modules to injective presheaves. For every integer $q > 0$ and every A-module M, the sheaf associated with the presheaf $\mathcal{H}^q(M)$ is zero.

Let $\underline{a}_{\mathcal{V}}$ denote the associated-sheaf functor for $\mathcal{V}$-presheaves, and let

$$\epsilon : C^{\sim}_{A} \hookrightarrow C^{\sim}_{A,\mathcal{V}}$$

be the inclusion functor from $\mathcal{U}$-sheaves into $\mathcal{V}$-sheaves. One has

$$\underline{a}_{\mathcal{V}}\mathcal{H}^0 = \epsilon$$

(II, 3.6), and since the functors $\underline{a}_{\mathcal{V}}$ and ϵ are exact,

$$\underline{a}_{\mathcal{V}}\mathcal{H}^q = 0$$

for every integer $q > 0$. For every $\mathcal{U}$-sheaf M and every $\mathcal{V}$-presheaf N, there is a functorial isomorphism

$$\mathrm{Hom}_{\widehat{C}_{A,\mathcal{V}}}(N, \mathcal{H}^0(M)) \simeq \mathrm{Hom}_{C^{\sim}_{A,\mathcal{V}}}(\underline{a}_{\mathcal{V}}N, \epsilon M).$$

When M is injective, ϵM is injective (1.9), and since $\underline{a}_{\mathcal{V}}$ is exact, the functor

$$N \longmapsto \mathrm{Hom}_{\widehat{C}_{A,\mathcal{V}}}(N, \mathcal{H}^0(M))$$

is exact. Hence $\mathcal{H}^0(M)$ is injective.

Theorem 3.2. Let (C, A) be a ringed $\mathcal{U}$-site, let R be a $\mathcal{U}$-presheaf of sets on C, and let M be a sheaf of A-modules. There exists a spectral sequence, functorial in R and in M:

$$H^{p+q}(\underline{a}R, M) \Longleftarrow E_2^{p,q} = H^p(R, \mathcal{H}^q(M)). \tag{3.2.1}$$

When C is not $\mathcal{U}$-small, the term $H^p(R, \mathcal{H}^q(M))$ is to be interpreted as the cohomology of the presheaf $\mathcal{H}^q(M)$ in the topos $\widehat{C}_{\mathcal{V}}$, where $\mathcal{V}$ is a universe containing $\mathcal{U}$ and such that C is $\mathcal{V}$-small (2.5.2).

By the definition of the associated-sheaf functor (cf. II), there is an isomorphism of functors

$$H^0(\underline{a}R, M) \simeq H^0(R, \mathcal{H}^0(M)).$$

The functor $\mathcal{H}^0$ carries injective objects to injective objects. The spectral sequence of composite functors (0.3) is the desired spectral sequence.

Corollary 3.3. Let X be an object of C and let

$$\mathcal{U} = (U_i \to X)_{i \in I}$$

be a covering family such that, for every $i \in I$, the morphism $U_i \to X$ is squareable. Then there is a spectral sequence, called the Cartan–Leray spectral sequence:

$$H^{p+q}(X, M) \Longleftarrow E_2^{p,q} = H^p(\mathcal{U}, \mathcal{H}^q(M)). \tag{3.3.1}$$

Let $R \hookrightarrow X$ be the sieve generated by $\mathcal{U}$. Since this sieve is covering, the sheaf associated with R is the sheaf associated with X (II, 5.2). Taking 2.4.1 into account, one has

$$H^{p+q}(\underline{a}R, M) = H^{p+q}(X, M).$$

The corollary follows from 2.3.4.

Corollary 3.4. There exists a spectral sequence, functorial in the sheaf M and in the object X of C,

$$H^{p+q}(X,M) \Longleftarrow E_2^{p,q} = \check{H}^p(X,\mathcal{H}^q(M)). \tag{3.4.1}$$

As X varies in C, this spectral sequence gives a spectral sequence of presheaves:

$$\mathcal{H}^{p+q}(M) \Longleftarrow E_2^{p,q} = \check{\mathcal{H}}^p(\mathcal{H}^q(M)). \tag{3.4.2}$$

These spectral sequences give functorial edge morphisms

$$\Phi^q(M) : \check{\mathcal{H}}^q(M) \longrightarrow \mathcal{H}^q(M), \tag{3.4.3}$$

$$\Phi^q_X(M) : \check{H}^q(X,M) \longrightarrow H^q(X,M). \tag{3.4.4}$$

The morphisms Φ^q and Φ^q_X are isomorphisms for $q=0$ or $q=1$. The morphisms Φ^2 and Φ^2_X are monomorphisms. More generally, if the presheaves $\mathcal{H}^i(M)$ vanish for $0<i<n$, then $\Phi^q(M)$ and $\Phi^q_X(M)$ are isomorphisms for $0 \le q \le n$ and monomorphisms for $q=n+1$.

The first spectral sequence is obtained by taking the inductive limit in the spectral sequence (3.2.1) over covering sieves $R \hookrightarrow X$, using (2.4.5.2). The second spectral sequence follows immediately from (2.4.5.3). The sheaves associated with the presheaves $\mathcal{H}^q(M)$ are zero for $q>0$ (3.1). Hence

$$\check{\mathcal{H}}^0(\mathcal{H}^q(M)) = 0 \qquad (q>0)$$

(II, 3.4 and II, 3.2). The assertions on the morphisms Φ^q and Φ^q_X follow at once.

Corollary 3.5. Let (E,A) be a ringed topos and let M be an A-module, a left module to fix ideas. Denote by $\mathcal{M}$ the underlying abelian sheaf of M. The functor

$$M \longmapsto \mathcal{M}$$

is exact, and therefore, for every object X of E, the canonical isomorphism

$$H^0(X,M) \xrightarrow{\sim} H^0(X,\mathcal{M})$$

extends to morphisms

$$H^q(X,M) \longrightarrow H^q(X,\mathcal{M}), \qquad q \ge 0.$$

These morphisms are isomorphisms.

For every object Y of E, by (2.4.5.4) one has

$$\check{H}^p(Y,M) = \varinjlim_{\mathcal{U}} H^p(\mathcal{U},M), \qquad \check{H}^p(Y,\mathcal{M}) = \varinjlim_{\mathcal{U}} H^p(\mathcal{U},\mathcal{M}),$$

where the inductive limit is taken over epimorphic families

$$\mathcal{U} = (Y_i \to Y)_{i\in I}.$$

It follows that the canonical homomorphism

$$\check{H}^p(Y,M) \longrightarrow \check{H}^p(Y,\mathcal{M})$$

is an isomorphism (2.3.4). Hence, by (2.4.5.3), the canonical homomorphism

$$\check{\mathcal{H}}^p(M) \longrightarrow \check{\mathcal{H}}^p(\mathcal{M})$$

is an isomorphism. If M is an injective A-module, one has $\check{\mathcal{H}}^p(\mathcal{M}) = 0$ for $p > 0$. Thus $\mathcal{H}^1(\mathcal{M}) = 0$, and by induction on p, using 3.4, $\mathcal{H}^p(\mathcal{M}) = 0$ for $p > 0$. Hence $H^p(X, \mathcal{M}) = 0$ for $p > 0$, by (2.4.2.2) and 2.5. Consequently the functor $M \mapsto \mathcal{M}$ carries injective objects to objects acyclic for $H^0(X, -)$, which gives the announced isomorphism.

Exercise 3.6. Let G be a topological group and let B_G be its classifying topos (IV, 2.5). Let E_G denote the object of B_G formed by the underlying topological space of G, endowed with the left translation action of the elements of G. The canonical morphism from E_G to the final object e_G of B_G is an epimorphism; it therefore gives a covering

$$\mathcal{U} = (E_G \to e_G).$$

For every abelian sheaf F on B_G, there is a spectral sequence

$$E_2^{p,q} = H^p(\mathcal{U}, \mathcal{H}^q(F)) \Longrightarrow H^{p+q}(B_G, F), \tag{3.6.1}$$

which we propose to study.

1) Show that, for every integer n, the topos

$$B_{G/E_G \times E_G \times \cdots \times E_G} \qquad (n \text{ factors})$$

is canonically equivalent to the topos (IV, 2.5)

$$\mathbf{Top}(G \times G \times \cdots \times G) \qquad (n-1 \text{ factors}).$$

For every integer n, denote by F_n the sheaf on the topological space $G \times \cdots \times G$ (n factors) induced by F.

2) Deduce that the term $E_2^{p,q}$ of (3.6.1) is canonically isomorphic to the p-th cohomology group of a complex of the form

$$H^q(e, F_0) \longrightarrow H^q(G, F_1) \longrightarrow H^q(G \times G, F_2) \longrightarrow \cdots,$$

which should be made explicit. Here e denotes the one-point topological space.

3) Deduce that the cohomology of the classifying topos B_G with values in locally constant sheaves is isomorphic to the corresponding singular cohomology of the spaces $G \times G \times \cdots \times G$, in cases where the cohomology of locally constant sheaves coincides with the corresponding singular cohomology, for example when G is locally contractible. For classifying spaces, consult [4].

4 Acyclic sheaves

Definition 4.1. Let (E, A) be a ringed topos, let F be an A-module, and let S be a topologically generating family of E. We say that F is S-*acyclic* if, for every object X of S and every integer $q > 0$,

$$H^q(X, F) = 0.$$

When $S = \operatorname{Ob} E$, the S-acyclic sheaves are called *flasque* sheaves.

4.2

Let (C, A) be a ringed $\mathcal{U}$-site and let F be a sheaf of A-modules on C. We say that F is C-*acyclic* if F is S-acyclic, where S is the family of sheaves associated with the objects of C.

Proposition 4.3. Let (C, A) be a ringed $\mathcal{U}$-site and let F be a sheaf of A-modules. Let

$$\mathcal{H}^0 : C_A^\sim \longrightarrow \widehat{C}_A$$

be the inclusion functor from A-modules to presheaves of A-modules; here these are $\mathcal{V}$-presheaves, with $\mathcal{V}$ a universe such that C is $\mathcal{V}$-small. The following conditions are equivalent:

(i) F is C-acyclic.

(ii) For every object X of C and every covering sieve $R \hookrightarrow X$,

$$H^q(R, \mathcal{H}^0(F)) = 0 \qquad \text{for all } q > 0.$$

(iii) For every object X of C,

$$\check{H}^q(X, F) = 0 \qquad \text{for all } q > 0.$$

(i) $\Rightarrow$ (ii) Since F is C-acyclic, the presheaves $\mathcal{H}^q(F)$ vanish for $q > 0$. The spectral sequence (3.2.1) then gives an isomorphism

$$H^q(R, \mathcal{H}^0(F)) \simeq H^q(X, F)$$

for every q. Hence $H^q(R, \mathcal{H}^0(F)) = 0$ for $q > 0$.

(ii) $\Rightarrow$ (iii) This is clear by passing to the inductive limit.

(iii) $\Rightarrow$ (i) One proves by induction on q that the presheaves $\mathcal{H}^q(F)$ vanish for $q > 0$, using (3.4.2).

4.4

It follows from criterion 4.3(ii) and from 3.5 that the property of being S-acyclic depends only on the underlying abelian sheaf. In particular, a sheaf of A-modules is flasque if and only if its underlying abelian sheaf is flasque.

Corollary 4.5. Let (E, A) be a ringed topos and let F be an A-module. The following properties are equivalent:

(i) F is flasque.

(ii) For every epimorphic family

$$\mathfrak{X} = (X_i \to X)_{i \in I},$$

one has

$$H^q(\mathfrak{X}, F) = 0 \qquad \text{for all } q > 0.$$

In 4.2, take C to be the topos E itself. The corollary then follows from the equivalence (i) $\Leftrightarrow$ (ii) of 4.2.

4.6

Injective sheaves are flasque. Flasque sheaves are S-acyclic for every topologically generating family S. Flasque sheaves are not necessarily injective, as is seen by taking E to be the topos of sets. S-acyclic sheaves are not necessarily flasque; see Exercise 4.15.

Proposition 4.7. Let (E, A) be a topos, let F be a flasque A-module, and let X be an object of E. For every subobject Y of X, the canonical homomorphism

$$H^0(X, F) \longrightarrow H^0(Y, F)$$

is surjective.

Suppose that Y is a subobject of X for which the morphism $H^0(X, F) \to H^0(Y, F)$ is not surjective. Let Z be the object obtained by gluing two copies X_1 and X_2 of X along Y. The object Z is covered by the two subobjects X_1 and X_2, and

$$X_1 \times_Z X_2 = Y.$$

Let $\mathfrak{X} = (X_1, X_2)$ be the resulting covering. One immediately sees that

$$H^1(\mathfrak{X}, F) \neq 0,$$

a contradiction.

4.8

The property described in 4.7 does not characterize flasque sheaves in the case of general topoi; see Exercise 4.15. It does characterize them, however, for topoi generated by their open subobjects, and in particular for topoi associated with topological spaces; see Exercise 4.16. In the case of topological spaces, the terminology *flasque* used here agrees with the terminology of [2]; see Exercise 4.16.

Proposition 4.9. Let

$$u : (E, A) \longrightarrow (E', A')$$

be a morphism of ringed topoi.

1) The functor u_* carries flasque A-modules to flasque A'-modules.

2) Let S and S' be topologically generating families of E and E' respectively, such that $u^*(S') \subset S$. The functor u_* carries S-acyclic A-modules to S'-acyclic A'-modules.

3) If u is a flat morphism (1.8), then u_* carries injective A-modules to injective A'-modules.

Let X be an object of E', let

$$\mathfrak{X} = (X_i \to X)_{i \in I}$$

be an epimorphic family, let F be a flasque A-module, and let $C^\bullet(\mathfrak{X}, u_*F)$ be the Čech complex of the covering $\mathfrak{X}$. There is a canonical isomorphism

$$C^\bullet(\mathfrak{X}, u_*F) \simeq C^\bullet(u^*(\mathfrak{X}), F),$$

using the adjunction between u_* and u^* and the fact that u^* commutes with fiber products. Moreover, u^* commutes with inductive limits, hence

$$u^*(\mathfrak{X}) = (u^* X_i \to u^* X)_{i \in I}$$

is an epimorphic family. Since F is flasque, one has

$$H^q(u^*(\mathfrak{X}), F) = 0 \qquad (q > 0),$$

and consequently

$$H^q(\mathfrak{X}, u_* F) = 0 \qquad (q > 0).$$

Thus $u_* F$ is flasque.

We prove 2). If S' is stable under fiber products, a proof analogous to the preceding one proves 2). In the general case we shall use the spectral sequence of the morphism u (5.3.2). Assertion 2) will not be used before 5.3.2. Let F be an S-acyclic sheaf. The sheaves $R^q u_* F$ are the sheaves associated with the presheaves

$$X \longmapsto H^q(u^* X, F).$$

Since S' is a topologically generating family and F is S-acyclic, one has $R^q u_* F = 0$ for all $q > 0$. The spectral sequence (5.3.2) then gives an isomorphism, for every object X of E',

$$H^q(X, u_* F) \simeq H^q(u^* X, F),$$

and hence for every X in S' one has $H^q(X, u_* F) = 0$.

We prove 3). If the morphism u is flat, the functor u^* on modules is exact (1.8). Hence its right adjoint u_* carries injective objects to injective objects (0.2).

Proposition 4.10. Let F be an A-module of the ringed topos (E, A), and let G be an injective A-module.

1) The functor

$$F \longmapsto \mathcal{H}om_A(F, G)$$

is exact.

2) The abelian sheaf $\mathcal{H}om_A(F, G)$ is flasque.

Proof. For 1), let

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

be an exact sequence. We must show that

$$0 \longrightarrow \mathcal{H}om_A(F'', G) \longrightarrow \mathcal{H}om_A(F, G) \longrightarrow \mathcal{H}om_A(F', G) \longrightarrow 0$$

is exact. It is enough to show that, for every object H of E, the sequence

$$\begin{aligned} 0 \longrightarrow \mathrm{Hom}_E(H, \mathcal{H}om_A(F'', G)) &\longrightarrow \mathrm{Hom}_E(H, \mathcal{H}om_A(F, G)) \\ &\longrightarrow \mathrm{Hom}_E(H, \mathcal{H}om_A(F', G)) \longrightarrow 0 \end{aligned}$$

is exact. But by (IV, 12.6) this last sequence is isomorphic to

$$0 \to \mathrm{Hom}_A(A_H \otimes_A F'', G) \to \mathrm{Hom}_A(A_H \otimes_A F, G) \to \mathrm{Hom}_A(A_H \otimes_A F', G) \to 0,$$

and the A-module A_H is flat (1.3.1).

For 2), let

$$\mathfrak{X} = (X_i \to X)_{i \in I}$$

be an epimorphic family of E, and let

$$C_\bullet(\mathfrak{X}) : \cdots \quad \begin{matrix}\longrightarrow\\[-6pt]\longrightarrow\\[-6pt]\longrightarrow\end{matrix} \quad \coprod_{i,j} \mathbb{Z}_{X_i \times_X X_j} \quad \begin{matrix}\longrightarrow\\[-6pt]\longrightarrow\end{matrix} \quad \coprod_i \mathbb{Z}_{X_i}$$

be the complex defined in 1.10. This complex is a flat resolution of $\mathbb{Z}_X$, which is itself flat (1.3.1).

Compute

$$H^q(\mathfrak{X}, \mathcal{H}om_A(F, G)) \simeq H^q(\mathrm{Hom}^\bullet_{\mathbb{Z}}(C_\bullet(\mathfrak{X}), \mathcal{H}om_A(F, G))) \simeq H^q(\mathrm{Hom}_A(C_\bullet(\mathfrak{X}) \otimes_{\mathbb{Z}} F, G))\,.$$

The complex

$$C_\bullet(\mathfrak{X}) \otimes_{\mathbb{Z}} F$$

is acyclic in degrees different from 0, since $C_\bullet(\mathfrak{X})$ is a flat resolution of a flat module. Therefore

$$\mathrm{Hom}_A(C_\bullet(\mathfrak{X}) \otimes_{\mathbb{Z}} F, G)$$

is acyclic in nonzero degrees.

Proposition 4.11. Let (E, A) be a ringed topos, and let F be a flasque (respectively injective) sheaf of A-modules.

1) For every object X of E, the $A|X$-module $j_X^* F$ is flasque (respectively injective).

2) For every closed subtopos Z of E, the sheaf of sections of F with support in Z *(IV, 14)* is flasque (respectively injective).

Assertion 1) follows from 4.5 when F is flasque. The functor j_X^* has an exact left adjoint $j_{X!}$ (IV, 11); therefore, when F is injective, $j_X^* F$ is injective (0.2). We prove 2). Let $i : Z \to E$ denote the inclusion morphism. The sheaf of sections of F with support in Z is the sheaf $i_* i^! F$ (IV, 14). The functor $i_* i^!$ is right adjoint to the exact functor $i_* i^*$ (IV, 14); hence it carries injective sheaves to injective sheaves.

Let U be the open complement of Z, let $j : U \to E$ be the inclusion morphism, and let F be a flasque sheaf. There is an exact sequence (IV, 14)

$$0 \longrightarrow i_* i^! F \longrightarrow F \longrightarrow j_* j^* F. \tag{4.11.1}$$

For every object X of E, one has

$$j_* j^* F(X) = F(X \times U),$$

and the morphism $F(X) \to j_* j^* F(X)$ induced by the last arrow of (4.11.1) comes from the canonical injection $X \times U \hookrightarrow X$. Since F is flasque, this morphism is surjective (4.7), and hence the last arrow of (4.11.1) is an epimorphism of presheaves. For every object X of E, the long exact cohomology sequence deduced from (4.11.1) gives

$$H^q(X, i_* i^! F) = 0 \qquad (q > 0),$$

and therefore $i_* i^! F$ is flasque.

4.12

The property that a sheaf be flasque (respectively injective) localizes (4.11.1). But it is not in general a local property; see Exercise 4.15. It is, however, a local property for topoi generated by their open subobjects, and in particular for topoi associated with topological spaces; see Exercise 4.16.

Exercise 4.13. Let X be a locally compact space and let F be a c-soft sheaf [2]. Using the local nature of softness (*loc. cit.*), show that the restriction of F to every paracompact open subset of X is a soft sheaf. Show that the paracompact open subsets form a basis for the topology of X. Deduce, denoting by S the family of paracompact open subsets of X, that the sheaf F is S-acyclic. Show that the sheaf of continuous functions on X is S-acyclic but is not flasque when X is not discrete.

Problem 4.14. Study totally acyclic topoi, i.e. topoi such that

$$H^q(X, F) = 0$$

for every $q > 0$, every object X, and every abelian sheaf F.

Exercise 4.15. Let G be a discrete group and let B_G be its classifying topos (IV, 2.4). Show that for every abelian sheaf F and every monomorphism $X \hookrightarrow Y$, the homomorphism

$$F(Y) \longrightarrow F(X)$$

is surjective. Let $E(G)$ be the group G, regarded as a homogeneous space under itself. It is an object of B_G. The topos $B_{G/E(G)}$ is equivalent to the punctual topos (IV, 8). The morphism $E(G) \to e$, where e is the final object of B_G, is an epimorphism. For every abelian sheaf F on B_G, the sheaf $F|E(G)$ is flasque. Deduce that the property of being flasque or injective is not local.

Exercise 4.16. A topos E is said to be *generated by its open subobjects* if the open subobjects of E, i.e. the subobjects of the final object e of E, form a generating family (I, 7). Such a topos has the following property:

(P) Every epimorphic family $X_i \to X$, $i \in I$, is refined by an epimorphic family $U_j \to X$, $j \in J$, where the $U_j \to X$ are monomorphisms.

a) Are there topoi satisfying property (P) which are not generated by their open subobjects? Topoi associated with topological spaces are generated by their open subobjects. If E is generated by its open subobjects (respectively has property (P)), then for every object X of E, the topos $E_{/X}$ is generated by its open subobjects (respectively has property (P)). Every subtopos of a topos generated by its open subobjects is generated by its open subobjects. Property (P) is not a local property.

b) Let E be a topos and let $\mathrm{Open}(E)$ be the category of open subobjects of E endowed with the induced topology. The inclusion functor $\mathrm{Open}(E) \to E$ is a morphism of sites, hence gives a morphism of topoi

$$\Pi : E \longrightarrow \mathrm{Open}(E)^{\sim}.$$

The functor Π^* is fully faithful. The morphism Π has, with respect to the 2-category of topoi generated by their open subobjects, a universal property which the reader should make explicit.

c) Let E be a topos generated by its open subobjects, and let $\mathrm{Point}(E)$ be the set of isomorphism classes of points of E. Show that $\mathrm{Point}(E)$ is small. Put a topology on $\mathrm{Point}(E)$ and define a morphism

$$\mathbf{Top}(\mathrm{Point}(E)) \longrightarrow \mathrm{Open}(E)^{\sim}$$

making $\mathbf{Top}(\mathrm{Point}(E))$ a subtopos of $\mathrm{Open}(E)^{\sim}$. For every topological space X, show that every morphism from $\mathbf{Top}(X)$ to E gives a morphism from $\mathbf{Top}(X)$ to $\mathbf{Top}(\mathrm{Point}(E))$. Show that a topos generated by its open subobjects is equivalent to a topos $\mathbf{Top}(X)$, with X a topological space, if and only if it has enough points. Show that there exist topoi generated by their open subobjects which are not equivalent to topoi $\mathbf{Top}(X)$ for any topological space X.

d) Let E be a topos satisfying property (P). An abelian sheaf F on E is flasque if and only if, for every monomorphism $X \hookrightarrow Y$, the homomorphism

$$F(Y) \longrightarrow F(X)$$

is surjective.

e) Let E be a topos satisfying property (P). Show that the property that a sheaf F be flasque is local in nature.

f) Let E be a topos generated by its open subobjects, and let e be the final object of E. An abelian sheaf F is flasque if and only if, for every open subobject U of e, the canonical morphism

$$F(e) \longrightarrow F(U)$$

is surjective.

Exercise 4.17 (flasque sheaves and change of universe). Let C be a $\mathcal{U}$-site, for example a $\mathcal{U}$-topos, and let $\mathcal{V}$ be a universe containing $\mathcal{U}$. Let

$$\epsilon : C^{\sim}_{\mathcal{U}} \longrightarrow C^{\sim}_{\mathcal{V}}$$

be the canonical inclusion between the corresponding categories of sheaves. Let F be a flasque $\mathcal{U}$-abelian sheaf on C. The aim is to show that ϵF is flasque. First observe that for every object X of $C^{\sim}_{\mathcal{U}}$ one has

$$H^q(\epsilon X, \epsilon F) = 0 \qquad (q > 0).$$

Every object Y of $C^{\sim}_{\mathcal{V}}$ admits an epimorphic family

$$\epsilon X_i \to Y, \qquad i \in I,$$

where the morphisms $\epsilon X_i \to Y$ are monomorphisms. It follows that

$$H^q(Y, \epsilon F) = \varprojlim_{\epsilon X \hookrightarrow Y}{}^{q} F(X).$$

To show that these $\varprojlim^q$ vanish for $q \neq 0$, one may reduce to the topos $\mathrm{Open}(C_{\mathcal{V}/Y})^{\sim}$ and use the fact that, for this topos, a sheaf is flasque if it is locally flasque.

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Continuation anchor. The next translated segment should begin with Exposé V, Section 5, "The $R^q u_*$ and the Cartan–Leray spectral sequence relative to a morphism of topoi", source line 1947 of `05/05.tex`.

SGA 4, Exposé V. Cohomology in Topoi

Draft translation, Sections 5–8

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Translator's status note. This fresh-only continuation covers Exposé V, Sections 5–8, corresponding to source lines 1947–4410 of the Orgogozo–Laszlo TeX snapshot. Terminology is normalized to current mathematical English: *ringed topos*, *direct image*, *Cartan–Leray spectral sequence*, *local Ext*, *cohomology with supports*, *family of supports*, *semi-simplicial object*, *skeleton/coskeleton*, and *hypercovering*. Numbering follows the source; several evident typographical slips in displayed formulae have been silently normalized.

5 The functors $R^q u_*$ and the Cartan–Leray spectral sequence relative to a morphism of topoi

5.0

Let

$$u : (E, A) \longrightarrow (E', A')$$

be a morphism of ringed topoi, and let u_* denote the direct-image functor. The same notation u_* will also denote the induced functor on modules. For modules, say left modules to fix ideas, the functor u_* is left exact. Its right derived functors are denoted

$$R^q u_*, \qquad q \geq 0.$$

Proposition 5.1.

1) For every A-module M, the sheaf $R^q u_* M$ is the sheaf associated with the presheaf

$$X' \longmapsto H^q(u^* X', M), \qquad X' \in \mathrm{Ob}(E').$$

2) The formation of the functors $R^q u_*$ commutes with restriction of scalars.

3) The formation of the functors $R^q u_*$ commutes with localizations. More precisely, for every object X' of E', if

$$u_{/X'} : E_{/u^* X'} \longrightarrow E'_{/X'}$$

denotes the morphism deduced from u by localization *(IV, 8)*, then, for every A-module M, there is a canonical isomorphism

$$R^q (u_{/X'})_* (M | u^* X') \simeq (R^q u_* M) | X', \qquad q \geq 0. \tag{5.1.1}$$

Let

$$u_*^\wedge : E^\wedge \longrightarrow E'^\wedge$$

be the direct-image functor for $\mathcal{U}$-presheaves, so that $u_*^\wedge M = M \circ u^*$. Since u^* and u_* are adjoint, one has an isomorphism

$$u_* \simeq \underline{a}\, u_*^{\wedge 0},$$

where $\underline{a}$ is the associated-sheaf functor for E'. Since the functors $\underline{a}$ and $u_*^\wedge$ are exact, one has

$$R^q u_* \simeq \underline{a}\, u_*^\wedge \mathcal{H}^q,$$

which is another way of stating 1). Assertion 2) then follows from 1) and 3.5. By the definition of the localized morphism $u_{/X'}$, there is the canonical isomorphism (5.1.1) for $q = 0$. The general case follows by observing that localization functors *(IV, 8)* are exact and send injective objects to injective objects (4.11).

Proposition 5.2. Let $u : E \to E'$ be a morphism of topoi, and let S' be a generating family of E'. The sheaves M which are acyclic for the functors

$$H^0(u^* X', -), \qquad X' \in S',$$

are acyclic for the functor u_*. In particular, flasque sheaves are acyclic for u_*.

This follows from 5.1.1).

Proposition 5.3. Let $u : E \to E'$ be a morphism of topoi, and let M be an abelian group object of E. There is a spectral sequence

$$E_2^{p,q} = H^p(E', R^q u_* M) \Longrightarrow H^{p+q}(E, M). \tag{5.3.1}$$

More generally, for every object X' of E', there is a spectral sequence

$$E_2^{p,q} = H^p(X', R^q u_* M) \Longrightarrow H^{p+q}(u^* X', M). \tag{5.3.2}$$

By definition of the direct- and inverse-image functors, there is an isomorphism

$$H^0(X', u_* M) \simeq H^0(u^* X', M).$$

The functor u_* sends injective objects to flasque sheaves (4.6 and 4.9). The spectral sequences displayed above are therefore the spectral sequences of composed functors (0.3).

Proposition 5.4. Let $u : E \to E'$ and $v : E' \to E''$ be two morphisms of topoi. There is a spectral sequence

$$R^p v_*\, R^q u_* \Longrightarrow R^{p+q}(v \circ u)_*.$$

One has $v_* u_* \simeq (vu)_*$, and u_* sends injective objects to flasque sheaves (4.9), hence to objects acyclic for v_* (5.2). Thus this is the spectral sequence of composed functors (0.3).

6 Local Ext and cohomology with supports

6.0

Let (E, A) be a ringed topos, and let M be an A-module, say a left module to fix ideas. The functor

$$N \longmapsto \mathcal{H}om_A(M, N),$$

from left A-modules to abelian groups, is left exact (IV, 12). Its right derived functors are denoted

$$\mathcal{E}xt_A^q(M, N). \tag{6.0.1}$$

In particular,

$$\mathcal{E}xt^0_A(M,N) = \mathcal{H}om_A(M,N). \tag{6.0.2}$$

By definition, one has canonical isomorphisms (2.1 and IV, 12)

$$H^0(E, \mathcal{E}xt^0_A(M,N)) = \mathrm{Ext}^0_A(E; M, N) = \mathrm{Hom}_A(M,N). \tag{6.0.3}$$

More generally, for every object X of E, one has (2.2)

$$H^0(X, \mathcal{E}xt^0_A(M,N)) = \mathrm{Ext}^0_A(X; M, N) = \mathrm{Hom}_{A|X}(M|X, N|X). \tag{6.0.4}$$

Proposition 6.1.

1) The formation of the functors $\mathcal{E}xt^q_A$ commutes with localization. More precisely, for every object X of E one has functorial isomorphisms

$$\mathcal{E}xt^q_A(M,N)|X \simeq \mathcal{E}xt^q_{A|X}(M|X, N|X). \tag{6.1.1}$$

2) The sheaf $\mathcal{E}xt^q_A(M,N)$ is isomorphic to the sheaf associated with the presheaf

$$X \longmapsto \mathrm{Ext}^q_A(X; M, N).$$

3) There is a spectral sequence

$$\mathrm{Ext}^{p+q}_A(E; M, N) \Longleftarrow E_2^{p,q} = H^p(E, \mathcal{E}xt^q_A(M,N)). \tag{6.1.2}$$

More generally, for every object X of E, there is a spectral sequence, functorial in X and in the arguments M and N,

$$\mathrm{Ext}^{p+q}_A(X; M, N) \Longleftarrow E_2^{p,q} = H^p(X, \mathcal{E}xt^q_A(M,N)). \tag{6.1.3}$$

By definition, (6.1.1) is an isomorphism for $q = 0$ (IV, 12). The general case follows from the fact that localization functors are exact and send injective modules to injective modules (2.2). The functor

$$N \longmapsto \mathcal{H}om_A(M,N) = \mathcal{E}xt^0_A(M,N)$$

sends injective modules to flasque sheaves, hence to sheaves acyclic for $H^0(X, -)$ (4.10). The spectral sequences (6.1.2) and (6.1.3) are therefore spectral sequences of composed functors, using (6.0.3) and (6.0.4). When X varies in E, (6.1.3) gives a spectral sequence of presheaves and hence, after passage to associated sheaves, a spectral sequence of sheaves. Since the sheaves associated with the presheaves $X \mapsto H^p(X, -)$ vanish for $p \neq 0$ (3.1), this spectral sequence degenerates and gives the isomorphism asserted in 2).

Proposition 6.2. The functors $(M,N) \mapsto \mathcal{E}xt^q_A(M,N)$, $q \geq 0$, form a δ-functor in the variable M and in the variable N. Explicitly, for every exact sequence $0 \to N' \to N \to N'' \to 0$ (respectively $0 \to M' \to M \to M'' \to 0$), one has long exact sequences

$$\cdots \to \mathcal{E}xt^q_A(M,N') \to \mathcal{E}xt^q_A(M,N) \to \mathcal{E}xt^q_A(M,N'') \xrightarrow{\delta} \mathcal{E}xt^{q+1}_A(M,N') \to \cdots, \tag{6.2.1}$$

respectively

$$\cdots \to \mathcal{E}xt^q_A(M'',N) \to \mathcal{E}xt^q_A(M,N) \to \mathcal{E}xt^q_A(M',N) \xrightarrow{\delta} \mathcal{E}xt^{q+1}_A(M'',N) \to \cdots. \tag{6.2.2}$$

It follows from the general properties of the functors Ext^q that, for every object X of E, the functors $(M,N) \mapsto \mathrm{Ext}^q_A(X; M, N)$, $q \geq 0$, form a δ-functor in each argument. The assertion follows by letting X vary and taking the associated sheaf (6.1).

6.3

Let (E, A) be a ringed topos, let M be an A-module, let Z be a closed subtopos of E *(IV, 9)*, and let U be its open complement. We denote by $H^0_Z(E, M)$ the group of sections of M whose support is contained in Z (IV, 14), and by $\mathcal{H}^0_Z(M)$ the subsheaf of M defined by the sections of M "with support in Z" (IV, 14). The functors $H^0_Z(E, -)$ and $\mathcal{H}^0_Z(-)$ are left exact (IV, 14). Their derived functors are denoted

$$H^q_Z(E, -), \qquad \mathcal{H}^q_Z(-),$$

respectively, and are called the *groups* (respectively *sheaves*) *of cohomology of M with supports in Z*.[1]

One has canonical isomorphisms (IV, 14)

$$H^0_Z(E, M) \simeq \mathrm{Hom}_A(A_Z, M) \simeq \mathrm{Ext}^0_A(E; A_Z, M), \tag{6.3.1}$$

$$\mathcal{H}^0_Z(M) \simeq \mathcal{H}om_A(A_Z, M) \simeq \mathcal{E}xt^0_A(A_Z, M), \tag{6.3.2}$$

and hence isomorphisms, for every $q \geq 0$,

$$H^q_Z(E, M) \simeq \mathrm{Ext}^q_A(E; A_Z, M), \tag{6.3.3}$$

$$\mathcal{H}^q_Z(M) \simeq \mathcal{E}xt^q_A(A_Z, M). \tag{6.3.4}$$

Notice that, since A_Z is a bimodule, the sheaves

$$\mathcal{E}xt^q_A(A_Z, M) \simeq \mathcal{H}^q_Z(M)$$

are endowed canonically with A-module structures.

For every object X of E, denote by $Z_{/X}$ the closed subtopos of $E_{/X}$ deduced from Z by localization; it is the complement of $U \times X$. By definition, one has canonical isomorphisms

$$H^0(X, \mathcal{H}^0_Z(M)) \simeq \mathcal{H}^0_{Z_{/X}}(E_{/X}, M|X). \tag{6.3.5}$$

Put

$$H^q_Z(X, M) = H^q_{Z_{/X}}(E_{/X}, M|X). \tag{6.3.6}$$

Taking account of (6.3.2), one has canonical isomorphisms

$$H^q_Z(X, M) \simeq \mathrm{Ext}^q_A(X; A_Z, M). \tag{6.3.7}$$

Proposition 6.4.

1) The formation of the functors $\mathcal{H}^q_Z$ commutes with localization. More precisely, for every object X of E and every A-module M, one has canonical isomorphisms

$$\mathcal{H}^q_Z(M)|X \simeq \mathcal{H}^q_{Z_{/X}}(M|X). \tag{6.4.1}$$

2) The sheaf $\mathcal{H}^q_Z(M)$ is the sheaf associated with the presheaf

$$X \longmapsto H^q_Z(X, M).$$

[1] Compare SGA 2, Exposé I for the case of ordinary topological spaces, and the exposé of Hartshorne cited later in the original source.

3) There is a spectral sequence

$$H_Z^{p+q}(E, M) \Longleftarrow H^p(E, \mathcal{H}_Z^q(M)). \tag{6.4.2}$$

More generally, for every object X of E, there is a spectral sequence

$$H_Z^{p+q}(X, M) \Longleftarrow H^p(X, \mathcal{H}_Z^q(M)). \tag{6.4.3}$$

This is just Proposition 6.1 translated by means of the dictionary of 6.3.

Proposition 6.5. With the notation of 6.3, let $j : U \to E$ be the canonical morphism. For every A-module M, there is an exact sequence of sheaves

$$0 \to \mathcal{H}_Z^0(M) \to M \to j_*(M|U) \to \mathcal{H}_Z^1(M) \to 0, \tag{6.5.1}$$

and, for $q \geq 2$, isomorphisms

$$\mathcal{H}_Z^q(M) \simeq \mathcal{E}xt_A^{q-1}(A_U, M) \simeq R^{q-1} j_*(M|U). \tag{6.5.2}$$

Moreover, there is a long exact sequence

$$\cdots \to H_Z^q(E, M) \to H^q(E, M) \to H^q(U, M) \to H_Z^{q+1}(E, M) \to \cdots, \tag{6.5.3}$$

Note. This exact sequence makes precise the role of the global cohomological invariants $H_Z^q(E, N)$ as the cohomology groups of E *modulo* the open U, with coefficients in M. and more generally, for every object X of E, there is a long exact sequence

$$\cdots \to H_Z^q(X, M) \to H^q(X, M) \to H^q(X \times U, M) \to H_Z^{q+1}(X, M) \to \cdots. \tag{6.5.4}$$

By the definition of A_Z, there is an exact sequence (IV, 14)

$$0 \to A_U \to A \to A_Z \to 0. \tag{6.5.5}$$

The functors $\mathcal{E}xt_A^q(A, -)$ vanish for $q > 0$. One has

$$\mathcal{H}om_A(A_U, M) \simeq j_*(M|U)$$

(IV, 14), and therefore (2.2)

$$\mathcal{E}xt_A^q(A_U, M) \simeq R^q j_*(M|U).$$

Finally, $\mathcal{E}xt_A^q(A_Z, M) \simeq \mathcal{H}_Z^q(M)$ by (6.3.4). The long exact sequence (6.2.2) gives in this case (6.5.1) and (6.5.2). The exact sequences (6.5.3) and (6.5.4) follow from the dictionary of 6.3 and from the long exact sequence of the δ-functor $\mathrm{Ext}_A^q(X; -, M)$, $q \geq 0$, associated with (6.5.5).

Proposition 6.6.

1) Flasque sheaves are acyclic for the functors $\mathcal{H}_Z^0$ and $H_Z^0(X, -)$.

2) The functors $\mathcal{H}_Z^0$ and $H_Z^0(X, -)$ commute with restriction of scalars.

Let M be a flasque sheaf. The exact sequence (6.5.4) gives equalities $H^q_Z(X, M) = 0$ for every X and every $q \geq 2$, and an exact sequence

$$0 \to H^0_Z(X, M) \to H^0(X, M) \to H^0(X \times U, M) \to H^1_Z(X, M) \to 0.$$

Since M is flasque, the morphism $H^0(X, M) \to H^0(X \times U, M)$ is surjective (4). Hence $H^1_Z(X, M) = 0$, and M is acyclic for $H^0_Z(X, -)$. Passing to associated sheaves, it follows that M is acyclic for the functor $\mathcal{H}^0_Z$ (6.4). It is clear that the functors $\mathcal{H}^0_Z$ and $H^0_Z(X, -)$ commute with restriction of scalars; since flasque sheaves are acyclic for these two functors, assertion 2) follows.

Proposition 6.7. Let (E, A) be a ringed topos, let Z be a closed subtopos of E, let U be its open complement, and let M, N be two A-modules. There are isomorphisms, functorial in M and N and compatible with change of closed subtopos,

$$H^0_Z(\mathcal{H}om_A(M, N)) \simeq \mathcal{H}om_A(M, \mathcal{H}^0_Z(N)) \simeq \mathcal{H}om_A(M \otimes_A A_Z, N).$$

This follows from (IV, 14), taking (6.3.2) into account.

6.8

Put

$$\mathcal{H}om_{A,Z}(M, N) = \mathcal{H}^0_Z(\mathcal{H}om_A(M, N)). \tag{6.8.1}$$

The functor $N \mapsto \mathcal{H}om_{A,Z}(M, N)$ is left exact. Its derived functors are denoted $\mathcal{E}xt^q_{A,Z}(M, N)$; these are the *sheaves $\mathcal{E}xt$ with supports in Z*. Thus

$$\mathcal{E}xt^0_{A,Z}(M, N) = \mathcal{H}om_{A,Z}(M, N). \tag{6.8.2}$$

Put $M_Z = M \otimes_A A_Z$. It follows from IV, 14 that M_Z is the direct image on E of the inverse image of M on Z. Therefore, taking account of 6.7, there are isomorphisms

$$\mathcal{E}xt^q_{A,Z}(M, N) \simeq \mathcal{E}xt^q_A(M_Z, N). \tag{6.8.3}$$

Now pass to global invariants. Put

$$\mathrm{Hom}_{A,Z}(M, N) = \mathrm{Ext}^0_{A,Z}(E; M, N) = H^0_Z(E, \mathcal{H}om_A(M, N)). \tag{6.8.4}$$

The group $\mathrm{Hom}_{A,Z}(M, N)$ is the subgroup of the group of morphisms from M to N whose support is in Z (IV, 14), i.e. which are zero on U. More generally, for every object X of E, put

$$\mathrm{Ext}^0_{A,Z}(X; M, N) = H^0_Z(X, \mathcal{H}om_A(M, N)). \tag{6.8.5}$$

The functors $N \mapsto \mathrm{Ext}^0_{A,Z}(X; M, N)$ are left exact. Their derived functors are denoted $\mathrm{Ext}^q_{A,Z}(X; M, N)$. These are the *groups* Ext *with supports in Z*. The definitions (6.3.6), (6.8.1), and (6.8.5), together with the isomorphisms of 6.7, give isomorphisms

$$\mathrm{Ext}^0_{A,Z}(X; M, N) \simeq \begin{cases} H^0(X, \mathcal{H}om_{A,Z}(M, N)), \\ \mathrm{Ext}^0(X; M, \mathcal{H}^0_Z(N)), \\ \mathrm{Ext}^0(X; M_Z, N). \end{cases} \tag{6.8.6}$$

The last isomorphism in (6.8.6) gives isomorphisms

$$\operatorname{Ext}^q_{A,Z}(X; M, N) \simeq \operatorname{Ext}^q_A(X; M_Z, N). \tag{6.8.7}$$

Proposition 6.9.

1) There are two spectral sequences, functorial in M and N and compatible with change of closed subtopos,

$$\mathcal{E}xt^{p+q}_{A,Z}(M,N) \Longleftarrow \begin{cases} E_2^{p,q} = \mathcal{H}^p_Z(\mathcal{E}xt^q_A(M,N)), \\ {}'E_2^{p,q} = \mathcal{E}xt^p_A(M, \mathcal{H}^q_Z(N)). \end{cases} \tag{6.9.1}$$

2) There are three spectral sequences, functorial in X, M, N and compatible with change of closed subtopos,

$$\operatorname{Ext}^{p+q}_{A,Z}(X; M, N) \Longleftarrow \begin{cases} E_2^{p,q} = H^p_Z(X, \mathcal{E}xt^q_A(M,N)), \\ {}'E_2^{p,q} = H^p(X, \mathcal{E}xt^q_{A,Z}(M,N)), \\ {}''E_2^{p,q} = \operatorname{Ext}^p_A(X; M, \mathcal{H}^q_Z(N)). \end{cases} \tag{6.9.2}$$

3) The sheaves $\mathcal{E}xt^q_{A,Z}(M,N)$ are canonically isomorphic to the sheaves associated with the presheaves
$$X \longmapsto \operatorname{Ext}^q_{A,Z}(X; M, N).$$

When N is injective, the sheaf $\mathcal{H}om_A(M,N)$ is flasque (4.10), hence acyclic for $\mathcal{H}^0_Z$ (6.6). The first spectral sequence in (6.9.1) is the spectral sequence of the composed functors in (6.8.1). Likewise, when N is injective, $\mathcal{H}^0_Z(N)$ is injective (4.11), and the second spectral sequence in (6.9.1) is the spectral sequence of composed functors deduced from 6.7. The spectral sequences in (6.9.2) are spectral sequences of composed functors deduced from the definition (6.8.5) and from the first two isomorphisms of (6.8.6). Finally, letting X vary in the second spectral sequence of (6.9.2) and taking associated sheaves, one obtains a spectral sequence of sheaves which degenerates by 3.1 and gives the isomorphisms of 3).

Proposition 6.10. The functors $(M,N) \mapsto \mathcal{E}xt^q_{A,Z}(M,N)$, $q \geq 0$, and $(M,N) \mapsto \operatorname{Ext}^q_{A,Z}(X; M, N)$, $q \geq 0$, are δ-functors in each of the variables M and N. If M_U denotes the sheaf $M \otimes_A A_U$ *(cf. IV, 11)*, then there is a long exact sequence

$$\cdots \xrightarrow{\delta} \mathcal{E}xt^q_{A,Z}(M,N) \to \mathcal{E}xt^q_A(M,N) \to \mathcal{E}xt^q_A(M_U,N) \xrightarrow{\delta} \mathcal{E}xt^{q+1}_{A,Z}(M,N) \to \cdots, \tag{6.10.1}$$

and an analogous exact sequence for the groups Ext.

The functors $\mathcal{E}xt^q_A(-,-)$ and $\operatorname{Ext}^q_A(X;-,-)$ form δ-functors in each variable (6.2), and the functor $M \mapsto M_Z$ is exact because A_Z is flat (1.3.3). The first assertion follows from the isomorphisms (6.8.3) and (6.8.7). The exact sequence (6.10.1), and the analogous exact sequence for the groups Ext, is the long exact sequence of the δ-functor

$$\mathcal{E}xt^q_A(-,N), \quad q \geq 0 \quad (\textit{respectively } \operatorname{Ext}^q_A(X;-,N), \quad q \geq 0)$$

relative to the exact sequence

$$0 \to M_U \to M \to M_Z \to 0,$$

taking (6.8.3) and (6.8.7) into account.

6.11

Let us briefly indicate how these results may be extended to the case of families of supports. Let E be a topos. For every object X of E, denote by $\mathrm{Fer}(X)$ the set of closed subtopoi of $E_{/X}$. This set is in bijection, by passage to complements, with the set of open subtopoi of $E_{/X}$, i.e. with the set of subobjects of X (IV, 8). The presheaf $\mathrm{Fer} : X \mapsto \mathrm{Fer}(X)$ is in fact a $\mathcal{U}$-sheaf, as is checked immediately; hence it is representable. In other words, there is an object of E, again denoted Fer, and a closed subtopos Z_{Fer} of $E_{/\,\mathrm{Fer}}$ such that, for every X, every element Z of $\mathrm{Fer}(X)$ is deduced from Z_{Fer} by base change along a morphism

$$u_Z : X \to \mathrm{Fer}$$

uniquely determined by Z.

Definition 6.12. A *family of supports* of E is a subset Φ of the set of closed subtopoi of E having the following properties:

(S1) the union of a finite family of elements of Φ belongs to Φ;

(S2) every closed subtopos of E contained in an element of Φ is an element of Φ.

6.12.1

Let E be a topos, let Φ be a family of supports of E, and let X be an object of E. Denote by $\Phi(X)$ the smallest family of supports of $E_{/X}$ containing the closed subtopoi of $E_{/X}$ deduced from closed subtopoi in Φ by the base change $X \to e$, where e is a final object of E. The functor $X \mapsto \Phi(X)$ is a subpresheaf of the sheaf Fer, hence is separated.

6.12.2

A family Φ of supports of E is said to be *of local character* if it has the following property:

(CL) For every epimorphic family $(X_i \to e)_{i\in I}$, where e is the final object of E, the sequence of sets

$$\Phi \longrightarrow \prod_i \Phi(X_i) \rightrightarrows \prod_{i,j} \Phi(X_i \times X_j)$$

is exact.

Let $a\Phi$ denote the sheaf associated with the presheaf $X \mapsto \Phi(X)$ (6.12.1). Condition (CL) is equivalent to the condition that the canonical morphism

$$\Phi \to a\Phi(e)$$

be a bijection (cf. the construction of the sheaf associated with a separated presheaf in Exposé II). To verify (CL), one may therefore restrict to a final family of epimorphic families $(X_i \to e)_{i\in I}$.

Examples 6.12.3.

1) Let Z be a closed subtopos of E. The set of closed subtopoi of E contained in Z is a family of supports of E. It is of local character.

2) Let T be a topological space and let Φ be a paracompactifying family of supports on T [7]. In general, Φ is not of local character.

3) Let T be a topological space and let p be an integer. The family Φ_p of closed subsets of T of Krull codimension $\geq p$ is a family of supports of T. It is of local character.

4) Suppose E is a topos with the following property: every epimorphic family $(X_i \to e)_{i\in I}$ is refined by a finite epimorphic family. Then every family of supports of E is of local character. Indeed, using 1), it is enough to show that a filtered inductive limit Φ_λ of families of local character is a family of local character. This follows immediately by passing to the inductive limit in the exact sequence of sets

$$\Phi_\lambda \to \prod_i \Phi_\lambda(X_i) \rightrightarrows \prod_{i,j} \Phi_\lambda(X_i \times X_j),$$

where $(X_i \to e)_{i\in I}$ is a finite epimorphic family. By 6.12.2 and the hypothesis on E, one may restrict to finite epimorphic families to verify (CL), and filtered inductive limits commute with finite projective limits (I, 2); hence exactness of these sequences of sets is preserved by passage to the inductive limit.

5) If E is a coherent topos (VI, 2.3), then, by the preceding discussion, every family of supports of E is of local character.

6.13

Let (E, A) be a ringed topos, let Φ be a family of supports of E, let N be an A-module, say a left module to fix ideas, and let X be an object of E. Put

$$\mathcal{H}^0_\Phi(N) = \varinjlim_{Z\in\Phi} \mathcal{H}^0_Z(N), \tag{6.13.1}$$

$$H^0_\Phi(E, N) = \varinjlim_{Z\in\Phi} H^0_Z(E, N). \tag{6.13.2}$$

If M is a left A-module, put

$$\mathcal{E}xt^0_{A,\Phi}(M, N) = \mathcal{H}om_{A,\Phi}(M, N) = \varinjlim_{Z\in\Phi} \mathcal{E}xt^0_{A,Z}(M, N), \tag{6.13.3}$$

$$\mathrm{Ext}^0_{A,\Phi}(X; M, N) = \mathrm{Hom}_{A|X,\Phi(X)}(M|X, N|X) = \varinjlim_{Z\in\Phi} \mathrm{Ext}^0_{A,Z}(X; M, N). \tag{6.13.4}$$

There are canonical isomorphisms

$$\mathcal{E}xt^0_{A,\Phi}(M, N) \simeq \mathcal{H}^0_\Phi(\mathcal{E}xt^0_A(M, N)), \tag{6.13.5}$$

and

$$\mathrm{Ext}^0_{A,\Phi}(X; M, N) \simeq H^0_\Phi(X, \mathcal{E}xt^0_A(M, N)). \tag{6.13.6}$$

When Φ is the family of closed subtopoi contained in a closed subtopos Z, there are isomorphisms

$$\begin{cases} \mathcal{H}^0_\Phi \simeq \mathcal{H}^0_Z, \\ H^0_\Phi \simeq H^0_Z, \\ \mathcal{E}xt^0_{A,\Phi}(-,-) \simeq \mathcal{E}xt^0_{A,Z}(-,-), \\ \mathrm{Ext}^0_{A,\Phi}(X;-,-) \simeq \mathrm{Ext}^0_{A,Z}(X;-,-). \end{cases} \tag{6.13.7}$$

The inductive limits defining the preceding functors are filtered. Hence these functors are left exact. Their right derived functors are denoted

$$\mathcal{H}^q_\Phi, \quad H^q_\Phi, \quad \mathcal{E}xt^q_{A,\Phi}(-,-), \quad \mathrm{Ext}^q_{A,\Phi}(X;-,-). \tag{6.13.8}$$

They too are computed by inductive limits of the functors $\mathcal{H}^q_Z$, H^q_Z, and so on, as Z runs through Φ.

From the isomorphisms (6.13.5) and (6.13.6), by passing to the inductive limit in the first spectral sequence of (6.9.1) and in the first spectral sequence of (6.9.2), one obtains two spectral sequences:

$$\mathcal{E}xt^{p+q}_{A,\Phi}(M,N) \Longleftarrow E_2^{p,q} = \mathcal{H}^p_\Phi(\mathcal{E}xt^q_A(M,N)), \tag{6.13.9}$$

$$\mathrm{Ext}^{p+q}_{A,\Phi}(X;M,N) \Longleftarrow E_2^{p,q} = H^p_\Phi(X, \mathcal{E}xt^q_A(M,N)). \tag{6.13.10}$$

6.14

With the notation of 6.13, and without any hypothesis on Φ, there is a functorial morphism

$$\theta : H^0_\Phi(E,N) \to H^0(E, \mathcal{H}^0_\Phi(N)). \tag{6.14.1}$$

This morphism is always injective, but is not in general an isomorphism. However, if Φ is *of local character*, then θ is an isomorphism, as is checked immediately. Similarly, if Φ is *of local character*, there is an isomorphism

$$\theta' : \mathrm{Ext}^0_{A,\Phi}(E;M,N) \simeq H^0(E, \mathcal{E}xt^0_{A,\Phi}(M,N)). \tag{6.14.2}$$

Proposition 6.15. Let (E,A) be a ringed topos such that E is coherent *(VI)*, let Φ be a family of supports of E, and let M, N be two A-modules. There are two spectral sequences:

$$H^{p+q}_\Phi(E,N) \Longleftarrow E_2^{p,q} = H^p(E, \mathcal{H}^q_\Phi(N)), \tag{6.15.1}$$

$$\mathrm{Ext}^{p+q}_{A,\Phi}(E;M,N) \Longleftarrow E_2^{p,q} = H^p(E, \mathcal{E}xt^q_{A,\Phi}(M,N)). \tag{6.15.2}$$

These spectral sequences are deduced from (6.4.2) and from the second spectral sequence of (6.9.2) by passage to the inductive limit over the closed subtopoi Z in Φ, taking account of the fact that the cohomology of a coherent topos commutes with inductive limits of sheaves (IV, 5).

6.16

Let us mention, without proof, that the second spectral sequence of (6.9.1) and the third spectral sequence of (6.9.2), with X the final object of E, may be extended to the case of families of supports, provided the topos E is coherent and the module M is perfect [13].

Finally, one may also generalize the notion of a family of supports by introducing presheaves of families of supports, sheaves of families of supports, and the corresponding cohomology groups and sheaves.[2]

[2] Cf. [12], Chapter IV, §1 (Lecture Notes in Mathematics 20, Springer) for the development of these notions in the form of a fugue with variations.

7 Appendix: Čech cohomology

Developing an idea due to P. Cartier, this section shows how, in an arbitrary topos, one may compute the cohomology of a sheaf by means of coverings. For the classical theory of paracompact spaces, see [7]; for another method applying to certain topoi, in particular to the étale topos, see [14].

7.1 Skeleton and coskeleton

7.1.0

Let Δ be the category of standard simplexes: its objects are the ordered sets

$$\Delta_n = [0, \ldots, n],$$

and its morphisms are increasing maps. Let E be a $\mathcal{U}$-topos, and let $\Delta(E)$ be the category of presheaves on Δ with values in E, in other words the category of *semi-simplicial objects* of E. Denote by $\Delta[n]$ the full subcategory of Δ defined by the objects Δ_p, $p \leq n$, by

$$i_n : \Delta[n] \to \Delta$$

the inclusion functor, and by $\Delta E[n]$ the category of presheaves on $\Delta[n]$ with values in E, in other words the category of *semi-simplicial objects of E truncated at order n*. By I, 5.1, there is a triple of adjoint functors (I, 5.3)

$$\begin{tikzcd} \Delta E[n] \arrow[r, shift left=4, "i_{n!}"] \arrow[r, shift right=4, "i_{n*}"'] & \Delta E \arrow[l, "i_n^*"'] \end{tikzcd}$$

where i_n^* is the restriction functor to $\Delta[n]$, and where i_{n*} and $i_{n!}$ are respectively its right and left adjoints. Note that ΔE and $\Delta E[n]$ are $\mathcal{U}$-topoi (IV, 1.2), and that (i_n^*, i_{n*}) is a morphism of topoi (IV, 3.1) which is an embedding of $\Delta E[n]$ into ΔE (I, 5.6 and IV, 9.1.1).

Definition 7.1.1. The functor

$$\mathrm{sk}_n = i_{n!} i_n^*$$

(respectively $\mathrm{cosk}_n = i_{n*} i_n^*$) from ΔE to ΔE is called the *skeleton functor of order n* (respectively the *coskeleton functor of order n*).

Skeleton and coskeleton functors are in constant use in the theory of semi-simplicial sets [3].

Proposition 7.1.2.

1) The adjunction morphism $\mathrm{sk}_n \to \mathrm{id}$ is a monomorphism. The canonical morphisms

$$\mathrm{sk}_n \mathrm{sk}_m \to \mathrm{sk}_n \quad (\textit{respectively}\ \ \mathrm{cosk}_n \to \mathrm{cosk}_n \mathrm{cosk}_m)$$

are isomorphisms when $n \leq m$ (respectively $n \geq m$).

2) Let $u : E \to E'$ be a morphism of topoi. The functor u^*, extended to semi-simplicial and truncated semi-simplicial objects, commutes with the functors $i_{n!}$, i_n^*, sk_n, and cosk_n.

The first assertion follows immediately from the definitions. To prove 2), it suffices to observe that, by I, 5.1, the functors $i_{n!}, \ldots, \mathrm{cosk}_n$ are computed by means of inductive limits and finite projective limits.

7.2 An acyclicity lemma

7.2.0

Let M be an abelian group of E. The *homology* of a semi-simplicial object K of E, with coefficients in M, is defined in the usual way. First form $A(K)$, the free abelian semi-simplicial group generated by K, and then form the tensor product $M \otimes_{\mathbb{Z}} A(K)$; this gives an abelian semi-simplicial group of E. One then considers the associated complex, obtained by taking the alternating sum of the boundary operators, and takes its homology. The homology objects are denoted

$$H_i(K, M).$$

They are abelian groups in E. When $M = \mathbb{Z}_E$, the free abelian group generated by the final object of E, we write simply $H_i(K)$. The groups $H_i(K)$ are called the *homology groups* of K. A semi-simplicial object K is called *acyclic* if, for every abelian group M of E, the groups $H_i(K, M)$ vanish for $i > 0$; equivalently, if the groups $H_i(K)$ vanish for $i > 0$.

The formation of homology commutes with inverse images by morphisms of topoi.

The aim of this subsection is to prove the following lemma.

Lemma 7.2.1. Let E be a topos, let $K_\bullet$ and $L_\bullet$ be two semi-simplicial objects of E, let

$$v_\bullet : K_\bullet \to L_\bullet$$

be a morphism of semi-simplicial objects, and let $n \geq 0$ be an integer. Assume that v has the following properties:

1) for every integer $p < n$, $p \geq 0$, the morphism $v_p : K_p \to L_p$ is an isomorphism;

2) the morphism $v_n : K_n \to L_n$ is an epimorphism;

3) the canonical morphisms $K_\bullet \to \operatorname{cosk}_n(K_\bullet)$ and $L_\bullet \to \operatorname{cosk}_n(L_\bullet)$ are isomorphisms.

Then, for every integer p, the morphism v_p is an epimorphism, and the morphism $v_\bullet$ induces an isomorphism on homology objects.

Remark. The proof in fact shows that, with a suitable definition of homotopy in topoi, the morphism $v_\bullet$ is a homotopy equivalence.

Proof. One may assume that E is the topos of sheaves on a small site C (IV, 1), and hence that E is a subtopos of a topos E' having enough fiber functors, for example the presheaf topos $C^\wedge$. Let

$$a^* : E' \to E$$

be the inverse-image functor for the embedding morphism $E \hookrightarrow E'$. Let

$$v_\bullet[n] : K_\bullet[n] \to L_\bullet[n]$$

be the morphism of truncated semi-simplicial objects obtained by truncating $v_\bullet$ at order n. Let $L_\bullet[n]'$ be the image, in E', of $K_\bullet[n]$ by $v_\bullet[n]$, and let

$$v_\bullet[n]' : K_\bullet[n] \to L_\bullet[n]'$$

be the morphism induced by $v_\bullet[n]$. Put

$$L! = i_{n*}L_\bullet[n]', \qquad K! = i_{n*}K_\bullet[n], \qquad v! = i_{n*}v_\bullet[n]',$$

where here i_{n*} is relative to E'. By 1), the morphism $v! : K! \to L!$ has properties 1), 2), and 3) of the lemma. Moreover, by 2), 3), and 7.1.2, the morphism $a^*v!$ is none other than $v_\bullet$. It is therefore enough to prove the lemma for $v!$. Thus one may assume that E has enough fiber functors, and by using these fiber functors one is reduced to the case where E is the category of sets.

First observe that the hypotheses of the lemma on the morphism $v_\bullet$ are stable under every base change $M_\bullet \to L_\bullet$ where $M_\bullet$ is a coskeleton of order n. Hence the fiber $P_\bullet$ of $v_\bullet$ at an arbitrary base point of $L_\bullet$ is of the form $i_{n*}P_\bullet[n]$, where $P_\bullet[n]$ is a non-empty complex truncated at order n of the form

$$P_n \to e \to e \to e \to \cdots \to e. \tag{7.2.2}$$

For every integer i, let $\delta\Delta_i$ be the semi-simplicial boundary of the simplex Δ_i. Every morphism from $\delta\Delta_i$ to $P_\bullet$ extends to a morphism from Δ_i to $P_\bullet$. Indeed, this is evident if $i \geq n$, since $P_\bullet$ is a coskeleton of order n, and it is clear for $i < n$ from the description (7.2.2). Since $P_\bullet$ is a Kan complex, $P_\bullet$ is homotopically trivial [6]. Note that $v_\bullet$ is a fibration in the sense of Kan [6]. Since the fibers of this fibration are homotopically trivial, $v_\bullet$ induces an isomorphism on the homotopy groups of $K_\bullet$ and $L_\bullet$ at every base point of $K_\bullet$. By the Hurewicz theorem, this implies that $v_\bullet$ induces an isomorphism on homology [6]. This proves the second assertion of the lemma. To prove the first assertion, note that every morphism from a complex truncated at order n into $L_\bullet[n]$ lifts to a morphism into $K_\bullet[n]$. Hence, by 3), every morphism from a complex into $L_\bullet$ lifts to a morphism into $K_\bullet$. Thus $v_\bullet$ is surjective.

7.3 Hypercoverings

7.3.0

Let C be a site belonging to the universe $\mathcal{U}$, and suppose that fiber products and products of two objects are representable. Denote by $C^\wedge$ the topos of $\mathcal{U}$-presheaves on C, and by $C^\sim$ the topos of $\mathcal{U}$-sheaves on C. An object K of $C^\wedge$ is called *semi-representable* if it is isomorphic to a direct sum of representable presheaves.

Let $\mathrm{SR}(C)$ be the full subcategory of $C^\wedge$ defined by the semi-representable objects. In $\mathrm{SR}(C)$, projective limits indexed by finite non-empty categories are representable, and the inclusion functor $\mathrm{SR}(C) \hookrightarrow C^\wedge$ commutes with these finite projective limits. By a remark already made, it follows that if $K_\bullet$ is a semi-simplicial object truncated at order n in $C^\wedge$ and all its objects are semi-representable, then the prolongation $i_n^+(K_\bullet)$ has the same property. Hence, if $K_\bullet$ is a semi-simplicial object of $C^\wedge$ all of whose objects are semi-representable, then, for every n, the object $\mathrm{cosk}_n(K_\bullet)$ has the same property.

7.3.1 Definitions and notation

7.3.1.1 Let $p > 0$ be an integer. A semi-simplicial object $K_\bullet$ of $C^\wedge$ is called a *hypercovering of type p* of C, or, when no confusion can result, a hypercovering of type p, if it has the following properties:

HR1) For every integer $n \geq 0$, K_n is semi-representable.

HR2) The canonical morphism $K_\bullet \to \operatorname{cosk}_p(K_\bullet)$ is an isomorphism.

HR3) For every integer $n \geq 0$, the canonical morphism of presheaves

$$K_{n+1} \to (\operatorname{cosk}_n(K_\bullet))_{n+1}$$

is a covering isomorphism of presheaves (II, 6.2). The canonical morphism of presheaves $K_0 \to e$, where e is the final object of $C^\wedge$, is a covering morphism of presheaves.

7.3.1.2 A hypercovering of type p is a hypercovering of type q for every $q \geq p$. A semi-simplicial object $K_\bullet$ will be called a *hypercovering* of C if it has properties HR1) and HR3).

7.3.1.3 We denote by HR_p (respectively HR), and call the *category of hypercoverings of type p* (respectively the *category of hypercoverings*), the following category:

a) the objects of HR_p (respectively HR) are the hypercoverings of type p (respectively the hypercoverings);

b) if $K_\bullet$ and $L_\bullet$ are two objects of HR_p (respectively HR), a morphism from $K_\bullet$ to $L_\bullet$ is a morphism $v_\bullet : K_\bullet \to L_\bullet$ of semi-simplicial objects of $C^\wedge$.

7.3.1.4 Let C be a site in which fiber products are representable, and let X be an object of C. A hypercovering of X (respectively a hypercovering of type p of X) means a hypercovering of the site C/X (respectively a hypercovering of type p of C/X). The category of hypercoverings of X (respectively hypercoverings of type p of X) is defined similarly. There is a sequence of inclusion functors

$$\cdots \mathrm{HR}_p \hookrightarrow \mathrm{HR}_{p+1} \hookrightarrow \cdots \hookrightarrow \mathrm{HR}\,.$$

These functors are fully faithful. We denote by HR_∞ the inductive limit of the categories HR_p.

7.3.1.6 For every integer $n \geq 0$, let Δ_n^c denote the standard semi-simplicial object of dimension n with values in the category of sets, i.e. the functor on Δ represented by the ordered set $[0, n]$. We also denote by $\underline{\Delta}_n^c$ the presheaf on C with values in semi-simplicial sets which is constant with value Δ_n. Thus $\underline{\Delta}_n^c$ is a presheaf of semi-simplicial sets. The two canonical injections of Δ_0 into Δ_1 define two morphisms of semi-simplicial presheaves

$$\underline{\Delta}_0^c \underset{e_1}{\overset{e_0}{\rightrightarrows}} \underline{\Delta}_1^c,$$

and consequently, for every semi-simplicial presheaf $K_\bullet$, two canonical injections

$$K_\bullet \underset{e_1}{\overset{e_0}{\rightrightarrows}} K_\bullet \times \underline{\Delta}_1^c.$$

Two morphisms

$$K_\bullet \underset{u_1}{\overset{u_0}{\rightrightarrows}} L_\bullet$$

of semi-simplicial presheaves are said to be *homotopic* if there exists a morphism

$$v : K_\bullet \times \underline{\Delta}_1^c \to L_\bullet$$

such that the diagrams

$$\begin{tikzcd}
K_\bullet \arrow[rr, "e_i"] \arrow[dr, "u_i"'] & & K_\bullet \times \underline{\Delta}_1^c \arrow[dl, "v"] \\
& L_\bullet &
\end{tikzcd} \qquad i = 0, 1$$

commute. The morphism v is called a *homotopy joining* u_0 *to* u_1. The relation that u_0 and u_1 are homotopic morphisms from $K_\bullet$ to $L_\bullet$ is not, in general, an equivalence relation. However, the equivalence relation generated by it is compatible with composition of morphisms. This allows us to define the category of semi-simplicial presheaves up to homotopy by quotienting by the equivalence relation generated by homotopy.

7.3.1.7 In this way one defines the categories $\mathcal{HR}_p$, $\mathcal{HR}_\infty$, and $\mathcal{HR}$, the *categories of hypercoverings up to homotopy.*

Theorem 7.3.2. Let C be a site satisfying the conditions of 7.3.0.

1) The category $\mathcal{HR}_p^0$ (respectively $\mathcal{HR}^0$) is filtered *(I, 2.7)*.

2) Let $K_\bullet$ be an object of HR_p (respectively of HR), let n be an integer with $0 \leq n \leq p$ (respectively $0 \leq n$), and let
$$u : X \to K_n$$
be a covering morphism of presheaves. There exists an object $L_\bullet$ of HR_p (respectively of HR) and a morphism $f : L_\bullet \to K_\bullet$ such that
$$f_n : L_n \to K_n$$
factors through u.

3) The semi-simplicial sheaves associated *(II, 3.5)* with hypercoverings are acyclic (7.2.0) in strictly positive degrees. The zeroth homology sheaf is isomorphic to the sheaf associated with the constant sheaf $\mathbb{Z}$.

Proof. We first prove 3). Let $K_\bullet^\sim$ be the semi-simplicial sheaf associated with $K_\bullet$. The associated-sheaf functor commutes with cosk_n (7.1.2.2)). Hence $K_\bullet^\sim$ has the following properties:

a) the canonical morphism $K_0 \to e$ is an epimorphism of sheaves;

b) for every integer $n \geq 0$, the canonical morphism
$$K_{n+1} \to (\mathrm{cosk}_n(K_\bullet))_{n+1}$$
is an epimorphism of sheaves.

For $n \geq 0$, put $L_{\bullet,n} = \mathrm{cosk}_n K_\bullet^\sim$. Then, for every n, there is a sequence of morphisms of semi-simplicial sheaves

$$K_\bullet^\sim \xrightarrow{u_n} L_{\bullet,n} \xrightarrow{v_n} L_{\bullet,n-1} \cdots \to L_{\bullet,0} \xrightarrow{v_0} e_\bullet,$$

and, for every j, $0 \le j \le n$, the morphism v_j has the properties of 7.2.1. Moreover, the morphism u_n induces an isomorphism on components of degree $\le n$. It follows from 7.2.1, by induction on n, that $K_\bullet$ is acyclic.

To prove 1) and 2), we introduce terminology.

Definition 7.3.3. Let $f : X_\bullet \to Y_\bullet$ be a morphism of semi-simplicial presheaves. One says that f is *special of type* p (respectively *special*) if:

1) The objects $X_\bullet$ and $Y_\bullet$ are canonically isomorphic to their coskeleta of order p, and $\operatorname{cosk}_p(f) = f$ (respectively no condition is imposed on f, $X_\bullet$, or $Y_\bullet$).

2) For every integer n such that $0 \le n \le p$ (respectively $0 \le n$), the morphism Φ_{n+1} appearing in the diagram below is covering:

$$\begin{tikzcd}
X_{n+1} \arrow[rr, "f_{n+1}"] \arrow[dr, "\Phi_{n+1}"] \arrow[dd] & & Y_{n+1} \arrow[dd] \\
& P_{n+1} \arrow[ur] \arrow[dl] & \\
(\operatorname{cosk}_n X_\bullet)_{n+1} \arrow[rr, "(\operatorname{cosk}_n f)_{n+1}"'] & & (\operatorname{cosk}_n Y_\bullet)_{n+1}.
\end{tikzcd}$$

Here the vertical arrows are defined by the canonical morphisms $X_\bullet \to \operatorname{cosk}_n X_\bullet$ and $Y_\bullet \to \operatorname{cosk}_n Y_\bullet$; the object P_{n+1} is the fiber product, and Φ_{n+1} is the unique arrow making the diagram commute.

3) The morphism $f_0 : X_0 \to Y_0$ is covering.

A semi-simplicial presheaf $K_\bullet$ is said to be special of type p (respectively special) if the canonical morphism

$$K_\bullet \to e_\bullet \qquad (e_\bullet \text{ is the final semi-simplicial object})$$

is special of type p (respectively special), i.e. if $K_\bullet$ satisfies conditions HR2) and HR3) (respectively HR3)) of 7.3.1.1.

Lemma 7.3.4.

1) The composite of two special morphisms (respectively special morphisms of type p) is special (respectively special of type p).

2) Let $K_\bullet$ be a special semi-simplicial presheaf (respectively special of type p), let $X_\bullet \xrightarrow{f} Y_\bullet$ be a special morphism (respectively special of type p), and let $u : K_\bullet \to Y_\bullet$ be a morphism of complexes. Then the fiber product

$$P_\bullet = K_\bullet \times_{Y_\bullet} X_\bullet$$

is special (respectively special of type p).

The proof of this lemma is left to the reader.

7.3.5. Proof of assertion 2) of 7.3.2

One may first assume that X is semi-representable. For every semi-simplicial object $L_\bullet$, let $\mathrm{Hom}_{(X)}(L_\bullet, K_\bullet)$ denote the set of morphisms of semi-simplicial objects equipped with a factorization

$$L_n \to X \xrightarrow{u} K_n.$$

The functor $\mathrm{Hom}_{(X)}(-, K_\bullet)$ is representable. Indeed, this functor sends inductive limits to projective limits, and $\Delta(C^\wedge)$ is a topos. We denote by P an object representing it. Similarly, for an object Z of $C^\wedge$, denote by $j_n^+(Z)$ the object representing the functor

$$L_\bullet \longmapsto \mathrm{Hom}(L_n, Z)$$

on $\Delta C^\wedge$; its construction by $\varprojlim$ is made explicit in III, 1.1. Here one observes that this construction involves only finite projective limits. Thus j_n^+ sends semi-representable presheaves to semi-simplicial presheaves with semi-representable components, and covering morphisms $Z' \to Z$ to special morphisms (7.3.3).

By the definition of $P_\bullet$, there is a cartesian square

$$\begin{tikzcd}
P_\bullet \arrow[r] \arrow[d] & j_n^+(X) \arrow[d, "j_n^+(u)"] \\
K_\bullet \arrow[r] & j_n^+(K_n).
\end{tikzcd}$$

By what has just been observed, the semi-simplicial objects $j_n^+(X)$ and $j_n^+(K_n)$ are semi-representable, and the morphism $j_n^+(u)$ is special of type n. It remains only to apply 3.4 and the formalities of 7.3.0.

7.3.6.0

Let $M_\bullet$ and $N_\bullet$ be two semi-simplicial presheaves. The functor

$$L_\bullet \longmapsto \mathrm{Hom}(L_\bullet \times M_\bullet, N_\bullet)$$

is representable. The semi-simplicial presheaf representing it will be denoted

$$\mathcal{H}om_\bullet(M_\bullet, N_\bullet).$$

The degree-n component of this object will be denoted $\mathcal{H}om_n(M_\bullet, N_\bullet)$. There is a canonical isomorphism

$$\mathcal{H}om_n(M_\bullet, N_\bullet) \xrightarrow{\sim} \mathcal{H}om_0(M_\bullet \times \Delta_n^c, N_\bullet).$$

Lemma 7.3.6. Let

$$\begin{tikzcd}
M_\bullet \arrow[r, hook] & N_\bullet \\
\mathrm{sk}_n\, \Delta_{n+1}^c \arrow[r, hook, "u"] \arrow[u] & \Delta_{n+1}^c \arrow[u]
\end{tikzcd}$$

(u the canonical injection)

be a cocartesian diagram of semi-simplicial presheaves. For every hypercovering $L_\bullet$, the morphism

$$\mathcal{H}om_0(N_\bullet, L_\bullet) \to \mathcal{H}om_0(M_\bullet, L_\bullet)$$

is covering.

Proof. There is a cartesian diagram

$$\begin{tikzcd}
\mathcal{H}om_0(N_\bullet, L_\bullet) \arrow[r] \arrow[d] & \mathcal{H}om_0(M_\bullet, L_\bullet) \arrow[d] \\
\mathcal{H}om_0(\Delta^c_{n+1}, L_\bullet) \arrow[r, "(*)"] & \mathcal{H}om_0(\operatorname{sk}_n \Delta^c_{n+1}, L_\bullet).
\end{tikzcd}$$

It is therefore enough to show that the morphism $(*)$ is covering. But $(*)$ is isomorphic to the morphism

$$L_{n+1} \to (\operatorname{cosk}_n L_\bullet)_{n+1},$$

which is covering by hypothesis.

7.3.7. Proof of assertion 1) of 7.3.2

The product of two objects of HR_p (respectively HR) is again an object of HR_p (respectively of HR), by 7.3.4. To prove that the category $\mathcal{HR}^0_p$ (respectively $\mathcal{HR}^0$) is filtered, it is enough to show that, given two morphisms in HR_p (respectively HR)

$$K_\bullet \underset{u_1}{\overset{u_0}{\rightrightarrows}} L_\bullet,$$

there exists a morphism $v : M_\bullet \to K_\bullet$ in HR_p (respectively in HR) such that the morphisms

$$M_\bullet \underset{u_1 v}{\overset{u_0 v}{\rightrightarrows}} L_\bullet$$

are homotopic, i.e. such that there is a $w : M_\bullet \times \Delta^c_1 \to L_\bullet$ making the diagrams

$$\begin{tikzcd}
M_\bullet \arrow[r, "e_i"] \arrow[d, "v"'] & M_\bullet \times \Delta^c_1 \arrow[d, "w"] \\
K_\bullet \arrow[r, "u_i"] & L_\bullet
\end{tikzcd} \qquad i = 0, 1 \tag{x}$$

commute.

For every semi-simplicial object $M_\bullet$ of $C^\wedge$, not necessarily a hypercovering, let $F(M_\bullet)$ be the set of pairs (v, w) with $v : M_\bullet \to K_\bullet$ and $w : M_\bullet \times \Delta^c_1 \to L_\bullet$ such that the diagrams (x) commute. The functor $M_\bullet \mapsto F(M_\bullet)$ is contravariant in $M_\bullet$, sends inductive limits to projective limits, and hence is representable by a semi-simplicial object $F_\bullet$. Evidently $F_\bullet$ is the vertex of a cartesian diagram

$$\begin{tikzcd}
F_\bullet \arrow[r] \arrow[d] & \mathcal{H}om_\bullet(\Delta^c_1, L_\bullet) \arrow[d, "\pi"] \\
K_\bullet \arrow[r, "{u_0, u_1}"'] & L_\bullet \times L_\bullet,
\end{tikzcd}$$

where π is defined by the two inclusions of Δ_0^c into Δ_1^c. To prove the assertion, it is therefore enough to show that $F_\bullet$ is an object of HR_p (respectively of HR). By 7.3.4.2) and the formalities of 7.3.0, it is enough to show that

a) $\mathcal{H}om_\bullet(\Delta_1^c, L_\bullet)$ is semi-representable;

b) the morphism π is special of type p (respectively special).

First, one verifies immediately that $\mathcal{H}om_\bullet(\Delta_1^c, L_\bullet)$ is a semi-representable semi-simplicial object. Indeed, its components are computed by finite projective limits from the L_n, and are therefore semi-representable. Let us verify b). If $L_\bullet$ is special of type p, first one shows that the morphism

$$\mathcal{H}om_\bullet(\Delta_1^c, L_\bullet) \to \operatorname{cosk}_p \mathcal{H}om_\bullet(\Delta_1^c, L_\bullet)$$

is an isomorphism; this verifies property 1) of 7.3.3. Next, the morphism

$$\pi_0 : \mathcal{H}om_0(\Delta_1^c, L_\bullet) \to L_0 \times L_0$$

is isomorphic to the canonical morphism

$$L_1 \xrightarrow{(d_0,d_1)} L_0 \times L_0,$$

which is covering by hypothesis, since $L_\bullet$ is a hypercovering.

It remains to verify property 2) of 7.3.3, i.e. to verify that, for all n, in the diagram

$$\begin{tikzcd}
\mathcal{H}om_{n+1}(\Delta_1^c, L_\bullet) \arrow[rr, "\pi_{n+1}"] \arrow[dr, "\Phi_{n+1}"'] \arrow[dd] & & L_{n+1} \times L_{n+1} \arrow[dd] \\
& P_{n+1} \arrow[ur] \arrow[dl] & \\
(\operatorname{cosk}_n \mathcal{H}om_\bullet(\Delta_1^c, L_\bullet))_{n+1} \arrow[rr, "(\operatorname{cosk}_n \pi)_{n+1}"] & & (\operatorname{cosk}_n L_\bullet)_{n+1} \times (\operatorname{cosk}_n L_\bullet)_{n+1}
\end{tikzcd}$$

where P_{n+1} is the fiber product, the morphism Φ_{n+1} is covering. A straightforward adjoint-functor chase shows that

$$P_{n+1} \xrightarrow{\sim} \mathcal{H}om_0(\operatorname{sk}_{n+1}(\Delta_{n+1}^c \times \Delta_1^c), L_\bullet),$$
$$\mathcal{H}om_{n+1}(\Delta_1^c, L_\bullet) \xrightarrow{\sim} \mathcal{H}om_0(\Delta_{n+1}^c \times \Delta_1^c, L_\bullet),$$

and that Φ_{n+1} is isomorphic to the morphism

$$\mathcal{H}om_0(\Delta_{n+1}^c \times \Delta_1^c, L_\bullet) \to \mathcal{H}om_0(\operatorname{sk}_{n+1}(\Delta_{n+1}^c \times \Delta_1^c), L_\bullet)$$

coming from the injection

$$\operatorname{sk}_{n+1}(\Delta_{n+1}^c \times \Delta_1^c) \hookrightarrow \Delta_{n+1}^c \times \Delta_1^c.$$

Now there is a sequence of subobjects of $\Delta_{n+1}^c \times \Delta_1^c$,

$$\operatorname{sk}_{n+1}(\Delta_{n+1}^c \times \Delta_1^c) = M_0 \hookrightarrow M_1 \hookrightarrow \cdots \hookrightarrow M_k = \Delta_{n+1}^c \times \Delta_1^c,$$

such that, for every $0 \leq i \leq k$, the morphism

$$M_i \hookrightarrow M_{i+1}$$

fits into a cocartesian diagram

$$\begin{tikzcd}
M_i \arrow[r, hook] & M_{i+1} \\
\operatorname{sk}_{n+1} \Delta^c_{n+2} \arrow[u] \arrow[r, hook] & \Delta^c_{n+2} \arrow[u]
\end{tikzcd}$$

One adds one after another the non-degenerate simplexes of dimension $n+2$ of $\Delta_{n+1} \times \Delta_1$. By 7.3.6, the morphism Φ_{n+1} is a composite of covering morphisms, hence is itself covering. This completes the proof of 7.3.2.

7.4 The isomorphism theorem

7.4.0

Let F be a presheaf of abelian groups on a site C satisfying the condition of 7.3.0. For every hypercovering $K_\bullet$, put

$$H^q(K_\bullet, F) = H^q(\operatorname{Hom}_{C^\wedge}(K_\bullet, F)).$$

The $H^q(K_\bullet, F)$ form a δ-functor on the category of abelian presheaves on C, but in general they are not the derived functors of $H^0(K_\bullet, F)$.

Put

$$\check{H}^q_r(C, F) = \varinjlim_{\mathrm{HR}_r} H^q(-, F) \qquad (r \text{ may be infinite}), \tag{7.4.0.1}$$

and

$$\check{H}^q(C, F) = \varinjlim_{\mathrm{HR}} H^q(-, F). \tag{7.4.0.2}$$

The $\check{H}^q_r(C, F)$ (respectively $\check{H}^q(C, F)$) still form a δ-functor, because the category $\mathcal{HR}^0_r$ (respectively $\mathcal{HR}^0$) is filtered (7.3.2). Since the categories $\mathcal{HR}_p$ are not necessarily $\mathcal{U}$-categories, these δ-functors take values in the category of $\mathcal{V}$-abelian groups, for a suitable universe $\mathcal{V}$.

Assume now that the presheaf F is a sheaf. Since the semi-simplicial sheaf associated with a hypercovering is acyclic (7.3.2), there is a spectral sequence, functorial in F and in $K_\bullet$,

$$H^p(K_\bullet, \mathcal{H}^q(F)) \Longrightarrow H^{p+q}(C^\sim, F). \tag{7.4.0.3}$$

Passing to the limit gives spectral sequences

$$\begin{aligned}
\check{H}^p_r(C, \mathcal{H}^q(F)) &\Longrightarrow H^{p+q}(C^\sim, F), \\
\check{H}^p(C, \mathcal{H}^q(F)) &\Longrightarrow H^{p+q}(C^\sim, F).
\end{aligned} \tag{7.4.0.4}$$

Theorem 7.4.1. Let C be a site satisfying the condition of 7.3.0, and let F be an abelian sheaf on C.

1) The spectral sequences (7.4.0.4) define isomorphisms

$$\check{H}^q_r(C,F) \xrightarrow{\sim} H^q(C^\sim,F), \qquad q \le r+1,$$

and a monomorphism

$$\check{H}^{r+2}_r(C,F) \longrightarrow H^{r+2}(C^\sim,F).$$

2) There is an isomorphism of δ-functors

$$\check{H}^q_\infty(C,F) \xrightarrow{\sim} \check{H}^q(C,F) \xrightarrow{\sim} H^q(C^\sim,F), \qquad \text{for all } q.$$

3) Let G be a presheaf of abelian groups and let F be the associated sheaf. There are isomorphisms of functors

$$\check{H}^q_r(C,G) \xrightarrow{\sim} H^q(C^\sim,F), \qquad q \le r-1,$$

and a monomorphism

$$\check{H}^r_r(C,G) \longrightarrow H^r(C^\sim,F).$$

When G is a separated presheaf, this latter morphism is an isomorphism, and there is a monomorphism

$$\check{H}^{r+1}_r(C,G) \longrightarrow H^{r+1}(C^\sim,F).$$

4) There are isomorphisms

$$\check{H}^q_\infty(C,G) \xrightarrow{\sim} \check{H}^q(C,G) \xrightarrow{\sim} H^q(C^\sim,F), \qquad \text{for all } q.$$

Proof. It is enough to show that, if N is an abelian presheaf whose associated sheaf is zero, then

$$\check{H}^q_r(C,N) = 0 \quad \text{for } q \le r \qquad (\textit{respectively } \check{H}^q(C,N) = 0 \text{ for all } q),$$

which is done immediately using 3.2.

Remark 7.4.2.

1) Note that, for sites satisfying the condition of 7.3.0, hypercoverings of type 0 are ordinary coverings, and the functors $\check{H}^q_\infty$ are none other than the functors $\check{H}^q$ introduced in (2.4.5.1).

2) Let C be a site with representable finite projective limits such that, for every object X of C and every covering family $(X_i \to X)_{i\in I}$, there is an $i_0 \in I$ such that $X_{i_0} \to X$ is covering. Then one can show that the hypercoverings of type p whose components are representable are cofinal in $\mathcal{HR}_p$. Similarly, hypercoverings $K_\bullet$ for which there exists an integer p such that $K_\bullet$ is of type p, and whose components are representable, are cofinal in $\mathcal{HR}$. This can be proved by following 3.5. These hypercoverings with representable components therefore suffice, in the case under consideration, to compute the cohomology of the sheaves associated with presheaves.

8 Appendix: Local inductive limits (by P. Deligne)

The editor recommends that the reader, as a rule, avoid reading this appendix. It presents a technique which sometimes makes it possible to extend to topoi without enough points assertions which are made trivial by the existence of points. This technique gives a sheaf-theoretic variant of a theorem of D. Lazard asserting that flat modules over a ring are inductive limits of finite free modules.

8.1 Locally filtered categories

8.1.0

If $\mathfrak{B}$ is a category and $p : \mathfrak{A} \to \mathfrak{B}$ is a category over $\mathfrak{B}$, we shall use the following notation. For $U \in \mathrm{Ob}(\mathfrak{B})$, $\mathfrak{A}_U$ is the fiber category $p^{-1}(U)$. For $f : U \to V$ in $\mathfrak{B}$ and objects $\lambda, \mu \in \mathfrak{A}$ with $p(\lambda) = U$ and $p(\mu) = V$, put

$$\mathrm{Hom}_f(\lambda, \mu) = p^{-1}(f), \qquad p : \mathrm{Hom}(\lambda, \mu) \to \mathrm{Hom}(U, V).$$

Definition 8.1.1. Let $\mathfrak{S}$ be a site. A *locally cofiltered category* over $\mathfrak{S}$ (or *left locally filtered category*) is a category $p : \mathfrak{L} \to \mathfrak{S}$ over $\mathfrak{S}$ satisfying:

($\mathfrak{L}0$) for every $f : U \to V$ in $\mathfrak{S}$ and every $\mu \in \mathrm{Ob}(\mathfrak{L}_V)$, locally on U there is an object $\lambda \in \mathrm{Ob}(\mathfrak{L}_U)$ with $\mathrm{Hom}_f(\lambda, \mu) \neq \varnothing$;

($\mathfrak{L}1$) for every $U \in \mathrm{Ob}(\mathfrak{S})$ and every finite family (λ_i) of objects of $\mathfrak{L}_U$, there is a covering $f_j : U_j \to U$ and objects $\mu_j \in \mathrm{Ob}(\mathfrak{L}_{U_j})$ such that, for all i, j, one has $\mathrm{Hom}_{f_j}(\mu_j, \lambda_i) \neq \varnothing$;

($\mathfrak{L}2$) for every $f : U \to V$ in $\mathfrak{S}$ and every double arrow $(\phi_0, \phi_1) : \lambda \rightrightarrows \mu$ above f, there is a covering $f_j : U_j \to U$ and arrows ψ_j with target λ, lying above f_j, such that $\phi_0\psi_j = \phi_1\psi_j$.

8.1.1.1 This definition simplifies when $\mathfrak{L}$ is fibered over $\mathfrak{S}$. Then axiom ($\mathfrak{L}0$) is satisfied, and ($\mathfrak{L}1$) and ($\mathfrak{L}2$) may be stated as follows:

($\mathfrak{L}'1$) for every $U \in \mathrm{Ob}(\mathfrak{S})$ and every finite family λ_i of objects of $\mathfrak{L}_U$, locally on U there is an object μ mapping to all the λ_i;

($\mathfrak{L}'2$) for every $U \in \mathrm{Ob}(\mathfrak{S})$, every double arrow $(\phi_0, \phi_1) : \lambda \rightrightarrows \mu$ in $\mathfrak{L}_U$ can, locally on U, be equalized by an arrow with target λ.

8.1.2

Let $\mathfrak{S}$ be a site, let $\mathfrak{L}$ be a category over $\mathfrak{S}$, and let e be a stack over $\mathfrak{S}$. Denote by $\Gamma\mathsf{e}$ the category of global sections $\mathcal{H}om^{\mathrm{cart}}_{\mathfrak{S}}(\mathfrak{S}, \mathsf{e})$ of e. If $\mathfrak{S}$ has a final object S, then $\Gamma\mathsf{e}$ is, up to the usual convention, just e_S.

Definition 8.1.2.1. The category of *local projective systems of objects of* e*, indexed by* $\mathfrak{L}$, is the category

$$\mathcal{H}om_{\mathfrak{S}}(\mathfrak{L}, \mathsf{e}).$$

Let c denote the composite functor, "associated constant local projective system",

$$\Gamma\mathsf{e} = \mathcal{H}om^{\mathrm{cart}}_{\mathfrak{S}}(\mathfrak{S}, \mathsf{e}) \hookrightarrow \mathcal{H}om_{\mathfrak{S}}(\mathfrak{S}, \mathsf{e}) \xrightarrow{u \mapsto u \circ p} \mathcal{H}om_{\mathfrak{S}}(\mathfrak{L}, \mathsf{e}).$$

When the dependence on $\mathfrak{L}$ has to be made explicit, we write $c_{\mathfrak{L}}$.

Definition 8.1.3. The *local projective-limit functor*, denoted $\varprojlim_{\lambda \in \mathfrak{L}}$, is the partially defined functor right adjoint to c.

Thus

$$\mathrm{Hom}(X, \varprojlim_{\lambda \in \mathfrak{L}} X_\lambda) \simeq \mathrm{Hom}(c(X), (X_\lambda)_{\lambda \in \mathfrak{L}}).$$

Definition 8.1.4. An $\mathfrak{S}$-functor $F : \mathfrak{L} \to \mathfrak{M}$ between locally cofiltered categories over $\mathfrak{S}$ is called *cofinal* if it satisfies:

($C\mathfrak{F}1$) for every $U \in \mathrm{Ob}(\mathfrak{S})$ and every $\mu \in \mathrm{Ob}(\mathfrak{M}_U)$, there is a covering $f_j : U_j \to U$ and objects $\lambda_j \in \mathrm{Ob}(\mathfrak{L}_{U_j})$ such that $\mathrm{Hom}_{f_j}(F(\lambda_j), \mu) \neq \varnothing$;

($C\mathfrak{F}2$) for every $f : U \to V$ in $\mathfrak{S}$ and every double arrow $(\phi_0, \phi_1) : F(\lambda) \rightrightarrows \mu$ above f, there is a covering $f_j : U_j \to U$ and arrows ψ_j with target λ in $\mathfrak{L}$, with $p(\psi_j) = f_j$, such that $\phi_0 F(\psi_j) = \phi_1 F(\psi_j)$.

The composite of two cofinal functors is cofinal; equivalences of locally filtered categories are cofinal. The proof is left to the reader.

Proposition 8.1.5. Let $F : \mathfrak{L} \to \mathfrak{M}$ be a cofinal functor between locally cofiltered categories over $\mathfrak{S}$, let e be a stack over $\mathfrak{S}$, and let $(X_\mu)_{\mu \in \mathfrak{M}}$ be a local projective system indexed by $\mathfrak{M}$. Then the morphism of functors, in $X \in \Gamma\mathsf{e}$ and with values in sets,

$$F^* : \mathrm{Hom}(c_{\mathfrak{M}}(X), (X_\mu)_{\mu \in \mathfrak{M}}) \longrightarrow \mathrm{Hom}(c_{\mathfrak{L}}(X), (X_{F(\lambda)})_{\lambda \in \mathfrak{L}})$$

is an isomorphism. Hence

$$F^* : \varprojlim_{\mathfrak{M}} X_\mu \longrightarrow \varprojlim_{\mathfrak{L}} X_{F(\lambda)}$$

is an isomorphism, the two sides being defined or undefined simultaneously.

The proof uses the following lemma, left to the reader.

Lemma 8.1.6. If $F : \mathfrak{L} \to \mathfrak{M}$ is cofinal, then, for every $f : U \to V$ in $\mathfrak{S}$ and every pair of arrows above f,

$$\phi_i : F(\lambda'_i) \to \mu, \qquad i = 1, 2,$$

there is a covering $f_j : U_j \to U$, objects $\lambda'_j \in \mathrm{Ob}(\mathfrak{L}_{U_j})$, and commutative diagrams

$$\begin{tikzcd}
& F(\lambda'_1) \arrow[dr, "\phi_1"] & \\
F(\lambda'_j) \arrow[ur, dashed, dash] \arrow[dr, dashed, dash] & & \mu \\
& F(\lambda'_2) \arrow[ur, "\phi_2"'] &
\end{tikzcd}$$

lying above $U_j \xrightarrow{f_j} U \to V$.

For $U \in \mathrm{Ob}(\mathfrak{S})$ and $\mu \in \mathfrak{M}_U$, condition $(C\mathfrak{F}1)$ gives a covering $f_j : U_j \to U$ and arrows $\phi_j : F(\lambda_j) \to \mu$ above f_j. If

$$\psi = (\psi_\lambda) \in \mathrm{Hom}(c_{\mathfrak{L}}X, (X_{F(\lambda)})_{\lambda \in \mathfrak{L}})$$

is the image of

$$\phi = (\phi_\mu) \in \mathrm{Hom}(c_{\mathfrak{M}}X, (X_\mu)_{\mu \in \mathfrak{M}}),$$

then the diagrams

$$\begin{tikzcd} & X_{F(\lambda_j)} \arrow[dl, "\psi_{\lambda_j}"'] \arrow[dr, "\phi_j"] & \\ X|U_j \arrow[rr, "\phi_\mu|U_j"'] & & X_\mu|U_j \end{tikzcd}$$

commute. Thus the ϕ_μ are locally determined by ψ; since e is a stack, F^* is injective. For surjectivity, start from ψ and construct ϕ with $F^*\phi = \psi$. The preceding construction gives local composites

$$\phi_{\mu,j} = \phi_j \circ \psi_{\lambda_j} : X|U_j \to X_\mu|U_j.$$

By Lemma 8.1.6 and $(\mathfrak{L}0)$ these morphisms agree locally on overlaps, hence glue to a morphism $\phi_\mu : X|U \to X_\mu$. The compatibility of the ϕ_μ with arrows in $\mathfrak{M}$ follows again locally, and gives the desired morphism of functors.

If E is an ordered set, we shall also denote by E the category with object set E and with $\mathrm{Hom}(i,j)$ reduced to one element or empty according as $i \le j$ or not.

Proposition 8.1.7. Let $\mathfrak{L}$ be a locally cofiltered category over a site $\mathfrak{S}$. There is an ordered set E and a functor $F : E \to \mathfrak{L}$ such that:

(i) E, viewed as a category over $\mathfrak{S}$ by means of $p \circ F : E \to \mathfrak{S}$, is locally cofiltered;

(ii) the functor F is cofinal.

Let $\mathfrak{L}'$ be the product of $\mathfrak{L}$ with the category defined by the ordered set $\mathbb{Z}$. The projection $\mathfrak{L}' \to \mathfrak{L}$ is cofinal, so it is enough to prove the proposition for $\mathfrak{L}'$. If X is a finite set of arrows in $\mathfrak{L}'$, let X^0 be the set of endpoints of arrows of X. Let E be the set of finite non-empty subsets X of $\mathrm{Fl}(\mathfrak{L}')$ such that there is an object λ_X of $\mathfrak{L}'$ satisfying:

a) no arrow of X, except 1_{λ_X}, has target λ_X;

b) if $\mu \in X^0$, then $1_\mu \in X$ and $\mathrm{Hom}(\lambda_X, \mu) \cap X \neq \varnothing$;

c) every diagram of arrows in X of the form

$$\begin{tikzcd} \lambda_X \arrow[r] \arrow[dr] & \mu \arrow[d] \\ & \mu' \end{tikzcd}$$

is commutative.

Order E by the relation opposite to inclusion. If $X \in E$ and $\mu \in X^0$, there is a unique arrow from λ_X to μ lying in X. If $X, Y \in E$ and $X \supset Y$, let $\lambda_{X,Y}$ be the unique arrow in X from λ_X to λ_Y. The assignments $X \mapsto \lambda_X$ and $(X \supset Y) \mapsto \lambda_{X,Y}$ define a functor $E \to \mathfrak{L}'$.

Lemma 8.1.8. The category E is locally cofiltered over $\mathfrak{S}$, and the functor $\lambda : E \to \mathfrak{L}'$ is cofinal.

One verifies the axioms in sequence. For ($\mathfrak{L}0$), given $f : V \to U$ and $X \in E_U$, choose $\lambda \in \mathrm{Ob}(\mathfrak{L}'_V)$ and $\phi \in \mathrm{Hom}_f(\lambda, \lambda_X)$, with $\lambda \notin X^0$; put $X' = X \cup \{1_\lambda\} \cup Y$, where Y consists of the composites

$$\lambda \xrightarrow{\phi} \lambda_X \xrightarrow{\psi} \mu, \qquad \psi \in X.$$

Then $X' \in E$, $\lambda_{X'} = \lambda$, and $\mathrm{Hom}(X', X) \neq \varnothing$. For ($\mathfrak{L}1$), given a finite family X_i over U, choose after refinement arrows from objects λ_j to the λ_{X_i}, and refine again so that arrows landing in common endpoints are equalized. Adding to $\bigcup_i X_i$ the identity of λ_j and all composites from λ_j to endpoints in the X_i gives objects $X_j \in E$ satisfying ($\mathfrak{L}1$). Axiom ($\mathfrak{L}2$) is trivial because there are no non-trivial double arrows in the ordered category E. Condition ($C\mathfrak{F}1$) is trivial because λ is surjective. Finally, ($C\mathfrak{F}2$) follows by the same equalization construction: for a double arrow $\lambda_X \rightrightarrows \lambda$, refine and add the equalizing arrows to X, so that the induced arrow $X_j \to X$ equalizes the given pair.

8.1.9.0 Suppose $\mathfrak{S}$ is the site of non-empty open subsets of the one-point topological space. A locally cofiltered category over $\mathfrak{S}$ is then just a left filtered category, i.e. one whose opposite is filtered in the sense of I, 2.7. Cofinal functors are the usual cofinal functors between left filtered categories (I, 8). Proposition 8.1.7 becomes:

Proposition 8.1.9. For every cofiltered category $\mathfrak{L}$, there is a cofiltered ordered set E and a cofinal functor $E \to \mathfrak{L}$.

In this special case, the proof of 8.1.7 reduces essentially to the proof of 8.1.9 given in I, 8.

8.1.10

Let $q : \mathfrak{T} \to \mathfrak{S}$ be a morphism of sites, and let $p : \mathfrak{L} \to \mathfrak{S}$ be a locally cofiltered category over $\mathfrak{S}$. Define a category $\mathfrak{L}^*$ as follows. An object is a quadruple (V, U, ϕ, λ) with $V \in \mathrm{Ob}(\mathfrak{T})$, $U \in \mathrm{Ob}(\mathfrak{S})$, $\phi \in \mathrm{Hom}(V, q^*U)$, and $\lambda \in \mathrm{Ob}(\mathfrak{L}_U)$. An arrow

$$(V, U, \phi, \lambda) \to (V', U', \phi', \lambda')$$

is a triple (f_1, f_2, f_3) with $f_1 : V \to V'$, $f_2 : U \to U'$, $f_3 : \lambda \to \lambda'$, satisfying

$$q^* f_2 \circ \phi = \phi' \circ f_1, \qquad p(f_3) = f_2.$$

The functor $(f_1, f_2, f_3) \mapsto f_1$ makes $\mathfrak{L}^*$ a category over $\mathfrak{T}$.

Lemma 8.1.11. The category $\mathfrak{L}^*$ is locally cofiltered over $\mathfrak{T}$.

Axiom ($\mathfrak{L}0$) is trivial. For ($\mathfrak{L}1$), start with a finite family $L_i = (V, U_i, \phi_i, \lambda_i)$. The ϕ_i define a morphism from V to the sheaf $q^* a \prod U_i = a \prod q^* U_i$; hence, locally on V, they factor through an object q^*U_j mapping to all q^*U_i. After applying ($\mathfrak{L}1$) in $\mathfrak{L}$ and refining, one obtains objects L_j mapping to all L_i. The verification of ($\mathfrak{L}2$) is similar: a double arrow gives a morphism into the kernel sheaf of a double arrow in $\mathfrak{S}$; after a local refinement, and then applying ($\mathfrak{L}2$) in $\mathfrak{L}$, the double arrow is equalized.

Proposition 8.1.12. Let $t^* : \mathfrak{S}^* \to \mathfrak{S}$ be a locally cofiltered category over $\mathfrak{S}$, and endow $\mathfrak{S}^*$ with the topology induced from that of $\mathfrak{S}$ by t^*. Then t^* is an equivalence of sites $t : \mathfrak{S} \to \mathfrak{S}^*$.

By construction, pullback by t^* sends sheaves on $\mathfrak{S}$ to sheaves on $\mathfrak{S}^*$. Axioms ($\mathfrak{L}0$) and ($\mathfrak{L}1$) applied to the empty family imply that every sufficiently small object of $\mathfrak{S}$ lies in the image of t^*, i.e. that the image of t^* is a covering sieve in $\mathfrak{S}$. By the comparison lemma (III, 4), the inclusion into $\mathfrak{S}$ of the full subcategory defined by $t^*(\mathrm{Ob}\,\mathfrak{S}^*)$ is an equivalence of sites; hence one may suppose that t^* is surjective on objects.

Let $\mathfrak{F}$ be a sheaf on $\mathfrak{S}^*$. The proof constructs, functorially in $\mathfrak{F}$, a presheaf $\mathfrak{F}_0$ on $\mathfrak{S}$ whose direct image is $\mathfrak{F}$. The essential point is that if two local presentations of an arrow in $\mathfrak{S}$ are chosen in $\mathfrak{S}^*$, the induced maps on values of $\mathfrak{F}$ agree after a common refinement, by ($\mathfrak{L}2$) and the sheaf condition. This gives a well-defined transition morphism $\overline{f}$ for every arrow f of $\mathfrak{S}$, independent of all choices. A further refinement argument, using ($\mathfrak{L}0$), ($\mathfrak{L}1$), and ($\mathfrak{L}2$), shows that $\overline{f}$ is functorial in f. Thus every sheaf $\mathfrak{F}$ on $\mathfrak{S}^*$ is the direct image of a uniquely determined presheaf $\mathfrak{F}_0$ on $\mathfrak{S}$, and morphisms of sheaves are induced by unique morphisms of such presheaves. Hence t_* is fully faithful. Finally, if $f_j : U_j \to U$ is a covering in $\mathfrak{S}$, it is the image of a covering in $\mathfrak{S}^*$; therefore $\mathfrak{F}_0(U)$ injects into $\prod_j \mathfrak{F}_0(U_j)$, and compatible local sections glue because they already glue in $\mathfrak{S}^*$. Thus $\mathfrak{F}_0$ is a sheaf, completing the proof.

8.1.13.0 Let $q : \mathfrak{T} \to \mathfrak{S}$, $p : \mathfrak{L} \to \mathfrak{S}$, and $\mathfrak{L}^*$ be as in 8.1.10, and let e be a stack over $\mathfrak{T}$. The stack $q_*\mathsf{e}$ over $\mathfrak{S}$ is the stack defined by

$$(q_*\mathsf{e})_U = \mathsf{e}_{q^*U}.$$

If $(X_\lambda)_{\lambda\in\mathfrak{L}}$ is a local projective system indexed by $\mathfrak{L}$ and with values in e, it defines a local projective system $(X_L)_{L\in\mathfrak{L}^*}$ by

$$X_L = \phi^* X_\lambda \qquad \text{for } L = (V, U, \phi, \lambda).$$

The reader is left to verify the following statement.

Proposition 8.1.13.1. With the preceding notation, one has

$$\varprojlim_{\lambda\in\mathfrak{L}} X_\lambda \simeq \varprojlim_{L\in\mathfrak{L}^*} X_L,$$

the two sides being defined or undefined simultaneously.

Let $\mathfrak{S}$ be a site and let $p^* : \mathfrak{S}^* \to \mathfrak{S}$ be a locally filtered category over $\mathfrak{S}$. If e is a stack over $\mathfrak{S}$, then $p_*\mathsf{e}$ is a stack over $\mathfrak{S}^*$ by 8.1.13.0; moreover the identity functor $\mathfrak{S}^* \to \mathfrak{S}^*$ is a locally filtered category over $\mathfrak{S}^*$, endowed with the induced topology, and every local inductive system indexed by $\mathfrak{S}^*$ defines a local inductive system over $\mathfrak{S}^*$ indexed by $\mathfrak{S}^*$ itself.

Proposition 8.1.14. With the preceding notation, one has

$$\varinjlim_{\lambda\in\mathfrak{S}^*/\mathfrak{S}} X_\lambda \simeq \varinjlim_{\lambda\in\mathfrak{S}^*/\mathfrak{S}^*} X_\lambda,$$

the two sides being defined or undefined simultaneously.

8.2 Local inductive limits in categories of sheaves

8.2.0

Let $p : \mathfrak{T} \to \mathfrak{S}$ be a morphism of sites. Let e be the following stack over $\mathfrak{S}$. An object of e is a pair $(U, \mathfrak{F})$ consisting of an object U of $\mathfrak{S}$ and a sheaf $\mathfrak{F}$ on p^*U. An arrow

$$f : (U, \mathfrak{F}) \to (V, \mathfrak{G})$$

is a pair consisting of an arrow $f_0 : U \to V$ and a morphism of sheaves on p^*U,

$$f_1 : p^*(f_0)^*\mathfrak{G} \to \mathfrak{F}.$$

The functor from e to $\mathfrak{S}$ is given by $f \mapsto f_0$. Notice that e_U is the *opposite category* of the category of sheaves on p^*U.

Let $\mathfrak{L}$ be a locally cofiltered, left locally filtered, category over $\mathfrak{S}$.

Definition 8.2.1. With the preceding notation, the category of local inductive systems, indexed by $\mathfrak{L}$, of sheaves on $\mathfrak{T}$, is the category

$$\mathcal{H}om_{\mathfrak{S}}(\mathfrak{L}, \mathsf{e})^{\mathrm{op}}.$$

The "projective-limit" functor of 8.1.3 is here called the *local inductive-limit functor*. It is a functor from $\mathcal{H}om_{\mathfrak{S}}(\mathfrak{L}, \mathsf{e})^{\mathrm{op}}$ to $(\Gamma\mathsf{e})^{\mathrm{op}} = \mathfrak{T}^{\sim}$.

Theorem 8.2.2. The local inductive-limit functor, from the category of inductive systems of sheaves on $\mathfrak{T}$ indexed by $\mathfrak{L}$ to the category of sheaves on $\mathfrak{T}$, is everywhere defined, commutes with finite projective limits, and commutes with arbitrary inductive limits.

Propositions 8.1.13, 8.1.14, and 8.1.12 reduce the proof to the case $\mathfrak{S} = \mathfrak{T} = \mathfrak{L}$. A local inductive system is then the data, for each $U \in \mathrm{Ob}(\mathfrak{S})$, of a sheaf $\mathfrak{F}_U$ on $\mathfrak{S}/U$ and, for every arrow $f : V \to U$ in $\mathfrak{S}$, a morphism $f^*\mathfrak{F}_U \to \mathfrak{F}_V$, functorial in f.

For such systems, the inductive limits are computed as follows.

Lemma 8.2.3. Let $U \mapsto \mathfrak{F}_U$ be a section of the category of presheaves over the objects of $\mathfrak{S}$. Then

$$a\big(U \mapsto \Gamma(U, \mathfrak{F}_U)\big) \simeq \varinjlim_{U} a\mathfrak{F}_U,$$

where a denotes the associated-sheaf functor.

By definition, the right-hand side represents the functor which assigns to a sheaf $\mathfrak{F}$ on $\mathfrak{S}$ the set of coherent systems of arrows $\phi_U : \mathfrak{F}_U \to \mathfrak{F}|U$. Such an arrow ϕ_U is in turn a coherent system of arrows $\phi_g : \mathfrak{F}_U(V) \to \mathfrak{F}(V)$, one for each $g : V \to U$. Applying 8.1.11 to the identity functor of $\mathfrak{S}$, and then using cofinality of the functor $U \mapsto 1_U$, gives

$$\varinjlim_{U} \mathfrak{F}_U(U)^{\sim} \simeq \varinjlim_{U} a\mathfrak{F}_U.$$

Returning to the definitions,

$$\varinjlim_{U} \mathfrak{F}_U(U)^{\sim} = a\big(U \mapsto \Gamma(U, \mathfrak{F}_U)\big).$$

This proves the lemma. It also shows, in the reduced case, that the functor $\varinjlim$ is everywhere defined and is the composite of the associated-sheaf functor and the functor

$$(\mathfrak{F}_U) \longmapsto (U \mapsto \Gamma(U, \mathfrak{F}_U)).$$

Both commute with finite projective limits, and the functor $\varinjlim$ also commutes with arbitrary inductive limits by its definition as an adjoint.

Lemma 8.2.3 gives a systematic procedure for proving properties of the local inductive-limit functor from the analogous properties of the associated-sheaf functor. Thus one obtains:

Proposition 8.2.4. Let $\mathfrak{T}$, $\mathfrak{S}$, and $\mathfrak{L}$ be as above. Let $\mathfrak{A}_\lambda$ be a local inductive system of sheaves of rings on $\mathfrak{T}$, and let $\mathfrak{L}_\lambda$ (respectively $\mathfrak{M}_\lambda$) be a local inductive system of sheaves of right (respectively left) modules over $\mathfrak{A}_\lambda$, all three indexed by $\mathfrak{L}$. Then

$$\varinjlim(\mathfrak{L}_\lambda \otimes_{\mathfrak{A}_\lambda} \mathfrak{M}_\lambda) \simeq (\varinjlim \mathfrak{L}_\lambda) \otimes_{\varinjlim \mathfrak{A}_\lambda} (\varinjlim \mathfrak{M}_\lambda).$$

Reduce to the case $\mathfrak{T} = \mathfrak{S} = \mathfrak{L}$, and note that the two sides coincide with the sheaf generated by the presheaf

$$\mathfrak{L}_U(U) \otimes_{\mathfrak{A}_U(U)} \mathfrak{M}_U(U).$$

Proposition 8.2.5. Let $\mathfrak{T}_1 \xrightarrow{q} \mathfrak{T}_2 \to \mathfrak{S}$ be two morphisms of sites, and let (F_λ) be a local inductive system of sheaves on $\mathfrak{T}_2$ indexed by a locally cofiltered category $\mathfrak{L}$ over $\mathfrak{S}$. Then

$$q^* \varinjlim_{\lambda \in \mathfrak{L}} F_\lambda \simeq \varinjlim_{\lambda \in \mathfrak{L}} q^* F_\lambda.$$

This follows immediately from the fact that q^* admits a right adjoint q_*.

Let $\mathfrak{S}$ and $\mathfrak{T}$ be two sites in which finite projective limits are representable. Let $p : \mathfrak{T} \to \mathfrak{S}$ be a morphism of sites such that $p^* : \mathfrak{S} \to \mathfrak{T}$ commutes with finite projective limits, and let $\mathfrak{L}$ be the locally cofiltered category over $\mathfrak{T}$ whose objects are triples (V, U, ϕ) with $V \in \mathrm{Ob}(\mathfrak{T})$, $U \in \mathrm{Ob}(\mathfrak{S})$, and $\phi \in \mathrm{Hom}(V, p^*U)$. It is obtained by applying 8.1.11 to $1_{\mathfrak{S}} : \mathfrak{S} \to \mathfrak{S}$.

Proposition 8.2.6. For every sheaf $\mathfrak{F}$ on $\mathfrak{T}$, one has

$$\mathfrak{F} \simeq \varinjlim_{\lambda \in \mathfrak{L}} \phi^* \phi_* \mathfrak{F},$$

where, by abuse of notation, ϕ denotes the morphism induced by p from $\mathfrak{T}/V$ to $\mathfrak{S}/U$.

Let ϕ' be the inverse-image functor in the sense of presheaves. From 8.2.3 one obtains

$$\varinjlim_{\lambda \in \mathfrak{L}} \phi'^* \phi_* \mathfrak{F} = \varinjlim_{\lambda \in \mathfrak{L}} a\phi' \phi_* \mathfrak{F} = \varinjlim_{\lambda \in \mathfrak{L}} \mathfrak{F}(V)^\sim = \varinjlim_{V} \mathfrak{F}(V)^\sim = \mathfrak{F}.$$

Proposition 8.2.7. Let $p : T \to S$ be a morphism of topoi, let $\mathcal{O}_S$ be a sheaf of rings on S, and let $\mathcal{O}_T = p^* \mathcal{O}_S$. For a right $\mathcal{O}_T$-module $\mathfrak{M}$ to be flat over $\mathcal{O}_T$, it is necessary and sufficient that the functor

$$\mathfrak{N} \longmapsto \mathfrak{M} \otimes_{\mathcal{O}_T} p^* \mathfrak{N} \tag{8.2.7.1}$$

from left $\mathcal{O}_S$-modules to abelian sheaves on T be exact.

Necessity is trivial. Conversely, suppose that (8.2.7.1) is exact. One may suppose p is defined by a morphism of sites $p : \mathfrak{T} \to \mathfrak{S}$ satisfying the hypotheses used above. Let $V \in \mathrm{Ob}(\mathfrak{T})$, $U \in \mathrm{Ob}(\mathfrak{S})$, and $\phi : V \to p^*U$; again denote by ϕ the induced morphism of sites $\mathfrak{T}/V \to \mathfrak{S}/U$.

Lemma 8.2.8. The functor

$$\mathfrak{N} \longmapsto \mathfrak{M}|V \otimes_{\mathcal{O}_V} \phi^*\mathfrak{N},$$

from $\mathcal{O}_S|U$-modules on $\mathfrak{S}/U$ to $\mathcal{O}_T|V$-modules on $\mathfrak{T}/V$, is exact.

One reduces to the case $V = p^*U$. Let $j : \mathfrak{S}/U \to \mathfrak{S}$ and $j' : \mathfrak{T}/p^*U \to \mathfrak{T}$ be the inclusion morphisms. Then

$$j'_!(\mathfrak{M}|V \otimes_{\mathcal{O}_V} \phi^*\mathfrak{N}) = \mathfrak{M} \otimes_{\mathcal{O}_T} p^* j_!\mathfrak{N}.$$

Lemma 8.2.8 follows because $j_!$ and $j'_!$ are exact and faithful.

Let Q be an $\mathcal{O}_T$-module. By 8.2.6,

$$Q = \varinjlim_{\phi : V \to p^*U} \phi^*\phi_* Q|V.$$

Therefore, by 8.2.4,

$$\mathfrak{M} \otimes_{\mathcal{O}_T} Q = \varinjlim_{\phi : V \to p^*U} \mathfrak{M}|V \otimes_{\mathcal{O}_V} \phi^*\phi_* Q|V.$$

By Lemma 8.2.8, the functors

$$Q \longmapsto \mathfrak{M}|V \otimes_{\mathcal{O}_V} \phi^*\phi_* Q|V$$

are exact on the left. Hence $\mathfrak{M} \otimes_{\mathcal{O}_T} Q$ is left exact in Q, and therefore exact.

Corollary 8.2.9. Let $f : (T, \mathcal{O}_T) \to (S, \mathcal{O}_S)$ be a morphism of ringed topoi. Then the inverse image by f of a flat sheaf of right modules is a flat sheaf of modules.

Factoring f through $(T, f^*\mathcal{O}_S)$, one reduces to the case $\mathcal{O}_T = f^*\mathcal{O}_S$. Let $\mathfrak{M}$ be a flat sheaf of modules on S. By 8.2.7, to check that $f^*\mathfrak{M}$ is flat it is enough to check exactness of the functor

$$\mathfrak{N} \longmapsto f^*\mathfrak{M} \otimes_{\mathcal{O}_T} f^*\mathfrak{N} = f^*(\mathfrak{M} \otimes_{\mathcal{O}_S} \mathfrak{N}),$$

which follows since f^* is exact.

Lemma 8.2.10. Let $(S, \mathcal{O}_S)$ be a ringed topos, let $\mathfrak{F}$ be a sheaf of left modules locally of finite presentation, let $\mathfrak{G}$ be a sheaf of bimodules, and let $\mathcal{H}$ be a flat sheaf of left modules. Then the canonical arrow

$$\mathcal{H}om(\mathfrak{F}, \mathfrak{G}) \otimes \mathcal{H} \longrightarrow \mathcal{H}om(\mathfrak{F}, \mathfrak{G} \otimes \mathcal{H})$$

is an isomorphism.

The question is local on S, so one may suppose that $\mathfrak{F}$ admits a finite presentation

$$\mathfrak{F}_1 \to \mathfrak{F}_0 \to \mathfrak{F} \to 0.$$

The first row of the following diagram is exact because $\mathcal{H}$ is flat:

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{H}om(\mathfrak{F}, \mathfrak{G}) \otimes \mathcal{H} \arrow[r] \arrow[d] & \mathcal{H}om(\mathfrak{F}_0, \mathfrak{G}) \otimes \mathcal{H} \arrow[r] \arrow[d] & \mathcal{H}om(\mathfrak{F}_1, \mathfrak{G}) \otimes \mathcal{H} \arrow[d] \\
0 \arrow[r] & \mathcal{H}om(\mathfrak{F}, \mathfrak{G} \otimes \mathcal{H}) \arrow[r] & \mathcal{H}om(\mathfrak{F}_0, \mathfrak{G} \otimes \mathcal{H}) \arrow[r] & \mathcal{H}om(\mathfrak{F}_1, \mathfrak{G} \otimes \mathcal{H}).
\end{tikzcd}$$

The assertion follows. Taking $\mathfrak{G} = \mathcal{O}_S$, one obtains:

Lemma 8.2.11. Every morphism from a sheaf of modules locally of finite presentation to a flat sheaf of modules factors, locally on S, through a free sheaf $\mathcal{O}_S^n$, $n \in \mathbb{N}$.

Theorem 8.2.12 (D. Lazard). For a sheaf of modules $\mathfrak{M}$ on a ringed site $(S, \mathcal{O}_S)$ to be flat, it is necessary and sufficient that it be a local inductive limit of finite free modules.

Sufficiency follows from 8.2.2 and 8.2.4. For every $U \in \mathrm{Ob}(\mathfrak{S})$, let $\mathfrak{L}_0(U)$ (respectively $\mathfrak{L}_1(U)$) be the category of finite free (respectively finitely presented) sheaves of modules on U, endowed with a linear map to $\mathfrak{M}|U$. For every arrow $f : V \to U$ in $\mathfrak{S}$, let f^* be the restriction functor from $\mathfrak{L}_i(U)$ to $\mathfrak{L}_i(V)$. These functors define a category $\mathfrak{L}_i$, fibered over $\mathfrak{S}$, whose fibers are the opposite categories of the categories $\mathfrak{L}_i(U)$.

The reader checks that $\mathfrak{L}_1$ is locally filtered over $\mathfrak{S}$ and that

$$\mathfrak{M} = \varinjlim_{(M,f) \in \mathfrak{L}_1} M.$$

If $\mathfrak{M}$ is flat, the inclusion of $\mathfrak{L}_0$ into $\mathfrak{L}_1$ is fully faithful and, by 8.2.11, satisfies $(C\mathfrak{F}1)$ of 8.1.4. It therefore automatically satisfies $(C\mathfrak{F}2)$, and $\mathfrak{L}_0$ is locally filtered. By 8.1.5,

$$\mathfrak{M} = \varinjlim_{(M,f) \in \mathfrak{L}_0} M,$$

which proves the theorem.

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SGA 4, Exposé VI

Finiteness Conditions. Fibred Topoi and Sites. Applications to Questions of Passage to the Limit

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0. Introduction

In the present exposé we study two kinds of questions which are closely related.

First, we make a systematic study of finiteness conditions for topoi. We successively study the notions of *quasi-compactness*, *quasi-separatedness*, and *coherence* (quasi-compactness plus quasi-separatedness), inspired by the analogous notions familiar in scheme theory. We treat first the case of an object X of a topos and that of an arrow $u : X \to Y$ in a topos (§1), then the topos E itself (§2), and finally a morphism of topoi $f : E \to E'$ (§3). Section 4 studies these notions for a topos obtained by gluing two topoi (Exposé IV, §9). For the following sections, the essential point to retain from these rather technical developments is the definition of a *coherent topos*: a topos equivalent to a topos of the form $C^{\sim}$, where C is a small site in which finite projective limits are representable and every covering family admits a finite covering subfamily; and the definition of a *coherent morphism* $f : E \to E'$ of coherent topoi: a morphism which, up to equivalence, is associated with a morphism of sites $f : C \to C'$, with C, C' as above and with the corresponding functor $f^* : C' \to C$ left exact.

It is useful to note that, with the finiteness notions used here, and especially with coherent topoi, we deliberately depart, for the first time, from the spaces familiar to analysts and "topologists" in the ordinary sense. For example, the topos associated to a separated topological space X is coherent only if X is a finite set. Thus this exposé is entirely oriented toward the kinds of topoi which arise in algebraic geometry and algebra. In fact, all topoi used so far in these disciplines (except, of course, in algebraic geometry by transcendental methods over the field of complex numbers) are locally coherent.

Second, we develop two *inductive-limit passage theorems* for cohomology of topoi. One says that the functors $H^i(E, -)$ on a coherent topos commute with inductive limits of abelian sheaves. The other says, essentially, that if a topos X is represented as a filtered projective limit of coherent topoi X_i, with coherent transition morphisms $f_{ij} : X_j \to X_i$, then the cohomology of X, with coefficients in an arbitrary abelian sheaf, is computed as the inductive limit of the cohomologies of the X_i. To give this statement a precise meaning, the main point is to define the notion of a *projective limit of topoi*; this is done in §§7–8. It is a useful geometric notion, probably beyond the context of projective limits of schemes, which had served M. Artin as the model for the initial definition. For example, we show how the various notions of "localization at a point" (of a noetherian space, of a scheme for its usual topology, and even for its étale topology) can be unified by defining the localization as the projective limit of the neighborhoods of that point, in perfect accord with the intuitive idea of localization at a point.

The two limit-passage theorems will be restated in the following exposé in the particular framework of étale cohomology of schemes. In that case they are constantly used throughout the rest of the Seminar, and practically everywhere one has worked with étale cohomology. A reader willing to admit these two theorems in the special form in which they are stated in Exposé VII, and interested only in applications to étale cohomology, may therefore omit the present exposé.

It is true that the notion of a *constructible object* of a topos, namely an object whose structural morphism $X \to e$ to the final object is coherent, will also play an important role throughout the rest of the Seminar. In the present exposé, however, it plays only a secondary role, if only because in the case of a coherent topos it coincides with the notion of a coherent object, which is the main notion here. The notion is developed independently in Exposé IX for étale sheaves on schemes, and in that setting it has useful special features which underlie the presentation in IX. Since the redaction of IX preceded the present redaction of VI by five years and is nevertheless perfectly usable, we content ourselves at the end of IX with proving the equivalence of the constructibility notion used there with the one introduced here.

1. Finiteness conditions for objects and arrows of a topos

Definition 1.1. Let C be a site. An object X of C is called *quasi-compact* if, for every covering family $(X_i \to X)_{i\in I}$, there exists a finite subset $J \subset I$ such that the subfamily $(X_i \to X)_{i\in J}$ is still covering.

Since a $\mathcal{U}$-topos E is always regarded as a site, this definition applies in particular to objects of E. The following result reduces the site case to the topos case.

Proposition 1.2. Let C be a $\mathcal{U}$-site, let $\epsilon : C \to C^{\sim}$ be the canonical functor, and let X be an object of C. Then X is quasi-compact as an object of the site C if and only if $\epsilon(X)$ is quasi-compact as an object of the topos $C^{\sim}$.

Proof. Assume X quasi-compact. Let $(F_i \to \epsilon(X))_{i\in I}$ be an epimorphic family. For each i, choose an epimorphic family $\epsilon(X_{ij}) \to F_i$. After refining these families, the composites $\epsilon(X_{ij}) \to \epsilon(X)$ may be assumed to come from morphisms $X_{ij} \to X$. Since the family of all $\epsilon(X_{ij}) \to \epsilon(X)$ is epimorphic, the corresponding family $X_{ij} \to X$ is covering. By quasi-compactness of X, a finite subfamily is still covering; hence the corresponding finite subfamily of the $F_i \to \epsilon(X)$ is epimorphic. Conversely, if $\epsilon(X)$ is quasi-compact, any covering of X gives an epimorphic covering of $\epsilon(X)$, and the result is immediate.

Thus quasi-compactness in sites reduces to quasi-compactness in topoi.

Proposition 1.3. Let C be a site and let $(X_i \to X)_{i\in I}$ be a finite covering family whose X_i are quasi-compact. Then X is quasi-compact.

Proof. Using 1.2, reduce to the case where C is a topos. If $(X'_j \to X)_{j\in J}$ is an epimorphic family, then after base change to each X_i it admits a finite epimorphic subfamily. Since I is finite, the union of these finite index sets is finite, and the corresponding subfamily already covers X.

Corollary 1.4. Let $(X_i)_{i\in I}$ be a family of objects of a topos E, and let $X = \coprod_i X_i$. Then X is quasi-compact if and only if every X_i is quasi-compact and all but finitely many X_i are initial.

Proof. The sufficiency follows from 1.3. Conversely, a finite subfamily must cover X, which forces all omitted summands to be initial. To see that each X_i is quasi-compact, write $X \simeq X_i \amalg X'_i$ and extend coverings of X_i to coverings of X.

Remark 1.5.1. Whether an object X of a topos E is quasi-compact depends only on the induced topos $E_{/X}$: it means that the final object of $E_{/X}$ is quasi-compact. Similarly, an object X of a site C is quasi-compact if and only if the final object of the induced site $C_{/X}$ is quasi-compact.

An object X of a topos E is quasi-compact if and only if every filtered increasing family $(X_i)_{i\in I}$ of subobjects of X whose union is X contains an $X_i = X$. This condition depends only on the ordered set of subobjects of X, that is, on the opens of $E_{/X}$.

Let $(X_i)_{i\in I}$ be a generating family of E. From the definitions and 1.3 it follows that, for X to be quasi-compact, it is necessary that X admit a covering by finitely many of the X_i, equivalently that X be a quotient of a finite sum of objects from the generating family. This condition is sufficient if all X_i are quasi-compact. Replacing the family by a subfamily, one may suppose $I \in \mathcal{U}$. Since the set of quotients of an object of E is small, the full subcategory $E_{\mathrm{qc}} \subset E$ of quasi-compact objects is equivalent to a category belonging to $\mathcal{U}$.

Example 1.6.1. Let X be a topological space. An open U is quasi-compact as an object of the site $\mathrm{Open}(X)$ if and only if U is quasi-compact for the induced topology. More generally, a sheaf F on X is quasi-compact as an object of $\mathrm{Top}(X)$ if and only if the étale space associated with F is quasi-compact.

For the category **Sch** of schemes and any of the standard topologies considered later, such as the Zariski, étale, fppf, or fpqc topology, a scheme is quasi-compact for that topology if and only if it is quasi-compact in the usual sense.

Definition 1.7. Let E be a topos and let $f : X \to Y$ be an arrow of E. The arrow f is called *quasi-compact* if, for every arrow $Y' \to Y$ with Y' quasi-compact, the object

$$X' = X \times_Y Y'$$

is quasi-compact. It is called *quasi-separated* if its diagonal

$$X \longrightarrow X \times_Y X$$

is quasi-compact. It is called *coherent* if it is both quasi-compact and quasi-separated.

Spelling out the definition, $f : X \to Y$ is quasi-separated precisely when, for every pair of arrows $g_1, g_2 : X' \rightrightarrows X$ over Y, with X' quasi-compact, the equalizer $\mathrm{Ker}(g_1, g_2)$ is quasi-compact.

Proposition 1.8. Let E be a topos.

(i) Every isomorphism in E is coherent. Composites of quasi-compact, respectively quasi-separated, respectively coherent, morphisms have the same property.

(ii) These three properties are stable under base change.

(iii) If $f : X \to Y$ and $g : Y \to Z$ are morphisms and gf is quasi-separated, then f is quasi-separated.

(iv) If $g : Y \to Z$ is quasi-separated and $gf : X \to Z$ is quasi-compact, respectively coherent, then f is quasi-compact, respectively coherent.

Proof. The proofs are the standard ones from algebraic geometry. For quasi-separatedness of a composite one uses the cartesian square involving the diagonals

$$\begin{tikzcd}
X \arrow[d] \arrow[dd, bend right=60, "\Delta_{X/Z}"'] & \\
X \times_Y X \arrow[r] \arrow[d] & Y \arrow[d, "\Delta_{Y/Z}"] \\
X \times_Z X \arrow[r] & Y \times_Z Y.
\end{tikzcd}$$

For (iv), use the graph $\Gamma_f : X \to X \times_Z Y$ and the cartesian diagram comparing it with $\Delta_{Y/Z}$. The remaining part reduces to the fact that a monomorphism is quasi-separated.

Corollary 1.8.1. Every monomorphism is quasi-separated.

Corollary 1.8.2. Let $f : X \to Y$ and $f' : X' \to Y'$ be two S-morphisms in E. If f and f' are quasi-compact, respectively quasi-separated, respectively coherent, then so is

$$f \times_S f' : X \times_S X' \longrightarrow Y \times_S Y'.$$

Remark 1.9.1. Let E be any category with fiber products, and let Q be a class of objects stable under isomorphism, playing the role of quasi-compact objects. One may define Q-quasi-compact, Q-quasi-separated, and Q-coherent morphisms by the preceding formulas. Proposition 1.8 and Corollary 1.8.1 remain valid; so does Corollary 1.8.2 when binary products exist.

Let S be an object of E, and let $f : X \to Y$ be a morphism in $E_{/S}$. By 1.5, f is quasi-compact, quasi-separated, or coherent in $E_{/S}$ if and only if the underlying morphism in E has the same property. In particular, an arrow $f : X \to Y$ is quasi-compact, quasi-separated, or coherent precisely when the structural morphism of the object (X, f) of $E_{/Y}$ to the final object is so. One also says that X is quasi-compact, quasi-separated, or coherent *over* Y.

An object X of a topos E is called *constructible* if it is coherent over the final object. Every morphism from a constructible object to another is coherent. The full subcategory E_{cons} of constructible objects is stable under finite projective limits.

The notions in 1.7 are useful mainly for topoi admitting a generating family of quasi-compact objects. Under this assumption they are local on the base.

Proposition 1.10. Assume E has a generating subcategory C consisting of quasi-compact objects, and let $f : X \to Y$ be a morphism in E.

(i) f is quasi-compact if and only if, for every $S \to Y$ with $S \in \operatorname{Ob} C$, the object $X \times_Y S$ is quasi-compact.

(ii) If $(Y_i \to Y)_{i \in I}$ is a covering family, then f is quasi-compact, respectively quasi-separated, respectively coherent, if and only if all base changes

$$f_i : X_i = X \times_Y Y_i \longrightarrow Y_i$$

have the corresponding property.

Proof. For (i), reduce testing against an arbitrary quasi-compact $Y' \to Y$ to a finite cover of Y' by objects of C, and use 1.3. For (ii), it suffices to treat quasi-compactness, since quasi-separatedness is the quasi-compactness of the diagonal and coherence is the conjunction. Again reduce to testing after a finite cover by objects of C.

Corollary 1.11. Let $g : Y \to Z$ be a morphism in E, and let $(f_i : X_i \to Y)_{i \in I}$ be a covering family with target Y.

(i) If I is finite and all gf_i are quasi-compact, then g is quasi-compact.

(ii) If the f_i are quasi-compact and all gf_i are quasi-separated, then g is quasi-separated. If moreover I is finite and the gf_i are coherent, then g is coherent.

Corollary 1.12. Let $(X_i)_{i \in I}$ be a finite family of objects of E, let $X = \coprod_i X_i$, and let $f_i : X_i \to Y$ be morphisms with associated $f : X \to Y$. Then f is quasi-compact, respectively quasi-separated, respectively coherent, if and only if the f_i have the corresponding property. In the quasi-separated case, the finiteness hypothesis on I may be omitted.

Definition 1.13. An object X of a topos E is called *quasi-separated* if, for every quasi-compact object S of E, every morphism $S \to X$ is quasi-compact. Equivalently, for quasi-compact S, T and morphisms $S \to X$, $T \to X$, the product $S \times_X T$ is quasi-compact. An object is *coherent* if it is quasi-compact and quasi-separated.

Proposition 1.14. Let $f : X \to Y$ be a morphism in a topos E.

(i) If Y is quasi-compact, then f quasi-compact implies X quasi-compact. If Y is quasi-separated, then X quasi-compact implies f quasi-compact. If Y is coherent, then f is quasi-compact if and only if X is quasi-compact.

(ii) If Y is quasi-compact, respectively quasi-separated, respectively coherent, and f has the same corresponding property, then X has that property.

Proof. The quasi-compact statements are formal from the definitions. For the quasi-separated case in (ii), let S be quasi-compact and $g : S \to X$. Since Y is quasi-separated, fg is quasi-compact; since f is quasi-separated, it follows that g is quasi-compact.

Corollary 1.15. Every subobject of a quasi-separated object is quasi-separated. For a family $(X_i)_{i \in I}$, the sum $X = \coprod_i X_i$ is quasi-separated, respectively coherent, if and only if the X_i are so; in the coherent case, all but finitely many X_i must be initial.

Corollary 1.16. Under the hypotheses of Proposition 1.10(ii), if the Y_i are coherent, then f is quasi-compact if and only if the objects

$$X_i = X \times_Y Y_i$$

are quasi-compact.

Corollary 1.17. Let E admit a generating family of quasi-compact objects, and let $(g_i : Y_i \to Y)_{i \in I}$ be a covering family, respectively finite covering family, with the Y_i coherent. Then Y is quasi-separated, respectively coherent, if and only if every g_i is quasi-compact, equivalently if and only if for all i, j the object

$$Y_i \times_Y Y_j$$

is quasi-compact.

Corollary 1.17.1. Let E be as in 1.17, let X be coherent, let $\mathcal{R} \rightrightarrows X$ be an equivalence relation, and put $Y = X/\mathcal{R}$. The following are equivalent:

$$Y \text{ is coherent}, \qquad X \to Y \text{ is quasi-compact}, \qquad \mathcal{R} \text{ is quasi-compact}.$$

Corollary 1.18. Let E have a generating subcategory C of quasi-compact objects, and let $Y \in E$. Then Y is quasi-separated if and only if every morphism $X \to Y$, with $X \in \operatorname{ob} C$, is quasi-compact.

Corollary 1.19. Let E admit a generating subcategory of quasi-compact objects. Let $(g_i : Y_i \to Y)_{i \in I}$ be a covering family with the Y_i quasi-separated and the g_i quasi-compact. Then Y is quasi-separated.

Proof. For any quasi-compact X and morphism $X \to Y$, after base change to Y_i the objects $X \times_Y Y_i$ are quasi-compact; since Y_i is quasi-separated, the induced morphisms to Y_i are quasi-compact. Proposition 1.10 then gives the result.

Remark 1.20.1. A reader accustomed to the terms quasi-compact and quasi-separated in algebraic geometry might expect further relations between the absolute notions for objects of E and the relative notions for arrows, such as: if $f : X \to Y$ has quasi-compact separated source X, then f is quasi-separated. Such properties require restrictive finiteness hypotheses on E, stronger than those in Proposition 1.10; they will be studied in the next section.

It again follows from 1.5 that quasi-separatedness of an object X depends only on the induced topos $E_{/X}$; it means that the final object of $E_{/X}$ is quasi-separated, or equivalently that in $E_{/X}$ the product of two quasi-compact objects is quasi-compact.

1.21. The case of sites

Let C be a $\mathcal{U}$-site, E the topos of $\mathcal{U}$-sheaves on C, f an arrow of C, and X an object of C. We say that f is a *quasi-compact morphism* (respectively quasi-separated morphism) of the site C if $\epsilon(f)$ is a quasi-compact (respectively quasi-separated) morphism of the topos E. Similarly, X is called a *quasi-separated object of the site* C if $\epsilon(X)$ is quasi-separated in E. When C is a $\mathcal{U}$-topos with its canonical topology, $\epsilon : C \to E$ is an equivalence, so this agrees with the preceding definitions.

The definition is unchanged when replacing $\mathcal{U}$ by a larger universe $\mathcal{V}$, at least when C has a topologically generating family of quasi-compact objects. Indeed, one reduces to $\mathcal{U} \subset \mathcal{V}$, so that E is a $\mathcal{V}$-site, and compares the corresponding $\mathcal{V}$-topos E' of sheaves on E. The quasi-compact-object case is already known from 1.2; quasi-compactness of arrows follows from the criterion 1.10(i) applied in E and in E' to the same quasi-compact generating family, quasi-separatedness of arrows follows by passing to diagonals, and quasi-separatedness of objects follows from the criterion 1.18.

Continuation anchor. The next untranslated point is SGA 4, Exposé VI, Section 1.22, source line 714 of `06/06.tex`.

SGA 4, Exposé VI

Finiteness Conditions. Fibred Topoi and Sites. Applications to Questions of Passage to the Limit

A. Grothendieck and J.-L. Verdier

1.22. Examples

Return to Example 1.6.2, the category (**Sch**) endowed with one of the topologies T_i. For a suitable universe $\mathcal{V}$, this is a $\mathcal{V}$-site, and the subcategory of affine schemes is a topological generating category consisting of quasi-compact objects. We leave to the reader the verification that an arrow $f : X \to Y$ of this site is quasi-compact, respectively quasi-separated, if and only if it is a quasi-compact, respectively quasi-separated, morphism of schemes in the usual sense. Likewise, an object of the site is quasi-compact, respectively quasi-separated, if and only if it is a quasi-compact, respectively quasi-separated, scheme in the usual sense. In fact, as here, one should say *coherent morphism of schemes* and *coherent scheme*, instead of "quasi-compact and quasi-separated morphism of schemes" and "quasi-compact and quasi-separated scheme", as is done in the second edition of EGA I.

1.22.1

For a given scheme $X \in \mathcal{U}$, one may also consider the topos $\mathrm{Top}(X)$ associated with the underlying topological space, or the topos $\mathrm{Top}(X_{\mathrm{et}})$, respectively $\mathrm{Top}(X_{\mathrm{fppf}})$, associated with the site of X-schemes which are etale, respectively locally of finite presentation, belong to $\mathcal{U}$, and are endowed with the etale, respectively fppf, topology. This is a $\mathcal{U}$-topos admitting a generating family consisting of coherent objects, corresponding to the affine objects of the defining site.

With this understood, the final object of this topos is quasi-compact, respectively quasi-separated, respectively coherent, if and only if the scheme X is quasi-compact, respectively quasi-separated, respectively coherent. Likewise, a morphism in the site of etale spaces over X, respectively in the site X_{et} of etale schemes over X, respectively in the site X_{fppf} of schemes locally of finite presentation over X, is quasi-compact, respectively quasi-separated, respectively coherent, if and only if it is a morphism of schemes which is quasi-compact, respectively quasi-separated, respectively coherent, in the usual sense.

Theorem 1.23. Let E be a $\mathcal{U}$-topos and let X be an object of E.

(i) The object X is quasi-compact if and only if, for every filtered inductive system $(Y_i)_{i\in I}$ of E, with $I \in \mathcal{U}$, the natural map

$$\varinjlim \mathrm{Hom}(X, Y_i) \longrightarrow \mathrm{Hom}(X, \varinjlim Y_i) \tag{1.23.1}$$

is injective.

(ii) Assume that E admits a generating subcategory C consisting of quasi-compact objects. If X is coherent, then for every datum as in *(i)* the map *(1.23.1)* is bijective.

Proof. For (i), first prove necessity. Suppose X quasi-compact. Let $i \in I$, and let $f_i, g_i : X \rightrightarrows Y_i$ be such that their composites f, g with $Y_i \to Y = \varinjlim Y_j$ are equal. We must show that there exists $j \geq i$ such that the composites f_j, g_j with $Y_i \to Y_j$ are already equal. Since in a topos filtered inductive limits commute with finite projective limits, and in particular with kernels, one has

$$\operatorname{Ker}(f, g) = \varinjlim_{j \geq i} \operatorname{Ker}(f_j, g_j).$$

By hypothesis $\operatorname{Ker}(f, g) = X$. Thus X is the union of the filtered increasing family of its subobjects $\operatorname{Ker}(f_j, g_j)$; since X is quasi-compact, it is equal to one of them. Hence $f_j = g_j$ for some $j \geq i$.

Conversely, to prove that X is quasi-compact, it is equivalent to prove that every filtered increasing family $(X_i)_{i \in I}$ of subobjects of X whose inductive limit is X contains an $X_i = X$. Since in a topos every monomorphism is effective, one has

$$X_i = \operatorname{Ker}\big(f_i, g_i : X \rightrightarrows X \amalg_{X_i} X\big).$$

Put $Y_i = X \amalg_{X_i} X$, and consider the inductive system formed by the Y_i. If Y is its inductive limit, then

$$Y \simeq X \amalg_{\varinjlim X_i} X = X \amalg_X X = X.$$

For a fixed i, the two arrows f_i, g_i therefore define the same element of $\operatorname{Hom}(X, Y)$. By the assumed injectivity there exists $j \geq i$ such that $f_j = g_j$, that is, $X_j = X$.

For (ii), it remains to prove surjectivity: every morphism $X \to Y = \varinjlim Y_i$ comes from a morphism $X \to Y_i$ for suitable i. Since C generates E and the Y_i cover Y, we can cover X by objects X_α of C so that each composite $X_\alpha \to X \to Y$ factors through some Y_i. Since X is quasi-compact, the family (X_α) may be taken finite, hence all these factorizations may be taken through a fixed Y_i, say as morphisms $g_{\alpha i} : X_\alpha \to Y_i$.

Since X is quasi-separated, the objects $X_\alpha \times_X X_\beta$ are quasi-compact. For each pair (α, β), the two composites from $X_\alpha \times_X X_\beta$ to Y_i induced by X_α and X_β become equal after composition with $Y_i \to Y$. By (i), after replacing i by some $j \geq i$, they are already equal. Since there are only finitely many pairs (α, β), one may choose a single such j. The morphisms $g_{\alpha j}$ then glue to a morphism $g_j : X \to Y_j$, whose composite with $Y_j \to Y$ is the given morphism $X \to Y$. This proves the theorem.

Remark 1.23.2. The hypothesis on E in (ii) may be replaced by an additional hypothesis on X: namely, that X admits a fundamental system of covering families by quasi-compact objects.

Corollary 1.24. Let E be a topos, and let C be a strictly full subcategory of E consisting of coherent objects. Using the fact that $\mathcal{U}$-indexed inductive limits are representable in E, one obtains a canonical functor

$$\operatorname{Ind}(C) \longrightarrow E. \tag{1.24.1}$$

This functor is fully faithful. If E admits a generating subcategory consisting of coherent objects and if C is stable under direct factors, then the following conditions are equivalent:

(i) The functor *(1.24.1)* is essentially surjective, hence an equivalence of categories.

(ii) C is a generating subcategory of E, stable under finite inductive limits.

(iii) The category C is equal to the full subcategory E_{coh} of E consisting of all coherent objects of E, and the necessary condition *1.23(ii)* for an object of E to be coherent is also sufficient.

Let E_{PF} be the full subcategory of E defined in I, 8.7.6. By I, 8.7.5 a), assertion 1.23(ii) is equivalent to the second inclusion in

$$C \subset E_{\mathrm{coh}} \subset E_{PF},$$

and the inclusion corresponding to full faithfulness is precisely that (1.24.1) be fully faithful. Since by hypothesis E_{coh} is a generating subcategory of E, the same is a fortiori true of E_{PF}. Moreover finite projective limits are representable in E, so I, 8.7.7 applies. Condition (iii) above can also be written $C = E_{\mathrm{coh}} = E_{PF}$, hence is equivalent to $C = E_{PF}$, which is condition (ii bis) of loc. cit. Similarly, condition (ii) above is condition (iii bis) of loc. cit. The equivalence of (i), (ii bis), and (iii bis) in loc. cit. therefore gives the equivalence of (i), (ii), and (iii) here.

The proof, more precisely the invocation of I, 8.7.7(ii bis), also gives part a) of the following corollary.

Corollary 1.24.2. Let E be a topos admitting a generating subfamily consisting of coherent objects, and let E_{PF} be the strictly full subcategory of the objects X of E such that the covariant representable functor $\mathrm{Hom}(X, -)$ commutes with filtered inductive limits. Then:

a) The natural functor
$$\mathrm{Ind}(E_{PF}) \longrightarrow E$$
is an equivalence of categories.

b) An object X of E belongs to E_{PF} if and only if it is isomorphic to a direct factor, equivalently the image of a projector, of the cokernel of a double arrow $X_1 \rightrightarrows X_0$, with X_1 and X_0 coherent.

c) The subcategory E_{PF} of E is stable under finite inductive limits and is equivalent to a small category.

It remains to prove b) and c). The first assertion of c) is trivial by I, 2.8, and implies the sufficiency in b) by 1.23(ii). For necessity, use 1.23(i), which shows that X is quasi-compact. Hence there exists an epimorphism $X_0 \to X$, with X_0 coherent, since E_{coh} generates E and is stable under finite sums. Let $R = X_0 \times_X X_0$, so that X is the cokernel of $R \rightrightarrows X_0$. By a), $R = \varinjlim R_i$ is a filtered inductive limit; put $X'_i = \mathrm{Coker}(R_i \rightrightarrows X_0)$. Since $X \in \mathrm{Ob}\, E_{PF}$, for sufficiently large i the morphism $u_i : X'_i \to X$ admits a section $v : X \to X'_i$. Put $w = vu_i : X'_i \to X'_i$. Then $w^2 = w$, and
$$X \simeq \mathrm{Coker}(w, \mathrm{id}_{X'_i} : X'_i \rightrightarrows X'_i).$$
Since R_i is again quasi-compact, cover it by $X_1 \to R_i$ with X_1 coherent; this gives $X'_i \simeq \mathrm{Coker}(X_1 \rightrightarrows X_0)$, proving b). Finally, since $E_{PF} \subset E_{\mathrm{qc}}$, the smallness assertion in c) follows from 1.5.3.

Corollary 1.25. Let E be a topos such that the full subcategory C of E consisting of coherent objects is generating. Consider the canonical functor

$$\varphi : \mathrm{Ind}(C) \longrightarrow E.$$

The following conditions are equivalent:

(i) The functor φ is essentially surjective, i.e. every object of E is a filtered inductive limit of coherent objects.

(i bis) The functor φ is an equivalence of categories.

(ii) Every finite inductive limit in E of coherent objects is coherent.

(ii bis) The cokernel in E of a double arrow of coherent objects is coherent.

(iii) The converse of *1.23(ii)* holds.

This is a special case of 1.24, since C is stable under finite sums, which implies the equivalence of (ii) and (ii bis), and since we have the following lemma.

Lemma 1.25.1. Every direct factor X' of a coherent object X of a topos E is coherent.

Indeed, X' is quasi-compact as a quotient of X, and quasi-separated as a subobject of X.

Corollary 1.26. Let E be a topos satisfying the equivalent conditions of *1.25.* Then every induced topos satisfies them as well.

This follows immediately from criterion (ii), using 1.20.2 and the fact that if E has a generating subcategory consisting of coherent objects, then the same is plainly true for every induced topos.

Remark 1.27. Important examples of topoi satisfying the conditions of 1.25 include noetherian topoi (2.14), and the Zariski or etale topos of a scheme (cf. IX, note p. 42). But even when E is a *coherent* topos (2.3), meaning that the full subcategory C of coherent objects is generating and stable under finite projective limits, it is not always true that C is *perfect* (2.9.1), i.e. satisfies the equivalent conditions of 1.25; see Exercise 1.28 below.

Exercise 1.28. Let E be a topos and let $p_1, p_2 : X_1 \rightrightarrows X_0$ be a double arrow in E. Define an increasing sequence of subobjects R_n $(n \geq 0)$ of $X_0 \times X_0$ by

$$R_0 = \mathrm{Sup}(\mathrm{Im}(p_1, p_2), \mathrm{Im}(p_2, p_1)),$$

and, for $n \geq 1$,

$$R_n = \mathrm{pr}_{13}(\mathrm{pr}_{12}^{-1}(R_{n-1}) \cap \mathrm{pr}_{23}^{-1}(R_{n-1})),$$

where pr_{ij}, $1 \leq i < j \leq 3$, denote the evident projections from the triple product of X_0 to the double product.

a) Show that one obtains an increasing sequence of subobjects of $X_0 \times X_0$, that $R = \varinjlim R_n$ is an equivalence graph in X_0, and that the canonical morphism

$$X = \mathrm{Coker}(p_1, p_2) \longrightarrow X_0/R$$

is an isomorphism.

b) Deduce that if X_0 is quasi-compact and $X = \mathrm{Coker}(p_1, p_2)$ is quasi-separated, hence coherent, then the sequence (R_n) is stationary. Conversely, if this sequence is stationary and if X_0 is coherent and X_1 quasi-compact, then X is coherent. For the latter point, write

$$R_n \simeq (R_{n-1}, \mathrm{pr}_2) \times_{X_0} (R_{n-1}, \mathrm{pr}_1)$$

and deduce that R_n is quasi-compact over X_0 for all n, then use 1.19.

c) Construct an example of two maps of sets $p_1, p_2 : X_1 \rightrightarrows X_0$ for which the sequence (R_n) is not stationary.

d) Let C be a small category in which finite inductive and finite projective limits are representable, and in which every morphism factors as an effective epimorphism followed by a monomorphism. Repeat the construction of the R_n for a double arrow (p_1, p_2) of C. Endowing C with the canonical topology, show that the canonical functor $\epsilon : C \to C^\sim$ is exact, hence commutes with formation of the R_n.

e) Take for C a small full subcategory of (**Set**), stable up to isomorphism under finite projective and finite inductive limits, and containing the sets X_0 and X_1 of c). Deduce that the topos $C^\sim$, which is coherent, does not satisfy the equivalent conditions of 1.25.

f) Let k be a field, and let C be the fppf site of schemes locally of finite presentation over k (SGA 3, IV, 6.3). Show that there exists a reduced affine scheme X over k admitting an automorphism f which is not of finite order; for example take the affine plane with $f(x) = x$, $f(y) = x + y$. Show that if $p_1, p_2 : X_1 \rightrightarrows X_0$ is a double arrow in C such that the corresponding double arrow of sets

$$p_1(\bar{k}), p_2(\bar{k}) : X_1(\bar{k}) \rightrightarrows X_0(\bar{k})$$

gives a non-stationary sequence (R_n), then the same is true of the double arrow of $C^\sim$ defined by p_1, p_2. Hence, if X_0 is of finite type over k, the cokernel in $C^\sim$ of (p_1, p_2) is not coherent. Deduce that the cokernel in $C^\sim$ of (f, id_X) is not coherent, so that $C^\sim$ does not satisfy the equivalent conditions of 1.25. More generally, if S is a non-empty scheme and C is the fppf site of schemes locally of finite presentation over S, then $C^\sim$ does not satisfy the conditions of 1.25. Notice that, contrary to appearances, $C^\sim$ is noetherian only when S is empty, as follows from the preceding discussion and 2.14.

Definition 1.30. Let E be a topos. An object X of E is called a *pre-noetherian object*[1] of E if it satisfies the following two equivalent conditions:

(i) Every subobject of X is quasi-compact.

(ii) Every increasing sequence of subobjects of X is stationary.

1.30.1

The equivalence of (i) and (ii) is clear. These conditions again depend only on the induced topos $E_{/X}$, and even only on the ordered set of opens of $E_{/X}$, isomorphic to the ordered set of subobjects of X. They are stable under passage to a subobject of X.

1.31

As in 1.3 and 1.4 one proves the following facts. If $(X_i \to X)$ is a finite covering family with the X_i pre-noetherian, then X is pre-noetherian; in particular a quotient of a pre-noetherian object of E is pre-noetherian. If $(X_i)_{i \in I}$ is a family of objects of E, then their sum X is pre-noetherian if and only if all but finitely many of the X_i are isomorphic to $\varnothing_E$, and all the X_i are pre-noetherian.

[1] For the stronger notion of a *noetherian* object, see 2.11 below.

1.32

Plainly a pre-noetherian object of a topos E is quasi-compact. If E admits a generating family consisting of pre-noetherian objects, the converse holds: the pre-noetherian objects of E are then precisely its quasi-compact objects.

Assume that E admits a generating family $(X_i)_{i \in I}$ consisting of quasi-compact objects. Then the following three conditions are equivalent:

(i) The X_i are pre-noetherian.

(ii) Every quasi-compact object of E is pre-noetherian.

(iii) Every monomorphism in E is quasi-compact.

We have just seen the equivalence of (i) and (ii); (ii) $\Rightarrow$ (iii) is immediate from the definitions, and (iii) $\Rightarrow$ (ii) follows from Definition 1.30(i).

Note that (iii) implies that *every morphism of E is quasi-separated*. Hence every object of E lying over a quasi-separated object of E is itself quasi-separated.

Example 1.33: the classifying topos of a group. Let G be a group in $\mathcal{U}$, and let B_G be its classifying topos, i.e. the category of sets in $\mathcal{U}$ on which G acts on the left. The non-empty connected objects of B_G are precisely the non-empty transitive G-sets, and every object E of B_G is the sum of its connected components, namely the non-empty minimal subobjects of E, or the orbits of G on E. Thus E is quasi-compact in B_G if and only if the set $\pi_0(E)$ of orbits is finite, and in that case E is even pre-noetherian. Such objects, indeed even the single non-empty connected object G_s, form a generating subcategory of B_G.

The final object e of B_G is quasi-separated, i.e. the product of two quasi-compact objects of B_G is quasi-compact, if and only if $G_s \times G_s$ is quasi-compact, which is plainly equivalent to G being finite. More generally, an object E of B_G is quasi-separated if and only if the stabilizers of its points are finite subgroups of G, equivalently if it is isomorphic to a sum of homogeneous spaces G/H with H finite. Indeed, by 1.15 one may assume E connected; if $x \in E$ has stabilizer G_x, then B_G/E is equivalent to B_{G_x}, and the final object is quasi-separated if and only if G_x is finite.

It follows from this criterion that every object of B_G lying over a quasi-separated object is quasi-separated; a fortiori, the product of two quasi-separated objects of B_G is quasi-separated. The coherent objects of B_G are the objects isomorphic to finite sums of homogeneous spaces G/H, with H finite. Since G_s is coherent, the coherent objects already form a generating subcategory of B_G. Moreover this subcategory is stable under fiber products. In the terminology of 2.11 below, B_G is a locally noetherian topos, but it is quasi-separated only if G is finite, in which case B_G is even noetherian.

One sees at once that a morphism $f : E' \to E$ is quasi-compact if and only if the induced map $\pi_0(f) : \pi_0(E') \to \pi_0(E)$ on orbit sets is proper, i.e. has finite fibers. This recovers the fact that a monomorphism is quasi-compact, hence every morphism is quasi-separated. Therefore coherent morphisms in B_G are simply the quasi-compact morphisms just characterized.

Let E be an object of B_G. It is easy to check that E is a filtered inductive limit of coherent objects of B_G if and only if the stabilizers of the points of E are ind-finite groups, i.e. filtered inductive limits of their finite subgroups. In particular, if G is not ind-finite, for example $G = \mathbb{Z}$, the final object of B_G is not an inductive limit of coherent objects. More precisely, B_G satisfies the equivalent conditions of 1.25 if and only if its final object is a filtered inductive limit of coherent objects, equivalently if and only if G is ind-finite.

2. Finiteness conditions for a topos

Proposition 2.1. Let E be a $\mathcal{U}$-topos. The following conditions are equivalent:

(i) There exists a full generating subcategory C of E, consisting of quasi-compact objects, and stable under fiber products.

(ii) There exists a $\mathcal{U}$-site C, every object of which is quasi-compact, in which fiber products are representable, and such that E is equivalent to the category of sheaves $C^{\sim}$.

Assume these conditions hold. Then in *(i)*, respectively *(ii)*, one may choose C $\mathcal{U}$-small. Moreover, for every $X \in \operatorname{ob} C$, the object X, respectively $\epsilon(X)$, of E is coherent.

The implication (i) $\Rightarrow$ (ii) follows from IV, 1.2.1, and the fact that one may take C $\mathcal{U}$-small follows immediately from the argument of IV, 1.2.3, applied to a small full subcategory generating in the topological sense. It remains to prove (ii) $\Rightarrow$ (i) and the last assertion of 2.1, which follow from the next result.

Corollary 2.1.1. Let C be a $\mathcal{U}$-site in which fiber products are representable and every object is quasi-compact. Let $E = C^{\sim}$ and $\varepsilon : C \to E$ be the canonical functor. Then, for every $X \in \operatorname{ob} C$, $\varepsilon(X)$ is a coherent object of E. The full subcategory of E consisting of coherent objects of E lying over some $\varepsilon(X)$ is stable under fiber products, and is of course generating in E.

We already know that $\epsilon(X)$ is quasi-compact. Let us prove that it is quasi-separated. Let F and G be quasi-compact objects of E, together with morphisms $F \to \varepsilon(X)$, $G \to \epsilon(X)$. We must show that $F \times_{\epsilon(X)} G$ is quasi-compact. Covering F and G by finitely many objects of the form $\epsilon(Y_i)$, respectively $\epsilon(Z_j)$, and proceeding as in 1.2 to reduce to the case in which the composites $\epsilon(Y_i) \to \epsilon(X)$, $\epsilon(Z_j) \to \epsilon(X)$ come from morphisms $Y_i \to X$, respectively $Z_j \to X$, one reduces by 1.3 to the case where $F \to \epsilon(X)$ and $G \to \epsilon(X)$ are the images under ϵ of morphisms $Y \to X$, $Z \to X$. But fiber products are representable in C, and ϵ commutes with them; hence the assertion reduces to quasi-compactness of $\epsilon(Y \times_X Z)$, already known for every object of the form $\epsilon(U)$.

Now let $G \to F$, $H \to F$ be morphisms of coherent objects of E, and assume there exists a morphism $F \to \epsilon(X)$, for some $X \in \operatorname{ob} C$. We prove that $G \times_F H$ is coherent. Since G and H are quasi-compact and F is quasi-separated, this fiber product is already quasi-compact. It remains to see that it is quasi-separated, equivalently by 1.15 that $G \times_{\epsilon(X)} H$ is quasi-separated. This follows from the more general fact below.

Corollary 2.1.2. Under the hypotheses of 2.1.1, for every quasi-separated object G of E, every morphism $G \to \epsilon(X)$ is quasi-separated. If $H \to \epsilon(X)$ is a second morphism, with H quasi-separated, then $G \times_{\epsilon(X)} H$ is quasi-separated.

The first assertion implies the second: by base change $G \times_{\epsilon(X)} H$ is quasi-separated over H, and since H is quasi-separated, it is quasi-separated. To prove that $G \to \epsilon(X)$ is quasi-separated, we use the following lemma.

Lemma 2.1.3. Let E be a topos and let $f : X \to Y$ be a morphism in E. Assume that E admits a generating family of quasi-compact objects and that there exists a covering family $(g_i : X_i \to X)_{i \in I}$ of X such that the X_i are quasi-compact and, for every pair i, j, the object $X_i \times_Y X_j$ is quasi-separated. Then, if X is quasi-separated, f is quasi-separated.

Indeed, the family

$$(X_i \times_Y X_j \to X \times_Y X)_{(i,j)\in I\times I}$$

is covering. Thus to show that the diagonal $\Delta_{X/Y} : X \to X \times_Y X$ is quasi-compact it suffices to check this after base change by each $X_i \times_Y X_j \to X \times_Y X$. The resulting morphism is the canonical morphism

$$X_i \times_X X_j \longrightarrow X_i \times_Y X_j.$$

Since X is quasi-separated and the X_i are quasi-compact, the source is quasi-compact; since the target is quasi-separated by hypothesis, this morphism is quasi-compact.

We now finish the proof of 2.1.2. Let $f : G \to \epsilon(X)$ with G quasi-separated. Cover G by objects $\epsilon(X_i)$ such that the composites $\epsilon(X_i) \to \epsilon(X)$ come from morphisms $X_i \to X$. By 2.1.3, since the $\epsilon(X_i)$ are quasi-compact, it suffices to verify that the fiber products

$$\epsilon(X_i) \times_{\epsilon(X)} \epsilon(X_j) = \epsilon(X_i \times_X X_j)$$

are quasi-separated. But we have already seen that every object of E of the form $\epsilon(U)$ is coherent. This proves 2.1.2 and completes the proof of 2.1.

Proposition 2.2. Let E be a topos containing a full generating subcategory C consisting of coherent objects. The following conditions on E are equivalent:

(i) Every quasi-separated object of E is quasi-separated over the final object.

(i bis) For every morphism $f : X \to Y$ in E, X quasi-separated implies f quasi-separated.

(i ter) Every object X of C is quasi-separated over the final object of E, i.e. the diagonal morphism $X \to X \times X$ is quasi-compact.

(ii) The product of two quasi-separated objects of E is quasi-separated.

(ii bis) The product in E of two objects of C is quasi-separated.

These conditions imply that the full subcategory E_{coh} of E consisting of coherent objects is stable under fiber products; a fortiori, E satisfies the equivalent conditions of 2.1.

Plainly (i bis) implies (i), and the converse follows from 1.8(iii). We have (i) $\Rightarrow$ (ii) by an argument already used: if X and Y are quasi-separated, then, by (i), X is quasi-separated over e; hence $X \times Y$ is quasi-separated over Y by base change, and is therefore quasi-separated by 1.14(ii). The same argument shows (i ter) $\Rightarrow$ (ii bis), while (i) $\Rightarrow$ (i ter) and (ii) $\Rightarrow$ (ii bis) are trivial. It remains to prove (ii bis) $\Rightarrow$ (i). Let X be a quasi-separated object of E. Cover X by objects X_i of C; since the X_i are quasi-compact, Lemma 2.1.3 applies to the morphism $X \to e$. We are reduced precisely to checking that the $X_i \times X_j$ are quasi-separated, which is the hypothesis (ii bis).

Remark 2.2.1. Condition (i ter) is also equivalent to the following condition (i quater): for every double arrow $g_1, g_2 : Y \rightrightarrows X$ in C, its kernel in E is quasi-compact, equivalently coherent.

This follows by applying criterion 1.10(i) to the diagonal morphism $X \to X \times X$ considered in (i ter).

Definition 2.3. A topos E satisfying the equivalent conditions of *2.2*, respectively of *2.1*, is called an *algebraic topos*, respectively a *locally algebraic topos*, or *locally coherent topos*. An object X of a topos E is called an *algebraic object* of E if the induced topos $E_{/X}$ is algebraic. A topos E is called *quasi-separated*, respectively *coherent*, if it is algebraic and its final object is quasi-separated, respectively coherent.

2.3.1

It is in algebraic topoi that the finiteness notions developed in Section 1 have fully satisfactory properties, analogous to the properties of the notions bearing the same names in the category of schemes. All locally algebraic, equivalently locally coherent, topoi encountered in practice are in fact algebraic, so the practical interest of the former notion seems at present rather limited. Locally algebraic non-algebraic topoi can nevertheless be constructed; see Exercise 2.17 f).

2.4. General remarks

Here are several general remarks concerning the notions introduced in 2.3; they should convince the reader that the terminology introduced is coherent.

2.4.1

Under the conditions of 2.1(i), respectively 2.1(ii), for every $X \in C$, the object X, respectively $\epsilon(X)$, is a coherent algebraic object of E. In other words, the induced topos $E_{/X}$, respectively $E_{/\epsilon(X)} \simeq (C_{/X})^{\sim}$, is algebraic, hence coherent. Equivalently, when C admits a final object, E itself is algebraic, hence coherent.

2.4.2

If E is algebraic, then every induced topos $E_{/X}$ is algebraic. For E to be locally algebraic, equivalently locally coherent, it is necessary and sufficient that there exist a family $(X_i)_{i \in I}$ of objects covering the final object such that the X_i are algebraic, respectively coherent, i.e. such that the induced topoi $E_{/X_i}$ are algebraic, respectively coherent. Sufficiency is immediate from the definitions and necessity follows from 2.4.1. Of course, an algebraic topos is locally algebraic; and if E is locally algebraic, every induced topos $E_{/X}$ is locally algebraic.

2.4.3

If X is an algebraic object of a topos E, then every object X' of E lying over X is again algebraic. In particular, every object of an algebraic topos is algebraic, and conversely.

2.4.4

The full subcategory of E consisting of the quasi-separated objects which are algebraic is stable under fiber products and subobjects, hence also under kernels. If E is algebraic, this category is just the category of quasi-separated objects of E, and is also stable under products of two objects. If E is quasi-separated, and only then, this category is stable under arbitrary finite projective limits, equivalently it contains the final object of E.

The full subcategory C of E consisting of coherent objects which are algebraic is stable under fiber products. If E is locally algebraic, equivalently locally coherent, then E is algebraic if and only if C is also stable under kernels of double arrows; E is quasi-separated if and only if, in addition, C is stable under products of two objects. Finally, E is coherent if and only if C is stable under arbitrary finite projective limits, equivalently if it contains the final object of E.

2.4.5

A topos E is locally coherent, respectively algebraic, respectively quasi-separated, respectively coherent, if and only if it admits a full generating subcategory C consisting of quasi-compact objects, automatically coherent by 2.1, and stable under fiber products, respectively under fiber products and kernels, respectively under fiber products and products of two objects, respectively under arbitrary finite projective limits. Equivalently, E is equivalent to a topos $C^{\sim}$, where C is a $\mathcal{U}$-site all of whose objects are quasi-compact and in which fiber products and products of two objects, respectively all finite projective limits, are representable. For necessity, take C to be the category of objects of E which are coherent and algebraic, and apply 2.4; for sufficiency, apply 2.2(i ter). In this statement one may of course again suppose C $\mathcal{U}$-small.

2.4.6

Let E be a topos. If E is algebraic, an object X of E is quasi-separated, respectively coherent, if and only if the induced topos $E_{/X}$ is so. When E is no longer assumed algebraic, one can say only that $E_{/X}$ is quasi-separated, respectively coherent, if and only if X is quasi-separated, respectively coherent, and algebraic. This is harmless, since presumably these notions will not have to be used outside the case where E is algebraic.

2.4.7

For the notion of a coherent, respectively quasi-separated, object of a topos E to have satisfactory properties, one must restrict to objects which are also algebraic, a condition automatically satisfied when E is algebraic. Since we will presumably never have to work with locally algebraic topoi which are not algebraic, the question of revising the terminology introduced in 1.13, by reserving it possibly for algebraic objects only, is not very pressing.

On the other hand, we have refrained from defining the notion of a quasi-compact topos. In the spirit of the present section, one would have to call quasi-compact those algebraic topoi whose final object is quasi-compact. We hesitated to introduce such usage because, if X is a topological space, it would not be true that X is quasi-compact if and only if the associated topos $\mathrm{Top}(X)$ is. In this connection note that $\mathrm{Top}(X)$ is locally coherent if and only if X admits a basis of opens consisting of quasi-compact opens U such that, for $U_j, U_k \subset U_i$, the intersection $U_j \cap U_k$ is still quasi-compact. This condition is satisfied by the topological spaces underlying schemes, but rarely by spaces encountered by analysts, even when those spaces are compact.

For ease of reference, we spell out the following.

Proposition 2.5. Let E be an algebraic topos and let $f : X \to Y$ be a morphism in E. If Y is quasi-separated, respectively coherent, then X is quasi-separated, respectively coherent, if and only if f is quasi-separated, respectively coherent.

Sufficiency was already proved in 1.14(ii), and holds without any hypothesis on E. Necessity in the quasi-separated case holds even without assuming Y quasi-separated; it is precisely the definition 2.3 via 2.2(i bis). It remains to prove that if Y and X are coherent, then f is coherent. Since f is already quasi-separated, it suffices to see that f is quasi-compact, which follows because X is quasi-compact and Y quasi-separated.

Corollary 2.6. Let E be an algebraic topos, let $f : X \to Y$ be a morphism in E, and let $(Y_i \to Y)_{i \in I}$ be a covering family with the Y_i quasi-separated, respectively coherent. Then f is quasi-separated, respectively quasi-compact, respectively coherent, if and only if for every $i \in I$ the object

$$X_i = X \times_Y Y_i$$

is quasi-separated, respectively quasi-compact, respectively coherent, in E.

Combine 1.10(ii) with 2.5 in the quasi-separated and coherent cases; the quasi-compact case was already proved in 1.16. Notice that necessity in 2.6 holds without any hypothesis on the topos E.

Corollary 2.7. Let E be a coherent topos and let X be an object of E. Then X is quasi-compact, respectively quasi-separated, respectively coherent, in E if and only if it is quasi-compact, respectively quasi-separated, respectively coherent, over the final object.

In particular, X is constructible if and only if it is coherent.

Corollary 2.8. Let E be a locally coherent topos, and let $f : X \to Y$ be a morphism in E. Then f is quasi-separated if and only if, for every quasi-separated object Y' of $E_{/Y}$, the inverse image

$$f^*(Y') = X \times_Y Y'$$

is quasi-separated in $E_{/X}$, equivalently in E.

Necessity holds without any condition on E: if Y' is quasi-separated, then $X' = X \times_Y Y' \to Y'$ is quasi-separated by base change, hence X' is quasi-separated. Conversely, there exists a covering family $Y_i \to Y$ by quasi-separated algebraic objects. By 1.10(ii), to check that f is quasi-separated it suffices to check that the morphisms $f_i : X_i = X \times_Y Y_i \to Y_i$ are quasi-separated. But the hypothesis on f^* implies that the X_i are quasi-separated, hence the f_i are quasi-separated because $E_{/Y_i}$ is algebraic.

Remark 2.8.1. Statements 2.5 and 2.6 remain valid if one replaces the hypothesis "E algebraic" by the more general hypothesis "Y is algebraic", as is immediate by applying the original statement to the induced topos $E_{/Y}$, which is algebraic. Similarly, in 2.8 it is enough to suppose that $E_{/Y}$, rather than E, is locally coherent.

Proposition 2.9. Let E be a topos and C a subcategory of E. The following conditions are equivalent:

(i) E is locally coherent, respectively coherent, satisfies the equivalent conditions of 1.25, and C is the full subcategory E_{coh} of E consisting of coherent objects.

(ii) C is a strictly full generating subcategory of E, stable under finite inductive limits and fiber products, respectively also under finite projective limits, and consists of quasi-compact objects.

The implication (i) $\Rightarrow$ (ii) follows directly from the definitions and from condition 1.25(ii). Conversely, if C is full, generating, consists of quasi-compact objects, and is stable under fiber products, respectively under finite projective limits, then by definition E is locally coherent, respectively coherent. If moreover C is strictly full and stable under finite inductive limits, then $C = E_{\mathrm{coh}}$ by 1.24, (ii) $\Rightarrow$ (iii), and condition 1.25(ii) is satisfied.

Definition 2.9.1. A $\mathcal{U}$-topos E is called *perfect* if it is coherent and satisfies the equivalent conditions of 1.25, i.e. if the subcategory E_{coh} of coherent objects of E is stable under finite inductive limits. It is called *locally perfect* if there exists a family $(X_i)_{i\in I}$ of objects of E covering the final object such that each induced topos $E_{/X_i}$ is perfect.

2.9.2

It follows immediately from this definition that a perfect, respectively locally perfect, topos is coherent, respectively locally coherent. Examples of perfect, respectively locally perfect, topoi include the noetherian, respectively locally noetherian, topoi introduced in 2.11 below; the Zariski and etale topoi of a coherent, respectively arbitrary, scheme (IX, note p. 42). As a counterexample, the fppf topos of a scheme S is locally coherent, and even coherent if S is a coherent scheme, i.e. quasi-compact and quasi-separated, but is not locally perfect if $S \neq \varnothing$, by Exercise 1.28 f).

In a locally perfect topos, the notion of constructible object is particularly stable, as 2.9.3 below shows; this is one reason why the notion seems interesting. Another is that, like coherent or locally coherent topoi, perfect or locally perfect topoi are stable under formation of the closed subtopos complementary to an open, and behave especially simply with respect to gluing of topoi.

Proposition 2.9.3. Let E be a locally perfect topos. Then the full subcategory E_{cons} of E consisting of constructible objects is stable under finite inductive limits, and of course also under finite projective limits.

Since constructibility is local on E, we reduce to the case where E is perfect. Then $E_{\text{cons}} = E_{\text{coh}}$, and the conclusion follows from Definition 2.9.1.

Corollary 2.9.4. Let E be a locally perfect topos. Then E is perfect if and only if E is coherent. For every object X of E, the induced topos $E_{/X}$ is locally perfect.

Problem 2.9.5. It is conceivable that perfect topoi are special enough to admit a quite explicit *structure theory*, following a model proposed by M. Artin at the time of the oral seminar. Call a topos *finite* if it is equivalent to a topos of the form $\widehat{C}$, where C is a finite category, and call a topos *profinite* if it is a filtered projective limit of finite topoi, in the sense of Section 6 below. Since a finite topos is noetherian, hence perfect, and a filtered projective limit of perfect topoi is perfect, a profinite topos is perfect. The question is whether conversely every perfect topos is profinite.

This is equivalent to asking whether every finite subcategory of E_{coh} is contained in a full subcategory F_0 of E_{coh} equivalent to a category F_{coh}, where F is a finite topos. If the answer were negative, one would have to find manageable additional intrinsic conditions on a perfect topos ensuring that it is profinite. Finally, one would still have to study the structure of profinite topoi in terms of a suitable notion of “Karoubian profinite category”, inspired by IV, 7.6 h). This study has not yet been carried out even in the special case where E is the Zariski or etale topos of a well-behaved coherent scheme X, and should then give invariants finer than the compact space X_{cons} associated with X.

In the same vein, we mention the following question. Let E be a perfect topos, let E' be the topos induced on an open of E, and let E'' be the corresponding closed subtopos. If E' and E'' are finite topoi, is E also finite?

Proposition 2.10. Let E be a topos. The following conditions are equivalent:

(i) E admits a full generating subcategory, stable under fiber products, consisting of pre-noetherian objects of E.

(ii) E admits a full generating subcategory consisting of pre-noetherian quasi-separated objects, equivalently pre-noetherian coherent objects.

(iii) E is locally coherent and every quasi-compact object of E is pre-noetherian.

(iii bis) E is locally coherent and admits a generating family consisting of pre-noetherian objects of E.

The equivalence of (iii) and (iii bis) follows from 1.32. It is trivial that (i) $\Rightarrow$ (iii bis) and (iii) $\Rightarrow$ (i), so (i), (iii), and (iii bis) are equivalent. Finally, (i) $\Rightarrow$ (ii) follows from the last assertion of 2.1, and it remains to prove (ii) $\Rightarrow$ (i). This implication follows because the full subcategory of E consisting of pre-noetherian quasi-separated objects, if it is generating, is stable under fiber products: such a fiber product is quasi-compact by definition, hence pre-noetherian by 1.32; and it is quasi-separated by the last assertion of 1.32.

Definition 2.11. A topos E is called *locally noetherian* if it satisfies the equivalent conditions of 2.10. It is called *noetherian* if, in addition, its final object is coherent, equivalently pre-noetherian and quasi-separated. An object X of a topos E is called a *noetherian object* of E if the induced topos $E_{/X}$ is noetherian.

2.12

If E is locally noetherian, then so is every induced topos $E_{/X}$. Conversely, if there exists a family (X_i) covering the final object such that the $E_{/X_i}$ are locally noetherian, then E is locally noetherian. If C is a full generating subcategory of E consisting of pre-noetherian objects and stable under fiber products, then for every $X \in C$, $E_{/X}$ is noetherian, i.e. the objects of C are in fact noetherian. Indeed, if E is locally noetherian and $X \in \operatorname{Ob} E$, then $E_{/X}$ is noetherian if and only if X is coherent. It follows that E is locally noetherian if and only if the final object of E can be covered by noetherian objects X_i.

2.13

In a locally noetherian topos E, every monomorphism is quasi-compact and every morphism is quasi-separated. A fortiori, E is an algebraic topos, not merely locally coherent, i.e. locally algebraic. Hence E is noetherian if and only if it is locally noetherian and coherent. In other words, if E is locally noetherian, then the noetherian objects of E are precisely the coherent objects of E.

2.14

Let E be a locally noetherian topos. If E is quasi-separated, i.e. if its final object is quasi-separated, then every object of E is quasi-separated. Thus the pre-noetherian objects of E are identical with the noetherian, equivalently coherent, objects of E. Since a quotient of a pre-noetherian object is pre-noetherian, it follows that E satisfies the equivalent conditions of 1.25, because it satisfies condition 1.25(ii bis). In particular, if C is the full subcategory of E consisting of noetherian, equivalently coherent, objects, then the natural functor

$$\operatorname{Ind}(C) \longrightarrow E$$

is an equivalence of categories. We conclude that *a noetherian topos is perfect*; hence a locally noetherian topos is locally perfect.

If one assumes only that E is locally noetherian, it is still true that every object of E is an inductive limit of its pre-noetherian subobjects, since in a topos admitting a generating subcategory of quasi-compact objects, every object is a filtered inductive limit of its quasi-compact subobjects. But since a pre-noetherian object of E need not be coherent, this statement has only very limited usefulness; and Example 1.33 shows that the conditions of 1.25 need not hold.

Example 2.15: topological spaces. Let X be a topological space. The following conditions are equivalent: (i) X is noetherian, i.e. every open subset of X is quasi-compact, equivalently every increasing sequence of opens of X is stationary; (ii) the topos $\mathrm{Top}(X)$ is noetherian; (iii) the final object X of $\mathrm{Top}(X)$ is noetherian; (iv) the final object X of $\mathrm{Top}(X)$ is pre-noetherian.

Indeed, (ii) $\Leftrightarrow$ (iii) $\Rightarrow$ (iv) $\Leftrightarrow$ (i) is trivial. Moreover (i) implies that the generating subcategory $\mathrm{Open}(X)$ of $\mathrm{Top}(X)$, stable under finite projective limits, consists of pre-noetherian objects; hence (ii). Similarly, X is locally noetherian, i.e. a union of noetherian opens, if and only if $\mathrm{Top}(X)$ is locally noetherian.

2.15.1

Thus Definitions 1.30 and 2.11 are compatible with the accepted terminology for topological spaces. On the other hand, in the general case we have chosen to give the term "noetherian object" a stronger meaning than the more naive notion of 1.30, so that the properties attached to the notion, rather than only the grammatical structure of the definition, agree with the intuition attached to noetherian topological spaces and noetherian schemes.

The reader will find further examples in the following exercises.

Exercise 2.16: spaces with operators and classifying topoi. Let X be a topological space on which a discrete group G acts, giving a topos $E = \mathrm{Top}(X, G)$. Interpret the objects X' of this topos as etale spaces over X with group of operators G, and note that the induced topos $E_{/X'}$ is canonically equivalent to $\mathrm{Top}(X', G)$.

a) The final object of E is quasi-compact if and only if the quotient topological space X/G is quasi-compact.

b) Let P be the object of E defined by the trivial etale space $X \times G$ with diagonal action of G. Show that the induced topos $E_{/P}$ is equivalent to $\mathrm{Top}(X)$. Deduce that E is locally coherent, respectively locally noetherian, if and only if $\mathrm{Top}(X)$ is locally coherent, respectively locally noetherian. Show that if E is locally coherent, i.e. locally algebraic, then it is even algebraic, using the generating family consisting of the objects T_U, where U is a coherent open of X, $T_U = G \times U$ with action $g \cdot (u, g') = (u, gg')$, and structural morphism $p_U : T_U \to X$ defined by $p_U(u, g) = g \cdot u$.

c) Assume E algebraic, i.e. $\mathrm{Top}(X)$ algebraic. Show that the structural morphism $P \to e$ is quasi-separated. Show that E is quasi-separated if and only if $\mathrm{Top}(X)$ is quasi-separated, equivalently the intersection of two quasi-compact opens of X is quasi-compact, and G is finite or X is empty.

d) Show that the structural morphism $P \to e$ is quasi-compact if and only if G is finite. Show that E is coherent, respectively noetherian, if and only if $\mathrm{Top}(X)$ is so and G is finite or X is empty.

e) Let E be a topos and G a group of E, hence a classifying topos B_G. Study the finiteness conditions in B_G, taking the preceding analysis as a guide. Do the same for a strict pro-group $\mathcal{G} = (G_i)_{i \in I}$ of E. In particular, if E is coherent, respectively noetherian, and the G_i are coherent, then $B_{\mathcal{G}}$ is coherent, respectively noetherian.

Exercise 2.17: topoi of the form $\widehat{C}$. Let C be a category equivalent to a category belonging to $\mathcal{U}$, and let $E = \widehat{C}$ be the category of presheaves on C. Through the canonical functor $\varepsilon : C \to \widehat{C}$, identify C with a full subcategory of E.

a) C is a generating subcategory of E consisting of quasi-compact objects. These objects are pre-noetherian if and only if, for every object X of C, every sieve of X is generated by a finite family of morphisms $X_i \to X$ of C. The final object of E is quasi-compact, respectively pre-noetherian, if and only if C has a final subcategory with finitely many objects, respectively every sieve of C is generated by a subcategory of C with finitely many objects.

b) The topos E admits a generating subcategory consisting of coherent, equivalently quasi-separated, objects if and only if the objects of C are coherent in E; equivalently, for any two arrows $Y \xrightarrow{f} X$ and $Z \xrightarrow{g} X$ in C, the object $Y \times_X Z$ of E is quasi-compact. Explicitly, one must be able to find a finite family of commutative squares in C,

$$\begin{tikzcd} & T_i \arrow[dl] \arrow[dr] & \\ Y \arrow[dr, "f"'] & & Z \arrow[dl, "g"] \\ & X & \end{tikzcd},$$

such that every other commutative square of the same form comes from a morphism $T \to T_i$. Notice that if $(F_i)_{i \in I}$ is a generating family of E, every $X \in \operatorname{ob} C$ is isomorphic to a direct factor of some F_i.

c) The topos E is locally coherent, i.e. generated by coherent algebraic objects, if and only if the objects X of C are coherent and algebraic in E. This means that condition b) holds, and also the following condition: for every double arrow $f, g : Y \rightrightarrows Z$ of C over an object X of C, the kernel of this double arrow in E is quasi-compact. Equivalently, there exists a finite family of commutative diagrams in C

$$\begin{tikzcd} T_i \arrow[r] & Y \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & Z \end{tikzcd}$$

such that every other commutative diagram of this form comes from a morphism $T \to T_i$. For E to be algebraic, the same condition is required for arbitrary double arrows (f, g) of C, not necessarily over an object X of C. For E to be quasi-separated, condition b) must hold and, in addition, for two objects $X, Y \in \operatorname{ob} C$, the product $X \times Y$ in E must be quasi-compact. For E to be coherent, the two preceding conditions must hold, and C must have a final subcategory with finite underlying set.

d) Give an example in which the objects of the generating subcategory C of E are pre-noetherian, but E has no generating family consisting of coherent objects. For this take C to be the category whose objects are X, T_i, $i = 0, 1, \ldots$, and e, the final object. Besides identities and the structural morphisms to e, the only morphisms are $u_i : T_i \to X$, $v_i : X \to T_i$, subject to $u_i v_i = \mathrm{id}_X$, and finally $p_i = v_i u_i$, satisfying necessarily $p_i^2 = \mathrm{id}_{T_i}$. One checks that the only sieves of X or of a T_i are the two trivial sieves, and that e has exactly one non-trivial sieve. Hence by a), the objects of C are pre-noetherian in E. However, $X \times X$ in E is not quasi-compact, since it does not satisfy the criterion in b).

e) Give an example in which the generating subcategory C of E consists of coherent objects of E, but E is not locally coherent. For this take C to be the category whose objects are distinct objects e, the final object, Y, Z, and T_i, $i = 0, 1, \ldots$, with as only morphisms, besides the morphisms to the final object and identities, morphisms $u_i : T_i \to Y$ and a double arrow $f, g : Y \rightrightarrows Z$ subject to $fu_i = gu_i$. With some patience, one checks that C satisfies condition b): all fiber products of objects of C are isomorphic either to $\varnothing_E$ or to objects of C, except for $Z \times_e Z$ and $Y \times_e Z$, which are covered by two elements of C. But plainly $\mathrm{Ker}(f, g)$ does not satisfy condition c).

f) Give an example in which C is stable under fiber products, so that E is a locally coherent, i.e. locally algebraic, topos, but E is not algebraic. Take the example of d) and the full subcategory C' of E consisting of Y, Z, T_i for $i > 0$, and $\varnothing_E$.

g) Assume C finite. Prove that E is noetherian; its noetherian, equivalently quasi-compact, objects are the contravariant functors $F : C^\circ \to \mathbf{Set}$ such that for every object X of C, $F(X)$ is a finite set; and the fiber functors on E send noetherian objects to finite sets.

h) Assume every morphism $f : X \to X$ of C factors as a composite ip, where i is a monomorphism and p admits a right inverse. Let X be an object of C, let $I(X)$ be the set of its subobjects in the sense of C, regarded as an ordered set, and let $S(I(X))$ be the set of subsets U of $I(X)$ such that if $X' \in U$ and $X'' \leq X'$, then $X'' \in U$. Show that the ordered set of subobjects of X in E is isomorphic to $S(I(X))$. Deduce that if $I(X)$ is finite, then X is pre-noetherian in E. In particular, if in addition to the factorization condition above, C is stable under fiber products, respectively finite projective limits, and if for every $X \in \mathrm{ob}\, C$ the set $I(X)$ of subobjects of X in C is finite, then E is locally noetherian, respectively noetherian.

i) Let C be the category of finite sets, or alternatively of non-empty finite sets, in $\mathcal{U}$. Then $E = \widehat{C}$ is the category of augmented simplicial sets, respectively of ordinary simplicial sets. Show that E is a noetherian topos, using h). Taking a pro-object $(X_i)_{i \in I}$ of C which is not essentially constant, show that E admits fiber functors which do not send noetherian objects to noetherian objects, i.e. finite sets.

j) Consider the following diagram of implications among properties of a topos E:

$$
\begin{array}{ccccc}
E \text{ noetherian} & \Rightarrow & E \text{ locally noetherian} & & \\
\Downarrow & & \Downarrow & \searrow & \\
E \text{ coherent} & \Rightarrow & E \text{ algebraic} & & \\
& & \Downarrow & & E_{\text{preNoeth}} \text{ generates } E \\
& & E \text{ locally coherent, i.e. } E_{\text{coh.alg}} \text{ generates } E & \Rightarrow & \\
& & \Downarrow & & \\
& & E_{\text{coh}} \text{ generates } E & & \\
& & \Downarrow & & \\
& & E_{\text{qc}} \text{ generates } E. & &
\end{array}
$$

Show that all implications in this diagram are strict, and that there are no further implications among the notions considered except those obtained by composition in the diagram. Here

E_{coh}, E_{qc}, E_{preNoeth}, and $E_{\mathrm{coh.alg}}$ denote respectively the full subcategories of E consisting of coherent, quasi-compact, pre-noetherian, and coherent algebraic objects. It suffices to restrict to topoi of the form $\widehat{C}$, using the results stated in d), e), and f).

k) Solve the following question, whose answer the writer of these lines does not know: can $E = \widehat{C}$ be locally noetherian without the sets $\mathrm{Hom}(X, Y)$, for $X, Y \in \mathrm{ob}\, C$, being finite?

Exercise 2.18. Let E be a topos.

a) Let X be a pre-noetherian object of E. Show that X is isomorphic to a finite sum of connected objects of E. Argue by contradiction that every increasing sequence of partitions of X is stationary.

b) Let X be an object of E, and let $(f_i : X_i \to X)_{i \in I}$ be a covering family of X. Show that if each X_i is isomorphic to a sum of connected objects of E, then the same is true of X. Reduce to the case where the X_i are connected, then to the case where they are subobjects of X, and consider on the index set I the equivalence relation R generated by the relation $X_i \cap X_j \neq \varnothing_E$. If $J = I/R$, show that X is the sum of the objects $X(j) = \mathrm{Sup}_{i \in j} X_i$.

c) Deduce from b) that if $(X_i)_{i \in I}$ is a generating family of E, then E is locally connected if and only if each X_i, $i \in I$, is isomorphic to a sum of connected objects of E.

d) Deduce from a) and c) that if E admits a generating subcategory consisting of pre-noetherian objects, in particular if E is locally noetherian, then E is locally connected, and a fortiori is isomorphic to the sum topos of a family of connected topoi, i.e. topoi whose final object is connected. In particular one can associate to E, for every morphism $f : P \to E$, where P is a "non-empty connected and simply connected" topos, a fundamental pro-group $\pi_1(E, f)$. Notice that if E is a non-empty locally noetherian topos, one can always find such an f, with P the punctual topos, by Deligne.

Continuation anchor. The next untranslated point is SGA 4, Exposé VI, Section 3, "Finiteness conditions for a morphism of topoi," source line 2074 of `06/06.tex`.

SGA 4, Exposé VI

Finiteness Conditions. Fibred Topoi and Sites. Applications to Questions of Passage to the Limit

A. Grothendieck and J.-L. Verdier

3. Finiteness conditions for a morphism of topoi

Definition 3.1. Let $f : E' \to E$ be a morphism of topoi. We say that f is *quasi-compact* (respectively *quasi-separated*) if, for every quasi-compact (respectively quasi-separated) object X of E, the object $f^*(X)$ is quasi-compact (respectively quasi-separated). We say that f is *coherent* if it is both quasi-compact and quasi-separated.

3.1.1

If f has one of the preceding properties, then every morphism of topoi isomorphic to f has the same property. It is also immediate that the composite of two quasi-compact (respectively quasi-separated, respectively coherent) morphisms of topoi is again quasi-compact (respectively quasi-separated, respectively coherent).

Proposition 3.2. With the notation of 3.1, let $(X_i)_{i\in I}$ be a generating family of quasi-compact (respectively coherent) objects of E. In the respective coherent case, assume that E' admits a generating family consisting of quasi-compact objects. Then f is quasi-compact (respectively coherent) if and only if, for every $i \in I$, the object $f^*(X_i)$ is quasi-compact (respectively coherent).

The necessity is trivial. For sufficiency in the quasi-compact case, let X be a quasi-compact object of E. There is a finite epimorphic family of morphisms $X_i \to X$. Pulling back, one obtains a finite epimorphic family $f^*(X_i) \to f^*(X)$, with the $f^*(X_i)$ quasi-compact by hypothesis; hence $f^*(X)$ is quasi-compact.

In the coherent case, it remains to prove quasi-separatedness. If X is quasi-separated in E, this may be expressed by the existence of an epimorphic family $X_i \to X$ such that the fibre products $X_i \times_X X_j$ are quasi-compact. After applying f^*, we get a covering family $f^*(X_i) \to f^*(X)$ such that the fibre products

$$f^*(X_i) \times_{f^*(X)} f^*(X_j)$$

are quasi-compact, since by the quasi-compact part already proved f^* sends quasi-compact objects to quasi-compact objects. Since the $f^*(X_i)$ are coherent, one concludes again by the criterion 1.17.

Corollary 3.3. Let C and C' be two $\mathcal{U}$-sites in which fibre products are representable and every covering family admits a finite covering subfamily. Let $g : C \to C'$ be a morphism of sites. Then the morphism of topoi

$$f : C^{\sim} \to C'^{\sim}$$

defined by g is coherent.

Indeed, $\varepsilon_C(C)$ (respectively $\varepsilon_{C'}(C')$) is a generating family of $C^{\sim}$ (respectively $C'^{\sim}$) satisfying the corresponding hypotheses of 3.2.

Proposition 3.4. Let E be a locally coherent topos, and let $f : X \to Y$ be an arrow of E, whence a morphism of induced topoi $E_{/X} \to E_{/Y}$. This morphism of topoi is quasi-compact (respectively quasi-separated, respectively coherent) if and only if the arrow f is quasi-compact (respectively quasi-separated, respectively coherent).

The quasi-compact case follows directly from the definitions, without any hypothesis on E. The quasi-separated case is precisely 2.8, and the coherent case is the conjunction of the two preceding cases.

Proposition 3.5. Let $f : E' \to E$ be a morphism of locally coherent topoi. Let $(X_i)_{i \in I}$ be a generating family in E consisting of coherent algebraic objects. The following conditions are equivalent.

(i) For every quasi-compact arrow u of E, the arrow $f^*(u)$ is quasi-compact.

(i bis) For every arrow $u : X_i \to X_j$, the arrow $f^*(u)$ is quasi-compact.

(ii) For every coherent object Y' of E', every coherent algebraic object Y of E, and every arrow $v : Y' \to f^*(Y)$, the corresponding morphism of topoi

$$f_v : E'_{/Y'} \to E_{/Y}$$

obtained by composing the localization morphism $E'_{/Y'} \to E'_{/f^*(Y)}$ deduced from v with the morphism $E'_{/f^*(Y)} \to E_{/Y}$ induced by f, is coherent.

(ii bis) The same condition as in (ii), but restricted to a set of data $v_\alpha : Y'_\alpha \to f^*(Y_\alpha)$ such that the Y'_α are algebraic and cover the final object of E'.

(ii ter) Given a generating family $(X'_{i'})_{i' \in I'}$ of E' consisting of coherent objects, and, for every $i' \in I'$, an index $i \in I$ and an arrow $v_{i'} : X'_{i'} \to f^*(X_i)$, the morphism of topoi $E'_{/X'_{i'}} \to E_{/X_i}$ defined by $v_{i'}$ is coherent for every i'.

Since every arrow $X_i \to X_j$ is quasi-compact, (i) implies (i bis). Conversely assume (i bis) and prove (i). Let $u : X \to Y$ be a quasi-compact arrow of E. In terms of an epimorphic family $X_i \to Y$, the hypothesis on u says that the objects $X \times_Y X_i$ are quasi-compact; equivalently, for every i there is a finite epimorphic family of arrows $X_j \to X \times_Y X_i$. Pulling back by f^*, we get an epimorphic family $f^*(X_i) \to f^*(Y)$, and, for every i, a finite epimorphic family

$$f^*(X_j) \to f^*(X \times_Y X_i) \simeq f^*(X) \times_{f^*(Y)} f^*(X_i).$$

The composite with the second projection to $f^*(X_i)$ is quasi-compact by (i bis). Hence the projection

$$f^*(X) \times_{f^*(Y)} f^*(X_i) \to f^*(X_i)$$

is quasi-compact for every i, and consequently $f^*(u)$ is quasi-compact.

It remains to prove (i) $\Rightarrow$ (ii) and (ii bis) $\Rightarrow$ (i), since (ii) $\Rightarrow$ (ii bis) is trivial and (ii ter) is a special case of (ii bis). The implication (i) $\Rightarrow$ (ii) is immediate: an object X of $E_{/Y}$ is quasi-compact (respectively quasi-separated) if and only if its structural morphism $X \to Y$ (respectively its diagonal $X \to X \times_Y X$) is quasi-compact, and the same criterion applies after pulling back by f. Finally assume (ii bis), and let $u : X \to Y$ be quasi-compact in E. Since the Y'_α cover the final object of E', it is enough to prove that the restrictions $f^*(u)|Y'_\alpha$ are quasi-compact. These restrictions are precisely $f^*_{v_\alpha}(u|Y_\alpha)$, reducing the assertion to the following corollary.

Corollary 3.6. Let $f : E' \to E$ be a coherent morphism of locally coherent topoi. Then, for every quasi-compact arrow u of E, the arrow $f^*(u)$ is quasi-compact.

By the implication (i bis) $\Rightarrow$ (i) already proved, it is enough to prove this when $u : X \to Y$ has X and Y coherent. But this follows at once from the hypothesis on f, which implies that $f^*(X)$ and $f^*(Y)$ are coherent.

Definition 3.7. Let $f : E' \to E$ be a morphism of locally coherent topoi. We say that f is *locally coherent* if it satisfies the equivalent conditions of 3.5.

Here "locally" means locally upstairs.

3.7.1

This notion again depends only on the isomorphism class of the morphism of topoi f, and is manifestly stable under composition of morphisms of topoi. Moreover, by 3.5, every coherent morphism is locally coherent; conversely, if E' and E are coherent, every locally coherent morphism is coherent. Criterion 3.5(ii bis) immediately gives the following criterion.

Corollary 3.7. Let $f : E' \to E$ be a morphism of locally coherent topoi. Let $(Y_i)_{i\in I}$ and $(Y_i')_{i\in I}$ be families of objects of E and E', and for each i let $v_i : Y_i' \to f^*(Y_i)$, whence a morphism of topoi

$$f_{v_i} : E'_{/Y_i'} \to E_{/Y_i}.$$

Assume that the Y_i' cover the final object of E'. Then f is locally coherent if and only if each f_{v_i} is locally coherent.

Applying this to the identity morphism of an induced topos gives:

Corollary 3.8. Let E be a locally coherent topos and let $v : X \to Y$ be an arrow of E. Then the morphism of induced topoi $E_{/X} \to E_{/Y}$ defined by v is locally coherent.

Remark 3.9. It seems that all algebraic morphisms of topoi encountered in practice are locally coherent; for this reason the term "locally coherent" is probably not destined for very frequent use. Equivalently, the coherent morphisms of topoi encountered in practice are coherent. Nevertheless one can construct non-coherent morphisms between coherent topoi, and in particular a non-coherent morphism from the punctual topos to a noetherian topos E; cf. 2.17(i).

Examples 3.10: topoi associated with schemes. Return to Example 1.22, the category (Sch) with one of the topologies T_i of SGA IV 6.3. For every scheme X, let X_{T_i} be the $\mathcal{V}$-topos defined by the induced site $(\mathrm{Sch})_{/X}$. A morphism of schemes $f : X \to Y$ defines a morphism of topoi $f_{T_i} : X_{T_i} \to Y_{T_i}$, and by 3.4 and 1.22 this morphism is quasi-compact (respectively quasi-separated, respectively coherent) if and only if the morphism of schemes f is quasi-compact (respectively quasi-separated, respectively coherent) in the usual sense of EGA IV. In any case, f_{T_i} is locally coherent by 3.8.

One may also associate with a scheme X the $\mathcal{U}$-topoi $\mathrm{Top}(X)$ and $\mathrm{Top}(X_{\mathrm{et}})$, as in 1.22.1; to every morphism of schemes $f : X \to Y$ one then associates morphisms $\mathrm{Top}(f)$ and $\mathrm{Top}(f_{\mathrm{et}})$ of the corresponding topoi. Let $T(f)$ be one of these two morphisms. By 3.2 one verifies immediately that $T(f)$ is always locally coherent, and that it is quasi-compact (respectively quasi-separated, respectively coherent) if and only if f is quasi-compact (respectively quasi-separated, respectively coherent) in the usual sense.

These observations show again that the terminology introduced here is compatible with the accepted terminology in scheme theory, and therefore deserves acceptance even from the most reluctant reader.

Exercise 3.11: coherent topoi and pretopoi.

a) A $\mathcal{U}$-*pretopos* (or simply pretopos) is a category C satisfying the following conditions:

   1) finite projective limits in C are representable;
   2) finite sums in C are representable, disjoint, and universal;
   3) equivalence relations in C are effective, and every epimorphism in C is universally effective;
   4) C is equivalent to a category belonging to $\mathcal{U}$.

   Show that if E is a coherent $\mathcal{U}$-topos, then the full subcategory E_{coh} of coherent objects is a $\mathcal{U}$-pretopos, and the inclusion $E_{\text{coh}} \to E$ is left exact, commutes with finite sums, and commutes with passage to quotients by equivalence relations. Show that the topology induced by E on $C = E_{\text{coh}}$ is the *precanonical topology*, namely the topology whose covering families $X_i \to X$ are those which admit a finite subfamily covering for the canonical topology of C. Hence E can be reconstructed, up to equivalence, from the pretopos $C = E_{\text{coh}}$ as the topos $C^{\sim}$, where C is endowed with its precanonical topology.

b) Let C be a $\mathcal{U}$-pretopos, endowed with its precanonical topology. Show that every morphism f in C factors as $f''f'$, with f' an epimorphism and f'' a monomorphism. Show that $E = C^{\sim}$ is coherent, and that the canonical functor $\varepsilon : C \to E$ induces an equivalence of C with E_{coh}. First show that ε is fully faithful, left exact, and commutes with finite sums and quotients by equivalence relations, and then show that a coherent subobject in E of an object of C lies in the essential image of C. Thus a $\mathcal{U}$-pretopos C can be recovered, up to equivalence of categories, from its associated $\mathcal{U}$-topos $E = C^{\sim}$.

c) Let C and C' be $\mathcal{U}$-pretopoi. Show that a functor $\varphi : C' \to C$ is a morphism of sites from C to C', for the precanonical topologies, if and only if φ is left exact and commutes with finite sums and quotients by equivalence relations.

d) With the notation of c), assume that C and C' are associated with two coherent $\mathcal{U}$-topoi E and E' as in a). Show that the canonical functor

$$\text{MorSite}(C, C') \to \mathcal{H}om_{\text{top}}(E, E')$$

   induces an equivalence from the first category, described in c), onto the full subcategory of the second consisting of coherent morphisms of topoi $E \to E'$. A quasi-inverse sends a coherent morphism $f : E \to E'$ to the functor $E'_{\text{coh}} = C' \to E_{\text{coh}} = C$ induced by f^*.

e) Let C be a $\mathcal{U}$-pretopos and $E = C^{\sim}$. Express directly, in exactness properties of C, the equivalent conditions of 1.25. Show that an object X of C is noetherian in E if and only if every increasing sequence of subobjects of X in C is stationary; hence E is noetherian if and only if every object of C satisfies this condition.

Exercise 3.12. A topos E is called *finite* if there exists a finite category C such that E is equivalent to $\widehat{C}$.

a) Show that E is finite if and only if E has enough essential points and the category $\text{Point}_{\text{ess}}(E)$ of essential points of E is equivalent to a finite category.

b) Assume E finite. Prove that every point of E is essential. Prove that the 2-functors $E \mapsto \mathrm{Point}(E)$ and $C \mapsto \widehat{C}$ establish 2-equivalences between the 2-category of finite topoi and the 2-category of Karoubian categories equivalent to finite categories. Prove also that every morphism of finite topoi is essential.

c) Assume $E = \widehat{C}$, with C equivalent to a finite category, and let F be a coherent topos. Recall that morphisms of topoi $f : E \to F$ correspond to functors $u : C \to \mathrm{Point}(F)$. Show that f is coherent if and only if it sends coherent points of E to coherent points of F, equivalently if $u(X)$ is a coherent point of F for every $X \in \mathrm{ob}\, C$. Deduce that every morphism from a finite topos to a finite topos is coherent.

d) Let X be a sober topological space. Show that $\mathrm{Top}(X)$ is finite if and only if X is a finite set.

4. Finiteness conditions in a topos obtained by gluing

The present section will not be used later in the Seminar.

4.1

Let E be a topos and U an open of E, that is, a subobject of the final object e of E. Consider the corresponding open and closed subtopoi of E:

$$E' = E_{/U}, \qquad E'' = E_{/U^c}.$$

We denote by

$$j : E' \to E, \qquad i : E'' \to E$$

the canonical morphisms of topoi. Recall that j is quasi-compact (respectively coherent) if and only if the inclusion $j : U \to e$ is a quasi-compact (respectively coherent) morphism in E. On the other hand, $j : E' \to E$ is always quasi-separated, because $j : U \to e$ is so. We shall give criteria for the closed subtopos E'' to be coherent and for the inclusion $i : E'' \to E$ to be coherent.

Proposition 4.2. With the notation of 4.1, the inclusion morphism $i : E'' \to E$ is quasi-compact; that is, for every quasi-compact object X of E, the object $i^*(X)$ of E'' is quasi-compact. Moreover, if X is pre-noetherian, then $i^*(X)$ is pre-noetherian.

This follows from Definition 1.1 and the following lemma.

Lemma 4.2.1. The map $Y \mapsto i^*(Y)$ induces an isomorphism of ordered sets between the set of subobjects of X containing the subobject $X_U = X \times U$ of X, and the set of subobjects of $i^*(X)$.

Indeed, since i_* is conservative (even fully faithful) and left exact, a morphism $u : Y \to Z$ in E'' is a monomorphism if and only if $i_*(u)$ is. Identifying E'', through i_*, with the full subcategory of E described in IV 9.7, subobjects in E'' of an object Z are the subobjects Y of Z in E which still belong to E'', equivalently which contain the copy of U. For $Z = i_*i^*(X) = X_{\complement U}$, described by the pushout

$$X_{\complement U} = X \amalg_{X_U} U,$$

a subobject of Z is the same as a pair consisting of a subobject $Y_1 \subset X$ and a subobject $Y_2 \subset U$, with equal inverse images in X_U. The condition that this subobject contain U says $Y_2 = U$, and the remaining condition on Y_1 is precisely $X_U \subset Y_1$. This proves the claimed order isomorphism.

Corollary 4.3.

a) Let X be an object of E. To prove X quasi-compact it suffices that $i^*(X)$ and $j^*(X)$ be quasi-compact; this condition is also necessary if $j : U \to e$ is quasi-compact.

b) Let Y be an object of E''. To prove Y quasi-compact it suffices that $i_*(Y)$ be quasi-compact; this condition is also necessary if U is quasi-compact.

For a), sufficiency follows from Definition 1.1 and the conservativity of the pair (i^*, j^*); necessity follows from the quasi-compactness of i and j. For b), since $Y \simeq i^*i_*(Y)$, sufficiency follows from 4.2. Necessity follows from a), using $i^*i_*(Y) \simeq Y$ and $j^*i_*(Y) \simeq U$.

Corollary 4.4. Let Y be an object of E''. If U is quasi-compact, or if E admits a generating family consisting of quasi-compact objects, then for Y to be quasi-separated (respectively coherent) it suffices that $i_*(Y)$ be quasi-separated (respectively coherent). If U is quasi-separated (respectively coherent) and $j : U \to e$ is quasi-compact, then for Y to be quasi-separated (respectively coherent) it is necessary that $i_*(Y)$ be so.

The coherent case follows from the quasi-separated case together with 4.3(b). Suppose that $i_*(Y)$ is quasi-separated. We must prove that if Y' and Y'' are quasi-compact objects over Y, then $Y' \times_Y Y''$ is quasi-compact. If U is quasi-compact, it suffices, by 4.3(b), to prove that

$$i_*(Y' \times_Y Y'')$$

is quasi-compact; this follows because $i_*(Y')$ and $i_*(Y'')$ are quasi-compact, because i_* commutes with fibre products, and because $i_*(Y)$ is quasi-separated. If instead E has a generating family of quasi-compact objects, write $i_*(Y')$ as a filtered inductive limit of its quasi-compact subobjects X'_α. Then $Y' \simeq i^*i_*(Y')$ is a filtered inductive limit of the $i^*(X'_\alpha)$, hence, since Y' is quasi-compact, is equal to one of them. Similarly $Y'' = i^*(X''_\beta)$. Thus

$$Y' \times_Y Y'' = i^*(X'_\alpha \times_{i_*(Y)} X''_\beta),$$

which is quasi-compact by 4.2.

Conversely, assume U quasi-separated and $j : U \to e$ quasi-compact. If Y is quasi-separated, then $i_*(Y)$ is quasi-separated: for quasi-compact objects X', X'' over $i_*(Y)$, it suffices by 4.3(a) to check that the images under i^* and j^* of $X' \times_{i_*(Y)} X''$ are quasi-compact. The first assertion follows from 4.2 and the quasi-separatedness of Y; the second image is $j^*(X') \times j^*(X'')$, which is quasi-compact because j is quasi-compact and U is quasi-separated.

Corollary 4.5. If E admits a generating subcategory consisting of quasi-compact (respectively pre-noetherian) objects, then so does E''.

This follows from 4.2 and the following lemma.

Lemma 4.5.1. If $(X_\alpha)_\alpha$ is a generating family in E, then $(i^*(X_\alpha))_\alpha$ is a generating family in E''.

Indeed the family of functors $Y \mapsto \mathrm{Hom}(i^*(X_\alpha), Y)$ on E'' is conservative, since $\mathrm{Hom}(i^*(X_\alpha), Y) \simeq \mathrm{Hom}(X_\alpha, i_*(Y))$ and i_* is conservative.

Proposition 4.6. With the notation of 4.1:

a) If E admits a generating family consisting of coherent objects, then the same is true for the closed subtopos E'', and the inclusion $i : E'' \to E$ is coherent.

b) Assume $j : U \to e$ quasi-compact. If E is locally coherent (respectively coherent, quasi-separated, locally noetherian, noetherian, locally perfect, perfect), then so is E'', and the inclusion $i : E'' \to E$ is coherent.

For a), by 4.5 the topos E'' has a generating family of quasi-compact objects, so by 3.2 it is enough to show that $i^*(X)$ is coherent for every coherent X of E. Using compatibility of the formation of the complement of an open with localization, one reduces to the case where X is the final object of E. Then it is enough to prove that the final object of E'' is coherent. By 4.4 this follows from the fact that $i_*(e_{E''}) = e_E$ is coherent.

For b), under each of the stated hypotheses E has a generating family of coherent objects, so a) already gives that i is coherent and that E''_{coh} generates E''. If E is coherent, the stability of E''_{coh} under finite projective limits follows from the criterion 4.4 (which applies since now U is coherent and $j : U \to e$ is quasi-compact), together with the left exactness of i_*. By localization, the locally coherent case follows. If E is quasi-separated, then the final object of E'' is quasi-separated by 4.4; products of quasi-separated objects in E'' are quasi-separated by the same criterion and the fact that i_* commutes with products.

Since a topos is locally noetherian (respectively noetherian) if and only if it is locally coherent (respectively coherent) and has a generating family consisting of pre-noetherian objects, 4.5 gives the noetherian assertions. If E is perfect, then E'' is already coherent, and it remains to check criterion 1.25(i): every object Y of E'' must be a filtered inductive limit of coherent objects. Since $i_*(Y)$ has such a presentation by coherent objects X_α, one has $Y \simeq i^* i_*(Y)$, a filtered inductive limit of the coherent objects $i^*(X_\alpha)$. Localizing gives the locally perfect case.

Remark 4.6.1. In fact in 4.6(b) the assumption that $j : U \to e$ be quasi-compact is unnecessary except perhaps in the quasi-separated case. It is enough, by the proof, to see that E coherent implies E'' coherent. Since U is a filtered inductive limit of its quasi-compact subobjects U_α, Section 7 will show that E'' identifies with the projective-limit topos of the closed subtopoi E''_α complementary to the $E_{/U_\alpha}$. These are coherent topoi with coherent transition morphisms; hence the limit E'' is coherent.

Corollary 4.7. Assume that E_{coh} is generating, and let X be an object of E.

a) For X to be quasi-separated it is necessary that $i^*(X)$ and $j^*(X)$ be quasi-separated; this condition is also sufficient if $j : U \to e$ is quasi-compact.

b) If $j : U \to e$ is quasi-compact, then X is coherent if and only if $i^*(X)$ and $j^*(X)$ are coherent.

c) If E is coherent and $j : U \to e$ is quasi-compact, then E is perfect if and only if E' and E'' are perfect.

Necessity in a) follows from the quasi-separatedness of j and of i, the latter by 4.6(a). For sufficiency, let X' and X'' be quasi-compact objects over X; it suffices by 4.3(a) to prove that the images under i^* and j^* of $X' \times_X X''$ are quasi-compact. This follows because i and j are quasi-compact and because $i^*(X)$, $j^*(X)$ are quasi-separated. Assertion b) is the conjunction of a) and 4.3(a). Necessity in c) was already proved in 4.6(b); sufficiency follows because, for a coherent topos, perfection is equivalent to stability of coherent objects under finite inductive limits, and b) gives this stability after gluing.

Exercise 4.8.

a) Resolve the following question, to which the writer confesses not to know the answer: with the notation of 4.1, if E is quasi-separated (respectively algebraic), is E'' so as well?

b) Assume that E is equivalent to a topos of the form $\widehat{C}$, where C is equivalent to a $\mathcal{U}$-small category. Prove directly, using 2.17(c) and IV 9.24, that if E is locally coherent (respectively coherent, algebraic, quasi-separated, locally noetherian, noetherian), then the same is true of E''.

4.9

Keep the notation of 4.1. Recall that the datum of a pair (E, U), consisting of a topos E and an open U of E, is essentially equivalent to the datum of a triple (E', E'', f), where E' and E'' are topoi and

$$f : E' \to E'' \tag{4.9.1}$$

is a *gluing functor*, that is, a left exact accessible functor. Starting with (E, U), the associated gluing functor is

$$f = i^* j_*. \tag{4.9.2}$$

We shall express, in terms of E', E'', and f, the fact that E is coherent (respectively perfect) and that U is quasi-compact, equivalently that E and E' are coherent. We already know that this implies E'' coherent. Thus the problem is the following: given two coherent topoi E', E'' and a gluing functor f, under what conditions is the glued topos E coherent (respectively perfect)? The perfect case reduces to the coherent case, since E is perfect if and only if E, E', and E'' have the corresponding coherent and perfect properties; however the final criterion takes a cleaner form when E' is already assumed perfect.

4.9.3

First suppose E coherent. Then the full subcategory E_{coh} of coherent objects $X = (X', X'', u : X'' \to f(X'))$ is reconstructed as the subcategory E_0 consisting of those X for which $X' = j^*(X)$ and $X'' = i^*(X)$ are coherent. A necessary condition for E to be coherent is therefore that this subcategory E_0 be generating. This condition is also sufficient: by 4.3(a), E_0 consists of quasi-compact objects; it is plainly stable under finite projective limits, and one concludes by 2.4.5.

Recall also that giving the gluing functor f is equivalent to giving a sheaf on E'' with values in $\mathrm{Pro}(E')^0$,

$$G \in \mathrm{Sh}(E'', \mathrm{Pro}(E')^0), \qquad G : E''^0 \to \mathrm{Pro}(E')^0, \tag{4.9.4}$$

or equivalently a functor

$$g : E'' \to \mathrm{Pro}(E') \tag{4.9.5}$$

which commutes with inductive limits. If C is a generating subcategory of E'', endowed with the induced topology, this datum is also equivalent to a sheaf on C with values in $\mathrm{Pro}(E')^0$,

$$G_C \in \mathrm{Sh}(C, \mathrm{Pro}(E')^0), \qquad G_C : C^0 \to \mathrm{Pro}(E')^0, \tag{4.9.6}$$

or to a functor

$$g_C : C \to \mathrm{Pro}(E') \tag{4.9.7}$$

satisfying the expected left exactness conditions. Of course g_C is just the restriction of g to C. The case most useful here is $C = E''_{\mathrm{coh}}$, giving

$$G_0 \in \mathrm{Sh}(E''_{\mathrm{coh}}, \mathrm{Pro}(E')^0), \qquad G_0 : E''^0_{\mathrm{coh}} \to \mathrm{Pro}(E')^0, \tag{4.9.8}$$

$$g_0 = G_0^0 : E''_{\mathrm{coh}} \to \mathrm{Pro}(E'). \tag{4.9.9}$$

Either of these data is again equivalent to the original gluing functor f.

4.9.10

Assume that the generating full subcategory C of E'' consists of quasi-compact objects and is stable in E'' under finite sums, quotients by equivalence relations, and fibre products. This is the case, for instance, for $C = E''_{\mathrm{coh}}$. If P is a category with finite projective limits, then a functor $G_C : C^0 \to P$ is a sheaf with values in P if and only if the opposite functor $g_C = G_C^0 : C \to P^0$ commutes with finite sums and quotients by equivalence relations. This describes explicitly the sheaves 4.9.6, and hence the gluing functors $f : E' \to E''$.

Proposition 4.10. Let E' and E'' be coherent topoi, and let $f : E' \to E''$ be a gluing functor, that is, a left exact accessible functor. Let E be the topos obtained by gluing. Let E'_{coh} denote the full subcategory of coherent objects of E', and let E'_{PF} denote the full subcategory of objects X such that the covariant representable functor $\mathrm{Hom}(X, -)$ commutes with small filtered inductive limits. Consider the following conditions.

(i) E is coherent.

(ii) For every object X' of E', there is a family of morphisms $v_\alpha : Y'_\alpha \to X'$, with coherent sources, such that the family $f(v_\alpha) : f(Y'_\alpha) \to f(X')$ is covering in E''.

(ii bis) f commutes with small filtered inductive limits, and (ii) holds for every $X' \in E'_{PF}$.

(ii ter) The canonical functor

$$\pi : \mathrm{Point}(E'') \to \mathrm{Pro}(E') \tag{4.10.1}$$

defined by f factors, up to isomorphism, through $\mathrm{Pro}(E'_{\mathrm{coh}})$:

$$\pi_0 : \mathrm{Point}(E'') \to \mathrm{Pro}(E'_{\mathrm{coh}}). \tag{4.10.2}$$

(iii) f commutes with small filtered inductive limits.

(iii bis) The functor g_0 in 4.9.9 factors, up to isomorphism, through a functor

$$E''_{\mathrm{coh}} \to \mathrm{Pro}(E'_{PF}) \tag{4.10.3}$$

via the fully faithful functor $\mathrm{Pro}(E'_{PF}) \to \mathrm{Pro}(E')$.

(iii ter) The canonical functor π in 4.10.1 factors, up to isomorphism, as

$$\pi_1 : \mathrm{Point}(E'') \to \mathrm{Pro}(E'_{PF}). \tag{4.10.4}$$

(iv) The functor g_0 in 4.9.9 factors, up to isomorphism, through a functor

$$E''_{\mathrm{coh}} \to \mathrm{Pro}(E'_{\mathrm{coh}}). \tag{4.10.5}$$

Then one has the diagram of implications

$$\text{(iv)} \Rightarrow \text{(i)} \Leftrightarrow \text{(ii)} \Leftrightarrow \text{(ii bis)} \Leftrightarrow \text{(ii ter)} \Rightarrow \text{(iii)} \Leftrightarrow \text{(iii bis)} \Leftrightarrow \text{(iii ter)}. \tag{4.10.6}$$

Corollary 4.11.

a) If E' is perfect, then all the conditions in 4.10 are equivalent. In particular, E is coherent if and only if f commutes with filtered inductive limits, or equivalently if and only if g_0 in 4.9.9 factors, up to isomorphism, through $\mathrm{Pro}(E'_{\mathrm{coh}})$.

b) The topos E is perfect if and only if E' and E'' are perfect and f (respectively g_0) satisfies the condition in a).

Indeed, a) follows from the fact that E' perfect means $E'_{\mathrm{coh}} = E'_{PF}$, so that (iii bis) implies (iv). Assertion b) follows from the preliminary remarks in 4.9.

4.12. Proof of 4.10

a) Unwinding the equivalent condition to (i) obtained in 4.9.3, the full subcategory E_0 of E must be generating. Thus, for every object

$$X = (X', X'', u : X'' \to f(X'))$$

of E, the family of all morphisms

$$v = (v', v'') : Y = (Y', Y'', v : Y'' \to f(Y')) \to X,$$

with $Y \in E_0$, that is with Y' and Y'' coherent, must be epimorphic. Since (i^*, j^*) is conservative, this means that the corresponding family $Y' \to X'$ is epimorphic in E', and the family $Y'' \to X''$ is epimorphic in E''. The first assertion is clear by taking objects Y lying over U, that is, with $Y'' = \varnothing$, because E'_{coh} generates E'. It remains to express the second assertion for arbitrary X.

b) Assume condition (ii) for the object X' considered. Then there is an epimorphic family $Y''_\beta \to X''$ whose composites with $u : X'' \to f(X')$ lift through morphisms $Y''_\beta \to f(Y'_{\varphi(\beta)})$. Since E''_{coh} generates E'', the Y''_β may be chosen coherent. The resulting commutative diagrams

$$\begin{tikzcd} Y''_\beta \arrow[r] \arrow[d] & f(Y'_{\varphi(\beta)}) \arrow[d] \\ X'' \arrow[r] & f(X') \end{tikzcd}$$

provide the desired epimorphic family in E. Thus (ii) implies (i). Conversely, assuming (i), apply the condition above to

$$X = j_*(X') = (X', f(X'), \mathrm{id} : f(X') \xrightarrow{\sim} f(X')).$$

Since the corresponding $Y'' \to f(X')$ form an epimorphic family, the family of the $f(v') : f(Y') \to f(X')$ is a fortiori epimorphic. Hence (i) and (ii) are equivalent.

c) Condition (iv) means that, for every object X'' of E''_{coh}, the pro-representable functor

$$g(X'') : X' \longmapsto \mathrm{Hom}(X'', f(X'))$$

on E' is pro-represented by a pro-object whose components lie in E'_{coh}. Equivalently, in the filtered category $E'_{/g(X'')}$ of pairs (X', u), where $u : X'' \to f(X')$, the full subcategory of those

(X'_α, u_α) with X'_α coherent is cofinal. Thus for every $u : X'' \to f(X')$ there is $v'_\alpha : X'_\alpha \to X'$ with X'_α coherent such that u factors through $f(v'_\alpha)$. It is then clear that (iv) implies (ii), since for fixed X' the family of all arrows $u : X'' \to f(X')$ with coherent source X'' is epimorphic. Hence (iv) implies (i).

d) Let p be a point of E''. By definition $\pi(p)$ is the pro-object of E' pro-representing the functor $h : X' \mapsto f(X')_p$. Saying that it comes from $\mathrm{Pro}(E'_{\mathrm{coh}})$ means that, in the category of pairs (X', u) with $u \in f(X')_p$, the subcategory with X' coherent is cofinal. Since a coherent topos has enough points, this condition, for all points p, is equivalent to the assertion that for every X' the family $f(X'_0) \to f(X')$, with X'_0 coherent, is epimorphic. This is exactly (ii). Thus (ii) is equivalent to (ii ter).

e) The condition that f commute with filtered inductive limits is equivalent, since the family of fibre functors of E'' is conservative, to the same condition for the composite functors $h : X' \mapsto f(X')_p$. This is equivalent to saying that the pro-object $\pi(p)$ pro-representing h belongs to the essential image of $\mathrm{Pro}(E'_{PF})$, by the following lemma.

Lemma 4.12.1. Let E' be a topos for which E'_{coh} is generating, and let $X' \in \mathrm{Ob}\,\mathrm{Pro}(E')$. Then X' belongs to the essential image of $\mathrm{Pro}(E'_{PF})$ if and only if the functor $h : E' \to \mathbf{Set}$ pro-represented by X' commutes with filtered inductive limits.

Since a filtered inductive limit of functors commuting with filtered inductive limits has the same property, one direction follows directly from the definition of E'_{PF}. Conversely, suppose that h commutes with filtered inductive limits. In the filtered category $E'_{/h}$ of pairs (X', u) with $u \in h(X')$, the full subcategory consisting of pairs with $X' \in E'_{PF}$ is cofinal: indeed every object X' is a filtered inductive limit of objects X'_i of E'_{PF}, and hence $h(X')$ is the filtered inductive limit of the $h(X'_i)$.

It remains only to prove (ii bis) $\Rightarrow$ (ii) and (iii) $\Leftrightarrow$ (iii bis). The first implication is immediate, since every object of E' is a filtered inductive limit of objects of E'_{PF}. For the second, the family of functors

$$Y'' \longmapsto \mathrm{Hom}(X'', Y''), \qquad X'' \in E''_{\mathrm{coh}},$$

is conservative and consists of functors commuting with filtered inductive limits. Hence f commutes with filtered inductive limits if and only if the composite functors $X' \mapsto \mathrm{Hom}(X'', f(X'))$ do so, which by Lemma 4.12.1 is equivalent to $g_0(X'')$ lying in the essential image of $\mathrm{Pro}(E'_{PF})$.

Remark 4.13. It is useful to spell out, in sheaf-theoretic language, condition 4.10(iv). Consider the commutative diagram of functors

$$\begin{tikzcd} \mathrm{Pro}(E'_{\mathrm{coh}})^0 \arrow[r, hook, "\alpha"] \arrow[d, "\beta_0"'] & \mathrm{Pro}(E')^0 \arrow[d, "\beta"] \\ (E'_{\mathrm{coh}})^\vee & (E')^\vee \arrow[l, "\gamma"] \end{tikzcd} \tag{4.13.1}$$

where the $\vee$'s denote categories of set-valued functors, and γ is restriction; the other arrows are the usual fully faithful functors. The inclusion α need not commute with projective limits, even finite ones, so a sheaf with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ may become only a presheaf after inclusion into $\mathrm{Pro}(E')^0$. Nevertheless, if a presheaf F with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ has αF a sheaf with values in $\mathrm{Pro}(E')^0$, then F itself is a sheaf. This follows because the sheaf condition is a left exactness property, and β, β_0, and γ commute with small projective limits.

Furthermore, from the characterization 4.9.10 of sheaves on E''_{coh} with values in an arbitrary category P, it follows that if $\alpha : P \to Q$ is such that $\alpha^0 : P^0 \to Q^0$ commutes with finite sums and quotients by equivalence relations, then composition with α sends sheaves to sheaves. Applying this to the inclusion $\mathrm{Pro}(E'_{\mathrm{coh}}) \to \mathrm{Pro}(E')$, one concludes that a functor

$$\varphi : {E''}^{0}_{\mathrm{coh}} \to \mathrm{Pro}(E'_{\mathrm{coh}})^0 \tag{4.13.2}$$

is a sheaf on E''_{coh} with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ if and only if $\alpha \circ \varphi$ is a sheaf with values in $\mathrm{Pro}(E')^0$. Thus the category of gluing functors $f : E' \to E''$ satisfying 4.10(iv) is canonically equivalent to the category of such sheaves φ. Such a sheaf canonically determines a gluing functor, up to unique isomorphism, and the topos E obtained by gluing is coherent. If E' is perfect, all coherent topoi obtained from E' and E'' by gluing arise in this way.

Since E'_{coh} is equivalent to a small category and is stable under finite projective limits, $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ identifies with the full subcategory of $\mathrm{Hom}(E'_{\mathrm{coh}}, \mathbf{Set})$ consisting of left exact functors. Hence giving a sheaf φ as in 4.13.2 is equivalent to giving a functor

$$F : {E''}^{0}_{\mathrm{coh}} \times E'_{\mathrm{coh}} \to \mathbf{Set} \tag{4.13.3}$$

which is a sheaf in the first variable and left exact in the second; equivalently, to giving a left exact functor

$$f_0 : E'_{\mathrm{coh}} \to E''. \tag{4.13.4}$$

If f is the associated gluing functor satisfying 4.10(iv), then f_0 is canonically isomorphic to the restriction of f to E'_{coh}. Thus, under the hypotheses considered, f_0 determines f, up to canonical isomorphism.

Corollary 4.14. When E' is a perfect topos, the category of gluing functors $f : E' \to E''$ which give a coherent glued topos E is equivalent, via $f \mapsto f_0 = f|E'_{\mathrm{coh}}$, to the category of left exact functors $f_0 : E'_{\mathrm{coh}} \to E''$, or equivalently to the category of sheaves on the site E''_{coh} (equivalently on the topos E'') with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$.

The first assertion also follows directly from 4.10, since $E' \simeq \mathrm{Ind}(E'_{\mathrm{coh}})$: for any category P with small filtered inductive limits, functors $E' \to P$ commuting with filtered inductive limits are equivalent, by restriction, to arbitrary functors $E'_{\mathrm{coh}} \to P$. If P has finite inductive limits, such a functor is left exact precisely when its restriction is.

Remarks 4.15.

a) The proof of 4.10 shows that conditions equivalent to (ii ter) and (iii ter) are obtained by replacing $\mathrm{Point}(E'')$ by any subcategory which defines a conservative family of fibre functors.

b) Taking E'' to be the punctual topos, one sees immediately that (ii ter) and (iii ter) are equivalent only when $E'_{\mathrm{coh}} = E'_{PF}$, that is, when E' is perfect. Thus the condition “E coherent” is equivalent to “f commutes with filtered inductive limits” only when E' is perfect. On the other hand, more generally taking E'' coherent of the form $\widehat{C}$, one sees that in this case condition (i) is equivalent to (iv) for every E'. In fact, in the general case the writer has not managed to construct an example where E is coherent but condition (iv) fails. Shameful!

5. Commutation of the functors $H^i(X,-)$ with filtered inductive limits

Theorem 5.1. Let E' be an algebraic topos, E a locally coherent topos, and $f : E' \to E$ a coherent morphism of topoi. Then, for every integer q, the functors $R^q f_*$ commute with filtered inductive limits of abelian sheaves.

Corollary 5.2. Let E' be a coherent topos. For every integer q, the functor $H^q(E',-)$ commutes with filtered inductive limits of abelian sheaves.

Corollary 5.3. Let E be a topos (respectively an algebraic topos), and let X be an algebraic and coherent (respectively coherent) object of E. For every integer q, the functor $H^q(X,-)$ commutes with filtered inductive limits of abelian sheaves.

Corollary 5.2 follows from 5.1 by taking E to be the punctual topos and $f : E' \to E$ the unique morphism. Conversely, 5.2 implies 5.3 by applying it to the localization morphism $j_X : E_{/X} \to E$, using the canonical isomorphism

$$H^q(X,F) \simeq H^q(E_{/X}, j_X^* F),$$

the fact that j_X^* commutes with inductive limits, and the coherence of $E_{/X}$. Finally, 5.3 implies 5.1: if $C = E_{\text{coh.alg}}$ is the generating subcategory of algebraic coherent objects, then $R^q f_* F$ is the sheaf associated with the presheaf

$$X \longmapsto H^q(f^*X, F), \qquad X \in C.$$

Since f is coherent, f^*X is coherent for every $X \in C$; by 5.3 the presheaf above commutes with filtered inductive limits in F, and sheafification also commutes with inductive limits.

It remains to prove 5.2. We already know that $H^0(E',-)$ commutes with filtered inductive limits. By a classical result it is enough to prove that a filtered inductive limit of sheaves acyclic for $H^0(E',-)$ is again acyclic. This follows from the following lemma, applied to $C = E'_{\text{coh}}$.

Lemma 5.4. Let E be a locally coherent topos and let C be a full generating subcategory of E, stable under fibre products, such that every object of C is quasi-compact. A filtered inductive limit of C-acyclic sheaves is C-acyclic.

Let $(F_i)_{i\in I}$ be a filtered inductive system of C-acyclic sheaves, and let $F = \varinjlim F_i$. Every $Y \in C$ is coherent, hence $F(Y) = \varinjlim_i F_i(Y)$. To prove F C-acyclic it is enough to show $\check{H}^q(Y,F) = 0$ for $q > 0$ and $Y \in C$. For a finite covering family $\mathfrak{X} = (X_\alpha \to X)$ by objects of C, the associated Cech complex uses only finite products of groups, and therefore

$$C^\bullet(\mathfrak{X}, F) = \varinjlim_i C^\bullet(\mathfrak{X}, F_i).$$

Filtered colimits are exact in abelian groups, so $H^q(\mathfrak{X},F) = \varinjlim_i H^q(\mathfrak{X}, F_i) = 0$ for $q > 0$. Passing to the limit over such coverings gives $\check{H}^q(X,F) = 0$.

Corollary 5.5. Let E be a coherent topos, and let $(X_i)_{i\in I}$ be a filtered decreasing family of coherent subobjects of the final object. Let Φ be the family of closed subtopoi contained in the closed complement of one of the X_i. Then Φ is a family of supports of E, and for every integer q the functors $H^q_\Phi(E,-)$ and $\mathcal{H}^q_\Phi$ commute with filtered inductive limits.

Let Z_i be the closed complement of X_i. For every abelian sheaf F one has the exact sequence

$$0 \to H^0_{Z_i}(E, F) \to H^0(E, F) \to H^0(X_i, F) \to H^1_{Z_i}(E, F) \to \cdots .$$

By 5.2 and 5.3 the functors $H^q(E, F)$ and $H^q(X_i, F)$ commute with filtered inductive limits in F. Hence so does $H^q_{Z_i}(E, F)$, and passing to the inductive limit over the Z_i gives the same assertion for $H^q_\Phi(E, F)$. The assertion for $\mathcal{H}^q_\Phi$ follows by localization and sheafification.

Definition 5.6. Let (E, A) be a ringed topos and let q be an integer. An A-module G is said to be of *finite q-presentation* if there is a family (X_i) of objects covering the final object of E such that, on each $E_{/X_i}$, there is an exact sequence of modules

$$L_q \to L_{q-1} \to \cdots \to L_1 \to L_0 \to G_{/X_i} \to 0, \tag{5.5.1}$$

where $L_p = A^{n_p}_{/X_i}$ for integers n_p.

Proposition 5.7. If G is a sheaf of finite q-presentation, then for every $i \leq q - 1$, the functor $\mathcal{E}xt^i_A(G, -)$ commutes with filtered inductive limits of A-modules.

The question is local on E. We may therefore assume an exact sequence as in 5.5.1. Put $L_\bullet = L_q \to \cdots \to L_0$. Then

$$\mathcal{E}xt^i_A(G, F) = \mathcal{H}^i(\mathcal{H}om_A(L_\bullet, F)),$$

and $F \mapsto \mathcal{H}om_A(L_\bullet, F)$ commutes with inductive limits.

Corollary 5.8. Let (E, A) be a ringed topos, let G be a sheaf of A-modules of finite q-presentation, and let X be an algebraic and coherent object of E. The functors

$$F \longmapsto \mathrm{Ext}^i_A(X; G, F)$$

for $\frac{i(i-1)}{2} \leq q - 1$ commute with filtered inductive limits in F.

This follows from 5.7 and 5.3 by the spectral sequence V 6.1.3.

5.9

Let (E, A) be a ringed topos, Φ a family of supports of E, G an A-module of finite r-presentation, and X an algebraic coherent object of E. Passing to the inductive limit over the closed subtopoi in Φ in the last spectral sequences of V 6.9.1 and V 6.9.2 gives spectral sequences

$$\begin{cases} {}'E_2^{pq} = \varinjlim_{Z \in \Phi} \mathrm{Ext}^p_A(X; G, H^q_Z F) \Rightarrow \mathrm{Ext}^{p+q}_{A,\Phi}(X; G, F), \\ {}''E_2^{pq} = \varinjlim_{Z \in \Phi} \mathcal{E}xt^p_A(G, H^q_Z F) \Rightarrow \mathcal{E}xt^{p+q}_{A,\Phi}(G, F). \end{cases} \tag{5.9.1}$$

By 5.7 and 5.8 there are canonical isomorphisms

$$\begin{cases} {}'E_2^{pq} \simeq \mathrm{Ext}^p_A(X; G, H^q_\Phi F), & \frac{p(p-1)}{2} \leq r - 1, \\ {}''E_2^{pq} \simeq \mathcal{E}xt^p_A(G, H^q_\Phi F), & p \leq r - 1. \end{cases} \tag{5.9.2}$$

Corollary 5.10. Use the hypotheses and notation of 5.5. Let A be a ring of E and G an A-module of finite q-presentation. For every integer i such that $\frac{i(i-1)}{2} \leq r - 1$, the functors

$$F \mapsto \mathrm{Ext}^i_{A,\Phi}(E; G, F), \qquad F \mapsto \mathcal{E}xt^i_{A,\Phi}(G, F)$$

commute with filtered inductive limits.

This follows from 5.5 and 5.7 by the first spectral sequences of V 6.9.1 and V 6.9.2.

6. Inductive and projective limits of a fibred category

6.0

We assume the reader familiar with the theory of fibred categories. This section is mainly intended to fix terminology and notation. Unless otherwise stated, all categories considered here belong to a fixed universe $\mathcal{U}$.

Let $\mathcal{F}$, $\mathcal{G}$, and $\mathcal{E}$ be categories, and let $\pi : \mathcal{F} \to \mathcal{E}$, $\pi' : \mathcal{G} \to \mathcal{E}$ be functors. We write $\mathcal{H}om_{\mathcal{E}}(\mathcal{F}, \mathcal{G})$ for the category whose objects are functors $u : \mathcal{F} \to \mathcal{G}$ over $\mathcal{E}$, that is, with $\pi' u = \pi$, and whose morphisms are $\mathcal{E}$-morphisms of functors, namely natural transformations sent by π' to identity morphisms.

For an object ξ of $\mathcal{E}$, the objects X of $\mathcal{F}$ with $\pi(X) = \xi$ are called the objects of $\mathcal{F}$ over ξ. If $f : \xi \to \eta$ is a morphism of $\mathcal{E}$, morphisms in $\mathcal{F}$ over f are those mapped to f. For X over ξ and Y over η, $\mathrm{Hom}_f(X, Y)$ denotes the set of morphisms $X \to Y$ over f.

Definition 6.1.

1) A morphism $m : X \to Y$ of $\mathcal{F}$ is *cartesian* if, for every morphism $p : Z \to Y$ over $\pi(m)$, there exists a unique morphism $q : Z \to X$ over $\mathrm{id}_{\pi(X)}$ such that $mq = p$.

2) The category $\mathcal{F}$ is *prefibred* by π over $\mathcal{E}$ if for every morphism $f : \xi \to \eta$ of $\mathcal{E}$, every object of $\mathcal{F}$ over η is the target of a cartesian morphism over f.

3) The category $\mathcal{F}$ is *fibred* by π over $\mathcal{E}$ if it is prefibred and if the composite of two composable cartesian morphisms of $\mathcal{F}$ is cartesian.

6.1.1

For $\xi \in \mathcal{E}$, the *fibre category* of $\mathcal{F}$ at ξ, denoted $\mathcal{F}_\xi$, is the subcategory whose objects lie over ξ and whose morphisms lie over id_ξ. For $X, Y \in \mathcal{F}_\xi$, write $\mathrm{Hom}_\xi(X, Y)$ for the morphisms in this fibre.

6.1.2

Let $f : \xi \to \eta$ be a morphism of $\mathcal{E}$. An object Y over η is the target of a cartesian morphism over f if and only if the functor

$$Z \longmapsto \mathrm{Hom}_f(Z, Y), \qquad Z \in \mathrm{ob}(\mathcal{F}_\xi),$$

is representable. When $\mathcal{F}$ is prefibred, this gives, up to unique isomorphism, a functor $f^* : \mathcal{F}_\eta \to \mathcal{F}_\xi$, called inverse image for f. After choosing inverse-image functors for all arrows of $\mathcal{E}$, $\mathcal{F}$ is fibred if and only if $f^* g^*$ is again an inverse-image functor for every composable pair f, g. There is then a canonical isomorphism

$$C_{f,g} : f^* g^* \to (gf)^* \tag{6.1.2.1}$$

satisfying the cocycle relation

$$C_{f,hg}(f^* C_{g,h}) = C_{gh,f}(C_{f,g} h^*). \tag{6.1.2.2}$$

6.1.3

Conversely, given categories $\mathcal{F}_\xi$ for each $\xi \in \mathcal{E}$, functors $f^* : \mathcal{F}_\eta \to \mathcal{F}_\xi$ for each $f : \xi \to \eta$, and isomorphisms $C_{f,g} : f^* g^* \to (fg)^*$ satisfying the cocycle condition, one constructs, in an essentially unique way, a category $\mathcal{F}$ fibred over $\mathcal{E}$ with these fibres and inverse-image functors.

6.1.4

For two categories $\mathcal{F}$, $\mathcal{G}$ over $\mathcal{E}$, write

$$\mathcal{H}om_{\mathrm{cart}/\mathcal{E}}(\mathcal{F},\mathcal{G})$$

for the full subcategory of $\mathcal{H}om_{\mathcal{E}}(\mathcal{F},\mathcal{G})$ consisting of functors which send cartesian morphisms of $\mathcal{F}$ to cartesian morphisms of $\mathcal{G}$. Its objects are called *cartesian functors*.

Let C, C' be categories and S a set of morphisms of C. Write $\mathrm{Hom}_{S^{-1}}(C, C')$ for the set of functors $C \to C'$ sending every morphism of S to an isomorphism. Let $(\mathbf{Cat})$ be the category of categories belonging to the universe, with functors as morphisms.

Let $\mathcal{F}$ be a category over $\mathcal{E}$, and let S be its set of cartesian morphisms. The category $\mathcal{F}$ defines two set-valued functors on $(\mathbf{Cat})$:

$$\begin{aligned} C &\longmapsto \mathrm{Hom}_{S^{-1}}(\mathcal{F}, C), \\ C &\longmapsto \mathrm{Hom}_{\mathrm{Cart}\,/\mathcal{E}}(\mathcal{F}, C \times \mathcal{E}), \end{aligned}$$

where $C \times \mathcal{E}$ is viewed over $\mathcal{E}$ by the second projection.

Proposition 6.2. These two functors $(\mathbf{Cat}) \to (\mathbf{Set})$ are canonically isomorphic and are representable.

For the first assertion, note that the cartesian morphisms of $C \times \mathcal{E}$ are exactly the morphisms $m \times f$ with m an isomorphism of C. For representability, one formally adjoins inverses to the morphisms of S, constructing the free category on the objects and morphisms of $\mathcal{F}$ and formal inverses of arrows of S, then quotienting by the relations of $\mathcal{F}$ and by $s^{-1}s = \mathrm{id}$, $ss^{-1} = \mathrm{id}$. The resulting category is denoted $\mathcal{F}(S^{-1})$; the canonical functor $Q : \mathcal{F} \to \mathcal{F}(S^{-1})$ represents $\mathrm{Hom}_{S^{-1}}(\mathcal{F}, -)$.

Definition 6.3. When $\mathcal{F}$ is fibred over $\mathcal{E}$, the functor $\mathrm{Hom}_{S^{-1}}(\mathcal{F}, -)$ is denoted $\varinjlim_{\mathcal{E}^0} \mathcal{F}$ and called the inductive-limit functor of $\mathcal{F}$ over $\mathcal{E}^0$. The representing category is also denoted $\varinjlim_{\mathcal{E}^0} \mathcal{F}$ and called the inductive-limit category of $\mathcal{F}$ over $\mathcal{E}^0$.

6.4.0

Assume $\mathcal{F}$ fibred over $\mathcal{E}$, choose inverse-image functors for all arrows of $\mathcal{E}$, and let

$$Q : \mathcal{F} \to \varinjlim_{\mathcal{E}^0} \mathcal{F}$$

be the canonical functor. The inclusions of the fibres, composed with Q, give for every $\xi \in \mathcal{E}$ a functor

$$u_\xi : \mathcal{F}_\xi \to \varinjlim_{\mathcal{E}^0} \mathcal{F},$$

and for every arrow $f : \xi \to \eta$ a diagram commuting up to canonical isomorphism

$$\begin{tikzcd} \mathcal{F}_\eta \arrow[dr, "u_\eta"] \arrow[dd, "f^*"'] & \\ & \varinjlim_{\mathcal{E}^0} \mathcal{F} \\ \mathcal{F}_\xi \arrow[ur, "u_\xi"'] & \end{tikzcd} .$$

Thus $\varinjlim_{\mathcal{E}^0} \mathcal{F}$ is the inductive limit, in the sense of pseudofunctors, of the pseudofunctor $\mathcal{E}^0 \to \mathbf{Cat}$ sending ξ to $\mathcal{F}_\xi$ and $f : \xi \to \eta$ to f^*. Even if $\xi \mapsto \mathcal{F}_\xi$ is an actual functor, this category is not generally the ordinary inductive limit in the sense of I 2.

Proposition 6.4. Let $\mathcal{F}$ be a fibred category over $\mathcal{E}$. Assume that $\mathcal{E}$ has the following properties:

L1) every span can be embedded in a commutative square;

L2) if $u, v : \bullet \rightrightarrows \bullet$ become equal after postcomposition with some morphism t, then they become equal after precomposition with some morphism w.

For example, if $\mathcal{E}^0$ is pseudo-filtered, these properties hold. Then the set S of cartesian morphisms of $\mathcal{F}$ has the following properties:

Fr1) S is stable under composition, and all isomorphisms lie in S.

Fr2) every diagram $A \to B \leftarrow C$, with the arrow $C \to B$ in S, can be completed to a commutative square whose new arrow to A is in S.

Fr3) if u, v become equal after postcomposition with an arrow of S, then they become equal after precomposition with an arrow of S.

The proof is left to the reader.

Proposition 6.5. Let $\mathcal{F}$ be a category and S a set of morphisms satisfying Fr1–Fr3. For every object X of $\mathcal{F}$, let $S(X)$ be the category of arrows in S with target X. Then $S(X)$ is cofiltered, and the localization $\mathcal{F}(S^{-1})$ can be described as follows:

a) The objects are the objects of $\mathcal{F}$.

b) For objects X, Y,

$$\mathrm{Hom}_{\mathcal{F}(S^{-1})}(X, Y) = \varinjlim_{S(X)} \mathrm{Hom}_{\mathcal{F}}(\,\cdot\,, Y).$$

c) Composition is represented by the usual calculus of right fractions: choose representatives $t : Z \to X$ in S and $f' : Z \to Y$ for f, and $t' : W \to Y$ in S, $g' : W \to Z'$ for g; use Fr2 to dominate the common target, and represent gf by the resulting composite with denominator in S.

d) If $Q : \mathcal{F} \to \mathcal{F}(S^{-1})$ is the canonical functor, then

$$Q^* : \widehat{\mathcal{F}(S^{-1})} \to \widehat{\mathcal{F}},$$

$F \mapsto FQ$, is fully faithful and injective on objects.

e) Its left adjoint

$$Q_! : \widehat{\mathcal{F}} \to \widehat{\mathcal{F}(S^{-1})}$$

is left exact. Thus Q itself is left exact when finite projective limits are representable in $\mathcal{F}$.

The proof is referred to [1].

Exercise 6.6. Let $\mathcal{F}$ be a category and S a set of morphisms satisfying Fr1–Fr3. For every object X, let $S(X)$ be the category of morphisms in S with target X.

a) Prove that $S(X)$ is cofiltered.

b) Let $J_S(X)$ be the set of sieves on X containing a sieve generated by a morphism of S. Show that the $J_S(X)$ define a topology T on $\mathcal{F}$.

c) For T, the morphisms of S are bi-covering.

d) If $\mathcal{J}^2(X)$ is the category of bi-covering morphisms with target X, then $S(X)$ is cofinal in $\mathcal{J}^2(X)$.

e) For a presheaf G on $\mathcal{F}$, if a denotes associated sheaf for T, then

$$aG(X) \simeq \varinjlim_{S(X)} G(\,\cdot\,).$$

A presheaf is a sheaf if and only if it sends all morphisms of S to isomorphisms.

f) Choose sheafification so that its restriction to $\mathcal{F}$ is injective on objects. Let $\underline{\mathcal{F}}$ be the full subcategory of sheaves associated to representable presheaves. Then $\underline{\mathcal{F}}$ is isomorphic to $\mathcal{F}(S^{-1})$. Its objects generate the topos $\mathcal{F}^{\sim}$, and the induced topology on $\underline{\mathcal{F}}$ is the chaotic topology.

g) The canonical functor $Q : \mathcal{F} \to \underline{\mathcal{F}}$ is a morphism from the site $\underline{\mathcal{F}}$ with chaotic topology to $\mathcal{F}$ with the topology T, and induces an isomorphism of the associated topoi.

h) A functor $u : \mathcal{F} \to C$ is continuous if and only if it sends all morphisms of S to isomorphisms.

i) Such a functor factors uniquely through Q:

$$\begin{tikzcd}
\mathcal{F} \arrow[rr, "u"] \arrow[dr, "Q"'] & & C \\
& \underline{\mathcal{F}} \arrow[ur, "v"'] &
\end{tikzcd}.$$

j) Let $\mathrm{Pro}_S\,\mathcal{F}$ be the full subcategory of $\mathrm{Pro}\,\mathcal{F}$ consisting of pro-objects whose transition morphisms lie in S. Show that Q factors as

$$\mathcal{F} \hookrightarrow \mathrm{Pro}_S\,\mathcal{F} \to \mathcal{F}_{S^{-1}},$$

and that the second functor has a fully faithful left adjoint $A : \mathcal{F}_{S^{-1}} \to \mathrm{Pro}_S\,\mathcal{F}$. For $X \in \mathcal{F}$, $AQ(X)$ is the pro-object $(Y \to X \in S(X)) \mapsto Y$.

Proposition 6.7. Let $\mathcal{E}$ be cofiltered and $\mathcal{F} \to \mathcal{E}$ a fibred category.

1) For $\xi \in \mathcal{E}$, let $\mathcal{F}/\xi$ be the fibred category over $\mathcal{E}/\xi$ obtained by base change. The canonical functor

$$\varinjlim_{\mathcal{E}/\xi} \mathcal{F}/\xi \to \varinjlim_{\mathcal{E}} \mathcal{F}$$

is an equivalence.

2) More generally, if $\mathcal{E}' \to \mathcal{E}$ is cofinal and $\mathcal{F}' \to \mathcal{E}'$ is obtained from $\mathcal{F}$ by base change, then

$$\varinjlim_{\mathcal{E}'} \mathcal{F}' \to \varinjlim_{\mathcal{E}} \mathcal{F}$$

is an equivalence of categories.

Proof: exercise.

Exercise 6.8.

1) Let $g : \mathcal{E}^0 \to \mathbf{Cat}$ be a functor and $\mathcal{G} \to \mathcal{E}$ the corresponding fibred category. Show that there is a canonical functor $\varinjlim_{\mathcal{E}^0} \mathcal{G} \to \varinjlim_{\mathcal{E}^0} g$. Show that it need not be an equivalence; one may take g constant with value the final object of $\mathbf{Cat}$ and $\mathcal{E}$ the category associated with a group.

2) If $\mathcal{E}^0$ is pseudo-filtered, show that the canonical functor above is an equivalence.

Proposition 6.9. Let $\mathcal{F} \to \mathcal{E}$ be a fibred category. The functor $(\mathbf{Cat}) \to (\mathbf{Set})$

$$C \longmapsto \mathrm{Hom}_{\mathrm{Cart}\,/\mathcal{E}}(C \times \mathcal{E}, \mathcal{F})$$

is represented by the category $\mathcal{H}om_{\mathrm{Cart}\,/\mathcal{E}}(\mathcal{E}, \mathcal{F})$.

Indeed, if $F : C \times \mathcal{E} \to \mathcal{F}$ is a cartesian $\mathcal{E}$-functor, then for every $X \in C$ the functor $\xi \mapsto F(X, \xi)$ is a cartesian functor from $\mathcal{E}$ to $\mathcal{F}$, depending functorially on X. This identifies $\mathrm{Hom}_{\mathrm{Cart}\,/\mathcal{E}}(C \times \mathcal{E}, \mathcal{F})$ with $\mathrm{Hom}(C, \mathcal{H}om_{\mathrm{Cart}\,/\mathcal{E}}(\mathcal{E}, \mathcal{F}))$.

6.10

The category $\mathcal{H}om_{\mathrm{Cart}\,/\mathcal{E}}(\mathcal{E}, \mathcal{F})$ is called the category of *cartesian sections* of the fibred category $\mathcal{F}$. It is sometimes also called the projective-limit category of $\mathcal{F}$ over $\mathcal{E}^0$, and denoted $\varprojlim_{\mathcal{E}^0} \mathcal{F}$.

6.11

Choose a cleavage of $\mathcal{F} \to \mathcal{E}$, that is, for every arrow $f : \xi \to \eta$ in $\mathcal{E}$ choose a base-change functor $f^* : \mathcal{F}_\eta \to \mathcal{F}_\xi$. For $\xi \in \mathcal{E}$, let

$$v_\xi : \varprojlim_{\mathcal{E}^0} \mathcal{F} \to \mathcal{F}_\xi$$

be evaluation at ξ. For every $f : \xi \to \eta$ there is, up to canonical isomorphism, a commutative diagram

$$\begin{tikzcd} \varprojlim_{\mathcal{E}^0} \mathcal{F} \arrow[r, "v_\xi"] \arrow[dr, "v_\eta"'] & \mathcal{F}_\xi \arrow[d, "f^*"] \\ & \mathcal{F}_\eta . \end{tikzcd}$$

Thus $\varprojlim_{\mathcal{E}^0} \mathcal{F}$ is the projective limit, in the sense of pseudofunctors, of $\xi \mapsto \mathcal{F}_\xi$. Even when this is an actual functor, this pseudofunctorial projective limit is not generally equivalent to the ordinary projective limit in the sense of I 2.

7. Fibred topoi and sites

7.1. Fibred topoi

Definition 7.1.1. Let $\mathcal{U}$ be a universe. A $\mathcal{U}$-topos fibred over a category I is a fibred category

$$p : F \to I$$

whose fibres are $\mathcal{U}$-topoi, and whose inverse-image functors relative to morphisms $f : i \to j$ of I are inverse-image functors $f^* : F_j \to F_i$ of morphisms of topoi $F_i \to F_j$.

7.1.2

Equivalently, a $\mathcal{U}$-topos fibred over I is a fibred category $p : F \to I$ whose fibres are $\mathcal{U}$-topoi and whose inverse-image functors are exact and commute with inductive limits.

7.1.3

Choose, for every arrow $f : i \to j$ of I, an inverse-image functor $f^* : F_j \to F_i$. For every composable pair $i \xrightarrow{f} j \xrightarrow{g} k$ there is a canonical isomorphism $c_{f,g} : f^*g^* \to (gf)^*$ satisfying the usual cocycle condition. Choose also for every $f : i \to j$ a right adjoint $f_* : F_i \to F_j$ to f^*. By adjunction one obtains canonical isomorphisms

$$c'_{g,f} : g_*f_* \to (gf)_* \tag{7.1.3.1}$$

satisfying the analogous cocycle condition. Thus one constructs a fibred category over I^0,

$$p' : F' \to I^0, \tag{7.1.3.2}$$

well-defined up to unique I^0-isomorphism. Its fibre over i is identified with F_i, and its inverse-image functors are the direct-image functors of the original morphisms of topoi.

7.1.4

Let $p : F \to I$ be a $\mathcal{U}$-topos fibred over I. Choosing, for each $f : i \to j$, a morphism of topoi $F_i \to F_j$ whose inverse image is the fibred inverse-image functor, and choosing the compatible cocycles, is called a *bisplitting* of the fibred topos. Such a choice amounts to a pseudofunctor $I \to (\mathcal{U}\text{-topoi})$. Conversely every such pseudofunctor gives, essentially uniquely, a $\mathcal{U}$-topos fibred over I with bisplitting. In practice a fibred topos will often be denoted $(F_i)_{i\in I}$, with the bisplitting implicit; the constructions below do not depend on this auxiliary choice.

Definition 7.1.5. Let $p : F \to I$ and $q : G \to I$ be $\mathcal{U}$-topoi fibred over I. A morphism $m : F \to G$ of fibred topoi consists of an I-functor $m^* : G \to F$ and, for every object i of I, a right adjoint $m_{i*} : F_i \to G_i$ to the induced functor $m_i^* : G_i \to F_i$. Each adjoint pair (m_i^*, m_{i*}) is required to be a morphism of topoi $F_i \to G_i$.

7.1.6

Let $m : F \to G$ be a morphism of $\mathcal{U}$-topoi fibred over I. The construction 7.1.3 associates fibred categories $F' \to I^0$, $G' \to I^0$. Choosing bisplittings, the I-functor m^* gives transition morphisms

$$b_f : f^* m_j^* \to m_i^* f^* \tag{7.1.6.1}$$

compatible with cocycles:

$$m_i^*(c_{f,g})\, b_f^*\, f^*(b_g) = b_{gf} c_{f,g}. \tag{7.1.6.2}$$

Passing to adjoints gives morphisms

$$b'_f : f_* m_{i*} \to m_{j*} f_* \tag{7.1.6.3}$$

with analogous relations. Hence one constructs an I^0-functor

$$m_* : F' \to G' \tag{7.1.6.4}$$

whose restrictions to fibres are the m_{i*}. The resulting functor is independent, up to canonical isomorphism, of the auxiliary choices.

7.1.7

Let $\mathrm{Hom}_{\mathrm{top},I}(F,G)$ be the set of morphisms of fibred topoi from F to G. It is the set of objects of a category $\mathcal{H}om_{\mathrm{top},I}(F,G)$: a morphism $m \to n$ is an I-morphism of functors $n^* \to m^*$, equivalently, by adjunction, an I^0-morphism $m_* \to n_*$. Thus one has two functors

$$\mathcal{H}om_{\mathrm{top},I}(F,G) \to \mathcal{H}om_I(G,F)^0, \tag{7.1.7.1}$$
$$\mathcal{H}om_{\mathrm{top},I}(F,G) \to \mathcal{H}om_{I^0}(F',G'). \tag{7.1.7.2}$$

They send a morphism $(m^*, (m_{i*}))$ respectively to its inverse-image functor $m^* : G \to F$, and to its direct-image functor $m_* : F' \to G'$. These functors are fully faithful. Their essential images are, respectively, the I-functors $G \to F$ which fibrewise are exact and commute with inductive limits, and the I^0-functors $F' \to G'$ which fibrewise have an exact left adjoint. This justifies the common abuse of specifying a morphism of fibred topoi by either its inverse or its direct image; the morphism is then defined only up to unique isomorphism.

A morphism is called *cartesian* if m_*, equivalently m^*, is a cartesian functor. The corresponding full subcategory $\mathcal{H}om_{\mathrm{top},\mathrm{Cart}\,/I}(F,G)$ embeds fully faithfully as

$$\mathcal{H}om_{\mathrm{top},\mathrm{Cart}\,/I}(F,G) \to \mathcal{H}om_{\mathrm{cart}/I}(G,F)^0, \tag{7.1.7.3}$$
$$\mathcal{H}om_{\mathrm{top},\mathrm{Cart}\,/I}(F,G) \to \mathcal{H}om_{\mathrm{cart}/I^0}(F',G'). \tag{7.1.7.4}$$

7.1.8

Morphisms of $\mathcal{U}$-topoi fibred over I compose as ordinary morphisms of topoi do. For a universe $\mathcal{V}$ with $\mathcal{U} \in \mathcal{V}$, one obtains a category $(\mathcal{V}\text{-}\mathcal{U}\text{-}\,\mathrm{Top}_{/I})$ whose objects are $\mathcal{U}$-topoi fibred over I with fibre categories in $\mathcal{V}$, and whose morphisms are morphisms of $\mathcal{U}$-topoi fibred over I. The usual functoriality of composition and the considerations of IV 3.3 and IV 3.4 extend without change. When I is the punctual category, this reduces to the usual theory of morphisms of $\mathcal{U}$-topoi.

7.1.9

Let $p : F \to I$ be a $\mathcal{U}$-topos fibred over I, and let $\phi : J \to I$ be a functor. The base-changed fibred category $F_J \to J$ is again a $\mathcal{U}$-topos fibred over J. The construction $F \mapsto F'$ of 7.1.3 is compatible with this base change. Base change is functorial, indeed 2-functorial, and for fibred topoi F, G over I one has commutative diagrams

$$\begin{tikzcd} \mathcal{H}om_{\mathrm{top},I}(F,G) \arrow[r] \arrow[d] & \mathcal{H}om_I(G,F)^0 \arrow[d] \\ \mathcal{H}om_{\mathrm{top},J}(F_J,G_J) \arrow[r] & \mathcal{H}om_J(G_J,F_J)^0, \end{tikzcd} \tag{7.1.9.1}$$

and

$$\begin{tikzcd} \mathcal{H}om_{\mathrm{top},I}(F,G) \arrow[r] \arrow[d] & \mathcal{H}om_{I^0}(F',G') \arrow[d] \\ \mathcal{H}om_{\mathrm{top},J}(F_J,G_J) \arrow[r] & \mathcal{H}om_{J^0}(F'_{J^0},G'_{J^0}), \end{tikzcd} \tag{7.1.9.2}$$

where the vertical functors are base change and the horizontal functors are respectively inverse image and direct image.

7.2. Fibred sites

7.2.1

A *$\mathcal{U}$-fibred site C* over a category I is a fibred category $p : C \to I$ whose fibres are endowed with topologies making them $\mathcal{U}$-sites, such that for every arrow $f : i \to j$ of I, the inverse-image functor $f^* : C_j \to C_i$ is a morphism of sites from C_i to C_j.

7.2.2

Let $p : C \to I$, $q : D \to I$ be $\mathcal{U}$-fibred sites. A *morphism of $\mathcal{U}$-fibred sites from C to D* is an I-functor $m : D \to C$ such that, for every object i of I, the induced functor $m_i : D_i \to C_i$ is a morphism of sites from C_i to D_i. Write $\mathrm{MorSite}_I(C,D)$ for the set of such morphisms. A *continuous functor from D to C* is an I-functor inducing continuous functors on fibres. One obtains fully faithful inclusions

$$\mathcal{M}\mathrm{orSite}_I(C,D) \hookrightarrow \mathcal{C}\mathrm{ont}_I(D,C)^0 \hookrightarrow \mathcal{H}om_I(D,C)^0. \tag{7.2.2.1}$$

Cartesian morphisms are defined as in 7.1.7, giving a diagram of full subcategories

$$\begin{tikzcd} \mathcal{M}\mathrm{orSite}_{\mathrm{Cart}\,/I}(C,D) \arrow[r, hook] \arrow[d, hook] & \mathcal{H}om_{\mathrm{Cart}\,/I}(D,C)^0 \arrow[d, hook] \\ \mathcal{M}\mathrm{orSite}_I(C,D) \arrow[r, hook] & \mathcal{H}om_I(D,C)^0. \end{tikzcd} \tag{7.2.2.2}$$

7.2.3

Morphisms of $\mathcal{U}$-fibred sites compose. Thus one defines the category $(\mathcal{V}\text{-}\mathcal{U}\text{-Site}/I)$, whose objects are $\mathcal{U}$-sites with fibres belonging to a universe $\mathcal{V}$ containing $\mathcal{U}$. This category has a natural 2-categorical structure, and the composition of morphisms is functorial with respect to morphisms between morphisms. For every $\phi : J \to I$ and every fibred site $C \to I$, the base change $C_J \to J$ is canonically a $\mathcal{U}$-fibred site; base change is 2-functorial.

7.2.4

Let $p : C \to I$ be a fibred category whose fibres are $\mathcal{U}$-sites, and assume finite projective limits are representable in the fibres. Then p is a $\mathcal{U}$-fibred site if the inverse-image functors are left exact and carry covering families to covering families; this condition is also necessary when the fibre topologies are no finer than the canonical topology. All fibred sites used in the Seminar will be of this type. Similarly, for fibred sites $p : C \to I$, $q : D \to I$ with finite projective limits in the fibres, a cartesian functor $m : D \to C$ whose fibre functors are left exact and carry covering families to covering families is a morphism of fibred sites from C to D; conversely under the same canonical-topology hypothesis. All morphisms of fibred sites used here will be of this type.

7.2.5

A $\mathcal{U}$-topos fibred over I is a $\mathcal{U}$-fibred site when its fibres are endowed with their canonical topologies, and this convention will be used below. If $m : F \to G$ is a morphism of fibred topoi, then its inverse-image functor $m^* : G \to F$ is a morphism of fibred sites from F to G. Hence there is a functor

$$(\mathcal{V}\text{-}\mathcal{U}\text{-}\,\mathrm{Top}_{/I}) \to (\mathcal{V}\text{-}\mathcal{U}\text{-}\mathrm{Site}_{/I}),$$

not in general faithful. It naturally extends to a 2-functor which, for two fibred topoi F, G, induces an equivalence

$$\mathcal{H}om_{\mathrm{top},I}(F, G) \to \mathcal{M}\mathrm{orSite}_I(F, G). \tag{7.2.5.1}$$

Composed with the inclusion into $\mathcal{H}om_I(G, F)^0$, this is precisely the inverse-image functor 7.1.7.1.

7.2.6

Let $p : C \to I$ be a $\mathcal{U}$-fibred site. Choose inverse-image functors $f^* : C_j \to C_i$. Passing to sheaf categories, f^* extends to a functor $\mathrm{Top}(f)^* : C_j^\sim \to C_i^\sim$, where $\mathrm{Top}(f) : C_i^\sim \to C_j^\sim$ is a morphism of topoi, giving a commutative square up to isomorphism

$$\begin{tikzcd} C_j \arrow[r, "f^*"] \arrow[d, "\varepsilon_j"'] & C_i \arrow[d, "\varepsilon_i"] \\ C_j^\sim \arrow[r, "\mathrm{Top}(f)^*"] & C_i^\sim . \end{tikzcd} \tag{7.2.6.1}$$

Together with the canonical cocycles, these data define a fibred category

$$p^\sim : C^{\sim/I} \to I, \tag{7.2.6.6}$$

whose fibres are the topoi $C_i^\sim$, and a cartesian functor

$$\varepsilon_{C/I} : C \to C^{\sim/I}. \tag{7.2.6.7}$$

The fibred topos $C^{\sim/I}$ is independent, up to canonical isomorphism, of all choices and is called the *$\mathcal{U}$-fibred topos associated* with the $\mathcal{U}$-fibred site C.

7.2.6.8 The formation of the associated $\mathcal{U}$-fibred topos commutes with base change of the base category.

Proposition 7.2.7. Let E be a $\mathcal{U}$-topos fibred over I, and C a $\mathcal{U}$-fibred site over I. The functor which sends a morphism of fibred topoi $m : E \to C^{\sim/I}$ to the composite $m^*\varepsilon_{C/I}$ induces an equivalence of categories, preserving cartesian objects,

$$\mathcal{H}om_{\mathrm{top},I}(E, C^{\sim/I}) \xrightarrow{\sim} \mathcal{M}\mathrm{orSite}_I(E, C).$$

If finite projective limits are representable in the fibres of C, the corresponding fully faithful functor

$$\mathcal{H}om_{\mathrm{top},I}(E, C^{\sim/I}) \to \mathcal{H}om_I(C, E)^0$$

has as essential image the I-functors $g : C \to E$ which are fibrewise left exact and carry covering families in fibres to epimorphic families in the corresponding fibres.

This is exactly IV 4.9.4 in the fibred setting; its proof generalizes immediately.

7.3. Examples

Example 7.3.1: the fibred topos of morphisms of a topos. Let E be a $\mathcal{U}$-topos, let $\mathrm{Fl}(E)$ be the category of morphisms of E, and let $p : \mathrm{Fl}(E) \to E$ send a morphism to its target. Then $(\mathrm{Fl}(E), p)$ is a $\mathcal{U}$-topos fibred over E. The fibre over X is $E_{/X}$, and the inverse image along $f : X \to Y$ is fibre product.

For every functor $\phi : I \to E$ one obtains by base change a $\mathcal{U}$-topos fibred over I, denoted

$$p_I : \mathrm{Fl}(E)_I \to I, \tag{7.3.1.1}$$

whose fibre over i is $E_{/\phi(i)}$. If $m : \phi_1 \to \phi_2$ is a morphism of functors $I \to E$, then there is a unique cartesian morphism of fibred topoi

$$\mathrm{loc}(m) : \mathrm{Fl}(E)_1 \to \mathrm{Fl}(E)_2 \tag{7.3.1.4}$$

which on fibres is the localization morphism

$$\mathrm{loc}(m(i)) : E_{/\phi_1(i)} \to E_{/\phi_2(i)}. \tag{7.3.1.3}$$

For composable morphisms m, n of such functors there are canonical isomorphisms

$$c(m, n) : \mathrm{loc}(n)\,\mathrm{loc}(m) \simeq \mathrm{loc}(nm), \tag{7.3.1.5}$$

satisfying the usual cocycle condition.

Example 7.3.2: the fibred site of morphisms of a site. Let C be a $\mathcal{U}$-site in which fibre products are representable. Let $\mathrm{Fl}(C)$ be the category of morphisms of C, and let $p : \mathrm{Fl}(C) \to C$ send a morphism to its target. With the induced topologies on the fibres, this is a $\mathcal{U}$-fibred site over C. If $\varepsilon_C : C \to C^{\sim}$ is the canonical functor, its natural extension to arrow categories gives a cartesian functor

$$\mathrm{Fl}(C) \to \mathrm{Fl}(C^{\sim})_C$$

over C, hence a morphism of fibred topoi

$$\mathrm{can} : \mathrm{Fl}(C^{\sim})_C \to \mathrm{Fl}(C)^{\sim/C}. \tag{7.3.2.3}$$

This morphism is a C-equivalence of $\mathcal{U}$-fibred topoi. After base change by any $\phi : I \to C$, one obtains an I-equivalence

$$\mathrm{can}_I : \mathrm{Fl}(C^\sim)_I \to \mathrm{Fl}(C)_I^{\sim/I}. \qquad (7.3.2.5)$$

If $m : \phi_1 \to \phi_2$ is a morphism of functors $I \to C$, then the localization morphisms of fibred sites $\mathrm{loc}(m) : \mathrm{Fl}(C)_1 \to \mathrm{Fl}(C)_2$ are compatible, up to isomorphism, with their counterparts after passing to $C^\sim$.

Exercise 7.3.3: small and large fibred sites. This exercise continues IV 4.10.6. Let $\mathcal{M}$ be the full subcategory of $\mathrm{Fl}(S)$ defined by the morphisms in $\mathcal{M}$, and let $p : \mathcal{M} \to S$ send a morphism to its target.

6°) Show that $(\mathcal{M}, p)$ is a fibred site over S, and, when fibre products are representable in S, the inclusion $\mathcal{M} \to \mathrm{Fl}(S)$ is a cartesian morphism from the fibred site $\mathrm{Fl}(S) \to S$ to $\mathcal{M} \to S$.

7°) Let $\mathcal{M}^{\sim/S} \to S$ be the associated fibred topos. Show that there is a canonical morphism $g : \mathrm{Fl}(S^\sim)_{/S} \to \mathcal{M}^{\sim/S}$ of fibred topoi over S, inducing on each fibre over $X \in S$ the morphism $(\mathrm{Prol}_X, \mathrm{Res}_X)$. Although each $g_X : S^\sim_{/X} \to S(X)^\sim$ has a section, show that g need not have a section in general.

7.4. The total topology of a fibred topos or site

Definition 7.4.1. Let $p : C \to I$ be a $\mathcal{U}$-fibred site, and for $i \in \mathrm{ob}\, I$ let $\alpha_{i!} : C_i \to C$ be the inclusion of the fibre. The *total topology* on C is the least topology making all the functors $\alpha_{i!}$ continuous. The category C with this topology is the *total site.*

Proposition 7.4.2. Let $p : C \to I$ be a $\mathcal{U}$-fibred site, let T be the total topology on C, and let T_i be the topology of the fibre C_i.

1) If X lies over i, a family $(X_\beta \to X)$ is covering for T if and only if it is refined by a covering family $(Y_\gamma \to X)$ for T_i inside C_i.

2) The topology induced by T on C_i is T_i.

3) For every i, the functor $\alpha_{i!} : C_i \to C$ is cocontinuous.

Property 3 follows from 1 and III 2.1. To prove 1, let $J(X)$ be the set of sieves described in 1. These $J(X)$ satisfy the axioms of a topology T'. It is the least topology making the covering families of the fibres covering in C, hence is no finer than T. It remains to prove that each $\alpha_{i!}$ is continuous for T', equivalently that the topology induced on C_i is T_i. This reduces to the following lemma.

Lemma 7.4.2.2. With the preceding notation, let $X \in C_i$ and let $u : R \to X$ be a morphism of presheaves on C_i. The following are equivalent:

(i) $R \to X$ is bi-covering for T_i.

(ii) $\alpha_{i!}(R) \to \alpha_{i!}(X)$ is bi-covering for T'.

For (i) $\Rightarrow$ (ii), covering is clear. If two maps $m, n : Y \rightrightarrows \alpha_{i!}R$ over an object Y lying over j become equal after applying $\alpha_{i!}u$, then using the formula

$$\mathrm{Hom}_{\widehat{C}}(Y, \alpha_{i!}R) \simeq \coprod_{w \in \mathrm{Hom}(j,i)} \mathrm{Hom}_{\widehat{C_j}}(Y, w^*R),$$

they correspond to maps into f^*R equal after f^*u. Since inverse-image functors are morphisms of sites, f^*u is bi-covering, so the two maps become equal after a covering of Y. Conversely, if $\alpha_{i!}u$ is bi-covering, cocontinuity of $\alpha_{i!}$ shows $\alpha_i^*\alpha_{i!}u$ bi-covering; the cartesian square

$$\begin{tikzcd} R \arrow[r] \arrow[d, "u"'] & \alpha_i^*\alpha_{i!}R \arrow[d, "\alpha_i^*\alpha_{i!}u"] \\ X \arrow[r] & \alpha_i^*\alpha_{i!}X \end{tikzcd}$$

then implies that u is bi-covering.

Remark 7.4.3.

1) Proposition 7.4.2 can be stated and proved in a slightly more general, but apparently useless, setting: the fibres have topologies and the inverse-image functors are continuous, without requiring them to be morphisms of sites.

2) The total topology of a fibred site is also the finest topology making the functors $\alpha_{i!} : C_i \to C$ cocontinuous.

3) If I is equivalent to a $\mathcal{U}$-small category, then the total site C is a $\mathcal{U}$-site: a small topological generating family is obtained by collecting such families in the fibres. The associated topos of sheaves on the total site is denoted $\mathrm{Top}(C)$ and called the *total topos*.

4) Since $\alpha_{i!} : C_i \to C$ is cocontinuous, it gives a morphism of topoi $\alpha_i : C_i^\sim \to \mathrm{Top}(C)$. Since $\alpha_{i!}$ is also continuous, the inverse-image functor $\alpha_i^* : \mathrm{Top}(C) \to C_i^\sim$ admits a left adjoint, also denoted $\alpha_{i!} : C_i^\sim \to \mathrm{Top}(C)$, extending the inclusion $C_i \to C$.

Proposition 7.4.4. Let $p : C \to I$ be a $\mathcal{U}$-fibred site over a small category I.

1) A presheaf F on C is a sheaf for the total topology if and only if, for every $i \in I$, its restriction $F|C_i = F \circ \alpha_{i!}$ is a sheaf on C_i.

2) A functor $m : C \to D$ to a site D is continuous from the total site C to D if and only if every restriction $m|C_i : C_i \to D$ is continuous.

If F is a sheaf on the total site, its restriction to every fibre is a sheaf because $\alpha_{i!}$ is continuous. Conversely, if all restrictions are sheaves, then by the description of covering sieves in 7.4.2 the sheaf condition on each fibre implies the sheaf condition on the total site. The assertion on continuous functors follows immediately from the sheaf criterion and the definition of continuity.

7.4.5

Let $p : C \to I$ be a $\mathcal{U}$-fibred site over a category equivalent to a small category, and let $f : i \to j$ be an arrow of I. Let $f : C_i^\sim \to C_j^\sim$ also denote the morphism of topoi determined by the fibred-site structure. With the notation of 7.4.3, one has a diagram of topoi

$$\begin{tikzcd} C_i^\sim \arrow[r, "\alpha_i"] \arrow[d, "f"'] & \mathrm{Top}(C) \\ C_j^\sim \arrow[ur, "\alpha_j"'] & \end{tikzcd} \tag{7.4.5.1}$$

which is not commutative in general. We define a canonical morphism of morphisms of topoi

$$\rho_f : \alpha_i \to \alpha_j \circ f. \tag{7.4.5.2}$$

At the inverse-image level this amounts to a natural morphism

$$\beta_f^* : f^*\alpha_j^* \to \alpha_i^*, \tag{7.4.5.3}$$

or, by adjunction, to

$$\gamma_f : \alpha_j^* \to f_*\alpha_i^*. \tag{7.4.5.4}$$

For a sheaf F on the total site and $Y \in C_j$, one has $\alpha_j^*F(Y) = F(\alpha_{j!}Y)$, while $f_*\alpha_i^*F(Y) = F(\alpha_{i!}f^*Y)$. The canonical cartesian morphism

$$m_Y : \alpha_{i!}f^*Y \to \alpha_{j!}Y \tag{7.4.5.5}$$

therefore yields

$$\gamma_f(F)(Y) = F(m_Y) : \alpha_j^*F(Y) \to f_*\alpha_i^*F(Y), \tag{7.4.5.6}$$

and hence the required morphism of functors. For composable $f : i \to j$, $g : j \to k$, with $c'_{g,f} : g_*f_* \to (gf)_*$, the compatibility relation is

$$c'_{g,f} \circ g_*(\gamma_f) \circ \gamma_g = \gamma_{gf}. \tag{7.4.5.8}$$

7.4.6

Let $C^{\sim/I} \to I$ be the fibred topos associated with C, and let $(C^{\sim/I})' \to I^0$ be the fibred category of direct images. The preceding construction sends a sheaf F on $\mathrm{Top}(C)$ to the family $F_i = \alpha_i^*F \in C_i^\sim$, together with morphisms

$$\gamma_f(F) : F_j \to f_*F_i,$$

satisfying the compatibility 7.4.5.8. Thus one obtains a functor

$$\Theta_C : \mathrm{Top}(C) \to \mathcal{H}om_{I^0}(I^0, (C^{\sim/I})'). \tag{7.4.6.1}$$

Proposition 7.4.7. If $C \to I$ is a $\mathcal{U}$-fibred site and I is equivalent to a small category, then Θ_C is an equivalence of categories.

A quasi-inverse is as follows. Given a section σ of $(C^{\sim/I})'$ over I^0, define a presheaf on C by

$$X \longmapsto \sigma(p(X))(X).$$

For a morphism $m : X \to Y$ over $f : i \to j$, factor m through the cartesian morphism $f^*Y \to Y$, and use the structural morphism $\sigma(j) \to f_*\sigma(i)$ to obtain the restriction map. This presheaf is a sheaf on the total site by 7.4.4, and the construction is functorial. Direct comparison with the definition of Θ_C shows that it is a quasi-inverse.

Remark 7.4.8. It follows in particular that $\mathcal{H}om_{I^0}(I^0, (C^{\sim/I})')$ is a $\mathcal{U}$-topos. This can also be checked directly using the usual axioms for topoi and the existence of a small generating category.

7.4.9

Let $p : C \to I$, $q : D \to I$ be two $\mathcal{U}$-fibred sites over a small category, and let m be a morphism of fibred sites from C to D. Let

$$m^{\sim/I} : C^{\sim/I} \to D^{\sim/I}$$

be the corresponding morphism of associated fibred topoi, and let

$$m_*^{\sim/I} : (C^{\sim/I})' \to (D^{\sim/I})'$$

be its direct-image functor. Passing to sections gives

$$\mathrm{Hom}_{I^0}(I^0, m_*^{\sim/I}) : \mathcal{H}om_{I^0}(I^0, (C^{\sim/I})') \to \mathcal{H}om_{I^0}(I^0, (D^{\sim/I})'). \tag{7.4.9.1}$$

On the other hand m, being continuous between total sites, gives a direct-image functor $m_* : \mathrm{Top}(C) \to \mathrm{Top}(D)$. Hence there is a diagram

$$\begin{tikzcd} \mathrm{Top}(C) \arrow[r, "\Theta_C"] \arrow[d, "m_*"'] & \mathcal{H}om_{I^0}(I^0, (C^{\sim/I})') \arrow[d, "\mathrm{Hom}_{I^0}(I^0, m_*^{\sim/I})"'] \\ \mathrm{Top}(D) \arrow[r, "\Theta_D"] & \mathcal{H}om_{I^0}(I^0, (D^{\sim/I})'). \end{tikzcd} \tag{7.4.9.2}$$

Proposition 7.4.10. The diagram 7.4.9.2 is commutative up to isomorphism. If m is cartesian, then m is a morphism from the total site C to the total site D.

The commutativity is a direct verification. If m is cartesian, it is enough, using 7.4.7, to show that the left adjoint to $\mathrm{Hom}_{I^0}(I^0, m_*^{\sim/I})$ is left exact. Fibrewise, write $m_i^* : D_i^{\sim} \to C_i^{\sim}$ for the inverse image. For each $f : i \to j$, adjunction and cartesianity give a canonical morphism

$$\mathrm{can}_f : m_j^* f_* \to f_* m_i^*. \tag{7.4.10.4}$$

If τ is a section of $(D^{\sim/I})'$, put $\sigma(i) = m_i^*(\tau(i))$, and define the transition morphism of σ by composing

$$m_j^*\tau(j) \xrightarrow{m_j^*\gamma_f(\tau)} m_j^* f_*\tau(i) \xrightarrow{\mathrm{can}_f} f_* m_i^*\tau(i). \tag{7.4.10.6}$$

This defines a section of $(C^{\sim/I})'$. A formal adjoint-functor chase shows this construction is the left adjoint in question. Its left exactness follows because finite projective limits of sections are computed fibrewise and all functors used in the construction commute with finite projective limits.

Corollary 7.4.11. Let $p : C \to I$ be a $\mathcal{U}$-fibred site over a small category, let $C^{\sim/I} \to I$ be its associated fibred topos, and let $\varepsilon_I : C \to C^{\sim/I}$ be the canonical morphism of fibred sites. Then ε_I induces an equivalence

$$\mathrm{Top}(\varepsilon_I) : \mathrm{Top}(C) \to \mathrm{Top}(C^{\sim/I})$$

of the corresponding total topoi.

This follows from 7.4.9.2 and the fact that ε_I is an equivalence on associated fibred topoi. Consequently, questions about the total topos may replace fibred sites by the associated fibred topoi.

Lemma 7.4.12. Let $p : C \to I$ be a fibred site over a small category I, and let i be a final object of I. The functor $\alpha_{i!} : C_i^{\sim} \to \mathrm{Top}(C)$ is exact and fully faithful. The pairs

$$\beta_i = (\alpha_{i!}, \alpha_i^*) : \mathrm{Top}(C) \to C_i^{\sim}, \qquad \alpha_i = (\alpha_i^*, \alpha_{i*}) : C_i^{\sim} \to \mathrm{Top}(C)$$

are morphisms of topoi, and the composite $\beta_i\alpha_i : C_i^{\sim} \to C_i^{\sim}$ is isomorphic to the identity.

For every $j \in I$, let $f_j : j \to i$ be the unique morphism. Using the equivalence Θ_C, the object $\alpha_{i!}(X)$ corresponds to the section $j \mapsto f_j^* X$, while α_i^* evaluates a section at i. Hence $\alpha_i^*\alpha_{i!} \simeq \mathrm{id}$, so $\alpha_{i!}$ is fully faithful. It is left exact because the f_j^* are left exact. The remaining assertions follow.

Theorem 7.4.13.1. Let $p : C \to I$ and $q : D \to I$ be $\mathcal{U}$-fibred sites over a small category I, and let $m : D \to C$ be an I-functor. For m to be a morphism from the fibred site C to the fibred site D, it is enough that m be a morphism from the total site C to the total site D.

Assume $m : D \to C$ is a morphism of total sites. We must show that for every i, $m_i : D_i \to C_i$ is a morphism of sites from C_i to D_i. First m_i is continuous: if $R \hookrightarrow X$ is a covering sieve in C_i, then, by the commutative square with the inclusions $\alpha_{i!}$ and by continuity of m and $\alpha_{i!}$, the morphism $\alpha_{i!}m_{i!}(R) \to \alpha_{i!}m_i(X)$ is bi-covering, and Lemma 7.4.2.2 implies the desired bi-covering condition in the fibre.

It remains to reduce to the case of fibred topoi. Since the m_i are continuous, they extend to sheaf categories as functors $m_i^* : D_i^{\sim} \to C_i^{\sim}$. These assemble into a cartesian functor $m^* : D^{\sim/I} \to C^{\sim/I}$. The square

$$\begin{tikzcd}
D \arrow[r, "m"] \arrow[d, "\varepsilon_I"'] & C \arrow[d, "\varepsilon_I"] \\
D^{\sim/I} \arrow[r, "m^*"] & C^{\sim/I}
\end{tikzcd}$$

commutes up to isomorphism. Since the vertical functors induce equivalences on total topoi, and since m is a morphism of total sites, it is enough to prove the assertion when C and D are fibred topoi.

In that case, each m_i sends the final object of D_i to the final object of C_i. Indeed, under the equivalence Θ_D, a sheaf on the total topos is a section, and m sends a section σ to $i \mapsto m_i\sigma(i)$. Since m is left exact, it sends the final section to the final section.

For arbitrary i, replace the fibred topoi by their localizations over the final object e_i of the fibre. This reduces to the case where i is final in I. In that case, using Lemma 7.4.12, m_i identifies with the composite $\alpha_i^* m^{\sim} \alpha_{i!}$. All factors are left exact, hence m_i is left exact and therefore is a morphism of sites from C_i to D_i.

Exercise 7.4.14. Let $X = (X_i)_{i \in I}$ be a fibred topos over I, and let T be a topos. Show that the categories $\mathcal{H}om_{\mathrm{top}}(X_i, T)$ are the fibres of a fibred category denoted $\mathcal{H}om_{\mathrm{top},I}(X, T \times I)$. Show that its category of sections over I is canonically equivalent to $\mathcal{H}om_{\mathrm{top},I}(\mathrm{Top}(X), T)$, where $\mathrm{Top}(X)$ is the total topos of X.

Exercise 7.4.15. Let $X = (X_i)_{i\in I}$ be a fibred topos over I. Define a cartesian morphism m from X to the constant fibred topos with fibre (**Set**). Show that the total topos of this constant fibred topos is canonically equivalent to $\widehat{I}$, and deduce a morphism of topoi

$$\mathrm{Top}(m) : \mathrm{Top}(X) \to \widehat{I}.$$

For an object $F = (F_i)_{i\in I}$ of $\mathrm{Top}(X)$, show that

$$\mathrm{Top}(m)(F) = (i \mapsto \Gamma(X_i, F_i)).$$

Deduce from 7.4.12 that if F is an injective abelian object, then each F_i is injective abelian, and hence

$$R^q \,\mathrm{Top}(m)(F) = (i \mapsto H^q(X_i, F_i)).$$

Show that the Cartan-Leray spectral sequence of m is

$$H^{p+q}(\mathrm{Top}(X), F) \Leftarrow \varprojlim_{I}{}^{(p)} H^q(X_i, F|X_i).$$

8. Projective limits of fibred topoi

8.1. Generalities

Definition 8.1.1. Let $\mathcal{U}$ be a universe and let $p : F \to I$ be a $\mathcal{U}$-topos fibred over I. A pair (C, m), consisting of a $\mathcal{U}$-topos C and a Cartesian morphism

$$m : C \times I \longrightarrow F$$

of $\mathcal{U}$-topoi fibred over I, is called a *projective limit* of the fibred topos F if, for every $\mathcal{U}$-topos D, the functor

$$\mathcal{H}om_{\mathrm{top}}(D, C) \to \mathcal{H}om_{\mathrm{top},\mathrm{Cart}\,/I}(D \times I, F) \tag{8.1.1.1}$$

obtained by base change and then by composition with m is an equivalence of categories.

8.1.2

With the notation of 8.1.1, let D be a $\mathcal{U}$-topos and let $m_1 : D \times I \to F$ be a morphism of $\mathcal{U}$-topoi fibred over I. Translating the definition, one sees that there is a morphism of topoi $n : D \to C$, together with an isomorphism of fibred morphisms

$$\gamma : m_1 \xrightarrow{\sim} m \circ (n \times \mathrm{id}_I).$$

If (n', γ') is another such pair, there is a unique isomorphism $\eta : n \to n'$ compatible with γ and γ'. Hence, if (D, m_1) is itself a projective limit of F, then n is an equivalence of $\mathcal{U}$-topoi; the additional commutation isomorphism determines the pair up to canonical isomorphism. Existence will be proved in 8.2.3 when I is cofiltered.

8.1.3

We write

$$\varprojlim_I F, \qquad \varprojlim_I F_i, \qquad \overleftarrow{F}$$

for a projective limit of the fibred topos F, and write

$$\mu : (\varprojlim_I F) \times I \to F$$

for the canonical morphism. For each object $i \in I$, the induced morphism on the fibre is denoted

$$\mu_i : \overleftarrow{F} \to F_i.$$

Choose a cleavage and an opcleavage for F, so that to every arrow $f : i \to j$ in I there corresponds a morphism of topoi $f. = (f^*, f_*) : F_i \to F_j$, with coherence isomorphisms $c_{f,g} : g.f. \simeq (gf).$. Since μ is Cartesian, for every $f : i \to j$ there is an isomorphism

$$b_f : f.\mu_i \xrightarrow{\sim} \mu_j, \tag{8.1.3.1}$$

and the family (b_f) satisfies the usual cocycle compatibilities.

Equivalently, if D is a $\mathcal{U}$-topos, if $m : D \times I \to F$ is a fibred morphism, if $m_i : D \to F_i$ are the fibrewise morphisms, and if

$$b'_f : f.m_i \xrightarrow{\sim} m_j$$

are the transition isomorphisms, then there is a morphism $n : D \to \overleftarrow{F}$ and isomorphisms

$$\gamma_i : m_i \xrightarrow{\sim} \mu_i n$$

compatible with the transitions. Conversely, data $\overleftarrow{F}$, μ_i, and b_f, satisfying this universal property, determine a unique fibred morphism $\mu : \overleftarrow{F} \times I \to F$, and $(\overleftarrow{F}, \mu)$ is the projective limit.

If $\mathcal{V}$ is a universe containing $\mathcal{U}$, this is precisely the 2-categorical projective limit of the pseudo-functor defined by the chosen transition morphisms $f.$. This should not be confused with a strict projective limit. Even when the pseudo-functor is an honest functor, the 2-categorical limit and the naive strict limit need not agree.

8.1.4–8.1.7

Let $p : F \to I$ and $q : G \to I$ be two $\mathcal{U}$-topoi fibred over I, and let $m : F \to G$ be a morphism of fibred topoi. If F and G have projective limits, the universal property gives a morphism

$$\overleftarrow{m} : \overleftarrow{F} \to \overleftarrow{G}, \tag{8.1.4.1}$$

together with the evident commutation isomorphism between $m \circ \mu_F$ and $\mu_G \circ (\overleftarrow{m} \times \mathrm{id}_I)$, unique up to a unique isomorphism.

For a base change $\phi : I' \to I$, write $F_\phi \to I'$ for the induced fibred topos. If $\overleftarrow{F}$ and $\overleftarrow{F_\phi}$ exist, the universal property gives a canonical morphism

$$m_\phi : \overleftarrow{F} \to \overleftarrow{F_\phi},$$

again unique up to a unique isomorphism.

Now let $p : F \to I$ be a $\mathcal{U}$-site fibred over a small category I, and let $F^{\sim/I} \to I$ be the associated fibred topos. Let D be a $\mathcal{U}$-topos. There are natural functors

$$\mathcal{H}om_{\mathrm{top,Cart}\,/I}(D \times I, F^{\sim/I}) \to \mathrm{MorSite}_{\mathrm{Cart}\,/I}(D \times I, F), \tag{8.1.6.1}$$

$$\mathrm{MorSite}_{\mathrm{Cart}\,/I}(D \times I, F) \to \mathcal{H}om_{\mathrm{Cart}\,/I}(F, D \times I)^0, \tag{8.1.6.2}$$

and, by the universal property of the inductive limit of the category fibred over I^0,

$$\mathcal{H}om_{\mathrm{Cart}\,/I}(F, D \times I)^0 \to \mathcal{H}om(\varinjlim_{I^0} F, D)^0. \tag{8.1.6.3}$$

Let $\Pi : F \to \varinjlim_{I^0} F$ be the canonical functor and let Θ denote the composite of these functors.

Proposition 8.1.7. The functor Θ is fully faithful. Its essential image consists of those functors

$$n : \varinjlim_{I^0} F \to D$$

for which $(n \circ \Pi, p) : F \to D \times I$ is a morphism from the total site $D \times I$ to the total site F.

This follows from 7.2.7, 7.2.2, and the universal property of 6.3, together with the criterion 7.4.13.1 for morphisms of total sites.

8.2. Construction of the projective limit when the index category is cofiltered

Lemma 8.2.1. Let I and I' be cofiltered categories, and let $\phi : I' \to I$ be a functor such that $\phi^0 : I'^0 \to I^0$ is cofinal. Let $p : F \to I$ be a $\mathcal{U}$-topos fibred over I, let D be a $\mathcal{U}$-topos, and let $m : D \times I \to F$ be a Cartesian morphism of fibred topoi. If m' is the base change of m to I', then (D, m) is a projective limit of F if and only if (D, m') is a projective limit of F'.

The proof is the expected, somewhat lengthy, verification, or equivalently the same statement for projective limits of pseudo-functors.

Lemma 8.2.2. Let $p : F \to I$ be a $\mathcal{U}$-site fibred over a small cofiltered category, let $\overrightarrow{F} = \varinjlim_{I^0} F$, and let $\pi : F \to \overrightarrow{F}$ be the canonical functor. Put on $\overrightarrow{F}$ the least topology for which π is continuous. For a $\mathcal{U}$-site D and a functor $n : \overrightarrow{F} \to D$, the following conditions are equivalent.

i) n is a morphism of sites $D \to \overrightarrow{F}$.

ii) $n \circ \pi$ is a morphism of sites, where F carries the total topology.

iii) $(n \circ \pi, p) : F \to D \times I$ is a morphism from the total site $D \times I$ to the total site F.

iv) For every object $i \in I$, the composite $F_i \xrightarrow{\alpha_{i!}} F \xrightarrow{n \circ \pi} D$ is a morphism of sites.

The equivalence of (i) and (ii) is 6.6. The equivalence of (iii) and (iv) is 7.4.13. To see that (iii) implies (ii), it is enough to note that the first projection $D \times I \to D$ is a morphism of sites; under the description $(D \times I)^{\sim} \simeq \mathcal{H}om(I^0, D^{\sim})$, its prolongation to sheaves is the functor taking a diagram $\sigma : I^0 \to D^{\sim}$ to $\varinjlim_{I^0} \sigma(i)$, and filtered colimits are exact. The implication (ii) implies (iv) reduces, by localization and then by passage to associated topoi, to the fact that the canonical section $i \mapsto \alpha_{i!}(e_i)$ becomes final after applying $n \circ \pi$.

Theorem 8.2.3. Let $p : F \to I$ be a $\mathcal{U}$-site fibred over an essentially small cofiltered category.

1) Let $\overrightarrow{F}$ be the site whose underlying category is $\varinjlim_{I^0} F$, equipped with the least topology for which $\pi : F \to \overrightarrow{F}$ is continuous. Then $\overrightarrow{F}$ is a $\mathcal{U}$-site and

$$(\pi, p) : F \to \overrightarrow{F} \times I$$

is a morphism from the fibred site $\overrightarrow{F} \times I$ to the fibred site F.

2) The morphism of fibred topoi

$$\mu : \overrightarrow{F}^{\sim} \times I \to F^{\sim/I}$$

deduced from (π, p) makes $(\overrightarrow{F}^{\sim}, \mu)$ a projective limit of $F^{\sim/I}$.

Thus one may write

$$\varprojlim_{I} F^{\sim} = \overrightarrow{F}^{\sim}. \tag{8.2.4.1}$$

The topology on $\overrightarrow{F}$ is also the least topology making all $F_j \to F \to \overrightarrow{F}$ continuous. The site $\overrightarrow{F}$ is called the *projective-limit site* of the fibred site F.

The proof reduces first to the case where the indexing category is small, by replacing I by a full subcategory whose opposite is cofinal. The first assertion is Lemma 8.2.2 applied to $D = \overrightarrow{F}$ and $n = \mathrm{id}$. For the universal property, one compares the diagram of functors

$$\mathcal{H}om_{\mathrm{top}}(D, \overrightarrow{F}^{\sim}), \quad \mathrm{MorSite}(D, \overrightarrow{F}), \quad \mathcal{H}om(\overrightarrow{F}, D)^0,$$

with their Cartesian analogues over I. The comparison functors are either the standard equivalences for sites and associated topoi, base-change functors, composition with μ or (π, p), or the equivalence supplied by the universal property of $\varinjlim_{I^0} F$. It remains only to compare the essential images of two fully faithful functors into $\mathcal{H}om_{\mathrm{Cart}\,/I}(F, D \times I)^0$; this is exactly Lemma 8.2.2.

8.2.8–8.2.12. Sheaves on the projective limit

Let $p : F \to I$ be a $\mathcal{U}$-topos fibred over I, and suppose that $\varprojlim_I F_i$ exists. Passing from the Cartesian morphism $\mu : \varprojlim_I F_i \times I \to F$ to direct-image functors gives a Cartesian functor

$$\mu_* : \varprojlim_{I} F_i \times I^0 \to F'.$$

Hence there is a canonical functor

$$\Theta : \varprojlim_{I} F_i \to \varprojlim_{I, f_*} F_i = \mathcal{H}om_{\mathrm{Cart}\,/I^0}(I^0, F'). \tag{8.2.8.1}$$

Theorem 8.2.9. Assume that I is essentially small and cofiltered. Then Θ is an equivalence of categories. If I is small and $\varprojlim_I F_i$ is identified with $\overrightarrow{F}^{\sim}$, the composite

$$\overrightarrow{F}^{\sim} \xrightarrow{\Theta} \mathcal{H}om_{\mathrm{Cart}\,/I^0}(I^0, F') \hookrightarrow \mathcal{H}om_{I^0}(I^0, F') \simeq F^{\sim}$$

is precisely the direct image by the morphism of sites $\pi : F \to \overrightarrow{F}$.

The content is often summarized by the formula

$$\varprojlim_{I, f_*} F_i^{\sim} \simeq \varprojlim_{I, f.} F_i^{\sim} \simeq (\varinjlim_{I^0, f^*} F_i)^{\sim}. \tag{8.2.11.1}$$

The theorem says that a sheaf on the projective-limit topos is known by the system of its direct images in the topoi F_i, that is, by a Cartesian section over I^0 of F'. Conversely, a compatible system $(X_i, X_j \simeq f_* X_i)$ comes from an essentially unique sheaf on the limit.

For the proof, one reduces to I small. The functor $\pi : F \to \varinjlim_{I^0} F_i$ is a morphism of sites. Its direct image is fully faithful, with essential image the sheaves on $\overrightarrow{F}$ sending Cartesian morphisms to isomorphisms. Under the equivalence $F^\sim \simeq \mathcal{H}om_{I^0}(I^0, F')$, these sheaves are precisely the Cartesian sections. Moreover, for $X \in \overrightarrow{F}^\sim$, the corresponding section is $j \mapsto \alpha_j^* \pi_* X$, and this is the same as $j \mapsto \mu_{j*} X$. This identifies π_* with the composite displayed above.

8.3. Topology of the projective-limit site: coherent topoi

Throughout 8.3, let $p : F \to I$ be a $\mathcal{U}$-site fibred over an essentially small cofiltered category, and assume that the inverse-image functors commute with fibre products. For every object X of a fibre F_i, choose a set $\mathrm{Cov}(X)$ of covering families satisfying (PT0) and (PT1), generating the topology of F_i, and stable under inverse image along the transition functors.

For an object Y of $\overrightarrow{F}$, define $\mathrm{Cov}(Y)$ to consist of all families $(Y_\alpha \to Y)_\alpha$ obtained, up to isomorphism in $\overrightarrow{F}$, from a covering family $(X_\alpha \to X)_\alpha \in \mathrm{Cov}(X)$ in some fibre F_i.

Proposition 8.3.3. The families belonging to $\mathrm{Cov}(Y)$, for $Y \in \overrightarrow{F}$, generate the topology of the projective-limit site. The assignment $Y \mapsto \mathrm{Cov}(Y)$ satisfies (PT0) and (PT1).

Lemma 8.3.4. Let $i \in I$, and let $n : X' \to X$ be a pullbackable morphism of F_i such that every pullback $f^*(n)$ along $f : j \to i$ is pullbackable. Then $\pi(n)$ is pullbackable in $\overrightarrow{F}$. In particular the canonical functors

$$\pi_j : F_j \hookrightarrow F \xrightarrow{\pi} \overrightarrow{F}$$

commute with fibre products.

Indeed, every morphism to $\pi(X)$ in $\overrightarrow{F}$ may be represented, after inverting a Cartesian morphism, by a morphism in a suitable fibre. Replacing it by the appropriate inverse image of n, the required fibre product is reduced to the fibrewise fibre product. The compatibility of the functors π_j with fibre products then follows from the formula for morphisms in the inductive limit and from the fact that filtered colimits commute with finite fibre products in sets.

The proof of Proposition 8.3.3 is then formal. The preceding lemma gives (PT0). Stability under base change is reduced to the corresponding statement in one fibre, using the fact that π_i commutes with fibre products. Finally, a sheaf for the topology generated by the displayed families is exactly a presheaf M on $\overrightarrow{F}$ for which every $M \circ \pi_i$ is a sheaf on F_i. This is the same as being a sheaf for the least topology making $\pi : F \to \overrightarrow{F}$ continuous.

Proposition 8.3.6. If, for every fibre, $X \mapsto \mathrm{Cov}(X)$ is a pretopology and all covering families in it are finite, then the families $\mathrm{Cov}(Y)$ are finite and form a pretopology on $\overrightarrow{F}$.

Only (PT2), stability under composition, has to be checked. Given a covering $(Y_\alpha \to Y)$ represented in one fibre and coverings of the Y_α, one first replaces the isomorphisms in $\overrightarrow{F}$ by genuine morphisms after pulling back along suitable arrows in I. Since the index category is cofiltered and the covering family is finite, all data may be brought to a single fibre. A further base change reduces to the case where the structural arrow of I is the identity. The final point is the following lemma.

Lemma 8.3.11. Let $j \in I$ and let $m : X' \to X$ be a morphism in F_j such that $\pi(m)$ is an isomorphism. Then there is a morphism $\ell : j' \to j$ such that $\ell^*(m)$ is an isomorphism.

A quasi-inverse to $\pi(m)$ can be represented by a span $X \xleftarrow{s_1} Y_1 \xrightarrow{q_1} X'$ with s_1 Cartesian. After another Cartesian pullback, the equality $\pi(mq_1) = \pi(s_1)$ becomes an equality in a fibre. Factoring through Cartesian morphisms gives a one-sided inverse for a pullback of m; applying the same argument to this inverse gives a two-sided inverse after a further pullback. Since the covering families are finite, one may choose a common pullback for all α, and the verification of (PT2) reduces to the (PT2) axiom in a single fibre.

Theorem 8.3.13. Let $p : F \to I$ be a $\mathcal{U}$-topos fibred over an essentially small cofiltered category. Assume that every fibre F_i is coherent and that, for every arrow $f : i \to j$, the morphism $f. : F_i \to F_j$ is coherent. Then $\varprojlim_I F_i$ is coherent and every canonical morphism

$$\mu_i : \varprojlim_I F_i \to F_i$$

is coherent.

Moreover, if F_{coh} denotes the full subcategory of F consisting of objects coherent in their fibres, then F_{coh} is fibred over I, and $\overrightarrow{F}_{\mathrm{coh}}$ is canonically equivalent to the category of coherent objects of $\varprojlim_I F_i$.

For each i, the site $(F_i)_{\mathrm{coh}}$, with the induced topology, defines F_i. Coherence of the transition morphisms implies that inverse images carry coherent objects to coherent objects, and hence defines a fibred site $F_{\mathrm{coh}} \to I$. By 8.2.3,

$$\varprojlim_I F_i \simeq (\overrightarrow{F}_{\mathrm{coh}})^{\sim}. \tag{8.3.13.1}$$

Finite limits are representable in the coherent sites, and finite covering families form a pretopology. Proposition 8.3.6 shows that the limit site has finite covering families and quasi-compact objects; by 2.4.5 the resulting topos is coherent. The morphisms μ_i are induced by morphisms of coherent sites $(F_i)_{\mathrm{coh}} \to \overrightarrow{F}_{\mathrm{coh}}$, hence are coherent.

For the final assertion, let X be coherent in $(\overrightarrow{F}_{\mathrm{coh}})^{\sim}$. Quasi-compactness gives, after moving to a sufficiently small index, a finite epimorphism $\mu_i(Y') \to X$ with $Y' \in (F_i)_{\mathrm{coh}}$. The corresponding equivalence relation is again coherent. After one more base change it is represented by an actual equivalence relation in $(F_i)_{\mathrm{coh}}$; its quotient is coherent in F_i, and its image is X. Thus every coherent object of the limit comes from $\overrightarrow{F}_{\mathrm{coh}}$.

Corollary 8.3.14. Under the hypotheses of 8.3.13, let $F \to I$ and $G \to I$ be coherent fibred topoi with coherent transition morphisms, and let $m : F \to G$ be a Cartesian morphism such that every fibre morphism $m_i : F_i \to G_i$ is coherent. Then the induced morphism $\overleftarrow{m} : \overleftarrow{F} \to \overleftarrow{G}$ is coherent.

This follows by applying 8.3.13 to the fibred sites of coherent objects and using 3.3.

Exercise 8.3.15. In the notation of 8.3.13, show that if the F_i are algebraic and the transition morphisms are coherent, then $\varprojlim_I F_i$ is algebraic and $(\varprojlim_I F_i)_{\mathrm{coh}} \simeq \overrightarrow{F}_{\mathrm{coh}}$.

8.4. Example: local topoi

Let $X = (X_i)_{i \in I}$ be a topos fibred over I. Extending it to pro-objects, one obtains a fibred topos over $\mathrm{Pro}(I)$ by

$$X_\alpha = \varprojlim_{i \in I_\alpha} X_i \tag{8.4.1.1}$$

for every pro-object $\alpha = (i_\alpha)$ of I.

Let E be a topos and let $\mathrm{Fl}(E) \to E$ be the fibred topos of 7.3.1. Extending it to $\mathrm{Pro}(E)$, and restricting along the fully faithful functor $\mathrm{Point}(E) \to \mathrm{Pro}(E)$, gives a fibred topos

$$\mathrm{Loc}(E) \to \mathrm{Point}(E).$$

The fibre at a point p is called the *localization of E at p* and is denoted $\mathrm{Loc}_p(E)$. By definition,

$$\mathrm{Loc}_p(E) = \varprojlim_{X \in \mathrm{Vois}(p)} E_{/X}, \tag{8.4.2.1}$$

where $\mathrm{Vois}(p)$ is the category of neighbourhoods of p.

If $m : p' \to p$ is a morphism of points of E, the universal property gives a point $\theta_p(m)$ of $\mathrm{Loc}_p(E)$, hence a functor

$$\theta_p : \mathrm{Point}(E)_{/p} \to \mathrm{Point}(\mathrm{Loc}_p(E)). \tag{8.4.3.1}$$

This functor is an equivalence of categories. Therefore $\mathrm{Point}(\mathrm{Loc}_p(E))$ is canonically equivalent to the category of generizations of p. There is also a canonical equivalence

$$\mathrm{Loc}_{\theta_p(m)}(\mathrm{Loc}_p(E)) \xrightarrow{\sim} \mathrm{Loc}_{p'}(E),$$

compatible with the canonical morphisms to $\mathrm{Loc}_p(E)$.

These localized topoi are mainly useful for topoi arising in algebraic geometry, or at least for topoi with finiteness properties. For a separated topological space, the localizations are punctual topoi, and the situation is essentially trivial. If $E = \mathrm{Top}(X)$ is the Zariski topos of a scheme X, if $x \in X$, and if $Y = \mathrm{Spec}(\mathcal{O}_{X,x})$, then

$$\mathrm{Loc}_x(E) = \mathrm{Top}(Y).$$

Étale examples occur later in Exposé VIII.

If E is locally coherent, the topoi $\mathrm{Loc}_p(E)$ are coherent; it is enough in 8.4.2 to restrict to algebraic coherent neighbourhoods and then apply 8.3.13. If $m : p' \to p$ is a morphism of points, the induced morphism $\mathrm{Loc}_{p'}(E) \to \mathrm{Loc}_p(E)$ is coherent.

A topos X is called *local* if the global-sections functor $\Gamma(X, -)$ is a fibre functor. The corresponding point is called the *centre* of the local topos. When E is locally coherent, the localized topoi $\mathrm{Loc}_p(E)$ are local; the centre of $\mathrm{Loc}_p(E)$ is canonically $\theta_p(\mathrm{id}_p)$, and its image in E is p. The topos $\mathrm{Loc}_p(E)$, with its canonical morphism to E, solves the 2-universal problem of mapping local topoi into E with the centre lying over p.

Several natural questions remain. If X is local, the final object has a maximal open $U \neq e$; is the closed complement, which is again local and has no nontrivial open, necessarily punctual? If X is obtained by gluing a topos U with the punctual topos through a functor $\Phi : U \to \mathbf{Set}$, what conditions on Φ make X local?

8.5. Sheaves on a filtered projective limit of topoi

Let $F = (F_i)_{i \in I}$ be a $\mathcal{U}$-topos fibred over a small cofiltered category, and choose a cleavage and an opcleavage. Let

$$\overleftarrow{F} = \varprojlim_I F_i, \qquad \mu_i : \overleftarrow{F} \to F_i. \tag{8.5.1.1}$$

Let $\mathrm{Top}(F)$ be the total topos. The canonical functor $F \to \overrightarrow{F}$ defines a morphism

$$Q : \overleftarrow{F} \to \mathrm{Top}(F). \tag{8.5.1.2}$$

Under the identifications

$$\overleftarrow{F} \simeq \mathcal{H}om_{\mathrm{Cart}\,/I^0}(I^0, F'), \qquad \mathrm{Top}(F) \simeq \mathcal{H}om_{I^0}(I^0, F'),$$

the morphism Q is the inclusion of Cartesian sections into all sections. Thus, for $M \in \overleftarrow{F}$,

$$Q_*M = (i \mapsto \mu_{i*}M). \tag{8.5.1.3}$$

For each i there is a canonical morphism $\alpha_i : F_i \to \mathrm{Top}(F)$, and although the triangle $\overleftarrow{F} \to F_i \to \mathrm{Top}(F)$ need not commute, one always has the canonical isomorphism

$$\alpha_i^* Q_* \simeq \mu_{i*}. \tag{8.5.1.6}$$

Proposition 8.5.2. For an object $(j \mapsto M_j)$ of $\mathrm{Top}(F)$, there is a functorial isomorphism

$$Q^*(j \mapsto M_j) \simeq \varinjlim_{I^0} \mu_j^* M_j. \tag{8.5.2.1}$$

An object of $\mathrm{Top}(F)$ consists of objects $M_j \in F_j$ and transition maps

$$\beta_f : M_j \to f_* M_i \tag{8.5.2.2}$$

or, by adjunction, $\beta'_f : f^*M_j \to M_i$, satisfying the usual cocycle condition. Combining β'_f with the transition isomorphisms $b_f : f.\mu_i \simeq \mu_j$ gives morphisms

$$t_f : \mu_j^* M_j \to \mu_i^* M_i, \tag{8.5.2.6}$$

with $t_f t_g = t_{gf}$. Hence the displayed filtered colimit is defined. For $N \in \overleftarrow{F}$, adjunction gives

$$\mathrm{Hom}(\varinjlim_{I^0} \mu_i^* M_i, N) \simeq \varprojlim_{I^0} \mathrm{Hom}_{F_i}(M_i, \mu_{i*}N),$$

where the right hand side is precisely the set of morphisms from the section $(i \mapsto M_i)$ to Q_*N. This proves the proposition.

Proposition 8.5.3. Assume that for every arrow $f : i \to j$ in I, the functor $f_* : F_i \to F_j$ commutes with small filtered colimits. Then, for every section $(i \mapsto M_i)$ of F',

$$Q_*Q^*(i \mapsto M_i) \simeq (i \mapsto \varinjlim_{f:j \to i} f_* M_j). \tag{8.5.3.1}$$

The transition maps in the right hand side are induced by the maps β_f and the coherence isomorphisms $c_{f,g}$. Under the stated commutation hypothesis the resulting section is Cartesian, and the natural map from $(i \mapsto M_i)$ to it is universal among maps to Cartesian sections.

Corollary 8.5.4. Under the hypotheses of 8.5.3, the functor Q_* commutes with small filtered colimits.

This follows by writing an object N as Q^*Q_*N, using the exactness of Q^*, computing colimits in $\mathrm{Top}(F)$ fibre by fibre, and then applying 8.5.3.

Corollary 8.5.5. Under the same hypotheses, for every $i \in I$ and every $M_i \in F_i$,

$$\mu_{i*}\mu_i^* M_i \simeq \varinjlim_{f:j \to i} f_* f^* M_i. \tag{8.5.5.1}$$

After replacing I by $I_{/i}$, one may suppose that i is final. Then μ_i factors through Q and the retraction of $\mathrm{Top}(F)$ to the final fibre, and the formula is an instance of 8.5.3.

Corollary 8.5.6. The functors $\mu_{i*} : \overleftarrow{F} \to F_i$ commute with filtered colimits.

Indeed μ_{i*} is the composite of Q_* with restriction to the fibre i.

Corollary 8.5.7. Assume 8.5.3. Let $i \in I$ and let $X \in F_i$ be such that $\mathrm{Hom}(X, -)$ commutes with filtered colimits on F_i. Then $\mathrm{Hom}(\mu_i^* X, -)$ commutes with filtered colimits on $\overleftarrow{F}$, and for every $M_i \in F_i$,

$$\mathrm{Hom}(\mu_i^* X, \mu_i^* M_i) \simeq \varinjlim_{f : j \to i} \mathrm{Hom}(f^* X, f^* M_i). \tag{8.5.7.1}$$

For every section $(j \mapsto M_j)$ of $\mathrm{Top}(F)$,

$$\mathrm{Hom}(\mu_i^* X, Q^*(j \mapsto M_j)) \simeq \varinjlim_{f : j \to i} \mathrm{Hom}(f^* X, M_j). \tag{8.5.7.2}$$

The functor $\mathrm{Hom}(\mu_i^* X, -)$ is $\mathrm{Hom}(X, \mu_{i*} -)$; the preceding corollaries and the hypothesis on X prove the colimit statement and the formulas.

Proposition 8.5.8. Let $G \to I$ be another $\mathcal{U}$-topos fibred over I, and let $m : F \to G$ be a Cartesian morphism of fibred topoi. The square

$$\begin{tikzcd} \overleftarrow{F} \arrow[r, "Q"] \arrow[d, "\overleftarrow{m}"'] & \mathrm{Top}(F) \arrow[d, "m^{\sim}"] \\ \overleftarrow{G} \arrow[r, "Q"] & \mathrm{Top}(G) \end{tikzcd} \tag{8.5.8.1}$$

commutes up to canonical isomorphism. For every $N \in \overleftarrow{F}$,

$$\overleftarrow{m}_* N \simeq \varinjlim_{I^0} \mu_j^* m_{j*} \mu_{j*} N. \tag{8.5.8.2}$$

The commutativity is immediate from the definitions. Since $N \simeq Q^* Q_* N$, the formula follows from 8.5.2 applied to the section $j \mapsto m_{j*} \mu_{j*} N$.

Proposition 8.5.9. Assume, in addition to the hypotheses of 8.5.8, that every m_{i*} and every transition functor f_* on F commute with filtered colimits. Then $\overleftarrow{m}_*$ commutes with filtered colimits and, for every section $(i \mapsto M_i)$ of $\mathrm{Top}(F)$,

$$\overleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{I^0} \mu_j^* m_{j*} M_j. \tag{8.5.9.1}$$

The proof starts from 8.5.8 and 8.5.3:

$$\overleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{i} \mu_i^* m_{i*} \big(\varinjlim_{f : j \to i} f_* M_j \big).$$

Using the commutation of m_{i*} with filtered colimits and the canonical isomorphisms $m_{i*} f_* \simeq f_* m_{j*}$, this becomes a colimit over the category of arrows of I. The functor from I to the arrow category which sends i to id_i is cofinal on opposites, giving the displayed formula.

Corollary 8.5.10. Under the conditions of 8.5.9, for every $i \in I$ and $M_i \in F_i$,

$$\overleftarrow{m}_* \mu_i^* M_i \simeq \varinjlim_{f:j\to i} \mu_j^* m_{j*} f^* M_i. \tag{8.5.10.1}$$

Remark 8.5.11. Recall that the direct image by a coherent morphism between coherent topoi commutes with filtered colimits. Thus 8.5.3–8.5.10 apply in particular to coherent fibred topoi and coherent Cartesian morphisms between them.

8.6. Ringed fibred $\mathcal{U}$-topoi

A $\mathcal{U}$-topos fibred over I is *ringed* if it is equipped with a sheaf of rings on its total site. After choosing a cleavage and an opcleavage, this is the same as giving rings A_i in the fibres F_i, together with morphisms $A_j \to f_* A_i$ for every arrow $f : i \to j$, subject to the usual compatibility conditions. Thus the transition morphisms $f_. : F_i \to F_j$ become morphisms of ringed topoi, and the data amount to a pseudo-functor from I to the 2-category of ringed topoi. Conversely, such a pseudo-functor reconstructs the ringed fibred topos.

The notion of projective limit extends immediately to ringed $\mathcal{U}$-topoi. If the underlying fibred topos has a projective limit $\varprojlim_I F_i$, and if the inductive system of rings $i \mapsto \mu_i^* A_i$ has a colimit in the category of rings of the limit topos, then

$$(\varprojlim_I F_i,\ \varinjlim_{I^0} \mu_i^* A_i)$$

is the corresponding projective limit in ringed topoi.

The formulas of 8.5 remain valid for abelian sheaves, and for modules provided inverse images are interpreted as inverse images of modules. A ringed fibred topos $(F, A) \to I$ is said to be flat on the right, respectively on the left, if for every $f : i \to j$ the morphism $f_. : (F_i, A_i) \to (F_j, A_j)$ is flat on the right, respectively on the left. If I is essentially small, then

$$Q : (\varprojlim_I F_i, \varinjlim_{I^0} \mu_i^* A_i) \to (\mathrm{Top}(F), A)$$

is flat on the right, respectively on the left, without any flatness assumption on the transition morphisms. If the ringed fibred topos is flat on the right, respectively left, then each μ_i is flat on the same side. Similarly, a Cartesian morphism of ringed fibred topoi whose fibre morphisms are flat on one side induces a projective-limit morphism flat on that side.

8.7. Cohomology of sheaves on a projective limit of topoi

In 8.7 let I be a small cofiltered category, let $(F, A) \to I$ and $(G, B) \to I$ be ringed $\mathcal{U}$-topoi fibred over I, and let

$$m : (F, A) \to (G, B)$$

be a morphism of ringed fibred topoi. Assume that, for every arrow $f : i \to j$, the derived functors $R^n f_*$ on modules commute with filtered colimits, and that for every object i, the derived functors $R^n m_{i*}$ commute with filtered colimits. This is the case, for example, when the fibres and the relevant morphisms are coherent. Write

$$\overleftarrow{A} = \varinjlim \mu_i^* A_i, \qquad \overleftarrow{B} = \varinjlim \mu_i^* B_i.$$

For modules, write μ_i^{-1} for inverse image as abelian sheaves, and reserve μ_i^* for inverse image of modules.

Lemma 8.7.2. If $(j \mapsto M_j)$ is an injective A-module of $\mathrm{Top}(F)$, then $Q^*(j \mapsto M_j)$ is acyclic for $\overleftarrow{m}_*$, and every M_i is flasque.

The flasqueness follows by restricting the flasque module $(j \mapsto M_j)$ to the localized fibred topos over $I_{/i}$, and then using the retraction from the total topos to the fibre. For acyclicity, put $N' = Q^*(j \mapsto M_j)$. Formula 8.5.3 gives

$$\mu_{i*}N' \simeq \varinjlim_{f:j\to i} f_*M_j.$$

Using the hypotheses, N' has the property that each $\mu_{i*}N'$ is acyclic for m_{i*} and for all transition functors. This property is stable under the standard dévissage criterion for acyclicity: in an exact sequence $0 \to N' \to N \to N'' \to 0$, the cokernel of $Q_*N' \to Q_*N$ is again a Cartesian section and hence is Q_* of an object of the projective limit; then 8.5.8, exactness of μ_j^{-1}, and exactness of filtered colimits give the required epimorphism after applying $\overleftarrow{m}_*$.

Theorem 8.7.3. Under the hypotheses of 8.7.1, for every $n \in \mathbb{N}$, the functor $R^n\overleftarrow{m}_*$ commutes with filtered colimits. Moreover, for every A-module section $(j \mapsto M_j)$,

$$R^n\overleftarrow{m}_*\, Q^*(j \mapsto M_j) \simeq \varinjlim_{I^0} \mu_j^* R^n m_{j*} M_j. \tag{8.7.3.1}$$

Since $Q^{-1}A = \overleftarrow{A}$, inverse image for modules agrees here with inverse image for abelian sheaves followed by extension of scalars. Lemma 8.7.2 computes the derived functor on a resolution by injective sections, giving the formula; commutation with filtered colimits follows at once.

Corollary 8.7.4. For every $n \in \mathbb{Z}$ and every module N on $\overleftarrow{F}$,

$$R^n\overleftarrow{m}_* N \simeq \varinjlim_{I^0} \mu_j^* R^n m_{j*}\mu_{j*}N. \tag{8.7.4.1}$$

Indeed $N \simeq Q^*(j \mapsto \mu_{j*}N)$.

Corollary 8.7.5. For every object $i \in I$, every integer n, and every A_i-module M_i,

$$R^n\overleftarrow{m}_*\mu_i^* M_i \simeq \varinjlim_{f:j\to i} \mu_j^* R^n m_{j*} f^* M_i. \tag{8.7.5.1}$$

After base change along $I_{/i} \to I$, one may suppose that i is final; then $\mu_i^* M_i$ is the inverse image under Q of the section $(f : j \to i) \mapsto f^*M_i$.

Corollary 8.7.6. For every $i \in I$ and every integer n,

$$R^n\mu_{i*}Q^*(j \mapsto M_j) \simeq \varinjlim_{f:j\to i} R^n f_* M_j, \tag{8.7.6.1}$$

$$R^n\mu_{i*}N \simeq \varinjlim_{f:j\to i} R^n f_*\mu_{j*}N, \tag{8.7.6.2}$$

and

$$R^n\mu_{i*}\mu_i^* M_i \simeq \varinjlim_{f:j\to i} R^n f_* f^* M_i. \tag{8.7.6.3}$$

The functors $R^n\mu_{i*}$ commute with filtered colimits.

This is obtained by applying 8.7.3–8.7.5 to the constant fibred topos with fibre F_i, after the base change $I_{/i} \to I$.

Corollary 8.7.7. Let $i \in I$ and let $X \in F_i$ be such that all functors $H^n(X, -)$ commute with filtered colimits. Then, for every n, there is a canonical isomorphism, functorial in the A_i-module M_i,

$$H^n(\mu_i^* X, \mu_i^* M_i) \simeq \varinjlim_{f:j\to i} H^n(f^* X, f^* M_i), \tag{8.7.7.1}$$

and the functors $H^n(\mu_i^* X, -)$ commute with filtered colimits. In particular, if the functors $H^n(F_i, -)$ commute with filtered colimits, then the functors $H^n(\overleftarrow{F}, -)$ commute with filtered colimits, and

$$H^n(\overleftarrow{F}, \mu_i^* M_i) \simeq \varinjlim_{f:j\to i} H^n(F_j, f^* M_i). \tag{8.7.7.2}$$

The proof compares the two Cartan–Leray spectral sequences

$$E_2'^{p,q} = \varinjlim_{f:j\to i} H^p(X, R^q f_* f^* M_i) \Rightarrow \varinjlim_{f:j\to i} H^{p+q}(f^* X, f^* M_i)$$

and

$$E_2''^{p,q} = H^p(X, R^q\mu_{i*}\mu_i^* M_i) \Rightarrow H^{p+q}(\mu_i^* X, \mu_i^* M_i).$$

Formula 8.7.6 identifies the E_2-terms, so the abutments agree. The filtered-colimit assertion follows from the analogous assertions for $H^n(X, -)$ and for $R^n\mu_{i*}$.

Remark 8.7.8. When the F_j are coherent and the transition morphisms are coherent, the hypotheses made on X and F_i in 8.7.7 are satisfied for coherent X and coherent F_i, by 5.2. In the same situation $\mu_i^* X$, respectively $\overleftarrow{F}$, is coherent by 8.3.13.

Corollary 8.7.9. Let $i \in I$, let M_i be a left A_i-module, and let L_i be a right A_i-module admitting a resolution

$$P_{i,k} \to P_{i,k-1} \to \cdots \to P_{i,0} \to L_i \to 0$$

where each $P_{i,k}$ is a finite direct sum of modules of the form $A_{i|X}$, with X satisfying the hypotheses of 8.7.7. If, in addition to the hypotheses of 8.7.1, the fibred topos F is flat on the right, then, for $n \leq k-1$, there are canonical isomorphisms

$$\operatorname{Ext}^n_{\overleftarrow{A}}(\overleftarrow{F}, \mu_i^* L_i, \mu_i^* M_i) \simeq \varinjlim_{f:j\to i} \operatorname{Ext}^n_{A_j}(F_j, f^* L_i, f^* M_i). \tag{8.7.9.1}$$

If $P_{i,\bullet}$ is the chosen resolution of L_i, flatness gives resolutions $f^* P_{i,\bullet}$ of $f^* L_i$ and $\mu_i^* P_{i,\bullet}$ of $\mu_i^* L_i$. The two spectral sequences obtained from these resolutions converge to the two sides of 8.7.9.1; on the E_1-page the comparison is the isomorphism 8.7.7.1, hence it is an isomorphism on abutments.

9. Appendix. Criterion for the existence of points

by P. Deligne

Proposition 9.0. Every locally coherent topos S has enough points.

Since the assertion is local on S, one may suppose that S is defined by a site $\mathscr{S}$ in which finite projective limits are representable and in which every covering family admits a finite subcover. It is enough to show that if a morphism of sheaves $u : \mathcal{F} \to \mathcal{G}$ is not a monomorphism, then there is a point x of $\mathscr{S}$ such that u_x is not injective. Thus there are an object $U \in \mathscr{S}$ and two sections $s, s' \in \mathcal{F}(U)$, with $s \neq s'$, and with $u(s) = u(s')$. Replacing $\mathscr{S}$ by $\mathscr{S}/U$, the assertion reduces to the following lemma.

Lemma 9.1. Let s and s' be two distinct global sections of a sheaf $\mathcal{F}$ on a site $\mathscr{S}$ satisfying the preceding hypotheses. Then there is a point x of $\mathscr{S}$ such that $s_x \neq s'_x$.

Let $P = (U_i)_{i \in I}$ be a projective system in $\mathscr{S}$, indexed by a filtered ordered set. For every sheaf $\mathcal{H}$ on $\mathscr{S}$, set

$$P(\mathcal{H}) = \varinjlim_i \mathcal{H}(U_i),$$

and for every object V of $\mathscr{S}$, set

$$P(V) = \varinjlim_i \operatorname{Hom}(U_i, V).$$

These functors commute with fibre products. If $P(V)$ sends coverings to surjective families of sets, then it defines a morphism of sites from the punctual topos to $\mathscr{S}$, and $P(\mathcal{H})$ is the associated fibre functor.

If $P = (U_i)_{i \in I}$ and $Q = (V_j)_{j \in J}$ are two such projective systems, we say that Q *refines* P if I is an ordered subset of J and P is the restriction of Q to I. There are then natural maps $P(\mathcal{H}) \to Q(\mathcal{H})$ and $P(V) \to Q(V)$.

For such a P, write s_P and s'_P for the images of s and s' in $P(\mathcal{F})$. Lemma 9.1 follows from the next lemma, applied to the projective system indexed by $\{0\}$ and reduced to the final object of $\mathscr{S}$.

Lemma 9.2. For every P with $s_P \neq s'_P$, there is a refinement Q of P, with $s_Q \neq s'_Q$, such that, for every covering $(V_i \to V)$ and every element $f \in Q(V)$, the element f lies in the image of one of the maps $Q(V_i) \to Q(V)$.

We first prove a finite form.

Lemma 9.3. Let $P = (U_i)_{i \in I}$ satisfy the hypotheses of Lemma 9.2. Let $(V_k \to V)_k$ be a finite covering in $\mathscr{S}$, and let $f \in P(V)$. There is a refinement Q of P, with $s_Q \neq s'_Q$, such that the image of f in $Q(V)$ lies in the image of one of the $Q(V_k)$.

Choose $i_0 \in I$ and a morphism $f' : U_{i_0} \to V$ representing f. For $i \geq i_0$, put

$$U_{i,k} = U_i \times_V V_k.$$

Since (V_k) covers V, the family $(U_{i,k} \to U_i)_k$ covers U_i. Therefore

$$\mathcal{F}(U_i) \to \prod_k \mathcal{F}(U_{i,k})$$

is injective, and after passing to the filtered colimit, finite products commuting with filtered colimits, one gets an injection

$$P(\mathcal{F}) \to \prod_k \varinjlim_i \mathcal{F}(U_{i,k}).$$

Hence for some k, the images of s_P and s'_P are distinct in $\varinjlim_i \mathcal{F}(U_{i,k})$. Let $I_0 = \{i \in I \mid i \geq i_0\}$, and form an ordered set $J = I \amalg I_0$ by placing the new copy of I_0 beneath the old terms in the evident way. Index by J the objects U_i and $U_{i,k}$. The resulting system Q refines P, still separates s and s', and the image of f in $Q(V)$ comes from $Q(V_k)$.

Lemma 9.4. Let P satisfy the hypotheses of Lemma 9.2. There is a refinement Q of P, with $s_Q \neq s'_Q$, such that, for every finite covering $(V_k \to V)$ and every $f \in P(V)$, the image of f in $Q(V)$ lies in the image of one of the $Q(V_k)$.

Let E be the set of triples consisting of an object V, a finite covering (V_k) of V, and an element $f \in P(V)$. Choose a well-ordering on E, and adjoin a largest element ∞. By transfinite induction construct refinements Q_e of P, with $s_{Q_e} \neq s'_{Q_e}$, such that Q_e refines $Q_{e'}$ for $e' \leq e$, and such that the condition corresponding to $e = (V, (V_k), f)$ is satisfied at stage e. At successor stages apply Lemma 9.3; at limit stages take the union of the preceding systems. Because filtered colimits are used in the definition of $Q(\mathcal{F})$, the inequality of the two sections is preserved at limit stages. Then Q_∞ proves the lemma.

Lemma 9.4 is now iterated. Starting with $Q_0 = P$, construct refinements Q_{n+1} of Q_n such that $s_{Q_n} \neq s'_{Q_n}$ and such that every covering condition for elements of $Q_n(V)$ is solved in Q_{n+1}. Let Q be the projective system obtained by taking the union of all the index sets. Then Q satisfies Lemma 9.2. The associated functor is therefore a point of $\mathscr{S}$, and by construction it separates s and s'. This proves Lemma 9.1 and hence Proposition 9.0.

The construction also gives the following cardinal refinement.

Corollary 9.5. Let $\mathscr{S}$ be a site in which fibre products are representable and in which every covering admits a finite subcover. Let c be the cardinal $\sup(\aleph_0, \operatorname{card}(\mathrm{Fl}\,\mathscr{S}))$. Then there is a very dense set X of points of $\mathscr{S}$ such that

i) $\operatorname{card}(X) \leq 2^c$;

ii) if $x \in X$ and $U \in \mathrm{Ob}\,\mathscr{S}$, then $\operatorname{card}(U_x) \leq c$.

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SGA 4, Exposé VII

Étale Site and Étale Topos of a Scheme

A. Grothendieck

In the present exposé and the following one we develop the most elementary properties relative to the étale topology and étale cohomology. The developments of the present exposé concern certain properties which, in essence, remain valid for other quite different topologies, such as the fppf topology. In the following exposé we shall develop properties rather special to the étale topology, depending on the very particular nature of étale morphisms.

We partially follow here three oral lectures of J. E. Roos, which he was unable to write up, especially in the proof of VIII, 6.3.

Unless expressly stated otherwise, it will be understood that the schemes considered in the present exposé and the following ones are elements of the fixed universe $\mathcal{U}$.

1. The étale topology

1.1

We denote by (Sch) the category of schemes, as elements of the fixed universe $\mathcal{U}$. Recall that a morphism $f : X \to Y$ in (Sch) is called *étale* if it is locally of finite presentation, flat, and if for every $y \in Y$ the fibre X_y is discrete and its local rings are finite *separable* extensions of $k(y)$. Equivalently, f is locally of finite presentation and, for every Y-scheme Y' which is affine in the absolute sense and every closed subscheme Y'_0 defined by a nilpotent ideal, the map

$$\mathrm{Hom}_Y(Y', X) \longrightarrow \mathrm{Hom}_Y(Y'_0, X)$$

is bijective. For the principal properties of this notion we refer to EGA IV, §§17 and 18, and provisionally to SGA 1, I and IV.

1.2

The *étale topology* on (Sch) is the topology generated by the pretopology for which, for each $X \in$ (Sch), $\mathrm{Cov}\, X$ consists of the families

$$(X_i \xrightarrow{u_i} X)_{i \in I}$$

with $I \in \mathcal{U}$, such that the u_i are étale and $X = \bigcup_i u_i(X_i)$ set-theoretically.

It is convenient to associate to every X the full subcategory $\mathrm{Et}_{/X}$ of $(\mathrm{Sch})_{/X}$ consisting of the arrows $X' \to X$ which are étale. We endow it with the topology induced, in the sense of Exposé III, by the étale topology of (Sch); this is also called the étale topology on X. We denote the resulting site by $X_{\text{ét}}$, the étale site of X. Strictly speaking, it would be preferable to reserve the notation $X_{\text{ét}}$ for the topos $\widetilde{X_{\text{ét}}}$ defined by this site; for practical reasons we keep here the notation used in the original seminar.

Every morphism in $\mathrm{Et}_{/X}$ is étale. It follows that a family $(X'_i \xrightarrow{u_i} X')_{i \in I}$ in $X_{\text{ét}}$ is covering if and only if it is surjective, i.e.

$$\bigcup_i u_i(X'_i) = X'.$$

As follows immediately from 3.1, $X_{\text{ét}}$ is a $\mathcal{U}$-site, so that the results of Exposés I–VI apply. The topos $\widetilde{X_{\text{ét}}}$ of $\mathcal{U}$-sheaves on $X_{\text{ét}}$ is the *étale topos* of X.

1.3

In what follows, unless expressly stated otherwise, all topological notions considered in $\mathrm{Et}_{/X}$ or in (Sch) are understood with respect to the étale topology. Moreover, in language and notation we shall often write X instead of $X_{\text{ét}}$ or $\mathrm{Et}_{/X}$. Thus a "sheaf on X" will mean a sheaf on the site $X_{\text{ét}}$. We denote the category of these sheaves by $\widetilde{X_{\text{ét}}}$; it is a $\mathcal{U}$-topos. If F is an abelian sheaf on X, its cohomology groups will be denoted $H^i(X, F)$, which in Exposé V would be written $H^i(X_{\text{ét}}, F)$.

1.4

Let $f : X \to Y$ be a morphism of schemes. The inverse-image functor

$$Y' \longmapsto Y' \times_Y X$$

induces a functor

$$f^*_{\text{ét}} : \mathrm{Et}_{/Y} \longrightarrow \mathrm{Et}_{/X},$$

which commutes with finite projective limits and sends covering families to covering families. Hence it is a morphism of sites

$$f^*_{\text{ét}} : X_{\text{ét}} \longrightarrow Y_{\text{ét}},$$

and therefore gives a functor on categories of sheaves

$$f_*^{\text{ét}} : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{Y_{\text{ét}}}, \qquad f_*^{\text{ét}}(F) = F \circ f^*_{\text{ét}}.$$

Moreover $f_*^{\text{ét}}$ admits a left adjoint

$$f^*_{\text{ét}} : \widetilde{Y_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$

extending the functor just defined on sites, and commuting with arbitrary inductive limits and finite projective limits. In other words $f^*_{\text{ét}}$ defines a morphism of topoi

$$f_{\text{ét}} : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{Y_{\text{ét}}}.$$

For a composite $X \xrightarrow{f} Y \xrightarrow{g} Z$, there are canonical isomorphisms

$$(gf)_*^{\text{ét}} \simeq g_*^{\text{ét}} f_*^{\text{ét}}, \qquad (gf)^*_{\text{ét}} \simeq f^*_{\text{ét}} g^*_{\text{ét}}, \qquad (gf)_{\text{ét}} \simeq g_{\text{ét}} f_{\text{ét}}.$$

Thus one obtains a pseudo-functor from (Sch) to the category of topoi in a universe $\mathcal{U}'$ containing $\mathcal{U}$.

1.5. Notation

We shall often write f^* and f_* instead of $f^*_{\text{ét}}$ and $f_*^{\text{ét}}$. If $f : X \to Y$ is a morphism of schemes, the functors $R^i f_*^{\text{ét}}$ will therefore be denoted simply by $R^i f_*$. With this notation the Leray spectral sequence for f and an abelian sheaf F on X is

$$H^*(X, F) \Longleftarrow E_2^{p,q} = H^p(Y, R^q f_*(F)).$$

Likewise, for two morphisms $f : X \to Y$ and $g : Y \to Z$, one has the Leray spectral sequence for the composite functor,

$$R^*(gf)_*(F) \Longleftarrow E_2^{p,q} = R^p g_*(R^q f_*(F)).$$

1.6

When $f : X \to Y$ is itself étale, X may be regarded as an object of $Y_{\text{ét}}$, and there is a canonical isomorphism of sites

$$X_{\text{ét}} \xrightarrow{\sim} (Y_{\text{ét}})_{/X}.$$

Under this isomorphism, the functor $f^*_{\text{ét}} : Y' \mapsto Y' \times_Y X$ identifies with the internal base-change functor in $\mathrm{Et}_{/Y}$. Consequently

$$f^* : \widetilde{Y_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$

is isomorphic to the restriction functor

$$f^*(F) \simeq F \circ i_{X/Y},$$

where $i_{X/Y} : X_{\text{ét}} \to Y_{\text{ét}}$ is the evident functor which regards an étale X-scheme $u : X' \to X$ as an étale Y-scheme by means of $fu : X' \to Y$.

1.7. Universes

If $X \neq \varnothing$, then $X_{\text{ét}}$ is not an element of the chosen universe $\mathcal{U}$. Nevertheless, as already noted, one can find a full subcategory C of $X_{\text{ét}}$, itself an element of $\mathcal{U}$, satisfying the hypotheses of the comparison lemma of Exposé III, so that restriction induces an equivalence

$$\widetilde{X_{\text{ét}}} \simeq \widetilde{C}.$$

Thus the étale topos is equivalent to a topos defined by a site $C \in \mathcal{U}$. Equivalently, $X_{\text{ét}}$ is a U-site, and the results of Exposés I–VI apply to it.

2. Examples of sheaves

a) Let $F \in \mathrm{Ob}((\mathrm{Sch})_{/X})$ be a scheme over X, and for every X' over X put

$$F(X') = \mathrm{Hom}_X(X', F).$$

The resulting functor $(\mathrm{Sch})^{\circ}_{/X} \to \mathbf{Set}$ is a sheaf for the étale topology, and even for the finer fpqc topology studied in SGA 3, IV. In other words, the étale topology on (Sch) is less fine than the canonical topology; this is essentially the content of SGA 1, VIII, 5.1, and also of SGA 3, IV, 6.3.1. A fortiori its restriction to $X_{\text{ét}}$ is a sheaf; it will again be denoted by F when no confusion can arise.

One must remember that, if $X \neq \varnothing$, the functor $F \mapsto \varphi(F)$, from schemes over X to sheaves on $X_{\text{ét}}$, is neither fully faithful nor even faithful; one cannot reconstruct F, up to isomorphism, from the sheaf $\varphi(F)$. For instance, when $X = \operatorname{Spec} k$ with k algebraically closed, the sheaf remembers only the underlying set of F. Compare Exposé IV.

The functor

$$(\mathrm{Sch})_{/X} \longrightarrow \widetilde{X_{\text{ét}}}$$

so obtained commutes with finite projective limits. Hence, if F is a group scheme, or an object with similar finite-limit structure, the associated sheaf is a sheaf of groups, and so on. Its restriction

$$\mathrm{Et}_{/X} \longrightarrow \widetilde{X_{\text{ét}}}$$

is just the canonical Yoneda functor; it identifies $\mathrm{Et}_{/X}$ with a full subcategory of the étale topos. We generally identify an object of $X_{\text{ét}}$ with the corresponding sheaf, denoted $\widetilde{X'}$ or simply X'.

Evidently

$$H^0(X,F) = \mathrm{Hom}_{(\mathrm{Sch})_{/X}}(X,F).$$

If F is a group prescheme over X, commutative if necessary so that $H^1(X,F)$ is within the framework of Exposé V, the usual arguments show that $H^1(X,F)$ is canonically isomorphic to the set of isomorphism classes of principal homogeneous sheaves of sets under F, or F-torsors, on $X_{\text{ét}}$. If F is affine over X, SGA 1, VIII, 2.1, together with standard arguments, shows that $H^1(X,F)$ is also the set of classes of schemes P over X, endowed with a right action of F, which are principal homogeneous bundles over X for the étale topology, equivalently representable F-torsors.

Remarks 2.1. Let $\varphi_X(Z)$ denote the sheaf on $X_{\text{ét}}$ associated with a scheme Z over X, and let $f : Y \to X$ be a morphism of schemes. There is an evident homomorphism, functorial in Z,

$$f^*\varphi_X(Z) \longrightarrow \varphi_Y(Z_Y), \qquad Z_Y = Z \times_X Y,$$

or, equivalently, an evident homomorphism

$$\varphi_X(Z) \longrightarrow f_*\varphi_Y(Z_Y),$$

given on $X' \in \mathrm{Ob}\, X_{\text{ét}}$ by

$$\mathrm{Hom}_X(X',Z) \longrightarrow \mathrm{Hom}_Y(X'_Y, Z_Y).$$

In general this homomorphism is not an isomorphism: formation of the étale sheaf associated with a relative scheme does not commute with inverse-image functors. It is, however, an isomorphism when Z is étale over X. Likewise $\varphi_X : \mathrm{Sch}_{/X} \to \widetilde{X_{\text{ét}}}$ is not faithful when $X \neq \varnothing$, though its restriction to $X_{\text{ét}}$ is fully faithful, since the topology of $X_{\text{ét}}$ is less fine than its canonical topology.

b) The preceding functor also commutes with direct sums indexed by sets in $\mathcal{U}$. In particular, if I_X denotes the constant X-scheme defined by a set I, the associated sheaf is just the constant sheaf $I_{X_{\text{ét}}}$. Since $I \mapsto I_X$ also commutes with finite projective limits, it carries groups to groups, commutative groups to commutative groups, and so on. If G is an ordinary commutative group, we simply write $H^i(X,G)$ for $H^i(X,G_X)$.

If, for example, G is finite, then G_X is finite and hence affine over X, and the final observation in (a) gives an interpretation of $H^1(X,G)$ as the set of classes of principal Galois coverings with group G. When X is connected and endowed with a geometric point a, the fundamental group $\pi_1(X,a)$ gives the canonical isomorphism, for G commutative,

$$H^1(X,G) \simeq \mathrm{Hom}(\pi_1(X,a),G).$$

One may say that, on passing from Zariski cohomology to the étale topology, one has done what is necessary to obtain the correct H^1 for a finite constant coefficient group. It is a remarkable fact, proved later in this seminar, that this is also enough to obtain the correct $H^i(X,G)$ for every torsion coefficient group, at least when G is prime to the residue characteristics of X.

c) Let F be an $\mathcal{O}_X$-module on X for the Zariski topology. With the notation of SGA 3, I, 4.6, define

$$W(F) : (\mathrm{Sch})^{\circ}_{/X} \longrightarrow \mathbf{Set}$$

by

$$W(F)(X') = \Gamma(X', F \otimes_{\mathcal{O}_X} \mathcal{O}_{X'}).$$

This is again a sheaf for the étale topology, and even for the fpqc topology, by SGA 1, VIII, 1.6 and SGA 3, IV, 6.3.1. Its restriction to $X_{\text{ét}}$, again denoted $W(F)$, satisfies

$$H^0(X, W(F)) = \Gamma(X, F) = H^0(X, F).$$

If F is quasi-coherent, more is true; see 4.3.

3. Generators of the étale topos. Cohomology of an inductive limit of sheaves

Proposition 3.1. Let X be a scheme and let C be a full subcategory of $X_{\text{ét}}$ such that, for every affine étale X-scheme X', C contains an object isomorphic to X'. Then C is a topological generating family of the site $X_{\text{ét}}$, and hence a generating family of the topos $\widetilde{X_{\text{ét}}}$. Moreover restriction

$$\widetilde{X_{\text{ét}}} \longrightarrow \widetilde{C}$$

is an equivalence of categories, where C is endowed with the topology induced from $X_{\text{ét}}$.

This is immediate from the comparison lemma.

Corollary 3.2. Assume that X is quasi-separated. Let the restricted étale site of X be the full subcategory C of $X_{\text{ét}}$ consisting of schemes étale and of finite presentation over X, with the topology induced from $X_{\text{ét}}$. Then:

(i) C is stable under fibre products, and is a site of finite type if X is quasi-compact;

(ii) restriction $\widetilde{X_{\text{ét}}} \to \widetilde{C}$ is an equivalence of categories.

Assertion (i) is trivial. Assertion (ii) follows from 3.1, since quasi-separatedness of X implies that an étale X-scheme X' which is quasi-compact, for example affine, is quasi-compact over X; if it is also locally of finite type over X, it is of finite presentation over X.

Proposition 3.3. Assume that X is quasi-compact and quasi-separated. Then the functors $H^q(X_{\text{ét}}, F)$, on the category of abelian sheaves on $X_{\text{ét}}$, commute with filtered inductive limits.

Indeed, by 3.2(ii) one may replace $X_{\text{ét}}$ by the restricted site C. Since X is quasi-compact, $X \in \mathrm{Ob}\, C$, and the conclusion follows from 3.2(i) and VI, 6.1.2(3).

4. Comparison with other topologies

4.0

First note that the examples of sheaves on $X_{\text{ét}}$ considered in §2 are naturally restrictions of sheaves defined on $(\text{Sch})_{/X}$, endowed with its étale topology, or even with the fpqc topology. In general let

$$u^* : X_{\text{ét}} \longrightarrow (\text{Sch})_{/X}$$

be the inclusion functor. It is continuous and commutes with finite projective limits, and hence defines a morphism of sites

$$u : (\text{Sch})_{/X} \longrightarrow X_{\text{ét}},$$

with the corresponding direct-image functor

$$u_* : \widetilde{(\text{Sch})}_{/X} \longrightarrow \widetilde{X_{\text{ét}}}, \qquad u_*(F) = F \circ u,$$

and a left adjoint

$$u^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{(\text{Sch})}_{/X}.$$

In complete rigor one must take account of universes here, since $(\text{Sch})_{/X}$ is a $\mathcal{U}$-category but not a $\mathcal{U}$-site. One may either work in a larger universe or replace $(\text{Sch})_{/X}$ by a suitable full U-small subcategory stable under finite projective limits and containing a comparison subcategory as in 3.1.

Proposition 4.1.

(i) The functor u^* is fully faithful. Thus, for every sheaf G on $X_{\text{ét}}$, the canonical homomorphism $G \to u_* u^* G$ is an isomorphism.

(ii) If G is an abelian sheaf on $X_{\text{ét}}$ and F is an abelian sheaf on the big étale site $(\text{Sch})_{/X}$, then the canonical comparison maps

$$H^*(X_{\text{ét}}, G) \xrightarrow{\sim} H^*((\text{Sch})_{/X}, u^*G), \qquad H^*(X_{\text{ét}}, u_*F) \xrightarrow{\sim} H^*((\text{Sch})_{/X}, F)$$

are isomorphisms.

Indeed, the functor u is a comorphism, and one applies Exposé III. Thus, for the study of sheaf cohomology, it is essentially equivalent to work with the small étale site $X_{\text{ét}}$ or with the big étale site $(\text{Sch})_{/X}$.

4.2

It is also useful to introduce on (Sch), and hence on each $(\text{Sch})_{/X}$, topologies other than the étale topology. The most useful among these are those defined in SGA 3, IV, 6.3. The coarsest is the *Zariski topology*, whose covering families are the surjective families of open immersions; it is less fine than the étale topology. The finest of these is the faithfully flat quasi-compact topology, abbreviated fpqc, the least fine topology for which Zariski coverings and faithfully flat quasi-compact morphisms are covering; it is finer than the étale topology. As already noted, the examples of sheaves on $(\text{Sch})_{/X}$ considered in §2 are already sheaves for the fpqc topology.

4.2.1

One must beware, however, that if F is an abelian sheaf on $(\mathrm{Sch})_{/X}$ for the fpqc topology, or for a topology such as fppf considered in SGA 3, the cohomology groups of F for the fpqc topology are not always isomorphic to those for the étale topology. This may already occur when $X = \operatorname{Spec} k$ and F is represented by an algebraic group over k, even for H^1. In general the étale topology gives the correct cohomology groups for coefficient groups which are étale, or more generally smooth, group schemes over X; but this is no longer true for group schemes such as radicial groups, for which the étale topology is too coarse and must be replaced by the fpqc or fppf topology.

4.2.2

As an example of the relation between cohomology for different topologies, consider the Zariski and étale topologies. Let X_{Zar} be the site of Zariski opens of X. There is a canonical inclusion functor

$$u^* : X_{\mathrm{Zar}} \longrightarrow X_{\text{ét}},$$

defining a morphism of sites

$$u : X_{\text{ét}} \longrightarrow X_{\mathrm{Zar}},$$

and hence direct and inverse image functors

$$u_* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X_{\mathrm{Zar}}}, \qquad u^* : \widetilde{X_{\mathrm{Zar}}} \longrightarrow \widetilde{X_{\text{ét}}}.$$

Geometrically, the pair (u_*, u^*) is a morphism of topoi

$$u : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X_{\mathrm{Zar}}}.$$

It yields a morphism of cohomological functors

$$H^*(X_{\mathrm{Zar}}, u_*F) \longrightarrow H^*(X_{\text{ét}}, F)$$

and the Leray spectral sequence

$$H^*(X_{\text{ét}}, F) \Longleftarrow H^p(X_{\mathrm{Zar}}, R^q u_* F),$$

for every abelian sheaf F on $X_{\text{ét}}$. This spectral sequence summarizes the general relation between étale and Zariski cohomology. For coefficient sheaves such as constant sheaves it is generally far from trivial; indeed in general $R^q u_* F \neq 0$, already over the spectrum of a field. However:

Proposition 4.3. Let F be a sheaf of modules on the prescheme X for the Zariski topology, and let $F_{\text{ét}}$ be the corresponding sheaf on $X_{\text{ét}}$. There is a natural morphism of cohomological functors

$$H^*(X, F) \longrightarrow H^*(X_{\text{ét}}, F_{\text{ét}}).$$

If F is quasi-coherent, this morphism is an isomorphism.

The left hand side is $H^*(X_{\mathrm{Zar}}, u_* F_{\text{ét}})$. It is enough to show

$$R^q u_*(F_{\text{ét}}) = 0 \quad (q > 0).$$

Using the standard computation of $R^q u_*$ and the fact that affine opens form a base of the topology of X, this follows from the following corollary.

Corollary 4.4. If X is affine and F is quasi-coherent, then

$$H^q(X_{\text{ét}}, F_{\text{ét}}) = 0 \quad (q > 0).$$

Let $C = X_{\text{ét}}$, and let C' be the full subcategory of affine objects. By 3.1,

$$H^q(C, F_{\text{ét}}) \simeq H^q(C', F|_{C'}).$$

Thus it suffices to prove that $F|_{C'}$ is C'-acyclic, or equivalently that it satisfies the criterion of Exposé V, 4.3: for every $X' \in \operatorname{Ob} C'$ and every covering family $\mathcal{R} = (X'_i \to X')_i$ in C', one has $H^q(\mathcal{R}, F) = 0$ for $q > 0$. We may suppose the family finite, and then replace the X'_i by their sum; hence the covering consists of a single surjective étale morphism $X'' \to X'$ of affine schemes. We are reduced to proving that, if $f : X \to Y$ is a surjective étale morphism of affine schemes and F is a quasi-coherent module on X, then

$$H^q(X/Y, F) = 0 \quad (q > 0).$$

This is proved in the theory of faithfully flat descent.

Remarks 4.5. In fact, for the last vanishing it is enough that $X \to Y$ be surjective and flat, not necessarily étale. The preceding proof therefore also gives analogous isomorphisms with the étale site replaced by $(\mathrm{Sch})_{/X}$ endowed with any of the topologies finer than the Zariski topology considered in SGA 3, IV, 6.3, for example the fpqc topology.

5. Cohomology of a projective limit of schemes

5.1

Let I be an increasing filtered preordered set, and let

$$\mathfrak{X} = (X_i)_{i \in I}$$

be a projective system of schemes, with affine transition morphisms $u_{ij} : X_i \to X_j$ for $i \geq j$. Recall from EGA IV, 8, that under these hypotheses the projective limit

$$X = \varprojlim_i X_i$$

exists in the category of schemes, indeed in the category of locally ringed spaces. It may be constructed as follows. Choose $i_0 \in I$. For $i \geq i_0$, write over X_{i_0}

$$X_i = \operatorname{Spec} \mathcal{A}_i,$$

where $\mathcal{A}_i$ is a quasi-coherent algebra; the $\mathcal{A}_i$ form an inductive system of such algebras. With

$$\mathcal{A} = \varinjlim_i \mathcal{A}_i,$$

one may take

$$X = \operatorname{Spec} \mathcal{A}.$$

Moreover, if $\operatorname{sp} S$ denotes the underlying topological space of a scheme S, then the canonical map

$$\operatorname{sp} X \longrightarrow \varprojlim_i \operatorname{sp}(X_i)$$

is a homeomorphism, and the canonical morphism of sheaves of rings

$$\varinjlim_i \gamma_i^{-1} \mathcal{O}_{X_i} \longrightarrow \mathcal{O}_X$$

is bijective, where $\gamma_i : X \to X_i$ is the canonical continuous map.

5.2

Informally, the content of EGA IV, 8 and 9 may be summarized as follows: when X_{i_0} is quasi-compact and quasi-separated, every schematic datum on X which is of finite presentation over X is equivalent to a datum of the same nature over some X_i, for sufficiently large i. Thus one proves:

(a) If $i_1 \in I$ and X'_{i_1}, X''_{i_1} are two schemes of finite presentation over X_{i_1}, put, for $i \geq i_1$,

$$X'_i = X'_{i_1} \times_{X_{i_1}} X_i, \qquad X' = X'_{i_1} \times_{X_{i_1}} X,$$

and define X''_i and X'' similarly. Then the canonical map

$$\varinjlim_{i \geq i_1} \mathrm{Hom}_{X_i}(X'_i, X''_i) \longrightarrow \mathrm{Hom}_X(X', X'')$$

is bijective.

(b) For every scheme X' of finite presentation over X, there exist an index $i_1 \in I$, a scheme X'_{i_1} of finite presentation over X_{i_1}, and an X-isomorphism

$$X' \simeq X'_{i_1} \times_{X_{i_1}} X.$$

5.3

The preceding result, and analogous results of EGA IV, 8, can be expressed in the language of inductive limits of fibred categories introduced in Exposé VI. Let $\mathcal{F}$ be the category of morphisms $f : T \to S$ of finite presentation of schemes, and let

$$\pi : \mathcal{F} \longrightarrow (\mathrm{Sch})$$

be the target functor. This is a fibred functor, and its fibres are equivalent to $\mathcal{U}$-small categories. Pulling back the fibred category $\mathcal{F}/(\mathrm{Sch})$ along the functor

$$i \longmapsto X_i : \mathrm{cat}(I) \longrightarrow (\mathrm{Sch})$$

gives a fibred category

$$\pi_{\mathfrak{X}} : \mathcal{F}_{\mathfrak{X}} \longrightarrow \mathrm{cat}(I),$$

whose fibre at i is the full subcategory $\mathcal{F}_{X_i}$ of $(\mathrm{Sch})_{/X_i}$ consisting of schemes of finite presentation over X_i; the change-of-base functor from i to j is the usual base change $X'_i \mapsto X'_i \times_{X_i} X_j$. Associating to X'_i its inverse image over X,

$$\varphi(X'_i) = X'_i \times_{X_i} X,$$

we get a natural functor $\varphi : \mathcal{F}_{\mathfrak{X}} \to \mathcal{F}_X$ which sends cartesian morphisms to isomorphisms. Hence it factors uniquely through a canonical functor

$$\psi : \varinjlim_{\mathcal{F}_{\mathfrak{X}}/\,\mathrm{cat}(I)} \mathcal{F}_{\mathfrak{X}} \longrightarrow \mathcal{F}_X.$$

The result of EGA IV, 8 recalled above says precisely that ψ is an equivalence of categories.

5.4

Statement (a) of 5.2 can be refined in various ways by introducing a class $\mathcal{M}$ of morphisms of schemes, stable under base change, and asserting that for a given X_{i_1}-morphism $u_{i_1} : X'_{i_1} \to X''_{i_1}$, the corresponding morphism $u' : X' \to X''$ has property $\mathcal{M}$ if and only if, for some $i \geq i_1$, the morphism $u_i : X'_i \to X''_i$ has property $\mathcal{M}$. This holds, for example, when $\mathcal{M}$ is one of the following classes: proper, projective, quasi-projective, affine, finite, quasi-finite, radicial, surjective, flat, smooth, étale, or unramified morphisms. The corresponding statements are found in EGA IV, §8 for the first cases, §11 for flatness, and §17 for the last three cases.

5.5

Replace $\mathcal{F} \to (\mathrm{Sch})$ by the fibred subcategory $\mathcal{G}$ consisting of étale morphisms of finite presentation. Its fibre $\mathcal{G}_S$ over a scheme S is the category of schemes étale and of finite presentation over S. We form similarly the fibred category $\mathcal{G}_{\mathfrak{X}}$ over $\mathrm{cat}(I)$, whose fibre at i is $\mathcal{G}_{X_i}$. By 3.2(ii), the étale topoi $\widetilde{(X_i)_{\text{ét}}}$ and $\widetilde{X_{\text{ét}}}$ are canonically equivalent to $\widetilde{\mathcal{G}_{X_i}}$ and $\widetilde{\mathcal{G}_X}$, where each $\mathcal{G}_S$ carries the topology induced from $S_{\text{ét}}$. By 3.2(i), each $\mathcal{G}_{X_i}$ is a site of finite type, and equivalent to a $\mathcal{U}$-small site. With these topologies, $\mathcal{G}$ becomes a fibred site over $\mathrm{cat}(I)$.

Lemma 5.6. The canonical functor

$$\varinjlim_{\mathcal{G}_{\mathfrak{X}}/\,\mathrm{cat}(I)} \mathcal{G}_{\mathfrak{X}} \longrightarrow \mathcal{G}_X$$

defines an equivalence, respecting the topologies, from the inductive-limit site of the restricted étale sites of the X_i to the restricted étale site of $X = \varprojlim_i X_i$.

The categorical equivalence is the assertion recalled in 5.4 for étale morphisms. The assertion about topologies follows in the same way by taking $\mathcal{M}$ to be the class of surjective morphisms of schemes.

Theorem 5.7. Let $\mathfrak{X} = (X_i)_{i \in I}$ be a projective system of schemes indexed by an increasing filtered preordered set. Assume that the transition morphisms $u_{ij} : X_j \to X_i$ are affine, so that $X = \varprojlim X_i$ is defined as in 5.1, and that the X_i are quasi-compact and quasi-separated. Let F be an abelian sheaf on the fibred site $\mathcal{G}$, i.e. a functor $\mathcal{G}^\circ \to \mathbf{Ab}$ whose restriction F_i to each $\mathcal{G}_i$ is a sheaf on X_i. Let

$$F_\infty = \varinjlim_i u_i^*(F_i)$$

be the sheaf induced on X, where $u_i : X \to X_i$ is the canonical morphism. Then the canonical homomorphisms

$$\varinjlim_i H^n(X_i, F_i) \longrightarrow H^n(X, F_\infty)$$

are isomorphisms.

In view of 5.6, this is the conjunction of VI, 8.7.3 and VI, 8.5.2. We shall mainly use the following particular case.

Corollary 5.8. With the preceding hypotheses and notation for $(X_i)_{i \in I}$, X, u_{ij}, and u_i, let $i_0 \in I$, and let F_{i_0} be an abelian sheaf on X_{i_0}. For $i \geq i_0$, let $F_i = u_{i_0 i}^* F_{i_0}$, and let $F_\infty = u_i^* F_{i_0}$ on X. Then the canonical homomorphisms

$$\varinjlim_i H^n(X_i, F_i) \longrightarrow H^n(X, F_\infty)$$

are isomorphisms.

The following case is often useful and belongs to 5.7, but not to 5.8.

Corollary 5.9. With the hypotheses and notation of 5.7, let G_{i_0} be a commutative group scheme locally of finite presentation over X_{i_0}. For $i \geq i_0$, put $G_i = G_{i_0} \times_{X_{i_0}} X_i$, and likewise $G_\infty = G_{i_0} \times_{X_{i_0}} X$. Then the canonical homomorphisms

$$\varinjlim_i H^n(X_i, G_i) \longrightarrow H^n(X, G_\infty)$$

are isomorphisms.

Indeed, G_{i_0} defines a sheaf on $(\mathrm{Sch})_{/X_{i_0}}$. We may suppose that i_0 is an initial object of I. Composition with the canonical functor $\mathcal{G} \to (\mathrm{Sch})_{/X_{i_0}}$ gives a contravariant functor F on $\mathcal{G}$, whose restriction to $\mathcal{G}_i$ is canonically the sheaf on X_i defined by G_i. Hence F is a sheaf on $\mathcal{G}$, and 5.7 applies. The hypothesis that G_{i_0} is locally of finite presentation over X_{i_0} ensures that the resulting F_∞ is the sheaf defined by G_∞, by EGA IV, 8.8.2(i) and Lemma 5.6.

Corollary 5.10. With the hypotheses and notation of 5.7, let F be a sheaf of commutative groups on X. Then the canonical homomorphisms

$$\varinjlim_i H^n(X_i, u_{i*}F) \longrightarrow H^n(X, F)$$

are isomorphisms.

The statements 5.7–5.10 generalize, by VI, 8.7.9, to corresponding statements for Ext^i of sheaves of modules; we shall not repeat these special forms here. Instead we record, for later reference, variants of 5.8 and 5.9 in terms of the functors $R^n f_*$.

Corollary 5.11. Let I be an increasing filtered preordered set. Let $(X_i)_{i \in I}$ and $(Y_i)_{i \in I}$ be two projective systems of schemes with affine transition morphisms, let $X = \varprojlim_i X_i$, $Y = \varprojlim_i Y_i$, and let $(f_i)_{i \in I}$ be a projective system of quasi-compact and quasi-separated morphisms $f_i : X_i \to Y_i$. Let $i_0 \in I$, and let F_{i_0} be a sheaf on X_{i_0}. For $i \geq i_0$, let F_i be its inverse image on X_i; similarly let $f : X \to Y$ be induced by the f_i, and let F be the inverse image of F_{i_0} on X. Then the canonical homomorphism

$$\varinjlim_i v_i^*(R^n f_{i*} F_i) \longrightarrow R^n f_* F$$

is an isomorphism, where $v_i : Y \to Y_i$ is the canonical morphism.

The question is local on Y_{i_0}, and one may suppose Y_{i_0} affine and i_0 initial. Then all the Y_i and X_i are quasi-compact and quasi-separated, and the assertion follows from VI, 8.7.3, reducing to 5.8 and using the computation of $R^n f_*$ as sheaves associated with presheaves.

Corollary 5.12. The same statement as 5.11 holds when F_{i_0} is a commutative group scheme locally of finite presentation over X_{i_0}, and F_i, F are its inverse images in the sense of base change for schemes.

Thus 5.9, respectively 5.12, is not a special case of 5.8, respectively 5.11; see §2(a) and Remark 2.1.

Corollary 5.13. With the notation of 5.11, let F be a sheaf of abelian groups on X. Then the canonical homomorphisms

$$\varinjlim_i v_i^*(R^n f_{i*}(u_{i*}F)) \longrightarrow R^n f_* F$$

are isomorphisms, where $u_i : X \to X_i$ and $v_i : Y \to Y_i$ are the canonical morphisms.

The proof is essentially the same as before.

Remarks 5.14.

a) The preceding limit-passage results remain valid for H^0, respectively for direct images f_*, with sheaves of sets rather than abelian sheaves, and for H^1, respectively $R^1 f_*$, with sheaves of not necessarily commutative groups; the non-commutative H^1 and $R^1 f_*$ are defined as usual in terms of torsors or principal homogeneous bundles. This can be checked directly in the general context of Exposé VI. It should also be possible, and useful, to give a variant for the non-commutative H^2 of "links" studied by Giraud.

b) The limit-passage results for fibred sites developed in Exposé VI only assume that the base category is filtered; it is not necessary that it come from a filtered preordered set. An arbitrary filtered index category is also the natural framework for the statements developed in the present section. We used the more restrictive framework only in order to give exact references to EGA IV, where, unfortunately, limit-passage questions from §8 onward are stated with index categories defined by filtered preordered sets. We shall nevertheless use henceforth, without further comment, the corresponding EGA results and the results of the present section for arbitrary filtered index categories, since the proofs in loc. cit. apply essentially without change.

SGA 4, Exposé VIII

Fiber Functors, Supports, Cohomological Study of Finite Morphisms

A. Grothendieck

1. Topological invariance of the étale topos

Theorem 1.1. Let $f : X' \to X$ be an *integral, surjective, radicial* morphism. Then the base-change functor

$$f^* : X_{\text{ét}} \longrightarrow X'_{\text{ét}}$$

is an equivalence of sites, that is, an equivalence of categories which is continuous, as is every quasi-inverse functor. Consequently the functors

$$f_* : \widetilde{X'_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}},$$
$$f^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X'_{\text{ét}}}$$

are quasi-inverse equivalences of categories.

The first assertion is well known when f is of finite presentation or when f is flat (SGA 1, IX, 4.10 and 4.11); the general case reduces to the finite-presentation case. Indeed, it is already known that the functor f^* is fully faithful (SGA 1, IX, 3.4). It remains to prove that f^* is essentially surjective, i.e. that every Z' étale over X' "comes from" a Z étale over X. Using the fact that f is a universal homeomorphism, together with the full faithfulness of f^*, one is reduced to the case where X, X', and Z' are affine, say the spectra of rings A, A', and B'. Writing A' as the filtered inductive limit of its finite-type A-subalgebras A'_i, the algebra B' comes from an étale algebra B'_i over some A'_i (cf. EGA IV, 8). Thus one is reduced to the case where A' is integral and of *finite type* over A, hence finite over A. Then A' is A-isomorphic to A''/J, where $A'' \simeq A[t_1, \ldots, t_n]$ and J is an ideal of A''. Writing J as the inductive limit of its finite-type subideals J_i, and putting $A'_i = A''/J_i$, one again has $A' = \varinjlim_i A'_i$, so that B' comes from an étale algebra B'_i over some A'_i (loc. cit.). On the other hand, since $\operatorname{Spec} A' \to \operatorname{Spec} A$ is surjective, the same is true of $\operatorname{Spec} A'_i \to \operatorname{Spec} A$; furthermore one checks easily that, since $\operatorname{Spec} A' \to \operatorname{Spec} A$ is radicial, the same is true of $\operatorname{Spec} A'_i \to \operatorname{Spec} A$ for i sufficiently large. For example, apply EGA IV, 1.9.9 to $S = \operatorname{Spec} A$: if $S_i \subset S$ denotes the set of points s for which the fibre of $X_i = \operatorname{Spec} A'_i$ at s has separable rank one, then the S_i are constructible subsets of S (EGA IV, 9.7.9), form an increasing family, and their union is S, by the hypothesis on $S' = \operatorname{Spec} A' \to S$ and the fact that the fibres of the S'_i are noetherian schemes. Hence $S = S_i$ for i sufficiently large. This reduces us to the case where A' is one of the A'_i, hence of finite presentation over A.

This proves the first assertion of Theorem 1.1. The fact that f_* and f^* are quasi-inverse equivalences follows immediately.

Corollary 1.2. Let F be an abelian sheaf on X, and let F' be its inverse image on X'. Then the canonical homomorphisms

$$H^i(X, F) \longrightarrow H^i(X', F') \tag{*}$$

are isomorphisms. Likewise, if F' is an abelian sheaf on X' and F is its direct image on X, the canonical homomorphisms $(*)$ are isomorphisms.

Examples 1.3. Here are some frequent applications of Theorem 1.1.

a) X' is a subscheme of X with the same underlying set as X, i.e. it is defined by a nil-ideal I.

b) X is a scheme over a separably closed field k, k' is an algebraic closure of k, and $X' = X \otimes_k k'$.

c) Let X be a geometrically unibranch scheme, for example an algebraic curve over a field k having only "cuspidal" singularities, excluding in particular ordinary double points. If X' is the normalization of X_{red}, then by definition $X' \to X$ is radicial, so that Theorem 1.1 applies.

Remarks 1.4. A morphism f as in Theorem 1.1 is a *universal homeomorphism*, i.e. it is a homeomorphism and remains one after every base extension. Conversely, if f is a universal homeomorphism, then f satisfies the hypotheses of Theorem 1.1 (cf. EGA IV, 8.11.6 when f is of finite presentation, and EGA IV, 18.12.11 in general). This explains the title of the present section.

Concerning Example 1.3(c), note explicitly that if X is an algebraic curve, say over an algebraically closed field k, having at least one singular point which is not unibranch, for instance an ordinary double point, and if X' denotes the normalization of X_{red}, then the conclusion of Corollary 1.2 already fails for H^1 with constant coefficients; compare SGA 1, I, 11(a), and SGA 1, IX, 5.

2. Sheaves on the spectrum of a field

Proposition 2.1. Let k be a field, let $\bar{k}$ be a separable closure of k, let $\pi = \mathrm{Gal}(\bar{k}/k)$ be its topological Galois group, and put $X = \operatorname{Spec} k$, $\bar{X} = \operatorname{Spec} \bar{k}$, on which π acts on the left. Consider the canonical functor

$$i : X_{\text{ét}} \longrightarrow \widetilde{X_{\text{ét}}}$$

(defined by $i(X')(X'') = \mathrm{Hom}_X(X', X'')$), and the canonical functor

$$j : \widetilde{X_{\text{ét}}} \longrightarrow \mathcal{B}_\pi$$

from the category of sheaves on $X_{\text{ét}}$ to the category $\mathcal{B}_\pi$ of discrete sets endowed with a continuous left action of π (cf. IV, 2.7), defined by

$$j(F) = \varinjlim_{\alpha} F(\operatorname{Spec} k_\alpha),$$

where k_α runs through the finite subextensions of $\bar{k}$. Then i and j are equivalences of categories. Consequently the topos $\widetilde{X_{\text{ét}}}$ is equivalent (IV, 3.4) to the classifying topos $\mathcal{B}_{\pi^\bullet}$.

The composite functor

$$ji : X_{\text{ét}} \longrightarrow \widetilde{X_{\text{ét}}} \longrightarrow (\pi\text{-}\mathbf{Set})$$

is an equivalence of categories by the fundamental theorem of Galois theory, in the form of SGA 1, V. Since $\mathcal{B}_\pi$ is plainly a topos (II, 4.14), the same is true of $X_{\text{ét}}$. Moreover a family $(X_i \to X)$ of morphisms in $X_{\text{ét}}$ is covering if and only if it is surjective, i.e. if and only if its image in $\mathcal{B}_\pi$ is surjective, or equivalently covering for the canonical topology of $\mathcal{B}_\pi$. This shows that the topology of $X_{\text{ét}}$ is indeed its canonical topology. Hence i is an equivalence, and therefore so is j.

Corollary 2.2. The functor j of Proposition 2.1 induces an equivalence between the category of abelian sheaves on $X = \operatorname{Spec} k$ and the category of Galois π-modules.

Indeed, the latter are precisely the abelian-group objects of the topos $\mathcal{B}_\pi$. Similarly, by transport of structure one obtains the following.

Corollary 2.3. Let F be an abelian sheaf on $X = \operatorname{Spec} k$, and let $M = j(F)$ be the associated Galois π-module. Then there is a canonical isomorphism of δ-functors in F:

$$H^*(X, F) \simeq H^*(\pi, M),$$

where the right hand side denotes Galois cohomology, studied for example in Serre's course *Cohomologie galoisienne*.

Corollary 2.4. Assume that k is separably closed. Then the functor

$$\Gamma : \widetilde{X_{\text{ét}}} \longrightarrow (\mathbf{Set})$$

is an equivalence of categories. If F is an abelian sheaf on $X = \operatorname{Spec} k$, then $H^i(X, F) = 0$ for $i \neq 0$. In other words, the topos $\widetilde{X_{\text{ét}}}$ is a punctual topos (IV, 2.2).

Corollary 2.5. Let k' be a separably closed extension of a separably closed field k, put $X = \operatorname{Spec} k$ and $X' = \operatorname{Spec} k'$, and let $u : X' \to X$ be the canonical morphism. Then the functor

$$u^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X'_{\text{ét}}}$$

is an equivalence of categories.

3. Fiber functors associated with geometric points of a scheme

3.1

Let X be a scheme. We shall call a *geometric point* of X any X-scheme ξ which is the spectrum of a separably closed field Ω. Giving such a point is therefore equivalent to giving an ordinary point $x \in X$ and a separably closed extension Ω of the residue field $k(x)$. The most frequent case for us will be that where Ω is a separable closure of $k(x)$, which we shall generally denote by $k(\bar{x})$, and the corresponding geometric point of X by $\bar{x}$. For fixed $x \in X$, the fields $k(\bar{x})$, respectively the geometric points $\bar{x}$, are determined only up to non-unique $k(x)$-isomorphism, respectively non-unique X-isomorphism.

Remarks 3.2. In most geometric questions it is more natural to restrict to algebraically closed Ω. The different convention used here is special to the study of the étale topology. It is the convention adopted in the definition of the fundamental group in SGA 1, V, 7. Note however that the developments of loc. cit. are equally valid with the convention adopted here, since the property of Ω used there is that every étale covering of $\operatorname{Spec} \Omega$ is trivial, precisely the assertion that Ω is separably closed.

Definition 3.3. Let X be a scheme, let ξ be a geometric point of X, and let $u : \xi \to X$ be its structural morphism. The *geometric fiber functor* associated with ξ, denoted $F \mapsto F_\xi$, is the composite functor

$$\widetilde{X_{\text{ét}}} \xrightarrow{u^*} \widetilde{\xi_{\text{ét}}} \xrightarrow{\Gamma_\xi} (\mathbf{Set}).$$

Equivalently, by Corollary 2.4, a geometric point ξ of X defines a point of the topos $\widetilde{X_{\text{ét}}}$ (IV, 6.1), and one takes the associated fiber functor.

3.4

Since the functor Γ_ξ of sections over ξ is an equivalence of categories (2.4), knowing the fiber functors $F \mapsto F_\xi$ is essentially equivalent to knowing the inverse-image functors u^*. Moreover it follows immediately from Corollary 2.5 that if ξ is a geometric point of X and there exists an X-morphism $\xi' \to \xi$, i.e. if ξ' corresponds to a separably closed extension Ω' of Ω, then the canonical homomorphism $F_\xi \to F_{\xi'}$ is a functorial isomorphism

$$F_\xi \simeq F_{\xi'}.$$

Thus, for the study of fiber functors, one may, after replacing Ω by the separable algebraic closure of $k(x)$ in Ω, restrict to the case where $k(\xi)$ is a separable closure $k(\bar{x})$ of $k(x)$. The fiber functors corresponding to geometric points of X localized at the same point $x \in X$ are isomorphic, non-canonically.

In accordance with the conventions of 3.1, we shall denote by $F_{\bar{x}}$ a fiber functor corresponding to a chosen separable closure $k(\bar{x})$ of $k(x)$.

We also record an evident transitivity property, which explains the technical usefulness of not restricting exclusively to geometric points defined by separable closures of residue fields. Let $f : X' \to X$ be a morphism of schemes, and let $u' : \xi' \to X'$ be a geometric point of X'. Then $u = fu' : \xi' \to X$ is a geometric point of X, which we denote by ξ. For these geometric points one has a functorial isomorphism in $F \in \operatorname{Ob} \widetilde{X_{\text{ét}}}$:

$$f^*(F)_{\xi'} \simeq F_\xi.$$

This follows from the transitivity formula for inverse-image functors,

$$(fu)^* \simeq u^* f^*.$$

Theorem 3.5. Let X be a scheme.

a) For every geometric point ξ of X, the fiber functor $F \mapsto F_\xi$, from $\widetilde{X_{\text{ét}}}$ to $(\mathbf{Set})$, commutes with arbitrary inductive limits, indexed by categories in $\mathcal{U}$, and with finite projective limits. In particular it sends sheaves of groups, respectively abelian sheaves, and so on, to groups, respectively abelian groups, and so on.

b) As x runs over the points of X, the fiber functors $F_{\bar{x}}$ form a conservative family: a morphism $u : F \to G$ of sheaves on X is an isomorphism if and only if, for every $x \in X$, the corresponding morphism $u_{\bar{x}} : F_{\bar{x}} \to G_{\bar{x}}$ is an isomorphism.

Taking account of the exactness properties in (a), we immediately record the following formal consequences of (b).

Corollary 3.6. Let $u : F \to G$ be a morphism of sheaves. Then u is a monomorphism, respectively an epimorphism, if and only if for every $x \in X$, the map $u_{\bar{x}} : F_{\bar{x}} \to G_{\bar{x}}$ is injective, respectively surjective.

Corollary 3.7. Let $u, v : F \to G$ be two morphisms of sheaves on X. Then $u = v$ if and only if, for every $x \in X$, one has $u_{\bar{x}} = v_{\bar{x}} : F_{\bar{x}} \to G_{\bar{x}}$. In particular, if u, v are two sections of F, then $u = v$ if and only if $u_{\bar{x}} = v_{\bar{x}}$ in $F_{\bar{x}}$ for every $x \in X$.

Corollary 3.8. Let $F \to G \to H$ be a sequence of two morphisms of sheaves on X. The sequence is exact if and only if, for every $x \in X$, the corresponding sequence $F_{\bar{x}} \to G_{\bar{x}} \to H_{\bar{x}}$ is exact.

We leave to the reader the deduction of these corollaries (cf. IV, 6.5.2), although they are also proved in part in the proof below.

Assertion (a) follows from the exactness properties already noted in Exposé VII, 1.4 for every inverse-image functor, such as $u^* : \widetilde{X_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$, and from the fact that $\widetilde{\xi_{\text{ét}}} \to (\mathbf{Set})$ is an equivalence. To prove (b), note the following formula, which gives a way to *compute* fiber functors.

Proposition 3.9. Let $\xi \xrightarrow{u} X$ be a geometric point of X, and let C_ξ be the category of "ξ-pointed étale X-schemes", i.e. pairs (X', u'), where $X' \in \operatorname{Ob} X_{\text{ét}}$ and $u' : \xi \to X'$ is an X-morphism. Then the opposite category C_ξ° is filtering, and for a variable sheaf F, respectively presheaf P, on X, there is a canonical functorial isomorphism

$$F_\xi \simeq \varinjlim_{C_\xi^\circ} F(X'), \qquad (aP)_\xi \simeq \varinjlim_{C_\xi^\circ} P(X'),$$

where aP denotes the sheaf associated with P.

Indeed, first note that for every presheaf P on $\xi_{\text{ét}}$, if aP is its associated sheaf, the natural homomorphism

$$P(\xi) \longrightarrow aP(\xi)$$

is an isomorphism; this follows immediately, for example, from the explicit computation of aP as $L(LP)$ (II, 3.19). Let $\varphi : \widetilde{X_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$ be the inverse-image functor. Then $u^* = \varphi^*$ is, by III, 2.3(d), the composite $a\varphi^\bullet i$, where $i : \widetilde{X_{\text{ét}}} \to \widehat{X_{\text{ét}}}$ is the inclusion functor and $a : \widehat{\xi_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$ is the associated-sheaf functor. By the preceding remark,

$$u^*(F)(\xi) = (\varphi^\bullet i(F))(\xi).$$

Thus it remains to apply the explicit expression for $\varphi^\bullet$ given in the proof of the corresponding general formula. The analogous assertion for a presheaf P is proved in the same way, using the functorial isomorphism $\varphi^* a(P) \simeq a\varphi^*(P)$. The fact that C_ξ is filtering, which remains true for *any* X-scheme ξ, not necessarily a geometric point, follows immediately from the existence of finite projective limits in C_ξ; this in turn follows from the fact that $X_{\text{ét}}$ is a subcategory of $(\mathrm{Sch})_{/X}$ stable under finite projective limits.

Now let $u : F \to G$ be a morphism of sheaves on X such that, for every $x \in X$, the map $u_{\bar{x}} : F_{\bar{x}} \to G_{\bar{x}}$ is a monomorphism. We prove that u is a monomorphism. It suffices to show that if $X' \in \operatorname{Ob} X_{\text{ét}}$ and $\varphi, \psi \in F(X')$ have the same image under u, then $\varphi = \psi$. Replacing X by X' and using the transitivity of fiber functors, we may suppose $X = X'$. For every $x \in X$, we have $u(\varphi)_{\bar{x}} = u(\psi)_{\bar{x}}$, hence $u_{\bar{x}}(\varphi_{\bar{x}}) = u_{\bar{x}}(\psi_{\bar{x}})$, and therefore $\varphi_{\bar{x}} = \psi_{\bar{x}}$. By Proposition 3.9, there exists $X'_x \in \operatorname{Ob} X_{\text{ét}}$, whose image contains x, such that φ and ψ have the same inverse image on X'_x. As x varies over X, the X'_x form a covering family of X; hence $\varphi = \psi$.

This proves that if all $u_{\bar{x}}$ are monomorphisms, then so is u. Suppose in addition that all $u_{\bar{x}}$ are epimorphisms. We prove that u is an epimorphism, hence an isomorphism. It remains to show that for every $X' \in \operatorname{Ob} X_{\text{ét}}$ and every $\psi \in G(X')$, there exists $\varphi \in F(X')$ with $u(X')(\varphi) = \psi$. Again we may suppose $X' = X$. By Proposition 3.9, for every $x \in X$ there exists $X'_x \in \operatorname{Ob} X_{\text{ét}}$, with image containing x, and an element $\varphi_x \in F(X'_x)$ whose image in $G(X'_x)$ is the inverse image of ψ. Since u is a monomorphism, the φ_x agree on the fibre products $X'_x \times_X X'_y$, and hence come from a well-determined element $\varphi \in F(X)$. Then $u(\varphi) = \psi$, since the inverse images on the X'_x agree. $\square$

Scholium 3.10. Theorem 3.5 implies that the collection of fiber functors for the étale topology has the same essential properties as in the usual theory of sheaves on a topological space. We shall use Theorem 3.5 and its corollaries very frequently, and for this reason we shall usually refrain from referring to them explicitly.

3.11

For every $x \in X$ we shall also, by abuse of language, denote by x the X-scheme $\operatorname{Spec} k(x)$, with its usual structural morphism

$$i_x : x = \operatorname{Spec} k(x) \longrightarrow X.$$

It gives rise to a canonical functor

$$i_x^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{x_{\text{ét}}}.$$

If F is a sheaf on X, put $F_x = i_x^*(F)$. Thus F_x is a sheaf on $x = \operatorname{Spec} k(x)$, and as such can also be identified with an étale scheme over x, by Proposition 2.1. It depends functorially on F, and the functor $i_x^* : F \mapsto F_x$ again commutes with arbitrary inductive limits and finite projective limits, by Exposé VII, 1.4.

If $\bar{x} = \operatorname{Spec} \overline{k(x)}$, the fiber functor $F \mapsto F_{\bar{x}}$ is canonically isomorphic to the composite of $F \mapsto F_x$ with the functor j of Proposition 2.1, taking into account that the latter is isomorphic to the fiber functor associated with the geometric point $\operatorname{Spec} \bar{k}$ of $\operatorname{Spec} k$. It follows immediately from this and Theorem 3.5 that the family of functors $(F \mapsto F_x)$, for $x \in X$, is still conservative. If $k(x)$ is separably closed, i.e. $x = \bar{x}$, then the notations F_x and $F_{\bar{x}}$ are strictly contradictory, the first denoting an object of $\widetilde{x_{\text{ét}}}$, the second a set; but by Proposition 2.1 this is an inessential contradiction.

3.12

The preceding remarks show that, if $\bar{x} = \operatorname{Spec} \overline{k(x)}$, then for every sheaf F on X the group $\pi_x = \operatorname{Gal}(\overline{k(x)}/k(x))$ acts naturally on the left on the fiber set $F_{\bar{x}}$. Thus $F \mapsto F_{\bar{x}}$ may in fact be regarded as a functor

$$\widetilde{X_{\text{ét}}} \longrightarrow \mathcal{B}_{\pi_x},$$

whose knowledge is essentially equivalent, by Proposition 2.1, to that of the functor $F \mapsto F_x$.

This shows, in an important special case, that contrary to what happens for sheaves on topological spaces, fiber functors generally have nontrivial automorphisms. In general, Theorem 7.9 below will determine all homomorphisms from one fiber functor to another.

Remark 3.13. The following is a sometimes useful generalization of Theorem 3.5(b). Let E be a subset of X such that, for every étale morphism $f : X' \to X$, the inverse image $E' = f^{-1}(E)$ is very dense in X' (EGA IV, 10.1.3), i.e. every open subset U of X' containing E' is equal to X'. Then the fiber functors $F_{\bar{x}}$ on $\widetilde{X_{\text{ét}}}$, for $x \in E$, already form a conservative family. Consequently Corollaries 3.6, 3.7, and 3.8 remain valid when one restricts to $x \in E$. This is seen immediately by repeating the proof of Theorem 3.5(b). The preceding result applies in particular in the following situations.

a) X is a Jacobson scheme (EGA IV, 10.4.1), for example locally of finite type over a field or over $\operatorname{Spec} \mathbb{Z}$, and E is the set of closed points of X.

b) X is locally of finite type over a scheme S, and E is the set of points of X which are closed in their fibre.

c) X is locally noetherian, and E is the set of $x \in X$ such that $\overline{\{x\}}$ is a finite set, i.e. a semi-local scheme of dimension ≤ 1; compare EGA IV, 10.5.3 and 10.5.5.

4. Strictly local rings and schemes

4.1

Recall that, following Azumaya, a local ring A is called *henselian* if it satisfies the following equivalent conditions.

(i) Every finite A-algebra B is a direct product $\prod_i B_i$ of local rings. The B_i are then necessarily isomorphic to the localizations $B_{\mathfrak{m}_i}$, where the $\mathfrak{m}_i$ run through the maximal ideals of B.

(ii) Every A-algebra B which is the localization of a finite-type A-algebra C at a prime ideal $\mathfrak{p}$ over the maximal ideal $\mathfrak{m}$ of A, and such that $B/\mathfrak{m}B$ is finite over $k = A/\mathfrak{m}$, is finite over A.

(iii) As in (ii), but assuming C to be étale over A at $\mathfrak{p}$, and $k \to B/\mathfrak{m}B$ to be an isomorphism.

(iv) Hensel's lemma, in its classical form, holds for A: for every monic polynomial $F \in A[T]$, denoting by $\dot{F} \in k[T]$ its reduction, and for every factorization

$$\dot{F} = \dot{F}_1 \dot{F}_2$$

of $\dot{F}$ as a product of two coprime monic polynomials, $\dot{F}_1$ and $\dot{F}_2$ lift, necessarily uniquely, to $F_1, F_2 \in A[T]$ such that $F = F_1 F_2$.

For the proof of these equivalences and the general properties of henselian rings, see EGA IV, 18.5.11. Recall that if A is any local ring, one proves, following Nagata, that there is a local homomorphism

$$A \longrightarrow A^h$$

with A^h henselian, universal for local homomorphisms from A to henselian local rings. The ring A^h is called the henselization of A. The construction in EGA IV, 18.6, different from Nagata's, consists in putting

$$A^h = \varinjlim_i A_i,$$

where A_i runs through the A-algebras essentially of finite type and essentially étale over A, i.e. localizations of an étale A-algebra, such that $A \to A_i$ is local and the residue extension $k(A_i)/k(A)$ is trivial. This construction shows in particular that A^h is flat over A, and that if A is noetherian then so is its henselian closure.

Definition 4.2. A local ring A is called *strictly local* if it satisfies one of the following equivalent conditions.

(i) A is henselian and its residue field is separably closed.

(ii) Every local homomorphism $A \to B$, where B is the localization of an étale A-algebra, is an isomorphism.

A scheme is called a *strictly local scheme* if it is isomorphic to the spectrum of a strictly local ring.

The equivalence of (i) and (ii) follows immediately from 4.1. In geometric form, 4.2(ii), respectively 4.1(iii), says that for every scheme X étale over $Y = \operatorname{Spec} A$, and every point x of the fibre X_y, respectively every point of X_y rational over $k(y)$, where y is the closed point of Y, there exists a section of X over Y passing through y, necessarily unique.

Definition 4.3. Let X be a scheme, let $\mathcal{O}_{X_{\text{ét}}}$ be the sheaf on $X_{\text{ét}}$ defined by

$$\mathcal{O}_{X_{\text{ét}}}(X') = \Gamma(X', \mathcal{O}_{X'}),$$

compare Exposé VII, 2(c), and let $u : \xi = \operatorname{Spec} \Omega \to X$ be a geometric point of X. The *strictly local ring of* X *at* ξ, denoted $\mathcal{O}_{X,\xi}$, or simply $\mathcal{O}_\xi$, is the fibre of the sheaf $\mathcal{O}_{X_{\text{ét}}}$ at the point ξ. Its spectrum is called the *strict localization of* X *at* ξ.

By Proposition 3.9, the fibre $\mathcal{O}_{X,\xi}$ has the description

$$\mathcal{O}_{X,\xi} = \varinjlim \Gamma(X', \mathcal{O}_{X'}), \tag{*}$$

where X' runs through the filtering opposite category of étale X-schemes pointed by ξ, i.e. endowed with an X-morphism $\xi \to X'$. By transitivity of inductive limits, one may replace $\Gamma(X', \mathcal{O}_{X'})$ on the right by $\mathcal{O}_{X',x'}$, where x' is the image of ξ in X', and obtains

$$\mathcal{O}_{X,\xi} = \varinjlim_i A_i, \tag{**}$$

where A_i runs through the algebras essentially of finite type and étale over $A = \mathcal{O}_{X,x}$, endowed with a k-homomorphism $k(A_i) \to \Omega$. Here x denotes the image of ξ in X. The transition homomorphisms among the A_i are the local homomorphisms of A-algebras $A_i \to A_j$ making the corresponding triangle commute,

$$\begin{tikzcd} k(A_i) \arrow[r] \arrow[dr] & k(A_j) \arrow[d] \\ & \Omega, \end{tikzcd}$$

which implies that if there exists an admissible homomorphism from A_i to A_j, it is unique. Thus the inductive limit $(**)$ is a limit over an increasing filtered ordered set.

4.4

The description $(**)$ shows that the strictly local ring of X at ξ depends essentially only on the usual local ring $A = \mathcal{O}_{X,x}$, where $x = u(\xi)$, and on the extension Ω of $k = k(A)$. It is also called the *strict henselization* of A, relative to the chosen separably closed extension Ω of k, and is denoted A^{hs}. We refer to EGA IV, 18.8 for the general properties of A^{hs}. We merely point out that the construction above again shows that A^{hs} is flat over A, and noetherian if A is noetherian. On the other hand, one easily proves (loc. cit.) that the local homomorphism

$$A \longrightarrow A^{hs}$$

endowed with the k-homomorphism

$$k(A^{hs}) \longrightarrow \Omega$$

solves the universal problem, relative to the data A and the extension Ω of k, of finding local homomorphisms

$$A \longrightarrow A'$$

with A' strictly local, endowed with a k-homomorphism

$$k(A') \longrightarrow \Omega.$$

In particular, A^{hs} *is* a strictly local ring, justifying the terminology in Definition 4.3.

4.5

Keep the notation of Definition 4.3 and choose an affine open neighbourhood U of x. In Proposition 3.9 one may plainly restrict to those X' which are affine over U. These then form a pseudo-filtered projective system of affine schemes, and we are in the general situation of Exposé VII, 5 (cf. VI, 5.12(b)). By the expression $(*)$ for $\mathcal{O}_{X,\xi}$, one has an X-isomorphism

$$\varprojlim_{\substack{X' \text{ étale} \\ \text{affine over } U \\ \xi\text{-pointed}}} X' \simeq \operatorname{Spec} \mathcal{O}_{X,\xi}, \tag{***}$$

which makes precise the intuitive geometric meaning of $\operatorname{Spec} \mathcal{O}_{X,\xi}$ as the *limit of the étale neighbourhoods of* ξ.

Proposition 4.6. Let X be a strictly local scheme, and let x be its closed point, hence a geometric point of X. Then the functor $F \mapsto \Gamma(X, F) : \widetilde{X_{\text{ét}}} \to (\mathbf{Set})$ is canonically isomorphic to the fiber functor $F \mapsto F_x$, and therefore commutes with arbitrary inductive limits, not necessarily filtered.

Indeed, by Definition 4.2(ii), the final object X of the category of x-pointed étale X-schemes considered in Proposition 3.9 dominates all the other objects. Hence $\varinjlim F(X')$ is canonically isomorphic to $F(X)$. □

In particular, the induced functor Γ on the category of abelian sheaves is exact. Therefore:

Corollary 4.7. Under the hypotheses of Proposition 4.6, for every abelian sheaf F on X,

$$H^q(X, F) = 0 \qquad \text{for } q \neq 0.$$

Corollary 4.8. Let X be a scheme, let ξ be a geometric point of X, let $\bar{X} = \operatorname{Spec} \mathcal{O}_{X,\xi}$ be the corresponding strictly localized scheme, let F be a variable sheaf on X, and let $\bar{F}$ be its inverse image on $\bar{X}$. Then there is a functorial isomorphism

$$F_\xi \simeq \Gamma(\bar{X}, \bar{F}).$$

Indeed, ξ may also be regarded as a geometric point of $\bar{X}$, and it suffices to apply Proposition 4.6 and the transitivity of fibres (3.4).

5. Application to computing the fibres of $R^q f_*$

5.1

Let

$$f : X \to Y$$

be a morphism of schemes, and let F be an abelian sheaf on X. We want to determine $R^q f_*(F)$. By Theorem 3.5, this amounts, up to minor qualifications, to knowing the geometric fibres $R^q f_*(F)_{\bar{y}}$, for $y \in Y$. Now by Exposé V, 5.1, $R^q f_*(F)$ is the sheaf associated with the presheaf

$$\mathcal{H}^q : Y' \longmapsto H^q(X \times_Y Y', F)$$

on Y. Therefore, by Proposition 3.9,

$$R^q f_*(F)_{\bar{y}} \simeq \varinjlim H^q(X \times_Y Y', F), \tag{5.1.1}$$

where the inductive limit is taken over the filtering opposite category of the Y' étale over Y pointed by $\bar{y}$. Choosing an affine open neighbourhood U of y, one may restrict in the limit to the cofinal category of the Y' which are affine and lie over U.

Introduce

$$\bar{Y} = \operatorname{Spec}(\mathcal{O}_{Y,\bar{y}}) = \varprojlim Y'$$

(cf. 4.5), and

$$\bar{X} = X \times_Y \bar{Y}.$$

In the projective system of the $X \times_Y Y'$, obtained from that of the Y' by base change along $X \to Y$, the transition morphisms are affine. Thus one is under the general hypotheses of Exposé VII, 5.1, and by the construction of $\varprojlim$ in loc. cit. there is a canonical isomorphism

$$\bar{X} = \varprojlim (X \times_Y Y').$$

Let $\bar{F}$ be the inverse image of F on $\bar{X}$. One obtains a canonical homomorphism

$$\varinjlim_{Y'} H^q(X \times_Y Y', F) \longrightarrow H^q(\bar{X}, \bar{F}),$$

and hence, by comparison with (5.1.1), a canonical homomorphism, functorial in F,

$$R^q f_*(F)_{\bar{y}} \longrightarrow H^q(\bar{X}, \bar{F}). \tag{5.1.2}$$

Assume now that $f : X \to Y$ is quasi-compact and quasi-separated. Then the same is true of $X \times_Y Y' \to Y'$, and since Y' is affine the schemes $X \times_Y Y'$ are quasi-compact and quasi-separated. Using (5.1.1) and the limit-passage theorem of Exposé VII, 5.8, we obtain the following.

Theorem 5.2. Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of schemes, let F be an abelian sheaf on X, let y be a point of Y, let $\bar{y}$ be the geometric point above y associated with a separable closure $k(\bar{y})$ of $k(y)$, let $\bar{Y} = \operatorname{Spec} \mathcal{O}_{Y,\bar{y}}$ be the corresponding strict localization, put $\bar{X} = X \times_Y \bar{Y}$, and let $\bar{F}$ be the inverse image of F on $\bar{X}$. Then the canonical homomorphism (5.1.2) is an isomorphism:

$$R^q f_*(F)_{\bar{y}} \simeq H^q(\bar{X}, \bar{F}).$$

This statement practically reduces the determination of the fibres of a sheaf $R^q f_*(F)$ to the determination of the cohomology of a prescheme over a strictly local scheme, and will be used constantly below. Technically, Theorem 5.2 explains the important role of henselian rings and strictly local rings in the study of étale cohomology.

Remark 5.3. The statement remains valid for $R^0 f_*(F) = f_*(F)$, with F an arbitrary sheaf of sets. Also, Theorem 5.2 remains valid for $R^1 f_*(F)$ when F is a sheaf of groups, not necessarily commutative, using the usual definition of $R^1 f_*(F)$ (cf. Giraud's thesis).

5.4

Assume now that f is a *finite* morphism. Then $\bar{f} : \bar{X} \to \bar{Y}$ is finite, and since $\bar{Y}$ is strictly local it follows that $\bar{X}$ is a finite sum of strictly local schemes $\bar{X}_i$. Therefore, using Corollary 4.7,

$$H^q(\bar{X}, \bar{F}) = 0 \qquad \text{for } q \neq 0. \tag{5.4.1}$$

On the other hand, the components $\bar{X}_i$ of $\bar{X}$ correspond to the points $\bar{x}_i$ of $\bar{X}$ above the closed point $\bar{y}$ of $\bar{Y}$, i.e. to the points of

$$\bar{X}_{\bar{y}} = X_y \otimes_{k(y)} k(\bar{y}).$$

These points may also be regarded as geometric points of X, and $\bar{X}_i$ is then nothing other than the strict localization of X at $\bar{x}_i$. Thus, by Corollary 4.8,

$$H^0(\bar{X}_i, \bar{F}) \simeq F_{\bar{x}_i},$$

and hence

$$H^0(\bar{X}, \bar{F}) = \prod_i F_{\bar{x}_i}, \tag{5.4.2}$$

a functorial isomorphism in the sheaf of sets F; here it is not necessary to restrict to abelian sheaves. Taking Theorem 5.2 and Remark 5.3 into account, formulas (5.4.1) and (5.4.2) give the following.

Proposition 5.5. Let $f : X \to Y$ be a finite morphism of preschemes, and let y be a point of Y. For every sheaf F on X, there is a functorial isomorphism

$$f_*(F)_{\bar{y}} \simeq \prod_{\bar{x} \in X_y \otimes_{k(y)} k(\bar{y})} F_{\bar{x}}.$$

Consequently formation of $f_*(F)$ commutes with every base change $y' \to Y$. If F is an abelian sheaf, then

$$R^q f_*(F) = 0 \qquad \text{for } q > 0.$$

The first formula in Proposition 5.5 is in fact independent of Theorem 5.2 and of the general limit-passage theorem of Exposé VII, 5.7. It implies, by Theorem 3.5, that $F \mapsto f_*(F)$ is an exact functor on the category of abelian sheaves, again giving $R^q f_*(F) = 0$ for $q \neq 0$. Here is a slight variant of Proposition 5.5.

Corollary 5.6. Let $f : X \to Y$ be an integral morphism. Then for every abelian sheaf F on X,

$$R^q f_*(F) = 0 \qquad \text{for } q \neq 0.$$

Moreover, the functor f_* on sheaves of sets commutes with every base change $Y' \to Y$.

Indeed, by Theorem 5.2 we are reduced to proving that, when Y is strictly local, $H^q(X, F) = 0$ for every abelian sheaf F on X. Write $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, with B an integral A-algebra. Writing $B = \varinjlim B_i$, where the B_i run through the finite-type subalgebras of B, which are even finite over A, we have $X = \varprojlim_i X_i$, where $X_i = \operatorname{Spec} B_i$. By Exposé VII, 5.13, it is enough to prove that $R^q f_*(F_i) = 0$ for $q \neq 0$ with F_i an abelian sheaf on X_i, and this follows from Proposition 5.5. The final assertion of Corollary 5.6 is proved by the same reduction to Proposition 5.5.

Corollary 5.7. Let $f : X \to Y$ be an immersion. Then for every sheaf F on X, the canonical morphism

$$f^* f_*(F) \longrightarrow F$$

is an isomorphism.

Since f factors as the product of an open immersion and a closed immersion, and Corollary 5.7 is trivial for an open immersion (IV, No. 3), one is reduced to the case where f is a closed immersion. It is enough to prove that for every $x \in X$, the corresponding homomorphism on geometric fibres

$$(f^* f_*(F))_{\bar{x}} \longrightarrow F_{\bar{x}}$$

is bijective. By transitivity of fibres (3.4), the left hand side is just the fibre $f_*(F)_{\bar{x}}$, so it remains to check that the canonical homomorphism

$$f_*(F)_{\bar{x}} \longrightarrow F_{\bar{x}}$$

is bijective. This follows immediately from Proposition 5.5.

Remark 5.8. Using Proposition 5.5 and proceeding as in Corollary 5.6, one easily proves that if $f : X \to Y$ is integral and F is a sheaf of groups on X, not necessarily commutative, then $R^1 f_*(F)$ is the final sheaf on Y; compare Remark 5.3.

6. Supports

Let U be a Zariski open subset of the scheme X. Then $U \in \operatorname{Ob} X_{\text{ét}}$, and in fact U is a subobject of the final object X of $X_{\text{ét}}$. Hence it defines a subobject $\widetilde{U}$ of the final object of $\widetilde{X_{\text{ét}}}$, i.e. an open subtopos of the étale topos of X (IV, 8.3).

Proposition 6.1. The preceding map $U \mapsto \widetilde{U}$ is an isomorphism from the ordered set of open subsets of X, in the Zariski sense, onto the set of open subobjects of the étale topos $\widetilde{X_{\text{ét}}}$.

Since $X_{\text{ét}} \to \widetilde{X_{\text{ét}}}$ is fully faithful, the map $U \mapsto \widetilde{U}$ plainly preserves order, i.e. $(U \subset V) \Leftrightarrow (\widetilde{U} \subset \widetilde{V})$, so in particular it is injective. To prove surjectivity, let F be a subsheaf of the final sheaf $\widetilde{X}$, and consider the objects of $X_{\text{ét}}/F$, i.e. the étale schemes X' over X such that $F(X') \neq \varnothing$. Since $X' \to X$ is étale, it is an open map (EGA IV, 2.4.6); in particular its image $\operatorname{Im}(X')$ is open. Let U be the union of the opens $\operatorname{Im}(X')$ for $X' \in \operatorname{Ob} X_{\text{ét}}/F$. Since the family $X' \to U$, for $X' \in \operatorname{Ob} X_{\text{ét}}/F$, is surjective, hence covering, one concludes that $U \in \operatorname{Ob} X_{\text{ét}}/F$, hence $X_{\text{ét}}/F = X_{\text{ét}}/U$, and therefore $F = U$. □

Taking Proposition 6.1 into account, we may therefore speak without ambiguity of an open of X, without specifying whether we mean this in the usual Zariski sense or in the sense of the étale topology.

Corollary 6.2. Let U be an open subset of X, and let $j : U \to X$ be the canonical immersion. Then the functor

$$j^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{U_{\text{ét}}}$$

induces an equivalence of categories

$$\widetilde{X_{\text{ét}}}/\widetilde{U} \longrightarrow \widetilde{U_{\text{ét}}}.$$

Indeed, for every site C in which finite projective limits exist, and for every subobject U of the final object e of C, the functor $j : C \to C_{/U}$ defined by $j(S) = S \times U$ is a morphism of sites and $j^* : \widetilde{C} \to \widetilde{C/U}$ induces an equivalence $\widetilde{C}/\widetilde{U} \to \widetilde{C/U}$. It remains only to combine this general fact with the canonical isomorphism $U_{\text{ét}} \simeq X_{\text{ét}}/U$.

Pictorially, Corollary 6.2 says that the operations of restricting to an open, in the usual sense for schemes on the one hand and in the sense of topoi on the other, are compatible. The following is the analogous compatibility for restriction to a closed subset.

Theorem 6.3. Let X be a scheme, let Y be a closed subscheme of X, let $U = X - Y$ with its induced structure, and let $i : Y \to X$ and $j : U \to X$ be the canonical immersions. Then the functor

$$i_* : \widetilde{Y_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$

is fully faithful. Moreover, for $F \in \operatorname{Ob} \widetilde{X_{\text{ét}}}$, the sheaf F is isomorphic to one of the form $i_*(G)$ if and only if $j^*(F)$ is isomorphic to the final sheaf on U.

Proof. Since i_* and i^* are adjoint, the full faithfulness of i_* is equivalent to the assertion that the functorial homomorphism

$$i^*(i_*(G)) \longrightarrow G$$

is an isomorphism; this is just Corollary 5.7. On the other hand, if $G \in \operatorname{Ob} \widetilde{Y_{\text{ét}}}$, then by Corollary 6.2 one checks trivially that $j^*(i_*(G))$ is the final sheaf on U.

Conversely, suppose $F \in \operatorname{Ob} \widetilde{X_{\text{ét}}}$ is such that $j^*(F)$ is the final sheaf. We prove that F is isomorphic to a sheaf of the form $i_*(G)$, or equivalently that the canonical homomorphism

$$F \longrightarrow i_* i^* F$$

is an isomorphism. It is enough to check this on all geometric fibres. If $x \in U$, this is precisely the hypothesis on F. If $x \in Y$, transitivity of fibres reduces us to checking that the homomorphism on fibres induced by

$$i^*(F) \longrightarrow i^*(i_* i^*(F))$$

is an isomorphism. But this homomorphism is an isomorphism by Corollary 5.7, applied to i and to i^*F. This completes the proof.

Corollary 6.4. The functor i_* induces an equivalence between the category of abelian sheaves on Y and the category of abelian sheaves on X whose restriction to $U = X - Y$ is zero.

In accordance with common usage, we shall therefore often identify an abelian sheaf on Y with an abelian sheaf on X which is zero on U.

6.5

By Proposition 6.1 we know how to interpret the opens of the topos $\mathcal{E} = \widetilde{X_{\text{ét}}}$ as ordinary opens U of X. By Theorem 6.3, with this identification, the residual topos $\mathcal{E}_{\complement U}$ of IV, 3 is canonically equivalent to $\widetilde{Y_{\text{ét}}}$, where $Y = X - U$ is endowed with any induced scheme structure. We may therefore apply to this situation the results of IV, 3, especially the description of sheaves F on X in terms of triples (F', F'', u), where F' is a sheaf on Y, F'' a sheaf on U, and u a homomorphism from F' to $i^* j_*(F'')$. We shall freely use below the notation $i^!$, $j_!$ of IV, 3, denoting functors

$$j_! : (\widetilde{U_{\text{ét}}})_{\text{ab}} \longrightarrow (\widetilde{X_{\text{ét}}})_{\text{ab}},$$
$$i^! : (\widetilde{X_{\text{ét}}})_{\text{ab}} \longrightarrow (\widetilde{Y_{\text{ét}}})_{\text{ab}},$$

where the subscript ab denotes the category of abelian sheaves. These functors give rise to the two exact sequences of IV, 3.7.

6.6

In view of the preceding developments and of the general terminology introduced in IV, 8.5, it is natural to introduce, for an abelian sheaf F on X, or a section φ of such a sheaf, the notion of the *support* of F, respectively of φ, as the closed subset, in the usual Zariski sense, complementary to the largest open over which the object in question vanishes.

In the present situation it is also appropriate to replace the general notation of Exposé V, 4.3, $H^q(\mathcal{E}_{\complement U}, F)$, $\mathcal{H}^q_{\complement U}(F)$, by the notation $H^q_Y(X, F)$, $\mathcal{H}^q_Y(F)$. The first denotes an abelian group, and the second an abelian sheaf on Y, or equivalently an abelian sheaf on X which is zero on U. Thus $\mathcal{H}^q_Y = R^q i^!$, and so forth. We refer to Exposé V, 6 for the general properties of these functors.

7. Specialization morphisms of fiber functors

7.1

We saw in No. 4 how to associate to every geometric point ξ of the scheme X a strictly local X-scheme
$$\bar{X}(\xi) = \operatorname{Spec} \mathcal{O}_{X,\xi},$$
which in fact depends only on the geometric point above the image x of ξ defined by the separable closure $\bar{k(x)}$ of $k(x)$ in $\Omega = k(\xi)$. We shall often restrict below to geometric points $\xi = \operatorname{Spec} \Omega$ which are *separable algebraic* over X, i.e. such that Ω is a separable closure of $k(x)$, in other words $\xi = \bar{x}$. Call an X-scheme Z a *strict localization of* X if it is X-isomorphic to a scheme of the form $\operatorname{Spec} \mathcal{O}_{X,\bar{x}}$. Then
$$\xi \longmapsto \operatorname{Spec} \mathcal{O}_{X,\xi} = \bar{X}(\xi)$$
is an equivalence from the category of separable algebraic geometric points over X, with only isomorphisms as morphisms, to the category of X-schemes which are strict localizations of X, again with only isomorphisms as morphisms.

The latter assertion is no longer true if one allows all X-morphisms, because in the second category there may be X-morphisms which are not isomorphisms. We therefore make the following definition.

Definition 7.2. Let ξ and ξ' be two geometric points of the scheme X. A *specialization arrow* from ξ' to ξ is an X-morphism between the corresponding strict localizations
$$\bar{X}(\xi') \longrightarrow \bar{X}(\xi).$$
We say that ξ is a specialization of ξ', or that ξ' is a generization of ξ, if there exists a specialization arrow from ξ' to ξ.

Specialization arrows compose in the evident way. Thus, taking specialization arrows as morphisms, the geometric points of X form a category, equivalent, as is the full subcategory of separable algebraic geometric points over X, to the full subcategory of $(\mathrm{Sch})_{/X}$ consisting of the strict localizations of X.

Lemma 7.3. Let X be a scheme, let Z be an X-scheme isomorphic to a pseudo-filtered projective limit of étale X-schemes X_i with affine transition morphisms (Exposé VII, 5.1), let ξ' be a geometric point of X, and let $Z' = \bar{X}(\xi')$ be the corresponding strict localization. Then:

a) the restriction map
$$\operatorname{Hom}_X(Z', Z) \longrightarrow \operatorname{Hom}_X(\xi', Z)$$
is bijective;

b) the two sides are nonempty if and only if the image x' of ξ' in X lies in the image of Z;

c) if T is a Z-scheme, then T is a strict localization of Z if and only if it is a strict localization of X.

Proof. For (a), one immediately reduces to the case where Z is one of the X_i, i.e. where Z is étale over X; after the base change $Z' \to X$ one may suppose $Z' \simeq X$, and the conclusion follows from Definition 4.2(ii). For (b), note that if x' is the image of a point z of Z, then $k(z)$ is necessarily a separable algebraic extension of $k(x')$, hence admits a $k(x')$-homomorphism into $k(\xi')$. Assertion (c) is an easy exercise left to the reader.

Definitions 7.2 and Lemma 7.3 imply the following.

Proposition 7.4. Let ξ and ξ' be two geometric points of the scheme X. Restriction defines a bijection from the set $\mathrm{Hom}_X(\bar{X}(\xi'), \bar{X}(\xi))$ of specialization arrows from ξ' to ξ, to the set of X-morphisms from ξ' to $\bar{X}(\xi)$.

Corollary 7.5. The geometric point ξ is a specialization of ξ' if and only if the same is true for their images x, x' in X, i.e. if and only if x belongs to the closure of $\{x'\}$.

Since $\operatorname{Spec} \mathcal{O}_{X,\xi} \to \operatorname{Spec} \mathcal{O}_{X,x}$ is faithfully flat (4.4), it is surjective, and one only has to apply the second assertion of Lemma 7.3.

Corollary 7.6. For every scheme X, let $\mathcal{P}t(X)$ be the category of geometric points over X, or equivalently of separable algebraic geometric points over X, with specialization arrows as morphisms. Let ξ be a geometric point of a scheme X, and let $\bar{X}(\xi)$ be the corresponding strict localization. Then there is an equivalence of categories

$$\mathcal{P}t(\bar{X}(\xi)) \xrightarrow{\sim} \mathcal{P}t(X)_{/\xi},$$

which associates to every geometric point ξ' of $\bar{X}(\xi)$ the corresponding geometric point over X, together with the specialization morphism to ξ deduced from the structural morphism $\xi' \to \bar{X}(\xi)$ by Proposition 7.4.

In other words, giving a generization ξ' of the geometric point ξ is essentially equivalent to giving a geometric point of $\bar{X}(\xi)$. By Lemma 7.3(c), the corresponding homomorphism $\bar{X}(\xi') \to \bar{X}(\xi)$ makes $\bar{X}(\xi')$ the strict localization of $\bar{X}(\xi)$ relative to ξ'.

7.7

For every geometric point ξ of X and every sheaf F on X, interpret the fibre F_ξ as $\Gamma(\bar{X}(\xi), \bar{F}(\xi))$, where $\bar{F}(\xi)$ is the inverse image of F on $\bar{X}(\xi)$ (Corollary 4.8). Then every specialization arrow

$$u : \xi' \longrightarrow \xi$$

induces a homomorphism, functorial in F,

$$u^* : F_\xi \longrightarrow F_{\xi'},$$

called the *specialization homomorphism* associated with u. It is clear from the transitivity of inverse images of sheaves that for a composite specialization arrow one has

$$(wu)^* = u^* w^*.$$

7.8

Recall that if $\mathcal{E}$ is a topos, a *fiber functor*, or by abuse of language a point of $\mathcal{E}$, is a morphism from the final topos (**Set**) to $\mathcal{E}$, i.e. a functor

$$\varphi : \mathcal{E} \longrightarrow (\mathbf{Set})$$

which commutes with arbitrary inductive limits and finite projective limits. The set of fiber functors of $\mathcal{E}$ is regarded as the set of objects of a full subcategory of $\mathrm{Hom}(\mathcal{E}, (\mathbf{Set}))$, called the *category of fiber functors* of the topos $\mathcal{E}$; its opposite is called the *category of points of* $\mathcal{E}$, denoted $\mathcal{P}t(\mathcal{E})$, cf. IV, 6.1.

When $\mathcal{E} = \widetilde{X_{\text{ét}}}$, where X is a scheme, we defined in 3.3 and 7.7 a functor

$$\mathcal{P}t(X) \longrightarrow \mathcal{P}t(\widetilde{X_{\text{ét}}}). \tag{*}$$

Here the left hand side is defined in Corollary 7.6. With this notation we have:

Theorem 7.9. Let X be a scheme. The functor $(*)$ is an equivalence from the category of geometric points over X, with specialization arrows as morphisms, to the category of points of the étale topos $\widetilde{X_{\text{ét}}}$, the latter being the opposite of the category of fiber functors on $\widetilde{X_{\text{ét}}}$.

Since this theorem will not be used later in the seminar, we give only a sketch of proof, in which we take some liberties with questions of universes.

a) *The functor in question is fully faithful.* For every $X \in \mathrm{Ob}\, X_{\text{ét}}$, let $X^\vee : \widetilde{X_{\text{ét}}} \to (\mathbf{Set})$ be defined by

$$X^\vee(F) = \mathrm{Hom}(X, F) = F(X).$$

Note that the fiber functor

$$\mathcal{E}_\xi : F \longmapsto F_\xi$$

may be written as

$$\mathcal{E}_\xi \simeq \varinjlim_i X_i^\vee,$$

where the X_i are affine schemes étale over X, indexed by a certain filtering category (cf. 4.3 and 4.5). Thus for every functor $\varphi : \widetilde{X_{\text{ét}}} \to (\mathbf{Set})$,

$$\mathrm{Hom}(\mathcal{E}_\xi, \varphi) \simeq \varprojlim_i \mathrm{Hom}(X_i^\vee, \varphi).$$

On the other hand, one has the well-known bifunctorial isomorphism

$$\mathrm{Hom}(X_i^\vee, \varphi) \simeq \varphi(X_i).$$

When φ is of the form $\mathcal{E}_{\xi'}$, hence

$$\varphi \simeq \varinjlim_j (X'_j)^\vee,$$

there is a natural bijection

$$\mathrm{Hom}(\mathcal{E}_\xi, \mathcal{E}_{\xi'}) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_X(X'_j, X_i). \tag{*}$$

On the other hand, in the category of schemes,

$$\bar{X}(\xi) = \varprojlim_i X_i, \qquad \bar{X}(\xi') = \varprojlim_j X'_j,$$

so that

$$\mathrm{Hom}_X(\bar{X}(\xi'), \bar{X}(\xi)) \simeq \varprojlim_i \mathrm{Hom}_X(\bar{X}(\xi'), X_i).$$

Since X_i is locally of finite presentation over X,

$$\mathrm{Hom}_X(\bar{X}(\xi'), X_i) \simeq \varinjlim_j \mathrm{Hom}_X(X'_j, X_i),$$

and therefore

$$\mathrm{Hom}_X(\bar{X}(\xi'), \bar{X}(\xi)) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}_X(X'_j, X_i). \tag{**}$$

Comparing $(*)$ and $(**)$ gives the desired conclusion, up to a compatibility check left to the reader.

b) *The functor in question is essentially surjective.*

7.9.1

First observe that if C is a site in which finite projective limits are representable, and $\widetilde{C} = \mathcal{E}$ is the corresponding topos, then every fiber functor φ on $\mathcal{E}$ can be represented as a filtered inductive limit of functors of the form

$$\check{Y}(F) = F(Y) = \mathrm{Hom}(\widetilde{Y}, F), \qquad Y \in \mathrm{Ob}\, C,$$

where $\widetilde{Y}$ is the sheaf associated with Y, as follows. Consider the category $C_{/\varphi}$ of pairs (Y, ξ), with $Y \in \mathrm{Ob}\, C$ and $\xi \in \varphi(\widetilde{Y})$, morphisms being defined in the evident way. There is an evident homomorphism

$$\varinjlim_{(Y,\xi)\in C^{\circ}_{/\varphi}} \check{Y} \longrightarrow \varphi. \tag{*}$$

Indeed,

$$\mathrm{Hom}\left(\varinjlim_{C_{/\varphi}} \check{Y}, \varphi\right) \simeq \varprojlim_{C_{/\varphi}} \mathrm{Hom}(\check{Y}, \varphi) \simeq \varprojlim_{C_{/\varphi}} \varphi(\widetilde{Y}).$$

There is a canonical element of $\varprojlim_{C_{/\varphi}} \varphi(\widetilde{Y})$, assigning to each (Y, ξ) the element $\xi \in \varphi(Y)$. Since finite projective limits exist in C, and the functor $Y \mapsto \varphi(\widetilde{Y})$ commutes with them, finite projective limits exist in $C_{/\varphi}$. A fortiori $C^{\circ}_{/\varphi}$ is filtering. Using the fact that every sheaf F is an inductive limit of sheaves of the form $\widetilde{Y}$, and that φ commutes with inductive limits, one easily concludes that $(*)$ is bijective, giving the asserted representation.

7.9.2

Also note that if $\mathcal{E}$ is a topos and Y an object of $\mathcal{E}$, giving a fiber functor ψ for $\mathcal{E}_{/Y}$ is equivalent to giving a pair (φ, ξ), where φ is a fiber functor for $\mathcal{E}$ and $\xi \in \varphi(Y)$. To $\psi : \mathcal{E}_{/Y} \to (\mathbf{Set})$ one associates (φ, ξ), where φ is the composite $Z \mapsto \varphi(Z) = \psi(Z \times Y)$, and where $\xi \in \varphi(Y) = \psi(Y \times Y)$ is the image of the unique element of $\psi(Y)$ by the diagonal morphism of $Y \times Y$.

7.9.3

Return now to the site $C = X_{\text{ét}}$, and let $\varphi : \widetilde{X_{\text{ét}}} \to (\mathbf{Set})$ be a fiber functor on the étale topos of X. Restricting φ to the full subcategory of $\widetilde{X_{\text{ét}}}$ formed by the opens of this topos, or equivalently by Proposition 6.1 by the opens U of X, gives a map

$$\varphi_0 : \mathcal{U}(X) \longrightarrow \{0, 1\}$$

commuting with arbitrary suprema and finite infima. Here $\varphi(U)$ is either empty or reduced to one point, and $\varphi_0(U)$ is defined to be 0 or 1 accordingly.

Using the fact that every irreducible closed subset of X has a unique generic point, one sees easily that φ_0 is defined by a unique point $x \in X$, by the condition

$$\varphi_0(U) = 1 \quad \text{if and only if} \quad x \in U,$$

i.e.

$$\varphi(U) = \varnothing \quad \Longleftrightarrow \quad x \notin U.$$

Let $F \in \operatorname{Ob} \widetilde{X_{\text{ét}}}$. The image of F in the final sheaf X is an open U, and since

$$\varphi(F) \longrightarrow \varphi(U)$$

is surjective, $\varphi(F) \neq \varnothing$ if and only if $\varphi(U) \neq \varnothing$, i.e. if and only if $x \in U$. Let $C_{/\varphi}$ be the category of pairs (X', ξ), where $X' \in \operatorname{Ob} X_{\text{ét}}$ and $\xi \in \varphi(X')$. We noted in 7.9.1 that $C^{\circ}_{/\varphi}$ is connected and filtering. Moreover, by 7.9.2 and the preceding discussion applied to X' in place of X, to an object (X', ξ) there is associated a unique point $x' \in X'$ such that, for an étale morphism $X'' \to X'$, ξ lies in the image of $\varphi(X'') \to \varphi(X')$ if and only if x' lies in the image of X''. It follows that the scheme $\varprojlim X'$, taken over the category C of the (X', ξ), exists and is a strict localization Z of X, corresponding to a geometric point ξ and to a fiber functor $\mathcal{E}_\xi$. For every sheaf F,

$$\mathcal{E}_\xi(F) \simeq \varinjlim_{C_\varphi} F(X').$$

By 7.9.1, φ is isomorphic to $\mathcal{E}_\xi$, completing the proof of Theorem 7.9.

8. Two spectral sequences for integral morphisms

Proposition 8.1. Let $f : X \to Y$ be an integral surjective morphism, and let F be an abelian sheaf on Y. For every integer i, let $\mathcal{H}^i(F)$ be the presheaf on $(\mathrm{Sch})_{/Y}$ defined by

$$\mathcal{H}^i(F)(Z) = H^i(Z, F_Z),$$

where F_Z is the inverse image of F on Z. Then there is a spectral sequence, functorial in F,

$$H^*(X, F) \Longleftarrow E_2^{pq} = H^p(X/Y, \mathcal{H}^q(F)),$$

the *descent spectral sequence.*

Of course, for a presheaf G on $(\mathrm{Sch})_{/Y}$, the symbol $H^p(X/Y, G)$ denotes the p-th relative Cech cohomology group, defined by the complex $C^n(X/Y, G) = G(X^{n+1})$, where X^m denotes the m-fold power in $(\mathrm{Sch})_{/Y}$. To prove Proposition 8.1, let $p_n : X^{n+1} \to Y$ be the projection, and put

$$A^n = p_{n*}(\mathbb{Z}_{X^{n+1}}),$$

where $\mathbb{Z}_{X^{n+1}}$ denotes the constant sheaf $\mathbb{Z}$ on X^{n+1}. The A^n are the components of a simplicial abelian sheaf on Y, hence of a complex A^* of abelian sheaves on Y. There is an evident homomorphism

$$\mathbb{Z}_Y \longrightarrow A^0. \tag{*}$$

Lemma 8.2. The complex A^*, endowed with the homomorphism $(*)$, is a resolution of $\mathbb{Z}_Y$. More generally, for every abelian sheaf F on Y, the complex $A^* \otimes_{\mathbb{Z}} F$ is a resolution of F.

Proof. We may assume that Y is affine, say $Y = \operatorname{Spec} A$, and $X = \operatorname{Spec} B$, with B an integral A-algebra. Write $B = \varinjlim_i B_i$, where the B_i run over the finite-type, hence finite, A-subalgebras of B. Then $X = \varprojlim X_i$, with $X_i = \operatorname{Spec} B_i$. Using Exposé VII, 5.11, the augmented complex $A^*(X/Y)$ is the inductive limit of the augmented complexes $A^*(X_i/Y)$. This reduces us to the case where f is finite. It is enough to prove that for every geometric point $\bar{y}$ of Y, the complex $A^*_{\bar{y}}$ is a resolution of $\mathbb{Z}_{\bar{y}}$, and remains so after tensoring with F. If $X_{\bar{y}}$ is the fibre of X at $\bar{y}$, Proposition 5.5 shows that $A^*_{\bar{y}}$ is just the analogous complex $A^*(X_{\bar{y}}/\bar{y})$. This reduces us to the case where Y is the spectrum of a separably closed field k. Using Examples 1.3(a) and (b), we may even assume k algebraically closed and X reduced; thus X is of the form I_Y, where I is a finite set. Then $A^* \otimes F$ identifies with the trivial cochain complex of the indexing set I, with coefficients in $F_{\bar{y}}$, and is indeed a resolution of $F_{\bar{y}}$.

As a consequence of Lemma 8.2 we obtain a spectral sequence

$$H^*(Y, F) \Longleftarrow E_2^{pq} = H^p(H^q(Y, A^* \otimes F)),$$

and it remains to identify the initial term. There is a canonical homomorphism

$$A^n \otimes F \longrightarrow p_{n*}p_n^*(F),$$

which is an isomorphism, again by reduction to the finite case and passage to fibres. Using Proposition 5.5, it follows that

$$H^q(Y, A^n \otimes F) \simeq H^q(X^{n+1}, p_n^* F) = \mathcal{H}^q(F)(X^{n+1}),$$

giving the announced initial term in Proposition 8.1.

Remark 8.3.

a) If Y is covered by a locally finite family of closed subsets Y_i, and if $X = \coprod_i Y_i$, then the canonical morphism $f : X \to Y$ is finite, and the initial term of the spectral sequence in Proposition 8.1 is described as the cohomology of the complex defined by the groups $H^q(Y_{i_0 \ldots i_p}, F|Y_{i_0 \ldots i_p})$, where $Y_{i_0 \ldots i_p} = Y_{i_0} \cap \cdots \cap Y_{i_p}$. This is the analogue of Leray's spectral sequence for a locally finite closed covering of an ordinary topological space (TF, Chapter II, 5.2.4). The latter can also be generalized to a spectral sequence relative to a "finite" morphism, i.e. a proper morphism with finite fibres, by proceeding as in Proposition 8.1.

b) Note the analogy between the spectral sequence of Proposition 8.1 and Leray's spectral sequence for a covering (X_i) of Y, here by X_i étale over Y. The latter would be obtained formally by writing the spectral sequence of Proposition 8.1 for $X = \coprod_i X_i$. It is likely that these two spectral sequences admit a common generalization, valid whenever one has a family of morphisms $(X_i \to Y)$ which is a universal effective descent family for the fibred category of étale sheaves over a variable base scheme; compare No. 9 below. The analogous question also arises in ordinary topology, concerning a common generalization of the two kinds of Leray spectral sequences for coverings, assumed either open or locally finite closed. Since these lines were written, P. Deligne has developed a general theory of descent spectral sequences; see Exposé V bis.

Proposition 8.4. Let Y be a scheme, let π be a profinite group, let $(\pi_i)_i$ be the projective system of its finite discrete quotient groups, let $(X_i)_{i\in I}$ be a projective system of principal coverings of Y, with groups π_i, the transition homomorphisms $X_j \to X_i$ being compatible with the homomorphisms $\pi_j \to \pi_i$ on the operator groups, and put $X = \varprojlim X_i$ (cf. Exposé VII, 5.1), so that π acts on the Y-scheme X. Let F be an abelian sheaf on Y. Then there is a Hochschild-Serre spectral sequence, functorial in F,

$$H^*(Y,F) \Longleftarrow E_2^{pq} = H^p\big(\pi, \varinjlim_i H^q(X_i, F_{X_i})\big),$$

where F_{X_i} is the inverse image of F on X_i, and $H^p(\pi,-)$ denotes Galois cohomology.

The term E_2^{pq} displayed here is also, by definition of $H^p(\pi,-)$ and by transitivity of inductive limits, isomorphic to

$$E_2^{pq} = \varinjlim_i H^p(\pi_i, H^q(X_i, F_{X_i})).$$

It is therefore enough to find an inductive system of spectral sequences, depending on i,

$$H^*(Y,F) \Longleftarrow {}^iE_2^{pq} = H^p(\pi_i, H^q(X_i, F_{X_i})).$$

This reduces us to defining the spectral sequence when π is finite. Then $f : X \to Y$ is a covering morphism in $Y_{\text{ét}}$, so we may write the Leray spectral sequence of this morphism. Taking into account the canonical isomorphisms

$$X^n \simeq X \times^G G^n,$$

a well-known calculation shows that for every presheaf $\mathcal{H}$ on $X_{\text{ét}}$ which transforms sums into products, $C^*(X/Y,\mathcal{H})$ is precisely the complex of homogeneous cochains of G with coefficients in $\mathcal{H}(X)$. This gives the asserted form of the initial term.

A variant of this proof, avoiding any calculation, consists in viewing $E \mapsto X \times^\pi E$ as a morphism from the site of finite π-sets to the étale site of Y, and in writing the Leray spectral sequence of this morphism. Finally, when π is finite, the Hochschild-Serre spectral sequence can also be regarded as a special case of Proposition 8.1 relative to the morphism $f : X \to Y$.

Corollary 8.5. Assume Y quasi-compact and quasi-separated. Then the spectral sequence of Proposition 8.4 is written

$$H^*(Y,F) \Longleftarrow E_2^{pq} = H^p(\pi, H^q(X, F_X)).$$

Indeed, by Exposé VII, 5.8, there are canonical isomorphisms

$$\varinjlim_i H^q(X_i, F_{X_i}) \simeq H^q(X, F_X).$$

Corollary 8.6. Let Y be a henselian local scheme with closed point y, let F be an abelian sheaf on Y, and let F_0 be the induced sheaf on $Y_0 = \operatorname{Spec} k(y)$. Then the canonical homomorphisms

$$H^n(Y,F) \longrightarrow H^n(Y_0, F_0)$$

are isomorphisms.

Indeed, let $\bar{y}$ be a geometric point over y, corresponding to a separable closure $k(\bar{y})$ of $k(y)$. Since Y is henselian, the strict localization X of Y at $\bar{y}$ is the projective limit of the connected pointed Galois étale coverings X_i. Thus the hypotheses of Corollary 8.5 apply. Since $H^q(X, F_X) = 0$ for $q > 0$ by Corollary 4.7, one obtains isomorphisms

$$H^n(Y, F) \xleftarrow{\sim} H^n(\pi, F(X)),$$

where π is the Galois group of X over Y, isomorphic to that of $\bar{k(y)}$ over $k = k(y)$. Similarly, or by Corollary 2.3,

$$H^n(Y_0, F_0) \xleftarrow{\sim} H^n(\pi, F_0(X_0)),$$

where $X_0 = X \times_Y Y_0 \simeq \operatorname{Spec} k(\bar{y})$. The restriction homomorphism $F(X) \to F_0(X_0)$ is an isomorphism by Corollary 4.8, and the conclusion follows.

9. Descent of étale sheaves

This section will not be used later in the seminar and may be omitted on a first reading.

Proposition 9.1. Let $f : X' \to X$ be a surjective, respectively universally submersive (SGA 1, IX, 2.1), morphism of schemes. Then the functor $f^* : X_{\text{ét}} \to X'_{\text{ét}}$ is faithful and conservative (cf. Corollary 3.6), respectively induces a fully faithful functor from the category of sheaves on $X_{\text{ét}}$ to the category of sheaves on $X'_{\text{ét}}$ endowed with descent data relative to $f : X' \to X$.

The first point follows immediately from the corresponding assertions on the fibres (3.6, 3.7). The second, which can also be stated by saying that f is a descent morphism relative to the fibred category of étale sheaves over variable schemes, means that for two sheaves F, G on X, the natural diagram

$$\operatorname{Hom}(F, G) \longrightarrow \operatorname{Hom}(F', G') \rightrightarrows \operatorname{Hom}(F'', G'') \tag{*}$$

is exact, where F' and G', respectively F'' and G'', are the inverse images of F and G on X', respectively on $X'' = X' \times_X X'$. Taking F to be the final sheaf gives:

Corollary 9.2. Let $f : X' \to X$ be universally submersive. Then for every sheaf F on X, the diagram

$$\Gamma(X, F) \longrightarrow \Gamma(X', F') \rightrightarrows \Gamma(X'', F'')$$

is exact, where $X'' = X' \times_X X'$, and F', F'' are the inverse images of F on X', X''.

We prove Proposition 9.1 using the following lemma.

Lemma 9.3. Assume f universally submersive. Let F be a sheaf on X, denote by $S(F)$ the set of subsheaves of F, and define $S(F')$, $S(F'')$ similarly. Then the natural diagram of maps

$$S(F) \longrightarrow S(F') \rightrightarrows S(F'')$$

is exact.

Proof. The injectivity of $S(F) \to S(F')$ follows immediately from the fact that f^* is conservative. Indeed, if F_i, $i = 1, 2$, are two subsheaves of F such that $f^*(F_1) = f^*(F_2)$, then the inclusions $F_3 = F_1 \cap F_2 \to F_i$ are isomorphisms after applying f^*, hence are isomorphisms, and so $F_1 = F_2$.

It remains to prove that if G' is a subsheaf of F' such that $\mathrm{pr}_1^*(G') = \mathrm{pr}_2^*(G')$, then G' is the inverse image of a subsheaf of F. Choose an epimorphism $F_1 \to F$ in $X_{\text{ét}}$, with F_1 representable by an étale scheme over X. Let $F_2 = F_1 \times_F F_1$, and similarly define F_1', F_1'', F_2', F_2''. These sheaves give a diagram of sets

$$\begin{tikzcd}
S(F) \arrow[r] \arrow[d] & S(F') \arrow[r] \arrow[d] & S(F'') \arrow[d] \\
S(F_1) \arrow[r] \arrow[d, shift left] \arrow[d, shift right] & S(F_1') \arrow[r, shift left] \arrow[r, shift right] \arrow[d, shift left] \arrow[d, shift right] & S(F_1'') \arrow[d, shift left] \arrow[d, shift right] \\
S(F_2) \arrow[r] & S(F_2') \arrow[r, shift left] \arrow[r, shift right] & S(F_2''),
\end{tikzcd}$$

in which the columns are exact by the exactness properties in the categories of sheaves on X, X', and X'', respectively. A standard diagram chase shows that, to prove exactness of the first row, it is enough to prove it for the second and third rows. For the second row this follows from the fact that F_1 is representable, $X' \to X$ is universally submersive, and SGA 1, IX, 2.3. The sheaf F_2 is also representable: it is a subsheaf of $F_1 \times_X F_1$, which is representable, and one applies Proposition 6.1. Thus the third row is exact as well. $\square$

We now prove Proposition 9.1, namely that every morphism $u' : F' \to G'$, compatible with the descent data on F' and G', comes from a morphism $u : F \to G$. Put $H = F \times G$, so $H' = F' \times G'$ and $H'' = F'' \times G''$. The graph of u' is a subsheaf Γ' of H', whose two inverse images are equal to the graph of the same morphism $u'' : F'' \to G''$. Hence, by Lemma 9.3, Γ' comes from a subsheaf Γ of H. I claim that Γ is the graph of a morphism $u : F \to G$, i.e. that the morphism $p : \Gamma \to F$ induced by pr_1 is an isomorphism. Indeed, it becomes an isomorphism after base change by $X' \to X$, and we apply the part of Proposition 9.1 already proved. The morphism $u : F \to G$ then has the desired property. $\square$

Theorem 9.4. Let $f : X' \to X$ be a surjective morphism of schemes. Assume that one of the following hypotheses holds:

a) f is integral;

b) f is proper;

c) f is flat and locally of finite presentation;

d) X is discrete, for example the spectrum of a field.

Then f is an *effective descent morphism* for the fibred category $\mathcal{F}$ over (Sch) of étale sheaves on variable schemes. That is, f^* induces an equivalence from the category of sheaves on X to the category of sheaves on X' endowed with descent data relative to $f : X' \to X$.

9.4.1

Let F' be a sheaf on X', endowed with descent data relative to $f : X' \to X$. Define a functor

$$G : X_{\text{ét}}^{\circ} \longrightarrow (\mathbf{Set})$$

by

$$G(Y) = \mathrm{Ker}\big(F'(Y') \rightrightarrows F''(Y'')\big).$$

One immediately checks that G is a sheaf for the étale topology. Moreover there is an injective canonical homomorphism $G \to f_*(F')$, hence a homomorphism $f^*(G) \to F'$, and the latter is evidently compatible with descent data. It remains to examine whether this homomorphism is an isomorphism.

Note that G is also defined by exactness of

$$G \longrightarrow f_*(F') \rightrightarrows g_*(F''),$$

so for every base change $Y \xrightarrow{h} X$ which commutes with the formation of $f_*(F')$ and $g_*(F'')$, the sheaf $h^*(G) = G_Y$ identifies with the sheaf descended from $F'_{Y'}$ by $Y' \to Y$; and of course the homomorphism $f_Y^*(G_Y) \to (F_Y)'$ is obtained from $f^*(G) \to F'$ by base change. Now assume that the base change $Y = X' \to X$ commutes with the formation of $f_*(F')$ and $g_*(F'')$. Since $Y' = X' \times_X X'$ has a section over $Y = X'$, the morphism $Y' \to Y$ is an effective descent morphism for every fibred category, in particular for $\mathcal{F}$. It follows that $f_Y^*(G_Y) \to (F_Y)'$ is an isomorphism, and therefore $f^*(G) \to F'$ becomes an isomorphism after base change by $Y' = X' \times_X X' \to X'$. As the latter morphism has a section, $f^*(G) \to F'$ is itself an isomorphism. Thus the descent datum on F' is effective.

9.4.2

This proves Theorem 9.4 in case (a), by Corollary 5.6. Case (b) follows similarly from the fact that if f is proper then f_* commutes with every base change $Y' \to Y$; this will be proved in a later exposé, XII, 5.1(i).

9.4.3

In case (c), the question is local on X. We may assume X affine and that there exists an affine scheme X_1', of finite presentation, quasi-finite and faithfully flat over X, together with an X-morphism $X_1' \to X'$. This reduces us to the case where f is quasi-finite. After further localization on X, this time in the étale topology, there exists an open X_1' of X' such that $X_1' \to X$ is finite and surjective, reducing us to the finite surjective case already treated in (a).

9.4.4

In case (d), after localizing on X, we may suppose that X is a single point, and, taking Theorem 1.1 into account, even that it is the spectrum of a field k. Moreover f is universally open (EGA IV, 2.4.9), so the hypotheses of Proposition 9.1 apply, and the argument of 9.4.1 still applies, with a base change by $Y = \operatorname{Spec} k(x)$, for $x \in X$. In that case it is still true that f_* commutes with the base change $Y \to X$, as will be seen later (XVI, 1.4 and 1.5).

This completes the proof of Theorem 9.4.

SGA 4, Exposé IX

Constructible Sheaves; Cohomology of an Algebraic Curve

M. Artin

0. Introduction

Starting with the present exposé, in contrast with the preceding exposés and with the classical cohomological theory of topological spaces, we are forced, for the validity of the essential statements, to restrict to *torsion* sheaves, and often more precisely to torsion sheaves prime to the residual characteristics.

Let us note that in fact a "reasonable" theory of cohomology with integral, or even real, coefficients for varieties over an algebraically closed field of characteristic $p \neq 0$, analogous to the classical theory for $k = \mathbb{C}$, does not exist, as Serre observed. More precisely, there is no contravariant functor H^1, defined for instance on the category of smooth projective schemes over an algebraically closed field K of characteristic $p > 0$, with values in finite-dimensional vector spaces over $\mathbb{R}$, or over a subfield of $\mathbb{R}$, commuting with products, i.e. such that

$$H^1(X \times Y) \xleftarrow{\sim} H^1(X) \times H^1(Y),$$

and such that $\dim H^1(X) = 2$ when X is a connected curve of genus 1. Indeed, it would follow that, if X is an abelian variety over k, the map $u \mapsto u^1$ from $\mathrm{End}(X)$ to $\mathrm{End}(H^1(X))$ is additive, hence is a representation of the opposite ring of $\mathrm{End}(X)$. But in characteristic p, there exist elliptic curves X whose endomorphism ring is a maximal order A in a quaternion algebra over $\mathbb{Z}$ ([3], p. 198), and such an algebra plainly has no representation on a vector space of dimension 2 over $\mathbb{R}$, since $A \otimes_{\mathbb{Z}} \mathbb{R}$ is a division ring, namely Hamilton's quaternions.

1. The formalism of torsion sheaves

Let T be a topos and F an abelian sheaf on T. We can multiply by n in F, for $n \in \mathbb{Z}$; this multiplication is induced by the analogous operation in $(\mathbf{Ab})$. Denote by ${}_nF$ the kernel of this multiplication. Thus ${}_nF(X)$, for $X \in \mathrm{Ob}\, T$, is the group of sections of $F(X)$ whose order divides n. Let $\mathcal{P}$ denote the set of all prime numbers.

Definition 1.1. Let p be a set of prime numbers. We say that F is a sheaf of p-torsion, or a p-sheaf, if the canonical morphism

$$\varinjlim_{n} {}_nF \longrightarrow F$$

is bijective, where n runs through the positive integers such that $\mathrm{ass}(n) \subset p$, i.e. such that all prime divisors of n belong to p. If $p = \mathcal{P}$ is the set of all prime numbers, we simply say that F is torsion.

Proposition 1.2.

(i) The sheaf F is of p-torsion if and only if F is the sheaf associated with a presheaf with values in abelian groups of p-torsion. One may take

$$P = \overset{\text{presheaves}}{\varinjlim_{\mathrm{ass}(n) \subset p}} {}_nF.$$

(ii) If F is of p-torsion and $X \in \mathrm{Ob}\, T$ is quasi-compact, then $F(X)$ is a group of p-torsion.

(iii) If T is locally of finite type (VI, 1.1), then F is of p-torsion if and only if $F(X)$ is a group of p-torsion for each quasi-compact $X \in \mathrm{Ob}\, T$. In this case one has

$$\varinjlim_{\mathrm{ass}(n) \subset p} H^q(T/X; {}_nF) \xrightarrow{\sim} H^q(T/X; F)$$

for every quasi-compact X and every q.

(iv) If $u : T \to T'$ is a morphism of topoi and if F' is of p-torsion on T', then the inverse image u^*F' is a sheaf of p-torsion on T.

(v) If $u : T \to T'$ is a morphism of topoi, with T and T' locally of finite type and u quasi-compact, and if F is a sheaf of p-torsion on T, then the sheaves $R^q u_* F$ are of p-torsion on T', for all q.

Proof. For (i), if a sheaf F is of p-torsion, then F is plainly the sheaf associated with the presheaf P whose group of sections $P(X)$ is the subgroup of p-torsion elements of $F(X)$. Conversely, let $F = aP$ be the sheaf associated with P, where P takes values in abelian groups of p-torsion. From the exact sequence

$$0 \longrightarrow {}_nP \longrightarrow P \xrightarrow{n} P$$

one obtains an exact sequence

$$0 \longrightarrow a({}_nP) \longrightarrow F \xrightarrow{n} F,$$

hence $a({}_nP) = {}_nF$. Since the functor a commutes with inductive limits, the isomorphism

$$\varinjlim_{\mathrm{ass}(n)\subset p}^{\text{presheaves}} {}_nP \xrightarrow{\sim} P$$

gives

$$\varinjlim_{\mathrm{ass}(n)\subset p} {}_nF \xrightarrow{\sim} F.$$

For (ii), put $P = \varinjlim^{\text{presheaves}} {}_nF$. Then $P \subset F$, so P is a separated presheaf, and

$$aP(X) = \varinjlim \ker\Big(\prod_i P(X_i) \rightrightarrows \cdots\Big),$$

where, as usual, the limit is taken over the covering families $\{X_i \to X\}$. Since X is quasi-compact, it is enough to use finite covering families. For such families, $\prod_i P(X_i)$ is of p-torsion, and hence so is its kernel. Thus $aP(X) = F(X)$ is of p-torsion.

For (iii), to check that $\varinjlim {}_nF \to F$ is bijective, it is enough to check values on each X in a family of generators. Thus the first assertion reduces to (ii). The displayed isomorphism follows from the same passage to inductive limits in cohomology for locally finite type topoi. Assertion (iv) follows from (i), since $u^*(aP') = a(u^\bullet P')$. Assertion (v) is an easy consequence of (iii).

We shall also need a non-abelian variant. There are technical difficulties in the definition, and we content ourselves with giving the definition for the étale topology of a scheme.

Definition 1.3. Let p be a set of prime numbers. A group G is called an *ind-p-group* if every finite subset of G generates a finite subgroup of order n, with $\mathrm{ass}(n) \subset p$. Equivalently, the finite subgroups G_α of order n_α, with $\mathrm{ass}(n_\alpha) \subset p$, form a filtered system and $G = \varinjlim G_\alpha$. If $p = \mathcal{P}$, one says *ind-finite group* instead of ind-p-group.

One verifies immediately:

Proposition 1.4.

(i) A subgroup and a quotient of an ind-p-group is again an ind-p-group.

(ii) A filtered inductive limit of ind-p-groups is an ind-p-group.

(iii) A finite projective limit of ind-p-groups is an ind-p-group.

Definition 1.5. Let X be a scheme. A sheaf F of groups on X is called a *sheaf of ind-p-groups*, respectively a sheaf of *ind-finite groups*, if for every étale $U \to X$, with U quasi-compact, the group $F(U)$ is an ind-p-group, respectively an ind-finite group.

Proposition 1.6. Let F be a sheaf of groups on a scheme X.

(i) The sheaf F is a sheaf of ind-p-groups if and only if for every geometric point ξ of X, the fibre F_ξ is an ind-p-group.

(ii) If $f : X \to X'$ is a morphism and F' is a sheaf of ind-p-groups on X', then f^*F' is a sheaf of ind-p-groups.

(iii) If $f : X \to X'$ is quasi-compact and F is a sheaf of ind-p-groups on X, then f_*F is a sheaf of ind-p-groups.

Proof. For (i), suppose first that F is a sheaf of ind-p-groups, and let ξ be a geometric point of X. Then $F_\xi = \varinjlim F(X')$, where X' runs through a pseudo-filtered system of schemes étale over X (VIII, 3.9). The affine, hence quasi-compact, X' form a cofinal system, so F_ξ is an inductive limit of ind-p-groups, hence an ind-p-group by Proposition 1.4(ii). Conversely, assume that every fibre F_ξ is an ind-p-group, and let $U \to X$ be étale with U quasi-compact. Let $S \subset F(U)$ be finite. For every geometric point ξ of U, the image of S in F_ξ generates a finite group. Since a finite group is finitely presented, one easily concludes that there exists an étale U_ξ over U, pointed by ξ, such that S generates a finite group in $F(U_\xi)$ (cf. VIII, 4). A finite number of such U_ξ cover U, say $\{U_1, \dots, U_n\}$. Then the image of S in $\prod F(U_i)$ generates a finite subgroup, and since $F(U) \subset \prod F(U_i)$, the conclusion follows. Assertion (ii) is trivial from (i), and (iii) is immediate.

3. Kummer and Artin–Schreier theories

3.0

We denote by $(\mathbb{G}_m)_X$ the sheaf represented on X by the multiplicative group $\operatorname{Spec} \mathcal{O}_X[t, t^{-1}]$. Thus, for an *étale* morphism $U \to X$, its sections over U are the units $\Gamma(U, \mathcal{O}_U)^*$. Let $n \in \mathbb{N}$. The kernel of the n-th power map on $(\mathbb{G}_m)_X$ is the sheaf of n-th roots of unity, denoted $(\mu_n)_X$. There is therefore an exact sequence at the left

$$0 \longrightarrow (\mu_n)_X \longrightarrow (\mathbb{G}_m)_X \xrightarrow{n} (\mathbb{G}_m)_X. \tag{3.1}$$

If n is invertible on X, that is, prime to all residual characteristics of X, the n-th power map is surjective as a morphism of *étale* sheaves:

Kummer theory 3.2. If n is invertible on X, then

$$0 \longrightarrow (\mu_n)_X \longrightarrow (\mathbb{G}_m)_X \xrightarrow{n} (\mathbb{G}_m)_X \longrightarrow 0$$

is exact.

Indeed, for $u \in (\mathbb{G}_m)_X(U)$, with $U \to X$ *étale*, one must find an *étale* covering on which u becomes an n-th power. Since n is invertible on U, the equation

$$T^n - u = 0$$

is separable over U, so

$$U' = \operatorname{Spec} \mathcal{O}_U[T]/(T^n - u)$$

is *étale* over U. The morphism $U' \to U$ is surjective, hence covering, and on U' the section u is an n-th power.

When n is invertible on X, the sheaf $(\mu_n)_X$ is locally constant, in fact locally isomorphic to $(\mathbb{Z}/n\mathbb{Z})_X$, and is constant in important cases. Kummer theory therefore gives information about the cohomology of X with values in certain constant sheaves. By definition

$$H^0(X, (\mathbb{G}_m)_X) = \Gamma(X, \mathcal{O}_X^*),$$

and in degree one one has the following form of Hilbert's Theorem 90.

Theorem 3.3. There is an isomorphism

$$H^1(X, (\mathbb{G}_m)_X) \xrightarrow{\sim} \operatorname{Pic} X,$$

where $\operatorname{Pic} X$ is the group of isomorphism classes of invertible sheaves on X.

Equivalently, the canonical morphism

$$H^1(X_{\mathrm{Zar}}, (\mathbb{G}_m)_X) \longrightarrow H^1(X_{\text{ét}}, (\mathbb{G}_m)_X)$$

is bijective. This amounts to saying that

$$R^1\epsilon_*(\mathbb{G}_m)_X = 0,$$

where $\epsilon : X_{\text{ét}} \to X_{\mathrm{Zar}}$ is the evident morphism of sites (VIII, 4). Thus one is reduced to the affine case. Since H^1 may be computed by Čech cohomology (V, 2.5), and since for quasi-compact X it is enough to use quasi-compact coverings, the assertion follows at once from descent theory for invertible sheaves (cf. FGA, p. 190, or SGA 1, VIII, 1).

3.3.1 Cartier divisors

Assume that X is noetherian and has no embedded components, and let x_j be the maximal points of X. Let R_j be the local artinian ring of X at x_j, and let $i_j : \operatorname{Spec} R_j \to X$ be the inclusion. There is a canonical injection, both on $X_{\text{ét}}$ and on X_{Zar},

$$(\mathbb{G}_m)_X \longrightarrow \prod_j i_{j*}(\mathbb{G}_m)_{\operatorname{Spec} R_j}.$$

Its cokernel D is called the sheaf of Cartier divisors on $X_{\text{ét}}$, respectively on X_{Zar}. In EGA IV, 21, these are simply called divisors.

Corollary 3.4. Let X be noetherian and without embedded components, let $\epsilon : X_{\text{ét}} \to X_{\mathrm{Zar}}$ be the canonical morphism of sites, and let D be the sheaf of Cartier divisors on $X_{\text{ét}}$. Then $\epsilon_* D$ is the sheaf D_{Zar} of Cartier divisors on X_{Zar}. In particular

$$H^0(X_{\text{ét}}, D) \simeq H^0(X_{\mathrm{Zar}}, D_{\mathrm{Zar}}).$$

This follows by applying ϵ_* to the exact sequence

$$0 \longrightarrow (\mathbb{G}_m)_X \longrightarrow \prod_j i_{j*}(\mathbb{G}_m)_{\operatorname{Spec} R_j} \longrightarrow D \longrightarrow 0,$$

which remains exact because $R^1\epsilon_*(\mathbb{G}_m)_X = 0$ by Theorem 3.3.

In characteristic $p \neq 0$, the following result will sometimes replace Kummer theory for the study of p-torsion coefficients.

Artin–Schreier theory 3.5. If X is of characteristic $p > 0$, there is an exact sequence

$$0 \longrightarrow (\mathbb{Z}/p\mathbb{Z})_X \longrightarrow \mathcal{O}_X \xrightarrow{\wp} \mathcal{O}_X \longrightarrow 0,$$

where $\wp$ is the morphism of additive groups defined by $\wp(f) = f^p - f$.

The kernel of $\wp$ is the sheaf of sections of $\mathcal{O}_X$ which locally lie in the prime field, hence is constant. The map $\wp$ is surjective for the *étale* topology, since the equation $T^p - T - a$, with $a \in \Gamma(U, \mathcal{O}_U)$, is always separable in characteristic p, hence defines an *étale* covering of U. In applying this exact sequence one uses VII, 4.2.

Proposition 3.6.

(i) Let X be a scheme and let $i : x \to X$ be the inclusion of a point. Then

$$H^1(X, i_*(\mathbb{Z})_x) = 0,$$

where $(\mathbb{Z})_x$ denotes the constant sheaf of value $\mathbb{Z}$ on the point x.

(ii) If X is irreducible and if, for every geometric point $\bar{x}$ of X, the strict localization $X_{\bar{x}}$ is irreducible, in other words if X is geometrically unibranch, then

$$H^1(X, (\mathbb{Z})_X) = 0.$$

Proof. For (i), the Leray spectral sequence

$$E_2^{pq} = H^p(X, R^q i_*(\mathbb{Z})_x) \Longrightarrow H^{p+q}(x, (\mathbb{Z})_x)$$

gives an injection $H^1(X, i_*(\mathbb{Z})_x) \to H^1(x, (\mathbb{Z})_x)$. But, since $x = \operatorname{Spec} k(x)$, the group $H^1(x, (\mathbb{Z})_x)$ is zero: this follows from the fact that $H^q(x, F)$, for $q > 0$, is torsion (VII, 2.2 and CG), and from the exact sequences

$$0 \longrightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z} \longrightarrow 0.$$

For (ii), let $i : x \to X$ be the inclusion of the generic point. Under the stated hypotheses the canonical morphism $(\mathbb{Z})_X \to i_*(\mathbb{Z})_x$ is immediately seen to be bijective, and the assertion follows from (i). The converse to this bijectivity is also true.

Remarks 3.7. In Proposition 3.6(i) and (ii), $\mathbb{Z}$ may be replaced by any torsion-free abelian group. On the other hand, (ii) becomes false if the geometrically unibranch hypothesis is dropped; for example take an algebraic curve over a field with an ordinary double point and compute $H^1(X, \mathbb{Z})$.

4. The case of an algebraic curve

4.0

Let X be a noetherian scheme of dimension one. Let $R(X)$ be the ring of rational functions on X, and let $i : \operatorname{Spec} R(X) \to X$ be the inclusion. The ring $R(X)$ is artinian: it is the product of the local rings of X at its maximal points, so one may use the results of VIII, 2.

Let F be an abelian sheaf on $\operatorname{Spec} R(X)$, and consider the sheaves $R^q i_* F$, $q > 0$, on X. It follows at once from VIII, 5.2 that the fibre of $R^q i_* F$ at a geometric point lying over a maximal point of X is zero. Since X has dimension one, such a sheaf is of a very special kind.

Lemma 4.1. Let X be noetherian and let F be a sheaf on X. The following conditions are equivalent; a sheaf satisfying them will be called a skyscraper sheaf.

(i) The fibre of F at every geometric point $\bar{x}$ lying over a non-closed point $x \in X$ is zero.

(ii) Each section $s \in F(X')$, with $X' \to X$ *étale* of finite type, is zero except at finitely many closed points of X'.

(iii) One has an isomorphism
$$F \simeq \bigoplus_x i_{x*} i_x^* F,$$
where x runs through the closed points of X, and $i_x : x \to X$ is the inclusion.

Proof. If (i) holds and $z \in F(X')$, its support is a closed subset of X' formed of closed points, hence is finite because X' is noetherian; this proves (ii). Conversely, (ii) implies (i) by the calculation of fibres (VIII, 3.9). Condition (iii) plainly implies (i), since the functor of fibres commutes with direct sums and the support of $i_{x*} G_x$ is contained in $\{x\}$. Finally, assuming (ii), the canonical morphism
$$F \longrightarrow \prod_x i_{x*} i_x^* F$$
factors through the direct sum; the resulting morphism to $\bigoplus_x i_{x*} i_x^* F$ is checked directly to be bijective.

4.1.1

Applying the lemma to the sheaf $R^q i_* F$ introduced above gives
$$R^q i_* F \simeq \bigoplus_x i_{x*} i_x^* (R^q i_* F),$$
with x running through the closed points of X. By VIII, 5, the fibre of $R^q i_* F$ at a geometric point $\bar{x}$ of X is
$$(R^q i_* F)_{\bar{x}} \simeq H^q(\operatorname{Spec} R(X_{\bar{x}}), F),$$
where $X_{\bar{x}}$ is the strict localization of X at $\bar{x}$, and F also denotes the induced sheaf on $\operatorname{Spec} R(X_{\bar{x}})$. Assume now that the residue field of every closed point of X is separably closed. Then
$$H^p(X, R^q i_* F) \simeq \begin{cases} \bigoplus_x H^q(\operatorname{Spec} R(X_x), F), & p = 0, \\ 0, & p > 0, \end{cases}$$
for $q > 0$, where X_x denotes the strict localization at x.

Corollary 4.2. Let X be a noetherian scheme of dimension one such that, for every closed point x, the residue field $k(x)$ is separably closed. Let F be an abelian sheaf on $\operatorname{Spec} R(X)$. In the Leray spectral sequence
$$E_2^{pq} = H^p(X, R^q i_* F) \Longrightarrow H^{p+q}(\operatorname{Spec} R(X), F)$$

one has

$$E_2^{pq} = 0 \quad (p > 0, q > 0), \qquad E_2^{0q} = \bigoplus_x H^q(\operatorname{Spec} R(X_x), F) \quad (q > 0).$$

Consequently there is an exact sequence

$$\begin{aligned} 0 \to H^1(X, i_*F) \to H^1(\operatorname{Spec} R(X), F) \to \bigoplus_x H^1(\operatorname{Spec} R(X_x), F) \to H^2(X, i_*F) \to \\ \to H^2(\operatorname{Spec} R(X), F) \to \bigoplus_x H^2(\operatorname{Spec} R(X_x), F) \to \cdots . \end{aligned}$$

4.3

Assume further that the ℓ-cohomological dimension of $\operatorname{Spec} R(X)$ is at most one. Let $K^1, \ldots, K^m$ be the residue fields of the artinian ring $R(X)$, equivalently the residue fields at the maximal points of X. By the dictionary of VIII, 2,

$$\operatorname{cd}_\ell(\operatorname{Spec} R(X)) = \sup_i \operatorname{cd}_\ell(G(\overline{K}^i/K^i)).$$

Every residue field K_x of $R(X_x)$ is a separable extension, possibly infinite, of one of the K^i, hence

$$\operatorname{cd}_\ell(G(\overline{K}_x/K_x)) \leq \operatorname{cd}_\ell(G(\overline{K}^i/K^i))$$

by CG II, 4.1, Proposition 10. Applying Corollary 4.2 to an ℓ-torsion sheaf F, one obtains

$$\begin{cases} 0 \to H^1(X, i_*F) \to H^1(\operatorname{Spec} R(X), F) \to \bigoplus_x H^1(\operatorname{Spec} R(X_x), F) \to H^2(X, i_*F) \to 0, \\ H^q(X, i_*F) = 0 \quad (q > 2). \end{cases} \tag{4.3.1}$$

The hypothesis $\operatorname{cd}_\ell(\operatorname{Spec} R(X)) \leq 1$ is satisfied, for example, when X is an algebraic curve over a separably closed field, by Tsen's theorem. Exact sequences of this type were studied by Ogg and Shafarevich.

4.4

Assume that ℓ is invertible on X, and take $F = (\mathbb{G}_m)_{R(X)}$. From Theorem 3.3 and CG, Chapter I, Proposition 12, one has, under the same hypothesis $\operatorname{cd}_\ell(\operatorname{Spec} R(X)) \leq 1$,

$$H^1(\operatorname{Spec} R(X), \mathbb{G}_m) = 0, \qquad H^q(\operatorname{Spec} R(X), \mathbb{G}_m) \text{ has no } \ell\text{-torsion for } q \geq 2. \tag{4.4.1}$$

The same holds with $R(X)$ replaced by $R(X_x)$. Corollary 4.2 therefore gives

$$H^1(X, i_*(\mathbb{G}_m)_{R(X)}) = 0, \qquad H^q(X, i_*(\mathbb{G}_m)_{R(X)}) \text{ has no } \ell\text{-torsion for } q > 1. \tag{4.4.2}$$

For any sheaf F on X, Lemma 4.1 shows that the kernel and cokernel of the canonical homomorphism $F \to i_*i^*F$ are skyscraper sheaves. Since the closed residue fields are separably closed, these skyscraper sheaves have no higher cohomology. Thus

$$H^i(X, F) \longrightarrow H^i(X, i_*i^*F)$$

is an isomorphism for $i \geq 2$. Applied to $(\mathbb{G}_m)_X$, this yields

$$H^i(X, \mathbb{G}_m) \text{ has no } \ell\text{-torsion for } i \geq 2, \tag{4.5}$$

under the hypotheses of (4.4). Combining this with Kummer theory and Theorem 3.3 gives:

Theorem 4.6. Let X be noetherian of dimension one, let n be an integer invertible on X, suppose that $\mathrm{cd}_\ell(R(X)) \leq 1$ for every prime ℓ dividing n, and suppose that for every closed point x of X the residue field $k(x)$ is separably closed. Then

$$H^q(X, (\mu_n)_X) = 0 \quad (q > 2),$$

and for $q \leq 2$ the cohomology is given by the exact sequence

$$\begin{aligned} 0 \to H^0(X, (\mu_n)_X) \to \Gamma(X, \mathcal{O}_X^*) \xrightarrow{n} \Gamma(X, \mathcal{O}_X^*) \to H^1(X, (\mu_n)_X) \\ \to \operatorname{Pic} X \xrightarrow{n} \operatorname{Pic} X \to H^2(X, (\mu_n)_X) \to 0. \end{aligned}$$

Corollary 4.7. Let X be a complete connected curve over a separably closed field k, of characteristic prime to n. Then

$$H^0(X, (\mu_n)_X) \simeq \mu_n(k),$$
$$H^1(X, (\mu_n)_X) \simeq {}_n \operatorname{Pic} X,$$

and

$$H^2(X, (\mu_n)_X) \simeq (\operatorname{Pic} X)/n \simeq (\mathbb{Z}/n\mathbb{Z})^c,$$

where c is the number of irreducible components of X.

For the last assertion, recall that the Picard scheme of such an X/k has a decomposition

$$0 \longrightarrow A \longrightarrow \mathcal{P}ic_{X/k} \longrightarrow \mathbb{Z}^c \longrightarrow 0, \tag{4.8}$$

where A is a connected algebraic group over k. Multiplication by n on such an A is *étale*, hence surjective on k-valued points. Thus ${}_n(\operatorname{Pic} X) \simeq {}_nA(k)$ and $(\operatorname{Pic} X)/n \simeq (\mathbb{Z}/n\mathbb{Z})^c$. If X is smooth over k, then A is the Jacobian and ${}_n \operatorname{Pic} X \simeq (\mathbb{Z}/n\mathbb{Z})^{2g}$, where g is the genus. Since $\mu_n(k) \simeq \mathbb{Z}/n\mathbb{Z}$, the ranks of the cohomology groups are $1, 2g, 1$, as expected.

5. The trace method

5.1

Let X be a scheme, $f : X' \to X$ a finite *étale* covering, and F a sheaf on X. Adjunction gives a restriction morphism

$$F \xrightarrow{\text{res}} f_*f^*F.$$

Since f is finite, $H^q(X, f_*f^*F) \simeq H^q(X', f^*F)$, whence the usual restriction map

$$H^q(X, F) \xrightarrow{\text{res}} H^q(X', f^*F). \tag{5.1.1}$$

There is also a trace morphism

$$f_*f^*F \xrightarrow{\text{tr}} F, \tag{5.1.2}$$

which induces

$$H^q(X', f^*F) \xrightarrow{\mathrm{tr}} H^q(X, F). \tag{5.1.3}$$

It is characterized by being local for the *étale* topology and, when X' is the disjoint sum of d copies of X, by being the summation map $F^d \to F$. The uniqueness is immediate, since every finite *étale* covering is locally constant; existence follows by the usual descent argument. Hence

$$\mathrm{tr} \circ \mathrm{res} : F \to F$$

is multiplication by the local degree of f. If the degree is a constant d, the same is true on cohomology.

Corollary 5.2. Let $f : X' \to X$ be a finite *étale* covering of constant degree d. If F is a sheaf of r-torsion on X, with $(r, d) = 1$, then $H^q(X', f^*F) = 0$ implies $H^q(X, F) = 0$.

Indeed, multiplication by d is an isomorphism of F, hence of its cohomology, and the resulting isomorphism factors through zero.

5.2.1

Let $i : U \to X$ be an immersion, let $f : U' \to U$ be a finite *étale* covering of degree d, and suppose that f is induced by a finite morphism $g : X' \to X$ by base change:

$$\begin{tikzcd} U' \arrow[r, "i'"] \arrow[d, "f"'] & X' \arrow[d, "g"] \\ U \arrow[r, "i"] & X. \end{tikzcd}$$

For a sheaf F on U one has the composites

$$F \xrightarrow{\mathrm{res}} f_*f^*F \xrightarrow{\mathrm{tr}} F,$$

whose composite is multiplication by d. Since $i_*f_* = g_*i'_*$, this gives

$$i_*F \xrightarrow{\mathrm{res}} g_*i'_*f^*F \xrightarrow{\mathrm{tr}} i_*F, \tag{5.3}$$

and since $i_!f_* \simeq g_*i'_!$,

$$i_!F \xrightarrow{\mathrm{res}} g_*i'_!f^*F \xrightarrow{\mathrm{tr}} i_!F, \tag{5.4}$$

again with composite multiplication by d. Passing to cohomology gives the corresponding restriction–trace composites for i_*F and $i_!F$.

Proposition 5.5. Let X be quasi-compact and quasi-separated, let ℓ be a prime, and let H be a semi-exact functor to **Ab**, defined on the category of ℓ-sheaves on X, and compatible with filtered inductive limits. The following are equivalent:

(i) $H = 0$.

(ii) $H(f_*i_!(\mathbb{Z}/\ell\mathbb{Z})) = 0$ for every finite morphism $f : Y \to X$, and every open immersion $i : U \to Y$ with U of finite presentation over X. If X is noetherian, one may restrict to integral Y.

Proof. By Corollary 2.7.2 and Proposition 2.5 it is enough to test H on sheaves $i_!G$, where $i : U \to X$ is the inclusion of a subscheme of finite presentation over X, and G is an isotrivial locally constant ℓ-sheaf on U. Choose a principal covering $U'' \to U$, with group $\mathfrak{g}$, on which G becomes constant. Let A be the ℓ-group of its values, and let $\mathfrak{h}$ be an ℓ-Sylow subgroup of $\mathfrak{g}$. If $U' = U''/\mathfrak{h}$, then the degree d of $U' \to U$ is prime to ℓ. Extending $U' \to U$ to a finite morphism $X' \to X$ (EGA IV, 8.12.6), and applying (5.4), one reduces to proving the vanishing for the pullback of G to U'. As an $\mathfrak{h}$-module, A admits a filtration whose successive quotients are $\mathbb{Z}/\ell\mathbb{Z}$ with trivial action; the assertion follows by semi-exactness.

Proposition 5.6. Let X be noetherian, let ℓ be a prime, and let φ be a function with values in $\mathbb{N}$, defined on the integral schemes finite over X. Suppose $\varphi(Y_1) \leq \varphi(Y_2)$ whenever there is an X-morphism $Y_1 \to Y_2$, and that the inequality is strict if Y_1 is a proper closed subscheme of Y_2. For finite Y over X, put $\varphi(Y) = \sup \varphi(Y_i)$, where the Y_i are the irreducible components of Y, with their reduced induced structures. The following are equivalent:

(i) For every finite Y over X and every ℓ-sheaf F on Y, one has $H^q(Y, F) = 0$ for $q > \varphi(\operatorname{Supp} F)$.

(ii) For every integral finite Y over X, one has $H^q(Y, \mathbb{Z}/\ell\mathbb{Z}) = 0$ for $q > \varphi(Y)$.

Proof. Assuming (ii), apply Proposition 5.5 to each H^q, using the facts that cohomology commutes with inductive limits and with direct image under finite morphisms. One is reduced to the case where Y is integral finite over X, and $F = i_!(\mathbb{Z}/\ell\mathbb{Z})_U$ for a non-empty open immersion $i : U \to Y$. The exact sequence

$$0 \longrightarrow i_!(\mathbb{Z}/\ell\mathbb{Z})_U \longrightarrow (\mathbb{Z}/\ell\mathbb{Z})_Y \longrightarrow (\mathbb{Z}/\ell\mathbb{Z})_Z \longrightarrow 0,$$

with $Z = Y - U \neq Y$, and induction on φ, give the vanishing. The converse is immediate.

Corollary 5.7. Let X be an algebraic curve over a separably closed field, of characteristic different from ℓ. Then the cohomology of X with values in any ℓ-torsion sheaf F vanishes in degrees > 2. If X is affine, it vanishes in degrees > 1.

For finite integral $Y \to X$, put $\varphi(Y) = 2 \dim Y$, or $\varphi(Y) = \dim Y$ in the affine case. Proposition 5.6 reduces the assertion to the case $F = \mathbb{Z}/\ell\mathbb{Z}$ on an integral curve. Since $(\mathbb{Z}/\ell\mathbb{Z})_Y \simeq (\mu_\ell)_Y$, Theorem 4.6 gives the vanishing in degrees > 2. In the affine case one may pass to the normalization and then to a smooth compactification $\bar{X}$; at least one point is missing from $\bar{X}$, so $A(k) \to \operatorname{Pic} X$ is surjective, where A is the Jacobian of $\bar{X}$. As $A(k)$ is divisible by $\ell \neq \operatorname{char} k$, so is $\operatorname{Pic} X$; hence $H^2(X, \mu_\ell) = 0$ by Theorem 4.6.

Remark 5.8. For later reference, the proof of Proposition 5.5 actually proves the following. Let X be quasi-compact and quasi-separated, ℓ a prime, and C a strictly full subcategory of the category of constructible abelian ℓ-torsion sheaves, stable under direct summands and extensions. Suppose that for every finite morphism $f : Y \to X$ and every open immersion $i : U \to Y$ with U of finite presentation over X, one has

$$f_*(i_!(\mathbb{Z}/\ell\mathbb{Z})_U) \in C.$$

Then C contains all constructible ℓ-torsion sheaves.

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SGA 4, Exposé X

Cohomological Dimension: First Results

M. Artin

1. Introduction

Throughout this exposé, unless the contrary is expressly stated, the word "sheaf" means "abelian sheaf."

Let X be a scheme and let ℓ be a prime number. Recall that the *ℓ-cohomological dimension* of X, denoted $\operatorname{cd}_\ell X$, is the largest integer n, or ∞ if no such integer exists, such that $H^n(X, F) \neq 0$ for at least one ℓ-torsion sheaf F on X. If $X = \operatorname{Spec} A$, we write $\operatorname{cd}_\ell A = \operatorname{cd}_\ell \operatorname{Spec} A$. Taking the dictionary of VIII, 2 into account, this recovers the classical notion of Galois cohomological dimension in the case where $A = k$ is a field. Since plainly

$$\operatorname{cd}_\ell\Big(\coprod_i X_i\Big) = \sup_i \operatorname{cd}_\ell X_i,$$

the classical theory extends immediately to arbitrary artinian rings. This theory is set out in Serre's *Cohomologie galoisienne*, cited below as CG; we shall add a few refinements in Section 2.

For instance, let X be an irreducible algebraic variety of dimension n over a separably closed field k, and fix $\ell \neq \operatorname{char} k$. The theorem of Section 4 gives the bound $2n$, the "true" dimension suggested by the topological analogy when $k = \mathbb{C}$. This result is quite elementary from the theory for fields. In fact the bound $2n$ can be improved except in the complete case: there is a bound $2n-1$ whenever at least one point is missing from X. This is the stable bound in the sense that every X of dimension n contains an open subset of cohomological dimension $2n-1$. For affine varieties one obtains the still sharper bound n, a generalization of a Lefschetz theorem for hyperplane sections. These two latter assertions are deeper and will be proved only later, in XIV.

Cohomological dimension is not a local notion, unlike what happens for paracompact spaces. A henselian local ring with separably closed residue field has cohomological dimension zero, but after removing its closed point one may obtain arbitrarily large cohomological dimension, namely $2n-1$ in the most important cases when $n = \dim A$. This again agrees with the topological analogy supplied by the cohomology of $U - \{x\}$, where U is a small open neighbourhood of a point of a complex analytic space.

2. Auxiliary results on fields

Theorem 2.1. Let k'/k be a finitely generated extension of a field k, and let ℓ be a prime. Then

$$\operatorname{cd}_\ell k' \leq \operatorname{cd}_\ell k + \operatorname{trdeg}(k'/k).$$

If the inequality is strict and $\ell \neq \operatorname{char} k$, then $\ell = 2$, the field k is orderable, and k' is not orderable, i.e. -1 is a sum of squares in k'.

When $\ell = \operatorname{char} k$, one has $\operatorname{cd}_\ell k' \leq 1$ (CG II, 2.2), so the question of equality in Theorem 2.1 scarcely arises. Also $\operatorname{cd}_2 k = \infty$ if k is orderable: an algebraic closure $\bar{k}$ contains a subextension

$\widetilde{k}$, a maximal ordered subfield, with $[\bar{k} : \widetilde{k}] = 2$; plainly $\mathrm{cd}_2 \widetilde{k} = \infty$, and $\mathrm{cd}_2 \widetilde{k} \le \mathrm{cd}_2 k$ by CG I, Proposition 14.

Proof. The inequality, and equality when $\mathrm{cd}_\ell k < \infty$, are due to Tate (CG II, 4.1.2). It remains to show that $\mathrm{cd}_\ell k' < \infty$ and $\mathrm{cd}_\ell k = \infty$ imply that k is orderable and $\ell = 2$. It is enough to treat (a) finite extensions and (b) $k' = k(x)$ with x transcendental over k. In case (a), after assuming the extension separable, the absolute Galois group of k' is an open subgroup of that of k. A theorem of Serre gives equality of ℓ-dimensions except when the larger group contains an element of order ℓ. An algebraically closed field has no finite-index subfield except in the case $\ell = 2$ and a real closed situation (Artin–Schreier), giving the claim. In case (b), take the henselization R of $\operatorname{Spec} k[x]$ at $x = 0$, with fraction field K. If $\mathrm{cd}_\ell k(x) < \infty$, then $\mathrm{cd}_\ell K < \infty$; the result follows from the next theorem applied to R.

Theorem 2.2. Let R be a henselian discrete valuation ring, with fraction field K and residue field k. Let ℓ be a prime. Then

$$\mathrm{cd}_\ell K = \mathrm{cd}_\ell k + 1$$

in the following two cases:

(i) $\ell \ne \operatorname{char} k$;

(ii) $\ell \ne \operatorname{char} K$, the field k is perfect, and $\mathrm{cd}_\ell k < \infty$.

This theorem was recently proved by Ax. If $\ell = \operatorname{char} K$, one always has $\mathrm{cd}_\ell K \le 1$ (CG II, 2.2). What remains in unequal characteristic, when $\operatorname{char} k = \ell$ and k is imperfect, should be related to the absolute module of differentials Ω^1_k (EGA IV, 20.4.3).

Lemma 2.2.1. Let R be a henselian discrete valuation ring with fraction field K, and let $\widehat{K}$ be the fraction field of the completion $\widehat{R}$. Then

$$G(\overline{\widehat{K}}/\widehat{K}) \simeq G(\bar{K}/K).$$

Proof. Equivalently, the functor $L \mapsto L \otimes_K \widehat{K}$ from finite *étale* K-algebras to finite *étale* $\widehat{K}$-algebras is an equivalence. Essential surjectivity is obtained by approximating a separable irreducible equation over $\widehat{R}$ by one over R, using the usual Newton method. For full faithfulness it is enough to see that if K' is a field, then $K' \otimes_K \widehat{K}$ is a field. If R' is the normalization of R in K', then R' is finite over R, a discrete valuation ring, and its completion is $R' \otimes_R \widehat{R}$; the corresponding total ring of fractions is therefore again a field.

The proof of Theorem 2.2 is then the standard argument of Tate, together with the preceding lemma. In case (i), when $\mathrm{cd}_\ell k < \infty$, one reduces to the complete case in CG II, Proposition 12. To see that $\mathrm{cd}_\ell K < \infty$ implies $\mathrm{cd}_\ell k < \infty$, let $\widetilde{K}$ be the maximal unramified extension of K, with residue field separably closed, and put $G = G(\bar{K}/K)$, $H = G(\bar{K}/\widetilde{K})$, $\bar{G} = G/H = G(\bar{k}/k)$. Passing to an ℓ-Sylow subgroup of $\bar{G}$ and using the structure of inertia gives the required finiteness. Case (ii) follows similarly from CG II, Proposition 12 after completion.

Theorem 2.3. Let A be a strictly local ring, let k be its residue field, and let K be the total ring of fractions of A. If K is noetherian and if $\ell \ne \operatorname{char} k$, then

$$\mathrm{cd}_\ell K \le \dim A.$$

The proof proceeds by noetherian induction and reduction to the case of a domain. One uses normalization, passage to finite extensions, and Theorem 2.2 applied to henselian localizations at codimension-one points. This is the same dévissage as in the classical proof for excellent local rings; the strict locality ensures that no residue-field contribution remains. For example, a strictly henselian discrete valuation ring has fraction field of ℓ-cohomological dimension at most one for ℓ prime to the residue characteristic.

Corollary 2.4. Let A be a strictly local noetherian ring, let K be its total ring of fractions, and let ℓ be invertible in A. Then $\operatorname{cd}_\ell K \leq \dim A$.

Corollary 2.5. Let X be a noetherian scheme, let x be a point, and let $X_{\bar{x}}$ be a strict localization at a geometric point over x. If ℓ is invertible at x, the total ring of fractions of $X_{\bar{x}}$ has ℓ-cohomological dimension at most $\operatorname{codim} x$.

3. Fraction fields of a strictly local ring

3.0

Let A be strictly local with closed point s, and let $U = \operatorname{Spec} A - \{s\}$. The preceding estimates for fraction fields will be applied to the fibres of direct images from U to $\operatorname{Spec} A$. The basic point is that the strict localization at a geometric point sees only the corresponding henselian local algebra, so that the cohomology of the punctured local scheme is controlled by the cohomological dimension of its total fraction ring.

Proposition 3.2. Let A be strictly local noetherian of dimension n, and let ℓ be invertible in A. Then for every ℓ-torsion sheaf F on $U = \operatorname{Spec} A - \{s\}$, one has

$$H^q(U, F) = 0 \qquad (q > 2n - 1).$$

If F is the restriction of a sheaf from $\operatorname{Spec} A$, the bound improves in the expected low-dimensional cases.

Proof. The proof uses the Leray spectral sequence for the immersion $j : U \to \operatorname{Spec} A$. Its higher direct images are computed on strict localizations, and Corollary 2.5 gives the necessary vanishing of their fibres. Induction on the dimension of the support then gives the bound.

Lemma 3.3. Let A be strictly local noetherian and let $j : U \to \operatorname{Spec} A$ be the complement of the closed point. If ℓ is invertible in A, then the sheaves $R^q j_* F$ have support constrained by codimension: their stalks vanish at a point once q exceeds the codimension of that point in the relevant strict localization.

This is just the stalkwise form of Corollary 2.5 and is the technical input for the estimates in the next section.

4. Cohomological dimension when ℓ is invertible in $\mathcal{O}_X$

Theorem 4.1. Let X be a noetherian scheme of dimension n, and let ℓ be invertible on X. Suppose that the ℓ-cohomological dimension of every residue field of X is bounded by N. Then

$$\operatorname{cd}_\ell X \leq N + 2n.$$

More precisely, if an ℓ-torsion sheaf F is supported in dimension $\leq d$, then its cohomology vanishes above the corresponding bound $N + 2d$.

Proof. It is enough, by the constructible filtration theorem of IX, to treat constructible sheaves, then to proceed by induction on the support. On a dense open subset of each irreducible component the sheaf becomes locally constant and isotrivial; after a finite *étale* cover and using the trace method of IX, 5, one reduces to constant sheaves. The local terms are controlled by Proposition 3.2 and Lemma 3.3. The spectral sequence for the inclusion of the generic points gives the bound.

Corollary 4.2. Let X be of finite type over a field k, with $\ell \neq \operatorname{char} k$. Then

$$\operatorname{cd}_\ell X \leq \operatorname{cd}_\ell k + 2 \dim X.$$

In particular, if k is separably closed, $\operatorname{cd}_\ell X \leq 2 \dim X$.

Corollary 4.3. Let X be of finite type over a separably closed field, and let ℓ be invertible on X. Then $H^q(X, F) = 0$ for every ℓ-torsion sheaf F and every $q > 2 \dim X$.

Corollary 4.4. Let X be an algebraic curve over a separably closed field, with $\ell \neq \operatorname{char} k$. Then $\operatorname{cd}_\ell X \leq 2$; if X is affine, the sharper bound $\operatorname{cd}_\ell X \leq 1$ is supplied later by XIV.

Example 4.5. One cannot hope for an analogous local formula in terms only of the local rings of X. There are discrete valuation rings with algebraically closed residue field whose fraction fields have arbitrarily large cohomological dimension; they arise by taking formal curve arcs through a rational point of a smooth variety whose function field is prescribed.

4.6

These estimates will be used repeatedly in the sequel. Their force is that, for ℓ invertible, the relevant cohomological dimension is twice the geometric dimension, plus the cohomological dimension of the ground field.

5. Cohomological dimension in the case $\ell = p$

5.0

Using Artin–Schreier theory (IX, 3.5), one obtains a better bound for the p-cohomological dimension of a scheme of characteristic p. Let $\operatorname{cd}_{\mathrm{qc}} X$ be the largest integer n such that $H^n(X, F) \neq 0$ for some quasi-coherent $\mathcal{O}_X$-module F in the Zariski sense.

Theorem 5.1. Let X be a noetherian scheme of characteristic $p > 0$. Then

$$\operatorname{cd}_p X \leq \operatorname{cd}_{\mathrm{qc}} X + 1.$$

In particular, a noetherian affine scheme of characteristic $p > 0$ has p-cohomological dimension at most one.

Proof. Apply IX, 5.5. Since higher direct images under finite morphisms vanish for abelian sheaves, and likewise for quasi-coherent modules, one reduces to proving $H^q(X, i_!(\mathbb{Z}/p\mathbb{Z})_U) = 0$ for an open immersion $i : U \to X$ and $q > \operatorname{cd}_{\mathrm{qc}} X + 1$. Let Y be a closed subscheme with support $X - U$, defined by a coherent ideal J. The Artin–Schreier map $f \mapsto f^p - f$ induces a morphism $J \to J$, and the Artin–Schreier sequence gives

$$0 \longrightarrow i_!(\mathbb{Z}/p\mathbb{Z})_U \longrightarrow J \longrightarrow J \longrightarrow 0.$$

Since J is coherent, its cohomology vanishes above $\mathrm{cd}_{\mathrm{qc}}\, X$, giving the assertion.

Corollary 5.2. Let X be of finite type over a field k of characteristic $p > 0$. Then

$$\mathrm{cd}_p\, X \leq \dim X + 1.$$

If k is separably closed and X is quasi-projective, then

$$\mathrm{cd}_p\, X \leq \dim X.$$

The first assertion is clear from $\mathrm{cd}_{\mathrm{qc}}\, X \leq \dim X$. For the second, the top coherent cohomology groups are finite-dimensional k-vector spaces. In the exact Artin–Schreier sequence above, the induced endomorphism of the top cohomology is of the form $\epsilon - \mathrm{id}$, with $\epsilon(rv) = r^p\epsilon(v)$. The corresponding additive morphism of affine space has Jacobian $-I$, hence is *étale*; over a separably closed field it is surjective. The top obstruction vanishes.

6. Cohomological dimension for schemes of finite type over $\mathrm{Spec}\,\mathbb{Z}$

Proposition 6.1. Let X be quasi-finite over $\mathrm{Spec}\,\mathbb{Z}$. Suppose either $\ell \neq 2$, or every residue field of X of characteristic zero is totally imaginary. Then $\mathrm{cd}_\ell\, X \leq 3$.

This follows from Theorem 2.2(ii), Corollary 4.2, and the fact that a number field, totally imaginary when $\ell = 2$, has ℓ-cohomological dimension at most two (CG II, 4.4). Class field theory shows that equality holds if $X \to \mathrm{Spec}\,\mathbb{Z}$ is proper and surjective.

The results of Section 4 do not apply unchanged when X is merely of finite type over $\mathrm{Spec}\,\mathbb{Z}$, because the hypotheses at points of residual characteristic ℓ fail in general. The estimate of Theorem 5.1 must be used in that characteristic.

Theorem 6.2. Let X be of finite type over $\mathrm{Spec}\,\mathbb{Z}$, and let ℓ be a prime. If $\ell = 2$, suppose that no residue field of X is orderable. Then

$$\mathrm{cd}_\ell\, X \leq 2\dim X + 1.$$

More precisely, let F be an ℓ-torsion sheaf such that, for every point x with $F_{\bar{x}} \neq 0$,

$$\dim \overline{\{x\}} \leq \begin{cases} (N-1)/2, & \ell \neq \mathrm{char}\, k(x), \\ N-1, & \ell = \mathrm{char}\, k(x). \end{cases}$$

Then $H^p(X, F) = 0$ for $p > N$.

Proof. Proceed by induction on X, and reduce to constructible F by writing any sheaf as the filtered limit of its constructible subsheaves. The set of points where $F_{\bar{x}} \neq 0$ is then constructible, and induction handles the case where it is not dense. If the support is in characteristic ℓ, Theorem 5.1 gives the bound; if it is in characteristic $p \neq \ell$, Corollary 4.3 and the fact $\mathrm{cd}_\ell\, \mathbb{F}_p = 1$ give it.

For a reduced X, write $X = X_0 \cup X_1$, where X_1 is the union of the irreducible components of characteristic ℓ, and X_0 the union of the others. The exact sequence

$$0 \to F \to j_{0*}j_0^*F \oplus j_{1*}j_1^*F \to C \to 0$$

has C supported on $X_0 \cap X_1$, which is of characteristic ℓ and dimension at most $\dim X_1 - 1$. The induction and Theorem 5.1 reduce the assertion to the cases $X = X_0$ and $X = X_1$; the latter was already handled.

It remains to treat the case in which every maximal point has characteristic different from ℓ. Let $i : x \to X$ be the generic point of an irreducible component. Using the exact sequence comparing F with the sum of its generic-point extensions, induction reduces to $F = i_* G$. The spectral sequence

$$H^p(X, R^q i_* G) \Longrightarrow H^{p+q}(\operatorname{Spec} k(x), G)$$

finishes the proof. If $s = \dim X$, then $k(x)[i]$ has transcendence degree $s - 1$ over $\mathbb{Q}[i]$, a field of cohomological dimension two; Theorem 2.1 gives $\operatorname{cd}_\ell k(x) = s + 1$, under the non-orderability hypothesis when $\ell = 2$. The stalks of $R^q i_* G$ vanish when q exceeds the appropriate codimension, by Proposition 3.2 and Theorem 2.2. Applying induction to these higher direct images gives the required vanishing.

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SGA 4, Exposé XI

Comparison with Classical Cohomology: the Smooth Case

M. Artin

1. Introduction

In Section 4 we shall prove that *étale* cohomology with values in a locally constant torsion sheaf agrees with classical cohomology for a scheme X smooth over $\operatorname{Spec}\mathbb{C}$. The proof, essentially the one given in Giraud's Bourbaki exposé, is elementary once one uses Grauert's theorem recalled below (4.3(iii)), namely that every finite covering of $X(\mathbb{C})$ comes from an *étale* covering of X. We give it here even though a more complete result will be proved later using Hironaka's resolution of singularities. The good neighbourhoods constructed in Section 3 may also be useful in other situations.

It is not true that *étale* cohomology and classical cohomology agree for non-torsion sheaves. For example, $H^1(X_{\text{ét}}, \mathbb{Z}) = 0$ for every normal scheme X (IX, 3.6(ii)), whereas the corresponding classical cohomology need not vanish.

2. Existence of sufficiently general hyperplane sections

We shall use the following classical result, for which the original notes lacked a convenient reference.

Theorem 2.1. Let k be a field, and let $V \subset \mathbf{P}^n_k$ be a projective algebraic scheme purely of dimension r. Let V' be the open subscheme of points of V smooth over $\operatorname{Spec} k$, and let P be a rational point of $\mathbf{P}^n$. Then:

(i) A sufficiently general linear subspace L of prescribed dimension d in $\mathbf{P}^n$ meets V transversally at every point of $L \cap V'$.

(ii) Let $\{H_1, \dots, H_s\}$, $s \leq r$, be hypersurfaces of degree $d \geq 2$ through P, and put $Y = H_1 \cap \cdots \cap H_s$. If $\{H_1, \dots, H_s\}$ is sufficiently general, then Y meets V transversally at every point of $Y \cap V'$.

Here "sufficiently general" means that the relevant parameter space, a Grassmannian in (i) or a product of projective spaces in (ii), contains a dense open subset over which the assertion holds. Thus it does not assert the existence of a rationally defined member when the ground field is finite.

Proof. For (i), induction reduces immediately to the case of a hyperplane. After covering V' by finitely many opens, one may assume an affine situation $V \subset \mathbb{A}^n_k = \operatorname{Spec} k[x_1, \dots, x_n]$. The hyperplane has equation

$$\ell(x) = a_0 + \sum_{i=1}^{n} a_i x_i.$$

Since a smooth scheme is locally a complete intersection, after shrinking one may suppose that V is cut out near V' by equations $f_1, \dots, f_{n-r}$ with an invertible Jacobian minor. The condition that L fail to meet V transversally is then expressed by the existence of a point $x \in V'$ satisfying an explicit finite system of equations in the parameters a_i. The image of this closed condition is constructible by EGA IV, 9.5.1. It is enough to check the generic parameter, where the equations plainly have no solution. This proves (i).

For (ii), use the rational map $\mathbf{P}^n \dashrightarrow \mathbf{P}^N$ defined by the linear system of degree d hypersurfaces through P. For $d \geq 2$ it is the blow-up of P followed by a locally closed immersion off P. Away from P the assertion reduces to (i) applied to the image. At P the conclusion is immediate for sufficiently general hypersurfaces.

3. Construction of good neighbourhoods

Definition 3.1. An *elementary fibration* is a morphism of schemes $f : X \to S$ which can be embedded in a commutative diagram

$$\begin{tikzcd} X \arrow[r, "j"] \arrow[dr, "f"'] & \bar{X} \arrow[d, "\bar{f}"] & Y \arrow[l, hook', "i"'] \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

satisfying:

(i) j is an open immersion, dense in every fibre, and $X = \bar{X} - Y$;

(ii) $\bar{f}$ is smooth and projective, with geometrically irreducible fibres of dimension one;

(iii) g is an *étale* covering, and every fibre of g is non-empty.

Such an embedding of X, if it exists, is easily checked to be unique up to canonical isomorphism.

Definition 3.2. A *good neighbourhood relative to S* is an S-scheme X for which there exist S-schemes

$$X = X_n, \ldots, X_0 = S$$

and elementary fibrations $f_i : X_i \to X_{i-1}$, $i = 1, \ldots, n$.

Proposition 3.3. Let k be algebraically closed, let $X/\operatorname{Spec} k$ be smooth, and let $x \in X$ be a rational point. Then there is an open neighbourhood of x in X which is a good neighbourhood relative to $\operatorname{Spec} k$.

Proof. Assume X irreducible. By induction on $n = \dim X$, it is enough to find a neighbourhood U of x and an elementary fibration $f : U \to V$, with V smooth of dimension $n-1$. Indeed, a good neighbourhood of $f(x)$ in V then pulls back to one of x in U.

Embed X as an affine open in $\mathbb{A}^r$, let X_0 be its closure in $\mathbf{P}^r$, let $\bar{X}$ be the normalization of X_0, and let $Y = \bar{X} - X$ with reduced structure. Let $S \subset \bar{X}$ be the singular locus; then $S \subset Y$, $\dim \bar{X} = n$, $\dim Y = n - 1$, and $\dim S \leq n - 2$.

Choose a projective embedding of $\bar{X}$ by an invertible sheaf which is a sufficiently high power of a very ample sheaf. By Theorem 2.1 there exist hyperplanes $H_1, \ldots, H_{n-1}$ through x such that $L = H_1 \cap \cdots \cap H_{n-1}$ meets $\bar{X}$ and Y transversally. Then $\bar{X} \cap L$ is a smooth connected curve, and $Y \cap L$ is zero-dimensional. Choose an additional hyperplane H_0 such that H_0 meets $\bar{X} \cap L$ transversally and avoids $Y \cap L$.

Projection from the center $C = H_0 \cap \cdots \cap H_{n-1}$ gives a rational map $\mathbf{P}^N \dashrightarrow \mathbf{P}^{n-1}$. Blow up C, take the proper transform $\bar{X}'$, put $X' = X - \bar{X} \cap C$, and let $Y' = \bar{X}' - X'$. We get a diagram

$$\begin{tikzcd}
X' \arrow[r, "j"] \arrow[dr, "f'"'] & \bar{X}' \arrow[d, "\bar{f}'"] & Y' \arrow[l, "i"'] \arrow[dl, "g'"] \\
 & \mathbf{P}^{n-1} &
\end{tikzcd}$$

Near the point $v = f'(x)$, the fibre of $\bar{f}'$ is identified with the smooth curve $\bar{X} \cap L$; by Hironaka's lemma, smoothness near that fibre follows from smoothness at its generic point. The morphism g' is *étale* near v because L meets Y transversally, and each fibre is non-empty. After replacing $\mathbf{P}^{n-1}$ by a neighbourhood of v, the diagram gives an elementary fibration.

4. The comparison theorem

4.0

Let X be a scheme locally of finite type over $\operatorname{Spec} \mathbb{C}$, and put on $X(\mathbb{C})$ its usual topology. Denote by X_{cl} the site whose objects are local isomorphisms $U \to X(\mathbb{C})$, that is, maps of topological spaces which are locally homeomorphisms onto open subsets of $X(\mathbb{C})$. A family $\{U_\nu \to U\}$ is covering if the images cover U. Since open immersions are local isomorphisms, the site X_{cl} has the same associated topos as the usual topological space $X(\mathbb{C})$. Thus one may compute ordinary cohomology using X_{cl}.

If $f : X' \to X$ is *étale*, the analytic map $f(\mathbb{C}) : X'(\mathbb{C}) \to X(\mathbb{C})$ is a local isomorphism, by the Jacobian criterion and the implicit function theorem. Therefore the functor $X' \mapsto X'(\mathbb{C})$ defines a morphism of sites

$$\epsilon : X_{\mathrm{cl}} \longrightarrow X_{\text{ét}}. \tag{4.2}$$

Theorem 4.3.

(i) X is connected and non-empty if and only if $X(\mathbb{C})$ is connected and non-empty.

(ii) The functor ϵ is fully faithful.

(iii) (Grauert–Remmert theorem.) The functor ϵ induces an equivalence between finite coverings of $X(\mathbb{C})$, with its usual locally compact topology, and *étale* coverings of X.

We accept (i) as known; it follows, for example, from GAGA. Assertion (ii) follows easily from (i): for a morphism $\varphi : X'(\mathbb{C}) \to X''(\mathbb{C})$ over $X(\mathbb{C})$, after reducing to the connected affine separated case, its graph is a connected component of $(X' \times_X X'')(\mathbb{C})$, hence comes from a connected component of $X' \times_X X''$. This component is the graph of a morphism $X' \to X''$. For (iii), by descent one reduces to X connected normal and affine. Taking a finite morphism $X \to \mathbb{A}^n$, any finite connected covering of $X(\mathbb{C})$ becomes an analytic covering of $\mathbb{A}^n(\mathbb{C})$, extends to an analytic covering of $\mathbf{P}^n(\mathbb{C})$ by Grauert–Remmert, and the resulting coherent analytic algebra is algebraic by GAGA. Taking $Y = \operatorname{Spec} A$ gives the desired *étale* covering.

Theorem 4.4. Let X be smooth over $\operatorname{Spec} \mathbb{C}$.

(i) The category of constructible locally constant torsion sheaves on $X_{\text{ét}}$ is equivalent to the category of locally constant torsion sheaves with finite fibres on X_{cl}; the quasi-inverse functors are ϵ_* and ϵ^*.

(ii) If F is a locally constant torsion sheaf with finite fibres on X_{cl}, and if the induced sheaf on $X_{\text{ét}}$ is again denoted F, then
$$R^q\epsilon_*F = 0 \quad (q > 0).$$

(iii) With the notation of (i), the canonical map
$$H^q(X_{\text{ét}}, F) \longrightarrow H^q(X_{\text{cl}}, F)$$
is bijective for every q. In particular,
$$H^q(X_{\text{ét}}, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} H^q(X_{\text{cl}}, \mathbb{Z}/n\mathbb{Z})$$
for every q.

Proof. For (i), the sheaves in question are precisely those represented by finite *étale* coverings of X, respectively finite coverings of $X(\mathbb{C})$. The equivalence of coverings is Theorem 4.3(iii). Assertion (iii) follows from (ii) and the Leray spectral sequence for ϵ. It remains to prove (ii).

The sheaf $R^q\epsilon_*F$ is associated with the presheaf $X' \mapsto H^q(X'_{\text{cl}}, F)$ on $X_{\text{ét}}$. We must therefore show that every cohomology class becomes zero after passing to an *étale* neighbourhood of any prescribed point.

Lemma 4.5. Let $\xi \in H^q(X_{\text{cl}}, F)$, and let $x \in X(\mathbb{C})$. There exists an *étale* morphism $X' \to X$, whose image contains x, such that the image of ξ in $H^q(X'_{\text{cl}}, F)$ is zero.

Proof. The proof is by induction on $n = \dim X$. The problem is local for the *étale* topology, so one may assume F constant, and by dévissage even $F = \mathbb{Z}/n\mathbb{Z}$. After replacing X by a Zariski neighbourhood of x, Proposition 3.3 allows us to assume that X admits an elementary fibration $f : X \to S$. With the notation of Definition 3.1, a direct calculation gives $R^q j_{\text{cl}*}F = 0$ for $q > 1$, the sheaf $R^1 j_{\text{cl}*}F$ is extension by zero of a constant sheaf on Y_{cl}, and $j_{\text{cl}*}F$ is the constant sheaf $\mathbb{Z}/n\mathbb{Z}$ on $\bar{X}$. Moreover $R^q\bar{f}_{\text{cl}*}(j_{\text{cl}*}F)$ is computed fibrewise. The Leray spectral sequence implies that
$$R^0 f_{\text{cl}*}\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/n\mathbb{Z}, \quad R^1 f_{\text{cl}*}\mathbb{Z}/n\mathbb{Z} \text{ is locally constant torsion}, \quad R^q f_{\text{cl}*}\mathbb{Z}/n\mathbb{Z} = 0 \ (q > 1).$$
Thus the Leray sequence reduces to
$$\cdots \to H^q(S_{\text{cl}}, f_{\text{cl}*}F) \to H^q(X_{\text{cl}}, F) \to H^{q-1}(S_{\text{cl}}, R^1 f_{\text{cl}*}F) \to \cdots . \tag{*}$$
Let $s = f(x)$. By the induction hypothesis and (i), every class on S with finite locally constant torsion coefficients dies after an *étale* neighbourhood $S' \to S$ of s. Since the functors $R^q f_{\text{cl}*}$ commute with *étale* base change, the pullback $X \times_S S' \to X$ is an *étale* neighbourhood of x, and the exact sequence (*) kills first the boundary term and then the remaining term. The case $q = 1$ also follows directly from the classification of principal finite coverings and Theorem 4.3(iii).

Variant 4.6. Serre gave a more elegant proof of Lemma 4.5. A connected good neighbourhood X satisfies:

(i) $\pi_n(X(\mathbb{C})) = 0$ for $n > 1$;

(ii) $\pi_1(X(\mathbb{C}))$ is an iterated extension of finitely generated free groups.

Indeed, for an elementary fibration $X \to S$, the map $X(\mathbb{C}) \to S(\mathbb{C})$ is a locally trivial fibration whose fibre is a non-complete smooth connected curve; such a fibre has vanishing higher homotopy and free fundamental group. The long exact homotopy sequence gives the assertions by induction. Hence $X(\mathbb{C})$ is a $K(\pi, 1)$. A cohomology class becomes zero on the universal cover; it remains to show that it becomes zero on a finite cover. This is equivalent to the group π_1 being good in Serre's sense, i.e.

$$H^q(\widehat{\pi}_1, M) \xrightarrow{\sim} H^q(\pi_1, M)$$

for every finite continuous π_1-module M. Iterated extensions of finitely generated free groups are good (CG I, pp. 15–16, Exercises 1, 2).

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SGA 4, Exposé XII

The Proper Base Change Theorem

M. Artin

1. Introduction

In this exposé we give the fundamental theorem which says, in a slightly more precise form, that for a *proper* morphism $f : X \to Y$ of schemes and for an abelian torsion sheaf F on X, the fibre of $R^q f_* F$ at a geometric point ξ of Y is isomorphic to the cohomology

$$H^q(X_\xi, F|_{X_\xi})$$

of the fibre X_ξ.

For a proper morphism of paracompact spaces, the analogous result is well known and easy (Godement II, 4.11). For schemes it is much subtler. In particular, the torsion hypothesis on F is really necessary, unlike in the paracompact topological case.

The proof proceeds in several stages. By more or less formal dévissage one reduces to the case where Y is noetherian, the relative dimension of f is at most one, and F is constant. To treat that case one uses the specialization theorem for the fundamental group (5.9), a non-abelian variant of the theorem stated above; this specialization theorem can be proved directly in the special case required here, as is done in the next exposé.

2. An example

Let us show that the torsion hypothesis on F is necessary. Let X be a nonsingular surface over an algebraically closed field k, let Y be a nonsingular curve over k, and let $f : X \to Y$ be a proper morphism whose generic fibre is smooth and whose geometric fibre at a geometric point $\bar{y}$ of Y is irreducible and has one ordinary double point. Let F be the constant sheaf $\mathbb{Z}_X$.

Recall the following fact: if z is any point of a scheme Z and $i : z \to Z$ is the inclusion, then $H^1(Z, i_*\mathbb{Z}_z) = 0$ (IX, 3.6(i)). Since X is normal, hence unibranch, $\mathbb{Z}_X \simeq i_*\mathbb{Z}_x$, where $i : x \to X$ is the inclusion of the generic point. Thus $H^1(X, \mathbb{Z}_X) = 0$, and likewise $R^1 f_* \mathbb{Z}_X = 0$. But for the special geometric fibre X_0, with its ordinary double point, one has

$$H^1(X_0, \mathbb{Z}_{X_0}) \simeq \mathbb{Z}.$$

Indeed, if $i_0 : x_0 \to X_0$ is the inclusion of the generic point and Q is the singular point, the curve has two branches at Q. Hence the fibre of $i_{0*}\mathbb{Z}_{x_0}$ at Q is $\mathbb{Z} \times \mathbb{Z}$, while away from Q the curve is normal and $i_{0*}\mathbb{Z}_{x_0} \simeq \mathbb{Z}_{X_0}$. There is an exact sequence

$$0 \longrightarrow \mathbb{Z}_{X_0} \longrightarrow i_{0*}\mathbb{Z}_{x_0} \longrightarrow (\mathbb{Z})_Q \longrightarrow 0,$$

where the last term is extension by zero from Q. All three terms have $H^0 \simeq \mathbb{Z}$, and $H^1(X_0, i_{0*}\mathbb{Z}_{x_0}) = 0$; the long exact sequence gives the displayed isomorphism.

3. Reminders on non-abelian H^1

We briefly recall the non-abelian cohomological facts needed below. Most proofs are omitted; they are well known in various special cases and can be found in Giraud's thesis.

Let X be a scheme and F a sheaf of groups on X. For every *étale* X'/X, there is a pointed set $H^1(X', F)$, covariantly functorial in F and contravariantly functorial in X'. It commutes with finite products of sheaves. If $f : X \to Y$ is a morphism, define a sheaf of pointed sets $R^1 f_* F$ as the sheaf associated with the presheaf

$$Y' \longmapsto H^1(X \times_Y Y', F),$$

for $Y' \to Y$ *étale*.

Proposition 3.1. Let $0 \to F \xrightarrow{u} G$ be an injection of sheaves of groups, and let $C = G/F$ be the sheaf of homogeneous spaces under G. There is an exact sequence of sheaves of pointed sets

$$0 \to f_*F \to f_*G \to f_*C \xrightarrow{\partial} R^1 f_*F \xrightarrow{u^1} R^1 f_*G.$$

This is obtained by sheafifying, for every *étale* Y'/Y, the usual exact sequence

$$0 \to F(X') \to G(X') \to C(X') \xrightarrow{\partial} H^1(X', F) \to H^1(X', G),$$

where $X' = X \times_Y Y'$. The equivalence relation induced by ∂ is the one arising from the action of f_*G on f_*C; the equivalence relation induced by u^1 is described in the usual way by twisting.

Proposition 3.2. For morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$, and every $Z' \to Z$ *étale*, putting $Y' = Y \times_Z Z'$ and $X' = X \times_Z Z'$, there is an exact sequence

$$0 \to H^1(Y', f_*F) \to H^1(X', F) \to H^0(Y', R^1 f_*F).$$

After sheafification this gives an exact sequence of pointed sheaves on Z:

$$0 \to (R^1 g_*) f_*F \to R^1(gf)_*F \to g_*(R^1 f_*F).$$

Proposition 3.3. The functors $H^1(X', F)$, for X' *étale* over X, and $R^1 f_*F$, for arbitrary $f : X \to Y$, are effaceable. More precisely, there exists an injection $F \to G$ such that all pointed sets $H^1(X', G)$ and all sheaves $R^1 f_*G$ are zero. If X is noetherian and F is a constructible sheaf of ind-finite groups, one may choose G again a sheaf of ind-finite groups.

Proof. For the *étale* topology the standard Godement resolution suffices:

$$F \longrightarrow G = \prod_{x \in X} i_{x*} i_x^* F, \tag{3.4}$$

where x runs through the points of X and $i_x : \bar{x} \to X$ is a geometric point over x. For X'/X *étale*, the sections of G are products over the geometric points above those of X, and the map $F(X') \to G(X')$ is injective. Since over a geometric point every *étale* scheme is a disjoint sum of spectra of separably closed fields, the H^1 of each summand vanishes; Proposition 3.2 then gives $H^1(X', G) = 0$, and hence $R^1 f_*G = 0$ for every f. If F is constructible on noetherian X, finitely many geometric points already suffice for injectivity, and the finite product remains a sheaf of ind-finite groups by IX, 1.4 and IX, 1.6.

3.5

As in abelian cohomology (V, 2.4), non-abelian H^1 may be computed by the Čech procedure. We shall use this without further comment.

4. The base-change morphism

Let

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"] & Y \end{tikzcd}$$

be a cartesian square of schemes, and let F be a sheaf on X. There is a canonical base-change morphism

$$g^* R^q f_* F \longrightarrow R^q f'_* g'^* F. \tag{4.1}$$

For $q = 0$ it is induced directly by adjunction. For general q, choose an injective or effacing resolution and use the compatibility of pullback with the construction of direct images; in the non-abelian case $q = 1$ the same construction gives a morphism of pointed sheaves

$$g^* R^1 f_* F \longrightarrow R^1 f'_* g'^* F. \tag{4.2}$$

When Y' is the spectrum of a separably closed field over a geometric point ξ of Y, this morphism is the canonical map from the fibre of $R^q f_* F$ at ξ to the cohomology of the geometric fibre X_ξ.

Remark 4.3. The construction is functorial in the square and compatible with composition of base changes. It is also compatible with the exact sequences of Section 3 and with the usual long exact sequences in abelian cohomology.

Proposition 4.4. Suppose that the base-change morphism is an isomorphism for a class of sheaves stable under extensions and under the operations used in the exact sequences above. Then it remains an isomorphism after dévissage through these exact sequences. In particular, for torsion abelian sheaves it is enough to verify proper base change after reducing to constructible sheaves and then to constant sheaves on suitable finite covers.

Proof. For abelian sheaves this is the usual five-lemma argument in the long exact sequence of cohomology associated with a short exact sequence. For H^1 of groups one uses the exact sequence of pointed sets from Proposition 3.1. The compatibility of the boundary maps with base change is built into the construction of (4.1) and (4.2). Thus the property is preserved under kernels, cokernels when meaningful, extensions, and the reductions supplied by IX, 5.5.

5. Statement of the main theorem and some variants

Theorem 5.1. Let $f : X \to Y$ be a proper morphism of schemes, and let F be an abelian torsion sheaf on X. For every morphism $g : Y' \to Y$, form the cartesian square

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"] & Y. \end{tikzcd}$$

Then for every $q \geq 0$ the base-change morphism

$$g^* R^q f_* F \xrightarrow{\sim} R^q f'_* g'^* F$$

is an isomorphism. Equivalently, for every geometric point ξ of Y,

$$(R^q f_* F)_\xi \simeq H^q(X_\xi, F|_{X_\xi}).$$

The theorem is the proper base change theorem. It is first stated here with torsion coefficients; the torsion condition cannot be omitted, as the example of Section 2 shows.

Corollary 5.2. Under the hypotheses of Theorem 5.1, if F is constructible, then the formation of $R^q f_* F$ commutes with every base change. In particular, its fibres are computed on the corresponding geometric fibres.

Corollary 5.3. Let S be the spectrum of a henselian local ring, let s_0 be its closed point, and let $f : X \to S$ be proper. If X_0 is the closed fibre and F is a torsion sheaf on X, then

$$H^q(X, F) \xrightarrow{\sim} H^q(X_0, F|_{X_0})$$

for all q, whenever the relevant support and constructibility hypotheses reduce the assertion to Theorem 5.1.

Corollary 5.3 bis. The same assertion holds after replacing S by its strict henselization at a geometric point; this is simply the fibre formulation of Theorem 5.1.

Corollary 5.4. Let $f : X \to S$ be proper, with S henselian local and closed fibre X_0. For torsion sheaves satisfying the preceding hypotheses, the restriction morphism on cohomology from X to X_0 is an isomorphism. The same remains true for suitable non-abelian H^1 statements with finite or ind-finite structure groups.

Corollary 5.5. Let S be the spectrum of a henselian local ring, with closed point s_0, and let $f : X \to S$ be proper. Put $X_0 = X \times_S s_0$. Then:

(i) For a finite constant sheaf of sets F, the map $F(X) \to F(X_0)$ is bijective.

(ii) For a finite constant sheaf of groups G, the map

$$H^1(X, G_X) \longrightarrow H^1(X_0, G_{X_0})$$

is bijective.

(iii) For an abelian torsion sheaf F, the restriction map

$$H^q(X, F) \longrightarrow H^q(X_0, F|_{X_0})$$

is an isomorphism in the cases covered by Theorem 5.1.

For constant sheaves, (i) is the assertion that connected components are preserved by passage to the closed fibre. For finite constant groups, (ii) is the corresponding assertion for principal finite Galois coverings.

5.6

Let $F = T_X$ be a constant sheaf of sets on X, with value T. Then $F(X)$ is the set of locally constant maps from X to T. If $X_0 \subset X$ is a subscheme and T has at least two elements, the map $F(X) \to F(X_0)$ is bijective exactly when the map from clopen subsets of X to clopen subsets of X_0 is bijective. If X_0 has finitely many connected components, this is equivalent to bijectivity of $\Pi_0(X_0) \to \Pi_0(X)$. Thus Corollary 5.5(i) may be restated as follows.

Corollary 5.7. Let S be the spectrum of a henselian ring, with closed point s_0. Let $f : X \to S$ be proper and let X_0 be the closed fibre. Then the map

$$\Pi_0(X_0) \longrightarrow \Pi_0(X)$$

induced by $X_0 \to X$ is bijective.

In particular $\Pi_0(X)$ is finite. In the noetherian case this can be proved directly.

Lemma 5.8. Corollary 5.7 holds when S is noetherian.

Proof. Since X is noetherian, $\Pi_0(X)$ is finite. Replacing X by a connected component, assume X is connected and non-empty. The image of X in S is closed and non-empty, hence contains the closed point, so $X_0 \neq \varnothing$. Let

$$X \xrightarrow{f'} S' \xrightarrow{\pi} S$$

be the Stein factorization of f. Then S'/S is finite and S' is connected, while the fibres of f' are connected and non-empty. The space X_0 is the union of the fibres over the closed points of S'. It remains to show that S' is the spectrum of a local ring. Since S is henselian, the finite S-algebra defining S' is a product of local rings (VII, 4.1(i)); because S' is connected, only one factor occurs.

5.9.0

If $F = G_X$ is a constant sheaf of groups with finite value G, then $H^1(X, G_X)$ classifies principal Galois coverings of X with group G. Hence Corollary 5.5(ii), together with (i), is equivalent to the following specialization theorem for the fundamental group.

Theorem 5.9. With the notation of Corollary 5.5, suppose X is connected and non-empty. Then X_0 is connected and non-empty, and for every geometric point $\bar{x}_0$ of X_0, the morphism of fundamental groups

$$\Pi_1(X_0, \bar{x}_0) \longrightarrow \Pi_1(X, \bar{x}_0)$$

is an isomorphism.

Equivalently, by Galois theory for coverings:

Theorem 5.9 bis. Let S be the spectrum of a henselian ring, with closed point s_0, let $f : X \to S$ be proper, and let X_0 be the closed fibre. For a scheme Z, denote by $\mathrm{Et}(Z)$ the category of *étale* coverings of Z. Then the restriction functor

$$\mathrm{Et}(X) \longrightarrow \mathrm{Et}(X_0)$$

is an equivalence of categories.

Remark 5.10. Theorem 5.9 was already proved in SGA 1, X, 2.1 when S is the spectrum of a complete noetherian local ring. The proof given here and in the following exposé is of a quite different nature.

5.11

One may also formulate a non-commutative variant of Corollary 5.5(ii) using Giraud's two-dimensional cohomology. Assume S noetherian. Let L be a link on X, locally represented in the *étale* topology by a sheaf of ind-finite groups. Consider

$$\varphi^{(2)} : H^2(X, L) \longrightarrow H^2(X_0, L_0),$$

where L_0 is the restriction of L to X_0. Then

$$(\varphi^{(2)})^{-1}(H^2(X_0, L_0)') = H^2(X, L)', \tag{5.11.1}$$

where primes denote the subsets of neutral elements.

Equivalently, for every gerbe $\mathcal{G}$ on X with link L as above, if $\mathcal{G}_0$ denotes its restriction to X_0, the restriction functor on categories of sections

$$\mathcal{G}(X) \longrightarrow \mathcal{G}_0(X_0)$$

is an equivalence. Full faithfulness follows from Corollary 5.5(i), and essential surjectivity when $\mathcal{G}(X) \neq \varnothing$ follows from Corollary 5.5(ii). The remaining assertion is precisely that neutrality after restriction to X_0 implies neutrality over X, which is (5.11.1). The proof of (5.11.1) is not given here; one can show more generally, for noetherian X and a closed subscheme, that validity of Corollary 5.5(i) and (ii) implies (5.11.1), using the descent result VIII, 9.4(a).

SGA 4 Translation Project – Batch 024

Proper Base Change through Cohomology with Proper Supports

Draft English LaTeX translation; compile-checked, not yet line-proofed against the scan.

SGA 4, Exposé XII – continuation

The Proper Base Change Theorem

M. Artin

6. First reductions

We use the abbreviation (f, F, g) for the data of Theorem 5.1, in any one of the three cases considered there.

Lemma 6.1. For Theorem 5.1 to hold for fixed f and F, and arbitrary g, it is enough that it hold after replacing S by every strict localization $\bar{S}$ of a scheme S' locally of finite type over S, at a point closed in its fibre over S. More precisely, if $\bar{g} : \bar{s} \to \bar{S}$ is the inclusion of the closed point and $(\bar{f}, \bar{F})$ is obtained from (f, F) by base change, it is enough to know the theorem for $(\bar{f}, \bar{F}, \bar{g})$.

Proof. One may assume $S = \operatorname{Spec} A$ and $S' = \operatorname{Spec} A'$. Write A' as the filtered inductive limit of its finite-type A-subalgebras A'_i. The limit formalism of Exposé VII, 5.11 and 5.14, reduces the question to the case where g is of finite type. To prove that the base-change homomorphisms are isomorphisms, it is enough to test on geometric fibres at points of S' which are closed in their fibre over S (VIII, 3.13). If $\bar{s}'$ is such a geometric point, the residue field extension is algebraic, so $\bar{s}'$ may also be regarded as a geometric point $\bar{s}$ of S. Let $\bar{S}$ and $\bar{S}'$ be the corresponding strict localizations. There is a commutative square

$$\begin{tikzcd} \bar{s}' \arrow[r] \arrow[d] & \bar{s} \arrow[d] \\ \bar{S}' \arrow[r] & \bar{S}, \end{tikzcd}$$

whose upper horizontal arrow is an isomorphism. By VIII, 5.2 and 5.3, the map induced on the fibre at $\bar{s}'$ by the base-change morphism for (f, F, g) identifies with

$$H^i(\bar{X}, \bar{F}) \longrightarrow H^i(\bar{X}', \bar{F}'),$$

where $\bar{X}$ and $\bar{X}'$ are obtained by base change from $\bar{S}$ and $\bar{S}'$. This map fits into the evident square comparing the cohomology of the special fibres. The top arrow is an isomorphism, and the two vertical arrows are isomorphisms by hypothesis. Hence the bottom arrow is an isomorphism. $\square$

Thus, in proving Theorem 5.1 for a fixed situation $(F, f, -)$, we are reduced to the special case of Corollary 5.5 with S strictly local. We shall need a general criterion for a morphism $h : Y \to X$ to induce bijections

$$H^i(X, F) \longrightarrow H^i(Y, h^*F).$$

The following elementary statements are recorded in somewhat greater generality than is strictly needed later.

Lemma 6.2. Let C and C' be abelian categories, let $T^\bullet$ and $T'^\bullet$ be cohomological functors from C to C', and let $\varphi : T^\bullet \to T'^\bullet$ be a morphism of cohomological functors. Assume both functors vanish in negative degrees and that T^i is effaceable for $i > 0$. For an integer n, the following are equivalent:

a) φ^i is an isomorphism for $i \leq n$ and a monomorphism for $i = n + 1$.

b) If $n \geq -1$, φ^0 is a monomorphism and φ^i is an epimorphism for $i \leq n$.

c) If $n \geq 0$, φ^0 is an isomorphism and T'^i is effaceable for $0 < i \leq n$.

When these conditions hold, $\varphi^{n+1}(A) : T^{n+1}(A) \to T'^{n+1}(A)$ is an epimorphism if and only if $T'^{n+1}(A)$ is effaceable. If $C' = \mathbf{Ab}$ and $\xi \in T'^{n+1}(A)$, then ξ is in the image of $T^{n+1}(A)$ if and only if it is effaceable.

Proof. The implication (a) $\Rightarrow$ (b) and the implication (a) $\Rightarrow$ (c) are immediate. Since the positive T^i are right satellites of T^0, the converse implications are the usual uniqueness and effaceability criterion for universal δ-functors. Equivalently, one argues by induction on n, embedding an object A in an object B which effaces $T^{n+1}(A)$, and applying the five lemma to the long exact cohomology sequences associated with $0 \to A \to B \to C \to 0$. This also gives the final effaceability criterion. $\square$

Corollary 6.3. Suppose that $T^\bullet$ and $T'^\bullet$ extend to cohomological functors on $D^+(C)$, vanishing in negative degrees on complexes whose cohomology vanishes in negative degrees. Let $K^\bullet \in D^+(C)$ with $H^i(K^\bullet) = 0$ for $i < 0$.

(i) If condition (a) of Lemma 6.2 holds, then

$$T^i(K^\bullet) \longrightarrow T'^i(K^\bullet)$$

is an isomorphism for $i < n$, and a monomorphism for $i = n + 1$.

(ii) If $u : L^\bullet \to K^\bullet$ induces an injection on H^0, then surjectivity of

$$T^{n+1}(L^\bullet) \to T'^{n+1}(L^\bullet)$$

is equivalent to the condition that the image of $T'^{n+1}(L^\bullet)$ in $T'^{n+1}(K^\bullet)$ lie in the image of $T^{n+1}(K^\bullet)$. In the case $C' = \mathbf{Ab}$, this is the corresponding elementwise criterion.

Proof. For (i), truncate $K^\bullet$ at the least degree where its cohomology is nonzero and argue by descending induction, using the exact triangle

$$H^m(K^\bullet)[-m] \longrightarrow K^\bullet \longrightarrow K'^\bullet \longrightarrow .$$

The associated long exact sequences and the five lemma give the assertion. For (ii), use the exact triangle attached to a mapping cylinder of u, and the same diagram chase; the first two vertical arrows are isomorphisms, while the last two are monomorphisms by (i).

Lemma 6.4. Theorem 5.1 holds when f is finite.

This is exactly VIII, 5.5 and 5.8.

Proposition 6.5. Let $h : Y \to X$ be quasi-compact and quasi-separated. In (i), let I be a set with at least two elements; in (ii) and (iii), let $\mathcal{L}$ be a nonempty set of primes; in (iii), let n be an integer.

(i) The following are equivalent: for every sheaf of sets F on X, the map

$$H^0(X, F) \to H^0(Y, h^*F)$$

is injective (respectively bijective); equivalently, after every finite morphism $X' \to X$, the induced map on locally constant maps with value I, or equivalently on clopen subsets, is injective (respectively bijective). If X is locally noetherian, this can be stated as surjectivity (respectively bijectivity) of $\Pi_0(Y') \to \Pi_0(X')$ for all finite $X' \to X$. Under these conditions, pullback on torsors under any sheaf of groups is faithful (respectively fully faithful), and pullback on étale coverings is faithful (respectively fully faithful).

(ii) The following are equivalent: for every sheaf of ind-$\mathcal{L}$-finite groups F, the maps in H^0 and H^1

$$H^i(X, F) \to H^i(Y, h^*F), \qquad i = 0, 1,$$

are bijective; equivalently this may be tested after every finite $X' \to X$ on constant finite $\mathcal{L}$-groups. If $\mathcal{L}$ is the set of all primes, this is the same as saying that pullback induces an equivalence on categories of étale coverings after every finite base change; in the noetherian case, it may be phrased in terms of Π_0 and Π_1.

(iii) The following are equivalent: for every $\mathcal{L}$-torsion abelian sheaf F,

$$H^i(X, F) \to H^i(Y, h^*F)$$

is an isomorphism for $i \leq n$ and a monomorphism for $i = n + 1$; equivalently, for $n \geq -1$, it is injective in degree 0 and surjective in degrees $i \leq n$; equivalently, this can be tested after finite $X' \to X$ on the constant sheaves $\mathbb{Z}/\ell^\nu\mathbb{Z}$, $\ell \in \mathcal{L}$, $\nu > 0$.

Corollary 6.6. Assume the hypotheses of Proposition 6.5. Then the following lifting criteria hold.

(i) Under the injective form of 6.5(i), if $\xi \in H^0(Y, h^*F)$ becomes induced from X after a monomorphism $F \hookrightarrow G$, then ξ was already induced from X.

(ii) Under the bijective form of 6.5(i), the same assertion holds for classes $\xi \in H^1(Y, h^*F)$ of torsors under a sheaf of groups.

(iii) Under 6.5(iii) for a given n, the same assertion holds for $\xi \in H^{n+1}(Y, h^*F)$, where F is an abelian $\mathcal{L}$-torsion sheaf.

Proof. The abelian assertion is Lemma 6.2 (or Corollary 6.3 in the derived formulation) applied to sheaves of $\mathcal{L}$-torsion. For (i), form the pushout $H = G \amalg_F G$; since monomorphisms in a topos are effective, $F \to G \rightrightarrows H$ is exact, and a diagram chase in global sections gives the claim. For (ii), use the sheaf $\mathrm{Isom}_F(P, P')$: the hypothesis on H^0 implies that pullback on F-torsors is faithful or fully faithful. If a h^*F-torsor becomes the pullback of a G-torsor, it is described by a section of the quotient torsor Q/F; the bijectivity of H^0 lifts this section to X, hence descends the reduction of structure group.

For Proposition 6.5 itself, reduce constructible sheaves by IX, 2.7.2 and IX, 2.14 to finite direct images of constant sheaves along finite maps. In the abelian case use the commutation of cohomology with filtered inductive limits (VII, 3.3) and Lemma 6.2. In the non-abelian case the same reduction uses effaceability of non-abelian H^1 and the exact sequence of torsors. The interpretations by clopen subsets, connected components, and étale coverings are the standard Galois-theoretic translations.

Remark 6.7. The arguments proving 6.5 and 6.6 are formal. They could be isolated as abstract lemmas in which the sheaves $p_*G_{X'}$, for finite $p : X' \to X$, play the role of cogenerators. When X is noetherian, the tests in 6.5 may often be restricted to integral finite X', and, for universally Japanese X, even to normal integral X'. The proof of 6.6 and the implications from (a) to the other forms do not use quasi-compactness or quasi-separatedness; those hypotheses enter only through passage to filtered limits of sheaves.

Proposition 6.8 (descent lemma). With the notation of Proposition 6.5, let $p : \bar{X} \to X$ be a morphism, let $\bar{Y} = Y \times_X \bar{X}$, and let $\bar{h} : \bar{Y} \to \bar{X}$, $p_Y : \bar{Y} \to Y$ be the evident maps. Assume either that p is surjective, or that h is a closed immersion and $X = p(\bar{X}) \cup h(Y)$.

(i) If $\bar{h}$ satisfies the injective (respectively bijective) condition of 6.5(i), and in the bijective case if

$$h^* p_* \bar{F} \longrightarrow p_{Y*} \bar{h}^* \bar{F}$$

is injective for every sheaf of sets $\bar{F}$ on $\bar{X}$, then h satisfies the corresponding condition 6.5(i).

(ii) If $\bar{h}$ satisfies 6.5(ii) and the same base-change morphism is bijective for every ind-$\mathcal{L}$-group sheaf $\bar{F}$, then h satisfies 6.5(ii).

(iii) If $\bar{h}$ satisfies 6.5(iii), and, for every abelian $\mathcal{L}$-torsion sheaf $\bar{F}$ on $\bar{X}$,

$$h^* R^i p_* \bar{F} \longrightarrow R^i p_{Y*} \bar{h}^* \bar{F}$$

is bijective for $i \leq n-1$ and injective for $i = n$, then h satisfies 6.5(iii).

Proof. If necessary replace $\bar{X}$ by the disjoint union $\bar{X} \amalg Y$, so that p is surjective. Then every sheaf F on X injects into $p_* p^* F$. In case (i), reduce to sheaves of the form $p_* \bar{F}$ and compare the square of global sections on $X, Y, \bar{X}, \bar{Y}$; injectivity or surjectivity follows from the hypothesis on $\bar{h}$ and the base-change map. In case (ii), use the non-abelian exact sequence for H^1 attached to p and p_Y; the middle vertical map is bijective by the hypothesis on $\bar{h}$, while the terms measuring obstruction are controlled by (i) and by the base-change hypothesis. In case (iii), replace the injection $F \to p_* p^* F$ by the morphism $F \to Rp_* p^* F$ in D^+, apply Corollary 6.3, and compare hypercohomology through the base-change morphism to $Rp_{Y*} \bar{h}^* \bar{F}$. The composite is the map induced by $\bar{h}$, hence an isomorphism in the prescribed range.

Corollary 6.9. In the situation of 6.8, assume in addition that h itself satisfies the relevant condition of 6.5 and that the base-change morphisms for p are isomorphisms in the same range. Let F be a sheaf on X of the corresponding type, and let ξ be a class on Y. Then ξ comes from X if and only if its pullback to $\bar{Y}$ comes from $\bar{X}$.

This is exactly the sharpened form of the proof of 6.8: the hypercohomology comparison shows that the only obstruction is detected after the auxiliary cover p.

Corollary 6.10. Assume the injective form of 6.5(i). To obtain the bijective form of 6.5(i), or 6.5(ii), or 6.5(iii), it is necessary and sufficient that the following local lifting condition hold. For every sheaf F of the relevant type, every class ξ in the relevant cohomology of Y, and every nonempty closed subset $X' \subset X$, there exists a morphism $p : \bar{X} \to X$ such that:

1°) $p(\bar{X}) \subset X'$ and contains a nonempty open subset of X';

2°) the relevant base-change morphisms for p are isomorphisms in the necessary degrees;

3°) the pullback of ξ to $\bar{Y}$ lies in the image of the cohomology of $\bar{X}$.

Moreover, if the theorem is already known through degree n, the same criterion characterizes whether a class in degree $n+1$ descends.

Proof. One direction is immediate from 6.9. Conversely, suppose a class ξ does not descend. Let Φ be the set of closed subsets $X' \subset X$ on which the restriction of ξ still does not descend, ordered by reverse inclusion. The limit theorem VII, 5.8, shows that Φ is inductive; choose a minimal element. Applying the stated local lifting condition to this minimal closed set, and enlarging p by adjoining the complementary closed part if needed, we may assume p is surjective over it. Corollary 6.9 then forces descent on the minimal closed set, a contradiction.

Proposition 6.11 (transitivity lemma). With the notation of 6.8, but without assuming p surjective, suppose that h satisfies the relevant condition of 6.5. Suppose also that the corresponding base-change morphisms for p are monomorphisms or isomorphisms in the same range: for sheaves of sets in case (i), for ind-$\mathcal{L}$-groups in case (ii), and for abelian $\mathcal{L}$-torsion sheaves in case (iii). Then $\bar{h} : \bar{Y} \to \bar{X}$ satisfies the same condition.

Proof. For sheaves of sets, the diagram used in the proof of 6.8 is read backwards. For ind-$\mathcal{L}$-groups, efface the two obstruction terms in the exact sequence for H^1 attached to p, using Proposition 3.3; the base-change hypotheses identify these terms with the corresponding terms over X, where they are effaceable. In the abelian case, compare

$$R\Gamma(\bar{X}, \bar{F}) \longrightarrow R\Gamma(\bar{Y}, \bar{h}^* \bar{F})$$

with the composite

$$R\Gamma(X, Rp_* \bar{F}) \longrightarrow R\Gamma(Y, h^* Rp_* \bar{F}) \longrightarrow R\Gamma(Y, Rp_{Y*} \bar{h}^* \bar{F}).$$

The first arrow is controlled by the hypothesis on h and Corollary 6.3, and the second by the base-change hypothesis. ☐

Remarks 6.12–6.13. The proof in cases (i) and (iii) gives a more precise statement for a fixed sheaf $\bar{F}$: it is enough to know base change for $R^i p_* \bar{F}$ and the corresponding cohomological comparison for each $R^j p_* \bar{F}$. An analogous fixed-sheaf statement for case (ii) would involve non-abelian two-dimensional cohomology, which is why the proof above uses effaceability instead.

The principal example of the conditions in 6.5 is the situation of Corollary 5.5, proved in the next exposé using the reductions of the present one. Similar questions arise for henselian pairs (X, Y), for formal completions defined by an ideal, and for putative “étale localizations along a closed subset”. One may conjecture that henselian pairs satisfy the non-abelian and abelian forms of 6.5, and ask when a weaker condition on sections of étale covers already implies henselianity. These remarks anticipate later developments and will not be used in the proof.

7. A variant of Chow's lemma

We need the following form of Chow's lemma, stripped of noetherian and irreducibility hypotheses.

Lemma 7.1. Let S be quasi-compact and quasi-separated, and let $f : X \to S$ be separated of finite type, with $X \neq \varnothing$. There is a commutative diagram

$$\begin{tikzcd} X \arrow[dr, "f"'] & & \bar{X} \arrow[ll, "\pi"'] \arrow[dl, "\bar{f}"] \\ & S & \end{tikzcd}$$

where π is projective, $\bar{f}$ is quasi-projective, and there is a nonempty open $U \subset X$ such that π induces an isomorphism $\pi^{-1}(U) \simeq U$.

Proof. First note a permanence property: if the lemma holds for a closed subscheme $X' \subset X$ and for a dense open $V \subset X'$ which is also open in X, then it holds for X, by composing the projective morphism $\bar{X}' \to X'$ with the closed immersion $X' \to X$. Cover X by finitely many nonempty affine opens $U_1, \ldots, U_n$. Since S is quasi-compact and quasi-separated, each $U_i \to S$ is quasi-affine. If $U = \bigcap U_i$ is dense, the usual proof of Chow's lemma (EGA II, 5.6.1) applies. If the intersection is nonempty but not dense, replace X by the closure of that intersection. If the intersection is empty, choose a maximal nonempty partial intersection and replace X by its closure, reducing the number of open sets. Induction on the number of opens gives the result.

8. Final reductions

Lemma 8.1. To prove any of the assertions 5.1(i), (ii), (iii), it is enough to prove it when f is projective and S is noetherian.

Proof. Assume first that the theorem is known for projective f. By Lemma 6.1, it is enough to handle the case of Corollary 5.5 with S strictly local. Apply Corollary 6.10 with $Y = X_0$, the closed fibre. Given the closed set and class occurring there, use Lemma 7.1 to find $p : \bar{X} \to X$ whose image contains a nonempty open part and with $\bar{X}$ projective over S. The projective case supplies the required base-change isomorphisms and the cohomological lifting condition, so Corollary 6.10 gives the theorem.

It remains to reduce the projective case to noetherian base. Again by 6.1 we may take S strictly local and prove Corollary 5.5. Embed X in $\mathbb{P}^r_S$ and replace X by $\mathbb{P}^r_S$ using the compatibility already established in 4.4. If $S = \operatorname{Spec} A$, write A as a filtered limit of noetherian henselian local rings A_i mapping locally to A, and use VII, 5.7 to commute cohomology with this limit. The closed fibres are obtained by the corresponding filtered limit of residue fields. Thus the theorem over A follows from the theorem over the noetherian A_i.

Corollary 8.2. Theorem 5.1(i) is true.

Indeed, combine Lemma 8.1, the criterion 6.5(i) in the connected-component form, and Lemma 5.8.

Lemma 8.3. To prove any of 5.1(i), (ii), (iii), it is enough to prove it for f projective of relative dimension ≤ 1, with S noetherian.

Proof. By Lemma 8.1 we reduce to $X = \mathbb{P}^r_S$. Induct on r. For $r = 1$ this is the stated relative-dimension case. For $r \geq 2$, blow up the standard $\mathbb{P}^{r-2}_S \subset \mathbb{P}^r_S$. The blow-up $\bar{X}$ maps both to X and to $X_1 = \mathbb{P}^1_S$; over X_1 it is locally a projective $(r-1)$-space. The fibres of $\bar{X} \to X$ and $X_1 \to S$ have dimension at most one. Applying the dimension-one hypothesis to the former and to $X_1 \to S$, and the induction hypothesis to $\bar{X} \to X_1$, the descent and transitivity lemmas 6.8 and 6.11 give the result for $X \to S$.

Lemma 8.4. To prove Theorem 5.1(ii), it is enough to prove the specialization theorem for the fundamental group, Theorem 5.9 bis, when f is projective of relative dimension ≤ 1 and S is noetherian strictly local.

This is immediate from Lemma 8.3, Lemma 6.1, and criterion 6.5(ii).

Lemma 8.5. To prove Theorem 5.1(iii), it is enough to prove Theorem 5.1(ii) and the following Picard-surjectivity statement.

Proposition 8.6. Let S be henselian noetherian, let $f : X \to S$ be projective of relative dimension ≤ 1, and let X_0 be the closed fibre. Then

$$\operatorname{Pic} X \longrightarrow \operatorname{Pic} X_0$$

is surjective.

Proof. By 6.5(iii) and Corollary 8.2, it remains to prove surjectivity of

$$H^q(X, \mathbb{Z}/\ell^\nu\mathbb{Z}) \longrightarrow H^q(X_0, \mathbb{Z}/\ell^\nu\mathbb{Z})$$

for $q \geq 1$. Degree 1 follows from 5.1(ii). Since X_0 has dimension at most one over a separably closed field, the groups vanish for $q > 2$ if $\ell \neq \operatorname{char} k$, and for $q > 1$ if $\ell = \operatorname{char} k$. It remains to handle $q = 2$, $\ell \neq \operatorname{char} k$. Put $n = \ell^\nu$. Since S is strictly local, $\mu_n \simeq \mathbb{Z}/n\mathbb{Z}$ on S, hence on X. The Kummer sequences for X and X_0 give a diagram

$$\begin{tikzcd}
\operatorname{Pic} X \arrow[r] \arrow[d] & H^2(X, \mu_n) \arrow[r] \arrow[d] & H^2(X, \mathbb{G}_m) \arrow[d] \\
\operatorname{Pic} X_0 \arrow[r] & H^2(X_0, \mu_n) \arrow[r] & H^2(X_0, \mathbb{G}_m).
\end{tikzcd}$$

Since $H^2(X_0, \mathbb{G}_m) = 0$ by IX, 4.6, surjectivity of $\operatorname{Pic} X \to \operatorname{Pic} X_0$ implies surjectivity in degree 2, proving the lemma.

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SGA 4, Exposé XIII

The Proper Base Change Theorem: End of the Proof

M. Artin

1. The projective flat case

We recall Theorem 5.9 bis of Exposé XII in the case needed here.

Proposition 1.1. Let S be the spectrum of a noetherian henselian ring, let $f : X \to S$ be projective and flat, and let X_0 be the closed fibre. Then the restriction functor

$$\mathrm{Et}(X) \longrightarrow \mathrm{Et}(X_0)$$

is an equivalence of categories.

Full faithfulness follows already from XII, 6.5(i). It remains to show that every étale covering of X_0 comes from one of X.

Lemma 1.2. Let S be locally noetherian, let $f : X \to S$ be projective and flat, and let $Y \subset X$ be a subscheme. The subfunctor of the final functor on $(\mathrm{Sch})/S$ defined by the condition $Y' = X'$ after base change to S' is representable by a subscheme $Z \subset S$. If Y is open (respectively closed), then so is Z.

Proof. If Y is closed in an open $U \subset X$, the condition $U' = X'$ is represented by the complement of the proper image of $X - U$. We reduce to Y closed. Let $\mathcal{O}_X \twoheadrightarrow \mathcal{O}_Y$. The condition $Y = X$ is equivalent to the existence of a splitting $\mathcal{O}_Y \to \mathcal{O}_X$, or equivalently to lifting the identity section of $\mathcal{H}om(\mathcal{O}_X, \mathcal{O}_X)$ to a section of $\mathcal{H}om(\mathcal{O}_Y, \mathcal{O}_X)$. By EGA III, 7.7.8, these section functors are represented by vector bundles over S, and the inverse image of the identity section is a closed subscheme of S.

Lemma 1.3. Let S be locally noetherian, $f : X \to S$ projective and flat, and E locally free on X. The functor QE, assigning to S'/S the quotients A' of E' endowed with the structure of an $\mathcal{O}_{X'}$-algebra étale over X', is represented by a scheme locally of finite type over S.

Proof. The relative $\mathrm{Quot}(E)$ functor is represented by a scheme locally of finite type. The condition that the universal quotient be flat over X' is open, so we reduce to the following representability statement.

Lemma 1.4. Let S be locally noetherian, $f : X \to S$ projective and flat, and M locally free on X. The functor assigning to S'/S the algebra structures on M' which are étale over X' is represented by a scheme of finite type over S.

Proof. Bilinear multiplication laws on M are represented by the vector bundle associated with $\mathcal{H}om(M \otimes M, M)$. The associativity and commutativity conditions are equalities of sections of suitable vector bundles; Lemma 1.2 represents the locus where they hold. After imposing these, the existence of a unit is represented by a vector bundle of sections of M, again cut out by an equality of sections. Finally, for a commutative unital algebra structure, the condition of being étale is open. This proves Lemma 1.4 and hence Lemma 1.3.

1.5

Let Z/S represent the functor in Lemma 1.3. We examine smoothness of Z over S. Translating SGA 1, III, 3.1, into the language of QE, one obtains the following infinitesimal lifting criterion. Let B be an artinian local ring with maximal ideal $\mathfrak{m}$, let $J \subset B$ satisfy $J\mathfrak{m} = 0$, let $S' = \operatorname{Spec} B$ and $S'' = \operatorname{Spec} B/J$. Given a quotient A'' of E'' with an étale algebra structure over X'', it should lift to a quotient A' of E' with étale algebra structure over X'.

Since X' and X'' have the same underlying space, an étale algebra A' lifting A'' is unique up to unique isomorphism. The only obstruction is therefore to lifting the surjection $E'' \twoheadrightarrow A''$. The exact sequence

$$0 \to J \to B \to B/J \to 0$$

gives

$$0 \to J \otimes \mathcal{H}om(E', A') \to \mathcal{H}om(E', A') \to \mathcal{H}om(E'', A'') \to 0,$$

so the obstruction lies in $H^1(X', J \otimes \mathcal{H}om(E', A'))$. Since $J\mathfrak{m} = 0$, this is $J \otimes H^1(X_0, \mathcal{H}om(E_0, A_0))$, after compatibility of cohomology with base-field extension.

Corollary 1.6. Let Z/S represent the functor of Lemma 1.3, let $X_Z = X \times_S Z$, and let A_Z be the universal étale quotient algebra. Then Z is smooth over S at every point z_0 for which

$$H^1(X_0, \mathcal{H}om(E_0, A_0)) = 0,$$

where the subscript 0 denotes restriction to the fibre of X_Z/Z over z_0.

We now finish the proof of Proposition 1.1. Let $Y_0 \to X_0$ be an étale covering, corresponding to an étale algebra A_0. For n large, $A_0(n)$ is generated by global sections, hence there is a surjection

$$\mathcal{O}_{X_0}^N(-n) = E_0 \twoheadrightarrow A_0.$$

Choose n even larger so that

$$H^1(X_0, \mathcal{H}om(E_0, A_0)) = 0.$$

Let $E = \mathcal{O}_X^N(-n)$, and let Z/S be the representing scheme. The quotient A_0 determines a rational point z_0 of the closed fibre of Z, and by Corollary 1.6, Z is smooth at z_0. Since S is henselian, a smooth S-scheme with a point in the closed fibre has a section through that point. The corresponding quotient of E is an étale algebra on X lifting A_0, proving essential surjectivity.

2. The case of relative dimension ≤ 1

Proposition 2.1. With the notation of XII, 5.9 bis, if S is noetherian strictly local and $f : X \to S$ is projective of relative dimension ≤ 1, then

$$\mathrm{Et}(X) \longrightarrow \mathrm{Et}(X_0)$$

is an equivalence of categories. Together with XII, 8.5, this proves XII, 5.1(ii).

Lemma 2.2. Let $S = \operatorname{Spec} A$ be local noetherian, and let X/S be projective of relative dimension $\leq n$. There exists a projective flat S-scheme Z of relative dimension $\leq n$, and a closed immersion $X \hookrightarrow Z$ over S.

Proof. Embed X in $\mathbb{P}^N_S$. If J is the homogeneous ideal defining X, then over the closed fibre, of dimension $\leq n$, choose $N - n$ homogeneous equations in the image of J cutting out a subscheme of dimension n. Lift them to elements of J, and let Z be the closed subscheme they define in $\mathbb{P}^N_S$. By EGA IV, 11.3, Z is flat over S, and by EGA IV, 13.1.5, its relative dimension is n. □

Proof of Proposition 2.1. Assume X and X_0 connected. Embed X as a closed subscheme of a projective flat Z/S of relative dimension at most one. We have

$$\begin{array}{ccc} \mathrm{Et}(Z) & \longrightarrow & \mathrm{Et}(X) \\ \downarrow & & \downarrow \\ \mathrm{Et}(Z_0) & \longrightarrow & \mathrm{Et}(X_0). \end{array}$$

The right vertical functor is fully faithful by XII, 5.8. It remains to show essential surjectivity. The left vertical functor is essentially surjective by Proposition 1.1. Thus it is enough to show that every étale covering of X_0 is induced from one of Z_0. We are reduced to the following proposition over a separably closed field.

Proposition 2.3. Let k be separably closed, let Z be locally of finite type over k and of dimension ≤ 1, and let $X \hookrightarrow Z$ be a connected closed subscheme. Every étale covering of X is induced by an étale covering of Z.

Proof. The zero-dimensional case is trivial. Otherwise write $Z = X \cup Y$, where Y is the closure of $Z - X$, and put $V = X \cap Y$, a discrete locally artinian scheme. The finite surjective morphism $X \amalg Y \to Z$ is an effective descent morphism for étale coverings. Moreover

$$(X \amalg Y) \times_Z (X \amalg Y) = X \amalg Y \amalg V^{(1)} \amalg V^{(2)},$$

where the two copies of V are exchanged between the X- and Y-components by the projections. Given an étale covering $X' \to X$ of degree n, choose any étale covering $Y'' \to Y$ of degree n, for instance the trivial one. A descent datum is just an isomorphism between the two induced covers of V. Since k is separably closed, every étale covering of V is split, so such an isomorphism exists.

Remark 2.4. One can avoid the use of the full descent theorem by proving directly that $\mathrm{Et}(Z)$ is equivalent to the category of triples (X', Y', θ), where X' and Y' are étale coverings of X and Y, and θ identifies their pullbacks to V. This follows from the analogous gluing statement for quasi-coherent modules on a union of two closed subschemes.

3. An auxiliary result on the Picard group

Proposition 3.1. Let X be noetherian, and let $X_0 \subset X$ be closed such that: (a) $\dim X_0 \leq 1$; (b) for every nonempty connected closed subset $Y \subset X$, the intersection $Y_0 = Y \cap X_0$ is nonempty and connected. Let Z_0 be the union, over $x \in \mathrm{Ass}\,\mathcal{O}_X$, of the sets $\overline{\{x\}} \cap X_0$ when this intersection is reduced to one point.

(i) Every Cartier divisor D_0 on X_0 whose support avoids Z_0 is induced by a Cartier divisor D on X. If $D_0 \geq 0$, then D may be chosen ≥ 0.

(ii) If there is an affine open $U_0 \subset X_0$ containing the finite set $\mathrm{Ass}\,\mathcal{O}_{X_0} \cup Z_0$ (for instance if X_0 has an ample invertible sheaf), then $\mathrm{Pic}\,X \to \mathrm{Pic}\,X_0$ is surjective.

Proof. For (i), decompose any Cartier divisor on the one-dimensional scheme X_0 into the difference of effective Cartier divisors. At a point x of the support of an effective divisor, choose a local equation $f_x \in \mathcal{O}_{X_0,x}$ which is a non-zero-divisor and lift it to a section g_x of $\mathcal{O}_X$ on a neighbourhood U_x. By avoiding the associated points of $\mathcal{O}_X$, using the definition of Z_0 and the connectedness hypothesis, shrink U_x so that g_x is a non-zero-divisor. The closure of $V(g_x)$ has a connected component meeting X_0 only at x; that component defines a Cartier divisor on X inducing the desired local contribution. Summing over the finite support gives D.

For (ii), every invertible sheaf on X_0 is represented by a Cartier divisor whose support avoids Z_0: on the affine open containing $\mathrm{Ass}\,\mathcal{O}_{X_0} \cup Z_0$, the invertible sheaf is trivial after passing to the corresponding semilocal ring. Apply (i).

Corollary 3.2. Let S be the spectrum of a noetherian henselian ring, and let $f : X \to S$ be proper of relative dimension ≤ 1. If $X_0' \subset X$ is a closed subscheme with the same underlying space as the closed fibre X_0, then

$$\mathrm{Pic}\,X \longrightarrow \mathrm{Pic}\,X_0'$$

is surjective.

Apply Proposition 3.1 to (X, X_0'). The dimension condition is immediate, and connectedness of intersections follows from XII, 5.7. The affine-open hypothesis follows because a proper curve over a field is projective, hence has ample invertible sheaves. This completes the proof of XII, 5.1(iii).

Remark. If one does not wish to use projectivity of proper curves over a field, Corollary 3.2 may be used only in the projective case, which is enough for the application in XII. Conversely, accepting projectivity of proper algebraic curves, Corollary 3.2 also implies that X is projective over S under its hypotheses: lift an ample class from X_0 to X and apply EGA III, 4.7.1.

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SGA 4, Exposé XIV

A Finiteness Theorem for a Proper Morphism; Cohomological Dimension of Affine Algebraic Schemes

M. Artin

This exposé contains two important theorems whose proofs use the proper base change theorem.

1. Finiteness theorem for a proper morphism

Theorem 1.1. Let $f : X \to S$ be proper and of finite presentation. Let A be a left noetherian ring which is a $\mathbb{Z}/n\mathbb{Z}$-algebra for some n. Let F be a constructible sheaf of sets (respectively groups, respectively A-modules) on X. Then f_*F (respectively $R^1 f_* F$, respectively $R^q f_* F$ for every $q \geq 0$) is constructible.

Corollary 1.2. If X is proper over the spectrum of a separably closed field k, and F is a constructible torsion abelian sheaf on X, then the groups $H^q(X, F)$ are finite for all $q \geq 0$.

Remark 1.3. Properness is essential for abelian sheaves: take $F = \mathbb{Z}/p\mathbb{Z}$ on $\mathbb{A}^1_k$, with $p = \operatorname{char} k$, and either compactify into $\mathbb{P}^1_k$ or take the structural morphism. One expects broader finiteness results for coefficients prime to the residual characteristics over excellent bases; in characteristic zero this follows from resolution of singularities.

Proof of Theorem 1.1. The question is local on S. Write affine S as an inverse limit of schemes S_0 of finite type over $\operatorname{Spec} \mathbb{Z}$. The data descend to some S_0, and proper base change for $S \to S_0$ reduces constructibility to the case S of finite type over $\operatorname{Spec} \mathbb{Z}$.

Lemma 1.4. It is enough to prove Theorem 1.1 when S is of finite type over $\operatorname{Spec} \mathbb{Z}$.

Lemma 1.5. The assertion of Theorem 1.1 for sheaves of sets is true: if F is constructible, then f_*F is constructible.

Proof. Assume S noetherian. Embed F into a finite product $G = \prod_i \pi_{i*} C_i$, with $\pi_i : X_i \to X$ finite and C_i constant constructible (IX, 2.14). It is enough to treat $F = C_X$ constant finite. By proper base change, the geometric fibre of f_*F at $\bar{s}$ is $C^{\pi_0(X_{\bar{s}})}$. This is finite, and the function $s \mapsto \#\pi_0(X_{\bar{s}})$ is constructible by EGA IV, 9.7.9. Hence f_*F is constructible.

Lemma 1.6. Let S be noetherian, $f : X \to S$ proper, and F constructible, either as a sheaf of groups or of A-modules. If $S = \bigcup S_i$ is a finite union of subschemes and the restrictions $R^q f_{i*} F_i$ are constructible on each S_i, then $R^q f_* F$ is constructible on S.

This follows from proper base change, which identifies the restriction of $R^q f_* F$ to S_i with $R^q f_{i*} F_i$, and from the gluing criterion for constructible sheaves (IX, 2.8).

Lemma 1.7. Let S be noetherian and suppose

$$\begin{tikzcd} X \arrow[dr, "f"'] & & \bar{X} \arrow[ll, "\pi"'] \arrow[dl, "\bar{f}"] \\ & S & \end{tikzcd}$$

is a commutative diagram of proper morphisms. Let $j : U \hookrightarrow X$ be an open such that $\bar{U} = U \times_X \bar{X} \to U$ is an isomorphism, and let $Y = X - U$ with its reduced closed structure. If Theorem 1.1 holds for $\bar{f}$, for $Y \to S$, and for π, then it holds for f.

Proof. For modules, use

$$0 \to j_! j^* F \to F \to i_* i^* F \to 0,$$

where $i : Y \hookrightarrow X$. The term supported on Y is handled by the induction hypothesis on $Y \to S$. For $j_! j^* F$, one has

$$j_! j^* F \simeq \pi_* \bar{j}_! \bar{j}^* \pi^* F,$$

verified fibrewise by proper base change. Higher direct images under π of $\bar{j}_! \bar{j}^* \pi^* F$ vanish because π is an isomorphism over U and the fibres over Y carry no support. Leray then identifies the required direct images for f with those for $\bar{f}$, which are constructible. The set and group cases follow from the same dévissage, using the corresponding exactness statements for H^0 and H^1.

Lemma 1.8. Assume S is of finite type over $\operatorname{Spec} \mathbb{Z}$. It is enough to prove Theorem 1.1 when f is projective.

Proof. Apply the Chow lemma of XII, 7.1 to a nonempty closed part of X, and then Lemma 1.7. Noetherian induction on closed subsets of X gives the result.

Lemma 1.9. It is enough to prove Theorem 1.1 when S is of finite type over $\operatorname{Spec} \mathbb{Z}$ and $X = \mathbb{P}^r_S$.

For a projective X, embed it in $\mathbb{P}^r_S$; extend a constructible sheaf by zero, and use the compatibility of direct image with closed immersions.

Lemma 1.10. Theorem 1.1 is true.

Proof. We now argue by induction on r for $X = \mathbb{P}^r_S$. The case $r = 0$ is finite. For $r \geq 1$, stratify S so that the constructible sheaf is locally constant on a suitable stratification of $\mathbb{P}^r_S$, and apply the hyperplane/blow-up reduction already used in XII, 8.3: blowing up a standard $\mathbb{P}^{r-2} \subset \mathbb{P}^r$ reduces to morphisms of relative dimension 1 and to a projective bundle of smaller relative projective dimension. The proper base change theorem controls all restrictions to strata, while Lemma 1.6 glues the constructibility. This completes the proof.

Remark 1.11. The proof above is deliberately parallel to the reductions in Exposé XII. The noetherian hypotheses enter only through constructibility and noetherian induction; the conclusion is then transported back to arbitrary bases by the limit argument at the start.

2. A variant of dimension

For a sheaf F on a scheme of finite type over a field, let $d(F)$ denote the smallest integer d such that the support of F is contained in a closed subset of dimension $\leq d$; equivalently, after replacing F by a constructible subsheaf when needed, $d(F)$ measures the dimension of its support.

Definition 2.2. For a scheme X locally of finite type over a field, put

$$\delta(X) = \sup\{\operatorname{trdeg}_k k(x) \mid x \in X\},$$

where k is the ground field. For a torsion sheaf F, set $\delta(F) = \delta(\operatorname{Supp} F)$.

Proposition 2.3. Let $f : X \to Y$ be a morphism of schemes locally of finite type over a field. For a torsion sheaf F on X, one has

$$\delta(R^q f_* F) \leq \delta(F) - q$$

in the range where the cohomological dimension estimates below apply.

Proof. By constructible approximation and the finiteness theorem, reduce to constructible F. The statement is local on Y and is checked on strict localizations. The higher direct image at a geometric point is the cohomology of the corresponding fibre, whose dimension is controlled by $\delta(F)$. The assertion is then the standard dimension drop in cohomology with torsion coefficients.

Proposition 2.4. If X is affine and algebraic over a separably closed field, and if F is a torsion sheaf with $\delta(F) \leq d$, then $H^q(X, F) = 0$ for $q > d$, once the theorem is known in dimensions $< d$. This proposition is the induction form used in Section 4.

3. Cohomological dimension of affine algebraic schemes

Theorem 3.1. Let X be an affine algebraic scheme over a separably closed field, and let F be a torsion abelian sheaf on X. If $d(F) \leq d$, then

$$H^q(X, F) = 0 \qquad (q > d).$$

In particular $\operatorname{cd}(X) \leq \dim X$.

This is the affine Lefschetz theorem in the form needed here.

Corollary 3.2. If X is affine of dimension d over a separably closed field, then every torsion sheaf on X has vanishing cohomology in degrees $> d$.

Corollary 3.3. For any field k, and X affine algebraic over k, the cohomology of X is controlled by the cohomological dimension of k and the dimension of X, via the Hochschild-Serre spectral sequence after base change to a separable closure.

Corollary 3.4. Let $f : X \to Y$ be an affine morphism of schemes algebraic over a separably closed field. If F is torsion and $d(F) \leq d$, then

$$R^q f_* F = 0 \qquad (q > d),$$

and more precisely the support dimension of $R^q f_* F$ is at most $d - q$.

Corollary 3.5. If X is strictly local and is the strict localization of an algebraic scheme of dimension $\leq d$, and if $U \subset X$ is the complement of a principal divisor, then $\mathrm{cd}(U) \leq d$.

Remark 3.6. The formulation is stable under passage to strict localizations and is designed for induction in Section 4.

4. Proof of Theorem 3.1

Lemma 4.1. The theorem is clear for $d = 0$, and the case $d = 1$ follows from the already established cohomological dimension of affine curves.

Indeed, if $d = 0$, the support is finite and VIII, 5.5 applies. If $d = 1$, reduce to a constructible sheaf on a scheme of dimension at most one and use IX, 5.7.

Lemma 4.2. To prove Corollary 3.4 in dimension d, it is enough to treat the structural morphism $\mathbb{A}^1_{Y'} \to Y'$.

Proof. Embed the affine morphism in some affine space $\mathbb{A}^N_{Y'}$. Induct on N. For $N > 1$, factor the projection as

$$\mathbb{A}^N_{Y'} \to \mathbb{A}^{N-1}_{Y'} \to Y'.$$

The Leray spectral sequence, together with the induction hypothesis on N and on support dimension, reduces the assertion to the case of relative affine line.

Assertion 4.3(d). Let X' be strictly local of dimension $\leq d$, the strict localization of an algebraic scheme over a field. Let $f \in \Gamma(X', \mathcal{O}_{X'})$, and put $U = X' - V(f)$. Then $\mathrm{cd}(U) \leq d$.

Lemma 4.4. Assertion 4.3(d) implies Corollary 3.4(d).

Proof. Compactify $\mathbb{A}^1_{Y'}$ into $\mathbb{P}^1_{Y'}$ by adding the section at infinity. For $i : \mathbb{A}^1_{Y'} \hookrightarrow \mathbb{P}^1_{Y'}$, the Leray spectral sequence

$$H^p(\mathbb{P}^1, R^q i_* F) \Rightarrow H^{p+q}(\mathbb{A}^1, F)$$

reduces high-degree vanishing to the fibre of $R^q i_* F$ at the point at infinity. By VIII, 5.2 this fibre is the cohomology of a punctured strict localization; by Assertion 4.3(d), it vanishes in degrees $> d$. Proper base change for $\mathbb{P}^1$ handles the remaining terms.

Lemma 4.5. Corollary 3.4(d') for all $d' < d$ implies Assertion 4.3(d') for all $d' \leq d$.

Proof. Let X', f, and U be as in Assertion 4.3(d). Choose an algebraic model X_0 and write the strict localization as the limit of affine étale neighbourhoods X_i. A constructible torsion sheaf on U descends to some U_i, and VII, 5.8 identifies cohomology on U with the filtered colimit of cohomologies on the U_i. It is enough to kill classes after passing to a further neighbourhood. Consider the morphism $X_0 \to \mathbb{A}^1$ defined by f and base-change to the strict localization of $\mathbb{A}^1$ at the origin. The complement of the closed fibre is an affine algebraic scheme over the generic point and has dimension $\leq d-1$. By the induction hypothesis and Hochschild-Serre for the generic residue field, whose cohomological dimension is one, this complement has cohomological dimension $\leq d$. Hence all classes in degrees $> d$ die after this passage, proving the assertion and completing the induction.

SGA 4, Exposé XV

Acyclic Morphisms

M. Artin

Introduction

Let $g : X \to Y$ be a morphism of schemes. In the first section we study conditions ensuring that g is acyclic, meaning that for every torsion sheaf F on Y, the natural maps

$$H^q(Y, F) \longrightarrow H^q(X, g^*F)$$

are isomorphisms, and that this remains true after replacing Y by an étale $Y' \to Y$. A necessary condition is that the geometric fibres be acyclic, namely have trivial cohomology for constant torsion sheaves. We shall see that this condition, together with a local condition called local acyclicity, implies acyclicity.

The fundamental theorem is that a smooth morphism is locally acyclic for torsion sheaves prime to the residual characteristics. This implies the smooth base-change theorem developed in the following exposé. The present exposé is independent of Exposés XI–XIV; proper base change reappears from the next exposé onward in combination with the theorem here.

1. Global and local acyclicity

Proposition 1.1. For a morphism $g : X \to Y$, the following conditions are equivalent, with the parenthetical "respectively" referring to the bijective version.

(i) For every $Y' \to Y$ étale and every sheaf of sets F on Y', the adjunction map $F \to g'_* g'^* F$ is injective (respectively bijective).

(ii) For every such Y' and F, the map $H^0(Y', F) \to H^0(X', g'^* F)$ is injective (respectively bijective).

(iii) The same assertion holds for every locally quasi-finite $Y' \to Y$.

(iv) If g is quasi-compact and quasi-separated, it is enough in (ii) to test constant sheaves $I_{Y'}$ for one fixed set I with $\#I \geq 2$.

Lemma 1.2. Every locally quasi-finite morphism $Y' \to Y$ is, locally on Y' for the étale topology, finite over an étale neighbourhood of Y. More precisely, near any point of Y' there is a diagram

$$\begin{tikzcd} Y'_i \arrow[r] \arrow[d] & Y' \arrow[d] \\ Y_i \arrow[r] & Y, \end{tikzcd}$$

where the horizontal arrows are étale, the left vertical arrow is finite, and the Y'_i cover Y'.

Proof. Take a henselization of Y at the image point. The localization of the base change of Y' at the chosen point is finite over this henselian local scheme. Descend this finite open neighbourhood to a sufficiently small étale neighbourhood of Y by the standard limit theorem of EGA IV.

The implication from étale to locally quasi-finite in Proposition 1.1 follows from Lemma 1.2 and finite base change (VIII, 5.6). The equivalence with the test by constant sheaves is XII, 6.5(i).

Definition 1.3. A morphism satisfying the injective form of Proposition 1.1 is called (-1)-acyclic; one satisfying the bijective form is called 0-acyclic. It is universally (-1)-acyclic or universally 0-acyclic if every base change has the same property. A scheme is (-1)-acyclic if it is nonempty, and 0-acyclic if it is connected and nonempty.

Corollary 1.4. If $g : X \to Y$ is 0-acyclic, then for every locally quasi-finite $Y' \to Y$ and every sheaf of groups F on Y', the map

$$H^1(Y', F) \longrightarrow H^1(X', g'^* F)$$

is injective.

Corollary 1.5. If g is surjective, then g is (-1)-acyclic. Conversely, if g is quasi-compact and quasi-separated and (-1)-acyclic, then g is surjective. In that case (-1)-acyclicity is already universal.

Proposition 1.6. Let $\mathcal{L}$ be a set of primes and let $n \in \mathbb{N}$ (respectively $n = 1$ in the non-abelian case). The following are equivalent, with the non-abelian variant obtained by replacing abelian $\mathcal{L}$-torsion sheaves by ind-$\mathcal{L}$-finite groups and by restricting to degrees $0, 1$.

(i) For every étale $Y' \to Y$ and every such sheaf F on Y',

$$F \xrightarrow{\sim} g'_* g'^* F, \qquad R^q g'_* g'^* F = 0 \quad (1 \leq q \leq n).$$

(ii) For every étale $Y' \to Y$, the maps

$$H^q(Y', F) \to H^q(X', g'^* F)$$

are bijective for $q \leq n$ and injective for $q = n + 1$.

(iii) The same is true for every locally quasi-finite $Y' \to Y$.

(iv) If g is quasi-compact and quasi-separated, it is enough to require surjectivity after every locally quasi-finite base change for the constant sheaves $\mathbb{Z}/\ell^\nu\mathbb{Z}$, $\ell \in \mathcal{L}$, $\nu > 0$ (respectively for finite $\mathcal{L}$-groups in H^0, H^1).

Proof. The equivalence of (i) and (ii) is the Leray spectral sequence, and in degree one for groups the usual non-abelian exact sequence. The passage from étale to locally quasi-finite base changes uses Lemma 1.2 and the finite base-change isomorphism. Finally, (iv) is equivalent to (iii) by XII, 6.5.

Definition 1.7. A morphism satisfying Proposition 1.6 is called n-acyclic for $\mathcal{L}$ (respectively 1-aspherical for $\mathcal{L}$). It is acyclic for $\mathcal{L}$ if it is n-acyclic for all n. The universal versions are defined by requiring the same property after every base change. A scheme is n-acyclic for $\mathcal{L}$ if it is 0-acyclic and has $H^q(X, \mathbb{Z}/\ell\mathbb{Z}) = 0$ for all $\ell \in \mathcal{L}$, $1 \le q \le n$; it is 1-aspherical if $H^1(X, G) = 0$ for every finite $\mathcal{L}$-group G.

Remarks 1.8. If g is quasi-compact and quasi-separated, universal n-acyclicity can be checked after base changes locally of finite presentation. It is not known here whether an n-acyclic morphism is automatically universally n-acyclic. For $n = 0$, Definition 1.7 recovers Definition 1.3.

Proposition 1.9.

(i) If $X \xrightarrow{g} Y \xrightarrow{h} Z$ and g is 0-acyclic (respectively 1-aspherical, respectively n-acyclic), then for every sheaf F on Z of the relevant type, the direct images for hg and for h are canonically identified through the relevant degrees. Consequently h has the same acyclicity property if and only if hg has it.

(ii) A projective limit of quasi-compact quasi-separated morphisms with affine transition maps inherits 0-acyclicity, 1-asphericity, and n-acyclicity from the approximating morphisms.

The proof uses the Leray spectral sequence for (i), and the limit formalism of VII, 5, for (ii).

Corollary 1.9.1. If $g : X \to Y$ is quasi-compact and quasi-separated and is n-acyclic (respectively 1-aspherical), then every geometric fibre $X_{\bar{y}}$, for $\bar{y}$ algebraic over a point of Y, has the same property.

Proposition 1.10. For a morphism $g : X \to Y$, the following conditions are equivalent.

(i) For every pair of geometric points $\bar{x} \mapsto \bar{y}$, the induced morphism of strict localizations $\widetilde{X} \to \widetilde{Y}$ is 0-acyclic (respectively 1-aspherical, respectively n-acyclic).

(ii) In every cartesian diagram

$$\begin{tikzcd} X \arrow[d, "g"] & X' \arrow[l] \arrow[d, "g'"] & X'' \arrow[l] \arrow[d, "g''"] \\ Y & Y' \arrow[l] & Y'' \arrow[l] \end{tikzcd}$$

with $Y' \to Y$ étale and $Y'' \to Y'$ étale of finite presentation, the corresponding base-change maps

$$g'^{*} R^q i'_* F \longrightarrow R^q j'_* g''^{*} F$$

are isomorphisms in the prescribed range and monomorphisms in the next degree.

(iii) The same condition holds when $Y'' \to Y'$ is only quasi-finite and quasi-separated.

(iv) Equivalently, one may take $Y' \to Y$ locally quasi-finite and $Y'' \to Y'$ a quasi-compact open immersion.

Proof. For (i) ⇒ (iii), check the base-change map on geometric fibres, where VIII, 5.2 and 5.3 identify those fibres with cohomology of strict localizations; then use the acyclicity of $\widetilde{X} \to \widetilde{Y}$. For (ii) ⇒ (i), express a strict localization as the inverse limit of affine étale neighbourhoods and descend the given étale morphism and sheaf to one of them. The equalities of fibres furnished by VIII, 5.2 identify the desired cohomology maps. Finally, the passage between (ii), (iii), and (iv) uses the main theorem on quasi-finite morphisms: factor a quasi-finite separated morphism as an open immersion followed by a finite morphism; the finite part is handled by VIII, 5.5 and 5.8.

Definition 1.11. A morphism satisfying Proposition 1.10 is called locally 0-acyclic, locally 1-aspherical for $\mathcal{L}$, or locally n-acyclic for $\mathcal{L}$. The evident universal versions are obtained by imposing the same condition after arbitrary base change.

Corollary 1.12. If $g : X \to Y$ is locally of finite type, it is enough to test local acyclicity at geometric points $\bar{x}$ which are closed in their geometric fibre.

Proposition 1.13.

(i) If $X \xrightarrow{g} Y \xrightarrow{h} Z$, and g is surjective and locally 0-acyclic (respectively locally 1-aspherical, respectively locally n-acyclic), then h has the corresponding local property if and only if hg has it.

(ii) Projective limits with affine transition maps preserve local 0-acyclicity, local 1-asphericity, and local n-acyclicity.

Remarks 1.14. The case $n = -1$ is parallel: local (-1)-acyclicity means that the induced maps on strict localizations are surjective, or equivalently that the base-change map for $i'_*(\cdot)$

is a monomorphism. Flat morphisms are locally (-1)-acyclic. For morphisms locally of finite presentation over locally noetherian bases, this is closely related to universal openness.

Theorem 1.15. Let $g : X \to Y$ be quasi-compact and quasi-separated, let $\mathcal{L}$ be a set of primes, and let $n \in \mathbb{N}$ (respectively $n = 1$ in the non-abelian case). Assume g is $(n-1)$-acyclic and locally $(n-1)$-acyclic for $\mathcal{L}$. Let F be a sheaf on Y, abelian $\mathcal{L}$-torsion if $n \geq 1$ (respectively a sheaf of groups), and let $\xi \in H^n(X, g^*F)$. Then ξ comes from a unique element of $H^n(Y, F)$ if and only if, for every geometric point $\bar{y}$ of Y algebraic over a point of Y, its restriction to $H^n(X_{\bar{y}}, F|_{X_{\bar{y}}})$ lies in the image of $H^n(\bar{y}, F_{\bar{y}})$. For $n \geq 1$, this simply says that all those fibre restrictions are zero.

Corollary 1.16. If g is quasi-compact and quasi-separated and locally $(n-1)$-acyclic for $\mathcal{L}$, then g is n-acyclic for $\mathcal{L}$ (respectively 1-aspherical) if and only if all its geometric fibres are n-acyclic (respectively 1-aspherical).

Corollary 1.17. A morphism $g : X \to Y$ is locally n-acyclic for $\mathcal{L}$ (respectively locally 1-aspherical) if and only if, for every geometric point $\bar{x}$ of X, all geometric fibres of the induced morphism of strict localizations $\widetilde{X} \to \widetilde{Y}$ are n-acyclic (respectively 1-aspherical).

Corollary 1.18. If g is locally of finite type, the criterion of Corollary 1.17 may be checked only at geometric points $\bar{x}$ closed in their fibres.

Proof of Theorem 1.15. First reduce to the case Y strictly local. In the abelian case with $n \geq 1$, the Leray spectral sequence and the $(n-1)$-acyclicity of g give an exact sequence

$$0 \to H^n(Y, F) \to H^n(X, g^*F) \to H^0(Y, R^n g_* g^*F).$$

The last term can be tested on geometric fibres by VIII, 5.2. The case $n = 0$ and the non-abelian case $n = 1$ use respectively the same fibrewise argument for sections and the non-abelian exact sequence

$$H^1(Y, F) \to H^1(X, g^*F) \to H^0(Y, R^1 g_* g^*F).$$

Assume henceforth Y strictly local. Corollary XII, 6.10 reduces us to finding, for each maximal point of Y, a finite morphism $Y' \to Y$ whose image contains a nonempty open neighbourhood and such that the pullback of ξ descends. Passing to a separable closure of the residue field at the maximal point provides such finite integral neighbourhoods; the fibrewise hypothesis ensures that, after one of them, the class becomes zero over the corresponding open part (or comes from sections in degree 0). The complementary closed part is handled by the minimality argument of XII, 6.10.

For $n \geq 1$, after replacing Y by such a finite cover, we may suppose that $Y = U \cup Z$ with $\xi|_{X_U} = \xi|_{X_Z} = 0$. Let $i : U \hookrightarrow Y$, $j : Z \hookrightarrow Y$, and use

$$0 \to F_{U,Y} \to F \to F_{Z,Y} \to 0.$$

The class ξ lifts to cohomology with coefficients in $F_{U,Y}$, and after embedding $F_{U,Y}$ into i_*F_U and applying XII, 6.6, it is enough to prove that

$$H^n(X, i'_*F'_{U'}) \to H^n(U', F'_{U'})$$

is injective. The Leray spectral sequence for i' and local $(n-1)$-acyclicity identify $R^q i'_* F'_{U'}$ with pullbacks of $R^q i_* F_U$ for $q \leq n-1$. Global $(n-1)$-acyclicity then makes the relevant low-degree terms vanish, giving the desired injectivity. For $n = 1$ with non-abelian coefficients, the same argument uses the non-abelian exact sequence. For $n = 0$, glue sections over U and Z; compatibility can be checked after the surjective base change by g, and the local (-1)-acyclicity gives the necessary injectivity of the base-change map for i_*. □

Remark 1.18. There is a useful refinement when finitely many geometric points $\bar{y}_i \to Y$ detect the sheaf F, for example when Y is noetherian and F is constructible. In that case the criterion of Theorem 1.15 need only be checked at those points. Indeed, by XII, 6.6 one reduces to sheaves of the form $f_*M_{\bar{y}}$, and local acyclicity identifies their pullbacks and higher direct images with the corresponding sheaves on the fibre $X_{\bar{y}}$. The cohomological assertion then becomes precisely the assertion on that fibre.

2. Local acyclicity of a smooth morphism

Theorem 2.1. Let $g : X \to Y$ be smooth, and let $\mathcal{L}$ be the set of primes different from all residual characteristics of X. Then g is locally acyclic and locally 1-aspherical for $\mathcal{L}$.

Corollary 2.2. For the structural morphism $\mathbb{A}^m_Y \to Y$, the morphism is acyclic and 1-aspherical for the set of primes distinct from the residual characteristics of Y.

Proof. By induction on m, reduce to $m = 1$. Theorem 2.1 gives the local condition; Corollary 1.16 reduces global acyclicity to the fibres. For a separably closed field k, $\operatorname{Spec} k[t]$ has the required prime-to-p cohomological acyclicity by IX, 4.6. Prime-to-p 1-asphericity follows from the elementary fact that a finite tame Galois covering of $\mathbb{P}^1_k$ unramified away from infinity is trivial: the Hurwitz formula gives

$$2(g' - 1) = 2n(g - 1) + \deg \mathfrak{d}_{X'/X} \leq -2n + (n - 1),$$

forcing $n = 1$.

Proof of Theorem 2.1. The assertion is étale-local on X and Y. We may therefore take $X = \mathbb{A}^m_Y$, and then reduce by transitivity to $m = 1$: $X = \operatorname{Spec} \mathcal{O}_Y[t]$. By Corollary 1.18 it is enough to check the fibres of the induced morphism between strict localizations at a point $\bar{x}$ closed in its geometric fibre. After a finite local base change on Y, one may assume the residue fields of $\bar{x}$ and $\bar{y}$ coincide. Translating the coordinate, one may assume $\bar{x}$ is the point $t = 0$. Thus one is reduced to rings of the form $A\{t\}$, the henselization of $A[t]$ at the ideal $(\operatorname{rad} A, t)$.

Notation 2.3. For a henselian local ring A, denote by $A\{t\}$ the henselization of $A[t]$ at the prime ideal generated by $\operatorname{rad} A$ and t.

Corollary 2.4. If A is henselian and A' is a finite A-algebra, then

$$A'\{t\} \simeq A' \otimes_A A\{t\}.$$

Lemma 2.5. Let A be henselian with residual characteristic p. Every geometric fibre of $\operatorname{Spec} A\{t\} \to \operatorname{Spec} A$ is acyclic and 1-aspherical for $\mathcal{P} - \{p\}$.

Proof. Each geometric fibre is a projective limit of affine algebraic curves over a separably closed field. Hence its cohomological dimension is at most one by IX, 5.7 and VII, 5.7. It remains to prove 1-asphericity. Write A as a filtered limit of henselizations of finite-type $\mathbb{Z}$-algebras at prime ideals. Since $A\{t\} = \varinjlim A_\alpha\{t\}$, limit arguments reduce to the case where A is excellent. Nonemptiness follows from the section $t = 0$. Connectedness is reduced, after normalization in finite separable field extensions and using Corollary 2.4, to the fact that $A\{t\}$ is integral when A is normal. The remaining assertion is Lemma 2.6.

Lemma 2.6. Let A be henselian and excellent, and let $\bar{Z}$ be a geometric fibre of $\operatorname{Spec} A\{t\} \to \operatorname{Spec} A$. Every principal Galois cover of $\bar{Z}$ with group of order prime to the residual characteristic of A is trivial.

3. Proof of the principal lemma

After replacing A by a finite normal algebra, Lemma 2.6 reduces to the generic fibre of $\operatorname{Spec} A\{t\}$, with A normal. It is enough to prove the following form.

Lemma 3.1. Let A be normal henselian excellent with residual characteristic p, and let Z be the generic fibre of $\operatorname{Spec} A\{t\} \to \operatorname{Spec} A$. If $Z' \to Z$ is a principal finite Galois étale covering of group order prime to p, then $Z' \to Z$ is trivial.

Proof. Let K be the fraction field of A. After finite separable extension of K and normalization of A, the cover descends to a finite normal $A\{t\}$-algebra. The ramification is prime to the residual characteristic and is controlled by the divisor $t = 0$. The argument reduces, by normalization and localization, to the following purity-style lemma.

Lemma 3.2. Let B be a normal excellent local ring, $0 \neq t \in \operatorname{rad} B$, and let B' be a finite normal integral B-algebra containing B. Assume the induced extension is étale away from the closed point and splits completely on $\operatorname{Spec}(B/tB)$ away from the closed point. Then, after localization at the possible non-étale locus, one is reduced to the case $\dim B \geq 3$, where Lemma 3.3 applies.

Lemma 3.3. Let B be an excellent normal local ring, let $0 \neq t \in \operatorname{rad} B$, and assume $\dim B \geq 3$. Let B' be a finite normal integral B-algebra containing B, and assume B'/tB' is completely split over B/tB away from the closed point. Then B' is étale over B.

Proof. Let $X = \operatorname{Spec} B$ and let $U \subset X$ be the open where B' is étale. Since $(X - U) \cap V(t)$ is just the closed point, and $V(t)$ is defined by one equation, one has $\dim(X - U) \leq 1$. Because X is catenary and $\dim B \geq 3$, the complement has codimension at least two. For $X' = \operatorname{Spec} B'$, universal catenarity gives the same codimension estimate for the complement of $U' = X' \times_X U$, and normality gives

$$B' = \Gamma(X', \mathcal{O}_{X'}) \simeq \Gamma(U', \mathcal{O}_{U'}).$$

Equivalently, the finite algebra is the direct image of its étale restriction to U. Complete B. The completion is normal and faithfully flat, and direct image from U commutes with this base change. Thus it is enough to treat the complete case. In that case the local Lefschetz theorem of SGA 2, X, 2.1 and 2.5, applied with the depth estimate just obtained, shows that any such cover of U split on the punctured divisor is itself completely split. Hence it extends as a trivial étale covering over X, so B' is étale over B.

Remark 3.4. The proof of Lemmas 3.2 and 3.3 works under more general hypotheses. It is enough, for example, that the completion $\widehat{B}$ be normal, or even that the (tB)-adic completion be normal and a quotient of a regular ring.

4. Appendix: a criterion for local 0-acyclicity

The results of this section are not used later in the seminar.

Theorem 4.1. Let $g : X \to Y$ be flat with separable fibres. Assume X and Y locally noetherian (respectively assume g locally of finite presentation). Then g is universally locally 0-acyclic.

Proof. By standard limit arguments, reduce to the locally noetherian case and then to proving local 0-acyclicity. Corollary 1.17 reduces this to the assertion that the geometric fibres of a strictly local flat morphism with separable fibres are connected. This is EGA IV, 18.9.8.

Corollary 4.2. If Y is discrete, then every morphism $g : X \to Y$ is universally locally 0-acyclic.

Reduce to $Y = \operatorname{Spec} k$ with k perfect, then approximate X by schemes of finite type over k, replace by its reduction, and apply Theorem 4.1.

Corollary 4.3. Suppose g is surjective and satisfies the hypotheses of Theorem 4.1 or Corollary 4.2; in the first, non-finite-presentation form, assume also that g is either quasi-compact and quasi-separated or universally open. Then g is a universal effective descent morphism for the fibred category of étale sheaves on variable schemes.

Proof. Such a morphism is universally submersive, so VIII, 9.1 gives universal descent. Effectivity is local on the target, and after replacing X by a finite disjoint union of affine opens whose images cover the affine target, one may suppose X affine and quasi-compact over Y. The argument of VIII, 9.4.1 applies, using the smooth-base-change result that g_*F commutes with locally 0-acyclic base change.

Remarks 4.4. This gives the deferred proof of VIII, 9.4(d). A simpler proof in the case of field extensions reduces to $\operatorname{Spec} K \to \operatorname{Spec} k$, for which universal local 0-acyclicity follows from the prime-to-characteristic acyclicity discussed later. It is plausible that every quasi-compact, quasi-separated, surjective, locally (-1)-acyclic morphism is an effective descent morphism for étale sheaves. It is also plausible that under the hypotheses of Corollary 4.2 one has local acyclicity and local 1-asphericity for all primes distinct from the residual characteristics; in characteristic zero this follows from resolution of singularities.

SGA 4, Exposé XVI

The Smooth Base-Change Theorem and Applications

M. Artin

1 The smooth base-change theorem

Let

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ S' & \xrightarrow{g} & S \end{array}$$

be a cartesian square of schemes. The base-change morphism

$$g^* R^q f_* F \longrightarrow R^q f'_* g'^* F$$

has already been constructed in Exposé XII. The purpose of the present section is to prove that it is an isomorphism when the lower morphism g is smooth and the coefficients are torsion prime to the residual characteristics.

Theorem 1.1. Let $f : X \to S$ be any morphism, let $g : S' \to S$ be smooth, and let F be a torsion abelian sheaf on X whose torsion primes are invertible on S'. Then the canonical base-change homomorphism

$$g^* R^q f_* F \longrightarrow R^q f'_* g'^* F$$

is an isomorphism for every q. The corresponding assertions in degree zero for sheaves of sets and in degree one for sheaves of groups hold in the same prime-to-characteristic range.

Proof. The assertion is local on S' for the etale topology. Since g is smooth, etale-locally it factors as an etale morphism followed by a projection $\mathbb{A}^r_S \to S$. Etale base change is formal. Thus the essential case is the projection from affine space. By Exposé XV, a smooth morphism is locally acyclic for torsion primes invertible on the base. Applying local acyclicity to the strict localizations used to compute the stalks of $R^q f_* F$, one obtains the required isomorphism on every geometric fibre of S'. Since the assertion may be checked on stalks, the theorem follows. □

Corollary 1.2. If $S' \to S$ is etale, then the same base-change isomorphism holds for all torsion coefficients. In particular, higher direct images for the etale topology are compatible with etale localization on the base.

Corollary 1.3. If $S' \to S$ is a localization, a strict localization, or the inclusion of a geometric point obtained as a limit of smooth neighborhoods in the prime-to-characteristic range, the same comparison remains valid by passage to the limit.

Remark 1.4. The theorem is not a substitute for proper base change. Proper base change allows arbitrary change of base for proper f; smooth base change allows a special class of base changes for arbitrary f. The two results are complementary and are combined repeatedly in later arguments.

2 Specialization theorem for cohomology groups

Let S be a scheme, let $s \rightsquigarrow t$ be a specialization represented by a morphism from a strictly local scheme, and let $X \to S$ be a morphism. Smooth base change gives canonical specialization maps between the cohomology groups of the geometric fibres, whenever the specialization is realized through smooth neighborhoods and the coefficients are prime to the residual characteristics.

Theorem 2.1. Let $f : X \to S$ be a morphism and let F be a torsion sheaf on X, with torsion primes invertible on S. Suppose that S is locally noetherian and that two geometric points $\bar{s}$ and $\bar{t}$ lie in a common smooth specialization situation. Then there are canonical specialization homomorphisms

$$H^q(X_{\bar{s}}, F_{\bar{s}}) \longrightarrow H^q(X_{\bar{t}}, F_{\bar{t}})$$

functorial in F, compatible with cup products and long exact sequences, and invariant under refinement of the chosen smooth neighborhood.

The proof is a direct reading of the stalks of $R^q f_* F$. Smooth base change identifies the pullback of $R^q f_* F$ to a smooth neighborhood with the higher direct image after pullback. The strict local base has a closed point and a generization; evaluation at these points gives the specialization map.

Corollary 2.2. If $f : X \to S$ is smooth and proper, and F is finite locally constant with order invertible on S, then $R^q f_* F$ is locally constant finite. Hence the cohomology groups of the geometric fibres form a locally constant system on S.

Proper base change gives constructibility and fibrewise identification; smooth base change shows that the constructible sheaf has no specialization jumps on the smooth base. After shrinking to connected components, the finite stalks and specialization maps are constant.

Corollary 2.3. For a smooth proper family over a connected normal base, the prime-to-characteristic cohomology groups of the geometric fibres are canonically identified up to the monodromy action of the etale fundamental group of the base.

This is the etale analogue of the topological fact that a proper smooth map of complex manifolds is locally topologically trivial.

3 Relative cohomological purity

The next application is a relative purity statement. Let X be regular, let $Y \subset X$ be a regular closed subscheme of codimension c, and let $U = X \setminus Y$. For torsion coefficients prime to the residual characteristics, the local cohomology of X with support in Y is concentrated in degree $2c$ and is given by a Tate twist.

Theorem 3.1. Let $i : Y \hookrightarrow X$ be a regular immersion of codimension c, with X and Y locally noetherian, and let n be invertible on X. Then

$$R^q i^! \mathbb{Z}/n\mathbb{Z} = 0 \quad (q \neq 2c), \qquad R^{2c} i^! \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/n\mathbb{Z}(-c).$$

Equivalently,

$$H^q_Y(X, \mathbb{Z}/n\mathbb{Z}) = 0 \quad (q < 2c),$$

and the first non-zero local cohomology sheaf is the expected orientation sheaf.

Proof. The assertion is local for the etale topology on X. By the definition of a regular immersion, and after etale localization, the pair (X, Y) is compared to $(\mathbb{A}^c_Y, Y)$, where Y is embedded as the zero section. Smooth base change reduces the calculation to the case where Y is the spectrum of a separably closed field. Then the computation is the cohomology with supports of affine space at the origin. For $c = 1$ this is the Kummer sequence and the calculation of $\mathbb{G}_m$; the general case follows by iterating the one-dimensional calculation and using the Kunneth formula for supports. □

Corollary 3.2. If X is regular and $U = X \setminus Y$, with Y regular of codimension c, then the first possible obstruction to extending a prime-to-characteristic torsion cohomology class from U across Y occurs in degree $2c-1$. In codimension at least two, low-degree cohomology is unchanged by removing Y.

Remark 3.3. In later language this is absolute cohomological purity. In this exposé it is obtained in the relative form needed for base-change and finiteness applications.

4 Comparison theorem over the complex numbers

Let X be a scheme locally of finite type over $\mathbb{C}$, and let X^{an} be the associated complex analytic space. A sheaf F on $X_{\text{ét}}$ gives, by pullback along the natural functor from analytic neighborhoods to etale neighborhoods, a sheaf F^{an} on X^{an}.

Theorem 4.1. For every constructible torsion sheaf F on $X_{\text{ét}}$, the natural comparison maps

$$H^q(X_{\text{ét}}, F) \longrightarrow H^q(X^{\mathrm{an}}, F^{\mathrm{an}})$$

are isomorphisms for all q. The same is true for cohomology with supports and is compatible with long exact sequences, cup products, and proper or smooth base change.

The smooth case with locally constant finite coefficients was treated in Exposé XI. The general case follows by embedding X into a smooth scheme locally, stratifying so that F is locally constant finite on the strata, and using the exact sequence for an open stratum and its closed complement. Proper base change and smooth base change ensure compatibility with the two geometric operations needed in the induction.

Corollary 4.2. The etale cohomological dimension bounds proved algebraically agree over $\mathbb{C}$ with the corresponding classical bounds for the analytic topology.

5 Finiteness theorem for algebraic schemes over a separably closed field

Theorem 5.1. Let X be a scheme of finite type over a separably closed field k, and let F be a constructible torsion sheaf on X. Then the groups $H^q(X, F)$ are finite when F has finite fibres, and they vanish for all sufficiently large q. More precisely, the vanishing is bounded by the cohomological-dimension estimates of Exposés X and XIV.

Proof. Choose a compactification of X when available, or argue by noetherian induction using an open affine stratification. On affine strata the affine Lefschetz theorem gives the dimension bound. On proper pieces, the finiteness theorem for proper morphisms applies. The long exact sequence for an open and its closed complement then gives both finiteness and vanishing for X. For a non-algebraically closed but separably closed field, the same proof applies after using the invariance of the etale topos under purely inseparable extension. □

Corollary 5.2. Let $f : X \to S$ be a morphism of finite type with S noetherian and let F be constructible torsion on X. After stratifying S, the sheaves $R^q f_* F$ are constructible in the range controlled by the preceding theorem whenever f is proper, smooth, or is built from these cases by the standard open-closed decompositions.

This corollary summarizes the role of Exposé XVI: smooth base change supplies local constancy in smooth directions, proper base change supplies invariance under proper specialization, and the affine cohomological-dimension theorem supplies vanishing on open affine pieces.

SGA 4, Exposé XVII – Beginning

Cohomology with Proper Supports

P. Deligne

Introduction

The preceding exposés developed the basic finiteness, base-change, and acyclicity theorems for torsion etale cohomology. The present exposé begins the construction of cohomology with proper supports and of the functors which will ultimately be denoted $Rf_!$. The construction is not merely a formal variant of ordinary direct image. It requires a derived-category formalism capable of handling compactifications, gluing, and independence of choices.

The guiding principle is the following. If $f : X \to S$ is separated and of finite type, choose a compactification

$$X \xrightarrow{j} \bar{X} \xrightarrow{\bar{f}} S,$$

with j an open immersion and $\bar{f}$ proper. Then cohomology with proper supports should be computed by

$$Rf_!K = R\bar{f}_*j_!K.$$

One must prove that this object is independent of the compactification, functorial in f, compatible with base change, and compatible with the operations needed in duality. Exposé XVII supplies the formal machinery for this construction.

6 Terminological preliminaries

6.1 Categories, universes, and topoi

All categories and topoi are understood relative to the fixed universe conventions of the earlier exposés. A topos is a category equivalent to the category of sheaves on a site. Morphisms of topoi are written with the usual pair of adjoint functors (f^*, f_*), where f^* is left exact. For a ringed topos (X, A), modules always mean sheaves of A-modules on X.

6.2 Complexes

A complex K in an abelian category $\mathcal{A}$ is cohomologically indexed:

$$\cdots \to K^{n-1} \xrightarrow{d^{n-1}} K^n \xrightarrow{d^n} K^{n+1} \to \cdots, \qquad d^n d^{n-1} = 0.$$

The translated complex $K[1]$ is defined by $K[1]^n = K^{n+1}$ with differential $-d_K$. This sign is essential for the standard triangles and mapping cones to behave functorially.

6.3 Homotopy and derived categories

Let $C(\mathcal{A})$ be the category of complexes, $K(\mathcal{A})$ the homotopy category, and $D(\mathcal{A})$ the derived category obtained by inverting quasi-isomorphisms. The bounded variants are denoted $D^+(\mathcal{A})$, $D^-(\mathcal{A})$, and $D^b(\mathcal{A})$. A functor between abelian categories which is left exact, right exact, or exact gives rise, under the usual hypotheses on injective or flat resolutions, to derived functors between the corresponding derived categories.

6.4 Triangulated conventions

Distinguished triangles are written

$$K \longrightarrow L \longrightarrow M \longrightarrow K[1].$$

A short exact sequence of complexes gives such a triangle. Conversely, the cohomology long exact sequence attached to a distinguished triangle is the principal computational device. The octahedral axiom is used only in its standard consequence: mapping cones and compositions are compatible up to unique isomorphism in the derived category.

6.5 Truncations

For a complex K, the canonical truncations $\tau_{\leq n}K$ and $\tau_{>n}K$ fit into a distinguished triangle

$$\tau_{\leq n}K \to K \to \tau_{>n}K \to \tau_{\leq n}K[1].$$

They give the standard spectral sequences used throughout the exposé. In particular, if a cohomological functor is known on sheaves placed in one degree, the truncation filtration reduces statements about bounded below complexes to statements about their cohomology sheaves.

6.6 Flat and injective resolutions

In the categories of sheaves of modules on a ringed topos, there are enough injectives. Flat resolutions exist under the hypotheses used here and may be chosen functorially enough for derived inverse image and tensor product. When a functor is exact on injectives, or sends a chosen class of adapted objects to adapted objects, its derived functor is computed by resolving only once.

6.7 Notation for supports

If $j : U \hookrightarrow X$ is an open immersion and $i : Z \hookrightarrow X$ is the closed complement, $j_!$ denotes extension by the initial object outside U for sheaves of sets, and extension by zero for abelian sheaves or modules. The fundamental exact sequence is

$$0 \to j_!j^*F \to F \to i_*i^*F$$

with the usual interpretation in the abelian case; it gives the localization triangle

$$Rj_!j^*K \to K \to Ri_*i^*K \to .$$

6.8 Compactifiable morphisms

A morphism $f : X \to S$ is compactifiable if it factors as an open immersion $j : X \hookrightarrow \bar{X}$ followed by a proper morphism $\bar{f} : \bar{X} \to S$. For schemes, separated morphisms of finite type are compactifiable by Nagata's compactification theorem in the contexts in which it is used here. The construction of $Rf_!$ first depends on the choice of such a compactification and then proves independence.

7 Derived categories

7.1 Exact functors and the sign rules

1.1.1. Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories. Applying F degreewise gives a functor on complexes. If F is exact, it sends quasi-isomorphisms to quasi-isomorphisms and therefore induces a triangulated functor

$$D(\mathcal{A}) \longrightarrow D(\mathcal{B}).$$

It commutes with shifts and carries distinguished triangles coming from short exact sequences of complexes to distinguished triangles.

The compatibility with shifts is not literally automatic unless the sign convention for [1] is fixed. With the convention $d_{K[1]} = -d_K$, the natural isomorphism $F(K[1]) \simeq F(K)[1]$ is compatible with differentials.

***1.1.2.** For a morphism of complexes $u : K \to L$, the cone $C(u)$ is the complex

$$C(u)^n = L^n \oplus K^{n+1}, \qquad d(l, k) = (d_L l + u(k), -d_K k).$$

There is a distinguished triangle

$$K \xrightarrow{u} L \to C(u) \to K[1].$$

A morphism u is a quasi-isomorphism if and only if $C(u)$ is acyclic.

1.1.4. Let F and G be exact functors. A natural transformation $F \to G$ induces a natural transformation between the associated derived-category functors. If the transformation is an isomorphism on the abelian category, it is an isomorphism on the derived category.

1.1.4.2.1. Given two composable morphisms of complexes $K \xrightarrow{u} L \xrightarrow{v} M$, the cones $C(u)$, $C(v)$, and $C(vu)$ fit into the standard octahedron. All later compatibility statements for boundary morphisms are consequences of this octahedron.

1.1.5. If $0 \to K' \to K \to K'' \to 0$ is a short exact sequence of complexes, then there is a functorial distinguished triangle

$$K' \to K \to K'' \to K'[1].$$

The connecting morphism $K'' \to K'[1]$ is represented by the usual boundary map; the sign is the one determined by the cone convention above.

1.1.6. Let $F : \mathcal{A} \to \mathcal{B}$ be left exact. Its right derived functor $RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$ is constructed by replacing a bounded below complex by an injective resolution and applying F. If F sends injective objects to objects acyclic for another left exact functor G, then

$$R(GF) \simeq RG\, RF.$$

1.1.8. Dually, if F is right exact and enough flat or projective resolutions are available, one constructs LF on $D^-(\mathcal{A})$. In sheaf theory the important examples are derived tensor product and derived inverse image for modules over ringed topoi.

1.1.9. For a ringed topos (X, A), the tensor product of complexes is given by the total complex of the double complex $K^p \otimes_A L^q$. The commutativity constraint involves the Koszul sign

$$x \otimes y \longmapsto (-1)^{pq} y \otimes x,$$

for homogeneous elements $x \in K^p$, $y \in L^q$. With this convention, derived tensor product is symmetric monoidal.

***1.1.10.1.** If $u : K \to K'$ and $v : L \to L'$ are morphisms of complexes, the tensor product of cones is compatible with the cone of the tensor product morphism only after inserting the standard Koszul signs. These signs are the source of the signs in the projection formula and in the Kunneth morphism.

1.1.12. Let $f : (X, A) \to (Y, B)$ be a morphism of ringed topoi. The inverse image of modules is right exact and its left derived functor is denoted

$$Lf^* : D^-(B) \to D^-(A).$$

When the objects involved are flat, this is computed by the ordinary inverse image. The direct image f_* is left exact and has a right derived functor Rf_* on bounded below derived categories.

1.1.15. For $K \in D^-(B)$ and $L \in D^+(A)$ there is a natural projection morphism

$$K \otimes^L_B Rf_* L \longrightarrow Rf_*(Lf^* K \otimes^L_A L).$$

Under the usual boundedness and finite Tor-dimension hypotheses this morphism is an isomorphism. Establishing such isomorphisms for $Rf_!$ is one of the purposes of the later sections.

1.1.16. All constructions above are compatible with restriction to open subtopoi, with extension by zero, and with the localization triangles associated with an open-closed decomposition. These compatibilities are fixed once and for all by the cone and shift conventions; no further sign choices will be made later.

7.2 Derived functors

Let $\mathcal{A}$ be a triangulated category and let S be a saturated multiplicative system. If $A \in \mathrm{Ob}(\mathcal{A})$, we denote by A^+ and A^- the ind- and pro-objects

$$A^+ = \varinjlim_{A \xrightarrow{s} A'} A', \qquad A^- = \varprojlim_{A' \xrightarrow{s} A} A',$$

where s ranges over the filtered categories of arrows in S with source, respectively target, A. The functors $A \mapsto A^+$ and $A \mapsto A^-$ invert the elements of S, and they identify the localized category $\mathcal{A}(S^{-1})$ with full subcategories of the categories of ind- and pro-objects of $\mathcal{A}$. This is the device used here to define derived functors without choosing resolutions at the outset.

Definition 1.2.1. Let F be an exact covariant multi-functor from triangulated categories $(\mathcal{A}_i)_{0<i\leq n}$ to a triangulated category $\mathcal{A}$, and let S_i and S be saturated multiplicative systems in $\mathcal{A}_i$ and $\mathcal{A}$.

(i) The right derived functor RF, relative to the S_i and S, is the functor from the localized categories $\mathcal{A}_i(S_i^{-1})$ to the category of ind-objects of $\mathcal{A}(S^{-1})$ obtained by sending (A_i) to the image of $F(A_i^+)$.

(ii) The functor RF is said to be defined at (A_i) if the resulting ind-object of $\mathcal{A}(S^{-1})$ is essentially constant, that is, if it comes from an actual object of $\mathcal{A}(S^{-1})$. That object is then called the value of RF at (A_i).

(iii) The functor F is right derivable if RF is defined everywhere. In that case the same symbol RF denotes the resulting functor with values in $\mathcal{A}(S^{-1})$.

(iv) A family (A_i) is said to be adapted, or split, for RF if the canonical morphism $F(A_i) \to RF(A_i)$ is an isomorphism.

The left derived functor LF is defined dually, with values first in pro-objects. The same definitions apply when F is covariant in some variables and contravariant in others. When the derived functor is everywhere defined, this definition agrees with Verdier's definition of derived functors in localized triangulated categories.

Proposition 1.2.2. Let (A_1', A_1, A_1'') be a distinguished triangle in $\mathcal{A}_1$, and let $A_i \in \mathrm{Ob}(\mathcal{A}_i)$ for $i \neq 1$.

(i) If $(A_1', (A_i))$ and $(A_1'', (A_i))$ are split for RF, then $(A_1, (A_i))$ is split for RF.

(ii) If RF is defined at $(A_1', (A_i))$ and at $(A_1'', (A_i))$, then it is defined at $(A_1, (A_i))$, and

$$RF(A_1', (A_i)) \to RF(A_1, (A_i)) \to RF(A_1'', (A_i)) \to RF(A_1', (A_i))[1]$$

is a distinguished triangle in $\mathcal{A}(S^{-1})$.

The implication (ii) $\Rightarrow$ (i) follows at once from exactness of F: the comparison from the triangle obtained by applying F to the triangle obtained by applying RF is an isomorphism at two vertices, hence at the third.

Lemma 1.2.2.1. If $\Delta = (X, Y, Z)$ is a distinguished triangle in a triangulated category $\mathcal{A}$ and S is a saturated multiplicative system, then the triangle $\Delta^+ = (X^+, Y^+, Z^+)$ in the ind-category is a filtered inductive limit of distinguished triangles of $\mathcal{A}$.

Let $\mathcal{L}$ be the category whose objects are morphisms of distinguished triangles with source Δ and whose three component arrows belong to S. The usual calculus of fractions for saturated systems gives the following facts. If Δ' is another distinguished triangle and $f : \Delta \to \Delta'$ is a morphism in $\mathcal{A}(S^{-1})$, then after composing Δ' with a morphism of triangles whose components lie in S, the morphism f is represented by an actual morphism of triangles in $\mathcal{A}$. If two such representatives become equal in $\mathcal{A}(S^{-1})$, then they become equal after another such composition. These two assertions imply that $\mathcal{L}$ is filtered, and

$$\Delta^+ = \varinjlim_{s \in \mathcal{L}} \operatorname{target}(s).$$

This proves the lemma.

To prove Proposition 1.2.2(ii), choose objects X' and X'' representing $RF(A_1', (A_i))$ and $RF(A_1'', (A_i))$, and complete the degree-one morphism $X'' \to X'$ to a distinguished triangle (X', X, X''). After replacing the input objects by targets of morphisms in the relevant multiplicative systems, the given comparison maps are represented by sections. The axiom TR3 then completes them to a morphism of distinguished triangles. Applying $\operatorname{Hom}(Y, -)$ for arbitrary $Y \in \mathcal{A}(S^{-1})$, one obtains a morphism between two exact sequences. The four outside vertical arrows are isomorphisms by the representing property of X' and X'', hence the central arrow is an isomorphism by the five lemma. Therefore X represents $RF(A_1, (A_i))$, and the displayed triangle is distinguished.

Definition 1.2.3. Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories and let $F : \mathcal{A} \to \mathcal{B}$ be an additive covariant functor.

(i) The right derived functors of F are the functors

$$D^+(\mathcal{A}) \to \operatorname{Ind} D^+(\mathcal{B}), \qquad D(\mathcal{A}) \to \operatorname{Ind} D(\mathcal{B}),$$

derived from the extensions of F to complexes.

(ii) An object $A \in \mathcal{A}$ is right acyclic for F, or acyclic for RF, if the complex with A in degree zero is split for RF.

The main case is the one in which RF is everywhere defined, so that it may be regarded as a functor to $D^+(\mathcal{B})$, or to $D(\mathcal{B})$, not merely to the ind-category.

Proposition 1.2.4. The natural square

$$\begin{array}{ccc} D^+(\mathcal{A}) & \longrightarrow & D(\mathcal{A}) \\ \downarrow RF & & \downarrow RF \\ \operatorname{Ind} D^+(\mathcal{B}) & \longrightarrow & \operatorname{Ind} D(\mathcal{B}) \end{array}$$

is commutative.

Indeed, if K is zero in degrees $< n$ and $K \to L$ is a quasi-isomorphism, then the composite $K \to L \to \tau_{\geq n} L$ is again a quasi-isomorphism. Thus quasi-isomorphisms from K to bounded below complexes form a cofinal subcategory of all quasi-isomorphisms with source K. This proves the proposition formally. In particular, if $RF : D(\mathcal{A}) \to D(\mathcal{B})$ is everywhere defined, its restriction to $D^+(\mathcal{A})$ is the usual bounded-below derived functor.

1.2.5. The preceding definitions and Proposition 1.2.4 extend without change to bounded complexes of multi-functors, covariant in some variables and contravariant in others. The dual statements define left derived functors and compare $D^-(\mathcal{A})$ with $D(\mathcal{A})$.

Proposition 1.2.6. Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, let $\mathcal{P} \subset \mathrm{Ob}(\mathcal{A})$, and let $F : \mathcal{A} \to \mathcal{B}$ be additive. Suppose that every object of $\mathcal{A}$ is a quotient of an object of $\mathcal{P}$, and that whenever

$$0 \to P \to Q \to R \to 0$$

is exact with $Q, R \in \mathcal{P}$, one has $P \in \mathcal{P}$ and

$$0 \to FP \to FQ \to FR \to 0$$

is exact. Then every object of $\mathcal{P}$ is acyclic for LF.

The left resolutions of an object by elements of $\mathcal{P}$ are cofinal among all left resolutions. If a complex

$$\cdots \to A_2 \to A_1 \to A_0 \to A \to 0$$

with $A_i \in \mathcal{P}$ is acyclic, cut it into short exact sequences

$$0 \to \mathrm{Im}(d_{i-1}) \to A_i \to \mathrm{Im}(d_i) \to 0.$$

Induction on i shows that each image belongs to $\mathcal{P}$, and that applying F preserves these short exact sequences. Hence the image complex is acyclic.

Proposition 1.2.7. Let $F : \mathcal{A} \to \mathcal{B}$ be additive. If every object of $\mathcal{A}$ is a quotient, respectively a subobject, of an object acyclic for LF, respectively for RF, then

$$LF : D^-(\mathcal{A}) \to D^-(\mathcal{B}), \qquad RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$$

are everywhere defined. Moreover every bounded-above, respectively bounded-below, complex of acyclic objects is split for the corresponding derived functor.

For LF, quasi-isomorphisms from complexes of acyclic objects to a given bounded-above complex form a cofinal system. It remains to see that if $s : L \to K$ is such a quasi-isomorphism between complexes of acyclic objects, then $F(s)$ is a quasi-isomorphism. The cone of s is acyclic and still has acyclic terms. Its brutal truncations are bounded complexes of acyclic objects, hence split; passing to cohomology and letting the truncation bound tend to infinity gives the desired vanishing. The proof for RF is dual.

Proposition 1.2.8. Let $F^\bullet$ be a bounded-below complex of functors from $\mathcal{A}$ to $\mathcal{B}$. Assume that for every $K \in K^+(\mathcal{A})$ there exists a quasi-isomorphism $L \to K$ with $L \in K^+(\mathcal{A})$ split for all the functors RF^i. Then $RF^\bullet : D^+(\mathcal{A}) \to D^+(\mathcal{B})$ exists, and every bounded-below complex split for all the RF^i is split for $RF^\bullet$.

This is the usual spectral-sequence argument applied to the double complex obtained by resolving termwise. The dual statement holds for left derived functors and bounded-above complexes.

Definition 1.2.9. Let $F : D^b(\mathcal{A}) \to D(\mathcal{B})$ be a functor and let $d \subset \mathbb{Z}$ be an interval. We say that F has cohomological amplitude contained in d if, for every object $A \in \mathcal{A}$,

$$H^i F(A) = 0 \qquad (i \notin d).$$

If $d = [0, x]$ we say that F has cohomological dimension $\leq x$; if $d = [-x, 0]$, homological dimension $\leq x$. The functor is of finite dimension if some finite interval works.

Proposition 1.2.10. Under the hypotheses of Proposition 1.2.7, assume in addition that $RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$, respectively $LF : D^-(\mathcal{A}) \to D^-(\mathcal{B})$, is of finite dimension. Then every unbounded complex of acyclic objects is split, and the corresponding derived functor exists on $D(\mathcal{A})$.

This is Verdier's bounded-amplitude argument. Since cohomology of the image of a truncation can contribute only in a fixed finite band, the comparison between an unbounded complex and its truncations stabilizes degree by degree. Hence the functor on the unbounded derived category is obtained as the stable value of the functor on bounded truncations.

8 Fibred categories in derived categories

8.1 Introduction

The reader may skip this section on a first reading. Its purpose is to make precise the formalism underlying the "trivial duality theorem" and to remove the ambiguity, raised earlier by Artin, about the variance of derived functors under changes of ringed sites. We use the language of categories fibred over a base category, as in SGA 1, Exposé VI. A cleaved fibred category over a category C is equivalent to a pseudo-functor $C^\circ \to \mathbf{Cat}$; the analogous statement for cofibrations corresponds to covariant pseudo-functors.

Definition 2.1.1. If X is a ringed site, $D(X)$ denotes the derived category of the abelian category of sheaves of left modules on X. When the ring must be mentioned, we write $D(X, A)$.

If $f : X \to Y$ is a morphism of ringed sites, then one has

$$Rf_* : D^+(X) \to D^+(Y), \qquad Lf^* : D^-(Y) \to D^-(X).$$

Passing to opposite categories, the first becomes a functor $D^+(X)^\circ \to D^+(Y)^\circ$. Thus the collection of categories $D^+(X)$, as X varies, is naturally both fibred and cofibrated in different senses, depending on whether one follows inverse image, direct image, or their opposites.

2.1.2. Let C be the category of ringed sites under consideration. The assignment

$$X \longmapsto D^+(X)$$

together with Rf_* for morphisms $f : X \to Y$, defines a pseudo-functor from C to triangulated categories, whenever the relevant boundedness hypotheses for right derived direct images are satisfied. The assignment $X \mapsto D^-(X)$ with Lf^* defines a pseudo-functor from C° to triangulated categories.

The coherence isomorphisms are the canonical ones

$$R(gf)_* \simeq Rg_* Rf_*, \qquad L(fg)^* \simeq Lg^* Lf^*,$$

obtained from the composition theorem for derived functors. The point of spelling this out is that later formulas, for instance projection and base change, are identities not in an isolated derived category but in these fibred systems.

2.1.3. Given a cartesian square of ringed sites

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y, \end{array}$$

there are canonical base-change morphisms

$$Lg^* Rf_* K \longrightarrow Rf'_* Lg'^* K$$

for $K \in D^+(X)$ whenever both sides are defined. These morphisms are functorial in the square and compatible with composition of squares.

They are built by deriving the adjunction map $g^* f_* o f'_* g'^*$. The main theorems of the preceding exposés identify hypotheses under which these morphisms are isomorphisms: proper base change for proper morphisms, smooth base change for smooth morphisms and prime-to-characteristic coefficients, and finite or affine cohomological-dimension bounds for the other cases.

8.2 Fibred categories in triangulated categories

Definition 2.2.1. A category fibred in triangulated categories over C is a fibred category $\mathcal{D} \to C$ such that every fibre $\mathcal{D}_X$ is a triangulated category and every inverse-image functor attached to a cartesian arrow is exact. A cleaving is normalized if the inverse image attached to the identity of X is the identity functor and if the comparison isomorphisms for composition satisfy the usual associativity condition.

Equivalently, a normalized cleaved fibred category in triangulated categories is a pseudo-functor $C^\circ \to$ **TriCat**, where **TriCat** denotes the 2-category of triangulated categories and exact functors. Dually, a cofibrated category in triangulated categories is a pseudo-functor $C \to$ **TriCat**.

2.2.2. Let $\mathcal{D} \to C$ be fibred in triangulated categories. A morphism $u : AoB$ in the total category lying above $f : X \to Y$ may be written, after choosing a cleaving, as a morphism

$$A \longrightarrow f^*B$$

in the fibre over X. It is cartesian if this morphism is an isomorphism. Exactness of pullback means that cartesian liftings carry distinguished triangles to distinguished triangles.

This language will be used to say that a construction is compatible with all base changes, rather than restating the corresponding square for each occurrence.

2.2.3. Let $\mathcal{D} \to C$ and $\mathcal{E} \to C$ be fibred in triangulated categories. A morphism of fibred categories $\Phi : \mathcal{D} \to \mathcal{E}$ is exact if its restriction to every fibre is exact and if it carries cartesian arrows to cartesian arrows. A natural transformation between such morphisms is compatible with base change if it is a morphism in the 2-category of fibred categories over C.

Thus the projection formula, the Kunneth formula, and the base-change formula will be recorded as morphisms, or isomorphisms, of functors between fibred triangulated categories.

8.3 The trivial duality formula

Let $f : X \to Y$ be a morphism of ringed sites. Suppose for the moment that all objects involved are bounded in such a way that no ambiguity about derived functors occurs. There is a canonical morphism

$$Lf^*R\mathcal{H}om_Y(K, L)oR\mathcal{H}om_X(Lf^*K, Lf^*L).$$

Adjunction gives the corresponding morphism

$$R\mathcal{H}om_Y(K, Rf_*M)oRf_*R\mathcal{H}om_X(Lf^*K, M).$$

This is called the trivial duality morphism: it is a formal consequence of the adjunction (Lf^*, Rf_*), not of any properness or finiteness theorem.

Proposition 2.3.1. Under the standard boundedness hypotheses, for $K \in D^-(Y)$ and $M \in D^+(X)$, there is a functorial morphism

$$R\mathcal{H}om_Y(K, Rf_*M) o Rf_* R\mathcal{H}om_X(Lf^*K, M).$$

If K is represented by a bounded complex of flat modules, this morphism is computed by the ordinary sheaf Hom adjunction. It is compatible with composition of morphisms of ringed sites.

Corollary 2.3.2. If K is perfect, or more generally if it has finite Tor-amplitude and the usual boundedness conditions hold, the preceding morphism is an isomorphism. Equivalently, the derived projection formula

$$K \otimes_Y^L Rf_*M \to Rf_*(Lf^*K \otimes_X^L M)$$

is an isomorphism in the same range.

The proof is local on Y and reduces to a finite complex of finite free modules, where it is the elementary tensor-Hom adjunction. The formulation in fibred categories guarantees compatibility with later gluing constructions.

9 Gluing fibred categories and compactifiable morphisms

9.1 Introduction

The construction of $Rf_!$ for a non-proper morphism proceeds by choosing a compactification and setting $Rf_! = R\bar{f}_* j_!$. To prove independence of the compactification, and to obtain functoriality, one needs a gluing formalism. This section records it in the language of fibred triangulated categories.

9.2 Compactifiable morphisms

Definition 3.2.1. A morphism $f : X \to S$ is compactifiable if it admits a factorization

$$X \xrightarrow{j} \bar{X} \xrightarrow{\bar{f}} S$$

where j is an open immersion and $\bar{f}$ is proper. Such a factorization is called a compactification of f.

For schemes, separated morphisms of finite type are compactifiable in the noetherian situations considered here. The category of compactifications of a fixed compactifiable morphism is cofiltered after allowing domination: two compactifications are dominated by the closure of the diagonal image in their product over the base. This is the basic geometric fact behind independence of compactification.

Proposition 3.2.2. Let $f : X \to S$ be compactifiable. Given two compactifications $X \hookrightarrow \bar{X}_1 \to S$ and $X \hookrightarrow \bar{X}_2 \to S$, there is a third compactification $X \hookrightarrow \bar{X} \to S$ and proper morphisms $\bar{X} \to \bar{X}_i$ inducing the identity on X. If the original compactifications are compatible with a base change, the dominating compactification may be chosen compatibly after refinement.

The proof is the usual closure of the graph of X in $\bar{X}_1 \times_S \bar{X}_2$. Properness is stable under products and closed immersions, and the open immersion of X into the closure is induced by the diagonal.

9.3 Gluing

Let $i : Z \hookrightarrow X$ be a closed immersion and $j : U \hookrightarrow X$ the complementary open immersion. For abelian sheaves there is a short exact sequence

$$0 \to j_! j^* F \to F \to i_* i^* F \to 0.$$

In the derived category this becomes the localization triangle

$$Rj_! j^* K \to K \to Ri_* i^* K \to Rj_! j^* K[1].$$

Proposition 3.3.1. Let $X = U \cup Z$ be an open-closed decomposition. A functor $T : D(X) \to \mathcal{T}$ to a triangulated category is determined, up to unique isomorphism, by its restrictions to $D(U)$ and $D(Z)$, together with compatibility with the localization triangle. More precisely, if two exact functors agree after composition with $j_!$ and i_*, and the comparison respects the boundary morphism, then they agree on all of $D(X)$.

Indeed every $K \in D(X)$ sits in the localization triangle above. Applying the two functors and using the assumed agreement at the first and third vertices gives the agreement at the middle vertex by the triangulated five lemma; uniqueness follows from the same argument applied to morphisms.

Proposition 3.3.2. Suppose $f : X \to S$ is compactifiable and $X = U \cup Z$ is an open-closed decomposition. If the functors $Rf_!$ have been constructed for $f|_U$ and $f|_Z$, and if they satisfy base change and localization for those pieces, then there is a unique exact functor $Rf_! : D(X) \to D(S)$ fitting into the triangle

$$R(f|_U)_! j^* K \to Rf_! K \to R(f|_Z)_! i^* K \to .$$

It is compatible with proper direct image when f is proper.

This is the abstract gluing step. It will be applied inductively, using compactifications to replace open pieces by proper schemes and using the already established proper base-change theorem.

Proposition 3.3.3. Let $X \hookrightarrow \bar{X} \to S$ and $X \hookrightarrow \bar{X}' \to S$ be two compactifications, with $\bar{X}' \to \bar{X}$ proper and equal to the identity over X. Then for every $K \in D(X)$ there is a canonical isomorphism
$$R\bar{f}_* j_! K \simeq R\bar{f}'_* j'_! K.$$
These isomorphisms are transitive for further domination.

Let $p : \bar{X}' \to \bar{X}$ be the proper morphism. Since p is an isomorphism over X, proper base change for the open immersion shows that

$$j_! K \simeq Rp_* j'_! K.$$

Applying $R\bar{f}_*$ gives the result. Transitivity follows from the uniqueness of the adjunction morphisms and the associativity isomorphism $R(\bar{f}p)_* \simeq R\bar{f}_* Rp_*$.

10 Resolutions and application to the base-change arrow

10.1 Flat resolutions

The construction of the base-change arrow for $Rf_!$, and the proof of its independence from auxiliary choices, require resolutions which are well behaved with respect to inverse image, tensor product, and extension by zero. Flat resolutions provide this control for module-valued sheaves.

Definition 4.1.1. Let (X, A) be a ringed topos. A complex P of A-modules is called flat-adapted if each P^n is flat and if tensoring by P preserves quasi-isomorphisms in the range under consideration. A resolution $P \to K$ is flat-adapted if P is flat-adapted and the morphism is a quasi-isomorphism.

Proposition 4.1.2. Every bounded-above complex of modules on a ringed topos admits a flat-adapted resolution. These resolutions may be chosen functorially enough to define $- \otimes^L_A -$ and Lf^*, and complexes of flat-adapted objects are split for these left derived functors.

The proof uses the fact that every module is a quotient of a flat module, for example a sum of free modules. The class of flat modules satisfies the closure condition in Proposition 1.2.6: in a short exact sequence $0 \to P \to Q \to R \to 0$ with Q and R flat and with the sequence pure in the required sense, P remains adapted for tensor computations. The general statement is obtained by replacing "flat" by the standard class of K-flat objects.

4.1.3. If $f : (X, A) \to (Y, B)$ is a morphism of ringed topoi, a flat-adapted resolution $P \to K$ of a complex on Y computes

$$Lf^* K = f^* P.$$

Moreover, if Q is flat-adapted on X, then $f^*P \otimes_A Q$ computes $Lf^*K \otimes_A^L Q$.

These compatibilities are needed to make the projection morphism and the base-change morphism into morphisms of triangulated functors, rather than merely into maps after choosing arbitrary resolutions.

10.2 Godement flasque resolutions

Let X be a topos with enough points, or more generally work with a conservative family of fibre functors. The Godement construction associates to a sheaf F a cosimplicial flasque sheaf, and hence a resolution $F \to C^\bullet(F)$. It is functorial in F and compatible with inverse image to the extent needed for base-change maps.

Definition 4.2.1. A sheaf I on a topos X is flasque for the present purposes if the functor of sections over any object of the site has vanishing higher derived functors on I. A complex is flasque-adapted if applying global sections, or more generally direct image, to it computes derived direct image.

Proposition 4.2.2. Every bounded-below complex of abelian sheaves admits a functorial flasque-adapted resolution. If $f : X \to Y$ is a morphism of topoi, applying f_* to such a resolution computes Rf_*. The construction is compatible with restriction to open subtopoi and with finite products of sites.

For a single sheaf F, embed it functorially into the product of its skyscraper sheaves at the chosen points; iterate the cokernel. The resulting resolution is flasque, and exactness is checked on the conservative family of points. For complexes, apply the construction by the usual Cartan-Eilenberg procedure. The functoriality is strict enough for all comparison morphisms used later.

4.2.3. Let

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a cartesian square of topoi. Choosing functorial flasque resolutions gives a natural base-change morphism

$$g^*Rf_*K \to Rf'_*g'^*K.$$

This construction agrees with the morphism obtained by deriving the underived adjunction map.

The agreement follows because both constructions are induced by the same map on degree-zero sheaves and because the derived category is obtained by inverting quasi-isomorphisms. Functoriality with respect to pasting of squares follows from the functoriality of the Godement resolution.

10.3 The base-change theorem

Theorem 4.3.1. For the morphisms and coefficients in the range of the preceding base-change theorems, the canonical morphism

$$Lg^* Rf_* K \longrightarrow Rf'_* Lg'^* K$$

constructed by the resolution formalism coincides with the base-change morphism studied in Exposés XII–XVI. In the proper case, for torsion coefficients, it is an isomorphism; in the smooth case, for torsion prime to the residual characteristics, it is an isomorphism; and in the finite cohomological-dimension range it is an isomorphism or monomorphism in the stated degrees.

The proof is a comparison of constructions. The base-change morphism of Exposé XII was defined fibrewise and then glued by noetherian induction. The present morphism is defined by resolving and applying adjunction. On a strict localization of the base, the two maps are identified after applying the fibre functor, where both are the natural restriction map on etale cohomology. Since the relevant fibre functors are conservative, the two morphisms are the same. The assertions about being an isomorphism are exactly the theorems already proved.

Corollary 4.3.2. Let $f : X \to S$ be compactifiable, choose a compactification $X \xrightarrow{j} \bar{X} \xrightarrow{\bar{f}} S$, and set

$$Rf_! K = R\bar{f}_* j_! K.$$

For every base change $g : S' \to S$ there is a canonical morphism

$$Lg^* Rf_! K \to Rf'_! Lg'^* K.$$

It is independent of the chosen compactification. Under the proper-base-change hypotheses for $\bar{f}$, it is an isomorphism in the same range.

Independence follows from Proposition 3.3.3 and the transitivity of dominating compactifications. If a third compactification dominates two given ones, the two induced base-change morphisms agree after pullback to the dominating compactification; by proper base change for the domination morphism they therefore agree already on the original compactifications.

Corollary 4.3.3. The functor $Rf_!$, for compactifiable morphisms, is functorial in f. If $X \xrightarrow{f} Y \xrightarrow{g} Z$ are compactifiable, then there is a canonical isomorphism

$$R(gf)_! \simeq Rg_! Rf_!.$$

This isomorphism is compatible with base change and with the localization triangles for open-closed decompositions.

Choose compactifications of f and g, dominate the induced compactification of gf, and apply the proper functoriality of ordinary direct image. The possible choices are compared through further domination. The localization compatibility is checked by reducing to the case of a proper morphism and an open-closed decomposition, where it is the usual exact triangle for $j_!$ and i_*.

Remark 4.3.4. The theorem in this subsection is not yet the final construction of $Rf_!$. It gives the base-change arrow and the independence of compactification in the bounded situations where the required resolutions and cohomological dimensions are available. The next section of Exposé XVII begins the full construction of direct image with proper supports and its comparison, finiteness, Kunneth, and symmetric-power properties.

11 Direct images with proper supports

11.1 The fundamental construction

We now begin the construction indicated in Section 3. For a compactifiable morphism one wants a functor $Rf_!$ which agrees with Rf_* when f is proper and with extension by zero when f is an open immersion. The only possible definition is obtained by decomposing a compactifiable morphism into these two elementary cases and then proving that the result is independent of the decomposition.

5.1.1. Let X be a topos, let U be an open of X, and let $j : U \to X$ be the inclusion. The extension-by-the-empty-object functor for sheaves of sets on U is left adjoint to restriction j^*. The same notation $j_!$ is used for pointed sheaves, sheaves of groups, and modules over a ring sheaf. In the abelian and module cases this is extension by zero.

If $i : F \to X$ is the closed complement of U, then $j^*j_!$ is the identity and $i^*j_!$ is the constant functor with value the initial, equivalently zero, object. These properties characterize $j_!$. They imply exactness of $j_!$ in the abelian cases, the natural monomorphism $j_! \hookrightarrow j_*$, and the transitivity formula

$$(j_1 j_2)_! \simeq j_{1!} j_{2!}.$$

Lemma 5.1.2. Let $f : X \to Y$ be a morphism of topoi, let $V \subset Y$ be open, and let $U = f^{-1}V$. For the square

$$\begin{array}{ccc} U & \xrightarrow{f'} & V \\ \downarrow j' & & \downarrow j \\ X & \xrightarrow{f} & Y, \end{array}$$

there is a unique base-change isomorphism

$$f^* j_! \xrightarrow{\sim} j'_! f'^*$$

compatible with the usual base-change morphism $f^* j_* \to j'_* f'^*$.

The adjoint map $j^* f_* \to f'_* j'^*$ is plainly an isomorphism. Transposing it gives $j'_! f'^* \simeq f^* j_!$. Compatibility with the inclusions into direct image is checked after restricting to U, where it is tautological.

5.1.3. If X is ringed by A, exactness of $j_!$ gives an exact functor

$$j_! : D(U, j^*A) \to D(X, A).$$

The transitivity formula for open immersions and the adjunction formula

$$\mathrm{Hom}_{D(X)}(j_!K, L) \simeq \mathrm{Hom}_{D(U)}(K, j^*L)$$

hold in the derived categories.

5.1.4. Let $f : X \to Y$ be proper, let A be a torsion ring sheaf on Y, and ring X by f^*A. If the dimensions of the fibres of f are bounded by n, then $R^q f_* F = 0$ for $q > 2n$ and every f^*A-module F. Hence

$$Rf_* : D(X, f^*A) \to D(Y, A)$$

is defined on the unbounded derived category in the range used here. For a composite of proper morphisms, one has the transitivity isomorphism $R(gh)_* \simeq Rg_* Rh_*$.

5.1.5–5.1.6. Consider a commutative square

$$\begin{array}{ccc} U & \xrightarrow{j_1} & X \\ \downarrow g & & \downarrow f \\ V & \xrightarrow{j_2} & Y, \end{array}$$

where j_1, j_2 are open immersions and f, g are proper of bounded fibre dimension. For $K \in D(U)$ there is a canonical isomorphism

$$j_{2!} Rg_* K \xrightarrow{\sim} Rf_* j_{1!} K.$$

After restricting to V, the assertion becomes the proper base-change isomorphism for the cartesian square $V \times_Y X$. On the closed complement $Y - V$, both sides vanish: for the right side this follows from proper base change and the fact that the restriction of $j_{1!}K$ to the complementary closed fibre is zero. Thus the morphism is an isomorphism globally.

Theorem 5.1.8. Let S be ringed by a torsion ring sheaf A, and let (S) be the category of quasi-compact quasi-separated S-schemes with S-compactifiable morphisms. There is, in one and essentially only one way, a functor

$$Rf_! : D(X, A_X) \to D(Y, A_Y)$$

for every morphism $f : X \to Y$ in (S), together with transitivity isomorphisms

$$R(gf)_! \simeq Rg_! Rf_!,$$

satisfying the cocycle condition. On proper morphisms it is Rf_*; on open immersions it is $j_!$; and for a square as in 5.1.5 the compatibility is the isomorphism just described.

The existence is exactly the gluing theorem of Section 3 applied to the two subcategories of proper morphisms and open immersions. The uniqueness follows from the fact that every compactifiable morphism is, by definition, a composite of an open immersion and a proper morphism, and any two compactifications are dominated by a third. The compatibility for a dominated compactification is supplied by proper base change.

5.1.9. When $f : X \to \operatorname{Spec} k$, with k separably closed, one writes

$$R^q f_! K = H^q_c(X, K)$$

and calls this hypercohomology, or cohomology, with proper support. The notation $Rf_!$ is slightly abusive: it is not the right derived functor of an ordinary functor $f_!$ in general, although an ordinary $f_!$ for pointed sheaves and groups will be described later.

Variants 5.1.11–5.1.15. The same construction gives: an ordinary functor $f_!$ for pointed sheaves; a functor $R^1 f_!$ for sheaves of ind-finite groups, independent of compactification; a triangulated way-out functor

$$Rf_! : D^+_{\mathrm{tors}}(X) \to D^+_{\mathrm{tors}}(Y);$$

and the corresponding version for ringed schemes with torsion coefficient rings. These variants use only proper base change and the equality $j_! = j_*$ for an open immersion which is also proper.

5.1.16. Let $j : U \to X$ be open, let $i : F \to X$ be the closed complement, and let $f : X \to S$ be compactifiable. For $K \in D(X, A_X)$, the localization triangle gives

$$R(fj)_! j^* K \to Rf_! K \to R(fi)_! i^* K \to R(fj)_! j^* K[1].$$

Consequently there is a long exact cohomology sequence with proper supports. Over a separably closed field this is the usual exact sequence

$$\cdots \to H^q_c(U, G) \to H^q_c(X, G) \to H^q_c(F, G) \to H^{q+1}_c(U, G) \to \cdots .$$

5.1.17. For a proper morphism $u : X \to Y$ over S, and $K \in D(Y, A_Y)$, the adjunction map $K \to Ru_* u^* K$ gives a contravariant pullback on cohomology with proper supports:

$$u^* : Rg_! K \to Rf_! u^* K,$$

where $f = gu$. For $S = \operatorname{Spec} k$, this is the map

$$H^q_c(Y, K) \to H^q_c(X, u^* K).$$

11.2 The base-change theorem

5.2.1. Let $f : X \to Y$ be a morphism of topoi, let $V \subset Y$ be open, and let $U = f^{-1}V$. Write $j : V \hookrightarrow Y$, $j' : U \hookrightarrow X$, and $f' : U \to V$. Let A be a ring sheaf on Y, let A' be a ring sheaf on X, and let L be a bounded-above complex of (A', f^*A)-bimodules. There is a canonical isomorphism

$$L \otimes^L_{f^*A} f^* j_! K \xrightarrow{\sim} j'_!\Big(j'^* L \otimes^L_{f'^*A} f'^* K\Big)$$

for $K \in D^-(V, A)$. If L has finite Tor-amplitude over f^*A, the same formula is defined for arbitrary $K \in D(V, A)$.

The first ingredient is the base-change isomorphism $f^* j_! \simeq j'_! f'^*$ of Lemma 5.1.2. The second is the elementary compatibility of extension by zero with tensor products:

$$L \otimes^L_{f^*A} j'_! M \simeq j'_!\Big(j'^* L \otimes^L_{f'^*A} M\Big).$$

For complexes adapted to the relevant tensor functors this is an underived identity; the displayed formula follows by passage to derived categories.

5.2.2. Let $f : X \to S$ be a compactifiable morphism and choose a compactification $X \xrightarrow{j} \bar{X} \xrightarrow{p} S$. For a base change $g : S' \to S$, form the cartesian diagram

$$\begin{array}{ccccc} X' & \xrightarrow{g'} & X & \xrightarrow{j} & \bar{X} \\ \downarrow f' & & \downarrow f & & \downarrow p \\ S' & \xrightarrow{g} & S & = & S, \end{array}$$

with $X' \xrightarrow{j'} \bar{X}' \xrightarrow{p'} S'$ obtained by base change. If A and A' are torsion coefficient rings on S and S', and if L is a bounded-above complex of (A', g^*A)-bimodules, then the proper base-change theorem for p and the formula of 5.2.1 give an isomorphism

$$L \otimes^L_{g^*A} g^* Rp_* j_! K \ \xrightarrow{\sim}\ Rp'_* j'_!\Big(f'^* L \otimes^L_{g'^*A} g'^* K\Big).$$

Equivalently,

$$L \otimes^L g^* Rf_! K \xrightarrow{\sim} Rf'_!\Big(f'^* L \otimes^L g'^* K\Big)$$

after identifying $Rf_! = Rp_* j_!$ and $Rf'_! = Rp'_* j'_!$.

This is the base-change isomorphism for cohomology with proper supports attached to the chosen compactification. The construction is simply the composition of two already constructed isomorphisms: first move g^* past the proper direct image Rp_*, and then move extension by zero past the pullback and tensor product.

Lemma 5.2.3. The isomorphism of 5.2.2 is independent of the compactification $X \hookrightarrow \bar{X} \to S$.

It is enough to compare two compactifications when one dominates the other. Suppose $q : \bar{X}_1 \to \bar{X}_2$ is proper and induces the identity on X. The isomorphism attached to $\bar{X}_1$ is the composite of those attached to the open immersion j_1, the proper map q, and the proper map p_2; the isomorphism attached to $\bar{X}_2$ is the composite of those attached to j_2 and p_2. By adjunction for $j_{2!}$, equality can be checked after restriction to the open X, where both composites are the identity. Two arbitrary compactifications are dominated by a third, so the assertion follows.

Lemma 5.2.4. Let $X \xrightarrow{v} Y \xrightarrow{u} S$ be a composite of compactifiable morphisms. The base-change isomorphisms of 5.2.2 for u, v, and uv are compatible with the transitivity isomorphism

$$R(uv)_! \simeq Ru_!Rv_!.$$

In other words, the two natural ways of passing from

$$L \otimes^L g^* R(uv)_! K$$

to the corresponding expression on the base-changed composite agree.

Choose compatible compactifications of v and u, and hence of their composite. The assertion then reduces to the associativity of proper base change and to the transitivity formula for extension by zero. The general case follows by replacing the chosen compactifications by a common dominating system and using Lemma 5.2.3.

Further compatibility 5.2.4.1. The pullback just described is not an isolated operation. If $X \xrightarrow{u} Y \xrightarrow{v} Z$ are proper over S, the pullback for vu is the composite of the pullbacks for v and u. If an open immersion is inserted in the diagram, the same assertion remains true after replacing ordinary direct image by extension by zero. These compatibilities are immediate on a chosen compactification and therefore hold in the compactification-independent formalism by the uniqueness statement of Theorem 5.1.8.

5.2.4.2. Let $K \to L \to M \to K[1]$ be a distinguished triangle on X. For a compactifiable $f : X \to S$, the triangle

$$Rf_!K \to Rf_!L \to Rf_!M \to Rf_!K[1]$$

commutes with all pullbacks by proper morphisms over S and with the localization triangle for an open-closed decomposition of X. Thus the exact sequences with proper support are natural not only in the coefficient complex but also in the geometric pair.

Transition within 5.2. The construction so far gives the functor and its formal exactness. What is still missing is the strong base-change theorem: under the hypotheses used in ℓ-adic and torsion cohomology, the base-change morphism

$$g^* Rf_! K \to Rf'_! g'^* K$$

should be an isomorphism. Section 5.2 is devoted to this statement and to its reduction to the proper and open cases already isolated above.

Continuation anchor. The next batch should continue SGA 4, Exposé XVII at the continuation of Section 5.2, source line 4315 of `17_fr.tex`.

SGA 4 Translation Project – Batch 025

Exposé XVII completion and Exposé XVIII opening through Section 2

Draft English LaTeX translation; compile-checked and rendered.

SGA 4, Exposé XVII – continuation

Cohomology with Proper Supports and the Construction of the Functor $f_!$

P. Deligne, after A. Grothendieck

This batch continues at the proof following Lemma 5.2.4. The statement of Lemma 5.2.4 was already included in the preceding delivery; it asserted the compatibility of the base-change isomorphism for a composite $f = uv$ with the base-change isomorphisms attached separately to u and to v.

5.2. The base-change theorem, continued

Proof of Lemma 5.2.4. Choose a compactification of the pair (u, v). Thus one has a diagram

$$\begin{tikzcd}
X' \arrow[r, "g''"'] \arrow[d, "v'"'] & X \arrow[r, hook, "j_1"] \arrow[d, "v"'] & \bar{X} \arrow[r, hook, "j_2"] \arrow[d] \arrow[dl, "p_1"'] & \bar{\bar{X}} \arrow[dl, "p_3"] \\
Y' \arrow[r, "g'"'] \arrow[d, "u'"'] & Y \arrow[r, hook, "j_3"] \arrow[d, "u"'] & \bar{Y} \arrow[dl, "p_2"] & \\
S' \arrow[r, "g"'] & S & &
\end{tikzcd}.$$

The assertion is that the base-change isomorphism c_1 for f is the composite c_2 of the two isomorphisms for u and v. The isomorphism c_1 is obtained by composing the base-change isomorphisms attached to j_1, j_2, p_3, and p_2. The second isomorphism c_2 is obtained by composing those attached to j_1, p_1, j_3, and p_2. Hence the comparison reduces to the middle square: compare the composite attached to p_3 and j_2 with the composite attached to j_3 and p_1.

After the base change by g we obtain the Cartesian diagram

$$\begin{tikzcd}
\bar X' \arrow[rrr, hook, "\bar g''"] \arrow[drr, hook, "j_2'"] \arrow[dd, "p_1'"] & & & \bar X \arrow[drr, hook, "j_2"] \arrow[dd, "p_1"] & & \\
& & \bar{\bar X}' \arrow[rrr, "\bar{\bar g}"] \arrow[dd, "p_3'"] & & & \bar{\bar X} \arrow[dd, "p_3"] \\
Y' \arrow[rrr, hook, "g'"] \arrow[drr, hook, "j_3'"] \arrow[ddr, "u'"'] & & & Y \arrow[drr, hook, "j_3"] \arrow[ddr, "u"] & & \\
& & \bar Y' \arrow[rrr, "\bar g'"'] \arrow[dl, "p_2'"] & & & \bar Y \arrow[dl, "p_2"] \\
& S' \arrow[rrr, "g"'] & & & S. &
\end{tikzcd}$$

Put $L' = {p_2'}^* L$ and $K' = j_{1!}K$. The two composites to be compared are isomorphisms

$$c_1', c_2' : L' \overset{\mathbf{L}}{\otimes} (\bar g')^* R(j_3 p_1)_! K' \xrightarrow{\sim} R(j_3' p_1')_! \Big((j_3' p_1')^* L' \otimes (\bar g'')^* K' \Big).$$

By the elementary formula for extension by zero, it is enough to verify equality after restriction to the open $Y' \subset \bar Y'$, that is, after applying ${j_3'}^*$. On this open the assertion is immediate from the definitions of the base-change maps. This proves the compatibility for the chosen compactification. The usual comparison of two compactifications by a third one gives the result in general. $\square$

Lemma 5.2.5. Let

$$(S'', \mathcal{A}'') \xrightarrow{g_2} (S', \mathcal{A}') \xrightarrow{g_1} (S, \mathcal{A})$$

be morphisms of ringed schemes, the rings being sheaves of torsion rings, and let $f : X \to S$ be compactifiable. With the evident base-changed diagram

$$\begin{tikzcd}
X'' \arrow[r, "g_2'"'] \arrow[d, "f''"] & X' \arrow[r, "g_1'"'] \arrow[d, "f'"] & X \arrow[d, "f"] \\
S'' \arrow[r, "g_2"'] & S' \arrow[r, "g_1"'] & S,
\end{tikzcd}$$

where X, X', X'' are ringed by $f^*\mathcal{A}, f'^*\mathcal{A}', f''^*\mathcal{A}''$, respectively, the two ways of forming the base-change isomorphism for the composite $g_1 g_2$ agree. Equivalently, the diagram

$$\begin{tikzcd}
L(g_1g_2)^*Rf_! \arrow[rr, "\sim"] \arrow[d, equal] & & Rf''_!L(g'_1g'_2)^* \arrow[d, equal] \\
Lg_2^*Lg_1^*Rf_! \arrow[r, "\sim"] & Lg_2^*Rf'_!Lg'^*_1 \arrow[r, "\sim"] & Rf''_!Lg'^*_2Lg'^*_1
\end{tikzcd}$$

is commutative.

Proof. The assertion follows immediately from the analogous statements for $j_!$ and for Rp_* when p is proper. One writes a compactification $f = pj$, expands the two composites, and uses functoriality of proper base change and of extension by zero. This is a purely formal pasting argument. □

Theorem 5.2.6. The functor $Rf_!$ commutes with base change.

This summarizes the preceding lemmas: the base-change morphism is defined independently of the chosen compactification, is compatible with composition of compactifiable morphisms, and is functorial for iterated changes of base.

Variants 5.2.7. The same assertions hold in the following forms.

(i) The functors $f_!$ for sheaves of pointed sets, and the functors $R^1f_!$ for sheaves of groups, commute with base change.

(ii) The variants of $Rf_!$ defined for bounded-below torsion complexes also commute with base change.

(iii) Under the hypotheses of 5.2.2, if L is bounded and has finite torsion dimension, then the functor $Rf_! : D(X) \to D(S)$ commutes with base change. As observed at the point where the morphism was constructed, the extra boundedness and finite-torsion-dimension hypotheses on L are not essential for the construction itself.

Theorem 5.2.6 is essentially equivalent to the conjunction of the following fibre formula and tensor formula.

Proposition 5.2.8. Let $f : X \to S$ be compactifiable, and let $\mathcal{F}$ be a torsion sheaf on X. For every geometric point s of S, if X_s is the geometric fibre and $\mathcal{F}_s$ the induced sheaf on it, there is a canonical identification

$$(R^qf_!\mathcal{F})_s = H^q_c(X_s, \mathcal{F}_s).$$

Corollary 5.2.8.1. If the fibres of f have dimension at most d, then $Rf_!$ has cohomological dimension at most $2d$: for every torsion sheaf $\mathcal{F}$, one has $R^if_!\mathcal{F} = 0$ for $i > 2d$. Consequently $R^{2d}f_!$ is right exact on torsion sheaves.

Proof. By base change one reduces to the case where S is the spectrum of an algebraically closed field. Choose a proper compactification $j : X \hookrightarrow \bar{X}$ over S. Then $\dim \bar{X} = \dim X \leq d$, and for a torsion sheaf $\mathcal{F}$

$$H^i_c(X, \mathcal{F}) = H^i(\bar{X}, j_! \mathcal{F}) = 0 \qquad (i > 2d),$$

by the cohomological dimension theorem for noetherian schemes. □

Proposition 5.2.9. Let $f : X \to S$ be compactifiable, let $\mathcal{A}$ be a sheaf of torsion rings on S, let K be a bounded-above complex of left $f^*\mathcal{A}$-modules, and let L be a bounded-above complex of right $\mathcal{A}$-modules. There is a canonical isomorphism

$$L \overset{\mathbf{L}}{\otimes}_{\mathcal{A}} Rf_!(K) \simeq Rf_!\Big(f^*L \overset{\mathbf{L}}{\otimes}_{f^*\mathcal{A}} K\Big).$$

Theorem 5.2.10. Let $f : X \to S$ be compactifiable, $\mathcal{A}$ a sheaf of torsion rings on S, and $K \in D^-(X, f^*\mathcal{A})$. If K has finite Tor-dimension $\leq d$, then $Rf_!K$ has finite Tor-dimension $\leq d$.

Proof. It is enough to verify that for every right $\mathcal{A}$-module L on S,

$$\mathcal{H}^{-i}(L \overset{\mathbf{L}}{\otimes} Rf_!K) = 0 \qquad (i > d).$$

By Proposition 5.2.9 the left-hand side is

$$R^{-i}f_!\Big(f^*L \overset{\mathbf{L}}{\otimes} K\Big).$$

The hypothesis on K gives $\mathcal{H}^{-i}(f^*L \overset{\mathbf{L}}{\otimes} K) = 0$ for $i > d$, and the assertion follows since applying $Rf_!$ to a complex with no cohomology below degree $-d$ cannot create such cohomology. □

In the special case of constant noetherian rings one has a slightly more general statement.

Theorem 5.2.11. Let $f : X \to S$ be a morphism of sites, let A be a ring, let C be its centre, and let $d \in \mathbb{Z}$. Assume:

(i) $Rf_* : D^+(X, C) \to D^+(S, C)$ has finite cohomological dimension;

(ii) A is right noetherian;

(iii) S has enough points.

For every $K \in D^-(X, A)$, if K has Tor-dimension $\leq d$, then $Rf_*K \in D^-(S, A)$ has Tor-dimension $\leq d$.

Proof. By the finite cohomological dimension hypothesis, f_* is derivable on unbounded derived categories in the required range. By the pointwise Tor-dimension criterion, it suffices to test the stalks of Rf_*K. Thus, for every point s of S, and every finitely generated right A-module L, it suffices to prove

$$H^{-i}(L \overset{\mathbf{L}}{\otimes} Rf_*K) = 0 \qquad (i > d).$$

As in the proof of Theorem 5.2.10 this follows if the natural map

$$L \overset{\mathbf{L}}{\otimes} Rf_*K \longrightarrow Rf_*(L \overset{\mathbf{L}}{\otimes} K)$$

is an isomorphism. Take a resolution L^* of L by free finitely generated right A-modules. The functors involved are way-out to the right; hence it suffices to prove the assertion for the individual components of the resolution. In that case L is finite free, and the assertion is immediate. □

Remarks 5.2.12.

(i) The hypothesis that S have enough points is probably unnecessary.

(ii) If the noetherian hypothesis on A is omitted, the theorem remains valid provided the functors R^if_* commute with filtered inductive limits.

(iii) The assumption that the coefficient sheaf of rings on S be constant is essential. A counterexample is obtained by taking S to be the spectrum of a strictly local discrete valuation ring, taking f to be the inclusion of the generic point η, taking $\mathcal{A}$ to have value $\mathbb{Z}/n\mathbb{Z}$ at η and $\mathbb{Z}/n^2\mathbb{Z}$ at the closed point s, with $n > 1$ prime to the residue characteristics, and taking the module sheaf $f^*\mathcal{A}$.

5.3. The comparison theorem and the finiteness theorem

Let X be a scheme of finite type over $\mathbb{C}$. Associated with X are the ét-site $X_{\text{ét}}$, the classical or transcendental site X_{cl}, and a morphism of sites

$$\epsilon : X_{\text{cl}} \longrightarrow X_{\text{ét}}.$$

If $j : U \to X$ is an open immersion, the square

$$\begin{array}{ccc} U_{\text{cl}} & \longrightarrow & U_{\text{ét}} \\ \downarrow & & \downarrow \\ X_{\text{cl}} & \longrightarrow & X_{\text{ét}} \end{array}$$

identifies the topos of U_{cl} with the open subtopos of X_{cl} inverse image of the open subtopos $U_{\text{ét}} \subset X_{\text{ét}}$. If $p : Y \to X$ is proper, then the associated continuous map $p^{\text{an}} : Y^{\text{an}} \to X^{\text{an}}$ is proper and separated; by Chow's lemma this is reduced to the projective case.

Let $f : X \to S$ be a compactifiable morphism of schemes of finite type over $\mathbb{C}$, and write $f = pj$ with $j : X \hookrightarrow \bar{X}$ open and $p : \bar{X} \to S$ proper. The diagram of classical and ét-sites is

$$\begin{tikzcd}
X_{\mathrm{cl}} \arrow[rr, "\epsilon"'] \arrow[dr, "j_{\mathrm{cl}}"] \arrow[dd, "f_{\mathrm{cl}}"'] & & X_{\text{ét}} \arrow[dr, "j"] \arrow[dd, "f" near end] & \\
& \bar{X}_{\mathrm{cl}} \arrow[rr, "\epsilon"' near start] \arrow[dl, "p_{\mathrm{cl}}"] & & \bar{X}_{\text{ét}} \arrow[dl, "p"] \\
S_{\mathrm{cl}} \arrow[rr, "\epsilon"'] & & S_{\text{ét}}. &
\end{tikzcd}$$

The functor $p_{\mathrm{cl}*} j_{\mathrm{cl}!}$ is the usual direct image with proper support $f_{\mathrm{cl}!}$. Since the question is local on S, extension by zero carries flasque sheaves on X_{cl} to sheaves on $\bar{X}_{\mathrm{cl}}$ acyclic for $p_{\mathrm{cl}*}$, and hence

$$Rf_{\mathrm{cl}!} = Rp_{\mathrm{cl}*} j_{\mathrm{cl}!}.$$

By the construction of $j_!$ one has a base-change isomorphism

$$\epsilon^* j_! \xrightarrow{\sim} j_{\mathrm{cl}!} \epsilon^*.$$

By proper base change, one has a morphism

$$\epsilon^* Rp_* \longrightarrow Rp_{\mathrm{cl}*} \epsilon^*.$$

Composing these gives a morphism, for torsion complexes,

$$\epsilon^* Rf_! \longrightarrow Rf_{\mathrm{cl}!} \epsilon^*.$$

As in Lemma 5.2.3, this morphism is independent of the chosen compactification of f, and it satisfies the expected transitivity formula.

Theorem 5.3.3 (comparison theorem). The morphism

$$\epsilon^* Rf_! \longrightarrow Rf_{\mathrm{cl}!} \epsilon^*$$

is an isomorphism.

Proof. By construction it is enough to treat the case where f is proper. The two functors under consideration are triangulated and way-out. Thus one reduces to the case where the complex is a torsion sheaf placed in degree zero. The result is then the comparison theorem of Exposé XVI. □

Variants 5.3.4. If $\mathcal{A}$ is a sheaf of torsion rings on $S_{\text{ét}}$, the same argument gives an isomorphism between functors

$$D(X, f^*\mathcal{A}) \longrightarrow D(S, \epsilon^*\mathcal{A}).$$

In low degree one also obtains the usual variants for pointed sheaves and for sheaves of ind-finite groups.

Corollary 5.3.5. Let X be a separated scheme of finite type over $\mathbb{C}$, admitting a compactification by a proper $\mathbb{C}$-scheme, and let F be a torsion sheaf on X. Then

$$H^q_!(X, F) \simeq H^q_c(X_{\mathrm{cl}}, \epsilon^* F).$$

Theorem 5.3.6 (finiteness). Let $f : X \to S$ be a compactifiable morphism of finite presentation and let A be a left noetherian torsion ring. The functor

$$Rf_! : D(X, A) \longrightarrow D(S, A)$$

sends complexes with constructible cohomology to complexes with constructible cohomology.

Proof. The assertion is local on S, which may be assumed affine. By passage to the limit one reduces to the noetherian case. The complexes with constructible cohomology form a triangulated subcategory, and $Rf_!$ is way-out; it is therefore enough to prove that, for a constructible sheaf $\mathcal{F}$ of A-modules, the sheaves $R^q f_! \mathcal{F}$ are constructible. Choose a compactification $f = pj$. Then

$$R^q f_! \mathcal{F} = R^q p_*(j_! \mathcal{F}),$$

and $j_!\mathcal{F}$ is constructible. The assertion follows from the finiteness theorem for proper direct images. □

Corollary 5.3.8. Under the same hypotheses, if S is noetherian and K is a bounded complex of constructible A-modules on X, then the formation of $Rf_!K$ is compatible with restriction to locally closed strata and with passage to geometric fibres; in particular the ranks and finite stalk modules occurring in the cohomology sheaves of $Rf_!K$ vary constructibly on S.

This is the usual form in which the finiteness theorem is used below: after compactification the assertion is reduced to the proper case, and all local statements are checked after applying the base-change theorem.

5.4. The Künneth formula

Let

$$\begin{tikzcd} X' \arrow[d, "f"] & Z' \arrow[l, "p"'] \arrow[r, "q"] \arrow[d, "h"] & Y' \arrow[d, "g"] \\ X & Z \arrow[l] \arrow[r] & Y \end{tikzcd}$$

be a diagram over a scheme S, with the lower objects over S, and let $\mathcal{A}$ be a sheaf of torsion rings on S. In the proper case, for complexes K on X' and L on Y', one has the usual Künneth morphism

$$Rf_*K \overset{\mathbf{L}}{\boxtimes}_{\mathcal{A}} Rg_*L \longrightarrow Rh_*(K \overset{\mathbf{L}}{\boxtimes}_{\mathcal{A}} L).$$

When f and g are merely compactifiable, factor them into open immersions followed by proper morphisms. With $Z' = X' \times_S Y'$, extension by zero gives the canonical identification

$$j_{3!}(K \boxtimes_{\mathcal{A}} L) \ \xrightarrow{\sim}\ j_{1!}K \boxtimes_{\mathcal{A}} j_{2!}L.$$

Composing the inverse of this isomorphism with the proper Künneth morphism yields a Künneth morphism with proper supports

$$Rf_!K \overset{\mathbf{L}}{\boxtimes}_{\mathcal{A}} Rg_!L \longrightarrow Rh_!(K \overset{\mathbf{L}}{\boxtimes}_{\mathcal{A}} L),$$

for $K \in D^-(X', \mathcal{A})$, $L \in D^-(Y', \mathcal{A})$. The construction is independent of the chosen compactifications.

Theorem 5.4.3. Under the above hypotheses, the Künneth morphism for cohomology with proper supports is an isomorphism.

Proof. By construction it is enough to prove the theorem when the morphisms are proper. Consider a base change of the entire diagram

$$(S_1, X_1, Y_1, Z_1, X_1', Y_1', Z_1') \longrightarrow (S, X, Y, Z, X', Y', Z').$$

The base-change morphisms for $Rf_!$, $Rg_!$, and $Rh_!$ fit into a commutative square with the two Künneth morphisms:

$$\begin{tikzcd} C^*(Rf_!K \overset{\mathbf{L}}{\boxtimes} Rg_!L) \arrow[r, "\sim"] \arrow[d] & C^*Rf_!K \overset{\mathbf{L}}{\boxtimes} C^*Rg_!L \arrow[r, "\sim"] & Rf_{1!}C^*K \overset{\mathbf{L}}{\boxtimes} Rg_{1!}C^*L \arrow[d] \\ C^*Rh_!(K \overset{\mathbf{L}}{\boxtimes} L) \arrow[r, "\sim"] & Rh_{1!}C^*(K \overset{\mathbf{L}}{\boxtimes} L) \arrow[r, "\sim"] & Rh_{1!}(C^*K \overset{\mathbf{L}}{\boxtimes} C^*L). \end{tikzcd}$$

Hence it suffices to check the assertion after arbitrary base change to geometric points of Z. We may therefore assume S is the spectrum of an algebraically closed field and that the structural maps from X and Y to S are the relevant maps. The problem becomes the familiar map

$$Rf_!K \overset{\mathbf{L}}{\otimes}_{\mathcal{A}} Rg_!L \longrightarrow R(f \times g)_!(K \overset{\mathbf{L}}{\boxtimes}_{\mathcal{A}} L).$$

In the proper case the proof is the standard one: apply the projection formula and proper base change to the square defining $X \times_S Y$. The compatibility square shows that the reduction made above is legitimate, and the theorem follows. □

More generally, if $(f_i : X'_i \to X_i)_{i \in I}$ is a finite family of compactifiable morphisms of S-schemes, if $X = \prod_S X_i$, $X' = \prod_S X'_i$, and $f : X' \to X$ is the product, then one has a canonical isomorphism

$$\overset{\mathbf{L}}{\boxtimes}_{i \in I} Rf_{i!}(K_i) \xrightarrow{\sim} Rf_!\Big(\overset{\mathbf{L}}{\boxtimes}_{i \in I} K_i\Big).$$

Neither the definition of this isomorphism nor the definition of its two sides requires the choice of a total order on I.

5.5. Cohomology of symmetric powers

The main result of this section is Theorem 5.5.21. It computes, for suitable coefficients, the cohomology with proper supports of a symmetric power of a scheme in terms of the cohomology with proper supports of the original scheme. The topological analogue for semi-simplicial topology is classical: if X_* is a semi-simplicial set, the geometric realization of the n-th symmetric power $X_*^n/\mathfrak{S}_n$ maps canonically and continuously to the n-th symmetric power of the realization of X_*, and this map induces the expected isomorphism on cohomology.

We recall the algebraic ingredients used in the proof. Lazard's theorem says that every flat module over a ring A is a filtered inductive limit of finite free modules. Consequently filtered inductive limits of finite free, respectively finite projective, modules give precisely the flat, respectively flat Mittag–Leffler/projective-type, modules needed here; flatness is stable under extensions; and every short exact sequence of flat modules is a filtered inductive limit of split short exact sequences.

For an A-module M, write

$$TS^n(M) = (M^{\otimes n})^{\mathfrak{S}_n}$$

for symmetric tensors. If B is a commutative A-algebra, then $TS^n(B)$ is an A-algebra, and $\operatorname{Spec} TS^n_A(B)$ is the n-th symmetric power of $\operatorname{Spec} B$ over $\operatorname{Spec} A$.

Following Roby, a polynomial law $f : M \to N$ is a system of maps

$$f_{A'} : M \otimes_A A' \longrightarrow N \otimes_A A'$$

functorial in the commutative A-algebra A'. It is homogeneous of degree n if $f_{A'}(a'x) = a'^n f_{A'}(x)$. The functor

$$N \longmapsto \operatorname{Hom}^{n\text{-ic}}(M, N)$$

is represented by a module $\Gamma^n(M)$ with universal degree-n polynomial map $\gamma^n : M \to \Gamma^n(M)$. Similarly, multihomogeneous polynomial laws of multidegree $(n_i)_{i\in I}$ are represented by $\bigotimes_i \Gamma^{n_i}(M_i)$.

The functors TS^n and Γ^n commute with filtered inductive limits and carry finite free modules to finite free modules. Thus they preserve flat modules. The canonical degree-n map

$$m \longmapsto m \otimes \cdots \otimes m$$

from M to $TS^n(M)$ defines a natural map

$$\Gamma^n(M) \longrightarrow TS^n(M),$$

which is an isomorphism for finite free M, and hence also for flat M.

If $P_1 \to P_0 \to M \to 0$ is a presentation and $P_1' = P_1 \times P_0$, with maps $j_0, j_1 : P_1' \to P_0$, the sequence

$$\Gamma^n(P_1') \rightrightarrows \Gamma^n(P_0) \longrightarrow \Gamma^n(M) \longrightarrow 0$$

is exact. For a flat presentation this gives the description

$$TS^n(P_1') \rightrightarrows TS^n(P_0) \longrightarrow \Gamma^n(M) \longrightarrow 0.$$

If $M = M' \oplus M''$, there is a canonical decomposition

$$\Gamma^n(M) \simeq \bigoplus_{n'+n''=n} \Gamma^{n'}(M') \otimes \Gamma^{n''}(M'').$$

For a short exact sequence $0 \to M' \to M \to M'' \to 0$, the module $\Gamma^n(M)$ has a canonical decreasing filtration whose graded pieces are

$$\Gamma^i(M'') \otimes \Gamma^{n-i}(M'),$$

provided the sequence is split; by Lazard's theorem the same holds when M'' is flat. The functor Γ^n commutes with extension of scalars.

Dold–Puppe theory identifies co-semi-simplicial objects of an abelian category with non-negatively graded complexes through the normalized-chain functor N. This equivalence and its inverse preserve homotopies. Applying Γ^n levelwise to a co-semi-simplicial resolution and normalizing defines the derived functors $L\Gamma^n$ used below. If a property is stable under finite sums and direct factors, it holds for the semi-simplicial terms up to degree r if and only if it holds for the normalized complex up to degree r.

Lemma 5.5.4. For an exact sequence of complexes satisfying the flatness conditions needed to apply Dold–Puppe, the filtrations on Γ^n associated with the short exact sequence pass to the derived functors $L\Gamma^n$. In particular the associated graded pieces are the sums of the terms

$$L\Gamma^i(K'') \overset{\mathbf{L}}{\otimes} L\Gamma^{n-i}(K').$$

Proposition 5.5.6. If $K \to L$ is a quasi-isomorphism between suitable bounded complexes whose terms satisfy the required flatness hypotheses, then $L\Gamma^n(K) \to L\Gamma^n(L)$ is a quasi-isomorphism. The construction is compatible with finite direct sums, extension of scalars, and the canonical product maps.

The proof proceeds by replacing complexes by co-semi-simplicial objects, filtering by degeneracy degree and by the standard filtration attached to a short exact sequence, and then using the exactness properties of Γ^n recalled above. The key point is that the functors under consideration preserve the flat objects used for the resolutions.

Now let $p : X \to S$ be a quasi-projective morphism. For a sheaf F of $\mathcal{A}$-modules on X, define an external divided-power sheaf $\Gamma^n_{\text{ext}}(F)$ on $\operatorname{Sym}^n_S(X)$. Locally on affines, this is the sheaf associated with $\Gamma^n_A(M)$. The construction is functorial in F, commutes with localization, and is compatible with extension by zero for open immersions:

$$\Gamma^n_{\text{ext}}(j_!F) \simeq \operatorname{Sym}^n_S(j)_!\Gamma^n_{\text{ext}}(F).$$

Derived functors $L\Gamma^n_{\text{ext}}$ are obtained by resolving by complexes of flat modules of Tor-dimension ≤ 0. They satisfy the same formal properties as $L\Gamma^n$.

Let $u : X \to Y$ be a compactifiable morphism of S-schemes. Assume that the symmetric powers $\operatorname{Sym}^n_S(X)$, $\operatorname{Sym}^n_S(\bar{X})$, and $\operatorname{Sym}^n_S(Y)$ exist for a compactification $X \xrightarrow{j} \bar{X} \xrightarrow{\bar{u}} Y$. The symmetric Künneth morphism is the composite

$$L\Gamma^n_{\text{ext}} Ru_! \longrightarrow R\operatorname{Sym}^n_S(u)_! L\Gamma^n_{\text{ext}},$$

obtained by writing $Ru_! = R\bar{u}_* j_!$, applying the proper symmetric Künneth morphism for $\bar{u}$, then using the displayed compatibility with $j_!$. It is independent of the compactification, is transitive for composites, and is compatible with extension of scalars, ordinary base change, and radicial base change in the forms needed later.

Theorem 5.5.21 (symmetric Künneth formula). Let S be coherent, i.e. quasi-compact and quasi-separated, and ringed by a sheaf of commutative torsion rings $\mathcal{A}$ whose fibres are noetherian. Let every S-scheme be ringed by the inverse image of $\mathcal{A}$. Let $u : X \to Y$ be a quasi-projective morphism of S-schemes. For every integer $n \geq 0$ and every

$$K \in D^{b,\mathrm{Tor}\leq 0}(X, \mathcal{A}),$$

the symmetric Künneth morphism

$$k^n : L\Gamma^n_{\text{ext}} Ru_! K \longrightarrow R\operatorname{Sym}^n_S(u)_! L\Gamma^n_{\text{ext}} K$$

is an isomorphism in $D^{b,\mathrm{Tor}\leq 0}(\operatorname{Sym}^n_S(Y), \mathcal{A})$.

The quasi-projectivity hypothesis is used only to ensure that u has a compactification for which the symmetric powers of X, $\bar{X}$, and Y exist. With algebraic spaces it should be replaceable by compactifiability.

Lemma 5.5.22. Theorem 5.5.21 follows from the following special case. For every algebraically closed field k, every smooth projective curve X over k, and every prime ℓ, the maps

$$L\Gamma^n\Big(R\Gamma(X, \mathbb{Z}/\ell)\Big) \longrightarrow R\Gamma(\mathrm{Sym}^n_k(X), \mathbb{Z}/\ell)$$

are isomorphisms for all $n \geq 0$.

Here $R\Gamma$ means global sections, while $L\Gamma^n$ is the derived divided-power functor. The reduction proceeds by devissage. A short exact sequence of bounded complexes of Tor-dimension ≤ 0

$$0 \to K' \to K \to K'' \to 0$$

has the property that, if the symmetric Künneth morphisms are isomorphisms for two of the three complexes, then they are so for the third. The proof uses the filtration on $L\Gamma^n(K)$ with graded pieces $L\Gamma^{n-i}(K') \overset{\mathbf{L}}{\otimes} L\Gamma^i(K'')$, together with induction on n and the ordinary Künneth formula.

The problem is local on S, and any complex is built by successive extensions from complexes supported on affine locally closed pieces. Thus one reduces to the case in which S is affine and u has relative dimension at most one. Checking on geometric points of $\mathrm{Sym}^n_S(Y)$ reduces to $S = \operatorname{Spec} k$, with k algebraically closed. A further radicial and finite-support reduction reduces Y to a finite disjoint union of copies of $\operatorname{Spec} k$, and by decomposing over the components of Y one reduces to

$$Y = S = \operatorname{Spec} k, \; \dim X \leq 1.$$

The coefficient ring is then a noetherian ring A.

Proposition 5.5.23. Let A be a commutative noetherian ring, let $K \in D^-(A)$, and let $n \in \mathbb{Z}$. The following are equivalent:

(i) K has Tor-dimension $\leq n$;

(ii) for every $x \in \operatorname{Spec} A$ and every $i < -n$,

$$H^i(K \overset{\mathbf{L}}{\otimes}_A k(x)) = 0.$$

The nontrivial implication is proved by noetherian induction on closed supports. If a module N is supported in a closed subset F, reduce first to the case in which N is an A/I-module for the reduced ideal defining F. Then compare N with its fibre over a maximal point η of F, using the exact sequence

$$0 \to M_1 \to N \to N \otimes k(\eta) \to M_2 \to 0.$$

The modules M_1, M_2 are supported away from η, so the induction hypothesis applies to them, while the fibre term is a sum of copies of $k(\eta)$. This proves the vanishing for N.

Corollary 5.5.24. A morphism $u : K \to L$ in $D^-(A)$ is an isomorphism if, for every $x \in \operatorname{Spec} A$, the induced morphism

$$K \overset{\mathbf{L}}{\otimes}_A k(x) \longrightarrow L \overset{\mathbf{L}}{\otimes}_A k(x)$$

is an isomorphism in $D^-(k(x))$.

Apply Proposition 5.5.23 to the cone of u.

Continuing the reduction of Lemma 5.5.22, compatibility with extension of scalars and Corollary 5.5.24 reduce the coefficient ring to a field of characteristic $\ell > 0$. Shifting complexes and truncating reduce to sheaves placed in degree zero. Such a sheaf is a filtered inductive limit of constructible sheaves. The trace method reduces to sheaves of the form $j_! A_{X'}$ for $j : X' \to X$ quasi-finite, and transitivity replaces X by X'. Thus we reach the case

$$\dim X \leq 1, \qquad K = A_X.$$

After replacing X by its reduction and cutting off the singular finite set, one reduces to either a smooth curve or a finite set. A final devissage of the constant sheaf over connected components gives the smooth connected curve case; the finite connected case is trivial.

Lemma 5.5.26. Theorem 5.5.21 follows from the two following special cases of Lemma 5.5.22:

Case 1. $k = \mathbb{C}$.

Case 2. k has characteristic $p > 0$ and $\ell = p$.

If the characteristic of k is zero, the Lefschetz principle reduces to $k = \mathbb{C}$. If $\operatorname{char} k = p > 0$ and $\ell \neq p$, lift the smooth projective curve X to a curve X' over the Witt vectors $W(k)$. The morphisms

$$f'_n : \operatorname{Sym}^n_{W(k)}(X') \longrightarrow \operatorname{Spec} W(k)$$

are projective and smooth. By the specialization theorem, the sheaves $R^i f'_{n*}(\mathbb{Z}/\ell)$, as well as the terms $L^i\Gamma^n(Rf'_*(\mathbb{Z}/\ell))$, are locally constant on $\operatorname{Spec} W(k)$. Hence the assertion for the special fibre follows from the corresponding assertion for a geometric generic fibre, which is of characteristic zero.

For Case 1, compare the algebraic statement with the topological one by the comparison theorem. A complex smooth projective curve is triangulable. By the same devissage as before, it is enough to prove the topological statement for a closed simplex D of dimension $0, 1$, or 2. Then

$$R\Gamma(D, \mathbb{Z}/\ell) = \mathbb{Z}/\ell$$

in degree zero, and $\operatorname{Sym}^n(D)$ is contractible, so both sides of the symmetric Künneth map are $\mathbb{Z}/\ell$ in degree zero; the induced map on H^0 is the identity.

For Case 2, one compares the $\mathbb{Z}/p$-statement with a coherent statement for $\mathcal{O}$-modules. Let $p : X \to S$ be quasi-projective. A quasi-coherent sheaf on X is regarded as a sheaf of $\mathcal{O}_X$-modules on the ét-site. The functor Γ^n_{ext} exists for quasi-coherent modules and is compatible with affine computations:

$$\Gamma^n_{\mathrm{ext}}(\widetilde{M}) = \widetilde{\Gamma^n_A(M)}.$$

When p is flat, define $D^{b,\mathrm{Tor}_S \le 0}(X, \mathcal{O}_X)$ to consist of complexes which, as complexes of $p^{-1}\mathcal{O}_S$-modules, have Tor-dimension ≤ 0. The functor Γ^n_{ext} derives to

$$L\Gamma^n_{\mathrm{ext}} : D^{b,\mathrm{Tor}_S \le 0}(X, \mathcal{O}_X) \to D^{b,\mathrm{Tor}_S \le 0}(\mathrm{Sym}^n_S(X), \mathcal{O}).$$

For quasi-projective flat $u : X \to Y$ there is a coherent symmetric Künneth morphism

$$L\Gamma^n_{\mathrm{ext}} Ru_* K \longrightarrow R\,\mathrm{Sym}^n_S(u)_* L\Gamma^n_{\mathrm{ext}} K.$$

Proposition 5.5.34. Under these coherent hypotheses, the coherent symmetric Künneth morphism is an isomorphism.

Proof. The question is local on S, so assume S affine. Localizing on $\mathrm{Sym}^n_S(Y)$, one assumes Y affine. Resolving a quasi-coherent complex by a finite alternating Cech complex associated with an affine cover of X reduces by devissage to the case where $K = Rj_*K_1$ for an affine open immersion $j : X_1 \hookrightarrow X$. Transitivity reduces the proof to the affine case for X, Y, S. Then Ru_* and $L\Gamma^n$ are calculated directly on a bounded complex of flat quasi-coherent modules, and the assertion is the affine identity

$$\Gamma^n_{\mathrm{ext}} u_* \mathcal{F} \simeq \mathrm{Sym}^n_S(u)_* \Gamma^n_{\mathrm{ext}} \mathcal{F},$$

which both sides identify with $\widetilde{\Gamma^n_A(F)}$. □

In characteristic p, the Artin–Schreier sequence

$$0 \to \mathbb{Z}/p \to \mathcal{O} \xrightarrow{F-1} \mathcal{O} \to 0$$

relates the $\mathbb{Z}/p$-cohomology to coherent cohomology. The Frobenius $F : x \mapsto x^p$ acts p-linearly on $R\Gamma(X, \mathcal{O})$, on $R\Gamma(\mathrm{Sym}^n_k X, \mathcal{O})$, and therefore on $L\Gamma^n(R\Gamma(X, \mathcal{O}))$. The coherent Künneth isomorphism is compatible with F.

Lemma 5.5.36. Let k be a perfect field of characteristic $p > 0$, let $K \in C^{\ge 0}(\mathbb{Z}/p)$ be a finite-dimensional complex of $\mathbb{Z}/p$-vector spaces, and let $L \in C^{\ge 0}(k)$ be a finite-dimensional complex of k-vector spaces with a p-linear endomorphism F. Let $u : K \to L$ be a morphism such that $H^*(Fu) = H^*(u)$, and assume that

$$0 \to H^i(K) \to H^i(L) \xrightarrow{F-1} H^i(L) \to 0$$

is exact for all i. Then the sequences

$$0 \to H^i(\Gamma^n K) \to H^i(\Gamma^n L) \xrightarrow{F-1} H^i(\Gamma^n L) \to 0$$

are exact.

Since $Fu - u$ is null-homotopic, write $Fu - u = dH + Hd$. Perfection of k gives $FH_0 - H_0 = H$, so replacing u by the homotopic map $u - (dH_0 + H_0 d)$ lets us assume $Fu = u$. The functor sending a finite-dimensional k-space with p-linear F to its fixed part $V^S = \{v \mid Fv = v\}$ is exact; the hypothesis says $K \to L^S$ is a quasi-isomorphism. It remains only to note that fixed parts commute with Γ^n. This follows from the standard decomposition below.

Lemma 5.5.37. A finite-dimensional vector space V over a perfect field k, endowed with a p-linear endomorphism F, has a unique decomposition

$$V = V_0 \oplus V_n$$

with $V^S \otimes k \simeq V_0$, $F(V_n) \subset V_n$, and $F|_{V_n}$ nilpotent.

This completes the proof of the symmetric Künneth theorem.

6. The functor $f_!$

We now give a direct construction of the functor $f_!$ appearing in 5.1.11. This construction will then be used to define the trace morphism for flat quasi-finite morphisms of finite presentation.

6.1. A new construction of $f_!$

Let $f : X \to S$ be locally of finite type. A subset $P \subset X$ is said to be proper over S if it is the image of a closed subscheme of X proper over S.

Proposition 6.1.1. Let $f : X \to S$ be locally of finite type and quasi-separated.

(i) A subset of X proper over S is locally closed, and is closed if f is separated.

(ii) If f is separated, the union of two subsets proper over S is proper over S.

(iii) For every S-scheme U, let $\Phi_X(U)$ be the set of subsets of $U \times_S X$ proper over U. The functor Φ_X is a sheaf for the fpqc topology. If f is separated, Φ_X is a sheaf of filtered ordered sets under inclusion.

The first two assertions are immediate. The presheaf of all subsets of $U \times_S X$ is a sheaf for any topology whose coverings are surjective, and Φ_X is a subsheaf by the descent of properness and finite-type closed images.

If $\mathcal{F}$ is an ét-sheaf of pointed sets on X, the support of a section is the complement of the largest open set on which that section is equal to the distinguished section.

Definition 6.1.2. Let $f : X \to S$ be separated and locally of finite type, and let $\mathcal{F}$ be a pointed sheaf on X. Define $f_!\mathcal{F}$ to be the subsheaf of $f_*\mathcal{F}$ whose sections over an ét-scheme $U \to S$ are the sections of $\mathcal{F}$ on $U \times_S X$ with support proper over U.

Proposition 6.1.1 ensures that this is indeed a sheaf.

Lemma 6.1.3. If $f = gh$ is the composite of two separated morphisms locally of finite type, there is a unique isomorphism

$$f_! \simeq g_! h_!$$

compatible with the inclusions into f_* and $g_* h_*$.

When f is an open immersion, this functor is extension by zero. When f is proper, it is f_*. Hence this construction agrees with the earlier construction whenever both are defined.

Proposition 6.1.4. Let $f : X \to S$ be separated and locally of finite type.

(i) The functor $f_!$ is left exact and commutes with filtered inductive limits.

(ii) If f is quasi-finite, then $f_!$ is exact on abelian sheaves.

(iii) For every Cartesian square

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S, \end{tikzcd}$$

there is a base-change isomorphism $g^* f_! \simeq f'_! g'^*$, compatible with the usual base-change morphism $g^* f_* \to f'_* g'^*$.

Proof. For $U \in S_{\text{ét}}$,

$$f_!\mathcal{F}(U) = \varinjlim_{P \in \Phi(U)} \Gamma_P(\mathcal{F}),$$

where the limit is over the filtered system of supports proper over U. If U is quasi-compact and quasi-separated, the functor $\mathcal{F} \mapsto f_!\mathcal{F}(U)$ commutes with finite projective limits and filtered inductive limits, proving (i). Assertion (ii) follows from (iii) by testing exactness on geometric fibres. For (iii) the question is local on S. Write X as the filtered union of open subschemes of finite type over S, reduce to the finite-type case, and use Chow's lemma to find a projective surjection $p^0 : X^0 \to X$ with fp^0 compactifiable. With $X^1 = X^0 \times_X X^0$, the sequence

$$0 \to \mathcal{F} \to p^0_* p^{0*} \mathcal{F} \rightrightarrows p^1_* p^{1*} \mathcal{F}$$

is exact. Functoriality and left exactness reduce the assertion to sheaves of the form $p^0_* \mathcal{G}$, and then to the compactifiable case, already proved in 5.2.7. $\square$

Counterexample 6.1.6. In general $Rf_!$ is not the derived functor of $f_!$. If X is affine over an algebraically closed field k, then

$$f_!\mathcal{F} = \bigoplus_{x \in X(k)} H^0_x(\mathcal{F}),$$

so the derived functors of $f_!$ are

$$\bigoplus_{x \in X(k)} H^i_x(\mathcal{F}).$$

But if X is smooth, irreducible, and one-dimensional, then for ℓ prime to $\operatorname{char} k$

$$H^2_c(X, \mu_{\ell^n}) \simeq \mathbb{Z}/\ell^n\mathbb{Z}.$$

6.2. The trace morphism in relative dimension zero

The main result is the trace theorem 6.2.3. The technical preliminaries 6.2.1 and 6.2.2 allow the proof to reduce to finite morphisms; this reduction is also essentially visible from the local structure theorem for quasi-finite morphisms.

Let $f : X \to S$ be quasi-finite and separated. For $U \in S_{\text{ét}}$, define $\Psi_f(U)$ to be the category of decompositions of f^*U into open and closed pieces

$$f^*U = \coprod_{i \in I} V_i,$$

where I is a finite pointed set with base point 0, and V_i is finite over U for $i \neq 0$. A morphism from $(V_i)_{i \in I}$ to $(V'_j)_{j \in I'}$ is a map $\sigma : I' \to I$ of pointed sets such that

$$V_i = \bigcup_{j \in \sigma^{-1}(i)} V'_j.$$

We say that φ' refines φ if such a morphism exists.

For an abelian sheaf $\mathcal{F}$ on S, a decomposition $\varphi = (V_i)$ gives maps

$$\tau_\varphi : \mathcal{F}^{I^*}|_U \longrightarrow f_!f^*\mathcal{F}|_U, \qquad I^* = I \setminus \{0\},$$

by summing the adjunction maps along the finite components. If φ' refines φ, the transpose map ${}^t\sigma : \mathcal{F}^{I^*} \to \mathcal{F}^{I'^*}$ makes the corresponding triangle commute.

Proposition 6.2.2.

(i) The fibred category Ψ_f is locally filtered to the right over $S_{\text{ét}}$.

(ii) The maps τ_φ induce an isomorphism

$$\varinjlim_{\varphi \in \Psi_f(U)} \mathcal{F}^{I^*}|_U \xrightarrow{\sim} f_! f^* \mathcal{F}|_U.$$

If S is strictly local with closed point s, the scheme X admits a decomposition in $\Psi_f(S)$ with one component empty over s and all other components finite over S with one-point special fibre. This is final in $\Psi_f(S)$, and the assertion is immediate after taking the stalk at s. The general case follows by passage to strict localizations at geometric points.

Theorem 6.2.3. There is a unique rule which assigns to every separated, flat, quasi-finite morphism of finite presentation $f : X \to S$, and every abelian sheaf $\mathcal{F}$ on S, a trace morphism

$$\mathrm{Tr}_f : f_! f^* \mathcal{F} \longrightarrow \mathcal{F},$$

satisfying:

(Var 1) functoriality in $\mathcal{F}$;

(Var 2) compatibility with every base change;

(Var 3) compatibility with composition of such morphisms;

(Var 4) normalization: if f is finite locally free of constant rank n, the composite

$$\mathcal{F} \to f_* f^* \mathcal{F} = f_! f^* \mathcal{F} \xrightarrow{\mathrm{Tr}_f} \mathcal{F}$$

is multiplication by n.

The properties imply additivity on disjoint unions: if $X = X' \amalg X''$, then Tr_f is the sum of $\mathrm{Tr}_{f'}$ and $\mathrm{Tr}_{f''}$. For the construction, restrict in Ψ_f to decompositions for which each finite component is locally free of constant rank. This subcategory is cofinal. Giving a map $f_! f^* \mathcal{F} \to \mathcal{G}$ is the same as giving compatible maps $\mathcal{F}^{I^*} \to \mathcal{G}$ for all such decompositions. For the trace take the weighted sum

$$(s_i)_{i \in I^*} \longmapsto \sum_{i \in I^*} n_i s_i,$$

where n_i is the rank of $V_i \to U$. Compatibility under refinement is exactly the additivity of ranks. This gives existence. Uniqueness follows from the cofinality and the normalization condition.

There is also a local numerical description. If f is as above and $s \in S$, for a point x over s let $\mathcal{O}_s$ be the strict henselization at s, choose the corresponding local component $\mathcal{O}_x$, and let $n(x)$ be the rank over the generic point. The function $x \mapsto n(x)$ is a weighting of the fibres; the trace is the weighted sum.

The trace produces covariance of cohomology with proper supports for quasi-finite flat morphisms. If

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

with u separated, flat, quasi-finite, and of finite presentation, then for a sheaf $\mathcal{F}$ on Y

$$u_* : f_! u^* \mathcal{F} \simeq g_! u_! u^* \mathcal{F} \xrightarrow{g_!(\mathrm{Tr}_u)} g_! \mathcal{F}.$$

If g is compactifiable and K is a suitable torsion complex, this gives

$$u_* : Rf_! u^* K \longrightarrow Rg_! K,$$

and in the case $S = \operatorname{Spec} k$, k algebraically closed,

$$u_* : H^q_c(X, u^* K) \longrightarrow H^q_c(Y, K).$$

For an étale surjective map $u : X \to Y$, let X_n be the $(n+1)$-fold fibre product of X over Y. The simplicial augmentation $(X_n) \to Y$ gives, for a sheaf $\mathcal{F}$ on Y, augmented sheaves $\mathcal{F}_n = u_{n!} u_n^* \mathcal{F}$.

Proposition 6.2.9. If $u : X \to Y$ is étale, separated, of finite type, and surjective, then for every abelian sheaf $\mathcal{F}$ on Y:

(i) the sheaves $\mathcal{F}_n$ form a left resolution of $\mathcal{F}$;

(ii) the complex of nondegenerate chains of this simplicial sheaf is a left resolution of $\mathcal{F}$. If the geometric fibres of u have at most d points, this resolution has length at most d.

By base change it suffices to check the statement over a separably closed field, where it is the homology of a simplex.

For an open covering $(U_i)_{i \in I}$ of Y, ordered when finite, this construction recovers the alternating Cech resolution

$$(\mathcal{F}_n)_{\mathrm{ndeg}} = \bigoplus_{|P|=n+1} j_{P!} j_P^* \mathcal{F},$$

where $U_P = \cap_{i \in P} U_i$.

If $K \in D^-(Y, \mathcal{A})$, these resolutions give the extraordinary Cech resolution of K, and, in a triangle over S with g compactifiable, a spectral sequence

$$E_1^{-p,q} = R^q f_{p!}(u_p^* K) \Longrightarrow R^{-p+q} g_! K.$$

Equivalently,

$$E_2^{-p,q} = \check{H}^p(R^q f_{p!}(u_p^* K)) \Longrightarrow R^{-p+q} g_! K.$$

These are the localization spectral sequences "above."

Proposition 6.2.11. If f is étale, separated, and of finite type, then $f_!$ is left adjoint to f^*; the trace morphism is the adjunction counit.

For an object X of a site, the restriction functor to X has a left adjoint; the latter is generated by the presheaf

$$V \longmapsto \bigoplus_{\xi \in \mathrm{Hom}(V,X)} F(\xi).$$

In the étale case this left adjoint commutes with base change. Comparing it with the trace-defined $f_!$ on geometric fibres proves the assertion. This interpretation is specific to the étale site and should not be transplanted uncritically to the fpqf site.

6.3. Variant of the trace for continuous coefficients

This subsection proves assertions of Grothendieck concerning continuous coefficients and applies them to the construction of trace morphisms for suitable fppf sheaves.

Let A be a commutative ring. If L and M are finite projective A-modules, the map $u \mapsto \bigwedge^m u$ from $\mathrm{Hom}(L, M)$ to $\mathrm{Hom}(\bigwedge^m L, \bigwedge^m M)$ is polynomial of degree m. The associated construction is functorial and behaves well under base change. It is used to define maps on symmetric powers and norm-like maps attached to finite locally free morphisms.

For a finite locally free morphism $f : X \to T$ of rank n, there is a canonical section

$$t : T \longrightarrow \mathrm{Sym}^n_T(X)$$

which, on affine pieces, is defined by the characteristic polynomial or, equivalently, by the elementary symmetric functions of the finite locally free algebra. The construction is compatible with base change, with addition of finite locally free subschemes, and with composition.

Let G be a commutative group over S, represented in a setting in which torsors and symmetric powers behave well. If K is a G_X-torsor on X, one constructs a G-torsor K_n on $\mathrm{Sym}^n_T(X)$. It is obtained by descent from the external sum of the pullbacks of K to X^n, using the symmetric group action. The construction is compatible with base change, change of group, addition of torsors, addition of the integer n, and multiplication of the integer n. In formulas,

$$(g'_n)^* K_n \simeq (g'^* K)_n, \qquad r(K_n) \simeq r(K)_n,$$
$$(K' + K'')_n \simeq K'_n + K''_n,$$

and if $n = n_1 + n_2$, then for the canonical map

$$\alpha : \mathrm{Sym}^{n_1}_T(X) \times_T \mathrm{Sym}^{n_2}_T(X) \to \mathrm{Sym}^n_T(X)$$

one has
$$\alpha^* K_n \simeq \operatorname{pr}_1^* K_{n_1} + \operatorname{pr}_2^* K_{n_2}.$$
If $\alpha : \operatorname{Sym}^n_T(\operatorname{Sym}^m_T(X)) \to \operatorname{Sym}^{nm}_T(X)$ is the canonical multiplication map, then
$$\alpha^*(K_{nm}) \simeq (K_m)_n.$$

When $f : X \to T$ is finite locally free of rank n, the trace of a torsor is defined by
$$\operatorname{Tr}_f(K) = t^*(K_n),$$
where $t : T \to \operatorname{Sym}^n_T(X)$ is the canonical section. For a morphism locally free only locally on T, the same formula defines $\operatorname{Tr}_f(K)$ locally and the preceding compatibilities solve the descent problem. Thus Tr_f is compatible with base change, with change of group, with addition of torsors, and with composition:
$$\operatorname{Tr}_{fg}(K) \simeq \operatorname{Tr}_f(\operatorname{Tr}_g(K)).$$

For relative curves one obtains the useful divisor form. If $X \to S$ is a smooth quasi-projective curve, D_1, D_2 are relative divisors finite over S, and K is a G-torsor on X, then
$$\operatorname{Tr}_{D_1+D_2}(K) \simeq \operatorname{Tr}_{D_1}(K) + \operatorname{Tr}_{D_2}(K).$$

If
$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "p"'] & & Y \arrow[dl, "q"] \\ & S & \end{tikzcd}$$
with X, Y smooth quasi-projective curves over S and u quasi-finite, then for a divisor D on X finite over S
$$\operatorname{Tr}_{D/S}(u^* K) \simeq \operatorname{Tr}_{u_* D/S}(K),$$
and if u is finite, for a divisor E on Y finite over S,
$$\operatorname{Tr}_{E/S}(\operatorname{Tr}_{X/Y} K) \simeq \operatorname{Tr}_{u^* E/S}(K).$$

Proposition 6.3.28. Let A and B be commutative R-algebras and let $u : A \to B$ make B a finite locally free faithful A-module. Then the co-semi-simplicial complex of algebras
$$\left(TS^n_R(B^{\otimes_A (k+1)})\right)_{k \geq 0}$$
is, as a differential co-semi-simplicial $TS^n_R(A)$-module, a split resolution of $TS^n_R(A)$.

The associated complex depends only on B as an A-module with the map $A \to B$. Since $B = A \oplus L$ as an A-module after choosing a splitting locally, one writes an explicit homotopy operator.

Proposition 6.3.29. Let A be a ring, B^* a co-semi-simplicial A-algebra, and $A \to B^*$ the structural map. For extension of scalars to be faithful, respectively fully faithful, it is enough that

$$0 \to A \to B^0,$$

respectively

$$0 \to A \to B^0 \xrightarrow{j_0 - j_1} B^1,$$

be exact and remain exact after tensoring with any A-module.

From Propositions 6.3.28 and 6.3.29 one obtains that the required descent condition holds for every affine group G. More generally it should extend to groups represented by algebraic spaces; in particular one expects the flat algebraic-space case to satisfy the stronger condition.

Proposition 6.3.31. Let

$$0 \to G \xrightarrow{r} G' \xrightarrow{u} H \to 0$$

be an exact sequence over S. Suppose that G' and H, hence G, satisfy the basic representability condition, that G' satisfies the descent condition for the construction of K_n, and that G satisfies the first descent condition. Then G satisfies the second descent condition.

The proof is local for the étale topology on the source. It is enough to treat a G-torsor K for which the induced G'-torsor is trivial. Such a torsor is equivalently a trivial G'-torsor K' together with an isomorphism $H \simeq u(K')$. On $Z = \mathrm{Sym}^n_T(X)$, the induced isomorphism $H_Z \simeq u(K'_n)$ provides precisely the descent datum required for the construction.

7. Appendix

The appendix uses the notation and results of Exposé VI_B. Let $\mathrm{Sch}_!$ be the category whose objects are quasi-compact quasi-separated schemes and whose morphisms are separated morphisms of finite type. Let $\mathcal{E}$ be the bifibred category over $\mathrm{Sch}_!$, with fibres opposite to the corresponding ét-topoi. Fix an object R of $\mathrm{Sch}_!$, and let $\mathcal{E}_R$ be the induced bifibred category over $\mathrm{Sch}_{!R}$. Let $\mathcal{A}$ be a ring object of the topos of sections $\Gamma(\mathcal{E}_R^\circ)$, satisfying:

(i) $\mathcal{A}$ is a cocartesian section over $\mathrm{Sch}_{!R}^\circ$;

(ii) for every $X \in \mathrm{Sch}_{!R}$, $\mathcal{A}_X$ is a torsion sheaf of rings.

The goal is to define, for every morphism $f : X \to Y$ in $\mathrm{Sch}_{!R}$, a functor

$$Rf_! : D(X, \mathcal{A}_X) \longrightarrow D(Y, \mathcal{A}_Y),$$

which coincides with the previously defined functor when f is compactifiable and which makes the categories $D(X, \mathcal{A}_X)$ into a cofibre category over $\mathrm{Sch}_{!R}$. In modern terminology, Nagata compactification makes every morphism of $\mathrm{Sch}_!$ compactifiable; the appendix shows how to avoid using that theorem by relying instead on Chow's lemma.

7.0. Preliminaries

Lemma 7.0.3. Let

$$\begin{tikzcd} X' \arrow[r, "j'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "j"'] & Y \end{tikzcd}$$

be a commutative diagram of schemes, where f, f' are proper and j, j' are immersions. For every complex $K^\bullet \in D^+_{\mathrm{tors}}(X')$, the canonical morphism

$$j_! R^+ f'_* K^\bullet \longrightarrow R^+ f_* j_! K^\bullet$$

is an isomorphism.

Proof. By the lemma on way-out functors, it is enough to treat torsion sheaves in fixed cohomological degree:

$$j_! R^i f'_* F \longrightarrow R^i f_* j_! F.$$

Taking the closed images of j and j', reduce to the case where the immersions are dense open immersions. Properness then makes the square Cartesian. The proper base-change theorem, applied at every geometric point of Y, gives the result. □

Construction 7.0.4. Let D be a category and let X, Y be D-objects of $\mathrm{Sch}_{!R}$ such that, for every arrow $\alpha \to \beta$ of D, the transition maps $X_\beta \to X_\alpha$ and $Y_\beta \to Y_\alpha$ are proper. Let $j : X \to Y$ be a morphism with each $j_\alpha : X_\alpha \to Y_\alpha$ an immersion. The isomorphism of Lemma 7.0.3 for $i = 0$, together with its compatibilities, defines an exact functor

$$j_! : \mathrm{Mod}(\Gamma(\bar{X}), \mathcal{A}) \longrightarrow \mathrm{Mod}(\Gamma(\bar{Y}), \mathcal{A}),$$

and hence a triangulated functor on derived categories. For every object α of D, the square comparing this $j_!$ with the usual extension by zero $j_{\alpha!}$ on the fibre at α is commutative.

Construction 7.0.5. Given a square

$$\begin{tikzcd} X' \arrow[r, "i'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "i"'] & Y \end{tikzcd}$$

with i, i' immersions, f proper, and f' proper and surjective, one obtains a square of simplicial objects

$$\begin{tikzcd}
{[X'|Y']} \arrow[r, "j"] \arrow[d] & {[X|Y]} \arrow[d] \\
C^{\Delta}_{Y'} \arrow[r, "i"'] & C^{\Delta}_{Y},
\end{tikzcd}$$

where every j_n is an immersion. There is a canonical isomorphism

$$i_! \simeq R^+\bar{u}_{f*}\, j_!\, L^+\bar{u}^*_{f'}.$$

After replacing by closed images, one may suppose the immersions are dense opens. The square is then Cartesian and f is surjective. The base-change relation gives $L^+\bar{u}^*_f i_! \simeq j_! L^+\bar{u}^*_{f'}$; cohomological descent for u_f gives the displayed isomorphism.

Lemma 7.0.6 (Chow-type lemma). Let S be quasi-compact and quasi-separated and let $f : X \to S$ be separated of finite type. Then there exists a commutative diagram

$$\begin{tikzcd}
X' \arrow[rr, "p"] \arrow[dr, "f'"'] & & X \arrow[dl, "f"] \\
& S &
\end{tikzcd}$$

with p proper and surjective, and with f' quasi-projective. If f is already proper, p may be chosen projective; if f is an immersion one may impose the evident compatibility with the immersion. The construction is stable under the refinements required in the simplicial arguments below.

7.1. Definition of $R^+f_!$

Definition 7.1.1. A pseudo-compactification of a morphism $f : X \to Y$ in $\mathrm{Sch}_{!R}$ is a triple $(p, j, \bar{f})$ consisting of a proper surjection $p : X' \to X$, an immersion $j : X' \to \bar{X}$, and a proper morphism $\bar{f} : \bar{X} \to Y$, such that $fp = \bar{f}j$ and $\bar{f}j$ is the compactified form of fp.

Chow's lemma provides pseudo-compactifications. For such a triple, set

$$R^+f_!K = R^+\bar{f}_*\, j_!\, L^+\bar{u}^*_p K,$$

where $\bar{u}_p$ denotes the simplicial augmentation attached to the proper hypercover generated by p. The descent formalism makes this independent of refinements in the sense explained below.

Construction 7.1.3. If $(p, j, \bar{f})$ is a pseudo-compactification of $f : X \to Y$, the above formula defines a functor

$$R^+f_! : D^+(X, \mathcal{A}_X) \longrightarrow D^+(Y, \mathcal{A}_Y).$$

For a second pseudo-compactification dominating the first, there is a canonical comparison isomorphism between the two functors.

Construction 7.1.4. For two pseudo-compactifications of the same morphism f, choose a common refinement obtained by fibre product and a further Chow modification. The comparison isomorphisms through the common refinement are independent of all choices. Hence $R^+f_!$ is intrinsically defined on D^+.

The independence is proved by checking the cocycle condition for a triple of pseudo-compactifications. Lemma 7.0.3 handles extension by zero across immersions, and cohomological descent handles the proper surjections. The key technical lemmas are the commutativity of the comparison maps associated with two successive refinements and their compatibility with the descent augmentations.

Proposition 7.1.6. If f is compactifiable in the ordinary sense, the functor $R^+f_!$ just constructed is canonically isomorphic to the functor $R^+f_!$ of Section 5.

Proposition 7.1.7. For a composite $X \xrightarrow{u} Y \xrightarrow{v} Z$ in $\mathrm{Sch}_{!R}$, there is a canonical transitivity isomorphism

$$R^+(vu)_! \simeq R^+v_!R^+u_!.$$

It is compatible with the corresponding transitivity for a triple of composable morphisms.

Definition 7.1.8. The functor $R^+f_!$ is called the direct image with proper supports on bounded-below complexes.

Construction 7.1.9. Given a Cartesian square in $\mathrm{Sch}_{!R}$

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"'] & S, \end{tikzcd}$$

there is a base-change morphism

$$g^*R^+f_!K \longrightarrow R^+f'_!g'^*K.$$

It is defined by choosing compatible pseudo-compactifications of f and f', applying the base-change morphisms already available in the compactifiable case, and checking independence of choices by the comparison theorem for pseudo-compactifications.

7.2. Some properties of $R^+f_!$

Proposition 7.2.1. If the relative dimension of f is at most d, then $R^+f_!$ has cohomological dimension at most $2d$ on torsion sheaves. Equivalently, $R^if_!F = 0$ for $i > 2d$.

This is reduced by pseudo-compactification and cohomological descent to the compactifiable case, where it is Corollary 5.2.8.1.

Proposition 7.2.2. For morphisms in $\mathrm{Sch}_{!R}$, the functor $R^+f_!$ sends distinguished triangles to distinguished triangles and is compatible with the transitivity isomorphisms. Its formation is local for the ét-topology on the target.

Remark 7.2.3. The construction is intentionally arranged so that all properties are inherited from the compactifiable case after replacing a morphism by a pseudo-compactification. The price paid is the use of cohomological descent in verifying independence of choices.

Proposition 7.2.4. For every Cartesian square as in 7.1.9, the base-change morphism

$$g^*R^+f_!K \longrightarrow R^+f'_!g'^*K$$

is an isomorphism.

To prove this, choose pseudo-compactifications compatible with the square. The construction gives a morphism between the two cohomological descent spectral sequences

$$g^*R^q(f \circ \pi_1^p)_!(\pi_1^{p*}K) \Rightarrow g^*R^{p+q}f_!K$$

and

$$R^q(f' \circ \pi_1'^p)_!(\pi_1'^{p*}g'^*K) \Rightarrow R^{p+q}f'_!g'^*K.$$

The E_1-terms are isomorphisms by the compactifiable base-change theorem, hence so is the abutment.

7.3. Definition of $Rf_!$

We now extend $R^+f_!$ to a reasonable functor

$$Rf_! : D(X, \mathcal{A}_X) \longrightarrow D(Y, \mathcal{A}_Y)$$

for every morphism f of $\mathrm{Sch}_{!R}$.

Choose a pseudo-compactification (p_1, j_1, f_1) of f. For an $\mathcal{A}_X$-module F, take the canonical flasque resolution of

$$j_{1!}L\bar{u}_{p_1}^*F,$$

apply the direct image by f_1, and take the associated simple complex. One obtains a functorial complex

$$0 \to \mathcal{X}^0(F) \xrightarrow{S^1(F)} \mathcal{X}^1(F) \to \cdots \to \mathcal{X}^n(F) \xrightarrow{S^n(F)} \mathcal{X}^{n+1}(F) \to \cdots$$

which represents $R^+f_!(F)$ in $D^+(Y, \mathcal{A}_Y)$. If the relative dimension of f is at most d, then the cohomology of this complex vanishes in degrees $> 2d$; hence it is canonically represented by the bounded complex

$$0 \to \mathcal{X}^0(F) \to \mathcal{X}^1(F) \to \cdots \to \mathcal{X}^{2d-1}(F) \to \operatorname{Ker} S^{2d}(F) \to 0.$$

Lemma 7.3.2. The functors $\mathcal{X}^0, \mathcal{X}^1, \ldots, \mathcal{X}^{2d-1}$, and $\operatorname{Ker} S^{2d}$, are exact in F.

The exactness of the $\mathcal{X}^n$ follows from exactness of the canonical flasque resolution and from the fact that direct image preserves exact sequences of flasque sheaves. For $\operatorname{Ker} S^{2d}$, apply the long exact homology sequence to the truncated tails

$$\sigma_{\geq 2d+1}\mathcal{X}^\bullet(F).$$

Since the cohomology above degree $2d$ vanishes, the kernel term is exact.

Lemma 7.3.3. Let A and B be abelian categories, and let $J^\bullet$ be a bounded complex of exact functors $A \to B$. The construction which sends a complex $K^\bullet \in C(A)$ to the simple complex associated with the double complex $J^\bullet(K^\bullet)$ defines a triangulated functor $K(A) \to K(B)$ and preserves quasi-isomorphisms.

This follows by writing the construction as

$$C(A) \to C^b(C(B)) \to C(B),$$

where the last arrow is the simple-complex functor, and then applying the standard acyclicity criterion for bounded double complexes with exact rows.

Exercise 7.3.4. For every $K^\bullet \in D^+(X, \mathcal{A}_X)$, the canonical map

$$\tau_{\leq 2d}\mathcal{X}^\bullet(K^\bullet) \to \mathcal{X}^\bullet(K^\bullet)$$

induces a quasi-isomorphism on the associated simple complexes. One observes first that

$$\varinjlim_{k \geq 2d} (\tau_{\leq k}\mathcal{X}^\bullet(K^\bullet))_s \to (\mathcal{X}^\bullet(K^\bullet))_s$$

is an isomorphism, and then uses the fact that cohomology sheaves commute with filtered inductive limits.

Define

$$R^{(1)}f_! : D(X, \mathcal{A}_X) \to D(Y, \mathcal{A}_Y)$$

by the bounded exact-functor complex displayed above. By the preceding exercise, it may be computed directly from the unbounded flasque construction, and it is independent of the pseudo-compactification chosen. The verification amounts to repeating Constructions 7.1.3 and 7.1.4, and Proposition 7.1.6, without the superscript +. The original redaction notes that the details are straightforward but lengthy.

Definition 7.3.6. The functor

$$Rf_! : D(X, \mathcal{A}_X) \to D(Y, \mathcal{A}_Y)$$

constructed above is again called the direct-image functor with proper supports.

Final compatibility 7.3.7. The functor $Rf_!$ commutes with base change, and the tensor formula 5.2.9 and the finite Tor-dimension theorem 5.2.10 remain valid for every morphism in $\mathrm{Sch}_{!R}$. These properties justify the preceding construction and put it on the same formal footing as the compactifiable construction of Section 5.

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SGA 4, Exposé XVIII – beginning

The Global Duality Theorem

Opening through the integration formalism for torsors on curves.

Exposé XVIII. Introduction

This exposé is devoted to Poincare duality. The construction follows Verdier's model for separated locally compact topological spaces. One first proves abstractly that the functor $Rf_!$ of Exposé XVII admits a right adjoint, denoted $Rf^!$, and establishes a number of formal properties of this adjoint. Only after this is done does one compute it more explicitly for smooth morphisms.

The possibility of carrying out the construction rests on four properties of $Rf_!$. For the existence of $Rf^!$ and for its formal behavior one uses: first, that $Rf_!$ commutes with base change; second, that the functors $R^if_!$ commute with filtered inductive limits and vanish for i large; third, that $Rf_!$ can be computed at the level of complexes, that is, from an actual functor between categories of complexes. For the calculation of the adjoint when f is smooth, one uses the computation of $Rf_!((\mathbb{Z}/n\mathbb{Z})_U)$ for sufficiently small U.

Section 3.1, independent of Sections 1 and 2, constructs $Rf^!$. The construction uses special features of the situation, not merely the formal properties just listed. The original redaction explicitly remarks that this makes the section unpleasant and that not every detail of the mechanism was transparent to the redactor. Section 2 treats the calculation of $Rf_!((\mathbb{Z}/n\mathbb{Z})_U)$; it is formal from Section 1, where the cohomology of curves is studied. Section 1 proves more than will later be needed, because it also treats continuous coefficients. It is strongly influenced by Serre.

For later use, the essential points of Section 1 are the trace morphism for curves, treated in 1.1 and generalized in 2.9, and the effacement theorem 1.6.9. A reader willing to accept a transcendental argument may read 1.1 and then go directly to Sections 2 and 3. The main ingredients in 1.6.9 are Poincare duality for smooth curves over an algebraically closed field and the acyclicity lemma used in the proof of smooth base change. The result is generalized in 2.14. Section 3.2 gathers the consequences.

Verdier's method permits the definition of $Rf^!$ for compactifiable morphisms. The price paid for this generality is that the compatibility assertions to be checked are no longer the familiar ones. Some of them, even apparently trivial statements such as 3.2.3, have rather intricate proofs. The appendix to Exposé XVII also allows one to extend the definition to separated morphisms of finite type with coherent target, that is, quasi-compact and quasi-separated target.

1. Cohomology of curves

1.1. The trace morphism

Let $n \geq 1$ be an integer and let X be a scheme on which n is invertible. The sheaf μ_n of n-th roots of unity is a locally free module of rank one over the constant ring sheaf $\mathbb{Z}/n\mathbb{Z}$. For $i \in \mathbb{Z}$ one sets

$$\mathbb{Z}/n\mathbb{Z}(i) = \mu_n^{\otimes i}. \tag{1.1.1.1}$$

If $n = dn'$, exponentiation by d identifies

$$\mu_n \otimes_{\mathbb{Z}/n\mathbb{Z}} \mathbb{Z}/n'\mathbb{Z} \simeq \mu_{n'},$$

and hence gives canonical isomorphisms

$$\mathbb{Z}/n\mathbb{Z}(i) \otimes_{\mathbb{Z}/n\mathbb{Z}} \mathbb{Z}/n'\mathbb{Z} \simeq \mathbb{Z}/n'\mathbb{Z}(i). \tag{1.1.1.2}$$

If F is an abelian sheaf killed by n, put

$$F(i) = F \otimes_{\mathbb{Z}/n\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}(i). \tag{1.1.1.3}$$

This object is independent of the chosen annihilator. For a torsion sheaf prime to the residual characteristics of X, define

$$F(i) = \varinjlim_{m} {}_mF(i), \tag{1.1.1.4}$$

where m runs multiplicatively through integers invertible on X. The functors $F \mapsto F(i)$, the Tate twists, are exact. They commute with inverse image; for quasi-compact and quasi-separated morphisms they also commute with direct image. If $\mathcal{A}$ is a sheaf of rings killed by an integer invertible on X, they extend to exact functors on $D(X, \mathcal{A})$, commuting with the derived inverse and direct image functors.

Definition 1.1.2. A flat curve over a scheme S is a morphism $f : X \to S$ which is flat, separated, of finite presentation, and whose fibres are purely of dimension one.

The phrase "purely of dimension one" does not exclude the empty scheme. In contrast with the convention of EGA, a curve over a field is allowed here to be empty, but it is required to be separated. When S is the spectrum of a field we simply say "curve over S"; such a curve is quasi-projective.

Let now k be an algebraically closed field, let X be a smooth connected curve over k, and let F be a finite abelian group of order prime to $\operatorname{char} k$. The trace map is the canonical isomorphism

$$\operatorname{Tr}_X : H^2_c(X, F(1)) \longrightarrow F.$$

It is defined first for $X = \mathbb{A}^1_k$, using the standard computation of the compactly supported cohomology of the affine line, and then transported to arbitrary smooth curves by excision, normalization, and the compatibility of proper supports with finite maps. For a disconnected curve, the trace is the sum of the traces over the irreducible components.

Lemma 1.1.4. Let X be a smooth curve over an algebraically closed field and let $(U_i)_{i\in c}$ be non-empty opens, one in each irreducible component of X. Then the diagram

$$\begin{tikzcd} \bigoplus_{i\in c} H^2_c(U_i, F(1)) \arrow[rr, "\sim"] \arrow[dr, "\oplus \operatorname{Tr}_{U_i}"'] & & H^2_c(X, F(1)) \arrow[dl, "\operatorname{Tr}_X"] \\ & F & \end{tikzcd}$$

is commutative.

Lemma 1.1.5. Let $u : X' \to X$ be a finite flat morphism of smooth curves over an algebraically closed field. The trace maps are compatible with the cohomological trace attached to u:

$$\operatorname{Tr}_{X'} = \operatorname{Tr}_X \circ u_*.$$

The proof reduces to the case where u is finite etale over an open of X. On that open the assertion is a fibrewise summation. The complement is handled by the localization exact sequence for cohomology with proper supports.

Proposition 1.1.6. Let X be a smooth curve over an algebraically closed field and let F be a torsion group prime to the characteristic. The trace map identifies $H^2_c(X, F(1))$ with the group of locally constant functions on the set of connected components of X, with values in F. In particular, for connected X, it is an isomorphism onto F.

This is obtained by applying Lemma 1.1.4 to a dense affine open of each component and using the known calculation for the affine line, then passing to the limit over the prime-to-characteristic torsion.

Lemma 1.1.7. Let $j : U \hookrightarrow X$ be a dense open immersion of smooth curves over an algebraically closed field, with complement Z. The boundary homomorphism in the localization sequence

$$H^1(Z, F) \to H^2_c(U, F(1)) \to H^2_c(X, F(1))$$

is compatible with the local residue maps at the points of Z.

Lemma 1.1.8. Let X be the spectrum of a henselian discrete valuation ring, with generic point η and closed point s, and let F be prime-to-the-residue-characteristic torsion. The boundary map

$$H^1(\eta, F) \to H^0(s, F(-1))$$

is the usual tame residue map.

Lemma 1.1.9. The trace morphism for curves is uniquely characterized by the preceding compatibility with finite flat maps and with the residue maps for henselian traits.

1.2. The one-acyclicity of the affine line

The next goal is to prove an acyclicity assertion for the affine line which will later be used to efface cohomology classes. Let S be a scheme on which n is invertible. Write $p : \mathbb{A}^1_S \to S$ for the projection.

Theorem 1.2.2. For every sheaf F of $\mathbb{Z}/n\mathbb{Z}$-modules on S, the canonical map

$$F \longrightarrow Rp_*p^*F$$

is an isomorphism in degrees 0 and 1, and $R^1p_*p^*F = 0$. Equivalently, $\mathbb{A}^1$ is one-acyclic for prime-to-characteristic torsion coefficients.

The theorem is local on S, and it suffices to prove it after strict henselization at a geometric point. Thus one reduces to the case where S is the spectrum of a separably closed field. For a constant finite sheaf F, the assertion becomes the fact that every finite etale cover of $\mathbb{A}^1$ which is trivial over the geometric generic point in the relevant prime-to-characteristic range is already trivial. The proof is given by a standard compactification argument on $\mathbb{P}^1$ and by the description of tame ramification at infinity.

Lemma 1.2.3. Let S be strictly local and let F be a finite locally constant sheaf on $\mathbb{A}^1_S$ of order invertible on S. If F is trivial on the closed fibre, then it is trivial after a finite etale base change of S; for strictly local S, it is already trivial.

Lemma 1.2.4. Let X be a normal compactification of $\mathbb{A}^1_k$ over an algebraically closed field. A finite etale cover of $\mathbb{A}^1_k$ of prime-to-characteristic degree which is tamely ramified at infinity is controlled by the tame inertia group at infinity. In particular, the prime-to-characteristic part relevant to $R^1p_*p^*F$ vanishes for the affine line.

Combining the two lemmas gives Theorem 1.2.2 for finite locally constant coefficients. General constructible torsion sheaves are then handled by devissage, and arbitrary torsion sheaves by filtered colimits.

The original proof proceeds through a sequence of explicit reductions. First one reduces to affine S, then to S noetherian, and finally to strictly henselian traits and geometric points. The comparison morphism

$$F \longrightarrow p_*p^*F$$

is immediately an isomorphism on stalks. The only point is the vanishing of $R^1p_*p^*F$, which can be tested by finite etale torsors under finite groups. Such torsors extend, after normalization, to covers of $\mathbb{P}^1$ tamely ramified only at infinity, and the tame fundamental group calculation gives the vanishing.

Lemma 1.2.6. Let G be a finite group of order invertible on the base. Every G-torsor on $\mathbb{A}^1_S$ which is trivial on a geometric fibre is locally trivial on S, and over a strictly local base is trivial.

Lemma 1.2.8. The formation of the above vanishing is compatible with passage from a scheme to a limit of affine schemes. Thus the noetherian case implies the general coherent case.

Lemma 1.2.9. The morphism which represents the obstruction to algebraicity of the cohomology class is itself algebraic. Consequently the class that vanishes after every geometric specialization vanishes after an etale localization of the base.

In the language of the original redaction, the last steps verify the algebraicity of a morphism $\widehat{\varphi}$. One first checks the assertion on completed local rings, then descends from formal neighborhoods by finite presentation. This supplies the algebraic torsor whose existence was predicted by the formal calculation.

Variants 1.2.15. The preceding acyclicity remains valid after replacing constant coefficient groups by finite locally constant coefficient groups of order invertible on the base. It is also stable under finite products with a fixed base scheme and under the evident base-change operations.

1.3. Integration of torsors

Let $f : X \to S$ be a smooth projective curve. The integration formalism associates to a torsor on X and to a line bundle, or more generally to a divisor, a torsor on S. This is the relative form of the Weil reciprocity and norm formalism. It will be used to construct pairings and trace maps appearing in duality.

If D is an effective relative divisor finite over S, and if K is a torsor under a commutative group G on X, the trace along D gives a G-torsor

$$\mathrm{Tr}_{D/S}(K)$$

on S. If $\mathcal{L}$ is an invertible sheaf sufficiently ample so that it is represented by such divisors locally on S, one defines a torsor

$$\langle \mathcal{L}, K]$$

by choosing a divisor representing $\mathcal{L}$ and applying the trace. The independence of the divisor follows from the compatibility of trace with rational equivalence.

The construction is additive in the line bundle:

$$\langle \mathcal{L}_1 \otimes \mathcal{L}_2, K] \simeq \langle \mathcal{L}_1, K] + \langle \mathcal{L}_2, K],$$

and additive in the torsor:

$$\langle \mathcal{L}, K_1 + K_2] \simeq \langle \mathcal{L}, K_1] + \langle \mathcal{L}, K_2].$$

It is compatible with every base change $S' \to S$. If $u : S \to X$ is a section and D_u its image, then

$$u^*K = \mathrm{Tr}_{D_u/S}(K) \simeq \langle \mathcal{O}(D_u), K] - \deg(K)\,\mathcal{U},$$

where $\mathcal{U}$ denotes the neutral torsor.

Proposition 1.3.4. The two additivity isomorphisms above are compatible. For line bundles $\mathcal{L}_1, \mathcal{L}_2$ and torsors K_1, K_2, the two ways of identifying

$$\langle \mathcal{L}_1 \otimes \mathcal{L}_2, K_1 + K_2]$$

with the sum of the four torsors

$$\langle \mathcal{L}_i, K_j], \qquad i, j = 1, 2,$$

are equal.

This compatibility, together with the associativity and commutativity constraints, permits one to extend the definition by linearity from sufficiently ample line bundles to arbitrary line bundles.

Assume from now on that the relative curve is smooth and projective and that the group sheaf G satisfies the usual hypotheses guaranteeing effective descent for torsors and compatibility of trace with base change. For a G-torsor K on X, the rule

$$\mathcal{L} \longmapsto \langle \mathcal{L}, K]$$

defines a morphism from the Picard stack of X/S to the stack of G-torsors on S. It is additive, functorial, and compatible with base change.

Formulary 1.3.8. The functor $\langle -, K]$ sends tensor product of line bundles to sum of torsors, sends the trivial line bundle to the neutral torsor, and is compatible with pullback along $S' \to S$. For fixed $\mathcal{L}$, the functor $K \mapsto \langle \mathcal{L}, K]$ is additive in K.

Conversely, given such a morphism of stacks, one recovers a torsor on X by evaluating at the universal point. More precisely, for a section $t : T \to X_T$ over an S-scheme T, the line bundle $\mathcal{O}(t)$ gives a torsor $F(\mathcal{O}(t))$ on T. Taking $T = X$ and t the identity section yields a torsor $\Psi(F)$ on X.

Theorem 1.3.10. Under the hypotheses just stated, the functor

$$\Phi : K \longmapsto \langle -, K]$$

is an equivalence from the category of G-torsors on X to the category of additive, base-change-compatible morphisms from the Picard stack of X/S to the stack of G-torsors on S.

Proof. One reduces to the case where the fibres of X/S are connected and non-empty. Starting from a morphism F, define $\Psi(F)$ by evaluating F on $\mathcal{O}(t)$ for the universal section t. The additivity of F gives canonical identifications showing that $\Phi(\Psi(F)) \simeq F$. Conversely, for a torsor K, the torsor obtained from $\Phi(K)$ by evaluation at the universal point is canonically K. The two identifications are inverse by the compatibility in Proposition 1.3.4. $\square$

Example 1.3.11. For $G = \mathbb{G}_m$, one writes $\langle \mathcal{L}, \mathcal{M} \rangle$ instead of $\langle \mathcal{L}, \mathcal{M}]$. If $\mathcal{L}$ and $\mathcal{M}$ are invertible sheaves on X, then $\langle \mathcal{L}, \mathcal{M} \rangle$ is an invertible sheaf on S. If D is an effective relative divisor, then

$$\langle \mathcal{O}(D), \mathcal{M} \rangle \simeq N_{D/S}(\mathcal{M}).$$

If D and E are disjoint effective relative divisors, this gives a canonical trivialization

$$\sigma_{D,E} : \langle \mathcal{O}(D), \mathcal{O}(E) \rangle \simeq \mathcal{O}.$$

Proposition 1.3.12. Let D_1, D_2, E be effective relative divisors, with D_i disjoint from E. Let f be a rational function with $\operatorname{div}(f) = D_1 - D_2$, equivalently an isomorphism $\mathcal{O}(D_1) \simeq \mathcal{O}(D_2)$. Then the square comparing the trivializations

$$\langle \mathcal{O}(D_i), \mathcal{O}(E) \rangle \simeq \mathcal{O}$$

with multiplication by $N_{E/S}(f)$ is commutative. The analogous square with the two variables interchanged is also commutative.

Corollary 1.3.13 (Weil reciprocity)**.** Let D_1, D_2, E_1, E_2 be effective relative divisors, all intersections required below being empty. If $\operatorname{div}(f) = D_1 - D_2$ and $\operatorname{div}(g) = E_1 - E_2$, then

$$N_{E_1/S}(f) N_{E_2/S}(f)^{-1} = N_{D_1/S}(g) N_{D_2/S}(g)^{-1}.$$

Equivalently,

$$f(\operatorname{div} g) = g(\operatorname{div} f).$$

Corollary 1.3.14. For $G = \mathbb{G}_m$, the degree appearing in the torsor integration formalism is the ordinary degree of an invertible sheaf on the fibres.

The biadditive pairing extends from sufficiently ample line bundles to all invertible sheaves. One obtains the following form.

Formulary 1.3.16. For every smooth projective curve $f : X \to S$, there is a functor

$$(\mathcal{L}, \mathcal{M}) \longmapsto \langle \mathcal{L}, \mathcal{M} \rangle$$

from pairs of invertible sheaves on X to invertible sheaves on S, compatible with arbitrary base change. It is biadditive, symmetric, normalized by $\langle \mathcal{O}, \mathcal{M} \rangle \simeq \mathcal{O}$, and for an effective divisor D satisfies

$$\langle \mathcal{O}(D), \mathcal{M} \rangle \simeq N_{D/S}(\mathcal{M}).$$

It is compatible with rational equivalence of divisors and with the commutativity constraint induced by Weil reciprocity.

For a field k, this pairing admits an explicit expression. Let X be a smooth projective connected curve over k, and let f, g be non-zero rational functions whose divisors have disjoint supports after the usual separation of positive and negative parts. The commutator attached to the pairing is the tame-symbol product

$$\langle f, g \rangle = \prod_x (-1)^{v_x(f)v_x(g)} \left(\frac{f^{v_x(g)}}{g^{v_x(f)}} \right)(x),$$

the product being over the finite set of points where either divisor is supported.

Lemma 1.3.18. With the notation just introduced,

$$\langle f, g \rangle = \prod_x (-1)^{v_x(f)v_x(g)} \left(f^{v_x(g)} / g^{v_x(f)} \right)(x).$$

This is the usual tame-symbol formula. It implies the compatibility assertions of the formulary, in particular the reciprocity relation needed for the symmetry of the pairing.

1.4. Strictly commutative Picard stacks

Some elementary results on Picard stacks are needed in order to express the integration theorem of Section 1.3 as a universal-coefficient formula. The original notes point out that these results are suggested by a 1966 letter of Grothendieck to Verdier, and that the relevant associativity and commutativity conventions for bifunctors are those introduced by Mac Lane.

Let C be a category and let $F : C \times C \to C$ be a functor. An associativity datum on F is a functorial isomorphism

$$\sigma : F(F(X, Y), Z) \xrightarrow{\sim} F(X, F(Y, Z)).$$

It is required to satisfy the usual coherence condition: for every finite family $(X_i)_{i \in I}$, all parenthesized products formed from the X_i by means of F are canonically identified, and the identifications are compatible with further parenthesizing. Equivalently, this is Mac Lane's pentagon condition, although the source formulation uses the free semigroup generated by I.

Suppose in addition that one has functorial isomorphisms

$$\tau_{X,Y} : F(X, Y) \xrightarrow{\sim} F(Y, X).$$

The pair (σ, τ) makes F associative and strictly commutative when the preceding coherence is compatible with passage from the free semigroup to the free abelian semigroup on I. In concrete terms this is the conjunction of the pentagon, the two involutivity conditions for τ, the hexagon, and the extra strict condition that $\tau_{X,X}$ be the identity on $X + X$. The last condition is the point of the word "strictly"; it is not automatic in ordinary symmetric monoidal categories.

Definition 1.4.2. A strictly commutative Picard category $\mathcal{P}$ is a non-empty groupoid, endowed with a functor

$$+ : \mathcal{P} \times \mathcal{P} \to \mathcal{P},$$

associativity and commutativity isomorphisms

$$(X + Y) + Z \simeq X + (Y + Z), \qquad X + Y \simeq Y + X,$$

which make $+$ associative and strictly commutative, and such that for every object X the functor $Y \mapsto X + Y$ is an equivalence of categories.

If $(X_i)_{i \in I}$ is a family of objects in such a category, there is, unique up to unique isomorphism, a functorial way of assigning to every element $\underline{n} \in \mathbb{Z}^{(I)}$ an object $\Sigma(\underline{n})$, with isomorphisms

$$\Sigma(e_i) \simeq X_i, \qquad \Sigma(\underline{n} + \underline{m}) \simeq \Sigma(\underline{n}) + \Sigma(\underline{m}),$$

compatible with all associativity and commutativity constraints. This is the extension from finite sums to formal integer linear combinations.

A strictly commutative Picard category has a unique neutral object up to unique isomorphism. One may define it as an object e together with an isomorphism $e + e \simeq e$. There are then unique unit constraints

$$e + X \simeq X, \qquad X + e \simeq X,$$

compatible with associativity, and the symmetry exchanges them. The group $\mathrm{Aut}(e)$ is abelian, and translation by any object identifies $\mathrm{Aut}(e)$ with the automorphism group of that object.

Definition 1.4.5. A strictly commutative Picard stack $\mathcal{P}$ on a site $\mathcal{S}$ is a stack in groupoids on $\mathcal{S}$, equipped with a sum functor

$$+ : \mathcal{P} \times_{\mathcal{S}} \mathcal{P} \to \mathcal{P}$$

and with associativity and commutativity isomorphisms which, over every object U of the site, make $\mathcal{P}(U)$ a strictly commutative Picard category.

The adjective will usually be omitted below: "Picard stack" always means strictly commutative Picard stack. A Picard stack has a global neutral object e, and $\mathcal{A}ut(e)$ is an abelian sheaf.

An additive functor $F : \mathcal{P}_1 \to \mathcal{P}_2$ between Picard stacks is a morphism of stacks together with a functorial isomorphism

$$F(x+y) \xrightarrow{\sim} F(x) + F(y)$$

compatible with associativity and commutativity. A morphism of additive functors is a morphism of the underlying functors which is compatible with these structural isomorphisms. Such a morphism is automatically an isomorphism.

Given two Picard stacks $\mathcal{P}_1$ and $\mathcal{P}_2$, one obtains another Picard stack

$$\mathcal{HOM}(\mathcal{P}_1, \mathcal{P}_2)$$

whose objects over U are additive functors $\mathcal{P}_1|U \to \mathcal{P}_2|U$, and whose morphisms are the morphisms of additive functors. The sum of two functors is defined objectwise.

Similarly, a biadditive functor $\mathcal{P}_1 \times \mathcal{P}_2 \to \mathcal{P}$ is a functor additive in each variable, with the two additivity structures compatible in the evident square. The biadditive functors form a Picard stack

$$\mathcal{HOM}(\mathcal{P}_1, \mathcal{P}_2; \mathcal{P}).$$

The evaluation map

$$\mathcal{HOM}(\mathcal{P}_1, \mathcal{P}_2) \times \mathcal{P}_1 \to \mathcal{P}_2, \qquad (F, x) \longmapsto F(x),$$

is biadditive. There is also a two-universal biadditive functor

$$\mathcal{P}_1 \times \mathcal{P}_2 \to \mathcal{P}_1 \otimes \mathcal{P}_2,$$

meaning that, for every Picard stack $\mathcal{P}$, composition induces an equivalence

$$\mathcal{HOM}(\mathcal{P}_1 \otimes \mathcal{P}_2, \mathcal{P}) \xrightarrow{\sim} \mathcal{HOM}(\mathcal{P}_1, \mathcal{P}_2; \mathcal{P}). \tag{1.4.8.1}$$

If $u : \mathcal{S}_1 \to \mathcal{S}_2$ is a morphism of sites and $\mathcal{P}$ is a Picard stack on $\mathcal{S}_1$, the direct image $u_*\mathcal{P}$ is the Picard stack on $\mathcal{S}_2$ given by

$$(u_*\mathcal{P})(V) = \mathcal{P}(u^*V).$$

This construction is functorial in the two-categorical sense.

One also needs the stackification process. A Picard prestack on $\mathcal{S}$ is a prestack in groupoids carrying the same Picard operations fibrewise. If $j : \mathcal{P} \to a\mathcal{P}$ is the stack associated with such a prestack, there is a unique Picard-stack structure on $a\mathcal{P}$, up to essentially unique equivalence, for which j is additive. Moreover, for every Picard stack $\mathcal{Q}$, pullback by j gives an equivalence

$$\mathcal{HOM}(a\mathcal{P}, \mathcal{Q}) \xrightarrow{\sim} \mathcal{HOM}(\mathcal{P}, \mathcal{Q}). \tag{1.4.10.1}$$

The basic algebraic model for Picard stacks is provided by two-term complexes. Let $C^{[-1,0]}(\mathcal{S})$ be the category of complexes of abelian sheaves

$$K: \quad K^{-1} \xrightarrow{d} K^0$$

concentrated in degrees -1 and 0. To such a complex one associates a Picard prestack $\mathrm{pch}(K)$ as follows. Over U, the objects are sections of $K^0(U)$. A morphism from x to y is a section $f \in K^{-1}(U)$ such that $df = y - x$. Composition and addition are the additions in K^{-1} and K^0. Let

$$\mathrm{ch}(K)$$

be the Picard stack associated to this prestack. Then

$$\mathcal{H}^0(K)$$

is the sheaf associated to the presheaf of isomorphism classes of objects of $\mathrm{ch}(K)$, and

$$\mathcal{H}^{-1}(K) \simeq \mathcal{A}ut(e),$$

where e is the neutral object.

A morphism of complexes $f : K \to L$ induces an additive functor

$$\mathrm{ch}(f) : \mathrm{ch}(K) \to \mathrm{ch}(L).$$

The functor $\mathrm{ch}(f)$ is an equivalence if and only if f is a quasi-isomorphism. If $f, g : K \to L$ are two morphisms, then a morphism of additive functors $\mathrm{ch}(f) \to \mathrm{ch}(g)$ is the same thing as a homotopy $H : K \to L[-1]$ satisfying

$$g - f = dH + Hd. \tag{1.4.12.4}$$

Lemma 1.4.13. Every Picard stack is equivalent to $\mathrm{ch}(K)$ for some complex $K \in C^{[-1,0]}(\mathcal{S})$. Moreover, every additive functor

$$F : \mathrm{ch}(K) \to \mathrm{ch}(L)$$

may, after replacing K by a quasi-isomorphic complex K', be represented by a morphism of complexes $K' \to L$.

The proof is explicit. Choose local objects $k_i \in \mathcal{P}(U_i)$ which locally generate the stack. Put $K^0 = \bigoplus_i \mathbb{Z}_{U_i}$. The universal formal sums of the k_i define an additive functor $\mathrm{ch}(0 \to K^0) \to \mathcal{P}$. Define K^{-1} to be the sheaf of pairs consisting of a local section x of K^0 and an isomorphism from the image of 0 to the image of x. The differential records x. This produces a complex K and an equivalence $\mathrm{ch}(K) \simeq \mathcal{P}$. The assertion about additive functors is obtained by choosing local representatives for the image of generators and then pulling back K^{-1} along a local epimorphism onto K^0.

Proposition 1.4.15. The construction ch induces an equivalence of categories

$$D^{[-1,0]}(\mathcal{S}) \xrightarrow{\sim} \mathrm{Ch}^\flat(\mathcal{S}),$$

where $D^{[-1,0]}(\mathcal{S})$ is the full subcategory of the derived category consisting of complexes with cohomology only in degrees -1 and 0, and $\mathrm{Ch}^\flat(\mathcal{S})$ is the category of Picard stacks with morphisms given by isomorphism classes of additive functors.

The inverse equivalence is denoted by $\mathcal{P} \mapsto \mathcal{P}^\flat$. Thus a Picard stack is represented, up to the appropriate equivalence, by its associated two-term derived object.

If $L \in C^{[-1,0]}(\mathcal{S})$ has L^{-1} injective, then $\mathrm{pch}(L)$ is already a stack, and every additive functor $\mathrm{ch}(K) \to \mathrm{ch}(L)$ is isomorphic to $\mathrm{ch}(f)$ for an actual morphism of complexes $f : K \to L$. Consequently the construction ch may be regarded as a two-equivalence between Picard stacks and two-term complexes with injective term in degree -1, with two-morphisms given by homotopies.

Under this dictionary one obtains the following formulas:

$$\mathcal{HOM}(\mathcal{P}_1, \mathcal{P}_2)^\flat \simeq \tau_{\leq 0} R\mathcal{H}om(\mathcal{P}_1^\flat, \mathcal{P}_2^\flat), \tag{1.4.18}$$

$$(f_*\mathcal{P})^\flat \simeq \tau_{\leq 0} Rf_*(\mathcal{P}^\flat), \tag{1.4.19}$$

and

$$(\mathcal{P}_1 \otimes \mathcal{P}_2)^\flat \simeq \tau_{\geq -1}(\mathcal{P}_1^\flat \otimes^{\mathbf{L}} \mathcal{P}_2^\flat). \tag{1.4.20}$$

If G is an abelian sheaf, then $G[1]$, with G placed in degree -1, corresponds to the Picard stack of G-torsors. If $K \in C^{[-1,0]}(\mathcal{S})$, extensions

$$0 \to G[0] \to E \to K \to 0$$

form a Picard stack $\mathcal{EXT}(K, G)$, with addition given by the Baer sum. Sending an extension to the torsor over $x \in K^0$ consisting of lifts of x gives an equivalence

$$\mathcal{EXT}(K, G) \xrightarrow{\sim} \mathcal{HOM}(\mathrm{ch}(K), \mathrm{ch}(G[1])). \tag{1.4.23}$$

Finally, if F is an object of the site and $\mathbb{Z}^{(F)}$ is the abelian sheaf freely generated by the represented sheaf F, then giving an additive functor

$$\mathrm{ch}(\mathbb{Z}^{(F)}) \to C$$

is the same as giving an object of $C(F)$. If $f : \mathcal{S}/F \to \mathcal{S}$ is the canonical morphism of sites, this gives

$$\mathcal{HOM}(\mathrm{ch}(\mathbb{Z}^{(F)}), C) \cong f_* f^* C. \tag{1.4.24.1}$$

At the level of complexes the corresponding statement is

$$R\mathcal{H}om(\mathbb{Z}^{(F)}, K) \simeq Rf_* f^* K, \tag{1.4.24.2}$$

for $K \in D^+(\mathcal{S})$.

1.5. The universal-coefficients formula

Let $f : X \to S$ be a smooth projective curve. Denote by

$$\mathrm{PIC}(X/S)$$

the Picard stack on the big fppf site of S obtained as the direct image of the stack of invertible sheaves on X. In the notation of the preceding section,

$$\mathrm{PIC}(X/S) = f_* \operatorname{ch}(\mathbb{G}_m[1]), \qquad \mathrm{PIC}(X/S)^\flat = \tau_{\leq 0} Rf_*(\mathbb{G}_m[1]). \tag{1.5.1}$$

Every section t of X/S defines the invertible sheaf $\mathcal{O}(t)$. Thus one obtains a morphism of Picard stacks

$$\operatorname{ch}(\mathbb{Z}^{(X)}) \longrightarrow \mathrm{PIC}(X/S). \tag{1.5.1.3}$$

For every Picard stack C on S, this gives a functor

$$\mathcal{HOM}(\mathrm{PIC}(X/S), C) \longrightarrow \mathcal{HOM}(\operatorname{ch}(\mathbb{Z}^{(X)}), C) \simeq f_* f^* C. \tag{1.5.1.4}$$

In derived terms, if $K = C^\flat$, the same map is represented by

$$R\mathcal{H}om(\tau_{\leq 0} Rf_*(\mathbb{G}_m[1]), K) \longrightarrow Rf_* f^* K. \tag{1.5.1.6}$$

For $C = \operatorname{ch}(G[1])$, the theorem of Section 1.3 says exactly that this functor is an equivalence.

Theorem 1.5.2 (universal coefficients)**.** Let $f : X \to S$ be a smooth projective curve, let C be a Picard stack on S_{fppf}, and put $K = C^\flat$. Assume locally on S that K is represented in $D^+(S_{\mathrm{fppf}})$ by a two-term complex

$$G_{-1} \to G_0$$

whose terms satisfy the hypotheses imposed in Section 1.3 on coefficient groups. Then:

(I) The functor

$$\mathcal{HOM}(\mathrm{PIC}(X/S), C) \longrightarrow f_* f^* C \tag{1.5.2.1}$$

is an equivalence of Picard stacks.

(II) The morphism

$$\tau_{\leq 0} R\mathcal{H}om(\tau_{\leq 0} Rf_*(\mathbb{G}_m[1]), K) \longrightarrow \tau_{\leq 0} Rf_* f^* K \tag{1.5.2.2}$$

is an isomorphism.

The complex $\tau_{\leq 0} Rf_*(\mathbb{G}_m[1])$ therefore plays, for the curve X/S, the role played by homology in the classical universal-coefficient theorem. For a fixed C, the two assertions are equivalent under the correspondence between Picard stacks and two-term complexes. The assertion is local on S. It has already been proved when $C = \mathrm{ch}(G[1])$, by Theorem 1.3.10. For a two-term complex $G_{-1} \to G_0$, one compares the long exact cohomology sequence for Rf_*K with the corresponding sequence for the left hand side of (1.5.2.2), and applies the five lemma.

The formula is functorial for finite flat morphisms of curves. If

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f_1"'] & & Y \arrow[dl, "f_2"] \\ & S & \end{tikzcd}$$

is flat between smooth projective curves over S, then the norm map on Picard stacks and pullback of sections give an essentially commutative square

$$\begin{tikzcd} \mathbb{Z}^{(X)} \arrow[r] \arrow[d, "u"'] & \mathrm{PIC}(X/S) \arrow[d, "N_{X/Y}"] \\ \mathbb{Z}^{(Y)} \arrow[r] & \mathrm{PIC}(Y/S). \end{tikzcd}$$

For every Picard stack C this yields the expected compatibility of universal coefficients with pullback along u. At the derived level this is the commutativity of the square

$$\begin{tikzcd} R\mathcal{H}om(\tau_{\leq 0}Rf_{2*}\mathbb{G}_m[1], K) \arrow[r] \arrow[d] & Rf_{2*}f_2^*K \arrow[d, "u^*"] \\ R\mathcal{H}om(\tau_{\leq 0}Rf_{1*}\mathbb{G}_m[1], K) \arrow[r] & Rf_{1*}f_1^*K. \end{tikzcd} \tag{1.5.3.3}$$

For torsor coefficients G, the trace for u gives the transposed compatibility: the trace on cohomology corresponds to pullback/norm on the Picard object.

Locally for the etale topology on S, one may split non-canonically

$$\tau_{\leq 0}Rf_*\mathbb{G}_m[1] \simeq f_*\mathbb{G}_m[1] \oplus R^1f_*\mathbb{G}_m.$$

This gives, for an abelian sheaf G satisfying the standing hypotheses, an isomorphism

$$\mathcal{H}om(R^1f_*\mathbb{G}_m, G) \xrightarrow{\sim} \mathcal{H}om_S(X, G) \tag{1.5.4.1}$$

and an exact sequence

$$0 \to \mathcal{E}xt^1(R^1f_*\mathbb{G}_m, G) \to R^1f_*G \to \mathcal{H}om(f_*\mathbb{G}_m, G) \to 0. \tag{1.5.4.2}$$

The first map and the first term of the exact sequence are induced by the canonical map from X to its relative Picard scheme; the last map is the degree map. After the usual local decomposition $\mathrm{Pic}_{X/S} \simeq \mathrm{Pic}^0_{X/S} \times \mathbb{Z}^r$, the exact sequence may be written

$$0 \to \mathcal{E}xt^1(\mathrm{Pic}^0_{X/S}, G) \to R^1 f_* G \to \mathcal{H}om(f_*\mathbb{G}_m, G) \to 0. \tag{1.5.4.3}$$

Putting $K = \mathbb{G}_m[1]$ in Theorem 1.5.2 gives an autoduality theorem for $\tau_{\le 0} Rf_*\mathbb{G}_m[1]$. A representative complex has a canonical three-step filtration with successive quotients, up to unique quasi-isomorphism,

$$\mathrm{Pic}_{X/S} / \mathrm{Pic}^0_{X/S}, \qquad \mathrm{Pic}^0_{X/S}, \qquad (f_*\mathbb{G}_m)[1].$$

The corresponding dualities are the natural duality between the discrete component group and $f_*\mathbb{G}_m$, and the autoduality of the Jacobian $\mathrm{Pic}^0_{X/S}$.

The non-proper case is more delicate. Let

$$\bar{f} : \bar{X} \to S$$

be proper, of finite presentation, with fibres purely of dimension one; let $Y \subset \bar{X}$ be finite over S, defined by an ideal $\mathfrak{m}$, and let $X = \bar{X} - Y$ be smooth over S. For a torsion etale sheaf G, prime to the residual characteristics, one introduces $\mathbb{G}_m(\mathfrak{m})$, the sheaf of invertible functions on $\bar{X}$ equal to 1 along Y. A $\mathbb{G}_m(\mathfrak{m})$-torsor is the same as an invertible sheaf on $\bar{X}$ rigidified along Y. Relative divisors on X, finite over S, give such objects.

The technical input is that sufficiently positive rigidified line bundles have enough sections. If S is local with closed point s, $\mathcal{O}_{\bar{X}}(1)$ is relatively ample and rigidified along Y, and $\mathcal{M}$ is a rigidified invertible sheaf, then for $n \gg 0$ every section of $\mathcal{M}(n)$ over the fibre $\bar{X}_s$, equal to 1 on Y_s, lifts locally on S to a section equal to 1 on Y. The obstruction lies in $H^1(\bar{X}, \mathcal{N}(n))$ for a coherent sheaf $\mathcal{N}$, and vanishes for n large.

One then imposes the following general-position conditions on sections of a rigidified invertible sheaf $\mathcal{L}$. A section m should be equal to 1 on Y, and its zero scheme $Z(m) \subset X$ should be a relative divisor, hence finite over S. For two or three sections one requires the same condition for the affine pencils

$$\lambda m_1 + (1-\lambda) m_2, \qquad \lambda m_1 + \mu m_2 + (1-\lambda-\mu) m_3.$$

By a general-position argument, after passing to an etale neighbourhood of a geometric point of S, finitely many sections may be joined by auxiliary sections satisfying these conditions.

Let now K be a G-torsor on X. If (m_1, m_2) is such a good pair of sections of $\mathcal{L}$, trace along the divisor

$$\lambda m_1 + (1-\lambda) m_2 = 0$$

gives a G-torsor on the affine line over S. By the acyclicity lemma for affine space, this torsor is pulled back from S, and the specializations at 0 and 1 yield a canonical isomorphism

$$\mathrm{Tr}_{Z(m_1)/S}(K) \xrightarrow{\sim} \mathrm{Tr}_{Z(m_2)/S}(K). \tag{1.5.12.1}$$

This isomorphism is independent of the auxiliary choices. If E is a relative divisor on X, then replacing $\mathcal{L}$ by $\mathcal{L} \otimes \mathcal{O}(E)$ changes the trace torsor by adding $\mathrm{Tr}_{E/S}(K)$. Hence one can define, locally and then by descent, a torsor

$$\langle \mathcal{L}, K]$$

which is additive in $\mathcal{L}$.

Automorphisms of $\mathcal{L}$ act trivially on $\langle \mathcal{L}, K]$. Indeed, the sheaf $\bar{f}_* \mathbb{G}_m(\mathfrak{m})$ is connected by affine lines: any two sections are joined, over an open of the affine line with non-empty fibres, by $\lambda x + (1-\lambda) y$. Since every morphism from such an open to the torsion group G is constant, the automorphism action is trivial. Thus $\langle \mathcal{L}, K]$ depends only on the class of $\mathcal{L}$ in $R^1 \bar{f}_* \mathbb{G}_m(\mathfrak{m})$.

Consequently, for every section λ of $R^1 \bar{f}_* \mathbb{G}_m(\mathfrak{m})$ one obtains a torsor $\langle \lambda, K]$. This construction is additive in λ, hence gives an extension $e(K)$ of $R^1 \bar{f}_* \mathbb{G}_m(\mathfrak{m})$ by G. It is compatible with base change and with addition of torsors, so there is an additive functor

$$e : \bar{f}_* \operatorname{ch}(G[1]) \longrightarrow \mathcal{E}XT(R^1 \bar{f}_* \mathbb{G}_m(\mathfrak{m}), G).$$

Let $j : X \to R^1 \bar{f}_* \mathbb{G}_m(\mathfrak{m})$ be the map sending a point t to the class of $\mathcal{O}(t)$. Pullback by j gives the inverse functor.

Proposition 1.5.14. Under the hypotheses above, the functors e and j^* are inverse equivalences of Picard stacks. In particular there is an isomorphism

$$\mathrm{Ext}^1(R^1 \bar{f}_* \mathbb{G}_m(\mathfrak{m}), G) \xrightarrow{\sim} H^1(X, G).$$

This identification is compatible with finite flat morphisms of compactified curves. If $\bar{u} : \bar{X}_1 \to \bar{X}_2$ carries Y_1 into the inverse image of Y_2 and induces a finite flat morphism $X_1 \to X_2$, then trace on torsors corresponds to pullback of rigidified line bundles. If $\bar{u}$ itself is finite flat and $Y_1 = \bar{u}^{-1} Y_2$, then pullback of torsors corresponds to the norm of rigidified line bundles.

1.6. An effacement theorem

The present subsection, from Lemma 1.6.6 onward, may be read independently of Sections 1.2 through 1.5.

Lemma 1.6.1. Let X be a scheme, let $Y \subset X$ be a closed subscheme defined by an ideal $\mathfrak{m}$, and put $U = X - Y$, with immersions

$$U \xrightarrow{j} X \xleftarrow{i} Y.$$

Let $\mathbb{G}_m(\mathfrak{m})$ be the fpqc sheaf whose sections over an X-scheme are the invertible functions equal to 1 on the inverse image of Y. For n invertible on X, the sequence

$$0 \to j_! \mu_n \to \mathbb{G}_m(\mathfrak{m}) \xrightarrow{n} \mathbb{G}_m(\mathfrak{m}) \to 0$$

is exact on the big etale site of X.

The proof is the Kummer sequence for $\mathbb{G}_m$ together with the exact sequence relating functions on X, functions on Y, and functions congruent to 1 modulo $\mathfrak{m}$. Since the direct image from the closed subscheme is exact, a diagram with exact columns and the usual Kummer rows gives the assertion.

Let k be a field. The notes recall that, under standard representability hypotheses, a sheaf which occurs as the middle term of an exact sequence of fppf sheaves with representable neighbouring terms is representable and locally of finite type. If P is a connected commutative algebraic group of finite type over k, and n is invertible in k, then

$$0 \to {}_nP \to P \xrightarrow{n} P \to 0 \tag{1.6.2.1}$$

is exact. If G is a finite etale commutative group killed by n, then

$$\mathrm{Hom}({}_nP, G) \xrightarrow{\sim} \mathrm{Ext}^1(P, G). \tag{1.6.2.2}$$

Assume now that X is proper over $\operatorname{Spec} k$. The exact sequence

$$0 \to \mathbb{G}_m(\mathfrak{m}) \to \mathbb{G}_m \to i_*\mathbb{G}_m \to 0$$

gives

$$0 \to p_*\mathbb{G}_m(\mathfrak{m}) \to p_*\mathbb{G}_{m,X} \to (pi)_*\mathbb{G}_{m,Y} \to R^1p_*\mathbb{G}_m(\mathfrak{m}) \to \mathrm{Pic}_{X/k} \to \mathrm{Pic}_{Y/k}\,. \tag{1.6.3.1}$$

The terms are represented by group schemes of the expected finiteness type. Therefore $p_*\mathbb{G}_m(\mathfrak{m})$ and $R^1p_*\mathbb{G}_m(\mathfrak{m})$ are representable.

Definition 1.6.4. The group scheme representing $R^1p_*\mathbb{G}_m(\mathfrak{m})$ is denoted

$$\mathrm{Pic}_{\mathfrak{m},X/k}\,.$$

Its neutral component is $\mathrm{Pic}^0_{\mathfrak{m},X/k}$, and $\mathrm{Pic}^\tau_{\mathfrak{m},X/k}$ is the inverse image in $\mathrm{Pic}_{\mathfrak{m},X/k}$ of the torsion subgroup of the component group.

For algebraically closed k and n invertible in k, the Kummer sequence of Lemma 1.6.1 gives

$$H^1_c(U, \mu_n) \xrightarrow{\sim} {}_n\,\mathrm{Pic}^\tau_{\mathfrak{m},X/k}(k). \tag{1.6.4.1}$$

Let X now be a smooth curve over an algebraically closed field k, let $\bar{X}$ be its smooth projective compactification, and let $Y = \bar{X} - X$. For a finite abelian group G killed by an integer n prime to $\operatorname{char} k$, the preceding identifications, together with Proposition 1.5.14 and

(1.6.2.2), give a canonical isomorphism

$$\mathrm{Hom}(H^1_c(X,\mu_n),G) \xrightarrow{\sim} H^1(X,G). \tag{1.6.5}$$

If $u : X \to Y$ is finite between smooth curves, this isomorphism is compatible with trace on compactly supported cohomology and pullback on ordinary cohomology.

Lemma 1.6.6. Let X be a smooth curve over an algebraically closed field k, let n be invertible in k, and let $u : X' \to X$ be a tame etale covering. Then

$$\mathrm{Tr}_u : H^1_c(X',\mathbb{Z}/n) \to H^1_c(X,\mathbb{Z}/n)$$

is, non-canonically unless an isomorphism $\mu_n \simeq \mathbb{Z}/n$ has been chosen, the transpose of

$$u^* : H^1(X,\mathbb{Z}/n) \to H^1(X',\mathbb{Z}/n).$$

One proof is immediate from (1.6.5) after identifying μ_n with $\mathbb{Z}/n$. A second proof uses comparison with classical topology over $\mathbb{C}$, where the result is the usual transposition under Poincare duality; the characteristic-zero case follows by Lefschetz principle, and the positive-characteristic case is obtained by lifting the curve and the tame covering to the Witt vectors and applying base change to the locally constant sheaves $R^1 f_*\mathbb{Z}/n$ and $R^1 f_!\mathbb{Z}/n$.

Lemma 1.6.7. Let X be a smooth connected curve over an algebraically closed field k, let p be the characteristic exponent of k, and let n be prime to p. There exists a principal finite etale Galois cover $u : X' \to X$, of degree dividing a power of n, such that

$$\mathrm{Tr}_u : H^1_c(X',\mathbb{Z}/n) \to H^1_c(X,\mathbb{Z}/n)$$

is zero.

Indeed, take the largest connected abelian principal cover of X with Galois group killed by n. By construction, pullback kills $H^1(X,\mathbb{Z}/n)$. Lemma 1.6.6 then says that the trace on compactly supported cohomology is zero.

Lemma 1.6.8. Let $f : X \to S$ be smooth and let $x : S \to X$ be a section. There exists a Zariski open neighbourhood U of $x(S)$ in X such that the geometric fibres of $U \to S$ are connected.

Fundamental Lemma 1.6.9. Let $f : X \to S$ be a compactifiable smooth curve, let x be a geometric point of X, let s be its image in S, and let n be invertible on S. Then there exist an etale neighbourhood V of s in S and an etale neighbourhood U of x in $X_V = X \times_S V$, fitting into

$$\begin{tikzcd}
x \arrow[r] \arrow[d] & U \arrow[r, "u"] \arrow[dr, "f'"'] & X_V \arrow[r] \arrow[d, "f_V"] & X \arrow[d, "f"] \\
s \arrow[rr] & & V \arrow[r] & S,
\end{tikzcd} \tag{1.6.9.1}$$

such that

$$R^0 f'_!(\mathbb{Z}/n) = 0,$$

the trace map

$$\mathrm{Tr}_u : R^1 f'_!(\mathbb{Z}/n) \to R^1 f_{V!}(\mathbb{Z}/n)$$

is zero, and the curve trace

$$\mathrm{Tr}_{f'} : R^2 f'_!(\mathbb{Z}/n(1)) \to \mathbb{Z}/n$$

is an isomorphism.

For the first condition it is enough to arrange that f' be quasi-affine. For the last it is enough that the geometric fibres of f' be connected; Lemma 1.6.8 supplies this after an etale localization and shrinking. It remains to kill the trace in degree one. The problem reduces to the case of geometrically connected fibres. For a given geometric point t of S, Lemma 1.6.7 supplies, on the fibre X_t, an etale principal cover whose trace on $H^1_c(-, \mathbb{Z}/n)$ is zero. By the acyclicity theorem used in smooth base change, there is an etale neighbourhood $U \to X$ of x such that, over the fibre at t, the map $U_t \to X_t$ factors through that cover. The fibre at t of

$$\mathrm{Tr}_u : R^1 f'_!(\mathbb{Z}/n) \to R^1 f_!(\mathbb{Z}/n) \tag{1.6.9.2}$$

is then zero. As U varies, the images of these maps form a filtered decreasing system of subobjects of the constructible sheaf $R^1 f_!(\mathbb{Z}/n)$, and the stalk is eventually zero at every geometric point. By noetherian induction the system is eventually zero globally. This completes the proof of the fundamental lemma.

2. The trace morphism

This section proves the general trace morphism and the local effacement theorem which will be used in the global duality theorem. The proof of the principal trace theorem uses only the results of Section 1.

Lemma 2.1. Let $f : X \to S$ be compactifiable of relative dimension $\leq d$, and let $j : U \hookrightarrow X$ be an open subscheme of finite type whose complement has fibre dimension $< d$. For every torsion sheaf F on X, the canonical map

$$R^{2d}(fj)_! j^* F \xrightarrow{\sim} R^{2d} f_! F$$

is an isomorphism.

Let $i : Y = X - U \hookrightarrow X$. The localization exact sequence for $f_!$ contains

$$R^{2d-1}(fi)_! i^* F \to R^{2d}(fj)_! j^* F \to R^{2d} f_! F \to R^{2d}(fi)_! i^* F.$$

The two outside terms vanish by the cohomological-dimension estimate, since the fibres of Y have dimension $< d$.

Lemma 2.2. Let $f : X \to S$ be compactifiable of relative dimension $\leq d$. Let $u_i : X_i \to X$ be a family of etale, separated, finite-type morphisms, and write $X_{ij} = X_i \times_X X_j$. Then for every torsion sheaf F on X, the sequence

$$\bigoplus_{i,j} R^{2d}(fu_{ij})_! u_{ij}^* F \rightrightarrows \bigoplus_i R^{2d}(fu_i)_! u_i^* F \to R^{2d} f_! F \to 0$$

is exact.

This is the right exactness of $R^{2d} f_!$, together with the etale descent exact sequence for sheaves established in Expose XVII.

Lemma 2.3. Let $f : X \to S$ be flat and of finite presentation. The locus of X where f is Cohen-Macaulay is relatively of finite presentation and dense in every fibre.

The assertion is local on S, and one reduces to the noetherian case. Fibrewise, over a field, the local rings at maximal points are artinian, hence Cohen-Macaulay. Since the Cohen-Macaulay locus is open, the result follows.

Lemma 2.4. Let $f : X \to S$ be flat, of finite presentation, and Cohen-Macaulay, with S quasi-separated. Every point of X has a Zariski neighbourhood U admitting a quasi-finite flat morphism of finite presentation

$$U \to \mathbb{A}^d_S.$$

This is the local projection result of EGA IV 15.4.3.

Lemma 2.5. Let A be a Cohen-Macaulay local ring of dimension d, with maximal ideal $\mathfrak{m}$. If

$$\underline{u} = (u_1, \ldots, u_d), \qquad \underline{v} = (v_1, \ldots, v_d)$$

are two systems of parameters, then they may be joined by a finite sequence of systems of parameters in which each consecutive pair differs in only one parameter.

The proof is by induction on d. For $d < 2$ there is nothing to prove. For $d \geq 2$, choose $x \in \mathfrak{m}$ avoiding the finite union of associated primes of A/u_1 and A/v_1. Then x is regular modulo both u_1 and v_1, and one applies the induction hypothesis successively in A/u_1, A/x, and A/v_1.

Lemma 2.6. Let X be a Cohen-Macaulay scheme of finite type over an algebraically closed field k, and let

$$u, v : X \to \mathbb{A}^d_k$$

be two quasi-finite flat morphisms. Every closed point $x \in X$ has a Zariski neighbourhood U on which $u|U$ and $v|U$ can be joined by a finite sequence of quasi-finite flat morphisms to $\mathbb{A}^d_k$, each step changing only one coordinate.

After translating the targets, reduce to the case where both morphisms send x to the origin. The germs of quasi-finite flat maps to $\mathbb{A}^d_k$ sending x to 0 are the same as systems of parameters of $\mathcal{O}_{X,x}$. Lemma 2.5 applies. The general case is obtained by first joining u to $u - u(x)$ and v to $v - v(x)$ one coordinate at a time.

If

$$X \xrightarrow{g} Y \xrightarrow{f} S$$

are composable compactifiable morphisms, with relative dimensions $\leq e$ and $\leq d$, respectively, the composition spectral sequence gives by the maximum-cycle argument a canonical isomorphism

$$R^{2d} f_! R^{2e} g_! F \xrightarrow{\sim} R^{2(d+e)}(fg)_! F. \tag{2.7.2}$$

Let $a^d : \mathbb{A}^d_S \to S$ be affine d-space over a coherent base. If F is torsion and prime to the residual characteristics, define recursively an isomorphism

$$\mathrm{Tr}_{a^d} : R^{2d} a^d_! F(d) \xrightarrow{\sim} F. \tag{2.8.1}$$

For $d = 0$ it is the identity; for $d = 1$ it is the trace for curves from Section 1.1; for $d + 1$ it is obtained by writing $\mathbb{A}^{d+1}_S = \mathbb{A}^1_{\mathbb{A}^d_S}$ and composing the two traces. This trace is invariant under permutation of the coordinates, because the symmetric group is contained in the connected affine group acting trivially on both source and target after base change.

Theorem 2.9. Let (f, d, F) consist of a compactifiable morphism $f : X \to Y$, an integer d, and a torsion sheaf F on Y prime to the residual characteristics. Assume that f satisfies condition $(*)_d$: there is an open $U \subset X$ on which f is flat of finite presentation with fibres of dimension $\leq d$, and such that $X - U$ has fibres of dimension $< d$. Then there is a unique morphism

$$\mathrm{Tr}_f : R^{2d} f_! f^* F(d) \to F$$

with the following properties.

It is functorial in F; it is compatible with every base change $Y' \to Y$ with Y' quasi-compact and quasi-separated; it is compatible with composition via the isomorphism (2.7.2); and it is normalized as follows: for finite locally free f of rank n and $d = 0$, the composite $F \to f_* f^* F \to F$ is multiplication by n; for $f : \mathbb{A}^1_Y \to Y$ and $d = 1$, the trace is the isomorphism (2.8.1).

For $d = 0$ this is the trace of Expose XVII for quasi-finite flat morphisms. For affine space it is necessarily the trace (2.8.1). Suppose first that f admits a quasi-finite flat map $u : X \to \mathbb{A}^d_Y$. Define

$$t(f, u) = \mathrm{Tr}_{a^d} \circ R^{2d} a^d_! (\mathrm{Tr}_u).$$

This construction is functorial and compatible with base change. To see that it is independent of u, reduce by base change to an algebraically closed field and by Lemma 2.2 to a Zariski-local assertion on X. Lemma 2.6 then reduces to the case where two maps differ in only one coordinate. In that case one factors through $\mathbb{A}^{d-1}$ and uses the invariance of (2.8.1), together with the one-dimensional trace of Section 1.1.

For f flat, Cohen-Macaulay, of finite presentation and pure relative dimension d, choose a quasi-compact open cover on which such quasi-finite flat maps to $\mathbb{A}^d_Y$ exist. Lemma 2.2 glues the local traces to a unique trace Tr_f. For a general f satisfying $(*)_d$, shrink the good open U so that $f|U$ is Cohen-Macaulay and pure of dimension d, and use Lemma 2.1 to identify the top direct image for f with that for $f|U$. This gives existence and uniqueness, together with functoriality, base change, and the normalizations.

It remains only to verify full compatibility with composition. A formal associativity lemma says that, for three composable morphisms $X \to Y \to Z \to T$, if the composition law holds for the first two adjacent pairs, then the two possible triple compositions are equivalent. Thus one reduces to diagrams where both morphisms are described by quasi-finite flat maps into affine spaces. The needed commutativity then follows from the cartesian square

$$\begin{tikzcd} \mathbb{A}^e_S \arrow[r] \arrow[d] & \mathbb{A}^e_T \arrow[d] \\ S \arrow[r] & T \end{tikzcd}$$

with $S \to T$ quasi-finite flat, since after reducing to an algebraically closed field and then to an artinian local algebra, the finite trace is multiplication by the degree while the affine-space trace is stable under base change. This completes the proof of Theorem 2.9.

Proposition 2.10. The trace morphisms of Expose XVII for finite flat morphisms, the trace for smooth curves of Section 1.1, and the affine-space trace (2.8.1) are all special cases of Theorem 2.9.

The trace in Theorem 2.9 is an isomorphism precisely when the geometric fibres of f have exactly one irreducible component of dimension d, with multiplicity prime to n. The notes defer the direct proof, since it follows immediately from the global duality theorem.

Let $f : X \to S$ and $g : Y \to S$ satisfy $(*)_d$ and $(*)_e$. The Kunneth isomorphism gives

$$R^{2d} f_!(\mathbb{Z}/n) \otimes R^{2e} g_!(\mathbb{Z}/n) \xrightarrow{\sim} R^{2(d+e)}(f \times g)_!(\mathbb{Z}/n). \tag{2.11.1}$$

Proposition 2.12. After the appropriate Tate twists, the square

$$\begin{tikzcd} R^{2d} f_!(\mathbb{Z}/n)(d) \otimes R^{2e} g_!(\mathbb{Z}/n)(e) \arrow[r, "\sim"] \arrow[d, "\mathrm{Tr}_f \otimes \mathrm{Tr}_g"'] & R^{2(d+e)}(f \times g)_!(\mathbb{Z}/n)(d+e) \arrow[d, "\mathrm{Tr}_{f\times g}"] \\ \mathbb{Z}/n \arrow[r, "="] & \mathbb{Z}/n \end{tikzcd}$$

is commutative.

This is just the compatibility of the trace with composition, read through the asymmetric construction of the Kunneth morphism used in Expose XVII.

Let $\mathcal{A}$ be a sheaf of rings on S, and suppose $n\mathcal{A} = 0$ with n invertible on S. For $K \in D(S, \mathcal{A})$ there is the projection-formula isomorphism

$$K \otimes^{\mathbf{L}}_{\mathbb{Z}/n} Rf_!(\mathbb{Z}/n)(d) \xrightarrow{\sim} Rf_!(f^*K(d)). \tag{2.13.1}$$

The top trace may therefore be regarded as a morphism

$$Rf_!(f^*K(d)[2d]) \longrightarrow K. \tag{2.13.2}$$

Theorem 2.14. Let $f : X \to S$ be smooth, compactifiable, and pure of relative dimension d, and let n be invertible on S. For every geometric point x of X, with image s in S, there exist an etale neighbourhood V of s and an etale neighbourhood U of x in X_V,

$$\begin{tikzcd} U \arrow[r, "j"] \arrow[dr, "f'_V"'] & X_V \arrow[r] \arrow[d, "f_V"] & X \arrow[d, "f"] \\ & V \arrow[r] & S, \end{tikzcd} \tag{2.14.1}$$

such that the map

$$R^i f'_{V!}(\mathbb{Z}/n) \to R^i f_{V!}(\mathbb{Z}/n)$$

is zero for $i < 2d$, and

$$\mathrm{Tr}_{f'_V} : R^{2d} f'_{V!}(\mathbb{Z}/n)(d) \to \mathbb{Z}/n$$

is an isomorphism.

The proof uses a homological lemma. If $K_0 \to K_1 \to \cdots \to K_{2k}$ is a chain in a bounded derived category, each K_i has cohomology only in degrees $[0, k]$, and each transition induces zero on cohomology in degrees $< k$, then the total composite $K_0 \to K_{2k}$ factors through the top cohomology object $H^k(K_0)[-k]$. Iterating Theorem 2.14 in relative dimension d therefore gives a stronger factorization: after a suitable etale localization, the canonical map

$$Rf'_!(\mathbb{Z}/n)(d) \to Rf_!(\mathbb{Z}/n)(d)$$

factors through $\mathbb{Z}/n[-2d]$ via the trace.

The theorem is proved by induction on d. For $d = 0$ it is tautological, and for $d = 1$ it is exactly the fundamental lemma 1.6.9. For $d \geq 2$, localize so that f factors as

$$X \xrightarrow{h} Y \xrightarrow{g} S$$

with g and h smooth compactifiable of strictly smaller positive relative dimensions d' and d''.

Applying the inductive factorization to g and to the base changes of h, one obtains a diagram

$$
\begin{tikzcd}
U \arrow[r, "u"] \arrow[dr] & f^{-1}W \arrow[r] \arrow[d] & X \arrow[d, "h"] \\
V \arrow[r, "v"] \arrow[dr] & g^{-1}W \arrow[r] \arrow[d] & Y \arrow[d, "g"] \\
& W \arrow[r] & S
\end{tikzcd}
$$

for which the lower-dimensional trace maps are isomorphisms and the corresponding canonical maps factor through the trace objects. The composition diagram for $Rf_!$, together with the trace compatibility of Theorem 2.9, shows that the composite for f factors through $\mathbb{Z}/n[-2d]$. Hence the maps in degrees $< 2d$ are zero and the top trace is an isomorphism. This completes the proof.

Continuation marker. The next source anchor is Exposé XVIII, Section 3, “The global duality theorem,” source line 4691 of `18_fr.tex`.